

# TFPT — Changelog

All changes and new results, by date

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This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (vN), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the `introduction` document. Module numbers vN refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

## 2026-09-09 (CLXII): Source fields, full lattice dynamics and Clock/E8 bridge consolidated

The post-Round 27 research archive, including Rounds 28–50 and subsequent named branches, is integrated without rewriting source-pinned historical records. The contracts paper gives a consolidated derivation of local charged CAR fields from the original filled QWZ sea: both adjoints, integer charge, finite-energy generator convergence and the top-edge charge minimum  $q^2/2 - q/4$ . Neutral four-point collision control and the improved auxiliary-window estimate are included with their exact scope. Neither result supplies the microscopic half-charge  $E_8$  spin field.

For the separate, declared compact- $U(1)$  rotor/CAR parent, the new text records all-electric resummation, full all-time quasi-local automorphisms, the failure of point-norm continuity on the full bounded algebra, and neutral physical ground-state representations. Stronger coercivity and Wilson-loop spectral moments yield nonzero positive-energy weight, not a mass or a many-body gap theorem. The actual Clock phase classification, interaction provenance, eight-bit lattice cocycle refinement and scalar-symbol Weyl obstruction identify the still missing common-parent bridge. The sharp half-twist implementability obstruction is retained; renormalized inter-sector fields are the next construction target.

The related double-cover, inter-sheet and prime-event investigations are archived with their negative controls and separate arithmetic scope. The current charged-field regression chain contains 131 tests per mode (ordinary and optimized Python); these are execution controls, not independent mathematical review or physical completion. Frontier, introduction via this changelog, research overview, open-problem map, website metadata, PDF mirrors and reproducibility instructions are updated together. All T1–T8, TOE, empirical and RH markers are unchanged.

## 2026-09-07 (CLXI): Non-RH research through Round 27 integrated into repository, papers and website

The previously local theory-contract archive is shipped with its proofs, source-pinned checkers and immutable per-round validation snapshots. The intervening geometry, positive-parent, constraint-domain and charge transport branches retain their distinct hypotheses and additional off-shell inputs; they are not combined into a claimed microscopic TOE. The contracts paper now includes the full-reference argument for the original finite-regulator  $g = 0$  scalar/charge model:  $e^{-48\beta NJ} Z_D \leq Z \leq Z_{\text{abs}} \leq Z_D$ , retaining every charge, the E8 cocycle and the charge-dependent scalar determinants. This removes the artificial extra charge heat-trace factor from the sign, memory and hop bounds, but not the extensive exponential sign budget.

For  $N = 27, T = 3, \beta = 1, J = 0.1$ , the certified 1% omission planner improves from  $K4096/P962$  to  $K2670/P354$ ; refining the old inequalities to the same integer search gives  $K4035/P956$ . A declared weak-hopping example at  $L = 9, T = 33, \delta = 3/4, m = a = 1, J = 1/866052$  certifies  $R = 12$ : 25 of 33 quadratic time offsets, memory error at most 0.2045%, and combined omission error below 1% with  $K12235/P5$ . The full eliminated determinant remains a full-history function. The partition itself is not evaluated, and the naive history counts remain astronomical. Round 27 has two checkers with 23/25 exact groups and 18 input pins each, 37 isolated fixtures passing normally and under optimization, and 29 prerequisite reruns. No T1–T8, physical TOE, empirical or RH status is promoted.

The existing correction to `v472_quillen_detline_moduli.py`, its ledger description and the corresponding contract is included unchanged: the finite result is an occupied Berry line and integrated Chern response, not a globally nonvanishing Dai–Freed section or a continuum Quillen connection. Its seven checks pass. Frontier prose, public research text, document metadata, PDFs and generated mirrors are updated together.

## 2026-09-06 (CLX): QWZ zero-mode portability correction and a free 3D matter contract

Linux reproduction exposed an eigensolver-dependent choice of a pure zero-mode vacuum in `v1033_charged_disorder.py`. The numerical diagnostic now uses the basis-independent fixed-particle ground-space mixture and computes variational energy bounds over every pure zero-space filling. At  $N = 64$  the smooth  $N$ -scaled mixture excess is 6.904290, with full ground-space interval  $[5.333652, 8.474929]$ ; the former 8.475 value was filling-dependent, not a unique vacuum coefficient. The charged-disorder proof is amended explicitly. The same 35/35 typed checks pass without weakening an exact identity; the integration guards now report subprocess output on failure and include a degenerate-basis regression. All 20/20 integration tests pass in isolated Linux Python 3.11 and 3.12 environments. The registered suite remains 1028 modules, with 233 Round-7 checks.

A separate, unpromoted theory contract in `experiments/theory-contracts/free-scalar-3d/` supplies a finite-range symmetric free-scalar stress on the original staggered sites, links and plaquettes without changing the energy density or current. Its complete real-space derivation and 68/68 exact checks include all three momentum Ward identities, genuine negative controls, bilinear Bloch transitions, periodic conserved charges and the quadratic Weyl operator lift. This is not yet a new registered verification module or a TFPT parent construction. Interacting matter, nonlinear gravity feedback and the missing homogeneous gravity receiver remain open. No T1–T8, TOE or RH marker moves.

## 2026-09-05 (CLIX): non-RH round 7 — free quantum curvature, charged twists and interaction boundaries

Five modules `v1031–v1035` add 233/233 typed checks (183 exact, 13 source-contract, 37 floating) to the registered suite ( $1023 \rightarrow 1028$  modules), including the round-6 quantum prerequisite. `v1031_quantum_tensor_curvature.py` (69/69) constructs the zero-mean Weyl/Fock target and its distribution limit; `v1032_covariant_curvature.py` (34/34) supplies exact identities for the positive local Lorentz-covariant free curvature field with helicities  $\pm 2$ . Neither derives a common microscopic TFPT parent. `v1033_charged_disorder.py` (35/35) constructs auxiliary  $\mathbb{Z}_4$  flux-register corners and proves the quarter-twist mismatch  $h = 1/4 \neq h_\lambda = 1$  and the integer-dressing parity obstruction. `v1034_parent_mirror.py` (36/36) proves the factorized onsite lower gap  $1 + \sqrt{3}/2$  and refutes a proposed bare-hopping vacuum-relative form bound, not every interacting gap argument. The newly chosen bond 3/5 is distinguished from the source coefficient 3/10. `v1035_matter_coupling.py` (59/59) keeps the physical Fourier factors in the prescribed-source Ward identities and exposes the dynamical-source, nonlinear scalar and zero-mode limitations. Full written proofs, five narrowly typed ledger subclaims, paper-body citations and the public status mirrors are integrated. Source-reading `v1033/v1034` require the complete repository and

are not advertised as browser-only certificates. No T1–T8 or TOE marker moves; no RH claim. Fourteen representative exact Wolfram checks are authored and included in the master runner; local execution is deferred because the installed kernel requires license activation. The last verified 116/116 count is unchanged; no successful independent-engine run is claimed.

## **2026-09-05 (CLVIII): non-RH round 4 — uniform TEL-B norm and explicit T1–T8 boundaries**

Five modules graduate as v1026–v1030; the suite grows from 1018 to 1023 modules. Their standalone checks pass 353/353. `v1026_telb_round4_closure.py` (199/199, native python-flint 0.9.0) executes CF, DG, one-sided CROSS, 512 validated Acb eigensystems, sixteen Fourier modes, all infinite aliases and 154 numerical regressions. With v1022/v1025 inputs it proves  $\|R_N\|_{\text{HS}} < 2.995906 < 3$  at fixed  $M = 1$ ,  $N_y = 8$  for all even  $N \geq 16$ . The former finite 1.7833 smooth bound is withdrawn; the valid finite bound is 1.800594. The historical 2.11 and sharp Bound-B 0.62 targets are not proved. `v1027_signed_det_car_wall.py` (47/47) constructs canonical DET excitations and a signed  $z = 1$  wall for fixed classical gauge backgrounds, changing the flat local parent to an additive one; there is no hard bare-mirror gap or chiral continuum claim. `v1028_gauss_d4_cap.py` (36/36) counts full 192-mode site Gauss characters modulo five (one site 3; cube 0 on the specified nine-irrep link truncation), constructs a symmetric internal gap-one cap, and retains the symmetry and Majorana selection-rule obstructions. `v1029_local_tensor_constraints.py` (44/44) gives a local two-mode radiative target only after explicit global zero-mode removal; the unmodified periodic target is unstable. `v1030_alg2_sector_frame.py` (27/27) proves the conditional sector-connectivity/actual-word frame bridge, with joint-adjoint and high-output tails still required. The complete proof supplement is included in the contracts paper, with intro/frontier/MMST, ledger, website and reproducibility surfaces synchronized. An independent-engine representative Wolfram mirror is supplied but its execution is unavailable locally (no kernel found); no verified Wolfram count is increased. The static discipline census (v649) now recognizes genuine renamed imports of the shared check function; seven regression tests reject comment/string-only and unrelated-import false positives. Its 18-check audit and twelve deterministic replays pass without exemptions. The 68-check v1006 battery now requires the native norm certificate together with the explicitly open FE-GEN/ALG-EXH/MMST boundaries in the memorandum; three document-guard regression tests reject a missing premise or the old external-box text alone. The 31-check v1016 battery likewise replaces its historical scalar-increment document guard by the current norm certificate and open-boundary guard, with three negative regression tests; its numerical thresholds are unchanged. All three regression sets run in CI. The atlas generator removes trailing whitespace introduced by one-line truncation. All T1–T8 and the TOE contract remain open. No RH claim.

**2026-09-05 (CLVII) (non-RH T1–T8 Round 3 graduates as v1022–v1025; suite 1014  $\rightarrow$  1018, ledger rows 1178  $\rightarrow$  1184 through six narrow subcontracts; 156/156 module checks; NO marker move; TFPT.TOE.COMPLETE.01 and every T1–T8 gate remain [O]).** `v1022_telb_round3_certificates.py` (83/83; native python-flint==0.9.0, 256-bit Arb/Acb) replaces the float/Richardson TEL-B input by 5812 rational cells and the exact cut jump, proving  $\text{TV}_p(F'') \leq 15.817695962988924\dots < 15.928674$  and  $K_3^{\text{TV}} < 2.517464500835$ . Together with the stated analytic digamma/midpoint lemmas, its executable Arb arithmetic gives  $c_\varepsilon = 0.288547359238391\dots$ ,  $c_\zeta = 0.110677064900715\dots$  and  $c_{\text{potential}} = 0.399224424139106\dots < 0.400$ ; full C2b remains open exactly at CF  $\leq 0.139/\sqrt{N}$  and DG  $\leq 0.060/\sqrt{N}$ . Its exact one-mode  $M_2$  countermodel refutes the former ALG2 Vandermonde-to-generated-algebra implication: FE-GEN or an exact gauge/code-projected microscopic-algebra theorem is still required, and ALG1 remains conditional on the generated projection.

v1025\_telb\_c1\_c2a\_certificates.py (8/8; native 100-bit Arb/Acb) executes the existing analytic C1/C2a proofs:  $C_* \leq 0.1520453048\dots < 0.1525$  and  $N \sup_k |\phi_N(k)| \leq 0.508228779140631\dots < 0.509$  for every even  $N \geq 16$ . v1023\_derivative\_filtered\_wall.py (16/16) proves only the finite quasi-free singular-value block: the high continuation branch is separated, but the bare  $b$  subspace is not invariant, the low branch has  $z = 2$  rather than a Weyl cone, and no DET16/DET32, interacting-parent, measure/anomaly/index or 3+1D theorem follows. v1024\_parent\_internal\_boundaries.py (49/49) supplies finite parent-internal T6–T8 witnesses and no-gos: a positive seam/neutrino-pair intertwiner with projective full-Fock  $D_4$  lift but no mass texture; the uniform-gap and direct-TT locality obstructions but no graviton; and a gapped internal  $Z_5$  cap that is chosen,  $D_4$ -breaking and complement-trivial, hence no global  $\rho_0$ . T1 retains only finite axiom-core/W-bridge support while unconditional provenance and the dimension selector remain open; T3 remains at Hamiltonian-class level; T5 retains finite IR witnesses and the v1020 spectator obstruction while prime completion, OS reconstruction, confinement and scattering remain open. The common missing object is still one shared 3+1D parent satisfying T3–T8. Native-only handling is explicit in the website reproducer; v1022/v1025 are not represented as Pyodide certificates. NO RH CLAIM. Surfaces: tfpt\_research\_contracts, tfpt\_4\_frontier, introduction, tfpt\_2\_standard\_model, the MMST article/PDF, ledger, generated verification maps/index, website paper/DAG/gap/FAQ mirrors and changelog.

2026-09-05 (CLVI) (all-place Tate / finite-carrier / Rankin audit graduates as v1021; isolated suite delta 1013  $\rightarrow$  1014, isolated ledger-record delta 1177  $\rightarrow$  1178, one new row PRIME.ALLPLACE.TATE.INTERSECTION.01 [O]; NO marker move; Python-only / Wolfram mirror deferred). v1021\_all\_place\_tate\_rank\_audit.py (23/23): exact scalar Tate projector and good-place tensor/End splittings; the Tate identity remains inertia-fixed at  $p = 2$ ; completed zeta is  $2\xi(u)/(u(u-1))$ , not  $\xi(u)$ ; the symmetric-square prime coefficient is  $a_p^2 - p^3$ , not  $a_p^2 - 2p^3$ . Conditional on the cited external strict-floor premise, rank-nullity excludes representing the complete window form as one exact linear pullback through a fixed finite carrier; nonlinear methods, growing finite discretizations and direct limits are not excluded. An exact sparse counterexample corrects the historical Rankin inference:  $A(X) = O(X^q)$  with nonnegative coefficients does not imply  $b_n = O(n^{q-1+\epsilon})$ , and a maximum fitted on a finite window makes its same-window inequality tautological. Rankin’s actual 1939 argument uses a quantitative summatory error term, not a pole and positivity alone. The correction of record is placed outside the preserved historical docstrings of rankin\_positivity\_miniature\_probe.py and hecke\_orbit\_transfer\_probe.py; their valid Jacobi, Hecke and Rankin–Selberg identities remain, while T52 is narrowed to a path-space vacuum resolvent plus classical local Satake determinants, not a path-space Fredholm determinant. The global all-place intersection form and its independent positivity remain OPEN. NO RH CLAIM. Surfaces: note\_hilbert\_polya\_truncations; tfpt\_prime\_front; catalog part 19; concept map; dated RH balance; website T49 erratum; ScriptIndex and website changelog regenerated.

2026-09-04 (CLV) (r475-repair Lean audit of the selected-path arch tent rate; NO ledger row, NO marker move: PRIME.RDAGGER.ARCH\_IDENTITY\_AND\_RATE.01 is experiments-side). The named Prop SelectedArchErrorQuadraticRate with fixed constant  $\text{archRateConst} = 4(1 + 1/64)^2 = 4.1259765625$  is **FALSIFIED** in the small-support regime (steps = 1, meshExp = 6,  $x_0 = 1$ :  $\text{err}_{\text{arch}}/\Delta^2 = 8.0283458222 >$

4.1259765625; also 4.4488, 7.13–7.24); the endpoint-kink defect  $-\theta(1-\theta)/2$  is proved exactly. The  $O(\Delta^2)$  rate itself is not refuted — large-support ratios 3.9893 ( $k = 5-8$ ) and 3.8466 ( $k = 9-12$ ) from `arch_rate_probe.py` (9/9) and `verify_arch_rate.py` (6/6). Additive Lean repair `rh/lean/RH/SelectedArchErrorQuadraticRateClassical.lean`: `SelectedArchErrorQuadraticRateExists` ( $\exists C(f)$ ) is the correct interface; `selectedArchError_tendsto_zero_of_rateExists` proved; `SelectedArchWeightedInterpolationEstimate` OPEN. Lean build 3578 jobs green; five protected sorries unchanged. Catalog part\_18 KILLED/LOSSY\_CONSTANT. NO RH CLAIM. Surfaces: `rh/README.md`, `rh/INVENTORY.json`, `rh/lean/README.md`, website `changelog.ts` regenerated.

2026-09-04 (CLIV) (TEL-B reduction absorbed into the `SEAM.MMST.TYPEIII.CHARGED.01` contract text and ledger row; NO marker move, status stays [O]). The standalone note `articles/2026-09-04/telb_bound_b_sketch.tex` is superseded and removed: the contract is now the single home. Cover identity + sawtooth split  $R_N = R_N^{\text{sm}} + M_N$ ; bound (A) certified numerically for all  $N$  ( $\|R_N^{\text{sm}}\| \leq 1.7833$ , all-mode BV, one float64 TV constant); bound (B) reduced to  $\sum |\text{Res}_N|^2 \leq 0.2769$  (measured 0.1550), open items C1/C2a/C2b. Residual restated as TEL-B-EXTERNAL (2026-09-04 reduction); the  $A_R < 43$  dyadic-increment route is superseded (measured  $A_R = 0.1238$  at r645, no longer the route). Numerically  $2.57 < 3$ , conditional analytic  $2.9495 < 3$ ; analytically open. Evidence (experiments, not veri-citable): `mmst_telb_trace_decomposition_probe.py`, `mmst_telb_cover_split_probe.py`, `mmst_telb_bound_a_probe.py` (9/9), `mmst_telb_bound_b_probe.py` (17/20, G18–G20 open by design). Surfaces: `tfpt_research_contracts` (new TEL-B reduction subsection); ledger row claim text only; website `changelog.ts` regenerated.

2026-09-04 (CLIII) (spectator-sector obstruction graduates as v1020; suite 1012  $\rightarrow$  1013, ledger 1177  $\rightarrow$  1178, one new row `BULK.PRIME_COMPLETION.01` [O] and one existing row `SEAM.BULK4D.RECON.01` [O] annotated (NO marker move, NO status change); Python-only / Wolfram mirror deferred as for v974–v1019, engine still DEFERRED\_NO\_ENGINE). `v1020_spectator_sector_obstruction.py` (9/9,  $\sim 25$  s): finite free-fermion witness for the spectator-sector obstruction, re-derived (no probe imports) from the r644 audit harvest (`spectator_sector_obstruction_probe.py`, 9/9, 27.7 s, verdict `OBSTRUCTION_WITNESSED_AND_REPAIR_SEPARATES`, SPEC 07169b54..., FILE 7f58f498...). Stacking a decoupled trivial gapped sector  $C$  ( $M=3$ , Chern 0, gap  $\geq 1.095$ ) onto the seam model  $B$  ( $M=1$ ,  $N_Y=8$ , 16 modes) leaves every seam datum unchanged ( $A1-A5$ : Chern  $C_B=C_{B\oplus C}=-1$ ,  $C_C=0$ ; zero modes 2/0/2 with  $B$ -support 1.0; edge dispersion coincident to  $1.4 \times 10^{-15}$ ; seam-restricted covariance identical to 0.0 both grids; edge-profile overlap 1.0) and changes the bulk ( $A6$ : dim 16 vs 32,  $E_0(C)=-25.31$ , mid-strip rank 4 vs 8,  $\|\Gamma_0-\Gamma_\alpha\|_{\text{HS}}$  0.752536 vs 0.760202). Repaired criterion ( $B$ ): Krylov generatedness `spectator_dim(B $\oplus$ C) - spectator_dim(B)=16` at all 128 momenta; commutant dim 1 ( $B$ ) vs 17 ( $B\oplus C$ ) generically,  $\Pi_C=0\oplus 1$  in commutant (residual 0.0); selection unique ( $B$  selected,  $B\oplus C$  rejected). Honest: `spectator_dim(B)=10` at  $p=0$  only (no onsite term at  $M=1$ ,  $T_Y$  rank 1) — the criterion is “generated almost everywhere in  $p$ ”, not pointwise. `DIMENSION.UPLIFT.FIREWALL.01` F2 dimension typing: 2+1D free-fermion hypothesis, re-scoped 4D conclusion. Ledger: new contract `BULK.PRIME_COMPLETION.01` [O] (uniqueness in the category of seam-generated prime completions with trivial relative commutant; kill: a seam-generated completion with nontrivial relative commutant, or two inequivalent prime completions); the existing row `SEAM.BULK4D.RECON.01` is annotated with the obstruction note (B7 as worded is unsatisfiable:  $\partial(B \otimes C) = \partial(B)$  for any decoupled anomaly-

free  $C$  with trivial boundary; uniqueness re-scoped to `BULK.PRIME_COMPLETION.01` and stays [O]. The 2D modular-invariant uniqueness of the holomorphic  $(E_8)_1$  net is untouched and is NOT a 4D no-spectator theorem. NO RH CLAIM. Surfaces: `tfpt_research_contracts` (obstruction note in the `SEAM.BULK4D.RECON.01` paragraph + a new `BULK.PRIME_COMPLETION.01` paragraph, both citing v1020); website `ScriptIndex` regenerated, `changelog.ts` regenerated.

2026-09-03 (CLII) (CI repair + README currency; NO theory change, NO marker moves, suite stays 1012). (1) Core Verification workflow. `v553_exact_form_identities.py` was the sole blocker of the GitHub *Core Verification* workflow since 2026-07-29 (250 consecutive failures, identical traceback): the corrupted-Green-column control (S2.CTRL) pairs  $t_1$  with  $L_P t_1 = \mu_1^P t_1$ , so its  $2 \times 2$  Gram is rank one BY CONSTRUCTION ( $\sigma_2/\sigma_1 \sim 10^{-25}$ ); an LU solve on it is platform-dependent — finite round-off on macOS/Accelerate, exact-zero pivot  $\Rightarrow$  `LinAlgError` on x86-64 OpenBLAS. `span_value` now evaluates the Dirichlet functional on the span itself via the minimal-norm pseudo-inverse (`lstsq`); every printed number is unchanged (25/25; S2.SPAN  $9.0 \times 10^{-13}$ , S2.SINES 1.42–4.8, S2.CTRL 255– $7.7 \times 10^3$ ), verified under numpy 2.4.6 and 2.5.2. Consequence typed honestly: modules v554–v1019 had never run on GitHub CI before this fix – and the first pass beyond v553 exposed the second blocker: `v570_separation_floor.py` section 4 materialised the  $n \times n$  pair kernel by outer products ( $n$  up to 26091 atoms on the kernel subset: 5.4 GB per dense temporary,  $> 30$  GB peak), which OOM-killed the 16 GB runner (exit 143; reproduced in a 14 GB Linux container). The three pair-level readouts (Cauchy–Binet form, negative pair mass, best single pair) are now evaluated in row blocks (`pair_kernel_stats`); peak  $32.7 \rightarrow 3.5$  GB, 8/8 unchanged, only the Cauchy–Binet noise figure moves  $4.5 \times 10^{-14} \rightarrow 4.1 \times 10^{-14}$  (bar  $10^{-9}$ ). Third obstacle, structural: a hosted runner is  $\sim 2.3 \times$  slower than the reference machine (v570 reached after 28 min vs. 12 min), so the full suite ( $\sim 3$  h here) cannot finish inside GitHub’s 6-hour job limit; `run_all.py` gained `-shard K/N` (the  $K$ -th of  $N$  interleaved slices, index mod  $N$  – the slices partition the list exactly and the interleaving spreads the heavy modules v640–v680 / v770–v790 evenly; measured  $\leq 38$  min per slice of four on the reference machine) and `verify.yml` runs a  $2 \times 4$  matrix (Python 3.11/3.12  $\times$  shard 1–4) with unbuffered stdout, so a crash is reported at the module that caused it (the buffered CI log had pointed at v568 while v570 was the culprit). The local exit gate stays the plain python `run_all.py`. Fourth, in the first sharded pass: `v649_discipline_audit.py` D6.1 required the git-ignored local `.lake/build` directory among its “build artefacts” – a reference-machine fact, absent from every fresh checkout; the check now requires the COMMITTED build descriptors (`lakefile.lean`, `lake-manifest.json`, `lean-toolchain`) and reports the local build directory informationally (18/18; the fresh lake build is the Lean Proofs workflow’s job). Fifth, exposed by the interleaving itself: v93 and v99 compute module-level `mpmath` constants (e.g.  $(2/3)^6$ ) at `IMPORT` time with whatever global precision the previous module left behind (`tfpt_constants` sets 40 digits at import; v65, v35 and others lower `mp.dps` to 20–30 inside `run()` and leave it there), so their  $10^{-30}$  checks passed in the historical full order by accident and failed in shard 1/3 (reproduced: v65 then v93  $\Rightarrow$  1 FAIL). `run_all.py` now restores the canonical precision before every module, which makes the suite order-independent – every module sees exactly the state it sees standalone (where all 1012 are green). Sixth, two environment defects from the first sharded pass: the runner’s own `-shard` argument leaked into the `sys.argv` that the in-process embedded probes

parse (v955–v959 and the other round-31-convention modules: “unrecognized arguments”, exit 2, pattern gates red) – `run_all.py` now scrubs its options from `sys.argv` after parsing; and v777 imports `yaml` for its preregistration file, which `requirements.txt` did not list (`pyyaml` added). What REMAINS red on Linux after all of this is typed, not patched: four modules whose frozen bars or pins sit at platform round-off level – v916 C1 pins the SHA of the float certificate vector ( $k$ , norm and  $\lambda_{\min}$  reproduce to every printed digit, the byte hash does not), v966/v967 embed sealed probes whose CONTROL-world deviation is  $1.3 \times 10^{-5}$  against a  $10^{-6}$  bar (the main identities sit at  $10^{-16}$ ), and v990 W(c-frozen) has its control at  $1.1\text{--}2.2 \times 10^{-8}$  against a  $10^{-8}$  tolerance. Widening a sealed bar is a re-seal and is left to the owner; the reference platform for those pins is the Apple-Silicon/Accelerate stack. (2) Sync Audit workflow. `build.sh` audit runs the `rh/ gate` (`run_rh.py -fast`) after `audit_sync.py`; the workflow installed no dependencies, so 257 sealed probes failed on `numpy`. With the dependencies installed (incl. `numba`, on which the SHA-pinned probe r445 gates) the gate still failed on GitHub, and those failures are not repairable from the repository: the sealed probes are bound to the reference machine – r464/r475/r487 read the git-ignored `rh/lean/.lake` `mathlib` cache, and on Linux (`glibc` + `OpenBLAS`, `arm64` and `x86-64` alike) 19/257 sealed smoke gates miss their frozen control bars by round-off (e.g. r257 G21: controls  $9.9 \times 10^{-3}$  vs bar  $10^{-3}$ ), on `macos-latest` still 6/257. Sealed sources are not edited to fit a platform; `audit.yml` therefore runs exactly what the badge names – `audit_sync.py`, platform-neutral, green – and documents that the `rh/ gate` stays in the local `bash` `build.sh` audit on the reference machine (a reproducibility debt now typed, not hidden). (3) README currency. Prediction count aligned with the website (27 status-graded falsifiable test surfaces, 16-entry frozen core registry) in the README prose, the hero image, the terminal GIF and the `./verify` footer (all said 23); the document count corrected to the 12 published documents + the repo-only `tfpt_prime_front` (`docs/THEORY.md` list completed with S, N1, N2, P); the `-full` runtime corrected from “~4 min” to ~4–5 h (README, `verify`, `docs/VERIFICATION.md`); the Zenodo DOI corrected to the v5.4 record 10.5281/zenodo.21128999 (+ concept DOI 18328333; 20846087 is the v5.3 record); the SEAM.EQUIV.01 status-table row reworded per the 2026-07-22 route split (the “modulo a cited theorem” clause belongs to SEAM.EQUIV.MMST.01 [C]). (4) Lean Proofs workflow. Every run since the evening of 2026-08-03 ended in the runner’s shutdown signal (exit 143): the default target fans out over the 4 cores of the 16 GB runner, and since 2026-08-10 it also pulls in the 17 generated `WallLadder/RungKz*` kernel-decide certificates (21–360 GB each; by the project’s own README buildable only serially via `scripts/build_wall_ladder.sh`). The wall-dependent audit lines (six `#print` axioms, fifteen `#check`, three signature-lock examples) moved VERBATIM from `AxiomCheck/AuditCheck/AuditContract` into the new `TfptCarrier/WallLadderAudit.lean`; the new root `TfptCarrier/CIRoot.lean` imports everything except `{WallCertifiedHead, WallLadderAudit}` (drift-guarded by `audit.sh` check 8); CI now pre-builds the heavy modules serially (`QuarticHalf` needs `-M 15360`) and runs `scripts/audit.sh -core` – all eight audit checks pass (run 33727029128, 27 min). No proof content changed; the wall rungs stay certified off-CI (`AUDIT_TRANSCRIPT.txt`). (5) Still red, not fixable from the repo: the `tfpt5-shadow` mirror sync fails with HTTP 403 — the `SHADOW_REPO_TOKEN` PAT (set 2026-06-12) expired after its last successful push on 2026-07-12 and must be

re-issued.

2026-09-02 (CLI) (directed  $E_8$  readout + Coxeter–Euler completion graduate as v1018–v1019; suite 1010  $\rightarrow$  1012, ledger 1174  $\rightarrow$  1176, two new rows; Python-only / Wolfram mirror deferred as for v974–v1017, engine still DEFERRED\_NO\_ENGINE). (1) v1018\_e8\_directed\_readout.py (47/47): E8.DIRECTED.READOUT.01 [E] Identity, exact over  $\mathbb{Z}/\mathbb{Q}$ . Re-derived (no probe imports) from r609 (e8\_directed\_readout\_probe.py, 49/49, verdict E8\_READOUT\_SEALED(7/8), SPEC 3edfb824...). Seven exact cells: Seifert  $S + S^\top = A_{E_8}$ , charpoly( $-S^{-1}S^\top$ ) =  $\Phi_{30}$ , rank( $S - S^\top$ ) = 8; Hamming enumerator  $1+14y^4+y^8$ ,  $240 = 16+14\cdot 16$ ,  $S(3, 4, 8)$ ; srg(120, 56, 28, 24) eig  $56/8/-4$  mult  $1/35/84$ ;  $N(n) = 240\sigma_3(n)$  for  $n \leq 10$ ;  $A_P(n)/A_P(1) = \tau(n)$  for  $n \leq 8$  on three axes,  $E_4^3 - E_6^2 = 1728\Delta$ ;  $(\mathbb{Z}/30\mathbb{Z})^\times \cong C_2 \times C_4$  (orders  $\{1, 2, 2, 2, 4, 4, 4, 4\}$ ),  $D_\chi = L(s, \chi)L(s-3, \chi)$  for  $n \leq 200$ ;  $E_6 \oplus A_2$  index 3, Smith  $(1^6, 3, 3)$ , glue  $78/81/81$ . C7 Gauss-code transform stays OPEN (rem:c7-audit). NO RH CLAIM. (2) v1019\_coxeter\_euler\_completion.py (46/46, 4.5 s): E8.COXETER.EULER.COMPLETION.01 [E] Identity plus Numerical Euler products. Re-derived from r617 (SPEC 5aa3935b..., FILE 6ac6bab2..., 45/45, COMPLETION\_EXACT). Exact:  $\det(I - xC) = \Phi_{30}$ ,  $\text{Tr } C = -1$ ; classical Moebius  $\Phi_{30}$ ;  $U = 1 \oplus C$ ,  $\text{Tr } U = 0$ ,  $\det(I - xU) = 1 - x^2 - x^3 + x^6 + x^7 - x^9$ . Global  $Z_C$  and  $D_{E_8}$  as zeta quotients, abs. conv.  $\text{Re } s > 1/2$ . Numerical ( $X = 10^5$ ):  $D_{E_8}$  residual  $4.76 \times 10^{-4}$  at  $s = 0.75$ ,  $4.16 \times 10^{-12}$  at  $s = 1.5$ . Class: vanishing linear term generic from  $\text{Tr } C = -1$  ( $A_1 \rightarrow 1/\zeta(2s)$ ;  $A_2 \rightarrow 1/\zeta(3s)$ ;  $E_6 \rightarrow \zeta(4s)\zeta(6s)/[\zeta(2s)\zeta(3s)\zeta(12s)]$ ;  $B_2$  still contains  $1/\zeta(s)$ ;  $E_8$  selects only the divisor set of 30. Beurling-generic. Fence (verbatim, no marker upgrade): The trace-free completion is zero-free and pole-free in  $\text{Re } s > 1/2$ . The splitting into the scalar zeta channel and the Coxeter channel is open and RH-equivalent. No RH claim. NO-GO E8.COXETER.REGULARIZED\_SPLIT.NO\_GO.01:  $\det_2$  cannot isolate  $1/\zeta(s)$ . Surfaces: tfpt\_prime\_front §(i) + Prop. 1 (v1018) and a new paragraph after Prop. 1 (v1019); tfpt\_4\_frontier + introduction harvest notes; note\_e8\_gaussian\_code C6 citation (C7 stays open); website papers.ts / primeFront.ts / VerificationDag.

2026-09-01 (CL) (kernel-Loewner positivity at  $L = 0.3$  graduates as v1017; suite 1009  $\rightarrow$  1010, ledger 1173  $\rightarrow$  1174, one new Numerical/certified row (NOT [E]); Python-only / Wolfram mirror deferred as for v974–v1016, engine still DEFERRED\_NO\_ENGINE). v1017\_kernel\_loewner\_positivity.py (26/26): PRIME.RDAGGER.KERNEL\_LOEWNER.01 Numerical/certified float64 (NOT [E], NOT interval). Re-derived (no probe imports):  $Q_W(h) \geq 2.1 \times 10^{-3} \|h\|_2^2$  on  $\text{supp}(h) \subset [-0.3, 0.3]$  ( $2L = 0.6 < \log 2$ , prime term empty). G1 identity vs defining digamma ( $\sigma_A(0) = -5.3721834192256654$ , Dirichlet Rayleigh  $1.180 \times 10^{-2}$ , odd sector 0.2226,  $c_L$  in hull). G2 Loewner:  $I - \text{Re } R_x$  PSD after zero-extension to  $L^2(\mathbb{R})$ ;  $\kappa_w$  booked. G3 401-dim Legendre of a degree-48 Chebyshev surrogate + Bernstein HS tail charged 3 $\times$ : enclosed floor  $2.122 \times 10^{-3}$  (claimed  $c = 2.1 \times 10^{-3}$ ). Independent r495: translation identity 6/6 exact over  $\mathbb{Q}$  with strictly positive boundary-strip mass; doubled- $c_L$  budget  $-2.188$  (false-world). BOUNDARY: r496 NO\_GO(compact-tail@ $L = 0.8$ ) named — the compact-tail method does not scale past the prime-free zone; this row does NOT claim  $\lambda_*(L) \geq 0$  for general  $L$ . NO RH CLAIM. Surfaces: tfpt\_prime\_front new subsection after wave-14; tfpt\_research\_contracts wave-14 follow-on paragraph; tfpt\_4\_frontier + introduction harvest note; rh/paper/rh\_program.tex kernel-Loewner theorem + certification box at the end of rh\_program.tex §2; website papers.ts /



primeFront.ts / VerificationDag.

2026-08-31 (CXLIX) (Monday-morning harvest — axiom-core closure + state/gap batteries graduate as v1015–v1016; suite 1007 → 1009, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1014, engine still DEFERRED\_NO\_ENGINE). (1) v1015\_axiom\_core\_closure.py (11/11): character-blind P-dem response DERIVED —  $r = (1, 1, 1)$  exactly at collar sizes 12 and 16, democracy residual 0; mutants split  $(1, 0, 0)$  and  $(5/4, 1, 5/4)$ . Together with P-anch (v1014) BOTH W-bridge premises are derived at finite level; the axiom-core remainder is ZERO modulo the externalized MMST identification. AX.P2.01 stays an axiom (finite shadows; continuum identification is the same external leg). T1 note on TFPT.TOE.COMPLETE.01: the structure-postulate route is now fully premise-supported at finite level. (2) v1016\_state\_gap\_batteries.py (31/31):  $R_4$  seam-modular cap is the unique finite-level non-entropic selector ( $D_{tr} = 0.1434$ , SK RMS  $3.038 \times 10^{-3}$ ,  $3\pi/5$  saddle preserved; finite-proxy caveat; FTRANSFER.SK.RH00.01 stays [O]). TEL-B certified-tail BLOCKED ( $N^2 D_N$  530 → 19032, UV-supported remainder); strictly smaller external target  $A_R < 43$  with factor- $\sim 350$  measured headroom;  $N$ -uniform CAR nuclearity PROVED ( $C_\beta = (67.97, 29.73, 12.15)$ ) as ALG specialist input; ALG-EXH unchanged (SEAM.MMST.TYPEIII.CHARGED.01 stays [O]). DFP volume-uniform gap cited-verified for  $\mathbb{Z}_2/\mathbb{Z}_4$  open chains ( $a = 0.0233$ ,  $a/16 = 0.00146$ ); dynamical-link leg closed at that level; remaining rotor/wall/3+1D (CHIRAL4D.MIRROR.DET16.01 stays [C]). Surfaces: tfpt\_research\_contracts executed-update paragraphs (five items + L1 nuclearity note); tfpt\_4\_frontier + introduction Monday-morning note; MMST handoff HTML update-2026-08-31 block; website papers.ts / VerificationDag.

2026-08-30 (CXLVIII) (late-evening harvest — thermodynamic-dynamics theorem + detline/W-bridge refinements graduate as v1013–v1014; suite 1005 → 1007, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1012, engine still DEFERRED\_NO\_ENGINE). (1) v1013\_thermodynamic\_dynamics.py (12/12): the mandatory-dynamics leg of QFT4D.LATTICE.FUNDAMENTAL.01 is CLOSED at Hamiltonian-class level. Cited articles/2026-08-30/tfpt\_thermodynamic\_dynamics\_en.tex/.pdf: uniform Lieb–Robinson bound  $J = 12/5$ ,  $R = 2$ ,  $z = 4$ ,  $\kappa = 16043/450$ ,  $v_{LR} = 32086e/225 \sim 387.64$ ; norm-convergent  $\tau_t$ ; gauge-invariant quasilocal  $\mathcal{A}^G$  (Gauss caveat resolved on the invariant subalgebra); existence of ground and  $\beta$ -KMS states. Numeric twin: measured  $v = 2.0 \ll$  proved bound; analytic finite-volume bound dominates nested differences at buffers 8/16/32;  $1/r$  mutant ratio 4.84; Gauss-positive control commutes, violating embedding has commutator norm 1. Remaining: phase uniqueness, limit gap, IR universality. Decision typing UNCHANGED. (2) v1014\_bridge\_refinements.py (17/17): finite detline restriction isomorphism VERIFIED ( $c_{\text{phase}}$  constancy  $3.7 \times 10^{-16}$ , conjugation covariance exact,  $C = W_{\text{bulk}} = W_{\text{seam}} = +1$  / mirror  $-1$ ); orientation fixes only the conjugate branch ( $2 \rightarrow 1$ ); the residual constant  $U(1)$  is the  $A_0$  reference normalization — recorded on ALPHA.QUILLEN.EXACT.01. W-bridge: P-anch DERIVED ( $e_1 = R^{-1}(1, 1, 2)^T$  via the winding lock); P-dem PARTIAL (the  $\mathbb{Z}_4$  average annihilates the nontrivial character space); missing object = the character-blind determinant-response map — the same MMST/Quillen externalized leg as TEL-B/ALG-EXH/P1. AX.P2.01 stays an axiom. Convergence: every remaining axiom-core and continuum gap now runs through ONE externalized theorem. Surfaces: tfpt\_research\_contracts executed-update

paragraphs; `tfpt_4_frontier` + introduction late-evening note; website `papers.ts` / `VerificationDag`.

2026-08-30 (CXLVII) (evening TOE-gate wave — four executed witnesses graduate as `v1012`; suite  $1004 \rightarrow 1005$ , ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for `v974–v1011`). `v1012_toe_gate_witnesses.py` (40/40): T5 IR universality on the physical QWZ edge —  $c \rightarrow 1$  (limit 0.99993), clustering dichotomy  $\xi \cdot \text{gap} \approx 1.089$  vs power-law exponent 2.034, Lieb–Robinson cone  $v_{\text{LR}} = 5/3 \geq c$  with leakage 0.015, cubic curvature  $\rightarrow 1/6$  with volume; TYPED SPLIT: gauge holonomy + seam clock are not propagating modes, so the common- $c$  test needs dynamical gauge excitations. T8  $\rho_0$  eight-candidate table ALL unique; hard sectors kill the SK response ( $\sim 10^{-31}$ ); every response-alive set is exactly  $\rho_{\text{KMS}}$  even without  $\mu_4$  — entropic-proximity selection is VACUOUS at finite level; verdict NO\_SET\_WORKS with the double diagnosis; binding requirement = a non-entropic principle (Euclidean-cap / orientation-branch). T6  $\kappa$  six-fold NULL: six pre-declared D4-odd candidates, all parity-checked, all NULL (five at  $10^{-14}$ ;  $K_4 = 3.55 \times 10^{-3}$  is  $18.8 \times$  too large); the  $1.885 \times 10^{-4}$  cancellation requires the true 4D functional — third and final scaffold-level nu-mechanism bound this week. T7 interacting TT: conserved interaction-complete lattice stress  $\sim 10^{-15}$ , TT spectral positivity exact, sum rules  $\sim 10^{-15}$ , dominant positive- $Z$  pole persists under  $\phi^4$  (gapped);  $Z$  trend  $k_1$  suppressed 0.958 /  $k_2$  enhanced 1.033; free analytic match. Missing: TFPT content,  $k \rightarrow 0$  volume scaling, Ward identities. Ledger notes only on `TFPT.TOE.COMPLETE.01`, `FTRANSFER.SK.RH00.01`, `FLAV.NU.TEXTURE.MECHANISM.01`, `GRAV.SPIN2.EMERGENCE.01` (all stay [O]). Surfaces: `tfpt_research_contracts` executed-update paragraphs; `tfpt_4_frontier` + introduction evening-wave note; website `papers.ts` / `VerificationDag`.

2026-08-30 (CXLVI) (afternoon harvest — simplicity campaign + 3+1D ladder graduate as `v1010–v1011`; suite  $1002 \rightarrow 1004$ , ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for `v974–v1009`). (1) `v1010_simplicity_bridge_census.py` (42/42): derivation matrix over four frozen seeds, unique remainder min  $S_4 = 2$ ; anchor  $(1, 1, 2)$  is not among the three declared Milnor maps;  $\varphi_0$  four novel candidates NULL. Bridge  $Q$  CANONICAL — the bit-selector ladder forces  $c = 2$  and family  $n = 3$  forces  $Q$  column sums  $(9, 5, 1)$ ;  $C$ 's carrier-looking 5 is inherited from  $R$  column 3, not inserted (the `v1005` circularity flag is corrected as forced provenance). Bridge  $W$  CHOICE-but-conditionally-unique: spectrum  $\{5/2, 1, 1\}$  alone has 1078 solutions; democratic-image + carrier-anchor kernel leave exactly the corpus shift (count 1); remaining obligation = seam-geometric derivation of those two directional premises. Milnor strong all-structure bridge NONEXISTENT at  $\mathbb{F}_2$  (clock 2 vs 4, CP rank 2 vs 4) and at every upstairs home ( $\mathbb{Z}/4$ , Gaussian  $k = 2, 3$ , affine); only the coarse  $D$ -module+pairing bridge exists (witness constructed;  $|\text{Aut}(X)| = 36864$ ,  $|\text{Aut}(Y)| = 8$ ). Ring-internal identities of `v1004` (rank clock, socle CP, self-clock) unaffected. `SEAM.MILNOR.LOCALRING.01` stays [O] (canonical status SHARPENED); `AX.P2.01` stays an axiom (pencil rehabilitated,  $W$  premises remain). (2) `v1011_3p1d_ladder_battery.py` (32/32): B6–B9 compressed — minimal 3+1D viable (exact Gauss 32768, confinement  $W(A=1, g=2) \approx 0.063$  vs  $W(A=1, g=0.8) \approx 0.995$ , exact in-face isotropy,  $g_* \approx 1.915$ ); exact tree-fixing  $524288 \rightarrow 4096$  with tree-independence  $\sim 10^{-14}$ ; DET survives coupling; back-reaction; periodicity restores holonomy  $K$ ; charged channels  $\sim 2.014 \times$  at weak  $g$ ; charge-0 control matches B8 to  $10^{-10}$ ;  $K$  collapses above  $g \sim 1.6$ . BINDING: the

matching point of the coupling functional must sit in the deconfined/weak regime. QFT4D.LATTICE.FUNDAMENTAL.01 Decision typing UNCHANGED (Kronecker limitation removed); GAUGE.DETLINE.FIXPOINT.01 stays [O] with the matching-scale constraint recorded. Surfaces: tfpt\_research\_contracts executed-update paragraphs (MILNOR sharpening, P2 rehabilitation, QFT4D ladder, DETLINE matching-scale); tfpt\_4\_frontier + introduction afternoon note; website papers.ts / VerificationDag.

2026-08-30 (CXLV) (review wave 7 — four exploration strands graduate as v1006–v1009: the MMST lemma battery, the DET16 stability battery, the 2+1D master-object assembly scaffold, and the  $\rho_0$  constrained minimizer; suite 998  $\rightarrow$  1002, ledger 1173 no new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1005). (1) v1006\_mmst\_lemma\_battery.py (68/68): five lemmas in-house on SEAM.MMST.TYPEIII.CHARGED.01 [O] — L3 crossed-product; UGF  $K_G = \pi^2/4$  (measured plateau  $\sim 1.023$ , max measured  $K$  1.029399 at  $N = 16$ ); TEL-a  $N^{-2}$  with exact isometries ( $A_{\text{TEL}} = 5120(1 + \pi/2)$ , measured envelope  $3.420384 < 8$ ); integer D5+A3 pairings;  $\mathbb{Z}_4$  outerness in the limit + finite-inner correction. Residual boxed: TEL-B-EXTERNAL (Hankel  $C_R$ ; HS remainder sourced from the paper max 2.105079358 / fit 2.107565497 / v998 bound 2.11, not recomputed) + ALG-EXH (two Buchholz–Verch estimates). Display stays [O]. Cited articles/2026-08-30/mmst\_charged\_scaling\_limit\_en.tex (21 pp) + externalization\_mmst\_handoff\_v2\_en.html. (2) v1007\_det16\_stability\_battery.py (67/67): T1/T2 proved (v1002); T3 Michalakis–Zwolak cited-verified (LTQO,  $t_* > 0$ , gap  $\geq 1/2$ ). Counting line  $n = 2, 3$ ;  $n = 2$  closing scan through  $t = 4$ ; dynamical  $\mathbb{Z}_2/\mathbb{Z}_4$  uniquely gapped (full-probe min 0.728854416 pinned; compressed suite). SW  $1 - \alpha t^2 + O(t^4)$ ;  $c_{\text{geo}}/L$  bounded. 1/192 boxed (ordinary hopping not block-diagonal); BDL boxed (no volume-uniform gap remainder). CHIRAL4D.MIRROR.DET16.01 stays [C]; CHIRAL4D.NOMIRROR.01 unmoved. Cited articles/2026-08-30/det16\_mirror\_gap\_theorem\_en.tex. (3) v1008\_master\_assembly\_scaffold.py (22/22): 2+1D master-object scaffold-coherent — combinatorial  $K = 0$ ; frozen-link  $q^2 \sim 4$ ,  $K(q = 0) = 0$ ; unique-contractive  $|R'| < 1$ ; QWZ wall slope  $\approx -0.9989$ , DET gap  $\geq 1.5$  through  $t = 0.6$ ; assembly  $\mathbb{Z}_6$  (1, 2, 3, 0), clock [0, 1, 1, 2].  $L = 3$  spaces 262144 / 129024 out of suite. Decision QFT4D.LATTICE.FUNDAMENTAL.01 and GAUGE.DETLINE.FIXPOINT.01 [O] unmoved. (4) v1009\_rho0\_minimizer.py (13/13): finite uniqueness of the KMS compression (Klein); Hessian strictly positive; centre dim 1; identity  $\sim 8 \times 10^{-17}$ . TYPED: V1000\_CONSEQUENCE\_MISMATCH (compression kills the SK response vs frozen contrast  $3.550 \times 10^{-3}$ ) + CENTER\_FLATNESS\_PREMISE\_FALSE. Canonical formula needs a weaker admissible set. FTRANSFER.SK.RH00.01 stays [O]. Surfaces: tfpt\_research\_contracts executed-update paragraphs; tfpt\_4\_frontier wave-7 paragraph; introduction status card; website papers.ts / VerificationDag.

2026-08-30 (CXLIV) (review wave 6 — two exploration probes graduate as v1004–v1005 plus two registrations and display corrections: the Milnor–Frobenius–Gray finite algebra, the circularity-flagged relative pencil, the local-ring identification contract, the Weinberg–Witten spin-2 firewall, and honest rest/SM/gravity wording; suite 996  $\rightarrow$  998, ledger 1171  $\rightarrow$  1173: two new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1003). (1) v1004\_milnor\_frobenius\_gray.py (40/40): CLASSICAL (cite, no claim) — Milnor algebra of  $x^2 + y^3 + z^5 \cong \mathbb{Q}[y, z]/(y^2, z^4)$  dimension 8, spectral numbers =  $E_8$  exponents  $\{1, 7, 11, 13, 17, 19, 23, 29\}$ . NEW-IN-CORPUS

EXACT [E]-finite: rank clock  $(2 - a)(4 - b)$  table-sum  $30 = h(E_8)$ ;  $S = pqr \cdot \mu/8$ ; self-clock iff  $\mu = 8$ ;  $(2, 3, 5)$  unique among ordered spherical triples; Galois–Gray  $11/7/-1$  (binary Gray order  $0 \rightarrow 1 \rightarrow 3 \rightarrow 2$ ); CP = socle/Frobenius; integral dual-number bridge  $\mathbb{Z}[y, z]/(y^2, z^4) \otimes \mathbb{F}_2$  free rank 4 over  $D = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$ , basis-dependent. INTERPRETIVE [C]:  $PG(3, 2) = AG(3, 2) \sqcup PG(2, 2)$ . Mutants:  $(2, 3, 7)$   $S = 63 \neq 42$ ;  $A_8$  self-clock survives but exponents fail; wrong socle rank 4. NEW ledger row SEAM.MILNOR.LOCALRING.01 [O]: required canonical geometric/mechanical identification of the raw seam collar /  $E_8$  transition bus with  $\mathbb{F}_2[y, z]/(y^2, z^4)$  preserving dual-number action, Frobenius pairing, Gray torsor, deck action and CP (rank clock and socle-CP as outputs); kill = second inequivalent identification or  $\mu_4$ -equivariance failure; currently [O] because Milnor algebra + exponents are classical and the corpus map is unproven. (2) v1005\_relative\_pencil\_bridges.py (43/43): EXACT [E]-finite Hankel  $p_n p_{n+2} - p_{n+1}^2 = 2^{n+1} + \text{Jacobi spec } \{1, 2\}$  diag  $(4/3, 5/3)$  (v106/v119) and winding spec  $\{5/2, 1, 1\} + \text{fixed plane (v4)}$ . EXACT BUT CIRCULARITY-FLAGGED [C]: unique  $t = 2$ ,  $\lambda = 7/3$ , factorisation  $(3\lambda - 7)(\lambda^2 - 2\lambda + 2)$ , CM  $J_0^2 = -I$ , dual number mod 2, index  $25 = g_{\text{car}}^2$  SNF  $(1, 1, 25)$  — BUT carrier 5 already sits in the column sums of  $Q_2$  and  $C$ : NOT a P2 selector until  $Q$ ,  $C$ ,  $\varepsilon$  are derived from seam geometry independently; AX.P2.01 typing UNCHANGED. INTERPRETIVE [C]: Klein-four SNF  $(2, 2)$  only after  $D = 2d$  (side-blind nulls v521, not a theorem). Mutants:  $t = 0$  fails; perturbed  $C$  breaks factorisation;  $2 + 3^n$  breaks Hankel. (3) registrations (no marker moves). NEW GRAV.WEINBERG.WITTEN.01 [O]: binding firewall BEFORE any spin-2 marker move — the microscopic Hamiltonian family must NOT possess a fundamental Lorentz-covariant local conserved  $T_{\mu\nu}$ ; Lorentz, diffeomorphism Ward identities and the spin-2 mode must emerge in the SAME IR limit; acceptance = positive TT density, two helicities, massless pole, emergent diff Ward, universal soft coupling, named evaded Weinberg–Witten premise; kill = microscopic covariant local  $T_{\mu\nu}$  together with a massless composite spin-2 claim; child of T7, cross-link GRAV.SPIN2.EMERGENCE.01 (not back). (4) display corrections (honest wording, no claim changes). Status-at-a-glance: compiler rest  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  now sits NEXT TO the strict-physical-TOE rest (ten-component Rest<sub>TOE</sub> / eight TFPT.TOE.COMPLETE.01 gates, one line). SM completeness retyped: the discrete Standard-Model *structure* (charges, families, flavor operators) is derived under the axioms; 4D existence, chiral measure, mirror gap, couplings and neutrino dynamics are open. Gravity display: “parameter-free” headlines replaced by “the classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]”. Surfaces: tfpt\_research\_contracts (new sections MILNOR.LOCALRING, WEINBERG.WITTEN; SPIN2 firewall note; T7 child; v358/v359 keybox titles); tfpt\_4\_frontier wave-6 paragraph; introduction status card + SM/gravity lines; tfpt\_2 gravity heading + conclusion; website StatusMatrix / WhatTfptClaims / ClaimStack / GravityEmergence / OpenGates / VerificationDag.

2026-08-29 (CXLIII) (review wave 5 — two exploration probes graduate as v1002–v1003 plus three registrations: the DET16 finite-cluster hit, the two neutrino-texture structural nulls, the master scaling-limit theorem SEAM.MMST.TYPEIII.CHARGED.01, the gauge-stiffness step-scaling divergence, and the holomorphic-KMS document; suite  $994 \rightarrow 996$ , ledger  $1168 \rightarrow 1171$ : three new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v1001). (1) v1002\_spin10\_mirror\_projector.py (39/39): DET16 rank-one projector on the full  $2^{16}$  cluster —

$\Omega^\dagger \Omega = P_{16}$ ,  $\Omega \Omega^\dagger = P_0$ , defect 0;  $P_\phi$  hermitian idempotent  $2.8 \times 10^{-18}$ , rank 1;  $h_{\text{mir}}$  spectrum  $\{0^1, 1^{65535}\}$ , gap exactly 1; all 45  $\mathfrak{so}(10)$  Fock commutators = 0 ( $\det \rho_{16} = 1$ ); parity commutator 0;  $|0\rangle$  and  $|F\rangle$  singlets. Hopping stability on the 4-mode analogue: gap 0.90 (2 clusters) / 0.86 (3 clusters) at  $t = 0.2$ ; spectators untouched. NEW ledger row CHIRAL4D.MIRROR.DET16.01 Candidate [C] (exact finite-cluster gapping; open: dynamical gauge fields, domain-wall geometry, 4D volume theorem). (2) the Casimir kill, permanent. Number-preserving Casimir projector uniqueness fails at every cluster size: one  $16 = \{N=0:1, N=16:1\}$  (none at  $N=8$ );  $k=2$  half-filling multiplicity 27;  $k=3:1463$  (exact Weyl alternation,  $|W(D_5)|=1920$ ). Ledger note on CHIRAL4D.NOMIRROR.01: number-preserving Casimir projector route excluded; display stays [O]. (3) v1003\_nu\_structural\_nulls.py (24/24): charged-sector data  $h = (1, 1, 1)$ ,  $d = (64, 60, 64)$  [E]; Schur-texture NULL — all 8 preregistered  $(K, B)$  fail, D4 reflection forces  $\lambda_1 = \lambda_3$ , hierarchy  $(\varepsilon, 2\varepsilon, 3)$  structurally unreachable from D4-even seam data, required  $K \sim [3536.998, 1768.499, 1/3]$  matches none of 9 comparators; orientation-doubling structure theorem —  $\text{sgn } J$  odd sector generically lifts the degeneracy (exact rational witness); scale NULL — 12/12 fail, required cancellation  $1 - |c/a| = 1.885 \times 10^{-4}$  not supplied,  $\eta$ -branches isospectral. NEW ledger row FLAV.NU.TEXTURE.MECHANISM.01 [O]: both nulls are binding constraints (mechanism must be D4-odd AND supply a  $\sim 2 \times 10^{-4}$  even-odd cancellation; pure seam data excluded [X]-typed). FLAV.NUSCALE.05/.06 notes only; candidates unmoved. (4) registrations (no marker moves). NEW SEAM.MMST.TYPEIII.CHARGED.01 [O]: ONE scaling-limit theorem  $\Psi_{\lambda, N} \rightarrow \Psi_\lambda$  strongly on the finite-energy core + crossed-product continuity, from which  $B \cong (E_8)_1$ ,  $\mu = 1$ , type  $\text{III}_1$ , unique rotational KMS, unique CE, deck equidistribution and modular invariance follow as corollaries; children SEAM.SIMPLECURRENT.GENERATOR.01, SEAM.STATE.RPMIXING.01 continuum, ALPHA.QUILLEN.EXACT.01 rigidity, AX.P1.01 reading (listed on the parent, not back). Cited document articles/2026-08-29/holomorphic\_kms\_extension\_en.tex (13 pp): unconditional rotational-KMS uniqueness via KLM + Longo–Tanimoto GIVEN the identification; P1 = theorem-conditional on seam identification +  $\beta_{\text{angle}} = 2\pi$ ; axiom typing UNCHANGED. GAUGE.DETLINE.FIXPOINT.01: mechanism executed at fixed volume (unique fixed point, contraction  $-0.632$ , exact  $q^2$  and  $SU(2)$  Dynkin ratio 4.0) BUT the preregistered thermodynamic step-scaling  $L = 2.8$  is DIVERGENT ( $g_*$  oscillates  $0.65..1.69$ ,  $|R'| > 1$  at  $L = 4, 5, 8$ ); typed hurdle: the 1+1D toy has no plaquette self-interaction, the real contract needs  $\geq 2+1\text{D}$ ; display stays [O].  $v_{\text{geo}}$  boundary note: unit choice closed; dimensionless hierarchies  $(M_{\text{Pl}}/v_{\text{EW}}, M_R/M_{\text{scal}}, \Lambda_{\text{QCD}}/v_{\text{EW}}, T_{\text{reh}}/M_{\text{scal}})$  remain predictive obligations. Surfaces: tfpt\_research\_contracts (new sections MIRROR.DET16, TYPEIII.CHARGED, NU.TEXTURE.MECHANISM; DETLINE stiffness update; NOMIRROR Casimir kill); tfpt\_4\_frontier wave-5 paragraph; tfpt\_2 (texture-mechanism note); introduction status card  $v_{\text{geo}}$  hierarchy sentence.

2026-08-29 (CXLII) (review wave 4 — four exploration probes graduate as v998–v1001 plus the lattice-fundamental quasilocal-family amendment, the  $v_{\text{geo}}$  closed-metrology retype, the ALPHA.QUILLEN relative-determinant note, and the master contract TFPT.TOE.COMPLETE.01; suite 990  $\rightarrow$  994, ledger 1166  $\rightarrow$  1168: two new rows, existing contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v997). (1) v998\_seam\_modular\_closure.py (14/14): two-edge HS remainder bounded by 2.11 on the frozen plateau  $N_x \in (16, 24, 32, 48, 64, 96)$ ; lattice character  $\theta_{E_8}/\eta^8$  EXACT four orders  $[1, 0, 248, 0, 4124, 0, 34752]$ ;  $\Theta_{E_8}(\tau) = E_4(\tau)$  as Eisenstein series,  $q$ -expansion

match to  $\geq 8$  orders exact (Jacobi identity; the reviewer's C7 sharpening);  $j^{1/3}$  numeric transform at 60 digits with  $S$ -weight bookkeeping. Ledger notes on SEAM.EQUIV.MMST.01 and SEAM.SIMPLECURRENT.GENERATOR.01: C7 arithmetic shadow now exact; net-level C7 stays cited/open. (2) v999\_weak\_collision\_hp.py (14/14): the continuous-time dilation leg — generator extraction Richardson  $1.098 \times 10^{-10}$ ;  $\text{diag } L_{\text{GKSL}} = Q = \log B$  symbolic;  $e^Q = B$  exact; GNS splitting defect 0; locality  $1.355 \times 10^{-20}$ ; strong-collision singularity contrast rank 9/81. DYN.UNITARY.DILATION.01: the finite ladder is now COMPLETE (discrete/kernel/field-free/interacting-classical/quantum-coupled/continuous-time); open = the thermodynamic field limit only; contract stays [O]. (3) v1000\_spin2\_sk\_witnesses.py (26/26): lattice graviton — positivity exact, 2 TT helicities at all 5852 momenta ( $N = 6..12$ ),  $\omega = k - k^3/24$  exact; SK — KMS/FDT  $\sim 10^{-17}$ ,  $\rho_0$ -dependence contrast  $3.550 \times 10^{-3}$  vs static  $10^{-16}$ , saddle 5-fold degenerate with  $3\pi/5$  unique in the  $\mu_4$  lift [C]. GRAV.SPIN2.EMERGENCE.01 quadratic Hamiltonian half executed; FTRANSFER.SK.RHO0.01 mechanism executed; both contracts stay [O]. (4) v1001\_nu\_pentagon\_chain.py (14/14):  $U_e = I$  inventory theorem; misalignment  $U = U_{v9} R_{13}(\theta, \phi)$ ; pentagon double hit  $\phi = 288^\circ = 4(2\pi/5)$  frozen, all three measured angles  $\leq 0.56\sigma$  (honest max pull 0.557) at  $\theta = 2\pi/35$ , unique LEE survivor 2.7%; v3 chain SHA-16 a4c28732fa687620 with  $\Sigma = 0.0599$  eV,  $m_\beta = 9.0$  meV,  $m_{\beta\beta} \in [1.5, 3.8]$  meV,  $\delta_{\text{CP}} = 287.66^\circ$ ; v270- $\theta_{23}$  tension  $1.85\sigma$  TYPED; census null 0/1607. NEW ledger row FLAV.NUSCALE.06 Candidate [C] (pentagon-class misalignment; mechanism = the  $Q_+$ -to-flavor operator [O]; kills DESI/DUNE 287.7 vs 240/JUNO). NO seesaw closure; FLAV.NUSCALE.05 unmoved. (5) Amendments (typed; no marker moves). QFT4D.LATTICE.FUNDAMENTAL.01: a single finite box is not a world — the fundamental object is the quasilocal consistent family  $\{H_\Lambda\}$  with  $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$ ; continuum ( $a \rightarrow 0$ ) optional, thermodynamic limit and controlled IR universality mandatory (dated 2026-08-29). ANCHOR.VGEO.01/.02: display retyped from open-physics/primitive phrasing to CLOSED METROLOGY STRUCTURE ( $\mathbb{R}_+$  unit torsor; one dimensional calibration necessary by dimensional analysis — No-Unit v153/v725); display stays [O], no physical [E]. ALPHA.QUILLEN.EXACT.01: the correct object is the relative determinant  $Q(A) = \log[\det' \Delta_A / \det' \Delta_{A_0}] + 8 b_1 c_3^6 \log \varphi(A)$ ; BFK  $2^{-4}$  is  $A/\tau/\alpha$ -independent at fixed cut number so  $\delta \log 2^{-4} = 0$  — no physical compensation (consistent with COMPENSATION\_INTERNAL). NEW master contract TFPT.TOE.COMPLETE.01 [O]: eight gates T1–T8 with the binding separation  $\text{TOE}_{\text{Complete}} := \text{AND}(T1..T8)$  vs  $\text{TOE}_{\text{Validated}} := \text{Complete AND independent holdouts}$ ; every existing 4D/dilation/detline/SK/flavor/gravity contract is a child of one gate. Surfaces: tfpt\_research\_contracts (executed-update paragraphs at MMST/SIMPLECURRENT, DILATION, GRAV.SPIN2, SK.RHO0; new sections TOE.COMPLETE and NUSCALE.06; lattice-fundamental amendment; ALPHA relative-det note); tfpt\_4\_frontier + introduction status card /  $v_{\text{geo}}$  row; tfpt\_2 (NUSCALE.06 candidate paragraph).

2026-08-28 (CXLI) (the completeness wave — four exploration probes graduate as v994–v997 (suite 986  $\rightarrow$  990, ledger 1166  $\rightarrow$  1166: eight contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v993). (1) v994\_mmst\_criteria\_battery.py (11/11): 5 of 7 MMST identification criteria executed [E]-measured — measured  $c = 8.005$  (16 Majorana copies), conformal edge spectrum  $[0.506, 1.516, 2.523, 3.524]$ , current normalisation  $k = 1.000000000000$ , sector collapse  $\det D_8 = 4 \rightarrow \det E_8 = 1$ , and the flagship:  $\mu_4/\text{GSO}$ -projected lattice character sums reproduce the  $(E_8)_1$

vacuum character  $[1, 0, 248, 0, 4124, 0, 34752]$  three orders deep; C6 is now *written out* in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient  $2/\pi^2 = 0.20264$  vs measured 0.203, 0.18%); C7 modular invariance stays cited/open. Ledger notes on SEAM.EQUIV.MMST.01 (stays [C]) and SEAM.SIMPLECURRENT.GENERATOR.01 (stays [O]; open rest = C7 + net G2/G3/G4). (2) `v995_kms_subnet_rigidity.py` (10/10): all 33 v972 mixing directions killed by KMS alone on finite  $CAR_8 = M_{256}$ ; covariance provably redundant; unique  $E$ -invariant crossed-product extension  $196608 \rightarrow 0$ . SEAM.STATE.RPMIXING.01 finite KMS rigidity [E]; type-III/continuum stays [O]. (3) `v996_gauge_grammar_kill.py` (17/17): H0 structural kill — negative  $b_2/b_3$  flip the stationary-root signs ( $SU(2)$  inverse  $-151.2$ ,  $SU(3)$   $-115.9$ ); 64 conventions 0 hits;  $U(1)$  control 137.0359992168 survives all mutants; detline curvature  $\rightarrow 2\pi$  exponentially with rate  $0.6936 \sim \ln 2$  (numerical Bismut–Freed shadow). GAUGE.DETLINE.FIXPOINT.01 naive grammar excluded [E]; genuinely nonabelian structure required — typed;  $\alpha_s(M_Z)$  remains an external input; contract stays [O]. (4) `v997_spin2_smg_witnesses.py` (24/24): Barnes–Rivers quadratic decomposition —  $R+R^2$  clean (one massless TT pole, two helicities, scalaron at  $M_{\text{scal}}$  exact); the local  $a_4$  Weyl<sup>2</sup> truncation necessarily carries an opposite-residue spin-2 ghost — typed; FK SMG toy:  $N = 8$  unique Spin(7)-singlet ground state, Schmidt rank 1, volume-independent gap 14 at  $L = 1, 2, 3$ ;  $N = 4$  contrast  $2^L$ -degenerate; physical chiral edge keeps  $\pi/L$  exactly. GRAV.SPIN2.EMERGENCE.01 quadratic witnesses [E], content pinned on the untruncated form factor; CHIRAL4D.NOMIRROR.01 SMG toy-proved [E], 4D Spin(10) stays [O]; both contracts stay [O]. Canonical note (not a module) on FLAV.NUSCALE.05: NU\_TEXTURE\_CENSUS\_NULL 0/1607 + 0/200 — frozen  $3 \times 3$  inventory provably insufficient; mechanism [O] sharpened. Surfaces: `tfpt_research_contracts` (executed-update paragraphs at MMST/SIMPLECURRENT, RPMIXING, GAUGE.DETLINE.FIXPOINT.01, GRAV.SPIN2.EMERGENCE.01, CHIRAL4D.NOMIRROR.01, ALPHA.QUILLEN rigidity/ $\ln 2$  notes); `tfpt_2` (census-null note); `tfpt_4_frontier` + introduction card addendum. Exploration record: `articles/2026-08-28/big_picture_simplicity_first_en.md`.

2026-08-28 (CXL) (review wave 3 — the third external bird’s-eye review adjudicated and integrated: ONE new module v993 plus THREE new [O] contracts plus a v986 motivation retype plus the dual-rest formula; suite 985  $\rightarrow$  986, ledger 1163  $\rightarrow$  1166; NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v992). (1) `v993_minimal_defect_selector.py` (29/29): the FULL rank-8 ADE census (SNF of Cartan) with cyclic  $\mathbb{Z}_4$  glue on BOTH factors selects  $(D_5, A_3)$  uniquely ( $D_3 \cong A_3$  aliases collapsed) — the  $D+A$  architecture of v624 is an OUTPUT of (rank 8 + cyclic  $\mathbb{Z}_4$ ), not an INPUT; kill tests  $D_4 \oplus D_4 / A_7 \oplus A_1$  (product 16, fail cyclicity) and the  $\mathbb{Z}_3$  control  $E_6 \oplus A_2$ ;  $[\lambda]^2 = [v]$  exact (lattice shadow of  $U^2 = (-1)^F$ ); the  $\mu_4$  average is the UNIQUE trace-preserving finite conditional expectation (uniqueness needs the module property; kernel dim 36/720 without it). Ledger notes on AX.P1.01/AX.P2.01: axiom typing UNCHANGED; the equidistribution identification stays the open [C] of v813; cited-not-proved remain  $c_- = 8$  and holomorphic uniqueness of  $(E_8)_1$ . SEAM.SIMPLECURRENT.GENERATOR.01 canonical status grows the  $[\lambda]^2 = [v]$  parity-shadow line. (2) Three new research contracts, all [O], no scripts: GAUGE.DETLINE.FIXPOINT.01 (all three gauge couplings as unique stationary points of one nonabelian zeta-det/inflow functional; generalizes ALPHA.QUILLEN.EXACT.01; honest:  $\alpha_s(M_Z)$  is currently an external input); GRAV.SPIN2.EMERGENCE.01 (massless transversal spin-2 pole from the same spectral determinant; Einstein equation stays downstream/conditional;

GRAV.NONCIRCULAR.01 binding); FTRANSFER.SK.RH00.01 (cosmological transfers need  $(S, \rho_0)$ , not  $W[J]$  alone; typed candidate  $\theta_i = 3\pi/5$ ,  $\mu_4$  orientation picks  $k = 0$ ; FTRANSFER.GENERATING.01 covers equilibrium only). (3) v986 docstring retype: the  $\text{Tr } I = 3$  motivation is loose ( $\text{Spec}(I) = \{1, 1, 1\}$ ); the operator reading is the  $Q_+$  eigenvalue route  $\text{Spec}\{1, 2, 3\}$ ; texture data-constrains-texture; check logic and pass/fail UNCHANGED. (4) Dual rest: introduction status card (and the research-contracts keybox, with a brief frontier mirror) now carries  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (compiler) beside  $\text{Rest}_{\text{TOE}}$  (strict-physical-TOE accounting of the 2026-08-27/28 contract wave); nothing in the second formula is claimed closed. Exploration not promoted: the wave-3 Quillen zero-mode Gram identity is a reformulation ( $|\det \text{DFT}_4| = 16$ ), not a Quillen-metric object.

2026-08-28 (CXXXIX) (the promote round — six exploration probes graduate as v987–v992 (suite 979  $\rightarrow$  985, ledger 1162  $\rightarrow$  1163: one new DECISION row QFT4D.LATTICE.FUNDAMENTAL.01; eight contract rows updated, NO status-marker upgrades; Python-only / Wolfram mirror deferred as for v974–v986). (Decision) QFT4D.LATTICE.FUNDAMENTAL.01: TFPT's *physical* completeness criterion is the finite, local, unitary lattice quantum theory (Hamiltonian route: Gauss-law Hilbert space, hermitian local  $H$ ,  $T = e^{-aH}$  positive by construction, Lieb–Robinson cone); the exact continuum Osterwalder–Schrader limit (QFT4D.OS.RECON.01) is retyped as *mathematical reinforcement*, no longer the physical bottleneck; Poincaré invariance is an IR fixed-point property. Status: Decision (registered route/typing; no [E] claim, no marker move; QFT4D.OS.RECON.01 unchanged as the math programme). Anchors: Euclidean T2 kill of the overlap shortcut (exact;  $N_t$ -exact: violation absent at  $N_t = 2$ , appears at  $N_t = 4$  det-only  $\lambda_{\min} \approx -0.249$ ; Wilson control PSD everywhere) and the Hamiltonian route clearing T2 by construction (v989). (1) v987\_os\_dilation\_package.py (21/21): kernel OS continuation exact ( $B = P_1 + \frac{2}{3}P_2 + \frac{1}{3}P_3 > 0$ ,  $H = -\log B$  PSD with gap  $\log \frac{3}{2}$ , Wick anchor  $U(-in) = B^n$  for  $n = 1..6$ ; v971 negative kernel has no real half-step); size-uniform free band; interacting congruence family; quantum SWAP  $d(\theta) \sim \theta^{1.99}$ . DYN.UNITARY.DILATION.01 executed legs grow; continuous field-level OS + branch selection stay [O]. (2) v988\_psi\_lambda\_reduction.py (13/13): S3 crossed-product/tower exchange exact; S1/S2 quantitative skeleton measured (window HS bounded, global HS<sup>2</sup> logarithmic in the topological phase); remaining lemma reduced to cited quasi-free theorems + the MMST identification — reduction, *not* closure. SEAM.SIMPLECURRENT.GENERATOR.01 honest new typing: lattice shadow [E] (v983) + quantitative skeleton [E]-measured + reduction-to-cited-class [C] conditional. (3) v989\_lattice\_gate\_battery.py (22/22): T4 exact SM anomaly bookkeeping; T5/T6 overlap index/GW circle; Euclidean T2 kill + Hamiltonian T2 clearance + chiral Gauss census (single-exponent Gauss iff  $q_L = q_R$ ). TFPT4D.LATTICE.ACTION.01 + CHIRAL4D.NOMIRROR.01 executed harness + named kill; contracts stay [O]. (4) v990\_wj\_transduction\_mechanism.py (13/13):  $\mathbb{Z}_6$  content shadow +  $W[J]$  identities  $\partial W / \partial J = \langle O \rangle$ , Hessian = Kubo, mixed  $\partial^2 W / \partial J \partial \theta$  seam-coupled  $\approx -9.3 \times 10^{-3}$ , decoupled control exactly 0. FTRANSFER.GENERATING.01 + OBS.TRANSDUCTION.01 mechanism executed at finite level; contracts stay [O]. (5) v991\_detline\_bulk\_edge.py (11/11): bulk twist-torus curvature  $2\pi \cdot (+1)$  equals seam holonomy winding +1 (diff  $5.6 \times 10^{-16}$ );  $M = 3$  control zero both sides. SEAM.DETLINE.UNIFICATION.01 finite shadow [E]; continuum Bismut–Freed stays [O]. (6) v992\_joint\_likelihood\_v1.py (24/24):  $\chi^2 = 11.80/\text{dof } 9$  ( $p = 0.225$ ),  $\nu_{\text{eff}} = 1$ , TFPT at 0th percentile of 200 scrambled decoders, LOO stable, holdouts declared. PRED.JOINTLIKELIHOOD.01 v1 executed (numerical,



ledger type [N]); promotion conditions typed; contract stays [O]. Paper surfaces: `tfpt_research_contracts` (executed-update paragraphs at each of the six contracts plus the decision at QFT4D.OS.RECON.01 and TFPT4D.LATTICE.ACTION.01); `tfpt_4_frontier` (shorter decision mirror at the CONTRACT.QFT4D spine). Exploration note (not promoted): `experiments/nu-scalaron-falsification/` leptogenesis discriminator (decuple route  $\times 1.06$  in band, scenario route  $\times 0.09$  killed.).)

2026-08-28 (CXXXVIII) (the master-object wave, part 2 — the second external bird’s-eye review (“one 4D lattice + one seam simple current + one unitary dilation + one determinant line + one  $W[J]$ ”) adjudicated: its four machine-checkable candidate results verified and integrated as FOUR new modules (suite 975  $\rightarrow$  979, ledger 1161  $\rightarrow$  1162: one new row FLAV.NUSCALE.05; three contract rows updated, NO status-marker moves). (1) `v983_simple_current_generator.py`: the SEAM.SIMPLECURRENT.GENERATOR.01 generator is *identified* — the glue vector  $\lambda = (\omega_s, \omega_f)$  with  $\|\lambda\|^2 = 2$ ,  $h_\lambda = 1$ , order 4 in the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  of  $D_5 \oplus A_3$ ; isotropic glue  $\Rightarrow$  even unimodular rank-8 overlattice ( $E_8$ ); coset root census [52, 64, 60, 64] — the odd fusion powers of the ONE class carry the full 128 spinor sector; kill tests (vector-class / mismatched glue) fire; holomorphy shadow  $\det 1$  vs  $\det D_8 = 4$  at the lattice level; the analytic half (G1–G4) stays [O]. (2) `v984_markov_qca_dilation.py`: the DISCRETE leg of DYN.UNITARY.DILATION.01 realized constructively — the collision step  $V = C(U_B \otimes I)$  is exactly unitary, the Stinespring identity  $\Phi(\rho) = \sum_j P_j U_B \rho U_B^\dagger P_j$  holds symbolically,  $\text{diag } \Phi(\text{diag } p) = Bp$  and six fresh ancillas give  $B^6 = T$  exactly (radius-1 collision band); must-fail fires ( $|U_B^2|^2 \neq B^2$ ); typed remark AGAINST the review text: a reused mod-3 adder ancilla still reproduces the populations exactly; the continuous Hamiltonian/OS legs stay [O]. (3) `v985_quillen_channel_swap.py`: the v974 CHANNEL-SWAP face of ALPHA.QUILLEN.EXACT.01 resolved at the finite level by  $\mu_4$ -character grading — the deck swap  $r \rightarrow r+2$  maps the jump spectrum  $[0, 2, 4, 2]$  onto the mark pattern  $[[4, -2, 0, -2]]$  exactly (odd swap fails), the graded  $\det'$  difference is  $c_3$ -free =  $\log(\ln 2/4)$ , the channelwise unit ratio is uniform  $4/\ln 2$ ; continuum Bismut–Freed, KMS rigidity and the  $2^{-4}$  compensation stay [O]. (4) `v986_nu_scalaron_rung.py`: the SCALARON-TRACE RUNG  $M_R = N_{\text{fam}} M_{\text{scal}} = 3c_3^{7/2} \bar{M}_{\text{Pl}} = 9.18 \times 10^{13}$  GeV — a structured point INSIDE the v481 window,  $-1.8\%$  from the 1-loop requirement (the review claimed  $\sim 1\%$ ; the honest value is 1.8),  $\sim 20\times$  sharper than the declined integer rung; exact ladder position  $3\sqrt{c_3} = 0.598$ ,  $0.26\%$  from the closed  $3/5$  Clebsch door — adjacency TYPED, not consumed (v488 killed the Clebsch mechanism; the generation-blind scalaron trace is a different class, itself [O]); kill test  $\Sigma m_\nu = 0.059$  eV; new row FLAV.NUSCALE.05 [C]/[N], no seesaw closure. The remaining review content is typed, not consumed: the axion line ( $f_a = M_{\text{scal}}/128$ ,  $\theta_i = 3\pi/5$ ) and the Riemann–Roch carrier ARE the existing v185/v211/v189; the 4D action, chiral measure, OS package, bulk dictionary, determinant-line unification,  $W[J]$ , transduction matrix and joint likelihood stay open contracts exactly as registered on 2026-08-27. Paper surfaces: `tfpt_research_contracts` (three executed-update paragraphs), `tfpt_2_standard_model` (scalaron-trace rung paragraph).)

2026-08-27 (CXXXVII) (consolidation wave 14 — rounds r334–r357 frozen: the reviewer-authorized consolidation of the terminal, L2 and L\* discovery rounds graduates as FIVE new modules `v978_terminal_density_martingale.py` / `v979_cover_growth_k2.py` / `v980_lstar_margin_chain.py` / `v981_lstar_borodin_duality.py` / `v982_dirichlet_matched_frame.py` (suite 970  $\rightarrow$  975, ledger 1156  $\rightarrow$

1161; every module re-derives its exact layer FROM SCRATCH – sympy symbolic + pure Fractions, no probe imports – and re-runs the sealed round verdicts as exact decision logic on the frozen record aggregates with tipping mutants; the eighteen discovery probes stay sealed in experiments/tfpt-discovery/, pinned in rh/INVENTORY.json and re-verified by run\_rh.py; single-module tests green 24/24 + 28/28 + 26/26 + 22/22 + 23/23, each < 1 s, byte-identical on re-run). (1) v978 — the terminal density-martingale layer (r339 + r341), NEW PRIME.TERMINAL.DENSITY\_MARTINGALE.01 [E]:  $X(v) = d(v)/d(\text{root})$  is a martingale from MASS CONSERVATION ALONE; the moment dictionary  $E[X_\infty^2] = mM_2$ ,  $E[X_\infty^3] = m^2M_3$ ,  $\max X_\infty = mq_{\max}$  (the terminal target IS a martingale moment; the banked r306  $C_2 = 1.0694$  certifies the  $A = 2$  polylog form on the 57-rung set); the tilted tower  $E[X_\infty^3] = E_Q[\prod \Gamma]$  with steps  $\hat{p}_c = p_c R_c^3/\Gamma$  (the untilted reading breaks by an exact Fraction — caught); the exact hand-off  $E_{3h} \leq (mq_{\max})^2 \text{msh}$ ; the algebraic pair ceiling  $\Gamma \leq 4$  via  $4R - R^3 = R(2 - R)(2 + R) \geq 0$ ; the r339/r341 letters honest-negative on the budget side (the target grows  $8\times$  slower than the worst case;  $R^* = 3/2$  a tuning surface); the r324 MEASURED composition stays the end state. (2) v979 — the three-arm cover, the growth law and K2 (r344/r346/r351/r353/r355), NEW PRIME.L2.COVER\_GROWTH\_K2.01 [E] Frame-A-typed: the FAB identity  $F_{AB} = mq_{\max}/\log m$  exact; the data-free K1 formula  $R_{\text{ALG}} = 4^{1/3} \rightarrow$  grid  $8/5$  interior via exact cube comparisons; the r355 mesh identity  $h - \nu u \in (0, 3/2]$  exact; the first certifying cover  $(0/51, m_0^* = 10^{22.6})$ , the P02 predictor  $(51/51)$ , the growth ceiling  $C_{\text{FAB}} = 14.93$  (fresh tranche  $0/6$ ; polylog  $m_0^* = 10^{18.9}$ ) — all under the BINDING r353 Frame-A restriction (frame B breaks the ceiling +21% and kills the reserve floor at  $m \sim 760$ : the sliding coverage is FINITE, PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] retyped accordingly); K2 ( $F_{AB} g_{\text{rel}} \leq 11.87$ ) is the sole cross-family survivor, certified NU-free over three aspects (r355) — the gap geometry is the  $\nu$ -free coordinate of the lane. (3) v980 — the closed  $L^*$  margin-law chain (r345/r347/r348/r350/r352/r354), NEW PRIME.LSTAR.MARGIN\_CHAIN.01 [E]: the one-line identity  $m'_2(p' + q' - m'_2) = p'q' - c'^2$  (Vieta); the two-level theorem  $(121/250, 79/250, 72/250$  by hand;  $r'_2 = 4375/9559 =$  the r345 formula); the pinning theorem (orthonormal top-2 geometry  $\Rightarrow$  the dressed charpoly factors as  $(t - m)(t - mg_{21})$ :  $m'_2 =$  margin IDENTICALLY); the rate-equality theorem in exact ratio form; the  $\rho_K$  identities (toy  $1/8, 23/24$ ); the frozen censuses ALPHA\_CLOSED ( $\alpha = a_c + \delta$ , residual 0.033), RHOR\_REDUCED (one object,  $500\times$ ), PHI\_DICTIONARY\_GO ( $c_{\text{pred}}$  at  $5.5 \times 10^{-4}$ , corr 1.0 — computability, no closed form) and DELTA0\_UNRESOLVED (pool exhausted at  $10^{3.90}$ ): the lane is FROZEN as a specialist problem; the r343/r345 protocol lesson (curvature-honest flatness) recorded as method. (4) v981 — the Borodin particle-hole duality (r356), NEW PRIME.LSTAR.DUAL\_HOLE.01 [E] (the wave's strongest single theorem): the complementation  $\Pi_N^u + G \Pi_{N-1}^{u^\vee} G^{-1} = I$  bit-exact at half filling with the reciprocal dual weight  $u^\vee = 1/(uP^2)$  and a RATIONAL conjugator;  $L^* \Leftrightarrow R > \frac{1}{2}I$  with margin  $= 2 - 1/\lambda_{\min}(R)$  (spectral map symbolic, both directions realized); the half-filling law IS Borodin's rank condition (integer lemma); the dual weight is EXACTLY  $u^\vee \propto c_j(1 - x_j)/|f|$  — the r354 anti-correlation of the two  $\varphi$  blocks is DUALITY ALGEBRA BY DESIGN; the honest main verdict DUALITY\_REPARAM\_ONLY stands (no carrier, no compression: the lane goes final to the memo), with RESERVE\_LOCALIZED and the AC-class exclusion ( $a_{\rho_K} = 1.4222$  vs the Lubinsky-type constant prediction) banked census-side. (5) v982 — the matched Dirichlet frame (r357), NEW PRIME.WORLD.DIRICHLET\_MATCHED\_FRAME.01 [E] (census on 126 windows): the  $\chi$ -arch side DERIVED symbolically from  $\Lambda(s, \chi)$  ( $F_A^\chi(\xi) = -\log(\pi/q) + \text{Re } \psi((1 + 2a)/4 + i\xi/2)$ ;

trivial reduction = zeta; conductor shift =  $\log 3$  exact; parity kernel  $e^{-w/2} \rightarrow e^{-3w/2}$  at 30 digits); SECOND\_ARITHMETIC\_LIVES: the r330 death was a FRAME ARTIFACT — the wall holds on 42/42 windows for  $\chi \bmod 3$  AND  $\bmod 4$ , the living-world evidence goes from  $n = 1$  to  $n = 3$  arithmetics, the verbatim r330 battery retypes all four splits; mechanisms transfer while the zeta-graded signatures (exact pinning, saturated reserve,  $\varphi$  suppression  $439\times$  vs  $3\times$ , counting constants) do not: THE L\* MARGIN LEGISLATION IS A STATEMENT ABOUT HOW CLOSE ZETA'S WORLD SITS TO THE WALL; K2 holds 0/126 (the third independent test); the matched scramble breaks — the wall needs the arithmetic; kein Satz, NO GRH claim. Adjudication and surfaces. Block-B negatives consolidated as prose and row updates only (r334 capacity cap, r336/r340 world-blindness + Hall refutation with the wall-is-global-rescue find, r349 class law inside the v979 third arm, r353 FLOOR\_KILLS binding); PRIME.LSTAR.SUBORDINATION.01 [O] and PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] carry wave-14 notes, markers UNCHANGED; the wave-14 diary section in tfpt\_prime\_front, the wave-14 block in tfpt\_research\_contracts; rh/: INVENTORY via the guard generator, run\_rh.py MODULES  $\rightarrow$  v955–v982, README wave-14 block; website mirror + script index regenerated. The full-suite run\_all.py pass at 975 modules, bash build.sh audit and the manifest regeneration close the wave as the joint exit gate; NO RH claim, NO GRH claim, NO L\* claim anywhere in either direction; mincut base 4 / refined 5 UNCHANGED.)

2026-08-27 (CXXXVI) (the transport-lift wave — the two external-review rounds on the quantum structure of the seam transport graduate from the experiments tree into the load-bearing suite as v976\_seam\_lift\_birkhoff.py and v977\_transfer\_unistochastic\_wilson.py (suite 968  $\rightarrow$  970, ledger 1154  $\rightarrow$  1156: new rows TRANSFER.HIDDEN.CIRCULATION.01 and TRANSFER.COHERENT.WILSON.01, both “executed half [E] / selection open [O]”; the exploration probes transfer\_birkhoff\_circulation\_probe.py 42/42 and transfer\_coherent\_wilson\_probe.py 28/28 stay in experiments/tfpt-discovery/ as the discovery record). (1) v976 — the Birkhoff fibre and the entanglement-breaking no-go. The Birkhoff (random-permutation) fibre of the seam single step  $B$  (exact sixth root of the v221 kernel,  $B = \frac{1}{2}I + \frac{1}{18}P_{12} + \frac{2}{9}P_{13} + \frac{2}{9}P_{23}$  bit-exact) is exactly a one-parameter line  $t \in [0, 1/18]$ ; the hidden coordinate is the sign-character Fourier coefficient  $\hat{w}_t(\text{sgn}) = 6t$ , populations are exactly  $t$ -free, and no fibre member is chiral ( $w(C_{123}) = w(C_{132})$  forced). The one-step coherence spectrum is  $\{1, 6t, (2/3)^{\times 2}, (1/3)^{\times 2}\}$  and the triangle loop current is an exact eigen-observable  $\Psi_t(A) = 6t A$  — one step plus one current readout measures  $t$  without an ancilla. On the contrast qubit the Choi partial transpose has spectrum  $\{-3t/2, 1/6 + 3t/2, 1/3 + 3t/2, 1/2 - 3t/2\}$ : the deployed sixth-root spectrum sits exactly on the entanglement-breaking boundary because  $2/3 + 1/3 = 1$  (mutated spectra miss by  $\mp 1/12$ ); at the full three-level the family stays NPT at every tested iteration with  $\lambda_{\min} = -(1/3)(2/3)^n$ ,  $t$ -independent. The no-go that sharpens DYN.UNITARY.DILATION.01: the deployed Kraus dilation  $K_{ij} = \sqrt{T_{ij}} |i\rangle\langle j|$  is measure-and-prepare, hence entanglement breaking with quantum capacity 0 and  $C_0(T) = 0$  exactly — any dilation closing that contract must be a genuinely different object. Shannon KKT certificates:  $p^* = (1/2, 1/2, 0)$  is exactly capacity-achieving for both  $B$  and  $T$  ( $C(T) = 0.0084$  bit,  $C(B) = 0.4890$  bit, ratio 1.71 %) — the surviving classical message is binary on every scale. The defect–Birkhoff value coincidence (weights  $(9, 1, 4, 4)/18$  vs the v652 ladder  $g^2/18$ ) is recorded typed [C] with its honest multiplicity audit; the forcing functor is the named open contract TRANSFER.DEFECT.BIRKHOFF.01. (2) v977 — the unistochas-

tic  $SU(3)$  lift and the 5-12-13 Wilson code.  $B$  is unistochastic: an exact  $U \in SU(3)$  with  $|U_{ij}|^2 = B_{ij}$  entrywise, angles forced by  $B$  ( $\sin^2 \theta_{13} = 2/9$ ,  $\sin^2 \theta_{12} = 1/14$ ,  $\sin^2 \theta_{23} = 2/7$ ),  $\cos \delta = -4/\sqrt{65}$  unique, lift set  $= \{U, U^*\}$  up to rephasing; Jarlskog  $J = \pm 1/27$  exactly, and the real part of every Wilson plaquette is fixed by  $B$  through unitarity — the full rephasing-invariant Wilson data is classical up to the  $\mathbb{Z}_2$  orientation bit  $\text{sgn } J$ . The main plaquette is  $(-5 + 12i)/324$ , normalized  $(-5 + 12i)/13$  with  $5^2 + 12^2 = 13^2$  (atoms:  $5 = g_{\text{car}}$ ,  $12 = \dim \mathfrak{g}_{\text{SM}}$ ,  $13 = \Delta_Q = N(3+2i)$ ); all nine plaquettes close on four Gaussian phase types with norms exclusively  $\{10, 13, 130, 169\}$ . The center bridge:  $\text{disc } \chi_A = 65 = \det(A-I)^2 + (\det A/2)^2 = 4^2 + 7^2$  for the v530 atom matrix,  $e^{i\delta} = (-4 \pm 7i)/\sqrt{65} = (1 \pm 2i)(2 \pm 3i)/\sqrt{65}$ , and the v530 sibling quotients fail the identity — the bridge selects the atom quotient. Rationality selection:  $J_n$  is rational only at  $n = 1$  (the deployed step hits  $3(1 + 2b - 3a^2) = 1$  exactly, so  $J = 1/27$ ); census over 238 valid family points: exactly one full-code point = the deployed  $(2/3, 1/3)$  ( $0.42\% \ll$  the preregistered  $10\%$  kill). No fixed microstep:  $B^2 \neq |U^2|^2$  and a Dirichlet recurrence witness ( $n^* = 967909$ ) shows no single closed unitary generates the relaxing sequence — repeated steps require dephasing between steps, the six-step macro-gate  $U_6 \neq U_1^6$  (the coherent sharpening of clock-vs-walk). The PMNS near-miss stays a negative control ( $240.2551^\circ$  vs  $240^\circ$ ;  $-1/27 \neq -0.02965$ ; no PMNS derivation). Surfaces. Research-contracts document: the DYN.UNITARY.DILATION.01 section gains the executed lift-structure sharpening, the CCC reduced-channel prose and the two former “exploration-only” citations of rounds DCXCVI/DCXCVII are upgraded to \veri citations; origin\_theory gains the honest “what the classical transport does not determine” paragraph next to the v221 recoverability code; tfpt\_2 gains the  $65 = 4^2 + 7^2$  bridge corollary in the v530 center-quotient block. No status marker moves; both contracts stay open; no PMNS/CP/flavor claim.)

2026-08-27 (CXXXV) (the master-route wave — the external bird’s-eye review adjudicated and its promotable core integrated: ONE new verified module plus FIVE new ledger rows organize the entire 4D frontier into a single constructive route. (1) The conditional 4D dimension selector graduates as v975\_dimension\_selector\_4d.py (suite  $967 \rightarrow 968$ , ledger  $1149 \rightarrow 1154$ ; 16/16 sympy-exact,  $\sim 0.5$  s): under the named TFPT-compatible axiom set A1 ( $d > 2$ : propagating local gauge fields), A2 (dimensionless Yang–Mills coupling), A3 (real self-dual/anti-self-dual 2-form sectors: the Euclidean Hodge star an involutive endomorphism of 2-forms), A4 (chiral Weyl representations: even  $d$ ), the dimension  $d = 4$  is UNIQUE and MINIMAL on the exact scan window  $d = 2..12$  — and the selector is OVERDETERMINED: A2 alone  $= \{4\}$  ( $[g_{\text{YM}}] = (4 - d)/2$  forced symbolically from dimensionless-action power counting, unique zero 4) and A3 alone  $= \{4\}$  (the explicit  $6 \times 6$  star on  $\Lambda^2(\mathbb{R}^4)$ :  $*^2 = I$ , eigenvalues  $\pm 1$  at multiplicities  $3/3$  — the real  $\mathfrak{su}(2)_+ \times \mathfrak{su}(2)_-$  split), while A1 & A4 alone give  $\{4, 6, 8, 10, 12\}$  (chirality does NOT select 4 by itself — honest ablation ledger); Weyl chirality certified by exact Clifford algebras  $d = 2..7$  (even  $d$ :  $\gamma_*$  anticommutes,  $\gamma_*^2 = I$ , traceless; odd  $d$ : the product is CENTRAL and scalar); signature typed honestly (Lorentzian  $*^2 = -1$  on 2-forms — the real split is a EUCLIDEAN/OS-side statement); mutants caught (the wrong power counting  $[g] = (6 - d)/2$  selects  $\{6\}$ ; the “A4 alone” overclaim refused); FIREWALL:  $\mu_4$  and  $g_{\text{car}}$  enter NOWHERE (“ $\mu_4 \Rightarrow 4D$ ” stays forbidden per DIMENSION.UPLIFT.FIREWALL.01), and AXIOM PROVE-NANCE from  $\{c_3, g_{\text{car}}\}$  is the registered OPEN half — NEW ledger row DIMENSION.SELECTOR.4D.01 [C] (a conditional selector, not a derivation; no 4D construction contract advanced). (2) Four new research contracts register the genuinely new

content of the review (all [O], no scripts, no marker moves): SEAM.SIMPLECURRENT.GENERATOR.01 (the seam-gap compression: ONE normalized simple-current/spin-field intertwiner with convergence + energy bounds + locality + braiding + Q-system closure generates the ENTIRE 128-dim extension sector — the v973 residual  $N1 + N2 + N4$  collapses onto one convergence theorem; 1+1D in and out); TFPT4D.LATTICE.ACTION.01 (route step (2) of the firewall ordering made concrete: ONE fully explicit finite 4D lattice family  $S_a = S_{\text{Wilson}} + \bar{\psi} D_{\text{GW}} \psi + S_{\Phi} + S_{\text{seam/top}}$  with SEVEN machine-checkable finite gates T1–T7 — gauge invariance, positive transfer matrix, EXACT seam restriction, anomalies incl. global  $SU(2)$ , index/generations, mirror separation, nonvanishing connected 4-point — plus the twistor flat-direction observable test as a kill test); SEAM.DETLINE.UNIFICATION.01 (the determinant-line hypothesis:  $\text{Res}_{\text{seam}} \det D_{4D} \cong \det D_{\text{seam}}$  with connection AND bulk holonomy = the discrete orientation phase — would unify seam extension, anomaly freedom, generation index and CP orientation as projections of ONE object; the continuum Bismut–Freed identification is exactly the critical missing step); FTRANSFER.GENERATING.01 (the constructive successor of the FR.TRANSFER.01 guard: all four transfer bridges  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$  must arise from ONE generating functional  $W[J] = \log Z[J]$  — the guard stays binding, no upgrade permitted). (3) The programme section “the TFPT4D master route” in `tfpt_research_contracts` organizes the seven conditions (seam compatibility, RP, local unitary dilation, chiral anomaly freedom, one relevant direction = the ANCHOR.VGEO.02 torsor, approximate quantum-Markov recovery per Fawzi–Renner arXiv:1410.0664 and Chen–Rouzé arXiv:2504.02208, nontriviality), the dynamics correction (fundamental order: local unitary amplitudes  $\rightarrow$  decoherence  $\rightarrow$  Markov matrix; the DYN.MARKOV.EMBED.01 kernel result v971 stays the dephased shadow, the DCXCVII unistochastic/Jarlskog-sign finding cited as exploration-level motivation only), the programme ordering P1–P8 mapped onto existing + new rows, and the kill-criteria list (the route may die early); binding separation: RH and the PRIME.\* programmes stay separate — a green RH probe would be neither necessary nor sufficient for the 4D physics. `docs/OPEN_PROBLEMS.md` updated in the same change (the 4D frontier now points at the master route and the five new rows). The full-suite `run_all.py` pass, `bash build.sh` audit and the manifest regeneration follow as the joint exit gate; NO marker moves anywhere — the only new verified piece is the conditional selector [C])

2026-08-27 (CXXXIV) (seam/alpha promotion wave, part 3 of 3 — both named faces of the alpha target now carry computed witnesses: the certified content of the three finished alpha exploration probes graduates from the experiments tree into the load-bearing suite as `v974_alpha_faces_computed.py` (suite 966  $\rightarrow$  967, ledger 1148  $\rightarrow$  1149; embedding convention as in v960–v970: the THREE probes embedded BYTE-EXACT with full-source SHA-256 pins gated and byte-warded against the experiments tree, executed verbatim in isolated namespaces — `seam_bfk_conical_det` 21/21 SPEC d06b3bbe VERDICT match-modulo-local-factor, `laughlin_pump_em1` 20/20 SPEC f9b5cb0b VERDICT pump-quantized-matches-chern, `quillen_rigidity_chain` 18/18 SPEC b4ee0dd0 VERDICT chain-shadow-ok; plus the module-own exact section S0 (sympy/Fraction/mpmath, no probe imports): the BFK closed forms independently re-derived ( $\det'_{\zeta}(\Delta_{S^1}) = l^2$  and  $\det'_{\zeta}(|D|) = l$  from  $\zeta_R(0) = -\frac{1}{2}$ ,  $\zeta'_R(0) = -\frac{1}{2} \ln 2\pi$ ; Dirichlet  $\det = 2a$ ; the  $N = 4$  KMS reassembly as exact rationals in  $c_3$  —  $R_0 = 16c_3 \text{circ}(2, -1, 0, -1)$ ,  $\det' R_0 = 2^{16} c_3^3$ , glued product  $(2\pi)^2 = 1/(16c_3^2)$ , gluing constant  $2^{-4}$ ; spec multisets  $\{0, 2, 2, 4\}$ ,  $\text{Tr } B^k = 4^k + 2 \cdot 2^k$  vs the v484  $\text{Tr } C^k = 4^k + 2(-2)^k$ ,  $B + C = 2(I + S)$ ,

doublet ratio  $4/\ln 2$  with the  $c_3$  cancellation exhibited on a generic symbol; the v485 scale dictionary exact), the alpha anchor re-derived at 50 digits ( $8b_1 = 164/5 = (4/5)41$  exact Fractions,  $q(D_5) + q(A_3) = 2$ , UNIQUE root on the scanned log window,  $\alpha^{-1} = 137.03599921684071$ , split residual  $6.4 \times 10^{-58}$ ), the inflow bookkeeping exact (16 Majoranas = 8 complex copies,  $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$ ,  $k_Y = 5/3$ ,  $(1/k_Y)41/6 = 41/10 = b_1$ ) and a probe-free coarse FHS Chern certificate ( $C(M=1) = 1$ ,  $C(M=3) = 0$ , integer ward); tipping mutants: a single-byte hash mutation caught by the pin, the wrong KMS unit ( $2\pi \rightarrow 4\pi$ ) breaks the scale identity by exactly  $-2\ln 2$ , the  $b_1$  mutant breaks coefficients and split residual; single-module test green 17/17 checks in  $\sim 5$  s, byte-identical on re-run modulo WALL lines, green under both interpreters): (1) the BFK face executed — the Burghlelea–Friedlander–Kappeler Mayer–Vietoris determinant formula made executable on the seam circle cut at the four  $\mu_4$  marks: gluing constant EXACTLY  $2^{-N}$  (symbolic  $N = 1..4$ , general lengths, Laurent-polynomial identity + 50-digit numerics), massless reassembly  $\det'_\zeta(\Delta_{S^1}) = l^2$ , jump matrix = inverse mark Green matrix  $R(m) = (G_m|_{\text{marks}})^{-1}$  exactly; against the v484 mark matrix NO global proportionality (kernels in SWAPPED  $\mu_4$  channels: gauge zero mode vs sign rep) but channel-wise EXACT — the diagonal-zeta face of the merged target is thereby SHARPENED to the channel-swap question; the v485 scale channel reproduced exactly via the KMS unit  $2\pi = 1/(4c_3)$ . (2) EM1 computed — the cited inflow step “boundary level = bulk response” now carries a computed lattice witness on the v367 collar: pumped charge == edge spectral flow == Streda == Chern == 1 per flux quantum (Wannier winding residual 0.0, Streda  $\Delta N = C n_\phi$  exact integers, conjugation mutant sign-coherent across all four readings, wrong-band projector breaks quantization); single complex copy, the 16-copy collar as exact bookkeeping arithmetic only. (3) the finite Quillen identity two-sided — Berry curvature integral == Poincaré–Lelong divisor integral == +1 over the  $U(1)$ -twist torus (both references, both sizes), the difference an EXACT form ( $3.3 \times 10^{-16}$ ), transition winding == Chern (Dai–Freed); typed conditional on the quasi-free rigidity premise (measured, not proven — SEAM.STATE.RPMIXING.01); typed DAG: the remaining open content of ALPHA.QUILLEN.EXACT.01 = {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification} — no additional analytic unknown. LEDGER: NEW ALPHA.QUILLEN.FACES.01 [E] (the computed witnesses); UPDATED ALPHA.QUILLEN.EXACT.01 [O] (stays open — “why this functional” not closed), SEAM.CONTACT.UNIT.01/.02 (the BFK resummation named in v485 now executed), EM.WARD.01 and ALPHA.QUILLEN.INFLOW.01 (the EM1 step computed); NO marker moves anywhere;  $\alpha^{-1} = 137.0359992168$  stays [E] regardless. The full-suite run\_all.py pass, bash build.sh audit and the manifest regeneration follow as the joint exit gate)

2026-08-27 (CXXXIII) (seam/alpha promotion wave, part 2 of 3 — the seam route narrowing: the certified content of the three seam-route exploration probes graduates from the experiments tree into the load-bearing suite as v973\_seam\_route\_narrowing.py (suite 965  $\rightarrow$  966, ledger 1147  $\rightarrow$  1148; embedding convention as in v960–v970: the THREE discovery probes embedded BYTE-EXACT and executed verbatim with the printed SPEC SHA-256 hashes pinned and gated — mmst\_wzw\_coverage 19/19 SPEC e6f392cb, bcov\_measure\_uniqueness 20/20 SPEC 0875f39f, woit\_battery\_mmst 19/19 SPEC 451bd470; byte wards vs the experiments copies live; plus the module-own exact section S0 — the coverage arithmetic exact (the 240  $E_8$  roots enumerated as 112 integer + 128 half-integer spinor roots,  $248 = 120 + 128$  with  $120 = \binom{16}{2}$  bilinears,  $c = 16 \times \frac{1}{2} = 8 = \text{rank}$  with

$8 \leq 8 \leq 16$ ,  $\det \text{ Cartan } D_8 = 4 \rightarrow E_8 = 1$  by explicit simple-root Gram matrices, index  $2 \times 2 = 4 = |\mu_4|$ ), the uniqueness linear algebra as exact rationals on a unimodular-mixed 18-parameter space (block ranks EXACTLY (9, 8, 2, 2), joint 13, gauge 4 inside every kernel  $\Rightarrow$  moduli kernel EXACTLY 1; the full ablation ledger (1, 3, 1, 2, 3, 10) reproduced; the dropped- $C_4$  mutant opens the kernel to 3 — CAUGHT), the skeleton mechanisms exact (bipartite  $SHS = -H \Rightarrow$  every odd function of  $H$  — in particular  $\text{sign}(H)$  — vanishes identically on even distances, integer-exact on all odd powers, the next-nearest-hop mutant caught;  $PHP = -H$  on a two-leg strip  $\Rightarrow K_{\text{top}} = -K_{\text{bot}}$  exact), and the verdict bars as exact decision logic with tipping mutants (a fake COVERAGE\_FULL is REFUSED by the nonempty N1–N4 gate, the kernel-dim mutant refused, hash mutation caught); single-module test green 25/25 gates (21 module-own + 3 pattern gates + runtime) in  $\sim 1$  s, byte-identical on re-run): THE SEAM ROUTE NARROWING. (1) The MMST coverage carved (SEAM.EQUIV.MMST.01 [O] update): the cited Osborne–Stottmeister theorem (CMP 398 (2023) 219, arXiv:2107.13834) COVERS the bilinear half of the seam target — the  $\mathfrak{so}(16)_1$  fermion-bilinear WZW currents converge STRONGLY on the finite-particle core (Thm 5.1 displayed for  $u(1)$ , nonabelian via Cauchy–Schwarz Eq. 229 + the closing remark incl. Majorana algebras), Virasoro/Koo–Saleur (Thms A/4.11/4.16), correlators (Thms B/C), polynomial error estimates  $\text{Error}^2 \leq C 2^{-2\delta N}$ ,  $\delta \in [0, 2]$  — and does NOT cover four NAMED items: N1 the  $\mu_4$ /GSO simple-current extension  $SO(16)_1 \rightarrow (E_8)_1$  as a NET statement (the 128 weight-1 spinor currents are not bilinears), N2 the twist/spin-field sector (explicitly deferred), N3 the one-sided chiral edge net of a 2+1D bulk, N4 the OS/holomorphy selector  $\det K = 1$  vs 4; NEW MEASUREMENT: the edge-correlator convergence rate on the v367 collar is  $p = 3.48$  (Richardson 3.1–4.2, preregistered band  $[0.5, 4.0]$ ), error  $6.3 \times 10^{-2} \rightarrow 2.7 \times 10^{-6}$  over  $N_x = 16..256$ ; the  $M = 3$  control does NOT converge (min rel dev 0.836); HONEST: the ratio observable superconverges past  $\delta_{\text{max}} = 2$  — consistent with the cited UPPER bound, the band’s upper edge calibration-informed; the coverage gates are executable transcriptions of a human literature adjudication. (2) The twistor measure rigidity (SEAM.EQUIV.TWISTOR.01 [O] update): on the computable v508/v515 sector the 18-parameter deformation of the declared measure data (contact normalisation, 4 sector weights, residue normalisation, 4 linking charges, 4 centre positions; 4 gauge directions  $\Rightarrow$  14 moduli) has executed-constraint Jacobian ranks C1 clock 9 / C2 lockstep+integrality 8 / C3 charge 2 / C4 null-ideal 2, joint rank 13  $\Rightarrow$  the moduli kernel is EXACTLY 1 = the uniform sector-weight shift  $dv = (1, 1, 1, 1)$ , mapping to the untwisted Okubo square  $Q^{(0)} = 36(1, 2, 1, 0, 0)$ , annihilated by every executed certificate functional BY CONSTRUCTION (exactly flat to all orders); ablation ledger (1, 3, 1, 2, 3, 10); honest: NOT a BCOV derivation, the  $W_2 : W_{13} = 4:3$  Harvey–Moore tester typed NOT-TRANSCRIBABLE. (3) The cross-route OS skeleton (NEW SEAM.EQUIV.SKELETON.01 [E]): ALL five WOIT kill-test main invariants ( $\alpha$ : forced  $\eta = +i$ ,  $\Theta^2 = +1$ , bond RP;  $\beta_1$ : GSO 2-torsion;  $\beta_2$ : OS quotient + KMS  $\lambda_{\text{max}} > 1$ ;  $\beta_3$ : perp descent;  $\gamma$ : mirror flip, multiplicity-free, mark incidence) hold VERBATIM on the p+ip collar edge — 5/5; ALL divergences control-typed and tracing to ONE mechanism (the twistor lattice’s exact chiral checkerboard  $C(\text{even}) = 0$  vs the collar’s  $K(\text{even})$  up to 0.1736), independently confirming v519-R4’s lattice-placement typing from the second route; the collar’s top edge realizes the anti-chiral state EXACTLY ( $\max |K_{\text{top}} + K_{\text{bot}}| = 7.5 \times 10^{-16}$ ); the  $M = 3$  trivial collar loses the skeleton first at GAMMA.MARK.INCIDENT — the skeleton tracks the topological phase. LEDGER: NEW SEAM.EQUIV.SKELETON.01

[E] (executable skeleton, typed FULL/PARTIAL per test — an executable-level narrowing, NOT a route closure); UPDATED SEAM.EQUIV.MMST.01 [O] (residual now the NAMED N1–N4 list), SEAM.EQUIV.TWISTOR.01 [O] (measure rigid on the computable sector up to ONE named flat direction), SEAM.EQUIV.01 [O] (parent prose: the open structure concentrated on the same named extension question from both sides); NO marker moves — both routes and the parent stay [O]. Python-only per GATE.WOLFRAM.02; the full-suite run\_all.py pass, bash build.sh audit and the manifest regeneration follow as the joint exit gate)

2026-08-27 (CXXXII) (seam/alpha promotion wave, part 1 of 3 — the seam interaction front: the certified content of three exploration rounds of 2026-08-27 on the v529 Fidkowski–Kitaev seam toy and the v903 mixing slice graduates from the experiments tree into the load-bearing suite as v972\_seam\_interaction\_front.py (suite 964 → 965, ledger 1144 → 1147; embedding convention of v969/v970: the THREE discovery probes embedded BYTE-EXACT with pinned source SHA-256 hashes, byte-warded against experiments/tfpt-discovery/ and executed verbatim with the printed gate counts, VERDICT letters and SPEC SHA-256 prefixes pattern-gated — majorana\_rp\_class 20/20 SPEC bcdfcde6 VERDICT class-empty-on-basis, quartet\_avoiding\_os 18/18 SPEC 6ced7be3 VERDICT os-viable-on-avoiding, seam\_rp\_rigidity 25/25 SPEC d09cd983 VERDICT rp-admits-mixing; plus the module-own exact section S0 (sympy / GF(2) integer certificates, no probe imports) — the Jaffe–Pedrocchi DECIDABILITY LEMMA on a 4-Majorana toy (the 9 products  $\theta(M_a)M_b$  land on 9 pairwise-distinct crossing masks with unimodular coefficients, so the crossing matrix  $B$  is UNIQUE; the self-adjoint bond  $H_c = s i g_1 g_2$  extracts to  $B = \text{diag}(0, s, 0)$  and exactly ONE sign certifies — the wrong-sign mutant caught), the ODD-SPLIT ARITHMETIC (on  $\mathbb{Z}_{16}$  exactly the 2 mirror-fixed straddling quartets split 2+2 and the 4 off-center ones 1+3; the odd block  $[[0, b], [\bar{b}, 0]]$  has  $-B$  PSD iff  $b = 0$  — the obstruction; a 2+2 straddle needs only the good orientation; the flipped-sign-law mutant caught), the GF(2) GENERATION BOOKKEEPING (pair vectors on 16 points have rank exactly 15, the even algebra has  $2^{15} = 32768$  monomials; the within-halves single-cut restriction has rank exactly 14 — deficit EXACTLY 1, witnessed by the left-parity functional), the S1 WINDOW ARITHMETIC ( $33 = 24 + 9$ , the  $32 + 1$  split, rank-nullity  $33 - 16 = 17$ ; the kernel value  $-1/(1 + c^2)$  at  $c = \tanh(1/2)$  re-derived symbolically as  $-\cosh^2(1/2)/\cosh(1)$ , float pin  $-0.8240271368$ ; both wrong-form mutants caught) and FOUR VERDICT BARS with tipping mutants (a mutated embedded-source byte breaks the pinned SHA-256; the three live records compose to exactly the claim triple; a tipped gate count 19/20 and swapped verdict letters do NOT compose); single-module test green 20/20 gates in  $\sim 4$  s with both the venv and the system interpreter, byte-identical on re-run)): THE SEAM INTERACTION FRONT FROZEN. (1) The odd-split law (SEAM.INT.ODDSPLIT.01 [E], toy level): the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (arXiv:1406.1384; Wei–Li–Xiang arXiv:1601.01994) is made DECIDABLE on the v529 toy (unique  $B$ ), certifies the free FK Hamiltonian on all 8 bond cuts ( $\eta = -i$  forced), scores a PERFECT confusion matrix against measured RP over 35 cells (23/12/0/0), and finds the  $\mu_4$ -covariant quartic space (dim 464) EMPTY on the basis with the support-level exclusion pinning the full 1820-dim quartic space — THE DISCOVERY: the v529 straddle law REFINES to the ODD-SPLIT LAW (JP predicted RP survival on CENTER-straddled cuts,  $m = 4$  model, axes 7/15, 2+2 mirror split; direct measurement confirmed (37, 0, 0) at  $g = 0.5/1/2$ ; the v529 anchor ladder reproduced exactly). (2) OS viability on the avoiding family



(SEAM.INT.OSAVOID.01 [E], toy level): the clock-invariant  $m = 4$  member's quartet-avoiding cut family generates the FULL even seam algebra (rank 15, dim 32768; single-cut members  $m = 2/6$  typed deficit 1) and carries a POSITIVE OS/GNS structure at  $g = 1$  (min  $+4.4 \times 10^{-5}$ , GNS 37/37, distances monotone) while the straddled control genuinely fails ( $-6.9 \times 10^{-2}$ ); the  $\mu_4$  clock maps the avoiding pair  $\{3, 11\}$  onto itself exactly — the WOIT.OS.TWISTOR.01 kill-test-2 counterweight is NARROWED (an OS-viable cut geometry exists, selected by the toy's own clock), not removed. (3) The RP mixing census (SEAM.STATE.RPMIXING.01 [E], census): the v903 exclusivity is RAY- and REFLECTION-relative, not slice-wide — on the full 33-dim  $C_6$ -covariant mixing slice  $\theta_{abT}$  sees exactly ONE coordinate ( $a_J$ , kernel value  $-1/(1 + c^2) = -0.8240271368$  at  $c = \tanh(1/2)$ ; 32/33 invisible  $\Rightarrow$  a 32-dim MARGINAL RP hyperplane  $\{a_J = 0\}$ ) and the  $\theta_S$  Hermiticity-defect map has rank 16  $\Rightarrow$  a 17-dim first-order-neutral subspace with STRICT finite- $s$  witnesses ( $\lambda_{\min} = 0.1558$  at  $s = 1/8$  — the very direction  $\theta_{abT}$  kills); the v903 anchor reproduced ( $a_J = +0.04068013$ ). LEDGER: NEW SEAM.INT.ODDSPLIT.01 [E] + SEAM.INT.OSAVOID.01 [E] + SEAM.STATE.RPMIXING.01 [E]; UPDATES appended to SEAM.INT.FKTOY.01 (straddle law  $\rightarrow$  odd-split law), WOIT.OS.TWISTOR.01 (counterweight narrowed, contract stays Open) and SEAM.STATE.DERIVATION.01 (exclusivity typed ray-/reflection-relative). HONEST: NO marker moves anywhere; SEAM.EQUIV.01 stays [O]; the v529 MANDATORY FENCE (one toy, one interaction class, [C] flat-band parent) is inherited verbatim by all three results; JP is sufficient-only; the S1 census is sector-truncated ( $\deg \leq 2$  Grams +  $\deg \leq 4$  spots). Python-only per GATE.WOLFRAM.02. NO RH claim.)

2026-08-27 (CXXXI) (the external deep-review round — dimension / dynamics / gravity firewalls and the 4D construction ladder: the review diagnosis “chiral 1+1D seam net  $\not\equiv$  local, interacting 3+1D quantum world” is implemented as eight new contract IDs in `tfpt_research_contracts` plus one new load-bearing module. (1) DIMENSION.UPLIFT.FIREWALL.01 (standing rule, in force): no 1+1D theorem closes a 3+1D claim; every contract closure states the spacetime dimension of hypotheses AND conclusion; audit census recorded (SEAM.EQUIV.01 is 1+1D in and out — even fully proved it is a boundary building block, not the 4D theory; the v957 kinematic uplift does not close it; the anomaly-inflow bulk is invertible/topological, never to be silently equated with physical spacetime); the realistic route ordering (seam as boundary module  $\rightarrow$  genuine 4D Euclidean lattice  $\rightarrow$  OS package  $\rightarrow$  chirality/no-mirror  $\rightarrow$  only then BW/CHM + entanglement gravity  $\rightarrow$  bulk–seam equivalence) is registered as the programme skeleton. (2) SEAM.BULK4D.RECON.01 [O]: the previously implicit seam $\rightarrow$ bulk theorem made explicit (4D local algebras/Schwinger functions; locality/microlocality, Poincaré covariance + spectrum condition, spin-statistics + CPT, both helicities, clustering, nontrivial S-matrix, unique operator dictionary). (3) the dynamics split DYN.MARKOV.EMBED.01 / DYN.UNITARY.DILATION.01 (“three meanings of dynamics”: relaxation  $T^n$  / dissipative semigroup  $e^{t \log T}$  / unitary  $e^{-itH}$ ): the finite kernel half of MARKOV.EMBED is executed [E] as `v971_markov_embedding_generator.py` (15/15, sympy-exact) —  $Q = \log T$  of the v221 seam kernel is an EXACT classical Markov generator:  $e^Q = T$  by spectral calculus, row sums symbolically zero, off-diagonals exactly  $\{\ln(9/8), 2 \ln 3, 2 \ln 3\} = \{0.117783, 2.197225, 2.197225\} > 0$  (the reviewer's independently verified numbers), uniformization certificate  $Q + (4 \ln 3)I \geq 0$  entrywise  $\Rightarrow e^{tQ}$  positivity-preserving doubly stochastic for all  $t \geq 0$ ,  $\text{spec}(Q) = -\text{spec}(H_{\text{mod}})$  (gap  $= \Delta = 6 \ln(3/2)$ ), negative control: second eigenvalue  $-1/10$  gives  $\det = -1/200 < 0$

— provably NOT embeddable; the field level and the local size-consistent unitary dilation (Stinespring + Lieb–Robinson + OS continuation) stay [O]; binding precision: the shared F\_transfer relaxation shape is a universal CONTRACTION CLASS, not one physical clock (v723/v724/v777). (4) QFT4D.OS.RECON.01 [O] (the genuine 4D Euclidean lattice OS package) with the binding phrasing rule: the v381 all-orders leg is a *local perturbative construction under the stated EG/BRST hypotheses* (Duch weak adiabatic limit cited, hypotheses unverified for the concrete model; no confinement / mass gap / asymptotic charged states / nonperturbative Higgs vacuum / full S-matrix follows) — tfpt\_2 and tfpt\_5 phrasing restricted accordingly. (5) CHIRAL4D.NOMIRROR.01 [O] (explicit GW/overlap chiral lattice operator; local gauge-invariant measure, perturbative + global anomaly freedom, no Witten  $SU(2)$  anomaly, no mirrors, correct index/charges, stable continuum limit; Mastropietro arXiv:2303.02790 as calibration). (6) GRAV.NONCIRCULAR.01 (standing rule, in force): BW is a theorem WITHIN a local Poincaré-covariant QFT (Mund/Morinelli counterexample landscape cited), never a generator of Lorentz spacetime from a modular flow; every BW/CHM use declares assumption-vs-result; audit census: the entanglement-equilibrium Einstein derivation (v323/v358/v359) is typed ASSUMPTION (conditional on the 4D package, never citable as evidence for it), QG.AMB.01 correctly declared, the seam KMS  $2\pi$  chain dimension-typed, the intrinsic-BW programme the honest result-direction attempt. (7) OBS.TRANSDUCTION.01: the transduction principle (internal eigenvalue + nonvanishing projection  $BP_r \neq 0$  + physical coupling = observable signature; without the middle arrow neither non-observation falsifies nor observation confirms; minimal model machine-checked at exploration level in fo01, uniform functional blind to  $10^{-14}$ , character channel recovers 64/729 to  $10^{-10}$ ; the ACT+Planck birefringence proximity typed coincidence-level pending the shared-action proof). (8) PRED.JOINTLIKELIHOOD.01 [O]: evidence reorganized from hit-counting into overdetermined relation bundles (flavor sum rule;  $r = 3(1 - n_s)^2$ ,  $\alpha_s = -r/6$ ; one normalized axion action; character-resolved transfer) with the statistical demand (joint likelihood, atom-induced covariance, effective dof, prespecified null + formula grammar, real holdout); the v100 “13/13,  $< 10^{-30}$ ” statement retyped as a look-elsewhere quantifier under its declared null — the 2026-06-09 freeze protects FUTURE measurements, it does not retroactively make historically known data blind. Docs corrected in the same change: docs/OPEN\_PROBLEMS.md (the Einstein-derivation overstatement rewritten to the conditional Jacobson-route typing with the explicit does-NOT-follow list; “zero open internal mechanisms” fixed to match the contract scope fence — ledger wins; the all-orders phrasing restricted) and docs/FALSIFICATION.md (the null-model section retitled and rescope, freeze and kill tests unchanged). Ledger +9 rows; suite 963 modules; bash build.sh audit and the manifest regeneration are the joint exit gate)

2026-08-27 (CXXX) (consolidation wave 13 — the extraction repair and the terminal composition: the certified content of discovery rounds 323–327 (incl. the r326 Lean round and the r323 clean abort) graduates from the experiments tree into the load-bearing suite as v970\_extraction\_and\_composition.py (suite 962  $\rightarrow$  963, ledger 1133  $\rightarrow$  1135; wave-4 embedding convention: the FOUR discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — fa\_provenance 36/36 SPEC 9a6696f8 (r324-pre, sealed FIRST, then superseded mid-round by the binding reviewer course correction and BANKED), qmax\_m2\_

origin 36/36 SPEC dc36cacb (importing the embedded r324-pre), extraction\_order 18/18 SPEC 31277f91, group\_mass\_cap 34/34 SPEC 11e4fd40 (importing the embedded r324-pre and r324); full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 in pure Fractions with NO Lean call — the r324 interpolation  $M_3 \leq q_{\max} M_2$  and  $M_2 \leq q_{\max}$  exact (toy slack EXACTLY 1/36, DOUBLE equality at the flat profile; both reversed-mutant directions caught), the  $F_A$  identity  $F_A B \log m = m q_{\max}$  as exact constructional algebra ( $F_A$  DE-BLACK-BOXED as the rank-local normalization of  $m q_{\max}$ ; the dropped-gauge mutant caught), the r327 group ledger exact on the sealed toy (partition  $5 + 1 = A_1 = 6$ , cap chain  $m q_{\max} = 12/7 \leq n_g \times h_{gn} = 20/7$  slack  $8/7$ , two-ancestor bound  $5 \leq 2 \times 3$  slack 1; the drop-group and wrong-window-max mutants break by EXACTLY 1), the r325 onset formula  $\alpha^* = (n_g + 1)D_0/2$  as an EXACT TOY STABILIZATION (the tent-window read equals the exact atom sum  $35/16$  for every  $\alpha \geq 5/8$ , differs below, and is CONSTANT under dyadic mesh refinement  $D_0 \rightarrow D_0/4$ ; the nearest-neighbour mutant caught), the  $m_0^*$  composition arithmetic (the r324 log-space solver rebuilt verbatim landing  $10^{59.6}$ ; the exact exponent algebra  $158/1000 + 14/1000 = 172/1000 < 224/1000$  and  $1 - e/2 = 914/1000 \geq 888/1000$ ; the r328B chain-audit gate S0-T5d: the block-level need  $454/501$  and the chain bar  $94/501 \approx 0.188$  exact, the verdict survival  $172/1000 < 94/501$ , the non-sequitur  $0.888 \times 1.002 < 0.908$  caught, the chain-honest  $m_0^* = 10^{238}$  from the same solver, the false-fire window mutant caught; supercritical mutants return NONE and the DISCLOSED pre-freeze linear-space bug — a search capped at  $10^{18}$  that misses the root — is rebuilt and caught), the SIX SEALED VERDICT BARS as exact decision logic with tipping mutants at every live clause boundary, and the wave-13 composition gate; the r326 Lean round consumed as a REPORT and the r323 abort as a NOTE; single-module test green 26/26 gates (22 module-own +4 pattern gates) in  $\sim 2$  s, byte-identical on re-run): THE EXTRACTION REPAIR AND THE TERMINAL COMPOSITION FROZEN. (A) The clean abort (r323): the planned  $L^*$  indefinite-signature fork was aborted by the binding reviewer adjudication ( $L^*$  internally paused at allocation 0%) at the END OF THE DESIGN PHASE, BEFORE any write access — no probe, no measurement, no rh/ file, nothing counts as a round (post-abort suite 73/73 with 103/103 pins). (B) The terminal composition (r324, the pre-work banked as r324-pre; r327 the anatomy): the round ran under a binding mid-round course correction, disclosed in full — the original  $F_A$ -boundedness contract was executed and sealed FIRST (fa-bounded-distributional: KS  $0.0982 \leq 0.125$ ,  $p_{\max} = 0.090$ , envelope  $C_F = 3.357$ ; the honest tension disclosed: the per-rank  $p$  values at the named spikes are  $0.005/0.000/0.000$ ), then superseded (the cleaner decomposition is  $Q_{\max} \times M_2$ ); the binding contract fired pileup-grows-subcritical ( $e_{\text{tot}} = +0.172 < \text{CRIT} = 0.224$ , halves  $+0.141/-0.160$  both subcritical, the second FALLING — decided; by the reviewer contract a sufficient, first-class outcome): the pointwise  $M_2$  export REFUTED ( $C_2 = 2.2557$  fails 7/39 — exactly the banked spike set at  $+35..42\%$ ) while the distributional r302 export holds GO (KS  $0.023/0.016$ , tighter than the r302 core); the mechanism HALF AND HALF — the  $O(\log m)$  SCALE COUNT CERTIFIES pointwise ( $C_{\text{NSC}} = 2.0258$ , 0/39) but the per-scale mass  $O(1/m)$  is REFUTED ( $C_{\text{PIL}} = 9.3583$ , 11/39): the near-critical windows are ONE NEAR-MAXIMAL HEAVY SOURCE SCALE at the argmax block (kz53:  $s_0 = 13.78$  of  $G = 13.02$  at  $\text{nsc} = 5$ , the ladder median), not multiscale convergence; the composition typed MEASURED:  $\sum_j q_j^3 \leq 8.941 (\log m) m^{+0.172}/m^2 \Rightarrow N_3 \geq m^{0.914}/\sqrt{8.941 \log m}$ ,  $N_2 \geq N_3$  (the r306 exact chain)  $\Rightarrow N_2 \geq m^{0.888}$  for all  $m \geq m_0^* = 10^{59.6} \Rightarrow$  the full

chain to terminal\_positive\_main —  $m_0^*$  ASTRONOMICAL, the gap  $(274, 10^{59.6})$  IS THE DISCLOSED EXTRAPOLATION HYPOTHESIS, no cofinal claim (r328B independent chain audit, carried as a chain annotation with the record sealed: the bar 0.224 used the ATOM-level need 0.888, the block-level need is  $908/1002 \approx 0.9062$  — the chain-honest bar is 0.188, THE VERDICT SURVIVES  $0.172 < 0.188$ , the chain-honest  $m_0^*$  is  $10^{238}$ , and the going-forward bar is 0.188); r327 then located the spike structurally (cap-partial with anatomy SOURCE, median named ratio  $1.057 \geq 0.5$ ): on ALL 65 rungs the heaviest group of the argmax block is EXACTLY ONE  $\beta/\omega$  FOLD PAIR (one bulk + one window atom at one position, histogram  $[0, 65, 0]$ , median alignment 1.000 — the ancestors REINFORCE); at kz53 the single pair carries 88.8% of the argmax block at gap 0.076 — THE SPIKE IS LITERALLY ONE BULK/WINDOW COINCIDENCE (kz67 the exception shape: 6 groups); the  $\Lambda$ -pair cap REFUTED as a mid-ladder polylog ( $C_{LV} = 1.1838$  fails 19/39, named 0/4 — the exact two-ancestor algebra holds at slack 0.0, the CONSTANT does not close), the direct group cap fails at both A, THE GROUP COUNT CERTIFIES pointwise ( $C_{NG} = 2.6351$ , 0/39, named 4/4 — the THIRD certified  $O(\log m)$  count of the lane);  $e_{\text{route}} = +0.106 < 0.224$  but the halves STRADDLE ( $+0.411/-1.183$ ) — heavy-group-unbounded does NOT fire (the heavy-group coordinate DECAYS with depth: a shallow/mid phenomenon); the r324 MEASURED chain stays the end state, and the named direction is the COINCIDENCE GEOMETRY: where can the single heavy coincidence SIT (source geometry, not group count or multiplicity). (C) The extraction repair (r325) and the Level-C architecture (r326): r325 adjudicated the three reviewer variants for the R319 cofinality seam — elementwise-stabilization-go primary + co-letters canonical-mesh-rebuild-go + signed-defect-correction-go(sign) + anchor-only-insufficient: the repair is a QUANTIFIER STATEMENT, not a mesh theory — on the native v749 class (dyadic step-function autocorrelations, seed-sealed) ALL THREE channels (comb/arch/pole) of the canonical tower windows stabilize EXACTLY at the predicted finite anchor onset  $\alpha^* = (n_g + 1)D_0/2$  AND are CONSTANT under dyadic mesh refinement (comb  $3.3 \times 10^{-18}$ , arch  $1.5 \times 10^{-15}$ , pole  $2.0 \times 10^{-17}$ ) — NO mesh-cofinal ladder, NO transport,  $H_{\text{cof}}$  IS NOT NEEDED AND IS REPLACED AS THE TARGET ROUTE; variant B is the construction substrate of A (v749 T1.4b at rel dev 0.0), variant C the quantified density step (one-signed per-channel defect inside the rigorous interpolation envelope  $(D^2/8)\|F''\|K_{\text{ch}}$ , no wall margin anywhere); the anchor-only floor real ( $+1.77 \times 10^{-4}/+1.59 \times 10^{-4}$  — the R319 false-floor mechanism on own elements); r326 implemented the architecture in rh/lean/RH/Elementwise.lean (merged and green; lake with exactly 8 intentional sorry warnings, run\_rh -fast 75/75): sourceExact\_buildPrimeWindow PROVED (the wave-12 reviewer target “exactly one construction theorem” — SourceExact ELIMINATED as a free assumption from the extraction route; built windows provably carry their source), the comb stabilization elementwise PROVED (onset  $a_0(f)$  predefined from the element’s support alone, corpus gauge  $2\Lambda/\sqrt{n}$ , no mesh quantifier; honesty tie combRead\_eq\_window\_channel), weil\_nonneg\_of\_windowlocal PROVED (the EXTRACTION WITHOUT THE LADDER: one finite instantiation per element — Euclid anchor + native mesh) plus weil\_nonneg\_of\_bordered (composition through the named Prop BorderedCompressionBridge, no truth commitment); SORRY CENSUS  $5 \rightarrow 8$  with THREE NEW typed Level-C statements (arch\_elementwise\_stabilization + pole\_elementwise\_stabilization classical, specFamily\_sourceExact\_completion classical + definitional) — the wave-12 reservation “the Level-C distance appears

in no sorry” PARTIALLY DISCHARGED; the five legacy sorrys byte-identical. THE HONEST FORMULATION OF THE NEW CONNECTION (binding, verbatim): “two window-local lemmata (over the canonical family) + the elementwise architecture + cited classics (density, v912 continuity, the Weil criterion)  $\Rightarrow$  RH; open links: the arch/pole transcription (classical), the compression bridge, the source completion (definitional).” LEDGER: NEW PRIME.EXTRACTION.ELEMENTWISE.01 [E] (the R325 GO + the R326 implementation with the honest formulation verbatim); NEW PRIME.L2.RENYI3.MEASURED\_COMPOSITION.01 [O] (the r324 measured chain, the certified counts  $C_{NSC}/C_{NG}$ , the SOURCE anatomy, the coincidence-geometry direction); UPDATED PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] and PRIME.L2.RENYI3.PROVENANCE.01 [O] (the two-part provenance question executed —  $F_A$  de-black-boxed, r324-pre banked, the share route composed MEASURED), PRIME.REDTEAM.EXTRACTION\_AUDIT.01 [E] and PRIME.LSTAR.SUBORDINATION.01 [O] (the wave-13 Lean notes: census 5  $\rightarrow$  8 with the three-level assignment — Level C now named typed statements; lstar\_subordination byte-identical,  $L^*$  stays paused); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO change ships with this promotion — r326 is already merged and green (census 5  $\rightarrow$  8). Papers: the wave-13 diary section sec:elementwise-extraction in tfpt\_prime\_front, the wave-13 block + the wave-13 state notes in tfpt\_research\_contracts, the new section “Elementwise extraction and the measured composition (v970)” in rh/paper/rh\_program.tex (the Lean-theorem table with the proved kernel pieces prominent, the honest-formulation remark, the coincidence-geometry remark, the lane-status remark; the three-level remark and the Lean census section updated to 5  $\rightarrow$  8) + recompiled PDF; rh/README.md wave-13 block with the lane status (terminal: the MEASURED composition + the coincidence-geometry direction; extraction: the architecture stands, 3 typed Level-C rests;  $L^*$  paused; Lean: 8 sorries, all typed); rh/INVENTORY.json + make\_inventory.py extended additively (the v970 pin + the r324-pre/r324/r325/r327 consumption notes, all pre-existing pinned entries byte-exact; 110  $\rightarrow$  111 entries); run\_rh.py MODULES  $\rightarrow$  v955–v970, full RH suite green. The full-suite run\_all.py pass at 963 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate; NO RH claim anywhere in either direction)

2026-08-27 (CXXIX) (consolidation wave 12 — the red-team / fork morning: the certified content of discovery rounds 317–322 (incl. the r319 extraction-chain audit and the r320 Lean repair) graduates from the experiments tree into the load-bearing suite as v969\_forks\_and\_redteam.py (suite 961  $\rightarrow$  962, ledger 1131  $\rightarrow$  1133; wave-4 embedding convention: the FOUR discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — exception\_families 38/38 SPEC 04fbe5c0, indefinite\_fork 25/25 SPEC f2d98683, continuous\_coordinate 39/39 SPEC e68883ad (importing the embedded r317), antiphase\_sign\_law 25/25 SPEC 761b51d4 (importing the embedded r318); full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the exact two-sided concentration bracket  $q_{\max} \Phi_{H1} \leq \rho_2 \leq \Phi_{H1} = (F_A B)^2$  re-proved from scratch with both mutant directions caught, the U1–U3 red-team witnesses re-established in EXACT ARITHMETIC with no Lean call (U1: the  $B = -1$  window represents the empty spec against the terminal type while the retyped bridge refuses it and accepts the faithful  $B = 5/7$  witness; U2: the mesh-level-0 total node collision reduced exactly to the three rational  $e$ -facts

$2 \leq e$ ,  $3 \leq 2e$ ,  $e \leq 4$  on the certified series bracket  $685/252 \leq e \leq 685/252 + 1/35280$ , kill  $0 < 0$  exact; U3: the adversary  $Z_{\text{loc}} = |F| + 1$ ,  $\text{runs} = []$  demands  $(|F| + 1)^2 < 5/7$ , false for every  $F$  since  $(|F| + 1)^2 \geq 1 > 5/7$  exact), the r320 separation-discipline and canonical-split certificates (witness at anchor 2, atoms  $\{2, 3, 4\}$ , mesh level 4 — the first satisfiable level, via the integer facts  $2^7 < 3^5 < 2^8$  and the reduction  $1/8 < 2/5$  exact; the level-0 failure  $4/3 < 4$ ), the SIX SEALED VERDICT BARS as exact decision logic with tipping mutants at every live clause boundary, and the wave-12 composition gate; the r319/r320 Lean rounds consumed as REPORTS, their artifacts in rh/lean/ re-verified by lake build + the RH suite; single-module test green 22/22 gates (18 module-own + 4 pattern gates) in  $\sim 3$  s, byte-identical on re-run): THE RED-TEAM MORNING AND THE TWO FORKS FROZEN. (A) The extraction-chain audit and repair (r319/r320): the independent reconstruction of the chain from the two lemmata to Weil/RH found THREE KERNEL-CHECKED TYPE INCONSISTENCIES three levels above the boxes (U1 bridge $\times$ terminal via  $B = -1$ ; U2 bridge $\times$ L\* via the mesh-level-0 total node collision and  $p = X - 1$ ; U3 pair\_margin\_main forces its source predicate EMPTY — the r273 disease one level up) plus the mesh-vs-anchor COFINALITY SEAM (cofinal\_prime\_windows proves the anchor direction at fixed mesh, not the  $H_{\text{cof}}$  refinement direction — THE FALSE COFINALITY DIRECTION IS DOCUMENTED, NOT SOLVED); the honest chain verdict is BINDING: “two lemmata  $\Rightarrow$  RH” does NOT survive the literal reading — literally the two lemmata give ONLY window-local master positivity for the opaque, currently witness-less MainWindow; r320 REPAIRED the interface completely (bridge retype with  $u/B$  fidelity + separation discipline + budget\_pos; the additional arch/border finding sealed by the new opaque SourceExact; pair\_margin\_main retyped onto the canonical extraction with signRuns\_sum + canonical\_split PROVED; three permanent sorry-free guards, axiom census kernel-checked without sorryAx; the sorry-free witness witness\_represents; sorry census  $5 \rightarrow 5$  with two typed RE-TYPES, lake build green). (B) The reviewer adjudication (binding, adopted on every surface): the proof graph has THREE LEVELS and the two true holes are named WINDOW-LOCAL from wave 12 on — Level A (local window mathematics, PROVED:  $L^*(w) + T(w) \Rightarrow \text{MasterPositive}(w)$ , cleanly typed per r305/r320; “for every canonical finite window, L\* and terminal positivity imply augmented prefix positivity — this is a window-local theorem”), Level B (the universal window family — THE two arithmetic WINDOW-LOCAL holes:  $\forall w \in W_{\text{canonical}} : L^*(w) \wedge T(w)$ ), Level C (extraction to Weil/RH, OPEN: source-exact realization + admissible transport + convergence/stabilization in the correct directed order + density/continuity for the Weil criterion); two binding caveats: the sorry census does NOT measure the Level-C distance (the missing extraction architecture appears in NO sorry, because the correct theorem statements do not yet exist), and SourceExact — a good firewall — must NOT stand as a free assumption in a final proof: the named open Lean target is CanonicalPrimeWindow + exactly ONE construction theorem sourceExact\_buildPrimeWindow (load-bearing, not formalization polish); the typed per-layer reviewer assessment supersedes the wave-11  $4 \rightarrow 5$  scalar (finite dictionaries 9.5/10, formal window-local architecture 9/10, terminal mechanism 6/10, L\* mechanism 2/10, source/extraction architecture 2–3/10, complete RH proof path  $\sim 3/10$ , audit quality 9/10 — “stronger as a window-local research program; as a complete RH proof chain currently less far than previously assumed — the removal of a shortcut that did not exist”). (C) The fiber fork (r317/r321): fork (b) sealed whac-a-mole(kz53, kz83) BY CONTRACT (the blind gap rule recovers the REAL spike family  $B = \{kz55, kz67\}$

with finite certificates  $C_B = 1.0536$ , but class A stays empty at bar distance 0.017 and kz53/kz83 stay uncovered; the census find: the  $F_A$  top-3 are EXACTLY kz53/kz83/kz67 = 2.47/2.39/2.38 — one source-pure rank-local QMAX coordinate ranks the complete near-critical family on top, but as a CONTINUUM: the exception-family form is the wrong statement shape); fork (a) in the r317-sharpened form fired sliding-bound-go( $G_{SQ}$ ):  $\sum_j q_j^3 \leq 1.3056 F_A(w)^2 (\log m)^2 / m^2$  for  $m \geq 73$  with 0/39 mid-ladder test violations AND all four named r316/r317 violators INSIDE at reserves 7.0..9.6; the constant algebra-derived and source-pure ( $b = (\max_{\text{cal}} B)^2 = 1.1426^2$ , no target value in the calibration); honest:  $C_{\text{impl}} = 7.97$  is  $7.5\times$  looser than the r306 first-5 constant — FORM, not sharpness; the pure-algebra transfer does not close ( $B$  rising +0.122); scramble rejected only via the r317 class side condition; the provenance question now the two-part  $F_A/\text{qmax}$  split. (D) The base fork (r318/r322): r318 p2-main-specific — P1 BANKED AS LANGUAGE (the index statement a total restatement of  $L^*$ , 10/10 instances both truth values; the global inequality vacuous; the invariants world-blind), classical sign regularity DEAD on MAIN (premise-fails-on-main); r322 algorithm-artifact — the ( $D3, D4$ ) sign law and the r312 97.6/2.4 anatomy are LEAST-NORM-PROXIMAL DYKSTRA-BASIN properties, not psd-solution-set properties (sealed random starts carry modal pair  $(0, 2)$ ; the law not identity-forced — exact Fractions, positive forced value); the named residual: the SELECTION GEOMETRY. LEDGER: NEW PRIME.REDTEAM.EXTRACTION\_AUDIT.01 [E] (the audit + repair + the honest chain reading + the full reviewer adjudication verbatim, incl. “the false cofinality direction is documented, not solved”); NEW PRIME.L2.RENYI3.SLIDING\_BOUND.01 [O] (the theorem candidate with the disclosed looseness and the R324 lane note); UPDATED PRIME.L2.RENYI3.PROVENANCE.01 [O] (the fork executed both ways), PRIME.LSTAR.SUBORDINATION.01 [O] (wave-12 state + the binding lane pause:  $L^*$  internally PAUSED at allocation 0 %, reviewer decision — the selection geometry not pursued, reopening only under the six documented r322 conditions, the external memo candidate the two-weights sampling formulation  $\|J_N\| < 1$ ; the hole typed WINDOW-LOCAL), PRIME.LSTAR.ARCHITECTURE\_ADJUDICATION.01 [E] (wave-12 red-team note + the assessment-table supersession pointer); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO change ships with this promotion — r319/r320 are already merged and green (census 5, two typed retypes); the named open target CanonicalPrimeWindow + sourceExact\_buildPrimeWindow registered in README/paper; rounds R324 (terminal QMAX $\times M_2$ ) and R325 (extraction repair) IN FLIGHT, deliberately NOT consumed. Papers: the red-team/fork diary section sec:forks-redteam in tfpt\_prime\_front, the wave-12 block + the  $L^*$  wave-12 state in tfpt\_research\_contracts, the new section “Red team and the two forks (v969)” in rh/paper/rh\_program.tex (the honest-chain remark, the THREE-LEVEL remark with both caveats, the SourceExact-target remark, the sliding-bound theorem candidate, the basin-lesson remark, the typed reviewer-assessment table, the lane-status remark; the “two true holes” re-named window-local throughout) + recompiled PDF; rh/README.md wave-12 block with the lane status ( $L^*$  PAUSED; terminal sharpened to QMAX $\times M_2$ , R324 in flight; extraction fork R325 in flight; Lean: 5 sorries, the two true WINDOW-LOCAL holes) + the Level-C construction-theorem TODO; rh/INVENTORY.json + make\_inventory.py extended additively (the v969 pin + the r317/r318/r321/r322 consumption notes, all pre-existing entries byte-exact; 107 entries); run\_rh.py MODULES  $\rightarrow$  v955–v969, full RH suite green (86/86). The full-suite run\_all.py pass at 962 modules, the final bash build.sh audit and the manifest regeneration

follow as the joint exit gate; NO RH claim anywhere in either direction)

2026-08-26 (CXXVIII) (consolidation wave 11 — the architecture adjudication: the certified content of discovery rounds 305–316 (incl. 310b) graduates from the experiments tree into the load-bearing suite as v968\_architecture\_adjudication.py (suite 960  $\rightarrow$  961, ledger 1127  $\rightarrow$  1131; wave-4 embedding convention: the TEN discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — renyi3 27/27 SPEC 3bb365e153033ea8, fixed\_head full 28/28 / smoke 27/27 SPEC ec2bb0086f31ac73, block\_green 30/30 SPEC d51478504a645269, paired\_cone 33/33 SPEC f8d99877acb562b1, blockgreen\_nontriviality 34/34 SPEC fac7a8dfec9eb81f, blockgreen\_membership 32/32 SPEC 6c32f74962eef7af, renyi3\_proof\_fork 32/32 SPEC 6505dd10d5cce9fd, signed\_cubic\_flux 29/29 SPEC 841b319682af06c7, phi3\_functional 29/29 SPEC 92d35a3a4186d330, two\_regime\_bound 35/35 SPEC 5c28b12b1c3998c5; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the r306 HILL/LAGRANGE CHAIN (both Lagrange witnesses dev 0 EXACT, the power-mean slacks  $1/64$  and  $37/32768$  EXACT, the uniform equality case sharp, the flip mutant  $-1/64$  NEGATIVE caught) with the Rényi BRIDGE on the equality-tight rational witness (every comparison exact, the reversed-chain mutant caught), the r314 OPENING-FLUX LEMMA  $F_1 = g^3 - 3g^2 + 2g^3 \equiv 0$  on five rational witnesses plus the exact flux telescope (deviation the rational 0 on three toy blocks; the dropped- $2g^3$  mutant lands  $-2g^3 = -54/343$  EXACT), the r314 COLLISION COUNTING IDENTITY  $3p_1p_2 - 2p_3 = 8(n + 3n(n-1))$  EXACT on four mult-2 toy blocks (both sides 760; the mult-1 mutant breaks by the exact factor 8: 95 vs 760), an r311-CLASS EXACT RATIONAL FARKAS MINICERTIFICATE (PD target  $[[1, -1], [-1, 2]]$  in the span with the FORCED negative generator weight  $(2, 3, -1)$  — the sign wall in miniature; dual  $Y = [[0, 2], [2, 1]]$  with  $\langle Y, G_i \rangle \geq 0$  and  $\langle Y, Q \rangle = -2$  EXACT; the sign-flipped mutant target IS in the cone and the same  $Y$  fails on it — no overreach), the TEN SEALED VERDICT BARS as exact decision logic on the frozen record aggregates with tipping mutants at every live clause boundary, and the FOUR-LEVEL COMPOSITION GATE (a passing B2–B4 would hand the lane to architecture B, a firing r312 GO would keep Lane A open, a surviving two-regime bound would resolve the provenance — each tip changes the composed adjudication, the claim split is NOT hard-wired), no probe imports; single-module test green 30/30 gates (20 module-own +10 pattern gates) in  $\sim 50$  s, byte-identical on re-run up to the runtime line): THE ARCHITECTURE DAY FROZEN IN THE REVIEWER’S BINDING FOUR-LEVEL STRUCTURE. Level 1 — real formal theorems (Lean, rh/lean/, already built and green; NO Lean change ships with this promotion): lstar\_terminal\_implies\_master (r305:  $L^* + \text{terminal } q_N < 1 \Rightarrow$  full master positivity via Schur bordering + principal submatrices; augmented\_prefix\_positive and free\_window\_positivity now COROLLARIES — no third arithmetic axiom); the four closed Inertia theorems (psd\_base, positive\_prefix\_firewall incl. a from-scratch Sylvester criterion, sylvester\_pullback, half\_filling\_boundary); the real PrimeWindow construction (r310: nodes =  $\log p^k$ , weights =  $\Lambda(p^k)$  DERIVED via mathlib vonMangoldt); source exactness by rfl THROUGH derivation + mass conservation of ANY folding map (Finset.sum\_fiberwise — aggregation is a quotient, not a projection) + the STRONG stabilization form onto the Weil prime side (r310b); the r309 rank-one update identities; Hill monotonicity + the Rényi-3  $\Rightarrow N_2$  bridge; SORRY CENSUS  $9 \rightarrow 5$  with



clean bookkeeping — the two TRUE holes stand alone (lstar\_subordination, terminal\_positive\_main). Level 2 — certified finite statements: the Rényi-3 inequality  $\sum q^3 \leq 1.069 (\log m)^2 / m^2$  on 57 rungs (r306, growing reserve  $-0.322$ ;  $A = 1$  fails exactly at kz53/kz67 — polylog-SQUARED is the honest form); the fixed-head census 77/77 (r307, min  $2.19 \times 10^{-3}$ , amplification  $3.1 \times 10^4$ ); the block-Green identity on the finite objects (r308, toy  $t^2$  pin 4/7, f64  $4.0 \times 10^{-14}$ ) + strictness of the cone with EXACT RATIONAL FARKAS CERTIFICATES (r311, MINI#0 + SM1#0, 0/16 in-span PD samples decompose); paired-cone soundness on 20714 steps (r309, 0 violations, 56 % certified); fold multiplicity EXACTLY 2 + 93 % far-triple cancellation (r313); the signed cubic identity with VANISHING boundary term (r314, decided in exact Fractions ON THE REAL w9 DATA, every deviation the rational 0, f64  $4.5 \times 10^{-17}$  on all 57 rungs); the 21 small- $m$  certificates (r316,  $C_{\text{small}} = 1.0694$ ). Level 3 — negative architecture decisions (verbatim): fixed-head-dead (r307 head-grows: no fixed  $r \leq 8$ , best reserve  $6.5 \times 10^{-6}$  four decades under the bar — the one-atom anatomy a COORDINATE, not a DIMENSION); paired-cone-no-induction + paired-cone-world-blind (r309 B1–B4: B2 partial 56 %, B3 not reached, B4 missed — B does NOT overtake A); the r308 discriminator = the budget sign / a chordal restatement (r311: the  $\Delta$ -sparsity graph CHORDAL on every size, the trivial dual  $Y_t = e_t e_t^\top$  closes the stalls two-sidedly); coefficient-sign-wall — LANE A CLOSED AS A CONE LANGUAGE with the mechanism NAMED (r312: block-psd membership without rank-one-SDD membership, obstructed at the budget/border sign; Farkas mass on the border/budget row,  $t$ -share 0.583/0.725; the entire obstruction in 2.4 % SDD-forbidden negative cross-mixings); floquet-expanding (r313: med sr 1.0600, 0/57 contracting — no transfer-operator contraction form remains open at chain level); triple-type-majorants-wrong (r313); phi3-all-blind (r315: the entire break on kz55+kz67, FCIX 0.955/0.915 vs 0.629 — exactly where the flux cancellation dies); two-regime-dead (r316, with the anatomy: the obstruction family cuts ACROSS the FCIX stratum, kz53 at bulk-normal FCIX; the r306 first-5 protocol was LOAD-BEARING; kz55/kz67 near-one-block, TOP1 0.558/0.785 vs 0.18). Level 4 — open mechanisms (typed): the constructive PSD-G rule (closed as a lane, open as a question), the signed cubic flux theorem, the quantitative source functional, the crossing\_budget mathlib gap, the definitional source bridge,  $L^*$ , the terminal. LEDGER: NEW PRIME.LSTAR.ARCHITECTURE\_ADJUDICATION.01 [E] (the four-level structure verbatim); NEW PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01 [E] (ONLY the exact update identities: determinant dictionary, Sherman–Morrison chain,  $\pm(\eta \pm e_t)$  border split, local Schur reserve, signed source updates — per the reviewer’s binding claim split NOT one TRANSFER title); NEW PRIME.SOURCE.TENSOR.MECHANISM.01 [O] (the open question whether a source-pure cofinal control arises from them, registered AFTER r313/r316 with the documented negatives); NEW PRIME.L2.RENYI3.PROVENANCE.01 [O] (the pointwise GO as a certified finite statement + the open provenance question; trilogy state: identity exact, functional blind, two-regime dead; the R317 fork verbatim: a near-critical family coordinate OR a certified exception family — reviewer adjudication pending); UPDATED PRIME.LSTAR.CLOSED\_FUNCTIONAL.01 [O] (the Lane-A closure note), PRIME.LSTAR.SUBORDINATION.01 [O] (wave-11 state: the two formal holes, the source interface, the architecture adjudication), PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 (Lean restructuring note: sorries  $9 \rightarrow 5$ ); no other marker moves, mincut base 4 / refined 5 UNCHANGED. The universality-class framing typed: the base is fine-metric (metric-only, twin invariance everywhere), the fiber is

recursive (epstein holds, scramble breaks) — shared algebra, not necessarily a shared mechanism; the reviewer maturity raise  $4 \rightarrow 5$  (formal architecture 9.5) documented as a REVIEWER ASSESSMENT, not a machine-checked fact. Papers: the architecture-adjudication diary section `sec:architecture-adjudication` in `tfpt_prime_front` (the wave-10 “in flight” spots resolved historically), the wave-11 block in `tfpt_research_contracts`, the new four-level section “The architecture adjudication (v968)” in `rh/paper/rh_program.tex` (one subsection per level, the Closure/Source theorem tables, the universality-class remark, the no-go catalog extended by eight wave-11 rows, the Lean status brought to the 5-sorry census) + recompiled PDF; `rh/README.md` wave-11 block with the lane status (A closed as cone language; B closed, identities banked; C parked; fiber: the r306 GO stands, provenance open, R317 fork at the reviewer; Lean: 5 sorries, two true holes); `rh/INVENTORY.json` head-synced (100  $\rightarrow$  101 entries: the v968 pin + the r306–r316 consumption notes, all pre-existing entries byte-exact); `run_rh.py` MODULES  $\rightarrow$  v955–v968, full RH suite green. The full-suite `run_all.py` pass at 961 modules, the final `bash build.sh` audit and the manifest regeneration follow as the joint exit gate run by the coordinator; no round is in flight at this cut — R317 waits on the reviewer fork; NO RH claim anywhere in either direction)

2026-08-26 (CXXVII) (consolidation wave 10 — the cascade closure and the documented lane stop: the certified content of discovery rounds 301–304 graduates from the experiments tree into the load-bearing suite as `v967_l2_cascade_closure.py` (suite 959  $\rightarrow$  960, ledger 1126  $\rightarrow$  1127; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed `-smoke` stage with the printed SPEC SHA-256 hashes pinned and gated — `neff_target` 32/32 SPEC 6f8cc40420c57165, `unif_target` 30/30 SPEC 36df942427d45fe4, `atom_target` 26/26 SPEC 375e9f2b8e6c1697, `shortrange_law` full 37/37 / `smoke` 32/32 SPEC 2cc5d23fdaa75346; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 — the r303 MARGIN-INVARIANCE ALGEBRA (all four target margins  $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} = 3/50$  EXACT on the rational slope set through the three gated product identities r300/r301/r302; the halved-need mutant breaks the  $D \leftrightarrow \text{NEFF}$  invariance by EXACTLY  $\text{sl}_{L1} = 1/5$  — the exact root of the refuted  $\frac{1}{2}$ -conjecture; the record-decimal chain 0.055/0.908/0.094/0.888 EXACT with the disclosed 1/1000  $m_{\text{ATOM}}$  rounding artifact; the  $\sigma$  decomposition  $0.198 = 0.055 + 0.143$  EXACT; the  $\frac{1}{2}$ -conjecture refutation 0.099 vs 0.055, dev 0.044  $>$  the sealed 0.005), the r302 COHERENCE IDENTITY  $(1 + CV^2) \text{surv}^2 n_{\text{eff}}^{\text{atom}} = n_{\text{act}} \chi$  on two rational toys with the r301 count identity attached (the lag-0 double-count mutant breaks by 5/7 EXACT, the swapped- $D/Q$  mutant by 48/35 EXACT), the r304 CHI-LAG DECOMPOSITION  $\chi = 1 + 2 \sum_k T_k/Q = D/Q$  on both toys (5/7 and 5/19 EXACT; factor-1 mutant 1/7 EXACT), the r304 ZERO-SUM TAUTOLOGY as a gate  $(1 + 2 \sum_{\text{all}} \rho_k \equiv 0$  for mean-removed profiles; factor-1 and uncentered mutants 1/2 and 3/4 EXACT; the truncated net covariance  $NC(1) = 0.556$  EXACT,  $NC(16)$  recomposing to 0.710 within the disclosed 2/1000 rounding of the pinned  $0.712 < 1$ ; the PERIOD-4 COMB CERTIFICATE: every strong lag  $j \in (4, 16]$  sits at  $j = 4m$  (+) or  $j = 4m+2$  (–) and the sealed  $k_0$  rule finds NO  $k_0 \leq 8$ ), the  $-/-/+ /+$  SIGN-PATTERN CERTIFICATE with the kz18/kz23 Fractions pins and the honest rho-sign 41/42, the four sealed verdict bars as exact decision logic with tipping mutants at every live clause boundary, and the lane-stop composition

gate, no probe imports; single-module test green 27/27 gates (23 module-own +4 pattern gates) in 1.3 s, byte-identical on re-run): **THE CASCADE RETYPED, THE LANE STOP DOCUMENTED** — *the rejections and retypings ARE the frozen content.* (A) **THE r301 NEFF ANATOMY** (the wave-9 rest executed):  $n_{\text{eff}} = n_{\text{act}}/(1+CV^2)$  **EXACT** (ward  $\leq 6.8 \times 10^{-16}$  on 47 worlds) with the **PERFECT** count link  $n_{\text{act}} = m$  on 42/42 — the effective-carrier count **IS** the constructive level-2 block count (+1.002); the whole Rényi order family grows at **ONE** exponent (+0.963/ +0.926/ +0.894/ +0.738: **ECHT** anti-concentration, not a broadening tail); the margin is **jackknife-STABLE** (0/42 below **NEED**, no margin-fragile); sealed verdict **neff-split(B1|B3)**: the growth statement became the **BOUNDEDNESS** statement **unif-target** (slope  $(1+CV^2) \leq +0.094$ , measured +0.039, margin 0.055); **neff-derived** missed **ONLY** the sealed **CV-flat** clause (+0.039 > 0); **B2** closed honestly (the  $|dc|$ -weighted discrepancy is  $10\times$  the raw  $D_{\text{rank}}$  and near-flat: the r300 bridge lived on the **RAW** positions). (B) **THE r302 UNIF ANATOMY** (the **FIRST DERIVED** of the lane): the  $CV^2$  rise is **STRUCTURELESS**, the normalized block profile is **profile-stationary** (KS  $0.043 \ll 0.125$  over a  $3\times$  depth range) with an **EXACT**  $1/N$  transient onto  $m_2^\infty = 1.973$  (held-out dev 0.002; the r299 pointwise-cconv negative was the wrong convergence notion — convergence in **DISTRIBUTION**); the r301 depth flag resolved **EXACTLY** per half ( $0.982 = 0.929 + 0.053$ ,  $0.802 = 1.030 - 0.228$ : the  $CV^2$  head, not the count); the coherence identity **EXACT** with  $\chi = 0.63$  **DESTRUCTIVE** and falling ( $\rho_1 = -0.22$  its atom-level mechanism)  $\Rightarrow$  **unif-derived(B1+A2)** under the sealed clause, **ALL** hypotheses typed **MEASURED**; the relocated rest: atom-target (+0.888 needed, +0.942 measured). (C) **THE r303 REGRESS AUDIT** (the retyping round): **THE THREE 0.055 MARGINS ARE ONE ALGEBRAIC NUMBER** —  $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} = +0.0547$ , invariance devs  $\leq 9 \times 10^{-16}$ ; the core is  $S = \sigma^* - \text{sl}_D = +0.0547$  (the r299 **DIAG** margin); the  $\sigma$  level is **NOT** the core ( $\sigma^* - \sigma = +0.1976 = S +$  ratio surplus +0.1429 **EXACT**) and the  $\frac{1}{2}$ -conversion conjecture is **REFUTED**  $\Rightarrow$  regress-confirmed: the cascade **r297→r302** is **RETYPED** as an exact reduction dictionary (coordinate finding), not six proof steps; **THE HARD REVIEWER RULE** (binding from r303 on, verbatim): *a round counts as progress ONLY if it adds NEW information, not if it re-expresses the same slack in a new coordinate.* **THE FIRST CAUSAL COORDINATE**: the mechanism ladder on 1008 sealed marginal-preserving builds is **MONOTONE** in  $\rho_1$  ( $\chi$   $0.630 \rightarrow 0.764 \rightarrow 1.029 \rightarrow 1.342$ ; end margin  $+0.055 \rightarrow +0.057 \rightarrow +0.032 \rightarrow -0.044$ : **THE SIGN FLIP KILLS THE TARGET INEQUALITY**) — but  $\rho_1$  matching misses the destructive  $\chi$  level by 0.134  $\Rightarrow$  mixing-insufficient (the within-block structure beyond lag 1 named as the gap); bycatch theorem-grade:  $n_{\text{eff}}^{\text{atom}} = (L_1^a)^2/Q$  is a pure **MARGINAL** functional (invariance  $1.0 \times 10^{-15}$  on all 1008 builds — stationarity and mixing complementary, not redundant); rho-sign honest 41/42 (one rung positive: no raw per-rung theorem; Fractions-exact negative certificates kz18/kz23). (D) **THE r304 SHORT-RANGE LAW** (the reviewer stop case): the global lag profile  $\rho_{1..16}$  of the  $dc$  profile is a **STABLE, WORLD-SPECIFIC PERIOD-4 COMB** (med  $-0.222/-0.140/+0.089/+0.130$ , then strong lags at every  $k = 4m$  and  $4m+2$  up to 16; halves-stable 3/3), **NOT** a decaying tail — no  $k_0 \leq 8$  exists: longrange-structure + law-longrange fire on the **LETTER** of the sealed rule; the reviewer condition **SPLITS**: the net covariance  $NC(16) = 0.712 < 1$  **HOLDS** (net-negative) while summability-with-small-tail **FAILS** ( $SUM(16) = 1.563$ ); only the **TRUNCATED NC** carries content (the full sum is the gated zero-sum tautology); **THE STRUCTURAL COUNTERPOINT** (exact, ward  $4.7 \times 10^{-16}$ ): the  $\chi$ -relevant

within-block structure IS short-range (shares die by  $k \sim 4$ ; the r303 miss 0.134 sits at  $k \leq 3$ , dominated by lag 2 +0.181) — the long-range comb is INVISIBLE to  $\chi$ ; lag-8 matching reproduces the  $\chi$  LEVEL (0.652 vs 0.630,  $|d| = 0.022 \leq 0.05$ ) but breaks the SLOPES (0.028/0.027 > 0.02) — the exact COMPLEMENT of r303(a): the mechanism is TWO-SCALE (level from the short lags, slopes from the rung-wise  $\rho_1$  trend), no lag-matching family reproduces both; the graduated reviewer ladder is MONOTONE in  $\chi$  but NOT in the margin (honest negative: the r303 kill came from per-rung flipped targets — the margin responds to the TREND, not the level). THE CONSEQUENCE (the sealed reviewer stop case, map (iii)): THE GLOBAL-PROFILE MIXING ROUTE OF THE L2 LANE IS DOCUMENTED CLOSED — the honest stop state: L2 generic  $\Leftrightarrow$  anti-concentration of an explicit block field with LONG-RANGE (period-4 comb) structure; return later with new tools. What stands honestly: the exact identity dictionary (count, coherence,  $\chi$ -lag), the two-scale split,  $NC < 1$ , the  $-/-/+ /+$  sign pattern (Fractions-exact kz18/kz23), the marginal-functional invariance of  $n_{\text{eff}}^{\text{atom}}$  ( $10^{-15}$  on all 2100 + 1008 builds). LEDGER: NEW PRIME. L2.CASCADE\_CLOSURE.01 [E] (fine types Identity/Formal for the S0 skeleton + Numerical/Measurement for the ladders and censuses; the rejections and retypings ARE the frozen content); PRIME.L2.NEFF\_TARGET.01 [O] RETYPED to the documented stop state (the rest is no longer NEFF/UNIF/ATOM — all one slack  $S = +0.0547$  — but the stop state above, with the hard reviewer rule verbatim in the claim); PRIME.L2.REDUCTION\_CHAIN.01 [E] retyping note (the exact identities STAND, the reading is refined: the chain is a dictionary, not six proof steps); no other marker moves, mincut base 4 / refined 5 UNCHANGED. LEAN (rh/lean): NO new statements — the stop carries no new exact target form (the period-4 comb and the two-scale split are measured ladder aggregates, not exact finite identities, and a universally quantified ladder form would be refutable — the r273 guard pattern of RH/Counterexamples.lean); the hole stays lstar\_subordination. Papers: the cascade-closure diary section sec:cascade-closure in tfpt\_prime\_front (the wave-9 “r301/r303 in flight” spots resolved historically), the wave-10 block + the NEFF\_TARGET stop-state update in tfpt\_research\_contracts, the new “The cascade closure (v967)” section in rh/paper/rh\_program.tex (the margin invariance as its own theorem, the retyping as a methodological lesson — the sealing discipline caught the reparametrization trap; the causal  $\rho_1$  coordinate + the two-scale finding, the period-4 comb as the named open object, the documented lane stop with the clean equivalence state) + recompiled PDF; rh/README.md wave-10 block with the lane status update (L2: documented STOPPED; L\*: dormant since r296; both edges with a clean state); rh/INVENTORY.json + make\_inventory.py extended additively (89  $\rightarrow$  90 entries: v967 pin + the r301–r304 consumption notes, all pre-existing entries byte-exact); run\_rh.py MODULES  $\rightarrow$  v955–v967, full RH suite green. The full-suite run\_all.py pass at 960 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; no round is in flight at this cut; NO RH claim anywhere in either direction)

2026-08-26 (CXXVI) (consolidation wave 9 — the DENS fork and the L2 reduction chain: the certified content of discovery rounds 296–300 graduates from the experiments tree into the load-bearing suite as v966\_l2\_reduction\_chain.py (suite 958  $\rightarrow$  959, ledger 1124  $\rightarrow$  1126; wave-4 embedding convention: the five discovery probes embedded BYTE-EXACT and executed verbatim in their

sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — dens\_identity 39/39 SPEC ffb413c80cddc1d3, vdc\_chain\_provenance 27/27 SPEC e42a76eb8ae71422, window\_border\_transfer 28/28 SPEC 05e831bec627aee6, fejer\_decay 32/32 SPEC f432e94450ad545d, diag\_target 31/31 SPEC 55218b5db8ece163; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the r297  $\sigma^*$ -composition arithmetic ( $\delta'(\sigma^*(\text{pad})) = \text{pad}$  EXACTLY for every rational slope pair, the pad-dropped mutant composes to 0: the r297-m2 sharpness as an identity), the r298 decomposition identity  $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$  on the exact positional Fejér block form (the hand toy  $3 = 1 + 2$  with attribution  $1/1$ ; the  $|\Delta|$  and wrong-weight mutants break by exactly 4 and  $2/3$ ), the r300 participation identity  $D \times n_{\text{eff}} = L_1^2$  with ratio multiplicativity and the machine-checked DIAG $\leftrightarrow$ NEFF equivalence (record decimals  $-0.571$  in both decompositions, NEFF need  $0.908$  / margin  $0.055$  exact in BOTH coordinates; the signed-sum mutant breaks by exactly  $8/3$ ), the r300 kernel envelope  $F_H(s) \leq \min(H, 1/(Hs))$  exact-rational with equality witnesses on BOTH branches ( $H = 2$ :  $s = 0, \frac{1}{2}$ ;  $H = 3$ :  $s = 0, \frac{1}{4}$ ), and the five sealed verdict bars as exact decision logic with tipping mutants at every clause boundary, no probe imports; single-module test green 28/28 gates (18 module-own +5 pattern gates) in 1.3 s, deterministic on re-run): THE DENS FORK CLOSED HONESTLY, THE  $\delta' > 0.21$  TARGET OF THE GENERIC HALF REDUCED TO ONE INEQUALITY. (A) THE r296 DENS FORK (honest negative, the lane closed): the reviewer hard fork on the r292 curvature axis lands dens-world-blind under the sealed rules — THE COUPLING NUMBER  $\cos(e_{\text{top}}, \nabla \lambda_{\text{max}}) = +0.394$  sits BELOW the sealed 0.40 bar (noise max 0.331; T1 misses PART by 0.006, honest), the  $\lambda_3$  control couples HARDER ( $-0.574$ : a near-crossing eigenvalue-BAND property, not top-specific), every arithmetic candidate of the hash-sealed library stays  $\leq 0.38$ , and the only above-chance structure (the 4-dim moment-gradient subspace, 0.825) HOLDS on both dead controls (0.571/0.603) — construction-adjacent low-moment smoothness, the sealed world-downgrade fires (disclosed tension: the absolute collapse bar is conservative for subspace members, rule unmoved); PER THE PRE-ADJUDICATED REVIEWER FORK the DENS-identity lane is CLOSED and the resources move to the L2 front. (B) THE r297 TARGET INEQUALITY:  $\delta'(F2) > 0.21$  is EQUIVALENT to  $\sigma := \text{slope}(S_F/M_W^2) \leq \sigma^* = 2(\text{sl}_{c2} - 0.21) - \text{sl}_{\text{pref}} = -0.516$ ; measured  $\sigma = -0.714$  — THE TRUTH CLEARS THE TARGET WITH 0.198 SIGMA MARGIN (0.099 in  $\delta'$ ): *what is missing is not room, it is provenance*; all three classical provenance routes VALID 47/47 but SLOPE-FAIL  $\Rightarrow$  chain-provenance-partial(B3, slope miss 0.307 EXACT), with THE SURPRISE: the chain is EXACTLY ORTHOGONAL w.r.t. the WINDOW measure (devs 0.000 world-blind — the Fractions-proved Parseval sum rule attaches verbatim); the magnitude wall named (+0.22.. +0.24 imbalance growth). (C) THE r298 EXACT TRANSFER:  $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$  EXACT ( $8.8 \times 10^{-16}$  of scale, 47 worlds), SIGN-PRESERVING (the  $|\Delta|$  mutant breaks by  $\sim 1.7$  of scale LOUD — the r297 magnitude error certified as a must-fail); the linear window zero MEASURED (the drive is 100 % transfer at the linear level); sealed verdict transfer-dominant: the window main term is EMPTY (med  $-3.94$  dec, slope  $-1.386$ ), the cross term negligible ( $-1.4 \times 10^{-4}$ )  $\Rightarrow S_F \approx B(P\Delta, P\Delta)$  — the vdc input IS the Fejér energy of the difference measure itself (share 0.99,  $T > 0$  on 42/42; the two r297 gap halves unified, sp  $+0.69$ , broad-based). (D) THE r299 DECAY SPLIT: the exact spectral representation  $B = (1/L) \sum_k F_H(\theta_k) |\hat{\Delta}(k)|^2$  is LOWPASS (0.93/0.04/0.02); THE STRUCTURAL FIND: FULL-SUPPORT

overlap on 42/42 rungs —  $\Delta_{\text{fresh}} \equiv 0$ ,  $\Delta$  IS a pure c-value difference measure on ONE shared node set with NO pointwise convergence (cconv 0.86, +0.045: the decay is aggregate cancellation); the ET/Abel composition FAILS LOUD (composed +1.948 = 1.504 + 0.444 EXACT vs  $\sigma^* = -0.516$ ; mass-target missed by 2.46 — the r297 magnitude wall recurs at the frequency level); the live edge fires:  $\text{sl}_D = -0.571 \leq \sigma^*$  (margin 0.055) with FALLING ratio  $B/D$  (-0.168)  $\Rightarrow$  decay-split (diag-target + ratio flatness; cvalue-target on the B3 side); the O-sign class separates MAIN (o-pos, reinforcing,  $O < 0$  on 13/42 only) from both dead controls (o-neg) — the first world-separating sign class of the L2 lane, disclosed. (E) THE r300 DIAGONAL ANATOMY:  $\text{sl}_D = 2\text{sl}_{L1} - \text{sl}_{\text{neff}} = 2(+0.196) - (+0.963) = -0.571$  EXACT (dev  $6.7 \times 10^{-16}$ ) — THE ENTIRE DIAGONAL DECAY IS PARTICIPATION GROWTH (‘many small instead of few large’ as a measured identity;  $n_{\text{eff}} \sim N$ , sp +0.96); BOTH magnitude factorizations fail exactly as r297 predicted (B1 chain-norm CS  $-0.592 + 0.208 = -0.384$  EXACT AND the  $|dw|$  identity census BREAKS — the proven sum rule does NOT attach to the difference measure, the chain-norm shortcut closed honestly; B2  $\max \times L_1$   $-0.542 + 0.196 = -0.346$  EXACT, miss 0.170 — the max is fully atom-controlled,  $m_x/\max_{\text{atom}}$  med 1.07: the irreducible loss is the fill decay  $-0.225$ , INVISIBLE to every  $\max \times \text{mass}$  bound); sealed verdicts diag-split(B3/neff-target) + ratio-bounded-structural (the exact kernel-envelope majorant  $R_{\text{env}}$  med 1.61 FALLS at  $-0.122$  on 47/47 — the ratio half of the r299 rest pair settled at the structural level; the lobe-width heuristic refuted, sp  $-0.47$ ); the FILL class separates MAIN (fill-low) from both dead controls (fill-high) — the second world-separating class. THE CHAIN STATE (the frozen wave content):  $[\text{neff-target}] \Rightarrow \text{sl}_D \leq \sigma^* \Rightarrow \sigma \leq \sigma^* \Rightarrow$  the r297 target  $\Rightarrow$  the v964-S0 vdC theorem  $\Rightarrow \delta' > 0.21$  on the generic half (exceptions: 6 via the r287 F2 certificates, kz15 exact-finite via r270) — the L2 lemma stands reduced to ONE inequality: neff-target,  $\text{slope}(n_{\text{eff}}) \geq +0.908$  (measured +0.963, margin 0.055; the  $D_{\text{rank}}$  bridge sp  $-0.81$  real but correlational; round 301 in flight, NOT consumed). LEDGER: NEW PRIME.L2.REDUCTION\_CHAIN.01 [E] (fine types Identity/Formal for the S0 skeleton + Numerical/M Measurement for the route tables; honest negatives VERBATIM: the magnitude-route catalog r297/r299/r300, the ET failure +1.948, no c-value convergence, the DENS closure); NEW PRIME.L2.NEFF\_TARGET.01 [O] — the one remaining inequality with the named correlational bridge and the sealed kill list; CONTRACT UPDATES: PRIME.LSTAR.CLOSED\_FUNCTIONAL.01 [O] (the DENS fork closed — front (2) answered negatively at the identity level, the lane routed to L2), PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] (the L2 front reduced to neff-target); no other marker moves. LEAN (rh/lean): NO new statements — neff-target is a measured ladder-slope aggregate (a halves log-slope estimator over 42 rungs with a threshold derived from measured slopes), not an exact finite identity and not a matrix theorem, and a universally quantified ladder form would be refutable exactly like the pre-r273 universal edge forms (the machine-checked guard pattern of RH/Counterexamples.lean); the hole stays lstar\_subordination. Papers: the L2-chain diary section sec:l2-chain-arc in tfpt\_prime\_front, the wave-9 block + the VDC\_LEMMA/CLOSED\_FUNCTIONAL state updates in tfpt\_research\_contracts, the new “The L2 reduction chain (v966)” section in rh/paper/rh\_program.tex (the DENS closure paragraph, the reduction diagram, the exact identities as theorem environments — target equivalence, sign-preserving transfer, participation identity, kernel envelope — the magnitude no-go catalog as a table, the two world-separating classes as a remark, the NEFF target as an open problem) + recompiled PDF; rh/README.md wave-9

block (r301 in flight noted); rh/INVENTORY.json + make\_inventory.py extended additively (84 → 85 entries: v966 pin + the r296–r300 consumption notes; the sync strictly additive next to the in-flight r301 worker); run\_rh.py MODULES → v955–v966, full RH suite green (65/65 items, wall 489 s, 81/81 pinned entries byte-identical, ALL CHECKS PASSED). Mincut base 4 / refined 5 UNCHANGED (a reduction-plus-closure set moves no edge); the full-suite run\_all.py pass at 959 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; the parallel r301 NEFF worker is NOT consumed; NO RH claim anywhere in either direction)

2026-08-26 (CXXV) (consolidation wave 8 — the  $L^*$  curvature arc: the certified content of discovery rounds 290–295 graduates from the experiments tree into the load-bearing suite as v965\_lstar\_curvature\_arc.py (suite 957 → 958, ledger 1122 → 1124; wave-4 embedding convention: the six discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — profile\_functional 32/32 SPEC f953dd71f483d0fa, ridge\_anatomy 30/30 SPEC bb512c17927818f5, curvature\_form 36/36 SPEC 050821ff05e1be74, metric\_reconciliation 43/43 SPEC 33c44cc697d4c3df, f10\_stability 42/42 SPEC 88c6fd1e3b5a5d39, f10\_sp\_hardenig 44/44 SPEC 4d7d809597d69afb; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the r292 polarization identity  $H(u, v) = [d^2(u+v) - d^2(u-v)]/4$  with the wrong-prefactor must-fail, the r291 budget telescope (exact partition additivity of the side-selected first-order budget) with the dropped-atom must-fail, the r293  $\alpha = 1$  simple-zero doubling law ( $|h(t_0+2d)|/|h(t_0+d)| = 2$  exactly, the double-zero mutant gives exactly 4 LOUD), and the sealed r294/r295 decision bars as exact Fractions logic with tipping must-fails at every clause boundary, no probe imports; single-module test green 25/25 gates (19 module-own +6 pattern gates) in 1.6 s, deterministic on re-run): THE GEOMETRY MEASURED, THE FUNCTIONAL SEARCH SEALED HONESTLY NEGATIVE, F10 NOT PROMOTED. (A) THE r290 BASIN GEOMETRY: the working set around MAIN is a SOFT-SHOULDERED, STRONGLY ANISOTROPIC TUBE in the exact  $\theta_{eq}$  LAG coordinate (kill fraction 0.38/0.62/1.00 at  $5 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$ ; NEAR radius  $\sim 5e-4..2e-3$  gap-equivalent); the world-directed axes kill 5–50x EARLIER than random (onsets  $4 \times 10^{-5}..2.3 \times 10^{-4}$  vs the jitter threshold  $10^{-3}..3 \times 10^{-3}$ ); the SMOOTH (arithmetic) axis is a privileged killer (onset  $1.36 \times 10^{-4}$ ) yet ORTHOGONAL to the first-order wall gradient ( $\cos = -3 \times 10^{-5}$ ) — collective, not gradient-linear; the r280 ridge is REAL (min  $C = 185$  at factors 1..8, first death at 16). (B) THE r291 RIDGE ANATOMY: the lift is a FIRST-ORDER BUDGET phenomenon — one threshold in  $(1.280, 1.291]$  separates lift from no-lift over ALL 18 matched-dose cases (PRIME/HEAD/XIPOS lift;  $k_{min} = 9$ , top atoms 2, 3, 5, 13, 11, 4, 29, 7, 89; exactly ONE overdrive retraction TOP6 at factor 8, margin 9.09); NO FIXED POINT (plateau min  $C = 185$ , never 186, the axis decoheres); LIFT MAIN-SPECIFIC (EPSTEIN first-order flat,  $\theta_{up} = 5.03 \times 10^{10}$ , never lifts); SMOOTH COLLECTIVE-QUADRATIC ( $d^2 = -23.3$  vs  $-0.98$  random, ratio 23.7). (C) THE r292 CURVATURE SPECTROSCOPY: the wall Hesse form is NEGATIVE on all 29 sealed directions and RANK-1 DOMINATED in the L2 spectral metric (92.5 % of  $\sum |\lambda|$ ,  $\lambda_{top} = -0.418$ ) with the top axis a pure on-support DENS combination — NOT the SMOOTH axis ( $|\cos| = 0.07$ ) and NOT the ridge (flattest sector, L2-rank 28/29); the  $m^*$  threshold and the TOP6 retraction are near-flip nonlinearities INVISIBLE to jets up to third order ( $q_R$  median

$-0.911$  = second-order ASSIST;  $B_2 = 27.6/B_3 = 150.9$  over-predict); EPSTEIN second-order structureless ( $5.4 \times 10^{-15}$ ). (D) THE r293 METRIC RECONCILIATION: three metrics sealed BEFORE any evaluation (distortions 11/59/124, largest sector DENS); F10 beats the baseline OF ITS OWN METRIC for the first time in four rounds (home margin  $+0.024 = 0.884 - 0.860$  vs the mixed r292 margin  $-0.023 = 0.884 - 0.907$ : mix-is-cause, both RE-DERIVED in the gates) and carries partial information ( $+0.423$  test /  $+0.826$  train); ALL 8 selected flips are SIMPLE  $h_{184}$  ZEROS ( $\alpha = 1.000 \pm 0.003$ , 225 gated wall evaluations) — but mstar-no-law ( $m_{\text{dir}}^*$  scatters 1.1295..1.3931 straddling the r291 bracket, a fixed-dose artifact; spreads 3.0/3.0/3.8/10.2 vs bar 1.25); the TOP6 retraction IS a SECOND simple  $h_{184}$  zero at  $f_{\text{ret}} = 7.107$  (lift window 1.226..7.107), on the side the monotone budget prognosis is blind to. (E) THE r294 STABILITY ADJUDICATION (honest negative): the  $|\text{sp}|$  win of F10 replicates on ALL FIVE fresh seed-disjoint corpora (margins  $+0.003..+0.115$ , wins 5/5 re-derived) BUT the partial channel does NOT (median  $0.299 < 0.3$ , 2/5)  $\Rightarrow$  sealed verdict f10-fragile, PROMOTION NO; robustness typed train-robust ( $\sigma = 0.0101$ ) + rank2-carries (top DENS eigenaxis alone  $+0.855$ , DENS share 0.989 — the mechanism hint) + step-stable; window transport a KNIFE EDGE (w11  $+0.002$  / w7  $-0.009$ ) with DENS MECHANISM CONSTANCY (0.795/0.989/0.834); l2-not-dens. (F) THE r295 HARDENING ADJUDICATION (honest negative): TWENTY fresh corpora (192 forbidden seeds enumerated, zero overlap;  $H_{\text{tr}}$  hash gate 3447ed198a56) — F10 wins 14/20 (margin median  $+0.028$ , min  $-0.079$ ): the sealed three-clause HARDENED bar fails on TWO clauses  $\Rightarrow$  f10-sp-majority — the r294 5/5 was itself TOP-OF-DISTRIBUTION, exactly as the r293 partial was; a DOCUMENTED REGULARITY (19/25 combined census), NOT a theorem; PARTIAL20 median  $+0.279$ , IQR [ $+0.182, +0.401$ ]; partial-family-map: NO family STRONG (PATH  $+0.156$  / WORLD  $+0.245$  / FRAC  $+0.067$  NULL / DENS  $+0.104$ , pooled 2921); r293-luck (matched median  $+0.346$ : gain clause passes, level clause fails by 0.004); l2-via-conservation near-tautological ( $\eta_0$  share  $2.9 \times 10^{-30}$ ). THE FROZEN CONSEQUENCE: after FIVE sealed class families (local scalars best 0.263 / non-local best 0.471 / two-form mixed  $0.884 < 0.907$  / metric-reconciled F10 fragile / hardened  $|\text{sp}|$  form majority-only) NO closed predictive profile functional of the working set exists over the honest bar — F10 stays UNPROMOTED, the sealed r294/r295 rejections ARE the frozen content. LEDGER: NEW PRIME.LSTAR.CURVATURE\_ARC.01 [E] (fine types Identity/Formal for the S0 skeleton + Numerical/M Measurement for the geometry census; honest negatives VERBATIM: all-functionals-blind (5 class families), f10-sp-majority(14/20, kein Satz), mstar-no-law, r293-luck); NEW PRIME.LSTAR.CLOSED\_FUNCTIONAL.01 [O] — the closed-functional question with the sealed kill catalog and the named open fronts (loss-corpus forensics K05/K07/K19, the rank-2 DENS core, why L2 substantially); CONTRACT UPDATE: PRIME.LSTAR.SUBORDINATION.01 [O] carried at the wave-8 state (the characterization question functionally negative-sealed, the geometry measured; no round in flight at this cut). LEAN (rh/lean): NO new statements — the wave carries no new formal target form, the hole stays lstar\_subordination. Papers: the curvature-arc diary section in tfpt\_prime\_front (the wave-7 “r290 in flight” spots resolved historically), the wave-8 blocks in tfpt\_research\_contracts, the new “The curvature arc (v965)” section in rh/paper/rh\_program.tex (basin geometry, budget mechanism, curvature-spectroscopy table, the five-family functional kill catalog as a table, the honest F10 story as a case study in sealing discipline) + recompiled PDF; rh/README.md wave-8 block; rh/INVENTORY.json + make\_inventory.py



extended additively (78  $\rightarrow$  79 entries: v965 pin + the r290–r295 consumption notes); run\_rh.py MODULES  $\rightarrow$  v955–v965, full RH suite green (59/59 items, ALL CHECKS PASSED). Mincut base 4 / refined 5 UNCHANGED (a geometry-census-plus-rejection set moves no edge); the full-suite run\_all.py pass at 958 modules, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXXIV) (consolidation wave 7 — the L\* coherence census: the certified content of discovery rounds 286–289 graduates from the experiments tree into the load-bearing suite as v964\_lstar\_coherence\_census.py (suite 956  $\rightarrow$  957, ledger 1120  $\rightarrow$  1122; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — lstar\_margin\_scaling 29/29 SPEC 0a44ac4e3a76e809, l2\_deterministic\_cancellation 25/25 SPEC 761d88fa566faa14, destructive\_coherence 31/31 SPEC da46f7eef702ac37, arch\_kernel\_diophantine 42/42 record SPEC 91cdc2b194502168; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — the Fejér window-sum identity, the van der Corput inequality with its Cauchy–Schwarz proof route and  $H = 1$  equality witness, the Abel/Erdős–Turán bound, the wrong-prefactor and dropped-terminal must-fails, the 2-power resonance count  $\binom{8}{2} = 28$ , in pure Fractions; single-module test green 21/21 gates in 1.1 s, deterministic on re-run): THE MARGIN WARNING RESOLVED, THE vdC THEOREM NAMED, THE DIOPHANTINE ROUTE EXCLUDED. (A) THE r287 vdC THEOREM: the exact, CONSTANT-FREE, ARITHMETIC-FREE van der Corput inequality  $|\sum_j P_j|^2 \leq \frac{m+H-1}{H} \sum_{|h|<H} (1 - \frac{|h|}{H}) A(h)$  at the frozen window  $H = \lceil \sqrt{m} \rceil$  delivers  $\delta' = +0.309 > 0.21$  WORLD-BLIND (admissible: L2 is the generic half) and certifies 6/7 exceptions +38/42 rungs — the L2 lemma has its first NAMED classical theorem candidate; F1 discrepancy/Erdős–Turán dies loud ( $\delta' = -0.21$ , 0/7, detector FIRE +1.44 dec); the autocorrelation surprise: on 39/44 worlds the block sequence REINFORCES at root scale — the c3 cancellation is already absorbed into the  $P_j$  magnitudes, the frames win through the  $\sqrt{m}$  count economy, not lag structure. (B) THE r286 MARGIN-SCALING CLARIFICATION (the DCXX warning RESOLVED): 15 new anchors  $N_w = 942..1218$ , ALL margins mp-sign-safe positive (min  $+1.806 \times 10^{-8}$ , every anchor mp dps-30/45 staggered) — NO counterexample; the  $O(1)$  census offset holds beyond the old family cap (off distribution {0:1, 1:10, 2:2, 3:1, 4:1}, max +4); margin law  $\alpha \approx 3.05$  vs  $S$ , FLATTENING (curvature  $-1.60$ , the honest anti-extrapolation flag), driver = the cancellation sharpening  $c_w \rightarrow 1$ ; HARMLESS quantified: the margin falls SLOWER than the local eigenvalue loading speed;  $q_N$  reconciled (r258 FLAT reproduced — two different objects); extrap-calibrated 15/15. (C) THE r288 CARRIER MAP: the destructive coherence is carried by the ANTIPHASE NEXT-NEAREST (3–4 fold) ARCH–ARCH pairs ( $X_v = -0.0517$ ,  $z_v = -3.149$ ,  $C_{\text{off}} = -0.1046$ ; band 1–2 87% positive/in phase) with TOTAL control reversal (all four controls constructive at their own crossing); the chain zeros CO-MOVE with the atoms (turn rate 0.24 = static prediction 0.23) yet  $z_v$  flips already at dose 0.005: FINEST ALIGNMENTS far below the median phase resolution; honest negatives sampling-blind + source-separator-not-found + different-objects. (D) THE r289 metric-only ADJUDICATION (THE central refutation of the diophantine reading): the RATIONAL TWIN — every tent center  $\log n$  replaced by a rational  $p/q \times \Delta$  (denominators  $\leq 56801$ , position cost  $\leq 2.1 \times 10^{-9}$ ), destroying

EVERY exact log-linear-form relation — KEEPS THE FULL SIGNATURE IDENTICAL (min  $C = 184$ , crossing 185,  $z_v = -3.149$ , mp-confirmed at the  $1.7 \times 10^{-4}$  margin): the effect of the diophantine structure is NULL; the shuffle twins die purely metrically ( $\theta_{\text{eff}} = 0.12$ ,  $\sim 40\text{--}120\times$  beyond the measured coherence threshold  $10^{-3} \dots 3 \times 10^{-3}$  of the local gap); the only sub-gap entry point: the TENT-SPLIT FRACTIONS (completeness identity  $4.2 \times 10^{-14}$ ); the Baker need documented  $\sim 7900\times$  too weak AND unnecessary — the firewall reads the fraction profile metrically, not its arithmetic: the why-MAIN search continues in PROFILE space (the precise open front: which functional of the fraction profile forces the destructive coherence). LEDGER: NEW PRIME.LSTAR.COHERENCE\_CENSUS.01 [E] (fine types Identity/Formal + Numerical/Measurement; honest negatives explicit: sampling-blind, source-separator-not-found, diophantine-excluded(metric-only) with the rational twin verbatim); NEW PRIME.PORT.L2.VDC\_LEMMA.01 [O] — the vdC form as a provable chain statement (the remaining input: the chain origin of the  $P$ -variance scaling; kz15 exact-finite per r270); CONTRACT UPDATES: PRIME.LSTAR.SUBORDINATION.01 [O] carried at the wave-7 state (margin decay resolved harmless on 57/57 anchors, carrier class named, diophantine excluded — the open front is the profile functional; r290 profile\_functional\_probe.py in flight, NOT consumed), PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] records the vdC candidate for the H5/L2 front. LEAN (rh/lean): NO new statements needed — the hole stays lstar\_subordination (docstring refreshed to the wave-7 state); README census updated. Papers: the coherence-arc diary section in tfpt\_prime\_front, the wave-7 blocks in tfpt\_research\_contracts, the  $L^*$  section of rh/paper/rh\_program.tex extended by the coherence/twin findings with the metric-only twin table and the L2 section by the vdC theorem; rh/problem/lstar\_problem.tex gains a short “What is known (update)” paragraph (margin law + 57 anchors + the rational-twin fact: the problem is NOT diophantine) + recompiled PDF; rh/README.md wave-7 block; rh/INVENTORY.json + make\_inventory.py extended additively (v964 pin + the r289 sync repair — the helper now tracks the INVENTORY head); run\_rh.py MODULES  $\rightarrow$  v955–v964. Mincut base 4 / refined 5 UNCHANGED (a census-plus-adjudication set moves no edge); the full-suite run\_all.py pass, the final bash build.sh audit and the manifest regeneration follow as the joint exit gate run by the coordinator; the parallel r290 profile-functional worker is NOT consumed; NO RH claim anywhere in either direction)

2026-08-25 (CXXIII) (consolidation wave 6 — the  $L^*$  reduction dictionary: the certified content of discovery rounds 282–285 plus the  $L^*$  standalone problem graduates from the experiments tree into the load-bearing suite as v963\_lstar\_reduction\_dictionary.py (suite 955  $\rightarrow$  956, ledger 1118  $\rightarrow$  1120; wave-4 embedding convention: the four discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — representation\_contest 30/30 SPEC 0f9954b8f9727800, fullsource\_quasideterminateness 26/26 SPEC 4cf5ea53ffa61ee7, lstar\_two\_measure 30/30 SPEC 687112a83a767c81, christoffel\_decomposition 33/33 SPEC 9781e6d60aa5660a; full records sealed experiments-side, re-verified by rh/verification/run\_rh.py; plus the module-own exact section S0 — Hankel determinants, monic OP chains, frame congruence minors, full Cauchy–Binet gauge exhaustion, dual chains in pure Fractions with two declared float eigen cross-checks; single-module test green 19/19 gates in 1.0 s, deterministic on re-run): THE FREE-WINDOW QUESTION GETS ITS CANONICAL

FORM. (A) THE r283 A2 CHAIN as exact gates: the mu-frame congruence  $\min_k (D_\mu - G) = D_k(\tilde{\mu})$  at every  $k \leq S_+$  (one-nu-atom truncation must-fail LOUD); the frame contraction  $h_0..h_{m-1} > 0 \iff \lambda_{\max}(E_m) < 1$  with  $E$  the  $\nu$ -dressed  $\mu$ -CD kernel (both truth values); the crossing dictionary  $\lambda_{\max}$  crosses 1 EXACTLY at  $\min C + 1$  (capacity-ceiling typed where  $\min C = S_+$ ); the pigeonhole ceiling  $\min C \leq S_+$  re-proved in the frame (null-polynomial witness, exact on all five toys); capacity-as-counting REFUTED by the rank-one pair (same rank, same support, opposite fate; the deciding values are METRIC,  $1.25 \times 10^{-13}$  vs  $1.25 \times 10^2$ )  $\Rightarrow$  L\* IS THE CANONICAL REDUCTION: for every real polynomial  $p$  with  $\deg p < N_w$ ,  $\int p^2 d\nu < \int p^2 d\mu$  (equivalently  $\lambda_{\max}(E_{N_w}) < 1$ ) — a two-measure moment problem instead of a determinant cascade. (B) THE r285 DECOMPOSITION BOOKKEEPING exact: the Christoffel sandwich  $\max_k v_k K_n(y_k) \leq \lambda_{\max} \leq \text{trace}$  with its rational consequences at every degree; the multiplicative decomposition  $\lambda_{\max} = \text{maxdiag} \times (1 + \text{assist})$  with the budget equivalence sign-exact. (C) THE r282 FOUR-LANGUAGE ELIMINATION as named negative gates (contest-all-dead; the common deep reason made explicit: *every classical positivity language forces positivity exactly for the positive measure class* — MAIN is signed): SOS exists IFF the negative register is empty (the fake SOS misses  $h_2$  by exactly  $N_2 = 48360721965/70120631072$ ); the Kasteleyn orientation class is EXACTLY  $\{\pm \text{sign } w\}$  by full  $2^S$  exhaustion and is value-preserving IFF  $S_- = 0$  (defect =  $2 \times \text{negmass}$  exact:  $55/36, 44/35, 114/35, 0$ ); Hamiltonian-PSD =  $h > 0$  bookkeeping; the dual-pair sync holds PER THEOREM ( $h_n h_{S-1-n}^\# = 1$  exact, wrong dual loud at  $1.4 \times 10^2$ ). (D) TYPED MEASUREMENTS (record pins): the r284 one-atom-vs-collective dichotomy (w9  $n_{\text{DIAG}} = 187$  vs crossing 185 — MAIN rides the single-atom Christoffel edge to the last free degree, all four controls die collectively; extremal anatomy PR 1.89 on two shallow- $u$  ARCH atoms below the first prime, smallp-depleted, one-band; shielding-blind; Nyquist refuted at the discriminant), the r285 legs ((D) SEPARATES at window scale — controls'  $n_{\text{DIAG}} 50..91 \ll 184$ ; SUB-classical edge growth  $p = 0.38$ , classical coverage  $0/42$ : any (D) proof must be a DISCRETE bound; MAIN's wall assist the extreme ensemble low-outlier pct 0.00; and the first two positively MAIN-separating detectors of the program: assist-at-crossing 0.0195 vs dead 1.69..2.99, random-sign  $z = -3.15$  destructive vs dead  $+4.95.. +12.44$  all constructive), and the honest DCXX margin-decay finding ( $42/42$  ladder margins positive, family minimum  $1.4175 \times 10^{-7}$  at anchor  $z = 233$ ,  $S = 1717$ , vs the flagship  $1.68 \times 10^{-4}$  at  $z = 16$  —  $\sim 3$  orders of decay; the r286 margin-scaling round `lstar_margin_scaling_probe.py` is IN FLIGHT and NOT consumed). LEDGER: NEW PRIME.LSTAR.REDUCTION.DICTIONARY.01 [E] (fine types Identity + Numerical/Measurement, honest negatives explicit); NEW PRIME.LSTAR.SUBORDINATION.01 [O] — the L\* contract itself (the canonical form verbatim:  $\int p^2 d\nu_z < \int p^2 d\mu_z$  for  $\deg p < N_w$  on the admissible anchor family; state  $42/42$  + the open margin decay; the (D)+(C) anatomy; the standalone document `rh/problem/lstar_problem.tex` as external reference; kill lists inherited); CONTRACT UPDATES: FULLSOURCE.QUASIDEFINITENESS.01 consumed (round 283, mechanism-partial(A2), L\* established), REPRESENTATION.CONTEST.01 consumed (round 282, contest-all-dead); the BASEFIBER north star now carries the canonical L\* form. LEAN (rh/lean, build green): NEW `lstar_subordination` on VonMangoldtWindow (documented sorry, the canonical-form target) PLUS the *proved* direction `lstar_implies_free_window` (L\*  $\Rightarrow$  free-window positivity, finite algebra against the r283 chain). Papers: the L\*-arc diary section in `tfpt_prime_front`, the wave-6 blocks in `tfpt_research_contracts`,

the new “The  $L^*$  reduction” section in `rh/paper/rh_program.tex` (canonical form as a theorem environment with certification box, the (D)/(C) split, the four-language elimination table, the margin decay stated neutrally, the pointer to `rh/problem/`); `rh/README.md` reduction chain ( $L^*$  as the canonical stage); `rh/INVENTORY.json` + `make_inventory.py` extended additively (v963 pin + r285 sync repair); `run_rh.py` MODULES  $\rightarrow$  v955–v963. Mincut base 4 / refined 5 UNCHANGED (a reduction-plus-elimination set moves no edge); the full-suite `run_all.py` pass, the final bash `build.sh` audit and the manifest regeneration follow as the joint wave-4/5/6 exit gate run by the coordinator; the parallel r286 and L2-cancellation workers are NOT consumed; NO RH claim anywhere in either direction)

2026-08-25 (CXXII) (consolidation wave 5 — the half-filling pinning theory: the certified content of discovery rounds 279–281 graduates from the experiments tree into the load-bearing suite as `v962_halffilling_pinning_theory.py` (suite 954  $\rightarrow$  955, ledger 1115  $\rightarrow$  1118; wave-4 embedding convention: the three discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage with the printed SPEC SHA-256 hashes pinned and gated — oriented\_theorem 32/32 SPEC 9107709b4f4a65d1, budget\_localization 29/29 SPEC 7abf7a208bb45e43, halffilling\_pinning 28/28 SPEC a00815722a93a3cb; full records sealed experiments-side, re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 in pure Fractions — Stieltjes chains, Sturm chains, Vandermonde solves, node polynomials, no floats; single-module test green 13/13 gates in 1.4 s, deterministic on re-run): the reviewer demand “freeze rounds 279–281 as a SMALL MATHEMATICAL THEORY, not just as experiments” is executed — FOUR NAMED THEOREMS as gates: (T1) THE MOMENT COUNTING THEOREM (r281, Identity): the pivot  $h_n$  consumes  $m_0..m_{2n}$ , the  $S$  free moments are  $m_0..m_{S-1}$ , hence the FREE pivots are exactly  $h_0..h_{N_w-1}$  and  $h_{N_w}$  is the first forced one — half-filling IS the end of the free moment space, arithmetic-exact for ALL  $S = 2.2000$  plus the exact-rational freedom demonstration ( $dm = e_{S-1}$  moves  $h_{N_w-1}$  alone; wrong- $L$  must-fail loud): the reviewer question “why half-filling” is answered by counting; (T2) THE CROSSING BUDGET THEOREM (r279, Identity):  $\#(h_n < 0, n < S) = S_-$  EXACTLY (Jacobi minor-sign rule + Sylvester congruence), WORLD-BLIND — five frozen signed measures exact, real-side w9 104 / w13 98 / controls 141/94/6 / kz15 121 / kz52 551; (T3) THE TWO-SIDED PARITY THEOREM (r279, Identity):  $\text{sign } Q_n(x_j) = e_j(-1)^{S-1-j} \text{sign}(h_n)$  at every node and degree, union occupancy ODD in every weight-agreement gap and EVEN in every disagreement gap —  $h$ -blind, with the census bilanz  $c_n + c_{S-1-n}^\# = (S-1) - |D| + 2 \text{scD}(n)$  exact at every degree and by exhaustion over all 87376 sign-vector pairs  $k = 2..8$ ; (T4) THE MAIN WINDOW REDUCTION: given T1 + T2 the ENTIRE open statement of the wall is  $\min C \geq N_w$ , EXACTLY equivalent to  $(\forall n < N_w : h_n > 0)$  — one placement question, the north star in reinstform: WHY IS THE SIGNED PRIME MOMENT FORM QUASI-DEFINITE UP TO THE MAXIMALLY FREE ORDER? FOUR NAMED REFUTATIONS as negative gates (the negative information is load-bearing): no-universal-o1-pinning (r281 exact: the one-negative toy has offset  $N_w - 2$ , unbounded in  $S$ ; only provable ceiling: pigeonhole  $\min C \leq S - S_-$ ;  $C \leq 5$  is a 42-rung MEASUREMENT), no-extremality (r280 measured + exact witness: three mp-confirmed directions push the w9 crossing 184  $\rightarrow$  185 past half-filling; the 4-atom one-negative measure has  $\min C = N_w + 1$  exactly), no-generic-maslov-obstruction (r279 exact: 3 rational counterexamples — the

two-sided machinery provably carries no arithmetic), no-simple-offset-law (r281 measured: six sealed source-pure candidates,  $\max |sp| = 0.273 \ll 0.75$ , all world-blind). LEDGER: NEW PRIME.WALL.HALFFILLING\_PINNING\_THEORY.01 [E] (the theorem set + refutations, fine types Identity + Numerical/Measurement); PRIME.PORT.RHP.MIDPOINT.ORIENTED\_THEOREM.01 [O] updated (rounds 279–281 consumed: theorem share promoted, obstruction candidate refuted, the center precisified to the budget localization); FULLSOURCE.BASEFIBER.01 and METRIC\_FIREWALL.01 carry the reformulated north star; NEW open contracts PRIME.PORT.REPRESENTATION.CONTEST.01 [O] and PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01 [O] (the two rounds in flight at this cut, NOT consumed, with the reviewer forbidden lists). LEAN (rh/lean, build green,  $6 \rightarrow 8$  intentional sorry): the fog-free central hole free\_window\_positivity ( $\forall n < \text{cap} : 0 < h_n$  on MainWindow; via T4 the base half of the master theorem — master\_implies\_free\_window PROVED); T1 PROVED for real (moment\_counting\_free\_pivots + first\_forced\_pivot, pure arithmetic); T4 PROVED (main\_window\_reduction); T2 stated (crossing\_budget, sorry with v962 reference); Counterexamples.lean extended by the wave-5 guard upper\_pinning\_not\_universal — the EXACT one-negative instance with machine-checked Hankel minors (incl. the negative  $\det H_4 = -1962/13625$ ), a genuine moment computation, not a bookkeeping model — plus the proved arithmetic escape o1\_pinning\_escape. Papers: the pinning-decomposition diary paragraph in tfpt\_prime\_front, the north-star reformulation in tfpt\_research\_contracts, the small-theory subsection + north star in rh/paper/rh\_program.tex, the finalized reduction chain in rh/README.md; rh/INVENTORY.json extended additively (v962 pin + Lean state). Mincut base 4 / refined 5 UNCHANGED (a theorem-plus-refutation set moves no edge); the full-suite run\_all.py pass, the final bash build.sh audit and the manifest regeneration follow as the joint wave-4/5 exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXXI) (consolidation wave 4 — the terminal-surface closure and the midpoint-orientation dictionary: the certified content of discovery rounds 260–278 graduates from the experiments tree into the load-bearing suite as v960\_terminal\_surface\_closure.py and v961\_midpoint\_orientation\_dictionary.py (suite 952  $\rightarrow$  954, ledger +4; WAVE-4 EMBEDDING CONVENTION: all fourteen discovery probes embedded BYTE-EXACT and executed verbatim in their sealed -smoke stage — deterministic, seconds each — with the printed SPEC SHA-256 hashes pinned and gated and the byte-equality wards live; the full-mode records stay sealed experiments-side and are re-verified by rh/verification/run\_rh.py; v960 from ten probes, first-pass full records 31/31 + 26/26 + 26/26 + 28/28 + 29/29 + 29/29 + 28/28 + 28/28 + 24/24 + 24/24: terminal\_crossratio SPEC 36eee7b40bd897a8, terminal\_triangle SPEC db32d31f2e543a9c, cancellation\_adjudication SPEC 151a9176f089dd5b, drive\_local\_asymptotics SPEC 0c4df40812b9b76b, phase\_bulk\_bound SPEC f5578bb91f84ac2a, kz15\_boss SPEC 8834a1c259bdb605, universal\_pair\_theorem SPEC 66f61c8a436af90e, l2\_scaling\_anatomy SPEC 3fb76b65848814f8, euler\_mechanism SPEC 8cb1804e71dd87bb, kyp\_memory SPEC 88ebdb00867469fd; v961 from four probes, 32/32 + 25/25 + 29/29 + 31/31: wronskian\_dictionary SPEC 56e8a03efd1690e4, minimal\_firewall SPEC ed17d79fc037ab1a, maslov\_census SPEC 3858fd16bde0c9c0, metric\_stability SPEC 7031200f008d34d1; plus module-own exact sections S0 — v960: the two-branch reduction with the non-certifiable witness, the driven-recursion C1b triangle, the pair decomposition with the structural alternation gain, the interval-certificate logic with its width ward, the KYP

double obstruction o1/o2; v961: the base Casoratian identity  $p_n q_{n+1} - p_{n+1} q_n = h_n$  with  $c' = 1$  symbolic in  $z$ , the bordered-Wronskian telescope, the Hellmann–Feynman position gradient in exact dual-number arithmetic, the Jacobi interlacing/reality direction with the flipped-beta must-fail; single-module promotion tests green: v960 15/15 gates 8.6 s, v961 8/8 gates 2.4 s, both timing-normalized deterministic on re-run): (1) THE MILESTONE — THE 42-RUNG SURFACE CENSUS  $q_N < 1$  IS CERTIFIED ON ALL 42 RUNGS as a finite fact: 41 mechanism-certified target-blind (35 cheap via the r260/r263 two-branch theorem, exception set exactly  $\{kz15, kz20, kz22, kz36, kz38, kz39, kz52\}$ , both mains close WITHOUT cancellation; kz20/22/36/38/39/52 via the sealed phase bounds r268–r270, no detector fires) + kz15 EXACT-FINITE via the r270 interval certificate (outward-rounded intervals, dps 640, end width  $1.5 \times 10^{-92}$ , margin +0.02680, the dps-halving ward destroys it as demanded; EnclOK class, no mechanism) — a census-plus-certificate result, NOT a cofinal theorem: the window quantifier stays; (2) THE UNIVERSAL PAIR THEOREM (r271): H1–H4 structural/source-side, Lean-proved in rh/lean/RH/PairBound.lean; H5 = the margin  $|Z_{\text{local}}| + \varepsilon < \sqrt{5/7}$  is the SINGLE window-dependent hypothesis — the cofinal front, precisely typed by r272 (bound-coarseness: the truth margin RISES with  $N$ ; flip condition  $\delta' > 0.21$  of available 0.45, address c3) and r273 (perturbation-insensitive + the firewall map: THE WALL, not the cancellation rate, is what is arithmetically special about MAIN); (3) THE MIDPOINT-ORIENTATION DICTIONARY (r274): base Casoratian = the pivot chain itself, the  $h$ -free midpoint form IS the node polynomial (the r231 sign-blindness as an identity), the augmented telescope  $D_{n+1} = B - W_n^{\text{aug}}/W_n^{\text{base}}$  exact; and THE MASLOV CENSUS GO (r277): rule R2 = Jacobi interlacing/reality passes blind 42/42 with controls firing EXACTLY at flip+1, one-way, not  $h$ -equivalent — while the raw atom-Sturm census is honestly REFUTED as the winding quantity (MAIN breaks  $c_n \equiv n$  at 56/48 at positive  $h$ ; sturm-chain-verified not awarded); (4) HONEST NEGATIVES kept load-bearing: the eventual-triangle no-go (kz52 itself an exception), three identity routes dead with break loci, the retracted w13 Gram provenance, the KYP memory no-go FROM STRUCTURE (o1 fiber forcing + o2 descent window, exact-rational on 46/46 worlds; the admissible Riccati memory target-inverse, sp +1.00), the metric-firewall continuum with the perturbative-only stability law (exact Hellmann–Feynman gradients, u-profile predictive — the small primes 2, 3, 5 carry). NEW LEDGER ROWS: PRIME.PORT.COUPLEDTAU.SURFACE\_CLOSURE.01 [E] and PRIME.PORT.RHP.MIDPOINT.ORIENTATION\_DICTIONARY.01 [E] (fine types Identity/Formal + Numerical/Measurement, honest negatives explicit) plus the two successor contracts PRIME.PORT.RHP.MIDPOINT.ORIENTED\_THEOREM.01 [O] (the contraposition form: an independent index obstruction forbidding the R2 break before half-filling on MAIN; round 279 oriented\_theorem\_probe.py in flight, NOT consumed) and PRIME.PORT.WALL.METRIC\_FIREWALL.01 [O] (the global metric-firewall lemma; certified state perturbative-only); CONTRACT UPDATES: TERMINAL\_CROSSRATIO.01 [O] progressed (surface certified; cofinal front = H5 / lemma L2), PREFIX\_RESUMMATION.01 [O] progressed (r261 finite-combinatorial half closed negative; the named path = the oriented midpoint theorem), FULLSOURCE.BASEFIBER.01 [O] carries the formalized master target form augmented\_prefix\_positive (rh/lean/RH/Window.lean; per r276/r278 the MainWindow predicate must encode METRIC exactness). RH-WORKSPACE PIPELINE INTEGRATION (the documented DXCIX TODO): bash build.sh

audit now runs `rh/verification/run_rh.py -fast` as its own “RH workspace” section (must end RH SUITE: ALL CHECKS PASSED; verified 32/32 items, 52/52 pinned entries byte-identical after the wave-4 INVENTORY extension); `run_rh.py` MODULES extended to the six RH modules v955/v956/v958/v959/v960/v961; `rh/INVENTORY.json` +2 pinned module entries + fourteen consumption notes; `rh/paper/rh_program.tex` updated (the surface-closure and orientation sections, the extended no-go catalog, the wave-4 edge states) and rebuilt; `rh/README.md` status updated. Papers: the new surface-closure/orientation-arc diary subsection in `tfpt_prime_front` (stale “rounds 260/261 in flight” wording retired), the updated BASEFIBER/terminal-edge blocks + the new wave-4 theorem-set/contract block in `tfpt_research_contracts`. Mincut base 4 / refined 5 UNCHANGED (a surface census + certificate set moves no edge); the full-suite `run_all.py` pass, the final bash `build.sh` audit and the manifest regeneration follow as the wave-4 exit gate run by the coordinator; NO RH claim anywhere in either direction)

2026-08-25 (CXX) (the coupled-tau-terminal promotion and the CCC D4 execution record — the certified theorem block of discovery rounds 256–259 graduates from the experiments tree into the load-bearing suite as `v959_coupledtau_terminal_dictionary.py` (suite 951 → 952, ledger +3; four discovery probes re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with their printed SPEC SHA-256 hashes pinned and gated: `baseborder_factorial_probe.py` 21/21 SPEC c7aa9f46889414c4, `coupledtau_probe.py` 26/26 SPEC 73d8247f6de36a2b, `budget_anatomy_probe.py` 27/27 SPEC 28026ae6b0750098, `parametrix_pass_probe.py` 19/19 SPEC e79da4f43d63213a; plus a module-own exact section S0 — the recursion-to-Riccati implication, the bilinear tau form derived algebraically and gated, the terminal consequence gate in both truth values, the corner-ray  $t^*$  normal form, and the positive-prefix PSD theorem  $\det H_p = \prod h_i$  with an exact Schur-complement must-fail; single-module run 9/9 gates, 188 s): (1) THE POSITIVE-PREFIX FIREWALL + THE  $2 \times 2$  CAUSAL CENSUS (r256, effects-mixed-coupledtau): MAIN carries the positive prefix through its entire window ( $p_{\max} = 184 = N$ , zero negative modes — the *only* positive-prefix world) while the controls type indefinite-continuation at 21/25/27; every prefix kernel numerically PSD (worst  $5.7 \times 10^{-16}$ ); on the causally clean shared prefix  $n^* = 21$  the base rotation is null ( $-1.5 \times 10^{-6}$ ) and the border carries the prefix gap ( $-0.1759$ ); TWO CONTAMINATION DOWNGRADES ON RECORD: the round-254 scramble-compactness headline is base contamination ( $f_{\text{neg}} = 0.9311$ , the  $h < 0$  mode space at  $734 \times$  excess) and the r253/r254 Epstein fiber anomaly lives entirely in the indefinite continuation degrees  $k \geq 25$  — neither may be cited as a border/fiber discriminator any more; (2) THE COUPLED-TAU RECURSION IN BOTH FORMS (r257):  $\tau_{n+1} = a_n \tau_n$ ,  $\tau_{n+1}^{\text{aug}} = a_n \tau_n^{\text{aug}} + b_n \tau_n$  with  $a_n = c_n^2 h_n$ ,  $b_n = -(c_n F_n)^2$ , equivalently the Riccati drain  $D_{n+1} = D_n - F_n^2 / h_n$  (anchor  $D_N = 5/7 - \rho_{N-1}$  rel  $4.8 \times 10^{-11}$ , mp dps 220 through the scramble flip) AND the bilinear tau form  $\tau_n^{\text{aug}} \tau_{n+1} - \tau_{n+1}^{\text{aug}} \tau_n = (c_n F_n)^2 \tau_n^2$  (a manifest square — the drain increment is a positive tau cross-ratio); the base alone carries the sign, the border only nonnegative magnitude; the cocycle breaks exactly at the control flips 21/25/27; THE FROZEN MICRO-FALSIFIER PASSED BLIND for the first time, at the coefficient-field level (control flips 3/3 AND forced tail 0/2/2/3/1 on all five windows; the parametrix half stays open); (3) THE BUDGET TELESCOPE (r258):  $D_N = B - S_{N-1}$  exact (12 windows incl.  $N = 878$ , mp dps 220 at  $2.4 \times 10^{-16}$ ),  $B = S_{N-2} + 5/7$  by construction with the 5/7 typed floor-imported (the inverted r241 floor of the measured

$\min h_{N-1}/F_{N-1}^2 = 1.4278$ ; source-pure derivation open = the named terminal task); the consequence gate exact:  $\text{margin} > 0 \iff q_N = \rho_{N-1}/(5/7) < 1$  — the entire fiber positivity is ONE terminal inequality; typed numerical: the  $q$  census (42/42, flat, min margin 0.0139), the  $F$ - $h$  covariance +1.000 (a measured structural fact, NOT a law — the driver is the quotient itself); and (4) THE THREE HONEST NEGATIVES (r259, parametrix-failed): level crossing refuted (branches cross 5–16 degrees too early; MAIN w9 crosses 45 times with zero true flips), the one-swap no-go (fully overlapping gap distributions — round 255’s orientation-lowdim(1) REVISED on record: the missing information is not one-dimensional), the mass-majorant no-go (all three sealed majorants 0/42, median  $\times 490$ , break at  $0.97N$ ); the named obstacle: a RESUMMATION gap, not a selection gap. NEW LEDGER ROWS: PRIME.PORT.RHP.COUPLEDTAU.TERMINAL.01 [E] (fine types Identity/Formal + Numerical, honest negatives explicit) and the two terminal-edge successor contracts PRIME.PORT.COUPLEDTAU.TERMINAL\_CROSSRATIO.01 [O] (the last fiber edge = a positive tau cross-ratio: prove  $h_{N-1}/F_{N-1}^2 > 7/5$  on the cofinal ladder; reviewer proof-form requirements and kill list frozen — the bound must bind the ratio; the mass-majorant class stop-listed) and PRIME.PORT.FULLSOURCE.PREFIX\_RESUMMATION.01 [O] (the last base edge = a globally oriented prefix resummation: the determinant as a signed partition sum, completing the coefficient-field pass to a parametrix-level pass; level crossing and  $k$ -swap truncations stop-listed); PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 [O] carried forward at the post-round-259 state with the r256 causal decision (base/fiber separation artificial) and the precise localization recorded — mincut base 4 / refined 5 UNCHANGED, rounds 260/261 in flight on the successor contracts and NOT consumed. AND THE CCC D4 EXECUTION RECORD: the ledger row CCC.SEAM.CROSSOVER.01 [O] is updated — demand D4 (the preregistered disc search experiments/ccc-crossover-disc) EXECUTED 2026-08-24 with a ROBUST NULL on the Planck PR3 SMICA map (first data contact; freeze intact 3/3, frozen-kernel guards 5/5 before data contact; map SHA-256 pinned, 2013312960 bytes; seed 20260824, reruns bit-identical):  $p_{\text{global}} = 0.673$ ,  $p_{\text{counts}} = 0.446$ ,  $\text{BH-}q_{\text{min}} = 0.673$  vs the frozen  $q < 0.01$  threshold; 171 candidates with no sign excess (83/88); injection recovery 100% at  $200\,\mu\text{K}$  — a well-powered, real null; kill criteria K1–K6 all NA, replication legs not required for a null; NO type change, the contract stays [O] via D1–D3, the cyclic reading stays [C], the compiler core untouched (the kernel was output of the [E] kinematics, not input); results machine-readable in results.json. Papers: the new coupled-tau-terminal-arc diary section in tfpt\_prime\_front, the updated FULLSOURCE.BASEFIBER block + the new theorem-set/terminal-edge block and the D4 execution record in tfpt\_research\_contracts. The single-module promotion test is green (9/9 gates, probes byte-exact vs the experiments tree); the full-suite run\_all.py pass, bash build.sh audit and the manifest regeneration follow as the wave-3 exit gate of the same consolidation. No omega closed, no marker moved beyond the one [E] and two [O] registrations and the recorded D4 execution; NO RH claim anywhere; NO claim about CCC in either direction)

2026-08-24 (CXIX) (the bordered-RHP-dictionary promotion — the certified theorem block of the RHP campaign rounds 244–253 graduates from the experiments tree into the load-bearing suite as v958\_bordered\_tau\_readout\_dictionary.py (suite 950  $\rightarrow$  951, ledger +2; six discovery probes re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with their printed SPEC SHA-



256 hashes pinned and gated in the module: bordered\_hankel\_probe.py 23/23, bordered\_finite\_rank\_probe.py 29/29, border\_centering\_probe.py 24/24, targetreadout\_error\_probe.py 20/20, base\_gauge\_constant\_probe.py 22/22, schlesinger\_pairing\_probe.py 26/26; plus a module-own exact-rational PSD base theorem with a frozen signed must-fail witness): (1) the bordered Hankel end-object (bordered PSD = wall + budget) has an *exact* Riemann–Hilbert dictionary — the Uvarov border column carries  $F_n$  as its  $z^{-1}$  coefficient, and the budget is a CD-kernel functional of the terminal  $Y_n$  data alone,  $S_{n-1} = \iint K_n d\sigma d\sigma$ ; (2) the border is a *Schlesinger rank-1 insertion* over a canonical  $2 \times 2$  RHP with a one-dimensional budget cocycle ( $\det Y_3 = -h_{n-2}F_{n-1}$  newly derived; IIKS generator rank exactly 2; the irreducible- $3 \times 3$  steepest-descent campaign dropped without replacement); (3) the centering congruence with  $c = F_0/h_0$  eliminates the geometric rank-1 head algebraically, the  $\sigma^\circ$  dictionary and dual-norm identity  $Q_m = t^T H^{-1} t$  are exact, and the extensive tail compresses to ONE terminal CD readout ( $T = \iint [K_N - K_8] d\sigma^\circ d\sigma^\circ$ , dps-160 exact through the full depth  $N = 184$ , rel  $3.3 \times 10^{-160}$ ); (4) the three error-evaluation formulas are exact ( $\delta h_n = (R_1)_{12}$ ; the central rest identity; the error-kernel sandwich at  $1.1 \times 10^{-81}$ ); (5)  $R_1$  is a contour integral ( $\frac{1}{2\pi i} \oint (R(z) - I) dz$ , verified at  $3.2 \times 10^{-14}$ ) and the global-gauge thesis is refuted on all three prongs (the base is blocked by a *drifting, orientation-relevant* layer, not a constant); (6) the fiber target is the exact augmented tau quotient  $D = B^\circ - Q_7 - T = \tau^{\text{aug}}/\tau$  at full depth (three routes at  $10^{-11}$ , mp  $10^{-16}$ ), with the pole layers classified — rank-1 border poles removable by residue closure, the  $D$ -layer essentially singular ( $\exp(c/(z-s))$  type). The campaign contract PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 [O] is registered with the post-round-255 measured state (the measurement rounds 250/254/255 stay experiments-side, typed MEASUREMENTS: outer-model-fails, offdiag-extensive with the compactness inversion — SCRAMBLE collapses to 2 eigendirections, MAIN needs 28–30 — and orientation-lowdim(1): the base flip is a one-bit  $O(0.2 \text{ dec})$  selection event of an extensively accumulated quantity); the theorem set is registered as PRIME.PORT.RHP.BORDERED.READOUT.01 [E]. Mincut base 4 / refined 5 unchanged; no other marker moves; NO RH claim in either direction. Round 256 (baseborder\_factorial\_probe.py) was still in flight at this promotion cut and is not consumed. Papers: the prime-front bordered-dictionary arc subsection and the research-contracts blocks added.)

2026-08-24 (CXVIII) (the CCC-crossover promotion — the kinematics dictionary of the conformal-cyclic crossover graduates from the experiments tree into the load-bearing suite as v957\_ccc\_crossover\_kinematics.py (suite 949  $\rightarrow$  950; six discovery probes of rounds DL–DLV re-run GREEN AS TYPED at promotion and embedded BYTE-EXACT with pinned SHA-256 hashes: ccc\_crossover\_gates\_probe.py 28/28, ccc\_uplift\_4d\_probe.py 29/29, ccc\_fork\_kms\_holonomy\_probe.py 18/18, ccc\_kernel\_gamma\_kms\_probe.py 21/21, ccc\_transfer\_selector\_probe.py 14/14, ccc\_kappa\_freeze\_probe.py 7/7): (1) Penrose’s reciprocal gauge  $\hat{\Omega}\check{\Omega} = -1$  is the element  $\tau = \rho^2\sigma : z \mapsto -1/z$  of the PROVEN  $\mu_4$  Möbius stabiliser  $D_4$  —  $\tau$  reverses the clock (= the  $\mathbb{Z}_2$  sheet parity), swaps the aeon poles  $\{0, \infty\}$ , and the crossover locus  $|\hat{\Omega}| = |\check{\Omega}|$  is the unit circle carrying ALL FOUR marks; (2) conformal-factor rigidity: on the 2D seam the clock alone leaves exactly ONE constant-curvature modulus (Liouville-exact dilation family) and the reciprocal  $\mathbb{Z}_2$  kills it (unique zero at  $\lambda = 1$ ); on the 3D crossover the deck ALONE kills all conformal moduli (the  $so(4,1)$  deck-commutant is 4-dimensional  $su(2)+u(1)$ , ALL Killing, boost content

$3.3 \times 10^{-16}$ ; global uniqueness on the explicit Obata boost family, 200-axis scan bounded below by 0.53) — TFPT’s candidate answer to CCC’s open unique-conformal-factor problem,  $v_{\text{geo}}$  staying pure unit calibration; (3) the entropy reset of the cyclic [C] reading is a REDUCED CHANNEL: the v221 transport is CPTP not unitary ( $\|T^t T - 1\|_2 = 0.9999$ ), its Stinespring isometry explicit ( $V^\dagger V = 1_3$  exact),  $\|T^n - P_\star\|_2 = (2/3)^{6n}$  EXACTLY, the global overlap invariant and the environment retaining the full distinguishability — “unitarity + rank-1 attractor” is isometry + partial trace; (4) the kinematic  $2D \rightarrow 3D \rightarrow 4D$  uplift from corpus objects only: stereographic( $\text{Hopf}(v)$ ) =  $z_1/z_2$  EXACTLY (the seam sphere IS the Hopf base), the v492 deck  $\text{diag}(i, i^{-1})$  acts freely so the crossover  $X = S^3/\mathbb{Z}_4$  is the smooth lens boundary of the corpus  $A_3$  ALE  $XY = Z^4$ , clock  $\text{diag}(i, 1)$  and the reciprocal lift  $J$  normalise the deck and descend ( $J$  to  $\tau$ , inverting the clock), the four marks lift to four Hopf fibre circles paired by the deck into the TWO  $\mathbb{Z}_2$  sheet orbits with Gauss linking number 1.0000, the 4D bridge is the cone  $\mathbb{C}^2/\mathbb{Z}_4$  with the inversion  $x \mapsto x/|x|^2$  a conformal involution (pullback  $|x|^{-4} \delta$  symbolic) fixing the unit lens POINTWISE and swapping the aeon ends, and the Lorentzian shadow  $(1/\cos^2 T)(-dT^2 + g_{S^3})$  is Einstein  $R_{ab} = 3g_{ab}$  (symbolic Ricci) with a SPACELIKE round- $S^3$  crossover and  $\check{\Omega} = -\cos T$  vanishing to first order; (5) the KMS/modular closure of discrete  $\rightarrow$  dynamics at the free level: detailed balance EXACT,  $\text{spec}(-\ln T) = \{0, 6 \ln \frac{3}{2}, 6 \ln 3\}$  EXACTLY (the seam gap IS a modular energy),  $T = e^{-H_{\text{mod}}}$  (one clock tick = one unit of KMS time at  $\beta = 1$ ), OS Gram PSD, and the v526 normalisation  $\beta_{\text{angle}} = 2\pi = 1/(4c_3)$  replicated exactly = the Gibbons–Hawking normalisation on the crossover fibre; (6) the holonomy dictionary: the two clock-pole circles descend to the lens core geodesics with the full  $\mu_4$  monodromy  $\chi(D) = i$  while EVERY seam string carries exactly the 2-torsion  $\chi(D^2) = -1$ , and  $D^2 = -1$  is trivial on the base while flipping the spinor coordinates — the v522 statement “gaugeable datum = the 2-torsion  $\{1, (-1)^F\}$ ; full wrap = fermion parity” in geometric form (typed [C] correspondence); (7) the selector necessity (the  $\gamma$ -facing theorem): the admissible OS reflections of the v512 mark family  $M(\delta)$  form the pair-exchanging  $V_4$  for every  $\delta \neq \pi/2$  and close to  $D_4$  EXACTLY at  $\delta = \pi/2$  (the order-4 clock factorises into admissible reflections,  $\rho = \tau \circ s_{\text{mid}}$ , iff  $\mu_4$ ), and at Majorana operator level on the  $N = 16$  seam ring with fermionic reordering signs kept the free parent is  $\Theta_a$ -real on ALL 16 axes (v521 side-blindness replicated) while the interacting family is  $\Theta$ -real on exactly  $\{k \neq 4 : 2, k = 4 : 4\}$  axes and the clock-axis cuts ONLY at  $k = 4$ , asymmetric couplings breaking reality — the UNIQUENESS half of the v534 straddle selector ( $\delta = \pi/2$ , symmetric couplings) is thereby DERIVED as a  $\Theta$ -reality necessity (the survival half and the coupling sign stay v534’s nonperturbative toy facts, cited; reality is sign-blind, checked); (8) the frozen Gate-6 kernel: exactly two decay modes  $\ln(729/64)$  and  $\ln 729$  (ratio  $\ln 3 / \ln \frac{3}{2} = 2.709511$ ), the frozen defect-class overlap table  $(\pm 1/\sqrt{2}, +1/\sqrt{6}) / (0, -2/\sqrt{6})$  with the sheet-parity sign rule, disc radius  $\theta_{\text{max}} = \eta_{\text{rec}}/(\eta_0 - \eta_{\text{rec}}) \approx 1.16^\circ$ , and  $\kappa$  DERIVED from the v526 KMS normalisation up to a declared convention band —  $u_{\text{rec}} \leq 8.3 \times 10^{-5}$  ticks across the whole band, relic contrast  $\leq 3.3 \times 10^{-4}$ : the template degenerates ROBUSTLY to the sharp causal top-hat disc; three freeze hashes pinned and recomputed in-run. NEW LEDGER ROWS: CCC.SEAM.KINEMATICS.01 [E] (the certified dictionary) and CCC.SEAM.CROSSOVER.01 [O] (the dynamical-crossover research contract: D1 interacting algebra on the crossover = WOIT  $\gamma$  in the crossover geometry, D2 field-level Stinespring, D3 the cosmology fork decision — Starobinsky  $r = 12/N_\star^2 \in [0.0033, 0.0048]$  priced decidable by  $\sigma(r) \sim 10^{-3}$

experiments —, D4 the executed disc search; kill criteria frozen incl. the forbidden mark/Hawking-point count identification); new contract section in `tfpt_research_contracts`; the origin-theory Penrose-CCC bullet SHARPENED (asymptotic de Sitter does NOT imply conformal flatness —  $C^2 = 48M^2/r^6$  for Schwarzschild–de Sitter, machine-checked; the CCC smoothness/Weyl conditions are additional crossover data) and pointed at the certified kinematics; PREREGISTERED SEARCH scaffolded at `experiments/ccc-crossover-disc/` (frozen hypotheses `ccc_disc_v1.yaml` with kill criteria K1–K6, null battery and  $BH-q < 0.01$  + replication decision rule; freeze-check CLI 3/3 hashes intact; 5/5 guard tests; catalog entry §7b.9 of the experiments README) — verdict slot `data_limited` (preregistered); the Planck PR3 SMICA data pass executed separately at experiments level (first data contact logged in the project README). FENCES CARRIED EVERYWHERE: the cyclic/CCC reading stays [C] and is NOT upgraded by the promotion; `WOIT.OS.TWISTOR.01` stays [O] with all kill tests live (the module certifies kinematics + free level + the necessity direction ONLY); the deck fork is decided only CONDITIONALLY on the v522 wrap identification; a null disc search damages nothing in the compiler core. Research notes DL–DLV appended to `experiments/next.txt`; FULL `run_all.py` ALL CHECKS PASSED at 950 modules; `bash build.sh` release green; `bash build.sh audit` → AUDIT OK; manifests regenerated LAST. No omega closed beyond the two new rows; NO claim that CCC follows from TFPT)

2026-08-24 (CXVII) (the integrable-dictionary-arc promotion — the seven durable certified theorem sets of discovery rounds 224–226 and 228–231 graduate from the experiments tree into the load-bearing verification suite as TWO modules (suite 947 → 949, ledger 1099 → 1104); every source probe RE-RUN GREEN AS TYPED at promotion and embedded BYTE-EXACT with its printed SPEC SHA pinned and gated (all seven at 100% first pass, deterministic records, sealed specs; rounds 199–223 stay experiments-side as adjudication/classification rounds, and round 227, the Fermi-edge classification, stays experiments-side by design with its builders embedded as a read-only library): MODULE ONE, `v955_tau_iiks_toda_dictionary.py` (4/4 pattern gates, ~3 s; probes `tau_symbolic_probe.py` r224 note DL, 18/18, SPEC `fb04cd7ce9fb306b`; `lax_conditioned_probe.py` r225 note DLI, 15/15, SPEC `93fd9759db7e7b3c`; `hirota_sign_probe.py` r226 note DLII, 17/17, SPEC `d78e236bf633de7b`; the v881 library `port_integrable_kernel_probe.py` embedded read-only): THE FINITE INTEGRABLE DICTIONARY OF THE WALL — (1) THE TAU IDENTIFICATION EXECUTED (TAU-IKS-EXACT): the wall determinant IS the exact IKS Fredholm tau function at the finite-identity level on all five heavy rungs — Sylvester as a FULL characteristic identity  $\det(I - sQ_{\text{state}}) = \det(I - sE_{\text{node}})$  on the whole  $s$ -grid, the  $s$ -DRESSED Schur family  $\det(I - sE) = \det(I - sR) \det(I - sD_P(s))$  EXACT with the fixed-kernel alias (gap 0.06–0.20 at  $s = 0.5$ ) named and blocked, the confluent diagonal ( $\leq 3.2 \times 10^{-13}$  — the contract’s declared death spot dissolves), the IKS dressing identity ( $\leq 7.9 \times 10^{-13}$ ) with the dressed port generators SOURCE-EXPLICIT IN CLOSED FORM (the G31 surprise,  $\leq 1.5 \times 10^{-13}$ ), the Fredholm/JMU variational identity in TWO independent times ( $\leq 1.4 \times 10^{-9}$ ), and the relative tau transfer exact over  $-212.84/-195.50$  log-unit determinant collapses ( $\leq 3.4 \times 10^{-12}$ ); controls INVERTED per contract (Epstein/scramble SATISFY the algebra and differ in the VALUE  $\text{sign } \tau(1) = -1$ ); (2) THE LAX DEGREE ADJUDICATED (LAX1-H-EXACT): the word “2” was a hypothesis — the true minimal degree

in the closing direction is ONE and the closing direction is  $h$ , not  $s$  (rank-1  $h$ -step at  $2.9 \times 10^{-16}$ , explicit rank-2 relative displacement, blind next-generator prediction  $3.4 \times 10^{-16}$  and blind tau-step prediction  $3.2 \times 10^{-11}$  on the sealed rungs, zero curvature identically world-blind, mpmath wards dps 80/120; the  $s$ -time has NO fixed degree — basis-invariant  $\varepsilon_d$  plateau 0.34–0.60, CONDITIONING-ONLY — and across rungs NO common carrier: typed, not hidden); (3) THE SIGNED-TODA DICTIONARY (HIROTA-TODA-EXACT + WALL-EQUIVALENT):  $\tau_{w,n} = D_n(\mu - \nu)/D_n(\mu)$  (Hankel-determinant QUOTIENT),  $r_n = h_n(\mu - \nu)/h_n(\mu)$ , the Hirota coefficient source-pure from a scaled SIGNED Stieltjes recursion (worst  $1.7 \times 10^{-11}$ .. $4.0 \times 10^{-8}$  at full depths, dps-60/80 warded), and the sign adjudication:  $\hat{\gamma}_n > 0$  for all  $n \iff$  QUASI-DEFINITENESS of the signed defect measure  $\mu - \nu \iff$  the wall, with the explicit SIGNED-TODA  $X - Y$  structure — MAIN carries ALL 184  $r_n > 0$  (min +0.3666), the controls flip at  $n = 25/21/27$ ; HIROTA-CONE-GO not reached (the expected base verdict, now a theorem); and MODULE TWO, v956\_signedmoment\_halffilling\_duality.py (7/7 pattern gates,  $\sim 9$  s; probes holedual\_probe.py r228 note DLIV, 14/14, SPEC a823d4be3f0b1c06; pontryagin\_maxpos\_probe.py r229 note DLV, 17/17, SPEC b062906cb458da2a; jfraction\_probe.py r230 note DLVI, 17/17, SPEC 124cda6f00caeeca; rhp\_midpoint\_probe.py r231 note DLVII, 13/13, SPEC 90d2c5a8926d820a; read-only libraries port\_integrable\_kernel\_probe.py, hirota\_sign\_probe.py, fermiedge\_classify\_probe.py): THE HALF-FILLING GEOMETRY OF THE SIGNED MOMENT PROBLEM — (1) THE HALF-FILLING BOUNDARY (NOT-A-HOLE-EDGE + HALF-FILLING-BOUNDARY):  $N_w = \lceil \# \text{supp}(\tilde{\mu})/2 \rceil$  EXACT on all five windows and past the cap the wall dies IMMEDIATELY (first  $r$ -flip at  $N_w + 0/2/2/3/1$ ,  $O(1)$  offsets, two independent paths + mp dps-40 ward) —  $\mu - \nu$  is quasi-definite EXACTLY up to half-filling, THE WALL IS MAXIMAL; the complement identity and  $h$ -duality exact at dps 80 ( $1.3 \times 10^{-78}$  synthetic /  $6.6 \times 10^{-56}$  real w9 subset); R-DUAL-OBSTRUCTED typed; (2) THE FREE MOMENT WINDOW LAW (SIGNATURE-EXPLANATION-REFUTED + FREE-MOMENT-WINDOW-LAW + MAIN-EXHAUSTS-FREE-MOMENT-WINDOW): both sharp signature predictions FAIL on every window ( $n_{\text{flip}} = 184/153/170/367/592 \neq p = 263/211/237/503/816$ ); the corrected law — the largest Hankel window inside the  $S$  free moment parameters is  $n = (S + 1)/2 = N_w$ , MAIN survives THE ENTIRE FREE MOMENT WINDOW plus  $O(1)$  forced-tail degrees while the controls die at 11–15% of it ( $n = 25/21/27$ ); the inertia theorem exact in all parts (boundary inertia (184, 1, 0); mp dps-60 chain  $\text{ind}_-(H_{263}) = 43 \leq q = 104$ ); the maximal Lagrange contractor exact on the toy (the equivalence  $H_p > 0 \iff \|C\| < 1$  in BOTH truth values,  $10^{-53}$ .. $10^{-62}$ ) and ASTRONOMICALLY non-contractive at full degree ( $\sigma_{\max}(C) = e^{110.5}$ , MAXDEG-NOT-CONTRACTIVE); at window degree SAME-CONTRACTOR-EXACT ( $\lambda_{\max}(Q_{N_w}) = 0.999832$ .. $0.999999 < 1$  — the v881 testing law closes the circle); (3) THE FOUR-PATH DICTIONARY (FREEPREFIX-EXACT-WALL-EQUIVALENT + FULL-CHAIN-REVERSAL-EXACT + NO-SOURCE-REVERSAL): moment prefix = Jacobi chain = Euclidean chain = value chain, one object in four exact source-pure paths ( $m_k = h_0(J^k)_{00}$  exact in rationals; the r228 offsets ARE forced-coupling survival counts 0/2/2/3/1; the dps-300 Euclid prefix matches the real w9 value chain at  $2.5 \times 10^{-13}$ ; Epstein's FIRST WRONG EUCLIDEAN PIVOT exactly at its flip  $n = 25$ ); the dual signed measure carries the MIRRORED Jacobi chain ( $\alpha_m^\# = \alpha_{S-1-m}$  AND  $\beta_m^\# = \beta_{S-m}$ ) but is NOT a builder transformation — half-filling is not the center of a self-dual chain;  $\beta > 0$  stays the wall in EVERY coordinate; (4) THE MID-

POINT CONNECTION THEOREM (MIDPOINT-CONNECTION-EXACT + SIGNED-STOKES-WALL-EQUIVALENT, derived by residue calculus BEFORE the run and frozen): the dual FIK problem is an EXPLICIT  $L$ -gauge transform of the original,  $Y_{N-1+l}^\# = JY_{N-1-l+1}W$  with  $W = \text{antidiag}(1/L, L)$ , built from THE NODE POLYNOMIAL ALONE (exact in rationals; real w9 at  $9.1 \times 10^{-94}$ , dps 100); the gauge is  $h$ -FREE, so the wall's orientation sits EXCLUSIVELY in the  $h$ -chain the gauge does not see — the decisive Stokes multiplier is  $c_w\beta_n$  with  $c_w > 0$ ; honest negatives typed (NODE-LOG-POLE-REMOVER-ONLY; NO-SOURCE-CRITICAL-FILLING,  $t_c$  open; QUENCHED model supported at spread 0.44); THE MICRO FALSIFIER FROZEN: any future parametrix must blindly predict the forced tail 0/2/2/3/1 AND the control flips 25/21/27 from the same connection. THE CLAIM SPLITTING OF NOTES DLI/DLVII CARRIED TO THE LEDGER: new rows PRIME.PORT.TAU.FINITE.IIKS.01 [E] and PRIME.PORT.TAU.NOPOLE.COFINAL.01 [O] split out of the v883-registered umbrella PRIME.PORT.TAU.01 (which stays [O] for the fully symbolic arbitrary- $h$  statement, its ledger prose updated to record the finite-identity execution — NO silent upgrade), and PRIME.FREEMOMENT.JFRACTION.01 [E] + PRIME.JACOBI.DUAL.REVERSAL.01 [E] split from PRIME.FREEMOMENT.POSITIVEPREFIX.01 [O] (the wall itself in free-moment coordinates; half-filling is the LOCATION, its positive reachability the result to be proven); every wall statement in both modules is typed WALL-EQUIVALENT and consumed nowhere; surfaces: two veri cards in `tfpt_3`, the new integrable-dictionary-arc diary section in `tfpt_prime_front` with the live suite-count parentheticals refreshed (947  $\rightarrow$  949) and the TAU.01/LAX2.01 execution notes added at every [O] mention, the PRIME.PORT.TAU.FINITE.IIKS.01 / PRIME.PORT.TAU.NOPOLE.COFINAL.01 / freemoment contract blocks and the LAX2/HIROTA adjudication notes in `tfpt_research_contracts`, the endform article promotion section + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v956, README/Zenodo 949; FULL `run_all.py` ALL CHECKS PASSED at 949 modules; `bash build.sh audit`  $\rightarrow$  AUDIT OK; manifests regenerated LAST. The mincut stays base 4 / refined 5; no omega closed, no marker moved beyond the four [E]/[O] registrations and the recorded execution notes; NO RH claim anywhere)

2026-08-22 (CXVI) (the Bughunt-X-arc promotion — the six durable certified theorem sets of discovery rounds 192 and 194–198 graduate from the experiments tree into the load-bearing verification suite (suite 941  $\rightarrow$  947, ledger 1093  $\rightarrow$  1099), after the tenth adversarial bughunt (round 199, note DXXIII, `bughunt10_probe.py` 30/30, SPEC 5551aa7b967230f1, 11 findings, 1 MAJOR, 0 verdict flips, AND THE X4 HEADLINE: every exact claim of the arc reproduced in fully own code — the ACF/pole/arch laws to  $\leq 10^{-61}$ , the Fejér cap re-proved by an independent single-node dual route, the classical LMN/Matveev/Kolountzakis–Révész/Siegel citations web-verified verbatim) covered the arc and its corrections of record (KA/KB/KC/KE applied as appended CORRECTION-OF-RECORD blocks, KD note-only per precedent, note DXXIV) were ADOPTED VERBATIM on every new surface (in particular KA MAJOR: the round-197 nonnegativity gate class is retyped GRID-MEASURED with the verbatim scope rider — the record gate was grid-only with a sign-blind bar at the three deepest rungs, THE HEADLINE IS TRUE AND NOW STRONGER via Bughunt X's exact-rational Sturm chains at  $h = 4, 5, 13, 20$ ; KB MINOR: SUBCONE-SHARP-AT-Q2 reads “exact attainment at RATIONAL lags only, near-attainment elsewhere” — the

“f64 precision” wording is retired; KC: the “edge minimum IS the  $j_{r_0}$  ladder” identification is exact UN-NORMALIZED, the r197/r198 pair consistent; KE: the round-192 G11 symbolic leg was vacuous — the  $e \mid 2fk$  law’s machine content lives in the census gates; KD POSITIVE, note-only per precedent and recomputed in v951: the arch kernel IS the classical Weil archimedean kernel in the  $L \rightarrow \infty$  limit); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v949\_commensurability\_mechanism.py (11/11, POSITIVITY-IRRELEVANT-EXACTNESS-KILL; probe commensurability\_mechanism\_probe.py r192 note DXIV, 28/28, SPEC dbc14014899fb286 verified invariant under the appended KE block, record 276 s, re-run green at promotion 317.0 s): THE EXACT NODE MECHANISM — the smooth world’s whole  $2a$ -periodic oscillation collapses AT THE NODES to Möbius data (closed forms recomputed in-run), the structural-zero census is the  $e \mid 2fk$  law in exact integers ( $h = 4$ : 12/18,  $h = 8$ : 32/120), and THE FIRST BAKER-CLASS PRICING IN THE PROGRAM’S HISTORY is recomputed from the published constants (Liouville gap 3.79 dex, LMN 1995 gap  $7.099 \times 10^3$  dex, Matveev 2000 gap  $1.682 \times 10^9$  dex at the  $h = 4$  minimizer; full ladders to  $3.10 \times 10^4$  /  $1.18 \times 10^{10}$  dex: BAKER-TOO-WEAK); the dose is DIRECTION-BLIND (killed both ways at every dose rung + jitter, COMM-WALL-DEAD) and  $L_f(t = 1)$  is PSD at all four dose rungs WORLD-BLIND: the margin consumes position EXACTNESS, not incommensurability magnitude — the sought source functional must consume the exact positions wholesale; MODULE TWO, v950\_zero\_channel\_capacity.py (8/8, CHANNEL-CHARACTERIZED; probe zero\_channel\_capacity\_probe.py r194 note DXVII, 28/28, SPEC fa49271201ba30fb, record 170 s, re-run green at promotion 209.6 s): THE SHANNON CHARACTERIZATION OF THE ZERO-COMB DICTIONARY — typed EXPLORATORY INSTRUMENT CHARACTERIZATION at discovery and promoted as exactly that (NO storage claim, NO arithmetic novelty): the capacity surface reaches error-free full ternary ( $\log_2 3$ ) from  $\varepsilon = 0.01$ ; RESLIM LOW = 0.001 / MID = 0.003 WITH THE BUGHUNT-X F6 QUANTIFICATION ADOPTED ( $P = 0.0032$  under the 20000-trial null MC — marginal-but-nonnull single-reading, decisive with averaging  $0.042 \rightarrow 0.257 \rightarrow 0.938$ ; the sqrt- $n$  prediction held at 0.2325); LINEARITY-SCALE 0.01; parity-erasure recovery 1.000 WITH THE F7 COROLLARY TYPING (a corollary of the zero-SER cell, not independent evidence; the 108-case parity-code erasure theorem recomputed exactly in-run); P3 refuted honest (CHANNEL-GAIN-LOW > MID 0.508); the round-107 verdict cited NOT upgraded; MODULE THREE, v951\_cancellation\_functional.py (10/10, WALL-IS-WEIL-ACF-FORM-EXACT; probe cancellation\_functional\_probe.py r195 note DXVIII, 24/24, SPEC a50b85bb112513a1 verified UNTOUCHED by the Bughunt-X corrections, record 1036 s + 1061 s re-run, four pre-freeze amendments disclosed incl. A3 the substantive float64-direction-noise reversal, re-run green at promotion 1102.8 s): THE ACF LAW ( $x^T W(u) x = -2 \int_0^{L-u} A_x A_x(\cdot + u) dt$  — the v947 two quadrature slots are ONE OBJECT; verified at every atom of  $h = 4, 5$  on an own in-module wall build), THE POLE SQUARE ( $8 \sinh^2(a/2) (\int e^{-t/2} A_x)^2$ , sympy generic), THE ARCH LAW ( $c^* = (\gamma + \ln 2\pi)/2$  exactly, every mode of  $h = 4, 5$ ) — THE WALL IS A CLOSED TRUNCATED WEIL FORM of the PD test function  $g_x$ ; THE KD PIN RECOMPUTED IN-RUN (the arc’s external handoff object):  $\kappa + c^* \rightarrow -\frac{1}{2} [\text{Re} \psi(1/4 + i\omega/2) - \ln \pi]$  in the  $L \rightarrow \infty$  limit (BH10:  $6 \times 10^{-18}$ ; finite- $L$  corrections  $O(e^{-L/2})$  typed not-small); per-atom manifest positivity PROVABLY IMPOSSIBLE (Wiener–Khinchin); the pole alone carries positivity; the sign law

$n_+ = 0$  at all 14 rungs; priced honestly as an ALGEBRAIC RESTATEMENT (the relabeling barrier named, not crossed; WEIL-ALLTESTS  $\leftrightarrow$  RH a flagged loop consumed by nothing); MODULE FOUR, v952\_turan\_extremal.py (9/9, TURAN-BUDGET-TAU-GAP; probe turan\_extremal\_probe.py r196 note DXXI, 23/23, SPEC a6edc3f911e8f069 verified invariant under the appended KB block, record 767 s + 802 s re-run, re-run green at promotion 795.6 s): THE CLASSICAL PD-CONE THEORY PRICED for the first time in 196 rounds, TWO-SIDED — every cone bound NEGATIVE at every rung while the truth is  $\tau$ -deep positive (CONE GEOMETRY CANNOT SEE POSITIVITY; gapdex 11.35  $\rightarrow$  89.03 == the  $\tau$  ladder at slope  $-1.007$ ,  $R^2$  1.000 — the relabeling barrier named AT THE CONE LEVEL) yet the classical Fejér cap describes the wall’s drain capacity almost exactly (CONE-DRAIN-O1-CAPTURE 0.14–0.24 dex flat — the round’s real news); the Fejér coefficient theorem machine-proved in-module at  $n = 1, 2$  (construction + single-node dual certificates); THE KB RATIONAL-LAG IDENTIVITY RECOMPUTED ( $M_{\text{sub}}(L/2) == 1/2 == \cos(\pi/3)$  identically — the Fejér extremal lifts exactly at rational lags ONLY); ARCH-LEG-SUBCONE-ONLY named; the TURAN-CONE-POSITIVITY loop flagged consumed by nothing; MODULE FIVE, v953\_fewatom\_reduction.py (10/10, A-V0-NONNEGATIVE-STURM-CERTIFIED; probe fewatom\_reduction\_probe.py r197 note DXX, 25/25, SPEC 35fb341bb281b04b verified invariant under the appended KA/KC block, record 790 s + 811 s re-run, re-run green at promotion 830.7 s):  $A_{v_0} \geq 0$  AT ALL 14 RUNGS with THE KA SCOPE RIDER VERBATIM (the r197 gate class GRID-MEASURED; at  $h = 15/16/20$  the record minimum sits below the bar’s resolution — print evidence, not gate certificate; THE HEADLINE IS TRUE AND NOW STRONGER) and THE STURM CONTINUUM CERTIFICATES RECOMPUTED IN-MODULE at  $h = 4, 5$  (exact dyadic Fractions  $\rightarrow$  Chebyshev  $\rightarrow$  INTEGER primitive-PRS Sturm chain: zero roots in  $(-1, 1]$ ,  $P(\pm 1) > 0$  —  $A_{v_0} > 0$  on ALL of  $[0, L]$ , no floats; Bughunt X extends to  $h = 13, 20$  — THE BUGHUNT’S MATHEMATICAL DELIVERABLE FOLDED INTO THE PROMOTION, the bughunt round itself experiments-side per process precedent); THE ONE-LINE MECHANISM ( $A \geq 0 \Rightarrow$  every ACF sample  $\geq 0 \Rightarrow$  every atom negative — the v951 sign law reduced); the horizon bounded ( $Q_{99} \in \{2, 3\}$  incl. the holdout) WITH THE MANDATORY SCALE-RELATIVITY RIDER (the demand scale consumes every atom; NOT-A-COMPLEXITY-DROP-IN-H; any lossy tail bound certification-dead); the KC identification exact UN-NORMALIZED; OBJECT-A NAMED (the X6 targeting result): “prove  $A_{v_0(h)} \geq 0$  for all  $h$ ” is the ONE genuinely new tractable target — yields the sign law only, NO lever on wall positivity; and MODULE SIX, v954\_nodeless\_pf.py (10/10, PF-MODE-BASIS-DEAD-POLE-IS-THE-OBSTRUCTION; probe nodeless\_pf\_probe.py r198 note DXXII, 27/27, SPEC 7499c39a026d0d0f, record + deterministic re-run, amendment A1 disclosed, re-run green at promotion 817.1 s): THE PERRON-FROBENIUS ADJUDICATION — the r197 candidate route for Object-A DEAD IN THE MODE BASIS (off-diagonal censuses recomputed exactly at  $h = 4, 5$ :  $(15, 6)/(38, 17)$  mixed, checkerboard-mixed both parities; the trivial route killed both ways by the exact exhibit  $v = e_1$ ,  $A(L/2) = -1$ ), the obstruction LOCALIZED (the cone tests recomputed at  $h = 4$ : the wall operator violates on 11..14 of 20 Fejér members, THE NO-POLE OPERATOR ON ZERO — removing the pole alone repairs cone invariance: the piece that carries the positivity is the piece that breaks the cone), and the SURVIVING Z-STRUCTURE pinned (KERNEL-SPLIT-POLE-POSITIVE-HOPPING-Z: pole kernel negfrac 0.000 exactly, the wall’s kernel mixing ENTIRELY pole-vs-hopping); Object-A stays

open, GENUINELY FUNCTION-SPACE; the X6 discipline: “pole-vs-hopping balance” is the CLASS NAME of Object-W, not a new attack object. Ledger rows PRIME.COMMENSURABILITY.MECHANISM.01, PRIME.ZERO.CHANNEL.CAPACITY.01, PRIME.CANCELLATION.FUNCTIONAL.01, PRIME.TURAN.EXTREMAL.01, PRIME.FEWATOM.REDUCTION.01, PRIME.NODELESS.PF.01, all [E] NOT-RH-EVIDENCE active; the channel row carries the “instrument characterization, no storage claim” typing prominently; the arc’s terminal residue written everywhere in the canonical note-DII/DXVI/DXXIV form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k$  == LOOP, flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 = TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs: extension instantiated, TAILWPD world-front}; surfaces: six veri blocks in tfpt\_3, the new Bughunt-X-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed (941  $\rightarrow$  947; PDF recompiled), the endform article promotion section + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v954, README/Zenodo 947; one German research note appended to experiments/next.txt; FULL run\_all.py ALL CHECKS PASSED at 947 modules; bash build.sh release green; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. Round 193 (Bughunt IX) and round 199 (Bughunt X) stay experiments-side as process rounds per precedent — round 199’s Sturm-certificate deliverable is promoted INSIDE v953; the PRIME.RESIDUE.EXTERNAL.01 contract inventory deliberately NOT extended (the arc changes no residue statement). No omega closed, no marker moved beyond the six [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-21 (CXV) (the mechanism-arc promotion — the seven durable certified theorem sets of discovery rounds 184–191 graduate from the experiments tree into the load-bearing verification suite (suite 934  $\rightarrow$  941, ledger 1085  $\rightarrow$  1093), after the ninth adversarial bughunt (round 193, note DXV, 44/44, 9 findings, 1 MAJOR, 0 verdict flips, AND THE HEADLINE X5: every claimed-exact object of the arc reproduced in fully independent code with ZERO FAILED RECOMPUTES) covered the arc and its corrections of record (K1–K5, note DXVI) were applied as appended CORRECTION-OF-RECORD blocks and ADOPTED VERBATIM on every new surface (in particular K1 MAJOR: the round-186 structure token is written WALL-OFF-DIAGONAL-IS-ONE-FUNCTION-LOEWNER-EXACT everywhere — displacement rank 2 fixes ONLY the off-diagonal, the FULL wall is  $M_h \sim L_f + \text{diag}(\Delta)$  with source-side  $\Delta \neq 0$ , NOT the canonical Loewner matrix of the Loewner-1934 dictionary, and  $R_h(z)$  is the resolvent of  $L_f + \text{diag}(\Delta)$ , never the full-matrix claim; K2 MINOR: the ZB wiggle reads 1.07 in full precision (1.065464, twice replicated) — never the unsourced 1.14; K3 MINOR: the round-184 smoke lineage disclosed; K4: the glossary rule — every “source-side” qualified as either (eigenvector-free, given  $\tau$ ) or (root-free, ray-consuming)); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v942\_pi\_pattern\_transfer\_specificity.py (9/9, PI-TRANSFER-SPECIFICITY; probe pi\_pattern\_scan\_probe.py r184 note DIV, 35/35, SPEC 00fc85173fe07470, record 191 s, re-run green at promotion 212.2 s): THE DETECTOR-SPECIFICITY THEOREM — all five predefined pi-derived comb worlds land in the fake-world class (log10 tails  $-10.0.. -17.9$ , both holdout halves, ALL 100 non-Mangoldt instances, ZERO anomalies) while



the C-MANGOLDT positive control FIRES ON EVERY TRANSFERRED DETECTOR (spike  $C = 131.04$  vs null max 12.60; the structured controls prove the detectors CAN fire; pi's classical CF laws replicate as measured expectations) — TRANSFER VALIDITY PROVEN, the load-bearing methodology of the whole arc; typed INSTRUMENT CHARACTERIZATION, NOT ARITHMETIC NOVELTY, no pi-numerology claim, the P-BBP PROVISIONAL flag travels; MODULE TWO, v943\_mangoldt\_ablation\_localization.py (8/8, MANGOLDT-ABLATION-LOCALIZED; probe mangoldt\_ablation\_ladder\_probe.py r185 note DV, 26/27 with the single FAIL the disclosed control-side Beurling density band (5.74 vs  $[0.2, 5.0]$ ) — BH9-adjudicated an honest Poisson fluctuation, density-correct, the W4 headline unweakened; SPEC 504dcac5b2eac650, re-run green as typed at promotion 167.2 s, same single disclosed FAIL): THE ZERO-COUPLED SHARP LOCALIZATION — the additive property ladder W0 random  $\rightarrow$  W5 true Mangoldt switches BOTH arithmetic signatures on ONLY at W5 and SHARPLY (S1 jump 12.07 dex, S2 jump 61 $\times$ ); THE BEURLING HEADLINE ANSWER IS NO (full multiplicativity without Riemann-zero coupling produces NEITHER signature); necessity confirmed by every ablation, EVEN  $\pm 0.01$  POSITION JITTER KILLS BOTH; MODULE THREE, v944\_ground\_residue\_observability.py (11/11, GROUND-RESIDUE-OBSERVABILITY; probe ground\_residue\_obs\_probe.py r186 note DVI, 33/33, SPEC 48637c8898a1da5a verified invariant under the appended K1 correction block, record 782 s, amendment A1 disclosed, re-run green at promotion 792.7 s): THE EXACT RESIDUE IDENTITY  $\text{Res}_{z=\tau} R_h(z) = A_0^2 / \|d\|^2$  gated at ALL 14 rungs to  $\leq 7.8 \times 10^{-45}$ , the Feshbach scalar form with  $w'(\tau)$  EIGENVECTOR-FREE GIVEN  $\tau$  ( $\{\alpha, m, B\}$  source-linear, the CDLI adjudgate tie disclosed as prior art), STRICT INTERLACING  $\Leftrightarrow A_0 \neq 0$  both directions exact, GAP-IS-RESIDUE-IN-DISGUISE ( $\varepsilon \leq 2.55 \times 10^{-6}$ : not an independent handle), the honest descent EIGENVECTOR-DESCENDS-NOT-ELIMINATED (11.2–90.7 orders), the raw-gap floor RELABELED-AND-STOPPED, the  $\{l_0, l_2\}$  witness detector, and the K1-corrected off-diagonal Loewner law; MODULE FOUR, v945\_zero\_causal\_stratification.py (10/10, ZERO-CAUSALITY-DEMONSTRATED-WITH-STRATIFICATION; ONE JOINT MODULE for probes zero\_causal\_synth\_probe.py r187 note DVIII, 31/33 with the two FAILS the disclosed adjudication gates (the f-cell monotonicity violation and the r175-predicted S1 instrument death), SPEC c20e87eec6d158b9, re-run green as typed at promotion 819.0 s, and zb\_wiggle\_strat\_probe.py r190 note DXI, 23/23, SPEC 8639b3a78503a0f9 verified invariant under the appended K2 block, re-run green at promotion 64.5 s — r190 upgrades r187 BY REFERENCE, house style): THE CAUSAL ZERO-COUPLING DEMONSTRATION — comb worlds with controlled zero spectra synthesized through the r175 bridge run backwards; Z0 VALIDITY 0.0006 on  $C = 131.04$ ; the off-line dose surface monotone in  $\delta$  everywhere (first response  $(0.10, 10^{-2})$ , first death  $(0.05, 10^{-4})$ ); EXACT-ORDINATES-REQUIRED (Poisson AND GUE resamples both die); the stratification: the wiggle 1.07 deep inside every matched-dose selection spread (5.09/3.07/3.98), dose monotone in expectation, INV-FRAC 1.00 generic subset-overshoot, the per-zero leverage ladder ( $\gamma_2 = 21.02$  carries the most, no first-zero dominance, strongly subadditive), depth parity not-artifact; the r185 interpretation upgrades to DEMONSTRATED MECHANISM on the S2 channel — still NO claim about WHERE the true zeros lie; MODULE FIVE, v946\_census\_krein\_pencil.py (8/8, CENSUS-KREIN-PENCIL-PARTIAL-LIFT; probe census\_lift\_probe.py r188 note DIX, 72/72, SPEC 8ada6b97d56aca46, record 2066 s + 2060 s re-run, log lin-

eage disclosed in spec, re-run green at promotion 2067.1 s): THE EXACT KREIN  
 PENCIL —  $N_h(y) = (-1)^{K-1} \det(J) A_0 \det(\hat{A} - yJ)$  source-side with  $\sum \rho_k = -y_t$   
 exactly (dyadic-exact E1 at all six rungs, integer-Bareiss E2 at  $h \leq 13$ ); the  
 residue-sign ladder  $(1, 5)/(7, 3)/(6, 14)/(27, 14)/(45, 34)/(43, 73)$  MIXED at every true  
 rung; MIXEDNESS-TRANSFORM-INVARIANT (definite impossible in class);  
 MIXEDNESS-IS-ARITHMETIC (atoms-vs-no-atoms, wrong orientation, NOT a  
 sign source); H2 == KREIN STRONG DEFINITIZABILITY in operator form;  
 the bracket certified via ENVJ without computing a single root; roots never  
 consumed, tau-free; MODULE SIX, v947\_loewner\_pick\_dictionary.py (9/9,  
 LOEWNER-PICK-DICTIONARY-MISMATCH; probe loewner\_pick\_probe.py  
 r189 note DX, 28/28, SPEC a547448468899af9, record 1046 s + 1058 s re-run,  
 three pre-freeze amendments disclosed, re-run green at promotion 1082.9 s):  
 THE EXACT DICTIONARY — the wall is a DIAGONALLY-SHIFTED  
 one-function Loewner matrix  $\text{Raw}_h = L_f + \text{diag}(\Delta)$ , with THE NEW EXACT  
 COSINE-QUADRATURE DIAGONAL LAW  $\text{Raw}[k, k] - f'_{\text{prime}}(b_k) = 2a \text{pc}_k$   
 (both quadratures of the same atoms in different slots; sympy per-atom, mp  
 $\leq 6.0 \times 10^{-61}$ , BH9 own recompute  $\leq 6.3 \times 10^{-61}$ ); the pole leg a genuine Pick  
 function with EXACT Herglotz point mass  $2 \sinh^2(a/2) \delta_{-1/4}$ ; the Pick-class read-  
 ing REFUTED with margins ( $\lambda_{\min}(L_f) < 0$  at all 14 rungs, misses wall positivity  
 by 9.4–87.9 orders, refinement deepens); SMOOTH-PSD-AT-NODES via the  
 exact commensurate-sampling mechanism  $\sin(2a \omega_k) == 0$  (lattice geometry, not a  
 prime fingerprint); tau-screens both flat; the r45 LOEWNER-DEAD distinction  
 carried; and MODULE SEVEN, v948\_krein\_sign\_characteristic.py (10/10,  
 KREIN-SIGN-CHARACTERISTIC-LAW; probe krein\_definitizer\_probe.py  
 r191 note DXIII, 57/57, SPEC 332c1f48f48a6d82, record 1950 s + 1901 s re-run,  
 no post-freeze amendments, re-run green at promotion 1983.4 s): THE SIGN-  
 CHARACTERISTIC LAW — the kernel vector explicit,  $\varepsilon_i = -\text{sign}((F/A_0)'(y_i))$ ,  
 THE r188 RESIDUE LADDER IS THE KREIN SIGN-CHARACTERISTIC  
 MULTISSET (two-route proven; in the module on the full 32-cell exact Sturm  
 battery with no floats), THE PARITY SCAFFOLD (root-count parity per pole  
 gap from residue signs alone — the new source-side coordinate, verified at all  
 81 gaps), THE  $W(y)$ -CONGRUENCE-CLASS KILL BY THEOREM (the sign  
 characteristic is congruence-invariant: the mixed ladder kills the entire  $W(y) > 0$   
 class at once — the r188 upgrade path IMPOSSIBLE-IN-CLASS, remaining  
 doors named honestly), the Langer degeneracy exhibited (Cayley–Hamilton;  
 strong definitizability == H2 per rung), the realness-break witness detector  
 (the Krein invariant sees the witness class where the raw ladder is blind),  
 the strict definitizer ROOT-DEPENDENT (occupancy-minimality fails at  
 $h = 8, 13$  by exactly one interior doubled gap). Ledger rows PI.PATTERN.SCAN.01,  
 PRIME.MANGOLDT.ABLATION.01, PRIME.GROUND.RESIDUE.OBS.01, PRIME.ZERO.CAUSAL.  
 SYNTH.01, PRIME.ZB.WIGGLE.STRAT.01 (both on v945), PRIME.CENSUS.SPECTRAL.  
 LIFT.01, PRIME.LOEWNER.PICK.01, PRIME.KREIN.DEFINITIZER.01, all [E] NOT-RH-  
 EVIDENCE active; the mechanism-arc rows carry the NO-RH fences and the  
 “instrument characterization, not arithmetic novelty” typing prominently; the  
 arc’s terminal residue written everywhere in the canonical note-DII/DXVI  
 form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all  
 three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k$   
 == LOOP, flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 =  
 TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs:  
 extension instantiated, TAILWPD world-front}; surfaces: seven veri blocks  
 in tfpt\_3, the new mechanism-arc diary section in tfpt\_prime\_front with the

live suite-count parentheticals refreshed (934  $\rightarrow$  941; PDF recompiled), the endform article promotion paragraph + counts, the email draft promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v948, README/Zenodo 941; one German research note appended to experiments/next.txt; FULL run\_all.py ALL CHECKS PASSED at 941 modules; bash build.sh release green; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. Round 192 (the commensurability mechanism, note DXIV) STAYS experiments-side per precedent — it landed after the Bughunt-IX scope froze and waits for bughunt coverage; round 193 (Bughunt IX itself) stays as a process round. No omega closed, no marker moved beyond the eight [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-21 (CXIV) (the third externalization contract — the prime-front terminal residue packaged as a short, precise, self-contained EXTERNALIZATION CONTRACT PRIME.RESIDUE.EXTERNAL.01 for outside specialists in analytic number theory (large sieve, sampling/frame theory, de Branges spaces, explicit-formula methods), added as one new section in tfpt\_research\_contracts after the ALPHA.QUILLEN.EXACT.01 contract (before “Certifiability and recommended order”), plus a one-sentence pointer in the head keybox; DOCUMENTATION ONLY — no module touched, no number changed, no status marker moved, no ledger row added or edited, no gate closed or narrowed, NO RH claim (the contract identifier names the document, not a ledger row; the section states explicitly that it offers OPEN PROBLEMS and asserts no progress toward the Riemann Hypothesis in either direction): (1) the SELF-CONTAINED FINITE OBJECTS — rung  $h$ , modes  $\omega_k = k\pi/a$  with  $a = \frac{1}{2} \log h$  and  $K = \lceil 1.25 h \log h \rceil$ , the von Mangoldt atoms  $(\log q, \Lambda(q)/\sqrt{q})$  as the only arithmetic input, the wall matrix  $M_h = M_{\text{pole}} + M_{\text{arch}} - M_{\text{prime}}$  with the prime block linear in two atom transforms per mode (v935), the wall argmin ray  $d$ , the census polynomial  $N(y)$  (numerator of  $F(y) = c_0 + \sum (-1)^k c_k y/(y - b_k)$ ), the moment–Laurent form  $\Phi(z)$  (v924) and the jet mass  $\delta_h = |J_h|_{G_h}^2$  exactly (step A, v939); (2) the THREE MINIMAL OPEN STATEMENTS in the canonical note-DII form — R1 THE TRIPLE ( $\{H1 \wedge H2 \wedge H3\}$ -cofinal, one rung per dyadic block, all three at the same  $h$ ; H1: no census root with  $\text{Re } y \geq c^* y_t$ ,  $c^* = 1.10\text{--}1.15$ ; H2: census complete-real nonnegative; H3:  $y_t \leq 0.155 T_z^4$ ; the limsup form only mod the measured defect  $D = 0.0042$ ), R2 THE LOOP CHARACTERIZATION (census-forall- $k$ , honestly stated as RH-equivalent-in-currency via the four independent machine-detected proof-route cycles v928/v929/v930/v938; external value: a new independent sign source or a proof the loop is essential) and R3 THE H-PIN (the one  $\lambda$ -uniform edge of the pair  $\{L1, \text{WPD}\}$  with the  $\Omega$ -a/ $\Omega$ -b split and the two machine-exhibited linearity-escape maps — the eigenvector map and the census-root map, v937); (3) the COMPLETE TYPED HYPOTHESIS LIST — 11 PROVEN-FINITE rows citing v931/v932/v933/v934/v935/v936/v937/v938/v939/v940/v941 (theorem PF with the exponent the trace; the rate dictionary  $a = p/2 - 1$ ; the subsidy-empty pigeonhole; the unconditional Landau/Gonek pricing; the mode-level Landau bridge; the tau-free DK exclusion  $h \leq 13$  with the exact  $b^*(h)$  schedule; the WPD == H-pin consolidation; the exact census semigroup  $e^{-tT}$ ,  $T = 4y \partial^2 + 2 \partial$ ; step A exact; the sub-dof composition chain; the source-free Gram + projective linearity), 7 CITED-CLASSICAL rows (Landau 1912; Gonek 1985/1993 with Gonek 1984 flagged never consumed; Montgomery–Vaughan 1974; Ortega-Cerdà–Seip 2002; Nazarov–Turán 1993; Rodgers–Tao 2020 + Polymath 15 as cited ceilings),

1 OPEN row (R1–R3); (4) the HONEST MEASURED BLOCK — margins  $\geq 1.405$  with selector margins 2.14/3.05, the frozen alignment triple (descriptive band (0.38, 0.54), wall lock  $\mu_m = 0.2372 \log_{10} g_{\min}$  at  $R^2$  0.9916, deep excess 5.64–7.58 dex),  $\theta_\infty$  in  $[0.0766, 0.0977]$  typed OPEN-NONPERTURBATIVE-VARIATIONAL, and the refuter ladder ALWAYS PAIRED per BH8-F7 (measured 30–33 orders with the proven  $b^*(h)$  schedule 23.8/21.7/15.7/4.4 at  $h = 4/5/8/13$ , none at  $h \geq 16$ ); (5) the KILL-ATLAS SUMMARY TABLE — eleven named dead route classes (pointwise symbol minorants,  $\ell^1$  majorants, phase-discarding additive forms, Vieta pinch/power sums, moment/trace caps, Riccati/manifold transport, dBN pinch, global CBJ frame, sub-dof scoping, Carleson-as-floor, naive floor-power composition) each with its one-line death mechanism and round/witness; (6) the RUNNABLE FINITE MODELS — v931–v941 + the substrate v922–v930, the eleven frozen SPEC-hashed discovery probes (r171–r182), the frozen builder radius4\_an\_probe.py with the four control worlds, the  $2 \times 10^7$  verified-ordinate cache and the PT21 horizon, with exact paths; (7) the HONEST FORMALIZATION SURFACE — the only prime-front Lean content is the finite spacing/jet algebra (SpacingProduct.lean and companions); NOTHING of R1–R3 is formalised, stated plainly; (8) MUST-FAIL CONTROLS MF-P1–MF-P6 (Epstein/Scramble/Smooth refusal, the no-zero ancestry test, the predefinition test, and the loop firewall over the four flagged classes census-forall- $k$  /  $A_0$ -triangle / zero-verification-as-hypothesis / RH-conditional second moments); (9) ACCEPTANCE TESTS — an external result counts iff it delivers (a) an explicit all- $h$  all- $a$  bound, (b) a genuinely new sign source that separates AND independently orients, or (c) a proof that one of R1–R3 is RH-equivalent (recorded as closing the hardness question honestly, never as progress toward RH); the website paper mirror of tfpt\_research\_contracts (papers.ts) gains one section entry; tfpt\_research\_contracts.pdf, changelog.pdf and introduction.pdf recompiled and mirrored via bash build.sh website; maps and changelog.ts regenerated via bash build.sh gen; bash build.sh audit → AUDIT OK; manifests regenerated LAST. No omega closed, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; NO RH claim anywhere)

2026-08-20 (CXIII) (the post-bughunt-arc promotion — the six durable certified theorem sets of discovery rounds 177–182 graduate from the experiments tree into the load-bearing verification suite (suite 928 → 934, ledger 1079 → 1085), after the eighth adversarial bughunt (round 183, note DI, 20/20, 7 findings, 1 MAJOR, 0 verdict flips) covered the arc and its corrections of record (F1–F7, note DII) were applied as an appended CORRECTION-OF-RECORD block and ADOPTED VERBATIM on every new surface (in particular BH8-F1: the round-182 terminal adjudication is written SAME-CLASS-NOT-NEW-EDGE everywhere — a CLASS-level convergence, never an object identity — and BH8-F7: the round-177 refuter margin is always cited PAIRED, the proven  $b^*(h)$  schedule 23.8/21.7/15.7/4.4 orders above the  $10^{-25}$  bar at  $h = 4/5/8/13$  and none at  $h \geq 16$  alongside the measured 30–33 orders; per BH8-F2 the canonical rounds are 177 = manifold, 178 = L1/WPD, 179 = dBN, 180 = CBJ frame, 181 = CBJ sub-dof, 182 = alignment); every source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v936\_manifold\_invariance\_exclusion.py (10/10, MANIFOLD-INVARIANCE-EXCLUSION; probe manifold\_invariance\_probe.py r177 note CDXCV, 39/39, SPEC af86ec3b097ae8c4, record 1354.1 s, three amendments + one smoke fix disclosed, re-run green at promotion 1337.6 s): THE TAU-FREE DK EXCLU-

SION THEOREM —  $\sin \theta \leq (r_0/\tau)/(FG - a)$  with  $\tau$  CANCELLING in ratio form, direction-universal and witness-free, PROVEN  $h \leq 13$  at the frozen bar (measured to  $h = 20$ , the split kept) + the exact required-bar schedule  $b^*(h)$ ; the measured full-space refuter price ladder  $1.35 \times 10^5 \rightarrow 6.73 \times 10^8$  strictly growing with the mechanism identified as ALIGNMENT-SHARING; the BA3 scale-gauge theorem (not a universal catcher); the induction route priced dead AS A REDUCTION (TRANSPORT-RELABELS-NOT-REDUCES + LEVEL-CHEAPER-THAN-INCREMENT); MODULE TWO, v937\_l1wpd\_closure\_reduction.py (10/10, L1WPD-CLOSURE-REDUCTION; probe l1wpd\_closure\_probe.py r178 note CDXCIV, 29/29, SPEC 0e306146bebbd9be, record 228.9 s, no amendment, re-run green at promotion 224.4 s): WPD's  $\lambda$ -uniform content == the H-pin edge (AND-fire demonstrated at the declared edges — the pair {L1, WPD} carries ONE open  $\lambda$ -uniform edge, not two); the three-part bridge-expressibility verdict (source data bridge-expressible per census; the floor escapes linearity at the eigenvector and root maps, exact exhibits; the SW pricing vacuous — no closure); the v925 all- $x$  sharpening propagated with own replication ( $x_0 = 121/112$ , all 357  $D_{cs} > 0$ ); the r133-vs-r171 H1/H2/H3 name collision flagged; MODULE THREE, v938\_dbn\_heatflow\_census.py (11/11, DBN-HEATFLOW-CENSUS; probe dbn\_heatflow\_probe.py r179 note CDXCVI, 26/26, SPEC 67eaf86c7bfa7d84, record 672.4 s, one smoke fix disclosed, re-run green at promotion 659.4 s): THE EXACT CENSUS SEMIGROUP — the de Bruijn–Newman flow transports onto the derived census polynomial as a closed-form FINITE sum  $e^{-tT}$  (no zeros, no ODE) with the exact trace law  $\text{trace}(N_t) = \text{trace} + 2d(2d-1)t$  (measured deviation 0.0) and the factor-4 normalization lemma (window  $[0, 0.055]$  exact, BH8-verified); the dictionary THREEFOLD-INEQUIVALENT (every pinch claim dictionary-dependent); the missing half of the generator == the RH-conditional PAIR class (flagged loop, not consumed); theta-flow-inert floors ( $\geq 4021 \times$  the window); the  $\lambda_h$  criticality ladder  $2.39 \rightarrow 0.216$  strictly decreasing with THE LIMIT OPEN — honestly NOT an off-line-zero detector (Epstein slack larger); the witness preserved-not-expelled; MODULE FOUR, v939\_cbj\_frame\_adjudication.py (10/10, CBJ-FRAME-ADJUDICATION; probe cbj\_frame\_probe.py r180 note CDXCVII, 41/41, SPEC d7fbf2d956184674, record 1042.5 s, no amendment, re-run green at promotion 1005.5 s): STEP-A EXACT —  $\delta_h = |J_h|_{G_h}^2$  with remainder  $R = 0$  proven generically (the single most promotable theorem of the arc, certified at 14 rungs,  $\text{dev} \leq 1.9 \times 10^{-46}$ ); the cluster normal form (the basis-change determinant == the spacing product); the Montgomery–Vaughan pricing (killed exactly at linkage 1, carrying at  $3.15\pi$  with the exact rational margin  $1/21$  + the two-grid repair); the dof/seam/occupancy kill triple ( $n = 474$  vs rank 128, dof 112.8; seam 0.994;  $-0.7484$  dex/atom): the GLOBAL review form dead, the per-cluster floor carrying  $k$ -uniformly at fixed  $r$ ; the source-only selector ( $\hat{h}_k = 7/13/31/\dots/8191$  recomputed in-run, Bertrand-cofinal) underperforms the block average yet clears the SF2 demand (margins  $2.14/3.05$ , all rungs  $\geq 1.405$ ); the house Gram min-eig ladder  $1.1 \times 10^{-17} \rightarrow 5.2 \times 10^{-113}$ ; MODULE FIVE, v940\_cbj\_subdof\_blockfloor.py (9/9, CBJ-SUBDOF-BLOCKFLOOR-KILL; probe cbj\_subdof\_probe.py r181 note CDXCIX, 37/37, SPEC 2db82c76ce5f067c, record 867.9 s, one spec-text amendment disclosed (the sympy-exact Fejér constant  $1/6$ , BH8-verified), re-run green at promotion 772.1 s): the sub-dof-scoped floor statement posed exactly and honestly KILLED at the decisive measurement — the sub-dof mass fraction falls at  $+0.999$  dex/rung (recomputed) and saturates at zero, the scoped margin dies AT THE SELECTOR RUNGS ( $8.02 \times 10^{-3}$  at  $\hat{h}(B3) = 13$ , dead by  $\sim 125$ ); the unscoped r180 floor remains the

measured carrier (killed is the PROOF ROUTE through the well-conditioned subspace); CARLESON PRICED FOR THE FIRST TIME — the unconditional embedding WRONG-DIRECTION for a floor (Ortega-Cerdà-Seip, Ann. of Math. 155 (2002), BH8-web-verified) and the complete lower-class table with Nazarov–Turán (exponential in the term count) as the classical mirror of the measured occupancy law; FRACTION-WORLD-SEPARATING (floor value world-insensitive, mass location world-separating); the fixed- $r$  compactness sketch; MODULE SIX, v941\_alignment\_law\_factorization.py (11/11, ALIGNMENT-LAW-FACTORIZED; probe alignment\_law\_probe.py r182 note D, 37/37, SPEC e4cdb9a932093196 verified invariant under the appended correction block, record 775.6 s, five freeze retypings disclosed, re-run green at promotion 776.5 s): the sharpest unexplained regularity frozen as a MEASURED TRIPLE — the half-power envelope ( $\alpha$  inside the descriptive band  $(0.38, 0.54)$  at all 14 rungs, never a prediction per BH8-F4), the quarter-depth wall lock ( $\mu_m = 0.2372 \log_{10} g_{\min}$ ,  $R^2 0.9916$ ) and the deep excess  $(5.64\text{--}7.58 \text{ dex} == \text{the r181 kill mass})$ ; GRAM-SOURCE-FREE proven three ways; the key adjudication:  $s_i A_0$  LINEAR in  $d$ , the location law PROJECTIVE and  $A_0$ -free — the r178 obstruction does NOT bite; the one shared nonlinear link is the wall eigenmap (the law exactly as bridge-expressible as  $\delta$ , given  $d$ ); ALT-JETS-BREAK-ALIGNMENT; WITNESS-BREAKS-ALIGNMENT; the terminal adjudication SAME-CLASS-NOT-NEW-EDGE (four distinct formal statements over the same source object, the wall argmin ray  $d$ , in  $A_0$ -quotient form; no new residue edge via the machine-checked G44 exclusion, not via identity). Ledger rows PRIME.MANIFOLD.INVARIANCE.01, PRIME.L1WPD.CLOSURE.01, PRIME.DBN.HEATFLOW.01, PRIME.CBJ.CONFLUENT.FRAME.01, PRIME.CBJ.SUBDOF.BLOCKFLOOR.01, PRIME.ALIGNMENT.LAW.01, all [E] NOT-RH-EVIDENCE active; the arc's terminal residue written everywhere in the canonical note-DII form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$ ; the limsup form only mod  $D = 0.0042$ ) + {census-forall- $k == \text{LOOP}$ , flagged} + {H-pin = the one  $\lambda$ -uniform edge of the pair: L1 = TAIL proven + H-pin open; WPD( $a < \gamma_1^2$ )  $\Leftarrow$  H-pin; the WPD non- $\lambda$  legs: extension instantiated, TAILWPD world-front}; surfaces: six veri blocks in tfpt\_3, the new post-bughunt-arc diary section in tfpt\_prime\_front with the live suite-count parentheticals refreshed ( $928 \rightarrow 934$ ) and the superseded awaiting-bughunt prose rewritten, the endform article §8.22 + counts + inventory, the email draft seventh-promotion paragraph, the website PROMOTED card + DAG chips + honesty-banner range v535–v941, README/Zenodo 934; one German research note appended to experiments/next.txt (DIII); FULL run\_all.py ALL CHECKS PASSED at 934 modules; bash build.sh release green; bash build.sh audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. No omega closed, no marker moved beyond the six [E] rows, census cardinality 4 unchanged,  $\theta_\infty$  stays OPEN, the  $\lambda_h$  limit stays OPEN; the two bughunt rounds (176 and 183) stay experiments-side as process rounds per precedent. NO RH claim anywhere)

2026-08-20 (CXII) (the triple-cofinal-arc promotion — the five durable certified theorem sets of discovery rounds 171–175 graduate from the experiments tree into the load-bearing verification suite (suite  $923 \rightarrow 928$ , ledger  $1074 \rightarrow 1079$ ), after the seventh adversarial bughunt (round 176, note CDXCII) covered the arc with 0 verdict flips and its corrections of record (F1–F8, applied to the frozen records as appended CORRECTION-OF-RECORD blocks in round 177)

were ADOPTED VERBATIM on every new surface; every qualifying source probe RE-RUN GREEN AS TYPED before the surface edits (logs kept as \*.promo\_rerun.log): MODULE ONE, v931\_jetmass\_floor\_theorems.py (10/10, JETMASS-FLOOR-THEOREMS; probe jetmass\_floor\_probe.py r171 note CDLXXXVII, 36/36, SPEC 57de8b2a83677a9c, record 2056 s, two amendments disclosed, re-run green at promotion): THEOREM PF — the product floor  $F/A_0 \geq (1 - c^*/z)^{(1+\kappa)/c^*}$  from the census product identity + the Vieta trace at coefficient level + a concavity lemma, conditional EXACTLY on {H1 + H2 per rung}, THE EXPONENT IS THE TRACE (the r140 far-field law its  $\kappa \rightarrow 0$  shadow); H1 source-pure at all 25 rungs, H2 census complete-real incl. degree-116; the certified curve under the truth at  $\sim 100k$  tail zeros; the factorization [PF]  $\times$  [WF classical]  $\times$  [RATE] with JETLOCK-MEAS eliminated; the witness modus-tollens boundary; BH7-F3 definitional legs recomputed honestly; MODULE TWO, v932\_toproot\_theta\_statement.py (10/10, TOPROOT-THETA-STATEMENT; probe toproot\_theta\_probe.py r172 note CDLXXXVIII, 33/33, SPEC cf27df22aa5dffbf, record 2813 s, re-run green at promotion): TOPROOT stated SEQ-cofinal with three locked forms; the exact rate dictionary  $a = p/2 - 1$  proven at  $p = 3, 4, 5$ ; THEOREM TR-CAP proven and typed circular-for-TOPROOT; the  $A_0$ -triangle route the FOURTH flagged loop; admissibility recomputed in-run ( $p_{\max} = 10.2695$ , six powers of margin); H3 ( $y_t \leq 0.155 T_z^4$ , ancestors == {SOURCE}) certified 26/26 margin  $\geq 1.99$ , REFUTABLE in the inflation witness and not norm-continuous; MODULE THREE, v933\_h3\_cofinal\_adjudication.py (10/10, H3-COFINAL-ADJUDICATION; probe h3\_cofinal\_probe.py r173 note CDXC, 36/36, SPEC 876dafc977d3d8fc, re-run green at promotion 1339.5 s): saturation REAL with  $p_{\text{loc}} \rightarrow 4.024$  and  $\theta_\infty$  band  $[0.0766, 0.0977]$  under the bar (OPEN); pigeonhole exact yet subsidy-empty (quantifier gap  $D = 0.004183$  — the limsup form only mod  $D$ , BH7-F2); the witness excluded by the certified manifold {ground-state cert + two-sided lock + BA3 bridge  $\times 3.6 \times 10^8$ }, escalation-stable  $\times 10^5$ , H3 DK-norm-stable at measured residuals — SCOPED to the DK radius (gated  $h \leq 13$ , measured to  $h = 20$ , crossover  $h = 16$ , BH7-F4); all saturation analytics recomputed in-run from the cited ladder; MODULE FOUR, v934\_gonek\_pricing\_unconditional.py (8/8, GONEK-PRICING-UNCONDITIONAL; probe gonek\_pricing\_probe.py r174 note CDLXXXIX, 20/20, SPEC 3050678b352eaa9a, record 188 s, re-run green at promotion 244.2 s): Landau 1912 and Gonek 1985/1993 identified UNCONDITIONAL, Gonek 1984 machine-checked NOT-AN-ANCESTOR (the hidden loop detected, flagged, never consumed; externally re-verified in Bughunt VII); the envelope closed form with finite census-depth-insensitive ceiling; prefactor-insensitive absorption at any finite constant;  $\hat{c} \leq 0.086$  at snr up to  $2.7 \times 10^4$ , typed CONSTANT-EMPIRICAL-PER-CENSUS — the Gonek item LEAVES the residue; MODULE FIVE, v935\_thetainf\_landau\_bridge.py (9/9, THETAINF-LANDAU-BRIDGE; probe thetainf\_pin\_probe.py r175 note CDXCI, 28/28, SPEC 3044558e5fa52e01, record 577 s, re-run green at promotion 718.1 s): THE MODE-LEVEL LANDAU BRIDGE — the program’s first linear per-mode classical identity, residuals  $1.4 \times 10^{-5} \rightarrow 5.4 \times 10^{-5}$  at the 20M-zero census; the dictionary a proven bijection (pins nothing); the variational conditioning wall typed MEASURED-LAW extrapolation per BH7-F7 (“gemessene”, not “bewiesene” Verschwendung),  $T_{\text{req}} = 3.76 \times 10^{14} \rightarrow 1.75 \times 10^{77}$  vs  $T_{\text{PT}} = 3.0 \times 10^{12}$ ;  $\theta_\infty$  re-typed OPEN-NONPERTURBATIVE-VARIATIONAL. Ledger rows PRIME.JETMASS.FLOOR.01, PRIME.TOPROOT.THETA.01, PRIME.H3.COFINAL.01, PRIME.GONEK.PRICING.01, PRIME.THETAINF.PIN.01, all [E] NOT-RH-EVIDENCE

active; the arc's terminal residue written everywhere in the BH7-F1-corrected form: the TRIPLE  $\{H1 \wedge H2 \wedge H3\}$ -cofinal (one rung per dyadic block, all three at the same  $h$  — NOT  $H3$  alone; limsup only mod  $D = 0.004183$ ) + {census-forall- $k$  == LOOP, flagged} + {L1, WPD}; surfaces: five veri blocks in `tfpt_3`, the new triple-cofinal-arc diary section in `tfpt_prime_front` with the live suite-count parentheticals refreshed ( $923 \rightarrow 928$ ), the endform article and the email draft refreshed with the §8.21-class promotion record and the triple wording, README/Zenodo 928, the website PROMOTED card + DAG chips + honesty-banner range v535–v935; one German research note appended to `experiments/next.txt` (CDXCVIII); FULL `run_all.py` ALL CHECKS PASSED at 928 modules; `bash build.sh` release green; `bash build.sh` audit  $\rightarrow$  AUDIT OK; manifests regenerated LAST. No omega closed, no marker moved beyond the five [E] rows, census cardinality 4 unchanged, the quartic law stays MEASURED,  $\theta_\infty$  stays OPEN; rounds 176–180 stay experiments-side awaiting Bughunt VIII. NO RH claim anywhere)

2026-08-20 (CXI) (the externalization contracts — the two open non-RH kind-(A) external-math targets SEAM.EQUIV.01 and ALPHA.QUILLEN.EXACT.01 packaged as short, self-contained EXTERNALIZATION CONTRACTS for outside specialist mathematicians (conformal nets/VOA/AQFT for the seam; determinant lines/ $\eta$ -invariants/index theory for alpha), added as two new sections at the end of `tfpt_research_contracts` (before “Certifiability and recommended order”), plus a one-sentence pointer in the head keybox; DOCUMENTATION ONLY — no module touched, no number changed, no status marker moved, no gate closed or narrowed, NO RH claim: (1) the SEAM.EQUIV.01 contract states the minimal theorem (the scaling limit of the gapped  $\mu_4$ -equivariant CAR collar exists in local-net topology; the index-4 simple-current extension functor commutes with the limit; the OS reconstruction is the holomorphic  $(E_8)_1$  lattice net) broken into the four review-named lemmata L1 (uniform energy/nuclearity bounds — honestly typed: NO uniform nuclearity estimate exists in the corpus), L2 (convergence of local algebras, the cited Osborne–Stottmeister scaling-limit theorem with the  $\mu_4$ -equivariant structure retained), L3 (crossed product commutes with the limit; the finite index-4 Q-system is explicit, v125/v154/v469) and L4 (uniqueness of the holomorphic  $c = 8$  reconstruction;  $\det K = 1$  vs the same- $c$  rival  $SO(16)_1$ ), with the complete typed hypothesis list (12 PROVEN-FINITE rows citing v302/v113/v155/v160/v356/v366/v367/v368/v461/v376/v377/v378/v379/v125/v154/v216/v286, 5 CITED-CLASSICAL rows, 1 OPEN target), the runnable finite models, the HONEST Lean axiom surface (`SeamScalingLimit.lean` kernel-proves only the arithmetic hypotheses by decide; the analytic content is named cited axioms over opaque Props; no sorry; explicitly flagged: L1–L4 have ZERO Lean content beyond the composition audit), five must-fail controls (MF-S1:  $c = 8$  alone must NOT suffice — the red-team point; MF-S2: index-2 glue must yield  $SO(16)_1$ ; MF-S3: the trivial  $M = 3$  collar must give no chiral limit; MF-S4: non-order-4 clock breaks the marks; MF-S5: the v286 import firewall) and four acceptance tests; (2) the ALPHA.QUILLEN.EXACT.01 contract states the two-step theorem T1 (gap-stable continuum limit  $c_1(\det D_{\text{finite}}) \rightarrow c_1(\det_\zeta D_{\text{seam}})$  under norm-resolvent convergence with the Fermi gap uniformly open over the twist torus — the Dai–Freed section has no zero; finite side computed, v472) and T2 ( $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0 = \text{Bismut–Freed curvature} = \text{inflow response with the exact corpus coefficients } k_0 = |C| = 1, k_Y = 5/3, b_1 = 41/10, c_3 = 1/(8\pi), \text{exponent } -5/4 = -q(D_5))$ ), the two named open faces (the



analytic diagonal- $\zeta$ /multiplicity face v484/v485 and the topological EM1 lemma v48/v435), the typed hypothesis list (10 PROVEN-FINITE rows citing v3/v48/v341/v382/v434/v470/v435/v342/v367/v472/v484/v485/v391/v433, 5 CITED-CLASSICAL rows, 1 OPEN target), the runnable finite models, the HONEST Lean statement (there is NO Lean formalisation of this target — stated plainly), four must-fail controls (MF-A1: no experimental  $\alpha$  anywhere in the construction; MF-A2: deck/carrier/Chern-level ablations must destroy the CODATA match; MF-A3:  $M = 3 \Rightarrow C = 0$ ,  $M = -1 \Rightarrow C = -1$ ; MF-A4: the  $\{0, 3, 6\}$   $\pi$ -power/metric-independence partition) and the review’s three acceptance tests verbatim (no experimental  $\alpha$  input; unique physical fixed point 137.0359992168; ablations destroy the hit) plus one structural test; (3) the website paper mirror of `tfpt_research_contracts` (`papers.ts`) gains one section entry covering both contracts; `tfpt_research_contracts.pdf` and `changelog.pdf` recompiled and mirrored via `bash build.sh website`; `maps` and `changelog.ts` regenerated via `bash build.sh gen`; `bash build.sh audit`  $\rightarrow$  AUDIT OK; manifests regenerated LAST. Both targets stay [O]; NO RH claim anywhere)

2026-08-19 (CX) (the endgame-arc promotion — the five durable certified theorem sets of discovery rounds 162–169 graduate from experiments/ into the load-bearing suite as `v926_fullgap_growthlaw_theorems.py`, `v927_blockaverage_substrate_theorems.py`, `v928_depthblock_transfer_theorems.py`, `v929_sigmafloor_factorization.py` and `v930_window_instrument_mechanism.py` (suite 918  $\rightarrow$  923, ledger 1069  $\rightarrow$  1074); all five rows typed NOT-RH-EVIDENCE in either direction — identities, recipe-derived laws, finite/classical closures and one honest mechanism record: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at  $\{\text{TOPROOT, TLAWCAP-block, QSUBGAP-floor}\} +$  the  $a$ -walls — now carried WITH the r168/r169 terminal coordinates: the  $\lambda$ -uniform budget content factorizes as [census schedule, classical per  $k$ , LOOP if for-all- $k$ ]  $\times$  [SIGMA-FLOOR == DC-leg (classical per census)  $\times$  JET-MASS-FLOOR (the terminal  $\lambda$ -uniform residue, typed OPEN)]; every measured leg typed verbatim, the quartic law NOT claimed proven: (1) v926 THE QUARTIC/GROWTH-LAW FAMILY (round 162, note CDLXVII, ledger row PRIME.FULLGAP.GROWTHLAW.THEOREMS.01 [E]; source probe `fullgap_growthlaw_probe.py`, 28/28, SPEC 26bdb5a87f63c519, PINNED from run-of-record 4061.6 s + deterministic re-run 4237.5 s with timing-normalized diff EMPTY, both logs kept, per the multi-hour convention — its GL algebra re-gated in Bughunt V G19/G20 and recomputed in-run) — GL1–GL5 exact (the loo pinch: every leave-one-out ground matches  $\lambda_1$  within the  $\varepsilon^2$  budget; the arrowhead compression with the dual row; the one-row price equivalence + razor vacuity: D3 COLLAPSES TO THE RAZOR; the rate bookkeeping  $J t_r == 1 + \text{FULLGAP}$ ; the ladder/quartic chases + the razor composition: the QSUBGAP-floor flows from the  $\Theta$ -window through the razor); THE QUARTIC LAW  $J == \Theta T_z^4$  typed MEASURED with the Bughunt-V free-exponent/AIC adjudication RECOMPUTED IN-RUN ( $p = 3.9072 \pm 0.1136$ ; AIC prefers FIXED 4 (−30.87) over free (−29.79), rejects 3.5/4.5 at 3.6/5.2  $\sigma$ );  $\rho_j == 1.0000$  for ALL 189 left-out zone zeros (jet-COLLECTIVE, per-zero deflation REFUTED-MEASURED); the finite- $P$  moment model undershoots by 1.4–4.5 dex (the Y3/R4 rate-blind trace-cap vacuity class, one level higher); (2) v927 THE COFINAL BLOCK-AVERAGE SUBSTRATE (round 166, note CDLXXV, ledger row PRIME.BLOCKAVERAGE.SUBSTRATE.01 [E]; source probe `tau_blockaverage_probe.py`, 27/27, SPEC d86f42a04c69a4e2, RE-RUN GREEN-

AS-TYPED AT PROMOTION 2372.4 s with identical SPEC and empty timing-normalized diff) — BA1 (a certified block sum  $> 0$  forces  $\geq 1$  positive rung; dyadic hits COFINAL), BA2 (the determinant chain IS the  $\tau$  sign, Jacobi/Sylvester), BA3 ( $\tau \geq \text{zsum} - \text{OFF EXACT}$ , proven-mod-cited: PT21 + HSW22 + the r131 OFF recipe + ward-class cache, NO-LOOP), the Weyl split measured VACUOUS by 10.9–54.6 dex; the certified block enclosures (all lower ends  $> 0$  on B2/B3: POSITIVE-RUNG-PER-BLOCK CERTIFIED below the horizon), the wall dictionary  $\text{sign}(\tau) == \text{sign}(\text{wall}) == +1$  at ALL 25 rungs, the tlaw ladder with 19 NEW corpus-depth rungs; the AVG-BUDGET-WINDOW beyond the horizon typed OPEN; (3) v928 THE DEPTH-BLOCK TRANSFER FAMILY (round 168, note CDLXXXIII, ledger row PRIME.DEPTHBLOCK.TRANSFER.01 [E]; source probe depthblock\_transfer\_probe.py, 36/36, SPEC a4cd07144e0c5222, ZERO AMENDMENTS, RE-RUN GREEN-AS-TYPED AT PROMOTION 2371.6 s) — DT1 PROVEN (the step chain is exact algebra on top of BA3); DT2 THE DEMAND LAW PROVEN-BY-RECIPE (the round’s discovery:  $\varepsilon == \sqrt{h}(1+\eta)^2 G(T_{\text{PT}})/G(2\pi h)$  EXACT, the dyadic factor falls monotonically onto  $2^{3/2}$ ); THE CENSUS 3/2-LAW sympy-exact ( $T_{\text{req}}(h) \rightarrow (3\pi/\sigma_0) h^{3/2}$ : the budget route can NEVER become self-supporting); the budget horizon RECOMPUTED ( $h^* = 1.2566 \times 10^7$ ,  $k^* = 23.58$ :  $\sim 22$  dyadic blocks on PT21); the Bertrand entry interface sieve-verified in-run to  $10^6$ ; UNIVERSALIZING THE CENSUS MACHINE-DETECTED AS THE RH LOOP (the per- $k$  schedule acyclic); the step instantiated end-to-end (STEP(B2→B3) 9/9, STEP(B3→B4-partial) 11/11, SUBSTRATE-DIRECT, no self-supporting induction); ENTRY-NEW-CONFIRMED 4/4 (cone entry always at the largest new PRIME-POWER atom of the dyadic gap, excluding trailing 2-powers — the note-CDLXXXIII formulation, per the Bughunt-VI F1 correction of record; fakes die by NO-ENTRY); (4) v929 THE TERMINAL FACTORIZATION (round 169, note CDLXXXIV, ledger row PRIME.SIGMAFLOOR.FACTORIZATION.01 [E]; source probe sigmafloor\_probe.py, 35/35, SPEC a16a4db1055488d2, RE-RUN GREEN-AS-TYPED AT PROMOTION 2375.2 s; AMENDMENT 1 disclosed: the all-rung onset-rate fit typed CACHE-TOP-LIMITED) — SF1 THE SUPPLY FACTORIZATION  $\sigma == (1 - \text{slop}) \delta \text{DC EXACT}$  (identity devs  $\leq 2 \times 10^{-60}$  at all 25 rungs, F64-immune): the r168  $\sigma$ -flatness EXPLAINED AS TWO TRENDS ( $\delta O(1)$ -flat  $\times$  DC rising toward  $1/2$ ) — the last unexplained measured law of r168 DECOMPOSED; the DC leg typed DC-CLASSICAL (proven-mod-cited per census; Landau 1912/Gonek 1993 AS FORM, GONEK-CONSTANT-UNPRICED); THE JET-MASS-FLOOR NAMED AND TYPED OPEN (the terminal  $\lambda$ -uniform residue, arithmetic-pinned, per-rung classical); the onset-excess cap REFUTED as a floor (SF2); ANY explicit polynomial rate census-absorbable (SF4, sympy-exact); the pinning-supply route machine-flagged as the RH loop’s costume (SF5); the anatomy block floor certified on B2/B3 ( $\sim 11$  dyadic blocks PROVEN-MOD- $\{\text{measured lock, cited Gonek}\}$ ); the controls finding: the raw floor holds in EVERY fake world while the BA3 bridge is FALSE — the arithmetic content is the  $\{\text{floor, bridge}\}$  PAIR; (5) v930 THE WINDOW-INSTRUMENT MECHANISM RECORD (round 167, note CDLXXXI, ledger row PRIME.WINDOW.INSTRUMENT.MECHANISM.01 [E]; source probe window\_instrument\_probe.py, 46/46, SPEC bfdcf374504006e7, PINNED from run-of-record 4138.5 s + re-run 4047.0 s per the multi-hour convention, both logs kept; the mid-run owner-directive re-freeze disclosed; the entry law re-gated by the r168 batteries re-run green at this promotion) — WI1–WI6 exact (plant transport: the PSD premise survives positive band mass at any

strength; the ghost secular threshold  $\tau$ -priced; the perturbation law first-order symmetric; the budget-moment dictionary  $2E^2 == 8A_0^2 \sin^2 y_t^2 \Phi^2 / y^3$ ; the theta chases; BANDPOS(unbounded ladder)  $\iff$  RH — the loop identification); the Stieltjes-moment refutation RECOMPUTED from the pinned slots (Hankel NOT PSD, log-convexity break  $\geq 1.5$ ); THE MECHANISM FINDING pinned at its honest grade: NO PARTIAL COMB IS POSITIVE — cone entry only in the last atom’s tail with the tangent criterion from the prime source, all fakes die by NO-ENTRY (typed MEASURED, CHAIN-NOT-ASSEMBLED; the asymmetry law honestly DEMOTED to generic-cone); (6) SURFACES: five ledger rows (1069  $\rightarrow$  1074); veri blocks in tfpt\_3 (audit-counted body) + the new prime-front diary section “The endgame arc” with the suite-count refreshes (918  $\rightarrow$  923 across the live parentheticals); endform article §8.20 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v930; README/Zenodo 923; German note CDLXXXVI appended to next.txt (head numeral re-verified immediately before writing: the concurrent Bughunt-VI lane had taken CDLXXXV — one collision resolved by re-read); (7) GATES: the endgame chain was independently re-audited mid-promotion by Bughunt VI (round 170, note CDLXXXV, 22/22, 0 MAJOR / 2 MINOR / 4 NOTE, NO verdict flips, NO arrow retypes — the arrow table A1–A10 re-verified with own code) and its live-surface recommendations are adopted on these new surfaces (F1: the entry pattern cited only in the note-CDLXXXIII prime-power formulation; F2: “result lines bit-identical / timing-normalized empty” wording; F3: the cofinality legs re-coded against real objects in v927/v928); three of the five affordable source probes re-run green-as-typed BEFORE any edits with identical SPEC SHAs and empty timing-normalized diffs (27/27 at 2372.4 s, 36/36 at 2371.6 s, 35/35 at 2375.2 s; logs kept as \*.promo\_rerun.log), the two multi-hour probes pinned with disclosure (r162 re-gated in Bughunt V; r167 re-gated by the r168 batteries); v926–v930 standalone green (12/12, 12/12, 11/11, 12/12, 10/10, each  $\sim 1$  s); FULL run\_all.py ALL CHECKS PASSED at 923 modules; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last. NO RH claim in any direction — identities, recipe-derived laws and finite/classical closures say nothing about the truth of RH either way)

2026-08-19 (CIX) (the Bughunt-V corrections of record — the corrections that the round-164 adversarial audit (note CDLXIX, experiments/tfpt-discovery/bughunt5\_probe.py, 29/29 gates, SPEC b466fd5e6dcbc640, record run + deterministic re-run bit-identical, 0 MAJOR / 4 MINOR / 2 NOTE, NO round verdict flips, every load-bearing razor/quartic/ladder/census claim of rounds 149–162 and the promotion wave v922–v925 independently CONFIRMED) recommended for promoted and live surfaces, applied to the LIVE surfaces only (frozen probes and notes are never retro-edited — the correction is the record); ONE documented count correction and NO other numeric value change anywhere; v925 stays green with unchanged gate count (14/14) and unchanged verdict: (1) BH5-F2, THE COUNT CORRECTION (72  $\rightarrow$  73): v925\_edge\_cleanup\_closures.py (docstring E1 headline, the D1 check text and the D1 inline comment + website mirror, byte-identical) and the ledger row PRIME.EDGE.CLEANUP.CLOSURES.01 said “72 sliver + first-zero targets” / “72 unique targets” — the record log’s own per-block certified counts (1/1, 1/1, 1/1, 1/1, 18/18, 51/51) sum to 73, and the unique-target arithmetic confirms 73: 71 sliver zeros + the first zeros above the band at b8/b18 only (their slivers are EMPTY; at b5/b13/b24/b28 the first zero lies INSIDE the sliver and is already counted); all 73 certificates exist and

were always green — only the headline count changes, the gate logic was already correct (the module gated  $\Sigma == 73$  before this correction); the paper/website prose was already count-free (“per-zero”, “at every sliver + first-zero target”); (2) BH5-F6, THE ROUND-LABEL DRIFT (live-surface leakage only): the corpus convention fixes `anchor_epslock` = round 153 (r152 = the CDLVI promotion round, no promotable content) — corrected at the two live sites that had inherited the drift: the `tfpt_prime_front` scope paragraph (“the round-152/153 anchor coordinates” → “the round-153 anchor coordinates”) and the CVIII entry wording (“round-152/153/155” → “round-153/155”); the frozen `assembly_walls` docstring and note CDLXVI keep the drift as the record; (3) BH5-F1 CHECKED CLEAN on live surfaces: r150’s overstated “betapos == S1/jr by exact chase” (exactly:  $S1/jr - \text{betapos} == \text{betapos}^2 / (\text{FULLGAP} - \text{betapos}) > 0$ , magnitude  $2.7 \times 10^{-7}$  at  $x = 5$  — below r150’s 4-digit print) lives ONLY in the frozen r150 probe and note CDLIV; v923 already promotes the corrected GF5a law with the factor  $(\lambda_1 - \tau) / (\lambda_1 - \beta_0) \geq 1$  — no live edit needed; (4) BH5-F3 CHECKED CLEAN: the r160 rescue-instrument overstatement (“projected nonneg least squares” for a greedy eigenvalue-lifting loop; enum RESCUE-INFEASIBLE-IN-CLASS certifying only THAT trajectory) appears on NO live surface — r160 was never promoted, and the live scope paragraphs carry its verdict only as INFORMATIVE-NEGATIVE (the correct typing, confirmed by the audit); (5) BH5-F4/F5 CHECKED CLEAN: no leakage of “SPEC\_SHA moves once” or the b8-SCRARITH 0.2597 misattribution to any live surface (frozen-file wordings stay as the record); (6) GATES: v922–v925 re-run green BEFORE any edit (13/13, 20/20, 12/12, 14/14, ~1 s each); v925 re-run green AFTER the correction with unchanged gate count 14/14; `build.sh release` + EXIT GATE `build.sh audit` AUDIT OK; manifests last; German note appended to `next.txt`. NO RH claim in any direction — a bookkeeping correction says nothing about the truth of RH either way)

2026-08-19 (CVIII) (the spectral-balance/edge theorem-arc promotion — the four durable certified theorem sets of discovery rounds 152–161 graduate from experiments/ into the load-bearing suite as `v922_spacing_jet_sumrule_theorems.py`, `v923_spectral_balance_razor_theorems.py`, `v924_moment_laurent_rootladder.py` and `v925_edge_cleanup_closures.py` (suite 914 → 918, ledger 1065 → 1069); all four rows typed NOT-RH-EVIDENCE in either direction — identities, enclosures and finite/classical closures: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at {TOPROOT, TLAWCAP-block, QSUBGAP-floor(= the pair {SUSCAP2R, DELTA1FLOOR})} + the *a*-walls; every conditional or measured leg typed and carried verbatim; the family-wide exact algebra was Lean-hardened in the second Lean round (round 158, note CDLXII: five new Lean 4 modules, 28 fully proven theorems on the three standard axioms, build green) and each module re-checks the shipped Lean theorem names at every suite run: (1) v922 THE HPIN/SPACING THEOREM FAMILY (rounds 135 + 154, notes CDXXXIX + CDLVIII, ledger row PRIME.SPACING.JETSUMRULES.THEOREMS.01 [E]; source probes `hpin_floor_probe.py`, 33/33, SPEC 80366b4e62779398, PINNED from run-of-record 1837.6 s with THREE deterministic logs kept (D1/D2 re-gated by r154-G32 + Lean r158), and `nearalign_probe.py`, 31/31, SPEC f92b3fb59b142254, re-run green-as-typed at promotion, 1572.7 s) — the derivative identity  $E'_N(\mu_j) == 4 \sin(A\mu_j) F'(\mu_j^2)$  with the spacing-product form (the H-pin floor is a ZERO-SPACING PRODUCT), the jet sum rules of both

orders ( $\sum 1/F' == -A_2/A_0^2$ ;  $\sum F''/F'^3 == 2A_2/A_0^3$ ; the boundary-jet telescope IS the reciprocal-floor moment data), the  $A_0$ -cancellation (the demand ratio is  $A_0$ -FREE), the certified lattice tail, and the sign-uniform corollary carried as an OBSTRUCTION (census MIXED at every rung); SpacingProduct.lean proves the spacing product for GENERIC FINITE  $K$  over an arbitrary field; (2) v923 THE PINCH/ENCLOSURE/RAZOR FAMILY (rounds 157 + 161, notes CDLXI + CDLXV, ledger row PRIME.SPECTRAL.BALANCE.RAZOR.01 [E]; source probes spectral\_balance\_probe.py, 26/26, SPEC 3ed388698a138e31, re-run green-as-typed at promotion, 4077.8 s, non-timing diff EMPTY, and gfloor\_probe.py, 26/26, SPEC cc7837138d41add7, PINNED from run-of-record 4068.2 s + deterministic re-run identical per the multi-hour convention, both logs kept, its GF algebra recomputed in-module) — SB1–SB5 (the trace loop  $\tau \text{ TrH} == \text{tf}/\text{FULLGAP}$  with  $1 \leq \text{tf} \leq K - 1$  UNCONDITIONAL: DELTA1FLOOR  $\Leftrightarrow$  TRACEFLOOR two-sided, tail closed; the chi-cap; the enclosure  $\rho^2 \delta_1 \leq g \leq \delta_1$ ; the OVG identity; the share floor) + GF1–GF5 (THE RAZOR ENCLOSURE  $1/(s + 1/\delta_1) \leq g < 1/s$  with TWO-LEVEL EQUALITY at the lower end — the exact witness  $\rho^2 = 1/(1 + P\delta_1)$  realizes  $s == P = 10^6$  AND  $g ==$  the lower end EXACTLY, no algebra-only improvement exists: THE  $g$ -FLOOR IS THE  $s$ -CAP GIVEN THE  $\delta_1$ -FLOOR, measured width  $1.51 \times 10^{-4} \rightarrow 8.45 \times 10^{-8}$ ; the pinch law  $1 - sg == g^2 T_2 / \rho^2$ ; the one-sided share pricing; the Newton value law  $N_1 == 1/(H_t + s)$  with  $B(q_0) == -\rho^2 P'_c(q_0)$  and the jet-ratio chase  $J_g t_g == 1 + g$ ; the S1 absorption + twin squeeze: both final cores ONE statement-form at two weightings, the value-merge REFUTED — the J-family is BLIND to  $g$ ); W1/W2/W3 cited from v920, no double count; (3) v924 THE MOMENT-LAURENT/ROOT-LADDER FAMILY (round 156, note CDLX, ledger row PRIME.MOMENT.LAURENT.ROOTLADDER.01 [E]; source probe rootladder\_probe.py, 29/29, SPEC 02755d6b7ad0cfcb, re-run green-as-typed at promotion, 1873.5 s, zero amendments) — the escaped census-root ladder is the zero set of the moment-Laurent  $\Phi(z) = z - 1 + \sum_m J_{m+1} z^{-m}$  (self-similarity iff moment flatness, exact remainder), the QUARTER-CAP  $J_2 \leq 1/4 + z_0|\rho|$  PROVEN (measured  $z|\rho| \leq 0.0102$ : proven headroom over the 0.1506 saturation), the exact sum rules, the sign-pattern law, the cascade ( $T > 0$  pointwise SOURCE-PURE down to  $Y^* = 1.02\text{--}2.1 b_{\text{top}}$ : the escaped-root cap CERTIFIED), the two-sided split, and the witnesses (shape sector scale-blind; J-BOUNDEDNESS ARITHMETIC-PINNED); (4) v925 THE EDGE/CLEANUP CLOSURES (round 159, note CDLXIII, ledger row PRIME.EDGE.CLEANUP.CLOSURES.01 [E]; source probe edge\_cleanup\_probe.py, 34/34, SPEC f6c0318841bb3942, re-run green-as-typed at promotion, 1830.8 s) — SIMPLICITY FALLS (census row Z8 eliminated  $10 \rightarrow 9$  by LM-split subsumption + zero-cost avoidance; residual NONDEGENERATE-CELL witness-certified free), THE Q-SWAMP CLOSES WHOLESALE with the 357 Turing-class local band certificates RECOMPUTED IN-RUN at every suite run from the committed PT21-warranted verified\_zeros\_n7000.json cache ( $D_{\text{cs}} > 0$  at all integer  $x = 3..121 \times$  battery  $a \in \{1, 4, 16\}$ , worst margin 0.346 at  $(12, 1)$ ;  $x_0(\text{HSW}) = 121 + x_0(\text{BW25}) = 112$  recomputed; heights  $\leq 7264.75 \ll 3 \times 10^{12}$ , ward-typed): THEOREM A's conditionality SHARPENS to every integer  $x \geq 3$  (H-pin untouched), and the band-edge sliver closes below horizon per-zero (pinned; the certificate price rides  $e^{-\text{depth}}$ , machine-pinned); (5) NOT PROMOTED, deliberately: the round-160  $J_2$  prime-force adjudication (note CDLXIV, verdict INFORMATIVE-NEGATIVE — the single-lever search closed) stays experiments-side: multi-hour world-builds with no exact-algebra layer a module could recompute, residue reading still in

active motion; likewise the round-153/155 anchor/census coordinates; (6) SURFACES: ledger rows (1065  $\rightarrow$  1069); veri blocks in tfpt\_3 (audit-counted body) + the new prime-front diary section “The spectral-balance/edge theorem arc” with the suite-count refreshes (914  $\rightarrow$  918 across the live parentheticals); endform article §8.19 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v925; README/Zenodo 918; note CDLXVI; (7) GATES: all four affordable source probes re-run green-as-typed BEFORE any edits (identical SPEC SHAs; 31/31, 26/26, 29/29, 34/34; logs kept as \*.promo\_rerun.log), the two pinned probes disclosed (r135 three logs; r161 both logs); v922–v925 standalone green (13/13, 20/20, 12/12, 14/14, each  $\sim 1$  s); FULL run\_all.py ALL CHECKS PASSED at 918 modules; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last. NO RH claim in any direction — identities, enclosures and finite/classical closures say nothing about the truth of RH either way)

2026-08-18 (CVII) (the prime-front theorem-arc promotion — the four durable certified theorem sets of discovery rounds 128–151 graduate from experiments/ into the load-bearing suite as v918\_doublelimit\_reduction\_theorem.py, v919\_secular\_gw\_pinning.py, v920\_loop\_equivalence\_theorems.py and v921\_counting\_dominance\_closures.py (suite 910  $\rightarrow$  914, ledger 1061  $\rightarrow$  1065); all four rows typed NOT-RH-EVIDENCE in either direction — reductions, identities and counting closures: they close no omega, narrow no gate, move no other marker; the omega census stays at cardinality 4, the residue at {TOPROOT, ONSETCAP(=TLAWCAP), SUSCAP2R} + DELTA1FLOOR + the  $a$ -walls; every conditional or measured leg typed and carried verbatim; all candidates survived the adversarial Bughunt III (round 130) and/or Bughunt IV (round 149, note CDLIII, 0 MAJOR): (1) v918 THEOREM R (round 128, note CDXXX, ledger row PRIME.DOUBLELIMIT.REDUCTION.THEOREM.01 [E]; source probe doublelimit\_proof\_probe.py, 28/28, SPEC 0c77ef7ce43fd62e, re-run green-as-typed at promotion, 398.5 s; independently re-derived in Bughunt III) — the load-bearing reduction of the arc, proven with elementary inputs: L1 + WPD  $\Rightarrow$  (H-conv) + (H-trace) with the EXPLICIT RATE  $\max(1, C)(-4 \log(1 - t/4)) d_1$  via the exact series identity  $\log(R_{\text{fin}}/R_a)(t) = \sum_k t^k d_k/k$ ; with the round-122 NF-closure theorem:  $\text{RH} \Leftarrow [\text{NF-closure}] + [\text{Theorem R}] + \{\text{L1, WPD}\}$  on dense  $a$ ; Lemmas N/S/X and the exact  $C_{0k} \leftrightarrow \text{jet}$  dictionary travel with it; honest typing carried: WPD’s  $k$ -tail IS window positivity (exact witness  $k^*(B) = 29959/76013/145094$  from  $4w^* = 10001/10000$ , recomputed in-run), L1 stays obstructed at the node-law wall — proven is the REDUCTION and the COORDINATE, not the convergence; (2) v919 THE SECULAR IDENTITY + THE GUINAND–WEIL PINNING (round 131, note CDXXXIII, ledger row PRIME.SECULAR.GW.PINNING.01 [E]; source probe l1\_weylllaw\_probe.py, 31/31, SPEC 7d8cc2e5ca8108fb, re-run green-as-typed at promotion, 217.4 s; independently reproduced in Bughunt IV from the classical explicit formula,  $|\Delta \log_{10}| = 2.7 \times 10^{-7}$ ) —  $\det(M_\eta - z) = -z C(z^2)/A_0$  EXACT (recomputed on an own exact-rational instance, Fractions, dim 11: the operator nodes ARE the census zeros, node count EXACTLY  $K - 1$ );  $\tau = 2 \sum_\rho |E_N(\gamma_\rho)|^2$  within certified allowances (Guinand–Weil, cited), the certified smallness law, the certified gap law in the resolvability zone ( $x = 13$ : 21/21 zone zeros pinned, max gap  $2.9 \times 10^{-17} \leq 5.6 \times 10^{-8}$ ; edge layer MEASURED-ONLY); L1-TAIL closed in closed form (HSW22 Cor. 1.2 + PT21, recomputed against an own IBP integral); the crossover theorem  $T = 2\pi x$  exact — the corpus constant  $\text{INBAND\_F} = 1/\text{KFAC} = 0.8$  is a THEOREM; the boundary-jet law; the band

half of L1 (H-pin) stays OPEN, Z1 evidence typing stands; (3) v920 THE LOOP/EQUIVALENCE THEOREMS (rounds 140 + 142, notes CDXLIV + CDXLVI, ledger row PRIME.LOOP.EQUIVALENCE.THEOREMS.01 [E]; source probes jetlock\_bandmass\_probe.py, 33/33, SPEC 85e7ba691b3eef58, PINNED from run-of-record 3824.7 s + deterministic re-run 3783.4 s per the multi-hour convention (both logs kept, loop algebra re-gated in Bughunt IV and recomputed in-run), and tlawcap\_suscap2r\_probe.py, 26/26, SPEC 85971e173cd910f8, re-run green-as-typed at promotion, 1904.7 s) — the anti-double-counting layer, all recomputed exactly: BAND-MASS  $\Leftrightarrow$  TLAWCAP  $\Leftrightarrow$  EPS-LOCK modulo {JETLOCK, TAILVIS} (Theorem B2: band-mass is EPS-LOCK-COMPLETE), the onset IS the jet ratio and the jet ratio IS escaped census-root mass (J2/J3: JET-LOCK(poly)  $\Leftrightarrow$  TOPROOT; J4: the hardness is cancellation ALIGNMENT), the W1 pinch identity ( $s \times \text{gap} \equiv 1$  is NOT an identity:  $1 - g/\delta_1 \leq sg \leq 1$  exactly), QSUBGAP  $\Leftrightarrow$  SUSCAP2R  $\wedge$  DELTA1FLOOR (W2, both directions exact), DELTA1FLOOR  $\Leftarrow$  FULLGAP (W3, Cauchy interlacing, source-pure; measured TIGHT 1.0000), plus the exact DELTA1FLOOR-required witness ( $s \equiv 1$  while  $\text{gap} \rightarrow 0$  — the Bughunt-IV F1 correction of record carried in-module); reshape theorems: no omega closes, no future double-counting possible; (4) v921 THE COUNTING-CLASS CLOSURES (rounds 133 + 143, notes CDXXXVI + CDXLVII, ledger row PRIME.COUNTING.DOMINANCE.CLOSURES.01 [E]; source probes dominance\_proof\_probe.py, 40/40, SPEC 00b98f5eebf9c5fb, re-run green-as-typed at promotion, 339.3 s, and toproot\_tailvis\_probe.py, 27/27, SPEC 4fcd70beb7cf4f17, PINNED from run-of-record 4016.1 s + deterministic re-run 3921.4 s per the multi-hour convention (both logs kept, T1 re-derived independently in Bughunt IV)) — Theorems M/E (ball/edge counting, no sign reads), Theorem T UNCONDITIONAL ( $N_{\text{fin}}(T) \leq N_{\text{true}}(T)$  for all  $T \geq T^*(x)$ ; ladder 63.55/79.17/102.80/147.13 recomputed vs the classical  $\gamma_{K-1}$  anchors), Theorem C exact (MRB strictly weaker than DOM — per-node one-sidedness neither needed nor claimed), Theorem A CONDITIONAL on H-pin (closed forms recomputed:  $x_0 = 121$ , BW25  $x_0 = 112$ ; the Q-swamp strip  $D < 0$  on  $x \in \{13..89\}$  carried, not smoothed), Theorem T1 (TAILVIS existence is CLASSICAL from HSW22/RvM alone:  $L^* = O(20)$  falling — TAILVIS eliminated as a standalone arithmetic omega, counting class) and Theorem T3 (the round-140 horizon limitation at  $x = 24/28$  RESOLVED without new zeros; recomputed from the v919 pins to  $< 2\%$ ); the window carrier  $x \lesssim 2.3 \times 10^5$  disclosed (carrier class, not a new omega); (5) NOT PROMOTED, deliberately: the round-147 adjudgate elimination (AD1/AD2 + the cell lemma modulo simplicity) and the round-150/151 collapse-rate/multiblock coordinates stay experiments-side — their coordinates are still in active motion and the simplicity leg is open; (6) SURFACES: ledger rows (1061  $\rightarrow$  1065); veri blocks in tfpt\_3 (audit-counted body) + the new prime-front diary section “The prime-front theorem arc” with the promotion of record and the suite-count refreshes (910  $\rightarrow$  914 across the live parentheticals); endform article §8.18 + Anthropic email promotion paragraph; website PROMOTED feed card + DAG scripts + banner/meta v535–v921; README/Zenodo 914; note CDLVI; (7) GATES: all four affordable source probes re-run green-as-typed BEFORE any edits (identical SPEC SHAs; 28/28, 31/31, 40/40, 26/26), the two multi-hour probes pinned with disclosure; v918–v921 standalone green (13/13, 11/11, 17/17, 9/9, each  $\sim 1$  s); FULL run\_all.py ALL CHECKS PASSED at 914 modules; build.sh release + EXIT GATE build.sh audit AUDIT OK; manifests last. NO RH claim in any direction — reductions

and identities say nothing about the truth of RH either way)

2026-08-16 (CVI) (the Bughunt-III corrections of record — the four wording/label corrections that the round-130 adversarial audit (note CDXXXII, experiments/tfpt-discovery/bughunt3\_probe.py, 20/20 gates, 0 MAJOR / 4 MINOR / 3 NOTE, NO verdict flips, every promoted number of v916/v917 independently CONFIRMED) recommended for promoted and frozen surfaces, applied to the LIVE surfaces only (frozen probes and notes are never retro-edited — the correction is the record); NO numeric value changes anywhere, both modules stay green with unchanged gate counts (20/20, 13/13) and unchanged verdicts: (1) v916\_epstein\_weil\_violation.py (+ website mirror): the pin PIN\_EULER\_BOUND = 0.664 relabeled as the  $N = 2 \times 10^4$  TRUNCATION of  $\sum_{n \geq 4} r_Q(n) n^{-1.6} / r_Q(1)$  PLUS the declared Abel-model tail — NOT the “full-table value” (its old label); the true full sum is 0.645728 (bughunt 4-decade ladder  $N = 4 \times 10^3..4 \times 10^6$ , spread  $8 \times 10^{-7}$ ) and v917’s 0.650 is the same construction at  $N = 2 \times 10^5$ ; the pinned number 0.664 itself is UNCHANGED (a valid upper bound; both reads  $< 1$ , the D1 gate logic stands); (2) v917\_driver\_rate\_certification.py (+ website mirror + the ledger row PRIME.KR4.DRIVER.CERT.01): the localization census premise restated from the too-strong “every off-line zero below 45 is located” to the gated repair —  $\delta \geq 0.01$  census boxes + band-straddle completeness + the window-width argument:  $\max |v(a_0)| = 0.646 < 1$  on the census-blind strip, any  $|v| > 1$  zero must sit at  $\gamma \in (15.645, 15.703)$  INSIDE the straddle-boxed band, and blind-strip contributions to the depth diffs are negative or  $\leq 10^{-40}$  — hidden zeros can only MASK the fire, never fake it (the localized verdict SURVIVES; papers and website prose checked: the too-strong sentence appears on no other live surface); the 0.650 Euler-region read carries its truncation label now too; (3) the round-129 “round-124  $t_{\text{death}}$ ” misattribution (the collapse is ROUND 123): searched tfpt\_prime\_front, articles/, website — NO live surface carries the wrong round label, nothing to correct (the frozen round-129 probe and note CDXXXI stay as they are per the freeze convention; note CDXXXII documents the error); (4) the note-CDXXVII poly-filter class-label shorthand (the hull normalization  $|p| \leq 1$  on the spectral hull dropped): searched papers, website, ledger — the shorthand leaked to NO live surface, nothing to correct (the frozen note stays; the probe’s own hypothesis-carrying statement was always right). GATES: v914–v917 re-run green BEFORE any edit; v916/v917 re-run green AFTER (20/20, 13/13, verdicts unchanged); build.sh gen + release; EXIT GATE build.sh audit AUDIT OK; manifests last; note CDXXXIV (head numeral re-verified immediately before writing: the concurrent L1-Weyl-law lane had taken CDXXXIII — one collision resolved by re-read). NO RH claim in any direction)

2026-08-16 (CV) (the Epstein detection-arc promotion — the two durable certified results of discovery rounds 123 and 125 graduate from experiments/ into the load-bearing suite as v916\_epstein\_weil\_violation.py and v917\_driver\_rate\_certification.py (suite 908  $\rightarrow$  910); both rows typed NOT-RH-EVIDENCE in either direction — known falsehoods (the off-line zeros of  $Q = x^2 + 5y^2$  are Potter–Titchmarsh 1935 / Davenport–Heilbronn 1936), new certified instruments; the min-cut is unchanged (flows 4/4): a detector is a falsifier, not a prover: (1) THE CERTIFIED EPSTEIN WEIL-POSITIVITY VIOLATION (v916, 20/20, ledger row PRIME.KR4.EPSTEIN.COLLAPSE.01 [E]; source probe epstein\_collapse\_probe.py round 123, frozen SPEC bb51d1a3299f4f62, 27/30 with exactly the three PRE-DECLARED falsified predictions G18/G19/G21 failing, re-run green-as-typed at promotion, 998.5 s) — an explicit sha-pinned unit test



vector (len 1038, sha b01436d53d2cbbbee, REPRODUCED at every suite run together with the full quadratic form) whose tent-autocorrelation test function pairs **NEGATIVELY** with the source data  $-g''_Q$  built purely from  $r_Q(n)$  lattice counts:  $V = x^T T x = -1.9697 \times 10^{-2}$ , error budget  $\leq 4.3 \times 10^{-68}$ , two-precision (dps 80/110) agreement  $9.8 \times 10^{-79}$  — windowed Weil positivity for  $\xi_Q$  **FAILS** at  $t \leq 3.114$ ; the  $Q$  world dies at  $t_{\text{death}} = 2.988$  (recomputed eigenvalue crossing  $k^\dagger = 998$ , mesh mate stable,  $Q$ -scramble dies at 2.73) while **TRUE** zeta and the conductor-matched null  $\zeta_K = \zeta_{L-20}$  complete positivity through  $L = 8.0$  (pinned run-of-record; ZK anchor and  $L = 4.0$  prefix recomputed; the certificate vector reads **POSITIVE** on the matched null); the value channel is **CERT-NUM**, the off-line-zero interpretation step **COND-CLASSICAL** (Weil/Guinand explicit formula, cited); the three pre-declared falsified predictions travel verbatim (attribution **COLLECTIVE** — witness removal moves the death by only 0.018, all-located removal survives to 4.428; the law misses by  $0.505 > 0.5$ ; the census is falsified: **FOUR** off-line pairs below 45, the round-117 witness never  $\gamma$ -minimal, the 15.6682 **DRIVER** carries  $26\times$  its excess); (2) **THE CERTIFIED LOCALIZED RATE DETECTION OF THE DRIVER** (v917, 13/13, ledger row **PRIME.KR4.DRIVER.CERT.01 [C]**; source probe `driver_cert_probe.py` round 125, frozen SPEC bcc64cb371dff9c8, 32/33 with exactly the single **PRE-DECLARED G11** fail, re-run green-as-typed at promotion, 881.1 s) — the driver  $\rho = 0.9329696975 + 15.6682495313i$ , source-located from the  $\Lambda_Q$  sieve ( $\Delta\gamma = 6.6 \times 10^{-5}$ ), fires in the band-peeled rate channel at depth  $m^* = 14$  with certified  $\text{LB} = 1.000137 > 1$ , **LOCALIZED** from  $m_{\text{loc}}^* = 15$  (the violating window contains  $a^* = 245.6815$ : it is the driver); the band peel  $\gamma \in (7.90, 31.40)$  (winding-complete, every peel  $|\xi_Q|$ -verified, phantom peel doubly excluded) collapses the unpeeled price  $m_2 = 908$  by  $\times 65$  at sieve cost only  $N = 2 \times 10^5$ ; the new exact algebra is recomputed symbolically at every run ( $v^* = 1 + \varepsilon$  **EXACTLY REAL** at the matched pin, weight exactly 1,  $\text{LB} \rightarrow v^*$  to six digits by  $m = 25$ , certified depth 32,  $\text{Err} \approx 2.1 \times 10^{-12}$ ,  $\text{meas} \leq \text{cond}$ ) together with the quantified detection-cost law  $m^* \approx \ln(B_{\text{band}}/\varepsilon)/\ln(1/v_{\text{edge}})$ , sieve price polynomial in  $1/\varepsilon$  via the band, measured 3.19 certified depths per unit  $\ln N$ ; the grade is carried verbatim and never upgraded — **CERT-GIVEN-AUDIT** (rigorous Cauchy given the incomplete-gamma audit comparator plus the **DECLARED** tail/peel models) and **DISGUISE-ADJACENT** (the deep read is an Euler-computable truncated windowed Weil transform) — and the honest caveat is carried: the pre-declared **G11** gate (locator relocates 4/4) **FAILED**, the weak round-117 witness (excess  $2.9 \times 10^{-5}$ ) sits below the locator’s measured threshold at  $N = 2 \times 10^5$ , so the locator is typed a 3/4 **MEASURED** instrument and no 4/4 claim is promoted; (3) **SURFACES**: ledger rows **PRIME.KR4.EPSTEIN.COLLAPSE.01** + **PRIME.KR4.DRIVER.CERT.01** ( $1059 \rightarrow 1061$ ); veri blocks in `tfpt_3` (audit-counted body) + the new prime-front diary section “The Epstein detection arc” with the promotion of record, the suite-count refreshes ( $908 \rightarrow 910$  across the live parentheses) and the correction of record on the round-102 witness sentence (never  $\gamma$ -minimal); endform article + Anthropic email promotion paragraphs; website **PROMOTED** feed card + DAG nodes + banner/meta v535–v917; **README/Zenodo** 910; note **CDXXVIII** (head numeral re-verified immediately before writing: the concurrent selection lane had taken **CDXXVII** — one collision resolved by re-read); (4) **GATES**: both source probes re-run green-as-typed **BEFORE** any edits (identical SPEC SHAs, identical failing sets); v916/v917 standalone green (20/20 in  $\sim 12$  s, 13/13 in  $\sim 1$  s); **FULL** `run_all.py` **ALL CHECKS PASSED** at 910 modules; `build.sh` **release** + **EXIT GATE** `build.sh` **audit** **AUDIT OK**; manifests last. **NO RH** claim in any direction — both rows are falsifier instruments on a classical known-falsehood

world; zeta’s omega absorbs every finite certificate)

2026-08-15 (CIV) (the weekend promotion — the two durable proven results of the discovery weekend graduate from experiments/ into the load-bearing suite as `v914_pascal_region_theorem.py` and `v915_eulerpick_certified_floors.py`; suite 906  $\rightarrow$  908, both modules green standalone (18/18 in  $\sim 20$  s; 15/15 in  $\sim 40$  s) and registered in `run_all.py`, the script registry and the status ledger (new rows `PRIME.PASCAL.REGION.THEOREM.01` [E] and `PRIME.EULERPICK.CERTIFIED.FLOORS.01` [E], both typed as certified-finite results with their citations disclosed and explicitly NOT as RH-evidence rows); all four source probes re-run green before any edit (`moonshot_sol_probe.py` 12/12, 0.4 s; `moonshot_o3_probe.py` 16/16, 100.8 s, frontier verbatim; `eulerpick_ladder_probe.py` 31/31, 102.8 s; the round-100  $10^{13}$  compiled sieve NOT re-run — its `sieve4_run1.log` run-of-record is pinned per the PINNING DISCLOSURE convention); NO RH claim in any direction, no other marker moved, no gate closed or narrowed: (1) **v914 THE PASCAL-REGION THEOREM** (rounds 103 + 106):  $C_{n,k}(256) > 0$  for EVERY  $n \geq 0$  and every integer  $0 \leq k \leq 7,444,682,106,464,286,365,865$  ( $\sim 7.44 \times 10^{21}$ ), proven from cited external inputs alone — Platt–Trudgian 2021 Thm 1 ( $T = 3,000,175,332,800$ ), Hasanalizade–Shen–Wong JNT 235 (2022) Corollary 1.2 ( $|N(t) - M(t)| \leq 0.1038 \log t + 0.2573 \log \log t + 9.3675$ ,  $t \geq e$ ), Rosser 1941 Thm 19, the classical  $\gamma_1$  isolation; the module recomputes the COMPLETE round-103 four-region partition at every suite run ( $B_0(T) = 7.57565571105606088 \times 10^{-10}$  anchored; decisive ratios  $1.876981097339/1.176577026464/335.835/1416.754$ ; mechanism root  $k^* = 1.405443828790708 \times 10^{19}$ ), recomputes the quadratic verification-height law  $k_{\text{front}}(T) = (\lambda_\infty + O(1/\log T)) T^2/(a - 1/2)$  with  $\lambda_\infty = 0.226987054723979955$  (residual  $< 10^{-60}$ ), pins the 33-cell run-of-record frontier partition and re-proves a frozen 6-of-33 subsample with full 16384-packet endpoint inequalities INCLUDING the worst cell (endpoint ratio  $1.000001535385559$ ; the next  $2 \times 10^{-6}$  relative step fails — mechanism-exact); **THE ROUND-110 HSW ATTRIBUTION FLAG RESOLVED FROM THE PAPER ITSELF**: Cor. 1.2 IS the  $N(t)$  bound with exactly the constants consumed, Cor. 1.4 is the companion  $S(t)$  bound ( $\min\{\dots + 8.3675, 0.1095 \log t + 0.2042 \log \log t + 3.0305\}$ ) and is NOT used — the probe docstring’s “Cor. 1.4 is the corrected replacement” was the slip, the numbers were always right; the old (0.110, 0.290, 2.290) constants are the earlier Platt–Trudgian  $S(t)$  bound whose proof consumed the invalid Cheng–Graham estimate; HONEST TYPING carried in module, ledger and papers: verification-powered, FORM-LOCAL (zero-location content below  $\gamma_1 = 14.13$ ; says nothing about screw sections or Weyl disks), the  $k \rightarrow \infty$  quantifier untouched, no wall positivity, no zero table, no zeta evaluator — NOT evidence for or against RH; (2) **v915 THE CERTIFIED EULER–PICK FLOORS**  $N = 1.4$  (rounds 95 + 100): outward-rounded interval positivity certificates with the recomputed/pinned split disclosed exactly — RECOMPUTED in-run: the certificate self-tests, the certified digamma, the own sieve to  $1.6 \times 10^7$  with all three cited  $\psi$ -bounds warded against own data (Büthe 2018; Rosser–Schoenfeld 1962 Thm 12; Nair 1982), the cap- $4 \times 10^6$   $N \leq 2$  and cap- $1.6 \times 10^7$   $N \leq 3$  floors reproducing the round-95 record ( $\log_3 = 2.3643695 \times 10^{-10}$ ), the summation-error model of the compiled run re-derived rigorously from the pinned  $\pi(10^{13})$  plus an RS–Abel upper bound (no sieve needed), the Nair far-floor necessity (the crude  $\psi \geq 0$  floor blocks  $N = 4$  at ANY cap) with the corrected budget  $X_{\text{req}}(4) = 10^{11.52}$ , the  $N = 5$  knowledge wall, and a recomputed detection subset ( $N^*(5; 0.1) = 3$ ,  $N^*(8; 0.1) = 4$ ,  $N^*(8; 0.4) = 3$ ); PINNED from run-of-record: the cap- $10^{13}$  intervals

$N=1$   $[4.5917135 \times 10^{-2}, 4.5917136 \times 10^{-2}]$ ,  $N=2$   $[9.0288887 \times 10^{-6}, 9.0288888 \times 10^{-6}]$ ,  $N=3$   $[6.9310239 \times 10^{-10}, 6.9310252 \times 10^{-10}]$ ,  $N=4$   $[8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  (margin 0.749) with  $\pi(10^{13}) = 346,065,536,839$  reproduced exactly by the recorded instrument; HONEST TYPING (round-108 F3 carried verbatim, no overselling): the falsification channel is DISPROOF-SOUND but REACH-CAPPED at  $\gamma \sim 5-8$ , far below Platt–Trudgian  $3.0 \times 10^{12}$  — zero live falsification power in the classically certified range; “open and empty” is a theorem given the cited inputs; the channel stays open and EMPTY; (3) SURFACES: `tfpt_prime_front` gains the promotion-of-record notes and `\veri{}` citations in the owning sandbox-diary subsections (instrument tier; roadmap; frontier audits) with the freeze-time suite counts rephrased honestly (906 at freeze  $\rightarrow$  908 today); the end-form article and the Anthropic email draft gain matching promotion-of-record paragraphs; the website prime-front page and feed gain the promotion entry and tier upgrades (Euler–Pick floors and Form-A frontier now named as modules, reach caps kept verbatim), the DAG E8 node gains both scripts, README and Zenodo counts move to 908/v1-v915; `ScriptIndex.tsx`, `changelog.ts`, `suite.ts` and the maps regenerated via `bash build.sh gen`; full suite re-run `run_all.py`  $\rightarrow$  ALL CHECKS PASSED at 908 modules; `bash build.sh audit`  $\rightarrow$  AUDIT OK; manifests regenerated LAST. NO RH claim anywhere)

2026-08-15 (CIII) (the boundary-and-resolvent sync — discovery freeze rounds one-hundred-eleven and one-hundred-twelve synced into every live surface; experiments/ only — TWO frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction; both source probes re-run green before any edit (`unpriced_classes_probe.py` 26/26, 88 s; `resolvent_closure_probe.py` 36/36, 266 s): (1) ROUND 111 (note CDXI, contract PRIME.UNPRICED.CLASSES.01, `unpriced_classes_probe.py`, 26/26 gates, 88 s, SPEC `ec787fc5`, deterministic re-run bit-identical, verdict UNPRICED-ADJUDICATED — tally 25  $\rightarrow$  30 candidates, 0 PASS): the five argument classes the round-108 audit named as genuinely OUTSIDE the atlas enumeration are all adjudicated and killed, none surviving as an open lever; (U-A, THE HEADLINE — UA-RATE-IS-ZERO-CURRENCY) Zagier’s published horocycle RH-equivalence (Zagier 1979/81:  $m_y(f) = m(f) + o(y^{1/2})$  unconditional,  $\text{RH} \iff \text{error } O(y^{3/4-\epsilon})$ ; Sarnak CPAM 34 (1981); Verjovsky 1994) — a published RH-equivalence NEVER ADJUDICATED UNTIL NOW, confirmed absent from every atlas artifact; the instrument built (incomplete Eisenstein series  $\psi(t) = t^2 e^{-t}$ , horocycle integral  $= 3/\pi +$  scattering contour, warded two-route:  $\text{dev} \leq 1.5 \times 10^{-8}$ ,  $|\varphi(1/2 + it)| = 1$  to  $9.9 \times 10^{-32}$  — the unconditional 1/2 exponent is UNITARY, no spectral gap) measures the deep-window exponent 0.7451 (the RH-range 3/4 in the classically verified region) and, on planted-zero synthetic scattering worlds at  $\beta = 0.6/0.75/0.9$ , the exponents 0.6971/0.6222/0.5473 against the law  $1 - \beta/2$  ( $|\text{dev}| \leq 0.0029$ ) — THE HOROCYCLE RATE IS EXACTLY THE AFFINE REPARAMETRIZATION  $\theta = 1 - \Theta/2$  OF  $\sup \text{Re } \rho$ : the emptied E2 zero-free/density currency, alignment content linear  $\log y$  phases (the round-97 class,  $\times 8.2$ ); rigidity technology rate-free (Ratner) or non-horospherical (Lindenstrauss–Mohammadi–Wang CRAS 361 (2023): unspecified exponents, explicitly beyond the horospherical case — the modular horocycle IS horospherical, where the classical spectral 1/2 has stood since 1981); DISGUISE by rule R1; (U-B — UB-INAPPLICABLE-BY-TYPE) the zero set of Hardy  $Z$  is infinite (29 sign changes on  $[0, 100]$  measured = the RvM count) while o-minimal definable sets have finitely many connected

components —  $\zeta$  on the line is definable in NO o-minimal structure; the serious adjacent result verified and typed (Masser JNT 131 (2011): rational points on the GRAPH,  $\leq C(\log D/\log \log D)^2$  — count currency, position-free, unsigned); (U-C — UC-HOST-GAP-CONFIRMED)  $\lambda \geq 1/4$  with positive multiplicity forces the line (the Selberg-zeta precedent) but  $\lambda = 1/5$  with positive multiplicity yields the real pair  $1/2 \pm \sqrt{5}/10$  exactly — positivity alone does not force the line, it needs identity AND the  $1/4$  gap, which at the arithmetic quotients is Selberg’s own OPEN conjecture (Kim–Sarnak floor  $975/4096$ , deficit  $49/4096$  exact); for  $\zeta$  the host identity is the mountain (Connes’ trace formula  $\iff$  RH) — Stage 1 of the roadmap in another language, no new lever; (U-D — UD-MAGNITUDE-LAW-ONLY) every proven FHK-type deliverable (ABBRs 2019, FHK I+II, AOR moments; Harper RH-conditional) is a  $|\zeta|$ -magnitude statement IN LAW  $\rightarrow$  the emptied E1 class; the exception check closed exactly: iid uniform phases have ZERO signed expectation and a flip-invariant law — the randomized Euler product carries NO signed alignment observable; (U-E — UE-CANCELLATION-ONLY-INEFFECTIVE) Green–Tao  $\log^{-A}$  savings long-interval with Siegel–Walfisz-ineffective constants, MRTTZ 2023 qualitative  $o(1)$  on average, the charitable  $C = 1$  envelope table showing the ENTIRE content in the constant, NO effective constant for  $\Lambda$ -nilsequence correlations, and the sharpest signed corollary of  $|X| \leq \text{env}$  being  $X \geq -\text{env}$  — orientation exactly at envelope strength,  $\times 8.2$  short; THE CLOSING SENTENCE: the atlas stands at 30 candidates, 0 PASS, with the vocabulary boundary itself now adjudicated — every named argument class inside AND outside the original enumeration is formally priced; (2) ROUND 112 (note CDXII, contracts PRIME.SAFEPOINT.RESOLVENT.CLOSURE.01 + PRIME.CONNES.GROUNDSpace.BLOCK.01, `resolvent_closure_probe.py`, 36/36 gates, 262s, SPEC ea7863f2, deterministic re-run identical, two disclosed X1 tail-cut amendments; composite RESOLVENT-TRANSCRIPTION + CROSSCARRIER-BOUNDS + JENSEN-NO-DICTIONARY): THE IDENTIFICATION — arXiv:2511.22755’s operator (CCM’s rank-one  $D_{\log}^{(\lambda, N)} = D_{\log} - |D_{\log} \xi\rangle \langle \delta_N|$  with  $\xi$  the lowest eigenvector of the Weil form, and its  $\det_{\text{reg}}$  construction) is, MODE FOR MODE, the corpus’s own round-89 record extremal family (lattice  $\pi k/a$ ,  $a = \frac{1}{2} \log x$ ; the round-89 arbiter quotient IS the literal CCM operator); built explicitly at  $x = 3, 5$  and certified: kernel  $M\xi = 0$  to  $10^{-68}$ , spectrum = record zeros (10/10 at  $x = 5$ , rel  $2.7 \times 10^{-8}$ ), their  $\det_{\text{reg}}$  formula executable as a rank-one determinant identity to  $10^{-75}$  — THE CORPUS HAD INDEPENDENTLY BUILT CCM’S OPERATOR BEFORE THEIR PAPER; THE GROUNDSpace-BLOCK VARIANT ELIMINATES CCM’S SIMPLICITY AND EVENNESS HYPOTHESES AT CONSTRUCTION LEVEL ( $M_{\text{blk}} = D(I - P)$  via the biorthogonal block projector, basis-rotation-invariant to  $1.5 \times 10^{-16}$ , STABLE where the simple selection swings  $O(1)$  under  $10^{-10}$  perturbations, the exterior/matrix-Wronskian identity  $\det(M_{\text{blk}} - z) = \det(D - z) \det(I_2 - W(z))$  to  $5.2 \times 10^{-15}$ , PD quotient metric; the remaining realness step on  $E/K$  is the named unproven block-CCM lemma, finite-dimensional and zeta-free); THE KILL: the pins converge 16/16 (slope  $-0.65$ ,  $\tau$ -screen 0/16) BUT THE Z1 SCREEN FIRES — band pins equal cache partial sums to rel  $7.7 \times 10^{-12}/9.4 \times 10^{-12}$ , the exact round-89 conviction INHERITED BY THE LITERAL CCM CARRIER; side-by-side at matched depth the Krein benchmark extrapolates genuinely (max dev  $2.9 \times 10^{-4}$ ) while the CCM carrier band-transcribes ( $1.3 \times 10^{-1}$ ) — THE KREIN CARRIER REMAINS THE ONLY NON-TRANSCRIBING ATTACK SURFACE IN THE CORPUS;

CROSSCARRIER-BOUNDS:  $\Delta_m = \text{Tr}[A_x^{m+1}]$  vs the round-102 Hausdorff  $b_m$ (256) — 17/17  $m$ -rows MONOTONE LOWER BOUNDS FROM BELOW, deficit = the RvM tail beyond the band edge (median ratio 0.786; 11/41 roots beyond the band kept in all sums — the operator mirror of the Nyquist excess): the Hausdorff field and the CCM operator are ONE OBJECT ONLY IN THE LIMIT — at finite depth the operator is a band-truncated basis of the field from below (all 45 operator cells  $n + k \leq 8$  nonnegative; a third carrier confirms via Krein  $\sigma$ -derivative moments to  $8.7 \times 10^{-4}$ – $2.7 \times 10^{-3}$  within the bias model); THE JENSEN WEDGE (arXiv:2608.08682:  $n^3 \log^2(n+2) \geq Kd^5 \Rightarrow$  hyperbolic, polynomially improving GORTW) verified verbatim but typed NO-Dictionary — the only corpus bridge (the Li transform at  $a = 1/4$ ) is alternating-signed, so by round-108 form-locality the wedge adds NO unconditional Hausdorff cells; THE FORM-LOGIC GATES verify the architecture’s selling point exactly (F1: one Stieltjes representation would generate all three positivity families simultaneously; F2: finite certificates are form-local — Pick fires at  $N = 5$  where Hausdorff stays positive to depth 40) — the logic is correct, AND PRECISELY THEREFORE WORTHLESS WITHOUT A NON-TRANSCRIBING CARRIER; the remaining analytic theorem is DOUBLY NAMED: CCM’s own §7–8 prolate approximation  $\iff$  the corpus’s round-90 Weyl-disk contraction, pointwise at the pins; (3) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The vocabulary boundary priced, and the resolvent closure adjudicated: the CCM operator is the corpus’s own record family”; the end-form article gains §8.16 with the abstract status sentences and the traceability rounds 111–112 (with the Zagier 1979/81, Sarnak CPAM 34 (1981), Verjovsky Kodai 17 (1994), Lindenstrauss–Mohammadi–Wang CRAS 361 (2023), Masser JNT 131 (2011), Kim–Sarnak, arXiv:2511.22755 and arXiv:2608.08682 citations); the Anthropic email draft gains one status paragraph (the tally-30 close + the CCM identification as the collaborator hook: their operator is the corpus’s record family, and the corpus’s machine eliminates their simplicity hypothesis); the website prime-front page gains chapter 40 (the live-updates chapter renumbered 40  $\rightarrow$  41) and one consolidated sandbox feed entry (verdict RESOLVENT-TRANSCRIPTION, new enum, script `resolvent_closure_probe.py`); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-15 (CII) (the corrections-and-frontier sync — discovery freeze rounds one-hundred-six through one-hundred-nine synced into every live surface, PRIMARILY A CORRECTIONS ROUND: the round-108 big-picture logic audit and the round-109 Bughunt II independently convicted the published “three equivalent lemma forms” language and two related headlines, and the corrected wordings are now carried on every surface per the round-87/97 correction-of-record convention (frozen probes and notes are NOT retro-edited — the correction is the record); plus the new content: the  $530\times$  Form-A frontier extension with its derived quadratic law, and the code-synthesis adjudication of the program’s founding intuition; `experiments/` only — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO round verdict flips, NO number wrong, NO RH claim in any direction: (1) THE CORRECTIONS OF RECORD (round 108, `bigpicture_logic_probe.py`, 31/31 gates, 2.5s, SPEC 889edf4a, deterministic re-run identical, verdict BIGPICTURE-FINDINGS(3): F1 FORMS-EQUIVALENCE-ONE-SIDED(+TRANSFER-NONE), F2 STAGE3-

ENTRY-FORM-MISMATCH, F3 FALSIFIER-REACH-CAP; and round 109, note CDIX, contract PRIME.BUGHUNT2.R86.R105.01, bughunt2\_r86\_r105\_probe.py, 30/30 gates, 11.6s, SPEC 759d0209, deterministic re-run identical, verdict BUGHUNT2-FINDINGS(7, max MAJOR): 1 MAJOR / 6 MINOR / 0 FATAL — the two audits INDEPENDENTLY CONVERGENT on the same headline finding): (K1) “three equivalent lemma forms” (paper §8.13/§8.14, prime-front 12.45/12.46, website chapters 37/38; the changelog C/CI wordings are frozen history and are corrected [HERE](#), not retro-edited) is corrected to “three RH-SUFFICIENT lemma forms” — the honest form  $\times$  direction table: the HAUSDORFF form is a GENUINE proven RH-equivalence (forward audited-sound via the H1c continuation chain, independently re-derived GAP-FREE in round 109, plus Zhang arXiv:2303.09396 Thm 2; backward elementary + gated), the WEYL-DISK and SIGN-POSITION forms are ONE-SIDED PENDANTS (form  $\Rightarrow$  RH established; RH  $\Rightarrow$  form OPEN — the round-90 lemma, measured 16/16 truth-tight but never proven, and the CF-realness/Connes-§6.6 hypotheses respectively), EVERY mutual equivalence that holds routes THROUGH RH (no direct form-to-form edge, gate J1.G5; the only direct bridge is the (AC) weld stopping at the exponential-autocorrelation cone), TRANSFER IS NONE — demonstrated by an explicit counterexample world: an off-line quadruple at  $\gamma = 20$  (above  $\gamma_1$ ) keeps EVERY Hausdorff cell positive ( $n + k \leq 50$ ,  $b_n$  to  $n = 240$ ) while the Euler–Pick section FIRES at  $N = 12$  (certified negative pivot at 200 dps) — and CERTIFIED REGIONS ARE FORM-LOCAL: the  $k \leq 10^{19}/7.4 \times 10^{21}$  Form-A theorems imply NOTHING about screw sections or Weyl disks, their zero-location content sitting below  $\gamma_1 = 14.13$  where Platt–Trudgian is infinitely stronger; the stale adjective “uniform” in Form B’s name is retired on every surface (the round-92 V0a correction: pointwise contraction at the pins suffices); (K2) THE FALSIFIER REACH CAP (round 108 F3) is attached wherever the surfaces said “standing disproof channels / open and empty” (roadmap §8.14, prime-front 12.46, website chapters 37/38): both instruments are DISPROOF-SOUND (the disproof direction consumes only the proven forward implication) but REACH-CAPPED far below Platt–Trudgian’s  $3.0 \times 10^{12}$  (Euler–Pick  $N \leq 4$  sees off-line zeros only at  $\gamma \sim 5$ –8; the Hausdorff field is provably blind above  $\gamma_1 = 14.13$  at the safe point) — “open and empty” is a THEOREM GIVEN THE CITED INPUTS, not an observation: zero live falsification power in the certified ranges; (K3) THE STAGE-3 INTERFACE FENCE (round 108 F2) is attached to “Stage 3 conditional-complete”: the only kernel-checked end-to-end extraction (v848 + CofinalWeil.lean) accepts ONLY the wall form  $H_{\text{cof}}$ , which is not among the three entry forms and is not producible from them without RH (test-class mismatch, absence-gated); the weakest link is the Stage-2 $\rightarrow$ 3 interface itself — finite acceptance gates (cells  $n + k \leq 86$ ) against infinite-quantifier entry forms, a jump no corpus machinery bridges (v912 bridges convergence, not positivity); (K4) THE MINOR PROSE FIXES (round 109): the screwind mesh-ladder misattribution corrected in prime-front 12.45 (0.222/0.184/0.156 at  $\delta = 0.012/0.006/0.003$  — the earlier 0.207 was the  $\delta = 0.008$  round-93 value 0.2065; MESH-DAMPED stands), plus on record without surface impact: one hardcoded-true gate in the round-104 tally (the only live one across all 17 probes), two decorative tautological conjuncts in the round-92 probe (the countable Nevanlinna–Pick step asserted-with-citation, honestly typed everywhere), the roadmap probe’s stale r95  $N = 3$  acceptance value (r100 tightened  $\log_3$  to  $6.9310239 \times 10^{-10}$ ; frozen probe, finding on record), two sieve4\_helper.c header slips (certificate UNAFFECTED: true worst per-term  $13.33u \ll K_{\text{TERM}} = 32u$ ); (K5) THE OCT

BOUNDARY SHARPENED (round 108 J2, absence-gated at 0 atlas hits each): wherever the surfaces state atlas completeness (paper §8.9/limitation 8, prime-front OCT subsection, website chapter 33), the vocabulary-relative fence now names the FIVE UNPRICED classes — ergodic/measure-rigidity positivity transfer, model-theoretic/o-minimal counting, bigger-group representation-theoretic positivity, GMC/Fyodorov–Hiary–Keating (reduces-in-spirit to magnitude, with an enumeration gap), higher-order Fourier uniformity of Möbius; AND THE CLEAN SIDE carried in the same breath (round 109 B1, own implementations): the round-103 theorem re-proved end to end ( $B_0 = 7.575655711 \times 10^{-10}$  exact match, all four partition ratios to  $10^{-9}$ , wall root  $k^* = 1.40544 \times 10^{19}$  consistent), the  $N = 4$  certification re-derived (Nair exact for  $n \leq 60$ , far-floor necessity confirmed, Cholesky logic sound), the Hausdorff H1c continuation GAP-FREE, the Lean theorems saying exactly what round 99 claims (axioms exactly [propext, Classical.choice, Quot.sound]), all 16 in-scope probes re-run green during the audit; (2) ROUND 106 (moonshot\_o3\_probe.py, 16/16 gates,  $\sim 104$  s, independently re-run green, verdict MOONSHOT3-EXTENDED(frontier = 7,444,682,106,464,286,365,865; quadratic law) + MOONSHOT3-PARTIAL(exact  $G$  / sign-regularity / fixed- $N$  theorem) + MOONSHOT3-OBSTRUCTED(PF orientation; downward Hankel transport)): the round-103 theorem EXTENDED 530-FOLD —  $C_{n,k}(256) > 0$  for ALL  $n \geq 0$  and  $0 \leq k \leq 7.4447 \times 10^{21}$ , proven from Platt–Trudgian 2021 plus the CORRECTED Hasanalizade–Shen–Wong bound (JNT 235 (2022), Cor. 1.2:  $|N(t) - M(t)| \leq 0.1038 \log t + 0.2573 \log \log t + 9.3675$ ; HONEST SOURCE CORRECTION: the commissioned (0.110, 0.290, 2.290) constants bound  $S(T)$ , not  $N(T)$ , and their Cheng–Graham input was later identified as invalid — replaced, not patched); mechanism: the retained  $q^k$  factor, a 16384-packet RvM lower bound below  $T - 1$ , a Gaussian-series tail enclosure above  $T$ , a 33-cell endpoint partition; worst gate lower/tail ratio 1.000001535385559, minimum packet count 25,397,045.4, the next  $2 \times 10^{-6}$  relative step FAILS — mechanism-exact; THE LAW:  $k_{\text{front}}(T) = (\lambda_\infty + O(1/\log T)) T^2/(a - 1/2)$  with  $\lambda_\infty = 0.226987054723979955$  — VERIFICATION HEIGHT  $qT$  BUYS  $q^2$  COLUMNS, the geometric ceiling  $k_{\text{geo}} = (T^2 + 1/4)/(a - 1/2)$  intrinsically unreachable (the route stops at  $0.227 k_{\text{geo}}$ ); NEW EXACT STRUCTURE: the closed-form two-variable generating function  $(X + Y - XY)G = XaF(a(1 - X)) + Y \frac{a}{1-Y} F(\frac{a}{1-Y})$  with the pole-divisor EQUIVALENCE (not an RH proof), total sign-regularity in the REVERSED- $k$  orientation (the standard Pólya-frequency orientation is WRONG — the  $2 \times 2$  minor manifestly negative), THE IMPORTANT CORRECTION: HANKEL POSITIVITY ALONE IS NOT FORM A (uniform measure on  $[2, 3]$ : every Hankel matrix positive definite yet  $C_{0,1} = -3/2$  — the localizers  $H^+$  and  $H - H^+$  required), the fixed- $N$  eventual-positivity theorem ( $b_m(a) = (\sqrt{a} \log a/8)(1/2)_m/m! + O_N(\sqrt{a})$ ), and the exact downward-transport obstruction (for two atoms  $\frac{d}{dt} \log \det H_2 = 4 - 3(y_1 + y_2)$ , either sign possible — fixed- $N$  eventual positivity cannot be propagated down to  $a = 256$ ); (3) ROUND 107 (note CDVII, contract PRIME.CODE.SYNTHESIS.01, code\_synthesis\_probe.py, 18/18 gates+wards, 2.2 s, SPEC 5a6c3851, deterministic re-run bit-identical, verdict CODE-READING-TRUE-DECODING-DIRECTION): the program owner’s founding intuition “primes as error-correcting code in TFPT” given its first formal adjudication under that name and VINDICATED AS MEASURED STRUCTURE — the primes ARE the unique codeword of an error-correcting code in the compiler’s own Gram grammar (ambient: non-negative atomic measures at tent mesh  $\delta = 0.006$ ; codeword: the prime-power comb; check family: nested Toeplitz sections, one check per depth; MINIMUM DISTANCE: the round-96 rigidity law RE-MEASURED INDEPENDENTLY,

$\ln \eta^*(r) = -1.0730r - 7.0783$ , identical to four digits; decoding radius: 6/6 impostors die at the exact round-96 exits; rate  $\rightarrow 0$ ); SAME GRAMMAR as the compiler's literal E8 code layer ([8, 4, 4] self-dual, weight enumerator {0:1, 4:14, 8:1}, Construction A  $\rightarrow$  240 roots, rebuilt exactly in-probe) — TFPT's one axiom INSTANTIATES at the arithmetic seam as this code, structure not progress; DIFFERENT DUAL TYPE with the decisive number measured: THE DUAL RANK GROWS WITH SLOPE 1.0000 ( $k_{\text{fail}}/D = 0.9960/0.9975/0.9983/0.9987$ ; syndrome rank  $5 \rightarrow 12$ ; a non-codeword passes every bounded-depth check family BIT-EXACTLY below its defect), adversarial LTC gap  $\rightarrow 0$  beyond any bounded family's support (local errors testable and depth-improving,  $0.9984 \rightarrow 1.0000$ ) — the code runs measure  $\rightarrow$  identification (DECODING) while RH needs codeword  $\rightarrow$  all checks (ENCODING): THE QUANTIFIER DOES NOT COMPRESS, and the reason is now a measured number; both surprise channels carried as live gates and EMPTY; (4) SURFACES SYNCED IN THIS ENTRY: tfpt\_prime\_front gains the sandbox diary subsection "The frontier extension, the code reading, and the campaign audited: the corrections of record" plus the corrected 12.45/12.46 wordings, the 12.45 mesh-ladder correction, and the OCT vocabulary fence with the five unpriced classes; the end-form article gains §8.15 with the form  $\times$  direction status table, the corrected §8.13 close and §8.14 fences, the abstract status sentence, new limitation 11, the limitation-8 fence, and the traceability rounds 106–109 (with the Hasanalizade–Shen–Wong citation); the Anthropic email draft gains one status paragraph (the corrected three-forms state, the frontier law, the code-reading result); the website prime-front page gains chapter 39 (the live-updates chapter renumbered  $39 \rightarrow 40$ ), the corrected chapters 37/38, the chapter-33 vocabulary fence, and one consolidated sandbox feed entry (verdict BIGPICTURE-FINDINGS, new enum, script bigpicture\_logic\_probe.py); website/lib/changelog.ts regenerated via bash build.sh gen; NO RH claim anywhere)

2026-08-15 (CI) (the roadmap — discovery freeze rounds one-hundred-three through one-hundred-five synced into every live surface: the campaign's complete machine-checked map compiled into the one artifact it never produced — a staged, milestone-gated research roadmap from today's frozen state to a complete RH proof, every stage falsifiable against the corpus's own instruments — fed by the largest proven positivity region of the campaign and the final classical pricing; experiments/ only — THREE frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction; the honesty fences travel first: A ROADMAP IS NOT PROGRESS ON RH, NO STAGE IS CLAIMED STARTED, AND THE MATHEMATICS OF STAGES 1–2 DOES NOT YET EXIST: (1) ROUND 103 (moonshot\_sol\_probe.py, verdict MOONSHOT-PARTIAL(Form A)): from cited external inputs only — Platt-Trudgian 2021 (all zeros to  $T = 3,000,175,332,800$  on the line), Rosser 1941, the classical  $\gamma_1$  isolation; no zero table, no zeta evaluator — THE LARGEST PROVEN POSITIVITY REGION OF THE CAMPAIGN:  $C_{n,k}(256) > 0$  for EVERY  $n \geq 0$  and every  $0 \leq k \leq 10^{19}$  (an infinite half-strip in  $n$ , ten quintillion complete  $k$ -columns; the scope inside the theorem: the  $k \rightarrow \infty$  quantifier is touched nowhere, nothing about RH is proven); the three suggested proof routes die with named mechanisms — the exact prime-side transmutation turns the positive Beta spectral window into a sign-changing generalized-Laguerre source window (the classical Laguerre zero theorem: no termwise-positive prime compar-



ison exists), the Killip–Simon/Szegő direction dies on pure-point entropy ( $w = 0$  a.e., entropy  $-\infty$ , floor  $e \geq 0$  only), the asymptotic- $a$  transport is nonuniform in  $(n, k)$ ; (2) **ROUND 104** (note CDV, contract PRIME.LEVINSON.CLASS.01, levinson\_class\_probe.py, 23/23 gates, SPEC 65ac12a9, run-of-record levinson\_class\_probe.run.log, verdict LEVINSON-ADJUDICATED(LEVINSON-FAIL; tally  $24 \rightarrow 25$ , 0 PASS)): the LAST formally unexamined classical theorem type about zero positions — positive proportion on the line (Selberg 1942; Levinson 1974  $\kappa \geq 1/3$ ; Conrey 1989  $> 2/5$ ; the standing record  $\kappa > 0.417293$ , Pratt–Robles–Zaharescu–Zeindler 2020) — priced against the wall budget: the literal (liminf) class constrains NOTHING at any finite height (exact finite-modification argument), the granted-effective reading breaches every deployed cell by 2–3 orders ( $382/1149/1059$  at  $h = 184/388/839$ ), and a proportion theorem must come within  $1 - \kappa^* < 4.8 \times 10^{-3}$  of ALL zeros ( $\kappa_{\text{ledger}}^* = 0.9952/0.9983/0.9988$ , each a LOWER bound) before it can feed the wall — with this round every named unconditional classical theorem type about zero positions is formally priced (zero-free regions, density, Weyl equidistribution, pair correlation, positive proportion); (3) **ROUND 105** (note CDVI, contract PRIME.ROADMAP.01, roadmap\_probe.py, 29/29 gates, SPEC 79092ebf, deterministic re-run identical, run-of-record roadmap\_probe.run1.log, verdict ROADMAP-COMPILED(stages: S0 DONE; S1a|S1b OPEN; S2 OPEN; S3 CONDITIONAL-KERNEL-CHECKED; gates 29/29)): every machine-groundable claim asserted — acceptance-gate values re-computed from cited constants or grep-warded verbatim in their owning frozen artifacts, the stage DAG acyclic with exactly the named open stages, the downstream chain’s artifacts present with verdicts grep-verified; **STAGE 0** (done, frozen): the atlas complete at 25 candidates 0 PASS, exactly one open edge E4, three equivalent lemma forms, two certified falsification instruments with open-and-empty channels (Euler–Pick floors  $N = 1$ :  $[4.5917135 \times 10^{-2}, 4.5917136 \times 10^{-2}]$  down to  $N = 4$ :  $[8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  at cap  $10^{13}$ ; Hausdorff certified depth 86, 5290 cells,  $\gamma^* 189/563$ ), the  $k \leq 10^{19}$  theorem re-derived inside the probe (decisive ratios  $1.176577/335.8/1416.8$ ; the  $n = 0$  mechanism wall re-located between  $1.40 \times 10^{19}$  and  $1.41 \times 10^{19}$ ), the rigidity law  $\ln \eta^*(r) = -1.073r - 7.08$ , the seven-property specification sheet, 27 kernel-checked theorems; **STAGE 1** (OPEN — new mathematics): structural positivity of Hodge-index type for Spec  $\mathbb{Z}$  — verified literature state: Deninger’s dynamical Lefschetz trace formula PROVED for smooth foliated flows (Álvarez López–Kordyukov, arXiv:2402.06671) but NO arithmetic host constructed; Connes–Consani Riemann–Roch on the scaling-site orbits and for  $\mathbb{Z}$  but NOT on the square of the scaling site; milestone (1a) an arithmetic dynamical system hosting the proven trace formula (periodic orbits  $\log p$ , weights  $\log p/p^{m/2}$ ), acceptance gate: reproduce the (AC)/Euler–Pick interface  $c^T P_N c = \langle -g'', f_c \star \tilde{f}_c \rangle$  at the 16 frozen  $\sigma$  of the SVPIN spec (minutes-scale, by machine); milestone (1b) OR Riemann–Roch on the square of the scaling site, acceptance gate: the intersection pairing on exponential autocorrelations must reproduce the certified  $\mathcal{P}_N$  ladder ( $N \leq 4$  floors as frozen acceptance values); **STAGE 2** (OPEN — new mathematics): the Hodge-index/Castelnuovo analog on the Stage-1 substrate, acceptance gates = the corpus’s measured consumption requirements: FAIL the surrogate worlds exactly at the measured depths ( $r = 0.264$  smooth by support,  $r = 0.744$  scramble by weights), be SIGNED and ALIGNMENT-CARRYING (the v913 class), SURVIVE the  $\tau$ -screen (bands  $0.30/0.70$ ), and reproduce the certified Hausdorff cells (depth 86) while STRICTLY EXTENDING the  $k \leq 10^{19}$  theorem’s reach; **STAGE 3** (CONDITIONAL-COMPLETE — assembly, already machine-held): from the Stage-2 inequality to localized Weil positivity in ANY ONE of the three

lemma forms, then the machine-held extraction — v848 (EXTRACTION-CHAIN-COMplete, hypothesis (H) isolated), v912 (unconditional rate  $O(D^2 \log(1/D))$ ), the Lean kernel tier (CofinalWeil/CofinalPredefinition, pickMatrix\_posSemidef, sv\_implies\_rh with the honesty lock skeleton\_not\_unconditional), the SVPIN skeleton (SVPIN-ROUTE-OPEN), the Hausdorff equivalence (HAUSDORFF-EQUIVALENCE-SOUND), the CCM same-currency interface (CCM-SAME-CURRENCY); THE FALSIFICATION SPINE (parallel, permanent): the two certified instruments keep running as standing disproof channels, both open and EMPTY (a single certified negative would be an RH-disproof candidate); the extension walls are KNOWLEDGE walls, not compute walls (Euler–Pick  $N = 5$  needs  $\psi$ -knowledge beyond  $10^{19}$ , floor/ $\lambda_5 = 4.7 \times 10^5$  at any cap; the Form-A  $n = 0$  mechanism exhausts at  $k \approx 1.405 \times 10^{19}$ ); THE HONEST PRICING: Stages 1–2 require mathematics that does not exist today — the roadmap proves interfaces and obstructions, never existence; Stage 3 is assembly; the corpus’s contribution to every stage is the same — acceptance gates that turn each milestone into a minutes-scale machine check, so any candidate, from anyone, can be adjudicated immediately; NO timeline promised; (4) SURFACES SYNCED IN THIS ENTRY: tfpt\_prime\_front gains the sandbox diary subsection “The road from here: a machine-gated path to a complete proof”; the end-form article gains §8.14 with the abstract status sentence and the traceability rounds 103–105; the Anthropic email draft gains one status paragraph (the roadmap as the collaboration offer — each milestone has a minutes-scale acceptance test); the website prime-front page gains chapter 38 (the live-updates chapter renumbered 38 → 39) and one consolidated sandbox feed entry (verdict ROADMAP-COMPILED, new enum, script roadmap\_probe.py); website/lib/changelog.ts regenerated via bash build.sh gen; NO RH claim anywhere)

2026-08-15 (C) (the instrument tier — discovery freeze rounds ninety-three and ninety-five through one-hundred-two synced into every live surface, INCLUDING one flagged correction of record to already-published text: the collective-rescue explanation, the certified Euler–Pick ladder and its  $N = 4$  extension, the rigidity law, the SP pricing round that falsified the operative round-88 predictor extension, the CCM adjudication, the Lean kernel round, the  $\alpha$ -induction depth record, and the Hausdorff safe-point instrument — the one open input now held in THREE equivalent lemma forms with TWO certified falsification instruments; experiments/ only — EIGHT frozen probes plus a Lean formalization round, several run as concurrent lanes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) THE CORRECTION OF RECORD (round 97, note CCCXCVIII, sp\_expsum\_pricing\_probe.py, 13/13 gates, deterministic 0.8 s twice, SPEC 55207ff7, verdict SP-OPERATIVE-FALSIFIED + SP-PRICED + SP-CLASSICALLY-OUT-OF-REACH( $\times 8.2$ ) + SP-CONDITIONAL-LADDER + SP-PREFIX-DEAD-IN-WINDOW + SP-LITERAL-NONVANISHING-IN-RANGE; carried per the round-87 convention — frozen probes and notes are NOT retro-edited, the correction is the record): the round-88 rounding-margin statement priced against the full classical exponential-sum arsenal AND its operative form MEASURED FALSE beyond  $x = 13$  — census extended from 79 points ( $x \leq 13$ ) to 18798 points ( $x \leq 890$ , cache edge  $\gamma_{7000} = 7264.75$ ), the operative (SP) FALSE from  $x = 21$  onward with 567  $\tau$ -screened counterexamples, THE ROUND-88 EXACTNESS A SMALL- $x$  ACCIDENT OF THE MARGIN BUDGET, mechanism ARITHMETIC-FREE (Fejér tracking-variance floor

$\text{rms} \rightarrow 1/(2\pi) = 0.159$ , measured 0.26, vs budget 0.5 — no taper in the family survives); pricing (all constants generous, deepest rung  $x = 890$ ): best unconditional miss  $\times 8.2$  (Trudgian  $S(t)$  ceiling 4.11), even RH itself (Carneiro–Chandee–Milinovich) 1.015 at the deepest rung; prefix/exceptional-set salvage DEAD IN-WINDOW (lowest violation  $k/K = 0.010$ , prefix depth stalls  $\leq 131$ ); HONEST FENCE: the deployed source-side approximant died, NOT TPL(i) — the eigenvector truth side unmeasured beyond  $x = 13$ , the published round-88 claim (exact at 79 points,  $x \leq 13$ ) remains factually true; the corrective clause added to the published §8.10 wording, §8.12 property 1, the prime-front round-88 subsection, the website chapter 34 fence, and a new paper limitation 10; (2) ROUND 93 (note CCCXCIV, `collective_rescue_probe.py`, 10/10 wards + 7/7 law gates, verdict COLLECTIVE-IS-SUPPORT): round 92’s exit/re-entry FULLY EXPLAINED — exact causal onset bound (bit-identical prefix, spread exactly 0) plus measured offset  $\sim 2$  lags, completed-stream  $\max |\alpha|$  CONSTANT at 0.183932 ( $r = 2.568\text{--}4.788$ , band maxima falling), all Lyapunov candidates killed (the fixed-prefix class EMPTY by causality), mass-restored control dies at the IDENTICAL lag 400 — mass-without-atoms does not rescue, in-window survival requires the exact arithmetic weights; (3) ROUND 95 (note CCCXCVI, `eulerpick_ladder_probe.py`, 31/31 gates, verdict EULERPICK-CERTIFIED( $N \leq 3$ ) + EULERPICK-GENERIC-DECAY + EULERPICK-DETECTION-LAW): the falsification ladder rebuilt to CERTIFIED standard (outward-rounded intervals end to end,  $N = 3$  certified positive at cap  $1.6 \times 10^7$ :  $[2.3643695 \times 10^{-10}, 1.1497752 \times 10^{-9}]$ ), the channel self-tested on a planted negative family and EMPTY; decay GENERIC (control decomposition within  $1.2\times$  of Euler, the true  $\gamma_1$  appearing nowhere); detection law  $N^*(\delta) = 6.1 + 1.95 \log_{10}(1/\delta)$ ,  $\gamma$ -BLIND beyond  $\gamma \approx 50$  (undetected through  $N = 36$ ); promotion honestly declined; (4) ROUND 96 (note CCCXCVII, `rigidity_inverse_probe.py`, 9/9 wards, verdict RIGIDITY-TIGHTENING): screw positivity to depth  $r$  pins the measure onto the prime comb exponentially tightly — ALL 17 adversarial impostors die (shifted-prime comb, golden quasi-prime comb at exact prime density, crystals, mass-matched tails, sub-mesh splits at every scale, jitters, a deep weight swap; all 15 single-atom deletions fatal within  $J \leq 8$  lags), moduli law  $\ln \eta^*(r) = -1.073r - 7.08$  with  $c_{\text{mod}} = 1.073$  vs  $c_\lambda = 0.814$  — CONDITIONING-PRICED up to an  $O(1)$  alignment factor, the theorem candidate typed (anchors measured, law fitted, all-depth OPEN); (5) ROUND 98 (note CCCXCIX, contract on arXiv:2310.18423, `ccm_pn_adjudication_probe.py`, 22/22 gates, verdict CCM-CHECKED + CCM-SAME-CURRENCY): the one external published RH reduction adjudicated — CCM’s published local terms reproduce  $\xi'/\xi(1/2 + \sigma)$  to rel  $2.3 \times 10^{-50}$  and their quadratic form reproduces the corpus EULER–PICK MATRICES ENTRYWISE ( $\tau$ -screen: RELOCATION); NEW FINITE INSTRUMENT: all 6 prime-bearing Galerkin sections of  $Q_n$  positive ( $\lambda_{\min} = 2.30 \times 10^{-3}$  at  $n = 2.5$  down to  $1.01 \times 10^{-5}$  at  $n = 13$ ; pole constraint load-bearing); controls die through the same construction —  $P(n)$  IS localized Weil positivity, any semilocal prolate candidate finitely testable against the (AC) interface today; (6) ROUND 99 (note CD, the Lean round: `TfptCarrier/EulerPick.lean`, `DeltaOneNoGo.lean`, `ParityLemma.lean`, `SVSkeleton.lean`; 27 kernel-checked headline theorems, axioms exactly [`propxt`, `Classical.choice`, `Quot.sound`], no sorry, no `native_decide`): Euler–Pick forward PSD at the SV nodes for every  $N$ , the  $\delta_1$  no-go counterexample package, the FULL Parity Lemma under ONE named simplicity hypothesis (alternating signs at canonical gap midpoints — exactly where zero simplicity enters), the SV skeleton with THE HONESTY

LOCK skeleton\_not\_unconditional (an instance with RH = False — packaging alone yields nothing); (7) ROUND 100 (note CDI, sieve4\_eulerpick\_n4\_probe.py, 22/22 gates, 2682s, cap  $X = 10^{13}$ , helper source SHA pinned, run-of-record sieve4\_run1.log, hard external ward  $\pi(10^{13}) = 346,065,536,839$  REPRODUCED EXACTLY):  $\lambda_{\min}(\mathcal{P}_4) \in [8.278338 \times 10^{-15}, 1.3840906 \times 10^{-14}]$  CERTIFIED POSITIVE (margin 0.749),  $N \leq 3$  floors tightened massively, the round-95 upgrade-path model CORRECTED (the beyond- $10^{19}$  tail floor was omitted; Nair 1982 —  $\psi(x) \geq (x-2)\ln 2$  — unblocks it, corrected  $X_{\text{req}}(4) = 10^{11.52}$ ), THE KNOWLEDGE WALL STARTS AT  $N = 5$  (the floor exceeds  $\lambda_5$  by  $4.7 \times 10^5$  at any cap — no sieve helps); (8) ROUND 101 (note CDII, contract PRIME.SCREW.INDUCTION.01, screwind\_induction\_probe.py, 10/10 wards + 5/8 law gates — the 3 fails are FINDINGS, typed; companion screwind\_meshdrift\_check.py; verdict SCREWIND-KICK-LAW(exact) + SCREWIND-STEP-IS-WALL + SCREWIND-DICHOTOMY(descriptive-only)): DEPTH RECORD — all sections positive to  $r = 9.0$  at BOTH meshes (corpus record 4.788),  $\sup |\rho| = 0.183932$  attained at  $k = 0$ , flatness carried by a cancellation of giants ( $\pm 288.5$  vs net RMS 0.023), THE EXACT KICK LAW at atom arrival (affine-exact, dev  $\leq 2.8 \times 10^{-16}$ , numerator pure arithmetic), G7 fails INFORMATIVELY: PRIME ARRIVALS STABILIZE (net arrival  $\alpha$  positive 33/39, atom-bin median  $|\rho|$  below background); THE WALL RELOCATES INTO THE CERTIFICATE, the named open piece — a source-certified two-sided energy floor with  $\kappa_+ < 1/2$  — IS localized positivity; (9) ROUND 102 (note CDIII, externally proposed contract PRIME.HAUSDORFF.SAFEPOINT.01, hausdorff\_safepoint\_probe.py, 40/40 gates, 143.6s, SPEC dacdfte2, composite verdict HAUSDORFF-EQUIVALENCE-SOUND + HAUSDORFF-REPRODUCED + HAUSDORFF-CERTIFIED-DEPTH(86;  $\gamma^*$  189/563) + HAUSDORFF-REPLICATION + HAUSDORFF-DETECTOR + HAUSDORFF-Z1-ZEROMEASURE + NOT-WORLD-NEW): the moment-field criterion (RH  $\Leftrightarrow$  all Pascal cells  $C_{n,k}(a) \geq 0$  at the safe point  $a = 256$ ) built, audited, adjudicated — equivalence skeleton SOUND (Hausdorff/Widder needs no extra boundedness, the one sharp continuation step closes, companions symbolically exact), the proposer's spec numerically faithful (561/561 cells, weakest ratio 1.000220), THE HEADLINE: CERTIFIED DEPTH 86 (5290 cells),  $\gamma^*$ -reach 189 at  $a = 256$  and 563 on the  $a = 1024$  arm vs Euler–Pick's certified  $N = 3$  ( $\gamma \sim 5-8$ ) — THE CHEAPEST CERTIFIED RH-CONSEQUENCE INSTRUMENT IN THE CORPUS; three-carrier replication (signpos/Krein/SV pins, no zero cache anywhere in construction); the detector honestly priced ( $\sim 1/\delta$  below  $\gamma_1$ , provably BLIND above — Epstein control concrete: off-line witness LOCATED at  $\rho = 0.69692705 + 36.37406369i$ , RH-for- $Q$  provably false, the  $Q$  Pascal field positive to depth 100 exactly as the phase model predicts); Z1 typing source-only in INPUT, zero measure in CONTENT; NOT-WORLD-NEW (Zhang arXiv:2303.09396 Thm 2 is the identical criterion at the central point — the novelty is the safe-point shift, the source-only decoder, the certified conditioning); THE NAKED REMAINING TASK: positivity of ALL  $C_{n,k}(a)$  — Weil positivity in Hausdorff currency, THE THIRD EQUIVALENT LEMMA FORM beside the standing two; (10) THE CONSOLIDATED CLOSE, carried on every surface: the campaign now holds THREE equivalent lemma forms of the one open input (uniform Weyl-disk contraction; ground-eigenvector sign positions; Hausdorff cell positivity), three agreeing source-only carriers, two certified falsification instruments (Euler–Pick  $N \leq 4$ ; Hausdorff depth 86), the rigidity law, and 27 kernel-checked theorems — and at the bottom,

unchanged, localized Weil positivity; (11) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The three lemma forms, the certified instruments, and a correction of record” plus the corrective clauses in the round-88 subsection and the specification-sheet P1; the end-form article gains §8.13 with the abstract status sentence, the §8.10/§8.12 corrections, limitation 10 and the traceability rounds CCCXCIV and CCCXCVI–CDIII; the Anthropic email draft gains one status paragraph (the three-lemma-form state, the two certified instruments, the CCM interface point); the website prime-front page gains chapter 37 (the live-updates chapter renumbered  $37 \rightarrow 38$ ), the chapter-34 correction pointer, and one consolidated sandbox feed entry (verdict HAUSDORFF-CERTIFIED-DEPTH); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-15 (XCIX) (the cohomological specification sheet — discovery freeze rounds ninety-one, ninety-two and the specification round synced into every live surface: two honest negatives, the Verblunsky-invariance kill with the Euler–Pick criterion and the (AC) identity pinned, and the campaign’s machine-checked characterization of the missing RH input compiled into a literature-facing requirements document for the cohomological programs; `experiments/only` — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 91 (note CCCXCII, the patterns round, run in a parallel session): the program’s first magnetar giant-flare QPO bed (MAG.QPO.LADDER.01, `magnetar_qpo_ladder_probe.py`, 8/8 gates, SPEC 4ee58b0f) tests SGR 1806–20 and SGR 1900+14 against the geometric (3/2) relaxation ladder with a 20000-draw log-uniform MC null, a 17-base look-elsewhere battery, a power gate (planted ladder  $p = 5 \times 10^{-5}$ ) and a specificity gate (integer harmonics  $p = 0.078$ ): base 1.5 gives  $p = 0.3163/0.3558$ , Fisher-joint  $p = 0.3584$  — VERDICT NULL, S15-typed (placebo base 1.80 best in both sources, documented as an LEE-loaded base-battery artifact candidate, NOT escalated; SGR 0526–66 DATA-LIMITED); the LEE-controlled unclaimed-observable scan (PHYS.UNCLAIMED.DECODER.01, `unclaimed_observable_scan_probe.py`, 5/5 after disclosed amendment A1 — gate G02 mis-specified as a minimality bar, amended to a membership bar, no verdict bar moved) returns SCAN-UNDECIDABLE with the structural finding as the content: the frozen compiler vocabulary is SATURATED at  $2\sigma$  tolerance (median placebo  $C_{\min} 3.66 \leq \text{control } C_{\max} 5.46$ ) — new predictions provably cannot come from numeric scanning and must be derived theory-side; (2) ROUND 92 (note CCCXCIII, contract PRIME.SCREW.VERBLUNSKY.INVARIANT.01, `vbk_invariant_probe.py`, 23/23 gates, deterministic 14.1/14.0 s, SPEC c730781e, verdict VERBLUNSKY-NO-INVARIANT(mechanism exhibited)): the  $\sigma$ -correction (pointwise Weyl-disk contraction on  $\sigma_r = 1 + 1/r$  accumulating at the interior point 1 suffices;  $\sigma \downarrow 1/2$  uniformity NOT load-bearing), the exact  $\delta_1$  no-go (pin convergence cannot prove section positivity backwards), THE EULER–PICK CRITERION audited and pinned ( $\text{RH} \Leftrightarrow P_N = [(P(\sigma_j) + P(\sigma_k))/(\sigma_j + \sigma_k)] \geq 0$  for all  $N$  with  $P(z) = \xi'/\xi(1/2 + z)$ ; source-only falsification ladder from  $\Gamma$ ,  $\pi$ ,  $n^{-s}$  and an own  $\Lambda$ -sieve:  $\lambda_{\min}(P_N) = 4.59 \times 10^{-2}$  down to  $2.44 \times 10^{-62}$  for  $N = 1..12$ , all positive,  $\kappa = 2.27 \times 10^{61}$  at  $N = 12$ ) and THE (AC) IDENTITY  $c^T P_N c = \langle -g'', f_c \star \tilde{f}_c \rangle$  gated symbolically exactly and numerically against the corpus-exact screw  $g$  ( $8.581810362 \times 10^{-2}$  vs  $8.578893345 \times 10^{-2}$ , rel  $3.4 \times 10^{-4}$ ) — Euler–Pick positivity IS Weil positivity on exponential autocorrelations, by identity; the kill: complete

prime packets are high-rank INDEFINITE Toeplitz updates of displacement rank two (ranks 26/90/130, inertias (13, 102, 13)/(45, 70, 45)/(65, 50, 65)), the induced  $\alpha$  step nonlinear and prior-state dependent ( $\times 316$ ), the TRUE packet stream itself EXITS the Schur disk after  $p = 2/3/5/7$  ( $r = 1.122/1.620/1.950/2.400$ ) and re-enters only collectively (after  $p = 11$ : 0.313; full: 0.184); control wards reproduce SMOOTH  $r = 0.264$  (diagnosed as a SUPPORT error — zero true prime mass before  $\log 2$ , PNT mass 0.282) and SCRARITH  $r = 0.744$  (weight  $\times 1.451$  at kept positions); the only exact survival condition is WALL-EQUIVALENT — clean kill, the remaining all-depth principle is Weil positivity in yet another currency; (3) THE SPECIFICATION ROUND (note CCCXCV; the intervening numeral CCCXCIV belongs to a concurrent discovery lane and is not part of this sync; contract PRIME.COHOMSPEC.SHEET.01, `cohomspec_probe.py`, 20/20 gates, 8.8 s, SPEC 423d934e, independently re-run deterministic, verdict COHOMSPEC-COMPILED(7 properties: 6 machine-grounded, 1 citation-typed)): the campaign’s measured characterization of the missing input compiled into a REQUIREMENTS DOCUMENT for any cohomology/intersection theory for  $\text{Spec } \mathbb{Z}$  (Deninger’s program; the Connes–Consani scaling site), each property grounded by grep ward or independent recomputation in its owning frozen artifact — (P1) the signed, alignment-carrying, unconditional, sub-spacing ordinate-position input of the one open edge E4 (v913, kill-atlas DAG, the 24/0/19 tally,  $\sim 1\%$  sub-spacing; sharpest corpus forms:  $D_{\max} = x - 3/8$  at  $e^{-11}$  per forced skip, min margin 0.0317 / load 4); (P2) shape-independence (deployed exponent  $-3.379$  recomputed by OLS as  $-3.3795$ ; alternatives shallower by  $\geq 2.2$ ); (P3) not magnitude/density/existence (Littlewood 214.7/1527.1; zero-density empty for  $A > 2$ ; border membership  $\det M = s \det B$  recomputed in exact Fractions); (P4) the Gram form on exponential autocorrelations must BE the Euler–Pick matrices — the concrete interface any cohomological trace formula must match, recomputed (ladder rel dev  $\leq 3.8 \times 10^{-9}$ , (AC) weld rel devs  $\leq 4.3 \times 10^{-11}$ ), testable at any finite  $N$  by machine today; (P5) positivity must fail on the control worlds exactly as measured ( $r = 0.264$  by support,  $r = 0.744$  by weights, both recomputed); (P6) survive the  $\tau$ -screen (bands 0.30/0.70, fifteen DISGUISE routes); (P7, citation-typed) the function-field precedent mapped to its break points, verified against the live literature — Deninger’s dynamical Lefschetz trace formula proved for smooth foliated flows (Álvarez López–Kordyukov, arXiv:2402.06671) with no arithmetic host constructed (Doc. Math. ICM 1998, arXiv:math/0505354); Connes–Consani Riemann–Roch on the scaling-site orbits  $C_p$  (Selecta Math. 2017) but not on the square; the semilocal/prolate line (arXiv:2006.13771, arXiv:2310.18423) and Connes’ 2026 extremal letter (arXiv:2602.04022, zero reality by Connes–van Suijlekom Thm 6.1, convergence open = the corpus’s REALROOT-DEPTH-UNDECIDED); (4) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “What any cohomology for  $\text{Spec } \mathbb{Z}$  must supply: a machine-grounded specification”; the end-form article gains §8.12 with the abstract status sentence and the traceability rounds CCCXCII, CCCXCIII and CCCXCV with the verified literature list; the Anthropic email draft gains one status paragraph (the collaborator point: a testable interface where there was an analogy); the website prime-front page gains chapter 36 (the live-updates chapter renumbered 36  $\rightarrow$  37) and one consolidated sandbox feed entry (verdict COHOMSPEC-COMPILED); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; HONESTY FENCE carried on every surface: the sheet is a requirements document derived from machine-checked no-gos — it proves nothing about Deninger’s program or Connes–Consani, claims no progress on RH, and

whether any theory can meet the specification is OPEN; NO RH claim anywhere)

2026-08-14 (XCVIII) (the Stieltjes–Vitali pin route and the Krein screw-function carrier — discovery freeze rounds eighty-nine and ninety synced into every live surface: the externally proposed countable-pin reduction of the real-root architecture, and the construction of its one priced missing ingredient — the Krein inverse-spectral realization of Suzuki’s screw function, the first pin carrier of the campaign to pass the Z1 screen; experiments/ only — TWO frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 89 (note CC-CLXXXIX, externally proposed contract PRIME.STIELTJES.VITALI.PIN.01, stieltjes\_vitali\_pin\_probe.py, 17/17 gates, 249 s, SPEC f450832f, independently re-run bit-identical, verdict SVPIN-ROUTE-OPEN): replace the full semilocal trace convergence by countably many positive resolvent scalars in the absolutely convergent Euler half-plane — the pins  $P_x(\sigma) = \sum_k 2\sigma/(\lambda_{x,k}^2 + \sigma^2)$  must converge to  $\xi'/\xi(1/2 + \sigma)$  at  $\sigma_r = 1 + 1/r$  (source-only right side via  $\Gamma$ ,  $\pi$ ,  $\Lambda(n)$ ,  $n^{-s}$ ); then Herglotz normal-family compactness (ONE pin suffices), Vitali on the pins accumulating at the interior point  $i$ , the identity theorem on  $\Im z > 1/2$ , analytic continuation on the connected  $\mathbb{C}_+ \setminus Z(\Xi)$  and the pole contradiction give (SV)  $\Rightarrow$  RH, the functional equation closing  $\beta < 1/2$ ; THE SKELETON IS SOUND gate by gate: weight-comparability constant 65.01 (bar  $2 \times 10^3$ ) executed per rung (normal-family ratio 0.033 on all four), the sign-convention ambiguity HARMLESS for the pins ( $\Xi(z) = \xi(1/2 - iz)$  vs  $\xi(1/2 + iz)$  give the identical target via  $\xi'/\xi(1-s) = -\xi'/\xi(s)$ ; numeric lock at  $\sigma = 1$ :  $4.61359285 \times 10^{-2}$  vs  $4.61359517 \times 10^{-2}$ , dev  $2.3 \times 10^{-8}$ ), spectrum honesty MEASURED not assumed (census 4/10/20/41 all real, no zero mode, min spacing  $\geq 1.3$ ), the degenerate Herglotz limit excluded by the pin bound; the ONE named condition: all- $x$  realness is the CF simplicity-plus-evenness hypothesis; the named non-gap: THE CHAIN MOVES THE ENTIRE BURDEN INTO (SV) ITSELF — no skeleton step is the wall, the wall is the premise; tail bookkeeping explicit, all three tails decreasing (true zero tail  $2.85 \times 10^{-2} \rightarrow 1.18 \times 10^{-2}$  within 2% of the  $\log T/T$  model; measured Nyquist-excess edge mass  $\leq 1.31 \times 10^{-3}$ , the window definition corrected in the spec after a smoke caught band spill; exact lattice tail  $1.35 \times 10^{-2} \rightarrow 8.0 \times 10^{-3}$ ); THE FINITE PRE-CHECK CONVERGES ON ALL 16 FROZEN  $\sigma$  VALUES ( $x = 3/5/8/13$ ;  $x = 21$  skipped and declared by the arbiter cost law):  $\Delta$  negative everywhere (missing tail dominates), slopes  $-0.62 \dots -0.65$ , measured drop 2.6 vs the a-priori RvM model 2.4 (bar 1.6), NO  $\sigma \downarrow 1/2$  rate degradation on this ladder (the expected boundary effect beyond reachable depth, stated honestly), the FULL divisor row converging FASTER (the lattice comb compensates  $\sim 2/3$  of the missing tail); CONTROLS ALL SEPARATE: mesh-CF MESH-POISONED through the same pins (median overshoot  $0.94 \rightarrow 6.32$  growing like  $\log x$  — the wrong limit, measured), SCRPOS/EPSTEIN/SMOOTH/SCRARITH separate at medians  $2.7 \times 10^5$ – $8.8 \times 10^5$  (band metric) and 12.4–45.9 (target metric) on 16/16 rows (SCRARITH throws one imaginary pair — the world breaks the SV realness hypothesis, reported), Christoffel weights structurally excluded by AST,  $\tau$ -screen 0/16 relocations at max slope 0.009 over  $\sim 47$  decades — NO DISGUISE; THE HONEST WEIGHT, Z1-typed: on the extremal family the measured pin convergence IS the transcription of cache partial sums (band pins vs cache rel  $7.7 \times 10^{-12}$  /  $9.4 \times 10^{-12}$  /  $2.1 \times 10^{-5}$ ) — instrument consistency, never source-side content; the naive Suzuki spectral readout is DEAD (SUZPIN-FORM-SPECTRUM: hat-compression eigenvalues are a form-spectrum ladder

$\sqrt{\mu} = 1.02/2.12/3.17 \dots$ , not the ordinates,  $\gamma_1 = 14.13$ ;  $P_{\text{suz}}(1)$  grows  $1.85 \rightarrow 2.47$  vs target 0.046) — the SECOND architecture of the campaign to survive every gate, the corpus’s logically sharpest open target, its priced remaining task exactly one: the Krein inverse-spectral realization of Suzuki’s screw function; three pre-freeze smoke fixes disclosed, zero post-freeze amendments; (2) ROUND 90 (note CCCXC, contract PRIME.SUZUKI.KREIN.REALIZATION.01, krein\_screw\_realization\_probe.py, 22/22 gates, 18 s, SPEC bd5fba32, independently re-run bit-identical, verdict SUZKREIN-CARRIER-OPEN(lemma: uniform Weyl-disk contraction)): the priced missing ingredient BUILT — THE CORRESPONDENCE:  $g$  extracted corpus-exactly (Suzuki eq. 1.3 with the v643 convention lock; the Lerch coefficient  $+1/4$  GATED against the origin expansion  $g \sim \frac{1}{2}t \log t + At$  with  $A = \frac{1}{2}(\log 2\pi - \psi(2)) = 0.707546$  — a  $-1$  misreading would fail by  $O(1)$ ), the screw property verified ( $G(t, s)$  PSD on all four windows,  $\lambda_{\min}/\lambda_{\max} \geq -8.1 \times 10^{-19}$ ), Krein’s accelerant/continual-Schur correspondence invoked with its hypothesis CERTIFIED (strict positivity of every Toeplitz section of  $-g''$ , certified by the completed Szegő run,  $\max |\alpha| = 0.156$ ), the normalization DERIVED not fitted (lag reads are exact second differences of  $g$ , two independent routes agreeing to  $1.8 \times 10^{-10}$ ; partition of unity gives the pin as  $\frac{1}{2}\langle -g'', e^{-\sigma|\cdot|} \rangle$ ; the Weil formula identifies  $\xi'/\xi(1/2 + \sigma)$  layer by layer, archimedean/pole/prime each gated); THE CONSTRUCTION: Levinson/Szegő IS the Toeplitz-structured discrete Krein/Gelfand–Levitan solve, the Verblunsky  $\alpha$  are the realized Hamiltonian data and CAUSAL (exact Toeplitz nesting — deeper windows only append), and THE REALIZED HAMILTONIAN SEES THE PRIMES:  $|\alpha|$ -spikes at  $r = \log q$  for  $q = 2/3/4/5/7$ , offset  $\leq 1$  bin, contrast 2.8–5.2; conditioning TRIVIAL (amplification  $\sim 23$  under  $10^{-25}$  perturbations — the feared  $e^{cx}$  explosion belongs to the trig-family spectral-resolution ladder, not the screw-kernel Toeplitz sections); the  $\delta$ -rough pointwise profile declared honestly, the  $\delta$ -stable objects (pins, disks, spike positions) gated; THE CONVERGENCE: all 16 frozen  $\sigma$  rows (verbatim from the SVPIN spec) CONVERGE, 16/16 truth-tight ( $\sigma = 0.6$ :  $\Delta = +7.30 \times 10^{-5} \rightarrow +6.33 \times 10^{-7}$  over  $L = \log 3 \dots \log 13$ , factor 115), Weyl-disk contraction MEASURED  $R \sim e^{-(\sigma+c)L}$  with  $c = 0.85 \dots 1.76$  median 1.36 (limit-point behaviour), and THE SURPRISE: the disk center RESUMS THE MISSING TAIL ( $|\hat{P} - S_{\text{trunc}}|/\text{tail-model median } 1.000$ ; at  $\sigma = 0.6$  the center beats the truncated source sum by a factor  $\sim 600$ ); THE CONTROLS DIE AT THE POSITIVITY LEVEL ITSELF — the strongest separation of the campaign: the prime-free SMOOTH accelerant LOSES SCREW POSITIVITY at window depth  $r = 0.264$  (same  $r$  at both mesh sizes — a continuum property), the golden-scrambled SCRARITH world at  $r = 0.744$  — THE TRUE PRIMES (positions AND weights) ARE NECESSARY FOR SCREW POSITIVITY at these windows; THE Z1 SCREEN PASSED FOR THE FIRST TIME: no cache transcription (min over all  $N \leq 7000$  of the max- $\sigma$  rel dev  $= 8.5 \times 10^{-3}$  vs bar  $10^{-6}$ ; AST firewall zero-free; SUZKREIN-EXTRAPOLATES(1.000)), explicitly distinct from DEBRANGES-COMB-BLIND in input (screw lag data, not wall-block moments), direction (inverse-spectral construction, not HB re-coordinatization) and adjudication (quantitative convergence + world controls, not structure existence), the dead SUZPIN form-spectrum readout never used; THE REPRICED LEMMA AND THE HONEST PRICE: the literal SVPIN trace-class norm is MEASURED FLAT (16/16 non-decaying; appended Verblunsky blocks grow — prime atoms arrive forever) — the correct currency is WEYL-DISK CONTRACTION, the minimal missing lemma stated exactly (“let  $g$  be Suzuki’s screw function with every finite section of  $-g''$  positive (= localized Weil positivity); then the Weyl



disks  $D_a(i\sigma)$  of the Krein realization contract with  $R_a(\sigma) \leq C(\varepsilon)e^{-(\sigma+c)\cdot 2a}$  uniformly on  $\sigma \geq 1/2 + \varepsilon$ , with limit point  $\xi'/\xi(1/2 + \sigma)$ "; suppliers: Krein's accelerant theorem, de Branges chain monotonicity; OPEN: the uniformity as  $\sigma \downarrow 1/2$ ) — AND THE ALL- $a$  SECTION-POSITIVITY HYPOTHESIS IS LOCALIZED WEIL POSITIVITY ITSELF: the carrier localizes the remaining task and fixes the currency, it does NOT remove the positivity input; disclosures: pre-freeze float64 calibration, two smokes (six fixes incl. a normalization slip caught by the table's own  $\Delta = -\text{tgt}$  pattern), a first-full-run abort at a circular liveness filter (repaired and disclosed in the spec, SPEC-hash change documented 1f898f95  $\rightarrow$  bd5fba32) — no bar, grid, ladder or verdict-bearing rule moved; (3) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection "The Stieltjes–Vitali pin route and the Krein screw-function carrier"; the end-form article gains §8.11 with the abstract status sentence and the traceability rounds CCCLXXXIX–CCCXC; the Anthropic email draft gains one status paragraph (the missing input now has TWO precisely stated lemma forms, and the Krein carrier is the first source-only host whose controls die at the positivity level); the website prime-front page gains chapter 35 (the live-updates chapter renumbered 35  $\rightarrow$  36) and one consolidated sandbox feed entry (verdict SUZKREIN-CARRIER-OPEN); `website/lib/changelog.ts` regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-14 (XCVII) (the real-root saga and the sign-position endpoint — discovery freeze rounds eighty-three through eighty-eight synced into every live surface: the real-root architecture and its three-lane conflict, the arbiter, the TPL round with the PARITY LEMMA and the counting law, the tested-simplicity round, the adversarial bughunt and the SIGN-POSITION PREDICTOR; experiments/ only — TEN frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction; standing convention: the round-87 bughunt CORRECTED three MAJOR misstatements in the round-83/84 records, the frozen probes and notes are NOT retro-edited — the correction is the record — and every surface synced by this entry carries only the corrected numbers: (1) ROUND 83 (note CCCLXXXII, three lanes: `semilocal_realroot_limit_probe.py` 16/16, SPEC 8eff2685, verdict REALROOT-ARCHITECTURE-OPEN; `stc_sol_convergence_probe.py` 4/4, verdict STC-OPEN; `stc_opus_convergence_probe.py` 20/20, verdict STC-DISGUISE + STC-FALSE for the mesh family): the first route of the campaign to genuinely leave the kill atlas — source-only finite SELF-ADJOINT operators (real spectra by construction, no wall positivity, no zero data consumed), limit identified with  $\Xi$  via the full trace formula on a determining even Paley–Wiener class; skeleton audited as executable gates (determination lemma sound in two classical steps with NO convergence rate consumed; two repairable gaps repaired: an origin-exclusion minorant derivable from R3 itself, and the source-only  $E''(0)$  pin  $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi$ ); census fully real 41/41 at  $x = 13$ , all six trace rows CONVERGE, world controls separate at median  $1.26 \times 10^6$ – $1.40 \times 10^6$  vs the pre-frozen bar 5, honest price  $\lambda_{\min} = 2.5 \times 10^{-54}$  at  $x = 13$  (internal precision  $\sim e^{-4\pi x}$ ); CORRECTED zero tracking (F1, stronger than first recorded): first-zero deviation  $-8.4 \times 10^{-16}$  at  $x = 8$  AND 13 (exactly the cache's own float64 quantization of  $\gamma_1$ ,  $+8.407 \times 10^{-16}$  vs the 30-digit literature value), mean displacement over all 15 matched zeros 0.0 at float64 resolution; CORRECTED  $\varepsilon$  finals (F6): up to 117% under the exact census ( $2.071 \times 10^{-8} \rightarrow 9.550 \times 10^{-9}$ ;  $3.708 \times 10^{-9} \rightarrow 6.290 \times 10^{-9}$ ; worst decay exponent  $-4.47 \rightarrow -4.96$ ), all six rows

staying CONVERGES; lane two hits 14.1347/21.0220/25.0109 at  $x = 8$  source-only and proves R3 ALONE implies Weil positivity ( $h = |\hat{g}|^2$  on a real spectrum) — full RH strength; lane three PROVES the Nyquist density obstruction (spec density  $\ell/2\pi$  exact and mesh-independent, rel dev  $2.22 \times 10^{-16}$  over a  $16 \times$  mesh sweep, vs  $\Xi$ 's log density — R3 AND R4 fail for every mesh at fixed window), unweighted trace diverges 4/4 to  $x = 144$ , weighted converges (honest finals  $9.3 \times 10^{-4}/3.3 \times 10^{-6}/2.3 \times 10^{-6}/6.2 \times 10^{-4}$ ; F4) but the weights are the Christoffel function of the zero measure — the unknown; honest node-vs-zero pairing at  $x = 55$ : 18-vs-3 one-sided, 36-vs-6 two-sided (F2); the three-lane conflict deliberately frozen unresolved; (2) ROUND 84 (note CCCLXXXIII, `realroot_arbiter_probe.py`, 17/17, 933 s, SPEC fc8a3a2b, independently re-run bit-identical, verdict REALROOT-DEPTH-UNDECIDED): the conflict resolved BY MECHANISM — different operator family, not different sums, not a bug, not primarily depth; all three lanes summed honest unweighted sums against the identically normalized source side; a new deep rung  $x = 21$  built at the honest price ( $K = 80$ , 128 dps  $\geq$  the declared floor 119.6,  $\lambda_{\min} = 1.25 \times 10^{-93}$ ): the mesh-CF  $(0, 30)$  count grows  $7 \rightarrow 22$  over  $x = 5..144$  tracking the uniform Nyquist prediction (measured deviations 10.2/10.6/10.9% at  $x = 13/21/34$  vs the frozen 30% tolerance; F3), the extremal/quotient family stays EXACTLY on the  $\Xi$  census at every measured depth —  $(0, 30) = 3$ ,  $(0, 60) = 13$ ,  $(0, 100) = 29$  at  $x = 21$ , minimum zero  $= \gamma_1$  to  $10^{-4}$ , all 50 Nyquist-excess zeros exiled to  $(100, 163]$ ; one unified table: mesh DIVERGES 4/4, extremal CONVERGES 4/4 on the same  $h$ ; NOT false-for-family, NOT quotient-escapes (the thinning is MEASURED, not proven; Connes §6.6 open); the Z1-TRANSFER identity names why a proof is the wall itself: the trig-Galerkin matrix IS the Gram matrix of zero evaluations,  $M[j, k] = 2 \sum_{\gamma} B_j(\gamma) B_k(\gamma)$ ,  $\lambda_{\min} = 2 \sum_{\gamma} E(\gamma)^2$  ( $6.31 \times 10^{-5}$  after the analytic cache-tail correction, 92% model share) — source-only in INPUT, zero-measure in CONTENT; falsification rungs priced ( $x = 34$ :  $\sim 1.5$  h;  $x = 55$ :  $\sim 10$  h), NO finite rung can verify — the surviving lemma carries full RH strength; (3) ROUND 85 (note CCCLXXXIV, three lanes: `tp_sol_probe.py` 7/7 TPL-OBSTRUCTED, `tp_opus_probe.py` 32/32 TPL-REDUCED + TPL-DICHOTOMY-MEASURED + TPL-OBSTRUCTED, `tp_da_probe.py` 9/9 TPL-CIRCULAR): energy sign and zero census are LOCALLY INDEPENDENT observables (off-line quartets:  $\Delta Q = -4\delta^2 E'(t)^2 < 0$  yet Rouché preserves the count;  $\lambda_{\min}$  flips to  $-9.1 \times 10^{-17}$  at  $1.25 \delta^*$  while the  $(0, 30)$  count stays 3); the ARCH block IS the smooth zero measure exactly ( $1.1 \times 10^{-6}$ ) — the entire  $e^{-cx}$  positivity is supplied by the PRIME BLOCK ALONE, no density-input argument can prove TPL(i);  $\|\text{PRIME}\|/\text{gap} \sim 10^{4.36x}$  — any prolate comparison must be an EXACT cancellation; THE PARITY LEMMA: TPL(i) IS — no analysis, a finite combinatorial rewriting — the statement that the ground eigenvector of one explicit matrix carries a prescribed sign pattern, verified exactly on three rungs; THE COUNTING LAW: the Nyquist surplus peaks at  $D_{\max} = x - 3/8$ , the tracking band ends exactly at the peak, marginal cost per forced skip one fixed factor  $e^{-11}$ , constant to 11% over 23 DECADES; the measured census support simultaneously CONVICTED CIRCULAR (the recorded counts ARE the smooth RvM values to  $< 1$ : 29.0023 vs 29; the instrument transcribes any fed measure including the prime-free world; outrunning classical verification needs dps  $\sim 2.1 \times 10^{12}$ ; the only off-line-sensitive channel is the wall scalar, slope 2.03, independently confirming the  $-4\delta^2 E'^2$  law); one sentence: the NUMBER of sign changes is forced by counting, the arithmetic substance sits EXCLUSIVELY in the POSITIONS, which the prime block alone must supply; (4) ROUND 86 (note CCCLXXXV, `simple_principle_probe.py`, 22/22, SPEC ef1bc6dd, verdict

**SIMPLE-PRINCIPLE-IS-TRANSCRIPTION):** four readings tested against pre-registered bars — one-axiom **STRONG-REFUTED** (five normalizations, zero hits on  $\log \pi$ , gaps 0.226–1.341; the difference dictionary exact: a gauge orbit, empty as a source), transcription **CONFIRMED** 3/3 (the register’s Mertens head is **EXACTLY**  $\prod_{p|210}(1-1/p) = 48/210$ , i.e. 2.1293 bit =  $\log_2(35/8)$  — the compiler pair itself; the  $E''(0)$  pin is the Hadamard identity to  $2.4 \times 10^{-8}$ ), fixed-point **REFUTED** (back-distance exactly  $1.0000 \delta$  — every admissible measure is a fixed point), Gram-seat square law  $\delta_* = \sqrt{\text{margin/curvature}}$  numerically **EXACT** on the Gram channel (ratio 0.9997, slope 2.0000; order split: position first-order at slope 0.9669, reality second-order) typed **PATTERN-ONLY** by the frozen **FOLDCOV** rule; the surviving principle: **TFPT’s positivity is always GRAM POSITIVITY OF NODE EVALUATIONS** — a theorem exactly where the nodes are compiler data, the arithmetic seam the single place whose nodes are not; two corollaries sharpen the **OCT** without moving it (the order split mechanically explains all 24 dead sign-source candidates; the  $\sim 1\%$  sub-spacing price is instrument-dependent,  $h^{-1.69}$  along the deployed ladder); (5) **ROUND 87** (note **CCCLXXXVI**, `bughunt_r79_r85_probe.py`, 24/24, 230 s, **SPEC 9346a32b**, verdict **BUGHUNT-FINDINGS(9, max MAJOR)**): adversarial audit of rounds 79–85 — **NINE** findings, 3 **MAJOR** / 6 **MINOR** / 0 **FATAL**, **NO** round verdict flips, every finding pinned by an executable gate; the three majors are the **F1/F2/F6** corrections carried above (the honest zero-tracking claim **STRONGER** than published); one systemic minor: **NINE** gates across the corpus hardcoded `check(..., True)` — never able to fail while counted in the  $N/N$  tallies, including the very line recording the round-80 “candidate #19 gate **FAIL**” verdict (the verdict right, the gate testing nothing); checked clean: every round-79–82 headline number reproduced on fresh re-runs, all ten probes re-ran green, no firewall violations; (6) **ROUND 88** (note **CCCLXXXVII**, `signpos_probe.py`, 19/19, 11.5 s, **SPEC b0003dcc**, independently re-run bit-identical, verdict **SIGNPOS-PREDICTOR-EXACT + SIGNPOS-TP-DEAD + SIGNPOS-RIGID + SIGNPOS-K-STABLE**): the campaign’s most concrete endpoint — a **SOURCE-ONLY** predictor of the sign-change positions (the parity of  $\text{round}(\theta(\omega_k)/\pi + 1 + S_x(\omega_k))$  at the Nyquist lattice,  $S_x$  the band-limited phase built from the prime-node comb) is **EXACT** at **ALL** 79 lattice points across  $x = 3/5/8/13$ , with measured first-order arithmetic content: the density alone fails at exactly **FOUR** lattice points and the prime-node comb repairs precisely those four — **ARITHMETIC LOAD 4**, the first arithmetically computable first-order-channel object of the campaign; scramble control fails 6/11 with  $\lambda_{\min} = -1.11$ ; **THE CATCH** named precisely: exactness for all  $x$  is half-integer-margin control of the band-limited  $S$  at the lattice points (measured minimum margin shrinking to 0.0317) — verbatim **TPL(i)** itself, the **RH** hardness relocated for the **THIRD** time to the same seat, now as a rounding-margin statement about one explicit band-limited function of the primes; Gantmacher–Krein **STRUCTURALLY DEAD** ( $\lambda_{\min}(\text{ARCH}) = -2.666$ ;  $M$  neither totally nonnegative nor Green’s/one-pair in any tested ordering; the minor census localizes nothing); the position map **RIGID** (modulus = the ordinate-to-lattice distance); the positions **K-STABLE** in the deployed band (the round-85 caveat discharged for the positions, the joint  $(x, K)$  declaration remaining for the margins); (7) **SURFACES SYNCED IN THIS ENTRY**: `tfpt_prime_front` gains the sandbox diary subsection “The real-root architecture, the Parity Lemma, and the sign-position predictor”; the end-form article gains §8.10 with the abstract status sentence, limitation 9 (the bughunt corrections and the hardcoded-gate finding as a methodology limitation) and

the traceability rounds CCCLXXXII–CCCLXXXVII; the Anthropic email draft gains one status paragraph; the website prime-front page gains chapter 34 (the live-updates chapter renumbered  $34 \rightarrow 35$ ) and one consolidated sandbox feed entry (verdict SIGNPOS-PREDICTOR-EXACT); website/lib/changelog.ts regenerated via `bash build.sh gen`; NO RH claim anywhere)

2026-08-14 (XCVI) (the kill-atlas consolidation arc — discovery freeze rounds seventy-nine through eighty-two synced into every live surface: the RH implication DAG, the 50-route kill atlas, the SAT adjudication, the untested-five adjudication, the bird’s-eye shape/direction/bootstrap round and the OBSTRUCTION COMPLETENESS THEOREM; experiments/ only — FOUR frozen probes, NO promotion, NO module, NO ledger row, NO marker moved, NO gate closed or narrowed, suite unchanged at 906, NO RH claim in any direction: (1) ROUND 79 (note CCCLXXVI, `kill_atlas_dag_probe.py`, 43/43 gates after the N12/N13 self-correction of entry XCV, verdict ATLAS-DAG-CONSISTENT + ONE-OPEN-EDGE-CONFIRMED + FOUR-CLASSICAL-CITATIONS-CARRIED + NO-SIGN-SOURCE-PASSES + LOCALIZATION-DID-NOT-MOVE-THE-EDGE): the RH implication chain as an explicit 11-node, 10-edge DAG, acyclic on the conjunctive hypergraph, typed 3 PROVEN / 2 CERTIFIED-FINITE / 4 CITED-CLASSICAL / 1 OPEN with EXACTLY ONE open edge E4 (the predefined mesh-cofinal ladder positivity); the honest non-PROVEN list E1/E2/E8/E4/E7/E9/E10 carried as an executable gate; the 50-route atlas assigns every killed route to exactly one edge (17 E4 / 21 E2 / 8 E10 / 2 E1 / 2 off-chain), every verdict token grep-verified inside its own artifact;  $\tau$ -screen 15 DISGUISE / 1 ADVANCE / 34 not-applicable with the single ADVANCE pinned NOT on E4; six coordinate systems ( $\tau_h > 0$ ,  $n_h - q_h > 0$ ,  $\sigma_h < 1$ , the anti-cancellation scalar, the  $P_{\text{err}}$  budget, v913’s best-conditioned (L)) pinned as ONE and the same edge — the localization campaign sharpened the edge’s type, closed the second open edge (v912) and corrected its statement (mesh order), and did NOT move it; (2) ROUND 80 (note CCCLXXVIII, `sat_projection_alignment_probe.py`, 35/35, SPEC 1b8b6f8e, verdict SAT-DISGUISE + SAT-LITTLEWOOD-EMPTIED + SELECTION-SIGN-MINED): the externally proposed signed-projection closure of E4 is adjudicated dead as submitted — the projection identity is EXACT (zero residual in exact Fractions) and EMPTY ( $A_2$ ’s column block unit-triangular with  $\det = 1$  as exact integers, so  $P = I$  and  $\|(I - P)y\|^2 \equiv 0$  identically; the ellipse inequality is  $s \geq 0$  VERBATIM; the “optimal signed profile”  $w^*$  is  $B$ -independent, zero arithmetic content); the  $\theta$ -average is verbatim v913’s (L) with its standing kills (HH-CLAIMS-WITHDRAWN, CLASSICAL-GAP  $2.470 \times 10^{11} / 1.990 \times 10^{14}$ ); the proposed  $\theta$ -selection is the kernel-checked sign-mined index (`signMinedIndex_not_familyNoninterfering`); the small-effective-rank premise is REFUTED (stable rank  $109.6 \rightarrow 1595.2$ ); candidate #19 FAILS the frozen gate (separates without independently orienting) — coordinate system number SEVEN for E4, gate tally  $18 \rightarrow 19$  candidates, 0 PASS / 14 FAIL / 5 UNTESTED; (3) ROUND 81 (note CCCLXXIX, `untested_sign_sources_probe.py`, 33/33, verdict SIGNSOURCES-ADJUDICATED): all five remaining UNTESTED candidate sign sources adjudicated FAIL with exact killing numbers — #20 G2-SEPARATES-NOT-ORIENTS (law exact at  $7.2 \times 10^{-16}$  on truth, defect energy ZERO on truth, class floors  $0.191 / -0.509 / -0.227$  below the need), #21 KREIN-CERT-IS-WALL (Douglas exact over  $\mathbb{Q}$ : contraction margin  $= 1 - \|C\|^2 = s/n$  VERBATIM), #22 U1-NAMED-TYPES-EMPTY (honest instrument on the full  $2 \times 10^7$  certified ordinate cache; smooth/jitter/shift surrogates violate by  $104\text{--}636\times$ ; the jitter

sweep prices the required ordinate precision at  $\sim 1\%$  OF THE MEAN ZERO SPACING; Weyl carries no rate, rate-carrying density is the emptied E2 class, pair correlation conditional), #23 U2-BEYOND-CLASSICAL (MIN-U2 named exactly,  $h$ -flat; a planted off-line pair violates it at  $\delta = 0.3635/0.2088/0.3533$  BELOW the classical zero-free widths  $0.4475/0.4443/0.4487$  — on the deployed windows it already excludes zeros no classical theorem excludes), #24 U5-ENUMERATED-DEAD (log-convexity violated by truth itself at  $-0.182$ ; downward-closure lemma: the zero comb is admissible in every enumerated class, so no upper-bound-type inequality can supply the required lower bound) — NEW TALLY: 24 candidates, 0 PASS, 19 FAIL, NO blanket UNTESTED left; (4) ROUND 82 (note CCCLXXX, `birdseye_shape_freedom_probe.py`, 35/35, 54 s, SPEC 8b78c201, verdict BIRDSEYE-OCT-CONSOLIDATED): (A) SHAPE-INDEPENDENCE-MEASURED — the deployed family is RH-saturating (margin-decay exponent  $-3.379$ ); five structurally different admissible families (Fejér, Gauss-tent, Hann,  $K \star K$  square, T5.2-style perturbation; Beurling–Selberg minorants typed INADMISSIBLE-BY-CAP) priced in the same Weil-square currency with an exact explicit-formula ward decay FAR shallower (exponents  $-0.46 \dots -1.09$ , min gap 2.29) — the maneuver exists but buys nothing: single-window ladders fail E7 density (54.5% projection residue), mesh-dense families inherit  $\lambda_{\min}(\Omega_{\text{EF}}) \leq s$  by the exact Schur bound ( $3.97 \times 10^{-7} \rightarrow 4.19 \times 10^{-12}$ ), and the localization is SHAPE-INVARIANT (square shapes degenerate the alignment, 15/15 position-blind worlds positive; off-line exclusion BELOW every classical zero-free width,  $\delta_{\min} = 0.055\text{--}0.26$  vs  $\sim 0.42$ , 5/5 beyond-classical; the deployed linear coordinate reproduces the  $\sim 1\%$  sub-spacing bar at  $107.1/4.36/0.21 \times$  margin for  $\varepsilon = 0.5/0.1/0.01$ ); (B1) EXISTENCE-IS-DISGUISE — the existence direction closed PERMANENTLY at the border-membership step in exact Fractions (a positive extension always exists;  $\det M = s \det B$  with PD  $\Leftrightarrow s \geq 0$  both ways; the Krein ellipsoid contains the arithmetic border with margin EXACTLY  $s$ ; the Carathéodory–Toeplitz one-step margin equals the next Schur scalar) — the gap between “an extension exists” and “the arithmetic one is it” IS E4; (B2) BOOTSTRAP-STILL-DEAD — the  $2 \times 10^7$ -ordinate cache certifies MIN-U2 at  $2/5$  cells and the  $\tau$  coordinate at  $0/5$ ;  $T_{\text{req}} \sim h^{4.77}(\tau) / h^{1.23}(\text{EF})$ , extrapolated  $\tau$ -coverage ends at  $h \approx 36$ , precision NOT binding, HEIGHT is; (C) THE OBSTRUCTION COMPLETENESS THEOREM, every premise machine-asserted: (i) exactly one open edge E4, (ii) the candidate map COMPLETE (24/0/19, no blanket UNTESTED), (iii) the missing input exactly characterized (signed, alignment-carrying, unconditional,  $\sim 1\%$  sub-spacing, minimal form MIN-U2 beyond-classical), (iv) shape-independent across the audited admissible families — plus the dual/existence direction closed and the bootstrap dead; SCOPE in the same breath: finite-rung (cells to  $h = 2854$ , ladder cited to 12632), atlas completeness EDITORIAL and typed as such, E7–E10 classical citations, NO-WITNESS counter-evidence carried unsmoothed (all certified reads positive to  $h = 12632$ , deepest  $2.79579794131272506 \times 10^{-15}$ , decay flattening) — a no-go about the TYPE of the missing input is not evidence about its EXISTENCE, and the theorem is NOT a statement that RH is unprovable; (5) SURFACES SYNCED IN THIS ENTRY: `tfpt_prime_front` gains the sandbox diary subsection “The kill atlas, the complete candidate map, and the Obstruction Completeness Theorem”; the end-form article gains §8.9 with the abstract status sentence, limitation 8 and the traceability rounds CCCLXXVI–CCCLXXX; the Anthropic email draft gains one status paragraph; the website prime-front page gains chapter 33 (the live-updates chapter renumbered  $33 \rightarrow 34$ ) and one consolidated sandbox feed

entry (verdict BIRDSEYE-OCT-CONSOLIDATED); website/lib/changelog.ts regenerated via `bash build.sh gen`; the stale `changelog.pdf / introduction.pdf` debt of entry XCV is cleared by the same rebuild; NO RH claim anywhere)

2026-08-14 (XCV) (editorial honesty correction, wording only — no number changed, no module touched, no status marker moved, no gate closed or narrowed, NO RH claim: the corpus-wide synthesis audit of note CCCLXXVI (discovery probe `kill_atlas_dag_probe.py`, 41/41, verdicts ATLAS-DAG-CONSISTENT + ONE-OPEN-EDGE-CONFIRMED + FOUR-CLASSICAL-CITATIONS-CARRIED, experiments/ only) reconstructed the RH implication chain as an explicit 11-node, 10-edge DAG and recorded that the corpus headline “the chain is reduced to exactly one unproven inequality” counts INEQUALITIES correctly, and counts correctly what the programme itself must supply, but drops four classical citations: the honest non-PROVEN list is E1/E2/E8/E4/E7/E9/E10 — two CERTIFIED-FINITE heads whose SCOPE is finite, the ONE OPEN inequality E4, and FOUR CITED-CLASSICAL edges (E8 the Suzuki/W1 identification, E7 density plus the equi-Lipschitz/Dini continuity modulus, E9 the faithful cap, E10 Weil’s criterion) — (1) WHERE THE QUALIFIER WAS ALREADY CARRIED, hence untouched: `tfpt_prime_front` states “theorem-grade modulo the named citations” at both extraction-chain sites and names Weil 1952, Bombieri 2000, Suzuki and Iwaniec–Kowalski Thm 5.12; the ledger row `PRIME.FORM.CONVERGENCE.THEOREM.01` types the W1 dictionary, Weil’s explicit formula and criterion, the admissible BV class and the DENSITY leg as CITED and adds that (H\_cof) is untouched and RH-hard; `changelog XCIII` carries the same pinning disclosure; the `v912` docstring says verbatim that edge 6 is NARROWED, not closed; and the website mirror repeats the qualifier in the prime-front feed and the papers index; (2) WHAT IS CORRECTED: the two article surfaces that stated the reduction without it — `articles/2026-08-11/paper_endform_and_inertia_bridge_en.md` (abstract, §8.6, §8.7, §9) and `articles/2026-08-11/email_anthropic_draft_en.md` (status paragraph and ask 3) — each gain ONE clause naming the four classical legs beside the one open inequality; (3) THE AUDIT’S FOUR OTHER FLAGGED ITEMS, CHECKED AGAINST EVERY LIVE SURFACE AND NOT LIVE, hence no edit: NOGO-COMPOSITE-VERIFIED occurs only as the stated provenance of the promoted NOGO-CORE-VERIFIED core, in a ledger row titled verbatim “A NO-GO / TYPING ROW AND EXPLICITLY NOT PROGRESS” and in a module docstring that says it is NOT progress toward a proof; WIRING-DEGENERATE and THETA-CONVENTIONAL occur only in the seam/compiler lane (`SEAM.STATE.WIRING.SELECTOR.01`, `v911`) and are listed among no RH-chain route kills; the superseded matrix-stage conditioning exponents and the withdrawn Li–Keiper Vinogradov–Korobov envelope exist only in experiments/ and never entered a paper body, a ledger row or the public mirror; NO RH claim anywhere)

2026-08-14 (XCIV) (promotion round six — THE SIGNED-AND-ALIGNMENT LOCALIZATION, a NO-GO/TYPING module that is explicitly NOT progress, plus the mesh-order wording debt closed at its last two sites: ONE module (`v913_signed_alignment_localization.py`, 24/24, verdict NOGO-CORE-VERIFIED, ~28s; suite 905 → 906; ledger 1056 → 1057 with the new row `PRIME.NOGO.SIGNED.ONLY.01` typed [O]/no\_go\_typing; discovery probe `signed_only_nogo_probe.py`, note CCCLXXIII, 87/87, SPEC 49530809, PROBE\_SHA d1c42a05) — (1) ITEM ONE, THE MESH-ORDER DEBT (wording only, no number changes, no marker move): note CCCLXX had logged two sites skipped during the

CCCLXIX correction pass; v848\_extraction\_chain.py stated the chain as “the canonical cofinal ladder” and “CONCLUSION: cofinal finite positivity (H) on the canonical ladder” in BOTH its docstring and the text it PRINTS at runtime, and run\_all.py carried the same wording in its registry line — all of them now say cofinal IN THE MESH-REFINEMENT ORDER  $D_j = 2^{-j}$ , because at a FIXED mesh  $D_0$  the read is exactly cap-independent, so a ladder cofinal only in the window/cap parameter is eventually constant and converges to  $W_C[K] + W_C[e_{D_0}]$ , giving the FALSE FLOOR  $Q_W \geq -|W_C[e_{D_0}]|$  (measured  $-2.114 \times 10^{-3}$  at  $D_0 = 1/32$ ,  $-2.128 \times 10^{-4}$  at  $D_0 = 1/128$ ); where the module explains why cofinality suffices it now carries the one-line reason in its proven form — were the limit negative, the per-element convergence of v912 would make the catch set  $\{j : Q_j(f) < 0\}$  a TAIL of the mesh order, and a cofinal set meets every tail; module re-run unchanged at 19/19 and re-mirrored byte-identically into website/public/verification by the build step; (2) ITEM TWO, WHAT THE NEW MODULE SAYS: any argument that closes the deployed budget inequality for the DEPLOYED BUDGET SHAPE must be (a) SIGNED — it must orient a quantity that is ODD under the comb sign flip — and (b) ALIGNMENT-CARRYING — it must relate the position of the prime atoms to the spectral directions of  $K$ ; in particular it can be NEITHER a magnitude bound on  $\psi(x) - x$  NOR a natural-grammar identity valid in every arithmetic world and therefore comb-blind; (3) THE PROMOTED CORE, exact and self-contained: the six kernel identities in exact arithmetic ( $\int K_D = 0$ ;  $\|K_D\|_\infty = 1$ ;  $\|K_D\|_1 = 4D/3$ ;  $D^{-1} \int K_D^2 = 2/3$ ;  $\int K_D e^{-t/2} = -(D/8)T_D^2$ ; and the load-bearing  $\|K'_D\|_1 = 4$  INDEPENDENT of  $D$  — the conversion constant of a magnitude hypothesis does not get cheap as the mesh refines), the geometric supply recomputed by TWO independent routes (closed form vs. direct quadrature, deviation  $1.6 \times 10^{-50} / 2.5 \times 10^{-49}$ ;  $B = -1.2648057425013541016$  at  $h = 184$  and  $-1.3147508968858830811$  at  $h = 388$ , negative and  $O(1)$ ), the Littlewood floor  $4 \log \log \log N_h$  (2.1654551 at  $N = 263$ , 4.1231791 at  $N = 14634429$ ) against a slack that shrinks with depth — ratio 214.7 on the module’s OWN conservative slack and 1527.1 on the deployed one, so the ENTIRE MAGNITUDE CLASS IS EMPTY; the provable neutrality of Selberg’s symmetry formula ( $\Lambda = \mu * \log$  and  $\mu * \log^2 = \Lambda \log + \Lambda * \Lambda$  are formal identities in the free module on  $\{\log p\}$ , information gain exactly 1.000); the exact three-term split with its congruence invariance and comb-flip parity (all three terms invariant under EVERY  $\text{diag}(1, M)$ , outer terms EVEN and the carrying term ODD, the  $F_4$  kernel  $A_2^T B^{-1} A_2$  itself a congruence invariant — so no unsigned, flip-invariant hull can bound the carrier below); and the Beurling–Nyman/Baez-Duarte target exactly in  $\mathbb{Q}$  ( $\lambda_{\min}(G_N) \leq (\log 2\pi - \gamma)/N$ : cap 0.420220 against the needed  $c_B = 0.5523$ , so there is NO  $N$ -uniform Gram floor and the deployed hypothesis is FALSE for  $N \geq 3$ , plus the explicit non-implying family  $\tilde{G} = AA^T + (4/5)^2 I$  with  $\tilde{d}^2 \geq 1/2$  forever); (4) THE BOUNDARY, RESPECTED EXACTLY: the deployed matrix-stage reads (S3.5–S3.12, S4, S5) STAY in experiments/ because they inherit the float64 entry-slack premise and the cited ladder; PINNING DISCLOSURE — PINNED here are the identities, the split algebra, Selberg neutrality, the Beurling–Nyman witnesses, the supply recomputation, the module’s own prime-power sieve, the frame rebuild with  $h$  PREDICTED (184..12632), the square-root-barrier placement and the frozen probe SPEC-SHA reproduced BEFORE any evaluation; CITED and NOT pinned are the deployed ladder and its certified reads (X1), the deployed entry slack (X2), the classical constants and Littlewood’s  $\Omega$ -theorem itself (X3), the float64 entry slack of the wall generators (X4), the CCCLXVIII demand sums (X5), the CCCLXII/CCCLXIII chain measurements and deployed constants

(X6) and the  $n_0$ -normalised margin exponents (X7); (5) SCOPE, STATED IN THE SAME BREATH: TEN classes are proven empty for this budget shape (E1–E10), FIVE are merely unexplored with NO emptiness claimed (U1–U5) and FIVE clauses are named open (O1–O5, among them that nothing is said against arguments legitimately consuming zero positions and that the reverse implication rests on four unestablished premises); the theorem is FINITE-RUNG ( $h = 184/388/839$  with the ladder to  $h = 12632$ ) and makes NO all- $h$  claim; it does NOT say that the Riemann Hypothesis is unprovable, that the wall route fails, or that no proof exists; (6) COUNTER-EVIDENCE CARRIED, NOT SMOOTHED: NO-WITNESS STANDS — the certified reads are POSITIVE at all NINE depths (0 negatives, 0 straddles) with deepest certified read  $2.79579794131272506 \times 10^{-15}$  at  $h = 12632$ ; the  $n_0$ -normalised decay FLATTENS with depth ( $-3.21092511$  globally against  $-2.67942161$  on the deepest step  $5746 \rightarrow 12632$ ) and the re-derived  $s/D^2$  step  $1393 \rightarrow 2015$  is POSITIVE ( $+0.50302892$ ), so the normalised drift is NOT monotone, and no re-derivable exponent comes near the frozen collapse bar  $-8.0$ ; a no-go about the TYPE of the missing input is not evidence about its EXISTENCE; (7) SURFACES: `tfpt_prime_front` gains the residual-obstruction subsection “The remaining obstruction is signed and alignment-carrying (localization)” with the probe’s own S9 wording, Paper 3 a new audit card with the `\veri` citation, the end-form article and the Anthropic email draft one honest paragraph each, and the website mirror is synced; the row supplies no positivity, no interval and no certificate, closes no gate, narrows no gate and moves NO marker anywhere — it must not be counted as progress nor as evidence for or against the Riemann Hypothesis; NO RH claim anywhere)

2026-08-13 (XCIII) (promotion round five — THE FORM-CONVERGENCE THEOREM: the extraction chain’s per-element convergence premise stops being a measurement and becomes an unconditional theorem, and a false cited premise it uncovered is corrected across the corpus: ONE module (`v912_form_convergence_theorem.py`, 35/35, verdict FORMCONV-PROVEN,  $\sim 12$ s; suite  $904 \rightarrow 905$ ; ledger  $1055 \rightarrow 1056$  with the new row PRIME.FORM.CONVERGENCE.THEOREM.01; discovery probe `form_convergence_theorem_probe.py`, note CCCLXV, SPEC `d911226f`, promoted VERBATIM with no downscoping)

— (1) THE THEOREM: for  $f$  real, supported in the deployed window and AFFINE ON EVERY GRID CELL, the finite Galerkin form converges to the Weil functional,  $Q_D(f) \rightarrow Q_W(f)$ , UNCONDITIONALLY and with the explicit envelope  $|Q_D(f) - Q_W(f)| \leq D^2[(8 \sinh(B/2) + \mathfrak{m}(B))A_2 + \Theta_0 A_0 + (A_0 + A_2)(\log(1/D) + \log B + 4) + 2A_0 e^{-B/2}(1/B + 2) + A_1(1 + D)] + (1 + D)(D^3/12)\kappa_3$ , i.e. rate  $O(D^2 \log(1/D)) = O(2^{-2j}j)$ , with  $\Theta_0 = 3 \log 2 + \pi/2 + \gamma + \log \pi = 5.372183419225665$  and, for continuous  $f$ , log coefficient exactly  $7/24$  (sup  $5/24$ ) — replacing the corpus typing MEASURED (rates  $-1.58..-1.84$  per level, `v848`) that meta-audit CCCXXXVII Q-2 had flagged as a measured premise read as settled; (2) THE MECHANISM, three exact steps: the deployed lag vector IS the capped Weil functional on the even tent ( $c_d = W_C[S_{dD,D}]$ ), so the finite form equals  $W_C[\widetilde{K}_D]$  EXACTLY on the piecewise-linear interpolant of  $k_d = Da_d$ ; the EXACT DEFECT LEMMA  $k_d - K(dD) = -(D^2/12)G(dD)$  holds for every lag because  $u \mapsto f(u)f(u + dD)$  is QUADRATIC on each cell and the midpoint error of a quadratic is exactly  $-(D^3/24)q''$  — ZERO deviation in exact rationals over 1474 lag identities and 780 strictly interior points; and the faithful cap  $C \geq b_K + D$  turns the error into  $W_C$  applied to ONE explicit function, bounded layer by layer with elementary majorants; the interpolation constant is the interior



sup of  $|K''|$ , NOT  $\|f'\|_2^2$  (equal for continuous  $f$ , different for jump elements: E09  $3 > 2$ , E10  $5/3 < 2$ ); sharpness is measured against a CLOSED-FORM leading term (ratio 0.575..1.066, drifting to 1 through  $j = 18$  — the law is genuine, not a loose envelope), and uniformity (V) holds on every TWO-SIDED support window, not on an  $L^2$  ball; (3) THE CORRECTED PREMISE (a correction of a wrong cited statement, NOT a downgrade of a result): the chain's cited " $C^0$ -continuity of  $Q_W$  at fixed support" is FALSE in the pure sup norm — must-fire control C5 exhibits the explicit even Lipschitz family  $e_n(w) = (1/n) \min(1, w/e^{-n^2})(1 - w/2)_+$  with  $e_n(0) = 0$  and  $\|e_n\|_\infty = 1/n \rightarrow 0$  while  $|A[e_n]| = 2.57, 4.28, 6.19, 8.14, 10.11, 12.09$  grows linearly — and the true hypothesis is uniform convergence PLUS an equi-Lipschitz (Dini) condition at the origin, which the admissible even compactly supported BV class supplies automatically (T5.2 verifies the corrected modulus of  $W_C$  on a declared 12-perturbation battery, worst true/bound 0.0226); the false premise is corrected wherever it was cited: `tfpt_prime_front`, `tfpt_research_contracts`, the `v848` module docstring and its website mirror, `run_all.py`, the ledger row `PRIME.EXTRACTION.CHAIN.01` and the Lean prose of `TfptCarrier/CofinalWeil.lean`; (4) CLASSICAL INPUTS, all 11 typed (T4.3): NO zero-free region, NO PNT, NO zeta zeros, NO fit — the only cited arithmetic input is Rosser–Schoenfeld 1962 Thm 12,  $\psi(x) < 1.03883x$  for all  $x > 0$ , consumed ONLY for the support-uniform constant and independently sieve-verified in-run to  $10^7$  (worst ratio 1.038821); PINNING DISCLOSURE (the CCCXXXVI lesson applied a priori): the `W1` measure dictionary, Weil's explicit formula and criterion, the admissible class with its DENSITY leg, the all- $x$  Rosser–Schoenfeld statement and the UNIVERSAL QUANTIFIER itself are CITED, not pinned by these gates; (5) LEAN (DELTA-A, honestly partial): the new kernel-checked module `TfptCarrier/CofinalEnvelope.lean` states the envelope premise as a named structure `FormEnvelope` (per-element constants, a DELTA-B window level, the explicit bound), DERIVES the convergence hypothesis from it (`tendsto_of_formEnvelope`), and restates `cofinal_weil`, `cofinal_weil_for_fixed_idx` and `cofinal_weil_predefined` with `hconv` REPLACED by the instantiated envelope — plus a non-vacuity lock proving the envelope STRICTLY STRONGER than bare `Tendsto` and an inhabited witness with nonzero error at the envelope rate; axioms `propext/Classical.choice/Quot.sound` only, no sorry, no native\_decide; the envelope itself is NOT discharged inside the kernel, so DELTA-A is REDUCED, not closed; DELTA-B (stated, not a gap): (H-grid), (H-cap) and (H-align) hold for fixed  $f$  and all large  $j$ , which is exactly what `Tendsto atTop` needs; papers (`tfpt_3` new audit card, `tfpt_prime_front` the extraction-chain premise corrected and the theorem cited, `tfpt_research_contracts` the contract's cited hypothesis corrected) and the website mirror synced; the chain's genuinely open item is UNCHANGED — (H\_cof), the PREDEFINED cofinal positivity inequality, is untouched and RH-hard; no marker moves; NO RH claim anywhere)

2026-08-13 (XCII) (cofinal-index noninterference hardening — the PREDEFINED requirement is moved from an overclaim in prose into an explicit, honestly external Lean contract; no marker move, no RH claim: `TfptCarrier/CofinalPredefinition.lean` adds (1) `cofinal_weil_for_fixed_idx`, the extraction implication for an arbitrary explicit index with `StrictMono`, `PSD` and form-convergence hypotheses — binder order prevents the theorem body from selecting the index but does not prove caller provenance; (2) the negative theorem `old_api_accepts_sign_mined_idx`, exhibiting that

the former mathematical payload accepts a selector branching on measured signs; (3) `signMinedIndex_not_familyNoninterfering`, which rejects that exposed selector under extensional family-noninterference; (4) the hardened `PredefinedCofinalHypothesis/cofinal_weil_predefined` API, requiring an explicit abstract `NoninterferenceContract.Predefined(A,idx)` certificate; and (5) the constantization pair, proving the exact residual limitation — after construction, a selected value can be represented by a constant selector with the same output, so extensional dependent types cannot recover algorithmic history. The concrete source boundary excluding the matrix family and its sign outputs therefore remains the named external audit premise unless the construction language or an equivalent effect/provenance system is formalized. The old `CofinalWeil.lean` core is relabelled as mathematical payload only; dependent Lean-module comments, audit locks, the Lean/verification READMEs, `tfpt_prime_front`, Paper 3’s audit card, the end-form article, ledger row `FORM.PRIME.COFINAL.WEIL.01`, registry text and the public prime-front feed are synchronized. No verification result, status marker, finite positivity claim or RH claim changes.)

2026-08-13 (XCI) (website/video maintenance — the `/prime-front` page gains the sigma-chain diary chapter and the explainer film its regional-theorem coda; NO theory change, NO marker moves, NO RH claim: (1) THE PAGE — new diary chapter “32 — The sigma chain and the regional theorem (sandbox probes)” covering the experiments freeze rounds 73–75 plus the morning package round, ALL SANDBOX per the page’s promoted-vs-sandbox convention (v911 remains the newest promoted module, every counter untouched): the identity  $\sigma = 1 - s/n$  with the two 0.665s dismantled as a provenance-proven coincidence; the derived per-cell exact-rational chain certifying every built wall-legal cell (151/151 surface + 59/59 deep to  $h = 6344$ , worst bound 0.727,  $O(1)$  margins); the joint Radau relation lifting definiteness to class level ( $\inf \lambda_1 + 0.008$  vs  $-574$ ); the SOS dual certificate (1111 exact rational certificates, 151/151 at  $\eta = 0.273$ ) sealed as the three-part theorem package with the 202-line `stdlib-only` checker (the audit entry point — re-proves the theorem in 2 s, teeth 3/3); the level-3 endform (legal sub-ladder to  $h = 8204$  with the first measured termination signal, the sign living in the seat-to-bulk coupling; B3 Lipschitz-certified, B2 proved-conditional, B1 one measured scalar; the razor margin  $+0.0104$  typed an irreducible measurement); the closed no-gos (Pick DEAD, finite-gap ILLDEFINED, shells JCONTRACT-DEAD); and the honest closing map in its frozen form — a REGIONAL finite positivity theorem plus a single open cofinal construction problem, the quantifier is RH, NOT “almost-RH”; the live-updates section renumbered  $32 \rightarrow 33$ , one new diary-feed entry (badge sandbox, verdict PACKAGE-SEALED); (2) THE FILM — new closing frame 10 “Das regionale Theorem und das kofinale Problem” extends the explainer  $2:39.1 \rightarrow 3:05.1$  (26.0 s window at 159.12 s, crossfade in): the finite positivity question as a REGIONAL THEOREM (every matrix whose entry data lie in the certified region  $K$  is positive definite; on-screen chip “151/151 cells · SOS proof object · 202-line `stdlib` checker · re-proves in 2 s”), the single remaining CONSTRUCTION problem (a predefined window sequence, cofinal, staying inside the region — keyword glow on “cofinal”) and the honest close as two dry statement beats: “The proof object is ready for audit — the RH quantifier is not”; full manual rebuild per the documented README sequence (EN+DE script line 10 — 1:1 for the de subtitle track, TTS re-run with all 10 lines at rate  $+0\%$  inside their frozen windows — frames 1–9 unchanged, captions rebuilt from 433 words,

9 transitions verified, render at 5554 frames, en/de subtitle tracks 34 → 37 cues, last cue ends 3:04.12); WEBSITE SYNC — the four public assets re-shipped (mp4 + en/de VTT, poster byte-identical), PrimeFrontVideo.tsx chapters + coda marker at 2:39, the component transcript verified character-identical to the caption track, JSON-LD duration PT2M39S → PT3M5S with uploadDate 2026-08-13, section header “ $2\frac{1}{2}$  minutes” → “3 minutes”; nothing under experiments/, verification/, articles/ or the papers/ledger touched (the concurrent research lane owns those))

2026-08-12 (XC) (promotion round four — the consolidation of rounds sixty-four through seventy-one: the FINITE wall closure with its transfer law, and the wiring freedom theorem: THREE modules (v909–v911: finite wall closure, finite zero transfer, wiring freedom; suite 901 → 904; ledger 1052 → 1055 with the rows PRIME.WALL.FINITE\_CLOSURE.01, PRIME.WALL.FINITE\_ZERO\_TRANSFER.01 and SEAM.STATE.WIRING.SELECTOR.01 closed) — (1) THE FINITE WALL CLOSURE (v909, 28/28, PRIME.WALL.FINITE\_CLOSURE.01): on the deployed finite ladder every reachable wall face is certified from cited inputs and the composed census  $B \wedge W1 \wedge W2$  holds on 39/39 matched surface + 8/8 deep rungs (frozen experiments runs of rounds 64–67, cited with their spec hashes) — the  $B$ -half via the v905 interval-certified class floor ( $\min c_B = 0.5523$ ), the  $W1$  face via the monotone-composition measure inequality (the two +8..+9 dex composition gaps of rounds 55–59 DISSOLVE into one exact Loewner step, composition price +0.29 dex) plus VERIFIED ZEROS AS EXACT DATA at the named  $j = 16$  seat (the truncated explicit formula with 7000 mpmath ordinates, every read tested against the independent prime side, the tail bracketed by the unconditional Rosser–Abel envelope: 39/39 surface, composed 67/67 with the cited deep 28/28), and the  $W2$  face via the recomposed certificate  $m_{\text{cert}} = FC + \hat{P}_{\text{arith}} - \text{TAILB} > 0$  (16/67 at the 7000-ordinate cache, 67/67 + 8/8 once the  $2 \times 10^7$ -ordinate certified cache pays the priced tail; Odlyzko + LMFDB/Platt, every ordinate below the Platt–Trudgian height  $T_0 = 3 \times 10^{12}$ , EXTERNAL-CITED); the promoted module re-verifies the load-bearing spine SELF-CONTAINED on a reduced frozen subset with its own committed 7000-ordinate cache re-warded in-run (Rosser corridor per index both sides,  $\gamma_1$  identity, independent  $|\zeta|$  spot checks  $2.1 \times 10^{-12}$ ; heart ward on every read, slack +2.1..+4.8 dex; closure 6/6; the off-line-impostor control fires at  $8.2 \times 10^4 \times$ ) and cites the full census rather than recomputing it; LANGUAGE FROZEN:  $W1/W2$  are algorithmically independent evaluations of the SAME localized Weil form — a strong mutual crosscheck, NOT two independent proofs; positivity certified on a finite family of Galerkin sections along the MEASURED critical direction (direction-conditional, UNIF-PATH caveat); a finite verified-zero sum can never prove RH; (2) THE TRANSFER LAW (v910, 10/10, PRIME.WALL.FINITE\_ZERO\_TRANSFER.01, SPEC 40a7b5c3): the zero-supply economics of the finished wall —  $T_{\text{req}}(h) \sim h^{2.8}$  ( $\log_{10} T_{\text{req}} = -3.045 + 1.228 \ln h$ ,  $R^2$  0.882) and the ratio to the window’s own spectral reach  $\pi/D_h$  GROWS (+0.897 dex/ln  $h$ ): verdict RATIO-GROWING ⇒ EXTERNAL-BATTERY / W-UNBOUNDED-PER-RUNG — the finite engine does NOT scale by buying zeros; the parametrized transfer law  $\text{ZerosCertified}(T) \wedge \text{TailEnvelope}(T) \Rightarrow M_h \succ 0$  for all deployed  $h \leq H(T)$  with the measured envelope  $H(\gamma_{7000}/\gamma_{2M}/\gamma_{2 \times 10^7}) = 254/1256/2806$ ,  $W1$  cheap (binds 3/75), the next  $h$ -decade at  $T \sim 3.4 \times 10^9$ , and  $H(T)$  the measuring rod any analytic per-window bound must beat; the module re-verifies the shallow end (12 rungs  $h$  142–254, anchors  $K = 250..7000$ , closure census 2/3/4/6/8/12, zero non-monotone, heart ward on all

492 cells,  $H$ -consistency 12/12; one disclosed control-rule amendment, fail-first preserved: the jitter bar restored to the upstream frozen  $\geq 5$ -reads rule); (3) THE WIRING FREEDOM THEOREM (v911, 30/30 + 26/26 + 21/21, verdicts WIRING-DEGENERATE + THETA-CONVENTIONAL + RP-FRAME-COVARIANT; three frozen probes of rounds 69–71 embedded byte-exact, exact integer/rational arithmetic; SEAM.STATE.WIRING.SELECTOR.01 CLOSED — the v908-registered contract answered): the exact Gröbner census of the orbit/edge constraint classes leaves EXACTLY ONE admissible three-dimensional wiring component modulo the rule gauge (the rotation cell  $C_{\text{rot}}$ ); the  $Z/X$  wirings are excluded gauge-invariantly by orientation propagation ( $\det u < 0$  identically); the deployed pure- $i$  point is an INTERIOR point, connected to the pure- $J$  ray by the integer gauge element  $(\oplus_5 J_2) \oplus I_6$  — pure- $i$  IS A DEPLOYMENT REPRESENTATIVE, NOT A COMPILER THEOREM; no compiler-side demand distinguishes the  $\theta_S$  sheet-swap frame within the 9-dim rule-gauge algebra (five demand classes all silent on the angle), the strict-collar RP Grams transform EXACTLY covariantly under the rule gauge (RP of a (frame, wiring) pair depends only on the relative angle on the  $\delta$ -circle), and the same deployed parent that fails strict collar RP at  $\theta_S$  (defect  $2t$ ) PASSES it exactly in the integer-conjugated frame: the rounds-58/59 obstruction is a FRAME statement, not a wiring no-go; the canonical sentence is frozen (sha 79950314) and “pure- $i$  is compiler-forced” is henceforth FORBIDDEN (equivalents without the frame qualifier included); the v898/v903 [O] premise unmoved; NEW COMMITTED DATA ARTIFACT: verification/verified\_zeros\_n7000.json (7000 mpmath dps-15 ordinates with disclosed in-module rebuild path and in-run wards); papers (tfpt\_prime\_front new subsection “The finite wall closure and the transfer law” + the rounds-60–63 open-fronts forward link, tfpt\_research\_contracts the wiring-selector block OPEN→CLOSED + the rounds-64–71 bridge + the frame qualifier on the v908 block, tfpt\_1 Wick keybox dated addendum rounds 69–71, tfpt\_3 four audit cards updated/added) and website mirrors synced (prime-front feed round entry, DAG, counters 901 → 904 / v535–v911, papers.ts, README, Zenodo, the reproducer now ships the zero cache); NO marker moves; NO RH claim anywhere)

2026-08-11 (LXXXIX) (reflux rounds sixty through sixty-three — the certificate round: the B-half of the wall end-form CLOSES as a certified surface theorem over the IDEAL source objects, the tail gets a mechanism map of five dead routes with named seats and two structural positives, the half-gap target is REGISTERED with a frozen constant and passes its FIRST BLIND HOLDOUT, and the round-60 seam questions close in equilibrium — no NESS needed: FOUR modules (v905–v908: B-floor ideal certificate, tail cartography, half-gap registered target, seam equilibrium wiring; suite 897 → 901; TWELVE frozen probes with all fail-first spec amendments preserved, embedded byte-exact and executed verbatim, plus two round-61 frame probes cited at docstring level; ledger 1040 → 1052 with 12 promoted rows + dated notes on the v903 ness-parent demand and the rounds-57–59 certifiable-B-floor demand) — (1) THE CERTIFIED IDEAL-OBJECT SURFACE FLOOR (v905, 21/21 + 25/25, bfloorpg-measured (certified-surface-floor-achieved(min  $c_B=0.5914$ )) + bfloorpgival-measured (ival-surface-floor-achieved(min  $c_B=0.5523$ , 0 refused / 0 failed / 0 skipped)), PRIME.PORT.BFLOOR.PG.01 + PRIME.PORT.BFLOOR.PG.IVAL.01): the chain  $B \succeq \frac{1}{2}P_G + c_{\text{dom}}I$  and  $P_G \succeq c_G I$ , every inequality decided by EXACT-RATIONAL LDL (the v897 certificate class) on all 39/39 reachable steps —  $P_G$  is Gershgorin-POSITIVE (the positive chain is classically certifiable; the single

non-classical link is the dominance decision; after three dead certificate rounds the missing piece was the CERTIFICATE CLASS, not a new congruence) — and, via the SPEC-v3 interval rollout (full pipeline as rigorous outward enclosures, fail-closed with first-class refusals, Ogita–Rump error-free-transformation hi-tier certifying exactly the six former width-refusals), over the IDEAL source objects:  $\min c_B = 0.5523$  on 39/39 (93.4% of the float reference),  $\tau$ -screen PASS ( $-0.226$ ), the smooth world REFUSED by the identical exact machine (0/35); HONEST SCOPE typed everywhere: a SURFACE statement —  $h$ -uniformity beyond  $h \sim 900$  open, all  $h$  beyond the surface open, the  $n > q$  half untouched and RH-hard; (2) THE TAIL MECHANISM MAP (v906, 14/14 + 10/10 + 20/20 + 22/22 + 24/24 + 22/22 from SIX frozen probes, tailsign-measured + tailrepair-measured + pgschur-measured + tailabel-measured + segsplit-measured + oscpair-measured, PRIME.PORT.TAILSIGN.01/.02 + PRIME.PORT.PG.SCHUR.01 + PRIME.PORT.TAIL.ABEL.01 + PRIME.PORT.TAIL.SEGSPLIT.01 + PRIME.PORT.TAIL.OSCPAIR.01; typed explicitly mechanism/negative cartography, NOT progress): FIVE dead routes with named seats (the tail sign is NET-ONLY —  $\sup q_v$  positive beyond every cut, the terminal lobe positive 67/67; the pointwise repair is SET-DIFFUSE and grows as  $X^{0.89}$  with an anti- $\tau$  head; the directional Löwner route loses TWO orders of magnitude by the floor-vs-inverse mechanism ( $q \leq \bar{q}$  exact 39/39 but  $n > \bar{q}$  dies 38/39, loss med 91.3); the cumulative/Abel route is exact ( $\psi$ /Riesz identities warded) but its honest integer envelope leaves the gap law  $e^{+1.744\alpha}$  — the measured hole is the OSCILLATION, the true remainder itself  $\sim 10^4 \mu_1$ ; the segment split is COSTLIER than global (gain  $< 1$  on 67/67, the ANCHOR PRICE; the named missing object: a DIFFERENCE envelope for  $\psi(y) - \psi(x) - (y - x)$  on short deep windows, Brun–Titchmarsh/Selberg class)) and TWO structural positives (headfloor-o1: the entire  $h$ -decay of the wall margin lives in the atom tail, the open statement collapses to the one-sided bound  $\text{tail}_B > -\text{head}_B$  between two  $O(1)$  objects; the 12-BIN REDUCTION: a tiny low-frequency window-harmonic carrier census, measurably NOT  $\gamma$ -concentrated at these depths, reduces the tail bound to  $\sim 12$  windowed  $\psi$ -fluctuation Fourier coefficients — honestly typed: the reduced margin carries the FULL decay, dimension reduced, difficulty not); (3) THE REGISTERED HALF-GAP TARGET + THE FIRST BLIND HOLDOUT (v907, 17/17 + 14/14, halfgap-registered (reg-frozen(1/2, no-adjust) / surface-pass(67/67, min margin +2.48e-03) / origin-open(4)) + deepholdout-scored (halfgap-holdout-pass(28/28, min margin +0.2232) / law-continues( $-1.925$ )), PRIME.PORT.HALFGAP.01 + PRIME.PORT.DEEP.HOLDOUT.01; ledger-typed a FALSIFICATION INSTRUMENT, explicitly NOT evidence for RH):  $n_h - q_h \geq \frac{1}{2} \mu_1(h)$  registered on the 67-rung surface with the constant EXACTLY  $\frac{1}{2}$  frozen under an explicit NO-ADJUST clause that is part of the object (registry sha ae292e55; four candidate origins of the  $\frac{1}{2}$  typed OPEN — an algebraic derivation is demanded before any claim upgrade; reparam-declared: the inequality is the wall margin reparametrized, its value is falsifiability, not novelty); the first genuine blind holdout PASSES on 28 new deeper faithful rungs ( $h$  1219–2854, bit-exact table fidelity,  $\sim 90\times$  margin headroom, NO depth erosion, the honest cut-ladder drift disclosed; B-floor/ $P_G$  persist at declared float level; the margin law  $-1.925$  continues over 95 rungs; controls discriminate 3/3) — a pass is survival of a falsification attempt, never evidence; (4) THE SEAM EQUILIBRIUM WIRING (v908, 22/22 + 25/25, exact arithmetic, margstab-measured (interior-empty / stratum-not-isolated / escape-noncovariant(price  $2\varepsilon$ ) / floor-exchange) + nessparent-measured (price-zero / two-seat-law / finite-ness-nogo / drive-rp-neutral / ness-not-forced), SEAM.CFIN.MARGINAL.STABILITY.01

+ SEAM.CFIN.NESS.PARENT.01 closed NOT-NEEDED): strict 2-cycle RP is IM-POSSIBLE on the whole  $C_6$ -covariant class (the seat law reduces exactly to  $\text{diag}(a_J, -a_J)$ , interior EMPTY; the only escape is covariance-breaking at exact linear price  $2\varepsilon$ , and the floor exchange  $\text{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$  puts the crossover exactly at the  $1/200$  floor); the strict-collar obstruction is a TWO-SEAT LINEAR LAW with kernel  $\{J, Z\}$ , the deployed wiring is pure-i = a maximally obstructed covariant direction (explaining the v903 ‘30 entries, magnitude  $2t$ ’ law exactly), and EQUILIBRIUM kernel witnesses  $V_J/V_Z$  carry the full canonical  $1/200$  mixing at ZERO entropy production (stationary  $\Rightarrow \sigma \equiv 0$  exactly; drive buys no RP) — the ness-parent demand closes as NOT-NEEDED and the successor contract SEAM.STATE.WIRING.SELECTOR.01 (is pure-i compiler-forced?) is registered; the v898/v903 [O] premise unmoved; papers (prime-front continuation section, research-contracts block, Paper 1 keybox addendum, Paper 3 audit cards) and website mirrors synced; NO marker moves; NO RH claim anywhere)

2026-08-10 (LXXXVIII) (numerical-robustness amendment to v772 against an upstream scipy 1.17.0 wrong-result bug + the runner-side version guard; NO gate/bar/semantics change, NO status change: the single failure of the 2026-08-10 full-suite run was root-caused to an UPSTREAM ENVIRONMENT BUG, not to the module or the theory — scipy  $\geq 1.15$  auto-detects matrix structure in `scipy.linalg.solve` and dispatches the complex-symmetric shift  $A - zI$  (real symmetric  $A$ , complex  $z$ ) of the v772 REC-d reconstruction Ward to the LAPACK ?sysv diagonal-pivoting driver, and scipy 1.17.0’s sysv path returns a WRONG solution on this matrix family (backward residual  $\|(A - zI)G - V\|$  up to  $9.6 \times 10^1$  at the two complex- $z$  cells, inflating the REC-d relative residual to  $2.6 \times 10^3$ ; real- $z$  cells unaffected; fixed upstream in scipy 1.17.1); the amendment is ONE LINE: the Ward’s independent dense solve now passes `assume_a="gen"`, explicitly pinning the LU (gesv) driver that WAS the unconditional default when the module was frozen (identical semantics, available on every scipy  $\geq 1.0$ ; under numpy 1.26/scipy 1.13 the identical call already used LU and the gate read  $5.2 \times 10^{-13}$ ) — all frozen bars, ladders,  $z$  points, preregistered fails (CAUCHY-6/GAP-7/GAP-8 part 1; EDGE-A part 2) and both pattern gates are byte-identically unchanged, REC-d stays a genuine two-path Ward (eigh route vs independent LU solve), and the amendment is fully disclosed in the module docstring; verified green under BOTH python 3.9.6/numpy 1.26.4/scipy 1.13.1 and python 3.14.3/numpy 2.4.2/scipy 1.17.0; the runner-side guard `scipy>=1.10,!1.17.0` added to `requirements.txt` (the affected release excluded for anyone installing the suite; the firewalled experiments/\* sandbox pins are separate environments and untouched); website mirror byte-synced (`website/public/verification/v772_qf_feshbach_effective.py`) and `website/lib/discipline.ts` regenerated; `bash build.sh audit`  $\rightarrow$  AUDIT OK after the change; WITH THIS the full-suite run of 2026-08-10 stands at 897/897 modulo the documented environment bug — the raw run under scipy 1.17.0 was 896/897 with the v772 REC gates as the SOLE casualty, every GAP value bit-identical to the preregistration — no other module touched, no ledger row moved)

2026-08-10 (LXXXVII) (reflux rounds fifty-seven through fifty-nine — the cartography round: the wall rewritten in tangent-Schur coordinates with the FIRST substantive  $\tau$ -screen pass, the pivot/moment-matrix relocation map typed explicitly as mechanism-not-progress, the seam RP/modular EXCLUSION with two registered dead readings, and the 12-bit health word

typed a pure diagnostic: FOUR modules (v901–v904: tangent-Schur/B-floor, wall relocation map, seam RP exclusion, healthcode12 diagnostic; suite 893  $\rightarrow$  897; ELEVEN frozen probes with all fail-first spec amendments preserved, embedded byte-exact and executed verbatim; ledger 1027  $\rightarrow$  1040 with 13 promoted rows + 1 dated RP note on SEAM.CFIN.KSMIX.01) — (1) THE TANGENT-SCHUR SPLIT (v901, 26/26 + 24/24 + 18/18, tangentschur-measured (safeblock(minB=0.679) / gap(min=0.052, med=0.888)) + bfloor-measured (bfloor-screen-pass(slope=-0.247) / certfloor-dead) + signfree-measured (decirc-achieved(ell-b)), PRIME.PORT.TANGENT.SCHUR.01 + PRIME.PORT.BFLOOR.01 + PRIME.PORT.SIGNFREE.NORMALIZATION.01): with the co-block  $B$  PD the wall update is  $PD \iff n - q > 0$  — two SOURCE-ONLY scalars (no  $\tau_{h+1}$ , no forward sign; all identities warded to  $8.4 \times 10^{-14}$ , PD equivalence boolean-exact 39/39); the co-block floor  $\lambda_{\min}(B) = 0.679$  is  $O(1)$  ladder-wide while  $\tau$  falls by factor 552 (the FIRST substantive  $\tau$ -screen pass, slope -0.247; coordinate honesty printed: the raw co-block scales like  $\tau$ , slope +0.753); the honest negative certfloor-dead: all four classical certified bounds negative on all 39 steps (best maximum -88 vs the true floor +0.679) — the floor is MEASURED, never certified; the failure-seat anatomy refutes the plan diagnosis (at all 12 v900 rank-1 failure points the co-block itself dies, the 2.5% off-mode restores it linearly, the seat is orthogonal to the critical direction); the sign-free normalization  $\ell = \det(S)^{1/8}$  de-circularizes the update coefficient — CIRCULARITY REMOVED, POSITIVITY NOT GAINED, both said explicitly; (2) THE RELOCATION MAP (v902, 22/22 + 23/23, edgechristoffel-measured (rewrite-exact / ratio-below-one(0.999950)) + gramradau-measured (m0-identity-exact(4.9e-13) / gram-completion-fails(x2:42, m:42, p:0) / radau-exact(3.8e-15) / locwall-seen(42/42)), PRIME.CASE.EDGE.CHRISTOFFEL.01 + PRIME.PORT.GRAM.COMPLETION.01 + PRIME.PORT.RADAU.WEIGHT.01; typed explicitly a relocation/mechanism map, NOT progress): the exact positive-weight rewrite  $1/d_{12} = 1 + \sum_m W_{h,m} Q(x_m)^2 - \beta$  ( $W > 0$  exactly at  $9.7 \times 10^{-14}$ ,  $\beta \geq 0$  termwise, independent slogdet route on all 37 rungs) with the relocation diagnosis  $1 - r \propto \tau$  (slope +1.008; the uniform  $q < 1$  target is the wall's own PD premise QUANTIFIED, no independent positivity source; the Rayleigh fence  $147\times$  tighter than the classical trace bound); the wall IS the moment matrix of the signed comb measure ( $\lambda_{\min}(M_0) = \tau$  at  $5.2 \times 10^{-9}$ ) but the  $W \succeq 0$  Gram completion fails on 42/42 rungs at the  $x = +1$  edge where the comb nodes accumulate (violation magnitude to  $-152\times$  the margin: NO source-only Gram representation at matched degree); the Gauss–Radau construction at the classical node is exact algebra ( $w^*$  IS the Christoffel function, med 0.9978, a pure  $\mu_+$  object; the P2 pivot is NOT a Radau weight); (3) THE SEAM RP/MODULAR EXCLUSION + THE TWO DEAD READINGS (v903, 25/25 + 23/23 + 20/20 + 20/20 + 23/23 + 18/18 from SIX frozen probes, seamstate-measured + seamnorm-measured + mediator-measured + twistedrp-measured + dilation-split + gappencil-measured, SEAM.STATE.DERIVATION.01 + SEAM.CFIN.MIXING.NORMALIZATION.01 + SEAM.CFIN.MINIMAL.MEDIATOR.01 + SEAM.CFIN.TWISTED.RP.01 + SEAM.CFIN.RP.DILATION.01 + SEAM.CFIN.GAP.PENCIL.01): strict reflection positivity forces  $t = 0$  under BOTH deployable spatial reflections (the odd-sector Gram eigenvalues are EXACTLY  $\pm|a_J|$  of the  $\{4, 5\}$  duad at  $2.2 \times 10^{-16}$ : OS positivity  $\iff a_J = 0 \iff$  the v898 mixing floor falls — REFLECTION POSITIVITY AND THE MIXING FLOOR ARE MUTUALLY EXCLUSIVE on this family);  $u \geq t$  forced ( $u = 0$  RP-excluded, the round-57 reading corrected); the  $2\pi$ -KMS locus of the boundary reduction is EXACTLY the point ( $t=0, \beta=2\pi$ ); the complete twisted-OS census is exclusionary (48  $\rightarrow$  16 in-

volute → 6 admissible → 0/6 with strict RP at  $t > 0$ ); the dilation route SPLITS (the v898 M2 parent is a  $\theta_{abT}$ -marginal witness ON the RP cone boundary whose Schur compression carries the full Pfaffian mixing with the exact rational identity  $t^2 \cdot 3m/(1 - m^2) = 1/200$ ; the strict collar route family-wide obstructed by an exact linear law, 30 defect entries all of magnitude  $2t$ ); the gap pencil is EXACT:  $\text{Pf}(A_{\text{dep}} + tA_{\text{int}}) = -(t-1)(3t^2-1)(9t^3+21t^2-t-1)$  with Sturm uniqueness on  $(0, 1/4]$ , kernel dimension exactly 2, the three critical surfaces projections of ONE variety, and NO root at  $1/8$ ; THE TWO DEAD READINGS as first-class negative ledger rows: ‘ $t = 1/8 = 2\pi c_3/|Z_2|$  is a compiler value’ is a DECORATION (55/55 scanned  $t$  pass all frozen v898 gates, allowed set  $(0, \infty)$ , pin-absent) and ‘ $N_{\text{fam}} = 3$  as minimal boundary-mediation rank’ is REFUTED (rank 1 suffices by the exact sign law + integer witness; the deployed mediator has rank 2 EXACTLY; what 3 pins is boundary purity = dimension bookkeeping); the [O] premise of SEAM.CFIN.WICKFUNCTOR.01/KMSMIX.01 is UNMOVED (rp-theta-open now carries the dated measured-exclusionary note); (4) THE HEALTH WORD (v904, 18/18, healthcode12-measured / diagnostic-only / trueword-allplus / softpivot-12(0/37) / v897-pivot10-reproduced / smoothclaim-refuted, PRIME.ARITH.HEALTHCODE12.01; ledger-typed DIAGNOSTIC, ZERO RH CONTENT in either direction): the true word is all-(+) on 37/37 (the certified Sylvester census repackaged, no new content); three plan claims corrected (the 12 factors live in the v899 12-window compression per rung, not in v897; the softest raw pivot is  $k = 2$ , not  $k = 12$ ; smooth tips flags  $\{2, 1\}$  first and never flag 3); the one surviving claim reproduced byte-exact (the v897 Epstein control refused at pivot index 10); the seeded syndrome table distinguishes four adversarial fingerprints — a classifier of manipulated combs, a tool, not a target; ALSO: the two parallel-session diary entries renumbered final (CIII→CXXV, CIV→CXXXIV; zero duplicate numerals among the 2026-08-10 notes); papers (tfpt\_prime\_front new Section “The tangent-Schur coordinates and the relocation map” + the part-intro chain extended, tfpt\_research\_contracts the rounds-57–59 registration blocks + the dated RP note in the v898 block, tfpt\_1 Wick keybox dated RP addendum, tfpt\_3 five audit cards + the dated note on the v898 card) and website mirrors synced (prime-front feed round entry, DAG + v901–v904, counters  $893 \rightarrow 897$  / v535–v904, README + Zenodo to 897/v904); NO marker moves; NO RH claim anywhere — every relocation typed as such, nothing gained positivity anywhere)

2026-08-10 (LXXXVI) (website/video maintenance — the /prime-front page brought up to the round-56 state and compressed, and the explainer film refreshed; NO theory change, NO marker moves, NO RH claim: (1) PAGE CURRENT — the stale hero counters replaced by single-source constants (suite scripts now read from the generated lib/suite.ts, the diary-run count computed from the feed data itself: 285 entries), the honesty banner and big-picture statistics interpolate the same sources (span v535–v900 inside the 893-script suite), the stale “314 probes” qualifier dropped from the sandbox-check statistic (kept as the dated lower bound 6638+); (2) PAGE COMPRESSED — the rendered text payload reduced from  $\sim 445$  KB to  $\sim 182$  KB and the built HTML from 1.30 MB to well under 0.9 MB with ZERO loss of honesty content: the four very long “preserved verbatim” status recaps (the T102–T176 sprint recap of the honesty banner, the end-of-series paragraph, the “Near” meaning-level paragraph, and the phase-2 full-proof narrative) moved word-for-word into a lazily loaded module (verbatimRecaps.tsx) behind collapsed on-demand expanders (VerbatimRecap.tsx), each replaced in place by a compact honest summary that



keeps every fence (“Sandbox; not RH evidence”); the diary feed now mounts full entry texts only on expand, and the inline feed window returned to its declared  $\sim 16$  newest entries (44 older entries moved verbatim to the lazy archive, 225  $\rightarrow$  269); (3) THE FILM REFRESHED — the one stale narration line of the 2026-08-03 explainer updated to the current suite state (“Almost nine hundred modules”; on-screen stat now “893 modules · machine-checked · 84 Lean files”), full TTS/captions/render re-run through the frozen frame windows (still 2:39, burned-in English captions, en+de subtitle tracks rebuilt from the same source; the component transcript verified character-identical to the caption track; JSON-LD uploadDate 2026-08-10); all other narration, timings and claims unchanged — the film still states “no proof of RH” explicitly)

2026-08-09 (LXXXV) (reflux rounds fifty-four through fifty-six — the reviewer’s “immediate” list delivered: the certificate rollout becomes INTERVAL-RIGOROUS (the informal  $\varepsilon_c$  lag-evaluation error model is RETIRED from the ladder), the channel-mixing state the block Wick functor awaits is CONSTRUCTED (KMS + exact Schur generation), the conditional diagonal contract becomes a NORM SQUARE in the frequency weights, and the normalized core update gets its exact two-dimensional anatomy with the honest negatives registered: FOUR modules (v897–v900: certified interval ladder, KMS/Schur mixing, Christoffel norm-square, core update anatomy; suite 889  $\rightarrow$  893; EIGHT frozen probes with all fail-first spec amendments preserved — v898–v900 embedded byte-exact and executed verbatim, v897 embedding the 66-minute ladder probe byte-exact with SHA wards, re-certifying all 15 tier-1 + 3 tier-2 rungs verbatim in-suite and freezing the full 42-rung census; ledger 1020  $\rightarrow$  1027 with 7 promoted rows + 4 dated notes on PRIME.PORT.TAIL.01 / PRIME.PORT.CERTIFIED.LADDER.01 / PRIME.CASE.PAIRCORR.CONTRACT.01 / SEAM.CFIN.WICKFUNCTOR.01) — (1) THE CERTIFIED INTERVAL LADDER (v897, probe verdicts ballhead-proven + ballladder-complete 10/10, 3960 s, PRIME.PORT.BALLLADDER.01):  $\sigma_h > 0$  with RIGOROUS `mpmath.iv` interval-arithmetic shifts on ALL 42 reachable rungs  $h = 142\text{--}878$  — 15 exact-rational integer Bareiss/Sylvester certificates ( $h \leq 300$ , floors  $4.5 \times 10^{-5}\text{--}2.1 \times 10^{-4}$ ) + 27 validated-precision Cholesky dps 120/200 certificates (floors  $5.2 \times 10^{-6}\text{--}8.4 \times 10^{-5}$ , pivot/bound ratios  $2.3\text{--}9.9 \times 10^{115}$ ); the GL-48 node-enclosure lemma (radius  $10^{-90}$ , definite sign changes, disjoint intervals, 2 in the interval weight sum), native outward-rounded interval transcendentals, exact midpoint/radius extraction, the rigorous shift  $2h \text{ rad}_{\max}$  ( $+ h/Q$  on the tier-1 grid) — no working-precision rounding in the tier-1 decision path; the Epstein control REFUSED at pivot 10 by the identical interval machinery; the informal conservative  $\varepsilon_c$  model of v884/v887 is RETIRED; the validated-precision tier label carried honestly, never upgraded; remaining named: the Lean composition (the asymptotic tail untouched, PRIME.PORT.TAIL.01); (2) THE CHANNEL-MIXING CANDIDATE (v898, 30/30, kmsmix-measured [kmsmix-found(u=1, t=1/8,  $\beta=1$ ), schurmix-generates(10/10), rp-theta-open], SEAM.CFIN.KSMIX.01 + SCHURMIX.01): the  $C_6$ -covariant channel-mixing KMS state EXISTS — the first point of the frozen grid passes ALL block-functor gates (CAR strict, exact covariance with the forced zeros, all 15 duad cross-blocks at the floor, grading 5/5 + 10/10, all 15 block Wick monomials with the canonical  $\text{Pf}_4$  signs of the v896 functor; robust at  $\beta = 2$ ; the hermitian commutant measured exactly,  $112 = 62 + 50$  with the 33-dim mixing slice); EXACT (sympy): the Schur/Feshbach elimination of the boundary gives  $A_{\text{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^T$  and GENERATES all 10 carrier duads from the bare diagonal state (uniform  $3J$  block, canonical

signs) — on both fronts (RH port, Wick functor) the real correlation lives in the ELIMINATED boundary; the frozen 15-signature table is the registered prediction; rp-theta-open the named open check; the [O] premise UNMOVED; (3) THE NORM-SQUARE CONTRACT (v899, 19/19 + 15/15 + 19/19, christratio-measured (winner-christoffel / extsource-decoupled(0.454)) + kernelsos-measured (index-large; rmin=1; not-absorbed) + edgedefect-measured (normsquare-achieved; kill=periodic; chain=carried(0.21)), PRIME.PORT.CHRISTOFFEL.RATIO.01 + PRIME.CASE.KERNEL.SOS.01 + PRIME.CASE.EDGEDEFECT.01): the EXACT deflated-Christoffel pivot identity  $1/d_{12} = 1 + v_*K_\sigma(y_*)$  (independent route  $2.0 \times 10^{-11}$ ; the winner identification by two orders of magnitude) with the honesty fence  $\lambda_{\min}(I - G) = \tau$  EXACT (a coordinate change, NOT an independent positivity source);  $c'/c$  SENS [0.223, 5.258],  $r_{12} > 0$  on 31/31; the pair-contract kernel's frequency weights  $W_{f,m} \geq 0$  EXACTLY with the ENTIRE negative content collapsing to ONE top-edge tent (closed-form boundary leakage,  $r_{\min} \leq 1$ ), and the PERIODIC full-weight fold KILLS it exactly — negative index 0 everywhere, 79/79 modified margins positive, the kz-9-alias-9 absorption failure  $1.381 \rightarrow 0.000$ , the boundary term  $\beta$  carried closed-form (max 0.21) — the conditional diagonal contract of v894 is a NORM SQUARE in the frequency weights (reviewer case 1 at the finite level); the arithmetic hypothesis itself REMAINS CONDITIONAL (pair-correlation class per v889); (4) THE CORE UPDATE ANATOMY (v900, 20/20 + 23/23, normcore-measured (family-bounded(2533) / update-exact(udim=2) / region-open) + graphregion-measured (coupling-free / graph-region-open / shadow-lawless), PRIME.PORT.NORMALIZED.CORE.01 + PRIME.PORT.GRAPH.REGION.01): the normalized deep core obeys  $X' = c(X + U)$  EXACTLY ( $8.3 \times 10^{-16}$ ) with  $c = \tau/\tau'$  and an effectively TWO-DIMENSIONAL input (one matrix mode at 97.5% energy); the honest negatives REGISTERED as the boundary conditions of any invariant-region theorem: no linear coupling law (both  $R^2$  BELOW the 200-shuffle null  $q_{95}$ ), the rank-1 truncation fails (the positivity is carried by the 2.5%-energy off-mode), naive product/graph invariant sets refused, the 2D shadow lawless — any region construction must keep the exact  $X + U$ , truncating no mode; papers (tfpt\_prime\_front new Section “The certified interval ladder and the norm-square contract”, tfpt\_1 Wick keybox extended with the constructed candidate, tfpt\_3 audit cards, tfpt\_research\_contracts round-54–56 registrations + dated notes) and website mirrors synced; NO marker moves; NO RH claim anywhere — the asymptotic tail and the one-sidedness stay open exactly as before)

2026-08-09 (LXXXIV) (reflux rounds fifty through fifty-three — the closure architecture: the exact hermitian congruence with a NON-DECAYING inheritance margin, the flag-chain induction, the honest Rouché kill, the  $8 \times 8$  reduction (the wall IS the fixed Schur core to seven digits), the explained margin scale, the assembled theorem skeleton with ONE bounded-margin statement open, the concrete conditional diagonal contract, the collectivity theorems, and the Wick functor arc resolved at block level: FIVE modules (v892–v896: closure architecture, relative margins, diagonal refinement, collective comb, Wick block functor; suite 884  $\rightarrow$  889; FOURTEEN frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 1012  $\rightarrow$  1023 with 10 promoted rows + dated skeleton updates) — (1) THE CLOSURE ARCHITECTURE (v892, 16/16 + 15/15 + 14/14 + 16/16): the exact congruence  $A_{h+1} = A_h^{1/2}(I + H_h)A_h^{1/2}$  ( $6.5 \times 10^{-14}$ ,  $H$  exactly hermitian) with the inheritance margin  $\eta_h = 1 + \lambda_{\min}(H_h)$  NON-DECAYING (min 0.0050, median 0.29, slope +0.108

while  $\tau$  collapses at  $-2.74$ ) — the open inequality is bounded-margin-shaped, not collapsing;  $\rho(W^{-1}\Delta C)$  was a misleading statistic (non-normal,  $\|M\|/\rho$  up to 26) — the hermitian  $H$  is the honest carrier; ALL 12 LDL flag quotients  $Q_k > 0$  on all 31 truth steps (soft-flag min  $2.06 \times 10^{-14}$ , sign-decidable at  $6.4 \times 10^9 \times$  the noise floor; the smooth world violates at the BULK flags first) — the shortest induction shape; the two-sided contour-norm (Rouché) route honestly killed ( $M \geq 1$  on 22/31 including off-cap; the argument principle itself verifies exactly); and THE WALL IS THE FIXED  $8 \times 8$  SCHUR CORE:  $\lambda_{\min}(S) w_{\text{core}}/\tau = 1 \pm 8.4 \times 10^{-8}$  with  $w_{\text{core}} \geq 0.999937$  on 41 rungs, conditional only on the TAU-RELATIVE exterior bound ( $\lambda_{\min}(R)/\tau$  trendless at 210–2200; the absolute margin shrinks  $h^{-2.865}$ ); (2) THE MARGIN SOURCES (v893, 20/20 + 18/18): SCALE-MATCHED — in the dangerous direction the increment and the operator collapse TOGETHER (slopes  $-3.716$  vs  $-3.456$ , residual correlation 0.974, 94 % shared variance): the non-decay of  $\eta$  is EXPLAINED (numerator and denominator are sections of ONE arithmetic object; open: only the  $O(1)$  scatter); the relative laws  $d_{12} = c\tau$  (trendless; the kz-21 outlier diagnosed as a COORDINATE-ARTIFACT in the round-53 sandbox — the kz-19 mechanism repeats, corrected spread  $124 \rightarrow 15$ : SOFTFLAG TAU-TIED) and  $\lambda_{\min}(R) \geq 210\tau$ ; COMPRESSIONS-DISTINCT (the port bulk floor is absolute, the core exterior tau-tied); THE PRINTED SKELETON: base certified +  $R \geq 210\tau$  + core  $\leftrightarrow \tau$  (7 digits) +  $\eta > 0$  — the ONE unconditionally open statement is the  $\tau$ -sign inheritance in bounded-margin form; (3) THE DIAGONAL REFINEMENT (v894, 11/11 + 13/13 + 12/12; the route stays CONDITIONAL per v889): the exact band-limited prime kernel  $K_{h,m}$  (regrouped at  $4.5 \times 10^{-15}$ ; the LaTeX contract statement frozen in-probe; all 79 finite margins positive); the endpoint bracket ( $F'(1) \geq$  deficit on 142/227 — the global linear collapse fails, deep small- $a$  cells pass monotonically); the inner-zone accounting (scale-normalized FLAT; the hard band NON-SHRINKING at  $\sim 20\%$  of aliases / 38–56 % of mass; stability on an  $a$ -graded mid shell); (4) THE COLLECTIVITY (v895, 18/18 + 12/12): the true avoidance law is ONE-SIDED ( $\max(-\mu_{\min}) = 0.9950 < 1$  on all 31 steps = PD inheritance itself; no norm-bound shape); the geometry/atom split is EXACTLY degenerate (the atom cutoff is slaved to the tent edge — no separable atom channel; the increment is the collective window re-test); the deep core IS the port core (the fixed aliases  $a_m = \pi^2 m^2$  to 0.1 %; ALL FOUR surgeries kill the coherence incl. the wrong- $\Lambda$  world — the signal is the ENTIRE multiplicative von Mangoldt comb); (5) THE WICK ARC (v896, 27/27 + 33/33 + 26/26): the six compiler roles exist in the deployed seam (sector law, grading,  $C_6$  intertwining exact) but the vacuum kernel is channel-diagonal; the PROVED scalar obstruction (the  $C_6$  2-cycle  $\{4, 5\}$  forces a duad zero; both relaxation witnesses constructed); the BLOCK RESOLUTION (commutant dimension 33; the canonical block covariance carries 15/15 blocks and 15/15 block Wick monomials with the canonical sign law, interleaving character +1 symbolic) and the decisive fact SEAM-DIAGONAL (the deployed seam has zero cross-blocks — the value level awaits a physical mixing mechanism; the [O] premise unmoved); papers and website mirrors synced; NO marker moves; NO RH claim anywhere)

2026-08-09 (LXXXIII) (reflux rounds forty-two through forty-nine — the great consolidation: the certificate rollout COMPLETES ( $\sigma_h > 0$  PROVEN on ALL 42 reachable rungs), the finite Pfaffian/ $K_6$  signature CLOSES (fermionic signs gauge-matched via the canonical character; the self-hosting counting theorem forces  $g_{\text{car}} = 5$ ; the flavor hexagon is the  $\text{Aut}(C_{\text{fin}})$  orbit), the four route decisions

of the diagonal half land (the wall dies 100 % through off-diagonal coherence; PNT-insufficient; the input is pair-correlation class), the surviving positivity architecture is measured (total positivity of the section family on 37/37 rungs; the wall margin IS the pole distance of the integrable  $s$ -flow; inheritance is a multi-factor statement), and the Möbius/carrier-invariance route is honestly killed with mechanisms: FIVE modules (v887–v891: certified ladder complete, finite Pfaffian signature, route decisions, total positivity flow, Möbius firewall kills; suite 879  $\rightarrow$  884; FOURTEEN frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 999  $\rightarrow$  1012 with 11 promoted rows + 2 dated route retypes) — (1) THE COMPLETE HEAD (v887, 7/7, certified-ladder-complete):  $\sigma_h > 0$  on ALL 42 reachable rungs (12 exact-rational integer Bareiss/LDL + 30 validated-precision mpmath Cholesky dps 120/200 with precision doubling; the module closes  $h = 859/878$  with floors  $\geq 8.352/9.187 \times 10^{-6}$  and pivot/bound ratios  $\sim 10^{115}/10^{195}$ ; the 40-rung parent rollout stays sandbox evidence, runtime-excluded; Epstein REFUSED at pivot 10; proven modulo the declared conservative evaluation error model; ball arithmetic and the Lean composition remain named); (2) THE FINITE SIGNATURE (v888, 27/27 + 19/19 + 42/42, exact integer/sympy): the  $S_6$  Specht channels  $(1, 0, 2/3)$  exact; the 15 doily lines = the 15 Plücker monomials of  $Gr(2, 5)$  with perfect sign match; the  $(16, 6, 2, 10)$  Hadamard anchor signature; the  $S_6 \rightarrow S_5$  vacuum selection (orbits  $[1, 5, 10]$  equal as sets to  $(\Lambda^0, \Lambda^4, \Lambda^2)$ ); THE FLAVOR HEXAGON IS THE  $\text{Aut}(C_{\text{fin}})$  ORBIT ( $g^2 = \sigma$  exact,  $g^3 =$  the  $W$ -edge sheet swap, the 6-orbit = the deployed transport edge for edge,  $(2/3)^6 =$  the frozen recovery value,  $0/600$  non-automorphisms factor); the  $K_6$  motif theorem ( $35 = 15$  perfect matchings + 10 vacuum triangles = secants + 10 vacuum-free = externals;  $q^* =$  vacuum incidence); THE FERMIONIC PFAFFIAN SIGNS gauge-matched to the Hodge table via the CANONICAL character  $\chi(i) = (-1)^{i+1}$  (triple-identified: vacuum-row Laplace;  $\Lambda^4/\Lambda^5$  volume form;  $C_6$  crossing character); THE SELF-HOSTING COUNTING THEOREM  $\binom{g+1}{2} = g!!$  with  $g = 5$  the only nontrivial solution and  $N_{\text{fam}} = (g - 2)!! = 3$  ([E] theorem; the Wick-compiler premise stays [O] — P2 narrowed to ONE premise, no marker move); the mod-30 anchor clock CLOCK-DISTINGUISHED (period-4 moduli exactly  $\{5, 10, 15, 30\}$ ); the operative-Coxeter-30 reading separately DEAD (sandbox, both sides); (3) THE ROUTE DECISIONS (v889, 8/8 + 12/12 + 10/10 + 12/12): the wall dies through PURE off-diagonal coherence (diagonal profile frozen at the kill point; separation  $> 10$  deep; honestly diag-intermediate);  $\theta^* = 0.700$  frozen (outside margin +0.0214 uniform on 42 rungs; inside law  $\sim 0.87/\log^2 h$ ); pnt-insufficient (the PNT-smooth reference itself violates the port margin on every rung; the truth's positivity is the mask-mass INTERACTION;  $\varepsilon_A$  is 6–970 $\times$  too large); and the signed-homotopy classification homotopy-indefinite-arithmetic (exact envelope identity at  $4 \times 10^{-14}$ ; the fixed-mask kernel is concavity-locked NSD; the diagonal route's arithmetic input is PAIR-CORRELATION CLASS — retyped from unconditional candidate to conditional reduction theorem); (4) THE SURVIVING ARCHITECTURE (v890, 10/10 + 7/7 + 14/14): TOTAL POSITIVITY of the section family (all 12 leading minors strictly positive on all 37 full-window rungs; Desnanot–Jacobi exact); the POLE-FREE  $s$ -corridor with  $s_h^* = 1/\lambda_{\max}(C_h)$  — the wall margin IS the pole distance of the integrable  $s$ -flow (slope  $-2.90$ ); the multi-factor determinant anatomy ( $Q_h = \prod(1 + \mu_i)$  exact; the truth keeps EVERY factor off  $-1$  while the smooth-mass world crosses on 22/28 steps and flips  $Q$  13 times; the one-scalar Möbius bridge honestly closed bridge-diffuse:  $|\alpha| \sim 10^{-3}$  is determinant-inert) —

POSITIVITY INHERITANCE, IF IT EXISTS, IS A MULTI-FACTOR/TOTAL-POSITIVITY STATEMENT; (5) THE KILL BATTERY (v891, 10/10+13/13+7/7): the round-CVI carrier-invariance was a NORMALIZATION ARTIFACT — fit-free gauge-free cross-ratios CR-DEAD (pooled median 0.128; anchors disagree; no cocycle), source-frozen gauges show REAL motion (GAUGE-MADE; the scalar cocycle does not track the wall, corr 0.129; Jacobi dead), and the six-world census shows the arithmetic separates AT THE FRAME (every control world including the smooth ones loses window subcriticality; scramble passes the Möbius bars BETTER than truth — rank-2 kinematics); the surviving arithmetic remnant is the deep-core 8-node cross-ratio coherence ( $4.3 \times 10^{-4}$  vs smooth  $2.7 \times 10^{-2}$ ) — prime-mass-driven, the named object; related sandbox kills recorded (static Cauchy identification dead; the Cauchy route can never certify:  $\text{dev}/\tau$  grows at slope +1.46); papers and website mirrors synced; NO marker moves; NO RH claim anywhere — the hardness decision stands: the open one-sidedness carries full RH weight)

2026-08-09 (LXXXII) (reflux rounds forty/forty-one — the flow corridor of the prime front: the determinant chain to the tau function lands with the PARAMETRIX DECISION (the Euler-product comb is GLOBALLY rigid — every smoothing-based parametrix is dead), the FIRST RIGOROUS POSITIVITY STATEMENT of the program (exact integer Sylvester certificates  $\sigma_h > 0$  on the head rungs), the complete flow/induction anatomy of the ladder (degree-2 Lax closure; NO monotone one-prime induction in any insertion order — positivity is a property of the COMPLETE comb), and the register-gapcode DISENCHANTMENT row: FOUR modules (v883\_tau\_chain\_parametrix.py, v884\_certified\_head\_positivity.py, v885\_ladder\_flow\_anatomy.py, v886\_register\_gapcode\_null.py; suite 875  $\rightarrow$  879; NINE frozen probes embedded byte-exact with all fail-first spec amendments preserved; ledger 989  $\rightarrow$  999 with promoted rows, the decided design constraints, and the open-contract registrations PRIME.PORT.LEADING.SIGN.01 + PRIME.PORT.TAIL.01) — (1) THE TAU CHAIN AND THE PARAMETRIX DECISION (v883, 5/5 + 3/3 + 3/3, tau-chain-exact + parametrix-measured (fluctuations-required) + zones-measured (both-required)): the Schur determinant identities  $\sigma_h = \det(I - E)/\det(B)$  and  $\det(I - E) = \det(I - R)\det(I - D_P)$  are machine-exact ( $4.8 \times 10^{-13}$  in log space) with positive bulk factors, hence  $\text{sign } \tau_h = \text{sign } \sigma_h =$  the wall with  $\tau_h := \det(I - D_P)$ , and the Jacobi variational identity  $\frac{d}{ds} \log \det(I - sD_P) = -\text{tr}[(I - sD_P)^{-1}D_P]$  holds on the true family; the  $\kappa/\rho$  readout:  $\text{slope}(\log \sigma) = -2.93$  vs  $2 \times$  the Mellin fluctuation slope  $-2.53$  ( $\sigma \sim \text{fluctuation}^2$  within 16 %); THE DECISION: the prime-power lattice with fully smooth quadrature masses ( $\tau = -78$  to  $-1706$ ) and even with EXACT local  $\pm \frac{1}{2}$ -log-unit mass sums ( $-78$  to  $-5537$ ) is ORDERS worse than the plain continuum ( $-0.11$  to  $-1.26$ ) against the true comb ( $+1.7 \times 10^{-4}$  to  $+6.7 \times 10^{-7}$ ), and the zone refinement kills BOTH ways (edge-only smoothing  $-3.7$  to  $-22.1$ ; interior-only  $-77$  to  $-7551$ ): the exact  $\Lambda$ -mass-position pairing is load-bearing EVERYWHERE — the multiplicative Euler-product structure is globally rigid, the unperturbed object of any RH analysis MUST carry the full von Mangoldt comb, and every small parameter lives in the flow/deformation direction; (2) THE CERTIFIED HEAD (v884, 5/5, certified-head):  $\sigma_h > 0$  PROVEN on kz 9 ( $h = 184$ ), 12 (151), 13 (168) — independent mpmath rebuild of the window lags (dps 60/100, conservative error model  $\varepsilon_c = 10 \times$  dps-disagreement, cross-ward  $1.1 \times 10^{-14}$  which CAUGHT a sign bug in the rebuilt Arch near-branch), matrix

shift  $\sim 2 \times 10^{-18}$  ( $\sim 14$  orders below the margins), then EXACT INTEGER Bareiss/LDL on the  $Q = 10^{20}$  grid: all pivots positive (Sylvester)  $\Rightarrow K \succ 0 \Rightarrow \sigma_h > 0$  via the exact v866 equivalence — no Lanczos, no floating point in the decisive step; certified floors  $\lambda_{\min} \geq 3.633/3.327/3.842 \times 10^{-4}$ ; the Epstein control through the IDENTICAL machinery is REFUSED at pivot index 10; typed honestly: proven MODULO the declared conservative transcendental-evaluation error model (validated-numeric standard), the ball-arithmetic closure and the 42-rung rollout are the named follow-ups — the head part of PRIME.PORT.TAIL.01 is delivered on three rungs; (3) THE FLOW/INDUCTION ANATOMY (v885,  $8/8 + 5/5 + 13/13 + 11/11$ , isoflow-measured (flow-open) + ladder-measured (perturbative + non-monotone) + primeladder-measured + lax2-measured (flow-noisy / lax2-partial)): the gauge-fixed IKS generator flow closes in DEGREE 2 (residual medians  $0.244/0.248$  with  $\{Y^2f, Y^2g\}$ ; degree 1 fails at  $0.498/0.457$ ) — an isomonodromy-compatible Lax structure, typed as structural precondition (controls close too; the arithmetic stays in the VALUE); the window-ladder step is marginally perturbative (accuracy median  $0.458$ ) but NON-MONOTONE in raw  $h$ ; the ONE-PRIME LADDER DECIDES: the pure-Arch start is catastrophically subcritical ( $\tau(0) = -9.9 \times 10^{13}$  at  $kz$  9;  $G_+$  not even PD deeper), the last sign crossing sits at  $u^*/U = 0.991-0.999$  in EVERY insertion order (ascending needs 97–99 % of the atoms, edge-first needs 100 %), per-atom increments only 54–64 % positive: POSITIVITY IS A PROPERTY OF THE COMPLETE COMB — v877's truncation reversal sharpened to atom resolution, no monotone one-prime induction exists, induction must run in a flow coordinate, not in comb content; and the raw degree-2 coefficient basis is ill-conditioned NOISE (roughness  $1.365$  vs  $0.7$ ; operator predictivity  $0.350$  vs  $0.3$ ; the integrated tau-transfer identity itself warded EXACT at  $5.6 \times 10^{-11}$ ) — the conditioned flow coordinate is the named open object (PRIME.PORT.LAX2.01); (4) THE GAPCODE DISENCHANTMENT (v886,  $24/24$ , gapcode-measured, an honest NEGATIVE/limit row per the frozen probe's own typing): the error-code reading of the 210-register is LITERALLY TRUE on the prime side (the register bits  $\{2, 3, 5, 7\}$  are the first four parity checks of the sieve; code rate  $48/210$ ; hard floor  $g_k \geq w(p_k \bmod 210)$  with 0 violations in  $5.76 \times 10^6$  pairs;  $I(\text{state}; \text{gap}) = 1.14$  bits) — AND the code is THE SIEVE: a zero-parameter wheel–Cramér null explains  $R^2 = 0.99839$  of the state profile, the arithmetic excess is  $4.02\%$  and shrinking, no knee at four checks, 210 sits ON the foreign-wheel trend ( $z = -0.61\sigma$ ), all eight anchor Walsh seats below  $2.5\sigma$  exactly as CRT predicts: everything the register knows about gaps, every 4-prime wheel knows — nothing moves with  $c_3$  or  $\varphi_0$ ; the register's uniqueness remains the arithmetic forcing  $\text{rad } |W(E_8)| = 7\#$  with  $\prod(p-1) = 48 = \Omega_{\text{adm}}$ ; papers: the prime-front rounds-40/41 section and contract registrations; website mirrors and generated surfaces synced; NO marker moves; NO RH claim anywhere — the demand-curve decision stands: the open one-sidedness carries full RH weight)

2026-08-09 (LXXXI) (reflux rounds thirty-eight/thirty-nine — the port compression of the wall: the four finite anchor theorems land, the Carleson wall becomes the EXACT operator geometry of ONE dressed port block (gauge, testing, Schur, integrable class), and the port source gets its universal law (the  $\sqrt{n/X}$  uniformization, the exact  $1 - e^{-1/2}$  edge mass, the Mellin–Cauchy kernel, the closed criticality budget): THREE modules (v880\_finite\_anchor\_theorems.py, v881\_carleson\_port\_geometry.py, v882\_port\_source\_mellin.py; suite  $872 \rightarrow 875$ ; EIGHT frozen probes embedded byte-exact with all fail-first spec amendments

preserved; ledger  $972 \rightarrow 989$  with 11 promoted rows + 6 open-contract registrations) — (1) THE FINITE ANCHOR THEOREMS (v880, 26/26, finite-anchor-closed): the closed census-free normal form  $q^*(x) = \sum_{i < j} x_i x_j + \sum_i x_i = \binom{|x|+1}{2} \bmod 2$  (weight-only; the v845 selector FORCES the linear part  $(1, 1, 1, 1)$  — the  $2^{16}$  search replaced by a theorem); the EXCLUSIVE anchor classification of  $PG(3, 2)$  lines (all 15 doily lines carry the weight multiset  $\{2, 1, 1\} =$  the anchor  $a = (1, 1, 2)$  with per-line Vieta  $(4, 5, 2)$ ,  $e_2 = g_{\text{car}}$ ,  $e_3 = |\mathbb{Z}_2|$ ; the 20 others split EXACTLY as 10 secants = the WRONG anchor  $(1, 2, 2)$  + 10 external lines — the anchor law characterizes the doily); the doily as the MULTIPLICATION TABLE  $\Lambda^2 E \times \Lambda^2 E \rightarrow \Lambda^4 E$  of the five-slot carrier (wedge nonzero EXACTLY on doily lines;  $5 \times 3 = 15$  factorizations); and QUADRATIC ONE-STEP EXHAUSTION forces  $g_{\text{car}} = 5$  uniquely among odd  $g$  — the theorem [E], the Calderón premise explicitly [O]: P2 narrowed, NOT eliminated, no marker move; (2) THE OPERATOR GEOMETRY (v881,  $11/11 + 5/5 + 8/8 + 5/5$ , pick-scalarized + testing-identified + port-schur-exact + integrable-confirmed; four sides of ONE object, counted once): the review’s literal rank-2 Pick target  $I - C^*C = \Gamma^*\Pi\Gamma$  is REFUTED as stated and replaced by the exact rank-2 JACOBI DISPLACEMENT  $[J, \Delta] = b_h(e_{h-1}r^\top - re_{h-1}^\top)$  (rel  $\leq 4.3 \times 10^{-14}$ , source-only, blind holdouts included); the GAUGE identity  $E = S(\sqrt{\tilde{\nu}} K_{\text{CD}} \sqrt{\tilde{\nu}})S$  — the wall operator is ORTHOGONALLY EQUIVALENT to the plain Carleson Gram, its diagonal the CLASSICAL testing functional  $\tilde{\nu}_m K_h(y_m, y_m)$  (the corpus wording updated: the arithmetic sits in the NODE AND WEIGHT distribution of the two measures, the critical cancellation is the coherent geometry of a Carleson Gram operator — not an irreducible kernel phase); the Schur/Haynsworth PORT REDUCTION  $I - E \succeq 0 \Leftrightarrow I - R \succeq 0 \wedge I - D_P \succeq 0$  with  $(1 - \lambda_{\max}(D_P))/\tau = 1.00$  on ALL rungs (THE PORT IS THE WALL: 13–54 nodes; bulk margin  $420\text{--}45000 \times \tau$ ; inertia integer-exact including both indefinite controls) — satisfying the round-37 NON-DECOMPOSITIONAL demand; and the theorem-grade surprise  $[Y, D_P]$  of EXACT rank 2 ( $10^{-14}$ ): the dressing PRESERVES the IKS integrable class — the dressed port operator is an integrable kernel with two explicit generators; (3) THE SOURCE LAW (v882,  $7/7 + 5/5 + 6/6$ , sqrt-uniformized + mellin-kernel-confirmed + atoms-identified (budget-closed)): under  $r = \sqrt{n/X}$  the weighted prime measure uniformizes to Lebesgue  $dr$  (Kolmogorov  $5 \times 10^{-4}$  at  $X = 10^7$ ; own SPF sieve; deployed tables = the true comb at 0.0033); the last-log-unit edge mass equals  $1 - e^{-1/2} = 0.393469$  (measured 0.3936) — the wall’s  $\sim 39\%$  port mass IS the classical PNT edge law; the port symbol is the Mellin–Cauchy kernel  $1/(1 + 2i\tau)$  and the FIT-FREE continuum prediction reproduces the deployed port numerators to 0.029 at the deepest rung (the deviation = the shrinking arithmetic fluctuation); the port numerators are the EXACT deployed-window prime sums ( $1.5 \times 10^{-13}$ ) with the explicit-formula reading at the alias frequencies, and the criticality BUDGET CLOSES ( $+1.405 - 1.674 + 0.284 = +0.015$ ):  $T \rightarrow 1$  is PNT growth exactly compensated by window geometry plus Christoffel — the wall is an ERROR-TERM statement about prime partial sums at the port frequencies; (4) THE HARDNESS DECISION and the OPEN CONTRACTS (ledger rows, sandbox-backed): PRIME.ERRORTERM.SCALE.01 DECIDED — injecting the exact off-critical zero signature, the wall margin equals perturbation ENERGY on the  $\tau$  law exactly (ratios 1.00–1.10;  $A^* \sim \sqrt{\tau}/\delta^2$  with the frozen convention  $\Delta\lambda = O(A^2\delta^4)$ ; the kill localizes THROUGH the port block): the reduction chain is an exact reformulation at RH strength, NO unconditional slack; new open contracts registered [O]: PRIME.PORT.LIMIT.01 (compressed operator-norm

convergence slope  $-2.80$ , criticality from below, atom-at-edge Weyl- $m$  class — the fixed-space limit theorem is the demand), PRIME.PORT.SCALAR.01 (the wall as ONE scalar inequality  $\sigma_h = [1 - \tilde{\nu}_m^* K_h(y^*, y^*)] - b^\top (I - E_{\text{rest}})^{-1} b \geq 0$ , exactness warranted on all 42 rungs), PRIME.PORT.COCYCLE.01 (the fixed  $12 \times 12$  window carries the wall exactly), PRIME.CASE.SUMRULE.01 (prove the Case  $C_0/C_1$  sum rule  $\Rightarrow$  unconditional  $T_h \leq 1$ ; the deployed chains are measured Killip–Simon class), PRIME.PORT.RIEMANNHILBERT.01 (the discrete  $2 \times 2$  IKS Riemann–Hilbert problem is WELL-POSED, resolvent closure  $10^{-12}$ ; the smooth PNT model VIOLATES the floor at finite depth — the prime discreteness is load-bearing: any steepest-descent program must keep the lattice in the unperturbed problem); papers: the prime-front rounds-38/39 section, tfpt\_1 finite-theorem keybox, tfpt\_3 audit cards, tfpt\_research\_contracts registrations; website mirrors and generated surfaces synced (875/v882); NO marker moves; NO RH claim anywhere)

2026-08-09 (LXXX) (reflux round thirty-seven — the partition closure: the compact closing round of the reopened Cotlar route — the analytic core is SAVED (the exact tridiagonal CD identity with the route’s FIRST  $h$ -uniform derived constants) and the decomposition chapter is CLOSED ENTIRELY (cells, packets, all frames): the cancellation is ATOMICALLY GLOBAL and the certificate demand is retyped NON-DECOMPOSITIONAL: ONE module (v879\_envelope\_partition\_closure.py; suite 871  $\rightarrow$  872; two probes embedded byte-exact, the contract chain 4621b899...  $\rightarrow$  5fd6bf61...  $\rightarrow$  27d9f0a3...  $\rightarrow$  2199e9b1... machine-verified at run time) — (1) THE ANALYTIC CORE (envelope\_derivation\_gap\_probe, 7 checks with the TWO frozen-honest FAILS T1.4/C.1 pattern-gated at exit 1, verdict envelope-empirical-only): multiplication by  $\cos \theta$  acts EXACTLY tridiagonally on the odd frame (the mirror symmetry closes the top boundary inside the space,  $A_{h-1,h-1} = -\frac{1}{2}$ ; the single rank-2 commutator defect at  $k = 0$ ; the Christoffel–Darboux telescoping identity at max residual  $1.3 \times 10^{-13}$  on every rung, battery + all five complete-comb holdouts), and the DERIVED pointwise bound  $|U_{ij}| \leq [C_{\text{int}} + \sqrt{(G_+^{-1})_{00}}(K_{-,i}^{-1/2} + K_{+,j}^{-1/2})]/|\cos \theta_i^- - \cos \theta_j^+|$  holds with ZERO entrywise violation and  $h$ -UNIFORM source constants —  $C_{\text{int}} \in [1.232, 2.328]$  (max/min  $1.89 \leq 2.0$ , log–log slope  $+0.106 \leq 0.15$ ) and  $\sqrt{(G_+^{-1})_{00}} \in [0.661, 0.899]$  from  $h = 168$  to 1445: THE FIRST ANALYTIC UNIFORMITY STATEMENT OF THE ROUTE, at entrywise grade; the two honest FAILS ARE the finding — the analytic envelope certifies NO cell decay (every Prüfer cell contains near-diagonal geometric pairs;  $K_1^{\text{an}}/K_2^{\text{an}}$  blow up as  $h^2$ ): the measured cell-envelope decay is CANCELLATION-CARRIED, provably not entrywise-derivable; and the frozen discrimination bar was pointed at the wrong constant ( $C_{\text{int}}$  itself discriminates: 1.41 truth / 3.59 Epstein / 12.8 scramble); the gap decomposition: no predeclared accounting below 3.568 (bar 1;  $C_1$ – $C_4$  damped censuses + cell variants  $n = 8/16/32$ ), diagonal share 15–24%,  $\sum_r \varepsilon_r = 6.9$ – $12.9$   $\alpha$ -uniform on clean complete-comb data (deep holdouts IN/ABOVE the battery band, slope  $+0.237$ ) — the round-36 gap ( $B_C \sim 4$  vs 1, local defects  $O(10)$ ) is STRUCTURAL to every cell frame: NO cell-partition Cotlar accounting of this family can certify; (2) THE TERMINAL STATEMENT (wave\_packet\_partition\_probe, 6/6, verdict partition-class-closed): the typed next object — a non-cell partition where the diagonal carries the mass — is built honestly and KILLED BY MEASUREMENT: both coherent-state designs (rank-Gabor on the node-rank phase space; chain packets through the exact discrete OP transform) are honest TIGHT frames on every rung ( $\|B\|_2 = \|C^G\|_2$  exact at  $10^{-8}$  rel; the gauge identity  $\sigma_{\max}(C^G)^2 = 1 - \lambda_1(\Delta)$  exact); the Gabor envelope DECAYS in phase space



(0.529  $\rightarrow$  0.007 at kz 243 — the first partition of the program to show decay) and the discrimination is massive (truth  $E_{\text{Schur}}$  4.985 vs Epstein 128.7 vs scramble 9006.6); BUT the diagonal does NOT carry under ANY pairing (nearest-packet share  $\leq 0.070$ , empirical-partner share  $\leq 0.183$  at injectivity  $\leq 0.77$  — diagonal mass  $\leq 18\%$ ) and the Schur constant GROWS (4.985  $\rightarrow$  9.222, slope +0.295; every functioning design  $\geq 3.5$  on every rung) — the cancellation that holds  $\|C\| \leq 1$  is ATOMICALLY GLOBAL, it lives in NO partition of this family; (3) THE BOOKING: ledger 971  $\rightarrow$  972 (PRIME.PARTITION.CLOSURE.01) with the dated round-37 notes on PRIME.CARLESON.PRIME.01 (the constraint set extended: the embedding certificate must be NON-DECOMPOSITIONAL — eigen-enclosure / Loewner-order / passivity, all three measured territory; the new carried assets typed: the tridiagonal identity, the  $h$ -uniform derived constants, their source-law persistence as  $h \rightarrow \infty$  as the named open object) and PRIME.PRUEFER.COMPENSATION.02 (from BOUNDED to CLOSED-AS-A-CLASS, with the stop-list entries: wave-packet / coherent-state partitions of the contractor family; entrywise bounds on the cell envelope); papers: the prime-front round-37 section “the partition closure” + the open-fronts chain extended, tfpt\_3 audit card, tfpt\_research\_contracts recorded block + the dated Carleson note; website mirrors and generated surfaces synced (872/v879); NO marker moves; NO RH claim anywhere)

2026-08-08 (LXXIX) (reflux round thirty-six — the complete-comb reversal: the deep-alpha negatives were ARTIFACTS, the preregistered Cotlar route REOPENS under its own frozen bars, the Prime Carleson chain is machine-exact, three exact compiler theorems land with kernel-checked mirrors, and the ledger gets its first QA fix plus a CI guard: FIVE modules (v874–v878; suite 866  $\rightarrow$  871) — (1) THE ARTIFACT (v877, 8/8, deep-alpha-artifact): the negative Krein floors at kz 142/177/243 ( $\lambda_1 = -331/-135/-215$ , two independent probes) were COMB-TRUNCATION artifacts — core.build\_window silently caps its atom table at  $\text{ATOM\_MAX} = 4 \times 10^5$  while those windows demand support to  $4.4 \times 10^5/8.0 \times 10^5/1.8 \times 10^6$ ; with the COMPLETE canonical comb the certified corner enclosures are strictly positive ( $+1.06 \times 10^{-5}/+1.52 \times 10^{-5}/+1.20 \times 10^{-5}$ ) and the Cholesky-certified full and Krein floors positive on all three rungs (extended builder = the IDENTITY on the controls, bit-equal atoms); NO negative canonical window exists; the certified ladder’s end at kz 121 was the atom-table DATA boundary, not a structural  $\alpha$ -wall; ( $H_{\text{cof}}$ ) unaffected; the silent truncation typed an infrastructure BUG with the registered follow-up (the builder must fail loudly beyond its table); (2) THE REVERSAL (v877, 9/9, cotlar-bounded-v2, the round’s headline): v2 re-executes the round-35 contract with cells, pairing, danger geometry and decision bars frozen IDENTICALLY (v1 spec SHA 4621b899... re-verified at run time; v2 spec SHA 5fd6bf61...; the single substantive change — complete combs at the deep holdouts — warded bit-identical elsewhere): the deep sums COLLAPSE INTO THE BATTERY BAND (kz 142:  $S_U 32.07 \rightarrow 5.65$ ,  $S_C 58.78 \rightarrow 4.10$ ; kz 177:  $14.29 \rightarrow 5.97$ ; kz 243:  $13.94 \rightarrow 5.87$ ), both channels BOUNDED by the frozen v1 rule (ratios  $0.95/0.94 \leq 1.2$ , log-log slopes  $0.006/0.007 \leq 0.10$ ) — the v1 kill was ENTIRELY the artifact; RUN 3 (gated on bounded, correctly withheld in round 35) unlocks and PASSES ( $K_1/K_2$   $h$ -stable on all five complete-comb holdouts: 8.36/8.14 and 40.35/40.47, bar  $1.2\times$ ); Epstein/scramble still break the contrast bars; the truncation audit re-types every affected finding DATED (pruefer v1 confounded  $\rightarrow$  re-decided; residual\_quadrature STRENGTHENED — its stalls detected the truncation correctly, a validated sign detector; the resonance  $\alpha$ -law addendum void; the poly-

gon holdout rows numeric-void; radau/preconditioner/cdcore/uvarov/skeleton unaffected); THE DATED CORRECTION on PRIME.KREIN.CONTRACTOR.01: the round-35 closing note is corrected — the v1 verdict was correct ON THE DATA AS MEASURED and v873's booking stands as the record of the preregistered v1 execution, but the stop-list entry is AMENDED (the class is NOT closed), the  $\alpha$ -not- $h$  law is retired (the truncation's signature), and the route reopens as PRIME.PRUEFER.COMPENSATION.02 with the remaining gap typed ( $B_C \sim 4$  vs 1; local defects 7–10); (3) THE CARLESON WAVE (v876,  $6/8 + 7/8 + 5/5$  with the three frozen-honest FAILs S3.1/S3.2/S2.1 pattern-gated at exit 1, NOT refit): the Carleson ward  $\lambda_{\max}(M_-) = \|C\|^2 = 1 - \tau$  at rel  $1.5 \times 10^{-13}/1.3 \times 10^{-15}$  makes arrows (i)–(iii) of the Prime Carleson chain MACHINE-EXACT (embedding constant  $\|C_h\|^2 = 1 - \tau_h$ ), the surviving analytic target registered as PRIME.CARLESON.PRIME.01 [O]; the polygon  $S_0 \geq |S_2|$  BREAKS on all 47 rungs at  $2\theta \approx 1.5\pi$  (typed: the WRONG compression of the Loewner order) and the holonomy is null with inverted contrast; the residual quadrature realizes  $\rho_h$  at FULL degree on all 44 floor-positive rungs ( $k^*/h$  median 1.000, slope 0.974; mp escalation past the 53-bit wall; a single atom just beyond the pole port; discrimination 7–9 $\times$ ); the resonance is THICK ( $\sim 1/5$  constant, excision gains nothing, soft direction disjoint — the wall is phase-distributed); (4) THE THEOREMS (v874  $14/14 + 11/11 +$  v875  $15/15$ , with the Lean mirrors TfptCarrier/FlavorFeedback.lean and TfptCarrier/GaussianShells.lean, 18 theorems, lake build green 3418 jobs, axioms propext/Classical.choice/Quot.sound, Lean manifest  $88 \rightarrow 90$  files): the FEEDBACK NORMAL FORM ( $P^{-1}R_sP = [[s+4, 0, 3], [0, 0, -2], [4, 1, 5]]$  exact symbolic in  $s$ ;  $\chi_{R_s} = (t - s - |\mu_4|)q_{\text{wind}} - 12t$  with  $12 = N_{\text{fam}}|\mu_4| = \dim g_{\text{SM}}$ ; the honest  $\mathbb{Z}_2^2$ -not- $\mu_4$  typing kernel-checked); the WINDING DECODER ( $q(\{2, 3, 5, 6\}) = (-4, -4, 2, 8)$ ;  $D(y) = (y+4)^2(y-2)(y-8)$  with budget coefficients (2, 48, 32, 256);  $|\mu_4|q(|Z_2|)/|R(E_8)| = -1/15 =$  the Walsh message eigenvalue — typed AUDIT theorem, not a functor); and the GLOBAL SHELL THEOREM ( $\hat{\Theta}_u/\Theta_L = -1/15$  for ALL odd shells from  $\text{Sp}(4, 2)$  transitivity +  $(1+i)$  doubling +  $\Theta_L = 240\sigma_3$  — the  $n \leq 16500$  census REPLACED as argument, retained as ward); (5) THE LEDGER QA FIX + THE GUARD (v878, 10/10, GATE.LEDGER.01): the first PRIMARY-KEY violation found and fixed — FORM.QGE0.03 was double-booked (2026-06-15 CohomologyGrading; 2026-06-18 SeamDeckClosure flat closure); the SeamDeckClosure row RENAMED FORM.QGE0.04 with a dated QA note and every flat-closure reference updated INCLUDING the two 2026-06-18 mentions in this changelog (corrected in place, documented here); the guard enforces unique claim ids, acyclic supersession, unique active supersession targets modulo a frozen 2-pair allowlist, and the dependency cross-reference cycle census frozen remove-only; NO marker moves; NO RH claim anywhere)

- **New modules (5).** v874\_feedback\_decoder\_lean.py (14/14 + 11/11 probe gates + 4/4 Lean mirror checks, FEEDBACK-NORMALFORM-EXACT + WINDING-DECODER-EXACT; probes flavor\_feedback\_probe.py + g31\_winding\_decoder\_probe.py embedded BYTE-EXACT and re-run verbatim; the Lean mirror TfptCarrier/FlavorFeedback.lean, 11 theorems); v875\_shell\_global\_lean.py (15/15 + 4/4, SHELL-GLOBAL-THEOREM; probe gaussian\_shell\_global\_probe.py; the Lean skeleton TfptCarrier/GaussianShells.lean, the ShellSystem hypothesis bundle + 7 consequence theorems); v876\_carleson\_polygon\_wave.py (6/8 + 7/8 + 5/5 with the three frozen-honest FAILs pattern-gated at exit 1; probes phase\_polygon\_probe.py, residual\_quadrature\_probe.py, pi\_resonance\_anatomy\_probe.py,  $\sim 9$  min); v877\_complete\_comb\_reversal.py (8/8 + 9/9; probes deep\_alpha\_sign\_probe.py + cotlar\_v2\_complete\_comb\_probe.py,  $\sim 17$  min; the

v1/v2 spec-SHA chain machine-warded); v878\_ledger\_integrity.py (10/10, the CI guard). Suite 866 → 871; all five standalone green; run\_all ALL CHECKS PASSED.

- **The reversal (the round’s deliverable).** The discipline machinery worked end to end on its own freshest negative: the residual horizon flagged the deep stalls, `deep_alpha_sign` typed the silent comb truncation at certificate grade ( $\text{ATOM\_MAX} = 4 \times 10^5$ ; complete-comb floors strictly positive — NO negative canonical window), and the IDENTICALLY frozen round-35 bars re-ran on complete data and REVERSED the closure: COTLAR-BOUNDED-V2, run 3 unlocked and passed, the route reopened as `PRIME.PRUEFER.COMPENSATION.02`. The correction is booked DATED on every surface (ledger, papers, website) while v873 and the round-35 records stand as what was measured; the remaining gap is typed, not hidden ( $B_C \sim 4$  vs 1; local defects 7–10). NO RH claim.
- **Ledger.** 13 new rows (`FLAVOR.FEEDBACK.NORMALFORM.01`, `E8.WINDING.DECODER.01`, `FORM.FLAVOR.FEEDBACK.01` [F], `E8.SHELL.GLOBAL.01`, `FORM.E8.SHELL.GLOBAL.01` [F], `PRIME.PHASE.POLYGON.01`, `PRIME.RESIDUAL.QUADRATURE.01`, `PRIME.PI.RESONANCE.ANATOMY.01`, `PRIME.CARLESON.PRIME.01` [O], `PRIME.DEEPALPHA.SIGN.01`, `PRIME.PRUEFER.COMPENSATION.02`, `PRIME.TRUNCATION.AUDIT.01`, `GATE.LEDGER.01`) + the dated round-36 CORRECTION notes on `PRIME.KREIN.CONTRACTOR.01` (the stop-list entry amended, the route reopened, the  $\alpha$ -not- $h$  law retired) and `PRIME.PRUEFER.COMPENSATION.01` (the v1 record stands as-measured) + dated notes on `E8.SIMPLEX.FOURIER.01` (census → theorem), `E8.G31.CLOCK_ALPHABET.01` (the decoder) and `PRIME.FLOOR.SKELETON.01` (the ladder end was a data boundary) + THE QA FIX: `FORM.QGEO.03` was double-booked since 2026-06-18 (a primary-key violation); the SeamDeckClosure flat-closure row is RENAMED `FORM.QGEO.04` with a dated QA note, all references updated (papers, scripts, registry, website, and the two 2026-06-18 entries of this changelog corrected in place — documented here); 958 → 971 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round-36 section “The complete-comb reversal” (`\veri{v876/v877/v878}`; the artifact, the reversal, the Carleson chain, the theorems crossref) + the DATED CORRECTION paragraph appended to the round-35 section + the open-fronts chain extended with round 36 and its round-35 clause corrected; `tfpt_1`: two new Gaussian-code keybox entries (the global shell theorem `\veri{v875}`; the winding decoder `\veri{v874}`) + the census sentence upgraded; `tfpt_2`: the feedback normal form added to the winding keybox (`\veri{v874}`); `tfpt_3`: five new audit cards + dated corrections on the v857/v873 cards; `tfpt_research_contracts`: the `PRIME.CARLESON.PRIME.01` [O] registration, three round-36 recorded blocks, the round-36 correction notes on the contractor and paircorr blocks, and the `GATE.LEDGER.01` QA record.
- **Website.** Prime-front feed round 36 (`primeFront.ts` + the verdict enum); counters 871/v878 (HonestyBanner, prime-front page metadata and hero); DAG +v874–v878; `papers.ts` (the research-contracts round-36 correction, the architecture –1/15 upgrade); `ResidualChain.tsx` carries the renamed `FORM.QGEO.04`; generated surfaces (`ScriptIndex`, `changelog.ts`, `suite.ts`, `discipline.ts`, maps) regenerated; README + Zenodo synced to 871/v878; manifests regenerated last (Lean manifest 88 → 90 files).
- **Infrastructure note (typed as a bug, follow-up registered).** `core.build_window` silently truncates the canonical comb at its atom-table cap; until the loud guard lands, every deep-window readout must check the demand  $e^{2\alpha} \leq \text{ATOM\_MAX}$  explicitly (the round-36 modules do).

2026-08-08 (LXXVIII) (reflux round thirty-five — the Cotlar decision: the first fully preregistered uniformity test of the program, EXECUTED: ONE module (v873, the executed round-34 preregistration `PRIME.PRUEFER.COMPENSATION.01`; suite 865 → 866) — the composite verdict cotlar-growing / route dead BY THE

CONTRACT'S OWN FROZEN RULES, booked exactly as frozen: (1) THE PROVENANCE CHAIN (machine-warded in the module's gate) — contract frozen + hashed (SPEC SHA-256 4621b899...7440811; 8/8 synthetic freeze wards; the decisive analysis locked behind the operational flag `pruefer_unfreeze.flag`, ABSENT at freeze) → committed byte-exact PRE-EXECUTION in the round-34 commit 526ca3eb (FILE SHA-256 93208a1b...4825d9fd — the proof of precedence) → the unfreeze flag created AFTER that commit with its audit line committed → executed with exactly TWO documented MECHANICAL fixes, both typed verbatim in the probe (the  $h \leq 900$  battery filter that the frozen spec's own battery definition mandates — the committed code omitted the spec's own text; the per-step joint positive renormalization against float overflow at near-breakdown chain tails — invariant for `atan2` and the wrapped increments in exact arithmetic, validated at  $\leq 6.6 \times 10^{-4}$  against a 60-digit `mpmath` reference), the FROZEN\_SPEC literal byte-identical throughout (SHA re-verified at every stage; executed FILE SHA-256 15d35b26...5ffb4cc) → re-executed verbatim at promotion (embedded BYTE-EXACT, RUN 1 + RUN 2, ~22 min); (2) RUN 1 (the anatomy, anchors kz 9/12/13/26/40) — the predeclared danger geometry is REAL: the frozen cells  $\{0, 15\} \cup \{7, 8\}$  capture 0.98/0.61/0.94/0.85/0.81 of  $\lambda_{\min}(T_1)$  — source-only Prüfer phase cells localize the negativity WITHOUT the computed soft eigenvector (kill 3 silent); but the LOCAL compensation fails ( $\sum_r \varepsilon_r = 6.88\text{--}10.08$ ) and the cross-block envelope does NOT decay (Spearman  $-0.17\text{..}0.00$  vs the frozen  $\leq -0.8$  bar, BIMODAL with the second peak exactly at the predeclared  $\pi$ -resonance cells  $d = 7\text{--}8$ ); (3) RUN 2 (the Cotlar decision, the 42-rung battery + the frozen deep holdouts kz 90/116/142/177/243) — the battery is FLAT ( $S_U$  5.55–6.30,  $S_C$  3.97–4.78) and the deep holdouts EXPLODE (kz 142:  $S_U$  32.07/ $S_C$  58.78; kz 177: 14.29/30.76; kz 243: 13.94/44.85), both channels GROWING per the frozen rule ( $\log\text{--}\log$  slopes  $0.209/0.408 \geq 0.20$ ), the contrast bars PASS (Epstein max rel 3927% / local-defect ratio 224×; scramble 314462% /  $1.26 \times 10^6$  — kill 4 silent), and the measured carry-forward law is typed: THE GROWTH TRACKS kz-DEPTH ( $\alpha$ ), NOT WINDOW SIZE  $h$  (kz 90/116 at  $h = 1430/1433$  stay INSIDE the battery band while kz 177 at SMALLER  $h = 1219$  explodes) — any uniform certificate must be  $\alpha$ -aware; RUN 3 correctly NOT executed (gated on BOUNDED); (4) THE KILL TABLE — kills 2 (power-law Cotlar growth) and 5 (the compensation stays global) FIRE, kills 1/3/4 do NOT; the blockwise Cotlar–Stein / phase-cell class on Prüfer phases is CLOSED and stop-listed; the surviving demand narrowed: a NON-STATIONARY phase bound on the explicit kernel with the  $\pi$ -resonance as the known critical structure; the honest FAILs pattern-gated, nothing refit — preregistration + frozen bars + committed hashes is the standard the program now holds itself to; NO marker moves; NO RH claim anywhere)

- **New module (1).** `v873_pruefer_cotlar_decision.py` (the executed preregistration: the probe `pruefer_compensation_probe.py` embedded BYTE-EXACT in its EXECUTED form and re-run verbatim in an isolated namespace — RUN 1 + RUN 2, ~22 min; the pattern gate encodes the frozen census INCLUDING the honest FAILs: the provenance wards P1–P3 (byte-equality vs the experiments source, the executed file SHA, the committed unfreeze-flag audit line), the spec-SHA ward G0, the danger-geometry capture bands, the local-dominance and envelope-decay FAILs, the battery/holdout Cotlar censuses, the frozen GROWING decision with slope bands, the  $\alpha$ -not- $h$  law, the contrast bars, the kill table, and the RUN-3-withheld ward). Suite 865 → 866; standalone green; the two mechanical fixes documented verbatim in the probe's own docstrings with the spec literal SHA unchanged.

- **The verdict (the round’s deliverable).** The program’s first fully preregistered uniformity test returns its own frozen verdict, and the negative is booked with the same machinery as a positive would have been: the blockwise Cotlar–Stein route on source-only Prüfer phase cells is DEAD — the Cotlar sums are  $h$ -stable on the battery but explode on the frozen deep holdouts as a power law in depth. What survives, typed: the danger-geometry positive (source-only cells find the dangerous blocks without the soft eigenvector), the truth/Epstein/scramble discrimination, the  $\pi$ -resonance as the named critical structure, and the  $\alpha$ -not- $h$  growth law as a constraint on every future candidate. NO RH claim.
- **Ledger.** 1 new row (PRIME.PRUEFER.COMPENSATION.01, the executed contract with the full provenance chain and the frozen census) + the dated round-35 closing note on PRIME.KREIN.CONTRACTOR.01 (the blockwise Cotlar/phase-cell class CLOSED by the pre-registered test’s own rules; the STOP-LIST entry: blockwise Cotlar–Stein on Prüfer phase cells; the demand narrowed to a non-stationary phase bound with the  $\pi$ -resonance critical structure and the  $\alpha$ -not- $h$  constraint; the danger-geometry positive and the discrimination recorded as carry-forward assets); 957  $\rightarrow$  958 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The Cotlar decision: the first fully preregistered uniformity test, executed” (`\veri{v873}`; the contract, the execution discipline, the verdict, what survives, and the honest note that preregistration + frozen bars + committed hashes is the program’s standard) + the round-34 section’s “not run” prose given its forward pointer + the open-fronts chain extended with round 35; `tfpt_3`: the audit card; `tfpt_research_contracts`: the round-35 dated notes on the PRIME.KREIN.CONTRACTOR.01 block and the contractor registration block (the execution record with the executed file SHA verbatim).
- **Website.** Prime-front feed: the round entry “reflux round thirty-five”; counters 865  $\rightarrow$  866 modules / `v535–v873`; DAG extended with `v873`; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 866 modules / `v873`.
- **Scope.** One module, one executed contract: the verdict is a route decision under the strictest protocol the program has used — every bar, kill and holdout frozen and hashed BEFORE execution, the frozen file committed pre-execution, the two mechanical fixes documented verbatim with the spec literal SHA unchanged, the honest FAILs pattern-gated and NOT refit, RUN 3 correctly withheld by the contract’s own gating; the unfreeze flag with its audit trail is committed with this round; NO marker moves; NO RH claim anywhere.

**2026-08-08 (LXXVII) (reflux round thirty-four — the Christoffel–Darboux wave: the kernel NAMED, the compensation identity EXACT, the measure-relation closure, and the preregistered next test: FOUR modules (`v869–v872`, from seven frozen probes; suite 861  $\rightarrow$  865) plus the FROZEN preregistration PRIME.PRUEFER.COMPENSATION.01 committed byte-exact and NOT run — the consolidation of the 2026-08-08 evening CD/Uvarov campaign in three waves plus one commit-time contract: (1) THE CD WAVE (the kernel identification) — the reproducing kernel  $K_+$  of the closed-form SourceContractor (`v866`) is EXACTLY the Christoffel–Darboux kernel of an explicit source measure  $\mu = \sum [2 \sin^2(\varphi/2) d_+(\varphi)/L] \delta_{2 \cos(D\tau)}$ : the displacement rank collapses from the growing  $\tau$ -coordinate budget (12–14, the  $\sqrt{L}$  law) to EXACTLY 2 in the multiplication coordinate  $x = 2 \cos(D\tau)$  with a hard  $\sigma_2/\sigma_3$  gap of  $6.6 \times 10^{12}$ – $4.8 \times 10^{13}$  on ALL 7 rungs including the blind holdouts `kz 40/49/60`, and the FIREWALLED chain-only reconstruction (Stieltjes–Lanczos source Jacobi chain + node/geometry data ONLY, enforced by the reconstruction signature) rebuilds  $C$  entrywise at  $8.1 \times 10^{-14}$ – $5.3 \times 10^{-13}$  — the whole contractor is three named objects: the arm weights, the phase/geometry factors, and ONE source Jacobi chain, with**

the honest boundary typed PRE-RUN and confirmed (the rank-2 collapse is WINDOW GEOMETRY — the controls collapse too, exactly as the derivation forces for any even density; the arithmetic lives in the MEASURE: control chains  $O(1)$  off at  $0.038/0.062$  vs the  $10^{-2}$  bar); the  $\mu_4$  quarter-turn Clark-phase hypothesis is falsified SHARPLY (the coupled pairs obey the EQUAL-PHASE law: offset  $-0.5..+0.6^\circ$  at  $\bar{R}$   $0.97-0.99$ , NOT  $90^\circ$ ; the Clark rank leg dies at  $E$ -rank  $85-99$ ; the phase coordinate itself source-only clean, circularity fence exact); the factorization  $C = D_- U D_+$  assembles EXACTLY ( $1.3 \times 10^{-14}$ ) with  $\|D_+\| \leq 0.9654$  by the Christoffel variational theorem,  $\|D_-\| \leq 0.8655$  measured, and  $U$  non-unitary BY THEOREM (the deployed grid  $2\times$ -oversamples the folded pairs: rows  $\geq 1.158$  — the  $\sqrt{2}$  over-normalization is structural); and the Gauss-node frame DECIDES the repair: exact where algebra promises ( $D_+^G = I$  at  $4.7 \times 10^{-11}$ , lossless bridge, frame-invariant port share — theorem P5) yet the unitary core does NOT exist (co-isometry defect  $O(1)$ ,  $0.93-1.30$ , cross-measure node misalignment at near-pole pairs), the surviving asset being the LADDER-WIDE damping bound  $D_-^G \leq 0.997041$  with  $34\times/2900\times$  discrimination and the named hypotheses (H-D)/(H-U)/(H-PORT) registered; (2) THE UVAROV WAVE (the measure-relation decision, two honest kills with typed residues) — NONE of the three classical spectral transformations (Uvarov/Christoffel/Geronimus) relates the two channel measures exactly or at fixed small rank at the source pole  $x_{\text{pole}} = 2 \cosh(D/2)$  (machinery warded exact on synthetics at  $9.2 \times 10^{-16}$ ; typed residue: Geronimus with SMALL NEGATIVE MASS uniformly closest at  $3.5-8.1\%$  Weyl residual, non-growing, discriminating — misses the  $5\%$  bar, no  $\lambda(D)$  law), and the co-isometry defect is DIFFUSE: full-rank, growing LINEARLY with  $h$  (Pearson  $+0.996$ ,  $\times 3.39$  first-to-last third), pole removal changes nothing, the closest-co-isometry bracket  $O(1)$  at near-full rank — NO finite-matrix compensation exists at this ingredient list (both closures stop-listed); (3) THE HEADLINE (the compensation identity) — the defect splits EXACTLY into two source-built terms,  $I - C^*C = T_1 + T_2$  with  $T_1 = I - U^*U$  (the INDEFINITE cross-measure Jacobi geometry) and  $T_2 = U^*(I - D_-^2)U$  (the PSD arithmetic Christoffel damping), warded at  $10^{-12}$  on all 42 rungs and transported losslessly into the Gauss frame; the EXACT global minimizer of the directional compensation ratio IS the soft direction (overlap  $\geq 0.9991$ ; min ratio  $= 1 + \tau/|v^*T_1v|$  to all printed digits), the rank-one pole deflation gives  $\Delta = \Delta_{\text{bulk}} + spp^*$  with the certified bulk floor at  $98-100\%$  of  $\lambda_2$  and  $s = \kappa\tau$  reproducing the v864  $\kappa$  law — THE WALL AS TWO EXPLICIT SOURCE NUMBERS (kz 9 at the soft direction:  $-0.0328 + 0.0330 = 1.68 \times 10^{-4} = \tau$ ), with the honesty line typed (the truth-side ratio  $\geq 1$  is implied by upstream certified positivity on these finite rungs — the identity buys the FORM: one two-term inequality at one named direction) and the discrimination anatomy measured (Epstein/scramble BREAK the damping's sign,  $\max D_- = 34.2/2917 > 1$ , and under-compensate on their OWN soft directions at worst ratios  $-175/-446513$ ); (4) THE PREREGISTRATION (committed, NOT run) — PRIME.PRUEFER.COMPENSATION.01: source-only Prüfer phases of the two arm chains, EXACTLY 16 fixed phase cells (boundaries  $k\pi/8$ , entrywise pairing primary, danger families predeclared), Cotlar–Stein decision rules and all success/kill criteria frozen BEFORE any deployed-data run, deep holdouts kz 90/116/142/177/243 never used by the CD wave, the decisive analysis locked behind an operational unfreeze flag,  $8/8$  synthetic freeze wards green; SPEC SHA-256 4621b899...7440811, FILE SHA-256 93208a1b...4825d9fd, committed byte-exact (any modification invalidates the preregistration); NO marker moves;

## NO RH claim anywhere)

- **New modules (4).** `v869_cdcore_clarkphase.py` (10 + 8 checks, zero fails, CDCORE-IDENTIFIED + CLARKPHASE-DEAD; probes `cdcore_probe.py` + `clarkphase_probe.py`: the frame factorization  $F = \text{diag}(\gamma) \text{diag}(s) V_{\text{poly}}$  machine-exact ( $3.0 \times 10^{-13}$ , Chebyshev 4th kind), the reproducing/projection wards ( $P^2 = P$ ,  $PF = F$  at  $10^{-14}$ ), the decisive displacement test (x-rank 2 with the  $10^{13}$  gap vs  $\tau$ -rank 12–14, Spearman +0.96, slope +0.17 vs  $\sqrt{L}$ 's +0.50), the firewalled CD reconstruction within certified budgets on construction AND holdouts, the entrywise weighted transport ( $6.8 \times 10^{-13}$ ), the leverage trace ward ( $\sum l_j = h$  at  $9.3 \times 10^{-16}$ ), the structural control collapse typed pre-run with the  $O(1)$  chain discrimination; the Clark falsification: passivity and spread wards, the E-rank leg (85–99 vs the  $\leq 8$  bar), the equal-phase offset law ( $\bar{R} \geq 0.97$ ), the exact circularity fence ( $\max |dz| = 0.0$  under comb perturbation), Epstein/scramble discriminate in E-rank/offset); `v870_christoffel_gauss_frame.py` (17/17 + 9/9, CHRISTOFFEL-PARTIAL + GAUSS-STILL-DEFECTIVE; probes `christoffel_transition_probe.py` + `gauss_node_unitary_probe.py`: the classical wards W1–W6 (trace, variational, Golub–Welsch, bridge, assembly, frame transfer), the row theorem P3, the payoff census (product bound  $\leq 1$  on 0/42 rungs — honest), the softport regression  $\beta(kz9) = 0.6057$  / ladder max 0.8591 with the sharpening question answered (pole share  $\sim 0.86$  in all three coordinate systems); the Gauss frame: tightness/Gauss–Christoffel/unit-rows exact,  $\sigma$ -spectra match the grid contractor,  $\tau$  lossless, port share frame-invariant, the co-isometry certificate 0.93–1.30 BREACHED (the measured content),  $D_-^G \leq 0.997041$  ladder-wide, the named hypotheses registered); `v871_pole_uvarov_crossdefect.py` (7 + 11, POLE-TRANSFORM-DEAD + CROSSDEFECT-DIFFUSE; probes `jacobi_pole_uvarov_probe.py` + `gauss_crossdefect_probe.py`: the machinery ward on synthetics ( $9.2 \times 10^{-16}$ ), the two-measure wards with the cdcore regression, the three-transformation census (all FULL/PARTIAL False), the Geronimus residue with the blind holdout transfer (ratio 0.97) and the  $2.58 \times 2.14$  discrimination; the defect anatomy: traceless ward ( $10^{-13}$ ), the rank-growth law ( $\times 3.39$ , Pearson +0.996), mode stability ( $0.123 \leq 0.15$ ), the  $U_0 + R$  split with the PSD base theorem and the  $\tau$  ward ( $\lambda_{\min}(\Delta_{\text{pos}}) = \tau$  to all digits)); `v872_damping_compensation.py` (10, COMPENSATION-EXACT; probe `cd_damping_compensation_probe.py`, spec v2 with the basis-free exact-minimizer amendment: the split ward ( $10^{-12}$ ), the unitary transport ( $7.7 \times 10^{-11}$ ), the exact Feshbach deflation ( $\lambda_{\min}(\Delta_{\text{bulk}}) = 0$  at  $2.6 \times 10^{-15}$ ), the  $\kappa$  regressions, the T2 sign theorem (zero negative entries), the exact global compensation census (min ratio 1.000008, implied by upstream positivity — the anatomy, not new evidence), the bulk floor ( $c/\lambda_2 \geq 0.98$ ), the four-way alignment table, the isolated scalar, and the discrimination anatomy). Suite 861  $\rightarrow$  865; all four standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all seven probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).
- **The preregistration (committed, NOT run).** `experiments/tfpt-discovery/pruefer_compensation_probe.py` — the FROZEN contract PRIME.PRUEFER.COMPENSATION.01: source-only Prüfer phases (structural AST firewall on the construction functions), EXACTLY 16 fixed phase cells with the entrywise pairing primary and the danger families DGR1/DGR2 predeclared, RUN 1/2/3 protocols (anatomy / Cotlar–Stein decision / uniformity candidate) with all bars, verdict rules and kills frozen BEFORE any deployed-data run, deep holdouts kz 90/116/142/177/243 (never used by the CD wave), the decisive analysis locked behind the operational flag file `pruefer_unfreeze.flag`, and 8/8 synthetic freeze wards green at freeze time. Committed BYTE-EXACT with this round: SPEC SHA-256 4621b89958531d6d3baae9fe762dbd10d23dc22eed77a57aa1c6b179a7440811, FILE SHA-256 93208a1b2edebde31388a87192055c9bf3e5a3de29af9fb3e34ddc904825d9fd; any modification invalidates the preregistration. Booked as a dated PREREGISTRATION note on PRIME.KREIN.CONTRACTOR.01 (the contract's next test); NO vN module, NO analysis run,

NO result claimed.

- **The compensation statement (the round’s deliverable).** The wall’s operator coordinates gain their classical names: the contractor’s kernel IS the Christoffel–Darboux kernel of one explicit source measure (the “non-monomial band-spreading letter” of rounds 32–33 is NAMED at kernel level — with the honest boundary that the naming is geometry, the arithmetic is the chain and the weights), the defect is the EXACT sum of an indefinite Jacobi cross-geometry and a PSD Christoffel damping, and the floor demand is ONE two-term inequality at ONE named direction with both terms explicit source objects ( $s = \kappa\tau > 0$ ). The classical escapes are closed with certificates and stop-listed: classical pole transformations at  $x_{\text{pole}}$  (typed Geronimus residue), finite-rank cross-defect corrections (linear rank growth),  $\mu_4$ -offset Clark kernels (the equal-phase law). The next test is PREREGISTERED, not run. NO RH claim.
- **Ledger.** 4 new rows (PRIME.CONTRACTOR.CDCORE.01, PRIME.CONTRACTOR.CHRISTOFFEL.01, PRIME.JACOBI.POLE.UVAROV.01, PRIME.CD.DAMPING.COMPENSATION.01) + the dated round-34 note on PRIME.KREIN.CONTRACTOR.01 (the kernel NAMED; the compensation identity; the Uvarov/finite-rank/Clark closures with the stop-list entries; the Prüfer preregistration with BOTH SHAs recorded verbatim) and the dated round-34 note on PRIME.FLOOR.PAIRCORR.01 (the wall as the two-term inequality at the soft direction, both terms source-built); 953 → 957 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The Christoffel–Darboux wave: the kernel named, the compensation exact, the anatomy of discrimination, and the preregistered next test” (`\veri{v869..v872}`) + the part-intro chain extended; `tfpt_3`: four new audit cards; `tfpt_research_contracts`: the round-34 dated notes on the PRIME.KREIN.CONTRACTOR.01 and PRIME.FLOOR.PAIRCORR.01 blocks including the preregistration record with both SHAs.
- **Website.** Prime-front feed: the round entry “reflux round thirty-four”; counters 861 → 865 modules / v535–v872; DAG nodes extended with v869–v872; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 865 modules / v872.
- **Scope.** The CD-wave modules are exact structural namings and typed route decisions under frozen preregistered protocols (the derivations typed BEFORE running; the structural non-discrimination of the rank collapse typed PRE-RUN — window geometry, not a kill; the non-unitarity of  $U$  a THEOREM, not a miss; the compensation census implied by upstream positivity and typed as anatomy); the Uvarov/crossdefect modules are honest kills with typed residues (machinery warded on synthetics first); the preregistration is a COMMIT-TIME contract — no analysis mode was executed, the file hash is the proof of freezing; NO marker moves; NO RH claim anywhere.

**2026-08-08 (LXXVI) (reflux round thirty-three — the wall’s operator form: the Krein coordinates, the SourceContractor formula, the Perron closure, and the quadrature reach: EIGHT modules (v861–v868, from sixteen frozen probes plus one Lean module; suite 853 → 861) — the consolidation of the 2026-08-08 Krein/port/formula campaign in five strands: (1) THE KREIN CHAIN — the deployed window form gets an EXACT Krein normal form  $Q_h(t) = \|B_+t\|^2 - \|B_-t\|^2$  (entrywise ward  $\leq 10^{-15}$ , both canonical cuts;  $B$  carries  $\sqrt{|\Lambda|}$  — linear in  $\Lambda$ , the round-32 grade barrier passed by construction) with the Douglas pencil identity  $1 - \|C_2\|^2 =$  the  $\tau$ -margin in the positive-side metric (rel  $\leq 3.9 \times 10^{-9}$ ): truth contracts on every rung, Epstein  $h=2$  explodes to  $\|C_E\| = 46.8$ , the scramble to 3693 — the discriminator IS the Douglas equivalence; the register wiring typed (the  $\mu$ -sign and the deck are TWO separate  $C_2$ s, the identification one faithful- $\mu_4$  character away with three candidate constructions machine-verified; the Redheffer/Mertens echo exact:  $\det R_n = M(n)$  with the vacuum column’s**



ENTIRE Smith deviation in the single invariant factor  $|M(n)|$ ; no resolvable floor–Mertens correlation,  $\min 4p = 0.378$ ); the defect theorem kernel-checked (TfptCarrier/KreinDefect.lean: contraction  $\Leftrightarrow$  PSD with the full-rank Douglas converse, SourceContractor the NAMED hypothesis with the circularity warning typed, krein\_cofinal\_weil composing contractors on a cofinal ladder to Weil nonnegativity — FORM.PRIME.KREIN.DEFECT.01 [F]); the defect has rank ONE ( $\lambda_1 =$  the  $\tau$ -margin exactly, separation  $3.9\text{--}14.1\times$  — the Redheffer Smith invariant, the vacuum code’s one-qubit price and the Krein soft port are ONE rank-one structure), the MONOMIAL CLOSURE THEOREM extends the word-census emptiness to EVERY word of ANY length (all source letters monomial, contractor monomial mass  $0.23\text{--}0.24 \Rightarrow$  miss  $\geq 0.874$ ; the 408 sign flips kill the corridor form, typed v2), and the 2-torsion weld law supplies the missing operator’s TYPE (the deck’s anticommutant is entirely faithful- $\mu_4$  — the deck welds only as  $J = MD$ ; all four candidates weld exactly, the control refuses exactly, the contractor connection null at  $\sqrt{2}$ ); (2) THE MACHINE — the 16-slot colligation architecture makes contraction a CONSEQUENCE (exactly unitary,  $\|D\| = 1/4$ ,  $\sigma_{\max} \leq 1 + 10^{-9}$  automatic, provably comb-blind) with the compiler-side discovery booked in Paper 1:  $\mathbb{F}_2^4$  IS the divisor lattice of 210 with Walsh–Hadamard carrying the lattice Möbius function and the scalar-limit determinant  $\prod_p(1 - 1/p) = 8/35$  EXACT; the identification gate fails honestly and localizes the gap — the contractor couples DIFFERENT  $|\tau|$  bands, the  $J$  flip couples equal  $|\tau|$ : the missing object is the grid-side band intertwiner, the explicit-formula kernel itself; (3) THE SCALAR REDUCTION — the exact Feshbach reduction turns the floor into ONE port impedance ( $\tau > 0 \Leftrightarrow s > 0$ ; four source ports pass; the pole port carries 84% of the soft mode with the exact skeleton  $\kappa^{-1} = \beta^2 + \lambda_1\rho$ , the closed-form  $a$ -term — the Poisson/Cauchy average of the signed density at  $s = 1/2$  — and  $\kappa - 1 \approx 6e^{-\alpha/2} \rightarrow 1$  at Pearson  $-0.99$ ; the backflow  $\leq 17$  bulk modes cancelling 99.9%: the classical shape EXISTS, the boundable term is exactly the arithmetic); the divisor-tower Weyl load repairs the hull’s comb-blindness at LOAD level ( $O(1)$  response) and the repaired phase readout discriminates at 1–2 orders (Epstein  $\kappa = 42.7$  / scramble 77.6 outside the  $1.5\times$  truth window; the typed caveat carried: Spearman( $s, \tau$ ) =  $-0.8$ , the discrimination is the load-bearing content); (4) THE FORMULA (the campaign’s headline) — the Douglas contractor factors EXACTLY as the target-free closed-form source expression  $C = W_-FG_+^{-1}F^HW_+$  ( $1.4 \times 10^{-14}$  on construction rungs AND blind holdouts; no zeros, no  $\tau$ , no fit) with CERTIFIED defect transfer on all seven rungs (the frozen Sylvester-12 kill fires as designed, correcting rank-12 to  $\sqrt{L}$  growth,  $k^* = 24.48$ ); the norm statement is the exact reformulation  $\|C\| \leq 1 \Leftrightarrow G_- \preceq G_+ \Leftrightarrow K \succeq 0$  — and the PERRON CLOSURE is proven:  $\rho(|C|) = 2.4\text{--}3.1 > 1$  everywhere, so NO absolute Schur/Hilbert-type inequality can ever certify — cancellation provably carries the arithmetic, the certificate must be PHASE-AWARE (the sharpest demand registered: PRIME.KREIN.CONTRACTOR.01 [O], one oscillatory-integral-grade bound beating 1 by the measured  $e^{\text{slope}\alpha}$  margin — it would discharge SourceContractor and with it ( $H_{\text{cof}}$ )); quadrature certificates for  $s > 0$  EXIST on every rung ( $m^* \leq 426$ , the Neumann frontier killed) but the depth law  $m^* \sim \sqrt{\text{cond}(G)}$  (Spearman  $+0.99$ ) is INTRINSIC — all three source preconditioners fail informatively, the  $m \approx 17$  prediction honestly failed,  $J_\infty$  named the free Chebyshev chain (Rakhmanov-generic typed): the certificate depth is the price of the cancellation; (5) THE AUDITS — the 210 register is canonical up to PROVEN gauge (Boolean layer generic 210/210 as warned;

the Euler determinants pin  $\{2, 3, 5, 7\}$  uniquely; the anchor prime 2 forced by ramification; two gauge classes = the chirality pair, then proven gauge by two exact no-go wards — the orientation functional vanishes identically via the Möbius complement, quadratic readouts provably blind by transposition; the 6-vs-7 note stands), and the  $17 = 12+3+1+1$  moving-sector identification is honestly buried at space level (principal angles  $\sim 90^\circ$ ); NO marker moves; NO RH claim anywhere)

- **New modules (8).** `v861_krein_normalform_lean.py` (37/37 + 8/8, zero fails, DECK-MU-PARALLEL + REDHEFFER-ECHO-EXACT + KREIN-CONTRACTOR-STABLE-NO-WORD, +4 Lean-mirror checks; probes `mobius_deck_redheffer_probe.py` + `krein_normalform_probe.py`: the two- $C_2$  wiring census with the three candidate constructions (diagonal characters  $j \in \{1, 3\}$ ; Galois  $\mu_K((p)) = \chi_4(p)$  exact on 609 + 619 primes; Hall chain parity row  $1 = \mu$ ),  $\det R_n = M(n)$  on 62 sizes with SNF  $(1, \dots, 1, |M(n)|)$  and the mod-2 shadow to  $2 \times 10^6$ , the Krein Gram ward entrywise on both cuts, the pencil identity, the Epstein/scramble discriminators, the 155-word census); `v862_defect_polar_weld.py` (8/8 + 22/22, DEFECT-ONE + MONOMIAL-CLOSURE-THEOREM + WELD-WITHOUT-CONTRACTOR; probes `krein_defect_polar_probe.py` + `mu4_clifford_weld_probe.py`); `v863_redheffer_colligation.py` (7/7, COLLIGATION-CONTRACTS-NOT-IDENTIFIES; probe `redheffer_colligation_probe.py`: unitarity  $10^{-12}$ , automatic contraction, the exact 8/35 scalar limit, the band-transfer census); `v864_softport_kappa_law.py` (5/5 + 6/6, SOFTPORT-FOUND + CAUCHY-RANK-SMALL + KAPPA-LAW-PARTIAL; probes `softport_cauchy_probe.py` + `pole_port_kappa_probe.py`); `v865_weyl_port_readout.py` (9 + 5/5 with the TWO frozen-honest FAILs S1.SPR + S4.RSP kept and pattern-gated, NOT refit, WEYL-PORT-BLIND + READOUT-REPAIRED; probes `divisor_weyl_port_probe.py` + `weyl_readout_repair_probe.py`); `v866_source_contractor_formula.py` (10 + 8/8 with the ONE frozen-honest FAIL S2.2 at exit code 1 per the frozen kill design, SYLVESTER12-ANGLES-FAIL + NORM-REFORMULATION; probes `displacement_sylvester12_probe.py` + `source_contractor_norm_probe.py`); `v867_radau_conditioning.py` (7 + 8 with the TWO frozen-honest FAILs S2.G2 + S3.G, RADAU-DEPTH-GROWS + CONDITIONING-INTRINSIC + J-INFINITY-NAMED; probes `softport_radau17_probe.py` + `preconditioned_port_probe.py`); `v868_divisor210_audits.py` (36/36 + 12/12 + 15/15, DIVISOR210-GAUGE-FAMILY(2) + MOVING12-DIMENSION-ONLY + CHIRALITY-DEGENERATE; probes `divisor210_canonicity_probe.py` + `moving12_soft17_probe.py` + `chiral_phase_probe.py`). Suite 853  $\rightarrow$  861; all eight standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all sixteen probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).
- **Lean (1 module promoted).** `TfptCarrier/KreinDefect.lean` — the Krein-defect theorem frame (14 declarations, 10 theorems; lake build green, 3416 jobs, no sorry, no `native_decide`, axioms `propext/Classical.choice/Quot.sound` only, verified by `#print axioms` on all ten): the defect representation (`defect / formAt_defect`), the main theorem `defect_psd_of_contraction` (via the exact factorization  $B_+^H(1 - C^H C)B_+$ ), the full-rank Douglas converse (the singular case a typed classical citation), the named-hypothesis structure `SourceContractor` (contractor + factorization + INDEPENDENT certificate; the circularity warning typed), and the cofinal composition `krein_cofinal / krein_cofinal_weil`. Imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round, 87  $\rightarrow$  88 files. Mirrored in v861 part C (numeric witnesses, v856/v849 precedent).
- **The operator form (the round's deliverable).** The wall now has exact operator coordinates: the Krein normal form with the Douglas pencil identity; the rank-one defect; the closed-form target-free contractor  $C = W_- F G_+^{-1} F^H W_+$  with certified defect transfer; the scalar pole-port

shadow with the closed-form leading term. And three PROOFS about the tool space: no absolute-value bound can ever certify (the Perron closure  $\rho(|C|) > 1$  everywhere); no source word of any length reaches the contractor (the monomial closure theorem); no fixed-depth quadrature family certifies uniformly (the intrinsic conditioning law). The sharpest demand is registered: ONE phase-aware bound on the closed-form kernel (PRIME.KREIN.CONTRACTOR.01 [O]) — it would discharge `SourceContractor` and, through `krein_cofinal_weil`, hypothesis ( $H_{\text{cof}}$ ). Stop list extended (source-word searches at all lengths; absolute-value kernel bounds; fixed-depth quadrature without conditioning control; chirality deciders via quadratic readouts). NO RH claim.

- **Ledger.** 10 new rows (PRIME.KREIN.NORMALFORM.01, FORM.PRIME.KREIN.DEFECT.01 [F], PRIME.KREIN.DEFECT\_ONE.01, PRIME.REDHEFFER.COLLIGATION.01, PRIME.SOFTPORT.FESHBACH.01, PRIME.WEYL.PORT.01, PRIME.SOURCECONTRACTOR.NORM.01, PRIME.KREIN.CONTRACTOR.01 [O], PRIME.SOFTPORT.RADAU.01, E8.DIVISOR210.CANONICITY.01) + the dated round-33 operator-form note on PRIME.RELATION.SKELETON.01 (the Krein coordinates; the three tool-space proofs; the extended stop list; the sharpened demand) and the dated round-33 note on PRIME.FLOOR.PAIRCORR.01 (the floor demand in Krein coordinates: the phases of the pair kernel, not its magnitudes); 943  $\rightarrow$  953 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The wall’s operator form: the Krein coordinates, the `SourceContractor` formula, the Perron closure, and the quadrature reach” (`\veri{v861..v867}`) + the part-intro chain extended; `tfpt_1`: the divisor-lattice-of-210 entry with its canonicity guard and the Redheffer/Mertens Smith-echo entry in the Gaussian code keybox (`\veri{v861/v862/v863/v868}`); `tfpt_3`: eight new audit cards; `tfpt_research_contracts`: the FORM.PRIME.KREIN.DEFECT.01 and PRIME.KREIN.CONTRACTOR.01 registration blocks, the round-33 canonicity record, and the dated round-33 notes on the PRIME.RELATION.SKELETON.01 and PRIME.FLOOR.PAIRCORR.01 blocks.
- **Website.** Prime-front feed: the round entry “reflux round thirty-three”; counters 853  $\rightarrow$  861 modules / v535–v868; DAG nodes extended with v861–v868; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 861 modules / v868.
- **Scope.** The Krein/port/formula modules are typed route decisions and reformulations under frozen preregistered protocols (the five frozen-honest FAILs are preregistered adjudications on record, bars NOT refit; the Perron closure and the monomial theorem are proofs about the TOOL SPACE, not bounds on the target; the formula is a REFORMULATION — the Loewner note typed); the canonicity module audits a deployed discovery (selection census, burial, proven-gauge wards — no new physical claim); the Lean module consumes and formalizes, never evaluates, positivity — `SourceContractor` stays a hypothesis; NO marker moves; NO RH claim anywhere.

**2026-08-08 (LXXV)** (reflux round thirty-two — the grade law, the connected current, and the E8 audits: FIVE modules (v856–v860, from seven frozen probes plus two Lean modules; suite 848  $\rightarrow$  853) — the consolidation of the 2026-08-08 morning campaign in three waves: (1) THE CONNECTED-CURRENT ARCHITECTURE — the global-centering KMS covariance of the Möbius commutator current  $\mathcal{L} = -[D, \log Z] = T(\Lambda)$  (exact at  $7.2 \times 10^{-15}$ ; Epstein  $h=2$  leaks 76% off the prime powers at RELATION level, before any state) is the FIRST design that EVADES the nuclear tax (the relapse kill does not fire: ratio  $\approx 0.3$  against the frozen band, geometry similarity 0.3–0.4 — genuinely different bookkeeping from the event-local Schur completions) and dies on the typed GRADE GAP instead: the covariance is the pair-correlation form (pair weights  $\lambda_m^2 = \Lambda(m)^2/m$ ,

QUADRATIC in the comb, sim 0.56–0.63; the centered main term IS the deployed smooth half-line subtraction at corr 0.99), the deployed window form is LINEAR, and routing the linear read through the correlator costs the variance itself:  $2.5\text{--}4.1 \times 10^5 \tau$  — an instrument, not a floor certificate; the phase-cell census is EMPTY on the entire deployed torus (0 positive cells of 220–1760 per anchor incl.  $\{2, 3, 5\}$  and  $\{2, 3, 5, 7\}$ ; grid  $\lambda_{\min}$  down to  $-3.48$ ; the phases control  $\sim 1\%$  of the demand; the independent tail kill:  $\lambda_{\min}(\text{tail})/\tau = -3.4 \times 10^3 \dots -4.8 \times 10^3$  — the split moves the demand onto an object FARTHER from PSD); and the conditional frame is kernel-checked: `TfptCarrier/CofinalCurrent.lean` — `ConnectedTail + PositiveHeadCell + PhaseRecurrence`  $\Rightarrow (H_{\text{cof}})$ , composed through `CofinalWeil` to Weil nonnegativity on the whole dense family, with the  $\delta = 0$  relaxation typed and the counterexample `connectedTail_not_sufficient` proven — hypothesis 2 measured EMPTY at the deployed frame: a named continuation contract, not a discharge (`FORM.PRIME.COFINAL_CURRENT.01 [F]`); (2) THE E8 AUDITS — the simplex-Fourier character theorem: the Gaussian census  $240 = 15 \times 16$  IS the spectral statement  $\hat{r} = (240, -16^{15})$  (the uniform-nonzero channel  $P = (J - I)/15$  with spectrum  $\{1, (-1/15)^{15}\}$  and two-step contraction  $1/225$ , exact in Fractions), the v817 packet numerator EQUALS the code Walsh transform at every level  $\leq 16$  (Eisenstein normal form  $15\Theta_{D8} = 7\Theta_L + 8\Theta_0$ ), and  $\hat{n}_2(n) = -1/15$  holds as an INTEGER IDENTITY at every odd  $n \leq 16500$  (all 1911 odd primes) — v817’s per-prime measurement is the prime restriction of a CHARACTER THEOREM of the code, typed honestly spectral-only (the gradings do not refine each other); the G31 clock alphabet:  $\text{Deg}(G_{31})/4 = \{2, 3, 5, 6\} = \{|Z_2|, N_{\text{fam}}, g_{\text{car}}, |R^+(A_3)|\}$  as a NORMAL FORM (gcd  $4 = |\mu_4|$ , lcm  $120 = |R^+(E_8)|$ , lcm/gcd  $30 = h(E_8)$ , audit identity  $16 \times 15 = 240$  REALIZED on the roots),  $(8, 12, 20, 24)$  the unique product-46080/sum-64 quadruple, the 607-group rank-4 kill scan ( $G_{31}$  the SOLE full-battery passer; impostors typed by name; honesty line: degree-only audits cannot see the reducible torus), the  $W(D_5) \times W(A_3)$  fence re-killed by COMPUTED centers 4 vs 1; the winding quadratic: the whole `tfpt_2` winding line is ONE adjugate quadratic  $q_{\text{wind}} = e_1^T \text{adj}(tI - R)\mathbf{1} = t^2 - g_{\text{car}}t + |Z_2|$  (irreducible, disc 17) with the triple lock  $s = 6$  (trace/det/Coxeter each alone),  $\Delta(6) = 39200 = 2 \cdot 140^2$  NOT a collision point (the sole  $2 \times \text{square}$  hit in the window) and the reality threshold  $s^* \approx 2.8250$  the unique  $\Delta$ -root in  $(2, 3)$  (deployed  $\approx 2.83$ ); (3) ROUND 33, THE GRADE THEORY (the day’s conceptual core) — `TfptCarrier/GradeNoGo.lean`: THE GRADE LAW kernel-checked — the entire direct-positivity graveyard reduces to ONE homogeneity no-go (a 1-homogeneous target can never be matched exactly by a 2-homogeneous Gram square: evaluate on a scaling ray at  $t = 1, 2$  — two lines, CLASS closure), the affine trilemma with the exact PSD tax (error = constant-Price + quadraticPrice) and the sharpened `affine_grade_no_go` (the background killed too), and the elevator lemma `tangent_psd_on_kernel_null` (derivatives of PSD families at kernel-null directions are positively signed — RATES, not amplitudes: the ONLY licensed grade-1 mechanism) — measured exchange rate  $10^4\text{--}10^5$  across the four graveyard instances (`FORM.PRIME.GRADE.NO_GO.01 [F]`); the instantiated elevator is REAL (PSD compound-Poisson Euler factors with no eigenvalue input, exact tangent within the certified Taylor bound, free null-sum positivity  $x^T \Psi x = w|X(\theta_q)|^2$ , rates Euler-indexed at rel dev 0.0) and dies honestly on SIGNED spectral densities on BOTH deployed sides: the comb Lévy density at exactly 50% negative mass (first flip  $\theta^* = \pi D/(2u_q)$ , prediction matched to  $\sim 1\%$ ) and the arch source on the DIGAMMA BAND ending at  $\tau^* = 6.27$  where  $\text{Re } \psi(\frac{1}{4} + i\tau/2) = \log \pi$  (the closure’s sharpest new number) — the Lévy/conditional-positivity elevator class CLOSED on the deployed sides (Epstein  $h=2$  refuses the

Euler indexing at construction grade, 82% off-pp); and the phase-lever theorem closes the finite-head phase-rescue FAMILY with certificates: granting every window atom a PRIVATE best phase, the budget never pays the arch deficit on any of the 67 rungs ( $d_0 - \text{budget}(\{2, 3\}) \geq 1.3894$ ;  $d_0 - \text{budget}(\{2, 3, 5, 7\}) \geq 0.6296$  — uniformly negative at every phase, no Kronecker rescue), three certified scope tiers (exact census / budget inequality / greedy sharp control), the D-frontier saturating at  $0.73 < 1$  (structural, not a boundary effect), the anti-vacuity ward firing; NO marker moves; NO RH claim anywhere)

- **New modules (5).** v856\_connected\_current\_lean.py (14/14 + 5/5, zero fails, CONNECTED-COVARIANCE-PARTIAL + PHASE-CELLS-EMPTY, +4 Lean-mirror checks; probes connected\_covariance\_probe.py + cofinal\_phase\_cell\_probe.py: the current exact  $(-[D, \log Z] = T(\Lambda)$  at  $7.2 \times 10^{-15}$ ;  $\chi_4/\zeta_{\mathbb{Q}(i)}$  pass in their own completions; Epstein  $h=2$  leaks 75.8% off-pp at relation level), states PSD with exact centering bookkeeping, the main subtraction = the deployed smooth subtraction (corr 0.991–0.995; the wrong half-weight loses the pair-weight carrier), the relapse kill does NOT fire (ratio2 0.29–0.31 outside  $[0.5, 20]$ ), the grade gap carries the identity residual (16–45 Ah units; variance price  $2.49\text{--}4.06 \times 10^5 \tau$ ); the torus census 0 positive cells with per-cell affineness exact, the Kronecker UFD lemma warded, the tail kill  $-3.4/ -4.8/ -3.7 \times 10^3 \tau$ , scramble 0/200); v857\_simplex\_fourier\_winding.py (26/26 + 20/20, SIMPLEX-FOURIER-EXACT + INTERTWINER-IDENTIFIED + WINDING-QUADRATIC-EXACT; probes simplex\_fourier\_probe.py + flavor\_winding\_quadratic\_probe.py: the Walsh law with ramification/equidistribution at every level, the packet-code intertwiner identity (both ladders  $[-16, 112, -448, 1136, \dots]$ ), the value theorem to 16500; the adjudgate quadratic with the triple lock, the locus census and four firing controls); v858\_g31\_clock\_alphabet.py (19/19, CLOCK-ALPHABET-SELECTIVE; probe g31\_clock\_alphabet\_probe.py: BFS closure 46080 exact, the 60-reflection census with zero quasi-reflections (v634 reproduced), the unique quadruple, the seven-gate battery over the 607-entry catalog, the family boundary to  $m = 120$ , X1/X2 controls, the computed-center fence kill); v859\_grade\_no\_go\_elevator.py (11/11, LOCAL-SIGN-FAILS, +4 Lean-mirror checks; probe euler\_schur\_semigroup\_probe.py: the mechanism real (Schoenberg PSD at all tested  $t$ , exact tangent, free null-sum positivity at  $10^{-10}$ , rates warded exact, constructor fence passed), the lag-form kill (cos-sim 0.004–0.099), the signed-density measurements (comb 0.500 exactly; arch flip bands ending 6.26–6.27 with the analytic  $\tau^* = 6.27$ ), scramble and Epstein controls); v860\_phase\_lever\_closure.py (7/7, PHASE-LEVER-THEOREM; probe phase\_lever\_theorem\_probe.py: the warded fast route, the census regression, the analytic lever bound dominating the exact sup ( $10.3\times$ ), the budget margins on all 67 rungs, the break-even census, the greedy exact-landscape control (S5b: still zero cells), the D-frontier saturation and the recalibrated anti-vacuity ward (v1.1, declared in the frozen spec)). Suite 848  $\rightarrow$  853; all five standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all seven probes re-run identically at promotion (2026-08-08); promoted VERBATIM with no downscoping (the round-31 embedding convention).
- **Lean (2 modules promoted).** TfptCarrier/GradeNoGo.lean — the grade barrier as a class-closure theorem (13 declarations, 10 theorems; lake build green, 3415 jobs, no sorry, no native\_decide, axioms propext/Classical.choice/Quot.sound only, verified by #print axioms on all ten): grade\_no\_go (the two-scalar core), gram\_two\_homogeneous / grade\_no\_go\_gram, the affine trilemma (affine\_gram\_expansion / affine\_gram\_tax / affine\_grade\_no\_go) and the elevator lemma tangent\_psd\_on\_kernel\_null. TfptCarrier/CofinalCurrent.lean — the conditional route to  $(H_{\text{cof}})$  (13 declarations, 7 theorems, axioms clean): HeadTailSplit, the three NAMED hypotheses with provenance typed, cofinal\_current (extraction from the recurrence), cofinal\_current\_weil (composition through the minimal H theorem), the  $\delta = 0$  relaxation, the Rayleigh reading, and connectedTail\_not\_sufficient. Both imported in TfptCarrier.lean; lean\_manifest.sha256 regenerated with this round, 85  $\rightarrow$  87 files.

Mirrored in v859 part A and v856 part C (numeric witnesses, v849/v843 precedent).

- **The grade law (the round’s deliverable).** Free positivity lives at QUADRATIC grade, the deployed target at LINEAR grade, and no exact match exists on any scaling ray — the four graveyard instances (Schur–Gram tax, covariance grade gap, wedge uniformity failure, phase-lever closure) are instances of ONE kernel-checked homogeneity law with measured exchange rate  $10^4$ – $10^5$ ; the affine escape is priced exactly; the only licensed grade-1 mechanism (rates) is proven abstractly, instantiated, and its deployed instantiation honestly killed by signed densities on both sides ( $\tau^* = 6.27$  typed analytically); the phase-rescue family is closed by theorem; and the two typed positive cores enter the record: the tax-evading zero-free pair-correlation instrument and the conditional kernel-checked frame three hypotheses  $\Rightarrow (H_{\text{cof}})$ . Stop list extended (finite-head phase rescues; amplitude-Gram constructions without grade accounting; Lévy/conditional-positivity elevators on the deployed sides). NO RH claim.
- **Ledger.** 7 new rows (PRIME.RELATION.CONNECTED\_COVARIANCE.01, FORM.PRIME.COFINAL\_CURRENT.01 [F], E8.SIMPLEX.FOURIER.01, E8.G31.CLOCK\_ALPHABET.01, FORM.PRIME.GRADE.NO\_GO.01 [F], PRIME.EULER.SCHUR.SEMIGROUP.01, PRIME.PHASE.LEVER.01) + the dated round-32 GRADE-LAW note on PRIME.RELATION.SKELETON.01 (the unifying diagnosis, the elevator closure, the extended stop list) and the dated round-32 instrument note on PRIME.FLOOR.PAIRCORR.01 (the pair-type object constructed zero-free; the round-31 transport note also restored to the contract block, mirroring the ledger); 936  $\rightarrow$  943 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The grade law: the graveyard unified by one homogeneity theorem; the licensed elevator and its honest death; the phase-rescue family closed” (`\veri{v856/v859/v860}`) + the part-intro chain extended + the Lean jobs count refreshed; `tfpt_1`: the simplex-Fourier character theorem and G31 clock alphabet entries in the Gaussian code keybox (`\veri{v857/v858}`); `tfpt_2`: the winding-quadratic addendum in the Winding Line block (`\veri{v857}`); `tfpt_3`: five new audit cards + the round-32 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the FORM.PRIME.GRADE.NO\_GO.01 and FORM.PRIME.COFINAL\_CURRENT.01 registration blocks, the round-32 E8 audit compressions record, and the dated round-32 notes on the PRIME.RELATION.SKELETON.01 and PRIME.FLOOR.PAIRCORR.01 blocks (Lean counts refreshed to 3415 jobs / 87 files).
- **Website.** Prime-front feed: the round entry “reflux round thirty-two”; counters 848  $\rightarrow$  853 modules / v535–v860 (the prime-front page metadata refreshed from its stale 841/v848); DAG nodes extended with v856–v860; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 853 modules / v860.
- **Scope.** The E8 audit modules (v857, v858) are exact-arithmetic corpus compressions and audits of deployed claims — normal forms, character identities and armed kill rules, no new physical claim; the architecture modules (v856, v859, v860) are typed route decisions under frozen preregistered protocols (the relapse expectation and the sign expectation both REFUTED BY MEASUREMENT and typed either way in the frozen specs — the kills moved to where the numbers put them); the Lean modules consume and formalize, never evaluate, positivity — all named hypotheses stay hypotheses; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXIV)** (reflux round thirty-one — the invariance atlas, the graded kernel, the code complex, and the minimal hypothesis kernel-checked: SEVEN modules (v849–v855, from fifteen frozen probes plus one Lean module; suite 841  $\rightarrow$  848) — the consolidation of the 2026-08-07 evening harvest in three waves: (1) THE F-PLAN — the compiler object is UNIQUE (all 14400 admissible tuples ONE  $Sp(4, 2) \times S_5$  orbit,  $\text{Aut}(C_{\text{fin}}) \cong C_6$ , unique up to NON-unique isomorphism) and the minimal H theorem is kernel-checked (`TfptCarrier/CofinalWeil.lean`: the  $\varepsilon/2$  implication proven with NO diagonal argument, the strict hierarchy

uniform  $\subsetneq$  pointwise  $\subsetneq$  cofinal — UniformMarginBound formally DEMOTED to an over-strong sufficient lemma; the load-bearing chain consumes EXACTLY ( $H_{\text{cof}}$ ); the TAX THEOREM closes the positive-dilation completion class at CLASS level ( $\text{tr } P + \text{tr } Q \geq 2\|R\|_*$  exact with zero-gap dual certificates, BASIS-INVARIANT — the v846 geometry-independence now a corollary; the exact optimum already  $\geq 4057.6 \times \tau$  on every rung, diverging  $\sim h^{0.24}$  vs  $h^{-3/2}$ ; the measured Schur designs only  $13.7\text{--}20.4\times$  above it — optimality could not have saved them); the index route closes by measurement (the density endpoint NOT PSD,  $n_- = 5.12$  — premise corrected; all 634 crossings upward, endpoint-determined; the velocity test at ratio 1.00: distance-to-crossing = metric margin / flow velocity); the corner cluster expansion is exact but its weights are POSITION-GEOMETRIC (route closed, two frozen-honest FAILS kept); the cone violations resolve into the GRADED KERNEL LAW: the strict census wanders ( $P_0 \sim X^{1.03}$ , max 244333 vs cap 100) but the depth- $10^{-2}$  G1 kernel is  $n \leq 73$  on ALL 67 rungs and the skin depth obeys  $D(X) = 0.754 X^{-0.548}$  ( $R^2$  0.99) — vanishing-depth skin, not kernel growth (PRIME.CELLCONE.GRADEDKERNEL.01 [O] registered; the tube field transports clean 0/355199 but is self-calibrating, four must-break controls kept as FAILS); (2) THE CODE COMPLEX — the Arf-vacuum syndrome theorem ( $a(t+U) = \hbar(u, v)$  on all 140 flats; the two-bit table 15/20/45/60; the LOCAL Hecke theorem  $12/16 \rightarrow 28/-4$  at all 15 points; the code pencil  $[15, 11, 3] \supset \{[15, 10, 4], 2 \times [15, 10, 3]\} \supset [15, 9, 4]$  with the canonicity lemma; the vacuum bit = the RM(2, 4) parity bit), the ovoid decoder ( $\text{SNF}(N) = \text{diag}(1^{10}, 0^5)$  torsion-free;  $N^+ = N^T/4 - J/36$  exact;  $P_5 = I/2 - B/4 + J/12$ ;  $\ker =$  the  $S_6$  standard rep 11/11; three rings:  $\mathbb{R}$  ovoid span /  $\mathbb{Z}$  torsion-free /  $\mathbb{F}_2$  code  $A_q = [15, 5, 6]$ ), the bent CSS pair ( $\hat{s}_q = -4s_q$  on all 16; the (16, 6, 2) difference set;  $[[15, 5, 3]] \rightarrow [[16, 4, 4]]$  with the vacuum transformation typed; THE  $-4$  TRIPLE POINT: Hecke = Gauss = Walsh =  $a_3$  as ONE machine-checked integer; the 16 bent translates the MUB partner of the v800 rays, no upgrade), and the von Mangoldt commutator theorem ( $\mathcal{L} = -M[D, Z] = -[D, \log Z] = T(\Lambda)$  EXACT three ways;  $\log Z$  RATIONAL, BCH truncates; the four-comb discipline at operator level with the Epstein leak  $[6, 14, 21, \dots]$  exact and repaired by  $a_A + a_B = 1 * \chi_{-20}$ ; the deployed carrier IS the first row — typed honestly as classical incidence algebra); the factorization complex (multiplicativity IS flatness at machine zero; Epstein curvature localized at class-group products with the  $\mathbb{F}_2$  Arf-lift obstruction PROVEN; gate 4 fails typed — the form is 0-chain-supported; the Pfaffian has no anchor-independent normalizer); (3) THE STRATEGIC ATTACKS — THE INVARIANCE ATLAS: the demand rewritten EXACTLY in Selberg-hierarchy coordinates and audited against on-range-verified elementary constants (the sieve reaches the  $\alpha \in 1..2$  band in SCOPE but ZERO coverage in GRADE, best factor  $205\times$ ; the demand at  $0.99\text{--}1.00 \tau_{\text{pnt}}$ , the factor-2 gate unclosable at truth — INPUT-CLASS-INVARIANT; the named sieve object  $d\text{Chain}_2$ ); the criteria align (the Li coefficients  $\lambda_n > 0$  to  $n = 20$  computed with NO zeros;  $d_N^2$  exact to  $N = 64$ ; BD-constant =  $2\lambda_1$  at  $4.9 \times 10^{-17}$ ; the NB span IS the spectral-mother geometry term by term; the Li transfer fails by CONE GEOMETRY) — no coordinate system holds hidden supply; the minimizer census (no frozen closed-form limit; the internal direction exists at 0.0217 rad and IS the comb block's soft direction at 0.0056 rad; the drift real, no identification claimed,  $-4/\pi$  a coincidence) — THE WALL IS INVARIANT ACROSS COMPRESSIONS, GEOMETRIES, CRITERIA AND INPUT CLASSES; the one object in four languages:  $e_1$  bound = form-factor band =  $d\text{Chain}_2$  = comb-soft direction; NO marker moves; NO RH

claim anywhere)

- **New modules (7).** `v849_cfin_unique_cofinal_lean.py` (22/22, CFIN-UNIQUE, +4 Lean-mirror checks; probe `cfin_uniqueness_probe.py`: the WLOG ladder exact (28 alternating forms one  $GL(4, 2)$  orbit;  $Sp(4, 2)$  transitive on the 6 Arf-1 refinements; 20 admissible  $\sigma$  per refinement, one orbit; 120 admissible equivariant  $\iota$ , one orbit); the census 14400, ONE orbit (orbit-stabilizer  $14400 \times 6 = 86400$  exact);  $\text{Aut}(C_{\text{fin}}) \cong C_6$  (orders  $[1, 2, 3, 3, 6, 6]$ , faithful into  $Sp(4, 2)$ , slot permutation determined); the Gaussian diagram typed (18-element centralizer torsor; the strict terminal reading FAILS honestly); controls fire (dot form 0 refinements; wrong Arf class: empty  $\iota$  set; no- $\sigma$ : canonicity dies  $16 \rightarrow 8 \rightarrow 4 \neq 1$ )); `v850_completion_tax_flow.py` (15 + 11, zero fails, TAX-GAP + FLOW-CROSSINGS; probes `psd_completion_tax_probe.py` + `spectral_flow_pivot_probe.py`: the nuclear-norm identity with zero-duality-gap certificate pairs on every anchor ( $\lambda_{\min} \geq -8.9 \times 10^{-15}$ ,  $\|W\|_{\text{op}} = 1.000000000000$ ,  $\langle W, R \rangle \text{ rel} \leq 1.6 \times 10^{-16}$ ; symbolic  $2 \times 2$  at 50 dps, gap  $2.7 \times 10^{-51}$ ), basis-invariant ( $\leq 1.1 \times 10^{-15}$ ); tax/ $\tau \geq 4057.6$  on all 67 rungs (max  $7.5 \times 10^5$ , slopes 1.99/1.86/0.46); scramble/Epstein fingerprints move the norm (0.54–0.96 / 0.12–0.22); the flow: 634 upward crossings,  $\text{SF} = n_-(0) - n_-(1)$  verified 67/67, last crossing hugs the deployed point (median gap  $1.6 \times 10^{-5}$ , slope  $-2.11$  vs  $\tau$ 's  $-1.65$ ), velocity ratio 1.00 (IQR 0.99..1.00); Jacobi pivot certification eigenproblem-free; scramble/Epstein/density controls typed); `v851_cluster_kernel_field.py` (15 + 10 + 7 with the SIX frozen-honest FAILS S3.2/S4.1/S2.C1/S2.C2/S4.C1/S4.C2 kept and pattern-gated, NOT refit; CLUSTERS-DIFFUSE + BOTH-PARTIAL + KERNEL-GRADED; probes `multiplicative_cluster_probe.py` + `cone_kernel_field_probe.py` + `graded_kernel_field_probe.py` over the read-only `cell_cone_transport_probe.py` parent: the expansion exact (sympy telescoping; Möbius elimination;  $2^{16}$ -subset ward 0.0; weights  $= -\text{tr}(B^{-1}E_i B^{-1}E_j)$ ) but  $E_2 = 0.883$  vs bar 3.0 and the scramble does not collapse it — position-geometric; strict censuses wander ( $P_0 \sim X^{1.03}/X^{1.02}$  at  $R^2$  1.00, max 244333/283607) while the G1 depth- $10^{-2}$  kernel is  $n \leq 73$  on all rungs; the causal tube field clean on all four rule $\times$ grouping combos (0/355199, cond  $\leq 57.3$ ) but self-calibrating; the skin law  $D(X) = 0.754 X^{-0.548}$  with verified  $X_0(\varepsilon)$ ; the G2 mechanism typed (1653 $\times$  / 39 $\times$ ); scramble depth control fires, Epstein anchor holds); `v852_arf_vacuum_ovoid.py` (24/24+23/23, ARF-VACUUM-SYNDROME-EXACT + OVOID-DECODER-EXACT; probes `affine_arf_vacuum_probe.py` + `doily_ovoid_decoder_probe.py`: the syndrome table, the local Hecke theorem, the code pencil with the canonicity lemma,  $\text{RM}(2, 4)$  restored with the vacuum bit as parity bit;  $N^T v_a = 0$ , Gram  $36I - 6J$ , SNF torsion-free with a det  $\pm 1$  minor, Penrose exact in  $\mathbb{Q}$ , the decoder exact on the predeclared set, the  $S_6$  standard rep, the three-ring correspondence; five controls fire); `v853_bent_css_mangoldt.py` (22/22 + 30/30, ARF-BENT-CSS-EXACT + MANGOLDT-COMMUTATOR-EXACT; probes `arf_bent_css_probe.py` + `mangoldt_incidence_probe.py`: the Walsh eigenvector at  $-4$ , the (16, 6, 2) set, the 31-word census, the two CSS codes with the 60 minimal words = the 60 isotropic planes, the  $-4$  triple point, the MUB partner (no upgrade);  $\mathcal{L} = -M[D, Z] = T(\Lambda)$  exact at  $N = 60$  (sympy) /  $N = 360$  (integer) /  $N = 10^4$  (float ward  $5.3 \times 10^{-15}$ ), the one-line proof termwise, powers = chains with nilpotency  $\lfloor \log_2 N \rfloor + 1$ ,  $\log Z$  rational with BCH truncation, the four-comb operator table with the Epstein leak exact and class-average repaired, the ideal-level Dedekind  $\Lambda_K$  with norm aggregation, the carrier bridge per-divisor exact,  $L^2 - [D, L] = T(\mu * \log^2)$ ); `v854_relation_hodge_pfaffian.py` (18/18, CURVATURE-DISCRIMINATES-ONLY; probe `relation_hodge_pfaffian_probe.py`:  $d_1 d_2 = 0$  exact,  $b_0 = 1$ ,  $b_1 = 16/21/22$  on the same-prime chords; the true comb exactly flat on ALL squares ( $\leq 1.8 \times 10^{-15}$ ); Epstein curvature 321/756/905 squares at the class-group products; the Arf lift PROVEN obstructed (GF(2) rank vs augmented); gate 4 fails typed ( $S_{\text{cob}} \sim 2.9\text{--}3.9 \times 10^4$ ); the Pfaffian log-spread infinite); `v855_invariance_atlas.py` (19 + 8 + 7 + 3, zero fails, SIEVE-SAME-GAP (spec v2, the S5.1 bar recalibration  $5.0 \rightarrow 2.0$  declared, measured 3.9) + ATLAS-SAME-WALL + PROFILE-DIVERGES + DIRECTION-TRANSIENT; probes `selberg_supply_probe.py` + `criteria_atlas_probe.py` + `minimizer_profile_probe.py` + `minimizer_direction_probe.py` over the read-only `spectral_flow_pivot_probe.py` frame:



the hierarchy decomposition exact ( $D_0 = d\text{Head}_2 - d\text{Chain}_2$  at  $10^{-9}$ ; levels 2/3 exact in Fractions); the envelopes verified on range and dominating (Chebyshev  $0.111x$  measured  $0.0956$ ); the ledger (best factor  $2.05 \times 10^2$ , route 2 up to  $1.1 \times 10^4$ ;  $|D_0|/\tau_{\text{pnt}} = 0.99..1.00$ ;  $\min |dS|_F/\text{GATE}_W = 2.28 > 1$ ); Brun–Titchmarsh typed UNCOVERED, the large sieve NOT-APPLICABLE per rung;  $\lambda_1 = 1 + \gamma/2 - \log(4\pi)/2$  at dev  $0.0$ , literature wards  $3 \times 10^{-10}$ ;  $d_N^2$  exact to  $N = 64$ ;  $\text{BD} = 2\lambda_1$ ; the NB span = the spectral-mother geometry (the exact  $\mu$  floor identity; the  $1/x$  mirror = J); the Li cone residual  $60.6\%$  vs  $5\%$ ; the minimizer census (RAY  $0.4530$  / ARCH  $1.0976$  rad; dispersion  $0.0217$ ; C03 tracking  $0.0056$ ; the Bonferroni demonstration 8 of 15 with the golden-ratio control rejected)). Suite 841  $\rightarrow$  848; all seven standalone green with pattern gates encoding the frozen censuses (check counts, FAIL ids, verdicts, exit codes) and the byte-exact provenance ward vs the embedded sources; all fifteen probes re-run identically at promotion (2026-08-07); promoted VERBATIM with no downscoping (the round-31 embedding convention: the frozen probe sources embedded byte-exact and executed verbatim in isolated namespaces — every printed spec SHA-256 reproduces).

- **Lean (1 module promoted; wording corrected by 2026-08-13 XCII).** `TfptCarrier/CofinalWeil.lean` — the minimal H theorem, kernel-checked (11 declarations; lake build green, 3411 jobs, no sorry, no `native_decide`, axioms `propext/Classical.choice/Quot.sound` only, verified by `#print axioms` on all eight theorems; imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round, 84  $\rightarrow$  85 files): `CofinalHypothesis` ( $H_{\text{cof}}$ ) stores the MATHEMATICAL CORE — a strictly monotone selected ladder with PSD rungs; it does not by field order enforce that the ladder was predefined or sign-independent. `limit_nonneg_of_cofinal_seq` — THE MINIMAL IMPLICATION, the  $\varepsilon/2$  argument proven; `ladderForm_nonneg / weil_nonneg_of_cofinal` — the dense-family form with NO diagonal argument (one PSD certificate per rung covers every element simultaneously); the STRICT HIERARCHY `uniformMarginBound_to_cofinal / pointwise_to_cofinal / cofinal_not_uniform` (witness  $1/(m+1)$ , reusing the v843 kernel gap) / `cofinal_not_pointwise` (witness  $\pm 1$  on the even ladder); the assembly `cofinal_weil`. THE FORMAL DEMOTION: the load-bearing chain of `PRIME.EXTRACTION.CHAIN.01` consumes EXACTLY the mathematical ( $H_{\text{cof}}$ ) payload; `UniformMarginBound` implies it and the converse fails kernel-checked — from this round an over-strong SUFFICIENT lemma, no longer the named wall. `PREDEFINED/noninterference` is the separate explicit external contract added by `CofinalPredefinition.lean` in XCII. ( $H_{\text{cof}}$ ) stays a hypothesis everywhere. Cited in v849’s part B (numeric witness mirror, v843/v837/v817 precedent).
- **The wall statement (the round’s deliverable).** THE INVARIANCE ATLAS: the wall is proven invariant across COMPRESSIONS (the diverging nuclear-norm tax closes the positive-dilation class — in any basis; the index is endpoint-determined; the corner cluster expansion is position-geometric), GEOMETRIES (basis invariance — the v846 TV-price geometry-independence now a corollary), CRITERIA (Weil = Li = Nyman–Beurling;  $\text{BD-constant} = 2\lambda_1$  at  $4.9 \times 10^{-17}$ ; the NB span IS the spectral-mother geometry; the Li transfer fails by cone geometry — certified partial data everywhere, one uniform statement missing everywhere) and INPUT CLASSES (statistics = sieve: the demand at truth exceeds the factor-2 gate on every rung — unclosable by ANY upper-bound class). The one object in four languages: the uniform  $e_1$  bound = the form-factor band  $\alpha \in 1..2$  = the bilinear chain fluctuation  $d\text{Chain}_2$  = the comb block’s soft direction ( $0.0056$  rad). The positive cores: the graded kernel law (depth, not sign) and the minimal kernel-checked hypothesis ( $H_{\text{cof}}$ ). Stop list extended (positive-dilation completions with fixed comb coupling in any basis; finite-dimensional index reformulations; corner cluster expansions; elementary upper-bound supply rows against the factor-2 gate). NO RH claim.
- **Ledger.** 9 new rows (`CFIN.UNIQUE.01`, `FORM.PRIME.COFINAL.WEIL.01` [F], `PRIME.PSD.COMPLETION.TAX.01`, `PRIME.CELLCONE.KERNELFIELD.01`, `PRIME.CELLCONE.GRADEDKERNEL.01` [O], `ARF.VACUUM.SYNDROME.01`, `ARF.BENT.CSS.01`,

PRIME.RELATION.HODGE.01, PRIME.CRITERIA.ATLAS.01) + four dated round-31 notes: the invariance-atlas closing note on PRIME.RELATION.SKELETON.01 (the four-axis invariance; the one object in four languages; the extended stop list), the transport note on PRIME.FLOOR.PAIRCORR.01 (the pair-type object =  $d\text{Chain}_2$  = the comb-soft direction), the demotion note on FORM.PRIME.EXCESS.SKELETON.01 ( $\text{UniformMarginBound} \rightarrow$  over-strong sufficient lemma), and the kernel-check note on PRIME.EXTRACTION.CHAIN.01 (step 2 kernel-checked; (H) formalized as  $(H_{\text{cof}})$ ); 927  $\rightarrow$  936 rows; NO marker moves.

- **Papers.** `tfpt_1`: the Arf-vacuum syndrome + code pencil, the bent CSS pair + the  $-4$  triple point, and the ovoid decoder entries in the Gaussian code keybox + the new  $C_{\text{fin}}$  uniqueness keybox ( $\text{Aut} \cong C_6$ ) after the normal-form keybox (`\veri{v849/v852/v853}`); `tfpt_prime_front`: the new round section “The invariance atlas: the wall is the same across compressions, geometries, criteria and input classes; the minimal hypothesis kernel-checked” (`\veri{v849/v850/v851/v853/v854/v855}`) + the part-intro chain extended; `tfpt_3`: seven new audit cards + the round-31 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the FORM.PRIME.COFINAL.WEIL.01 and PRIME.CELLCONE.GRADEDKERNEL.01 registration blocks, the round-31 code-layer compressions record, and the dated round-31 notes on the PRIME.RELATION.SKELETON.01, PRIME.MARGIN.LAW.01/FORM.PRIME.EXCESS.SKELETON.01 and PRIME.EXTRACTION.CHAIN.01 blocks.
- **Website.** Prime-front feed: the round entry “reflux round thirty-one”; counters 841  $\rightarrow$  848 modules / v535–v855; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 848 modules / v855.
- **Scope.** The code-layer modules (v849 part A, v852, v853 part A) are exact-arithmetic compressions and a uniqueness theorem — normal forms and set equalities, not new claims (the P2 residual  $R1'$  and the MUB reading stay typed, no upgrades); the no-go modules (v850, v851, v854) are typed route decisions under frozen preregistered protocols (exact identities [E]-grade; the six control FAILS preregistered-honest adjudications on record, bars NOT refit); the atlas module (v855) is an honest invariance mapping (the one bar recalibration declared in the frozen spec; no number invented for inapplicable supply rows); the Lean module consumes and formalizes, never evaluates, positivity —  $(H_{\text{cof}})$  stays a hypothesis; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXIII) (reflux round thirty — the three-architecture decision, the extraction chain complete, and the finite-compiler compressions: FIVE modules (v844–v848, from eight frozen probes; suite 836  $\rightarrow$  841) — the v5.4 strategy campaign consolidates: the three candidate proof architectures are DECIDED, all honest negatives with typed positive cores — (i) the relational Schur–Gram route dies STRUCTURALLY (three of four gates green incl. the exact  $\mu \log$  descent with the  $h=2$  Epstein refusal at construction grade; the corner identity fails in trace AND form by exactly the forced Cauchy–Schwarz diagonal price,  $\text{price}/\tau \sim 3.1 \times 10^4$ – $4.5 \times 10^4$  at the density-anchored floor) and the price is GEOMETRY-INDEPENDENT: the unitary spectral mother ( $J^2 = I$  exact; the deployed form IS the  $J$ -odd Toeplitz compression; the Euler product the exact operator algebra  $U_d U_m = U_{dm}$ ) gains 20–33 $\times$  at symbol grade but never at Gram grade — harvesting interference is Fejér–Riesz of the total symbol = the positivity itself; (ii) the sign-register wedge lift EXISTS as exact algebra (nonneg wedge weights per frame where the naive signed Lagrange provably cannot) but the frame-uniform law fails at  $\sim 15\%$  vs the  $4.4 \times 10^{-5}$  floor need — the commutant uniformity wall transported; (iii) the completed-cell Lorentz transport BREAKS the cone at the  $n=2$  cell on 67/67 rungs under both predeclared**

groupings (endpoint back in-cone = the certified  $\tau > 0$ ; ray cascade to  $(5, -3, 4)$  CONFIRMED, Kendall  $\pm 1.000$ ; INVERTED Epstein signature: the fake stays in-cone, the arithmetic sits IN the violations) — all three die on the SAME finer-than-statistical substance; AND the continuum extraction is separated from all RH semantics and CLOSED: the implication chain cofinal finite positivity (H)  $\Rightarrow$  Weil positivity  $\Rightarrow$  (Weil's criterion) the target is theorem-grade modulo named citations with a QUANTIFIER REDUCTION (no Mosco compactness, no uniform  $\delta$ , no diagonal argument; the pure-box class carries the EXACT Weil value at every finite level) — the wall is EXACTLY hypothesis (H) and nothing else (PRIME.EXTRACTION.CHAIN.01 [O] registered); plus two exact finite-compiler compressions: the message ladder  $M_n = 15 \cdot 2^n$  with  $h(E_8) = 30$  derived from the rebuilt root system, the Doily–Pascal rank theorem (the P2 integers as the singular-value data of the Cremona–Richmond incidence), and the finite compiler normal form  $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$  as ONE object with the code-to-matter kill built in; NO marker moves; NO RH claim anywhere)

- **New modules (5).** `v844_message_doily_rank.py` (23/23 + 16/16, MESSAGE-LADDER-EXACT + DOILY-PASCAL-RANK-EXACT; probes `binary_message_ladder_probe.py` + `doily_pascal_rank_probe.py`, promoted verbatim:  $M_n = 15 \cdot \Delta p_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$  against the ACTUAL rebuilt objects, one named structure per rung —  $M_1 = 30 = h(E_8)$  derived here from the rebuilt root system (8 simple roots, Cartan det 1, E8 degree sequence, highest root unique at height 29 with marks  $\{2, 2, 3, 3, 4, 4, 5, 6\}$ ),  $M_2 = 60$  Gaussian lines,  $M_3 = 120 = |R^+|$ ,  $M_4 = 240 = 15 \times 16$  census, coda  $248 = 15 \cdot 16 + 8$ ; wrong anchor  $(1, 2, 2)$  and wrong multipliers 14/16 miss all targets; AND the rank theorem:  $NN^T = B + 2I$  entrywise under the  $K_6$  duad model (all 225 cells),  $\text{charpoly}(NN^T) = (x - 9)(x - 4)^9 x^5$  exact — rank  $N = 10 = A_\Lambda$ ,  $\dim \ker N = 5 = g_{\text{car}}$  (the six ovoid indicators SPAN it), multiplicity  $9 = N_{\text{fam}}^2$ , top  $3 = N_{\text{fam}}$ , recovery  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}}$ , six-step  $(2/3)^6 = 64/729$  exact; NO  $(1/3)^6$  doily mode (v814 typed, no upgrade); wrong-pairing controls fire); `v845_cfin_normal_form.py` (28/28, SPEC v2 declared, NORMAL-FORM-ASSEMBLED; probe `finite_compiler_normal_form_probe.py`:  $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$  READ OFF the Gaussian quotient — the unique selector-picked refinement (16 refinements, Arf  $6 + 10$ ), the order-3 family cycle, the parity lift onto  $C_{\text{even}}(5)$  with  $\beta \equiv \hbar$  and  $q_{\text{wt}} \equiv q^*$ ; derived:  $|Sp(4, 2)| = 720$  full census,  $\text{Stab}(q^*) = 120$  faithful  $S_5$ , orbits  $[1, 5, 10]$ , Pascal  $(1, 10, 5) = \left(\binom{5}{0}, \binom{5}{2}, \binom{5}{4}\right)$ , the  $\bar{5}$  an OVOID of the doily, budget  $240 = 16 \cdot 15 / 248 = 16 \cdot 15 + 8$  against the actual census; the kill built in: the 128-spinor counting bound ( $\geq 7$  of 15 classes always adjoint-side; saturated  $8 + 7$ , zero mixed) + the weight-level kill (8 pairs, never 1) — ROOTCLASS-MIXED re-verified, NOT overturned; P2 narrowed to the v799 residual  $R1'$  ([C]), not eliminated; controls fire); `v846_schur_spectral_mother.py` (19 + 16 checks, zero fails, SCHUR-INDEFINITE + SPECTRAL-TV-UNIVERSAL; probes `relational_schur_gram_probe.py` + `spectral_mother_probe.py`: the Schur program closed as a typed structural negative — G1/G3/G4 green (manifest positivity; exact  $\mu$  log routing, Epstein bookkeeping defect 0.73–0.83 vs TRUE 0; Selberg safety), the G2 form gate fails structurally (the P0 deviation IS diag(price) at  $\leq 10^{-9}$ ; price/ $\tau \sim 3.1\text{--}4.5 \times 10^4$ ; trace fails by the same object); the spectral mother built as typed ( $J^2 = I$  dev 0;  $K$  the  $J$ -odd Toeplitz compression rel 0;  $U_d U_m = U_{dm}$  exact at integer commensurability): symbol-grade gain 20–33 $\times$ , Gram-grade best legal overhead  $34\text{--}55 = 5.7 \times 10^4\text{--}1.2 \times 10^5 \times \tau$  — THE TV PRICE IS GEOMETRY-INDEPENDENT); `v847_wedge_cellcone_transport.py` (13 + 9 checks with the TWO frozen-honest FAILS S5.SCR + S5.C3 kept and pattern-gated, NOT refit; WEDGE-PARTIAL + CONE-BROKEN + RAY-EDGE-CONFIRMED; probes `relational_lagrange_probe.py` + `cell_cone_transport_probe.py`: the naive signed Lagrange no-go EXACT (forced weights;  $P^-/\det \sim 2.0\text{--}3.2 \times 10^4$ ; the arch block NEGATIVE definite, eigen  $-2.285/-1.139$ ); the  $C_2$  register lift feasible on every anchor (LP resid  $\leq 8.9 \times 10^{-16}$ ); level C fails (design 0.170/0.151, held-out 0.153 vs floor  $4.4 \times 10^{-5}$ ; scramble  $\times 7.5$  vs  $\times 10$  bar — typed, not excused); the com-

pleted cells telescope exactly ( $3.4 \times 10^{-15}$ ;  $\Phi^\top J \Phi = cJ$  at  $2.2 \times 10^{-16}$ ;  $\min e_1 = 4.8546$  reproduces the envelope); both groupings exit on 67/67 rungs at  $n = 2$  (first-exit median cell 1–2; endpoint back in-cone 67/67); ray cascade  $r_{\text{end}} 0.5700 \rightarrow 0.8018$  / angle  $46.26^\circ \rightarrow 34.39^\circ$  (Kendall  $\pm 1.000$ ); incomplete contrast 67/67; equal-weight and wrong-norm controls fire 14/14; the Epstein comb does NOT fire (0/14) — the inverted signature kept); `v848_extraction_chain.py` (19/19, EXTRACTION-CHAIN-COMPLETE; probe `continuum_extraction_probe.py`, run-1→run-2 declared calibration (P2.0 exact-class certificate, v762 D2.0 precedent): the  $(D, X)$  ladder  $j = 4..11$  ( $D_j = 2^{-j}$ , 68–8704 lags,  $k_a = 28$  pinned); the pure-box class EXACT at every level (max err  $5.3 \times 10^{-14}$ ); identification rates  $-1.58..-1.84$  (median  $-1.818$ , dyadic prediction  $-2$ ); ledger geometric (med3 0.302, law  $-1.752$ ); X-direction EXACTLY stable (44 cutoff pairs at float zero); recovery constructive + Cauchy (exact strong- $L^2$  law in  $\mathbb{Q}$ ); F2/F3 band margins TYPED never gated; W1 regression wards ( $5.2 \times 10^{-13}$  /  $2.1 \times 10^{-6}$  /  $8.0 \times 10^{-8}$ ); controls fire (wrong-limit stall  $1.6 \times 10^6$ ; scrambled tower  $5.5 \times 10^5$ ). Suite 836  $\rightarrow$  841; all five run standalone green with pattern gates encoding the frozen verdicts (incl. the two expected-FAIL signatures); all eight probes re-run identically at promotion (2026-08-07); promoted VERBATIM with no downscoping (spec/rule SHA-256 values reproduce byte-exactly).

- **The wall statement (the round’s deliverable).** The three v5.4 architectures die on the SAME substance — the total-variation price (geometry-independent by the unitary mother), the frame-uniformity failure (the commutant wall transported) and the unrenormalized  $n = 2$  cell (the arithmetic IN the violations) are three coordinate presentations of the one finer-than-statistical datum the GUE-side rounds typed; the stop list extends (diagonal relational mothers; constant wedge laws; unrenormalized small- $n$  cell transports). On the other side the extraction is COMPLETE: cofinal finite positivity (H)  $\Rightarrow$  Weil positivity  $\Rightarrow$  the target, theorem-grade modulo named citations (Weil 1952, Bombieri 2000, Suzuki arXiv:2606.09096, Iwaniec–Kowalski Thm 5.12), with the quantifier reduction — the arithmetic wall is EXACTLY hypothesis (H), equivalently the uniform demand of PRIME.RELATION.SKELETON.01 restricted to the cofinal ladder, AND NOTHING ELSE. NO RH claim.
- **Ledger.** 5 new rows (MESSAGE.LADDER.01, NORMALFORM.CFIN.01, PRIME.SCHUR.GRAM.01, PRIME.WEDGE.LAGRANGE.01, PRIME.EXTRACTION.CHAIN.01 [O]) + the dated round-30 closing note on PRIME.RELATION.SKELETON.01 (the three-architecture decision; the extended stop list; the extraction separation) with its dependencies/evidence extended to v846–v848; 922  $\rightarrow$  927 rows; NO marker moves (the compressions are normal forms; the architecture modules are typed route decisions; the extraction registration is OPEN [O] with hypothesis (H) the isolated wall).
- **Papers.** `tfpt_1`: the message-ladder and Doily–Pascal rank entries in the Gaussian code keybox + the new  $C_{\text{fin}}$  normal-form keybox + the dated addendum in the P2 reduction-state keybox (`\veri{v844}`, `\veri{v845}`); `tfpt_prime_front`: the new round section “The three-architecture decision, and the extraction chain complete” (`\veri{v846/v847/v848}`) + the part-intro chain extended; `tfpt_3`: five new audit cards + the round-30 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the PRIME.EXTRACTION.CHAIN.01 registration block, the finite-compiler compressions record, and the dated round-30 closing note on the PRIME.RELATION.SKELETON.01 block.
- **Website.** Prime-front feed: the round entry “reflux round thirty”; counters 836  $\rightarrow$  841 modules / v535–v848; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 841 modules / v848.
- **Scope.** The compressions (v844/v845) are exact-arithmetic normal forms of existing corpus surfaces — compressions, not new claims (the  $(1/3)^6$  gap and the P2 residual stay typed, no upgrades); the architecture modules (v846/v847) are typed route decisions under frozen preregistered protocols (exact wards [E]-grade; the two control FAILS preregistered-honest adjudications on record, bars NOT refit); the extraction module (v848) never evaluates hypothesis (H) — the implication is measured/cited, the classical legs carry named citations, and the wall

stays OPEN [O]; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXII)** (reflux round twenty-nine — the margin as a doubly-derived law, and the wall kernel-checked: ONE module (v843, from one frozen probe plus one Lean module; suite 835  $\rightarrow$  836) — the certified margin series becomes a typed mathematical object:  $\tau = e_1(\alpha) h^{-3/2} \tau_{\text{pnt}}(\alpha)$  with the certified envelope constant reproduced EXACTLY across two independent coordinate systems ( $e_1 \in [4.855, 24.209]$  on all 67 rungs,  $\min \geq 4.335$  — the double-derivation ward;  $\tau$  vs  $\lambda_{\min}(A_h)$  at  $2.1 \times 10^{-12}$ ), the ONE preregistered-honest FAIL kept and pattern-gated (fit-free  $h$ -exponent median  $-1.341$  vs the frozen bar edge  $-1.35$ , NOT refit), the tower gives SCALE not recursion (ratio defects non-decaying), the excess is a GROWING cancellation (CI median 15.6, rising  $5.65 \rightarrow 29.46$  — the wall self-similar at cell level; no per-prime positive control: 22 287 of 351 122 aggregates nonpositive), the compiler connection is NULL (0/96, Bonferroni-honest) — and the finite-table limit becomes a KERNEL-CHECKED THEOREM: TftptCarrier/ExcessSkeleton.lean proves pointwise\_pos\_not\_uniform (per-rung positivity at every rung does not yield a uniform bound) and the bridge theorem excess\_floor consumes UniformMarginBound as the SINGLE NAMED HYPOTHESIS; the quantifier now sits on the ONE scalar series  $e_1 \geq 4.335$ ; same-class margin-law re-attacks stop-listed; NO marker moves; NO RH claim anywhere)

- **New module (1).** v843\_margin\_law\_excess\_lean.py (10 checks with the ONE preregistered-honest S2.2 FAIL pattern-gated + 4 Lean-mirror checks, verdict MARGIN-RECURSES-THE-WALL; discovery probe margin\_law\_probe.py, 2026-08-07, re-run identically at promotion, promoted VERBATIM with no downscoping,  $\sim 6$  s): Q1 THE LAW — the margin series recomputed in identified-corner coordinates ( $\tau = \lambda_{\min}(S + C) = \lambda_{\min}(A_h)$  at  $2.1 \times 10^{-12}$ ; cell ward  $\sum_j \text{cell}_j = v^T(A - B)v$  at  $10^{-8}$  on all rungs) reproduces the floor-strand envelope EXACTLY:  $e_1 = (\tau/\tau_{\text{pnt}})h^{3/2} \in [4.855, 24.209]$ ,  $\min \geq 4.335$  on all 67 rungs — two independently derived laws, one object;  $\tau_{\text{pnt}}$  carries the entire  $\alpha$ -decay (segment slopes  $-0.23..-0.30$ ),  $e_1$  the arithmetic; the fit-free  $h$ -exponent (21 near-equal-alpha rung pairs) measures median  $-1.341$ , 0.009 outside the frozen bar  $[-1.65, -1.35]$  — the honest miss kept, NOT refit (v829/v831 precedent); Q2 THE TOWER — the induced recursion on the dyadic mass subsequence  $4^m$  ( $m = 4..9$ ): ratio defects (0.6749, 0.5159, 0.0305, 0.3635) non-decaying — scale, not recursion; no quantifier reduction (the  $m = 12$  extrapolation typed HEURISTIC); Q3 THE CELLS — CI median 15.6 rising  $5.65 \rightarrow 29.46$ ; 22 287/351 122 per-prime aggregates nonpositive (deep  $C_2 = -0.623$ ) — the excess is NOT a positive sum over primes: the wall RECURSES into the cells; Q4 — compiler connection NULL (0/96 at  $10^{-3}$ ); CONTROLS — the scramble turns the margin negative on every rung; the  $h = 2$  Epstein routed margin negative AND below its own unrouted baseline at all anchors ( $-0.786/-1.100/-1.228$ , exact rational  $\Lambda_F$  routing). Suite 835  $\rightarrow$  836; standalone green with the pattern gate encoding the expected-FAIL signature.
- **Lean (1 module promoted).** TftptCarrier/ExcessSkeleton.lean — the excess-skeleton logical geography, kernel-checked (18 declarations; lake build green, 3410 jobs, no sorry, no native\_decide; imported in TftptCarrier.lean; lean\_manifest.sha256 regenerated with this round, 84 files): the two-giants decomposition (formAt\_add / excessAt\_eq\_comb — the excess carried EXACTLY by the comb block; the deployed structural block NOT assumed PSD), the interval-certificate lemma (pos\_of\_certified — the exact formal shape of every v842 certificate), the certified-negative discriminator (certified\_negative\_excludes — the Epstein rejection logic), the finite-ladder quantifier (ladder\_pos\_on\_finset), the bridge theorem excess\_floor with UniformMarginBound the SINGLE NAMED HYPOTHESIS (implying the per-rung IdentificationPositivity instances of SectorPositiveDescent; uniform\_to\_identification proven), and THE KERNEL-CHECKED GAP pointwise\_pos\_not\_uniform: margin sequences strictly positive at EVERY

rung with NO uniform lower bound exist (witness  $1/(m+1)$ ) — no finite certificate table, certified or not, discharges the uniform statement: the finite-table limit, until now a typing discipline, is a theorem. Cited in v843’s part B (numeric witness mirror, v837/v817 precedent).

- **The wall statement (the round’s deliverable).** The infinite quantifier sits on ONE scalar series:  $e_1(m)$ , certified  $\geq 4.335$  on every reachable rung, DOUBLY DERIVED (the consistency ward), self-similar at cell level, with no tower recursion and no compiler-constant shortcut; what any future route must supply is a UNIFORM lower bound on  $e_1$  — by the GUE-side findings a finer-than-statistical datum — with its arithmetic entry point declared. The bridge contract PRIME.FLOOR.PAIRCORR.01 [O] unchanged; same-class margin-law re-attacks (tower-recursion / cell-factorization of the excess) stop-listed per the measured self-similarity. NO RH claim.
- **Ledger.** 2 new rows (PRIME.MARGIN.LAW.01, FORM.PRIME.EXCESS.SKELETON.01 [F]) + the dated round-29 closing note on PRIME.RELATION.SKELETON.01 (the quantifier in its final coordinates; the self-similarity; the kernel-checked finite-table limit; the stop-list) + the dated round-29 margin-law note on PRIME.FLOOR.RATIO.01 (the envelope constant doubly derived — the consistency ward); 920  $\rightarrow$  922 rows; NO marker moves.
- **Papers.** `tfpt_prime_front`: the new round section “The margin as a doubly-derived law, and the wall kernel-checked” (`\veri{v843}`) + the part-intro chain extended; `tfpt_3`: one new audit card + the round-29 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the PRIME.MARGIN.LAW.01/FORM.PRIME.EXCESS.SKELETON.01 registration block + the dated round-29 notes on the PRIME.FLOOR.RATIO.01 and PRIME.RELATION.SKELETON.01 blocks.
- **Website.** Prime-front feed: the round entry “reflux round twenty-nine”; counters 835  $\rightarrow$  836 modules / v535–v843; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 836 modules / v843.
- **Scope.** The envelope ward, the regressions and the cell ward are [E]-grade machine checks on the frozen protocol; the law, the tower defects, the cell census and the compiler null are measured finite-level evidence; the one FAIL is a preregistered-honest adjudication on record (the bar NOT refit); the Lean statements are kernel-checked with `UniformMarginBound` deliberately NOT a theorem — the wall itself; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXXI) (reflux round twenty-eight — the wall gets coordinates: the corner-closure theorems, the GUE saturation, the relation carrier, and the certified skeleton: SIX modules (v837–v842, from eleven frozen probes plus one Lean module; suite 829  $\rightarrow$  835) — the corner route’s perimeter CLOSES as measured compression-class theorems (the 90-cell class map has no identity $\wedge$ visibility $\wedge$ placement cell; the quantifier extends to tower levels  $m \leq 3$ ; all 5276 subgroup expectations, the full pinching and Stinespring compressions obey the dichotomy; position-dependent carriers are EXTREMALLY PINNED —  $\hat{c}_{GL1} = -1$  locks the entire  $C_2$  mass, so identity-true carriers read zero on EVERY self-consistent comb), the zero side is QUANTIFIED (the bridge demand in Montgomery  $F(\alpha)$ -form with the named minimal missing input; the demand/GUE ratio saturates STRUCTURALLY at  $R_\infty = 1.11$  with the tight band pinned at unfolded  $\alpha \in 1..2$ ; the bootstrap loop is SHORT —  $g = 1/(k^2 R^2 c_{\text{sup}})$ , even ideal supply gives  $0.655 < 1$ : the wall’s self-conservation IS the saturation), and the route REOPENS with the relational input: the identification carrier EXISTS (PRIME.RELATION.MULT.01 [O]; all four gates incl. the self-consistency null; the  $h = 2$  Epstein  $x^2 + 5y^2$  separates at 200.0, Selberg-class-correct blindness), the identified corner’s excess is POSITIVE on all 67 rungs ( $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$ ), and the certified skeleton (PRIME.RELATION.SKELETON.01 [O]) encloses  $\tau_X$  STRICTLY POSITIVELY on all 67 rungs — the wall = the infinite quantifier over strictly-positive certified**

enclosures, sharpened demand: a UNIFORM lower bound on the excess margin; the corner/descent core kernel-checked in Lean (SectorPositiveDescent.lean, IdentificationPositivity the single named hypothesis); NO marker moves; NO RH claim anywhere)

- **New modules (6).** v837\_corner\_closure\_quantifier.py (19/19 + 17/17 + 3 Lean-mirror checks, DOORS-CLOSED + LEVEL2-CLOSED; probes hjelmslev\_open\_doors\_probe.py + hjelmslev\_level2\_corner\_probe.py, promoted verbatim: Door 1 — the GL1 corner reads the full lock-pair COMPRESSION, not a trace, so factor dressings are generically visible, but every visible dressing breaks the deployed-form identity by exactly its action on the lock pair (mass multiplier  $1.215 \times 10^{-1}$ , unital phase  $5.849 \times 10^{-2}$ ; the commutant invisible at  $2.7 \times 10^{-14}$ ; sweep dichotomy on all 24 draws: visibility  $\Leftrightarrow$  identity defect); Door 2 — the product law  $\hat{c}(\chi) = (-1)^e V_{\text{fac}}(w) M_{\text{fac}}(j; n)$  verified on the full 128-dim carrier Fourier for ALL  $128 \times 136$  character–event pairs (exact Fractions; 18-class collapse exact; the mass ward GENERALIZED: one position-blind V-mode + one  $\mu_4$ -mode per event, bit-identical under position scramble); THE CLASS MAP (90 cells): identity  $\Rightarrow$  NOT visible everywhere, carrier cells NONE; the quantifier extends to  $m \leq 3$  (rings  $\mathbb{Z}[i]/(1+i)^m$ , complete duals 16/256/4096; identity characters exactly  $\text{Ann}(\langle S_m \rangle)$ ,  $|S_m| = 7 \cdot 8^{m-1}$ ; Vfac spectra  $\{1, 0, -1/7\}$  — the jet refines position-blind data only); PART C: the Lean mirror of TfptCarrier/SectorPositiveDescent.lean (corner CP + trace split, descent transport, PSD limits, sector\_floor with IdentificationPositivity the single named hypothesis; lake build green, no sorry, imported in TfptCarrier.lean, v817 precedent)); v838\_corner\_expectation\_position.py (13/13 + 13/13, EXPECTATION-CLOSED + POSITION-CARRIER-TRADEOFF; probes conditional\_expectation\_probe.py + position\_carrier\_probe.py: ALL 5276 subgroups (74259 components) — identity( $E_H$ ) for EXACTLY 2 subgroups = the predicted criterion  $\text{Ann}(H) \subseteq \{1, \psi_0\}$ , 0 dichotomy violations, the full average reads ZERO; the pinching complete with the two pinned sectors as the ONLY PSD summands (the joint-use kill); Stinespring: the position data IS in the dilation yet the identity pins the read to the constant 1; the four position-carrier designs all fail the DECISIVE NULL (the self-consistent Epstein comb reads excess 0.0 identical to truth) — the fourth wall coordinate named EXTREMAL PINNING); v839\_paircorr\_bridge\_saturation.py (9/9 + 7/7, BRIDGE-FULLY-BEYOND + GUE-SATURATING; probes paircorr\_bridge\_map\_probe.py + gue\_saturation\_probe.py, promoted verbatim with no downscoping ( $\sim 0.8$  min per part, v830/v831 precedent; import remap to v823/v829/v830/v831 per the v831 precedent, shim symbols declared): the demand in Montgomery  $F(\alpha)$ -form on all 73 frames with the SCHOLARSHIP CORRECTION (Montgomery RH-conditional) carried in every supply row; the two-tier gap (unconditional misses by  $52\times$ ; the conditional window covers 43–86%; full-GUE rms closes 39/73); the NAMED OBJECT: unconditional  $F(\alpha, T) \leq C_F$  (0.33–1.7) out to  $\alpha \approx 3$ ; the saturation:  $R$  rises  $0.61 \rightarrow R_\infty = 1.114$  (deepest 20) inside  $[0.5, 1.5]$ ,  $\text{rms}(z) = 0.72$ , tight band unfolded  $\alpha \in 1..2$  (64% of the tower budget); the F-INSTRUMENT:  $\hat{F}(\text{win}) = 0.44 \pm 0.14$ , Poisson control  $0.94 \pm 0.46$  calibrates to 1, scramble fires); v840\_gue\_ablation\_loopgain.py (10/10 + 8/8, SATURATION-STRUCTURAL + LOOP-SHORT; probes gue\_ablation\_probe.py + bootstrap\_loop\_gain\_probe.py: all 6 source-native Guinand-admissible variants LAND at the plateau (1.058–1.199 vs 1.154, tol 0.35; 0 broke positivity), the tight band pinned in the UNFOLDED coordinate under  $D = 1/48, 1/64, 1/96$  and geometric- $h$  — a property of the tower class and the unfolded zeros, not the grid; the thinned comb breaks positivity ( $-0.105$ ), the scramble explodes ( $\times -954770$ ), Epstein positive-trivial ( $\times 9461$ ); the loop: supply  $c_{\text{hat}} = 0.88$  GUE-consistent, demand  $R$  climbing to 1.24, headline  $g(k=1) = 0.529$ ,  $g(k=2) = 0.132$ ,  $N_{\text{max}} = 0$  induction steps, IDEAL supply gives  $1/R^2 = 0.655 < 1$ ; the Poisson circularity ward passes ( $g_P = 0.296$ ; base case breaks 17.6%)); v841\_relation\_carrier\_ladder.py (18/18 + 13/13, RELATION-CARRIER-EXISTS + EXCESS-NONNEGATIVE; probes multiplicative\_relation\_probe.py + relation\_corner\_ladder\_probe.py: THE BREAKTHROUGH — the identification carrier EXISTS with the relational input:  $\mu * 1 = \delta$  and  $\Lambda = \mu * \log$  EXACT in the truncated convolution

frame; the four-comb discriminator exact (TRUE 0; s2s/ $\chi_4$  0 — Selberg-class-correct blindness; the  $h = 2$  Epstein  $x^2 + 5y^2$  reads 200.0 on 24 class-group-obstruction sites,  $21 = 3 \times 7$  first); ALL FOUR GATES pass incl. the self-consistency null (14/70 missing product relations, excess  $-0.297$  vs 0 at truth); the amplitude wiring fixes L-safety AUTOMATICALLY (H2 reads 196/392/496 exact on the anchors); the EXCESS positive on all 67 rungs (min  $+2.285$ , max  $+3.704$ ; scramble  $-1.224$ ) while  $\tau_X$  decays  $5.98 \times 10^{-4} \rightarrow 1.71 \times 10^{-5}$ ; `v842_excess_certified_skeleton.py` (9/9, SKELETON-CERTIFIED; probe `excess_certified_skeleton_probe.py`: strictly positive certified interval enclosures of  $\tau_X$  on ALL 67/67 rungs (widths  $5.4 \times 10^{-11}$ – $6.6 \times 10^{-9}$ , three to five orders below the margin; `chol_cert` discipline; midpoints reproduce the float ladder at  $4.9 \times 10^{-13}$ ); the  $h = 2$  fake certified-NEGATIVE on every anchor ( $-0.786/-1.100/-1.228$ , exact rational  $\Lambda_F$  routing); the scaling law typed: width slope  $+0.62$  vs margin slope  $-0.52$  per  $\alpha$ , horizon  $\alpha^* \approx 9.1$  (mass  $\sim 7.5 \times 10^7$ ) far beyond the ladder end 6.304 — certifiability not binding). Suite 829  $\rightarrow$  835; all six run standalone green with v757-precedent pattern gates; all eleven probes re-run identically at promotion (2026-08-07); the two zero-side modules promoted VERBATIM with no downscoping (measured  $\sim 1.5$  min each, v830/v831 runtime precedent).

- **Lean (1 module promoted).** `TfptCarrier/SectorPositiveDescent.lean` — the corner/descent core of the two hardest fronts, kernel-checked (lake build green, 3409 jobs, no `sorry`, no `native_decide`; imported in `TfptCarrier.lean`; `lean_manifest.sha256` regenerated with this round): corner compression by a hermitian idempotent is linear/idempotent/CP with the corner-trace split; descent towers with the measured-zero compatibility defect transport compatible positive families; entrywise limits of corner states are PSD (via `GramCompactness`); and the final theorem `sector_floor` passes `IdentificationPositivity` through as the SINGLE NAMED HYPOTHESIS — the kernel-checked logical geography of the wall: structural positivity proven, the identification datum deliberately NOT a theorem. Cited in v837’s part C (numeric witness mirror, v817 precedent).
- **The wall statement (the round’s deliverable).** The corner route’s perimeter is fully surveyed and CLOSED (register, tower levels, compressions, position carriers — four stop-list entries), the zero side is quantified (the demand SATURATES the GUE boundary structurally; the loop is SHORT: the wall conserves itself through the saturation), and the route REOPENS with the relational input: the identification carrier exists, the identified corner’s excess is positive on all 67 rungs, and the certified skeleton is strictly positive — the wall takes its sharpest form,  $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$ , and what remains is the INFINITE QUANTIFIER over strictly-positive certified enclosures: any future route must supply a UNIFORM lower bound on the excess margin, by the GUE-side findings a finer-than-statistical datum that no finite table settles. The bridge contract `PRIME.FLOOR.PAIRCORR.01` [O] unchanged. NO RH claim.
- **Ledger.** 6 new rows (`PRIME.CORNER.OPENDOORS.01`, `PRIME.CORNER.EXPECTATION.01`, `PRIME.FLOOR.BRIDGEMAP.01`, `PRIME.FLOOR.GUEABLATION.01`, `PRIME.RELATION.MULT.01` [O], `PRIME.RELATION.SKELETON.01` [O]) + the dated round-28 note on `PRIME.FLOOR.PAIRCORR.01` (the four closure coordinates complete; the saturation structural; the loop short; the relation route registered with the sharpened demand); 914  $\rightarrow$  920 rows; NO marker moves (the closures are typed route decisions, the registrations OPEN [O], the zero-side modules necessary-side consistency).
- **Papers.** `tfpt_prime_front`: the new round section “The wall gets coordinates: the corner closure theorems, the GUE saturation, the relation carrier, and the certified skeleton” (all six `\veri`) + the part-intro chain extended; `tfpt_3`: six new audit cards + the round-28 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the round-28 note on the `PRIME.FLOOR.PAIRCORR.01` block + four new blocks (the corner-closure quantifier closed at registration, the bridge/saturation/loop evidence block, and the two OPEN registrations `PRIME.RELATION.MULT.01` / `PRIME.RELATION.SKELETON.01`).
- **Website.** Prime-front feed: the round entry “reflux round twenty-eight”; counters 829  $\rightarrow$



835 modules / v535–v842; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 835 modules / v842.

- **Scope.** The closure modules (v837/v838) are typed route decisions under frozen preregistered protocols (exact wards [E]-grade, float window legs typed); the zero-side modules (v839/v840) are necessary-side demand/supply mappings and instrument calibrations with every supply row carrying its conditionality tag (Montgomery RH-conditional — the scholarship correction is part of the frozen protocol); the relation modules (v841/v842) register an OPEN route — the carrier exists and the skeleton is certified, the positivity along the ladder remains UNPROVEN and is exactly the wall; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXX)** (reflux round twenty-seven — the two-route decision at the wall, and three exact corpus compressions: FIVE modules (v832–v836, from seven frozen probes; suite 824 → 829) — the corner route’s STRUCTURAL half closes weight-generically (the character-corner identity  $T_{\text{GL1},X} = V^* e_{\text{GL1}} \pi(P_X) e_{\text{GL1}} V$  holds as polynomial algebra in free event weights with  $\hat{c}_j = -1$  exact for all events; the Gaussian Hjelmslev CP tower  $\mathbb{Z}[i]/(1+i)^m$ ,  $m = 1..5$ , is strictly projective at certified cb defect ZERO and carries the identity verbatim) — BUT identity and state corner are COMB-BLIND and the level- $m$  motion is the pure register-dilution law  $16^{1-m}$ : the arithmetic wall RELOCATES into the identification step and stays PRIME.FLOOR.PAIRCORR.01; the commutant SOS route is CLOSED DEFINITELY by exact rational certificates (unique trivial feasible point; forced  $G = \text{diag}(1, -1, -1)$  with dual point  $q(0, 1, 0) = -1$ ) — both routes stop-listed; plus the anchor affine normal form ( $T(x) = 2x - 2$ , the whole budget from ONE recursion), the flavor bidirectional checksum ( $Ra = (4, 10, 13) / R^T a = (6, 18, 8)$ , 1 of 12), the  $(1+i)$  ramification ladder (four corpus roles = ONE object) and the PG(3, 2) flag normal form ( $105 = 45 + 60$  as the  $\text{Sp}(4, 2)$ -orbit split, the doily obstruction typed); NO marker moves; NO RH claim anywhere)

- **New modules (5).** v832\_anchor\_flavor\_checksum.py (18/18 + 15/15, ANCHOR-AFFINE-EXACT + FLAVOR-BIDIR-CHECKSUM (selective); probes anchor\_affine\_probe.py + flavor\_bidir\_probe.py, promoted verbatim: the anchor power sums  $p_n = 2 + 2^n$  obey  $p_{n+1} = 2p_n - 2$  identically (sympy);  $T(x) = 2x - 2$  has the UNIQUE fixed point  $2 = |\mathbb{Z}_2|$  and the  $T$ -orbit of 4 is the compiler quintet (4, 6, 10, 18, 34),  $p_0 = 3 = N_{\text{fam}}$  prepended — the whole budget from one recursion:  $|R(E_8)| = p_1 p_2 p_3 = 240$ ,  $\dim E_8 = 240 + q_3 = 248$  (ladder identity  $p_4 - p_3 = p_3 - 2 = 8$ ),  $h(E_8) = p_2 p_3 / 2 = 30$ ,  $|R(D_5)| = 40$ ,  $\Omega_{\text{adm}} = 48$ ,  $10b_1 = 41$ ; the flavor matrix decodes the SAME anchor bidirectionally —  $Ra = (4, 10, 13) = (p_1, p_3, N(3+2i))$ ,  $R^T a = (6, 18, 8) = (p_2, p_4, p_3 - 2)$ ; of the 12 down-row candidates exactly ONE (the accepted (1, 5, 2)) passes the full identity set, the new one-number kill being the anchor contraction  $(Ra)_2 = 10 = p_3$  (siblings: 11, 12); controls fire incl. the sibling reproducing tfpt\_2’s failure numbers verbatim); v833\_gaussian\_ramification\_ladder.py (33/33, RAMIFICATION-LADDER-EXACT; probe gaussian\_ramification\_probe.py: the four documented roles of  $\pi_2 = 1+i$  re-certified from scratch as ONE ladder — norm doubling  $((1+J)^T(1+J) = 2I_8$  exact, zero class empty 0/240,  $240 = 15 \times 16$ , elementary divisors  $(1^4, 2^4)$ ), the 4-bit address (deck trivial), the non-split jet  $(\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2))$  unique among 24 bijections, deck  $= 1+\epsilon$ , exactly 0/65536 deck-equivariant sections, the 120 root pairs the  $q=1$  shell), and the metaplectic lift  $(\zeta_8 = (1+i)/\sqrt{2})$ ,  $(SH)^3 = \zeta_8 I$ ,  $(T_3 K_3)^3 = iI =$  the deck; exact  $\mathbb{Z}[\zeta_8]$  BFS census  $|C_2/\mu_8| = 11520$ , ZERO Galois-mixed classes,  $C_2 = K \sqcup \zeta_8 K$ ,  $\chi(H \otimes I) = 1$  the RM–CSS phase bit); controls: the split ring admits 0 isos, the split extension splits 65536/65536); v834\_pg32\_flag\_completion.py (26/26, PG32-FLAGS-EXACT + INCIDENCE-SPLIT; probe pg32\_flag\_completion\_probe.py:  $\binom{4}{2}_2 = 35$ ,  $140 = 35 + 105$ ,  $105 = 35 \times 3$  flags  $\cong$  point pairs; the RM layer rebuilt as set equalities ( $[15, 10, 4]$  enumerator exact, Grams  $22I + 6J / 6I + J / 28I + 7J$ ); the vacuum coordinate is ORIGIN RESTORATION; the kill typed as a

split:  $105 = 45 + 60$  is the  $\mathrm{Sp}(4, 2)$ -ORBIT decomposition on both sides (720 brute force), the polarity dictionary on the 60 non-isotropic rungs is bijective/equivariant/incidence-preserving, the 45 doily rungs are obstructed ( $(t+U) \cap U = \emptyset$  45/45, flag Gram  $18I+3J \neq 22I+6J$ ) — “105 checks = 105 flags” is a deck-equivariant bijection, NOT an incidence isomorphism (the typed reason the v822 dilation died)); v835\_corner\_hjelmslev\_tower.py (21/21 + 20/20, CORNER-IDENTITY-SYMBOLIC + HJELMSLEV-STRUCTURE-ONLY; probes character\_corner\_probe.py + hjelmslev\_cp\_tower\_probe.py: the corner identity holds as POLYNOMIAL ALGEBRA in free event weights (symbolic toy leg exact; all 3 deployed windows on two independent code paths); the entire equality step consumes ONE piece of arithmetic — the glue identity  $\sum_m \Theta_m = 240\sigma_3$  + the uniform  $H^*$  mass — forcing  $\hat{c}_j = -1$  EXACTLY for all 136 events (exact group-algebra Fourier, 128-dim register); honest scope: the identity-carrying object is NOT positive (1 of 12 sectors PSD), the positive GNS corner is NOT  $\tau_X$ , and the scramble proves identity + state corner COMB-BLIND while the lock data moves ( $\tau = +5.98 \times 10^{-4} \rightarrow -3.51$ ); the Hjelmslev ladder  $m = 1.5$  carries exact counts  $15 \cdot 8^{m-1}/35 \cdot 16^{m-1}/105 \cdot 32^{m-1}$ , strictly projective CP channels (certified cb defect identically ZERO), level-exact  $\hat{c}^{(m)} = -1$ , fence silent — but H5 decides AGAINST identification: the level- $m$  corner is comb-blind with pure dilution  $16^{1-m}$  (ratio 0.0625 exactly); the one live door: position-dependent Kraus data on the level- $m$  flags — register-side dressings provably cannot carry it (hjelmslev\_position\_kraus\_probe.py, POSITION-KRAUS-TRADEOFF, 13/13, probe-level, cited in the provenance)); v836\_commutant\_sos\_closure.py (29/29, COMMUTANT-SOS-INFEASIBLE; probe commutant\_sos\_probe.py, the declared SPEC v1→v2 witness-triple repair documented: the degree-2 SOS ansatz over the canonical five-dim abelian subalgebra  $\{\Pi_A(0), \Pi_A(1), \Pi_A(4), \Pi_A(7), \Pi_A(9)\}$  of the 39-dim commutant is closed DEFINITELY — (a) the exact reduction lemma collapses the SDP (4 blocks  $S_+^5$ , 60 unknowns, 85 constraints) to a rational LP with the TRIVIAL diagonal rewriting as unique feasible point (Vandermonde kernel  $\{0\}$ ): no cross-sector positive transfer; (b) the rank-3 det form has signature (1, 2) ( $\det S = t^2 - x^2 - y^2$ , compiler null vector  $q(5, -3, 4) = 0$ ), coefficient matching FORCES  $G = \mathrm{diag}(1, -1, -1)$  (not PSD) with the rational dual certificate  $q(0, 1, 0) = -1 < 0$ , and  $q$ -negative directions are achievable even on the nonneg cone (832/2799 truncation cuts); FENCE-CLEAN on the algebra layer, FENCE-HIT on the positivity branch — the nontrivial version is pair-correlation substance; controls fire (comb-specific identification, the v815 sector pattern, wrong-ratio inconsistency)). Suite 824 → 829; all five run standalone green and encode their frozen patterns per the v757 precedent; all seven probes (plus the position-Kraus companion) re-run identically at promotion (2026-08-07).

- **The wall statement (the round’s deliverable).** Both corner-era routes are DECIDED: the corner route delivers everything structural (identity, projector, strictly projective CP tower, level-exactness) and shows the structure is placement-blind — the arithmetic lives entirely in the IDENTIFICATION step (a positive  $P_{\mathrm{lock}}$  whose corner IS the deployed window form); the SOS route shows no positive algebraic rearrangement can manufacture the floor. The wall stays exactly PRIME.FLOOR.PAIRCORR.01 [O], relocated into identification, with both routes stop-listed per house discipline. NO RH claim.
- **Ledger.** 5 new rows (ANCHOR.AFFINE.01, GAUSSIAN.RAMLADDER.01, PG32.FLAGS.01, PRIME.CORNER.CHARACTER.01, PRIME.COMMUTANT.SOS.01) + the dated round-27 two-route note on PRIME.FLOOR.PAIRCORR.01 (corner structural half closed / wall relocated; SOS closed with the rational dual certificate; both stop-list entries; the position-Kraus trade-off finding); 909 → 914 rows; NO marker moves (the compressions are normal forms of existing corpus surfaces; the corner/SOS results are typed route decisions).
- **Papers.** tfpt\_1: the anchor affine normal form keybox (Theorem 8) + the affine pointer in the anchor-first refinement + the  $(1+i)$  ramification-ladder entry in the Gaussian code keybox ( $\backslash\mathrm{veri}\{v832\}$ ,  $\backslash\mathrm{veri}\{v833\}$ ); tfpt\_2: the bidirectional anchor checksum keybox after the spectral-selector block ( $\backslash\mathrm{veri}\{v832\}$ ); tfpt\_prime\_front: the new round section “The two-route decision at the wall” ( $\backslash\mathrm{veri}\{v834/v835/v836\}$ )

+ the part-intro chain extended; `tfpt_3`: five new audit blocks + the round-27 follow-up notes on the floor-contract prose; `tfpt_research_contracts`: the `PRIME.CORNER.CHARACTER.01/PRIME.HJELMSLEV.CPTOWER.01` and `PRIME.COMMUTANT.SOS.01` registration blocks + the dated round-27 note on the `PRIME.FLOOR.PAIRCORR.01` block; `note_e8_gaussian_code`: a dated pointer to the ramification ladder in the machine-verification section.

- **Website.** Prime-front feed: the round entry “reflux round twenty-seven”; counters  $824 \rightarrow 829$  modules / `v535–v836`; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `maps`); `README` and `Zenodo` updated to 829 modules / `v836`.
- **Scope.** The three compressions (`v832–v834`) are exact-arithmetic normal forms of existing corpus tables — compressions, not new claims; the corner/Hjlemslev/SOS results are typed route decisions under frozen preregistered protocols (exact wards `[E]`-grade, float window legs typed, the honest scope measured); the queued Lean follow-up (`SectorPositiveDescent.lean`) stays queued; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXIX)** (reflux round twenty-six — the typed circularity boundary: ONE module (`v831`, from one frozen probe; suite  $823 \rightarrow 824$ ) — the correlation-corrected alias second moment is EXACT (sympy-proven bilinear form  $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$ , identity-warded on all 73 battery frames), the round-25 premise “the phases carry the gap” is OVERTURNED (the AMPLITUDES carry it: the tower law decomposes exactly as  $-2.505 = -3.172$  amplitude law +  $0.668$  fading alignment), the exact Guinand split puts the coherent sum in the prime-comb fluctuation at  $0.01 \times$  the Poisson square-root scale (smooth  $-994$  / fluctuation  $+995$ , each  $\sim 1000 \times$  the residual, cancelling; the comb scramble explodes it  $10^6 \times$  — THE CANCELLATION IS THE ARITHMETIC) — and the analytic-envelope route of `PRIME.FLOOR.RATIO.01` is CLOSED HONESTLY at its typed circularity boundary: by Guinand the comb’s self-cancellation IS the zero-side floor statement itself, the required control is even deeper than Montgomery-type pair-correlation input, the envelope’s constant is arithmetic substance equivalent to the target — stop-list entry, no further analytic-envelope variants; the demand moves to the NEW bridge contract `PRIME.FLOOR.PAIRCORR.01` [O]; NO marker moves; NO RH claim anywhere)

- **New module (1).** `v831_prime_alias_second_moment.py` (15 checks with the SEVEN preregistered-honest premise-overturning FAILs `S1.SIGN/W2B/W2C/W2D/S4.GEOM/C1/C2` pattern-gated — the FAIL pattern IS the finding; verdict `ALIAS-CORRELATION-PAIRCORR`; discovery probe `prime_alias_second_moment_probe.py`, 2026-08-07, re-run identically at promotion; promoted VERBATIM with NO downscoping — measured runtime  $\sim 0.1$  min in-tree, the zero list is the `v823` cache + vectorized RS scan): the EXACT BILINEAR FORM (sympy-proven, `S0 3/3`): in the pole frame the zeros+pole gram is exactly  $M_2 = [[P_0 + (S_2 + S_c)/2, S_s/2], [S_s/2, (S_2 - S_c)/2]]$  with  $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$  — the correlation combination at the leading term is the coherent phase sum  $S_c$  (rotation-invariance ward at the conditioning-aware machine bar on all 73 frames, truncated-series dev  $3.8 \times 10^{-7}$ ); THE PREMISE OVERTURNED: the diagonal (random-phase) part  $\tau_{\text{diag}} = S_2/2$  slopes  $-3.172$  not  $-1$  (W2B), phase scramble  $-3.054$  (W2C), Poissonized zeros  $-3.654$  with  $q_{\text{poiss}}$  median  $0.34 > 0.5 q_{\text{true}}$  (W2D), and  $q = S_c/S_2$  changes sign across the battery ( $[-0.324, 0.552]$ , median  $0.19$ ; deep  $M = 1632$  suppression only  $\times 1.1$ ) — the full tower  $-2.505$  ( $R^2$   $0.98$ , W2A PASS) decomposes EXACTLY as amplitude law  $-3.172$  + fading alignment  $+0.668$ ; THE GUINAND SPLIT (exact, identity-route residue  $0.019$ , soft bar met):  $G_Z = (B - S_p - P) + (S_p - S)$  gives smooth share  $-994.15$  / prime-fluctuation share  $+995.15$ , cancelling to  $|S_c^{\text{fluc}}|/\sigma_{\text{sqr}} = 0.01$  (SQRT-CANCELLATION REGIME); the smooth surrogate does NOT carry it (`S4.GEOM`  $0.868$ ,

deficit-slope dev 2.274); comb scramble explodes the coherent sum  $\times 2.0 \times 10^6 - 1.7 \times 10^7$ , the Epstein  $x^2 + 5y^2$  comb  $\times 5.1 \times 10^3 - 1.3 \times 10^4$  — comb-specific arithmetic; THE TYPED CIRCULARITY BOUNDARY: proving the tower  $\tau$ -law requires bounding an oscillatory prime sum AT its square-root scale = variance-type, pair-correlation-equivalent substance (Montgomery-type on the zero side by the explicit formula), and by Guinand the comb’s self-cancellation IS the zero-side floor statement itself. Suite 823  $\rightarrow$  824; standalone green with the v757-precedent pattern gate.

- **Contracts.** NEW: **PRIME.FLOOR.PAIRCORR.01 [O]** — the floor  $\leftrightarrow$  pair-correlation bridge: the floor’s envelope constant is Guinand-EQUIVALENT to the zero-side floor statement; demands typed in both directions (FORWARD: an unconditional variance-type bound  $|\text{aniso}(S_p - S)| \leq C\sigma_{\text{sqrt}}$  uniform in  $h$  with the pair-correlation-grade entry point DECLARED — no hidden circularity; BACKWARD: instrument calibration of floor margins into effective pair-correlation statements, never a theorem about zeros); KILL: fluctuation share collapsing below 0.6 while the suppression persists, or the scale ratio rising systematically above 3 (one-point regime — the boundary mis-typed, the route reopens on PNT-grade input); CLOSE: the forward demand delivered with its entry point named; explicitly does NOT promise an unconditional proof (the proving direction contains pair-correlation-grade arithmetic, fenced). **PRIME.FLOOR.RATIO.01 [O] ANALYTIC-ENVELOPE ROUTE CLOSED** (dated closure note): the round-25 alias-phase blocker resolves INTO the typed boundary; NO further analytic-envelope variants (STOP-LIST entry per house discipline); kill criteria unchanged, still [O], promotion fenced.
- **Ledger.** 2 new rows (PRIME.FLOOR.ALIASMOMENT.01 + the contract row PRIME.FLOOR.PAIRCORR.01) + the dated round-26 closure note on PRIME.FLOOR.RATIO.01; 907  $\rightarrow$  909 rows; NO marker moves (the bilinear form and its expansion sympy-proven [E]-grade wards; the overturned premise and the sqrt-scale cancellation measured; the contracts [O]).
- **Papers.** **tfpt\_prime\_front**: the new round section “The typed circularity boundary: the analytic-envelope route closes, and the floor–pair-correlation bridge is named” (`\veri{v831}`); **tfpt\_3**: one new audit block + the round-26 follow-up notes on the floor-contract prose; **tfpt\_research\_contracts**: the PRIME.FLOOR.PAIRCORR.01 registration block, the dated closure note on PRIME.FLOOR.RATIO.01, and the typed-blocker note on the depth-kill/budget block.
- **Website.** Prime-front feed: the round entry “reflux round twenty-six”; counters 823  $\rightarrow$  824 modules / v535–v831; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 824 modules / v831.
- **Scope.** The bilinear form, its  $1/P_0$  expansion and the det identity are sympy-proven [E]-grade wards; the overturned premise, the exact Guinand split shares, the sqrt-scale ratio and the comb-scramble/Epstein fingerprints are measured finite-level evidence on the frozen battery; the seven FAILs are preregistered-honest adjudications on record (the FAIL pattern IS the finding, pattern-gated); the boundary is a TYPED BOUNDARY, not a theorem — the route closure is a discipline statement, not a positivity result; NO marker moves; NO RH claim anywhere.

**2026-08-07 (LXVIII)** (reflux round twenty-five — the exclusion detector and the depth kill: SIX modules (v825–v830, from eleven frozen probes; suite 817  $\rightarrow$  823) — the verified PSD rungs of the GL1 prime-comb tower become a hash-preregistered *window detector*: certified exclusion regions with rigorous Cholesky/Higham + witness-Rayleigh certificates to  $X = 24.81$ , a zero locator validated OUT-OF-SAMPLE on a disjoint untouched window (83% detection, 0% false positives, precision 0.086), and the capstone comb-native window verification whose census reconciles against the Riemann–von Mangoldt count ( $21+4$  typed misses = 25 = cache = RvM; new deepest certified rung  $X = 25.5$ , comb cap  $1.2 \times 10^{11}$ ) — registered as the new contract **PRIME.DETECTOR.WINDOW.01 [O]**; and the **PRIME.FLOOR.RATIO.01**

kill gates, played at full sieve depth, SURVIVE with growing margins while the  $\alpha^* \approx 11$  horizon falls as a float-convention artifact (citation grade to  $\alpha = 12.75$ ; envelope certified-explicit  $\rho \geq 4.335 h^{-3/2}$  on 73/73; the alias-phase proof blocker named) — every statement typed strictly weaker than classical verification; NO marker moves; NO RH claim anywhere)

- **New modules (6).** `v825_prime_exclusion_ladder.py` (26/26 + 21/21, LADDER-EXTENDED + SATURATION-TYPED: every verified PSD rung of the GL1 tower ( $D = 1/64$ ) carries a rigorous  $\lambda_{\min} > 0$  certificate — Cholesky PD verification with the Higham backward-error bound plus a Rayleigh-residual enclosure, e.g.  $\lambda_{\min}(X=24.81) = 1.078 \times 10^{-6} \geq 9.70 \times 10^{-7}$  CERTIFIED — and is inverted through the tent-read Guinand identity into quadruple-exclusion regions  $\{1/2 \pm \delta \pm i\gamma\}$  on the frozen rank-4 battery, every published boundary point re-proved by a witness-Rayleigh certificate; the census threshold  $\Xi_{\text{eff}} = X \delta_{\text{mb}}$  falls  $0.639 \rightarrow 0.457 \rightarrow 0.274 \rightarrow 0.218$  and PLATEAUS at  $0.218 \pm 3\%$  (gates recomputed from the frozen deep series; mechanism SUPPORT-ACTIVE-BUT-X-STALLED: support slope  $-0.68$ , rank-12 gain growing  $19.3 \rightarrow 25.0\%$  below the 30% bar); multi-zero double injection breaks without interference rescue; honest reach typed: Nyquist  $\pi/D = 201$ , region INSIDE the Platt-Trudgian strip — consistency re-derivation, not new territory); `v826_prime_exclusion_battery2.py` (16/16 + 13/13, FLOOR-LOWERED + RANGE-DECIDES-DECLINE: battery v2 designed a priori from instrument diagnostics only and SHA-256-hashed `fd39fb42..` BEFORE any evaluation (no zero datum;  $v1 \subset v2$ , pointwise dominance live-verified); old-grid floor  $\leq 0.0979$  with the grid-censoring caveat typed; the measurement-range extension (exact-subset grid, bit-for-bit ward) UNCENSORS the series  $0.1066/0.0982/0.0960/0.0913/0.0816$  at slope  $-1.39$  — the v825 plateau was the instrument, not the tower; benchmark  $\delta \geq 10^{-3}$  at  $X^* = 81.6$  (extrapolation, typed); side-finding: rank correlation  $-0.872$  of the uncensored boundaries with the distance to the nearest true ordinate, hash-verified non-circular); `v827_prime_zero_locator.py` (the ONE preregistered-honest C3 FAIL + 13/13, LOCATOR-NULL + LOCATOR-V2-RESOLVES: the width profile  $W(\gamma)$  PEAKS at true ordinates (explicit-formula redundancy: a real zero makes the injected quadruple redundant, the boundary rises);  $v1$  honestly failed its frozen gates (12/13 peaks but 61% false peaks at  $\ln 1.5$ ) — on record; the post-hoc separation (matched  $\geq 0.82$ , false  $\leq 0.73$ ) froze  $\ln 2.2$  for  $v2$ , preregistered `f57a2e7f..` BEFORE any test evaluation; out-of-sample on the disjoint window  $[60, 120]$ :  $20/24 = 83\%$  detection, 0% false positives, precision 0.086, all four controls (scramble/Epstein no peaks, shifted null below the structural re-lock ceiling, grid refinement 7/7 stable); depth law  $54 \rightarrow 62 \rightarrow 83\%$ ); `v828_prime_comb_window.py` (12/12, COMB-VERIFIES-WINDOW, the capstone: on  $\gamma \in (60, 120]$  at  $X = 24.8125$  ONE certified positivity object yields location (21/25 ordinates to  $\pm 0.25$ , max error 0.242, ZERO unmatched peaks), exclusion ( $\delta \geq \delta_{\text{mb}}(\gamma)$  pointwise, uniform bound 0.0496, witness-certified at six frozen samples) and the census (21+4 typed misses = 25 = cached verified count = rounded RvM main term  $25.237 \rightarrow 25$ ); miss autopsy complete (1 pair-merge below the grid resolution +3 prominence-limited; the depth-recovery prediction at the NEW deepest certified rung  $X = 25.5$ ,  $\lambda_{\min} = 8.07 \times 10^{-7} \geq 7.26 \times 10^{-7}$  CERTIFIED, verified as typed:  $0/3 + 0/1$ ); the scramble fails ALL THREE axes; WEAKER-THAN-CLASSICAL typing mandatory and carried verbatim: Turing proves  $N(T)$  and  $\delta = 0$  exactly, the comb leaves a  $\delta_{\text{mb}}$ -strip unresolved and can never close it at exponential comb cost — a necessary-side consistency demonstration; does NOT bear on RH); `v829_prime_floor_depth.py` (16/17 with the ONE preregistered-honest S5.HLAW FAIL + 12/12, ENVELOPE-HOLDS-DEEP + FAMILY-SCALES: the PRIME.FLOOR.RATIO.01 kill gates K1 ( $\rho h^{3/2} \geq c = 4.85$ , NO refit) and K2 ( $\cos^2 \theta \geq 1/2$ ) played at  $X = 18.375\text{--}25.5$  on uniform-grid tower frames whose  $h = 588\text{--}816$  sit INSIDE the deployed  $h$ -range while the depth doubles — the test decouples depth from dimension — and SURVIVED: margins  $\times 5.76/6.24/6.53/7.16/8.17/8.92$  growing monotonically,  $\min \cos^2 \theta = 0.849$ , non-decay +0.060 per log-unit (the  $h$ -law wins); the typed miss: the single pair collapses (gap  $40\times \rightarrow 673\times$ ) and the certified top-100 family takes over (0.977–0.981 of the floor,  $\rho_{\text{certfam}}$  within 2%);  $\alpha$ -coverage typed: citation grade only  $\alpha \in \{9.19, 10.36, 11.05\}$  — exactly the v830 blocker); `v830_prime_float_budget.py` (10/10 + 19 checks with the FIVE

preregistered-honest asymptotic FAILs A7/A8/A9/A10/S2.EXP, BUDGET-LINEAR-CLOSES-ALL + ENVELOPE-ASYMPTOTIC-PARTIAL: the deployed quadratic float convention  $100 \varepsilon \|A_2\|_F^2$  was four orders above the true tail term at depth; the explicit Higham-linear chain (compensated banded accumulation with TwoSum,  $N_{\text{SUB}} = 2^{16}$ ; the irreducible atom-position term  $\varepsilon X/D$ ; exact fsum dots; dps-40 spot anchors with  $\times 10$  caps) is  $7 \times -171 \times$  tighter and closes the citation-grade family gate 6/6 (old: 3/6) — citation grade extends to  $\alpha = 12.75$ , FULL SIEVE DEPTH; CONVENTION NOTE: the deployed v818 convention stays FROZEN in the existing suite, the linear budget is registered as convention for NEW modules only; the analytic part runs near-verbatim: family closed form exact on ALL 73 battery frames ( $3.6 \times 10^{-11}$ ), sympy-proven limit algebra, phase conspiracy excluded (dip floor 0.170), and THE EXPLICIT CERTIFIED ENVELOPE  $\rho \geq \rho_{\text{certfam}} \geq 4.335 h^{-3/2}$  on 73/73, non-decaying; the proof blocker typed: alias phase correlations — random-phase predicts  $\tau \sim h^{-1}$ , the tower measures  $h^{-2.5}$  — the envelope stands certified-explicit, not proven). Suite 817  $\rightarrow$  823; all six run standalone green and encode their frozen patterns (incl. the expected-FAIL patterns) per the v757 precedent; all eleven probes re-run identically at promotion (2026-08-07); the deep sieves (to  $1.2 \times 10^{11}$ ,  $\sim 17$ – $25$  min per probe) are downscoped in the suite modules to fast live sub-ladders at  $X \leq 18.375$  with the full-depth numbers recorded as frozen reference constants and every decision gate RECOMPUTED from them — predeclared in each module’s PROVENANCE.

- **Contracts.** NEW: **PRIME.DETECTOR.WINDOW.01 [O]** — the comb-native window detector (location + exclusion + census, hash-warded end to end, validated out-of-sample); OPEN: the depth-to-width law as theorem, the completeness mechanism, window extension beyond Nyquist (explicitly NOT promised); KILL: out-of-sample detection  $< 40\%$  on a deeper certified rung, a census contradiction on any PD-certified window, or a false positive surviving the grid-refinement ward; mandatory weaker-than-classical typing carried by every module of the strand. **PRIME.FLOOR.RATIO.01 [O] DEPTH-HARDENED** (dated contract-text update): kill gates played at the contract’s own preregistration and survived at  $X = 25.5$ ; the round-24  $\alpha^* \approx 11$  horizon REMOVED as a float-convention artifact (citation grade to  $\alpha = 12.75$ ; deployed v818 budget convention unchanged, the Higham-linear budget is convention for NEW modules only); the envelope now certified-explicit ( $c_{\text{cert}} = 4.335 h^{-3/2}$  on 73/73); the remaining proof blocker NAMED (alias phase correlations, random-phase  $h^{-1}$  vs tower  $h^{-2.5}$ ); kill criteria unchanged, still [O], promotion fenced.
- **Ledger.** 7 new rows (PRIME.EXCLUSION.LADDER.01, PRIME.EXCLUSION.BATTERY2.01, PRIME.EXCLUSION.LOCATOR.01, PRIME.EXCLUSION.WINDOW.01, PRIME.FLOOR.DEPTHKILL.01, PRIME.FLOOR.BUDGET.01 + the contract row PRIME.DETECTOR.WINDOW.01) + the dated depth-hardening update on PRIME.FLOOR.RATIO.01; 900  $\rightarrow$  907 rows; NO marker moves anywhere (certificates and preregistration hashes [E]-grade wards; exclusion maps, locator scores, envelope constants measured; the contracts [O]).
- **Papers.** **tfpt\_prime\_front**: the new round section “The exclusion detector: the certified ladder becomes a window-verification instrument, and the floor survives its kill gates at full sieve depth” (all six \code{veri} citations; the weaker-than-classical typing carried in the body); **tfpt\_3**: six new audit blocks + the two round-25 follow-up notes on the floor-contract prose; **tfpt\_research\_contracts**: the PRIME.DETECTOR.WINDOW.01 registration, the depth-kill/budget registration block, and the dated PRIME.FLOOR.RATIO.01 note.
- **Website.** Prime-front feed: the round entry “reflux round twenty-five”; counters 817  $\rightarrow$  823 modules / v535–v830; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 823 modules / v830.
- **Scope.** The certificates (Cholesky/Higham, witness-Rayleigh, preregistration hashes, sympy algebra, dps-40 anchors) are [E]-grade machine wards at float precision; the exclusion maps, locator scores, envelope margins and closure constants are measured finite-level evidence on frozen instruments; the honest negatives (LOCATOR-NULL, the S5.HLAW single-pair collapse,

the five asymptotic FAILs of the analytic envelope) are preregistered adjudications on record; the mandatory typing is everywhere: locations to  $\pm 0.25$  and exclusions to  $\delta \geq \delta_{\text{mb}}$  are strictly weaker than classical verification, every verified window lies inside the Platt–Trudgian strip, and the strand is necessary-side evidence — NO marker moves; NO RH claim anywhere.

**2026-08-06 (LXVII) (reflux round twenty-four — the floor-theorem skeleton: TWO modules (v823–v824, from four frozen probes; suite 815  $\rightarrow$  817) — the sector floor  $\det \hat{A}_2 = \lambda\tau$  becomes an EXACT machine-verified sum of squares over zero+pole rank-one carriers (Lagrange identity), the fixed pair (pole  $\times \gamma_1$ ) is certified strictly positive on all 14 rungs with the budget tightened  $\sim 10$  orders by the psd-remainder monotonicity chain, and the three-piece uniform certified lower-bound skeleton closes at CITATION grade for all  $h$  at the deployed  $\alpha$  up to the validity horizon  $\alpha^* \approx 11$  — PRIME.FLOOR.RATIO.01 stays [O], NARROWED: the floor bound is battery-relative and  $\alpha$ -bounded, necessary-side only, NO RH claim)**

- **New modules (2).** v823\_prime\_lagrange\_floor.py (7/8 with the ONE preregistered-honest old-budget FAIL S3.CERT + 9/9, LAGRANGE-CONCENTRATED + PAIR-CERTIFIED: every per-zero layer of the v692 master identity is EXACTLY rank-one — the odd-extension symmetry factors the common phase,  $v_\gamma = 2\sqrt{w(\gamma)}(S_1, S_2)$  with the real closed form  $S_i(\phi) = \sum_j t_{ij} \sin((h-j-\frac{1}{2})\phi)$  and the alias weight  $w(\gamma) = D \text{csinc}^2(\gamma D/2) \geq 0$  (layer dev  $\leq 1.2 \times 10^{-16}$ , pole rank-one  $\leq 4.8 \times 10^{-17}$ ); the LAGRANGE IDENTITY  $\det(G_Z + P) = \sum_{\text{pairs}} w_i w_j (a_i b_j - a_j b_i)^2$  holds at machine precision on all 14 rungs (ward  $\leq 1.4 \times 10^{-9}$ ); the POLE is the universal non-collinear leg (the pole family carries 1.0000 of the pair total, every top-100 pair is pole  $\times$  zero) and the moving zero leg obeys the ALIAS-COMB LAW (median comb distance 0.0007–0.0061 vs uniform 0.25); A0 typed honestly: the atom-side rank-one framing FAILS as frozen ( $0.10-0.23 < 0.5$ ) — the exact SOS realization lives on the zero+pole side; THE ROUTE CHANGE:  $R = \hat{A}_2 - G_Z - P$  is an exact float  $2 \times 2$  with  $\lambda_{\min}(R) = 1.6 \times 10^{-7} - 2.8 \times 10^{-6} \geq \text{margin}$  verified per rung, and the  $2 \times 2$  psd fact  $\det(B+R) \geq \det B$  ELIMINATES the tail estimate from the chain — BUD  $0.14-7.1 \rightarrow 2.4 \times 10^{-12} - 2.7 \times 10^{-9}$ , median 9.8 ORDERS tightened; the certified interval  $X^2(\text{pole}, \gamma_1) - \text{BUD}$  is strictly positive on 14/14 ( $\gamma_1$  pinned to `zetazero(1)` within  $10^{-11}$ ; dps-40 mpmath ward inside the interval); the fixed pair carries 0.0063–0.5832 of the floor with slope  $h^{+1.06}$  — a GROWING brick; controls fire (scramble  $\times 45.8-49.0$ , Epstein  $\times 1784-50029$  at det scale, collinear  $\det = 9.2 \times 10^{-16}$  with interval containing 0)); v824\_prime\_floor\_skeleton.py (12/12 + 9/9, FLOOR-PARTIAL + TAIL-CLOSED-ALL-H: PIECE 2 UNIFORM-VERIFIED — the ANALYTIC  $h \rightarrow \infty$  limit  $X_\infty(\alpha) = 16\alpha\pi^2 \sin(\alpha\gamma_1) \sinh(\alpha/2)$  [bracket] (Dirichlet closed form = direct sum at  $5.7 \times 10^{-13}$ , pipeline  $8.5 \times 10^{-13}$ , ZERO sign flips over 14  $\alpha \times 10$  synthetic  $h$ ; the explicit bound  $L(h) = X_\infty^2(\alpha)(1 - C(\alpha)/h)$  for  $h > h_0(\alpha) = C(\alpha) \leq 499.6$ ; honest limitation: 25  $\alpha$ -NODES of  $\sin(\alpha\gamma_1)$  on  $[2, 12]$  —  $h$ -uniform per  $\alpha$ , NOT ladder-uniform in  $\alpha$ ); PIECE 3 UNIFORM-VERIFIED — FAMILY EXHAUSTION: the certified top-100 (pole  $\times$  zero) family carries median 0.952 (min 0.931, max 0.974) of the floor, all carriers on-line by verified computation ( $\leq 2 \times 10^4$ ); the residue 0.015–0.049 with trend slope  $+0.010$  (truncation depth, not a structural carrier); the analytic family limit stays positive across ALL fixed-pair nodes (min  $1.62 \times 10^{-4}$  at  $\alpha = 2.68$ ) — the family never dies simultaneously; PIECE 1 CLOSED BY THE TAIL REPAIR: the crude coherence-discarding envelope closes only 4/14 ( $h \leq 434$ ; the autopsy names the coherence discard — measured band slack  $\times 2.7 \times 10^3 - 2.4 \times 10^5$ ); the PRODUCT-SUP envelope (exact Dirichlet closed form, grid-converged with doubling ward; the rigorous Lipschitz certificate of the sup is a NAMED elementary step, typed not claimed) + Abel summation against RvM counting with EXPLICIT unconditional Trudgian-grade constants (factor 2.13; the Vinogradov–Korobov region typed honestly USELESS, 0.985) closes ALL 14 rungs at the FIXED verified horizon  $T_{\text{ver}} = 3 \times 10^{12}$  with margins  $10^3-10^6$  (the secondary top-100 family gate closes 14/14 too); THE LAW CHANGE:  $\log h$  coefficient  $+1.98 \rightarrow +0.13$  — the  $h e^\alpha$  growth

was an ARTIFACT of the coherence discard, the sharpened law is  $h$ -FREE at fixed  $\alpha$ ; remaining growth  $\alpha$ -only ( $q = +1.79$ , the off-line allowance  $e^{2\alpha\delta}$  of the critical strip), re-crossing  $X_\infty^2$  near  $\alpha^* \approx 11.2$  — far beyond the deployed battery ( $\alpha \leq 6.146$ ) but finite and honestly typed; controls fire (synthetic off-line pairs break the psd chain 4/4 exactly at the off-line locus AND stay inside both envelopes; the scramble kills the family structure; the regression ward reproduces the old  $\{184, 210, 218, 434\}$  closure boundary)). Suite 815  $\rightarrow$  817; both run standalone green and encode their frozen patterns per the v757 precedent; all four probes re-run identically at promotion.

- **The net floor statement (the round’s deliverable, verbatim honesty).** On every rung of the deployed 14-rung battery,  $\lambda\tau = \det \hat{A}_2 \geq X^2(\text{pole}, \gamma_1) - \text{pert}(h) > 0$  at CITATION grade — no per-rung eigencheck, no zero-location input beyond the critical strip; inputs: the strip, verified on-line zeros to  $T_{\text{ver}} = 3 \times 10^{12}$  (Platt–Trudgian, cited), RvM counting with explicit unconditional constants (cited), the analytic pair limit, the product-sup per-zero envelope. The complete three-piece skeleton: exact SOS identity [E-grade wards] + analytic fixed-pair bound ( $h$ -uniform per deployed  $\alpha$ ) + certified 0.93–0.97 family exhaustion + psd remainder closed for all  $h$  up to  $\alpha^* \approx 11$ . This is NECESSARY-side evidence on a frozen battery — NOT RH; the floor bound is battery-relative and  $\alpha$ -bounded.
- **Contracts.** NEW: PRIME.FLOOR.LAGRANGE.01 (the exact Lagrange SOS identity + the pair certification) and PRIME.FLOOR.SKELETON.01 (the three-piece skeleton + the all- $h$  tail closure). PRIME.FLOOR.RATIO.01 [O] NARROWED (dated contract-text update): WHAT IS NOW CLOSED — the deployed-battery skeleton at citation grade (exact SOS, analytic fixed-pair bound with explicit  $h_0(\alpha) \leq 500$ , certified family exhaustion,  $h$ -free psd closure at  $T_{\text{ver}}$ ); WHAT REMAINS — (i)  $\alpha > \alpha^* \approx 11.2$  needs verified on-line DEPTH beyond  $T_{\text{ver}}$  (never location), (ii) the family version of the analytic bound at the  $\sin(\alpha\gamma_1)$  nodes, (iii) the product-sup Lipschitz certificate (named elementary step), (iv) the  $V_\infty$  positivity itself — the  $O(1)$  capture and the  $\rho > 0$  demand at infinity remain the contract; kill criteria unchanged; promotion of any  $\rho$  bound stays FENCED.
- **Ledger.** 2 new rows (PRIME.FLOOR.LAGRANGE.01, PRIME.FLOOR.SKELETON.01) + the dated narrowing update on PRIME.FLOOR.RATIO.01; 898  $\rightarrow$  900 rows; NO marker moves ([O] stays [O]; exact identities [E]-grade wards, certified intervals and shares measured/[C] per the ledger fine-typing).
- **Papers.** `tfpt_prime_front`: the new round section “The floor-theorem skeleton: the floor as an exact sum of squares, and the tail closed for all  $h$ ” (both `\veri` citations); `tfpt_research_contracts`: the two registrations + the PRIME.FLOOR.RATIO.01 narrowing note.
- **Website.** Prime-front feed: the round entry “reflux round twenty-four”; counters 815  $\rightarrow$  817 modules / v535–v824; generated surfaces regenerated (ScriptIndex, changelog.ts, maps); README and Zenodo updated to 817 modules / v824.
- **Scope.** The exact identity layer (rank-one factorization, Lagrange SOS) carries identity-grade machine wards at float precision; the certified intervals, shares and closure margins are measured finite-level evidence; the citation-grade chain uses ONLY the critical strip + cited verified depth + cited explicit RvM constants — no zero-location input, no RH-strength assumption; the honest boundaries (battery-relative,  $\alpha$ -bounded, the  $\alpha$ -nodes, the named sup certificate) are typed in the modules and the contract; NO marker moves; NO RH claim anywhere.

**2026-08-06 (LXVI) (reflux round twenty-three — one inequality: the floor reduces to the prime comb against its own density: NINE modules (v814–v822, from thirteen frozen probes; suite 806  $\rightarrow$  815) — the selection problem of the unified compactness contract is finite-level SOLVED by Mosco form convergence + Friedrichs minimality, the positive-descent master theorem applies to BOTH instances with the new prime summability number (kernel-checked Lean core),**



and the sector floor reduces to the SINGLE ratio inequality  $\rho = \tau/\tau_{\text{pnt}} > 0$  with a measured  $h^{-3/2}$  envelope (the new contract PRIME.FLOOR.RATIO.01 [O]); the K5 follow-up locates the deployed transport bit-exactly ( $T_{v221} = B^6$ , the dated SIXTHROOT correction) and types the six-step composition clock-only with the canonical  $T_{10}$  proposal; the Kraus tower gains the spread/commutant deciders and its 39-dim nonabelian commutant closure; the RM cascade lands (the 15-label channel = punctured  $\text{RM}(1,4)^*$  with the  $[[15,1,3]]$  CSS code and the closing self-reproduction cycle, the rule-A protocol set = the 105 minimal dual checks, the  $E_7$  vacuum completion  $112 + 21 = 133$  as a matrix identity); and the vacuum-completion reading of the continuum dies honestly in both transcriptions with the  $c^*$  pencil bound as the frozen fence)

- **New modules (9).** `v814_k5_sixstep_transport.py` (23/23, SIXSTEP-CLOCK-ONLY + TRANSPORT10-PROPOSED: the rate  $(2/3)^6$  arises canonically on THREE spatial routes (doily alternation, hexagon compression, hexagon circulation), each blocked by a typed obstruction (no unit mode — the norm obstruction [E neu]; the bipartite sign  $-2/3$ ; the wrong space  $\times 9$ ) — the deployed six is the CLOCK exponent ( $v124 \lambda_n = (1 - n/3)^6$ ,  $v486$  hand, fixed eigenbasis); sixth-power blindness [E neu]:  $C_6$  and  $2K_3$  have identical sixth powers, only per-step data distinguish walk from clock; THE CORRECTION (dated note on the  $v808$  row):  $T_{v221}$  (the exact spectral sum of  $v221$ 's transfer  $(1 - w)^6$ ) =  $B^6$  BIT-EXACTLY — the sixth-root verdict upgrades from SPECTRAL-ONLY to BITEACT-vs- $v221$ ; the canonical proposal  $T_{10} = P_{10}^6$  in the  $\Lambda^2$  basis passes ALL gates (spectrum  $\{1, (1/3)^6 \times 5, (2/3)^6 \times 4\}$  with multiplicities  $(5, 4) = (g_{\text{car}}, |\mu_4|)$ , equitable  $v774$  quotient =  $(Q_{\text{Pet}}/3)^6$  bit-exact) — a promotion-ready proposal, NO deploy claim); `v815_kraus_spread_commutant.py` (21/21 + 24/24, SPREAD-SIGMA-BROKEN + COMMUTANT-EXTENDS + AORB-NOT-MAXIMAL:  $\sigma$  acts on the 6 spreads as two 3-cycles, ZERO fixed (outer-class forced), no invariant coarsening — but the direct-sum protocol is  $\sigma$ -stable as a FAMILY with invariant witnesses  $\sum B_{45}(s) = 4I + 2B$  and per-orbit  $A_{\text{orb}}$  (integer spectrum  $\{0^5, 1^2, 4^5, 7^2, 9^1\}$ );  $[E, K] = 0$  lifts EXACTLY to channel / KMS tower / odd places (the  $C_2$ -character expectation unique per place) and the 600-dim register; the  $\{K, \sigma\}$  commutant is 39-dim nonabelian  $\mathbb{R} \oplus M_3(\mathbb{R}) \oplus M_3(\mathbb{C}) \oplus M_3(\mathbb{R}) \oplus M_1(\mathbb{C})$  with  $A_{\text{orb}}$  the canonical abelian 5-dim subalgebra, window-silent beyond the  $K$ -spectrum); `v816_prime_mosco_selection.py` (23/23 guards+controls, 6/6 legs, MOSCO-SELECTS: the decider table on identical rungs — moment oscillation 22.0829 (the failed selector, reproduced exactly) vs Friedrichs resolvents 0.0004 / ledger-adjusted Weyl 0.0843 / raw  $\mu$ -face 0.0000; cofinal-unique  $2.5 \times 10^{-5}$ ; M2 recovery exact  $1.2 \times 10^{-12}$ ; M1 margins +2.63 with the EDGE family typed a genuine boundary channel; the import diagnosis completed at form level — nothing whitened at the core, the whitened control reproduces 5.8772 exactly); `v817_positive_descent_master.py` (14/14, MASTER-BOTH-INSTANCES: the 7-hypothesis master theorem with the kernel-checked Lean core (`PositiveDescentMaster.lean`: telescoping + Cauchy tail, positivity closure, sector commutation);  $G_{\text{net}}$  re-verified (Choi PSD, rate 0.267/doubling); THE NEW NUMBER: the prime packet state defects are summable at  $0.701/\text{doubling} \approx 2^{-1/2}$  with  $\hat{m}_2(p) = -1/15$  exactly for every odd prime; control K4: positivity is a SEPARATE hypothesis); `v818_sector_floor_attack.py` (9/13 + 13/15 with exactly the preregistered FAILS, FLOOR-MECHANISM-PARTIAL + CAPTURE-BOUNDED / THEOREM-SHAPE / AMPLIFIER-MECHANISM:  $\text{onem} = F(r) \varepsilon K$  symbolically exact with the sign entirely in the vanishing weight; interlacing capture  $\tau/\lambda_{\min}(A_h)$  median 1.53 (range 1.04–4.36) — the 2-mode lock block is an  $O(1)$ -faithful witness of the full floor; capture angle  $\cos \theta$  median 0.990, scramble-robust (frame content); the rotation law closes symbolically with 100% staircase sign-match —  $r$ -monotonicity is now a calculus statement; the amplifier is carried by the smallest prime powers, envelope  $\rho h^{3/2} \in [4.85, 24.2]$  non-decaying; the contraction constant is density-level (Epstein  $s_E = 0.241$ )); `v819_prime_packet_rm14.py` (21/21, RM14-EXACT, Lean companion `PacketRM14.lean`: the 15-label channel row hull = punctured  $\text{RM}(1,4)^* = [15, 5, 7]$ , enumerator exact;  $\chi_{\text{NSR}}$  is ONE codeword (7+8 its two sides,  $C_2$  the affine bit); the cycle

$\text{RM}(1, 3) \rightarrow E_8 \rightarrow V \rightarrow \text{RM}(1, 4)^* \rightarrow \text{RM}(1, 3)$  closes with exactly 1344 equivalences; CSS  $[[15, 1, 3]]$  with dual  $[15, 10, 4]$ , logical distances  $(7, 3)$ , triorthogonality  $8/4/2$ ;  $\zeta_8$  typed [H]); `v820_prime_kraus_rm24.py` ( $16/16 + 14/14$ , KRAUS-RM24-RULE-CANONICAL + PLANEFRAMES-EXACT: the literal leg-support claim dead twice (weights  $1/2/3 < \text{distance } 4$ ; Sp orbits  $15+90$  vs  $45+60$ ); the canonical  $\omega$ -built rule identifies the rule-A protocol set with the 105 weight-4 checks of  $[15, 10, 4]$  exactly, Sp/ $\sigma$ /deck/KMS-covariant;  $A_3 = 12 / B_3 = 16$  certified as PLANE-COUNT statements ( $\sigma_3(3) = 28$ ,  $a_3 = -4$ ); the graded frame intertwiner  $V^*V = \sigma_3(n)I$ ,  $V^*\Gamma V = a_n I$  exact for  $n = 3, 5, 7$ ); `v821_prime_vacuum35.py` ( $22/22$ , VACUUM35-EXACT:  $140 = 105 + 35$ ; Grams  $22I+6J / 6I+J / 28I+7J$  entrywise; the  $E_7$  completion  $112 + 21 = 133$  as a matrix identity (v170 anchor [C])); Weitzenböck  $H_{105}^T H_{105} = 2B^2 + 14I$  as the canonical label-side left factor with the honest v758 scope; the  $\pm B$  cancellation inside one Gram; tight frame constant 28 on mean-zero;  $\text{RM}(2, 4)$  dual-containing); `v822_prime_vacuum_dilation.py` ( $17/19 + 12/14$  with the four FAILs being the preregistered PSD deciders, UNSHORTEN-DEAD + DILATION-DEAD: the 35-row Stinespring form fixes the shape exactly ( $6I+J$  ward, Weitzenböck bookkeeping integer) and buys  $\times 13.1$  — and both sectors STILL fail under both normalizations; THE GENERAL OBSTRUCTION QUANTIFIED:  $\text{PSD} \iff \text{coupling} \leq c^* = (2/21)\lambda_{\text{pencil}}(T_{\text{dep}}, T_{\text{cont}}^2)$ , measured deficits  $\times 3 \times 10^7$  (GL1) /  $\times 2 \times 10^2$  (f8) — the reading dies with the calibration, not the geometry; the continuum's completion role is already exhausted inside  $T_{\text{dep}}$ ;  $c^*$  is the frozen fence; module 7 / path-state NOT buildable).

- Contracts.** NEW: PRIME.MOSCO.SELECTION.01 (selection solved finite-level; the three infinite-level ingredients typed), TFPT.POSITIVE\_DESCENT.MASTER.01 (both instances; the Lean core; positivity a separate hypothesis), PRIME.FLOOR.RATIO.01 [O] — **the new main contract candidate**: the sector floor reduces to the single ratio inequality  $\rho(X) = \tau/\tau_{\text{pnt}} > 0$  with the measured  $h^{-3/2}$  envelope (constant  $\approx 4.85$ , non-decaying), direction owned by the density rotation law (calculus statement, hypotheses verified), capture angle-certified frame geometry ( $\cos \theta$  0.990, scramble-robust), the amplifier carried by the smallest prime powers; KILL = the envelope failing at depth or the capture angle collapsing; NO RH claim, PROMOTION FENCED. Also registered: K5.SIXSTEP.TRANSPORT.01, PRIME.AORB.REFINEMENT.01, PRIME.PACKET.RM14.01, PRIME.KRAUS.RM24.01 (carrying PRIME.C2.PLANEFRAMES.01), PRIME.VACUUM35.01, and PRIME.CONTINUUM.UNSHORTEN.01 as the vacuum-route CLOSURE row (both transcriptions dead, the  $c^*$  pencil bound the frozen fence, the exhaustion insight). Dated notes on CARRIER.PETERSEN.RADIAL.01 (the SIXTHROOT BITEXACT upgrade with the v221 evidence — the frozen SPECTRAL-ONLY stands as that probe's adjudication, ABEL-DEAD precedent — and the  $T_{10}$  proposal), PRIME.KRAUS.DOILY.01 (spread/commutant/aorb addenda), PRIME.Z1.COMPACTNESS.01 and PRIME.POSITIVE\_DESCENT.01 (selection solved via Mosco, unique limit via the master theorem, the floor the single remaining object  $\rightarrow$  cross-ref PRIME.FLOOR.RATIO.01). NO marker moves; the honest deaths recorded as preregistered adjudications; all declared repairs carried verbatim in PROVENANCE.
- Ledger.** 10 new rows (the nine module rows K5.SIXSTEP.TRANSPORT.01, PRIME.AORB.REFINEMENT.01, PRIME.MOSCO.SELECTION.01, TFPT.POSITIVE\_DESCENT.MASTER.01, SECTOR.FLOORATTACK.01, PRIME.PACKET.RM14.01, PRIME.KRAUS.RM24.01, PRIME.VACUUM35.01, PRIME.CONTINUUM.UNSHORTEN.01 + the contract row PRIME.FLOOR.RATIO.01) + 4 dated updates;  $888 \rightarrow 898$  rows; NO marker moves anywhere.
- Papers.** `tfpt_prime_front`: the new round section “One inequality: the floor reduces to the prime comb against its own density” (the Mosco selection, the master theorem, the floor mechanism and the new contract, the Kraus commutant tower, the vacuum-route closure); `tfpt_1`: the round-23 keybox entries (the clock-only six-step with the  $T_{10}$  proposal and the SIXTHROOT correction, the RM cascade with the  $[[15, 1, 3]]$  CSS code, the  $E_7$  vacuum completion); `tfpt_2`: the plane-frame Hecke match ( $\sigma_3(3) = 28$  and  $a_3 = -4$  as plane counts) and the RM-code reading of the 15-label channel join their sections; `tfpt_3`: nine new audit

blocks; `tfpt_research_contracts`: the registrations and dated updates above.

- **Website.** Prime-front feed: the round entry “reflux round twenty-three”; counters 806  $\rightarrow$  815 modules / v535–v822; papers.ts and VerificationDag updated; generated surfaces regenerated (ScriptIndex, changelog.ts, version.ts, maps); README and Zenodo updated to 815 modules / v822.
- **Scope.** The exact results (the bit-exact transport location, the commutant tower, the RM enumerators and Gram identities, the  $E_7$  completion) sit behind the standing fences (ROOTCLASS-MIXED intact; no matter semantics; the  $E_7$ /fingerprint readings [C]; the  $T_{10}$  and vacuum readings proposals/[H], not deployments); the analytic strand is measured finite-level evidence only — PRIME.FLOOR.RATIO.01 carries its kill criteria and its promotion fence, the honest kills (the vacuum route in both transcriptions, the four preregistered floor FAILs) are on record as adjudications; no marker moves; NO RH claim anywhere.

**2026-08-06 (LXV) (reflux round twenty-two — the K5 edge machine: five exact structures, two precise middles, and the honest kills: ELEVEN modules (v803–v813, from fifteen frozen probes; suite 795  $\rightarrow$  806) — the jet-code wave gives the Gaussian compiler its ring-theoretic normal form ( $W = L/2L$  free over the dual numbers, non-splitness = the no-origin obstruction), reconstructs the Hamming code from the NS/R bit alone ( $AGL(3, 2) = \text{Aut}(H_8)$ ), extends the CP intertwiner by the affine Gray register chart, and records two preregistered honest kills (the syndrome algebra and the analytic Clifford-activation ordering); the K5 wave certifies the Petersen carrier geometry whose radial sixth power IS the deployed transport spectrum, the doily incidence identity behind the recovery base rate, the cut-code reading of the Arf split, the Kraus context protocol with its exact commutant surprise, the flavor-filtration fingerprint — and closes the round with the P1 index-KMS identity  $c_3 = 1/(I \cdot \beta)$  and its convention audit)**

- **New modules (11).** `v803_e8_jetcode_affine_nsr.py` (18/18 + 15/15, JETCODE-EXACT+AFFINE-NSR-EXACT:  $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$  with  $\epsilon = 1+i$  (2 ramifies; unique iso; the split ring  $\mathbb{F}_2 \times \mathbb{F}_2$  fails all 24 bijections);  $W = L/2L$  FREE of rank 4 over the dual numbers (Hermite-certified  $\mathbb{Z}[i]$ -basis); the exact self-extension  $0 \rightarrow \epsilon W \rightarrow W \rightarrow V \rightarrow 0$  with  $\ker \epsilon = \text{im } \epsilon$ ; the deck IS the jet unit  $J = 1+\epsilon$ , re-deriving the v782 transition-bus torsor on all 240 roots; ZERO  $\mu_4$ -equivariant splittings among all 65536 sections — non-splitness IS the no-canonical-origin obstruction; the 120 root pairs = the  $q=1$  unit shell ( $240 = 15 \times 8 \times 2$ : root = (address, error jet, orientation));  $\text{Stab}(\chi_{\text{NSR}}) = AGL(3, 2)$  of order 1344 =  $\text{Aut}(H_8)$  with the SAME natural 8-point action, and affine functions on the NS/R torsor reconstruct  $RM(1, 3) = H_8$  with exactly 1344 equivalences — typed internal reconstruction, NOT a P2 removal); `v804_prime_carrier_gray.py` (23/23, GRAY-INTERTWINER-EXISTS: the affine Gray map encodes  $C_2 \times \mu_4$  into the R-class torsor respecting the v786 kill (a group iso is impossible — exponent 4 vs 2 — and not demanded); the fiber family deck-covariant EXACTLY with TRIVIAL measured  $\mu_4$ -cocycle (residual = a pure  $\mathbb{Z}_3$  line twist on the  $\sigma$ -fixed classes); the extended Stinespring map’s GL1 channel re-lands on the deployed Weil window at  $6.0 \times 10^{-16}$ ; one declared control repair); `v805_e8_syndrome_algebra.py` (13/13, SYNDROME-DEAD, the preregistered kill fires: max rank 222/256 over all four frozen phase conventions, the  $\pm r$  proportionality obstruction structural for the whole diagonal-phase family, both post-hoc escape directions score worse (215/160); survives:  $16 + 240 = 256$ , the 16 v112 neutral kernels = the diagonal MASA, same-class products  $A_0$ -valued with  $\mu_4$  coefficients exactly); `v806_hecke_local_clifford.py` (16/18 with the two FAILs the preregistered decider gates themselves, controls 4/4, LOCAL-CLIFFORD-BREAKS: all four  $p \bmod 8$  class sectors fail PSD — signed character combinations carry no completed  $L$ -function, the trivial mod-8 component is imprimitive; class 1 is the UNIQUE positivity-completable class (Euler repair exact,  $+1.4 \times 10^{-5}$ ); the margin ordering does NOT

track the proven activation depths (Spearman +0.40) — the Clifford activation reading stays decorative analytically, the arithmetic multirate theorem untouched; POSITIVE side-result:  $\chi_8/\chi_{-8}$  are two NEW PSD instances of the  $\Gamma_{\mathbb{R}}$  conductor rule (exact parities:  $\chi_8$  even,  $\chi_{-8}$  odd) — the rule map now covers the full mod-8 dual group; one declared bar re-freeze); `v807_lorentz_nullselector.py` (22 checks with the three honest  $h$ -order FAILs encoded +14/14, LORENTZ-COORDS-REVEAL+NULLSELECTOR-ALGEBRAIC:  $M(5, -3, 4) = 2(1, 2)^T(1, 2)$  exact — the compiler triple is a rank-one spinor square, Euclid  $(2, -1) \mapsto (5, -3, 4)$ ; THE REVEAL: the spinor slope is strictly monotone in the window width  $\alpha$  (66/66, Kendall 1.000, cross-ratio contraction CV 0.004) — the  $h$ -oscillation was  $\alpha$ -jitter; the approach  $2 - r \sim \alpha^{-0.23}$  stays glacial; the null selector  $|v_d|^2 = \det M_d = d(4 - d)$  forces  $d = 4$  on the RR/Serre functor vector with the anchor identities  $i+j = N_{\text{fam}}$ ,  $ij = |\mathbb{Z}_2|$ ,  $i^2+j^2 = g_{\text{car}}$ ,  $2ij = |\mu_4|$ ; THE BRIDGE:  $g^2 - N^2 = 4d = |H|^2$ , so null  $\iff$  glue index =  $d$  — independent of the v781 census, same output  $\{4\}$ ; honest downgrade: the seam family has no  $d$ -dependence to vary, the missing theorem typed); `v808_petersen_sixthroot.py` (19/19+16/16, PETERSEN-RADIAL-EXACT+SIXTHROOT-SPECTRAL-ONLY: the K5 edge machine — charges  $(-4, +1, +6)$  = the  $SU(5)$  decuple against the certified v774 table; Petersen  $SRG(10, 3, 0, 1)$  with  $|\text{Aut}| = 120$  census; distance shells  $(1, 3, 6)$  charge-pure; the flavor hexagon = the distance-2 shell edge-by-edge; equitable quotient with hypercharge eigenvector  $(6, -4, 1) \rightarrow -2$ ;  $\text{spec}(P^6) = \{1, 64/729, 1/729\}$  = the deployed transport spectrum EXACTLY; the 10-dim multiplicity prediction ( $\times 5 = g_{\text{car}}$  fast,  $\times 4 = |\mu_4|$  slow) typed [H neu];  $B = \frac{1}{18}[[13, 1, 4], [1, 13, 4], [4, 4, 10]]$  is the unique symmetric doubly stochastic PSD sixth root in the corpus basis,  $B^6 GL(3, \mathbb{Q})$ -conjugate to v486's  $M^6$  and the cusp-diagonal form but bit-equal to NEITHER — the corpus freezes the spectrum, not the basis); `v809_curve_code_doily.py` (21/21, DOILY-EXACT:  $NN^T = B + 2I$  entrywise,  $\text{spec} = \{9, 4^9, 0^5\}$ ;  $N/3$  has singular values  $\{1, 2/3, 0\}$  — the recovery base rate  $2/3$  as the unique nontrivial singular value of the canonical duad-to-syntheme incidence, the  $(2/3)^6$  identification typed pending the six-step composition; Petersen edges = syntheses 15/15; every census bridge factors exactly  $N_{\text{code}} = P_{\Delta}^T N_{\text{curve}}^T P_{\Gamma}$  — a duality, not an isomorphism; the Arf switch set  $T_6$  = the six within-block duads); `v810_cutcode_moment.py` (17/17 + 16/16, CUTCODE-EXACT+MOMENT-INCIDENCE-PARTIAL: the cut map is injective-isometric on all 256 cells and the Arf split  $16 = 1 + \bar{5} + 10$  IS the cut classification  $(0|5) + (1|4) + (2|3)$ ; Clebsch  $40 = 5 + 20 + 15$  over the Arf shells with within-10 = Petersen (census 120); roots/lines grade  $80 + 160 / 20 + 40$  — Arf/cut grading, NOT matter; the master moment  $\text{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$  from one symbolic identity; the  $210 = \binom{10}{4}$  map DECORATIVE per the frozen rule — the equivariant bijection killed by the exhaustive fixed-point obstruction, the partial  $80 \rightarrow 5 \times 16$  miss-slot map and the Newton tie survive as audit); `v811_prime_kraus_doily.py` (21/21, KRAUS-DOILY-PROTOCOL: the 105 Kraus legs = 15 diagonal + 90 off-diagonal; the canonical  $45 + 60 =$  reversal-quotient (the 90 legs double-cover the 45 flags 2:1) + fiber expansion ( $15 \times 4 = 60 =$  the v783 stabilizer rays); the  $Sp(4, 2)$  obstruction certifies no leg-subset partition — the literal partition is spread-relative = the ledger T3 census; the 45 transports quadratic ( $P_x P_y = \pm P_{x+y}$ ) spanning  $\text{End}(\mathbb{C}^4)$  at rank 16; THE SURPRISE:  $[B_{45}, B_{60}] = 0$  exactly —  $K = \frac{3}{7}E + \frac{4}{7}R$  is a direct-sum protocol with  $E$  a conditional expectation; measured: the certified spread is NOT  $\sigma$ -invariant — the named next falsifier); `v812_flavor_graph_filtration.py` (16/16, FILTRATION-FINGERPRINT-ONLY: the  $Q/K/L$  row budgets are exact, order-exact counts of the  $K_{3,2}/\text{prism}/K_5/H_8$  incidence chain with level-wise additivity mirroring  $L = K + Q$ ; NO entry-level reconstruction from any frozen candidate; the exhaustive false-positive rates 0.005–0.75% make the misses informative — typed audit fingerprint); `v813_p1_index_kms.py` (13/13, P1-IDENTITY-CLOSED:  $I \cdot \beta \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$  EXACT in formal arithmetic; the index is an independent  $c_3$ -free theorem (zero occurrences in all four sources, gated),  $\beta = 2\pi$  independently measured (v526; the radian pin is the one shared convention and cancels),  $c_3$  stays the P1 axiom the identity EXPLAINS; the finite evenness precursor measured ( $w_1 = w_3 = \frac{1}{4}$  exactly at every level, symmetric zero-sum offset  $\pm 2^{-(l+1)}$  — the no-anomaly shape); the sea state does NOT equidistribute (dev 0.188) — the bridge must normalize the modular response; controls:

index  $2 \rightarrow 1/(4\pi)$ , booked elsewhere by the corpus). Suite 795  $\rightarrow$  806; all eleven run standalone green and encode their frozen patterns per the v757 precedent (the two honest kills encode their expected-FAIL patterns explicitly); all fifteen probes re-run identically at promotion.

- **Contracts.** NEW: E8.RAMIFIED.JETCODE.01 [E] (carrying E8.AFFINE.NSR.01), PRIME.CARRIER.GRAY.01 (demands D1–D3: odd-place fiber charts, the  $\mathbb{Z}_3$  line twist as crossed-product structure, the infinite-level CP extension), E8.SYNDROME.ALGEBRA.01 (DEAD as frozen, successor criterion named: root-individual operator data), HECKE.LOCAL.CLIFFORD.01 (analytic ordering DEAD; next falsifier: primitive sector data, not class indicators), PRIME.LORENTZ.SPINOR.01 (the reveal + the named 1D monotonicity candidate), CARRIER.PETERSEN.RADIAL.01 [E/H] (carrying TRANSPORT.SIXTHROOT.01, the basis-freeze question named), CURVE.CODE.DOILY.01 (the  $(2/3)^6$  six-step target named), CARRIER.CUTCODE.01 [E] (carrying CARRIER.MOMENT.INCIDENT.01, the 210 decorative), PRIME.KRAUS.DOILY.01 (protocol-grade; the commutant/spread falsifier named), FLAVOR.GRAPH.FILTRATION.01 (fingerprint), P1.INDEX.KMS.01 registered OPEN [O] (the one open bridge: even modular-response normalization over the  $I$  deck sectors; kill = any asymmetric sector offset). Dated notes on ARF.BOUNDARY.CODE.01 (the affine NS/R reconstruction sharpens the attack surface;  $R1'$  unchanged), PRIME.CP.INTERWINER.01 (the Gray extension; D1–D3 unchanged), PRIME.SECTOR.CONTINUA.01 (the rule map covers the full mod-8 dual group) and E8.ONEOBJECT.01 (the null-selector addendum: a second independent  $d = 4$  mechanism on the same invariant  $4d$ ). NO marker moves; both kills recorded as preregistered adjudications; the declared repairs (Gray C2 scramble, local-Clifford C1b bar, Kraus report text) carried verbatim in PROVENANCE.
- **Ledger.** 11 new rows (E8.RAMIFIED.JETCODE.01, PRIME.CARRIER.GRAY.01, E8.SYNDROME.ALGEBRA.01, HECKE.LOCAL.CLIFFORD.01, PRIME.LORENTZ.SPINOR.01, CARRIER.PETERSEN.RADIAL.01, CURVE.CODE.DOILY.01, CARRIER.CUTCODE.01, PRIME.KRAUS.DOILY.01, FLAVOR.GRAPH.FILTRATION.01, P1.INDEX.KMS.01) + 4 dated updates; 877  $\rightarrow$  888 rows; measured/typed throughout; NO marker moves anywhere.
- **Papers.** `tfpt_prime_front`: the new round section “The K5 edge machine: five exact structures, two precise middles, and the honest kills” (the Kraus protocol, the Gray intertwiner, the local-Clifford negative, the doily and Lorentz strands, the P1 index identity); `tfpt_1`: the jet-code normal form and the affine NS/R reconstruction join the architecture narrative (keybox entries; the cut-code grading of Theorem 2’s Clebsch geometry); `tfpt_2`: the Petersen carrier geometry of the transport spectrum and the flavor-filtration fingerprint join their sections; `tfpt_3`: eleven new audit blocks; `tfpt_research_contracts`: the registrations and dated updates above.
- **Website.** Prime-front feed: the round entry “reflux round twenty-two”; counters 795  $\rightarrow$  806 modules / v535–v813; papers.ts and VerificationDag updated; generated surfaces regenerated (ScriptIndex, changelog.ts, version.ts, maps); README and Zenodo updated to 806 modules / v813.
- **Scope.** The exact results (the jet-code normal form, the affine NS/R reconstruction, the Petersen/doily/cut-code incidence identities, the Kraus protocol identities, the index-KMS identity) sit behind explicit fences (ROOTCLASS-MIXED intact everywhere; no matter semantics; the Hamming reconstruction is internal, not a P2 removal;  $c_3$  stays the P1 axiom — the identity explains, it does not derive); the honest kills (syndrome algebra, analytic Clifford ordering) and the precise middles (the 210 decorative, the filtration fingerprint, the spectral-only sixth root, the glacial  $\alpha$ -approach) are preregistered adjudications on record; no marker moves; no RH claim anywhere.

**2026-08-06 (LXIV) (reflux round twenty-one — the descent functor completes at finite level and the boundary code becomes three physical axioms: ELEVEN**

modules (v792–v802, from thirteen frozen probes; suite 784  $\rightarrow$  795) — the prime channel delivers all three demands of PRIME.POSITIVE\_DESCENT.01 at finite level (sector continua as one  $\Gamma_{\mathbb{R}}$  functor, the exact limit object with the first measured sector floor, the Stinespring carrier intertwiner) plus the halving-tails/ $k=5$  tower mechanism and the  $R$ -microstate honest negative; the seam/code strand forces Type II from three physical axioms, identifies the missing Clifford bit as the metaplectic  $S$ -lift at  $\tau = i$ , decides the canonical outer bridge and reads the carrier Pascal at character level; the clock strand turns the coset assignment into a theorem; and the  $G_{\text{net}}$  strand measures the martingale limit hypotheses as met)

- **New modules (11).** v792\_ftransfer\_kms\_coset.py (18 checks, KMS-COSET-THEOREM: the coset carried by a clock chart is a FUNCTION of its KMS/modular typing datum —  $S = -(\ln \lambda)^2/2$  with  $\lambda$  the time-1  $\text{PGL}_2$  holonomy multiplier per clock e-fold; Schwarzian-free, derivative-free classifier (non-circularity gated on its own source), blind assignment ( $F_{\text{pole}}/F_{\text{Boltzmann}}$  modular with  $|\ln \lambda| = \Delta = 6 \log(3/2)$ ,  $F_{\text{QCD}}$  autonomous,  $F_{\text{relic}}$  degenerate) and sympy-exact unblinding; controls incl. the v578 Möbius clock as the positive case — the executor  $\rightarrow$  groupoid  $\rightarrow$  coset chain of the CONTRACT.F.01 refinement is whole); v793\_f8sector\_conductor.py (15/15+16/16, F8SECTOR-PSD+CONDUCTOR-FUNCTORIAL: the  $f_8$  sector, broken at  $-132..-151$  under the register-trivial GL1 continuum, is PSD on the entire frozen ladder with its OWN weight-4/conductor-8/no-pole continuum ( $+1.52 \times 10^{-4} \rightarrow +3.20 \times 10^{-5}$ ); the continuum assignment is ONE closed rule  $\chi \mapsto q^{s/2} \prod_j \Gamma_{\mathbb{R}}(s + \mu_j)$  (duplication ward  $2.7 \times 10^{-15}$ ); the twist  $f_8 \otimes \chi_{-4}$  has conductor 16 by Atkin–Li + the Fricke ward ( $6.7 \times 10^{-16}$ ) and lands PSD with the rule continuum; four-sector coherence; the conductor datum load-bearing at exactly  $\ln 2$ ); v794\_cp\_extension\_gate.py (17/17, CPGATE-UNDECIDED-FALLING: restriction exactness/interlacing/uniform bounds measured (the frozen  $e^{X/2}$  pole prediction recalibrated, declared); the limit object = the per-sector positivity floor on  $V_{\infty} = \text{Z1-COMPACTNESS}$  sector-decorated (anchor  $3.882 \times 10^{-6}$  reproduced); deep trend: no crossing, GL1 power-law to  $3.9 \times 10^{-6}$  at  $X = 18.375$ ,  $\chi_4$  SATURATES at  $\sim 1.5 \times 10^{-6}$  — the first measured sector floor); v795\_halving\_tails\_k5.py (20/20+17/17, TAILS-IDENTIFIED+K5-MECHANISM-FOUND: exact identities  $v_2(D_p) = 6 + v_2(X_7) / 8 + v_2(X_1)$  with termwise-integral divisor sums; class-7 base  $\iff R_3(p) \equiv 16 \pmod{32}$ ;  $k$  deterministic on the 2-power residue-tower cells with zero exceptions at  $10^6$ ,  $k = 5$  only in cell (5, 3) of density  $1/64 \Rightarrow P = 1/128$  = the observed half mass; the class-7 mirror law  $v_2(D_p) = 3 + v_2(p+1)$ ; the governing-field derivation typed open); v796\_duad\_syntheme\_bridge.py (24/24, BRIDGE-CANONICAL: the Sylvester outer bridge  $\beta$  with  $\beta\sigma = \tau\beta$ ,  $\beta(q^*) = S^*$  unique up to the order-6 joint stabilizer;  $q_{\Theta}$  pulls back constantly to  $q_{\text{even}}^* = q^* + \hbar(\cdot, A)$  — the anchor shift resolves the v784 Arf mismatch;  $t_A \mapsto \rho = J$ ;  $\chi_{\text{NSR}}$  an exact two-variant torsor); v797\_r\_microstate\_identification.py (14/14, R-NO-ARROW-REALIZATION: the honest negative — no arrow-level realization of the  $R$ -grading; the count identity  $16B_n = 256R(n)$  exact but count-level only); v798\_seam\_clifford\_modular\_s.py (27/27, MODULAR-S-PARTIAL: the per-factor Hadamard IS the metaplectic  $S$ -lift at  $\tau = i$ , landing exactly in the missing coset  $\zeta_8 G_{31}$ ; the deck  $J$  = the metaplectic anomaly of the total Fourier (R2 placed); the strict  $q^*$  census EMPTY (0/46080), one  $G_{31}$ -conjugacy class of survivors, the  $\sigma$ -orbit Arf defects XOR to the anchor  $A$  (R3 named)); v799\_seam\_code\_typeii.py (19/19, TYPEII-FORCED: exhaustively over all 308,993 subspaces of  $\mathbb{F}_2^8$ , locality  $\iff$  self-orthogonal, integer spin  $\iff$  doubly even,  $S$ -closure/ $|\text{disc}| = 1 \iff$  self-dual = the 30  $\hat{e}_8$  copies; residual  $R1 \rightarrow R1'$  = the three physical axioms A1–A3); v800\_e8\_torsor\_fourier.py (30/30, TORSOR-FOURIER-PARTIAL: 120 canonical projective Fourier rays; the carrier block =  $(1+5+10) \boxtimes (1+3)$  under  $W(D_5) \times W(A_3)$  at character level — impossible at point level; the weight/character bases exactly mutually unbiased, ROOTCLASS-MIXED intact; the 45 deck-even modes = the 45 context Pauli observables); v801\_prime\_cp\_intertwiner.py (29/29, CP-INTERTWINER-EXISTS:  $\Phi = V^* \pi V$  from the 105 Kraus legs, arrow algebra = the full  $M_{15}$ ,

Choi exact-rational ( $105 \times 1/7$ ); the four automorphic channels of ONE map land on the deployed GL1 Weil window ( $6.0 \times 10^{-16}$ ) and the  $\Gamma_{\mathbb{R}}$ -rule windows, all PSD; F1 retyped as a negative Choi eigenvalue); `v802_gnet_martingale.py` (15/15, MARTINGALE-HYPOTHESES-MET: exact Haar filtration, defects summable at  $\sim 0.25$  per doubling, unique limit by telescoping, determinant witness  $\geq 0.99$ ; decimation converges to the WRONG degenerate limit — the coherent bond load-bearing). Suite 784  $\rightarrow$  795; all eleven run standalone green and encode their frozen patterns per the `v757` precedent; all thirteen probes re-run identically at promotion.

- **Contracts.** NEW: `SEAM.CODE.TYPEII.01` (R1  $\rightarrow$  R1': the seam satisfies A1 mutual locality, A2 integer conformal spin, A3 holomorphy/index one — the three-axiom equivalence certified exhaustively), `SEAM.CLIFFORD.MODULAR_S.01` (partial: coset positive, R2 solved as the metaplectic anomaly, R3 named through the Arf defects), `CURVE.CODE.OUTERSPIN.01` (decided CANONICAL; the `v784` inner kill stands), `PRIME.CP.INTERTWINER.01` (D1–D3 registered: dilation family, normality on the Z1-COMPACTNESS object, the limit identification — contains RH, fenced), `GNET.MARTINGALE.LIMIT.01` (hypotheses measured; steps 5–7 the typed analytic remainder). UPDATED with dated notes: `PRIME.POSITIVE_DESCENT.01` (all three demands now finite-level; the remaining analytic core restated as the per-sector positivity floor + D1–D3), `PRIME.Z1.COMPACTNESS.01` (the sector-decoration cross-reference), `CONTRACT.F.01/FTRANSFER.CLOCKS.01` (the coset theorem completes the fibered refinement), `HECKE.MULTIRATE2ADIC.01` (the tails identities, the residue tower, the mirror law; governing field typed open), `ARF.BOUNDARY.CODE.01` (all three residuals moved), `CURVE.CODE.2TORSION.01`, `E8.TRANSITIONBUS.01`, `E8.CLIFFORD2Q.01`, `HECKE.ARROW_MESSAGE.01`.
- **Ledger.** 11 new rows (`FTRANSFER.KMSCOSET.01`, `PRIME.SECTOR_CONTINUA.01`, `PRIME.CP.EXTENSION.01`, `HECKE.HALVINGTAILS.01`, `CURVE.CODE.OUTERSPIN.01`, `HECKE.R_MICROSTATE.01`, `SEAM.CLIFFORD.MODULAR_S.01`, `SEAM.CODE.TYPEII.01`, `E8.TORSOR.FOURIER.01`, `PRIME.CP.INTERTWINER.01`, `GNET.MARTINGALE.LIMIT.01`) + 10 dated notes; 866  $\rightarrow$  877 rows; honest deaths and declared repairs carried in the module provenance blocks (`[C]/[E]`/measured typing per the probes); NO marker moves anywhere.
- **Papers.** `tfpt_prime_front`: the new round section “The descent functor completes at finite level, and the boundary code becomes three physical axioms” (demands (1)–(3), the depth-class mechanization, the two honest adjudications) + the positive-protocol close retyped; `tfpt_1`: five round-21 keybox entries (outer bridge, Type II, metaplectic  $S$ -lift, torsor Fourier; the R1–R3 residual prose retyped); `tfpt_2`: the Type-II forcing, the martingale follow-up and the Clifford completion join their sections; `tfpt_3`: eleven new audit blocks + the stale next-step/residual prose retyped; `tfpt_4_frontier`: the coset-theorem note; `tfpt_research_contracts`: the five registrations and the dated updates above.
- **Website.** Prime-front feed: the round entry “reflux round twenty-one”; counters 784  $\rightarrow$  795 modules / `v535–v802`; `papers.ts`, `VerificationDag`, `OpenGates`, `HonestyBanner` and prime-front counters updated; generated surfaces regenerated (`ScriptIndex`, `changelog.ts`, `version.ts`, `maps`); `README` and `Zenodo` updated to 795 modules / `v802`.
- **Scope.** The exact results (the coset theorem, the  $\Gamma_{\mathbb{R}}$  rule instances, the tails/tower identities, the outer-bridge census, the Type-II equivalence chain, the metaplectic coset identities, the character-level carrier block, the Stinespring/Choi certificates) sit behind explicit fences; the honest caps and negatives (the empty strict census, the mutually-unbiased cap, the no-arrow-realization, the undecided-falling trend) are preregistered adjudications on record; sector PSD stays finite-level evidence — GRH-type statements unproven; the limit identification contains RH and is NOT claimed; no marker moves; no RH claim anywhere.

**2026-08-05 (LXIII)** (reflux round twenty — the largest promotion of the day: ELEVEN modules (`v781–v791`, from sixteen frozen probes; suite 773  $\rightarrow$  784) — the strategic round lands the  $d = 4$  forcing census with its honest clock kill, the

transition-bus torsor, the two-qubit Clifford identification and the curve/code 2-torsion partial; the prime channel lands the check32 theorem, the packet-480 kill and the arrow-ledger/broadcast pair; and the positive protocol lands the positive  $C_2$  lift, the multirate ladder with its constant-depth proof, the SNF/ $\mu_4$  adjudication and the positive-descent decider with the new open contract PRIME.POSITIVE\_DESCENT.01)

- **New modules (11).** `v781_one_object_clock_census.py` (21/21 + 23/23, ONE-OBJECT-PARTIAL+CLOCK-FIBER-DEAD: the TFPT glue pattern admits an even unimodular DIAGONAL glue of  $A_{d-1} \oplus D_{d+1}$  iff  $d = 4$  (exact census  $d = 2..12$ ; the two `v92` Lagrangians reproduced; explicit  $E_8$  certificate;  $d = 9$  dies at isotropy; controls fire) with the corollaries  $g_{\text{car}} = d+1 = 5$ ,  $N_{\text{fam}} = d-1 = 3$ ,  $|\mu_4| = 4 = d$ ; honesty gate:  $c_3 = 1/(2\pi d)|_{d=4} = 1/(8\pi)$  is a corpus-legitimate REWRITING, NOT a new derivation of P1; the  $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$  clock action on the 60 Gaussian lines is DEAD at three levels — full line-compatible group order 23040, max element order 12); `v782_e8_transition_bus.py` (35/35, TRANSITION-BUS-TORSOR: the 240 roots are 15 coordinate 8-frames whose pair fibers are canonical  $v^\perp$ -torsors; NO deck-equivariant  $V \times (V \setminus 0)$  labeling exists (proof-grade obstruction); the frozen symplectic predicate reproduces all 57600 inner products;  $240 = 5 \times 3 \times 16$  spread layer; the  $A_{15}$  comparison quantified); `v783_two_qubit_clifford.py` (32/32, CLIFFORD-PARTIAL(H4): the 60 Gaussian lines ARE the 60 two-qubit stabilizer rays entry-exactly over  $\mathbb{Z}[i]$  (60/60, support census  $4 + 24 + 32$ );  $G_{31} = \text{index-2 Clifford subgroup with } 46080 = \mu_4 \times \text{Pauli}_{16} \times Sp(4, 2)$ , classes = Pauli contexts, spreads = MUB pentads; the index bit is the Galois  $\sqrt{2}$ /Hadamard bit, not  $J \leftrightarrow -J$ ); `v784_curve_code_2torsion.py` (33/33, CURVE-CODE-PARTIAL: every CM generator period certified as an integral  $\mathbb{Z}[\zeta_{12}]$  recognition; the correspondence dies at two exact layers — deck syntheme-side  $[3, 3]$ -free vs code duad-side, and curve theta even/Arf-0 ( $1+6+9$ ) vs  $q^*$  odd/Arf-1 ( $1+5+10$ ); joint census 0 with positive control 6); `v785_hecke_check32.py` (20/20, CHECK32-THEOREM:  $f_8 \equiv E_{\text{odd}} \pmod{32}$  to  $q^{100000} = 25000 \times$  the Sturm bound 4 (one cited classical ingredient);  $a_p \equiv 1 + p^3 \pmod{32}$  on 9591/9591 odd primes with the cube-map decoder; mod-64 fails at  $q^3$ ; controls 91.7% and  $\sim 31/32$ ); `v786_prime_packet480.py` (28/28, PACKET480-ADDRESS-ONLY: the header ( $p \bmod 30$ ,  $a_p \bmod 32$ ) is exact 7-bit address arithmetic but the canonical intertwiner to the 128 spinor roots is IMPOSSIBLE — exponent 8 vs 2; 3 of 7 bits realizable; the 128 stays a fingerprint; kill fired as preregistered); `v787_hecke_arrow_broadcast.py` (48/48 + 21/21, ARROW-LEDGER-STRUCTURED+BROADCAST-EXACT: the 15 ramified arrows ARE the 15 polar labels (7 NS + 8 R), spread census (3, 1, 1, 1, 1),  $T/196 = (4/49)I + (45/49)\Pi_0$  exact with the 105 leg classes = the `v756` Kraus set,  $a_7 = +24$  sign-derived from the full 137600-line census;  $\bar{T}_n = M_n \otimes I_{16}$  on all odd shells 3..13 with  $B_n = 16R(n)$  arrow-exactly, new identities  $b = 2A_p$  and  $n_B = A_p B_p/2$ ; honest negative at  $p = 7$ ); `v788_positive_c2_lift.py` (42/42, C2LIFT-THEOREM:  $E_{\text{odd}} \pm f_8 = 2A/2B$  are POSITIVE theta series with  $B = 16R$  and  $a_n = \sigma_3(n) - 32R(n)$  — check32 becomes a kernel corollary; microstates 1, 4, 10, 24, 43; automaton censused on 41053 pairs + 70 prime powers; five-condition validator with joint detection 1.000000 and the tamper blind spot exactly the  $16\mathbb{Z}$  lattice); `v789_multirate_constdepth.py` (23/23 + 24/24, MULTIRATE-THEOREM/SPINLIFT-VERIFIED+CONSTDEPTH-THEOREM:  $v_2(1+p^3-a_p) \geq 5 + 2[\chi_{-4}] + [\chi_8]$  on all 78497 primes  $< 10^6$  with sharp witnesses 3/7/5/17; sifts Sturm-certified at level  $\leq 512$ ; mod-512 breaks at  $n = 9$ ;  $\text{Sh}(g) = -8E_{\text{odd}} + 256H$  exact with Kronecker factor  $\chi_8$ ; the constant depths PROVEN via the two named  $h = 1$  ingredients); `v790_snf_mu4_theta.py` (18/18 + 17/17, SNF-FINGERPRINT+MU4-PARTIAL: the  $256R(n)$  representation identity and the SNF ( $1^4, 2^4$ ) match are exact but the isometry census is EMPTY at every admissible level (kissing 240 vs 8; 0/20160; the naive 4+4 split inadmissible by theorem); the NEW exact identity  $c_{\text{sig}} = \frac{1}{15}E_4(q^2) - \frac{6}{5}E_4(q^4) + \frac{32}{15}E_4(q^8) - 8f_8$  — the cusp part of the quartic  $\mu_4$  character is exactly  $-8f_8$  with  $-8 = 52 - 60 = -\text{rank}(E_8)$ ; the  $C_2$  channel is pure Eisenstein; the all-shell quartic vanishing is FALSE); `v791_positive_descent.py` (30/30, DESCENT-PARTIAL: the packet GNS state on  $\mathbb{N}[C_2] \otimes \mathbb{N}[\mathbb{F}_2^4] \otimes \mathbb{N}[\mu_4]$  is manifestly positive in exact rationals; the naive



linear pushforward is NOT a state (F1); the GL1 descent is bit-identical to the deployed Weil window and its UNIQUE PSD sector of 24 (F2: the breaking tensor factor is the continuum/pole leg)). Suite 773 → 784; all eleven run standalone green and encode their frozen patterns per the v757 precedent; all sixteen probes re-run identically at promotion.

- **Lean.** `TfptCarrier/Check32.lean` (13 kernel theorems: the cube bijection on the 16 odd residues, the first-coefficient congruences as exact integer statements, the mod-64 failure witness) and `TfptCarrier/PositiveC2Lift.lean` (the positive-lift kernel with check32 as a corollary), both kernel-checked, lake build green (3406 jobs), Lean tree 80 modules; dated Lean notes carried in the `HECKE.CARRIER_CHECK32.01` and `HECKE.POSITIVE_C2_LIFT.01` ledger rows.
- **Contracts.** `PRIME.POSITIVE_DESCENT.01` registered OPEN [O] with the three demanded objects — (1) sector-adapted continua (the  $f_8$ -sector continuum as the finite-level falsifier), (2) GL1-sector PSD persistence in the Z1-COMPACTNESS frame (Weil positivity retyped as sector compression of a manifestly positive object), (3) the carrier intertwiner — plus stop conditions and inherited falsifiers; cross-referenced both ways with `PRIME.Z1.COMPACTNESS.01` (dated add-on note). Dated notes on `HECKE.GEOM.01` (the 2-adic skeleton now load-bearing), `HECKE.GEOM.HALFINT.01` (the ladder form of the Shimura lift; Kronecker factor =  $\chi_8$ ), `RTF.GNS.LEDGER.01` (the four-character envelope closed multiplicatively), `E8.GAUSSIAN.CODE.01` (the Clifford identification and the bus torsor), `PRIME.Z1LOOKAHEAD.01` (the packet-480 impossibility), `ARF.SPINORCOMPILER.01` (the  $q^*$  spread deployment and its Arf-parity boundary) and `ARF.ROOTCLASS.01` (the fence held through the round).
- **Ledger.** 11 new rows (`E8.ONEOBJECT.01`, `E8.TRANSITIONBUS.01`, `E8.CLIFFORD2Q.01`, `CURVE.CODE.2TORSION.01`, `HECKE.CARRIER_CHECK32.01`, `PRIME.PACKET480.01`, `HECKE.ARROW_MESSAGE.01`, `HECKE.POSITIVE_C2_LIFT.01`, `HECKE.MULTIRATE2ADIC.01`, `HECKE.SNF_THETA.01`, `PRIME.POSITIVE_DESCENT.01`) + 8 dated updates; 855 → 866 rows; measured/typed throughout, honest kills recorded as preregistered adjudications, the [C neu]/[E neu] typing carried, NO marker moves anywhere.
- **Papers.** `tfpt_1`: the four strategic entries join the code keybox (the  $d = 4$  forcing census with the P1 honesty gate, the transition-bus torsor, the Clifford identification, the curve/code partial); `tfpt_2`: the stabilizer-identification follow-up joins the  $G_{\text{net}}$  page; `tfpt_3`: eleven new audit blocks; `tfpt_prime_front`: the new section “The positive protocol: the prime channel as a positive routing code, and what the descent still needs” (check32 theorem, positive lift, multirate ladder + proof, arrow ledger + broadcast, the two preregistered adjudications, the descent and its three-object contract); `tfpt_research_contracts`: the `PRIME.POSITIVE_DESCENT.01` registration and the Z1-COMPACTNESS add-on note.
- **Website.** Prime-front feed: the round entry “reflux round twenty”; counters 773 → 784 modules / v535–v791; papers.ts narrative mirrors updated; README and Zenodo updated to 784 modules / v791.
- **Scope.** The exact results (the forcing census, the torsor structure theorem, the stabilizer identification, the check32/positive-lift/multirate theorems modulo their named classical ingredients, the broadcast factorization, the  $c_{\text{sig}}$  identity) sit behind explicit fences; the three fired kills (clock fiber, packet-480, all-shell quartic vanishing) are preregistered adjudications firing inside green machinery; the  $c_3$  link stays typed INTERPRETATION (P1 is not derived); ROOTCLASS-MIXED stays binding (no code→matter reading anywhere); the descent contract names what is still missing; no marker moves; no RH claim anywhere.

**2026-08-05 (LXII)** (reflux round nineteen: SEVEN promotions (v774–v780, from twelve frozen preregistered probes, all first-run adjudications reproduced identically at promotion; suite 766 → 773 modules) — the Arf compiler lands with its fired matter fence and the boundary-code uniqueness chain, the external-clock contract is EXECUTED on its expected K1 kill, remainder (a) of the (L1)

majorant closes in the corrected  $(1 + \ln M)$  form, the  $G_{\text{net}}$  GNS precursors cohere, and the sharpened-Z1 trilogy measures the z1-compactness unification)

- **New modules (7).** `v774_arf_spinor_compiler.py` (46/46, ARF-SPINOR-EXACT: the Arf/spinor compiler on  $V = L/(1+i)L$  — parity lift  $\iota \cong C_{\text{even}}(5)$  with  $\beta = \bar{h}$  exact, unique  $\sigma$ -selected  $q^*$  (census  $4 \rightarrow 2 \rightarrow 1$ ; the weight form IS  $q^*$ ), Arf census 6+10 with  $Sp(4, 2) \cong S_6$ ,  $16 = 1+\bar{5}+10$  as  $\text{Stab}(q^*) \cong S_5$  orbits,  $K_6$  duads,  $B = I + A_{KG(6,2)}$  with exact ovoid eigenvectors, hypercharge table  $X = (-2, -2, -2, 3, 3) \rightarrow$  the v310/v14 dictionary, moments  $(16, 0, 120, 0)$  with the chain 40/41/240/248, Pati–Salam 8+8 with  $B-L$  exact,  $\chi_{\text{NSR}} = \text{anchor bit}$ ;  $A_3^{\text{Fam}} \neq A_3^{\text{PS}}$  naming discipline); `v775_gaussian_class_d5_purity.py` (18/18, ROOTCLASS-MIXED — THE FIRED PHYSICAL KILL: no  $D_5 \oplus A_3/SU(5)$  convention makes even one class set pure,  $\geq 7$  adjoint-side classes for EVERY embedding by the 128-spinor-root bound, 0/56 splits; the code→matter reading incl. the “empty mailbox =  $\nu_R$ ” reading is fenced out of all main claims); `v776_boundary_hamming_uniqueness.py` (31/31, BOUNDARY-CODE-UNIQUE: census  $135 \rightarrow 30 \rightarrow 14 \rightarrow 2 \rightarrow 1$  without Hamming/ $E_8$  input, ONE deck-invariant  $\mathbb{Z}_2$  flag bit, Construction A  $\rightarrow E_8$  exact,  $g_{\text{car}} = 1+4 = 5$  as OUTPUT; residuals R1–R3 typed, NO P2-removal claim); `v777_ftransfer_clock_jets.py` (24+14 checks, 0 failures: FTRANSFER.CLOCKS.01 EXECUTED — the K1 kill fired exactly as the frozen prior expected, NO-COMMON-CONNECTION, transported Schwarzians sympy-exactly  $\{-\Delta^2/2, -\Delta^2/2, 0\}$ ; + the fibered CONTRACT.F.01 refinement FTRANSFER-FIBERED-CARRIES: discrete  $\text{PGL}_2$  base with  $\text{CR} = 4/3$  exact, constant seam cocycle  $\{0, \pm\Delta^2/2\}$ , thermal-vs-RG coset split); `v778_qgeo_gridsup_logsum.py` (19/19+19/19: remainder (a) of the v745 (L1) majorant FULLY closed in the corrected  $(1 + \ln M)$  form; the literal v715 all-refinements clause is FALSE at  $r_1 = 1$  and repaired; certified grid Fourier log-sum theorem with explicit constants; GATE.QGEO does not move); `v779_gnet_gns_arf.py` (27/27+27/27: GNS precursors COHERENT — exact Haar tower, commuting index-4 expectations, duality defect 0 with ratio 4; the exact dim law  $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1} \cos(\pi c/2)$  makes the  $\mathbb{C}[\mathbb{Z}_4]$  obstruction exactly  $2^{-\ell}$  and explains the  $\ell = 1$  anomaly; the honest negative: the 1+5+10 echo is non-geometric,  $K_{\text{matter}}$  does not govern transport); `v780_z1_compactness_trilogy.py` (gates 3/4+1/4+ legs 3/6 with exactly the preregistered fails: the N2 family carries contact/collision/coupling natively but not the widening cadence; the finite boundary triple is exactly Herglotz and dies as a whitened  $K=M/2$  import, cell-mass exactly 0.500 beyond the half-window; the raw variable-edge family is bounded (0.936) + equicontinuous (0.807) with summable  $\sim$ rank-one entry poles,  $c_{\text{mass}} 0.032 \ll 1$ ). Suite 766  $\rightarrow$  773; all seven run standalone green and encode their preregistered patterns per the v757 precedent.
- **Lean.** `TfptCarrier/ArfSpinorCompiler.lean` (23 kernel theorems: parity lift,  $\beta = \text{symplectic form}$ , selector uniqueness, Arf census, orbit partition, traceless/primitive  $X$ , moments 16/0/120/0 with the chain,  $\chi_{\text{NSR}} = \text{anchor bit}$ , PS split, must-fail dot-form control) and `TfptCarrier/CellCocycle.lean` (18 declarations: exact Banachiewicz/bordered-Schur update, PD cell  $\Rightarrow$  PSD correction  $\Rightarrow$  monotone Loewner flow, bounded-monotone entrywise convergence naming the missing uniform Loewner bound, Redheffer/linear-fractional composition over arbitrary rings), both kernel-checked, `lake build` green; dated Lean notes added to the PRIME.QFCOCYCLE.01 and PRIME.GRAM.DIAGONAL.01 ledger rows.
- **Contracts.** FTRANSFER.CLOCKS.01 moves PENDING  $\rightarrow$  EXECUTED (K1 kill as the frozen prior expected; freeze guards byte-verified; the four dictionary legs pass at  $571 \times / 1.15 \times 10^{-4} / 0.83\% / 0.0602$ ); CONTRACT.F.01 carries the dated jet-route-closure + fibered-refinement note (next named step: coset assignment from KMS/modular clock typing); ARF.BOUNDARY.CODE.01 registered OPEN [O] (derive evenness/self-duality from the seam axioms; residuals R1–R3; kills named); GATE.QGEO dated note (residue (a) closed in  $(1 + \ln M)$  form, the literal v715 clause corrected, (b)/(c) open, gate does not move); GATE.METRIC.08/.11 dated notes (the  $2^{-\ell}$  obstruction law as the finite precursor of the identification); PRIME.Z1.OPERATOR.01 and PRIME.KMS.INDUCTIVE\_STATE.02 carry the measured unification (compactness carried, selection

= import-faithful coupling as the single shared open object; merged target Z1-COMPACTNESS, cross-referenced, no row deletion).

- **Ledger.** 6 new rows (ARF.SPINORCOMPILER.01, ARF.ROOTCLASS.01, ARF.BOUNDARY.CODE.01, QGEO.GRIDSUP.01, GNET.GNSLIMIT.01, PRIME.Z1.COMPACTNESS.01) + dated updates on FTRANSFER.CLOCKS.01, CONTRACT.F.01, QGEO.CARL1.01, GNET.LOCALNET.01, GNET.PPINDEX.01, GATE.METRIC.08/.11, PRIME.Z1.OPERATOR.01, PRIME.KMS.INDUCTIVE\_STATE.02, PRIME.QFCOCYCLE.01, PRIME.GRAM.DIAGONAL.01; 849 → 855 rows; all measured/typed, NO marker moves anywhere.
- **Papers.** `tfpt_1/2`: the Arf compiler, the fired matter fence and the boundary-code uniqueness join the code sections (after the symplectic code lemma), and the  $G_{\text{net}}$  follow-ups gain the v779 continuum-strand paragraphs; `tfpt_3`: seven new audit blocks + dated notes on the v715/v745/v746/v773 blocks; `tfpt_prime_front`: the Arf-compiler round in the incidence programme, the QGEO ( $1 + \ln M$ ) note, and the new section “The sharpened-Z1 trilogy: compactness carried, selection open”; `tfpt_research_contracts`: the executed clock contract, the fibered CONTRACT.F.01 note, the ARF.BOUNDARY.CODE.01 registration, the GATE.QGEO/GATE.METRIC notes, the Z1/KMS unification notes and the `CellCocycle.lean` legacy; `tfpt_4_frontier`: the jet-route-closure note.
- **Website.** Prime-front feed: the round entry “reflux round nineteen”; counters 766 → 773 modules / v535–v780; research-contracts and F\_transfer mirrors updated (`papers.ts`, glossary, OpenGates, DAG nodes); README and Zenodo updated to 773 modules / v780.
- **Scope.** The exact results (Arf finite algebra + Lean, boundary-code census, sympy-exact clock kill and cocycle, the grid Fourier log-sum theorem, the dim law) sit behind explicit fences; the fired matter kill and the preregistered partials are criteria firing inside green machinery and stay on record; the honest residuals (R1–R3, residues (b)/(c), the import-faithful coupling) are named, not claimed; no marker moves; no RH claim anywhere.

**2026-08-05 (LXI)** (the qf offensive: **FIVE** promotions (v769–v773, from eight frozen preregistered probes, all first-run adjudications; suite 761 → 766 modules) — the diagonal Gram route CLOSES at a moving spectral edge, PRIME.GRAM.DIAGONAL.01 is closed per its own frozen stakes, and the sharpened PRIME.Z1.OPERATOR.01 takes over. **THE GATE-3 AUDIT PAIR:** v769 (merged, 5/5 gates + 34/168 cells, 14/14 + 14/14 guards+controls, qf-gauge/gate3prime-dead) — Gate 3’s full-ladder Cauchy demand on  $q_f$  was provably **STRONGER** than the diagonal contract needs: a deterministic  $\varepsilon$ -net subsequence (density 0.144) makes all 14  $q_f$  profiles Cauchy at the exact parent bars the full ladder failed, and the contract’s own  $E_Y$  corner objects are **BLIND** to  $q_f$  motion (corr +0.038; the 0.34 mode-entry jump at  $M = 888$  leaves the corners at routine level; 370 cross-cluster pairs, median  $\rho = 1.13$ ) —  $q_f$  is **REPRESENTATION GAUGE**, owning the  $\varepsilon$ -scale distribution of the Yosida deficit, not the value (VAR\_RATIO 0.021); the double limit is non-uniform (deficit  $X$ -slope grows 88× over four decades) but not certified singular; yet the corner-native weakened Gate 3’ **DIES** on its own predeclared  $1.6 \times 10^7$  comb ( $X = 16.5625$ ): the corner increments fall to  $X \sim 12.9$  and then **RISE** sustained (+0.10..+0.34 per  $X$  unit at  $\varepsilon = 10^{-3}$ , 0/56 cells, broad-spectrum) — the v764 precondition is **REFUTED** at the corner level. **THE TRANSPORT PAIR:** v770 (merged, gates 4/5 + 4/5 with exactly the preregistered decider/Z fails, 19/19 + 14/14, qf-bundle-partial/qf-settles-positive) — the Kato/polar transport of the rank-6 near-kernel is well-defined with wide margins (rank stable on 888..972, gap 6/7 = 0.34..0.56, angles  $\leq 2.72^\circ$ , TV flattening) and repairs the rigid frame for 14/14 functions (median med5 ratio 26.7 → 1.004); the residual non-Cauchy part is the frame-independent band-weight drainage

(radial share  $\sim 0.64$ ) — and the drainage was a TRANSIENT: on the  $10^8$  comb ( $X = 18.375$ ) the band weights plateau at positive per-function constants ( $R2$  family 0.2249–0.3583,  $R1$  family 0.0082–0.0793; zero-bar depth  $X^* \sim 227$ , i.e. never), the 6/7 gap bottoms at 0.1008 then recovers to 0.1397, the 7th/8th threshold crossings (992/ $\sim 1108$ ) are typed, the holonomy exponent is  $\sim 1$  (step refinement closed) — the Feshbach/boundary-triple treatment becomes MANDATORY. THE CENSUS: v771 (gates 6/12 with exactly the preregistered fails, 26/26 guards+controls, qf-representation-open) — the  $6 = 1+5$  hypothesis  $E_{qf} \cong E_7 \oplus E_{-2}$  FAILS the frozen census (the deployed scalar comb realizes only two characters of  $V$ ; no rank-6 lift exists; the near-kernel is a continuum-vs-odd-places balance, ro-dominant count 0/6 vs predicted 5) while the label side is exact and uniquely canonical ( $BB^T = 4I + 3J$ , spectrum (1,9,5); 27/27 wrong forms + 5/5 wrong labelings die on  $\sigma$ -sharpness); the honest residue: ONE rung-stable ram-odd contact direction ( $\cos = 0.997$ ) plus a second contact at  $\cos \sim 0.64$ ; the dimension match stays typed DECORATIVE. THE FIXED- $d$  PAIR: v772 (merged, gates 9/12 with exactly cauchy-6/gap-7/gap-8 failing + edge-a fail/edge-b pass/KM2 quiet, 28/28 + 20/20,  $\sim 180$  s at full frozen ladders per the v677 suite precedent, feshbach-partial/edge-keeps-moving; one declared guard-normalization correction of the G1.9  $z$ -independence Ward, no gate bar touched, repeat run identical) — the  $d = 6$  reduction is EXACT (reconstruction  $8.8 \times 10^{-13}$  against independent complex dense solves, 5 rungs  $\times 5 z$ ) and matrix-Herglotz certified, but its Cauchy tail flattens where mode 7 grazes the band;  $d = 7$  converges strongly ( $b_2 = +0.502$ ) but loses separation at the top rung;  $d = 8$  has an internal 8/9 crossing ( $87^\circ$ ); on the  $10^9$  comb ( $X = 20.6875$ , 160 s) the 7/8 edge closes into an AVOIDED CROSSING (gap 0.0039 at  $M = 1240$ , tail 0.0074 vs bar 0.10) while the transported  $d = 7$  block stays Cauchy straight through; the 9th mode enters at  $M = 1276$ ; the entry cadence WIDENS ( $\Delta X = 1.625/1.8125/2.625$ , mean 2.02, spread 0.381 — not regular at the frozen 0.25 bar) — fixed- $d$  Feshbach is closed PERMANENTLY: convergence was never the problem, band ownership is. THE CLOSURE DECIDER: v773 (gates 2/8 with exactly the six convergence-cell fails, 14/14 guards+controls, cocycle-domain-only; one declared implementation correction: Hotelling refinement of the carried cocycle state after a numerically destroyed first execution, coefficients/algebra/gates untouched, the eigh-based layer reproduced identically) — STRUCTURE WITHOUT LIMIT: the exact Möbius/Redheffer cell identity holds at  $9.7 \times 10^{-12}$  over 73 rungs  $\times 5 z$  (coefficients = the added source rows/columns; independent dense-solve Ward  $1.3 \times 10^{-13}$ ), every cell Schur block is PD (min eig  $D_k \in [0.0605, 0.0668]$  — a monotone Loewner flow), and the Nevanlinna/Stieltjes domain is preserved with ZERO breaches in all 730 rung $\times$ cell tests INCLUDING both typed mode-entry transitions ( $M = 992$ ,  $M = 1108$ ; transition increments unremarkable) — but no renormalized object converges at reachable depth (0/6 cells: the corner absorbs a flat  $\sim 0.1\%$  per rung, the band increments rise with the moving edge). THE CONTRACT MOVES: PRIME.GRAM.DIAGONAL.01 CLOSED per its own frozen stakes (dated closure note with the full decision cascade: Gates 1/2/4 positive from round LX, Gate 3 dead at the corner level via v769, Gate 5/v764 never executed, cocycle domain-only; survivors typed: the exact anti-alias theorem v760, the unconditional paired Abel bound v761, the canonical dense family v762, the Kato bundle frame and settled levels v770, the Herglotz-certified exact Feshbach reduction v772, the domain-preserving cell cocycle v773; NO new variants, the ten-item stop-list stays binding); PRIME.Z1.OPERATOR.01 stays OPEN [O] and is SHARPENED (dated note: a self-adjoint geometric bulk operator

with a canonical finite-dimensional boundary/threshold structure whose Weyl  $M$ -function delivers the qf block and whose relative trace formula reproduces the full Weil formula; mandatory measured constraints: the Hecke/E8 bulk + archimedean lift, the moving-edge cadence (entries 888/992/1108/1276,  $\Delta X$  widening  $1.625 \rightarrow 2.625$ , avoided crossing 0.0039), the ram-odd contact direction ( $\cos = 0.997$ ), the settled coupling levels ( $R2$  0.225–0.358,  $R1$  0.008–0.079), the cell-ordered domain-preserving prime transport; additional kills: cadence/level contradictions = dead on arrival); the PRIME.KMS.INDUCTIVE\_STATE.02 compactness note carries the dated follow-up (the  $q_f$  pointer is adjudicated; handover to the sharpened Z1). Papers: `tfpt_prime_front` gains the closure section “The qf offensive: the route closes at a moving spectral edge, and what survives”; `tfpt_research_contracts` carries the closure cascade + the Z1 sharpening; `tfpt_3` gains the five audit blocks. Both declared implementation corrections are carried verbatim in the module PROVENANCE blocks. No marker moves on evidence lines; NO RH claim anywhere)

2026-08-05 (Two working notes join the published document set)

- **note\_e8\_gaussian\_code (working note N1) published.** The Gaussian code bridge:  $E_8$  over  $\mathbb{Z}[i]$  via Construction A over the extended Hamming code  $[8, 4, 4]$  placed  $\mu_4$ -equivariantly, the canonical four-bit quotient  $L/(1+i)L \cong \mathbb{F}_2^4$  (the 240 roots distribute exactly  $15 \times 16$ ; the code returns as the message space), and the  $G_{31}$  quartic companion — exact-arithmetic probes  $26/26 + 22/22$  (promoted as v689/v690) and 65 kernel-checked Lean 4 theorems. No claim beyond the stated algebra.
- **note\_hilbert\_polya\_truncations (working note N2) published.** A computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert–Pólya candidate (v714, v716–v721, v727–v734); SHA-256-frozen truncation spectra hit 100% of the first 377 zeta zeros at tolerance 0.25 with no prime table and no zero data in the construction path; the remaining statement (Weil positivity in the limit) localized in four machine-verified equivalent languages. Explicit no-proof fence: no theorem about  $\zeta$ , no RH claim.
- **Structure sync.** Both notes registered in `build.sh` / `make_docs_map` / `make_manifest` / the document-set box, mirrored to the website (`papers.ts/release.ts`, new “working note” class) and added to the Zenodo deposit (description + upload list). The German-language big-picture overview (`big_picture_2026-08-02_de`) stays repo-only: the published set and website are English by house rule and no supplement pattern exists.

2026-08-04 (LX) (the diagonal gram round: NINE promotions (v759–v763, v765–v768; no v764 — the named unexecuted gate; suite  $752 \rightarrow 761$  modules) — the frequency front closes exactly, the paired tail is unconditional, and the wall moves into  $q_f$ . THE OFFENSIVE FOUNDATION: v766 (17/17, handoff-bulk-converges) — the  $\varepsilon$ -regulated resolvent  $G^\varepsilon = (A_X + \varepsilon I)^{-1}$  is the ADMISSIBLE operator-system evaluation; local Cauchy defects fall at  $b = 0.174$ – $0.310$  per  $X$  unit,  $\varepsilon$ -robust over three decades; the bare  $A^{-1}$  does NOT stabilize (+45%, named negative block); the layer decomposition is a CANCELLATION BALANCE (leading channel up to  $1.9 \times 10^{12} \times$  the total defect); v767 (6/6, handoff-gram-converges) — the growing-frequency PSD source Gram: Candidate A missed honestly (slope +0.471 territory), the predeclared Candidate B (anti-alias  $N_f = 2M+1$ ) converges (errors  $0.0855 \rightarrow 0.0377$  over  $\dim Q$   $738 \rightarrow 5734$ , slope  $-0.369$ ; PSD by construction, min symbol  $+7.4 \times 10^{-3}$ ); v768 (merged from two probes, gates  $11/13 + 4/4$ , tail-weil-partial+compat-eps3-converges) — the quadrature tail TERMINATES ALGEBRAICALLY ( $2.4 \times 10^{-13}$  uniform: no quadrature tail exists, every layer

error is structural), Weil identification on the frozen battery  $\kappa = 0.97768 \rightarrow 0.99028$  (agreement 0.0485), all layer slopes  $-0.365$  with atom+pole a  $10.5\times$  cancellation pair, and the two frozen  $\varepsilon = 10^{-3}$  endpoint fails of part 1 are DECIDED POSITIVELY by the part-2 preregistration on the deepest 36-rung ladder ( $b = 0.178/0.186$ , med5  $0.343/0.467$ ,  $b_2 = 0.069/0.096$ ; first wall sign at  $\varepsilon = 3 \times 10^{-4}$ :  $b_2 = -0.011$ , reported never gated). THE SIX-STRAND DECISION LINE: v759 (8/8, resolution-boundary) —  $E_{\text{total}}$  is  $q$ -invariant under  $N_f = q(2M+1)$  to  $\sim 10^{-13}$  on all three fixed windows: the remaining handoff error is boundary/cutoff/object-gluing, NOT quadrature; v760 (14/14, antialias-corrected) — the anti-alias claim becomes an EXACT theorem of the  $M$ -cell space (Parseval  $2 \times 10^{-14}$ , moments  $3.5 \times 10^{-13}$ , Gram  $5.9 \times 10^{-13}$ ; exact alias formula matched at every count; must-fail witness error exactly 1 at  $2M-2$ ); the sharp minimal count is CORRECTED to  $N_{f,\min} = 2M-1$  (spread- $S$ :  $M+S$ ), the deployed  $2M+1$  stays exact, nothing patched; v761 (merged,  $24+10$  checks, combined abel-paired-bound) — the Stieltjes/Abel split is exact (6/6 sympy certificates), the matrix identity holds to  $1.45 \times 10^{-11}$ ,  $E_{\text{origin}}$  closes via the W1/Suzuki bookkeeping (comb-independent, verified on the controls), and the UNCONDITIONAL bound  $\eta_K = |\det_K| + |\Theta_K(0)E(1)| + B_1 I_K$  with  $B_1 = 3.734$  frozen on the finite range holds for all 24 kernels on all 5 rungs with pure  $1/\log X$  decay — NO RH-strength input; the companion DERIVES the correct breaker standard (symmetric zero-violation gate + envelope hypothesis): Epstein breaks at  $B_1^E = 2.95 \times (7/24)$ , scramble  $262\times$ , truth passes  $24/24$ ; the parent's frozen abel-dead stays on record as the calibration lesson; honest typing: identity and slope-1 normalization are measure-generic, the certified  $\zeta$ -specificity is envelope quality + measure positivity; v762 (26/26, dense-core-canonical) — the canonical countable dense test family of the  $(D, X)$  tower: admissibility LOCATED per class (hats direct; the bare box fails the  $r^{-2}$  criterion and enters only correlated/tent-read — named caveat), density rates hat  $-1.566$  / box  $-0.503$  per level, exact  $\mathbb{Q}$  correlation closure on 8/8 type pairs, explicit stage map with  $I_{64}$  exact, and the 24 battery typed an initial diagnostic package (hat 8 IS family member hat(2,3), rank 35); v763 (merged, gates  $9/10 + 7/12$  preregistered, yosida-partial/qf-dead) — the bounded Yosida transform carries the handoff rates with NO  $1/\varepsilon$  bound and NO PD-margin assumption anywhere ( $|b_Y - b_G| \leq 10^{-13}$ , sandwich verified; the circularity kill did NOT fire), the near-kernel converges as an identified stable OBJECT, every fixed  $f$  has a monotone contraction-certified  $\varepsilon$ -limit — but  $q_f$  (the battery near-kernel pairing) is NON-CAUCHY at honest maximal depth ( $X = 15.1875$ , 279849 atoms,  $4\times$  rung density) and SURVIVES orthogonalization by structural subordination (89–99.9% of every battery function lies in the 6-mode near-kernel span): Q3 closes NEGATIVELY — THE  $q_f$  WALL is the round's named wall object (density and near-kernel avoidance collide on the same function class); v764 NOT EXECUTED — its precondition (theorem-capable bounds from v761 AND v763) failed via Gate 3; documented as the named open gate inside the new contract, no module fabricated; v765 (merged,  $16+8$  checks, redteam-green-combined) — firewall CLEAN (zero blacklist hits across all 21 source-path functions), baseline reproduced exactly, 8/8 controls break; RT6 decided: the symbol gate is the ONLY detector for excess-mode deck corruption (min symbol  $-0.45..-1.72$ , all other gates measurably blind), the deficit mode is caught by rate/ratio/ $\kappa$  — BINDING: certificates must always carry BOTH gate families (symbol/PSD and rate/ratio/identification). THE NEW CONTRACT: PRIME.GRAM.DIAGONAL.01 registered OPEN — PSD source matrices  $G_n$  (no target decomposition, no zeros, no positivity tests) with corner error  $\leq 2^{-n}$  against  $W$  on the canonical dense family  $\Rightarrow$  Weil positivity on the rational span by PSD limits

+ density; decision state recorded (Gates 1/2/4 positive, Gate 3 negative on the canonical class = the  $q_f$  wall, Gate 5 not executed, redteam green; kills that did NOT fire: no RH-strength input, no spectral gap; the 10-item stop-list binding). **LEAN ROUND 7: GramCompactness.lean** (8 theorems), **AntiAliasExact.lean** (19 general- $N$  theorems), **DenseWeilCore.lean** (42 theorems); lake build green (3402 jobs), manifest at 76 files. No marker moves; NO RH claim anywhere)

- **New modules** (9). v759\_handoff\_fixed\_window\_resolution.py (8), v760\_antialias\_exact.py (14), v761\_atom\_pole\_abel.py (24+10, one module from two probes per the v518/v668 precedent, the preregistered C.EPSTEIN miss encoded per the v757 pattern), v762\_dense\_weil\_core.py (26), v763\_yosida\_handoff.py (12+14 guards/controls, gates 9/10 + 7/12 preregistered), v765\_handoff\_redteam.py (16+8), v766\_handoff\_bulk.py (17), v767\_handoff\_frequency\_gram.py (6), v768\_handoff\_tail\_weil.py (12+8 guards/controls, gates 11/13 + 4/4) — all wired into run\_all.py (suite 752 → 761, ALL CHECKS PASSED); discovery probes remain in experiments/tfpt-discovery/ with frozen results.
- **Lean.** TfptCarrier/GramCompactness.lean (8 theorems — the abstract finite-stage Gram compactness/diagonal-extraction closure for the contract’s compactness step), TfptCarrier/AntiAliasExact.lean (19 general- $N$  theorems: discrete orthogonality, offset-invariant exact quadrature, Parseval, moment reconstruction, threshold witness, midpoint alias sign; axioms only propext/choice/Quot.sound), TfptCarrier/DenseWeilCore.lean (42 theorems: enumeration bijection, stage bounds, grid module closure; honest analytic remainder in the header); imports added to TfptCarrier.lean; lake build green (3402 jobs); lean\_manifest.sha256 at 76 files.
- **Contracts.** PRIME.GRAM.DIAGONAL.01 registered OPEN [O] in tfpt\_research\_contracts + ledger (target form, the round’s measured decision state, the not-fired kill criteria, the 10-item stop-list, and the unexecuted Gate 5 = the planned v764 documented as the named open gate); PRIME.KMS.INDUCTIVE\_STATE.02 gets the dated note: the compactness step now has MEASURED handoff evidence (v766/v767/v768) and the abstract Lean closure (GramCompactness.lean); the surviving open object is the  $q_f$  wall of v763.
- **Ledger.** 9 new module rows (PRIME.HANDOFFRES.01, PRIME.ANTIALIAS.01, PRIME.ABELPAIR.01, PRIME.DENSECORE.01, PRIME.YOSIDAQF.01, PRIME.HANDOFFREDTEAM.01, PRIME.HANDOFFBULK.01, PRIME.HANDOFFGRAM.01, PRIME.HANDOFFTAIL.01) + the contract row + the dated KMS note; finite convergences typed measured, exact results typed [E]; the honest deaths/PARTIALs (tail-weil  $\varepsilon = 10^{-3}$  first pass, QF-DEAD, parent ABEL-DEAD) documented as such; NO marker moves.
- **Papers.** tfpt\_prime\_front: the new round section “The diagonal gram round: the frequency front closes exactly, the paired tail is unconditional, and the wall moves into  $q_f$ ” (all nine \veri citations); tfpt\_research\_contracts: the PRIME.GRAM.DIAGONAL.01 contract block + the KMS dated note; tfpt\_3: the round’s audit block.
- **Website.** Prime-front feed: the round entry “the diagonal gram round”; counters 752 → 761 modules / v535–v768; README and Zenodo updated to 761 modules / v768.
- **Scope.** The exact results (anti-alias identities, Abel identity + unconditional  $\eta_K$ , dense-core algebra, three Lean modules) sit behind explicit fences; the PARTIALs and deaths are preregistered criteria firing inside green machinery and stay on record; no RH claim, no marker moves; the  $\varepsilon \rightarrow 0$  wall (PD persistence) and the battery-limited identification remain open exactly as quantified.

**2026-08-04 (LIX) (the incidence programme: SEVEN promotions (v752–v758, 134 new checks, all green; suite 745 → 752 modules) — the second external**

review, fully adjudicated: five exact wins, one decorative seven, and the keystone negative that names the wall as the missing factor. **THE CENTERPIECE:** v752 (36/36, proj-hamming-exact) — the projective Hamming incidence lemma on  $V = L/(1+i)L$ :  $h = H/4$  is  $\mathbb{Z}[i]$ -valued with hermitian Gram determinant 1 (the UNIMODULAR hermitian E8 lattice over  $\mathbb{Z}[i]$ ),  $\bar{h}$  alternating and nondegenerate, so  $(V, \bar{h})$  is symplectic over  $\mathbb{F}_2$ ; on the 15 points  $BB^\top = 4I + 3J$  EXACT with the design parameter set 2/3/4/7/15/16 and eigenvalues  $7^1(+2)^9(-2)^5$  (the multiplicities (1, 9, 5) are the symplectic signature);  $K^2 = (4/49)I + (45/49)\Pi_0$ ; NS/R is a POINT of the geometry ( $\chi_{\text{NSR}} = \bar{h}(\cdot, \text{FSUM})$ , the anchor bit;  $248 = 120 + 128$ ); **THE v737 RESOLUTION:** the seam-48 distribution 0:96/1:64/2:64/3:16 derived exactly ( $16 \times$  the  $3 \times 5$  label table) AND the  $5 \times 48$  coordinatization exists via exactly ONE  $\bar{u}$ -invariant spread (kill explained, not revised); two honest corrections to the review derivation documented; companion Lean module `ProjectiveHamming.lean` (32 theorems, kernel-checked, lake build green 3391 jobs). **FOUR MORE WINS:** v753 (25/25, polarity-match) — the 15 ramified Hecke hyperplanes ARE the  $\bar{h}$ -orthogonal spaces, canonical bijection  $y(\varphi) = \bar{G}^{-1}\varphi$ ,  $\sigma$ -functorial, deck and parity commute; canonicity census: exactly 1 of 28 nondegenerate alternating forms is  $\sigma$ -invariant  $= \bar{h}$ ; v754 (19/19, twostep-exact) — the real  $n \rightarrow 4n$  down-up pass IS exactly  $K^2$ , degree factor  $196 = (2 \cdot 7)^2$  closed, KMS half-weight exponent-exact, identical integer matrix on all 9 contexts,  $T = 4B^2$ , cone-preserving; honest boundary: naive label chains do not carry  $K^2$  (correspondence composition mandatory); v755 (13/13, schur-recursion-alive) —  $S_k \succeq 0$  on all 8 ram-odd stages of the tower nesting ( $\lambda_{\min}$  flat, long-double-hardened), Douglas trivial, controls break immediately; the Albert/Haynsworth induction frame stands; honestly: for  $\zeta$  data equivalent to tower PD — the remainder is the structural Gram certificate; v756 (7/7, kms-incidence-cp-covariant) — 105 explicit Kraus terms, all four compatibilities exact ( $\sigma$  at 0, half-weight  $10^{-16}$ ,  $\beta=1$  balance  $7 \times 10^{-18}$ , Gate-0) — step 3 of the master contract SOLVED on the ramified layer; local candidate state  $\tau_{\text{inc}} \otimes \tau_{\text{Toeplitz}}$ . **THE DECORATIVE SEVEN:** v757 (26 checks, one preregistered break, fano-disc7-decorative) — the preregistered four-condition Fano $\rightarrow$ discriminant(-7) test breaks exactly at condition 2: 2 splits in  $\mathbb{Q}(\sqrt{-7})$  so the 2-rank is  $\leq 2 < 3$ , the code  $\mathbb{F}_2$ -structures are arithmetically unrealizable; the three-cycle half is real (Fano  $\bar{\sigma}$  = the  $\times 2$  multiplier, C corner = fixed point, norm  $14 = \dim G_2$ ); the seven-correlation is permanently typed decorative. **THE KEYSTONE:** v758 (8/8, cert-continuum-root-dead) — the review certificate [continuum root  $\oplus$  Kraus] is NOT mountable, the wall IS the missing left factor (arch+pol PSD at no stage, Levinson break exactly between the  $n=2/n=3$  cells ladder-wide, already  $W_0$  indefinite); real remain the [E] building blocks + the corrected target shape: cell-wise cascade factorization, rescue strictly cell-ordered. **CONTRACTS:** PRIME.PD.PERSISTENCE.01 gets the big dated update (induction form registered / assembled certificate closed / cell-wise factorization the named open object); PRIME.KMS.INDUCTIVE\_STATE.02 carries the step-3-solved note. No marker moves; no RH statement anywhere)

- New modules (7). v752\_projective\_hamming\_incidence.py (36),  
v753\_ramified\_polarity.py (25), v754\_ramodd\_twostep.py (19),  
v755\_simpler\_schur\_recursion.py (13), v756\_kms\_incidence\_stinespring.py (7),  
v757\_fano\_disc7\_bridge.py (26, one preregistered break), v758\_simpler\_certificate.py (8). Suite 745  $\rightarrow$  752 modules, ALL CHECKS PASSED; the one [FAIL] line in the green run (v757 C2.d) is exactly the preregistered honest break. All seven probes re-run before promotion (outputs identical modulo runtimes).



- **Lean.** `TfptCarrier/ProjectiveHamming.lean` committed (32 theorems, kernel `decide/norm_num` only, no `sorry/native_decide`; lake build green, 3391 jobs); import added to `TfptCarrier.lean`; `lean_manifest.sha256` regenerated (73 files).
- **Contracts.** `PRIME.PD.PERSISTENCE.01`: the big dated update — (i) the induction form registered (the Schur ladder on the canonical tower, v755; open need = the structural Gram certificate per step, measured boundary data recorded), (ii) the assembled certificate a CLOSED route (the v758 keystone negative: the wall is the missing left factor), (iii) the corrected target shape (cell-wise cascade factorization) the named open object. `PRIME.KMS.INDUCTIVE_STATE.02`: dated note — step 3 (complete positivity) solved on the factorized ramified 15-label Toeplitz layer (v756); the remainder stays exactly the compactness step.
- **Ledger.** 7 new module rows (`E8.PROJHAMMING.01`, `PRIME.HECKEPOLARITY.01`, `PRIME.HECKETWOSTEP.01`, `PRIME.SCHURREC.01`, `PRIME.KMSSTINESPRING.01`, `QGEO.FANODISC7.01`, `PRIME.CERTROOT.01`) + dated notes on `PRIME.PD.PERSISTENCE.01`, `PRIME.KMS.INDUCTIVE_STATE.02` and `E8.SEAM48.01` (the v737 resolution); 828 → 835 rows.
- **Papers.** `tfpt_prime_front`: the new section “The incidence programme: five wins, one decorative seven, and the wall as the missing factor” (the  $BB^T = 4I + 3J$  formula prominent with the 2/3/4/7/15/16 design set, the v737 resolution, the keystone negative). `tfpt_1`/`tfpt_2`: the symplectic code lemma added to the code sections. `tfpt_3`: seven audit blocks.
- **Website.** Prime-front feed: the round entry “the incidence programme”; counters 745 → 752 modules / v535–v758; README and Zenodo updated to 752 modules / v758.
- **Scope.** The exact lemmas are finite census/incidence statements behind explicit fences; the two honest negatives (v757, v758) are preregistered criteria firing inside green machinery; excluded on purpose: `big_picture_*`, `note_*`, `articles/`, the news-checker note and the other worker files; no marker moves; no RH statement anywhere.

2026-08-04 (LVIII) (the news round: a pure data-refresh against the August-2026 literature — no new analysis of our own, honest in both directions. CONFIRMING: JUNO’s Neutrino-2026 update (207 days) lands at  $\sin^2 \theta_{12} = 0.3036 \pm 0.0064$  — the sharpest single measurement of the solar angle,  $+0.49\sigma$  from the frozen prediction 0.306747, central value moving *toward* it from the 59.1-day run; ICARUS’s first standalone result (no  $\nu_\mu$  disappearance, new 3+1 exclusions at 90% CL) adds another sterile null exactly as  $N_{\text{fam}} = 3$  requires; the sequentially-valid DESI reanalysis localises the evolving-dark-energy signal almost entirely in the LRG2 bin (without it the evidence collapses, mildly pro- $\Lambda$ CDM — the dissolution trend the  $w = -1$  watchdog predicts); two new independent birefringence analyses ( $\beta = 0.30^\circ \pm 0.05^\circ$  constant-model-preferred; PR4 field-level  $0.32^\circ \pm 0.12^\circ$ ) stay consistent with the  $0.2424^\circ$  seed; CMB-SPA  $H_0 = 67.19 \pm 0.38$  lands  $+0.11\sigma$  from the parameter-free budget 67.15 (correlated legs, never a new independent hit). WEAKENING, equally prominent: CMB-SPA (SPT-3G D1+ACT+Planck) gives  $n_s = 0.9679 \pm 0.0033$  — the central value sits *above* the TFPT band edge 0.967 for the first time (edge at  $-0.27\sigma$ , preferred branch 0.9611 at  $-2.06\sigma$ ; not yet the dated kill “robustly  $\geq 0.967$ ”, typed an honest mild weakening, score-card row tension + watch-flag); the DESI DR2 effective-mass posterior keeps the  $\Sigma m_\nu = 0.0588 \text{ eV}$  floor under  $+3.1\sigma$  model-dependent pressure (unchanged, on record). ONE BRIDGE DIES: the muon  $g-2$  seam-vertex bridge’s dated kill condition (2026-07-02) has FIRED — WP25 (Phys. Rep. 2025) adopts the lattice HVP as the SM baseline and the BMW hybrid evaluation (Nature 2026) confirms it, experiment–SM =  $0.5\sigma$ , so the converged  $\Delta a_\mu$  lies outside  $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ : the vertex identification is excluded in its present form,

the scorecard row is typed `kill_channel`, ledger `FR.MUONG2.01` carries the dated kill note, and `tfpt_4_frontier` gets the dated update paragraph + summary-table move `[C]→[X]` on the bridge — the exact compiler number  $\delta_2/(2\pi)$  and the compiler core are untouched (no core marker moves): the watchdog delivered exactly what it was built for. **OPERATIONAL:** the Vela GO-programme preregistration is **FROZEN** as `experiments/p9-forecast/hypotheses/p9_go_programmes_v1.yaml` ahead of the expected 2026/27 giant-glitch window (no glitch since 2024-04-29; long-component decision rule and the 8-TOAs/day cadence requirement inside; draft SHA recorded as provenance); CHIME/FRB Catalog-2 + RepCat4 data-availability notes recorded in `repeater-cascade` and `frb-ontology`. Scorecard `120 → 121` typed rows (consistent 49 → 47, null 27, parked 4, `kill_channel` 1); `predictions.ts` mirrors the `JUNO/ns/g-2` status; no verification module touched, no core marker moves)

- **Scorecard (regenerated).**  $\theta_{12}$  row: JUNO 207-day value  $0.3036 \pm 0.0064$  ( $+0.49\sigma$ ; supersedes the 59.1-day 0.3092).  $n_s$  row: CMB-SPA  $0.9679 \pm 0.0033$  added, status `CONSISTENT→TENSION` + watch-flag (central value above the 0.967 band edge). Flat-budget row: CMB-SPA  $H_0$  leg  $+0.11\sigma$  (correlated). Sterile watchdog: ICARUS null leg.  $w$  watchdog: LRG2-localisation note. Birefringence row: two new supporting legs.  $g-2$ : WP2025-lattice row `TENSION→KILL_CHANNEL` (kill condition fired), WP2020-dispersive row `PARKED` (baseline withdrawn by WP25).
- **Ledger.** `FR.MUONG2.01`: dated kill note “(6) KILL FIRED 2026-08-04” — bridge dissolved, exact value stays `[E]`, core untouched. No other row touched.
- **Papers.** `tfpt_4_frontier`: dated update paragraph in the muon section + summary-table bridge move `[C]→[X]` (core untouched); no other paper touched.
- **Experiments.** Vela/Auger GO-prereg frozen (`p9_go_programmes_v1.yaml`); CHIME Cat2/RepCat4 data notes; news-check note `experiments/news_check_2026-08-04.md` on record.
- **Website.** `predictions.ts`: JUNO 207-day confrontation,  $n_s$  watch-flag (+ CMB-SPA numbers), the dated news-round entry (verdict `MIXED`), audit counters 121 rows; `papers.ts`: muon section mirror updated (bridge dissolved).

**2026-08-04 (LVII)** (the full day wave: **TWELVE** promotions (v740–v751, 157 new checks, all green; suite 733 → 745 modules), the frozen external-clock preregistration, Lean round 6 (47 new theorems), and the retroactive record of the editorial commit. **THE KMS END-FORM:** v740 (8/8, kms-extension-dead) — the strong master contract form is algebraically dead twice:  $\beta = 1$  KMS is impossible on Gate-0’s nonzero-grade Laurent unitaries ( $T(u_p u_p^*) = 1$  vs  $T(u_p^* \sigma_i(u_p)) = 1/p$ , exact defects 1/2, 2/3, 4/5), and exact covariant \*-restrictions do not exist; local extended Gram positivity survives (55,296 monomials/window); v741 (9/9, kms-toeplitz-dead) — the Bost–Connes semigroup Toeplitz correction repairs the algebraic KMS step **COMPLETELY** (boundary error  $10^{-2} \rightarrow 10^{-9}$ , directed system), but the covariant inductive step stays dead (UCP covariance nonmonotone, preregistered); the master contract is refined to `PRIME.KMS.INDUCTIVE_STATE.02` (operator-system compactness; keep the proven local blocks, ask only for a weak-\* cluster state). **THE CHANNEL ANATOMY:** v742 (14/14, interference-collective) — the W3 margin is **ONE** collective scalar balance (arch + ram-odd vs split + inert), strictly monotone over 35 windows, with **TWO** exact Ward identities ( $\sum c = 0$  to  $1.8 \times 10^{-15}$ ; rigid block =  $\dim \cdot \dim \cdot m_0$ ); no pair reduction; scramble  $\times 2 \times 10^3$ – $3.9 \times 10^6$ ; v743 (7/7, schur-cone-dead) — exact Moebius update ( $2.9 \times 10^{-41}$ ) + exact  $T^*JT$  identity, then the one-liner kill: cone-preserving iff

layer-positive, and no prime layer is ( $c_0 = 0$ ) — layer-wise recursion is the wrong granularity level; v744 (15 checks, hecke-sos-channels-partial) — no canonical column choice factorizes (cos-half indefinite; PSD- $B^*B$  consumes 5–6 orders beyond the margin); the Ihara channel route fails coherently (channels grade positions, positivity lives in frequencies — structural, not  $\zeta$ -specific); finiteness measured:  $\dim 16$  vs 168–862 required frequencies. THE PROOF BLOCK: v745 (11/11, l1-sector-proven-modulo-elementary) — (L1) of the CAR majorant lemma proven modulo elementary steps for all 6 sectors (exact geometric sums instead of Euler–Maclaurin, leading term at  $6 \times 10^{-4}$ ,  $\gamma = 2$  derived, 39744+108 inequalities 0 violations,  $r_1 = 1$  sharp); GATE.QGEO does not move (L2 hard-technical, L3 medium remain). THE NET THEOREM: v746 (16/16, gnet-local-functor-alive) — net axioms exact (isotony/graded locality/clock covariance), Watatani index LOCALLY exactly 4 for every interval  $\ell \geq 2$ , Ramond = half-line/bond defect (solitonic; leakage  $\sqrt{2/N}$ ); 4 must-fails fire. THE ZUSE/WOLFRAM ADJUDICATION: v747 (24/24, mixed) — the tower as a confluent multiway system: diamond exact (CRT theorem), branchial = Grassmann  $[4, 2]_q$  at the prime places (the one new fingerprint),  $\Lambda$  recursion = causal path statistics; A2/A4 renamings, A1/A3 useful typings; v748 (15/15, equivalently-useful) — the  $\sigma$ -quotient is confluent everywhere (two-line lemma:  $\sigma$  is a multiway automorphism); only the half-orbit WEIGHTING breaks, exactly at the ramified place, same location as  $[D, A]$ , not the same object; rests as exploration + the one reference sentence in the introduction’s Zuse positioning. THE SIMPLIFICATION TRILOGY: v749 (8/8, tower-nested-reduces) — THE structural upgrade: the window quantifier reduces exactly to the nested  $(D, X)$  tower (X-nesting  $6.7 \times 10^{-18}$ , dyadic  $D$ -refinement exact — PD fine forces PD coarse; every deployed window a tower member at dev 0.0); the canonical truncation replaces the historical window family in PRIME.PD.PERSISTENCE.01; honest limits: incommensurable chains, falling margins; v750 (5/5, gluing-detector) — the gluing identity is an off-line detector at identity level (sees  $\delta = 0.01$ , sharper than the PD break  $\delta^* = 0.02$ ; envelope slope measures  $\delta$ ); the rigidity thesis honestly false (the identity holds unconditionally); v751 (15/15, ward-monotone-mixed) — running deterministic in direction (100%, parameter-free  $\zeta/L$  constants,  $\mp u/2$  drifts from  $p^2 \equiv 1 \pmod{4}$ , slope-Ward cancellation  $4 \times 10^{-4}$  — the margin inherits ZERO deterministic drift), but the balance sign is zero content (budget  $2.9 \times 10^4 \times$  margin) — the induction route is RH-circular. THE CLOCK PREREGISTRATION: experiments/ftransfer-clocks/committed (design YAML byte-frozen + contract note) and FTRANSFER.CLOCKS.01 registered in the contracts paper — four frozen external clocks (proper time / inverse temperature / RG time / e-folds), the one-class Riccati/Schwarzian prediction (the seam channels transport to  $-\Delta^2/2$  exactly), three kills (K1/K2/K3), honest prior “expected: K1 kill”; executor PENDING (the FTC.01 data interfaces need curated  $g_{*s}$ /PDG overlay tables first). LEAN ROUND 6: TraceLedger.lean (28 theorems — the finite Gauss-quadrature trace formula as an exact ledger kernel, monic-Jacobi intertwining certificate, sharp must-fail at  $m_6$ ) + PinningLemma.lean (19 theorems — L-PIN-1 Wilkinson + L-PIN-2 Kato–Temple/Temple positivity in sqrt-free rational form); lake build green (3390 jobs), no sorry/native\_decide. RETROACTIVE RECORD: commit 414471b2 (2026-08-04, between LVI and this round) was the pure editorial restructuring of tfpt\_prime\_front — no content changes, the  $\sim 248$ -page paper reorganized into parts with part intros; recorded

here for the dated trail. NO RH claim, NO marker moves)

- **New modules (12).** v740\_kms\_extension\_switch.py (8), v741\_kms\_toeplitz\_semigroup.py (9), v742\_channel\_interference.py (14), v743\_schur\_cone\_recursion.py (7), v744\_hecke\_channel\_columns.py (15), v745\_qgeo\_car\_l1\_sector\_lemma.py (11), v746\_phys\_gnet\_local\_functor.py (16), v747\_automaton\_multiway\_tower.py (24), v748\_automaton\_sigma\_orbifold\_confluence.py (15), v749\_simpler\_tower.py (8), v750\_simpler\_gluings.py (5), v751\_ward\_monotone.py (15) — 157 checks, each promoted verbatim from its discovery probe with numbers frozen from the promotion run (all twelve probes re-run before promotion; outputs identical modulo runtimes; sibling imports resolve against the promoted v696/v716/v718/v735/v740/v747 modules). The three [FAIL] lines in the green run (v740 K1.ii/K1.iii, v741 T2.ii) are exactly the preregistered honest kills.
- **Lean.** TfptCarrier/TraceLedger.lean (28 theorems) + TfptCarrier/PinningLemma.lean (19 theorems) committed (kernel-checked, no sorry/native\_decide; lake build green, 3390 jobs); imports added to TfptCarrier.lean; lean\_manifest.sha256 regenerated.
- **Contracts.** PRIME.KMS.INDUCTIVE\_STATE.02 registered (operator-system compactness end-form; .01 annotated as settled at the finite level); PRIME.PD.PERSISTENCE.01 restated on the canonical nested tower (PRIME.TOWERNEST.01 paragraph); FTRANSFER.CLOCKS.01 registered (executor pending); dated GATE.QGEO note (the (L1) proof block).
- **Ledger.** 12 new module rows + 2 new contract rows (PRIME.KMS.INDUCTIVE\_STATE.02, FTRANSFER.CLOCKS.01) + dated notes on PRIME.KMS.INDUCTIVE\_STATE.01, PRIME.PD.PERSISTENCE.01, GATE.QGEO.01 and Lean notes on PRIME.MOONSHOT.05/PRIME.K1CAPTURE.01; 813 → 827 rows.
- **Papers.** tfpt\_prime\_front: the new section “The simplification wave” (the tower upgrade, the two closed hopes, the KMS end-form, the channel anatomy). tfpt\_2: the  $G_{\text{net}}$  local-net paragraph (the algebraic-QFT counterpart of the OS package). tfpt\_3: 12 new audit blocks. introduction: the one Zuse-positioning reference sentence (the multiway reading measured out, equivalently useful).
- **Website.** Prime-front feed: the round entry “the full day wave”; counters 733 → 745 modules / v535–v751; README and Zenodo updated to 745 modules / v751.
- **Scope.** Five honest kills promoted as encoded expected outcomes (preregistered criteria fire inside green machinery); the exact lemmas are finite/census statements behind explicit fences; the clock contract is a preregistered search-target design typed [X] with the honest prior “expected: K1 kill”; no marker moves; no RH statement anywhere.

**2026-08-04 (LVI)** (the morning after: **FOUR** review strikes (v736–v739, 104 new checks, all green) — two exact structural lemmas, two honest kills, and two new research contracts. **STRIKE 1, EXACT** (v736, 26/26, orbit-packet-exact): the Gaussian ORBIT PACKET lemma — Burnside  $7 = (15+3+3)/3$  on the 15 nonzero classes; the canonical packet split  $240 = 5 \times 48$  (one fixed block + four moved blocks of 12 Gaussian lines each, label-free canonical; packets =  $1 + 4 = 5 = g_{\text{car}}$ ); the  $(1, 1, 2)$  anchor from the fixed-syndrome leader weights  $(0, 1, 1, 2)$  with leader pair  $e_6/e_7$  = the v638 anchor pair; the  $q = 16$  normal form  $240 = 16 \cdot 15$ ,  $248 = 240 + 8$ ; the CIRCULARITY FENCE is part of the lemma — P1/P2 come back as EXACT-INTERNAL-RECONSTRUCTION (reconstruction, NOT derivation), and the packet law is  $\sigma$ -CLASS-specific (160 free order-3 elements of G31 give Burnside 5); companion Lean module OrbitPacket.lean (34 theorems, kernel-checked, lake build green at 3388 jobs). **STRIKE 2, DEAD** (v737, 24/24 machinery, seam48-dead): the seam-48 intertwiner — the sharpest structural sharpening — dies on

its preregistered criterion: the  $48 \times 5$  incidence matrix is NOT identically 1 (entry distribution 0:96/1:64/2:64/3:16), representative-independent (0 of 2304 order-5 elements) with the must-fail control firing (0 of 25 random operators); valuable residues: the four moved blocks carry the v623 character table EXACTLY, the fixed block deviates exactly by the 12 lifted marks,  $u^5 = \pm J$  exact —  $240 = 48 \cdot 5$  remains a factorization WITHOUT coordinatization. STRIKE 3, EXACT (v738, 34/34, hecke-mod-ramified-channels): the Gaussian–E8 Hecke tower projects onto  $V = L/(1+i)L$  exactly and  $\sigma$ -functorially over ALL 5907 submodules (no kill); odd layers act as degree  $\cdot$  id (the fiber is Hecke-rigid); the ramified layer IS the 15 hyperplanes with the 2:1 deck; channels = the 7  $\sigma$ -orbit channels +1 trivial (resolving 7-vs-8); NS/R = the  $\sigma$ -fixed parity character (v722 re-derived); ram-odd ( $n = 2, 8, 32, \dots$ ) is the unique negative-pressure channel; honest boundary: the W3 margin is CROSS-CHANNEL CANCELLATION — no transfer-matrix reduction from the projection alone. STRIKE 4, DEAD (v739, 20/20, rank3-jet-dead): the review’s first-jet reading of the rank-3 wall is killed by its own  $\varepsilon^0$  test — the jet read itself is EXACT (the lock block as the first jet of ONE function to  $4.6 \times 10^{-16}$ , Loewner coordinates, odd-sector Schwarzian form  $[1 - r_{12}^2]_{\text{odd}} = (3/8) S[\tilde{g}](1) \varepsilon^4$ ), but the  $\varepsilon^0$  coefficient is identically 1 (not 0), the even sector exceeds the net entry ( $1.15\text{--}6\times$ ), and the pure jet block carries no collinearity (factor  $\geq 1.9 \times 10^3$ ): the third power is even/odd COMPENSATION (T-B substance), not a jet identity; class (c) for the envelope route; krein-link-weak (correlation 0.678 vs control 0.946). TWO NEW CONTRACTS: PRIME.KMS.INDUCTIVE\_STATE.01 (the review master contract — extend the v733 Gate-0 functor to a positive global KMS state at  $\beta = 1$  with  $T_h = \omega \circ \iota_h$ ; five separation steps individually named; kill = complete positivity failing already on a finite extension) and PRIME.PD.PERSISTENCE.01 (the universal-theorem form of the wall with the important correction  $\lambda_{\min} \geq 0$  instead of  $\geq \delta$  — the Ihara lesson: exact zero modes after rank saturation are admissible; target = cofinally infinitely many PD windows via state/Gram/cone-preserving recursion, NOT eigenvalue bounds; the review blocklist confirmed as permanent kills). NO RH claim, NO marker moves)

- **New modules (4).** v736\_orbit\_packet.py (26), v737\_seam48\_intertwiner.py (24 machinery), v738\_hecke\_mod\_ramified.py (34), v739\_rank3\_jet.py (20) — 104 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (all four probes re-run before promotion; outputs identical modulo runtimes; sibling imports resolve against the promoted v563/v683/v696 modules).
- **Lean.** TfptCarrier/OrbitPacket.lean committed (34 theorems, kernel decide/norm\_num only, no sorry/native\_decide; lake build green, 3388 jobs); import added to TfptCarrier.lean; lean\_manifest.sha256 regenerated with the new file.
- **Ledger.** 4 new module rows (E8.ORBITPACKET.01, E8.SEAM48.01, PRIME.HECKEMODRAM.01, PRIME.RANK3JET.01) + 2 new contract rows (PRIME.KMS.INDUCTIVE\_STATE.01, PRIME.PD.PERSISTENCE.01) + dated notes on E8.GAUSSIAN.CODE.01 (orbit packet closed, seam-48 dead), PRIME.Z1.MOONSHOT.01 (the ramified fiber typed exactly) and PRIME.RANK3FUNC.01 (the jet simplification dead); 807  $\rightarrow$  813 rows.
- **Papers.** tfpt\_prime\_front: the new section “The morning after: four review strikes” (the two exact structures, the two graves, the 7-channel structure, the documented exact jet/Loewner/Schwarzian forms, the two new contracts). tfpt\_research\_contracts: the two new contracts + the review blocklist confirmed as permanent kills. tfpt\_1/tfpt\_2: the orbit packet lemma in the code sections (the compact number-family table with the circularity fence; the seam-48 kill recorded). tfpt\_3: 4 new audit blocks.

- **Website.** Prime-front feed: the round entry “the morning after: four review strikes”; counters 729 → 733 modules / v535–v739; README and Zenodo updated to 733 modules / v739.
- **Scope.** Two honest kills promoted as encoded expected outcomes (preregistered criteria fire inside green machinery); the exact lemmas are census/reconstruction statements behind explicit fences (no P1/P2 derivation claim); no marker moves; no RH statement anywhere.

**2026-08-03 (LV)** (the closing slice: ONE promotion (v735, 10 construction checks, green) — the L1 route portfolio completed honestly, plus one number-fidelity fix on the website. **THE STATE-TRANSPORT ROUTE STAGNATES** (v735, 10/10, ucp-stagnates): for each window the simple-spectrum GNS shift generates the commutative spectral algebra  $C^*(J_h, 1) = \mathbb{C}^h$ , so a UCP map is exactly a row-stochastic transport kernel and the Choi SDP reduces to a finite transport LP — all eight adjacent maps along the nine-window ladder exist **CANONICALLY** as monotone quantile couplings (unitality/state preservation to  $3.1 \times 10^{-12}/1.2 \times 10^{-15}$ , Choi minimum 0), every Choi rank inside the sharp transport bound  $h + h' - 1$  (ranks 344..2688, log-log slope 0.991 vs bar 1.25: **LINEARLY** controlled sparse-transport ranks) — but the preregistered covariance gate **S2.3 FAILS HONESTLY** (aggregate shift/grade/phase defects 0.797..1.227..0.058: slope  $-0.487$ , tail not monotonically decreasing): mere state transport is **NOT** the missing L1 mechanism; the optimal maps are quantile Markov/conditional-expectation couplings, **NOT** \*-homomorphisms (multiplicativity defects  $\sim 5-8 \times 10^{-3}$  quantify the operator-system-vs- $C^*$  obstruction); the full noncommutative groupoid algebra was not tested. **THE PORTFOLIO**, complete: Sonin transfer dead (v732), UCP stagnates (v735), Gate 0 weak functor (v733), Suzuki bridge a measurement (v734) — what remains for L1 is atomicity/identification substance, not transport machinery. **QUALITY FIX** (website only): the prime-front feed entry for v727 misquoted the Carleman evidence as “band 0.34–0.41 vs the 0.5 threshold”; corrected to the docstring values — the measured Gauss/Freud growth band  $[0.638, 1.117]$  makes the Carleman sum diverge (all repository surfaces — ledger, registry, run\_all, papers — already carried the correct band; the remaining v727–v734 feed numbers spot-checked against the docstrings, no further discrepancy). **NO RH claim, NO marker moves**)

- **New module (1).** v735\_strat3\_ucp\_inductive.py (10 construction checks) — promoted verbatim from strat3\_ucp\_inductive\_probe.py (probe re-run before promotion; output identical modulo runtime; the sibling import rewired to the promoted v718 module).
- **Ledger.** 1 new row (PRIME.UCPLIMIT.01) + the dated LV closing-slice note on PRIME.Z1.MOONSHOT.01 (the L1 route portfolio complete; stays OPEN); 806 → 807 rows.
- **Papers.** tfpt\_prime\_front: one closing-slice passage in the keystone section (the portfolio sentence); tfpt\_research\_contracts: PRIME.Z1.MOONSHOT.01 update V; tfpt\_3: 1 new audit block.
- **Website.** Counters 728 → 729 modules/v535–v735; the Carleman number-fidelity fix in the keystone feed entry; README + Zenodo 729/v735; generated surfaces regenerated.

**2026-08-03 (LIV)** (the keystone round: **EIGHT** promotions (v727–v734, 90 new checks, all green) — **THE WALL IN FOUR EQUIVALENT LANGUAGES**, the K1 diagram closed classically, and honest graves for two strategy routes. **THE WALL, MAXIMALLY PRECISE** (v727, 11/11, l1-determined-wall-typed): determinacy of the limit moment problem is **CLASSICAL** (the raw  $\tau$ -measure

does not carry the limit; Gauss-damped Weil moments identify against the closed geometric side at  $1.4 \times 10^{-3}$  over  $m_0..m_{16}$ ; the measured Gauss/Freud band  $[0.638, 1.117]$  makes Carleman diverge) and the identification chain (a)–(d) contains RH NOWHERE (Weil 1952 unconditional); the seeming paradox resolves honestly — “ $T_h$  is PD” is a MEASUREMENT per  $h$ , not a theorem (scramble: Levinson death at depth 61); the one remaining statement is PD PERSISTENCE: infinitely many PD windows suffice for RH, each window a FINITE Levinson decision, measured 9/9 (min headroom 0.718); the converse is NOT shown; equivalently (Riesz–Haviland): the geometric side defines a positive measure on  $\mathbb{R}$ . THE FOURTH LANGUAGE (v734, 10/10, s1-canonical-suzuki-bridged): the Jacobi/Wheeler data define via an EXACT Krein correspondence (Weyl identity  $1.6 \times 10^{-15}$ , nilpotency  $3.1 \times 10^{-16}$ ) a canonical-system Hamiltonian per window with  $H_h \geq 0$  MEASURED on all nine (min  $l_k = 0.033..0.377$ ); the boundary phase carries the  $\gamma$ -side at scale EXACTLY 1; scramble breaks the Wheeler/Krein chain at depth 31; Suzuki (JFA 2021, GRH  $\equiv$  positivity of the canonical family, unconditional only  $\omega > 1$ ) becomes the fourth anchor — so the wall now reads: moments (Hankel-PD) / state (Levinson-PD) / symbol (Fejér  $\geq 0$ ) / Hamiltonian (Krein  $l_k > 0$ ), each finitely decidable per window, the  $h$ -quantifier IS the RH substance. THE K1 DIAGRAM CLOSES CLASSICALLY: the memo’s locality/symmetry mechanism for the  $\sim 900\times$  super-resolution is DEAD on two preregistered bars that die honestly (v728, 11/13, k1b-atom-pinning) — what carries the scale is ATOM PINNING of the raw measure (Nevai-type; lab rate  $K^{-1.84}$ , weight ratio  $\rightarrow 1.03$ , real correlate 0.64; off-line pairs break POSITIVITY instead of mislocating nodes); behind it the explicit PINNING LEMMA is certified (v730, 12/12: residual bound + Kato–Temple, one-line proofs, 0 violations at 598 peaks/535 matched zeros incl. ALL 377 of the largest window, SHA256-frozen; the certificate SATURATES at window width — sub-width precision needs atomicity beyond every window depth = L1, the sharpest localization); the gap layer: strong measured universality KILLED on the preregistered bar (v731, 5/5, gap-univ-killed: deviation plateaus 0.33; the uniform-in- $h$  Szegő condition breaks like  $h^{-0.64}$ ; a fixed-window CONDITIONAL candidate remains via Lubinsky 2009/Levin–Lubinsky 2008/Avila–Last–Simon 2010/Totik 2009) — so: capture theorem (v721) + pinning lemma (v730) + conditional gap rigidity (v731) + atom structure (= L1) = K1. TWO HONEST GRAVES: naive reflection positivity is DEAD on the truncations (v729, 5/5, rp-dead-on-truncations: the total window fails plain AND KMS-twisted RP at full scale; the FE symmetry buys exactly the atom-free Connes–Consani 2021 regime  $|u| < \log 2$  and no more; bulk universality EMERGING at gap points,  $0.304 \rightarrow 0.150$ ; the pinning lab reproduces the super-resolution scale, spacing/error up to  $211\times$ ); and the DIRECT transfer of the Connes–Consani Sonin compression is DEAD (v732, 21/21 construction checks, sonin-rank-grows: bad rank  $19 \rightarrow 86$ ; fenced — not the CC mechanism itself, retuning forbidden). PLUS THE FIRST EXACT BRIDGE INTO CONNES’ FRAMEWORK (v733, 13/13, gate0-weak-functor: the  $\sigma$ -descended groupoid maps by an exact coefficient-one graded  $*$ -algebra functor candidate into the  $S$ -local periodic-orbit subalgebra,  $S = \{\infty, 2, 3, 5\}$ , 6859 convolutions at deviation 0,  $\kappa \equiv 1$  at all three places; full-algebra extension typed OPEN). THE LITERATURE ANCHOR, as the dual council reads it: the wall has three names — Weil 1952 (explicit-formula positivity), Connes 1999/Connes–Consani 2021 (trace formula/atom-free regime), Suzuki 2021 (canonical-system positivity) — with Meyer 2005 the proven boundary line (unconditional spectral realization WITHOUT positivity); TFPT adds the

computable zeta-free truncation framework with emergent prime places and per-window decidability. NO RH claim, NO marker moves)

- **New modules (8).** v727\_l1\_identification.py (11), v728\_k1b\_superresolution.py (13; the two preregistered memo-bar deaths B1.1/B2.2 encoded as the expected promoted outcome), v729\_strat2\_rp\_universality.py (5), v730\_strat2\_pinning\_lemma.py (12), v731\_strat2\_gap\_universality.py (5), v732\_strat3\_sonin\_prolate.py (21), v733\_strat3\_gate0\_census.py (13), v734\_s1\_canonical.py (10) — 90 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (all eight probes re-run before promotion; outputs identical modulo runtimes; sibling imports rewired to the promoted v696/v714/v716–v721 modules).
- **Ledger.** 8 new rows (PRIME.L1IDENT.01, PRIME.K1BPIN.01, PRIME.RPUNIV.01, PRIME.PINLEMMA.01, PRIME.GAPUNIV.01, PRIME.SONIN.01, PRIME.GATE0.01, PRIME.S1CANON.01) + the dated LIV keystone note on PRIME.Z1.MOONSHOT.01 (L2 resolved classically as atom pinning + pinning lemma; L1 typed in four equivalent languages; the two strategy kills and the weak functor recorded; the literature anchor; stays OPEN); 798 → 806 rows.
- **Papers.** tfpt\_prime\_front: the new section “The keystone round: the wall in four equivalent languages” (the maximally precise wall statement, the four languages, the literature-anchor paragraph with Weil 1952/Connes 1999/Meyer 2005/Connes–Consani 2021/Suzuki 2021, the closed K1 diagram, the two graves, the weak functor); the two stale “open K1b question” spots in the moonshot-arc section now point forward. tfpt\_research\_contracts: PRIME.Z1.MOONSHOT.01 update IV — the contract at its final state of the day. tfpt\_3: 8 new audit blocks.
- **Website.** Prime-front feed: the round entry “the wall in four languages” (with the explicit no-proof fence); counters 720 → 728 modules / v535–v734; README and Zenodo updated to 728 modules / v734.
- **Scope.** No marker moves anywhere: PD persistence remains a MEASUREMENT per window (9/9 is not a universal quantifier); the two kills are honest deaths of strategy routes, cleanly fenced (the CC mechanism itself untouched); the weak functor is a finite-basis statement, the full-algebra extension open; no RH statement anywhere.

**2026-08-03 (LIII) (the evening bundling: ELEVEN promotions (v716–v726, 211 new checks, all green) — THE MOONSHOT ARC COMPLETES AT MEASUREMENT LEVEL (“the Hilbert–Pólya candidate, measured” — with the explicit no-proof fence), and the three physics interfaces move: the  $v_{\text{geo}}$  torsor closes in calibration form,  $G_{\text{net}}$  gets its first exact witnesses, and both thermal-time routes for  $F_{\text{transfer}}$  die. THE MOONSHOT (v716–v721): stage 2 GLUES (v716, 18/18, moonshot-stage2-glued: one object, ONE normalization — the free 3-scalar LSQ returns  $\kappa = (1, 1, 1)$  to  $< 10^{-12}$  on all five windows; the  $1/4$  in  $\text{Re } \psi(1/4 + i\tau/2)$  DERIVED as the  $\mu_4$  fixed-sector offset;  $\log \pi$  only through the declared self-dual UV cell; E8 forced AT THE GLUING —  $\mu_2$  tower 0.326 / wrong  $\mu_4$  class 1.000 / wrong twists 0.474 all rip); stage 3 is a STATE (v717, 21/21, state-on-truncations: constructive GNS vector state on every truncation via Levinson PD; the naive tower reading misses by  $|\Delta|/\text{W3-margin} = 104\text{--}1563$  — the trace-formula boundary IS the localized RH substance; KMS  $\beta = 1$  detailed balance measured; Ihara an exact state, Epstein breaks); stage 4 CONVERGES MEASURED (v718, 10/10: 100% of 377 frozen zero targets at tol 0.25 at  $h = 1433$ , rate  $-1.61$ , monotone GNS  $K$ -ladder  $0.4678 \rightarrow 0.0003$ , SHA256 prediction fire-wall, GUE light  $\langle r \rangle = 0.6178$  vs  $0.6189$ , scramble dies as a state at depth 61); the Satz-1 slice is EXACT (v719, 24/24, satz1-ledger-exact: the finite trace formula is exact Gauss quadrature at  $1.7 \times 10^{-11}$  and the term dictionary to the**



classical Weil formula CLOSES — pole  $1.7 \times 10^{-13}$  / atoms  $1.8 \times 10^{-15}$  / arch  $3.4 \times 10^{-16}$ ;  $\Delta \rightarrow \Gamma'/\Gamma$  term + prime term classically,  $\sigma = 1.000161$ ; E8 census over  $\{\mathbb{Z}^8, E8\}$  (Mordell 1938): ONLY the Gaussian E8 glues); the K collapse (v720, 14/14, k2k3-collapsed-classical: K3  $\rightarrow$  measured tightness (sigma-finite, vague topology forced), K2  $\rightarrow$  classical analysis on  $s > 1/2$  (unconditional Chebyshev majorant, exact slowdown law), the boundary EXACTLY  $s = 1/2$  — Mertens slope 1.9994); and the K1 capture lemma (v721, 14/14, k1-lemma-capture-only: capture proof-near from Markov–Stieltjes + symmetrized Christoffel + a closed capture condition; the certified radii sit  $\sim 900\times$  above the measured errors — the super-resolution is trace-formula content, the open K1b question). NET: everything a Hilbert–Pólya candidate needs exists at MEASUREMENT level; the remaining hard substance is (L1) identification at the line  $s = 1/2$  (no mass loss) + (L2) node convergence as a theorem; the contracts stay OPEN; NO RH claim. THE PHYSICS INTERFACES:  $G_{\text{net}}$  (v722, 9/9, ramified-sees-nsr: the NS/R grading IS the parity character of  $E8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$  — the Ramond projection is  $(1+i)$ -ADIC at the one ramified edge of the Hecke tower;  $15 \times 16$  classes parity-PURE, both controls fire; and v726, 57/57, t2-slice-go: optimal Pimsner–Popa constant EXACTLY  $1/4$  and Watatani index EXACTLY 4 on the CAR ladder via an explicit quasi-basis with FORCED weights,  $\mu_4$  DERIVED from the clock, the Ramond sector healed STATE-PRESERVINGLY by the Klein dressing — GNET.RAMIFIED.01 registered with kills; GATE.METRIC.08/10 typing unchanged);  $F_{\text{transfer}}$  (v723, 11/11, stt-killed + internal-pair-confirmed: the QCD 1-loop clock is PARABOLIC and not diagonalizable — no constant-rescaled modular flow reaches it; the pole multiplier exactly  $(2/3)^6$ ; and v724, 12/12, t3b-dead: conditioned modular flows of one internal mother state come out loxodromic/complex, the frozen solver classes do NOT arise — BOTH thermal-time routes closed, the external clock contract CONFIRMED and sharpened to  $\{\text{one unit}, \theta_i, C_p\}$ );  $v_{\text{geo}}$  (v725, 21/21, torsor closure pass: the interface is structurally CLOSED as an  $\mathbb{R}_+$  scale torsor — complete export table, dimension-matrix rank 1, all conditions  $\lambda$ -homogeneous, two-anchor consistency 0.11%, the calibration theorem stated; FENCE: no scale derivation (No-Unit); the one remaining dimensionless ratio NAMED:  $H_{\text{EW}} = \ln(\bar{M}_{\text{Pl}}/v_{\text{EW}}) = 37.1776$ ). PLUS LEAN ROUND 5: SineGramKeystone.lean + WatataniIndexFour.lean committed (lake build green, 3387 jobs, kernel-checked, no sorry/native\_decide) — the v701 sine-Gram keystone kernel and the  $\mu_4$  index-4 certificate in exact arithmetic)

- **New modules (11).** v716\_moonshot\_arch\_glue.py (18), v717\_moonshot\_state.py (21), v718\_moonshot\_spectral.py (10), v719\_moonshot\_traceformula.py (24), v720\_moonshot\_k2k3.py (14), v721\_k1\_node\_capture.py (14), v722\_phys\_ramified\_ns\_r.py (9), v723\_phys\_modular\_clock.py (11), v724\_phys\_t3b\_modular\_flows.py (12), v725\_phys\_vgeo\_torsor\_audit.py (21), v726\_phys\_car\_pp\_index.py (57) — 211 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (all eleven probes re-run before promotion; outputs byte-identical modulo runtimes).
- **Ledger.** 11 new rows (PRIME.MOONSHOT.02–.06, PRIME.K1CAPTURE.01, GNET.RAMIFIED.01, FTRANSFER.STT.01, FTRANSFER.COCYCLE.01, VGeo.TORSOR.01, GNET.PPINDEX.01) + dated LIII notes on PRIME.Z1.MOONSHOT.01 (the big update: stages 2–4, Satz-1 slice and K collapse done; kill audit K1/K3/K4 do not fire, K2 half-standing; remaining substance (L1)+(L2)+(L3); stays OPEN), ANCHOR.VGeo.02 and METROLOGY.CLOSURE.01 (torsor closure in calibration form; typing unchanged), FIRSTPRINCIPLES.BOUNDARY.01 (both thermal-time kills; anchor census  $\{\text{unit}, \theta_i, C_p\}$ ), GATE.METRIC.08 (index measured on the ladder; the identification half registered

under GNET.RAMIFIED.01) and GATE.METRIC.10 (the NS/R grading's arithmetic home); 787  $\rightarrow$  798 rows.

- **Papers.** `tfpt_prime_front`: the new section “The moonshot arc: the Hilbert–Pólya candidate, measured” (the four-stage narrative + the end ledger, no-RH fence stated twice) and the stage-2 fence in the L1/moonshot section updated. `tfpt_research_contracts`: PRIME.Z1.MOONSHOT.01 rewritten to the post-run state (stages done, kill audit, remainders), PRIME.Z1.OPERATOR.01 update III, the GATE.METRIC.08/10 follow-ups (v726/v722), the No-Unit keybox torsor paragraph (v725), and the F\_transfer thermal-time kill paragraph (v723/v724). `tfpt_3`: 11 new audit blocks + the stage-1/montage fences updated. `tfpt_1/tfpt_2`: the code-section follow-ups extended (E8 forced at the gluing; census) and the  $G_{\text{net}}/v_{\text{geo}}/F_{\text{transfer}}$  interface pointers added. `tfpt_4_frontier`: the first-principles boundary follow-up (clock kills + torsor).
- **Website.** Prime-front feed: the round entry + a separate moonshot highlight (“the Hilbert–Pólya candidate, measured” with the explicit no-proof fence); counters 709  $\rightarrow$  720 modules / v535–v726; the open-gates, status and glossary surfaces absorb the torsor/ $G_{\text{net}}$ /thermal-time state; README and Zenodo updated to 720 modules / v726.
- **Lean round 5.** `SineGramKeystone.lean` + `WatataniIndexFour.lean` committed (lake build green, 3387 jobs, kernel-checked, no sorry/native\_decide): the finite algebraic kernel of the v701 keystone (the product-to-sum identity  $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$  over an arbitrary commutative ring via Chebyshev recursion — the half-integer sine-moment form IS a Gram matrix,  $S = \sum_i 2w_i s_i s_i^T$ , hence PSD for nonnegative weights, with explicit Pythagorean-triple certificates; the FFT limit and the deployed window wiring honestly NOT formalized) and the  $\mu_4$  index-4 certificate of the v726 slice in its smallest faithful matrix model ( $M_4(\mathbb{Z}[i])$ : the clock-order shadow  $S^{12} = -1 \Rightarrow H^4 = 1$  forced by the NS spin structure, the explicit 13-element quasi-basis with  $\sum vv^* = 4 \cdot 1$  EXACTLY  $= N(1+i)^2$ , the Pimsner–Popa constant  $\lambda^* = 1/4$  sharp via an SOS certificate at the mixing minimizer,  $\mu_2$ /three-sector controls give index 2/3; the CAR/Fock layer and the Ramond dressing honestly NOT formalized); `TfptCarrier.lean` imports added; `lean_manifest.sha256` regenerated with the new files.
- **Exploration (firewalled).** `experiments/p9-forecast/` committed as a standalone preregistered forecast workspace (FRB/GW/PSR/UHECR detector kernels + hypothesis YAMLS; no ledger row, no paper claim, no website surface — experiments firewall). The K1b super-resolution probe and the L1 determinization probe are NOT in this round (workers still running; next round).
- **Scope.** No marker moves anywhere: the moonshot results are MEASUREMENTS (diagnostic SHA256 typing; the operator family stays zeta-free); PRIME.Z1.OPERATOR.01 and PRIME.Z1.MOONSHOT.01 stay OPEN; the  $v_{\text{geo}}$  closure is the calibration form, not a scale derivation (No-Unit stands); GATE.METRIC.08/10 typing unchanged (the Q-system identification is the registered open half); the external clock contract of F\_transfer is CONFIRMED, not internalized; no RH statement anywhere.

**2026-08-03 (LII) (the third daily bundling: FIFTEEN promotions (v701–v715, 158 new checks, all green) — the T-B census reaches 69/70, the Z1 continuum candidate is ASSEMBLED, and the moonshot's atom layer is groupoid-internal. HIGHLIGHT 1 (v712, 5/5 — the P4 strike): the v692 margin identity SCALARIZED along the transverse direction (the penalty pays only the transverse zero mass; median gain  $\times 4.23$  on the open ten) closes nine of the ten remaining windows: 69/70 COMPLETE FAMILY WINDOWS CLOSE at the cited  $3 \times 10^{12}$  (unconditional-modulo-citations); the single remainder is  $h = 5690$  with  $T^* = 1.6 \times 10^{14}$  (down from  $8.5 \times 10^{14}$ ); the deck lever honestly dead (triangle**

inequality). **HIGHLIGHT 2 (v713+v714):** the L1 MONTAGE (v713, 16/16, 11-assembled-measured) assembles the Z1 continuum candidate explicitly — cover limit  $6.2 \times 10^{-8}$ , point masses at the  $D^2$  rate, one UV slot, the 5b negative exposed as a normalization artifact, divergence exactly at  $s \leq 1/2$  (the critical line as the boundary of the construction, a consistency signature, not positivity), the missing theorems NAMED; and the MOONSHOT stage 1 (v714, 19/19, moonshot-go) shows the atom layer is GROUPOID-INTERNAL from the  $\mathbb{Z}[i]$ -E8 Hecke tower — primitive degrees = Gaussian prime norms exactly (no prime projector anywhere), de-divisorization at  $4 \times 10^{-13}$ ,  $\sigma$  descent 100 positions/100 masses/368 moments at 0.00; honestly fenced: the finite counting facts are E8-UNSPECIFIC, specificity is stage 2 of the newly registered contract PRIME.Z1.MOONSHOT.01. **HIGHLIGHT 3 (the chain arc S-A-S-G, v702–v711):** the lookahead question DECIDED (v702, 13/13, flow-verifier-not-generator: full comb verified  $913 \geq 898$ , autonomous reach 2–4 slots,  $g \sim 5$ –12 bits/slot information-theoretic); tolerance induction DEAD at RH grade (v703, 10/10); G1 open as a closed form (v704, 5/5: best law C2 median 0.9947, max 0.3130); the arch IS a deck-sector trace (v705, 7/7: exact tower traces  $3.4 \times 10^{-16}$ , scalar forced 1, must-fail  $\times 82$  — the geometry anchor); WEYL-NUL with the Schur-parameter identity ( $10^{-40}$ ) and the log 2 find (v706, 10/10); the section edge is an exact resolvent identity ( $3 \times 10^{-13}$ ) with the corridor position  $\sim 0.53$  open (v707, 9/9); the best position functional transports scalar-free (Levinson energy extremum, median 0.9986; outliers exactly on 16/64/81; v708, 7/7); the residual carries NO zero signature behind a temporal SHA256 firewall (v709, 10/10, layering-rejected); the two-stage law dies by its own bar but its Hecke stage heals the prime powers  $11.7\% \rightarrow 0.53\%$  circle-free from E8 counting (v710, 7/7); the first-birth error CONVERGES (promotion run  $\theta = 1.851$ ,  $R^2 = 0.692$ , bars 0.5/0.6; v711, 7/7) — net:  $G1 = [\text{E-edge}] + [\text{E-Hecke}] + [\text{M-rate}]$ . **THE REST:** the keystone identity (v701, 11/11:  $B = \text{sine-moment form}(c + \text{pole}) + \text{rank-1 pole square exact}$ ; the three gaps collapse onto ONE; triplication  $\mu_3$ -unique; V5 numerology dead) and the QGEO CAR route (v715, 22/22, qgeo-car-rates-summable: all 6+1 channels Schatten-Cauchy with summable rates, FH renormalization lattice-derived via Barnes  $G$  at  $8.8 \times 10^{-7}$ , RP cone stable; the majorant lemma (L1)–(L3) named — GATE.QGEO does not move). Between the bundlings, two website-side rounds landed separately: the visualization round (commit 52bcf00c: four animated schematics of the August-3 offensives on /prime-front, promoted and sandbox clearly separated, no new claims) and the prime-front explainer film (commit 76b092a0: a 2:39 German faceless-explainer video with burned-in captions, embedded on /prime-front; states “kein RH-Beweis” explicitly))

- **New modules (15).** v701\_big\_picture\_hunt.py (11), v702\_z1\_lookahead.py (13), v703\_chain\_tolerance\_scaling.py (10), v704\_chain\_mass\_law.py (5), v705\_chain\_deck\_sector.py (7), v706\_chain\_weyl\_mass.py (10), v707\_chain\_section\_edge.py (9), v708\_chain\_position\_functional.py (7), v709\_chain\_zero\_layer.py (10), v710\_chain\_two\_stage\_hecke.py (7), v711\_chain\_firstbirth\_scaling.py (7), v712\_rank3\_transverse\_deck.py (5), v713\_l1\_montage.py (16), v714\_moonshot\_hecke\_groupoid.py (19), v715\_qgeo\_car\_continuum.py (22) — 158 checks, each module promoted verbatim from its discovery probe with numbers frozen from the promotion run (v711: the worker-run point estimate  $\theta \sim 1.25/R^2 \sim 0.90$  vs the promotion run 1.851/0.692 documented; both far inside the preregistered bars).
- **Ledger.** 16 new rows (PRIME.KEYSTONE.01, PRIME.Z1LOOKAHEAD.01, PRIME.CHAINTOL.01, PRIME.CHAINMASS.01, QGEO.DECKSECTOR.01, PRIME.WEYLMASS.01, PRIME.SECTIONEDGE.01,

PRIME.POSFUNC.01, PRIME.ZEROLAYER.01, PRIME.TWOSTAGE.01, PRIME.FIRSTBIRTH.01, PRIME.RANK3TRANSV.01, PRIME.L1MONTAGE.01, PRIME.MOONSHOT.01, QGEO.CARLIMIT.01, plus the new contract PRIME.Z1.MOONSHOT.01); dated LII notes on PRIME.Z1.OPERATOR.01 (montage + moonshot; stays OPEN — the continuum theorems are missing), PRIME.RANK3DENSITY.01 (69/70,  $T^* = 1.6 \times 10^{14}$ ) and GATE.QGEO.01 (rates summable, lemma named — the gate does not move); 772  $\rightarrow$  788 rows.

- **Papers.** `tfpt_3`: 15 new audit blocks. `tfpt_research_contracts`: the PRIME.Z1.OPERATOR.01 update II (montage, moonshot, chain arc; formally OPEN), the new contract PRIME.Z1.MOONSHOT.01 (groupoid program; stage-2 gluing as the next gate; kill criteria K1–K4), the `prob:R1` update II (69/70), and the GATE.QGEO note. `tfpt_prime_front`: three new subsections (the chain arc S–A–S–G as an honest section narrative; the L1 montage and the moonshot; the T–B census at 69/70). `parity_toeplitz_classification`: the transverse-strike paragraph (69/70). `tfpt_1/2`: the Hecke-tower follow-up in the code sections (E8-unspecificity fence stated).
- **Website.** Prime-front feed entry (the third daily bundling); suite counters 694  $\rightarrow$  709; README and Zenodo description updated to 709 modules / v535–v715. (The two separate website rounds referenced above: 52bcf00c visualization, 76b092a0 explainer film.)
- **Scope.** No marker moves; the 69/70 closure is unconditional-MODULO-CITATIONS on the declared finite family (T–B itself stays the conjecture); PRIME.Z1.OPERATOR.01 stays OPEN (continuum theorems missing); v714 is honestly fenced (E8-unspecific finite counting facts — specificity is stage 2); GATE.QGEO does not move; no RH statement anywhere.

**2026-08-03 (LI)** (the afternoon bundling: **NINE** promotions (v692–v700, 140 new checks, all green) plus the **TWO** Lean modules committed — the T–B chain closes 60/70 windows and the Z1 series records its ground. **HIGHLIGHT 1** (the T–B chain, v692+v693): T–B(block) is **TYPED** (v692, 7/7:  $\hat{A}_2 =$  positive zero-Gram + psd rank-1 pole layer, exact per entry; the margin identity  $\det(G_Z + P) = \det G_Z + c_P(s_\perp^\top G_Z s_\perp)$  — the razor-thin margin **IS** the transverse zero mass, a **SUM OF SQUARES**; on-line zeros only help by  $2 \times 2$  psd superadditivity; “*E* kippt nicht” = a **QUANTIFIED PARTIAL-RH TAIL** statement, neither a 2c-envelope question nor a W3 renaming) and then **LARGELY CLOSED** (v693, 5/5: the explicit Ingham-form zero density arXiv:2507.15184 + Hasanalizade–Shen–Wong + halving + Platt–Trudgian built into a sharpened safe-side envelope — sinh difference pairing,  $V(0) = 0$ , on-line subtraction, PEN down  $\times 6.5$ –14.4: 60/70 complete windows close **UNCONDITIONALLY-MODULO-CITATIONS**; exact remainder  $9 \times T^* \approx 1$ – $3 \times 10^{13} + h = 5690$  at  $8.5 \times 10^{14}$ ; extension grows  $\sim e^{3.74a}$ ; Simonič honestly listed trivial-in-range). **HIGHLIGHT 2** (the Z1 series 5–5d, v695–v698, contract stays OPEN): the **MEASURE** comes from counting (v695, 25/25: seam+orbifold dead null-calibrated; the E8 counting route delivers the Weil measure **EXACTLY** — atoms 0.0, arch/pole via  $\Gamma$  duplication, 7/8 target spikes  $q = 0.000$ ; comb+pole positive-feasible, signedness = pole subtraction); the **CANONICAL OPERATOR** exists (v696, 18/18, z1-jacobi-opaque with the renaming clause **REFUTED**: CMV/Jacobi moments  $2 \times 10^{-13}$ ; features on prime-power slots  $q = 0.000$ , amplitude–mass link  $r = +0.76$ , positivity load-bearing 170/171 vs 2865); the **MASSES** transfer exactly (v697, 14/14, z1-uvarov-sequential-closed: atoms are lag insertions, duality proven;  $\Delta\alpha = w_1/E$  at  $5.6 \times 10^{-17}$ ; the **INVERTED** stabilization law — the  $\Gamma$  flow predicts the counting masses to  $\sim 10\%$ ); the **POSITIONS** are forced (v698, 15/15, z1-recursion-semi: windows 0.5–2 cells, jitter null 30/30, adversary –516; shooting recovers masses to 0.11% median; honest negatives: greedy saturation needs lookahead, residual noise-

like, E-transport recursive — the continuum reading is the remainder). **THE REST: v694** (20/20, closure-both-near-proof: retention closed with an  $O(k^2)$  certificate + straddle boundary case, separation law **FORM-CORRECTED** additive  $\alpha^* = C_{\text{cell}}/\delta + C''/\Delta\gamma$ ,  $C''' \leq 0.59$ , 0 exceptions in 97 tests; the two open parent configs detect at  $\alpha^* = 7.90$ ), **v699** (18/18, cone-dynamics-dead: prime updates are translations,  $\sim 99\%$  increments spacelike, no semigroup — the review-7.3 dynamic-cone route killed honestly; the leaf functional separates scramble/Epstein where det is blind), **v700** (18/18, orbit60-parked: all cascade routes dead, no involution of type  $2^{27}1^6$  in the line group of order 11520, Coxeter does not descend, no  $A_5$  signature). **PLUS LEAN ROUND 4: GaussianCodeBridge.lean + QuarticHalf.lean** committed (lake build green, 3385 jobs, kernel-checked, no sorry/native\_decide): the Gaussian-code-bridge algebraic core (SNF certificate,  $15 \times 16$  census with the zero class provably empty,  $\sigma$  semantics) and the quartic-half mechanism ( $\mu_4$ -orbit vanishing,  $\sum \langle \alpha, x \rangle^4 = 576q^2$  over the explicit 240 roots); Chevalley stays cited-not-formalized)

- **New modules (9).** v692\_rank3\_lockgram.py (7), v693\_rank3\_density\_close.py (5), v694\_interpolation\_lemma\_closure.py (20), v695\_z1\_trace\_operator.py (25), v696\_z1\_jacobi.py (18), v697\_z1\_uvarov.py (14), v698\_z1\_flow\_recursion.py (15), v699\_cone\_dynamics.py (18), v700\_orbit60\_census.py (18). All nine discovery probes committed alongside; the zero-comb caches reused repo-relative with local fallbacks.
- **Lean round 4.** Two new committed proof modules (imports added to TfftCarrier.lean; lake build green, 3385 jobs, no sorry, no native\_decide; lean\_manifest.sha256 regenerated WITH the new files): GaussianCodeBridge.lean (the v689 algebraic core:  $\mathbb{Z}[i]$ -module and SNF certificate with unimodular  $P, Q$  in both directions, the 240-root census  $15 \times 16$  with the zero class provably empty,  $\mu_4$  stability,  $\sigma$  as the order-3 family action with coordinate block  $= F_1 + F_2 + F_3$ ) and QuarticHalf.lean (the v690 provable part: the  $\mu_4$ -orbit factor  $1 + (-i)^d + (-1)^d + i^d = 0 \iff d \not\equiv 0 \pmod{4}$ , the per-pair weight law, the full vanishing half incl. the trivial extras, and the quartic power sum  $\sum_{\alpha} \langle \alpha, x \rangle^4 = 576 (\sum_i x_i^2)^2$  over the explicit 240 roots with holomorphic isotropy; the Chevalley basic-invariant step cited, not formalized).
- **Ledger.** 9 new rows (PRIME.RANK3LOCKGRAM.01, PRIME.RANK3DENSITY.01, PRIME.INTERPCLOSURE.01, PRIME.Z1MEASURE.01, PRIME.Z1JACOBI.01, PRIME.Z1UVAROV.01, PRIME.Z1FLOWREC.01, QGEO.CONEDYN.01, E8.ORBITLE.01); dated notes on PRIME.RANK3UNIF.01 (T-B typed and largely closed), PRIME.INTERP.01 (both blocks near-proof), PRIME.Z1.OPERATOR.01 (the 5–5d series status; stays OPEN), E8.GAUSSCODE.01/E8.QUARTICHALF.01 (formally verified in Lean; Chevalley cited-not-formalized).
- **Papers.** tfft\_3: 9 new audit blocks. tfft\_research\_contracts: the prob:R1 update inside the tenth slice (T-B typed + 60/70 closed with the exact  $T^*$  remainder list) and the PRIME.Z1.OPERATOR.01 series-5–5d update (measure from counting [E], canonical operator, masses+positions forced (measured); remaining: lookahead autonomy + continuum reading); Lean addenda on the two review contracts. tfft\_prime\_front: two new subsections (the T-B chain; the Z1 series). parity\_toeplitz\_classification: the T-B closure paragraph (v692–v693).
- **Website.** Prime-front feed entry (the afternoon bundling); suite counters  $685 \rightarrow 694$ ; README and Zenodo description updated to 694 modules / v700.
- **Scope.** No marker moves; the T-B closure is unconditional-MODULO-CITATIONS on 60/70 windows of the declared finite family (the remainder carries exact  $T^*$ ; extension beyond the family typed); PRIME.Z1.OPERATOR.01 stays OPEN (lookahead autonomy + continuum reading); the two kills (CONE-DYNAMICS-DEAD, ORBIT60-PARKED) are honest closures of review routes; no

RH claim anywhere.

**2026-08-03 (L)** (the great bundling: TEN promotions (v682–v691, 182 new checks, all green) — the five proof offensives of the day land in one round, and the master contract PRIME.UNIFPOS.01 is registered. **HIGHLIGHT 1** (v685, surface-closed, 8/8 — the theorem-level result):  $\det S > 0$  UNCONDITIONALLY on the entire declared window surface — the symbolic envelope  $K_{\text{env}}(a, h) \leq 0.9798 < 1$  for all 62 complete windows  $a \geq 3.434$ , the five below by finite certificates  $\kappa' \leq 0.108$ ; exact  $U_0$  rebooking  $M' = M + \Delta$ , inputs typed [E]/a-cited (Gram/Hutchinson  $\gamma_1$ ; Platt–Trudgian  $3 \times 10^{12}$ ); a SURFACE theorem on the declared family, NOT uniform in all  $a$  — T-B (the absorption margin) stays open, no RH statement. The chain below it: v683 (19/19, three functionals exact  $4.9 \times 10^{-16}$ , all entry-wise routes typed circular,  $\kappa_{\text{max}} = 0.0895$ ) and v684 (8/8, the zero side by the explicit formula — residuum  $\leq 6.9 \times 10^{-6}$ , 22 491 budget-certified zeros, unconditional  $\kappa_{\text{unc}} = 0.039\text{--}0.190 < 1$  class (a), the pretentious escape blocked  $\times 634$  by the classical zero-free strip). **HIGHLIGHT 2** (v691, hecke-sos-ihara-mechanism-extracted, 27/27): the target factorisation  $A = B^*B + P$  EXISTS EXACTLY in the Ihara lab ( $B$  by Chebyshev recursion,  $P \succeq 0 \iff$  Ramanujan), the deployed  $\zeta$  form is exactly the sine/defect HALF of the canonical split (index lemma), the Euler mechanism is measurable (resonance lattice exact 12/12, depth  $\times 1.98$ ), and the missing piece is named exactly: Z1, a Hilbert–Pólya-type operator (PRIME.Z1.OPERATOR.01, registered OPEN — offensive 5 running). **THE REST OF THE BUNDLE:** v682 (lk-split-dies, 20/20: the naive smooth  $L + K$  split is structurally dead — the pencil maximizer sits at  $\gamma_1 = 14.13$  found from primes alone, spike law  $\mu_{\text{max}} \approx 1 + 2a/\Omega$ ,  $\theta \sim 2.47a$ : the windows resolve individual zeros; one-sided/multi-resolution survive), v686 (geomsos-trace-exact-transport-open, 19/19:  $\Lambda(n)$  circle-free from  $E_8$  shell counting at  $7.1 \times 10^{-31}$ , cover-SOS canonical on dim 3, transport open at 91.4% top-3 mass; contract PRIME.GEOMSOS.01 with K1–K5), v687 (kernel-class-too-narrow, 18/18: the band-limited positive kernel class collapses exactly onto  $\{\lambda \text{Fejér}\}$ , class supremum  $0.0901684 < 0.10076$  — W2 stays closed via the v680/v681 route), v688 (interp-falsifier-constructive, 15/15: the in-kernel matched filter detects every off-line quadruple from  $2\alpha\delta \geq 1.974$ ,  $\times 2$  over the v677 mode map, masking adjudicated 48/48), v689 (gaussian-code-bridge-exact, 26/26:  $E_8(\mathbb{Z}[i])/(1+i) \cong \mathbb{F}_2^4$  by exact SNF, the  $240 = 15 \times 16$  census,  $\sigma$  as the RM(1,3) family permutation, must-fails fire), v690 (quartic-half-alive, 22/22: the vanishing half  $\{2, 14, 18, 30\}$  proven but honestly typed trivial ( $\mu_4$ -orbit factor); the substance:  $F_8/F_{12}/F_{20}/F_{24}$  algebraically independent G31 BASIC invariants, Chevalley cited, Jacobi carries 60/60 mirrors). **THE MASTER CONTRACT:** PRIME.UNIFPOS.01 (the Uniform Positivity Theorem — Suzuki’s localized Weil operator positive semidefinite for every  $a > 0$  from an operator/geometric representation defined without zeros and without RH-equivalent input; the machine-checkable intermediate form  $B_a = L_a + K_a$ ,  $|\langle K_a v, v \rangle| \leq \theta \langle L_a v, v \rangle$ , annotated with the v682 verdict; kill criteria: zero-defined  $C_a$  or RH-equivalent contractivity = renaming) as the tenth slice, plus the review contracts E8.GAUSSIAN.CODE.01/E8.QUARTIC.HALF.01/PRIME.W3.INTERPOLATION.01 and the OPEN PRIME.Z1.OPERATOR.01)

- **New modules (10).** v682\_lk\_split\_theta.py (20), v683\_rank3\_functionals.py (19), v684\_rank3\_zeroside.py (8), v685\_rank3\_uniformity.py (8), v686\_geometric\_sos.py (19), v687\_extremal\_kernel.py (18), v688\_interpolation\_detector.py (15), v689\_gaussian\_code\_bridge.py (26), v690\_quartic\_half.py (22),

v691\_hecke\_sos.py (27). All ten discovery probes committed alongside; the zero-comb caches (zero\_comb\_cache\_n2000.json, c1\_zero\_ext\_n2500.json, coverage\_hole\_zscan\_cache.npz) were already committed and are reused repo-relative with local fallbacks.

- **Ledger.** 15 new rows (PRIME.LKSPLIT.01, PRIME.RANK3FUNC.01, PRIME.RANK3ZERO.01, PRIME.RANK3UNIF.01, PRIME.GEOMSOS.01, PRIME.KERNELCLASS.01, PRIME.INTERP.01, E8.GAUSSCODE.01, E8.QUARTICHALF.01, PRIME.HECKESOS.01; contracts PRIME.UNIFPOS.01, E8.GAUSSIAN.CODE.01, E8.QUARTIC.HALF.01, PRIME.W3.INTERPOLATION.01, PRIME.Z1.OPERATOR.01); dated notes on PRIME.WEIL.OPERATOR.01, PRIME.W3STRUCT.01, PRIME.ZEROGAP.01, E8.CODESEM.01, E8.ST31.01.
- **Papers.** tfpt\_3: 10 new audit blocks. tfpt\_research\_contracts: the tenth slice (the master contract PRIME.UNIFPOS.01 with target statement, intermediate form, v682 annotation and kill criteria; the four review contracts registered). tfpt\_prime\_front: the great-bundling paragraph (rank-3 surface theorem, kernel collapse, falsifier, Hecke-SOS mechanism). parity\_toeplitz\_classification: the rank-3 chain paragraph (v683–v685). tfpt\_1/tfpt\_2: the Gaussian code bridge and the quartic half in the code sections.
- **Website.** Prime-front feed entry (the great bundling); suite counters 675 → 685; README and Zenodo description updated to 685 modules / v691.
- **Scope.** No marker moves; v685 is a SURFACE theorem on the declared window family, NOT a uniform-in- $a$  statement; T-B (the absorption margin, prob:R1) stays open; PRIME.Z1.OPERATOR.01 is registered OPEN (offensive 5 running); no RH claim anywhere.

**2026-08-03 (XLIX)** (the Monday-morning promotion round: **TWO** promotions (v680–v681, 24 new checks, all green) — **THE PINCH BREAKS AND THE HOLE CLOSES**: the pointwise W2 density map at the anchor window is now **GAPLESS**. **HIGHLIGHT 1** (v680, pinch-broken-split, 13/13): the v678 11.7% pinch was a **BOOKKEEPING ARTIFACT** of one-sided capture — centered capture (the same zero-gap theorem spent two-sided,  $\text{dist}(t, Z) \leq H_{\min}(t - 26)/2$ , machine-checked on 6595 comb grid points with 0 violations) **DOUBLES** the threshold to  $A_1 < 1/(2a_0) = 0.18034$ : the existing Bellotti–Wong constant 0.10076 passes with 79% headroom; and the Beurling–Selberg minorant **ELIMINATES**  $A_1$  from the counting chain entirely (mass loss exactly  $\pi/a_0$ , extremal per Logan/Littmann; explicit prime term  $P = 2.534$ , positive from  $t^* = 1.11 \times 10^7$ ; no Platt cap, every window incl.  $a \rightarrow \infty$ ; Weil identity on the comb at  $9.0 \times 10^{-5}$ ); the family reach  $[10, 870]$  is closed verification-backed ( $\min s_{\text{tot}} = 0.0063$  over all 5 windows  $\times$  lattice modes); the honest residue at that point: the  $O(1)$  coverage hole (2500,  $7.27 \times 10^6$ ) = 3.46 decades. **HIGHLIGHT 2** (v681, hole-closed-split-type, 11/11): the hole attacked on the three named paths and **CLOSED** with split typing — H3 as the main lever (the exact almost-periodic prime term with a hierarchical Lipschitz certificate moves the abstract entry from  $t^* = 1.11 \times 10^7$  down to  $t_x = 1.53 \times 10^5$ , a factor 72.5); the vectorized Riemann–Siegel scan with 223 949 budget-certified zeros (Gabcke remainder, correctness 603/603 against the certified strip, mpmath 12/12, found/expected 0.99774) carries  $(2515, 1.56 \times 10^5]$  with floor 0.01664; H1 honestly adjudicated (the Platt constant reproduces the hole boundary exactly — it cannot enter); **THE GAPLESS MAP**:  $s_{\text{tot}}(t; a_0) \geq 0.02259 \log(2 + t) - 0.5185$  for **ALL**  $t \geq 10$ , every region with named support and type (comb [E] / RS scan [E]-verification-consistent / exact-prime Weil chain abstract / Selberg uniform / Trudgian-2S unconditional from  $1.74 \times 10^{25}$ ); **THE W2 CHAIN AT THE ANCHOR COMPLETE**: density (v669) + frame-Gårding (v674/v678) + pointwise (v681);

## the calibration history documented with the pinned stage-1 duplicate bug)

- **New modules (2).** `v680_pinch_attack.py` (13), `v681_coverage_hole.py` (11). Both discovery probes committed alongside (`pinch_attack_probe.py`, `coverage_hole_probe.py`) together with the committed stage-1 scan cache `coverage_hole_zscan_cache.npz` (~1.2 MB; regenerated from scratch on a cache miss — a speed-up, not an input).
- **Ledger.** 2 new rows (`PRIME.PINCHBREAK.01`, `PRIME.HOLECLOSED.01`); dated notes on `PRIME.ZEROGAP.01` (the pinch is broken — the quantified target was a one-sided bookkeeping artifact; the residue moved to the coverage hole and was closed the same day), `PRIME.FEJERDENS.01` (the pointwise map at the anchor is gapless: the F3 obstruction resolved by two-sided bookkeeping + minorant replacement) and `PRIME.PACKETGARD.01` (the W2 chain at the anchor is complete: density + frame + pointwise, each with split typing); the ninth consolidated slice on `PRIME.WEIL.OPERATOR.01`.
- **Papers.** `tfpt_3`: 2 new audit blocks. `tfpt_research_contracts`: the ninth slice of `PRIME.WEIL.OPERATOR.01` (the W2 end-state after the pinch break and the hole closure: the pointwise density floor at the anchor window is a gapless split-type map — comb-certified / scan-certified / abstract-Weil / unconditional-asymptotic; remaining: deep family windows  $a > 4.43$  (Selberg-only entry), the formal Mosco writeup, the projection-norm form,  $a \rightarrow \infty$  (= W3/W4 territory); kill criteria unchanged; no RH statement).
- **Website.** Prime-front feed entry (the pinch-break round); suite counters 673  $\rightarrow$  675; sandbox counters 267  $\rightarrow$  269 probes and 5923  $\rightarrow$  5947 checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the pointwise map at the anchor is closed with MIXED typing — the abstract region is modulo the off-line window above  $3 \times 10^{12}$  until the unconditional Trudgian-2S entry; the deep family windows  $a > 4.43$ ,  $A_5(a)$ , the formal Mosco writeup and  $a \rightarrow \infty$  stay open; uniform W3 remains the conjecture); both verdicts are split-type by construction, the typing ledger is explicit; no RH statement anywhere.

**2026-08-02 (XLVIII)** (the sixth and final promotion round of the day: **TWO** promotions (v678–v679, 29 new checks, all green) — **THE DAY ENDS AT THE WALL WITH A DOOR HANDLE**: the last W2 gap now carries a concrete number (11.7% in one explicit constant), and the orbifold front reaches the continuum-convergence level. **HIGHLIGHT 1** (v678, zerogap-floor-only): the best explicit UNCONDITIONAL zero-gap theorem, documented and machine-verified — the  $S$ -difference route with  $S_{\text{bound}} = \min\{\text{Platt } |S| \leq 2.5167 \text{ to } 3.06 \times 10^{10}; \text{Trudgian 2014; Bellotti 2025 Cor. 1.5, asymptotic floor } 4\pi 0.10076 = 1.26619\}$ , verified on ALL 1999 Turing-certified comb gaps (0 violations on each chain separately, min air  $2.05\times$ , median  $5.24\times$ ); adaptive bands  $\delta_p = \kappa H_{\min}$  unconditionally hold  $\geq \kappa$  ordinates (0 misses)  $\Rightarrow$  the v674 FRAME packet floor is now UNCONDITIONALLY positive ( $0 < B_p \leq V_p$  everywhere, median discount 0.037) — the frame-Gårding chain is fully grounded in cited unconditional inputs; the projection form does NOT stabilize adaptively either (typed residual with inverted expectation, v642/v662/v674 pattern: 6/6 ( $\kappa, C$ ) cells miss,  $1/\log$  competitive, minimizer single-mode 0.898 — the pointwise obstruction is re-partition-invariant); **THE QUANTIFIED PINCH**: pointwise capture of a guaranteed zero needs the explicit  $S(t)$  constant  $A_1 < 1/(4a_0) = 0.09017$  — the best cited constant 0.10076 (Bellotti 2025) misses by 11.7% at every height: the last W2 gap as a concrete named target inequality at the frontier of explicit analytic number theory, independent of RH. **HIGHLIGHT 2** (v679, orb-os-continuum-slice, 19/19): the first continuum-OS slice of the seam orbifold B — correlator convergence with UNDERSTOOD rates (all six  $\sigma$ -channel observables at the uniform raw rate 0.659



= the Fisher–Hartwig  $2/3$  branch;  $\tau$ -channel contrast 1.298 vs  $4/3$ , the channel law  $\rho = 2(1 - \beta)^2 - 2\beta^2$ ; after one FH term the limits sit  $< 0.05\%$  at the CFT values; the  $\varepsilon$  channel exact); RP SURVIVES THE LIMIT ( $\lambda_{\min}$  extrapolations bounded away from zero:  $\sigma$   $1.36 \times 10^{-3}$  margin  $7\times$ ,  $\tau$   $2.80 \times 10^{-4}$  margin  $358\times$ , mixed  $1.34 \times 10^{-3}$  margin  $31\times$ ; must-fail  $\eta = -1$  flips); cluster exact in space AND Euclidean time ( $2\Delta = 0.222220$  vs  $2/9$  at  $8 \times 10^{-6}$ ; time gap = the  $\varepsilon$  level, cosh form, new time machinery); characters  $1/36, 1/9, 5/18, 4/9$  at  $10^{-12}$  with the measured  $N^{-2}$  rate — E0/E2/E3/E4 measured at the convergence level, E1 discrete, the formal-limit remainder named (GNS, uniform bounds); GATE.QGEO does not move)

- **New modules (2).** `v678_zero_gap_theorem.py` (10; the probe’s declared FAIL at the literal Z3.3 adaptive stabilization bar promoted as a typed-residual check with inverted expectation, `v642/v662/v674` pattern, numbers unchanged), `v679_orbifold_continuum_os.py` (19). Both discovery probes committed alongside (`zero_gap_theorem_probe.py`, `orbifold_continuum_os_probe.py`).
- **Ledger.** 2 new rows (`PRIME.ZEROGAP.01`, `QGEO.ORBOS.01`); dated notes on `PRIME.PACKETGARD.01` (the typed OPEN residue supplied: the frame floor is now unconditional), `PRIME.FEJERDENS.01` (the theorem family extended by the sharper Bellotti floor 1.26619), `PRIME.TURINGCERT.01` (all 1999 certified gaps verified against the explicit theorem) and `GATE.QGEO.01` (the continuum-OS data measured at the convergence level; the gate residual is now the formal limit construction + the `v622/v623` seam identification — NO marker move); the eighth consolidated slice on `PRIME.WEIL.OPERATOR.01`.
- **Papers.** `tfpt_3`: 2 new audit blocks. `tfpt_research_contracts`: the eighth slice of `PRIME.WEIL.OPERATOR.01` (the W2 end-state: frame-Gårding fully unconditionally grounded — zero-gap theorem + Fejér chain; the last gap carries a concrete number: the pointwise/projection form closes iff the explicit  $S(t)$  constant  $A_1$  improves from 0.10076 to  $< 0.09017$  (11.7%) — a named target at the frontier of explicit analytic number theory, independent of RH; kill criteria unchanged; no RH statement).
- **Companion note (no module).** The second citable short note `note_w3_detector_structure.pdf` (8 pages) stands ready uncommitted for the user: the W3 structure theorem (`v677`) + the two-lab validation (Epstein census, Ihara threshold map) + the  $C = 1$  quadrature mechanism (`v676`) as paper #2 of the prime-front line.
- **Website.** Prime-front feed entry (the zero-gap round); suite counters  $671 \rightarrow 673$ ; sandbox counters  $265 \rightarrow 267$  probes and  $5894 \rightarrow 5923$  checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the frame form is now unconditionally grounded, the pointwise A5(a) statement stays open with a quantified external target; uniform W3 remains the conjecture); `GATE.QGEO` stays open (convergence-level data, the formal limit named); the honest typed-residual FAIL (`ZEROGAP-FLOOR-ONLY`) is promoted as the finding, not hidden; no RH statement anywhere.

**2026-08-02 (XLVII)** (the fifth and final promotion round of the day: **FOUR** promotions (`v674–v677`, 76 new checks, all green) — **THE WALL IS SURVEYED**: two theorems, one frame-Gårding, and the honest typing “uniform W3 = the conjecture”. **HIGHLIGHT 1** (`v676`, `c1-quadrature-mechanism`):  $C = 1$  is a **DISCRETIZATION THEOREM** on the declared surface —  $q_r$  is the exact zero-side reading of the lock profile (Weil explicit formula on the complete comb, median residual  $1.1 \times 10^{-2}$  on 67 windows,  $q_{\text{pred}} > 0$  on 67/67), the  $h^{-1}$  is the DST normalization ( $q_r(N/2)/F_\alpha$   $h$ -flat at slope  $+0.04$ ), the constant is the zero-free RvM density integral (ratio 1.005; tertiles 0.66/0.43/0.37 predicted vs 0.61/0.45/0.39 measured), the lock sign via the `v591` pole killer; honest: the `v596` exponent  $-1.01$

was flip-contaminated (clean  $-1.28$ ; dated note on PRIME.LOCKPROJ.01), “norm  $\leq 1/h$ ” is a direction statement, and  $\sup \leq 1$  does NOT follow from the mean (the  $\sin^2$  sampling budget is the precise remainder, factor-2 headroom). **HIGHLIGHT 2 (v677, w3st-structure-theorem):** the equivalence chain as a THEOREM — odd-Toeplitz = Cantoni–Butler compression ( $1.5 \times 10^{-15}$ ), the sandwich theorem instead of naive DST diagonality (parity defect measured  $0.92..1.55$ ), the per-lag Weil dictionary unconditional with the master identity  $x^\top Ax = \sum T_x(\gamma) + P(x)$ ,  $T_x \geq 0$  on-line (alias comb) and cosh-amplified off-line terms ( $5.8 \times 10^{-16}$ ); the own Epstein winding census (exactly 12 off-line zeros,  $\zeta_K$  control 0) predicts the form break quantitatively ( $\lambda_{\min}$  ratio 0.803, factor-2 gate); the threshold map  $2\alpha s_{\min} \approx 1.4..2.6$  over 67 windows (the Ihara echo) — W3-on-the-family = theorem + detector certificate, UNIFORM W3 = the conjecture (Weil/Yoshida), typed as such. PLUS the frame-Gårding (v674, packet-average-only: the projection packet norm does NOT escape the  $1/\log$  drift — the central honest negative promoted as a typed-residual check with inverted expectation, v642/v662 pattern, minimizer single-mode tight 0.90 — BUT the frame inequality  $Q + C_0 G \succeq c_0 Y$  holds at every  $M$  with the v669 theorem constants,  $c_0 = 0.3058$ , margins  $+1.52.. + 2.29$ ,  $a$ -uniform, and the Mosco mechanism is numerically complete,  $8/8$ ; named remainder: within-packet equidistribution,  $\theta \rightarrow 0.20$ ) and the needle saturation (v675, needle-emergent-bulk: all four assembly candidates miss — jump set too dense  $p = 0.007$ , phase gradient wrong sign, lock rotation cleanly killed, pole weight significant but INVERTED — the needles are an emergent bulk property; W3 recommendation: MARGIN-regime form + Ihara calibration instead of a needle predicate))

- **New modules (4).** v674\_packet\_garding.py (21; the probe’s declared FAIL at the literal P2.1 stabilization bar promoted as a typed-residual check with inverted expectation, v642/v662 pattern, numbers unchanged), v675\_needle\_mechanism.py (21), v676\_c1\_mechanism.py (12), v677\_w3\_structure\_theorem.py (22). All four discovery probes committed alongside (packet\_garding\_probe.py, needle\_mechanism\_probe.py, c1\_mechanism\_probe.py with the committed c1\_zero\_ext\_n2500.json extension cache, w3\_structure\_theorem\_probe.py).
- **Ledger.** 4 new rows (PRIME.PACKETGARD.01, PRIME.NEEDLEMECH.01, PRIME.C1MECH.01, PRIME.W3STRUCT.01); dated notes on PRIME.LOCKPROJ.01 (the exponent correction:  $-1.01$  was flip-contaminated, clean  $-1.28$ ), PRIME.UNIFC.01 (the  $C = 1$  mechanism grounded), PRIME.GROUNDTRUTH.01 (the Ihara detection law echoed in the threshold map), PRIME.FEJERDENS.01 (the frame-Gårding connection) and PRIME.W3LAND.01 (the mechanism search saturates); the seventh consolidated slice on PRIME.WEIL.OPERATOR.01.
- **Papers.** tfpt\_3: 4 new audit blocks. tfpt\_research\_contracts: the seventh slice of PRIME.WEIL.OPERATOR.01 (W2 end-state: density plane closed, Gårding in frame form with theorem constants, Mosco complete, remainder = within-packet equidistribution; W3 contracted to “MARGIN-regime generic bound + needle risk map”,  $C = 1$  demystified-and-grounded as a quadrature theorem, the structure theorem types W3-on-the-family as theorem + certificate and uniform W3 as the conjecture; kill criteria unchanged).
- **Website.** Prime-front feed entry (the structure-theorem round); suite counters  $667 \rightarrow 671$ ; sandbox counters  $261 \rightarrow 265$  probes and  $5818 \rightarrow 5894$  checks; README and Zenodo description updated.
- **Scope.** No marker moves; W2/W3 stay open (the frame form holds with theorem constants, the pointwise A5(a) statement stays open; W3-on-the-family is a theorem + certificate, uniform W3 IS the conjecture — typed, no RH statement); the honest typed-residual FAIL (PACKET-AVERAGE-ONLY) and the fourfold needle miss (NEEDLE-EMERGENT-BULK) are promoted as

findings, not hidden; the  $C = 1$  statement carries no RH lever beyond the density and is typed as such.

**2026-08-02 (XLVI)** (the fourth promotion round of the day: **FIVE** promotions (v669–v673, 65 new checks, all green) — **THE W2 DENSITY PLANE CLOSES** (v669, the highlight of the round: the exact identity  $s_{\text{tot}} = 2\pi(F_a \star dN)$  via the Weil explicit formula on the tent test pair, verified  $2.9 \times 10^{-6}$  pointwise /  $7.2 \times 10^{-8}$   $t$ -averaged against the Turing-certified comb — the envelope peaks **ARE** the zeros ( $\gamma_{1..4}$  to 0.0002), the dips **ARE** the zero gaps — plus the **UNCONDITIONAL** chain RvM + Trudgian  $\Rightarrow \rho_{a,\delta} \geq 0.306 \log(2+t) - C_0$   $a$ -uniformly with 10/10 finite certificates on the zero-free grid, margins  $\geq 1.36$ ; the honest floor: the literal  $1/a$  and the plane-wave  $\pi/a$  widths sit **BELOW** the RvM threshold  $4\pi A_1 = 1.4074$ , the structural explanation of the Gårding drift — the remaining **W2** piece is the packet-to-point translation below the floor); **THE NEEDLE SEARCH NARROWS THREEFOLD** (v670 block deflation honestly **FAILED**: fidelity  $p_{90}$  only  $50.53 \rightarrow 49.35\%$  for  $K = 1 \rightarrow 8$ , leak median 1.000 everywhere — the coupling is bulk-spectral, not low-rank; v671 the Lehmer link is a clean **NULL**:  $h^3$ -partials all  $p > 0.1$ , placement control unremarkable — the needles are **NOT** zero-comb-driven; together with v667 not-in-the-BD-frame: the mechanism search moves frame-/assembly-side); and **THE LI DOUBLE** (v672 the **W1**  $\Rightarrow$  **Li** corollary:  $g_n = \frac{1}{2}e^{-|u|/2}L_{n-1}^{(1)}(|u|)$  derived exactly (Laguerre), the  $h = 1433$  window contains 30/32 finite **Li** coefficients at  $< 10\%$  form error ( $n = 1$  honestly 24%, a jump artifact;  $n = 5$  hair-thin 10.03%), pole cancellation to six digits, odd sector certified by  $\lambda_{\min} = +1.516 \times 10^{-3} > 0$ ; v673 the  $E_8 \leftrightarrow$  comb **Li** lock:  $\Lambda(E_4, s) = (2\pi)^{-s}\Gamma(s)\zeta(s)\zeta(s-3)$  completed with residues  $\pm 1/240$  — the  $E_8$  240 **IS** the residue normalizer —  $\lambda_n^L = 2\lambda_n^\zeta$  exact,  $E_8$ -arithmetic vs comb at budget usage 0.124, the naive single-map **Li** refused as non-tempered))

- **New modules (5).** v669\_fejer\_density.py (14), v670\_w3\_block\_deflation.py (18; verdict **BLOCK-DEFL-FAIL** promoted as the honest preregistered result), v671\_lehmer\_resonance.py (10; verdict **LEHMER-NULL**), v672\_li\_corollary.py (11), v673\_li\_e4.py (12). All five discovery probes committed alongside (fejer\_density\_bound\_probe.py, w3\_block\_deflation\_probe.py, lehmer\_resonance\_probe.py, li\_corollary\_probe.py, li\_e4\_probe.py).
- **Ledger.** 5 new rows (**PRIME.FEJERDENS.01**, **PRIME.BLOCKDEFL.01**, **PRIME.LEHMERNULL.01**, **PRIME.LICOROLLARY.01**, **PRIME.LIE4.01**); dated notes on **PRIME.GARDING.01/PRIME.GARDENV.01** (v669 explains the drift: per-mode control below the RvM floor is not certifiable at any height), **PRIME.W3LAND.01/PRIME.THETAPRED.01** (the threefold narrowing v670/v671), **PRIME.KEIPERLI.01** (the **Li** strand connects, v672/v673) and **PRIME.SHADOW.01** (the 240-residue identity + the **Li** lock); the sixth consolidated slice on **PRIME.WEIL.OPERATOR.01**.
- **Papers.** tfpt\_3: 5 new audit blocks. tfpt\_research\_contracts: the sixth slice of **PRIME.WEIL.OPERATOR.01** (**W2**: the density plane closed, remainder = the packet-to-point translation below the RvM floor; **W3**: narrowed threefold — not low-rank, not zero-comb-driven, not in the **BD** frame  $\Rightarrow$  frame-/assembly-side; new: the **Li** corollary and the  $E_4$  **Li** lock; kill criteria unchanged).
- **Website.** Prime-front feed entry (the Fejér-density round); suite counters  $662 \rightarrow 667$ ; sandbox counters  $256 \rightarrow 261$  probes and  $5753 \rightarrow 5818$  checks; **README** and **Zenodo** description updated.
- **Scope.** No marker moves; **W2/W3** stay open (the density plane closes at the wave-packet level only — the pointwise **A5(a)** Gårding inequality stays open); the honest **FAIL** verdict

(BLOCK-DEFL-FAIL) and the clean null (LEHMER-NULL) are promoted as findings, not hidden; positivity statements remain finite (30/32 Li coefficients,  $n \leq 32$ ), NOT Li's criterion; no RH statement.

**2026-08-02 (XLV)** (the closing bundle of the day: **NINETEEN** promotions in one pass (v650–v668, 301 new checks, all green) — the **ORBIFOLD FRONT COMPLETES AT THE LATTICE LEVEL** (modular data v650: nine  $Z[g, h]$  match  $|\theta/\eta|^2$ , S-covariance measured at a clean  $N^{-4}$  rate, T exact as the  $\mathbb{Z}_6$  Dehn shift, deck<sup>3</sup> =  $(-1)^F$  as exact  $\mathbb{Z}_6$  trace arithmetic; assembly v651:  $H^2(\mathbb{Z}_3, U(1)) = 0$  enumerated, of six candidates exactly  $B$  survives, T kills the naive  $\mathbb{Z}_3$  exactly; Arf hardening v652: the  $6^6$  classification leaves  $\{B, C3\}$ , the one-Fock-space pin forces  $B$ , the  $\mu_6$  ladder measures the C3-excised states directly; bond-defect theorem v653: twist = deck = bond defect as a lattice theorem in  $\mathbb{Q}(\zeta_6)$  with  $D_{\text{can}} = L^4$ ,  $D_{\text{can}}^3 = -1$ ); the ST31  $d/4$ -THEOREM (v654: every Springer-regular  $d$ -clock has  $x^{d/4} = \pm i \text{Id}$ , converse exact at  $d \in \{8, 20, 24\}$ , the unique  $d = 12$  exception is the compiler-clock class; “free  $\Rightarrow$  regular” killed); the W2/W3 SERIES (v655 Mosco preparation: resolvent Cauchy + uniform  $H_{\log}$ ; v656 margin bridge typed as lock identification; v657 the exact 2D formula  $r_{\text{id}} = (1 - q \tan^2 \theta)/(1 - \tan^2 \theta)$ , the deficit angle-driven; v658 the uniform bound typed FRAGILE ( $\alpha$ -fragility  $\pm 60\%$ ); v659 the landscape:  $\lambda_{\min} > 0$  on ALL 635 points, every peak a rotation artifact, MARGIN-regime share 0.0%; v660 the rotation predicate honestly FAILED (necessary, not sharp; 47/52 needles are diagonal crossings; successor block deflation); v661–v663 the Gårding arc: drift undecidable on 4 stages  $\rightarrow$  both remedies refuted, the drift lives in the total symbol  $\rightarrow$  THE ENVELOPE HOLDS measured  $a$ -uniformly,  $(c_0, C_0) \approx (0.021, 0.055)$ , “flat” was a  $t$ -range artifact, remaining task = a Fejér spectral-density bound); DISCIPLINE AND CONTROLS (v664 look-elsewhere: 42 slots / 19 hits / 6 kills, global  $p = 5.1 \times 10^{-3}$  conservative, NO single observation significant — re-types executed; v665 Keiper–Li on two independent routes, consistent at budget usage 0.124, injection detected from  $n = 1$ ; v666 Turing certificate: the comb complete below  $\gamma_{\max}$ , integrals at 5% of the Lehman bound; v667 Baez–Duarte control frame: Vasyunin  $1.7 \times 10^{-31}$ ,  $C_{\text{BD}}$  cross-anchor  $6.4 \times 10^{-8}$ , drift SHARED = intrinsic, needles NOT shared = frame-side; v668 ground truth + firewall, one module from both probes: true positivity has NO uniform margin,  $\delta(K) \rightarrow 0$  exactly, detection reach  $K^* \times s \approx 2\text{--}3$ , Fejér reads blind — and without the Euler product the machinery breaks by  $\sim 13$  orders of magnitude: THE central W3 recalibration); plus **LEAN ROUND 3** (three new certificate files, lake build green, 3383 jobs, no sorry) and the Nature-Comms primon-gas reference typed honestly in the prime-front paper)

- **New modules (19).** v650\_orbifold\_modular.py (17), v651\_orbifold\_assembly.py (17), v652\_orbifold\_arf.py (15), v653\_bond\_defect.py (18), v654\_st31\_degree8.py (30), v655\_w2\_mosco.py (7), v656\_margin\_link.py (11), v657\_rid\_alignment.py (15), v658\_w3\_uniform\_bound.py (11), v659\_w3\_landscape.py (14), v660\_theta\_predicate.py (20), v661\_garding.py (12), v662\_garding\_edgeband.py (16; the probe's declared FAIL at the literal symbol-stability bar promoted as a typed-residual check with inverted expectation, v642 pattern, numbers unchanged), v663\_garding\_envelope.py (18), v664\_look\_elsewhere.py (14), v665\_keiper\_li.py (13; shared zero cache zero\_comb\_cache\_n2000.json committed), v666\_turing\_cert.py (6), v667\_baez\_duarte.py (12), v668\_ground\_truth.py (35 = 22 + 13; ONE module merged verbatim from the two probes ihara\_ground\_truth\_probe.py + epstein\_firewall\_probe.py). All discovery probes committed alongside.

- **Ledger.** 19 new rows (QGEO.ORBMOD.01, QGEO.ORBASM.01, QGEO.ORBAREF.01, QGEO.BONDDEF.01, E8.ST31D8.01, PRIME.WEIL.MOSCO.01, PRIME.MARGINLINK.01, PRIME.RIDGEOM.01, PRIME.W3BOUND.01, PRIME.W3LAND.01, PRIME.THETAPRED.01, PRIME.GARDING.01, PRIME.GARDEDGE.01, PRIME.GARDENV.01, META.LOOKELSEWHERE.01, PRIME.KEIPERLI.01, PRIME.TURINGCERT.01, PRIME.BAEZDUARTE.01, PRIME.GROUNDTRUTH.01); dated notes on QGEO.TWISTOPE.01, QGEO.KLEINRP.01 and GATE.QGEO.01 (the orbifold front is complete at the lattice level; the [C] remainder is the continuum CFT statement + the seam identification — *no marker moves*), on E8.ST31.01/E8.ST31DEG.01 (v654 + Lean round 3), on EXTREV.LATTICE.01 ( $6 = p_2(a)$ : observation without statistical weight,  $p = 0.169$ ), E8.CODESEM.01 (number battery  $p = 0.593$  — the v646 finding quantified) and E8.CODE.01 (14/16 typed FORCED + the Lean code certificate), and the consolidated W2/W3 note on PRIME.WEIL.OPERATOR.01.
- **Lean round 3.** Three new committed proof modules, lake build green (3383 jobs, no sorry): `G31WordOrders.lean` ( $\chi_w \mid X^{24} - 1$  and  $\chi_u \mid X^{20} - 1$  via conjugate-quartic witnesses  $\chi\bar{\chi} = \Phi_{24}/\Phi_{20}$ , no proper divisor by companion-matrix kernel `decide` — all eigenvalues primitive), `HammingCode.lean` (self-duality, weight distribution  $\{0:1, 4:14, 8:1\}$ , coset-leader distribution  $\{0:1, 1:8, 2:7\}$ , covering radius 2), `SquareParity.lean` (the combinatorial kernel of the  $c = w^2$  parity kill at the multiset level, the permutation-level bridge honestly typed as not formalized).
- **Papers.** `tfpt_3`: 19 new audit blocks. `tfpt_research_contracts`: the fifth slice of PRIME.WEIL.OPERATOR.01 — the consolidated W2/W3 state (W2: envelope measured valid, Mosco prepared, drift intrinsic, remainder = Fejér spectral-density bound + symbol proof; W3: reduced to lock sign +  $q \tan^2 \theta \leq 1 - \delta$ , landscape mapped without positivity loss, ground-truth recalibrated — shrinking margins are the expected behavior of true positivity, the alarm signal is only  $\lambda_{\min} < -\text{floor}$ ; kill criteria unchanged). `tfpt_prime_front`: the closing-bundle recalibration paragraph + the external-context note on Wei/Zhai/Lu et al., “The Riemann Hypothesis manifested in dynamical quantum phase transitions” (Nat. Commun. 2026, DOI 10.1038/s41467-026-74935-8) — their primon-gas energies  $\log n$  with weights  $\Lambda(n)/\sqrt{n}$  are literally the TFPT atom table and the DQPT times are the zero comb; typed honestly as complementary verification, not RH progress.
- **Website.** Prime-front feed entry (the closing bundle); suite counters 643  $\rightarrow$  662; sandbox counters 236  $\rightarrow$  256 probes and 5452  $\rightarrow$  5753 checks; README and Zenodo description updated.
- **Scope.** No marker moves anywhere; GATE.QGEO does not move (the continuum statement stays open); W2/W3 stay open (measured/typed progress only); the two honest FAIL verdicts (THETA-PRED-FAIL, EDGEBAND-UNHEALED) are promoted as findings, not hidden; no positivity claim beyond the measured surfaces, no RH statement.

**2026-08-02 (XLIV)** (the discipline audit: the suite measures its OWN scientific process discipline and freezes it as checks (v649, META.DISCIPLINE.01) — ledger census 715 rows / 710 active with 214 documented kills / honest negatives / retypes, must-fail census 643 modules / 6040 static `check()` sites / 1469 negative-control keyword occurrences across 233 modules, preregistration census 47 contract rows (29 naming kill criteria before execution; executed chains FTR.PGL2.01 v533 $\rightarrow$ v632 and PRIME.WEIL.OPERATOR.01 K1–K3 $\rightarrow$ v643 verified in the ledger), deterministic replay 12/12 on a declared sample spanning v55–v648, five classical anchors recomputed against the literature (Jacobi  $r_4$ , Construction A  $\rightarrow E_8$ , Shephard–Todd G31 degrees, Suzuki archimedean constants, completed-zeta functional equation), 54 committed Lean proof modules — and the honest limit typed [C]: process discipline, not physics truth, the open gates stay open; PLUS the public mirror: a new /method page (“the verification framework”) built from a new

## generated data surface `website/lib/discipline.ts`)

- **New module.** `v649_discipline_audit.py` (18 checks,  $\sim 200$ s; discovery probe `discipline_audit_probe.py` 17/17, verdict DISCIPLINE-MEASURED): a read-only meta-module that parses the suite’s own files — ledger, module sources, contracts paper, Lean tree — and certifies the measured discipline metrics as frozen minimum bars. D1 ledger census (715 rows, 710 active, display classes E/C/O/X/Axiom/other = 169/222/282/37/2/3 at freeze; 214 rows with negative-evidence keywords: kill, ill-posed, honest negative, must-fail, retyped, superseded, dead, bingo, numerology); D2 must-fail census (643 modules, every one with checks; 6040 static `check()` call sites, clearing the published 5452 counter; 1469 must-fail/kill/scramble occurrences in 233 modules; typing marks  $[E] \times 3579$  vs  $[C] \times 1323$ ); D3 preregistration census (47 contract rows, 29 naming kill criteria; executed chains machine-verified: FTR.PGL2.01 preregistered `v533`  $\rightarrow$  executed `v632`, PRIME.WEIL.OPERATOR.01 kill tests K1–K3  $\rightarrow$  W1 theorem-closed `v643`, REG.FREEZE.01 frozen 2026-06-09); D4 reproducibility (deterministic replay 12/12 on the declared sample `v55`, `v91`, `v108`, `v205`, `v305`, `v405`, `v505`, `v555`, `v600`, `v634`, `v643`, `v648` — two fresh subprocess runs each, identical up to timing lines; 396 sympy-exact modules, 77 with explicit seeds); D5 external anchors (five classical recomputes against the literature: Jacobi 1834 four-square counts exact  $n = 1.40$ , Construction A  $[8, 4, 4] \rightarrow E_8$  with 240 minimal vectors, Shephard–Todd G31 degrees (8, 12, 20, 24), Suzuki  $\psi(\frac{1}{4})/L/A$  to 25+ digits, completed-zeta functional equation at  $3 \times 10^{-31}$ ); D6 Lean layer (54 committed `TfptCarrier` modules via `git ls-files`; `lake build` honestly *not* re-run in-suite); D7 the limit, typed  $[C]$  (process discipline, not physics truth; 278 class-O rows stay open incl. SEAM.THEOREM.01 and PRIME.WEIL.OPERATOR.01 W2–W4). Freeze discipline: every bar is the value measured at freeze time; a future census regression fails the module and forces an honest dated re-freeze.
- **Ledger.** New row META.DISCIPLINE.01 (canonical status DISCIPLINE-CERTIFIED); no other row changes, no marker moves anywhere.
- **Website.** New generated data surface `website/lib/discipline.ts` written by the new `verification/make_discipline_stats.py` (stdlib-only, hooked into `bash build.sh gen`; freshness enforced by `audit_sync.py` section A.generated). New page `/method` (“The verification framework”): the sandbox  $\rightarrow$  probe-with-kill-criteria  $\rightarrow$  suite  $\rightarrow$  ledger  $\rightarrow$  generated-mirrors pipeline as a static SVG; the measured D1–D6 numbers rendered from `discipline.ts`; the “buried coincidences” graveyard ( $|G31| = |W(D_5)| \cdot |W(A_3)|$  killed three ways `v634`; the bytecode  $p_n$  reading 0/11 `v646`;  $c = w^2$  dead by parity theorem `v647`; the fine-invariant  $\leftrightarrow C = 1$  bridge dead after  $h$ -control `v637`; the naive  $48 \times 5$  incidence `v629`; the ill-posed B-test `v624`; the same-day Lerch erratum `v643`); “what would falsify this” (the open contracts’ kill criteria); the E/C/O/X status typology with the measured distribution; and a sober AI-transparency paragraph (agents execute, the deterministic suite decides — `python3 verification/run_all.py` replays everything). Navigation entry, a compact “how we keep ourselves honest” band on the home page, and the `/method` link added to the prime-front honesty banner; suite counters 642  $\rightarrow$  643 on the affected pages, README and Zenodo description.
- **Scope.** The meta-module measures *process* discipline, not the truth of any physics claim; it is read-only and moves no marker; the `/method` page renders only generated or machine-checked numbers — no new claims anywhere.

**2026-08-02 (XLIII)** (the second round of the day: **THE LERCH ERRATUM** (honest, corrected the same day) plus six promotions (`v643`–`v648`) — the W1 chain (`v631`, `v640`–`v642`) had read Suzuki’s eq. (1.3) with Lerch coefficient  $-1$ ; the paper’s own §2.2 data lock  $+1/4$  (machine evidence `v643` C0.1), every measured number transfers VERBATIM via the exact identity  $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4 c_{\text{gal}}^{\text{sm}}(g)$ , and only the labels change: Suzuki’s smooth layer is  $+\rho$ , the true dictionary is the

SINGLE scalar  $+1/D$  on both layers (sign-compatible with positivity), and the origin constant vanishes,  $\kappa = 0$  exactly; ON THE CORRECTED READING: W1 becomes a MEASURE-LEVEL THEOREM (the projection lemma closes the  $L_0^2$  remainder — Suzuki’s mean-zero condition is automatic on the  $u$ -side;  $A_{\text{arch}} = -g''_{\text{smooth}}$  exactly at every lag,  $3.4 \times 10^{-52}$ ; form equality  $1.28 \times 10^{-10}$ ), W2 STARTS in honest slices (classical FEM density at rate; Rayleigh–Ritz monotone from above on all six nested sequences;  $\lambda_a = 0^+$  within  $\sim 10^{-9}$ , consistent with Thm 1.3, no sign statement; Mosco remainder typed), the  $\mathbb{Z}_3$  TWIST SECTOR IS REFLECTION POSITIVE under the parafermionic Klein-twist pairing (mirror  $\times$  charge conjugation,  $\omega^{kq}$  phases,  $\eta = (1, 1)$  unique among 36;  $N$ -stable 48..384; full mixed OS Gram PSD — the named v639 open item resolved at the lattice OS level), the RM(1,3) REVERSE READING is honestly two-part (anchor decomposition  $(0, 1, 1, 2)/\text{norm } 6$  exact with the  $\sigma$  action selecting the anchor sector; the bytecode  $p$ -sequence has NO code reading — the bingo is buried), the ST31 DEGREE ROUND anchors the  $\mu_4$  center as a POWER of both regular clocks ( $w^6 = \pm J$  with  $\chi_w = x^4 + ix^2 - 1$  exactly primitive  $\zeta_{24}$ ;  $u^5 = \pm J$  central;  $u^4$  free with census  $\{5:48\}$ ,  $48 = 240/5$ ) and KILLS  $c = w^2$  by a parity theorem, and the W3 SIGN-UNCERTAINTY TOOLBOX is typed: a real 25-digit dictionary to the critical strip (phase slope  $= L = \log \pi - \psi(\frac{1}{4})$ , the W1  $\delta_0$  weight) whose mass lever dies at  $d = 1$  ( $e^{-E^*} = 0.967$  where  $< \frac{1}{2}$  is needed;  $d \geq 21$  required) — while the W3 surface itself is empirically positive on all 67 complete windows ( $\min \lambda_{\min} = +8.26 \times 10^{-4}$ ; the incomplete combs are exactly the v618 flip set  $\{1219, 1292, 1445\}$ ))

- **The erratum (first, honest, dated).** The v631/v640–v642 chain read the Hurwitz–Lerch block of Suzuki’s screw function (eq. 1.3) with coefficient  $-1$ ; the correct coefficient is  $+1/4$ , locked by the paper’s own §2.2 data ( $A = \frac{1}{2}(\log 2\pi - \psi(2)) = 0.70754637$  reproduced only with  $+1/4$ ;  $g(0) = 0$ ; machine evidence v643 C0.1). All chain identities are correct identities of the kernel  $\tilde{g} = g - \frac{5}{4}$  Lerch, and all measured numbers transfer verbatim via  $c_{\text{gal}}^{\text{sm}}(\tilde{g}) = -4 c_{\text{gal}}^{\text{sm}}(g)$  — no check flips. Corrected labels: Suzuki’s smooth layer  $= +\rho$ ; the two-layer dictionary ( $-4D$  smooth,  $D^2$  atomic) collapses to the *one* scalar  $+1/D$ ;  $\kappa = 0$  exactly (the  $-\frac{5}{4}L\delta_0$  term of v640 was an artifact of the misreading). Documented as dated CONVENTION ERRATUM blocks in the four modules (all re-run green, numbers unchanged), erratum notes on the five PRIME.WEIL.\* ledger rows (CONTACT/DICT/BOUNDARY/PORTABLE/MATRIX), erratum sentences in the tfpt\_3 audit blocks, the contracts paragraph and the prime-front theory-open section, and the website mirrors (feed, W1 dictionary map, honesty banner).
- **New modules.** v643\_w1\_theorem.py (11 checks: convention lock/erratum evidence, projection lemma,  $A_{\text{arch}} = -g''_{\text{smooth}}$  exact,  $\kappa = 0$ , derived  $D^8$  moment brackets, form equality  $1.28 \times 10^{-10}$ , pole rank 1); v644\_w2\_form\_density.py (7 checks: classical FEM density at rate, Rayleigh–Ritz monotone on nested spaces,  $\lambda_a$  scan, typed Mosco/norm-resolvent remainder); v645\_klein\_rp.py (17 checks: dressed-string reality, charged Klein blocks PSD  $N$ -stable,  $\eta = (1, 1)$  unique of 36, must-fail controls, full mixed OS Gram); v646\_rm13\_reverse.py (20 checks: exact anchor identity, generic/specific split, bytecode negative, selective kill control); v647\_st31\_degree24.py (25 checks: regular 24/20 clocks with central powers, the parity-theorem kill of  $c = w^2$ ,  $u^4$  census  $\{5:48\}$ ); v648\_sign\_uncertainty.py (12 checks: 25-digit dictionary anchors, the  $d = 1$  mass-budget death, surface positivity 67/67, flip-set reproduction, typed verdict).
- **Ledger.** New rows PRIME.WEIL.THEOREM.01, PRIME.WEIL.W2.01, QGEO.KLEINRP.01, E8.RM13REV.01, E8.ST31DEG.01, PRIME.SIGNUNC.01; dated erratum notes on PRIME.WEIL.CONTACT.01, PRIME.WEIL.DICT.01, PRIME.WEIL.BOUNDARY.01, PRIME.WEIL.PORTABLE.01, PRIME.WEIL.MATRIX.01; dated notes on PRIME.WEIL.OPERATOR.01

(contract state: W1 theorem-closed, W2 started, W3 open with tool diagnosis, W4 unchanged), QGEO.TWISTOPE.01 (open item resolved), E8.CODESEM.01, E8.ST31.01, PRIME.UNIFC.01 (independent flip-set confirmation,  $\text{corr} = +0.704$ ).

- **Scope.** No marker moves; the erratum changes labels, not numbers (the transfer identity is exact and machine-checked); W1 closes as a theorem at the window level, W2 is started (not closed), W3 stays open (the sign-uncertainty route is closed as a typed negative — the toolbox needs a large parameter the  $\zeta$  strip does not have), W4 unchanged; GATE.QGEO does not move; no positivity claim beyond the measured surface, no RH statement.

**2026-08-02 (XLII)** (website: the home page catches up with the 2026-08 arcs — a new “geometric realization” section (the seam axioms force the  $\mu_3$ -cover  $y^3 = x^4 - 1$ ,  $N = 3$  pinned, the physical seam is the conformal seam, the covered seam and its  $\mathbb{Z}_3$  orbifold; static SVG map seam  $\rightarrow$  curve  $\rightarrow$  E8 with the 60-line/ST31 quotient) and a new “code and primes” section (E8 as an error-correcting code with the RM(1, 3) bit dictionary; primes as the shadow of the finished geometry; the W1 Weil-operator contact with its honest W2/W3 scope — no RH claim); claim-stack honesty updates)

- **Website only, no theory change.** New home sections GeometryArc (mirrors v617, v620, v622, v623, v628, v629, v633, v634, v639) and CodePrimesBand (mirrors v625, v626, v638 and v630, v631, v640–v642), placed between the claim stack and the honesty band; every card carries its display marker and the open items are stated in place.
- **Claim stack.** The declared-inputs layer records the conditional v624 reduction (three selectors force  $d = 4$ , so the two inputs reduce to  $d = 4$  plus  $\pi$  under the named architecture); the numerical layer now names the honest  $\sin^2 \theta_{13}$  stress ( $\approx 2.0\sigma$ , the largest single pull) instead of leaving it to the predictions page.
- **No marker moves; no new claims.** All content mirrors existing ledger rows; the bedrock (GATE.QGEO) and the RH-hard steps W2/W3 are stated open where the new sections mention them.

**2026-08-02 (XLI)** (the ten-front batch lands (v632–v642 promoted) — THE  $\text{PGL}_2$  CONTRACT EXECUTED: the four transfer solvers share ONE Möbius action on the disc  $-7$  norm line where they act at all (Koide reading exactly intertwined with the base translation,  $g_{\text{QCD}}$  exactly conjugate to it, Boltzmann = pole by the single-flow theorem; kill criterion NOT met); THE  $\mathbb{Z}[i]$ -E8 QUOTIENT AND ITS STABILIZER: the  $\mu_4$ -quotient of the 240 roots IS the hermitian-unimodular  $\mathbb{Z}[i]$ -E8 with the 60-hyperplane system of ST31,  $G_{31}$  is the FULL unitary stabilizer with Molien-unique degrees (8, 12, 20, 24) and the exact clock identities  $\sigma = c^4$ ,  $J = c^9$  — while the numerology  $|G_{31}| = |W(D_5)| \times |W(A_3)|$  is KILLED three ways (real structure:  $(Z_4 \circ 2^{1+4}).\text{Sp}_4(2)$ ); the compiler clock is NOT the regular 12-element); P CANONICAL AND CONSTRUCTED: the Lorentz congruence matrix is the unique minimal-Frobenius operator-compatible class in the full census AND, mod sign, the Frobenius-minimal integer congruence transporting the  $C_V$  null frame onto the canonical rank-one rays (finite 16-family); an HONEST NEGATIVE: after  $h$ -control the fine Hodge invariants carry NO independent information about the  $C = 1$  margin ( $\rho = -0.04$ ,  $p = 0.75$  — the direct window-level bridge route closes); THE CODE DICTIONARY: the unique equivariant Hamming placement is Reed–Muller RM(1, 3) on AG(3, 2) with information bits one per  $\mu_4$  pair, the  $\sigma$  3-cycle on the syndrome flag, and decode = projection verbatim (3840/3840); THE TWIST OPE: the interacting  $\mathbb{Z}_3$  orbifold slice stands



at the abelian vertex level ( $h_\sigma = 1/36$  three ways, exponent  $2\Delta = 2/9$ , crossing exactly symmetry-protected, closed four-point form; RP validated in the real class, the  $\mathbb{Z}_3$  Klein-twist pairing typed open); and THE W1 CLOSURE TRIO: the boundary equation  $A_{\text{arch}} = \frac{1}{4}g''_{\text{smooth}} - \frac{5}{4}(\log \pi - \psi(\frac{1}{4}))\delta_0$  EXACT with the derived  $1+1/(6d^2)$  moment law, the frozen dictionary PORTABLE on three fresh windows (atom constant  $D^2$  at machine precision), and the full-matrix operator level CLOSED (block norm  $2.7 \times 10^{-3}$  derived,  $6.3 \times 10^{-6}$  after measure-level re-binning; pole block rank 1; two typed lattice residuals) — plus Lean certificates for the W1 dictionary identities, the Lorentz congruence, and the  $G_{31}$  order identities)

- **New modules.** `v632_ftransfer_pgl2.py` (25 checks: substrate, discrete intertwining, native  $\text{PGL}_2$  classification, five must-fail controls); `v633_orbit60_quotient.py` (17 checks: hermitian-unimodular  $\mathbb{Z}[i]$ -E8,  $\sigma$  census  $19 \times 3 + 3$ , line group 11520, cascade killed); `v634_st31_structure.py` (56 checks: full stabilizer count, Molien uniqueness, Springer regulars, the numerology kill,  $\sigma = c^4/J = c^9$ , honest clock typing); `v635_p_canonicity.py` (18 checks: the 40-solution census, minimal-Frobenius class, fine-invariant separation); `v636_p_construction.py` (25 checks: null-cone census, boost normal form, the finite 16-family, must-fail  $\text{gl}_2$  ansatz); `v637_fine_c1_bridge.py` (8 checks: the preregistered honest negative); `v638_code_semantics.py` (27 checks:  $\text{RM}(1,3)$  dictionary, syndrome flag, decode = projection, the 30-census kill control); `v639_twist_ope.py` (25 checks: three weight routes, two-point, crossing, closed four-point, OPE data, differentiated RP); `v640_w1_boundary.py` (11 checks: the exact boundary equation, three boundary constants, the derived moment law); `v641_w1_portability.py` (10 checks: the frozen-dictionary kill test on  $h = 285/540/997$ ); `v642_w1_matrix.py` (8 checks: operator-level closure, pole-block rank 1, the two typed lattice residuals as inverted-expectation checks, numbers unchanged from the probe).
- **Lean.** Three new certificate files in `experiments/lean4-carrier-rigidity` (lake build green, no sorry): `WeilDictionary.lean` (the exact v631 W1-dictionary identities: Lerch collapse, structure theorem), `LorentzCongruence.lean` ( $P^\top J_{\text{det}} P = J_{\text{fix}}$  by kernel decide), `G31Orders.lean` (the  $G_{31}$ /glue order identities  $8 \cdot 12 \cdot 20 \cdot 24 = 46080 = 1920 \cdot 24$ ,  $60 = 7 + 11 + 19 + 23$ ).
- **Ledger.** New rows `FTR.PGL2.01`, `E8.ZIQUOTIENT.01`, `E8.ST31.01`, `PRIME.PCANON.01`, `PRIME.PCONSTRUCT.01`, `PRIME.FINEC1.01`, `E8.CODESEM.01`, `QGEO.TWISTOPE.01`, `PRIME.WEIL.BOUNDARY.01`, `PRIME.WEIL.PORTABLE.01`, `PRIME.WEIL.MATRIX.01`; dated notes on `PRIME.WEIL.DICT.01`, `PRIME.WEIL.OPERATOR.01`, `E8.INCIDENT.01`, `E8.CODE.01`, `QGEO.ORBCAS.01`, `PRIME.HODGECON.01`, `FTR.DISC7.NORM.01`, `EXTREV.LATTICE.01`.
- **Scope.** No marker moves; the honest negatives (cascade route, fine-invariant bridge,  $\mathbb{Z}_3$  mirror pairing,  $\text{gl}_2$  ansatz) are typed results that close routes, not gaps; W1 is theorem-capable at the window level with the  $L_0^2$  projection the last remainder; W2/W3 stay open and unmoved; no positivity claim, no RH statement; `GATE.QGEO` does not move.

**2026-08-02 (XL)** (the W1 dictionary lands (v631 promoted) — the v630 non-scalar residual IS the zeta pole term: the Lerch collapse  $[e^{-t/2}\Phi(e^{-2t}, 2, \frac{1}{4})]'' = 4e^{-t/2}/(1 - e^{-2t})$  (exact) yields the structure theorem  $g'' = -2 \cosh(t/2) - 4 \times$  (the TFPT arch density) — Suzuki's smooth layer is  $-4 \times$  the window density MINUS the pole block (TFPT's separate v591 rank-one); the closed per-lag ratio law  $-D[4 + (e^t + 1)(1 - e^{-2t})]$  verifies, after pole subtraction the smooth conversion is the SCALAR  $-4D$  (monotone to 1.0006), and the atom-layer constant is  $D^2$  exactly: W1 of `PRIME.WEIL.OPERATOR.01` closes at the measured level on both halves —  $\hat{A}_{\text{arch}} = -(1/4D) \text{Galerkin}_{\text{smooth}} + \text{pole rank-one}$ ; the theorem-level remainder and

the RH-hard steps W2/W3 stay typed open)

- **New module.** `v631_w1_dictionary.py` (7 checks: the exact Lerch collapse, the structure theorem, the route certificate, the closed ratio law, the scalar  $-4D$  closure, the  $D^2$  atom constant, the typed remainder).
- **Ledger.** New row `PRIME.WEIL.DICT.01`; resolution note on `PRIME.WEIL.CONTACT.01`.
- **Scope.** The implication map stays explicit:  $W1$  measured-closed  $\rightarrow$   $W2$  (open)  $\rightarrow$   $W3$  (open, RH-hard)  $\rightarrow$   $W4$  (classical given  $W2+W3$ ); closing  $W1$  does not move  $W3$ ; no positivity claim, no RH statement.

**2026-08-02 (XXXIX) (the six-front batch lands (v626–v630 promoted) — THE E8 CODE: Construction A on the self-dual extended Hamming code  $[8, 4, 4]$  yields E8 exactly (shells  $240/2160$ ; every single-bit error exhaustively correctable: “E8 is an error-correcting code” is a THEOREM); THE HODGE CHAMBER: all 67 complete windows land in the POSITIVE cone of the cover polarization lattice on ONE sheet (the measured half of the geometric positivity route; honest typing: chamber membership is density-layer, the fine  $C = 1$  arithmetic lives inside); THE ORBIFOLD CASIMIR: the covered seam’s deck classes  $\{1/6, 1/2, 5/6\}$  carry exactly the  $\mathbb{Z}_3$ -twist Casimir data (exact closed form, sector table  $(1/72, -1/24, 1/72)$ , twist gap  $1/18$ ); THE INCIDENCE CENSUS: the naive  $48 \times 5$  root incidence is KILLED ( $19 \times 12 + 3 \times 4$ ) with the sharp positive residue  $60 = D_{\text{start}}$  as the free clock-orbit count on the roots; and THE SUZUKI FIRST CONTACT: the TFPT atom table IS Suzuki’s prime measure literally (positions  $\log n$ , weights  $\Lambda(n)/\sqrt{n}$ , exact) —  $W1$ ’s atomic half CLOSED, the smooth conversion measured non-scalar (the preregistered residual, now with data))**

- **New modules.** `v626_e8_code.py` (7 checks: Hamming  $[8, 4, 4]$  self-dual, Construction A even unimodular = E8, shells, exhaustive syndrome decoding, typed compiler ties  $8/4/16/14$ ); `v627_hodge_chamber.py` (5 checks: exact transport  $z = P^{-1}y$ ,  $\det S > 0$  on  $67/67$ , one sheet, honest scramble typing); `v628_orbifold_casimir.py` (5 checks: exact closed form, exact  $1/N$  expansion  $\pi(6\nu^2 - 6\nu + 1)/12$ , sector table, twist gap  $1/18$ ); `v629_root_incidence.py` (6 checks: 240-root model, the kill,  $60 = D_{\text{start}}$  free clock orbits, 12  $\sigma$ -fixed roots, controls); `v630_suzuki_contact.py` (4 checks: the literal atom identity, tent structure, the measured non-scalar Galerkin profile, contract typing).
- **Ledger.** New rows `E8.CODE.01`, `PRIME.HODGECONE.01`, `QGeo.ORBCAS.01`, `E8.INCIDENCE.01`, `PRIME.WEIL.CONTACT.01`.
- **Scope.** No marker moves; no positivity claim beyond the measured surface; no RH statement; the interacting orbifold statement and the  $W1$  normalization dictionary stay the named residues.

**2026-08-02 (XXXVIII) (the prime-shadow audit lands (v625 promoted): the checkable core of “primes are the shadow of the finished geometry” is EXACT —  $\Theta_{E_8} = E_4$  with shell counts  $240 \sigma_3(n)$  (multiplicative over coprime addresses: the ‘address space’ reading IS unique factorization), the primes act as COMMUTING Hecke channels ( $T_p \Theta = (1+p^3)\Theta$  for  $p = 2, 3, 5, 7$ ;  $[T_2, T_3] = 0$ ) with the compiler’s theta as SIMULTANEOUS eigenvector, and  $L(E_4, s) = \zeta(s)\zeta(s-3)$ : the zeta function is the factorized shadow of the  $E_8$  counting function — primes enter AFTER the geometry; the speculative framings (eigenfrequencies, RH as coherence) stay typed hypotheses)**

- `v625_prime_shadow.py` (`PRIME.SHADOW.01`, 6 checks; probe 6/6, `PRIME-SHADOW-EXACT`). Theta from the glue decomposition  $(\theta_2^8 + \theta_3^8 + \theta_4^8)/2$ ; shells  $240\sigma_3(n)$  exact for

$n = 1..12$ ; multiplicativity census with the non-coprime must-fail ( $\sigma_3(4) = 73 \neq 81$ ); Hecke eigenvalues  $1 + p^3$  and commuting channels;  $L(E_4, 6) = \zeta(6)\zeta(3)$  to  $5 \cdot 10^{-8}$ . Honest scope: classical Jacobi/Hecke facts realized verbatim by the compiler's own objects — the direction of explanation (geometry first, primes as readout) fixed within the narrative.

- **Sync.** Ledger row + note on EXTREV.LATTICE.01; tfpt\_3 (audit block); diary; suite 618  $\rightarrow$  619 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-02 (XXXVII) (the external-review lattice audit lands (v624 promoted) — THE FOUR-MARK CLOSURE: under the  $D_{d+1} \oplus A_{d-1}$  glue architecture three independent selectors force  $d = 4$  (disc-group census, the norm equation  $(d+1)/4 + (d-1)/d = 2$ , the Pythagorean  $(3, 4, 5)$  triple of cycles/marks/carrier) — the two-input presentation reduces to ONE boundary degree  $d = 4$  plus  $\pi$ , conditional on the architecture; THE ANCHOR BYTECODE ( $\chi = (t-1)^2(t-2)$ , two-state recursion,  $240+8 = 248$ ); THE LORENTZ CONGRUENCE  $P^\top J_{\det} P = J_{\text{fix}}$  EXACTLY (index 6) — the prime-front determinant form and the cover polarization lattice are the SAME rational quadratic form; the review's  $B$ -test retyped ill-posed ( $B$  complex symmetric); and the RH architecture preregistered as PRIME.WEIL.OPERATOR.01 with verified Suzuki citations)**

- **v624\_external\_lattice\_audit.py** (EXTREV.LATTICE.01, 14 checks; probe 14/14, LATTICE-AUDIT-PASSED). Disc-group census with negative controls  $d = 3, 5, 8$ ; unique norm-equation solution (no other even target  $2k$ ,  $k \leq 10$ ); Pythagoras at  $d = 4$ ;  $(1, 1, 2)$  unique partition;  $c_3 = 1/(8\pi)$ ; bytecode identities exact; the congruence with  $\det P = -6$ ,  $36 = 6^2$ ,  $6 = p_2(a)$ ;  $B = M^\dagger J$  complex symmetric ( $\det 8$ ) — the literal integer- $R$  test cannot be posed; third  $\mathbb{Q}(\omega)$  ratio an exact square. AX.P2.01 keeps its typing.
- **PRIME.WEIL.OPERATOR.01 (new research contract).** The W1–W4 chain (exact Galerkin identification with Suzuki's localized Weil operator  $A_a$ ; form density; uniform discrete positivity — the RH-hard step, with the  $C = 1$  contraction reading as candidate mechanism; continuum passage via Yoshida's odd criterion), kill tests K1–K3, citations verified by web lookup (arXiv:2606.09096, arXiv:2607.24830). No RH statement.
- **Sync.** Ledger rows EXTREV.LATTICE.01 + PRIME.WEIL.OPERATOR.01 + notes on AX.P2.01/QGEO.EQORDER.01; tfpt\_research\_contracts (contract block); tfpt\_3 (audit block); diary; suite 617  $\rightarrow$  618 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXVI) (the covered seam lands (v623 promoted) — the lattice skeleton of the covering-level identification: the seam lifted to the  $\mu_3$ -cover is ONE 48-site NS circle for the uniform weights (the alternating pair DISCONNECTS it — a new independent kill), the lifted clock has EXACT ORDER 12 with  $L^4 = \text{deck}$  ( $r^4 = \omega$  as lattice combinatorics), the v622 kernel identity generalizes verbatim to  $N = 48$ , and the deck-character mode grid is the 12-grid ( $\{1/6, 1/2, 5/6\}$ ) with the untwisted sector equal to the base seam VERBATIM and the twisted pair carrying  $\omega$  on bilinears (the  $t = \omega/\bar{\omega}$  sheets); the fermionic deck lift satisfies  $\text{deck}^3 = -1$  (the spin double))**

- **v623\_covered\_seam.py** (QGEO.COVERSEAM.01, 9 checks; probe 9/9, COVERED-SEAM-SKELETON). Connectivity census ( $j = 1, 2$ : one circle; trivial and alternating: 3 components); the explicit 48-walk (three base turns, 12 crossings);  $L^4 = \text{deck}$  and  $L^{12} = \text{id}$  on all 48 points with proper order 12; the  $N = 48$  mode-sum identity; three 16-mode deck classes with the offset-1/2 class = the base NS frequencies exactly;  $\text{deck}^3 = -1$  on modes, bilinear eigenvalue

=  $\omega$ ; the 12 lifted marks in a single 12-cycle. Residue named: the interacting/CFT orbifold statement (twist-field OPE).

- **Sync.** Ledger row + note on QGEO.SEAMID.01; `tfpt_3` (audit block); diary; suite 616  $\rightarrow$  617 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXV) (the seam identification lands (v622 promoted) — THE PHYSICAL SEAM IS THE CONFORMAL SEAM: the v519 chiral seam kernel is EXACTLY the discrete NS (antiperiodic) chiral mode sum on the 16-site circle, including the even-distance zeros (the flat band IS the NS spectrum; the Ramond sum fails as must-fail control); the four mark-bond midpoints sit EXACTLY at  $\mu_4$  under the site map  $\theta_a = (2a + 1)\pi/16$  (no site touches a mark — the v519 “marks at bond midpoints” becomes literal geometry); the group dictionary is exact (clock =  $z \mapsto iz$ , RP reflections = diameter reflections); and the v534 STRADDLE LAW becomes geometry — straddled cuts = exactly the through-mark axes; the second v617 bedrock residue closes at the kernel + dictionary level, leaving the covering-level identification)**

- **v622\_seam\_identification.py** (QGEO.SEAMID.01, 10 checks; probe 10/10, SEAM-IDENTIFIED). Mode-sum identity for all  $d = 1..15$  with the closed form  $\sum_{j < 8} \sin((2j+1)x) = \sin^2(8x)/\sin(x)$ ; Ramond must-fail; mark midpoints =  $\mu_4$  exact; reflection axes  $(k+1)\pi/16$  with cuts  $\mapsto \{\pi/4, \pi/2, 3\pi/4, \pi\}$ ; through-mark axes = the straddled cuts  $\{7, 15\}$ ; v617’s bond reflection = cut  $k = 11$  with the v599  $(k, 5-k)$  pattern.
- **Sync.** Ledger row + notes on QGEO.SEAMFORCE.01 and GATE.QGEO.01 (bedrock state after v617/v620/v622; no status move); `tfpt_3` (audit block); diary; suite 615  $\rightarrow$  616 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXIV) (the follow-through round (v619–v621 promoted): the two  $C = 1$  violators are TRUNCATION ARTIFACTS of the prime-data cap (flip set = truncation set exactly, mechanism reproduced by injection at both scales,  $C = 1$  EXCEPTION-FREE on the complete-comb surface); the cyclic- $N$  census makes the bedrock residue CONDITIONAL (seam-admissible primitive-Lorentz orders =  $\{3, 5\}$  exactly,  $N = 6$  reduces to  $N = 3$ , and the compiler’s machine-checked 27/81 pin  $N = 3$  uniquely); and the interacting Klein–Landau semigroup CONTRACTS to the  $\mu_4$ -commensurate steps with the clock surviving as a positive transfer over the whole coupling grid)**

- **v619\_flip\_mechanics.py** (PRIME.FLIPMECH.01, 6 checks; probe 6/6, FLIPS-ARE-TRUNCATION). Atom demand  $u \leq 2\alpha$  vs data cap  $U_{\max} = 12.899$ : exactly the two v618 flip windows are truncated ( $\Delta u = 0.677/0.089$ ); no zero resonance (collective spectrum); injection reproduces both scales ( $-5e-5/ -1e-4$  vs observed  $-1.0e-4; -0.18/-0.22$  vs  $-0.29$ );  $C = 1$  with ZERO exceptions on the 67 complete windows (max 0.982). The flip windows are retired as data-boundary artifacts.
- **v620\_cyclic\_n\_census.py** (QGEO.NCENSUS.01, 9 checks; probe 9/9, N3-PINNED). Full  $\infty$ -ramification iff  $\gcd(N, 4) = 1$ ; genus table 1, 3, 3, 6, 7; primitive sheet dims; Chevalley–Weil signatures ( $N = 3$ : both sheets Lorentz;  $N = 6$ : no primitive Lorentz sheet – reduces to  $N = 3$ ); the pinning  $\det_{\mathbb{Z}}(1 - \text{deck}) = N^3$ : 27 (the v597 constant) vs 125; saturation 81 vs 625:  $N = 3$  unique.
- **v621\_interacting\_semigroup.py** (WOIT.GAMMA.SEMI.01, 6 checks; probe 6/6, CLOCK-SURVIVES-INTERACTING). Free v524 pattern reproduced at  $N = 16$  (even steps PD, odd steps zero one-particle diagonal); at  $g > 0$  Hermiticity survives exactly on  $\{4, 6, 7\}$  (the  $\mu_4$ -commensurate window; mechanism: the survivor is  $\alpha_4$ -invariant exactly, breaking translation

invariance to the quartet stabilizer); the clock step stays PD over  $g \in \{1/32..8\}$ : the reconstructed rotation group of the interacting toy is the clock tower.

- **Sync.** Three ledger rows + notes on PRIME.UNIFC.01, QGEO.SEAMFORCE.01, WOIT.GAMMA.TOY.01; tfpt\_3 (three audit blocks); tfpt\_prime\_front (flip-mechanism sentence); diary; suite 612  $\rightarrow$  615 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXIII) (the three-front round (v616–v618 promoted): the internal SU(2) SEPARATES kinematically (kill test (6)’s first slice —  $\rho_{\text{tw}} = \text{right-}j$  verbatim, the clock factorizes as left- $u$ o right- $u$  exactly, the deck is spacetime-side, and the complex choice breaks the internal factor to  $U(1)$  with the mark torsor as its  $\mu_4$ ); the conformal seam axioms FORCE the  $\mu_3$ -cover ( $\mathbb{Z}_4$ -Möbius rigidity + uniform weights + genus 3 + the seam reflections matching v599’s pattern — the bedrock residue narrows to the cyclic-3 choice and the physical $\leftrightarrow$ conformal identification); and the equidistribution constant  $C = 1$  is FROZEN on the declared surface with the sharpened sign-flip dichotomy (violator set = flip set exactly; honest correction of the XXIII note))**

- **v616\_su2\_internal\_kinematic.py** (WOIT.SU2.KIN.01, 17 checks; probe 17/17, SU2-KINEMATIC-SEPARATED). Right-sp(1) fiber-preserving, commutes exactly with the spacetime quaternion action, intersection  $\{0\}$ ; RHO4 = left- $e^{i\pi/4}$ o right- $e^{i\pi/4}$  exact; DECK4 = left- $i$ ; internal  $SU(2) \rightarrow U(1)$  by the complex choice;  $\sigma_{\text{std}}$  fixes right- $j$  and inverts the internal charge. Kill test (6)’s kinematic branch does not fire; the dynamical half stays live on  $\mathcal{A}_{\text{hol}}$ .
- **v617\_seam\_cover\_forcing.py** (QGEO.SEAMFORCE.01, 14 checks; probe 14/14, SEAM-FORCES-COVER). Order-4 Möbius multiplier forced primitive; harmonic cross-ratio orbit  $\{2, 1/2, -1\}$  (must-fail:  $4/3$  not in it); weight census  $81 \rightarrow 4 \rightarrow 2$  (mark equivalence kills the alternating pair);  $y^3 = x^4 - 1$  exact; Riemann–Hurwitz genus 3; bond reflection = the v599  $(k, 5-k)$  pattern with the two v519 cut points.
- **v618\_uniform\_constant.py** (PRIME.UNIFC.01, 6 checks; probe 6/6, UNIFORM-C1). 69 windows,  $q_{\text{model}}$  one-signed on the whole ladder;  $\varepsilon h \leq 0.982$  on all 67 lock-sign windows ( $C = 1$ ); tertiles 0.61/0.45/0.39; exactly two  $q_{\text{real}}$  sign-flip windows (1219, 1445) = the violator set exactly; scramble breaks by  $> 10^4$ . Honest correction: the XXIII note’s mechanism (“ $q_{\text{model}}$  crossing”) was wrong and the edge flip 1445 was missed.
- **Sync.** Three ledger rows + notes on WOIT.GAMMA.TOY.01, QGEO.COVER.01, PRIME.LOCKPROJ.01; tfpt\_3 (three audit blocks); tfpt\_research\_contracts (kill-test-(6) slice note); tfpt\_prime\_front ( $C=1$  frozen-module note); diary; suite 609  $\rightarrow$  612 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXII) (the interacting gamma slice lands at toy level (v615 promoted): the gamma battery executed ON the RP-surviving interaction class (the alignment bit  $\delta = \pi/2$ , positive coupling, v534) — THE INTERACTING MIRROR FLIP: at the forced twist  $\eta = +i$  the mirror state’s odd sector is STRICTLY NEGATIVE DEFINITE on every cut and every coupling  $g \in \{1/32..8\}$  (bounded away from zero: kill test (3) cannot fire on the surviving class — v608 upgraded from  $g = 0$  to the full grid); the mark transport exists interacting (kill test (7) does not fire); the odd transfer spectrum stays multiplicity-free; and the dead member has no RP home — the dynamical bit selection and the chirality data CO-LOCATE on the same member)**

- **v615\_gamma\_toy\_interacting.py** (WOIT.GAMMA.TOY.01, 11 checks; probe 11/11, GAMMA-TOY-LANDED). Chiral/mirror parents ( $I \pm iC$ , gap 1); v534 survivor anchor reproduced; sector-exact mirror flip (even (29, 0, 0) PD, odd (0, 8, 0) ND, max odd

eigenvalue  $\leq -1.5 \cdot 10^{-3}$ ;  $\eta = -i$  convention control; mark transport  $\alpha_4$  exact 16/16 with Gram and transport form Hermitian PD at every  $g$ ; odd transfer spectrum simple at every  $g$  (honest finding: strong-coupling near-doubling at  $g = 8$  without closing;  $g = 0$  top 1.641438 = numerically the v524 clock top); dead-member control ( $m=2, s=+$ ) indefinite. Kill tests (3)/(6)/(7) stay formally live on  $\mathcal{A}_{\text{hol}}$ ; kill test (6) untouched.

- **Sync.** Ledger row + note on WOIT.GAMMA.FREE.01; `tfpt_research_contracts` ( $\gamma$ -stage note); `tfpt_3` (audit block); diary; suite 608  $\rightarrow$  609 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXXI) (the Gauss–Manin transport round lands (v614 promoted) — THE DECK STEP IS THE BOUNDARY MONODROMY: transporting the segment cycles along the rigid quarter-rotation loop, the discriminant  $\lambda(\tau) = e^{2\pi i \tau}$  winds ONCE around 0 and the transported periods factorize EXACTLY,  $I_m(\tau) = e^{i\pi\tau(m+1)/2} e^{-2\pi i \tau j/3} I_m(0)$ ; at  $\tau = 1$  the deck factor is  $\omega = t^4$  on EVERY sheet (the exponent  $4 = |\mu_4|$  forced by the puncture count); the canonical transport matrix in the segment basis is EXACTLY  $\omega M$  — the v613 dictionary  $r \sim \omega M$  is CANONICAL, no conjugator freedom left, and  $(\omega M)^4 = \omega$  re-derives  $r^4 = \omega$ ; v613’s residue F1 closed at the transport level)**

- **v614\_transport\_deck.py** (QGEO.TRANSPORT.01, 12 checks; probe 12/12, TRANSPORT-CANONICAL). Exact rotation substitution identity; uniform deck  $\omega$  across all three omega-sheet rows (must-fail:  $t^3 \neq \omega$ ,  $t^5 \neq \omega$ );  $\text{char}(\text{transport}) = \text{char}(r)$  exact; 25-digit branch-tracked certificates at  $\tau = 1/3, 1/2, 2/3, 1$  (dps 80, max  $5 \cdot 10^{-28}$ ;  $\tau = 1$  at  $2.7 \cdot 10^{-81}$ ). The reading: the braid fixes the boundary, the rigid rotation does not — their homological difference is one boundary loop of local-system monodromy.
- **Sync.** Ledger row + note on QGEO.PERNORM.01; `tfpt_3` (audit block); diary; suite 607  $\rightarrow$  608 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXX) (the canonical period normalization lands (v613 promoted) — THE ANALYTIC BRIDGE CLOSES: the quarter rotation acts DECK-FREE on the four branch segments (all 12 wrap factors exactly 1), and the exact dictionary is  $r = \text{deck} \circ \text{rotation}$  —  $\text{char}(r) = \text{char}(\omega M)$  EXACTLY, explaining  $r^4 = \omega$  structurally, with the deck power unique; the period rows (hol  $dx/y^2$ , hol  $x dx/y^2$ , antihol  $\overline{dx/y}$ ) realize the rotation eigenframe; the Hodge weights are explicit Gamma monomials ( $h_1 = (3\sqrt{3}/16\pi)\Gamma(1/3)^2\Gamma(1/6)^2$ , complex Beta formula); and  $J = -h$  PER CHARACTER: the compiler polarization is MINUS the analytic Hodge form — Riemann–Hodge positivity realized, v612’s named step B5 executed)**

- **v613\_canonical\_periods.py** (QGEO.PERNORM.01, 19 checks; probe 19/19, PERIOD-NORMALIZED). Deck-free rotation on segments (40-digit certificates) with the must-fail control (raw period rows are NOT  $r$ -eigenvectors, residual 0.42 — the naive ansatz falsified before the honest construction);  $\text{char}(r) = \text{char}(\omega M)$  exact with explicit conjugator  $G$ ; character match antihol  $\leftrightarrow \zeta_{12}$ ; omega-sheet signature (2, 1) with the unique negative on the antiholomorphic line =  $J$ ’s unique positive line (v612): all three character-matched sign products negative. Residue named: the canonical fork-basis identification (basepoint bookkeeping for  $G$ ).
- **Sync.** Ledger row + notes on QGEO.POLFRAME.01 and QGEO.PERIODS.01; `tfpt_3` (audit block); diary; suite 606  $\rightarrow$  607 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIX) (the polarization frame lands (v612 promoted): the braid rotation has three simple eigenvalues  $\zeta_{12} \cdot \{1, -1, -i\}$ , the polarization  $J$  diagonalizes**

**EXACTLY** in the rotation eigenbasis (values  $(1 + \sqrt{3})/2$ ,  $(1 - \sqrt{3})/2$ ,  $-1$ ), and **THE UNIQUE POSITIVE (HODGE) DIRECTION IS THE PRIMITIVE  $\zeta_{12}$  EIGENSPACE** — the analytic Hodge structure and the finite rotation agree on where positivity lives; the canonical period normalization (Beta weights) is the named next step)

- **v612\_polarization\_frame.py** (QGE0.POLFRAME.01, 9 checks; probe 9/9, **POLARIZATION-FRAMED**). Simple spectrum,  $\det r = -1$ ; exact diagonalization (honest typing: diagonality follows from J-unitarity, the content is the values and the signature distribution); product of diagonal values  $= \det J / |\det W|^2 = 1/2$  exactly.
- **Sync.** Ledger row + note on QGE0.PERIODS.01; tfpt\_3 (audit block); diary; suite 605  $\rightarrow$  606 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVIII) (the periods round lands (v611 promoted): the analytic layer of the cover is CLASSICAL** — all five basic period integrals are exact Beta values with Gamma denominators on the  $\zeta_{12}$  grid ( $\Gamma(11/12), \Gamma(7/12), \Gamma(5/12), \Gamma(1/12)$ ); the  $\mu_4$  rotation acts on periods with the exact characters  $i^{k+1}$ ;  $\Omega_E = B(1/3, 1/6)/2$  exactly; and the cross-factor period ratio admits no low-degree integer relation (PSLQ null) — the two CM factors are period-independent, as the v610 splitting demands)

- **v611\_periods.py** (QGE0.PERIODS.01, 6 checks; probe 6/6, **PERIODS-CLASSICAL**). Five symbolic Beta identities; three 40-digit rotation-character certificates; the classical  $\omega$ -CM period; the PSLQ-null cross-factor consistency; the period dictionary named as the Hodge/RP resource.
- **Sync.** Ledger row + note on QGE0.LANDSCAPE.01; tfpt\_3 (audit block); diary; suite 604  $\rightarrow$  605 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVII) (the curve landscape lands (v610 promoted): the Jacobian SPLITS into a  $\mathbb{Q}(\zeta_{12})$ -CM abelian surface** — the field of deck AND marks together — times the  $\mathbb{Q}(\omega)$ -CM elliptic quotient  $E : y^3 = u^2 - 1$ , point-count-confirmed ( $a_p(C) = a_p(E)$  exactly for all ten tested  $p \not\equiv 1 \pmod{12}$ ); the integer-spectrum census honestly relativizes single-char hits (7894 integer entries — single spectra are cheap; the anchor class is the most frequent nontrivial family at 286; the compiler's distinction is the joint structure); two new canonical period resources named)

- **v610\_curve\_landscape.py** (QGE0.LANDSCAPE.01, 9 checks; probe 9/9, **LANDSCAPE-MAPPED**). Differential characters  $(12, 12, 6)$ ; splitting confirmed at 14 primes (ten exact zero-differences; surface traces  $(-8, 4, -20, -32)$  at  $p \equiv 1 \pmod{12}$ );  $a_p = 0$  for  $p \equiv 2 \pmod{3}$ ; census: anchor class 286 = max nontrivial,  $\{1, 1, 3\}$  at 254; v598/v599 char hits retroactively confirmed as correctly typed candidates.
- **Sync.** Ledger row + notes on QGE0.COVER.01/QGE0.DICT.01; tfpt\_3 (audit block); diary; suite 603  $\rightarrow$  604 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXVI) (the translator factorization lands (v609 promoted): the geometric identity of  $c$  is CLOSED** — the mirror intertwiner  $d$  ( $dA = \Gamma(A)d$ ) is unique, unimodular ( $\det d = 1$ ),  $\mathbb{Z}[\omega]$ -integral and  $\Gamma$ -involutive ( $\Gamma(d)d = 1$  exactly), and the v605 element factorizes exactly as  $c = \lambda K_1 d$  with  $K_1 = J^{-1} \overline{F_2}^T M^{-1}$ :

$\theta_{\text{corpus}} = \text{Jadj} \circ \text{inner}(K_1) \circ \text{inner}(d)$  — all three v605 residuals are now resolved (seam reversal v603, zero-mode rule v607,  $c$ -identity v609))

- **v609\_translator\_factorization.py** (QGEO.CFACTOR.01, 8 checks; probe 8/8, C-FACTORIZATION-LANDED). Intertwiner space 1-dim;  $\det d = 1$ ;  $d \in M_3(\mathbb{Z}[\omega])$ ;  $\Gamma(d)d = 1$ ;  $c = \lambda K_1 d$  exact ( $\lambda = 1/7 - 3\sqrt{3}i/7$ );  $\text{inner}(K_1 d)$  reproduces  $\varphi$  on the pair; honest scope:  $F_2$  carries the corpus datum by definition; the v606 arithmetic negative explains why the cyclotomic route had to fail.
- **Sync.** Ledger row + notes on QGEO.TRANSLATOR.01/QGEO.MODESEP.01; tfpt\_3 (audit block); diary; suite 602  $\rightarrow$  603 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXV)** (the free gamma slice lands (v608 promoted) — the first executed step of the WOIT  $\gamma$  milestone: mirror modes are STRICTLY OS-negative sector-exactly (every mirror fermion vector has negative OS norm — no mirror survives reconstruction, kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly  $\{1, \sqrt{2} - 1\}$ ), and the mark transport exists ( $\tau_2$  is the mark-A $\rightarrow$ B transport, PD — kill test (7) does not fire free);  $\gamma$  itself stays open on  $\mathcal{A}_{\text{hol}}$ ; side finding:  $14c$  is unipotent mod  $(1 - \omega)$ )

- **v608\_gamma\_free\_slice.py** (WOIT.GAMMA.FREE.01, 9 checks; probe 9/9, GAMMA-FREE-SLICE-LANDED; the v524 machinery imported read-only). Chiral  $(16, 0, 0)$ ; anti-chiral  $(8, 8, 0)$  with even  $(8, 0, 0)$  PD and odd  $(0, 8, 0)$  strictly ND; clock spectrum  $\{1, \sqrt{2} - 1\}$  multiplicity-free on the fermion domain; mark algebras PD  $(4, 0, 0)$  with  $\dim 4 = |\mu_4|$ , products generate the full 16-dim basis,  $\alpha_2$  transports mark A to mark B.
- **Sync.** Ledger row + notes on WOIT.OS.TWISTOR.01/WOIT.BETA2.OS.01; tfpt\_research\_contracts ( $\gamma$ -stage free-slice note); tfpt\_3 (audit block); diary (incl. the  $c$ -unipotency side finding); suite 601  $\rightarrow$  602 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIV)** (the selection rule lands (v607 promoted): the infinity torsion is a genuine  $\mathbb{F}_3$  space, the deck rotation acts with char  $(x + 1)(x^2 + 1) \bmod 3$  — EXACTLY ONE projective eigenline, and it IS the top mode; the three modes carry mutually exclusive canonical labels (seam =  $R$ -real, top = deck-eigen, zero = the only unlabeled) and  $S_0$  flips exactly the labeled pair: BOTH v590 demands expressed through cover structure (typed reading; the gate residual is the continuum derivation))

- **v607\_selection\_rule.py** (QGEO.SELRULE.01, 10 checks; probe 10/10, SELECTION-RULE-LANDED).  $\omega$ -multiplication acts as the identity on the torsion; the induced  $r$ -action is  $[[0, 1, 0], [1, 0, 2], [2, 2, 2]] \bmod 3$  with only fixed class 0;  $x^2 + 1$  irreducible over  $\mathbb{F}_3$  gives structural uniqueness of the eigenline ( $\lambda = -1$ ), census-confirmed; the top-mode class is that eigenline, the zero mode and the seam are not projectively fixed (nor under  $r^2$ ).
- **Sync.** Ledger row + notes on QGEO.MODESEP.01/QGEO.INVOL.01; tfpt\_3 (audit block); diary; suite 600  $\rightarrow$  601 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXIII)** (the mode-separation round lands (v606 promoted — the suite reaches 600 scripts): the infinity torsion SEPARATES the three  $V$ -eigenlines (all pure 3-torsion at infinity, classes pairwise distinct — the selection carrier the real structure could not provide; the canonical “+1” rule stays open); the arithmetic-identity route for  $c$  is closed (cubic field disc  $-783$ , Galois  $S_3$ , not abelian); the equidistribution conjecture is posed with the explicit constant  $C = 1$



in the prime-front paper; and the contract survey executes its bookkeeping (the disc-7 preregistration's discrete half marked executed via v571/v575/v578; the WOIT  $\gamma$  milestone documented as the designed next step))

- **v606\_mode\_separation.py** (QGEO.MODESEP.01, 7 checks; probe 7/7, MODES-SEPARATED). Unique intertwiner (rank 3, 1-param); all three eigenline lifts pure torsion; fixed-frame signatures  $(2, 0, 2)/(2, 2, 0)/(1, 2, 0)$ , pairwise distinct, none trivial;  $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$  irreducible with discriminant  $-783 = -3^3 \cdot 29$  non-square.
- **Sync.** Ledger row + notes on QGEO.TRANSLATOR.01/QGEO.SEAMMARKS.01 and the cross-reference on FTR.DISC7.NORM.01; **tfpt\_prime\_front** (the constant-1 statement in **sec:theory-open**); **tfpt\_3** (audit block); diary; suite 599  $\rightarrow$  600 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XXII)** (the audit-and-measurement close of the day: the gate audit documents the cover program as **MODEL-COMPLETE** on the operator half (ledger note on GATE.QGEO.01, no status move — the residual is the conformal seam identification, the zero-mode selection, and the identity of  $c$ ); honest negatives quantified ( $c$  without independent geometric identity; the zero-mode line is not  $R$ -real); **tfpt\_1** gains the consolidated “cover program at a glance” paragraph; and the prime-front uniformity sharpens:  $\varepsilon \cdot h \leq 0.98$  on all 68 floor-passed windows —  $|q_{\text{real}}/q_{\text{model}}| \leq h^{-1}$  uniformly on the declared surface (note on PRIME.LOCKPROJ.01))

- **No new modules.** Sandbox measurements and typed notes only:  $c^2$ ,  $c\Gamma(c)$ ,  $\Gamma(c)c$  outside the word algebra (curiosities documented:  $\text{char}(7\Gamma(c)c) = y^3 + 3y^2 + 6y - 1$  integer monic;  $|\det(14c)|^2 = 28^3$ ;  $\ker V$  not  $R$ -real and no  $0 \leftrightarrow 2$  swap; tercile medians of  $\varepsilon \cdot h$ : 0.61/0.44/0.39 (stable, slightly decreasing); the single blow-up ( $h = 1219$ ) is exactly the declared  $q_{\text{model}}$  zero-crossing exclusion.
- **Sync.** Ledger notes on GATE.QGEO.01 and PRIME.LOCKPROJ.01; **tfpt\_1** consolidation paragraph; diary; suite unchanged (599, green); mirrors regenerated.

**2026-08-01 (XXI)** (the translator round lands (v605 promoted): the corpus RP anti-automorphism IS the J-adjoint twisted by an explicit inner element  $c$  ( $\varphi = J_{\text{adj}} \circ \theta_{\text{pull}}$  is inner,  $c$  unique up to scale,  $14c$  integral over  $\mathbb{Z}[\omega]$  — the single-sheet negatives resolved, the  $\bar{\omega}$  detour unnecessary); the seam is a **NONZERO** pure 3-torsion class at infinity (torsion coordinates  $(2, 2, 0)$ ); the 81 survives frame-joining as the saturation defect (chain indices  $3^5, 3^4$ ); the  $J_{\text{fix}}$  duality certified with a Lean addendum)

- **v605\_translator.py** (QGEO.TRANSLATOR.01, 11 checks; probe 11/11, TRANSLATOR-LANDED).  $\varphi(A) = cAc^{-1}$  exact (1-dim intertwiner space,  $c$  invertible);  $14c$   $\mathbb{Z}[\omega]$ -integral (denominator  $14 = 2 \cdot 7$ ); seam lift zero in the free part,  $(2, 2, 0)$  in the torsion of  $\text{coker}(1 - T_\infty)$ ; order chain  $O_1 \subset O_{\text{joint}} \cap \text{span}_1 \subset P_1$  with indices 243 and 81;  $C_V^T J_{\text{fix}} = J_{\text{fix}} C_{V_{\text{dual}}}$  integer;  $c$ 's independent geometric identity typed open.
- **Lean.** CoverEmbedding.lean addendum: **Jfix\_symmetric**, **Jfix\_det** = 72, **CV\_dual\_certificate**, **CVdual\_cubic**, **dual\_joint\_trace**; lake build green (3375 jobs).
- **Sync.** Ledger row + notes on QGEO.EQORDER.01/QGEO.SEAMMARKS.01/QGEO.DUALFORMS.01; **tfpt\_3** (audit block); diary; suite 598  $\rightarrow$  599 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XX) (the equivariant-order round lands (v604 promoted):**  $R$  IS an anti-isometry of the polarization ( $M^\dagger JM = J^T$  exact — correcting a diary side remark),  $J$  restricts to the integer form  $J_{\text{fix}}$  with  $\det = 72 = 8 \cdot 9$ ; the four mark frames generate a FULL order of index  $27 = 3^3$  in  $M_3(\mathbb{Z}[\omega])$ ; and the 3-torsion localizes at infinity ( $\text{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$ , finite cusps torsion-free): 81 at the marks, 27 at infinity — the integral infinity/order-torsion bridge; translator negatives extended (the  $\bar{\omega}$ -sheet named))

- **v604\_equivariant\_order.py** (QGE0.EQORDER.01, 12 checks; probe 12/12, **EQUIVARIANT-ORDER-LANDED**). Anti-isometry exact;  $J_{\text{fix}} = [[16, 2, 4], [2, -2, 2], [4, 2, -2]]$ ; all 28 rotated words  $\mathbb{Z}[\omega]$ -integral, joint lattice full rank 18, Smith divisors  $(1^{15}, 3, 3, 3)$ ;  $\text{coker}(1 - T_\infty)$  divisors  $(1, 1, 1, 3, 3, 3, 0, 0)$ ; each finite cusp twist torsion-free; three objects share  $(\mathbb{Z}/3)^3 = 27$  (infinity monodromy, equivariant order,  $\det(1 - \text{deck})$ );  $F_2$  vs  $J_{\text{fix}}$ -menu and algebra-twist negatives quantified.
- **Sync.** Ledger row + notes on QGE0.SEAMMARKS.01/QGE0.COVER.01; tfpt\_3 (audit block); diary (incl. the exploration round XX:  $\infty$ -embedding,  $R$ -orbit classes, translator hunt); suite 597  $\rightarrow$  598 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XIX) (the seam-and-marks round lands (v603 promoted):** the seam line is a REAL line of the cover (unique common eigenline,  $R$ -fixed) and  $S_0$  reverses it — the v590 seam-reversal demand holds geometrically; the infinity cusp separates in the unreduced Burau module; the first per-mark  $\mathbb{Z}/3$  witness lands ( $E_1$ , class order 3 in the word-order cokernel); the corpus-transpose translator negatives are quantified)

- **v603\_seam\_marks.py** (QGE0.SEAMMARKS.01, 15 checks; probe 15/15, **SEAM-REAL-LANDED**). Seam:  $Uv = 0$ ,  $Vv = v$  unique,  $v = (0, (\sqrt{3} - i)/(\sqrt{3} + i), 1)$ ,  $Rv \sim v$ ,  $S_0v = -v$ ; infinity: unreduced Burau braid relations, invariant column  $(1, 1, 1, 1)$ , full twist eigenvalues  $\{1, \omega^{\times 3}\}$ ; marks:  $E_1$  integer restriction, in the word algebra with denominator lcm 3 (order-3 cokernel class);  $E_4$  trivial,  $E_2/E_3$  outside (naive global map fails, equivariant framing typed); translator:  $F_2 = g^T \Sigma g$  matches no canonical menu form (all typed open).
- **Sync.** Ledger row + notes on QGE0.DUALFORMS.01/QGE0.INVOL.01; tfpt\_3 (audit block); diary; suite 596  $\rightarrow$  597 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVIII) (the duality-and-forms round lands (v602 promoted):** the J-adjoint exchanges the two parabolic types exactly ( $(U_1, V_1^J) \sim \text{compiler pair}$ ); the dictionary completes in the J-projector language (the corpus anchor formula  $H = 1 + V(V - 1)/2$  holds verbatim on the curve); Weil positivity holds exactly (the Riemann bilinear relations on the  $\omega$ -sheet); the canonical bilinear form  $B = M^\dagger J$  is symmetric with  $\det B = 8$ ; the v600 integer core is Lean-certified (FORM.QGE0.04); the contracts paper gains the  $\mu_3$ -cover model paragraph)

- **v602\_duality\_forms.py** (QGE0.DUALFORMS.01, 17 checks; probe 17/17, **DUALITY-FORMS-LANDED**).  $U_1$  J-self-adjoint,  $V_1^J$  the exact product reversal,  $(U_1, V_1^J)$  conjugate to the compiler pair;  $H$ ,  $Q$ ,  $Q_+$ , ladder all exact in the J-projector language; Weil-twisted form positive definite ( $\{2\sqrt{2} - 2, 2, 2 + 2\sqrt{2}\}$ ); honest negatives:  $g^T g \neq \lambda B$ , naive mark-local map fails, seam normal not a deck eigenvector (all typed open).
- **Lean (FORM.QGE0.04).** TfptCarrier/CoverEmbedding.lean: the v600 integer core via determinant-free Smith certificates ( $W = TS$ ,  $PT = \text{diag}(1, 1, 1, 3, 3, 3, 3)$  with unimodular  $P$ ,  $S$ -primitivity via unimodular minor); kernel decide; lake build green (3375 jobs).

- **Sync.** Ledger row + notes on QGEO.CANON.01/QGEO.EMBED.01; tfpt\_research\_contracts ( $\mu_3$ -cover model paragraph after the bedrock chain), tfpt\_3 (audit block); diary; suite 595  $\rightarrow$  596 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVII) (the canonicity round lands (v601 promoted): the J-adapted cusp projectors give a manifestly  $D_4$ -equivariant presentation of the pair (mark choice = winding choice), which realizes the DUAL parabolic exactly (conjugate to the transposed compiler pair; the direct intertwiner is singular); the v590 Klein four-group lives integrally on the curve; the Weil-twisted RP comparison and the seam identification stay open)**

- **v601\_equivariant\_dual.py** (QGEO.CANON.01, 14 checks; probe 14/14, **EQUIVARIANT-DUAL-LANDED**).  $GJ_k = (1 - T_k)J^{-1}(1 - T_k)^\dagger J$  idempotent rank-1 J-projectors rotating exactly under  $D_4$ ; pair family  $U_k = 3GJ_k$ ,  $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ :  $\mathbb{Z}[\omega]$ -integral, compiler-exact, equivariant by construction; duality: conjugate to  $(U_c^T, V_c^T)$  exactly (must-fail to  $(U_c, V_c)$  documented); Klein group: all eight spectral-sign involutions integral,  $S_0$  explicit,  $AB = S_0$ ; honest: naive J-word-Gram inertia  $(3, 2, 2) \neq (4, 0, 3)$  (Weil twist named open).
- **Sync.** Ledger row + note on QGEO.EMBED.01; tfpt\_1/tfpt\_2 (canonicity clauses), tfpt\_3 (audit block); diary; suite 594  $\rightarrow$  595 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XVI) (the joint embedding lands (v600 promoted): the compiler pair  $(U, V)$  is REALIZED on the  $\mu_3$ -cover — a  $\Gamma$ -even degree-2 pair,  $\mathbb{Z}[\omega]$ -integral, seven-word rank 7, simultaneous conjugation to the compiler pair solved exactly, integer restrictions on the saturated  $\text{Fix}(R)$  lattice, and saturation index  $81 = 3^4$  with the exact v566 divisor pattern; canonicity typed open, the gate does not move)**

- **v600\_joint\_embedding.py** (QGEO.EMBED.01, 15 checks; probe 15/15, **JOINT-EMBEDDING-LANDED**). The pair  $U^* = -2E_1 + (G_1\Gamma(G_1) + \Gamma(G_1)G_1) + 3$ ,  $V^* = -E_3 + \frac{1}{2}(G_2\Gamma(G_2) + \Gamma(G_2)G_2) + 1$ : chars and joint traces compiler-exact ( $\text{tr}(UV) = 3$ ,  $\text{tr}(UV^2) = 6$ ,  $\text{char}(U+V) = \text{char } Q$ ); seven words independent; explicit invertible  $g$  with  $gU^*g^{-1} = U_c$ ,  $gV^*g^{-1} = V_c$ ; the real structure integer on  $\mathbb{Z}^6$  with explicit saturated fixed-lattice basis (SNF certificate); integer restrictions; Smith divisors  $(1, 1, 1, 3, 3, 3, 3)$ , index 81 — the v566 self-code order structure on the curve. Must-break control passes; canonicity open (existence claim only).
- **Sync.** Ledger row + notes on QGEO.REAL.01, QGEO.DICT.01, QGEO.COVER.01 and DIAMOND.PARA.SELFCODE.01; tfpt\_1/tfpt\_2 (realization sentences), tfpt\_3 (audit block); diary; suite 593  $\rightarrow$  594 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XV) (the real structure lands (v599 promoted): the anti-holomorphic  $D_4$  reflection on the  $\mu_3$ -cover exists and is unique up to scale with  $R^2 = +1$ ; its grading functor closes all three dictionary gaps at char level ( $V, H$  with the anchor spectrum  $(1, 1, 2)$ ,  $Q_+$ ); the  $\Gamma$ -decomposition carries 5, 9, 27; the joint algebra embedding stays open (quantified negative); and the invariant form has  $\det J = 8$  — a sheet-diamond determinant-list member — with Lorentz signature  $(1, 2)$ )**

- **v599\_real\_structure.py** (QGEO.REAL.01, 18 checks; probe 18/18, **REAL-STRUCTURE-LANDED**). Existence and uniqueness of  $R(v) = M\bar{v}$  (1-dimensional intertwiner space,  $R^2 = +1$  exact, four twist intertwinings  $RT_kR^{-1} = T_{5-k}^{-1}$ ); the grading functor  $\Gamma(A) = M\bar{A}M^{-1}$  locates  $V \sim G_2 + G_3 - \Gamma(G_3)$ ,  $H \sim 1 + G_2 - \Gamma(G_3)$  (the anchor spectrum) and  $Q_+ \sim 1 + G_2 + G_3 - \Gamma(G_3)$

exactly at char level; the  $\Gamma$ -even/odd cusp elements carry  $\{0, 1, 5\}$ ,  $\{0, 3, 9\}$ ,  $\pm\sqrt{5}$ ,  $\pm\sqrt{27}$  and one nilpotent; the joint negative quantified ( $4 \times 6$  candidates, zero pairs pass  $\text{tr}(UV) = 3$  and  $\text{char}(U + V) = \text{char } Q$ ; kill not triggered); the invariant hermitian form is unique with  $\det J = 8$  and signature  $(1, 2)$  — Strategy II's Hodge prediction at form level, pinned.

- **Sync.** Ledger row + notes on QGEO.DICT.01/QGEO.COVER.01; tfpt\_3 (audit block); diary; suite 592  $\rightarrow$  593 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XIII) (the cover dictionary reaches THREE exact entries (v598 promoted):  $U \sim G_2$ ,  $Q_- \sim G_1 - G_3$  — the  $\mu_4$  sheet coupling is the diamond-diagonal Gram difference — and the binary AnchorLadder  $(1, 2, 4) \sim 1 + G_1 + G_3$ ; the grading half is absent from 7296 canonical combinations, with the real structure as the named suspect)**

- **v598\_cover\_dictionary.py** (QGEO.DICT.01, 4 checks; probe 4/4, DICT-THREE-ENTRIES). All three entries exact (numeric prefilter, symbolic confirmation); supporting structure: opposite-twist products nilpotent,  $Q_-^2 \sim x(x - 3)^2$ ; negatives quantified (7296 combinations, 0 matches for  $Q_+/V/H/Q$ ); the Strategy-I kill criterion not triggered; GATE.QGEO does not move.
- **Sync.** Ledger row + note on QGEO.COVER.01; tfpt\_1/tfpt\_2 (dictionary sentences updated), tfpt\_3 (audit block); diary; suite 591  $\rightarrow$  592 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (XII) (dictionary round 2 honest negative (the  $Q = U + V$  double criterion over  $\sim 50$  canonical elements: no  $Q/V$  hit; kill not triggered, the flag-adapted search is the designed next session); the real-structure map documented as blocked on the dictionary; and the self-contained equidistribution problem statement added to tfpt\_prime\_front (sec:theory-open) — the v596 identity family posed TFPT-free for external consumption)**

- **No module.** Exploration and editorial only: ledger untouched except mirrors; the one dictionary entry stays  $U = G_2$  (2026-08-01 XI); the open problem now has a publication-grade statement (explicit couplings  $c_\gamma$ , closed constant  $K(\alpha, D)$ , the observed collective cancellation, and the uniformity question named as the open content).

**2026-08-01 (XI) (the dictionary's first entry: U IS LOCATED — the winding operator appears as a cusp Gram on the  $\mu_3$ -cover; exploration round, documented as a note on QGEO.COVER.01)**

- **The hit.** The cusp Gram  $(1 - T_2)(1 - T_2)^\dagger$  has characteristic polynomial  $x^2(x - 3) = \text{char}(U)$  exactly — the rank-one winding operator (trace 3 =  $N_{\text{fam}}$ ) as a Gram of a  $D_4$ -conjugate cusp twist: the fifth fingerprint. Additional exact structure: opposite-twist products  $(1 - T_1)(1 - T_3)$  are nilpotent ( $x^3$ ),  $G_1 + G_3$  has spectrum  $\{0, 1, 3\}$ . Honest negative:  $V/H/Q$  not located in the scanned canonical ranges (products/sums/ $\omega$ -twists of length  $\leq 2$ ) — the  $V$ -dictionary likely requires the parabolic flag (the seam-cusp filtration); named next step; kill criterion unchanged.
- **Sync.** Ledger note on QGEO.COVER.01; diary; mirrors regenerated.

**2026-08-01 (X) (Strategy I/II executed: the  $\mu_3$ -cover  $y^3 = x^4 - 1$  hits FOUR compiler fingerprints (v597 promoted) — rank-3 Eisenstein  $H_1$ , projective  $D_4$  via the deck, cusp-class group  $(\mathbb{Z}/3)^4$  of order 81, and the unique invariant hermitian form of LORENTZ signature  $(1, 2)$  matching Paper II's polarization: the first**

geometric candidate that could close GATE.QGEO and the RP slot with one construction)

- **v597\_mu4\_cover\_model.py** (QGEO.COVER.01, 6 checks; probe 6/6, FINGERPRINTS-HIT). Genus 3, five cusps ( $\mu_4 + \infty$ ), Chevalley–Weil  $H^{1,0} = (1, 2)$ ; Burau at  $t = \omega$  with verified braid relations,  $r^4 = \omega \cdot 1$ ,  $r^{12} = 1$ ; cusp classes  $(\mathbb{Z}/3)^5$ /one relation  $= (\mathbb{Z}/3)^4$ , order  $81 = 3^4$  with the per-mark  $\mathbb{Z}/3$  mechanism ( $\det_{\mathbb{Z}}(1 - \text{deck}) = 27$  complementarily); invariant form unique up to scale, eigenvalues  $\{-2, 2-2\sqrt{2}, 2+2\sqrt{2}\}$ , signature  $(1, 2)$ ; single-cusp Gram spectrum  $\{0, 0, 2\}$  all-integer. OPEN and named: the  $Q/V$  operator dictionary, the real-structure/RP map to the v572 Gram data; kill criteria live, not triggered; GATE.QGEO does not move.
- **Sync.** Ledger row + notes on QGEO.INVOL.01 and DIAMOND.PARA.SELFCODE.01; tfpt\_1/tfpt\_2 (candidate paragraphs), tfpt\_3 (audit block); diary; suite 590  $\rightarrow$  591 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (IX) (the C-D-E round: the 1D lock projection (v596 promoted, T199 — the near-identity along the closed direction, median  $7.7 \times 10^{-4}$ , coupling spectrum collective); the v590 involution group formalized in Lean with its Klein-four closure  $A \cdot B = S_0$ ; and the big-picture strategy survey (the open core funnels into GATE.QGEO + the RP slot; two new geometric strategies preregistered))**

- **v596\_lock\_projection.py** (PRIME.LOCKPROJ.01, 6 checks; T199 6/6, NEAR-IDENTITY-1D). Along the closed v591 direction (parameter-free):  $|q_{\text{real}}/q_{\text{model}}|$  median  $7.7 \times 10^{-4}$ , decay  $h^{-1.01}$ ; the coupling spectrum is collective (max single-zero share 5.9%,  $N_{50} = 56$ , heavy zeros 14.1/25.0/30.4/40.9); 400-zero truncation fails (v589 plateau); scramble breaks by six orders — the arithmetic content of Problem 7.1 is ONE explicit identity family, equidistribution-flavored.
- **Lean.** A, B, klein\_closure ( $A \cdot B = S_0$ ), involutions and  $V$ -commutation added to ParityWeightLaws.lean; lake build green (3374 jobs).
- **The big-picture survey (E).** The gate/contract census shows the open core is SMALL: GATE.METRIC.19 funnels the whole  $G_{\text{net}}$  line into GATE.QGEO + standard AQFT assembly; GATE.QGEO itself is narrowed (v574/v590) to the parabolic  $\mu_4$  realization; the RP slot is the one dynamical open; absolute scales and external interfaces are typed contracts. Two new strategies preregistered in the diary: (i) the  $\mu_4$ -cover homology model ( $H_1$  of the cyclic covers  $y^4=x^4-1$  /  $y^3=x^4-1$  as  $\mathbb{Z}[i]$ - resp.  $\mathbb{Z}[\omega]$ -modules of rank  $3/2$  — the compiler  $Q$  as a canonical homology operator; testable by finite computation); (ii) RP positivity as HODGE positivity (the seam kernel as the polarization form of the cover’s Hodge structure — the Riemann bilinear relations would make positivity a theorem; testable against the v572 Gram data via exact Beta-function periods).
- **Sync.** Ledger row + notes; tfpt\_prime\_front (48  $\rightarrow$  49 modules, sec:v596/sec:t199, 199 probes / 4944 checks); Paper II (“thirty modules”); tfpt\_3; diary; suite 589  $\rightarrow$  590 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VIII) (the 0-1-2-3-4-5 round: repository pushed (v576–v594, 111 files); the residual bookkeeping executed (v595 promoted, T198: the v593 residual was the  $\sigma$  MAPPING — identity-level closed forms 1.00000–1.00017, honest amendment of the v593 attribution); derivation sketches added to the theorem section; the determinant law formalized in Lean (det\_S\_closed, build green))**

- **v595\_mapping\_completion.py** (PRIME.MAPCLOSE.01, 4 checks; T198 4/4, MAPPING-CLOSES). The exact dictionary is  $\sigma = d/N$ , hence  $a_{\text{eff}} = ND/2$  with limits  $[u_0/(ND), 2\alpha/(ND)]$  ( $ND = 2\alpha + O(D)$  — why the  $\alpha$ -form carried a first-order residual); entries AND det match at

1.00000–1.00017 (1.00000 at  $h = 1445$ ); kernel piece  $9 \times 10^{-6}$  (= the genuine  $O(1/N^2)$ ); the v593 numbers stand, its residual attribution is superseded (note on `PRIME.CUTOFF.01`).

- **Manuscript-grade theory section.** `sec:theory-proofs` added to `tfpt_prime_front`: derivation sketches for all seven theorems (Dirichlet sums and branch merge; geometric ladders; factorization/separability; the direction and determinant laws; the exact dictionary; partial summation + Bütthe).
- **Lean.** `det_S_closed` and `det_denominators_factor` added to `ParityWeightLaws.lean` (the closed determinant law as a field identity; the  $E^2$  cancellation structural); full lake build green (3374 jobs).
- **Repository.** Commit 0701e1e pushed (v576–v594, 111 files, +19133 lines); the remaining hard opens (determinant-level zero theorem; OS/RP; the  $\mu_4$  geometric model) stay documented.
- **Sync.** Ledger row + amendment note; `tfpt_prime_front` (47  $\rightarrow$  48 modules, `sec:v595/sec:t198`, 198 probes / 4938 checks); Paper II (“twenty-nine modules”); `tfpt_3`; diary; suite 588  $\rightarrow$  589 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VII) (the full follow-up slate lands: the cutoff completes the density layer END-TO-END (v593, T196), the FIRST UNCONDITIONAL STATEMENT via Bütthe 2018 (v594, T197), the consolidated theorem section in `tfpt_prime_front`, and the Lean-4 formalization of the round’s algebraic cores (`ParityWeightLaws.lean`, builds green against Mathlib))**

- **v593\_cutoff\_completion.py** (`PRIME.CUTOFF.01`, 4 checks; **T196** 4/4, **CUTOFF-CLOSES**). The v592 finite- $N$  det gap (0.19–0.50) was the integration cutoff:  $\sigma_0$ -corrected closed integrals match entries to 0.7–1%, det to 0.2–1%; residual  $O(1/N^2)$ , shrinking — the density layer of Problem 7.1 is analytically closed end-to-end.
- **v594\_unconditional\_cert.py** (`PRIME.UNCONDCERT.01`, 5 checks; **T197** 5/5, **verdict UNCONDITIONAL ENTRY CERT**). Partial summation (machine-verified) +  $|\psi(x) - x| < 0.94\sqrt{x}$  (Bütthe 2018, unconditional to  $10^{19}$ ; external data)  $\Rightarrow \sup |\text{Osc}| \leq 17.4$  unconditional (measured 1.474); certificate margin  $565\times$  — the first unconditional statement of the program; the determinant level stays with the zero-oscillation theorem.
- **Theory section + Lean.** `tfpt_prime_front` gains `sec:theory` (“The closed-form theory: seven statements”, Theorems 1–7 with module and Lean citations); new Lean-4 module `ParityWeightLaws.lean` (branch merge, rank-one determinant, null vector, direction law + approach, sheet involution  $S_0^2 = 1/[S_0, V] = 0/\text{tr} = -1$ ) — full lake build green (3374 jobs).
- **Sync.** Ledger rows + notes on `PRIME.DETLAW.01` and `PRIME.ZEROCOMB.01`; `tfpt_prime_front` (45  $\rightarrow$  47 modules, `sec:v593/sec:v594/sec:t196`, 197 probes / 4934 checks); Paper II (“twenty-eight modules”); `tfpt_3`; diary; suite 586  $\rightarrow$  588 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (VI) (the continuum determinant law (v592 promoted, T195):  $\det S(a)$  closed with the  $e^{2a}$  coefficient identically zero and exact leading law  $640\pi^2 e^a/a^3$ ; the real dominance tracks the closed formula at correlation 0.9985 (the v570 growth derived); entry-level arithmetic certificate with margin  $47.6\times$ ; PLUS the Paper-II consolidation section (the deterministic–arithmetic dissection as one narrative))**

- **v592\_continuum\_det Law.py** (`PRIME.DETLAW.01`, 6 checks; probe **T195** 6/6, **DET-LAW-DERIVED**).  $e^{2a}$  coefficient  $\equiv 0$  (rank-one);  $e^a$  term with exact closed coefficient carries the determinant at window scales (0.995–1.0002); leading law converges monotonically; dominance correlation 0.9985; finite- $N$  gap typed (0.19  $\rightarrow$  0.50); entry certificate  $\sup |\text{Osc}| \times \text{TV}$ , margin

47.6×, conditional on the measured sup (1.47) — the unconditional sup IS the zero-oscillation theorem.

- **Paper-II consolidation (follow-up 5).** New subsection “The deterministic–arithmetic dissection (machine survey)” before the cross-reference table: the v582–v592 arc as one narrative (two layers; the density layer closed-form with the rank-one pole term and the two laws; the arithmetic layer identified as the zero comb with a structural recovery plateau; the two named remaining opens). Follow-ups 2/3/4 stay open with reasons documented (no forced modules).
- **Sync.** Ledger row + notes on PRIME.PENCIL.SEPFLOOR.01 and PRIME.POLERANKONE.01; `tfpt_prime_front` (44 → 45 modules, `sec:v592/sec:t195`, 195 probes / 4925 checks); Paper II (“twenty-six modules” + dissection section); `tfpt_3`; diary; suite 585 → 586 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (V) (the pole term is EXACTLY rank-one and the locking-direction law is derived (v591 promoted, T194):  $S^{\text{pole}} = -32\pi^2 a e^a g g^T$  with  $\det \equiv 0$  symbolically;  $v_2/v_1 = -(a^2 + 16\pi^2)/(2(a^2 + 4\pi^2))$  with limit exactly  $-1/2$  — the v577 null-ray conjecture DERIVED, the v586 drift explained; plus the honest B/C rounds: deeper zero sums plateau (structural, note on v589) and the grading shortcut for the involution demands is falsified (note on v590))**

- **v591\_pole\_rank\_one.py** (PRIME.POLERANKONE.01, 5 checks; probe T194 5/5, **POLE-RANK-ONE**). Denominators factor completely ( $D_{11}D_{22} = D_{12}^2$ ); pole part exactly separable, residuals exponential-free; direction law with limit  $-1/2$  and approach  $-6\pi^2/(a^2 + 4\pi^2)$ ; ladder match 0.04–1.4% ( $h = 1445$ :  $-1.2248$  vs  $-1.2253$ ); via v586 the law describes the real corpus locking direction; the density layer = the correction theory of a rank-one matrix.
- **The B and C rounds, honest.** Deeper zero sums (500/1000 zeros,  $\gamma \leq 1419$ ) do NOT improve the depth recovery (plateau  $\sim 31$ –37 at  $h = 184$ ; entries degrade): the bottleneck is sub-cell placement reconstruction, exponentially beyond finite sums — note on PRIME.ZEROCOMB.01. And NO  $V$ -commuting involution implements the corpus  $\mu_4$  grading (all six fail;  $\Sigma S_0$  is an infinite-order shear): the two v590 demands stay independent — note on QGEO.INVOL.01. The OS/RP positivity theorem (D) remains open (no quick construction; the v572/v584 design space is exhausted).
- **Sync.** Ledger row + notes on PRIME.NULLRAY.01 and PRIME.LOCKDIR.01; `tfpt_prime_front` (43 → 44 modules, `sec:v591/sec:t194`, 194 probes / 4919 checks); Paper II (“twenty-five modules”); `tfpt_3`; diary; suite 584 → 585 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (IV) (the sheet involution exists unconditionally (v590 promoted): the integral  $V$ -commuting involutions are exactly  $(\mathbb{Z}/2)^3$  (rigidity, both twins); the v574 type theorem + seam reversal + zero-mode fixing single out the explicit  $S_0$  with exact  $\Sigma$ -signature uniquely — GATE.QGEO.01’s involution-existence input relocates to DERIVED)**

- **v590\_involution\_existence.py** (QGEO.INVOL.01, 7 checks; probe 7/7, **EXISTENCE-UNCONDITIONAL**). All eight spectral sign choices on  $V$ ’s eigenspaces are integer, involutive and order-stabilizing on both twins; narrowing  $6 \rightarrow 3 \rightarrow 2 \rightarrow 1$ : the trace  $+1$  half dies by the v574 rank theorem,  $Se_3 = -e_3$  and  $+1$  on  $\ker V$  single out  $S_0 = [[1, -1, 0], [0, -1, 0], [0, 0, -1]]$ ; honest scope: the corpus RP reflection does not commute with  $V$  (different structure), the alt twin has the same classification (not a bit selector); the two demands’ continuum derivation ( $\mu_4/D_4/H^1$ ) stays the named open.
- **Sync.** Ledger row + note on QGEO.WINDING.SELF.01; `tfpt_1` (Stage-B passage), `tfpt_2` (new paragraph), `tfpt_3` (audit block); diary; suite 583 → 584 scripts; website/README/zenodo

counters; mirrors regenerated.

**2026-08-01 (III) (the zeta zeros enter the instrument (v589 promoted, T193): the arithmetic depth layer is IDENTIFIED as the explicit-formula zero comb — correlation 0.82 with the first 200 damped zeros, a factor 5.7 of the lock depth recovered at the smallest window; the zero-oscillation bound becomes a quantified target)**

- **v589\_zero\_comb.py** (PRIME.ZEROCOMB.01, 4 checks; probe T193 4/4, **ZERO-COMB-IDENTIFIED**). The residual mass oscillation of the atom table is the explicit-formula comb  $-4 \sum_{\gamma>0} \sin(\gamma u)/\gamma$  (dense-grid correlation 0.82, slope 1.28 = truncation deficit; pointwise 1.00 at  $u = 3$ ); adding the comb to the density model improves the  $h = 184$  lock ratio  $176 \rightarrow 31$  (factor 5.7; non-monotone in truncation — honest), deep windows out of reach ( $h = 540$ :  $2646\times$ ); the zero-oscillation bound is now the quantified target “control  $\sum \sin(\gamma u)/\gamma$  at resolution  $\pi/D$  with amplitude accuracy = lock depth”. No bound claimed, NO RH statement.
- **Sync.** Ledger row + notes on PRIME.CLOSEDELTA.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (42  $\rightarrow$  43 modules, `sec:v589/sec:t193`, 193 probes / 4914 checks); Paper II (“twenty-four modules”); `tfpt_3`; diary; suite 582  $\rightarrow$  583 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (II) (the deterministic layer of Problem 7.1 closes in one function (v588 promoted, T192):  $G_{ij}(\beta)$  carries the comb side at the pole exponent and the archimedean side as the TRIVIAL-ZERO LADDER; closed  $B$  to  $2 \times 10^{-5}$ , the closed defect to 0.2% on all 70 windows, census decay  $h^{-1.40}$  — the explicit formula’s skeleton, pole vs  $\Gamma$ -factor, in closed form)**

- **v588\_closed\_delta.py** (PRIME.CLOSEDELTA.01, 6 checks; probe T192 6/6, **CLOSED-DELTA-EXACT**).  $G_{ij}(\beta) = \sum_d e^{-\beta d} W_{ij}(d)$  machine-exact (finite geometric-trig series over the v587 formulas);  $c_{\text{ar}}(d) = -Df(dD)$  with  $f(w) = \sum_n e^{-(2n+1/2)w}$  verified 1.000002 (near field  $d \leq 32$  as explicit quadrature constants); closed  $B$ : entries  $2 \times 10^{-5}$ , det  $6 \times 10^{-5}$ , all 70 windows; closed defect vs the v585 deterministic layer: 0.9999–1.0023 on all 70 windows incl. deep/anomalous; census decay  $h^{-1.40}$  (= v585’s  $-1.43$ ). Remaining for a theorem: elementary asymptotics of  $G$  and the ladder; the arithmetic layer stays with the primes.
- **Sync.** Ledger row + notes on PRIME.WCLOSED.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (41  $\rightarrow$  42 modules, `sec:v588/sec:t192`, 192 probes / 4910 checks); Paper II (“twenty-three modules”); `tfpt_3`; diary; suite 581  $\rightarrow$  582 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-08-01 (I) (a NEW EXACT FORMULA closes the deterministic layer’s entries (v587 promoted, T191): the diagonal lag weight is the single closed expression  $W_{kk}(d) = (2/N)[(N-1-d) \cos(\omega_k d) + \sin(\omega_k(d+1))]/\sin \omega_k$ , machine-exact for all lags; the exact cell-sum reproduces the prime-free model’s entries and determinant to 0.03%)**

- **v587\_w\_closed\_form.py** (PRIME.WCLOSED.01, 6 checks; probe T191 6/6, **CLOSED-FORM-EXACT**). The diagonal companion of the v576 cross formula, machine-exact ( $1.9 \times 10^{-15}$ ) for  $h = 7$ –1000,  $k = 1$ –3, all lags; the building blocks are closed Dirichlet sums, the apparent piecewise break at  $d = m$  cancels on assembly, the reflection-edge suppression is an exact two-term cancellation; the exact cell-sum reproduces the v583 entries (1.0011–1.0036) and det (0.9999–1.0003); macro limit  $g_k(\sigma) = 2(1-\sigma) \cos(2\pi k\sigma) + \sin(2\pi k\sigma)/(\pi k)$  with the v579 kernel its edge-anchored version identically. The first executed step of the v585/v586 classical program; named next: the arch/ $B$ -side closed form, then the  $\delta_{\text{PNT}}$  expansion.



- **Sync.** Ledger row + notes on PRIME.CHEBLOEWNER.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (40 → 41 modules, `sec:v587/sec:t191`, 191 probes / 4904 checks); Paper II (“twenty-two modules”); `tfpt_3`; diary; suite 580 → 581 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIX) (the locking DIRECTION is density-fixed (v586 promoted, T190): real vs prime-free locking eigenvector  $0.030^\circ$  median on all 69 windows while the depth stays  $1350\times$  apart — THE PRIMES ARE A PURE DEPTH AMPLIFIER ALONG A DENSITY-FIXED DIRECTION; the v577 slow-decay puzzle dissolves as a comparator artifact)**

- `v586_pnt_lock_direction.py` (PRIME.LOCKDIR.01, 7 checks; probe T190 7/7, **DIRECTION-DENSITY-FIXED**). Census on all 69 regular windows: angle max  $0.331^\circ$ , median  $0.030^\circ$ , decay  $h^{-1.16}$ ; witness certificate — the quadratic form of the real  $B - S$  along the deterministic direction reproduces the minimal eigenvalue within  $[0.98, 2.9]$  (median 1.42): the arithmetic content of Problem 7.1 is ONE quadratic form along ONE deterministic vector; the deterministic direction drifts ( $-1.73 \rightarrow -1.19$ , limit  $-0.551$  under  $1/\log h$ , consistent with the null-ray  $-0.5$ , not settled) — resolving v577’s comparator artifact; scramble must-break  $6-25^\circ$  vs  $0.026^\circ$ .
- **Sync.** Ledger row + notes on PRIME.NULLRAY.01 and PRIME.LOCKSPLIT.01; `tfpt_prime_front` (39 → 40 modules, `sec:v586/sec:t190`, 190 probes / 4898 checks); Paper II (“twenty-one modules”); `tfpt_3`; diary; suite 579 → 580 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVIII) (the two-layer split of the locking cancellation (v585 promoted, T189): THE DOMINANCE IS DENSITY, THE LOCKING IS ARITHMETIC — the prime-free model misses the Paper-II defect by 2–4 orders (median  $1350\times$ ) while reproducing  $\det S$ ; layers  $h^{-1.43} \times h^{-1.08} = h^{-2.51}$ ; the Problem-7.1 budget splits into a classical density part and a zero-oscillation part)**

- `v585_pnt_locking_split.py` (PRIME.LOCKSPLIT.01, 6 checks; probe T189 6/6, **TWO-LAYER-SPLIT**). The v583 model through the normalised defect on all 70 windows: median ratio 1350, growing  $h^{1.08}$  — the depth of the lock is the primes; the one-mode geometry is density-level (coarse lock  $10^{-2}-10^{-4}$  without primes); density layer  $h^{-1.43}$  always positive (scatter 0.70), real  $h^{-2.51}$ ; real sign overshoots exactly at  $h = 1219, 1445$  (+ the typed 1292 anomaly), deterministically never. Budget: density  $\sim -1.4$  classical, primes  $\sim -1.1$  zero-oscillation.
- **Sync.** Ledger row + notes on PRIME.PNTMODEL.01 and PRIME.PENCIL.ONEMODE.01; `tfpt_prime_front` (38 → 39 modules, `sec:v585/sec:t189`, 189 probes / 4891 checks); Paper II (“twenty modules”); `tfpt_3`; diary; suite 578 → 579 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVII) (the interacting RP route of the alignment bit, executed: EXT-SOURCE COUPLING SEPARATES THE 81-TWINS DIRECTIONALLY IN THE UNIT BOX (v584 promoted) — exact rational flip thresholds with the v572 pencil factors as numerators; positive coupling  $J \leq 15/16 < 1$  heals exactly the compiler twin, the alt twin is rigid; response pole on the golden polynomial)**

- `v584_ext_source_response.py` (DIAMOND.BIT.EXTSOURCE.01, 8 checks; probe 8/8, **UNIT-BOX-SEPARATED**). The interacting kernel  $K(s, J) = K_0 + sK_{\Pi} + Jvv^T$  ( $v$  from the shared Ext arrows  $E_{31}/E_{32}$ ) has exact thresholds  $J_c(s) = -1/(v^T K(s)^{-1}v)$ : compiler  $(15s^2-16s-16)/(16(s^2-s-1))$  and  $(15s^2-16s-16)/16$ , alt  $(s^2+16s+16)/(16(s+1))$  and  $-(s^2+16s+16)/16$  — numerators exactly the v572 pencil factors; verified numerically (16 flip

events). In the crossed phase the compiler is healed by positive  $J \leq 15/16 < 1$  (either arrow); the alt twin cannot be flipped by any positive coupling in  $|J| < 1$  (E31 threshold  $\geq 1$ , equality exactly at  $s = 0$ ; E32 negative). Negative coupling never heals (direction consistent with v534's toy RP selection). Exact anchors  $15/16, 1, -1; J_c(s^*) = 0$  via Ext-visibility of the critical mode; response pole on  $s^2 - s - 1$  ( $-1/\varphi$ , next to  $s^*$ ) — observed. Honest: response-level, no PSD anywhere, the alignment-bit contract stays open.

- **Sync.** Ledger row + note on DIAMOND.BIT.RPFORM.01; `tfpt_1` (follow-up passage extended), `tfpt_2` (new interacting-route paragraph), `tfpt_3` (audit block); diary; suite 577  $\rightarrow$  578 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XVI) (the named next step executed the same day: THE SMOOTH FUNCTIONAL IS IN CLOSED, PRIME-FREE FORM (v583 promoted, T188) — the two-term classical law  $4e^{u/2} - 2(\zeta'/\zeta)(1/2)$  (no zeros) reproduces  $\det S$  on all 69 regular windows parameter-free to 1.016–1.143; the  $h = 1292$  sign flip is genuine prime-fluctuation content)**

- **v583\_pnt\_model.py** (PRIME.PNTMODEL.01, 8 checks; probe T188 8/8, PRIME-FREE-CLOSED-FORM). The two-term law holds on the atom table (oscillation band  $< 1.5$  around the parameter-free constant  $-2(\zeta'/\zeta)(1/2) \approx -5.372$ ); the prime-free grid model (density  $2e^{u/2}$ , hard cutoff, theoretical constant) reproduces  $\det S$  on all 69 regular windows to 1.016–1.143 (mean 1.039), the  $S$  entries to 0.995–1.001, the dominance growth with correlation 0.998; honesty: constant sensitivity  $\sim 0.38/\text{unit}$  — the residual band is the zero-oscillation; the anomalous  $h = 1292$  window typed (prime-free +1844 vs real  $-1465$ : the sieve-horizon sign flip IS prime-fluctuation content). FINAL RELOCATION:  $h$ -uniform dominance = prime-free two-integral inequality + zero-oscillation bound.
- **Sync.** Ledger row + notes on PRIME.DENSITYDOM.01 and PRIME.PENCIL.SEPFLOOR.01; `tfpt_prime_front` (37  $\rightarrow$  38 modules, `sec:v583/sec:t188`, 188 probes / 4885 checks); Paper II (“nineteen modules”); `tfpt_3`; diary; suite 576  $\rightarrow$  577 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XV) (the quantitative occupation question answered at its first level: THE DOMINANCE IS DENSITY-DRIVEN (v582 promoted, T187) — the smoothed comb carries 99.8–99.9% of  $\det S$  on every regular window, the scrambled control only 70%: the real comb is anomalously smooth, and the  $h$ -uniform dominance question relocates to PNT-level regularity of the smooth density functional)**

- **v582\_density\_dominance.py** (PRIME.DENSITYDOM.01, 5 checks; probe T187 5/5, DENSITY-DRIVEN). Replacing the atom lag coefficients by their moving average (declared windows 2–20%) and recomputing  $\det S$  through the exact lag-side identity: 0.998–0.999 at 2%, 0.987–0.993 at 5%, degrading smoothly to 0.81–0.89 at 20% on every regular window — the dominance is carried by the deterministic density profile, fluctuations contribute  $\sim 1\%$ ; the scrambled control captures only 0.70 (random placement has real fluctuation content — the actual comb is anomalously smooth at these scales); the anomalous  $h = 1292$  window deviates (1.24), boundary named. THE RELOCATION: the v570  $h$ -uniform dominance reduces — up to a measured  $\sim 1\%$  correction needing only a crude bound — to a positivity/growth statement about the smooth density functional (PNT-level regularity); the named next step is a closed-form bound on that functional.
- **Sync.** Ledger row + notes on PRIME.PENCIL.SEPFLOOR.01 and PRIME.OCCUPATION.01; `tfpt_prime_front` (36  $\rightarrow$  37 modules, `sec:v582/sec:t187`, 187 probes / 4877 checks); Paper II (“eighteen modules”); `tfpt_3`; diary; suite 575  $\rightarrow$  576 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIV) (Regime B’s arithmetic half gets its measured object (v580: the lag-side identity is exact and the primes POPULATE the geometric sign map — every significant occupation cell matches  $K_\infty$ ), and the review’s multilevel transport lemma is measured and killed (v581: drift small but growing  $h^{+0.36}$  against the  $h^{-1-\delta}$  bar) + backflow parts T185/T186 + full sync)**

- **v580\_occupation\_map.py** (PRIME.OCCUPATION.01, 5 checks; probe T185 5/5, OCCUPATION-FOLLOWS-KERNEL). The lag-side polarization identity —  $\det S$  as the atom-lag double sum  $\frac{1}{2} \sum c_{\text{at}} c_{\text{at}} D(W, W)$  — is exact (the T163 correlation theorem,  $10^{-9}$  on four windows); every significant cell of the binned occupation map sign-matches  $K_\infty(\sigma_1, \sigma_2)$  on every tested window (50/50, 67/67, 65/65, 65/65): the arithmetic does not fight the kernel, it populates it — the open Regime-B content is the AMOUNTS; mid-diagonal negative, cross-scale positive; scramble control clean.
- **v581\_locking\_transport.py** (PRIME.TRANSPORT.01, 4 checks; probe T186 4/4, TRANSPORT-KILLED). The Priority-4 lemma measured on 40 doubling pairs: the locking-direction drift is small ( $0.3\text{--}6.7^\circ$  per octave, median  $2.35^\circ$ ) but GROWS ( $h^{+0.36}$ ) against the proposed summability bar — the review’s own kill criterion fires; no fixed-direction scalar transport; future transports must carry the window-dependent macro direction.
- **Sync.** Two ledger rows + notes on PRIME.MACROKERNEL.01 and PRIME.NULLRAY.01; `tfpt_prime_front` (34  $\rightarrow$  36 modules, `sec:v580/sec:v581/sec:t185/sec:t186`, 186 probes / 4872 checks); Paper II (“seventeen modules”); `tfpt_3`; diary; suite 573  $\rightarrow$  575 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XIII) (the review’s remaining strands executed: the jet contract natively (v578, MOEBIUS-IN-OWN-CLOCK — no shared wall-time Schwarzian, each flow projective in its own clock), the two-scale kernel signs verified globally (v579, SIGNS-MATCH 332/332 — the geometric half of Regime B closes), and the Ext-resolvent marking honestly killed (sandbox:  $\chi_{\text{Ext}} \equiv 0$  for both twins, the radical sublattices are identical))**

- **v578\_native\_jets.py** (FTR.JETS.01, 6 checks; probe 6/6, MOEBIUS-IN-OWN-CLOCK). The exact v99 trajectory has  $\{q, t\} = -\Delta^2/2$  in wall time and Schwarzian 0 in its own exponential clock; the 1-loop QCD flow has Schwarzian 0 in its own RG time; the native quadruple shares no wall-time jet invariant — each instance is projective in its own clock, the obstruction is the clock (anchored by the external data); a future joint jet test needs external anchor-level jets, as v575 fenced.
- **v579\_macro\_kernel\_signs.py** (PRIME.MACROKERNEL.01, 6 checks; probe T184 6/6, SIGNS-MATCH).  $\text{sign } K_\infty(\sigma, \tau)$  matches the real two-lag kernel on every resolved cell (332/332 at  $h = 300$ ); the diagonal  $\det W^\infty(\sigma)$  is negative on  $\sim 72\%$  (same-scale coupling negative, analytically); the geometric half of Regime B closes, the  $\Lambda$ -weighted arithmetic occupation stays open.
- **Sandbox honest negative.** The review’s Ext-sensitive marked resolvent  $\chi_{\text{Ext}}(z)$  vanishes identically for both twins with the natural markings, and the radical sublattices of the two integral orders are IDENTICAL ( $\text{span}\{3E_{31}, 3E_{32}\}$ ): the review’s own kill criteria fire; the Gibbs pencil (v572) remains the one working marked object. Diary record, no module.
- **Sync.** Two ledger rows + notes; `tfpt_research_contracts` (jet block), `tfpt_prime_front` (33  $\rightarrow$  34 modules, `sec:v579/sec:t184`, 184 probes / 4863 checks), Paper II (“fifteen modules”), `tfpt_3`; diary; suite 571  $\rightarrow$  573 scripts; website/README/zenodo counters; mirrors regenerated.

**2026-07-31 (XII) (the second review's Toeplitz strand audited – everything EXACT (v576: the cross-weight formula, the Chebyshev–Loewner edge theorem with the rank-one first read, the Pythagorean null bridge  $(5, -3, 4)$  typed compression, the exact  $N^{-4}$  defect coefficient, the pair-kernel sign polynomial) and the null-ray census executed (v577: correlated on all 70 windows, honestly partial) + backflow parts T182/T183 + full sync)**

- **v576\_cheb\_loewner\_edge.py** (PRIME.CHEBLOEWNER.01, 9 checks; probe T182 9/9, **FORMULAS-EXACT**). The polarized cross-weight formula  $w_d^{(ij)} = (2/N)[\sin \omega_j \sin(dw_i) - \sin \omega_i \sin(dw_j)]/(\cos \omega_i - \cos \omega_j)$  is exact against the corpus definition (symbolic  $N \leq 11$ ,  $10^{-11}$  surface) — the  $K = 2$  read block closes analytically; the reflection edge is a scaled Loewner matrix of  $U_{s-1}$  (exact; rank  $\leq s-1$ , the first read rank ONE); the leading profile's Lorentz image is the null triple  $(5, -3, 4)$  — the same as  $(g_{\text{car}}, N_{\text{fam}}, |\mu_4|)$  via Euclid, typed COMPRESSION; the edge defect is the exact coefficient  $-\frac{3\pi^4}{175}(s^2 - 4)(11s^2 + 6)N^{-4}$  (locally better than the  $h^{-3}$  demand); the pair polynomial  $P(s, z)$  ( $P(2, 3) = 0$  exact, transition  $s/z \approx 0.503/1.987$ ) is the analytic prototype of the v573 pattern; macro kernels verified. Three-regime split: edge CLOSED at  $N^{-4}$ , macro and transition OPEN; Problem 7.1's quantifier untouched.
- **v577\_nullray\_census.py** (PRIME.NULLRAY.01, 5 checks; probe T183 5/5, **NULL-RAY-CORRELATED**). The locking mode aligns with the exact null ray  $(2, -1)$  at  $23.5\text{--}33.8^\circ$  on all 70 windows (below random; scramble destroys it,  $+21^\circ$ ), declining slowly ( $h^{-0.10}$ ): the clean success bar is honestly NOT met — a real partial carrier; the macro regime bends the direction; follow-up:  $K_\infty(\sigma, \tau)$ .
- **Sync.** Two ledger rows + notes on PRIME.PAIRBAND.01 and PRIME.PENCIL.SEPFLOOR.01; **tfpt\_prime\_front** (verified layer  $31 \rightarrow 33$ , subsections **sec:v576/sec:v577/sec:t182/sec:t183**, counters 183 probes / 4857 checks); Paper II (cross-reference rows, “fourteen modules”); **tfpt\_3** audit blocks; diary; suite  $569 \rightarrow 571$  scripts; README/zenodo/website counters; mirrors regenerated.

**2026-07-31 (XI) (the continuous half of the cross-ratio check executed and closed: UNIFLOW-DEGENERATE (v575 promoted) — the native exports are forced degenerate by the single-flow theorem, so the continuous half closes onto v425; both halves of the preregistered check are now executed)**

- **v575\_cr\_continuous\_uniflow.py** (FTR.CRCONT.01, 7 checks; probe 7/7, **UNIFLOW-DEGENERATE**). The frozen protocol exports, recomputed from primitives ( $F_{\text{pole}} = e^{-\Delta} = (2/3)^6$  via the v99 generator;  $F_{\text{Boltzmann}}$  the SAME contraction by the v425 single-flow theorem;  $F_{\text{QCD } b_3} = -7$ ;  $F_{\text{relic adiabatic } 1}$ ), form the quadruple  $\{(2/3)^6, (2/3)^6, 1, -7\}$  with a forced repetition — and a repeated quadruple gives  $\text{CR} \in \{0, 1, \infty\}$  in every corner assignment (theorem + all 24 permutations): the continuous half can never pass natively. It closes onto the single-flow reduction; the anchors  $(v_{\text{geo}}, M_R, C_p, \theta_i)$  stay **[O]** walls; the meaningful cross-ratio content remains the discrete half (v571). Contract markers unchanged.
- **Sync.** New ledger row FTR.CRCONT.01; update notes on FTR.CRDISC.01 and CONTRACT.F.01; contracts-paper protocol block extended with the execution result; suite  $568 \rightarrow 569$  scripts; diary; README/zenodo/website counters; mirrors regenerated.

**2026-07-31 (X) (closing the round: the winding form's content located exactly (v574  $\rightarrow$  11 checks — without it a four-column family survives, exactly one uniform: the compiler), and the half-sided RP restrictions shown blind (v572  $\rightarrow$  11 — the interacting construction is the one remaining RP route))**

- **v574 extended (probe 12/12).** Dropping the winding demand entirely (arbitrary first column), the self-code package leaves the four-column family  $\{(0, 1, 1), (0, 3, 3), (3, 3, 3), (5, 1, 5)\}$  — exactly

one uniform: the compiler. The winding demand *is* the uniformity choice (1 of 4). Final declared list of the reconstruction: involution existence, column uniformity,  $N_{\text{fam}} = 3$  (E8-glue).

- **v572 extended (probe 11/11).** The Gibbs kernels restricted to the Levi and radical sublattices have identical inertia on both twins across the sampled range: the RP-positivity selection resists every kinematic and half-sided design tried — the genuinely interacting construction (the v534 FK-quartet analogue) is the one remaining route, typed open.
- **Sync.** Ledger/registry/run\_all counts; tfpt\_1/tfpt\_2 passages; diary; mirrors regenerated.

**2026-07-31 (IX) (three final deepenings: the maximal ablation drops ( $E$ ) too (v574  $\rightarrow$  10 checks — the reconstruction needs only involution existence, the winding column with  $N_{\text{fam}} = 3$  and the self-code), the Gibbs-critical mode is identified as the seam-row Ext-carrier direction (v572  $\rightarrow$  10), and the per-atom profile of the dominance oscillates standing-wave-like (v573  $\rightarrow$  10))**

- **v574 extended (probe 11/11).** The maximal ablation: dropping the exponent demand ( $E$ ) entirely, the winding column  $3(1, 1, 1)$  plus integrality plus the self-code package (parabolic + ladder + saturation 81) still pin the compiler  $Q$  uniquely (29 ladder candidates, one hit). ( $E$ ), ( $M$ ), ( $L$ ) are ALL redundant at the declared bound — what stays declared is the involution existence, the winding form and  $N_{\text{fam}} = 3$  (independently anchored in the  $E_8$  glue,  $240 = 16 \times 5 \times 3$ ); GATE.QGEO.01 narrowed by five items, not closed.
- **v572 extended (probe 10/10).** The critical mode at the compiler twin's Gibbs temperature is identified: rank one, supported entirely on the seam row (the radical /  $A_2$ -Ext-carrier row), fixes the seam line, is a fixed point of the sheet flow ( $VX = X$ ), annihilated by the dominant projector — the Gibbs criticality lives exactly in the Ext-carrier slot: the typed resonance with the alignment bit deepens to a shared carrier.
- **v573 extended (probe 10/10).** The per-atom share of  $\det S$  oscillates across the window with a fixed sign pattern (positive at  $x \sim 0.3/0.8$ , negative past the midpoint), amplitudes growing with  $h$  while the net stays 1 — a standing-wave-like anatomy, measured with its ladder, no mechanism claim.
- **Sync.** Ledger/registry/run\_all; tfpt\_1/tfpt\_2 passages; tfpt\_prime\_front counters (4843 checks, T181 10/10); diary; mirrors regenerated.

**2026-07-31 (VIII) (two decisive deepenings: the anchor-Hamiltonian Gibbs pencil delivers the first DYNAMICAL distinction of the reconstruction bit (v572  $\rightarrow$  9 checks: the compiler twin is Gibbs-critical at  $s^* = (8 - 4\sqrt{19})/15$ , the alternative never), and the redundancy ablation shrinks the reconstruction inputs to three (v574  $\rightarrow$  9 checks: ( $M$ ) and ( $L$ ) fall relative to the self-code demands))**

- **v572 extended (probe 9/9).** Along the Gibbs family  $\omega_g \propto \text{tr}(e^{-gH})$  of the v566 anchor operator  $H = I + \Pi$ , the pencil determinant factors exactly differently on the twins — compiler  $\frac{6561}{4096}(s^2+16s+16)^2(15s^2-16s-16)$ , alternative  $-\frac{6561}{4096}(s^2+16s+16)^3$  ( $6561 = 81^2$ ) — and only the compiler's extra factor has a root inside the physical Gibbs range: the compiler twin's RP kernel degenerates at a finite critical temperature, the alternative twin's never does. Honest typing: a rank-drop marker, not a positivity selection; the interacting RP question stays open.
- **v574 extended (probe 10/10).** Redundancy ablation: relative to the self-code demands (parabolic + ladder + saturation 81) the monodromy budget ( $M$ ) and the seam-line demand ( $L$ ) are redundant at  $B = 6$  — dropping both widens the class to 1274 members and the self-code still picks exactly the compiler  $Q$ : the declared geometric input list shrinks to ( $S$ )-existence, ( $W$ )-form and ( $E$ ); GATE.QGEO.01 narrowed by four items, not closed.

- **Exploratory (diary only).** The pair-mechanism hunt on the reference window: no dominant pair exists ( $\max |\text{contribution}| \sim 0.006 \times \det S$  — consistent with the dead two-atom witness), and the fine structure by smaller atom shows the smallest primes (2, 3) dragging negative while medium primes (11..41) carry positive net.
- **Sync.** Ledger/registry/run\_all counts; tfpt\_1/tfpt\_2 passages; diary; mirrors regenerated.

**2026-07-31 (VII) (the four lines deepened in the same round: the dominance anatomy sharpened to heavy-pair window-scale coherence (v573  $\rightarrow$  9 checks), the RP blindness shown robust under the compiler flow (v572  $\rightarrow$  7), the Stage-B forcing decomposed by the spectrum ablation (v574  $\rightarrow$  8), and the continuous-half export protocol frozen on the contract surface)**

- **v573 extended (probe T181 9/9).** The excess is *scale-locked* (pairs  $\geq h/2$  carry it; bands 16..256 stay negative) and *weight-concentrated*: the top-10%-weight pairs alone net  $0.85\text{--}1.00 \times \det S$  on every selected window — essentially  $\det S$  itself; the lag-grid phase is flat (mechanism killed). The certificate target is typed precisely: heavy-pair coherence at window-scale separation.
- **v572 extended (probe 7/7).** The spectral blindness is robust under the compiler’s own flow: the  $V$ -semigroup-deformed kernels keep identical inertia on both twins for every declared state and every  $n = 1..3$  — a dynamical selection needs genuine interaction.
- **v574 extended (probe 9/9, full 35-spectrum scan).** The honest decomposition: freeing the exponent spectrum, the one surviving double-selfconsistent family is exactly  $\{1, 2, N\}$  with the compiler shape — the self-consistency forces the *sheet* spectrum  $\{1, 2\}$  and the shape, while the  $A_3$  exponent demand contributes exactly one datum,  $N = 3$ ; both factors load-bearing.
- **Contract surface.** The frozen export protocol for the continuous cross-ratio half preregistered (each instance exports the fixed-point multiplier of its native flow, cf. v425, before evaluation); design [X], markers unchanged.
- **Sync.** Ledger/registry/run\_all counts updated; papers (tfpt\_1, tfpt\_2, tfpt\_3, tfpt\_prime\_front counters 4842 checks, Paper II cross-reference row, tfpt\_research\_contracts); diary; website mirrors regenerated.

**2026-07-31 (VI) (all four review follow-ups executed: the cross-ratio check passes on its discrete half (v571,  $y_F = 53$  Möbius-forced), the RP form is integrally anchor-visible on the twins (v572), the dominance is typed long-range (v573, T181), and Stage B forces the winding normalisation (v574,  $N_{\text{fam}} = 3$  by double self-consistency) + ledger/paper/diary/website sync)**

- **v571\_cr\_discrete\_corners.py (FTR.CRDISC.01, 9 checks; probe 9/9, CR-PASSES).** The preregistered cross-ratio check (v566 S9), executed on its discrete half: the four solver corners are the in-repo kernels  $J, K, C, F = M(1, t)$ ,  $t = -2..1$  (v224), and the corpus’ one canonical declared solver reading — the Koide anchor response  $\ell(M) = a^T M \mathbf{1}$  (v183) — reads (26, 35, 44, 53):  $\text{CR} = 4/3$  exactly, and the fit-free reconstruction *forces*  $y_F = 53$  from the three upstream corners (a new exact cross-corner identity). Honesty theorem: every linear functional passes (the check separates linear from nonlinear exports); negative controls fail as preregistered (determinants 28/25, dominant eigenvalue 1.3222). The continuous-solver half stays open per CONTRACT.F.01.
- **v572\_rp\_bit\_form.py (DIAMOND.BIT.RPFORM.01, 6 checks; probe 6/6, ANCHOR-VISIBLE).** The kinematic RP form on the 81-twins: spectrally blind (identical inertia for every declared state; never PSD — no v534-style kinematic positivity selection exists, honest negative), but the *integral* Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor  $9 = N_{\text{fam}}^2$ ) — the bit is visible to the RP form only through anchor-coupled

states; trace discriminant  $-81^2$  on both twins; the dynamical RP question stays the open RP-slot.

- **v573\_pair\_band\_structure.py** (PRIME.PAIRBAND.01, 6 checks; probe T181 6/6, LONG-RANGE). Where the v570 dominance lives: every near band ( $< 16$  lag cells) and the diagonal drag negative on the declared subset; the far band carries the whole excess with stable net  $([1.55, 1.91] \times \det S, \text{gross growing to } 13\times)$ ; the scramble explodes  $\det S \times 49$  and collapses the far share — long-range coherence, arithmetic placement; a certificate must control far-pair coherence (near-diagonal budgets aim at the wrong sign).
- **v574\_stageb\_winding.py** (QGE0.WINDING.SELF.01, 7 checks; probe 7/7, FORCED). Stage B, first narrowing: the involution *type* is killed by a rank theorem (a two-dim-positive involution cannot carry the binary trace code),  $(E)$  bounds  $N \leq 3$ , the ladder self-consistency selects one member per  $N$  ( $28/16/14 \rightarrow 1/1/1$ ), and  $\text{sat} = N^4$  holds *only* at  $N = 3$  ( $81 = 3^4$ ;  $32 \neq 1$ ,  $512 \neq 16$ ): the double self-consistency pins  $(N, Q) = (3, \text{compiler } Q)$  —  $N_{\text{fam}} = 3$  relocates from declared input to consistency-forced; GATE.QGE0.01 narrowed by two items, not closed; no P2 reduction, markers untouched.
- **Sync.** Four ledger rows + update notes on CONTRACT.F.01, DIAMOND.BIT.SELECT.01, PRIME.PENCIL.SEPFLOOR.01, QGE0.QRECON.01; `tfpt_research_contracts` (CR block “executed, discrete half”), `tfpt_1/ tfpt_2` (RP-form and Stage-B passages), `tfpt_3` (pair-band audit block), `tfpt_prime_front` (verified layer  $30 \rightarrow 31$ , subsections `sec:v573/sec:t181`, counters 181 probes / 4839 checks), Paper II (cross-reference rows, “twelve modules”); diary entries; suite  $564 \rightarrow 568$  scripts; README/zenodo/website counters; mirrors regenerated.

**2026-07-31 (V) (the T179 open question closed on the surface: the spectator separation certified from closed scalars (FLOOR-CERTIFIED) + v570 promoted + backflow part T180 + ledger/paper/diary/website sync)**

- **Probe separation\_floor\_probe.py** (T180, 9/9, verdict FLOOR-CERTIFIED). The open question v569 named — does the spectator eigenvalue  $\lambda_1$  of the relative pencil stay separated from 1? — answered on the reachable surface by *reduction*: two one-line theorems ( $P = \det S / \det B > 0, T = \text{tr } Z > 0 \Rightarrow \lambda_1 \geq \sqrt{P}$ ;  $P \leq 0 \Rightarrow \lambda_1 \geq T$ ) turn the separation into a closed per-window floor, and on all 70 declared T170 frame-A windows the floor clears the margin,  $\min F = 2.130 \geq 1.05$  (measured  $\min \lambda_1 = 4.534$ ): the separation follows from  $\det S / \det B$  and  $\text{tr } Z$  alone. The dominance is large and grows with depth ( $P$  median 30.1, slope  $h^{0.90}$  on the surface ladder), and  $1 - T + P = \det(B - S) / \det B$  links trace and determinant dominance exactly. *Honest negative*: the cheap two-atom witness route is dead —  $\det S$  is itself a pair-level cancellation object (Cauchy–Binet exact; negative pair mass up to  $13.7 \times \det S$ ; best single pair  $0.13\text{--}0.34 \times \det B$  against the  $\geq 1.10$  needed) — so the  $h$ -uniform dominance stays open, typed as pair-level size-vs-cancellation. Must-break: the position scramble flips  $\text{tr } Z$  to  $-78.28$ .
- **Promotion v570\_separation\_floor.py** (PRIME.PENCIL.SEPFLOOR.01, 8 checks,  $\sim 13$  s). New ledger row; update note on PRIME.PENCIL.ONEMODE.01; suite  $563 \rightarrow 564$  scripts. **Named limits**: declared finite surface only — the  $h$ -uniform dominance  $\det S \geq (1 + c)^2 \det B$  is the same open question v569 named, in sharper closed-scalar form; Paper-II Problem 7.1 untouched; no rate, no bound, no RH statement.
- **Sync.** `tfpt_prime_front` (verified layer  $29 \rightarrow 30$  modules, subsections `sec:v570` and `sec:t180`, index T11–T180, counters 180 probes / 4833 checks); Paper II (cross-reference row, reproducibility “eleven modules”); `tfpt_3` audit block; diary entry in `experiments/next.txt`; README/zenodo/website counters 564; mirrors regenerated.

**2026-07-31 (IV) (the relative pencil on the declared Paper-II surface: the one-mode question of the self-code review answered (ONE-MODE, labels corrected))**

+ v569 promoted + backflow part T179 + ledger/paper/diary/website sync)

- **Probe lambda\_pencil\_probe.py (T179, 4/4, verdict ONE-MODE).** Priority 5 of the self-code review, executed on the real surface: the v566 S8 pencil identities  $\det(B-S) = \det B (1-\lambda_1)(1-\lambda_2)$ ,  $D(B, S) = \det B (\lambda_1+\lambda_2)$ ,  $\det S = \det B \lambda_1 \lambda_2$  are machine-tight on all 70 declared T170 frame-A windows (worst residual  $2 \times 10^{-14}$ ; the v563 surface imported read-only, bit for bit). *The labels are the reverse of the review guess:* the locking mode is  $\lambda_2$ , hugging 1 (median 1.000021), while the spectator  $\lambda_1$  is large and uniformly separated —  $\min |1-\lambda_1| = 3.53$ ,  $\lambda_1 \in [4.53, 180.64]$ , every window above 1. *The one-mode answer:* exactly one eigenvalue can approach 1, so the two-dimensional cancellation of Paper II reduces to the single scalar  $1-\lambda_2$  on the whole reachable surface; the  $h = 540$  reference window reads  $1-\lambda_2 = -2.27 \times 10^{-5}$  (reproducing  $\det \hat{A} = 4.22 \times 10^{-3}$ ), and the position scramble explodes  $|1-\lambda_2|$  by  $\times 1.9 \times 10^6$  while the identities keep holding: the locking is arithmetic placement.
- **Promotion v569\_lambda\_pencil\_onemode.py (PRIME.PENCIL.ONEMODE.01, 6 checks, ~2 s).** New ledger row; update notes on PRIME.PAPER2.READOUT.01 and DIAMOND.PARA.SELFCODE.01; suite 562  $\rightarrow$  563 scripts. **Named limits:** declared finite surface only — Paper-II Problem 7.1 (the uniform quantifier) is untouched, typed not bounded; whether  $\lambda_1$  stays separated as  $h$  grows is a new named open question, strictly weaker than Problem 7.1; no rate, no bound, no RH statement.
- **Sync. tfpt\_prime\_front** (verified layer 28  $\rightarrow$  29 modules, subsections **sec:v569** and **sec:t179**, index T11–T179, counters 179 probes / 4824 checks); Paper II (cross-reference row for Problem **prob:R1**, reproducibility “ten modules”); diary entry in **experiments/next.txt**; README/zenodo/website counters 563; mirrors regenerated.

**2026-07-31 (III) (the residual reconstruction bit located and given a canonical witness: the ladder self-consistency selects the compiler  $Q$  uniquely (SELECTOR-FOUND); the split geometry excluded; Priority 2 sharpened + v568 promoted + ledger/paper/diary/website sync)**

- **Probe bit\_selector\_probe.py (10/10, verdict SELECTOR-FOUND).** Priorities 2+3 of the self-code review, fused with the v567 residual bit. *The bit is integrally visible:* the two 81-twins ( $q_{32} = 2=|\mathbb{Z}_2|$  vs 0) generate *different* integral orders (equal index, equal Smith divisors, different lattices). *The selector:* the dominant mode of the sheet direction is  $v(c) = (1, 2, 2c)$  identically, and the *ladder self-consistency* — the dominant mode reproduces its own eigenvalue as the geometric ladder  $(2^0, 2^1, 2^2) = (1, 2, 4)$  — selects  $c = 2$ , the compiler  $Q$ , uniquely on all 28 members of the carrier-free class (no 81-pin, no column budget); the witness is prior structure (binary AnchorLadder steps  $2^n$ ,  $2 = |\mathbb{Z}_2|$ , the v566 trace code). *The split geometry located and excluded:* a module complement exists iff  $q_{32} = 1$ , and that member’s algebra collapses to dimension 5 — the  $A_2$  Ext arrow is nonzero on the whole reconstruction class; neither twin splits: the bit toggles *which* nonsplit integral order is realised, not split-vs-nonsplit. *Priority-2 candidates, honest both ways:* the  $\mu_4$ -grading fails (the order is not graded; the 81 stays a fingerprint, the kill condition does not fire), while left  $V$ -multiplication induces a canonical  $\mathbb{F}_3$ -action on the cokernel  $(\mathbb{Z}/3)^4$  — *computed:* its characteristic polynomial is  $x(x-1)^2(x+1) \bmod 3$  (eigenvalues  $\{0, 1, 1, -1\}$ , NOT the regular  $\mu_4$  pattern  $x^4-1$ ;  $U$  acts as zero; both twins identical) — after two candidates the 81 stays a fingerprint.
- **Promotion v568\_bit\_selector\_ladder.py (DIAMOND.BIT.SELECT.01, 11 checks, ~1 s).** New ledger row; update notes on QGEO.QRECON.01 and DIAMOND.PARA.SELFCODE.01; suite 561  $\rightarrow$  562 scripts. **The physical selection is not derived:** why the dominant mode should satisfy the ladder self-consistency is the relocated open question — the same structural slot as the RP-selection of the alignment bit (v528/v534); no marker moves, no P2 reduction.
- **Sync. tfpt\_1** (selector appended to the Stage-A passage), **tfpt\_2** (a dedicated bit paragraph



after the audit paragraph); diary entry in `experiments/next.txt`; README/zenodo/website counters 562; mirrors regenerated.

**2026-07-31 (II) (the No-Carrier-Reconstruction audit, Stage A: the anchor self-code is carrier-free, and  $Q$  reconstructs up to ONE binary sheet-coupling choice (RECONSTRUCTS-TO-FAMILY) + v567 promoted + ledger/paper/diary/website sync)**

- **Audit probe `carrier_free_q_reconstruction_probe.py` (10/10, verdict RECONSTRUCTS-TO-FAMILY).** Priority 1 of the self-code review executed. *The circularity located:* v11's uniqueness certificate for  $Q$  uses row sums (4, 5, 6), column sums (9, 5, 1) and  $\chi_Q = (t-1)(t^2-5t+3)$  — the 5 =  $g_{\text{car}}$  in the budgets and in  $\chi_Q$ ; all of it *forbidden* in the audit. Under the corpus' own carrier-free data (sheet involution, v10 winding normalisation,  $A_3$  exponent spectrum, monodromy budget, integrality, seam-line demand): exactly  $52 \rightarrow 28$  (parabolic) admissible reconstructions — and **all 28 code the same anchor** ( $\chi_{H'} = x^3 - 4x^2 + 5x - 2$ , spectrum  $(1, 1, 2) = a$ ): the anchor self-code sits *upstream* of the carrier budgets. The single family invariant  $81 = N_{\text{fam}}^4$  (saturation index) pins  $Q$  to a *two-element* residual family ( $q_{32} \in \{2=|\mathbb{Z}_2|, 0\}$ , difference  $2E_{32}$ ) on which every classical carrier-free certificate coincides (anticommutator spectrum  $\{6, -4, -2\}$ , commutator charpoly  $t(t^2+12)$ , determinant, saturation) — the one separating certificate is the carrier column budget (9, 5, 1): exactly the forbidden input. Ablations: dropping the winding input widens the bound-3 set  $28 \rightarrow 1376$  (real load); bound-stable at  $B = 9$ .
- **Promotion `v567_carrier_free_q_reconstruction.py` (QGEO.QRECON.01, 9 checks, ~1 s; integer prefilters take the probe's 555 s to 1.2 s).** New ledger row; update notes on FLAV.QGEO.01 and DIAMOND.PARA.SELFCODE.01; suite 560  $\rightarrow$  561 scripts. **Stage A only, no marker moves:** whether the geometric inputs themselves derive from  $\mu_4/D_4/H^1$  parabolic geometry is exactly GATE.QGEO.01 (the audit is relocated, not closed); the carrier-free selector of the residual bit is open, its resonance with the split/nonsplit alignment bit a typed observation; no P2 reduction, AX.P2.01 untouched.
- **Sync.** `tfpt_1` (Stage-A result appended to the self-code paragraph), `tfpt_2` (a dedicated audit paragraph after the self-code block); diary entry in `experiments/next.txt`; README/zenodo/website counters 561; mirrors regenerated.

**2026-07-31 (I) (the parabolic anchor self-code: an external structural review audited (SELFCODE-VERIFIED) — the anchor  $a = (1, 1, 2)$  as the spectrum of a sheet-direction operator, double-coded by flag/Levi/radical, unique in its parabolic family; the 81 as a saturated lattice index; the  $A_2$  bit carrier + v566 promoted + ledger/paper/diary/website sync)**

- **Audit probe `parabolic_selfcode_audit_probe.py` (23/23, verdict SELFCODE-VERIFIED).** An external review proposed exact statements about the v218 sheet-diamond direction operators  $U = Q \text{diag}(1, 0, 0)$ ,  $V = Q \text{diag}(0, 1, 1)$ ; every claim holds exactly against the corpus objects, *including the saturation check the review had not run* (the integral order stabilises at word length 4; the word-basis Smith form survives). *The parabolic theorem:*  $\mathcal{A} = \mathbb{Q}\langle I, U, V \rangle$  is the full  $2|1$  line-stabiliser algebra (dim 7, word determinant  $-81$ , multiplicatively closed, mutation-fragile). *The double self-code:*  $H = I + \frac{1}{2}V(V - I)$  has spectrum exactly  $\{1, 1, 2\} = a$ , and  $\chi_H = x^3 - 4x^2 + 5x - 2$  carries  $(4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$  and  $= (\text{flag}, \text{Levi}, \text{radical})$  of  $\mathcal{A}$  simultaneously. *The uniqueness theorem:*  $(x - m)(x^2 - mx + 1)$  with all-positive-integer roots forces  $m = 2$  —  $a = (1, 1, 2)$  is the only anchor this family can code. *The 81:* the saturated order has elementary divisors  $(1, 1, 1, 3, 3, 3, 3)$  — index  $3^4 = 81$ , cokernel  $(\mathbb{Z}/3)^4$  — equal to the v218 determinant-staircase sum ( $N_{\text{fam}}^4$ ); the mark-local mechanism stays open. *The  $A_2$  bit carrier:* the basic algebra is the  $A_2$  path algebra with  $\dim \text{Ext}^1 = 1$ : exactly two module

classes (split / one nonsplit) — a canonical carrier for the alignment bit; the selection stays with RP/geometry. *Re-encoding typed as such:*  $p_n(a) = 1 + \text{tr}(V^n)$  (AnchorLadder through  $V$ ; 240 and rank 8 as trace readouts). *Test design:* the cross-ratio check  $\text{CR} = 4/3$  for the four CONTRACT.F.01 solvers (universal-progression honesty wired; negative control 28/25). *Reformulation:* the Paper-II rank-3 form as  $(1, 2)$  Lorentz geometry with the relative-pencil identities (eigenvalue locking as the named prime-front instrument).

- **Promotion** `v566_parabolic_anchor_selfcode.py` (DIAMOND.PARA.SELFCODE.01, 14 checks,  $\sim 1$  s, exact). New ledger row; update notes on DIAMOND.AXIS.01 (self-code + 81-index cross-link) and CONTRACT.F.01 (the preregistered cross-ratio check added to the contract surface); suite 559  $\rightarrow$  560 scripts. **No P2 reduction and no marker moves:** the “ $g_{\text{car}} = \dim(\mathcal{A}/J)$  derived” route is conditional on the open *No Carrier Reconstruction* audit (rebuild  $Q/V$  without carrier-aware inputs) — the review’s Priority-1 workstream, recorded as the named open door; the 2..9 dimension table is typed as compression (no-free-pattern).
- **Sync.** `tfpt_1` (self-code paragraph after v23), `tfpt_2` (self-code + 81-index +  $A_2$ -carrier paragraph after the v218 block), `tfpt_research_contracts` (cross-ratio check in the CONTRACT.F.01 section); diary entry in `experiments/next.txt`; website counters and mirrors regenerated; README/zenodo 560.

**2026-07-30 (II) (WOIT  $\beta_3$  executed: the  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality typed against  $\sigma_{\text{std}}$  — the OS cut and the  $(2, 2)$  signature forced, the kill branch empty by algebra, the induced member = the  $\Theta_{\text{phys}}$  carrier (SUCCESS) + v565 promoted + contract/paper/diary/website sync)**

- **Discovery probe** `woit_os_beta3_pt_dual_probe.py` (25/25, 2.4 s, verdict SUCCESS per the frozen preregistration; contract wording verbatim: success = Woit’s Minkowski conjugation identified with the family-D conjugation on the quotient, kill = the duality forces a clock-centralising (family A) structure). *The class (block theorem):* a  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality is a nondegenerate Hermitian form  $\Phi$  on  $\mathbb{C}^4$ ; clock compatibility has entrywise residual  $(\lambda_j \lambda_k - 1)\Phi_{jk}$  and forces the four cross-eigenspace entries to zero: two  $2 \times 2$  Hermitian blocks,  $16 \rightarrow 8$  real parameters (the deck adds nothing; the wrong clock lift breaks the split). *The dichotomy and the forced  $(2, 2)$ :* the  $\rho_{\text{tw}}$  pairing  $M^\dagger \Phi M = \varepsilon \bar{\Phi}$  splits the class exactly in two — the Euclidean branch ( $\sigma_{\text{std}}$ ’s home) has seam-line norm  $a(|z|^2 + 1)$ , no null point, no OS cut ever; the Minkowski branch has norm  $a(|z|^2 - 1)$ : *the null locus meets the seam line exactly in the seam circle* for every member (the OS cut is forced, not imposed), and  $\Phi_1 = -\bar{\Phi}_i$  exactly (charpoly mirrored), so the total signature is  $(2, 2)$  for every nondegenerate member — Woit’s Minkowski signature is a theorem of clock +  $\rho_{\text{tw}}$  pairing, not an input. *The kill is empty by algebra:* every compatible duality inverts clock and deck (family D); family A would need  $\rho_4^2 = \text{scalar}$ , but  $\rho_4^2 = \text{diag}(-1, 1, -1, 1)$  — no solution, decided not scanned;  $s_{\Phi}^2 > 0$ : the Kramers-free class. *The identification:* every seam-preserving Minkowski member induces the same conjugation  $z \rightarrow -\bar{z}$  ( $\mu = -1$ , fixed points  $\pm i$ ), perpendicular to  $\sigma_{\text{std}}$ ’s  $z \rightarrow +\bar{z}$ ; under the declared RP-side placement (exact rational- $\pi$  arithmetic, offsets  $\mp \pi/2$ ) it is  $\theta_\perp$  — the mirror whose descent v524 certified:  $\Theta_{\text{phys}}$  is the quotient shadow of the duality (regressions re-verified: Gram PD  $(16, 0, 0)$ , anti-unitary with  $\eta_\perp = +i$ ,  $\Theta^2 = +1$  per sector, time reversal, torsor closure). Role separation complete:  $\sigma_{\text{std}} \sim$  the OS cut (the metric), the  $\Phi$ -twisted duality  $\sim$  the descending mirror. Free/quotient level only;  $\mathcal{A}_{\text{hol}}$  untouched; all seven kill tests stay live; no marker moves.
- **Promotion** `v565_woit_beta3_pt_duality.py` (WOIT.BETA3.DUALITY.01, 22 checks,  $\sim 2$  s, sympy-exact + two 40-digit inertia certificates). New ledger row; WOIT.OS.TWISTOR.01 updated ( $\beta_3$  executed,  $\gamma$  next); suite 558  $\rightarrow$  559 scripts.
- **Sync.** `tfpt_research_contracts`: title line ( $\alpha/\beta_1/\beta_2/\beta_3$  executed,  $\gamma$  next), the  $\beta_3$  executed block, the roadmap item, the status paragraph, the straddle-hurdle wording ( $\gamma$  and beyond);

tfpt\_3: new Step 32, story counters thirty-one → thirty-two (title, intro, preamble), the Woit-bridge paragraph gains the  $\beta_3$ -stage status; diary entry in `experiments/next.txt`; website: `papers.ts` WOIT milestone list + v565, `suite.ts` 558 → 559, `ScriptIndex/changelog` mirrors regenerated; `README/zenodo` counters 559.

**2026-07-30 (I) (T178 backflow: the frame door of the CP mirror decided — R1 by theorem, R2 by anchor, the degeneracy explained (FIBER-PINNED) + v564 promoted + v563 Paper-II readouts registered + ledger/paper/diary/website sync)**

- **Prime-front diary T178** (`cp_frame_carrier_probe.py`, contract `FRAME.CARRIER`, 31/31, 2.4 s, verdict **FIBER-PINNED**, sandbox; fifth reverse-flow probe — **T177's two open items are pure finite algebra and are *decided*, not measured**). *R1, natural half, negative by theorem*: the fiber acts on the frame by a scalar on the  $n$ -row, so every homogeneous frame block factorises exactly ( $P(\rho n) = \rho^m P(n)$ , checked on fully generic degree-1..4 polynomials with symbolic coefficients), and the metric-free invariants of  $(\mathbf{1}, d, n)$  have *one* generator (bracket enumeration exact: the six permutations =  $\pm\Delta$ , the other 21 vanish identically in a generic  $n$ ; Weyl 1946 FFT as the cited class boundary) — every scale-free ratio is constant, so no frame-coupled scale-free  $\pi/3$ -carrier exists without a metric: T177's obstruction sentence is now a theorem. *R1, metric half, non-canonical*: declared metrics admit frame-coupled Tier-2 carriers, but they split into two character classes  $((-1, -1, +1)$  vs  $(-1, -1, -1))$  and the admissible normalisers contain the continuum  $q_t = t n_1^2 + n_2^2 + n_3^2$  (values  $441 \sin(\pi/3)/(25t + 117)$ , strictly monotone) — no canonical frame factor; the factors 21/8, 21/2, 441/142 are normaliser-relative frame data with the anti-numerology fence wired as a check. *The single-sine lemma (why T177 ended DEGENERATE)*:  $\{|\operatorname{Im} \rho_n^k|\}$  has exactly one nonzero element iff  $n$  is crystallographic ( $n \in \{1, 2, 3, 4, 6\}$ ,  $\varphi(n) \leq 2$ ), by  $\sin(4\pi/n) - \sin(2\pi/n) = 2 \sin(\pi/n) \cos(3\pi/n)$  (only zero at  $n = 6$  on  $n \geq 5$ ) — the compiler's two clocks  $\mu_4$  ( $j = 1728$ ) and  $\mu_6$  ( $j = 0$ ) are exactly the two maximal single-sine fibers, so value-based pinning of functionals is impossible there by theorem: T177's verdict was *forced*. *R2 selected by the anchor*: the Jarlskog invariant carries the full character tuple  $(-1, -1, -1)$  under (conjugation, sheet flip, deck reflection), each certified at 40 dps — matching the sheet-odd orientation class only; the unique anchor-matching *canonical* carrier is the fiber-pure  $\operatorname{Im} \rho = \sin(\pi/3)$  (selected, conditional on the anchor — not derived). No marker moves; `REG.FREEZE.01` untouched; the CP residual is relocated with a proof, not closed.
- **Promotion v564\_cp\_fiber\_pin.py** (`CP.FIBER.PIN.01`, 22 checks,  $\sim 2$  s, **sympy-exact**). The exact cores as one load-bearing module: the generic Euler factorisation, the one-generator bracket enumeration, the character split and the  $q_t$  continuum (anti-numerology as a check), the single-sine lemma (identity + enumeration to  $n = 24 + \text{totient criterion}$ ), and the anchor tuple with the selection and the `CP.MU6.01` consistency control. New ledger row `CP.FIBER.PIN.01`; update note on `CP.CHANNEL.IDENT.01` (R1 decided / R2 selected, markers unchanged).
- **Housekeeping: v563\_paper2\_readouts.py** (`PRIME.PAPER2.READOUT.01`, 24 checks) **registered**. The Paper-II readout closure was written in the previous session but never entered `run_all/registry/ledger`; registered now with its own ledger row. Paper II (`parity_toeplitz_classification`): the cross-reference list of non-recomputed readings narrows from three to one (the fitted exponents of the rate measurement), four new cross-reference rows, reproducibility section now “nine modules v555–v563”.
- **Sync**. Diary entry Teil 178 in `experiments/next.txt` (counters 178 probes / 4820/4820 checks); `tfpt_prime_front`: verified layer 27 → 28 modules (+ v563 subsection), new T178 subsection (`sec:t178`), T177 rest-list pointer updated, probe index T11–T178 + T178 row, counters and phase-2 range labels; `tfpt_5_redteam`: Target-D keybox (the fiber location now a theorem) and the CP-location winbox (frame door decided, single-sine lemma, anchor selection)

citing v564\_cp\_fiber\_pin.py; suite 556 → 558 scripts; website mirrors regenerated.

**2026-07-29 (XXII) (prime-front phase 2, the endgame: T176 — the density curve sits at zero and the measurement programme closes (SITS-AT-ZERO) + v562 dense-limit identities promoted + paper/diary/website sync)**

- **Prime-front diary T176 (dense\_limit\_probe.py, contract DENSE.LIMIT, 24/24, 118.7 s, verdict SITS-AT-ZERO, sandbox; lands *after* T177 — part numbering by contract, series counters raised by T176's checks).** The big sieve: ATOM\_MAX  $1.2 \times 10^6 \rightarrow 2.5 \times 10^7$  (factor 20.8; cost 0.08 s / 59.7 MB peak — the sieve was never the binding constraint), 1 566 727 prime-power atoms, density ceiling 361 → 6120 (6118 reached), 14 532 cells on 692 anchors, 0 indefinite. *The extended curve* (eight ratio-4 bins, cluster-robust):  $+0.8886 \pm 0.1027 \rightarrow +0.5148 \rightarrow +0.1948 \rightarrow +0.3922$  (declared monotonicity violation, explained by curvature)  $\rightarrow +0.1520 \rightarrow +0.0579 \rightarrow +0.0376 \pm 0.0410$  (dens  $\sim 779$ ,  $0.9\sigma$ )  $\rightarrow +0.1046 \pm 0.0881$  (dens  $\sim 2512$ ,  $1.2\sigma$ ); pooled over the new bins  $+0.0496 \pm 0.0372$  ( $1.3\sigma$ ) — no sign change, no stabilisation above zero; density slope  $-0.0725 \pm 0.0090$  ( $8.0\sigma$  over 4.0 decades); *the plateau window narrows*  $0.264 \rightarrow 0.074$  ( $2\sigma$ ), a factor 3.6; all six old bins reproduce at  $0.0\sigma$  against T175. Both curvature channels survive ( $\log h$   $+0.1529 \pm 0.0258$ ,  $5.9\sigma$ ;  $\log$  dens  $-0.0434 \pm 0.0060$ ,  $7.3\sigma$ ); none of the six preregistered reparametrisations makes it one exponent. *R1 (T175) closed*: the Feynman–Hellmann form  $d\lambda_{\min} = v^\top d\hat{A} v$  reproduces the rebuild to  $3.37 \times 10^{-6}$  at the minimum of a declared step ladder with the U-shape exhibited; dDEL/DEL carries 84%. *The  $300\times$  harmonic anomaly explained as a pole*: PD survives 2% of the phase period ( $\delta_{PD} = 5.9 \times 10^{-3}$ , product  $5.8 = O(1)$ ), and the two lowest modes are already 13.4% apart at  $\delta = 0$  — the intervention runs along a near-degeneracy, it does not create one. *Anti-promotion*: the estimator is LADDER-DEPENDENT at the 0.20 level — every quoted bin deficit must cite its rung ladder. True zero / power-law approach / low plateau stay indistinguishable (one more decade of window would cost ATOM\_MAX  $\sim 2.5 \times 10^9$ ); **the phase-2 measurement programme closes as planned** — no follow-up probe; the work continues in the classification papers and the backflow lines.
- **Promotion v562\_dense\_limit\_identities.py (PRIME.DENSE.LIMIT.01, 16 checks,  $\sim 0.5$  s, deterministic, small sieve).** The exact cores per instance: the Feynman–Hellmann identity with the step-ladder U-shape check and must-break mutations (R1.CLOSED); the phase-pole mechanism (PD-survival window, near-degeneracy at  $\delta = 0$ , sign change bracketed); sieve consistency at construction level (complete-comb lag vectors bit for bit under the bigger table; the truncated must-break moves); and the curve estimator as a deterministic procedure *with the ladder caveat wired as a check and into the ledger row*. SITS-AT-ZERO typed as a MEASUREMENT with the ladder caveat; the plateau/zero distinction explicitly OPEN; no RH statement. Suite 555 → 556 scripts.
- **Sync.** Diary entry Teil 176 in experiments/next.txt (counters 177 probes / 4789/4789 checks; phase-2 closure noted); prime-front paper T176 subsection (the extended curve as a table, the sieve result, the pole finding, the ladder caveat; keybox “After T176: the measurement programme closes”) + probe-index row + abstract counters + the final “Where phase 2 now stands” paragraph; website: prime-front feed entry T176 + v562 promotion entry, status counters, ScriptIndex / changelog mirrors regenerated.

**2026-07-29 (XXI) (T177 backflow: the CP mirror — two invariants carry  $\pi/3$ , four CP ledger rows narrowed, none upgraded (DEGENERATE) + v561 CP channel identities promoted + ledger/paper/diary/website sync)**

- **Prime-front diary T177 (cp\_invariant\_probe.py, contract CP.INVARIANT, 49/49, 0.2 s, verdict DEGENERATE, sandbox; the reverse flow resumed while T176 runs).** The gauge-degree toolkit of T173/T174 turned on the four CP ledger rows (REDTEAM.D.01, E8.CHAN.01, CP.MOD.01, DUAL.FRAME.01). All four constructions rebuilt exactly from the

compiler primitives ( $\delta = \pi/3 + 3\lambda_C^2 = 1.198231638$  rad certified against v88/v220;  $\det(\mathbf{1}, d, n) = 21$ ;  $J = 3.327 \times 10^{-5}$  with the CKM unitary to  $10^{-41}$ ). *New exact identity*: on the hexagonal unit  $\rho = e^{i\pi/3}$  the phase channel carries  $\sum_k \rho^{d_k} = 4\rho$  ( $\arg = \pi/3$  exactly, sheet-even — the  $E_8$  degrees are even) against the magnitude channel  $\sum_k \rho^{m_k} = 4$  ( $\arg = 0$  exactly, sheet-odd — the exponents are odd). *Tier-1 invariance is empty by algebra* (both kernels commute with conjugation, so every  $\pi/3$ -carrier is conjugation-odd); *two* inequivalent Jarlskog-class invariants carry  $\pi/3$  (sheet-even channel angle vs sheet-odd orientation), neither coupled to the frame — what is frame data is the oriented volume 21 and the sign, not the angle. The Jarlskog benchmark passes the invariance criteria and fails only the  $\pi/3$ -carrier test (its content is  $\delta$ ), validating the criterion instead of the value.

- **Four ledger rows narrowed, NO marker moves.** REDTEAM.D.01:  $\arg \prod c = 0$  exactly on both sectors; the residual is a *fiber* datum, not a magnitude datum (branch blindness = order divides  $N_{\text{fam}} = 3$ ). E8.CHAN.01: the exact hexagonal realisation + parity typing recorded; the phase-channel angle is sheet-blind (cannot distinguish quark from lepton). CP.MOD.01: scope narrowed — the invariant part of the frozen reading is the leading term  $\arg \rho = \pi/3$ , the full  $\delta$  is branch/frame-relative (no character; REG.FREEZE.01 untouched). DUAL.FRAME.01: the orientation *reading* is degree +1 in the frame normalisation ( $21 \sin(\pi/3)$  is frame data); the invariant core is the fiber datum  $\text{Im } \rho = \sin(\pi/3)$ , the sign a declared orientation. Prose narrowed in `tfpt_5_redteam` (Target D keybox + the located-residual box).
- **Promotion v561\_cp\_channel\_identities.py (CP.CHANNEL.IDENT.01, 25 checks, ~1 s, sympy-exact).** The exact cores as one load-bearing module: the hexagonal channel identity with parity typing and mutation controls ( $e^{i\pi/5}$ , mutated lists) plus the honest caveat wired as a check; Tier-1 emptiness as algebra; the frame removal ( $\text{Im}(\rho \det)/\det = \text{Im } \rho$  exactly, degree +1 account); the two Tier-2 characters as verified homomorphisms; the Jarlskog anchor ( $J$  passes rephasing/CP-oddness, fails only the carrier test). Narrowing statements only — no CP derivation claimed, R1/R2 explicitly OPEN. Suite 554  $\rightarrow$  555 scripts.
- **Sync.** Diary entry Teil 177 in `experiments/next.txt` (counters 176 probes / 4765/4765 checks; T176 and Paper II run in parallel); prime-front paper T177 reverse-flow subsection + probe-index row + counters; website: prime-front feed entry T177 + v561 promotion entry, status counters, DAG node, `ScriptIndex` / changelog mirrors regenerated.

**2026-07-29 (XX) (prime-front diary T175 (the placement phases are real and causal but non-smooth, the heterogeneity dissolves into the error bar, and the densest reachable bin is consistent with zero — PHASES-RESIST) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T175 (phase\_placement\_probe.py, contract PHASE.PLACEMENT, 38/38, 20.2 s, 1504 cells on 188 anchors — every prime power in  $[40, 1090]$  is an anchor, no selection, no frame rule anywhere — crossed with the T174 ladder  $h = 160 \dots 1280$  plus the declared floor  $h = 128$ ; verdict PHASES-RESIST, sandbox).** (1) *The phases.* The preregistered anchor invariant  $\theta_p = \text{frac}(\log p/\alpha)$  is unusable and the probe says so out loud (a smooth monotone function of  $\alpha$ , Rayleigh  $Z = 1296$  — a regression on it would be an  $\alpha$ -regression in disguise); only the cell phase  $\varphi_p(h) = \text{frac}(h\theta_p)$  is free and equidistributed (largest  $Z = 1.41$  on 1316 cells), with the generating identity  $\text{frac}(\log p/D) = \text{frac}(h \text{frac}(\log p/\alpha))$  certified at  $1.1 \times 10^{-13}$  (Weyl 1916). *The phases are real in log R*:  $F = 10.80$  against a within-rung-scrambled null with 95% point 1.69, +10.3% held-out residual variance under a rank-parity split, a composite placebo family at  $F = 0.81/-0.0275$ , amplitudes monotone in  $p$  (0.018 at  $p = 5$  to 0.141 at  $p = 19$ ). *But they are not the heterogeneity channel*: residual  $\chi^2/\text{dof} = 2.52$  from 3.14 (target  $< 2$  missed), the held-out excess scatter gets *worse* (−15.4%), the deficit untouched (0.18 $\sigma$ ); the exploratory scan excludes every other anchor statistic (all improve in sample and worsen held out). *What resolves the heterogeneity is the error bar*: with jackknife

slope errors the excess scatter goes  $0.1224 \rightarrow 0.0000$  and the robust pull spread  $1.41 \rightarrow 1.22$  (model-free interleaved half-split  $0.2722$  against  $sd = 0.2789$ ) — T174's  $\chi^2$  was substantially an OLS independence assumption, not a physical spread; *the  $5\sigma$  deficit is unaffected* (its bar never came from  $se_n$ ). (2) *The dense limit*. The corner is exhausted ( $\text{dens} \leq 361$  forced by the sieve cap and  $h \geq 128$ ; 359 reached); the deficit( $\text{dens}$ ) curve over three decades (6 declared ratio-4 bins, cluster-robust) falls  $+0.982 \pm 0.213 \rightarrow +0.576 \rightarrow +0.207 \rightarrow +0.194 \rightarrow +0.088 \rightarrow -0.057 \pm 0.132$  — monotone within  $2\sigma$  at every step, density slope  $6.1\sigma$ , *the densest reachable bin consistent with zero*; the extrapolation fence is explicit: a true zero, a power-law approach and a plateau below  $0.132$  are all consistent — undecidable under this sieve. The sharpened confound test finds *both* channels significant ( $\log h +0.2435 \pm 0.0670$ ,  $\log \text{dens} -0.0631 \pm 0.0172$ ; curvature  $a_2 = -0.137 \pm 0.029$ ) — T174's density driver survives but is no longer unique; the deficit is not a single exponent. Unplanned load-bearing finding: mass-preserving alternative assemblies are *inadmissible* (lump 0%, wide 46%/0% PD survival); the admissible continuous deformation gives sensitivities  $+0.384$  (sparse) vs  $-1.054$  (dense), ratio 0.4 — the sparse corner is *not* an assembly artefact. (3) *Rule proposal + intervention*.  $\text{dens} \geq 1$  proposed for the chain convention ( $+0.722 \pm 0.094$  unresolved vs  $+0.099 \pm 0.019$  resolved,  $6.5\sigma$  separation, 6% of cells; explicitly not retroactive); comb scramble destroys 12/12; *the new phase intervention* (with an exact  $\delta = 0$  rebuild control): displacing the six family atoms inside their cells gives  $d \log R / d\delta = 879$ , linear to a factor 1.011 over four declared decades — *the placement phase is causal for  $R$* , the first interventional statement of the series — and it overturns the *shape* of the D1 result: the fitted first harmonic predicts only 2.89 (a factor  $\sim 300$  too small), positive definiteness dies for  $\delta \geq 10^{-3}$ ,  $\text{GAP} = 1.6 \times 10^{-9} - 4.8 \times 10^{-6}$  — the Schur floor is a near-degeneracy and  $\log R$  is necessarily *non-smooth* in the phases. Candidates P175-1–P175-8 PENDING, no promotion. The rest: the closed-form  $d\lambda_{\min} = v^T d\hat{A} v$  response (R1); *the dense-corner zero needs a larger sieve, nothing else* (R2); the curvature vs the exponent reading (R3); an  $h$  floor for the positivity margin (R4); the last 22% of the pull spread (R5). T176 (DENSE.LIMIT, `dense_limit_probe.py`) is running — the larger sieve. Diary counters 175 probes / 4716 checks (series (CLXXXVI)).

- **Paper + website sync.** `tfpt_prime_front`: new phase-2 subsection (T175: the cell-phase construction with the invariant trap said out loud, the causality finding as key result, the error-bar resolution of the heterogeneity as the honest correction, the density curve with the zero-consistent end bin and the extrapolation fence, keybox “After T175” with the larger sieve as the only road and T176 running), section title T126→T175, abstract counters 175/4716, probe index extended (T175 row, running totals, T176 running); `experiments/next.txt` T175 entry (counters 175 Proben / 4716 Checks, Serie (CLXXXVI)); website: `primeFront.ts` feed entry T175 (verdict enum PHASES-RESIST added), status surfaces Teile 11–175 / 4716 checks (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`). NO promotion; NO marker moves; not RH evidence.

**2026-07-29 (XIX) (v560 — the frame/deficit identities promoted; prime-front diary T174 (the gauge route is exhausted by theorems, and the deficit is measured frame-free at  $5.0\sigma$  — PARTIAL-CANCEL) + paper + website sync; NO marker moves)**

- **New module v560 (PRIME.FRAME.DEFICIT.01).** Promotion of the theorem cores of discoveries T172 (contract FRAME.BEYOND), T173 (FRAME.RATE) and T174 (CANCEL.IDENTITY) as one module (`v560_frame_deficit_identities.py`, 24 checks,  $\sim 0.5s$ ) — *the gauge route of phase 2, closed as algebra*, the companion to v559's sixteen-link map, recomputed on a small declared surface (a  $4 \times 3$  (anchor,  $h$ ) rectangle of gap-blind frame-B cells, prime-power anchors  $n = 41 \dots 557$  crossed with the declared ladder  $h = (160, 226, 320)$ , comb COMPLETE on every cell, PD low block on 12/12; one non-prime-power instance  $n = 210$ ; one truncated-comb stress instance  $n = 1009$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$ ; the builder's signature is  $(n_{\text{zone}}, \alpha, M)$  — no frame label is an argument of any observable). (1) *The identity*

*transfer + the sieve-horizon discriminator* (T172): the KMS/Schur pair ( $3.2 \times 10^{-16}$ ), the lag-sum/correlation identity ( $1.4 \times 10^{-16}$  of the larger half; Abel 1826), the R4-free det collapse (at 0.020 of the conditioned bar  $64\varepsilon/|1 - r_{12}^2|$ ) and the additive split ( $7.6 \times 10^{-16}$ ) re-derived EXACTLY on a frame-B and a non-prime-power instance; the truncated-comb instance turns INDEFINITE ( $\lambda_{\min} = -9.6 \times 10^4$ ) against 12/12 PD complete-comb cells, the identity links holding either way (none uses positivity) — indefiniteness sits at the declared sieve horizon, not at a frame or zone. (2) *The  $q = 1 - s$  identity + the calibration* (T173):  $hD = \alpha$  per cell forces the OLS slopes to obey  $q = 1 - s$  to  $2.9 \times 10^{-14}$  (the  $h^{-3}$  target was the  $q = 1$  idealisation of a  $\Theta(D^3)$  law, as an identity); the gauge-degree account comes out as integers —  $G2 = t/\lambda_{\max}(B_{LL})$  and the delivery  $1 - r_{12}^2$  both (0,0), so  $G2$  is the unique doubly gauge-invariant arithmetic member;  $G3 = \mu_1^P G2$  identically (the offset-2 explained);  $(D/\alpha)^3 h^3 = 1$  a tautology. (3) *The blindness theorems + the additive limit* (T174): the amplitude  $c \rightarrow \kappa c$  is a gauge of both sides and of  $R$  over twelve decades (worst 0.026 of the per-cell conditioned bar); the scalar  $\mu_1^P$  — *the entire  $h^{-2}$  channel of the demand* — cancels exactly on both sides; the delivery is invariant under the full positive diagonal group while a non-scalar diagonal moves the demand by 0.23–1.40; and neither the arch term nor the comb term alone has a positive Schur floor on a single cell ( $0/12 + 0/12$  against  $12/12$  for the sum): no factorisation exists, so T173's P6 is NOT promoted to a theorem — only its multiplicative half. (4) *The KMS shape band* (unconditional, declared  $\varepsilon = (K\pi/(2h+1))^2/6$ ): the  $k^2$ -swap moves  $\log \text{GAP}$  by  $\leq 0.312$  of the certified band while the delivery does not move at all. (5) *The placement discriminator*: the value-multiset scramble destroys the floor on 24/24 draws. Each statement carries a mutation control that fails loudly. THE FRAME-FREE DEFICIT IS A MEASUREMENT, NOT A CLAIM OF THE MODULE ( $+0.1111 \pm 0.0222$  stays sandbox MEASURED); R1 typed OPEN (frame-free form); NO RH statement. Ledger row PRIME.FRAME.DEFICIT.01; registered in `run_all.py` and `script_registry.csv`; suite counter  $553 \rightarrow 554$  scripts.

- **Prime-front diary T174 (cancellation\_identity\_probe.py, contract CANCEL.IDENTITY, 37/37, 6.5s, 84 rectangle + 1020 power-check + 147 anchor-ladder + 178 leg windows; verdict PARTIAL-CANCEL, sandbox).** The cancellation question turned into blindness algebra: both sides of the T173 deficit are scale-free functionals of ONE raw Gram matrix, and the ledger reads — amplitude gauge THEOREM ( $\leq 0.028$  of the per-cell conditioned bar over twelve decades); *the complete  $\mu_1^P \approx h^{-2}$  channel THEOREM* ( $\leq 0.009$  — exactly absent from the deficit); full positive diagonal group THEOREM (delivery exact; the demand keeps the SHAPE — the asymmetry is the whole anatomy); KMS shape against  $k^2$  CERTIFIED (unconditional band,  $\leq 0.0075$  in deficit units = 4.8% of T173's number); arch/comb mixture DOES NOT CANCEL (additive and exact; no factorisation; neither term alone has a positive Schur floor on one of 84 cells); frame rule THEOREM ( $R = R(\alpha, M)$  identically — a sampling map, two frames on one cell agree bit for bit). Explained along the way:  $G2$  is the unique doubly gauge-invariant arithmetic member of T173's family, and the  $G3$ – $G2$  offset of exactly 2 is the  $\mu_1^P$  exponent (measured  $h^{-1.9975}$ ). The key result is a direct measurement: on the preregistered  $12 \times 7$  ( $\alpha, h$ ) rectangle the  $h$ -direction carries the deficit ( $a = -0.281 \pm 0.066$ ,  $4.3\sigma$ ; anchor direction flat); the exact between/within split puts it in the GRID response ( $3.3\times$  the anchor selection, whose sign is mostly wrong — the selection reading tested and rejected); the driver is the COMB DENSITY per lag cell (monotone  $+0.281 \rightarrow +0.062$  across the density cuts; on legs  $\log \nu$  and  $\log \text{dens}$  collinear at  $r = -0.921$  — T173's  $2.8\sigma \log \nu$  driver named and discharged); and the dense corner, exhausted (1069 cells, 151 anchor clusters, cluster-robust), gives  $\text{DEFICIT}(\text{frame-free}, \text{dens} \geq 3) = +0.1111 \pm 0.0222$  —  $5.0\sigma$  from zero,  $0.4\sigma$  from T173's  $+0.155 \pm 0.102$ . P6 does NOT become a theorem (only the multiplicative half) but is less load-bearing: the deficit is measured on a D-rule-free surface, and the fixed- $\nu$  invariance is explained (fixed  $\nu$  = fixed density regime). Stress green (scramble 24/24 destroys the block and moves  $R$  by  $10^2$ – $10^5$ ; jackknives stable; horizons declared). Candidates P9–P15; the rest: the per-anchor scatter ( $\chi^2/\text{dof} = 3.02$ ; candidate: the placement phases  $\text{frac}(\log p/D)$ ), the admissibility question ( $0.11$  vs  $\geq 3$  atoms per cell: factor 2.5), the sparse corner. T175

(`PHASE.PLACEMENT`, `phase_placement_probe.py`) is running — the phases and the dense limit. Diary counters 174 probes / 4678 checks (series (CLXXXV)).

- **Paper + website sync.** `tfpt_prime_front`: new phase-2 subsection (T174: the cancellation ledger as a compact table, the  $\mu_1^P$ -channel theorem, the  $G2$ /offset-2 explanation, the sampling-map reading of the frame rule, the frame-free rectangle measurement as key result with the density ordering, keybox “After T174” with the phases / dense limit and T175 running), the verified layer extended to twenty-six modules (new subsection for v560), section title T126→T174, abstract counters 174/4678, probe index extended (T174 row → v560, running totals, T175 running); companion blocks for v560 in `tfpt_1_architecture_e8`, `tfpt_3_e8_audit_bootstrap` and `tfpt_research_contracts`; `experiments/next.txt` T174 entry (counters 174 Proben / 4678 Checks, Serie (CLXXXV)); website: `primeFront.ts` feed entries T174 + v560 promotion (verdict enum PARTIAL-CANCEL added), status surfaces Teile 11–174 / 4678 checks / 26 modules v535–v560 (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`), suite counter 554 (`suite.ts`). NO marker moves; not RH evidence.

**2026-07-29 (XVIII) (prime-front diary T172 (the map is portable, the numbers are not — PARTIALLY-PORTABLE) + T173 (the demanded rate is itself a frame datum; the deficit  $+0.155 \pm 0.102$  is the number of the phase and no frame closes it — DEFICIT-VARIES) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T172** (`frame_beyond_probe.py`, contract `FRAME.BEYOND`, 44/44, 1.3 s, 54 windows on four surfaces plus 3 truncated-comb stress windows,  $h = 128$ –1444; verdict **PARTIALLY-PORTABLE**, **sandbox**). T171’s inherited question — how far does the certified map carry beyond frame A? — answered by a recipe inventory (four degrees of freedom: the frame rule  $D$ , the subdivision  $\nu$ , the zone family; the atom set is NOT free —  $\Lambda$  lives on prime powers, “beyond” means different *anchor* arithmetic) and a transfer table: *13 of the 16 links transfer unchanged — the whole identity tower plus both direction lemmas — and none breaks*; the three that shift are exactly the number-carrying ones (the two window certificates and the rate of link 16). Indefiniteness is *localised at the declared sieve horizon* ( $\lambda_{\min} = -3.7 \times 10^5$ – $-6.2 \times 10^6$  on 3/3 cut-comb stress windows against 0/54 complete-comb; the identity links keep holding there), not at a frame or zone. Constants map: 8/11 stay in frame A’s band, 3 wander on the congruence surface — a certificate on a new surface is a NEW certificate; the procedure and the directions transfer (54/54). R1’s near-degeneracy persists on every surface (angle  $< 10^{-2}$  on 54/54, decay on 9/9 legs) while the rate is frame-bound ( $h^{-1.83}$ – $h^{-2.87}$ ; zone arithmetic secondary, spread 0.40 vs 1.03); sharpest new result, the placement finding: scrambling atom positions at a fixed  $\Lambda$ -value multiset raises the angle by a median  $6.6 \times 10^4$  and removes the decay — the collapse belongs to the actual prime-power placement. Candidates P1–P4 PENDING (no promotion — bundling with T174 planned). Diary counters 172 probes / 4601 checks (series (CLXXXIII)).
- **Prime-front diary T173** (`frame_rate_probe.py`, contract `FRAME.RATE`, 40/40, 2.2 s, 284 windows plus 109 scramble twins over eleven preregistered frame data (A at  $\nu \in \{3, \dots, 16\}$ , gap-blind B, smoothed C),  $h = 128$ –1447; verdict **DEFICIT-VARIES**, **sandbox**). The demand side itself put on the measurement table:  $q = 1 - s$  is an *exact identity* ( $1.9 \times 10^{-15}$ , from  $hD = \alpha$ ) and  $q = 0.667$ – $0.777 < 1$  on every frame datum — *T171’s  $h^{-3}$  target was the  $q = 1$  idealisation of a  $\Theta(D^3)$  law*. The gap functional is calibrated instead of guessed (six preregistered candidates; the relative I5 gap  $t/\lambda_{\max}(B_{LL})$  hits the chain’s 3 on frame A, median 3.033, and is the only dimensionless one;  $(D/\alpha)^3$  exposed as a tautology). The deficit, read as one slope of the per-window ratio  $R = \text{GAP}/(1 - r_{12}^2)$ : pooled  $+0.155 \pm 0.102$  ( $\times 1.54$  over-scatter inflation,  $3.0\sigma$  above zero),  $1.8\times$  flatter than either side ( $\chi^2/\text{dof}$  REQ 4.87 / DEL 4.17 / difference  $2.38 = 2.4\sigma$  — constancy rejected), split-stable to 0.214 where both single sides drift  $\sim 0.5$ , *invariant under the anchor-to-grid rule at fixed  $\nu$*  ( $\chi^2/\text{dof} \leq 0.64$  —



theorem candidate for that half only), not invariant under  $\nu$  ( $2.8\sigma$ , confounded with arithmetic content under the cap;  $q$  the weakest driver at  $0.2\sigma$ ). Honest correction to T172 (P7): in the fixed- $h^{-3}$  convention the wider surface gives  $+0.015 \pm 0.084$  — consistent with zero — so T172's  $C$ -growth was neither sign- nor split-stable (its verdict is untouched). *R1 now stands frame-free*: the delivered near-degeneracy closes slower than the relative I5 gap it must pay, by  $h^{+0.155 \pm 0.102}$ ; no frame closes it (best 2 s.e. upper edge  $+0.170$  at B; 8/11 legs individually consistent with zero — pooled with the penalty, not claimed per leg) — frame shopping is over, by numbers. Stress: scramble removes the decay on 4/4 declared legs; jackknife  $\leq 0.026$ ; shuffle null  $3.7\sigma$  away. Candidates P5–P8 PENDING (no promotion — bundling with T174 planned). T174 (CANCEL.IDENTITY, `cancellation_identity_probe.py`) is running — the cancellation identity. Diary counters 173 probes / 4641 checks (series (CLXXXIV)).

- **Paper + website sync.** `tfpt_prime_front`: two new phase-2 subsections (T172: the four-degrees-of-freedom inventory, the 13/16 transfer, the sieve-horizon localisation, the placement finding as key result; T173: the  $q = 1 - s$  identity as display, the calibration, the compact deficit table, the honest T172 correction, the frame-free R1 statement as key result, keybox “After T173” with the cancellation identity and T174 running), section title T126→T173, abstract counters 173/4641, probe index extended (T172 + T173 rows, running totals, T174 running); `experiments/next.txt` T172 + T173 entries (counters 172/4601 (CLXXXIII) and 173/4641 (CLXXXIV)); website: `primeFront.ts` feed entries T172 + T173 (verdict enums PARTIALLY-PORTABLE and DEFICIT-VARIES added), status surfaces Teile 11–173 / 4641 checks (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`). NO promotion (T172s P1–P4 and T173s P5–P8 PENDING — bundling with T174 planned); NO marker moves; not RH evidence.

**2026-07-29 (XVII) (v559 — the phase-2 capstone promoted; prime-front diary T171 (the map is complete: all sixteen links in one connected run, eight no-gos as classified, the precision ledger closed — MAP-COMPLETE) + paper + website sync; NO marker moves)**

- **New module v559 (PRIME.PHASE2.CAPSTONE.01).** Promotion of discovery T171 (contract FINAL.MAP, verdict MAP-COMPLETE) as the phase-2 capstone (`v559_phase2_capstone.py`, 35 checks,  $\sim 0.3$  s) — *the capstone verifies the map and claims nothing new*: T126–T170 reduced the I5 uniformity question through sixteen exact reformulations to ONE object (R1), and this module is the load-bearing version of T171's connected run, everything recomputed on v551–v558's declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$ ; the  $16 \times 16$  low block positive definite on 12/12 — a numerical fact, never routed through the Weil criterion). *The chain, sixteen links in one pass* (13 THEOREM / 3 CERT): T152 Schur two-block (KMS  $1.3 \times 10^{-15}$ ; holds at  $0.5\lambda_{\min}$ , fails at  $1.02\lambda_{\min}$  — sharp; Haynsworth  $3.8 \times 10^{-15}$ ); T151 Sobolev  $b_k \leq C_S k$  (worst ratio 0.4816); T153  $\Psi$ -collapse with the Z-law ( $6.3 \times 10^{-16}$  /  $9.0 \times 10^{-17}$ ); T154 Ritz ceiling from below ( $g_{32}/s = 0.9520$ – $0.9849$ ); T155  $12 \times 12$  floor, sharp ( $t = 0.2295$ – $0.3719$ ); T156  $F(P, r)$  ( $L(16) = 6.6$ – $15.6$ ); T158 Thomson/ladder ( $3.1 \times 10^{-12}$ ; 36/36 trials below); T160 sampling (closed Dirichlet weights = correlation weights  $1.1 \times 10^{-15}$ ; sampled  $\Lambda$ -mass  $9.9 \times 10^{-15}$ ); T163 Abel swap with the TV floor (overshoot  $\times 5.2 \times 10^4$ – $1.1 \times 10^6$ ); T164 gauge/quantifier ( $\Psi(x^*) = 1/g_{16}$  to  $7.0 \times 10^{-12}$ ; 48/48 perturbed vectors larger); T165  $P_{\text{pr}} = 1$  at  $x^*$ ; T166 cascade  $1/(1 - R_{16}^2)$ , diagonal-invariant; T167  $K = 2$  exactness (milder  $\times 1.62$ – $5.20$ ); T169 det collapse R4-free ( $\leq 4.2 \times 10^{-12}$ , no positivity of  $A_h$  consumed); T170 rank-3 ( $\{-2, -1, +1\}$  exact;  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ ); T170 R1 shape via the  $2 \times 2$  Lagrange identity ( $\sin \angle = 1.169 \times 10^{-4}$ – $2.467 \times 10^{-3} \leq 10^{-2}$  per window; *rate and quantifier typed OPEN*). *The no-go battery*, 8/8 failing as classified (size budget  $\times 2.7 \times 10^3$ – $1.6 \times 10^4$ ; symbol infimum  $< 0$  on 12/12; sector invariance; perturbation scale; atom band dominates; sieve operator-norm  $\times 8.8 \times 10^3$ – $7.5 \times 10^4$ ; scramble  $\times 679$  median with rank-3 untouched on 60 draws;  $L_P$  gain = 1 exactly). *The precision*

*ledger*: needed  $2.752 \times 10^{-4}$  at the deepest window — finer than the RH yardstick  $h^{-0.5}$  by  $215\times$  and the best unconditional  $h^{-0.996}$  by  $13\times$  on this surface; both self-similarity identities  $\leq 6.8 \times 10^{-13}$ ; mutation controls fail loudly. ZERO new uniform-in- $m$  statements; the one open phase-2 object R1 explicitly typed OPEN; NO RH statement (RH\_DELTA = 0.5 a yardstick in exactly one role); Weil fence documented (link 14 R4-free by construction). Ledger row PRIME.PHASE2.CAPSTONE.01; registered in `run_all.py` and `script_registry.csv`; suite counter 552  $\rightarrow$  553 scripts.

- **Prime-front diary T171** (`final_map_probe.py`, contract FINAL.MAP, 43/43, 0.6 s, 12 windows over two frames  $\nu = 4/6$ ,  $h = 142\text{--}1445$ , lever arm  $10.2\times$ ; verdict MAP-COMPLETE, sandbox). The capstone stands — the map is complete: all sixteen links of the reduction chain, from the I5 floor to the final R1 shape, reproduce in ONE connected run (13 THEOREM / 3 CERT; honest exclusion said out loud: on  $\nu = 4$ ,  $h = 1445$  the low block is indefinite,  $\lambda_{\min} = -343$ , so the ladder links carry 11/12 while the identity links carry 12/12); all eight classified no-go routes fail on instances exactly as classified (size budget  $h^{+1.32}$ ; symbol infimum negative on 12/12; sector change an invariance  $\leq 2.1 \times 10^{-9}$ ; Kato  $2.0 \times 10^4\text{--}1.9 \times 10^7$  off scale; atom band  $h^{+0.79}$  faster; sieve  $\delta = +1.044$  against 3.0; scramble type-flip  $h^{-2.57} \rightarrow h^{+0.21}$ ,  $\times 11421$ ;  $L_P$  gain = 1 exactly); and the precision ledger closes: needed  $2.284 \times 10^{-7}$  at  $h = 1444$  (rate  $h^{-3.0}$ ) against the RH yardstick  $h^{-0.5} = 2.63 \times 10^{-2}$  — short by  $1.2 \times 10^5\times$  — and the best unconditional  $h^{-0.996} = 7.13 \times 10^{-4}$  — short by  $3.1 \times 10^3\times$ ; both self-similarity identities check ( $4.2 \times 10^{-10}$  /  $9.9 \times 10^{-10}$ , sign-agnostic incl.  $\nu_2 < 0$  on 1/12). At link 16 the RATE is frame-dependent ( $\nu = 4$ :  $h^{-1.86}$ ;  $\nu = 6$ :  $h^{-2.87}$ ) while the COLLAPSE is not. Balance of the quotable parts: 40 THEOREM / 5 CERT-UNIF / 9 CERT-WINDOW / 16 MEASURED / 9 NO-GO (T126–T161 honestly unaggregated). The file adds ZERO new uniform-in- $m$  statements — exactly why it is promotable: phase 2 is now a certified map with ONE open object (R1, classified as a near-degeneracy, not a size); R2/R3 booked, R4 removed. T172 (FRAME.BEYOND, `frame_beyond_probe.py`) is running — how far the map carries beyond frame A. Diary counters 171 probes / 4557 checks (series (CLXXXII)).
- **Paper + website sync.** `tfpt_prime_front`: new capstone subsection (T171) with the sixteen-link table, the no-go list and the precision-ledger display, a closing paragraph of the phase-2 section, the verified layer extended to twenty-five modules (new subsection for v559), section title T126 $\rightarrow$ T171, abstract counters 171/4557, probe index extended (T171 row, running totals, T172 running); companion blocks for v559 in `tfpt_1_architecture_e8`, `tfpt_3_e8_audit_bootstrap` and `tfpt_research_contracts`; `experiments/next.txt` T171 capstone entry (counters 171 Proben / 4557 Checks); website: `primeFront.ts` feed entries T171 + v559 promotion (verdict enum MAP-COMPLETE added), status surfaces Teile 11–171 / 4557 checks / 25 modules v535–v559 (`page.tsx`, `HonestyBanner.tsx`, `ModuleLadder.tsx`), suite counter 553 (`suite.ts`). NO marker moves; not RH evidence.

**2026-07-29 (XVI) (v558 — the bilinear/rank identities of phase 2 promoted; prime-front diary T170 (the bilinear form collapses by a rank-3 theorem, R1 classified as a near-degeneracy — SIEVE-RESISTS) + paper + website sync; NO marker moves)**

- **New module v558 (PRIME.BILINEAR.RANK.01).** Promotion of the closed theorem cores of T168 (contract LAGRANGE.MINORS), T169 (contract TSTAR.RATIO) and T170 (contract BILINEAR.SIEVE) as one deliberately narrow module (`v558_bilinear_rank_identities.py`, 27 checks,  $\sim 0.2$  s), everything recomputed on v551–v557’s declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table): (1) the Lagrange/Wronskian anatomy — the lag-sum identity  $\hat{a}_{ij} = t_i^T A_h t_j = \sum_d c_d W_d^{ij}$  with the closed correlation weights ( $1.6 \times 10^{-15}$ ; the  $1/\sqrt{\mu}$  factors cancel, T169-TH3), Euclidean orthogonality  $t_1 \cdot t_2 = 0$  ( $4.4 \times 10^{-16}$  — the near-parallelism is created *entirely* by the arithmetic metric), the Wronskian telescope  $(\mu_1 - \mu_2)u_r v_r = W_{r-1} - W_r$

with  $W_{-1} = W_{h-1} = 0$  ( $1.9 \times 10^{-17}$ ) and the maximal-minor identity  $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$  ( $4.6 \times 10^{-16}$ ; Lagrange 1773); (2) the exponent account  $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$  identically ( $2.5 \times 10^{-13}$ ) with the unconditional Gershgorin certificate  $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$  (12/12, loss 1.1885–1.1947; the  $|\hat{a}_{12}|$ -free mutation fails on every window); (3) the T169-TH7 det-collapse in closed form,  $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$  ( $8.7 \times 10^{-13}$ ) — the loop T167 scalar  $\rightarrow$  T168 factor  $\rightarrow$  T169 candidate closes as an identity, with the usable bound  $\nu_2 \leq 2 \det \hat{A} / (\hat{a}_{11} + \hat{a}_{22})$  tight to  $< 2$  exactly; (4) the exact bilinear von Mangoldt form (T170 TH1–TH3) —  $\hat{A} = B - S$  against the direct block ( $1.7 \times 10^{-14}$ ), the polarisation  $\det \hat{A} = \det B - D(B, S) + \det S$  ( $1.0 \times 10^{-10}$ ) and  $\det S = \sum_{n,m} \Lambda(n) \Lambda(m) K(n, m) / \sqrt{nm}$  with  $K = \frac{1}{2} D(X_n, X_m)$  through the *explicit* kernel matrix ( $7.5 \times 10^{-16}$ ; Cauchy–Binet 1812/1815), the wedge reading to the measured rank-one defect  $4.6 \times 10^{-3}$ – $3.1 \times 10^{-2}$  (honest CERT); (5) the rank-3 theorem (T170-TH4: polarisation matrix eigenvalues  $\{-2, -1, +1\}$  exactly,  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ , nonzero spectrum =  $\text{spec}(M P^T P)$  — the bilinear form *is* the rank-3 polynomial  $S_{11}S_{22} - S_{12}^2$  in three linear  $\Lambda$ -sums) and the Type II collapse (T170-TH5: Vaughan’s identity coefficient-exact for every  $U = V \in \{3, 4, 5, 6\}$ , Type II kernel 99.9%-energy rank  $[2, 2, 2, 2]$  against generic  $[20, 15, 11, 9]$ ); (6) the R4-free chain identity  $1 - r_{12}^2 = \det \hat{A} / (\hat{a}_{11} \hat{a}_{22})$  (T170-TH6 — the Weil fence never approached), the  $L_P$  no-go ( $\hat{a}_{12} = 1.1 \times 10^{-17}$  identically, no decay) and the scramble discriminator (rank-3 *untouched* on all 60 draws while the value of  $1 - r_{12}^2$  moves by a median factor 834.8: the arithmetic sits in the *joint values* of  $(S_{11}, S_{22}, S_{12})$  against  $B$ ) as must-breaks; every statement with a mutation control; the one open object R1 (the collinearity/near-degeneracy rate) explicitly typed OPEN; NO RH statement. Ledger row PRIME.BILINEAR.RANK.01; registered in `run_all.py` and `script_registry.csv`; suite counter 551  $\rightarrow$  552 scripts.

- **Prime-front diary T170 (bilinear\_sieve\_probe.py, contract BILINEAR.SIEVE, 40/40, 1.8 s, 70 frame-A windows  $h = 142$ – $1445$  on the T163–T169 union recipe, 67 arithmetically complete; verdict SIEVE-RESISTS, sandbox).** T169’s standard shape cashed in: the bilinear form is written down *exactly* —  $\hat{A} = B - S$  (TH1,  $3.5 \times 10^{-15}$ ),  $\det \hat{A} = \det B - D(B, S) + \det S$  (TH2,  $1.3 \times 10^{-10}$ ),  $\det S$  a genuine double von Mangoldt sum against the closed wedge kernel  $K = \frac{1}{2} D(X_n, X_m)$  (TH3,  $1.5 \times 10^{-14}$ ) — and on the reference window ( $h = 540$ ) the three pieces read 6.18,  $-198.8$ ,  $192.6$  and sum to  $4.22 \times 10^{-3}$ : a *five-order cancellation* between three  $\sim 200$ -sized pieces, with absolute target  $1.10 \times 10^{-5}$ . The route table (target  $\delta \geq 3.0$ ): truth  $+2.747$  MEASURED; Cauchy–Schwarz  $-0.000$ ; large sieve/duality  $+0.669$ ; Montgomery–Vaughan 1973  $-0.384$ ; the exact split  $+0.996$  (best); Vaughan Type I+II  $+0.669$  — all THEOREM. The collapse is structural, by two theorems: the **rank-3 theorem** (TH4:  $D$  is the polarisation of  $\det$  on  $2 \times 2$  symmetric matrices, rank 3, signature  $(1, 2)$ , so  $\text{rank } K \leq 3$  for every  $h$  and  $X$  —  $\sigma_4/\sigma_1 = 2.0 \times 10^{-15}$ , effective rank 1.38–2.71 against  $N \sim 10^3$ : the form *is*  $S_{11}S_{22} - S_{12}^2$  in three linear  $\Lambda$ -sums) and the **Type II collapse** (TH5: the closed weights see  $n$  only through  $\log n$ , so Vaughan Type II blocks have effective rank  $O(1)$  for every  $U, V$  — 2 singular values carry 99.9% against 153 generic,  $U = 12$ – $70$  scanned). Honest T161 comparison: GAINED — 32 frequencies reduce to THREE linear functionals, by theorem; NOT GAINED — the precision, which binds ( $5.3 \times 10^{-5}$  per sum at  $h = 285$ , sharpening  $h^{-3}$ , the RH yardstick  $1.1 \times 10^3$  too coarse;  $\ell^1$  mass  $41\times$  over the signed value). The chain becomes R4-free for free (TH6:  $1 - r_{12}^2 = \det \hat{A} / (\hat{a}_{11} \hat{a}_{22})$  an identity, sharper than T169-TH9 at  $-2.747$  vs  $-2.771$ ; the Weil fence never approached); end to end the best unconditional bound carries  $h^{-0.996}$  against the required  $\leq h^{-2.50}$  — missing by 1.50 *in the exponent*, and the shortfall is a theorem. Stress: the scramble leaves rank-3 and the unconditional bounds unchanged while the truth moves 2.76 in the exponent — **the arithmetic sits entirely in the joint values of  $(S_{11}, S_{22}, S_{12})$  against  $B$** ; flat-weight breaks;  $L_P$  exact; the  $\nu$ -decoupling is reported unsmoothed (frames  $\nu = 5/8/11$  fall at  $-1.70/-1.27/-1.74$  — T169’s  $h^{-2.95}$  is frame-A, the shortfall is not). Balance 6 THEOREM / 1 CERT-UNIF / 5 NO-GO. **R1 finally classified:**  $\det \hat{A} / (|\text{row}_1| |\text{row}_2|)$  is the sine of the angle between the two rows of  $\hat{A}$ , falling  $h^{-2.711}$  — R1 asks for an unconditional certificate that two explicit finite  $\Lambda$ -sum vectors become collinear

at rate  $h^{-3}$ : a *near-degeneracy*, not a size, beyond any size-bounding tool; needs beyond-RH *joint* precision. R2/R3 booked in T169; R4 off the path. T171 (`final_map_probe.py`, contract FINAL.MAP) is running — the capstone map of the whole phase.

- **Docs + website sync.** `tfpt_prime_front` gains the T170 subsection (the exact bilinear decomposition as a display, the rank-3 theorem and the Type II collapse as key results, the route table with its  $\delta$  numbers, the five-order cancellation, the final R1 classification as a row-angle near-degeneracy, the honest T161 comparison, the “After T170” keybox with R1-final and T171 running) and the v558 subsection in the verified layer (now *twenty-four* modules); the phase-2 section title moves to T126–T170, the probe index gains the T170 row and the running totals move to 170 probes / 4514 checks; companion v558 blocks in `tfpt_1`, `tfpt_3` and `tfpt_research_contracts`; `experiments/next.txt` gains the entry (series (CLXXXI)); website: `primeFront.ts` feed entries T170 + v558 promotion, status surfaces Teile 11–170 / 4514 checks / 24 modules v535–v558, T171 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XV) (prime-front diary T168+T169 (the self-similarity of the hardness is *proved*, the one open object becomes a bilinear von Mangoldt sum — MINORS-RESIST / RATIO-RESISTS) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T168** (`lagrange_minors_probe.py`, contract LAGRANGE.MINORS, 39/39, 2.3 s, the same 63-window union as T163–T167, recipe untouched; verdict MINORS-RESIST, sandbox). The determinant attacked as a sum of squares via the Lagrange identity — and the sum of squares is *real*: the  $h \times h$  arithmetic kernel  $A$  is **positive definite on all 63 windows** ( $\lambda_{\min} = 5.0 \times 10^{-6}$ – $8.5 \times 10^{-4}$ , 8–12 orders above the solver floor), so the Lagrange identity is a genuine sum of strictly positive squares ( $\rho_{\text{LAG}} = 1.000$  exactly, no sign cancellation) — hard-fenced as a statement about one finite matrix per window; the Weil criterion is never tested, assumed or reverse-inferred. The anatomy is exact:  $Q = X^T A X$  ( $3.9 \times 10^{-15}$ ), the mode vectors are **Euclidean-orthogonal** ( $r_{12} = 0$  to  $6.2 \times 10^{-17}$  — the near-parallelism is created *entirely* by the arithmetic metric), the Wronskian telescopes exactly ( $W_r - W_{r-1} = (\mu_1 - \mu_2)u_r v_r$ ,  $1.7 \times 10^{-13}$ ), the minor sup is closed and  $m$ -free ( $\sim N/\pi^2$ , attained to 0.770), Cauchy–Binet verified as algebra. The key measurement splits by basis: in the closed basis the minors are *maximal* ( $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$  exactly  $\sim h^4$  — the Wronskians carry zero arithmetic, all smallness sits in the PSD kernel at one closed vector); in the eigenbasis the sum is *thin* (90% from 4–101 pairs, top pair always ( $\lambda_{\min}, \lambda_{\max}$ ) at 6.8–41.7% alone, raw  $\kappa_w = 0.30$ – $0.94$ ). The exponent ledger closes arithmetic-free:  $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$  with  $\nu_1 \sim h^{+0.81}$ ,  $\hat{a}_{11} \sim h^{+0.76}$ ,  $\hat{a}_{22} \sim h^{+0.91}$  — every factor but one at the right power; best  $m$ -free-shaped bound  $h^{-0.548}$  union /  $h^{-0.880}$  frame  $A$ , short by  $h^{2.45}/h^{2.04}$ ; the open factor is identified exactly:  $\nu_2$  with minimiser  $t^* = Q_{12}/Q_{11}$ , needed to  $6.05 \times 10^{-3}$  median /  $2.82 \times 10^{-4}$  worst ( $h^{-1.146}$ ) — by T168-TH7 that factor *is* the target rewritten (ONE-TERM-MISSING deliberately not rendered). Scramble breaks harder than T167 (8/8 windows lose  $A \succ 0$  and the Gram property,  $n_{\text{neg}}$  up to 73);  $L_P$  control exact. **The hardness is self-similar under exact reformulation** — determinant  $\rightarrow$  PSD compound at a closed vector  $\rightarrow$  near-null Rayleigh value  $\rightarrow$  one ratio of two closed lag sums, same accuracy price. Balance 8 THEOREM / 0 CERT-UNIF / 3 CERT-WINDOW / 11 MEASURED / 7 NO-GO.
- **Prime-front diary T169** (`tstar_ratio_probe.py`, contract TSTAR.RATIO, 41/41, 3.0 s, same union surface; verdict RATIO-RESISTS — the self-similarity *proved*, sandbox). T168’s one open factor attacked as a preregistered candidate table: every genuinely closed candidate misses by  $O(1)$  and *diverges* from the threshold at  $h^{+1.99}$  (the block is atom-dominated,  $|\text{atom/arch}| = 3.5$ – $43$  resp.  $14$ – $461$ , and  $\hat{a}_{11}^{\text{arch}} < 0$  on 63/63 — the archimedean half is not even PSD); **the only candidate that meets the threshold,  $K7 = \sqrt{\hat{a}_{22}/\hat{a}_{11}}$  (63/63), does so**

by a new theorem (T169-TH7) because it puts  $\det \hat{A}$  back into the numerator:  $R(K7)$  collapses in closed form to  $2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$  ( $1.1 \times 10^{-9}$ ) — the  $t^*$ -language is provably exactly as hard as T167’s determinant: the loop closes as an identity (T167 scalar  $\rightarrow$  T168 factor  $\rightarrow$  T169 candidate = the same object). The real gains: **R3 is booked CERT-UNIF** —  $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$  (Gershgorin, unconditional, uniform in  $h$ ), 63/63, losing only 1.10–1.19 — the CERT-UNIF column moves off zero; R2 is honestly downgraded from cheap to OPEN (the absolute-value budget cannot even certify positivity); the chain is **rebuilt R4-free** ( $1 - r_{12}^2 \leq [\max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|] \cdot R(\hat{t}) / (\hat{a}_{11}\hat{a}_{22})$ , valid 63/63, frame-A trend  $h^{-2.948}$ , i.e.  $\varepsilon = 0.052$  inside the 0.5 carry window) so the Weil-shaped positivity of  $A_h$  never enters it — the Weil fence is respected by construction (uniform-in- $h$  positivity of  $A_h$  is RH-equivalent-shaped and is never routed or claimed); and by T169-TH3 the  $1/\sqrt{\mu}$  factors cancel, so the cond  $\sim 10^8$  ladder leaves the file. Stress:  $L_P$  exact (the  $t^*$ -recipe is a near-degeneracy artefact — its tightness *measures*  $\det \hat{A}$ ), scramble 8/8, flat-weight null 8/8, parity-pair control only  $6.7\times$  (honestly PARTIAL). Balance 10 **THEOREM** / 2 **CERT-UNIF** / 1 **CERT-WINDOW** / 4 **NO-GO**. The rest: **R1, for the first time in standard analytic shape** — an  $m$ -free bound on  $\det \hat{A}$  at  $h^{-3+\varepsilon}$  as a bilinear von Mangoldt sum against closed Dirichlet weights (equivalently the diagonal ratio to  $1.2 \times 10^{-2} / 5.6 \times 10^{-4}$ ); R2 = R1; R3 closed; R4 not needed. T170 (bilinear\_sieve\_probe.py, contract BILINEAR.SIEVE) is running the unconditional sieve toolbox (Montgomery–Vaughan, Vaughan decomposition) on the bilinear form — T161’s beyond-RH triage applied to *linear* sums; for *bilinear* forms the large sieve gives unconditional square-root cancellation.

- **Docs + website sync.** tfpt\_prime\_front gains the T168 subsection (the Lagrange identity and the PSD finding with the hard Weil fence, the basis split of the key measurement, the closed exponent ledger, the self-similarity statement) and the T169 subsection (the candidate table, the collapse theorem TH7 with the loop closure as the key result, the CERT-UNIF gain, the R4-free chain with  $\varepsilon = 0.052$ , the standard-shape restatement, the “After T169” keybox with R1 in bilinear shape and T170 running); the phase-2 section title moves to T126–T169, the probe index gains the T168+T169 rows and the running totals move to 169 probes / 4474 checks; experiments/next.txt gains both entries (series (CLXXIX), (CLXXX)); website: primeFront.ts feed entries T168+T169, status surfaces Teile 11–169 / 4474 checks, T170 running. NO promotion (T168/T169 candidates PENDING — bundling with T170 planned). Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XIV) (v557 — the cascade/vector identities of phase 2 promoted; prime-front diary T167 (three dresses are one inequality, the closed vector exact at  $K = 2$  — VECTOR-RESISTS) + paper + website sync; NO marker moves)**

- **New module v557 (PRIME.CASCADE.VECT.01).** Promotion of the closed theorem cores of T166 (contract SCHUR.CASCADE) and T167 (contract NULL.VECTOR) as one deliberately narrow module (v557\_cascade\_vector\_identities.py, 23 checks,  $\sim 0.1$  s), everything recomputed on v551–v556’s declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table): (1) the cascade identities — prefix (one Cholesky, all sixteen rungs,  $1.1 \times 10^{-12}$  on 192 comparisons), minor form  $1/g_K = \det G_{[1..K]} / (\mu_1^P \det G_{[2..K]})$  on the raw arithmetic Gram block ( $2.7 \times 10^{-12}$ ; Schur 1917) and regression form  $g_K/g_1 = 1/(1 - R_K^2)$  ( $8.8 \times 10^{-13}$ ); (2) the diagonal invariance of the gain under  $B \rightarrow DBD$  ( $2.7 \times 10^{-12}$  on 144 random diagonals — the gain belongs to the arithmetic Gram block alone); (3) the exact  $K = 2$  identity — the closed vector  $(1, -Q_{21}/Q_{22})$  attains  $1/g_2 = Q_{11}(1 - r_{12}^2)$  exactly and  $B_{11}g_2 = 1/(1 - r_{12}^2)$  identically ( $1.2 \times 10^{-12}$ ): the fast-null-vector is FREE at  $K = 2$ , dress (c) collapses onto dress (b); (4) the pivot identity  $u^T Q_K u = 1/g_K + \delta^T Q_K \delta$  for  $\delta_1 = 0$  ( $2.0 \times 10^{-12}$  on 252 evaluations; the  $\delta_1 = 0.1$  mutation breaks by exactly the closed cross term  $2\delta_1$ ) with the exact entry threshold and the unification

$(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$  — ONE inequality, not three; (5) the trial theorem (288 trials, every candidate a certified lower bound), the  $L_P$  no-go (cascade gain exactly 1, deviation 0.0) and the scramble type-change ( $B_{11}$  itself loses positivity on 49/60 declared draws, a majority on 12/12 windows) as must-breaks; every statement with a mutation control; the one open scalar R1 (an  $m$ -free upper bound on  $1 - r_{12}^2$ ) explicitly typed OPEN; NO RH statement. Ledger row PRIME.CASCADE.VECT.01; registered in `run_all.py` and `script_registry.csv`; suite counter 550  $\rightarrow$  551 scripts.

- **Prime-front diary T167** (`null_vector_probe.py`, contract NULL.VECTOR, 39/39, 1.4 s, the 63-window union of T166 rebuilt bit for bit; T163–T166 reproduced as fits; verdict VECTOR-RESISTS with a strong reduction, sandbox). The most constructive dress attacked and closed *as a construction*: 22 families preregistered, all evaluated through the pivot identity  $u^T Q u = 1/g_K + \delta^T Q \delta$  (10206 evaluations,  $3.2 \times 10^{-9}$ ) — **gram2 is EXACT at  $K = 2$**  ( $\rho = 0$ , a theorem: the closed vector attains  $1/g_2$  on 63/63,  $B_{11}g_2 = 1/(1 - r_{12}^2)$  an identity, certified gain exponent  $h^{+2.921}$  on frame  $A = h^{3-0.079}$ ,  $h^{+2.246}$  union). Perturbation theory dies structurally: Jacobi diverges (radius 1.0–12.7) but **the Kato series CONVERGES (radius 0.067) — to the WRONG object** (the exact bottom eigenvector is itself a useless trial,  $\rho = 6.04$ , overlap 0.083, relative gap  $5.4 \times 10^{-7}$ ) — closed off in every order. Two thresholds separated (envelopes on 189 perturbations each); **monotonically sharper in  $K$  — mildest at  $K = 2$**  ( $1.90\times$ ,  $h^{+0.353}$ ), **against the hypothesis and exactly where the vector is free**; at  $K = 6$  no closed family hits (separation  $h^{+1.138}$ , ratio 264 $\times$ ); anchor  $\text{eps}_{\text{ent}}(2) \rightarrow (1 - r_{12}^2)/4$  — T166's  $3.57 \times 10^{-4}$  is literally the quarter. End-to-end gain 2207– $7.84 \times 10^6$  certified; the deficit is the higher rungs, not the vector; **the scramble breaks harder than T166 saw: 8/8 windows lose positivity of the  $2 \times 2$  diagonal itself** ( $Q_{11} = -1.35 \times 10^5 \dots -1.02 \times 10^4$ ) — **a change of TYPE, not a factor**; anti-fitting cheat only  $h^{-0.823}$ ; off-recipe  $1 - r_{12}^2 \sim h^{-2.740}$  vs  $h^{-2.921}$ . **The unification as an identity**:  $\text{eps}_{\text{ent}}(K) = \rho(1/g_K)/S_K$  to machine precision — the determinant ratio (dress a),  $1 - r_{12}^2$  (dress b) and the entry threshold (dress c) are ONE inequality, not three. Balance 6 THEOREM / 0 CERT-UNIF / 2 CERT-WINDOW / 7 MEASURED; ONE-TERM-MISSING deliberately NOT rendered (the one missing term IS dress (b)); rest: R1, the only item — an  $m$ -free upper bound on  $1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C h^{-3+\epsilon}$  (required  $1.25 \times 10^{-5}$  median,  $3.19 \times 10^{-8}$  worst); redirected: work at  $K = 2$ . T168 (`lagrange_minors_probe.py`, contract LAGRANGE.MINORS) is running — the Lagrange identity, the determinant as a sum of squares.
- **Docs + website sync.** `tfpt_prime_front` gains the T167 subsection (the pivot identity and the gram2 exactness as displays, the Kato-converges-to-the-wrong-object lesson, the threshold monotonicity against the hypothesis, the unification identity as the key result, the type-change scramble, the “After T167” keybox with R1 as the only item and T168 running), the v557 subsection in the verified layer, the module table row, the phase-2 section title moves to T126–T167, the probe index gains the T167 row and the running totals move to 167 probes / 4394 checks; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v557 companion blocks; `experiments/next.txt` gains the T167 entry (series (CLXXVIII)); website: `primeFront.ts` gains the T167 and v557 feed entries, status surfaces Teile 11–167 / 4394 checks, 23 modules v535–v557, T168 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT; ledger row PRIME.CASCADE.VECT.01.

**2026-07-29 (XIII) (prime-front diary T166 (the cascade gain is a Gram-determinant cancellation, one inequality in three dresses — CASCADE-RESISTS) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T166** (`schur_cascade_probe.py`, contract SCHUR.CASCADE, 30/30, 2.1 s, the two surfaces rebuilt as a *union*: 27 frame-A windows  $h = 142$ –1320 plus 36 decoupled- $\nu$  windows over 6 zones,  $\nu \in \{4, 5, 6, 8, 11, 16\}$ ,  $h = 96$ –1371 — 63 windows

- with a  $14\times$  lever arm on which zone depth and resolution are independent knobs ( $\log h$  vs  $\log \nu$  correlated 0.33); **T165's three numbers reproduced on frame A, the union reading**  $h^{+2.759}/h^{+0.193}/h^{+2.566}$ ;  $\kappa = 0.038821$  verified at every jump point in  $[100, 4 \times 10^5]$ ; **verdict CASCADE-RESISTS, sandbox**). The one open object of T165 — the cascade lower bound  $g_{16}/g_1 \geq ch^{3-\varepsilon}$  uniform in  $m$  — attacked head-on and dissected completely. *The anatomy*: four identities, all theorems re-verified on 63/63 windows — the prefix property (one Cholesky yields all sixteen rungs), the minor form (Schur 1917), the **regression form**  $g_K/g_1 = 1/(1 - R_K^2)$  (the open object *is* a near-collinearity: mode 1 explained by modes 2...16 up to a residual  $h^{-3+\varepsilon}$ ), and the **diagonal invariance** of the gain under  $B \rightarrow DBD$  — it does not see the KMS  $1/\sqrt{\mu}$  normalisation at all: the  $h^3$  is a property of the arithmetic Gram block  $G = TAT^\top$  alone. The rung profile is extreme: rung 2 alone carries a median 59% of the gain, rungs 2–5 72%,  $K_{\text{half}} \in \{2..5\}$  on 63/63; rung 2 in closed form has  $1 - r_{12}^2 = 1.3 \times 10^{-7} - 4.5 \times 10^{-4} \sim h^{-2.92}$  (frame A) — one rung multiplies  $g$  by  $2207 - 7.8 \times 10^6$  where the  $m$ -free budgets are sharp only to a factor 21–26.
- **The cancellation finding — a third mechanism.** No entry of the  $2 \times 2$  bottom block cancels (each keeps 60–99% of its larger half) and neither half alone is collinear (arch  $1 - r^2 = 0.98$ , atoms 0.29) — yet the sum is degenerate to seven digits: **the cancellation lives in the  $2 \times 2$  Gram determinant** — the polarisation split  $\det(G_{\text{ar}} + G_{\text{at}}) = \det G_{\text{ar}} + \det G_{\text{at}} + \Pi$  (exact to  $6.7 \times 10^{-14}$ ) has pieces 2.6–630 cancelling to  $1.0 \times 10^{-3} - 1.7 \times 10^{-2}$ , a ratio  $2.6 \times 10^{-6} - 5.1 \times 10^{-4} \sim h^{-1.45}$  — the same arch-against-atom mechanism as T159/T160, one level up, in a quadratic functional of the lag sums. Pinned to the arithmetic twice: the  $L_P$  no-go breaks completely (gain  $\equiv 1.0000000000$  exactly), and the anti-fitting scramble (uniform random atom positions, same total mass) destroys the effect by a factor 4569.
  - **The best lower bound, the verdict, and two honest contract corrections.** Every route is an instance of the trial-vector theorem ( $1/g_K = \mu_1^{P-1} \min\{u^\top Gu : u_1 = 1\}$ ); the best closed route (two-rung,  $\beta$  from the atom block) certifies gain  $h^{+1.319}$  against the target  $h^{+3.110}$  — **an exponent gap of  $h^{+1.791}$ , not a constant: the one missing inequality is the cancellation** (required  $\beta$  precision  $3.57 \times 10^{-4}$ , all closed candidates miss by 12.4–2540); sharpest restatement  $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ , first flat at  $K = 6$  — exactly one uniform inequality missing, in three dresses (the det-ratio bound, the closed  $\beta$ , a closed near-null vector of  $A$  in the low modes). Corrections recorded honestly: form F2 (gain  $\geq ch^{3-\varepsilon}$ ) is strictly *weaker* than  $\inf_m g_{16} > 0$  ( $B_{11} \sim h^{+3.110}$ , F1 needs  $\varepsilon \leq -0.110$ ); the confinement route was a sub-surface artefact ( $h^{+2.714}$  on frame A  $h \leq 780$  vs  $h^{+1.963}$  on the full union — exposed by raising the eigen-horizon to the cap). Errand closed:  $U_{\text{ref}}(\text{union}) = 1.15 \times 6.4781 = 7.45$  supersedes T164's 4.90 and T165's 5.7327 on the union (off-recipe rule declared, CERT-WINDOW, never CERT-UNIF); the union supremum is zone-depth-driven (multi-covariate gain  $\sim h^{+2.92}\nu^{-1.53}$ ). Balance +5 THEOREM / +3 CERT-WINDOW / +4 MEASURED, deliberately no new CERT-UNIF; candidates P166.1...P166.9 all PENDING (no promotion — bundling with T167 planned). T167 (`null_vector_probe.py`, contract `NULL.VECTOR`) is running the most constructive dress.
  - **Docs + website sync.** `tfpt_prime_front` gains the T166 subsection (regression form, diagonal invariance and the Gram-minor ratio as displays, the Gram-determinant key result with the T159/T160 connection, the scramble factor, the two contract corrections, the “After T166” keybox), the phase-2 section title moves to T126–T166, the probe index gains the T166 row and the running totals move to 166 probes / 4355 checks; `experiments/next.txt` gains the T166 entry (series (CLXXVII)); website: `primeFront.ts` feed entry T166, status surfaces Teile 11–166 / 4355 checks, T167 running. Firewall unchanged: sandbox, no RH evidence, no zero data; every number typed THEOREM/CERT-UNIF/CERT-WINDOW/MEASURED/FIT.

**2026-07-29 (XII) (v556 — the gauge/ $P_{\text{pr}}$  identities of phase 2 promoted; prime-front diary T165 (the successor closes by certificate, one quantifier remains —**

## ETA-RESISTS) + paper + website sync; NO marker moves)

- **New module v556 (PRIME.GAUGE.PPR.01).** Promotion of the closed theorem cores of the discovery parts T164 (`sector_change_probe.py`) and T165 (`alignment_eta_probe.py`) — 23 checks,  $\sim 0.3$  s, everything recomputed on v551–v555’s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (Chebyshev 1852 / Rosser–Schoenfeld 1962), nothing cited from the sandbox; the companion to v555, which certified the *price structure* of the pairing — this module certifies its *gauge structure*. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; the T164/T165  $h$ -exponents ( $1/g_{16} \sim h^{+0.061}$ , the ascent price  $\sim h^{+3.3}$ , the split  $+3.110 - 3.049 = +0.061$ ) are sandbox fits and are *not* consumed; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made; the ONE open object after T165 ( $\inf_m g_{16}(m) > 0$ , the cascade-gain quantifier) stays open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The gauge theorem (T164, S2.1) [E].**  $x_1 = 1$  fixes  $a_1 = 1/\sqrt{\mu_1}$ , i.e. only the *scale* of  $a$ ;  $Q$  and TV are homogeneous of degree two in  $a$ ; hence the demand ratio  $Q/\text{TV}$  of the same  $a$ -direction — and with it  $\delta_{\text{bnd}}$  — is the *same number in every sector*: unchanged to  $2.9 \times 10^{-11}$  on 2520 sector-by-trial-vector combinations (5 weight families  $\times$  7 shifts  $\times$  6 taper widths on 12 windows), with  $\text{TV} \geq |w_0|$  in every sector and the pure shift scaling the certified value by exactly  $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$  ( $2.2 \times 10^{-11}$  on 432 points); candidate (i) settled in closed form — full-space  $\mu_0 = 0$  exactly, strictly worse. Mutation: the wrong transport map (the  $x$ -coordinates instead of the  $a$ -direction) moves the ratio by  $2.0 \times 10^2$ – $5.4 \times 10^3$ .
- **(2) The  $h^2$  conservation law (T164, S2.2) [E].** floor  $\times$  transfer  $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$  *identically* at every shift on the declared grid (72 window-by-shift points,  $2.2 \times 10^{-16}$ ): a sector shift can flatten the floor but pays the identical factor back as a transfer — the  $h^2$  is conserved, never removed; it lives in the *ratio*, not in the floor. Mutation: the wrong transfer misses by 0.9995 at the largest shift on every window.
- **(3) The power-one spend as an identity (T164, S1) [E].** The  $r$ -ceiling  $r \leq U/L$  with  $L = \lambda_1(A)/\mu_1^P = 1.1019$ – $1.5845$  (computed per instance from the full window) is *linear* in the entry ceiling, so  $d \log r / d \log U = +1$  exactly (central difference at  $U = 1/g_{16}$ , within  $1.1 \times 10^{-14}$  on 12/12): an  $h^\varepsilon$  ceiling costs  $h^{-\varepsilon}$  of  $r$ -margin for every  $\varepsilon > 0$  — no free tolerance in the consumption step (the measured power of the full T156 loss margin stays sandbox). Mutation: the quadratic consumption reads slope 2.0000 and is recognised.
- **(4) The  $P_{\text{pr}}$  identity and the predicate equivalence (T165, T1) [E].** The gauge-invariant dashboard satisfies  $P_{\text{pr}} = g_{16} \cdot R \cdot (\text{TV}/(t_1 v)^2)/\mu_1^P$  with no inequality anywhere — four factors, each a named object of the chain: the R-E-A quantifier  $g_{16}$ , the demand  $R = Q/\text{TV}$ , the T163 floor  $\text{tvf} \geq 1$ , and the Kac–Murdock–Szegő 1953 scale  $1/\mu_1^P = 943.6$ – $8609.6 \sim h^2$  — exact to  $4.3 \times 10^{-16}$  on all 192 battery vectors (knob settings, random sixteen-vectors *and* random full-window vectors), gauge-invariant under  $v \rightarrow tv$  to  $1.5 \times 10^{-11}$ ; the predicates  $R > 2\kappa$  and  $P_{\text{pr}} > 2\kappa g_{16} \text{tvf}/\mu_1^P$  agree as booleans on all 192 — the demand half of  $R2''$  and the crossing condition are *one* predicate, so R-F was never independent and its  $m$ -uniform form *is* R-E-A — and every crossing vector pays  $P_{\text{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$ – $142.8$ , above the preregistered tolerance  $\Theta_{\text{TOL}} = 10$  on 12/12: **the R-F conjunction is empty by an identity plus a floor, a certificate and not a failed search** (T165’s ascent stays sandbox, no search is run). Mutation:  $(t_1 v)$  unsquared scales like  $t$  and moves by  $1.0$ – $1.0 \times 10^4$ .
- **(5) The Schur-cascade direction and the exponent closure (T164/T165) [E].**  $s = \mu_1^P t_1^\top A^{-1} t_1 = 0.1696$ – $0.2472 \geq g_{16} \geq g_1$  window by window ( $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  — the T158 ladder is a chain of upper bounds with the route-(0) rung the loosest),  $P_K = \hat{a} s = 254.8$ – $7.7 \times 10^3 \geq 1$  (Kantorovich 1948); the exponent closure is an *identity of the lists* — per instance



$\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$  to  $1.6 \times 10^{-15}$ , the least-squares fits closing to  $1.3 \times 10^{-15}$  by linearity (T165's  $+3.110 - 3.049 = +0.061$  is this identity; the exponent values are sandbox fits, not consumed — what is promoted is that the certification lives in the cascade gain  $g_{16}/g_1$ , the object of the one open quantifier). Mutations: the  $g_8$  closure breaks by 0.10–0.24; a corrupted sixteen-block makes the Cholesky *refuse* on 12/12. Plus the monotone no-go staircase (damage  $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$  over  $\nu = 4/2/1/0.5$ , exact null control) and exact Dirichlet/parity controls.

- **Prime-front diary T165** (`alignment_eta_probe.py`, contract `ALIGNMENT.ETA`, 30/30, 15.1 s, the **T163/T164 surface rebuilt and re-verified, window ladder up to  $h = 1445$ , verdict ETA-RESISTS in the strong form, sandbox**). The successor question R-F closes *by certificate, not by failed search*: the new gauge-invariant identity  $P_{\text{pr}} = g_{16} \cdot R \cdot (\text{TV}/(t_1 v)^2) / \mu_1^P$  (machine-checked to  $3.3 \times 10^{-16}$ ) makes demand and price *one* equation, so with T163's floor every crossing vector pays  $\geq 2\kappa g_{16} / \mu_1^P$  — the old KMS  $h^2$  — and the two R-F clauses cannot be chosen independently. The anatomy: the free optimum is a *low-harmonic* object (93.7–96.8% of its energy in the first 32 parity modes against a 2.4–22.5% baseline), essentially orthogonal to the PNT main term (cosine  $-0.070 \dots - 0.031$ ), with heavy edges on prime-power cells (lift 1.36–2.12); the price decomposes *exactly* (to 0.0):  $h^{+3.261} = +1.997$  (KMS — the same  $h^2$  since T152)  $+1.279$  (demand overshoot the chain never asked for)  $+0.093$  (floor)  $-0.108$  ( $g_{16}$ ); the bar-tight counterfactual stays  $h^{+1.982}$ .  $\eta(\text{Cap})$  is two-sided: the ceiling is a theorem, and at  $\text{Cap} = 10$  it falls short of the requirement by  $5.5\text{--}392\times$  on every window, 0/27 ladder points reach the bar. The equivalence:  $R > 2\kappa$  and T163's crossing predicate are the *same* predicate (105/105 vectors)  $\Rightarrow$  R-F is strictly *stronger* than the R2'' demand, and the clauses collide — **R-F is empty; its  $m$ -uniform form is R-E-A**. T2 localises the quantifier further: the  $m$ -free budgets bound  $B_{11}$  sharply (factor 21–26) — but the wrong object:  $1/g_1$  grows  $h^{+3.110}$ ,  $1/g_{16}$  is flat  $h^{+0.061}$ , exponents close exactly; the whole certification lives in the 15 ladder increments — “bound  $g_{16}/g_1$  from below by  $h^{3-\varepsilon}$ ”, one scalar per window, a cancellation and hence provably beyond absolute-value budgets. T3, the decoupled surface ( $\nu$  as a knob: 25 admissible windows over 6 zones,  $\nu \in \{4, 5, 6, 8, 11, 16\}$ , collinearity  $0.87 \rightarrow 0.68$ ): **the quarter-bar drift is zone depth, not window length (factor 19)** — and two honest costs reported openly:  $\sup 1/g_{16}$  rises to 5.7327, exceeding T164's  $U_{\text{ref}} = 4.90$  by 17% (the constant was frame-A-specific and is retired as a surface constant), and the  $h$ -coefficient of  $1/g_{16}$  is no longer negligible at free  $\nu$ . End-to-end on the union (52 windows,  $h = 142\text{--}1320$ ): every crossing vector has  $P_{\text{pr}} \geq 54.6\text{--}5865 > \Theta_{\text{TOL}}$ , closed for every  $h > 84.9$ . Stress: T145 coarsening monotone (1.0/2.8/12.1/27.5, up to  $102\times$ ); surrogate  $8.6 \times 10^3\text{--}6.9 \times 10^4$ ; Dirichlet/KMS exact. Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED; 6 new PENDING candidates. **The honest bottom line: exactly ONE genuine open object remains** —  $\inf_m g_{16}(m) > 0$ , **a quantifier over a certified flat list; no missing term, no missing device, no sector**. T166 (`schur_cascade_probe.py`, contract `SCHUR.CASCADE`) is running at exactly that quantifier. NOT RH evidence; no marker of any pre-existing contract moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T165 subsection (the  $P_{\text{pr}}$  identity as the key display, the exact price decomposition, the R-F  $\iff$  R-E-A equivalence, the decoupled surface with the zone-depth answer and the retired  $U_{\text{ref}}$  reported honestly) and the After-T165 keybox (the one cascade-gain quantifier, T166 running), the v556 subsection in the verified layer (now twenty-two modules), the phase-2 section retitled T126–T165, the T165 row in the probe index and the updated counters (165 probes, 4325 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v556 companion blocks; `experiments/next.txt` gains the Teil-165 entry; `website/lib/primeFront.ts` gains the T165 and v556 feed entries; the prime-front status surfaces move to Teile 11–165 / 4325 checks / 22 modules (v535–v556); suite counter  $549 \rightarrow 550$  scripts. Ledger row `PRIME.GAUGE.PPR.01 [E]`; **no marker of any pre-existing contract moves**.

**2026-07-29 (XI) (prime-front diary T164 (both arms of the successor decided: the  $O(1)$  gate discharges to a single quantifier, sectors are a gauge — TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B)) + paper + website sync; NO promotion, NO marker moves)**

- **Prime-front diary T164** (`sector_change_probe.py`, contract `SECTOR.CHANGE`, 28/28, 31.7 s, the T163 surface rebuilt and re-verified: 28 log-spaced prime-power zones  $h = 142 \dots 1445$  with declared caps, 27 windows inside the cond horizon,  $|\psi(x) - x| \leq \kappa x$  with  $\kappa = 0.038821$  verified at every jump point in  $[100, 4 \times 10^5]$ ; verdict **TOLERANCE-CARRIES (arm A) / SECTOR-RESISTS (arm B), sandbox**). Both arms of T163's successor R-E are decided. *Arm A (the tolerance analysis)*: the station chain is instrumented with a power at every station — the T158 ladder really is a chain of upper bounds on the entry ceiling ( $s \geq g_{16} \geq g_1$  window by window), and at the heart the T156 kernel spends the ceiling at power *exactly one*:  $d \log(1 - F)/d \log U = -1.0005 \dots -1.0000$  with  $d \log r/d \log U = +1.0000$  exactly, so a ceiling growing like  $h^\varepsilon$  costs a margin decaying like  $h^{-\varepsilon}$  for every  $\varepsilon > 0$  — the declared one-sided rule returns  $\varepsilon^* = 0.50$  and the probe *disowns* it at once as the  $\varepsilon = 0$  tailwind ( $h^{+0.391}$ , bought from the measured drift of  $L$ ) divided by power one: no structural slack. But the sharp question is answered the other way: the single constant  $U_{\text{ref}} = 4.9008 = \max_h 1/g_{16}$  carries all 9 sub-surface windows out of sample,  $1/g_{16} = 1.7527\text{--}5.3286$  is flat ( $h^{+0.061}$  over a  $9 \times$  lever arm, split halves  $-0.010/+0.103$ ), and every  $g_K$  is strictly increasing (Schur 1917) — **the  $O(1)$  gate is not open as a quantity: it is discharged window by window by a Cholesky identity**, and what remains is exactly one quantifier,  $\sup_m 1/g_{16}(m) < \infty$  — the  $m$ -freedom of a certified flat list, the same grammatical form as every other open uniformity item and no longer an existence question about trial vectors. *Arm B (the sector change)*, *negative by a theorem*:  $x_1 = 1$  fixes only the *scale* of  $a$ ;  $Q$  and TV are homogeneous of degree two, hence  $Q/\text{TV}$  and  $\delta_{\text{bnd}}$  are the *same numbers in every sector* — the entry normalisation is a **gauge** (machine-checked to  $8.8 \times 10^{-10}$  over 5670 sector-by-vector combinations, declared horizon  $2.6 \times 10^{-8}$ ); the full space is strictly worse ( $\mu_0 = 0$  exactly, no finite floor); a shifted or weighted sector does flatten the floor ( $h^{+0.000}$  at  $\kappa_s = 4$ ) but pays the transfer factor  $1/\theta \sim h^{+1.997}$ , and floor  $\times$  transfer  $= 1/\mu_1^P$  *identically* — the two exponents sum to  $+1.997$  at every shift on the declared grid to  $10^{-9}$ : the  $h^2$  lives in the ratio, never in the floor. The genuine surprise, stated and immediately priced: a normalisation-free first-order ascent on  $Q/\text{TV}$  over the full window reaches 2.38–74.57 against the crossing bar  $2\kappa = 0.0776$  —  $31\text{--}960 \times$  over on 9/9 windows ( $\delta_{\text{bnd}} = -0.124 \dots -0.090$ , alignment efficiency  $\eta = 7.9\text{--}11.1\%$  of the Abel ceiling  $\|C\|_\infty = 30.3\text{--}930.6 \sim h^{+1.185}$ ) — and T163's TV floor *holds on the unconstrained optimiser* (TV  $\mu_1^P = 8.30\text{--}11.72 \geq 1$ , a family T163 never searched), at a price  $h^{+3.299}$ , worse than T163's crossing price  $h^{+1.91}$ : dropping the normalisation relocated the difficulty without reducing it.
- **R-B''' narrowed, R-D settled in type, R-F named**. R-B''' is a *naming* problem — two different quantities carried one name: the T162  $a$ -weighted quarter-bar object is  $0.1001\text{--}0.1623 \sim h^{-0.172}$  (T163's  $-0.172$  reproduced to 0.0004, under the 0.25 bar on all 27 windows with 35% margin), the raw arch/atom cancellation is a different number ( $h^{-2.04}$ ); a two-variable regression puts  $-0.046$  of the drift on  $\log h$  against  $-0.477$  on  $\log \alpha$  ( $10 \times$  more on the zone scale), and on this recipe  $\alpha$  and  $h$  are coupled by construction — narrowed, not closed: a surface with independent  $\alpha$  and  $h$  is needed. R-D is settled in *type*: under the shift  $s \rightarrow s/\theta$ ,  $\hat{a} \rightarrow \theta \hat{a}$ ,  $L \rightarrow \theta L$ , so  $P_K$ ,  $r$  and  $F(P_K, r)$  are unchanged to  $3.5 \times 10^{-16}$  on 54 combinations — every object the chain consumes downstream is a ratio of equally-homogeneous forms, hence no fifth device can come from a sector change ( $\rho = 1.0036\text{--}1.0140 > 1$  flat, T161, quoted and untouched). The one genuinely new item is **R-F**: is there a  $v$  with  $Q(v)/\text{TV}(v) > 2\kappa$  and  $Q(v)$  comparable to  $1/s$  — equivalently, can the alignment efficiency  $\eta$  stay above  $2\kappa/\|C\|_\infty$  at bounded price? No sector, no normalisation, no prime. Stress: the uniform surrogate breaks by  $1.5 \times 10^3\text{--}4.9 \times 10^7$ , the T145 no-go staircase is monotone ( $1.0 \rightarrow 9.1 \rightarrow 25.4 \rightarrow 129 \rightarrow 434$ , exact null control at  $\nu = 4$ , damage  $\geq 26.1$  at the coarsest setting on every window), the gauge break

is  $3.4 \times 10^{-2}$ , Dirichlet/parity/orthonormality controls  $\leq 2 \times 10^{-14}$ , and the random- $v$  pairing identity holds to  $3.8 \times 10^{-16}$ . Balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 6 MEASURED, zero load-bearing fit rows; candidates T164-Q1...Q9 all PENDING (no promotion — to be bundled with T165). NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T164 subsection (the power-one measurement, the quantifier collapse as the key result, the gauge theorem with the product identity, the unconstrained ascent with the TV-floor hold, the R-B''' narrowing with the  $\alpha/h$ -coupling lesson) and the After-T164 keybox (R-E-A/R-F/R-B'''/R-D, T165 running), the phase-2 section retitled T126–T164, the T164 row in the probe index and the updated counters (164 probes, 4295 checks); `experiments/next.txt` gains the Teil-164 entry; `website/lib/primeFront.ts` gains the T164 feed entry; the prime-front status surfaces move to Teile 11–164 / 4295 checks. T165 (`alignment_eta_probe.py`, contract ALIGNMENT.ETA) is running at the alignment successor R-F. No ledger row (sandbox); **no marker of any pre-existing contract moves.**

**2026-07-29 (X) (v555 — the Pareto/total-variation identities of phase 2 promoted; prime-front diary T163 (the front resists as a theorem — the  $h^2$  is the parity spectral gap, not the primes, FRONT-RESISTS) + paper + website sync; NO marker moves)**

- **New module v555 (PRIME.PARETO.TV.01).** Promotion of the closed theorem cores of the discovery parts T162 (`third_split_probe.py`) and T163 (`pareto_front_probe.py`) — 23 checks,  $\sim 0.4$ s, everything recomputed on v551–v554's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (Chebyshev 1852 / Rosser–Schoenfeld 1962, reproduced to  $4.2 \times 10^{-7}$ ), nothing cited from the sandbox; the companion to v554, which certified what the sixteen-form *pairs against* — this module certifies the *price structure* of that pairing. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; the T163  $h$ -exponents ( $P_{\text{aff}} \sim h^{3.05}$ ,  $P_{\text{cross}} \sim h^{1.91}$ , the TV exponent  $\sim h^{2.00}$ ) are sandbox fits and are *not* consumed; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 (R-E the successor with both arms prime-free; R-B''' the  $h$ -uniform positivity margin; R-D a fifth R1'' device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The exchange law and the crossing criterion (T163, R1.2) [E].**  $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}(x)/P(x))/\log X$  is an *identity* at all 612 grid points of the declared Fejér knob (50 geometric  $\sigma$  on  $[1, 4096]$  plus  $\infty$ , 12 windows;  $5.1 \times 10^{-16}$  relative, admissibility  $Q(x) > 0$  everywhere), and the crossing criterion  $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16} \text{TV}$  holds as an exact equivalence at every point — price and demand are two coordinates of *one* object. Mutation: the wrong constant moves the law by  $6.5 \times 10^{-2}$ –3.4.
- **(2) The total-variation floor (T163, R2.2) [E].** (T1)  $w_0(x) = \|a(x)\|^2$  ( $4.9 \times 10^{-16}$ ); (T2)  $\text{TV}(x) \geq |w_0(x)|$  by telescoping with  $w_M = 0$ ; (T3)  $x_1 = 1$  forces  $\|a\|^2 \geq 1/\mu_1^P$ , hence  $\mu_1^P \text{TV}(x) = 5.00$ –28.84  $\geq 1$  on all 708 trial vectors built, against the closed floor  $1/\mu_1^P = 1/(4 \sin^2(\pi/N)) = 943.6$ –8609.6  $\sim h^2$ ; (T4) every sub- $\frac{1}{2}$  grid point pays  $P > 2\kappa g_{16}/\mu_1^P = 15.4$ –142.8 (all 54 respect it) — a flat-price sub- $\frac{1}{2}$  demand is impossible in the parity sector on this surface: **R-C''' is closed negatively, by an inequality.** Negative control: the *inadmissible* vector  $0.01x^*$  undercuts the floor ( $2.5$ – $2.9 \times 10^{-3}$ ) and is recognised — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953).
- **(3) The chain-derived price axis (T163, R1.0) [E].**  $\sigma = 1$  gives  $x = e_1$  exactly with  $P = g_{16}/g_1$  (the  $K = 1$  rung of the T158 Cholesky ladder, the T157 route-(0) certificate;  $4.1 \times 10^{-16}$ ),  $\sigma = \infty$  gives  $g_{16}Q(x^*) = 1$  (the  $K = 16$  top rung;  $1.2 \times 10^{-12}$ ); the ladder carries all

16 terms strictly positive with monotone partial sums, the knob is monotone in both coordinates on every window (the knob curve *is* the Pareto front). Mutation: a corrupted sixteen-block makes the Cholesky *refuse* on 12/12.

- **(4) The Mellin cell moments and the Abel step (T162, C2/C5) [E].** The  $k$ -th Mellin rung  $\sum_d w_d C_d(-(2k + \frac{1}{2}))$  with  $C_d(s)$  the closed tent integral (Mellin 1896) agrees with an independent per-cell Gauss–Legendre quadrature to  $2.8 \times 10^{-13}$  on every rung  $k = 1 \dots 4$  (shifted tent centres break by  $\geq 1.3 \times 10^{-2}$ ); the boundary-free Abel identities hold at levels  $k = 1 \dots 3$  ( $8.3 \times 10^{-17}$ ; Abel 1826), the dropped-boundary level-1 form agrees because the gauge identity kills the boundary term exactly, the optimal level is *one* for T162’s closed reason ( $32\pi/\alpha = 40.6\text{--}59.4 > 1$ , measured ratio 30.8–69.1), and a gauge violation breaks by its closed prediction  $v_0 c_0 M$  to  $1.2 \times 10^{-15}$  (the positive control).
- **(5) The Lerch/Frullani log-moment (T162, R-A') [E].**  $L = \sum_d \Psi_d w_d$  agrees on three routes per window: direct; double-Abel *with* the Wronskian boundary term ( $2.8 \times 10^{-14}$ ; dropping it breaks by  $2.2 \times 10^{-4}\text{--}1.3 \times 10^{-2}$ ); the Lerch/Frullani integral with the  $d = 1$  term peeled closed ( $1.8 \times 10^{-15}$ ) — the peel constant  $2 \log 2$  reproduced to  $2.2 \times 10^{-16}$  and  $\Psi_1 = -\log 2$  exactly (Frullani 1828 / Lerch 1887 / Clausen 1832). Plus the monotone no-go staircase (damage  $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$  over  $\nu = 4/2/1/0.5$ , exact null control) and exact Dirichlet/parity controls.
- **Prime-front diary T163 (pareto\_front\_probe.py, contract PARETO.FRONT, 32/32, 1.9 s, 28 log-spaced prime-power zones  $h = 142 \dots 1445$  with declared caps, 27 windows inside the cond horizon, verdict FRONT-RESISTS, sandbox).** The survey of the Pareto front, and the front resists *as a theorem*: the price axis is chain-derived ( $P_{\text{aff}} = g_{16} B_{11} \sim h^{+3.049}$ , the T157 route-(0) certificate), the Fejér knob has both endpoints in the chain and is measured monotone in both coordinates (the knob curve *is* the front), no flat price crosses the yardstick (0/27 at caps 1.25/2/10 and every flat rung tier; only the full  $P_{\text{aff}}$  reaches 27/27), the crossing price  $P_{\text{cross}} \sim h^{+1.906}$  sits at 0.0066–0.2047 of  $P_{\text{aff}}$  — 5–152× strictly inside the already-accepted certificate, margin widening — and the price anatomy is an *identity*:  $\delta_{\text{bnd}} = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}/P)/\log X$  exactly at all  $27 \times 50$  grid points, the crossing exponent decomposing additively (TV +2.013,  $g_{16}$  –0.061). The four-line theorem (T1–T4) gives every admissible trial vector the total-variation floor  $\text{TV} \geq 1/\mu_1^P \sim h^2$  (verified on all 1674 built, slack 5.0–23.2), so the crossing price is bounded below by a quantity growing like  $h^2$ : the  $h^{2.86}$  of T162, the  $h^{1.91}$  of the crossing and T161’s granularity are *all* the reciprocal smallest parity eigenvalue — **the spectral gap of the parity Laplacian meeting the entry normalisation  $x_1 = 1$ : it was never in the primes**. The mode sweep  $K = 16 \rightarrow 64$  confirms it from the other side (price dividend, worse demand, TV exponent unmoved at  $h^{+1.81}$ ); R-B'' reduces to an  $h$ -uniformity question (the chain never needed the sign: duality plus an evaluated number, headroom  $\geq 10^{14.9}$ ); R-D is a type mismatch (operator smoothing degrades the floor by the same  $h^2$ ). Balance 11/1/4/10, zero fit rows. T164 (sector\_change\_probe.py), running, tests the successor R-E. NOT RH evidence; no marker moves.
- **Docs + website sync.** tfpt\_prime\_front gains the T163 subsection with the exchange-law display, the T1–T4 floor theorem as the key result, the chain-derived price axis, the monotone no-go staircase and the After-T163 keybox (R-E/R-B'''/R-D, T164 running), the v555 subsection in the verified layer (now twenty-one modules), the phase-2 section retitled T126–T163, the T163 row in the probe index and the updated counters (163 probes, 4267 checks); tfpt\_1, tfpt\_3 and tfpt\_research\_contracts gain the v555 companion blocks; experiments/next.txt gains the Teil-163 entry; website/lib/primeFront.ts gains the T163 and v555 feed entries; the prime-front status surfaces move to Teile 11–163 / 4267 checks / 21 modules (v535–v555); suite counter 548  $\rightarrow$  549 scripts. T164 (sector\_change\_probe.py) is running at the sector-change successor. Ledger row PRIME.PARETO.TV.01 [E]; **no marker of any pre-existing contract moves**.

**2026-07-29 (IX) (v554 — the sampling/harmonics identities of phase 2 promoted; prime-front diary T161 (the circularity triage: the hardness sits in the split, not in the primes, CLOSURE-RESISTS) + paper + website sync; NO marker moves)**

- **New module v554 (PRIME.SAMPLING.HARM.01).** Promotion of the closed theorem cores of the discovery parts T160 (`pairing_probe.py`) and T161 (`classical_closure_probe.py`) — 21 checks,  $\sim 0.5$  s, everything recomputed on v551–v553’s declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ) plus a declared  $\Lambda$ -battery of six log-spaced cut-offs  $X = 10^3 \dots 3.2 \times 10^5$  for the PNT item (finite von-Mangoldt sums only, no matrix anywhere), nothing cited from the sandbox; the companion to v553, which certified the algebra of the sixteen-form — this module certifies *what that form pairs against*, and it closes no open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement; finite  $\Lambda$ -sums are allowed and used, *no RH statement* is made, and the  $\delta$  triage of T161 (required cancellation depth against RH strength) is a documentation item, *not* a module claim; the open terms after T161 (R-A’ the prime-free log-moment; R-B’ the  $m$ -free  $16 \times 16$  Gram fraction bound; R-C’ one split of the pairing with  $\delta_{\text{eff}} < \frac{1}{2}$ ; R-D a fifth R1’’ device) stay open and typed open; each statement carries at least one mutation control that must fail loudly.
- **(1) The sampling identity (T160, P2) [E].** The atom half of the pairing equals  $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n}) \hat{w}(\log n)$  (plus one reflected term below the grid;  $\hat{w}$  the linear interpolant of the lag weights), exact to  $5.6 \times 10^{-15}$  of the absolute atom scale on 12/12 windows: identically a finite combination of sums  $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$  at 32 explicit frequencies. Mutation: a one-index grid shift breaks by  $5.1 \times 10^{-5} - 3.1 \times 10^{-3}$ .
- **(2) The moment laws and the TV theorem (T160, P1/P4) [E].**  $m_0 = 0$  ( $1.0 \times 10^{-16}$  of  $\|w\|_1$ ),  $m_1 = -[S_0^2 + 2 \sum_j P_j^2]$  ( $9.2 \times 10^{-16}$ ) and  $m_2 = -[2S_1 - (M-1)S_0]^2$  ( $3.4 \times 10^{-14}$ , a perfect square), both strictly negative on every window; U3 as a theorem with *checked* hypotheses ( $c^{\text{arch}} \leq 0$  and monotone on  $d \geq 1$ , 12/12):  $\text{TV}(c^{\text{arch}}) = 3.9817 - 4.8991 \leq |c_0| + 2 \sup < 6$ , closed and window-independent. Mutations: a one-index prefix shift breaks  $m_1$  by  $\geq 1.25 \times 10^{-2}$ , an  $(M-2)$  mis-normalisation breaks  $m_2$  by  $\geq 6.1 \times 10^{-2}$ , the full atom-including kernel violates the monotone hypothesis on 12/12.
- **(3) The harmonic-frequency theorem and the PNT main term (T161, P3) [E].**  $t_j(2\alpha) = 2\pi j$  *exactly* ( $2.8 \times 10^{-14}$ ) on every window and every battery cut-off — the 32 frequencies of the sampling identity are precisely the Fourier harmonics of the log-window  $[0, 2\alpha]$ ; at those frequencies  $\theta = 2\pi j$  collapses the partial-summation main term to the closed, parameter-free prediction  $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$  (de la Vallée Poussin 1896; the Mellin factor of  $x^{-1/2}$ ), matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12, the subtraction reducing the aggregate residual to  $0.008 - 0.466 \leq 0.5$  on every cut-off. Mutations: a wrong Mellin pole worsens the aggregate residual by  $\times 1.8 - \times 95.8$ ; a mass-matched *uniform* fake measure misses by 0.23–0.51 — the agreement is a property of the von-Mangoldt measure, not of any measure with the same mass.
- **(4) The head split and the scale-free Bernstein rate (T161, P1) [E].**  $c_d^{\text{arch}} = \Psi_d + D\hat{G}(dD, D)$  with  $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$  closed,  $D$ -free and  $m$ -free (the triangle-smeared simple pole; the  $\log D$  cancels identically), exact to  $3.2 \times 10^{-16}$  of  $\sup |c^{\text{arch}}|$  at *every* lag of every window, the two  $\Psi$  representations agreeing to  $2.9 \times 10^{-14}$ ; and  $\rho^* = (3 + \sqrt{5})/2 = 2.618034$ : the root of  $x^2 - 3x + 1$ , returned exactly by the  $D \rightarrow 0$  limit of the peeled interval  $[0.4\alpha, 2\alpha]$  and invariant under  $\alpha \times 10$  — scale-free, every window’s  $\rho(D) = 2.588 - 2.608$  below it (Bernstein 1912). Mutations: dropping  $\Psi$  misses  $c^{\text{arch}}$  by 0.99–1.00; the wrong interval ratio moves the limit to 3.0.
- **(5) The off-diagonal fraction bound and the sign refutation (T161, P2) [E].** The gauge identity licenses one boundary-free Abel step,  $Q^{\text{arch}} = \sum_{k,l} a_k a_l \sum_{d \geq 1} (\Delta c^{\text{arch}})_d R_{kl}^1(d)$

exactly ( $2.0 \times 10^{-13}$ ), and the certified per-window inequality is  $|\text{off-diagonal block}| \leq \frac{1}{4}|Q^{\text{arch}}|$  (measured 0.171–0.212 on 12/12) — with the *refuted* sign law wired as the negative control: the arch half is negative while its off-diagonal block is positive on 12/12, and 176/240 off-diagonal pairs violate the per-pair form on every window — nothing single-signed is promoted. Mutation: a one-index  $R$ -kernel shift breaks by  $\geq 6.8 \times 10^{-5}$ . Stress: the head split misses the T145 no-go reconstruction by 2.39; the Dirichlet cosine-sum identity (including the degenerate branch), the KMS eigenpairs and  $\text{odd\_toeplitz}(c^L) = L_P$  (zero tolerance) are exact.

- **Prime-front diary T161** (`classical_closure_probe.py`, `contract CLASSICAL.CLOSURE`, 35/35, 1.2 s, 19 **log-spaced prime-power zones**  $h = 50 \dots 1445$ , **verdict CLOSURE-RESISTS**, **triage verdict NEEDS-BEYOND-RH-STRENGTH**, **sandbox**). The two classical rests plus the circularity triage, and the triage returns its most consequential answer: *the chain is not circular*. R-A closes  $m$ -free —  $A = D\hat{A}$  exactly, only the  $1/s$  head binds, the scale-free Bernstein rate  $\rho^* = (3 + \sqrt{5})/2$ , the closed head split (no peeling), and an explicit degree schedule  $K(h) = O(\log h)$  (a *fixed* degree is refuted with a number; the residual is the prime-free log-moment, outside the polynomial ladder). R-B is refuted in all four readings (3168/4320 pairs; the aggregate by *sign*) and replaced by the certified fraction bound  $0.1035\text{--}0.1713 \leq \frac{1}{4}$ , flat. The triage: the 32 frequencies are exactly the Fourier harmonics of the log-window (theorem), the measured cancellation is fully the PNT main term, and the required depth is  $\delta = 1.1482\text{--}1.8809$  against RH strength  $\frac{1}{2}$  — in absolute terms 0.012–0.31 of the last term of the sum, below the boundary term of every partial summation, so no strengthening of the  $\psi(x) - x$  input reaches it: the chain needs *more* than RH-strength input would supply, which localises the  $h^2$  in the *split* rather than in the primes; a re-split moves the demand to  $\delta_{\text{eff}} = 0.9839\text{--}1.3767$  — it moves, but stays above  $\frac{1}{2}$  on 18/18. Balance 20/10/4/9/3. T162 (`third_split_probe.py`), running, searches for one split with  $\delta_{\text{eff}} < \frac{1}{2}$ . NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T161 subsection with the  $\delta$  table, the head-split display and the After-T161 keybox, the v554 subsection in the verified layer (now twenty modules), the phase-2 section retitled T126–T161, the T161 row in the probe index and the updated counters (161 probes, 4205 checks); the big-picture section states the split-design reading soberly; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v554 companion blocks; `experiments/next.txt` gains the Teil-161 entry; `website/lib/primeFront.ts` gains the T161 and v554 feed entries; the prime-front status surfaces move to Teile 11–161 / 4205 checks / 20 modules (v535–v554); suite counter  $547 \rightarrow 548$  scripts. T162 (`third_split_probe.py`) is running at the third-split search. Ledger row PRIME.SAMPLING.HARM.01 [E]; **no marker of any pre-existing contract moves**.

**2026-07-29 (VIII) (v553 — the exact-form identities of phase 2 promoted; prime-front diary T159 (the gauge identity and the Z-matrix law, cancellation is the weight, FORM-RESISTS) + paper + website sync; NO marker moves)**

- **New module v553 (PRIME.EXACT.FORM.IDENT.01).** Promotion of the theorem cores of the discovery parts T158 (`schur_entry_probe.py`) and T159 (`exact_form_probe.py`) — 25 checks,  $\sim 0.2$  s, everything recomputed on v551/v552's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; no-go ladders  $m = 48 \dots 288$  on *both* T145 forms; parity and Dirichlet controls at  $m = 64, 128, 256$ ), nothing cited from the sandbox; the companion to v552, which certified the four angle instruments — this module certifies the *algebra of the sixteen-form itself*, and it closes neither open term. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the two open terms after T159 — the one explicit pairing inequality for  $R2''$ , needing a correlation bound instead of sizes; a third device for the atom off-band block for  $R1''$  — stay open and typed open); each statement carries at least one mutation control that

must fail loudly.

- **(1) The Thomson dual form and the positive Cholesky ladder (T158, P1) [E].**  $s = (B^{-1})_{11} = \max_x (2x_1 - x^\top Bx)$  (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted: 8 random trial vectors per window never exceed  $s$ , so every trial is an upper bound on  $1/s$  — exactly the variational structure Cauchy–Schwarz lacks (v552 wired that refutation as a control; here the direction itself is the check). The ladder  $g_K = \sum_{j \leq K} y_j^2$ ,  $y = L^{-1}e_1$  from *one* Cholesky of the leading  $16 \times 16$  block: all terms strictly positive, partial sums strictly monotone,  $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  literally on 12/12, tight to 1.0642–1.1522 against the declared cap 1.5 (T158 measured  $\leq 1.2738$  at  $h = 254 \dots 1393$ , sandbox); mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12.
- **(2) The two-dimensionality (T158, P2) [E].**  $\text{span}\{t_1, A^{-1}L_P t_1\}$  attains  $s$  exactly ( $g/s = 1$  to  $9.0 \times 10^{-13}$  on 12/12) for the theorem reason  $L_P t_1 = \mu_1^P t_1$  (KMS 1953, eigenpair residual  $8.1 \times 10^{-13}$ ): the entry is a two-dimensional quantity the moment one Green column is granted. Negative controls: the two-sine span (no Green column) misses by 1.42–4.8, the corrupted Green column (no solve) by  $255\text{--}7.7 \times 10^3$ .
- **(3) The gauge identity and the TmH signature (T159, P4/P7) [E].** The reflection-odd section  $A_{rs} = c_{|r-s|} - c_{M-1-r-s}$  annihilates constant lag vectors *entrywise with zero tolerance* ( $\text{odd\_toeplitz}(\mathbf{1}) = 0$  exactly — the mechanism in one line), hence the gauge identity  $\sum_d w_d = 0$  holds to  $1.1 \times 10^{-14}$  of  $\sup |w|$  for the per-window optimiser and 3 random sixteen-vectors: the form is *blind* to the lag mass of  $c$ , and the lag mass is exactly where the archimedean and atom halves of the sixteen-form are  $h^2$ -sized and cancel. The premise — the Toeplitz-minus-Hankel signature  $A_{r,s} - A_{r+1,s+1} = f(r+s)$  — holds to  $1.4 \times 10^{-16}$ . Mutations: a one-index Hankel shift breaks the identity by  $2.7 \times 10^{-7}\text{--}4.2 \times 10^{-5}$ ; the T145 form  $R = aa^\top + \varepsilon I$  has signature defect  $7.4 \times 10^{-2}$  — not a parity section, the gauge identity is not even *defined* for it.
- **(4) The exact lag sum and the two closed scalars (T159, P2/P5) [E].**  $x^\top B_{LL} x = \sum_d c_d w_d$  with  $w_d = T_d - H_d$  against the direct assembly to  $2.7 \times 10^{-16}$  of the absolute scale  $\sum_d |c_d w_d|$  — and the honest price *measured*: relative to the cancelled total the same identity holds only to  $1.4 \times 10^{-11}$  on this small surface (the sandbox at  $h \leq 1293$  loses 4.0–8.1 digits; that depth *is* the  $h^2$  obstruction). The two closed scalars:  $w_0 = \sum_k x_k^2 / \mu_k^P$  (the  $N^2/4\pi^2$  factor itself) to  $4.2 \times 10^{-16}$  and  $2w_0 - w_1 = \|x\|^2$  to  $7.4 \times 10^{-16}$  (KMS 1953). Mutations: a 1% perturbation of the largest lag breaks the lag sum by  $2.2 \times 10^{-3}$ ; a one-index  $\mu$  shift breaks  $w_0$  by 0.60.
- **(5) The Z-matrix law and the Collatz–Wielandt floor (T159, U1/U2) [E].**  $B_{HH}^{\text{arch}}$  is a symmetric Z-matrix *raw* — every off-diagonal entry  $\leq 0$  with no tolerance on 12/12; the closed sign-based floor  $\min_k (B_{HH}^{\text{arch}} \mathbf{1})_k = 0.9093\text{--}1.1284 \geq t = 0.25$  with the theorem direction verified per instance against the exact  $\lambda_{\min}(B_{HH}^{\text{arch}}) = 0.9138\text{--}1.1432$  (Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8 on every window. The honest limit as content: the atom block obeys *no* sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is *negative* ( $-48.3 \dots -0.79$ ) against a positive exact bottom — admissible and vacuous,  $R1''$  stays open. Mutation: the checkerboard similarity  $S = \text{diag}((-1)^k)$  — an isometry,  $\lambda_{\min}$  moves by 0 exactly — destroys the raw law (fraction  $1.0000 \rightarrow 0.494\text{--}0.498$ ): the Z law is an entrywise fact, not a spectral one, and T159's checkerboard reading (S2) is refuted. Stress: the T145 no-go breaks on three axes of this module's structure; on the  $c(l) = 1/(1+l)$  reconstruction the ladder theorem survives while  $1/g_{16}$  explodes ( $1187 \rightarrow 7.11 \times 10^4$ ) — the instrument reports the collapse instead of hiding it; parity: for  $A = L_P$  the machinery returns  $s = 1$  with every ladder partial sum constant in  $K$ ; Dirichlet controls exact.
- **Prime-front diary T159 (exact\_form\_probe.py, contract EXACT.FORM, 41/41, 2.9 s, 23 of 24 prime-power zones  $h = 142 \dots 1293$ , the sign surface on 18 windows  $h = 142 \dots 963$ , verdict FORM-RESISTS, sandbox).** The cancellation executed algebraically: seven machine-checked identities to  $10^{-12}$  against the absolute scale (the cancellation eats 4.0–8.1

decimal digits, reported as a measurement), among them the closed 256-term Dirichlet-kernel form of the geometric weights and the *gauge identity*  $\sum_d w_d = 0$  exactly — the form is blind to the lag mass, exactly where the  $h^2$  halves live. The honest core answer is negative: the  $h^2$  does *not* telescope away, it is the common *weight* of both halves (both grow  $h^{+2.104}$ ), and four routes are closed with exponents — T158's own fixed sixteen-vector refuted at  $h^{+3.510}$ , one sine at  $h^{+3.066}$ , the preregistered ansatz family at  $h^{+3.001}$ , Abel- $\ell^1 \times \sup$  at every rung (best  $h^{+3.604}$ ). The structural win:  $B_{HH}^{\text{arch}}$  is exactly a symmetric Z-matrix with a closed sign-based Collatz–Wielandt floor 0.819–1.165 comfortably above  $t = 0.25$  — but the atom block obeys no sign law, and the full-block criterion is vacuous. Balance 6/3/4/1  $\rightarrow$  13 theorem / 6 cert-unif / 4 cert-window / 1 measured; a numerical horizon declared as its own measurement ( $\text{cond}(B_{LL}) > 10^{12}$  from  $h = 1292$ ); one fully explicit pairing inequality remains, under attack via correlation structure in T160 (`pairing_probe.py`), running. NOT RH evidence; no marker moves.

- **Docs + website sync.** `tfpt_prime_front` gains the T159 subsection with the After-T159 keybox, the v553 subsection in the verified layer (now nineteen modules), the phase-2 section retitled T126–T159, the T159 row in the probe index and the updated counters (159 probes, 4124 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v553 companion blocks; `experiments/next.txt` gains the Teil-159 entry; `website/lib/primeFront.ts` gains the T159 and v553 feed entries; the prime-front status surfaces move to Teile 11–159 / 4124 checks / 19 modules (v535–v553); suite counter 546  $\rightarrow$  547 scripts. T160 (`pairing_probe.py`) is running at the pairing bound. Ledger row PRIME.EXACT.FORM.IDENT.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (VII) (v552 — the four fixed-size angle instruments of phase 2 promoted; prime-front diary T157 (the confinement theorem and the one-entry reduction, ANGLES-RESIST) + paper + website sync; NO marker moves)**

- **New module v552 (PRIME.ANGLE.INSTR.01).** Promotion of the instrument cores of the discovery parts T155 (`bottom_floor_probe.py`) and T157 (`angle_floor_probe.py`) — 18 checks,  $\sim 0.4$ s, everything recomputed on v551's declared surface (the 12 deepest prime-power zones admitting a frame-A window inside the cap,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; a no-go size ladder  $m = 48 \dots 288$ ; parity and Dirichlet controls at  $m = 64, 128, 256$ ), nothing cited from the sandbox; the companion to v551, which certified the fixed-size ceiling step — this module certifies the four fixed-size *instruments* the angle campaign produced, and none of them closes an angle. Scope fence: per-instance identities and certified inequalities with the direction in the name — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the two open terms after T157 — an  $m$ -free lower bound on  $\hat{g}_1^2$ , equivalently an  $m$ -free upper bound on the one Schur diagonal entry  $1/s$ ; and an  $m$ -free  $R1''$  domination with a non-shrinking margin — stay open and typed open); each statement carries at least one mutation control that must fail loudly.
- **(1) The complement-floor certificate (T155) [E].**  $\min_{v \perp W} v^T L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^T)M^{1/2})$  with  $M = \text{diag}(\mu_{13}^P - \mu_k^P)$  and  $G = T_{12}Q_W$  — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every  $m$  and every subspace, direction checked: the certificate never exceeds the exact size- $m$  complement bottom (12/12) and agrees to  $1 - \text{ratio} \leq 1.01 \times 10^{-3}$  against the declared cap  $2 \times 10^{-3}$  on this small surface (T155 measured  $\leq 2.2 \times 10^{-7}$  at  $h = 142 \dots 1293$ , sandbox). Exactness controls both ways:  $W = \text{span}\{t_1..t_8\}$  returns  $\mu_9^P$  and the *worthless* configuration  $W = \text{span}\{t_2..t_{13}\}$  returns  $\mu_1^P$ , both to  $2.0 \times 10^{-11}$  — the instrument reports its own worthlessness; mutation: one corrupted overlap row moves the certificate by  $\geq 0.549$ , only downward.
- **(2) The sine-block confinement (T157) [E].** From the certified pencil floor  $A \succeq tL_P$  and the certified ladder ceiling  $\lambda_1 \leq S\mu_1^P$  alone,  $\|\gamma_H\|^2 \leq \lambda_1/(t\mu_{17}^P) \leq (S/t)/\rho_{17} = 0.0153\text{--}0.0245 < 1$  on



12/12: the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a *per-instance theorem* (measured tail  $6.7 \times 10^{-6}$ – $3.0 \times 10^{-5}$ ; T146’s measured “98 per cent” replaced by one line). Negative control: the pencil ceiling  $\kappa$  in place of the ladder  $S$  makes the bound vacuous ( $17.6$ – $477 > 1$  on 12/12); mutation: the inflated floor  $\kappa L_P$  is refuted at  $e_1(A)$  by  $3.3 \times 10^3$ – $1.0 \times 10^5$ .

- **(3) The Schur-entry identity (T157) [E].**  $s = \mu_1^P t_1^\top A^{-1} t_1 = (B^{-1})_{11}$  exactly ( $1.1 \times 10^{-11}$ ) and  $(B^{-1})_{LL} S_L = I$  to  $5.8 \times 10^{-11}$  with  $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$  the *same* fixed-size  $16 \times 16$  Schur complement the chain already forms:  $1/s = 4.0460$ – $5.8962$  is an  $O(1)$  number carried by *one* entry, so T156’s R2’’ loss ceiling  $r \leq 1/(Ls)$  is a statement about one number. Negative control (T157’s refutation wired as a check): the inversion-free Cauchy–Schwarz ceiling is valid and useless, off by  $250$ – $7.27 \times 10^3$  — the cancellation in  $b^\top B_{HH}^{-1} b$  is nearly complete and Cauchy–Schwarz cannot see it; mutation: a one-index shift breaks the identity by  $1.55$ – $2.15$ .
- **(4) The adaptive Lipschitz ceiling (T157) [E].** Interval bisection with the *local* Lipschitz constant certifies  $\inf f^{\text{arch}}/(4 \sin^2(\theta/2)) \geq 0.25$  on 12/12 windows in 519–1201 symbol evaluations (certified floor  $0.2505$ – $0.3062$ , never above the fine-grid minimum); the global constant is documented insufficient (a uniform grid would need up to  $61.4\times$  more points); mutation: fed a target above the true grid minimum the certificate *refuses* on 12/12. Stress: on the T145 no-go family the Schur floor survives ( $t = 0.05$ ) while the confinement goes vacuous ( $\times 52.0$ ) and  $(S_L)_{11}$  explodes ( $\times 60.0$ ) against the flat real band; parity control exact to  $4.9 \times 10^{-14}$ ; the Dirichlet residual hits the predicted  $2/\sqrt{N}$  to  $7.4 \times 10^{-5}$ .
- **Prime-front diary T157 (angle\_floor\_probe.py, contract ANGLE.FLOOR, 32/32, 3.9 s, 16 prime-power windows  $h = 142 \dots 1293$ , the alignment surface on 12 ( $h = 50 \dots 1077$ ), verdict ANGLES-RESIST, sandbox).** Neither angle falls, but both change shape: the  $p_1$  floor becomes a theorem times one measured fixed-size number — the sine-block confinement  $\|\gamma_H\|^2 \leq (S/t)/\rho_{17} = 0.0165$ – $0.0293$  proves  $e_1(A)$  lives 97.1–98.4% inside the first sixteen parity sines, and the resolvent route pins  $p_1 \geq \hat{g}_1^2(1 - \text{tail}) = 0.1968$ – $0.3228$ , 97.9% of the measurement (the classical Rayleigh angle bound is empty on 16/16, its block version empty at every  $J$ , the angle-free Cauchy–Schwarz route loses exactly the Kantorovich  $P = 5.6 \times 10^2$ – $1.0 \times 10^6$ ); the structural gem  $1/s = (S_L)_{11} = 2.3359$ – $6.2049$  flat makes the whole first term a bound on *one* diagonal entry of the  $16 \times 16$  Schur complement the chain already forms (Cauchy–Schwarz off by  $5.4 \times 10^2$ – $5.2 \times 10^5$ ); the arch half is uniformly certified for the first time (executed adaptive Lipschitz ceiling, 12/12, cost  $h^{0.85}$ ) with the two extremals at opposite ends of the band (the pointer: use the  $\theta$ -growth, not the infimum at  $\pi$ ); the alignment term stays a per-window domination ( $1.0003$ – $1.0907$ , margin  $7.3 \times 10^{-4}$ , shrinking trend); balance 9 theorem / 2 cert-unif / 2 cert-window / 3 measured (all three now fixed-size in the statement); the no-go breaks on *five* axes. T158 (schur\_entry\_probe.py) is running at the cancellation-seeing bound. NOT RH evidence; no marker moves.
- **Docs + website sync.** tfpt\_prime\_front gains the T157 subsection with the After-T157 keybox, the v552 subsection in the verified layer (now eighteen modules), the phase-2 section retitled T126–T157, the T157 row in the probe index and the updated counters (157 probes, 4047 checks); tfpt\_1, tfpt\_3 and tfpt\_research\_contracts gain the v552 companion blocks; experiments/next.txt gains the Teil-157 entry; website/lib/primeFront.ts gains the T157 and v552 feed entries; the prime-front status surfaces move to Teile 11–157 / 4047 checks / 18 modules; suite counter  $545 \rightarrow 546$  scripts. T158 (schur\_entry\_probe.py) is running at the cancellation-seeing bound. Ledger row PRIME.ANGLE.INSTR.01 [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (VI) (v551 — the fixed-size Ritz ceiling certificate of phase 2 promoted; prime-front diary T154 (the ceiling closes exactly at fixed size, ALIGN-RESISTS)**

+ paper + website sync; NO marker moves)

- **New module v551 (PRIME.RITZ.CEIL.01).** Promotion of the certificate core of the discovery part T154 (`green_align_probe.py`) — 16 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v550, which certified the bottom ladder  $\lambda_k(A) \leq S\mu_k^P$  by per-window  $\text{LDL}^\top$  counts of *size*  $m$  — this module certifies the *fixed-size* replacement T154 found for the ceiling step. Scope fence: per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all*  $D$  (the two open inputs after T154 — R1', an  $m$ -free floor for  $\lambda_{\min}(B_{HH})$  on the bulk parity block, and R2', an  $m$ -free lower bound for the  $L_P$  floor on the complement of the eight bottom Ritz directions — stay open and typed open; every floor consumed is a per-window certificate of size  $m$ ). Battery: the 12 *deepest* prime-power zones admitting a frame-A window inside the cap ( $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ,  $D = 7.248 \times 10^{-3} \dots 1.815 \times 10^{-2}$  — mirroring T154's own surface selection, declared rather than tuned: on the shallowest zones  $n \leq 11$  more than one bottom direction leaves the sine block and the fixed-depth certificate is *not* claimed there), a no-go size ladder  $m = 48 \dots 288$ , parity controls at  $m = 64, 128, 256$ , and 3 Dirichlet control windows; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The direction lemma (T154) [E].** For orthonormal  $Q$  and  $W = Q^\top A Q$ ,  $\lambda_k(A) \leq \theta_k$  for *every*  $k \leq \dim$  — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument at all; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module. Negative control: the *reversed* reading (Ritz values as lower bounds) is recognised as false on 12/12 windows,  $\theta_k$  overshooting  $\lambda_k$  by a factor  $\geq 18.6$  somewhere on every window.
- **(2) The sixteen-column certificate (T154) [E].**  $\text{span}\{t_1..t_8\} + A^{-1}L_P \text{span}\{t_1..t_8\}$  has *fixed* dimension 16, independent of  $m$  (one Green step by Cholesky back-solves, no inverse formed), and its ceiling  $K^F = \max_k \theta_k / \mu_k^P$  (8 completed Choleskys of matrices  $\leq 8 \times 8$ ) equals  $1.1019\text{--}1.5845$  with  $K^F \geq \max_k \lambda_k / \mu_k^P$  on every window *and*  $|K^F / K_{\text{bot}} - 1| \leq 9.9 \times 10^{-7}$  against the declared cap  $10^{-4}$  on 12/12, where  $K_{\text{bot}}$  is the inertia-certified true ladder constant (8 completed  $\text{LDL}^\top$  counts of size  $m$  per window; count = direct spectral count on all 96 pairs): the fixed-size ceiling *attains* the truth, and the size- $m$  counts are retired from the ceiling step (controls: the parity sines alone overshoot by 41.6–739 — T152/T153's  $O(m^2)$  overshoot reproduced at fixed size — and corrupting the Green columns breaks the attainment on 12/12).
- **(3) The one-direction anatomy (T154) [E] (angles measured).** The median of the seven aligned principal angles between the bottom eight of  $A$  and the bottom eight of  $L_P$  is  $0.03\text{--}0.14^\circ$  while the largest is  $83.79\text{--}89.70^\circ$  on every window (Björck–Golub 1973) — the misalignment is carried by *one* nearly orthogonal direction, and that single direction *is* the collapse price  $\lambda_1(A)/(t\mu_1^P) = 4.408\text{--}7.042$ , recovered in full per window by one completed Cholesky of  $A - \gamma I$  (size  $m$ , certified, not fixed size, labelled as such); control:  $A = L_P$  through the same instrument returns  $K_{\text{bot}} = K^F = 1$  and zero misalignment (deviations  $< 10^{-5}$ , the inertia ladder's backoff step) with the price exactly  $1/t$ .
- **(4) The no-go discriminator (T154) [E].** On the T145 no-go reconstruction  $c(l) = 1/(1+l)$  the odd section is positive definite and the Schur floor survives ( $t = 0.05$  on all 4 sizes) but  $K_{\text{bot}}$  explodes ( $\times 35.4$ ) and the sixteen-column ceiling explodes *with* it ( $\times 57.5$ ) over  $m = 48 \dots 288$  — the instrument does not report flatness where flatness is false and manufactures no closure; the Courant–Fischer direction is violated nowhere (16/16 positive-definite cases); the Dirichlet control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite —  $\lambda_1 = -5.60 \dots -3.55$  certified on 3/3 — so no ceiling exists there: the reflection half is load-bearing for positivity.

- **Prime-front diary T154** (`green_align_probe.py`, contract `GREEN.ALIGN`, 29/29, 5.0 s, 12 prime-power windows  $h = 50 \dots 1077$ , verdict `ALIGN-RESISTS`, `sandbox`). The ceiling half of T153's R2 closes exactly at fixed size ( $K^F = 1.1019\text{--}1.9964$  flat, agreement  $5.17 \times 10^{-7}$  on 12/12); the floor half is refuted at fixed size ( $\|R\|^2/(\beta\lambda_1) = 6.3 \times 10^3\text{--}5.6 \times 10^9$ ,  $x^{4.50}$ ) and recovered at full size (one Cholesky per window recovers the whole collapse price 4.408–7.985; per-window end to end  $4.45 \times 10^{-2}\text{--}3.13 \times 10^{-1}$ , inside the target band, while the  $m$ -free-in-shape number stays  $1.01 \times 10^{-2}\text{--}3.92 \times 10^{-2}$ ); the obstruction is named (seven bottom directions agree to  $0.15\text{--}1.35^\circ$  median, *one* sits at  $82.93\text{--}89.79^\circ$ ) and the missing fixed-size ingredient is priced (the  $L_P$  floor on the Ritz complement, arithmetic-free, flat  $5.93\text{--}8.25 \mu_1^P$ , worth 91–100% of the price); the Kato route to R1 fails by the same geometry (loss  $1.97\text{--}121$  attained  $24.9\text{--}89.8^\circ$  apart; the Hankel-reflection culprit refuted twice). Two open terms remain (R1'/R2', both uniformity terms); T155 (`bottom_floor_probe.py`) is running at exactly those two floors. NOT RH evidence; no marker moves.
- **Docs + website sync.** `tfpt_prime_front` gains the T154 subsection with the After-T154 keybox, the v551 entry in the verified layer (now seventeen modules), the phase-2 section retitled T126–T154, the T154 row in the probe index and the updated counters (154 probes, 3951 checks); `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v551 companion blocks; `experiments/next.txt` gains the Teil-154 entry; `website/lib/primeFront.ts` gains the T154 and v551 feed entries; the prime-front status surfaces move to Teile 11–154 / 3951 checks / 17 modules; suite counter  $544 \rightarrow 545$  scripts. T155 (`bottom_floor_probe.py`) is running at the two uniformity floors. Ledger row `PRIME.RITZ.CEIL.01` [E]; **no marker of any pre-existing contract moves.**

**2026-07-29 (V) (v550 — the odd-sector identities of phase 2 promoted; prime-front diary T151 (the one inequality closes by rerouting off the ladder, ODD-CARRIES) + paper + website sync; NO marker moves)**

- **New module v550 (PRIME.ODD.SECTOR.IDENT.01).** Promotion of the identity/certificate core of the discovery part T151 (`odd_ladder_probe.py`) — 19 checks,  $\sim 0.2$  s, everything recomputed on small frame-A windows, nothing cited from the `sandbox`; the companion to v549, which certified the parity compression and the quadratic ladder certificate  $\nu_k^P \leq Ck^2$  — this module certifies the *reroute* T151 found: off the  $\nu$  ladder, through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in  $k$ . Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open input after T151 — the  $m$ -freeness of the bottom pencil ratio  $R = K_{\text{bot}}/\kappa$  for the odd parity sector of a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), parity control models at  $m = 241$  (the certified  $(1, 1)$  pencil) and  $m = 2001$  (the  $\pi$  price), and a no-go size ladder  $m = 64 \dots 256$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The grid step-over and the honest negative (T151) [E].** The parity sines live on the *shifted* grid  $\theta_k = 2\pi k/N$ ,  $N = 2m+1$ , which never contains  $\theta = 0$ , with  $\mu_k^P = 2 - 2\cos\theta_k$  *exactly* ( $2.2 \times 10^{-16}$ ; eigen-residual  $2.3 \times 10^{-13}$ ); the symbol has  $f(0) = -47.6 \dots -6.3 < 0$  (T148's honest negative, re-read and never re-derived) with a nonempty negative window  $[0, \theta_c)$ , and  $\theta_c/\theta_1 = 0.371\text{--}0.485 \leq 0.9$  on *every* window: the odd grid *steps over* the negative window — odd-sector positivity is a grid fact, stepped over rather than cancelled. The honest negative beside it, promoted as content (T150's rest item 3 answered *negatively*): the symbol-only pencil floor  $\kappa_{\text{sym}}(1 - \rho)$  is vacuous ( $\rho \geq 1$ ) on 12/12 windows and the symbol moves by  $0.45\text{--}3.27$  relative across *one* mode spacing — no pointwise symbol statement is admissible at the bottom mode's resolution: the local model is a *matrix* statement, not a symbol statement (controls: the Dirichlet corner breaks the eigenpairs at 1.0, a one-index shift misses by  $6.5 \times 10^{-3}$ ).

- **(2) The certified bottom ladder (T151) [E].**  $\lambda_k(A) \leq S\mu_k^P$  for  $k \leq 8$  is certified per window by 8 completed LDL<sup>T</sup> inertia counts (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list),  $S = 1.1019\text{--}6.4725$ : the odd sector’s bottom spectrum *is* the parity-Laplacian ladder up to a printed  $O(1)$  factor — order 2 in  $k$ , with no trace of  $f(0) < 0$  in it (the Rayleigh floor  $L_P \geq \mu_1^P I$  absorbs it); the count equals the direct spectral count on all 96 (window,  $k$ ) pairs (control:  $S \times 10^{-3}$  certifies no eigenvalue below it anywhere).
- **(3) The discrete Sobolev step and the  $\pi$  price (T151) [E].**  $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\text{last}}^2$  exactly ( $7.6 \times 10^{-15}$ ),  $\|\nabla\psi\|_2^2 \leq \langle\psi, L_P\psi\rangle$ , and  $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\nabla\psi\|_2$  on the true window modes, the parity sines and random unit vectors — elementary, licensed by the odd sector’s virtual node  $\psi_{-1} = 0$  (Hardy–Littlewood–Pólya 1934 / Agmon 1965 the continuum address); calibration: on the exact parity model the pencil is  $(1, 1)$  (certified  $|\kappa - 1| = 1.1 \times 10^{-12}$  at  $m = 241$ ) and the Sobolev per-mode price is  $\pi$  to  $5.1 \times 10^{-8}$  at  $m = 2001$  — the honest,  $m$ -free cost of leaving the  $\ell^2$  world (controls: dropping the boundary term breaks the identity at  $4.8 \times 10^{-2}$ , the Neumann energy undercounts a first-node vector by 0.50).
- **(4) The closed archimedean minimum (T151; T150’s rest item 2 closed and attained) [E].** The archimedean lag kernel has two shape properties, checked per window and never assumed —  $c_i^{\text{arch}} < 0$  for  $i \geq 1$  and  $c_i^{\text{arch}}$  non-decreasing on  $i \geq 1$  — and given those,  $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$  *exactly* (max rel 0.0; value 2.1719–3.4674), attained at  $r = 0$ : a closed two-term functional of the smooth kernel, no matrix, no eigenvector, no loss (control: the full atom-including kernel violates the monotone shape on every window and the closed formula misses its minimum by  $\geq 8.7 \times 10^{-2}$ ).
- **(5) The linear per-mode bound and the no-go discriminator (T151) [E].** With  $\kappa = 0.1939\text{--}0.3599$  the certified pencil floor (*one* completed Cholesky of  $A - \kappa L_P$  per window, the floating-point floor carried as  $\text{fl}/\mu_1^P$ ; the direction re-read on the actual bottom eigenvalue) and  $K_{\text{bot}} = S$  from item (2), the chain  $b_k := 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \leq C_S k$  holds with  $C_S = 12.081\text{--}26.953 \leq 2\pi\sqrt{K_{\text{bot}}/\kappa}$  on the bottom 8 modes of every window — *linear* in  $k$  where the quadratic  $\nu$  ladder grew, every  $m$ -power cancelled, nothing additive anywhere (the  $\Theta(D^3)$  bottom that killed the Weyl / Bauer–Fike / Davis–Kahan route never enters) — and the bound dominates the truth  $m\|\psi_k\|_\infty^2$  and the parity Dirichlet energy on every read mode; the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.7362\text{--}18.6607$  is the *one* scalar the chain still owes, printed per window and typed open in  $m$ . The discriminator: on the T145 no-go form the ratio *explodes* on the size ladder ( $1.686 \times 10^3 \rightarrow 2.667 \times 10^4$ , factor 15.8 over  $m = 64 \dots 256$ ) while the exact parity model holds  $R = 1$  to  $1.1 \times 10^{-12}$  — a route that did not fail there would prove a false statement (controls: the ladder  $\times 10^{-3}$  fails against the true sups, the inflated floor  $1.5\kappa$  makes the Cholesky *refuse* on 12/12).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the symbol-only route stays certified vacuous; TV(log  $\Lambda$ ) stays withdrawn as a hypothesis (T149),  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T151 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the address of the symbol question — the computed answer is that the pointwise symbol language does *not* extend to the bottom mode), Weyl 1912, Parlett, Hardy–Littlewood–Pólya 1934 / Agmon 1965, Sylvester 1852 / Bunch–Kaufman 1977, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Wilkinson 1968 / Higham 2002, Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.ODD.SECTOR.IDENT.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T151, odd-carries).** The T151 probe (odd\_ladder\_probe.py, contract ODD.LADDER, 27/27, 43.4s, 72 prime-power windows  $m = 96 \dots 1491$ ) closes the one inequality by rerouting off the ladder: the odd grid steps over the symbol’s negative window ( $\theta_c/\theta_1 = 0.328\text{--}0.407$  on 72/72), the bottom spectrum is certified against the parity Laplacian

( $\lambda_k \leq S\mu_k^P$ ,  $S = 1.1019\text{--}2.3870$ ), and the discrete Sobolev step yields the *linear* per-mode bound  $b_k \leq C_S k$  with a *non-growing* constant ( $C_S = 11.5137\text{--}19.5731$ , trend  $x^{0.020 \pm 0.007}$ ) where the quadratic ladder grew ( $13.81\pi$  at  $x^{0.258}$ ); end to end moves to  $2.01 \times 10^{-2}\text{--}3.52 \times 10^{-2}$  of the true gap (gain up to  $3.18\times$ ), the bottleneck relocating from  $Q^*$  to  $\Psi$ . Two of T150's three rest items settled: the minimum of the archimedean diagonal is closed *and attained exactly* ( $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}} = 2.5928\text{--}4.8981$ , deviation 0.0), and the symbol route is a computed dead end (the local model is a matrix statement — the pointwise symbol language fails at the bottom mode's resolution, margin vacuous on  $72/72$ ). One scalar remains a fit: the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.3634\text{--}9.7108$  at  $x^{0.037 \pm 0.015}$  — and the T145 no-go breaks exactly there ( $x^{1.986}$ ). Verdict ODD-CARRIES: all six preregistered gates met, every layer typed (theorem / certified / elementary-verified), the one fit named. `experiments/next.txt` part T151 (series totals 151 probes / 3852 checks); `tfpt_prime_front.tex` gains the T151 subsection with the “After T151” keybox, the v550 entry in the verified layer (now sixteen modules), the phase-2 section retitled T126–T151 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v550 companion blocks; `website/lib/primeFront.ts` gains the T151 and v550 feed entries; suite counter  $543 \rightarrow 544$  scripts. T152 (`pencil_ratio_probe.py`) is running at exactly the remaining scalar — the  $m$ -freeness of the bottom pencil ratio — and is *not* part of this change.

**2026-07-29 (IV) (v549 — the gauge/parity identities of phase 2 promoted; prime-front diary T150 (the parity mechanism named, ONE-TERM-MISSING) + paper + website sync; NO marker moves)**

- **New module v549 (PRIME.GAUGE.PARITY.IDENT.01).** Promotion of the identity/certificate core of the discovery parts T149 (`weight_smoothing_probe.py`) and T150 (`mode_ladder_probe.py`) — 21 checks,  $\sim 0.3$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v548, which certified the factorisation  $\Gamma = \sqrt{Q^*}S_w$  and the Liouville machinery in the Jacobi gauge — this module certifies the gauge *freedom* around that machinery and the parity *structure* underneath it. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open input after T150 — an  $m$ -free constant  $C$  in the odd-parity-sector ladder  $\nu_k \leq Ck^2$  for a sign-changing-symbol Toeplitz section — stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 48 gauge assemblies (Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal), the full-section parity items on the 6 windows with  $M \leq 300$ , and a parity control model at  $m = 2001$  (vector operations only); each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The gauge freedom (T149) [E].** For *any* positive diagonal  $\tilde{\Lambda}$  the whitening  $\tilde{E} = \tilde{\Lambda}^{-1/2}A\tilde{\Lambda}^{-1/2}$  is an exact congruence, so  $\lambda_{\min}(A) \geq \text{cert } \lambda_{\min}(\tilde{E}) \cdot \min_j \tilde{\Lambda}_j$  is a valid certified lower bound for every gauge — an identity plus one Rayleigh step — and the family maximum is valid too (verified on all 48 assemblies); the factorisation identity  $\Gamma = \sqrt{Q^*}S_w$  holds *gauge-invariantly* (max rel  $2.4 \times 10^{-16}$ ); the const gauge (the geometric mean of the Jacobi diagonal) has  $\text{TV}(\log \tilde{\Lambda}) = 0$  *exactly* on every window — T149's elimination, not reduction, of T148's blocking scalar — with the certified entrywise sandwich  $\sigma = c_2/c_1 = 1.3868\text{--}2.5805$  (controls: a 1%  $Q^*$  perturbation breaks the identity at  $4.99 \times 10^{-3}$ , and on the shifted indefinite form  $A - 1.05\lambda_{\min}I$  the completed-Cholesky floor *refuses* on 12/12 windows).
- **(2) The two-regime discriminator (T149, the refutation as a check) [E].** The flutter-smoothing gauge (moving geometric mean, preregistered half-width  $m/16$ ) removes  $\text{TV}(\log \Lambda)$  by  $\geq 2.88\times$  on every window while moving the Liouville-*transported* bottom-block smoothness functional  $\nu_{\tilde{L}}$  by at most 0.0088 — and even the matched-amplitude *rough* re-gauge (lattice-scale oscillation) moves it by at most 0.0046 (bar 0.01 for both): the roughness  $\nu_{\tilde{L}}$  responds to is *not* in the multiplier — T149's withdrawal of the TV hypothesis, wired as a per-instance check

(control: the same functional read *without* the transport, on the whitened modes  $\psi$  which carry the multiplier, jumps by 0.41–9.76 under the same rough re-gauge — the invariance is carried by the transport, not by numbness of the functional).

- **(3) The explicit zone diagonal (T150) [E].**  $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$ , assembled lagwise with no matrix anywhere, equals the lumped-pair diagonal on every window (max rel  $5.9 \times 10^{-16}$ ) — the whitening diagonal is an *explicit* functional of the zone's lag arithmetic and of nothing else; the lag vector splits *exactly* as  $c = c^{\text{arch}} + c^{\text{atom}}$  with  $c^{\text{atom}} \leq 0$  entrywise, and the arithmetic mass carries the closed budget  $\|c^{\text{atom}}\|_1 \leq \sum_j \mu_j \leq 4B\sqrt{N}$  with  $B = 1.038821$  the *verified* maximum of  $\psi(n)/n$  over the whole atom table (Chebyshev 1852, computed and never assumed; Abel summation carries the closed form). The honest negative as content: the archimedean section alone is *indefinite* on 12/12 windows (completed  $\text{LDL}^T$ , up to 5 negative eigenvalues) while the full section has zero — the atoms are *co-responsible* for the positivity, so the additive perturbation reading (Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970, cited, computed dead by T150 at 2.4–5.8 orders, *not* used) is structurally empty and only the multiplicative / Loewner gauge step transfers (control: dropping the positive part in the zone formula breaks the identity at  $4.25 \times 10^{-1}$ ).
- **(4) The parity compression (T150, the named mechanism) [E].** With  $U_-$ ,  $U_+$  the antisymmetric / symmetric parity isometries of the full window,  $U_-^T T_M(c) U_- = T - H$  and  $U_+^T T_M(c) U_+ = T + H$  *exactly* (max rel  $3.9 \times 10^{-16}$ ) with cross block  $U_+^T T_M U_- = 0$  ( $1.4 \times 10^{-16}$ ) on all 6 windows with  $M \leq 300$ : the reflection-odd section used since T106 *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector (Basor–Ehrhardt 2009: the parity sectors are the natural objects of the Toeplitz+Hankel algebra, cited as the address and never as an authority); the completed  $\text{LDL}^T$  counts *localise* the negative inertia — the full section's negative eigenvalues (forced by the sign-changing symbol,  $f(0) = -29.2 \dots - 6.3 < 0$ ; T148's honest negative *re-read*, never re-derived) sit entirely in the *even* sector while the odd sector counts zero:  $f(0) < 0$  is explained rather than worked around (control: the sign-flipped basis does not reproduce  $T - H$ ,  $3.99 \times 10^{-1}$ ).
- **(5) The parity-sine calibration and the ladder certificate (T150) [E].** The parity sines  $t_k(r) = \frac{2}{\sqrt{N}} \sin(2\pi k(r+1)/N)$ ,  $N = 2m+1$ , are the exact eigenpairs of the parity Laplacian (reflecting corner 3, forced by antisymmetry) at  $\mu_k^P = 4 \sin^2(\pi k/N)$  — eigen-residual  $2.3 \times 10^{-13}$  — and the smoothness ladder *calibrates* on the exact model:  $\nu_k^P = \pi k^2$  for  $k = 1 \dots 12$  at  $m = 2001$  (max deviation  $3.0 \times 10^{-5}$ ), so the ladder form  $\nu_k \leq Ck^2$  is the right question and  $\pi$  its exact floor; on every window, in the const gauge ( $\varphi = \psi$  exactly — the ladder of the *pure* Toeplitz-minus-Hankel section), the parity coefficient licence holds,  $\nu_k^P \leq Ck^2$  with the per-window certified constant  $C = 11.70\text{--}17.28$ , and the derived per-mode ceiling  $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k+1)^2$  is a certified instance inequality — an  $m$ -free  $C$  is exactly the one open input and is *not* claimed (controls: the ladder  $\times 10^{-3}$  fails everywhere, a one-index calibration shift misses by 0.75).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ;  $\text{TV}(\log \Lambda)$  stays withdrawn as a hypothesis (T149),  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T149/T150 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953 (in the parity sector), Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the parity sectors as the algebra's natural objects — the address, never an authority; the sign-changing symbol is exactly what Basor–Ehrhardt does *not* cover, and that gap is the open input), Sylvester 1852 / Bunch–Kaufman 1977, Chebyshev 1852 (verified on the table), Abel, Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970 (cited, computed dead, *not* used), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.GAUGE.PARITY.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T150, one-term-missing).** The T150 probe (`mode_ladder_`

`probe.py`, contract `MODE.LADDER`, 36/36, 48.0s, 72 prime-power windows  $m = 50 \dots 1393$ ) names the mechanism: the form is exactly the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the sign-changing symbol's entire negative inertia sits in the *even* sector (certified 72/72) — T148's honest negative is explained rather than worked around. The flutter amplitude becomes a certified form functional ( $0.0606\text{--}0.1993$ , trend  $x^{-0.165 \pm 0.015}$ , T149's fit reproduced fit-free), the whitening diagonal an explicit zone functional, and the arithmetic atoms turn out to be co-responsible for the positivity (the archimedean section alone has 2–7 negative eigenvalues); the additive perturbation route is dead by 2.4–5.8 orders while the multiplicative gauge step closes. One gate remains: the ladder constant still grows —  $C \leq 43.391 = 13.81\pi$  certified per stratum, trend  $x^{0.258 \pm 0.009}$  against the flatness bar 0.25 — and the gap from  $\pi$  is exactly the arithmetic leakage of the bottom mode ( $\ell^1$  mass 1.41–1.79). Verdict ONE-TERM-MISSING: one named inequality left —  $\nu_k \leq Ck^2$  with  $m$ -free  $C$  for the odd parity sector at a sign-changing symbol. `experiments/next.txt` part T150 (series totals 150 probes / 3825 checks); `tfpt_prime_front.tex` gains the T150 subsection with the “After T150” keybox, the v549 entry in the verified layer (now fifteen modules), the phase-2 section retitled T126–T150 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v549 companion blocks; `website/lib/primeFront.ts` gains the T150 and v549 feed entries; suite counter  $542 \rightarrow 543$  scripts. T151 (`odd_ladder_probe.py`) is running at exactly the remaining inequality — the odd-sector ladder — and is *not* part of this change.

**2026-07-29 (III) (v548 — the Green/Szegő identities of phase 2 promoted; prime-front diary T148 (lifting statement, ONE-INPUT-MISSING) + paper + website sync; NO marker moves)**

- **New module v548 (PRIME.GREEN.SZEGO.IDENT.01).** Promotion of the identity/certificate core of the discovery parts T147 (`green_asymptotic_probe.py`) and T148 (`szego_bottom_probe.py`) — 21 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v547, which certified the a-priori level constant  $c_0^{\text{ap}}$  — this module certifies the two scalars that constant is made of. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the one open lifting input —  $\nu_L \leq F(\text{form})$  with  $F$   $m$ -free, equivalently bounded  $\text{TV}(\log \Lambda)$  or any weaker smoothness of the multiplier — stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 7 random positive definite forms for the form-independent items, and a weight-class model ladder  $m = 64 \dots 288$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The factorisation identity (T147) [E].**  $\Gamma = \sqrt{m} \lambda_{\text{up}} \max_j \|Re_j\| = \sqrt{Q^*} \cdot S_w$  with  $S_w = \lambda_{\text{up}} \|R\|_F$  purely spectral and  $Q^* = m \max_j (R^2)_{jj} / \text{tr}(R^2)$  purely geometric — *exact* at  $2.2 \times 10^{-16}$  on all 12 windows and 7 random forms: the all- $D$  question splits into two named scalars and into nothing else; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true  $\Gamma$  everywhere ( $S_w = 1.1186\text{--}1.5777$ ,  $Q^* = 2.0879\text{--}2.8634$ ; controls: a 1%  $Q^*$  perturbation breaks the identity at  $5.0 \times 10^{-3}$ , perturbing the largest Green column breaks the factorisation).
- **(2) The bottom-block equidistribution constant (T147) [E].** With  $\Pi_B$  the spectral projector onto  $B = \{k : \lambda_k \leq 10\hat{\lambda}\}$  (preregistered rule,  $|B| = 3 \dots 5$ ),  $\sum_j (\Pi_B)_{jj} = |B|$  *exactly* ( $3.3 \times 10^{-16}$ ), hence  $Q_B \geq |B|$  is a theorem ( $\max \geq \text{mean}$ ); the *measured*  $Q_B/|B| = 1.5091\text{--}1.7980$  lies in the preregistered band  $[1, 2.2]$  against the Szegő/Widom sharp reference  $Q_B \leq 2|B|$  — a measurement on a finite list, typed as such (controls: the rank- $|B|$  coordinate projector gives  $Q_B = m$  and overshoots by  $\geq 3.9$ , a 1% basis scaling breaks the trace identity).
- **(3) The second-difference  $\ell^1$  ceiling (T148, the lever) [E].** From the exact Dirichlet eigen-pair  $L_0 s_k = \mu_k s_k$  and  $\mu_k \geq 4k^2/(m+1)^2$ : the coefficient bound  $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^p \psi\|_1 / \mu_k^p)$

for  $p = 1, 2$  against the *true* coefficients on every window bottom mode and on random unit vectors (worst excess  $\leq 0$ ), the *m-free* closed form  $\|a\|_1 \leq 2\sqrt{\nu} + 1$  with  $\nu = (m+1)^{3/2} \|L_0\psi\|_1 / (2\sqrt{2})$  — every power of  $m$  cancelled — and the chain  $m\|\psi\|_\infty^2 \leq 2\|a\|_1^2$ ; calibration: for the Dirichlet bottom mode  $\|a\|_1 = 1$  *exactly* and  $\nu = 3.1415 \rightarrow \pi$  at every size (controls: the ceiling  $\times 10^{-3}$  fails on every window, a one-index  $\mu$  shift breaks the coefficient identity at  $1.5 \times 10^{-1}$ ).

- **(4) The layer-cake  $S_w$  certificate (T148) [E].** The completed  $\text{LDL}^\top$  inertia count at every rung of the preregistered ladder  $\tau_i = \rho_i \lambda_{i_0}$  ( $\rho = 1.5 \dots 1024$ ,  $\lambda_{i_0}$  a certified positive floor) equals the direct spectral count *exactly* on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), and the layer cake certifies  $S_w \leq 1.3288\text{--}1.8220$  per window (true  $1.1186\text{--}1.5777$ , slack  $\leq 1.216$ ); nothing here says *m-free* (control: dropping the bottom layer breaks the bound on every window — the bottom eigenvalue is the weight).
- **(5) The Liouville transform and the weight-class discriminator (T148) [E].** The chain  $m\|\psi\|_\infty^2 \leq \kappa_\Lambda m\|\hat{\varphi}\|_\infty^2 \leq 2\kappa_\Lambda \|a(\hat{\varphi})\|_1^2 \leq 2\kappa_\Lambda \text{ceil}(\hat{\varphi})^2$  with  $\varphi = \Lambda^{-1/2}\psi$  (the exact pencil equivalence, a change of variables and not a perturbation) holds stepwise on every bottom-block mode;  $\kappa_\Lambda = \max \Lambda / \min \Lambda = 1.3868\text{--}2.5805$  is a priori (a ratio of two diagonal entries of the form). The discriminator: on a model section with a genuine order-2 zero ( $f(0) = 0$ ,  $f''(0)/2 = 1$  exact), over the ladder  $m = 64 \dots 288$  at matched  $\kappa_\Lambda$  (mismatch 0.0013), the bounded-variation multiplier keeps  $\nu$  flat (max/min 1.003) while the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning (control: the first Liouville step  $\times 10^{-3}$  fails on every mode, provably — the step slack is bounded by  $\kappa_\Lambda^2$ ).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the symbol-level order-2 hypothesis is *refuted* (T148:  $f(0) < 0$  on every surface window) and enters nowhere as a theorem;  $\sigma_{\text{tot}}$  stays retired as a route; every T147/T148 exponent is a fit that stays in the sandbox. Kac–Murdock–Szegő 1953, Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the mechanism’s address, never an authority), Sylvester 1852 / Bunch–Kaufman 1977, Euler 1735, Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Davis–Kahan 1970 (cited, computed vacuous, *not* used), Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.GREEN.SZEGO.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T148, one-input-missing).** The T148 probe (`szego_bottom_probe.py`, contract `SZEGO.BOTTOM`, 31/31, 13.0s, 48 windows  $m = 96 \dots 1077$  in four  $D$ -strata, stress ladders to  $m = 1500$ ) dissects the lifting statement: the second factor  $S_w$  *closes* on the surface ( $\text{LDL}^\top$  layer cake,  $S_w \leq 1.9587$  certified against true  $1.282\text{--}1.683$ , trend flat, per  $D$ -stratum), at the price of an honest negative — the arithmetic Toeplitz symbol has  $f(0) < 0$  on 48/48 windows, so positive definiteness is a finite-section effect of the minus-Hankel part and the lumping and the KMS order-2 hypothesis survives only as a measurement ( $\alpha = 1.643\text{--}1.989$ ) — while the lifting statement resists with its hypothesis isolated to one named scalar (TV(log  $\Lambda$ ), proved to be the roughness and not the conditioning by the weight-class experiment); the first end-to-end number arrives: the certified chain delivers 8.36–15.86% of the true gap on all 48 windows (median 11.74%), and the a-priori-shaped chain is still a lower bound at a factor 10.5. Verdict ONE-INPUT-MISSING: exactly one scalar side ( $Q^*$ ) lacks an *m-free* certified statement, target  $\nu_L \lesssim 34$  against 282 measured. `experiments/next.txt` part T148 (series totals 148 probes / 3759 checks); `tfpt_prime_front.tex` gains the T148 subsection with the “After T148” keybox, the v548 entry in the verified layer (now fourteen modules), the phase-2 section retitled T126–T148 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v548 companion blocks; `website/lib/primeFront.ts` gains the T148 and v548 feed entries; suite counter  $541 \rightarrow 542$  scripts. T149 (`weight_smoothing_probe.py`) is running at exactly the remaining input — whether a smoothed whitening removes the TV roughness — and is *not* part of this change.



2026-07-29 (II) (v547 — the level-lemma identities of phase 2 promoted; prime-front diary T146 (level lemma, LEVEL-CARRIES) + website sync; NO marker moves)

- **New module v547 (PRIME.LEVEL.LEMMA.01).** Promotion of the a-priori core of the discovery part T146 (level\_lemma\_probe.py) — 20 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v546, whose item (5) certified  $S1'$  with a  $c_0$  *measured* at the minimiser's own layer cake — this module supplies the a-priori side, every factor a functional of the form alone. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the asymptotic delocalisation of the Green columns, T146's remainder, stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 9 random positive definite forms for the form-independent items, and a stress ladder  $m = 64 \dots 288$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The union identity for nested cakes (T146) [E].** For a nested decreasing chain with positive coefficients,  $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T \|\psi_t\|_1 - \|\psi_t\|^2$  with  $T = \sum_j c_j$  and  $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$  — form-independent and  $O(m)$ , the reason the cake base is movable at all; exact (0.0) against T145's  $K \times K$  double sum on all 12 window minimisers and 9 random vectors, *and* against the union sizes computed from the sets themselves with no nestedness used (controls: one shuffled proper level set  $3.2 \times 10^{-2}$ , a 1% perturbation of the heaviest coefficient  $4.8 \times 10^{-3}$ ).
- **(2) The layer-cake counting lemma at a general base (T146) [E].**  $G(b) \leq 2b^2 \|v\|_\infty \|v\|_1 / \|v\|^2 + \varepsilon(b)$  for *any* vector at all 6 preregistered bases (margin  $\geq 1.08$  on the battery and on every window minimiser); at  $b = 2$  the leading constant is the classical dyadic 8 *exactly*, and the base is *free* because the chain consumes only the lower domination  $|v| \leq \psi_t$  — the a-priori constant gains  $4/b_*^2 = 3.7612$  over base 2 on every window (controls: the bound  $\times 10^{-3}$  fails on every draw and base, and the one-level-down cake breaks the domination on every draw — the +1 in the cake exponent is load-bearing).
- **(3) The resolvent delocalisation bound (T146) [E].**  $\psi = \lambda R \psi$  to  $9.0 \times 10^{-14}$  — an *identity*, so no Davis–Kahan transfer step appears anywhere — with  $\lambda_{\text{up}} = \min_j \text{Rayleigh}(Re_j) \geq \lambda_{\text{min}}$  (Rayleigh–Ritz; window ratio 1.0592–1.2883), the coordinate bound  $|\psi_j| \leq \lambda \|Re_j\|$  (Cauchy–Schwarz) and the column norms certified from above with the linear-solve residual paid for, giving  $\|\psi\|_\infty \leq \lambda_{\text{up}} \max_j \|Re_j\|$  on every window (within 1.061–1.337 of the truth) and  $\|\psi\|_1 \leq \sqrt{m} \min(1, \Gamma_1)$ . The lever: the trivial ceiling  $\max_j \|Re_j\| \leq 1/\lambda_{\text{min}}$  would give  $\Gamma \leq \sqrt{m} = 5.1$ –16.9 (useless); the certified columns sit a factor 2.85–10.17 *below* it (preregistered bar 2.0), giving  $\Gamma = 1.8356$ –2.2797 — read off the form, never off the minimiser (controls: a 1%  $\lambda$  perturbation breaks the identity at  $1.0 \times 10^{-2}$ , the sup bound  $\times 10^{-3}$  fails on every window).
- **(4) The composite level lemma (T146) [E].**  $c_0^{\text{ap}} = 2b^2 \Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042$ –4.8488 dominates the measured level constant on all 12 windows (slack 1.986–2.418) and stays under the preregistered ceiling  $16 = 4 \times \text{Maz'ya's Dirichlet value } 4$ ; the certified window inequality  $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}} \Psi_{\text{abs}}$  holds on 12/12 (both sides by completed Cholesky, the density the  $|R|$  one with Charikar's cited constant), closing the chain  $\hat{\lambda} \geq 1/(c_0^{\text{ap}} \Psi_{\text{abs}})$ , gap  $\geq \hat{\lambda}/\kappa_{\text{up}}$  at loss factor 0.0423–0.1272 of the exact gap. The built-in discriminator: on T145's no-go form  $R = aa^T + \varepsilon I$ ,  $a_i = i^{-1/2}$ ,  $\Gamma$  grows  $3.676 \rightarrow 6.798$  over  $m = 64 \dots 288$  (ratio  $1.85 \geq$  the preregistered 1.5) and  $c_0^{\text{ap}} = 11.10$  *exceeds* every window value — the bound must be worse there, since a candidate that did not fail on the no-go would prove a false statement — while the theorem half survives on every no-go size and the Dirichlet path-Laplacian control stays flat (max/min 1.0063; control:  $c_0^{\text{ap}} \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; only the sign-free / Green constant is bounded a priori, the energy-route  $\sigma_{\text{tot}}$  stays *measured*, and every

T146 exponent is a fit that stays in the sandbox. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Miclo 1999, Davis–Kahan 1970 (cited, deliberately *not* used), Perron–Frobenius (not applicable to  $E$ , not used), Fukushima–Ōshima–Takeda 1994, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row **PRIME.LEVEL.LEMMA.01 [E]; no marker of any pre-existing contract moves**, and this is not RH evidence.

- **Diary, paper and website (T146, level-carries).** The T146 probe (`level_lemma_probe.py`, contract **LEVEL.LEMMA**, 30/30, 7.6 s, 64 windows  $m = 26 \dots 1173$ , 9/9 preregistered verdict conditions) closes L1 on the measurement surface as a chain of theorems and certified window inequalities with no step reading the minimiser: four hand-written profiles lose by 5–9 orders of magnitude, the Green column survives ( $\lambda_{\text{up}}/\hat{\lambda} = 1.0497\text{--}1.3764$ ), Davis–Kahan is instrumented and discarded (informative on 0/64; the spectrum bottom is a near-degenerate block,  $\lambda_2/\hat{\lambda} = 1.2542\text{--}3.4471$ ), the cake base proves free (the dyadic 8 falls to  $2b^2 \approx 2$ , gain 3.76 at zero cost) and  $c_0^{\text{ap}} = 3.9042\text{--}4.8488$  on 64/64 windows lands at the size of Maz'ya's classical Dirichlet value 4, closing the chain at loss factor 0.0422–0.1586; the one genuine remainder is  $D$ -uniformity for *all*  $D$  — the asymptotic delocalisation of the Green columns. `experiments/next.txt` part T146 (series totals 146 probes / 3689 checks); `tfpt_prime_front.tex` gains the T146 subsection with the “After T146” keybox, the **v547** entry in the verified layer (now thirteen modules), the phase-2 section retitled T126–T146 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the **v547** companion blocks; `website/lib/primeFront.ts` gains the T146 and **v547** feed entries; suite counter  $540 \rightarrow 541$  scripts. T147 (`green_asymptotic_probe.py`) is running at exactly the all- $D$  asymptotics and is *not* part of this change.

**2026-07-29 (I) (v546 — the capacity-chain identities of phase 2 promoted; prime-front diary T145 (Maz'ya proof, ONE-STEP-MISSING) + website sync; NO marker moves)**

- **New module v546 (PRIME.CAPCHAIN.IDENT.01).** Promotion of the identity-shaped core of the capacity cycle T142 (`conductance_profile_probe.py`), T143 (`sharp_capacity_probe.py`), T144 (`capacity_inequality_probe.py`) and T145 (`mazya_proof_probe.py`) — 26 checks,  $\sim 0.5$  s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and *no level lemma* (T145's L1 stays open, unclaimed). Battery: 12 windows ( $n = 4 \dots 139$ ,  $h = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ , 32–5102 positive off-diagonal edges); the covering kernel is positive definite on 11, where items (1)–(2) run; each identity carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The capacity decomposition (T142) [E].**  $K^{-1} = D^T J^{-1} D + x x^T / \text{cap}$  with  $J = D K D^T$  (assembled entrywise,  $3.8 \times 10^{-15}$ ),  $x = K^{-1} \mathbf{1}$  and  $\text{cap} = \mathbf{1}^T K^{-1} \mathbf{1}$ , as a matrix identity ( $1.4 \times 10^{-13}$ ); under the congruence  $u = K^{1/2} v$  the Dirichlet half is an *orthogonal projection* ( $P^2 = P$  at  $1.9 \times 10^{-12}$ ,  $\Omega = \lambda_{\max}(P) = 1$  to  $5.1 \times 10^{-12}$  — the T142 normalisation, where T141 had guessed 20.7–2724), its complement *is* the capacity rank-one and it annihilates  $K^{-1/2} \mathbf{1}$  (controls: dropping the rank-one  $3.4 \times 10^{-1}$ , a 1% cap perturbation  $3.4 \times 10^{-3}$ ).
- **(2) The exact capacity–Rayleigh identity (T143) [E].** With  $H = \text{diag}(s) + L_N$  ( $4.7 \times 10^{-16}$ ),  $L_N = D^T B D$  for the signed crossing kernel ( $4.9 \times 10^{-16}$ ) and  $\rho(W) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ), the assembled quotient  $[d^T (J^{-1} - B) d + (x^T v)^2 / \text{cap} - \sum_k s_k v_k^2] / [d^T J^{-1} d + (x^T v)^2 / \text{cap}]$  equals  $v^T (K^{-1} - H) v / v^T K^{-1} v$  for *every*  $v$  ( $9.5 \times 10^{-14}$  on 24 random vectors per window), reproduces  $1 - \rho(W)$  at the exact top eigenvector ( $1.7 \times 10^{-9}$  under the *declared* 1/gap conditioning bar  $5.4 \times 10^{-8}$ ), and every random quotient stays  $\geq 1 - \rho(W)$ ; on the constant vector the whole denominator *is* the capacity term (controls: dropping it,  $\text{rel} = 1$ ; the  $|N|$  crossing kernel,

$2.6 \times 10^{-1}$ ).

- **(3) Cholesky nestedness (T144) [E]**.  $\text{cap}_E([a, a+j])$  equals the prefix sum of squared forward-substitution entries of *one* triangular solve: 3404 interval capacities from 36 solves at  $2.0 \times 10^{-14}$  against independent Schur solves, exhaustive on the median window (every prefix,  $2.8 \times 10^{-14}$ ), monotone in  $b$  by construction (control: the reversed ordering computes *suffix* capacities and deviates by  $\geq 4.6 \times 10^{-1}$ ).
- **(4) The whitening certificate (T144) [E]**.  $\lambda_{\max}(W) \leq \kappa_{\text{up}}$  (completed Cholesky)  $\leq$  the Gershgorin ceiling on every window ( $\kappa_{\text{up}} = 1.2245\text{--}1.4236$  against  $\leq 2.1213$ ); the licensed chain direction  $\lambda_{\min}(E)/\kappa_{\text{up}} \leq \lambda_{\min}(A, A_B)$  holds on every window — a denominator comparison, untouched by T142’s numerator obstruction — and the Cholesky refuses at the shrunk shift  $0.995 \lambda_{\max}(W)$ .
- **(5) The M4 split and the certified chain (T145) [E]**. The mass half of Maz’ya’s step M4:  $\sum_k f_k^2 \leq \psi^2$  *pointwise* (a theorem for nonnegative truncations, exact slack  $\geq 0$ ), the row-wise implication on rows with  $(E1)_i \geq 0$ , measured mass ratio  $\leq 0.2995$  and the full step  $\sigma_{\text{tot}} = 0.2197\text{--}0.4425 < 1$  on every window although the Markov hypothesis is absent. Licence 4 is stated with  $|R|$  and *not*  $R^+$ , with the witness  $R = [[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$ ,  $x = (1, -1)$  ( $3.0 > 2.0$ ) built in as a counterexample check. The sign-free chain is ordered on every window, and S1’ is *certified per window on both routes*:  $\text{cert\_lam\_max}(R) \leq 12 \sigma_{\text{tot}} \Psi_{\text{pos}}$  (energy,  $c_0 = 2.6369\text{--}5.3099$ ) and  $\leq G_{\text{dy}} \Psi_{\text{abs}}$  (sign-free,  $c_0 = 2.5154\text{--}6.1449$ ); the better route gives  $c_0 = 2.5154\text{--}4.2265$  against the classical Dirichlet value 4. The all-sets density bound carries Charikar’s cited constant, bracketed by exhaustive enumeration of all 4095 subsets of a small block (control:  $c_0 \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status**. Per instance on small windows only; nothing uniform in the zone index or in  $D$ ; the  $c_0$  of item (5) is *measured* at the minimiser’s own layer cake and the a-priori level lemma (T145’s L1) stays open. Maz’ya 1985 (the strong-type half *not* used as a theorem anywhere — it is the thing the cycle transcribes), Muckenhoupt 1972, Fukushima–Ōshima–Takeda 1994, Miclo 1999, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.CAPCHAIN.IDENT.01 [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T145, one-step-missing)**. The T145 probe (`mazya_proof_probe.py`, contract MAZYA.PROOF, 33/33, 8.0 s, 64 windows  $m = 26 \dots 1173$ ) transcribes Maz’ya’s classical proof step by step onto the non-Markovian form: M1, M2, M3 are theorems, the Markov property sits in exactly one line (M4), that line splits, its mass half is a theorem and dominates ( $\sigma_{\text{tot}} = 0.215\text{--}0.443 < 1$ ),  $c_0$  becomes explicit (2.248–4.227, fit  $D^{0.028 \pm 0.017}$ , zone-uniform) and S1’ is certified on 64/64 windows; an explicit no-go proves a level-profile hypothesis is *necessary*; S2’ retired, R5 downgraded, R4 closed — the mathematical rest is *one* named lemma, L1. `experiments/next.txt` part T145 (series totals 145 probes / 3659 checks); `tfpt_prime_front.tex` gains the T145 subsection, the v546 entry in the verified layer (now twelve modules) and the updated probe index; `website/lib/primeFront.ts` gains the T145 feed entry; suite counter  $539 \rightarrow 540$  scripts. T146 (`level_lemma_probe.py`) is running at exactly L1 and is *not* part of this change.

**2026-07-28 (III) (v545 — the Hardy-core identities of phase 2 promoted; prime-front diary T140/T141 + website sync; NO marker moves)**

- **New module v545 (PRIME.HARDY.IDENT.01)**. Promotion of the identity blocks of the discovery parts T140 (`signed_band_probe.py`) and T141 (`discrete_hardy_probe.py`) — 36 checks,  $\sim 1$  s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: *identities*, closed-form quantities and the *validity* — never the *size* — of two certified

upper-bound shapes, plus one negative typing. No fit is promoted, no  $D$ -exponent, no zone-uniform statement, and no bound that beats the target. Battery: 11 windows ( $n = 8 \dots 139$ ,  $h = 58 \dots 285 \leq 300$ , core  $m = 57 \dots 284$ , 193–5102 positive off-diagonal edges, 10–87 loaded stripes),  $D = 7.578 \times 10^{-3} \dots 2.388 \times 10^{-2}$  (a factor 3.15, so item (8) is a comparison and not a single point), exact  $\rho(W) = 0.997759\text{--}0.999981$ ; the explicit  $n_e \times n_e$  signed Gram is formed only on the 5 windows with  $n_e \leq 900$ .

- **(1)–(3) The form lifting, the exact finite core and the energy reordering [E].** With  $M$  the edge–interval incidence matrix and  $C = \text{diag}(\sqrt{\Delta})M$ , the T139 telescope lifts from the *entries* of the signed Gram to its *form*:  $\text{Gram} = CHC^\top$  at  $3.0 \times 10^{-15}$ , hence  $\text{rank}(\text{Gram}) \leq h - 1$  (measured 57–183 against 193–882 edges; control: unweighted incidence 1.6). The covering kernel in *closed geometric form*,  $K_{rs} = W([r \wedge s, r \vee s])$ , equals  $M^\top \Delta M$  ( $1.3 \times 10^{-15}$ ) and the spectral radius is an *exact* eigenvalue of the finite core,  $\rho(W) = \lambda_{\max}(K^{1/2} H K^{1/2}) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ) =  $\lambda_{\max}(\text{Gram})$  ( $7.8 \times 10^{-16}$ ) — three independent routes, no truncation (control: the factorwise product  $\lambda_{\max}(K)\lambda_{\max}(H)$  misses by  $7.8 \times 10^2$ ). The energy reordering  $H = \text{diag}(s) + L_N$  holds at  $4.7 \times 10^{-16}$ : a mass term plus a long-range Dirichlet form, its weights positive on 0.676–0.872 of the off-diagonal area (a measurement, no law).
- **(4) The checkerboard split [E].** At group length exactly  $b$ ,  $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$  *entrywise* (0.0) on all 9 (window,  $b$ ) pairs over  $b = 4, 8, 16$ , and the resulting three-step Weyl sequence dominates  $\lambda_{\max}(\text{Band}_b)$  with 0 exceptions (worst ratio 1.6940) at a step count independent of the up to 9 groups — three steps instead of  $O(nb)$  (control:  $L = b/2$  loses entries,  $3.5 \times 10^{-1}$ ).
- **(5)–(6) The four Hardy identities and the closed Muckenhoupt quantity [E].** (i) the  $K^+$  Rayleigh form  $\rho(W) = u^\top H u / u^\top K^+ u$  at  $u = K^{1/2} v$  ( $6.6 \times 10^{-13}$ ); (ii) the signed crossing kernel  $L_N = D^\top B D$  with  $B_{kl} = \sum_{r \leq k \wedge l, k \vee l \leq s} N_{rs}$  — the same closed covering formula as  $K$ , on the increment grid ( $4.9 \times 10^{-16}$ ); (iii)  $Q = B_+ \mathbf{1}$  ( $1.4 \times 10^{-15}$ ): the Cauchy–Schwarz weight T140 introduced by hand *is* the row sum of  $B_+$ , so that step is the Gershgorin diagonal step on  $B_+$ ; (iv)  $DKD^\top = L_\Delta$  on the interior nodes ( $2.7 \times 10^{-14}$ ) with diagonal  $(\Delta \mathbf{1})_{k+1} = 0.4886\text{--}15.15$ , the endpoint edge mass — a *local* quantity and not  $\lambda_{\max}(K)$  (control: the crossing kernel of  $|N|$  does not reproduce  $L_N$ ,  $2.6 \times 10^{-1}$ ; the signs are load-bearing). The two-weight quantity  $A_{M0} = \max_k Q_k (\Delta \mathbf{1})_{k+1}$  is computed by two closed routes agreeing to  $8.3 \times 10^{-15}$ ; its value (65.5–4244) is *reported*, its size and zone behaviour are not promoted (control:  $\lambda_{\max}(K)$  in place of the local mass inflates it by  $\geq 358$ ).
- **(7) Two certified shapes, promoted as *valid* bounds only [E].** Every Loewner step is certified by a completed Cholesky ( $-1.5 \times 10^{-14}$ ,  $-2.3 \times 10^{-10}$ ). The additive Hardy chain dominates  $\rho(W)$  on 11/11 windows (factor 3.074–10.758); the exact two-term Weyl split dominates too (1.691–2.503) and lies *below* the chain everywhere, so it is the floor of the whole additive family; the genuinely joint shape  $\Lambda(H, Y) \cdot \Omega(Y)$  with a positive definite Jacobi Hardy form  $Y$  dominates on 11/11 with the closed-form geometric profile (8.815–74.872 at  $\Omega = 8.33\text{--}36.2$ ) and on 11/11 with the per-zone tridiagonal read-off of  $K^+$ . Promoted is the *dominance* on these instances and nothing about the factors. Controls: the same  $Y$  without its mass term is singular — the constant vector lies in its kernel ( $5.7 \times 10^{-17}$ ) while  $H$  does not vanish there — so the certificate *refuses* on 11/11; and dropping the *positive* part instead of the negative one falls below  $\rho(W)$  on 11/11.
- **(8) The sign-versus-absolute separation — a negative typing [X].** Over the measured  $D$  range the exactly bounded object (the signed Dirichlet share) spreads by  $1.36\times$ , while every absolute variant spreads by at least  $54.94\times$  (Loewner drop 62.03, Cauchy–Schwarz 54.94, product 56.92), and dropping the positive part of  $H$  alone costs  $9.96\text{--}464.63\times$ . The corresponding power laws are *fits* and labelled as fits:  $D^{-0.138 \pm 0.094}$  (signed) against  $D^{-0.776 \pm 0.352}$  (certified Muckenhoupt) and  $D^{-2.748}$ ,  $D^{-2.820}$ ,  $D^{-2.918}$  (absolute). Promoted is the instance comparison of the measured spreads, not the exponents.
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or

in  $D$ ; item (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*; item (7) promotes validity and not size — neither shape beats the target. Abel summation, Cauchy–Schwarz, Weyl 1912, Gershgorin 1931, Haynsworth 1968, Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979, Muckenhoupt 1972, Bradley 1978, Opic–Kufner 1990, Bennett 1987/1991, Gantmacher–Krein 1950/1960, Karlin 1968, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.HARDY.IDENT.01 [E] + [X]; **no marker of any pre-existing contract moves**, and this is not RH evidence.

- **Diary, paper and website.** experiments/next.txt part T141 (HARDY-RESISTS, 22/22; series totals 141 probes / 3547 checks); tfpt\_prime\_front.tex gains the T141 subsection, the v545 entry in the verified layer and the updated probe index; website/lib/primeFront.ts gains the T141 feed entry and the v545 promotion entry. T142 (conductance\_profile\_probe.py) is running at the one open ingredient and is *not* part of this change.

**2026-07-28 (II) (v543 + v544 — the lumped  $M$ -matrix pair and the long-lag support structure promoted; prime-front diary T136/T137 + website sync; NO marker moves)**

- **New module v543 (PRIME.MMATRIX.IDENT.01).** Promotion of the identity/theorem block of the discovery part T136 (35 checks,  $\sim 1$  s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* per-instance result; no fit, no  $D$ -exponent, no uniform-in-zone claim, no floor value. Battery: 20 seam windows ( $n = 4 \dots 59$ ,  $\rho = 1.25 \dots 4.00$ ,  $h = 25 \dots 299$ ; every factorised or diagonalised matrix  $\leq 300$ ) giving 40 symmetric matrices (pole-free odd sections + border Schur blocks), 39 with positive off-diagonals, plus 312 shifted ladder rungs and 8 whitened two-grid rungs.
- **(1)–(2) The lumped pair and the congruence [E].** With  $\Delta$  the positive off-diagonal part and  $L_\Delta = \text{diag}(\Delta \mathbf{1}) - \Delta$ , the pair  $A_B = A + L_\Delta$  is Stieltjes (off-diagonals exactly  $\leq 0$ ),  $L_\Delta$  is a graph Laplacian hence PSD with zero row sums, so  $A_B \mathbf{1} = A \mathbf{1}$  ( $1.6 \times 10^{-15}$ ) and  $A_B \succeq A$  — an *upper* bound on  $\lambda_{\min}(A)$  and a floor *never* (control: lumping the negative off-diagonals destroys Stieltjes,  $2.0 \times 10^{-1}$ ). The congruence  $A = A_B^{1/2}(I - W)A_B^{1/2}$ ,  $W = A_B^{-1/2}L_\Delta A_B^{-1/2} \succeq 0$ , holds as a *matrix* identity ( $4.4 \times 10^{-15}$ ) with  $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$  ( $4.4 \times 10^{-15}$ ), and the *declared circularity*  $\lambda_{\max}(W) < 1 \iff A \succ 0$  is checked in *both* directions (0 exceptions forward; 39/39 shifted controls in the indefinite range give  $\lambda_{\max}(W) \geq 1$ ) — the Loewner sandwich is vacuous by itself, and that is the content.
- **(3)–(4) The anchor and the empty shift ladder [E].** With  $x = A_B^{-1} \mathbf{1} \geq 0$  and  $A_B^{-1} \geq 0$  entrywise (Ostrowski 1937, *checked*),  $\lambda_{\min}(A_B) \geq 1/\max_r x_r$  on all 39 rows (min slack  $3.0 \times 10^{-3}$ , sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue (control: a positive off-diagonal makes it false,  $1.9000 > 0.1000$ ). Lumping commutes with diagonal shifts,  $(A - sI)_B = A_B - sI$  ( $2.5 \times 10^{-16}$  on 312 rungs), and  $(A_B - sI)^{-1} \geq A_B^{-1} \geq 0$  entrywise (Berman–Plemmons 1979, violation 0), so  $\max_r x_r(s)$  and the a-priori margin  $\|L_\Delta\|_\infty \max_r x_r(s)$  are non-decreasing along the whole ladder: the shifted ladder is *structurally empty* — a theorem, not a measurement.
- **(5)–(6) Varga’s identity and the a-priori Jacobi radius [E].** For the regular splitting  $A_B = D_0 - N_0$  (all three hypotheses checked on 39/39),  $\rho(J) = \tau/(1 + \tau)$  with  $\tau = \rho(A_B^{-1}N_0)$  at  $4.8 \times 10^{-15}$  ( $\rho(J) = 0.7347\text{--}0.9860$ ); the Collatz–Wielandt bracket at the anchor vector gives  $\min_r (Tx)_r/x_r \leq \tau \leq \max_r (Tx)_r/x_r$  (0 exceptions) and hence an *a-priori*  $\rho(J) \leq 0.9861 < 1$  with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. Controls: a non-regular splitting breaks the identity ( $1.1 \times 10^{-1}$ ); at a sign-changing test vector the quotient bound is false on 39/39.
- **(7) The  $k$ -scanned M18 majorant — a negative result [X].** The whitened mass identity  $\|L^{-1}\hat{s}\|^2 = \hat{s}^\top U^{-1}\hat{s}$  holds at  $7.2 \times 10^{-16}$  and the triangle chain is a *valid* majorant at every  $k$

separately (39 (rung,  $k$ ) pairs over  $k = 2 \dots 64$ , 0 violations), so the minimum over the ladder is legitimate — and it does *not* reach the preregistered T134 bar  $1/2$  on any rung (best 3.080; best over the three values T134 tried, also 3.080) while the *measured* bad mass is  $\leq 0.0120$ . The localisation term alone already exceeds the bar (0.542): the loss is the bound, and what is missing is a localisation statement. The bar is *not* moved and nothing is upgraded.

- **New module v544 (PRIME.LONGLAG.SUPP.01).** Promotion of the structure block of the discovery part T137 (24 checks,  $\sim 0.2$ s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons).
- **(1)–(4) The support is a comb stripe set, and its amplitude is certified [E].** The kernel splits exactly as  $c = c_{\text{arch}} - c_{\text{comb}}$  with  $c_{\text{comb}} \geq 0$ , and the archimedean section alone has *all* off-diagonals  $\leq 0$  on 20/20 windows: every positive off-diagonal of  $A$  is comb-generated (control: the full section carries  $+2.0 \times 10^{-1}$ ). Every one of the 20555 positive edges has its Hankel index  $M - 1 - r - t$  inside the atom-hat index set — per-window fraction exactly 1.000000 — so  $\text{supp}(\Delta)$  is a union of anti-diagonal stripes at prime-power atom positions (the converse is reported, *not* claimed). Each stripe is a perfect matching (0 endpoint collisions), so its edge vectors  $b_e = e_r - e_t$  are *exactly* orthogonal and  $L_\Delta$  restricted to a stripe is a direct sum of  $2 \times 2$  blocks (control: edges across two stripes overlap). And  $A_{rs} = A_{rs}^{\text{arch}} - c_{\text{comb}}[|r - s|] + c_{\text{comb}}[\text{anti}]$  gives the amplitude certificate  $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$  on all 20555 edges (excess  $-1.9 \times 10^{-6}$ , sharpest ratio 0.99998532) — no eigenvalue, no norm; control: the Toeplitz lag in place of the Hankel index breaks it on 20415/20555. Honestly undecided and *not* promoted: at  $h \leq 300$  the hats never overlap, so the tighter one-atom refinement coincides with the certificate here.
- **(5) The Green function, bracketed two-sidedly [E].** For the lumped comparison  $B = D_0 - N_0$ ,  $J = D_0^{-1}N_0 \geq 0$ , the anchor identity  $q_J = \max_r (Jx)_r / x_r = 1 - \min_r 1/(d_r x_r) < 1$  ( $8.2 \times 10^{-16}$ ) needs no measurement; then the Neumann partial sum to  $K = 8$  is an *entrywise* lower bound for  $B^{-1}$  ( $-2.1 \times 10^{-4}$ ) *because the discarded terms are nonnegative*, and partial sum  $+(x_r/x_s)q_J^{K+1}/((1 - q_J)d_s)$  is an entrywise upper bound ( $-1.4 \times 10^{-1}$ ), on 39 rows (20 pole-free sections + 19 border Schur blocks). Control: dropping the tail makes the upper bound false. This delivers item (2) of the T136 rest list.
- **(6) The absolute-value envelope family is certified dead [X].** With  $E = B^{-1}L_\Delta$ , Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are all  $\geq \rho(|E|)$ , so one rigorous *lower* bound decides the whole family at once: a Collatz–Wielandt quotient from below at a power-iterated weight gives  $\rho(|E|) \geq 1.3141 \geq 1$  on 20/20 rows (the cheapest four members all  $\geq 1.5937$ ). No member of the family can ever certify  $\lambda_{\max}(W) < 1$  — a certificate of *death*, not a measurement. The *signed* radius  $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$  on the same rows: the absolute value alone destroys the cancellation, and what survives is a sign-preserving bound that does not exist yet.
- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing uniform in the zone index, in  $D$  or in the gap. The T136  $D$ -exponents (anchor  $D^{-0.565}$ , margin gap  $D^{+2.72}$ , slack  $D^{0.116}$ ) are fits and stay in the sandbox; the structure bound assembled from the support and amplitude facts does *not* beat the margin (T137: 0 of 35 windows) and is *not* promoted; entrywise positivity of the *target* inverse at depth stays open. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Frobenius 1912–Perron 1907 / Collatz 1942–Wielandt 1950 / Gershgorin 1931 / Neumann series / Haynsworth 1968 / Wilkinson 1968–Higham 2002 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Diary Teile 136/137 (sandbox).** `m_matrix_pair_probe.py` (contract M.MATRIX.PAIR, 30/30, verdict ONE-CARRIES): the  $\rho(K)$  item closes with Varga’s regular splitting and the Collatz–Wielandt bracket (sharp 1.00–1.03 on 900/900), the exact three-way bookkeeping  $\lambda_{\min}(A) \sim D^{2.27} = \text{anchor } D^{-0.56} \times \text{gap } D^{2.72} \times \text{slack } D^{0.12}$  attributes the whole degradation

to the margin (which is only  $2.5 \times 10^{-6}$ – $2.5 \times 10^{-2}$  wide while every cheap bound on  $\rho(E)$  sits at 1.82–3.11), and M17 closes *negatively* (the bad pencil subspace is delocalised, participation ratio 0.54–0.98, its dimension tracking the atom count at  $n_{\text{bad}}/n_{\text{at}} = 1.64$  rather than the grid, so  $\sigma_b(k) \geq 0.646$  everywhere); promotion list 15  $\rightarrow$  22 ready. `long_lag_probe.py` (contract LONG.LAG, 22/22, verdict BOTH-RESIST):  $L_\Delta$  is fully comb-generated (the archimedean half has a positive off-diagonal on 0 of 40 windows), the structure bound does *not* beat the margin (0/35, overshoot  $24\text{--}7.1 \times 10^5$  gaps, second differences decaying only algebraically as  $\text{dist}^{-0.63}$  where Demko–Moss–Smith would need a geometric rate), the envelope family is certified dead from below ( $\rho(|E|) \geq 1.3218$  on 35/35), and the Green bracket closes two-sidedly (lower recovers 0.011–0.406 of the minimum entry, upper overshoots by 2.49–12.24). Series counter: 137 probes, 3438/3438 sandbox checks. Sandbox only for the probes: no ledger row, no marker move, not RH evidence.

- **Ledger/papers/website.** Two new ledger rows: PRIME.MMATRIX.IDENT.01 and PRIME.LONGLAG.SUPP.01; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *ten* modules; phase 2 extended to T126–T137); website mirrors (/prime-front feed entries T136/T137, status surfaces Teile 11–137 / 3438 checks, verification DAG, papers data) updated in the same change.

**2026-07-28 (I) (v542 — the margin-chain identities of phase 2 promoted; v379 hardened to an exact positive-mixture Gram certificate; prime-front diary T132/T133 + website sync; NO marker moves)**

- **New module v542 (PRIME.MARGIN.IDENT.01).** Promotion of the *identity/theorem block* of the discovery parts T128–T131 into one narrow load-bearing module (44 checks,  $\sim 0.2$  s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: identities and *per-instance* theorems only; no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ( $n = 3 \dots 97$ ,  $\rho = 1.25 \dots 4.00$ ,  $h_c = 24 \dots 214$ ,  $h_f = 36 \dots 299$ ; every inverted or diagonalised matrix  $\leq 300$ ), 43 graded/fine grid pairs ( $m \leq 172$ ) and 400 randomised profiles; each identity is a numerical residual against a *preregistered* tolerance plus at least one mutation control that must fail by  $\geq 10^{-3}$  relative.
- **(1)–(2) Border geometry and the (T) identity [E].**  $\sum_i (1 - g_i) = t_l + t_r$  from the exact interval geometry (max rel  $5.5 \times 10^{-13}$ ), so the  $\kappa$  weights are a probability vector and the flat border vector gives  $\kappa_{\text{flat}} = 1$  *exactly* (control: a 1% perturbation of  $g_i$  breaks it by  $7.9 \times 10^{-2}$ );  $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$  at  $2.2 \times 10^{-16}$ , with  $\tau$  independently recomputed from the *full* odd overlap matrix ( $1.1 \times 10^{-13}$ ); dropping any of the three correction terms breaks the identity.
- **(3)–(4) The profile identity and the curvature chain [E].**  $p_{N-1} = 2 - p_0 + 2E$  at  $2.3 \times 10^{-16}$  (instances) /  $1.4 \times 10^{-15}$  (400 random profiles), hence the equivalence  $2E > p_0 \iff p_{N-1} > 2$  with 0 exceptions and both branches realised (60/400); the Abel/Hölder chain  $\kappa \leq 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \bar{\Delta p}| + (p_{\text{max}} - p_{N-1})$  holds per instance with 0 violations (min slack  $1.0 \times 10^{-1} / 3.7 \times 10^{-2}$ ), while the *same* chain without the curvature term is violated on 88/400 profiles — the curvature term is load-bearing, not decorative.
- **(5)–(6) Assembly and the C ea/Strang defect [E].** The matrix-free two-scale assembly equals  $J^\top Q J$  ( $5.2 \times 10^{-16}$ ) and the graded overlap operator equals  $J$  ( $5.2 \times 10^{-14}$ ), reducing to the uniform odd overlap when both partitions are uniform;  $S_{\text{graded}} - S_{\text{uniform}} = R^\top X^{-1} R$  as a *matrix* identity on all 43 grids ( $2.1 \times 10^{-14}$ ) with a PSD defect on every grid, so the compression error is one-sided *by* the identity (controls: unnormalised merge weights  $8.4 \times 10^{-1}$ ; dropping the  $-B$  term in  $R$   $4.0 \times 10^{-1}$ ).
- **(7)–(9) Sandwich, Perron implication, counting chain [E].** The secular sandwich  $\varepsilon / (\|u\|^2 + \varepsilon / \mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon / \|u\|^2$  holds on all 48 pole-free sections at which  $A > 0$  and  $\varepsilon > 0$  are *checked* (relative width  $\leq 1.4448$ ), with Albert’s sign equivalence verified in *both* directions

(6 rescaled controls with  $\varepsilon < 0$  all give  $\lambda_{\min}(Q) < 0$ );  $S^{-1} > 0$  entrywise (measured 48/48)  $\Rightarrow$  sign-constant *and* simple ground vector as an *implication* (tridiagonal control: hypothesis fails, conclusion fails with it);  $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$  and  $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$  with 0 exceptions on a non-vacuous battery ( $s_2$  up to 31).

- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing is uniform in the zone index; the  $\kappa$  law is *not* claimed (T129 found it false in general), only the chain underneath it; (7) assumes nothing — positivity of the pole-free section at depth (the open M25d) is not extended and no floor value is carried; (8) has a *measured* hypothesis; (6) says nothing about the *size* of the defect (the open T130 exponent). Abel / Hölder / Schur–Haynsworth 1968 / Albert 1969 / C  a 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Hardening of v379 (SEAM.S3.RP.01).** T133 (CERT.FLOOR) certified that the  $10 \times 10$  Hankel matrix v379 assembles has, *as a double-precision object*, a negative direction at  $-7.75 \times 10^{-18}$  (60 digits) — inside the module’s old `min_eig > -1e-10` tolerance. The mathematical matrix is fine and the exact rescue is now in the module: the RP Gram is built explicitly as a *positive mixture*  $M = V^T \text{diag}(1/1600)V$ ,  $V_{ki} = e^{-\varepsilon_k \tau_i}$ , so  $x^T M x = \sum_k w_k (v_k \cdot x)^2$  is a *sum of squares* and  $M \succeq 0$  holds exactly; the mixture identity is zero at the rounding level ( $1.1 \times 10^{-15} \leq 64u|M|$ ) and the sum-of-squares route is nonnegative on 510/510 test directions where the assembled float form is negative on 2. The eigenvalue check is *retained and relabelled a diagnostic measurement*. Ledger row SEAM.S3.RP.01 reworded accordingly (method text: exact positive-mixture Gram identity, eigenvalue check as diagnostic); **marker stays [E] — no upgrade**. 7 checks (was 5), `run_all` green.
- **Diary Teile 132/133 (sandbox).** `bd_seam_probe.py` (contract BD.SEAM, 21/21, verdict SPECTRUM-ONLY): the model boundary-Dirac and the marks kernel share their spectrum to  $7.5 \times 10^{-13}$  but are *not* the same operator — the gap sits in the killing mass 0.1746 ( $N$ -stable), the killing measure equals the marks profile pointwise, the Markov remainder is exactly zero with a closed jump-weight form (odd lags positive, even lags zero — which explains T126’s 86.1%), and the  $D_4$  lattice symmetries leave the triad exactly invariant; ceiling: the DtN is a model DtN, the gate is *not* closed. `cert_floor_probe.py` (contract CERT.FLOOR, 23/23, verdict MIXED, 840 control validations): the v379 finding above plus a recommendation table — the hardening is implemented in this same change. Series counter: 133 probes, 3352/3352 sandbox checks. Sandbox only: no ledger row for the probes, no marker move, not RH evidence.
- **Ledger/papers/website.** New ledger row PRIME.MARGIN.IDENT.01; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *eight* modules; phase 2 extended to T126–T133); website mirrors (`/prime-front` feed entries T132/T133, status surfaces Teile 11–133 / 3352 checks, verification DAG, papers data) updated in the same change.

**2026-07-27 (IV) (prime front: T111 deep ladder maps three walls, crossing measured at  $n \approx 462$  — CROSSING-CONFIRMED; paper + diary + website sync; NO marker moves)**

- **Diary Teil 111 (sandbox).** `experiments/tfpt-discovery/deep_zone_stress_probe.py` (contract DEEP.ZONE.STRESS, 23/23 checks, 140s, verdict CROSSING-CONFIRMED): the probe stops extrapolating and measures — the T110 grid is reproduced bit-exactly, then the ladder is extended to every prime-power zone  $n \leq 1000$  plus a thin tail to 3001 (199 zones; chain ladder  $n = 2 \dots 521$  with 117 handovers). The ratio  $m_k/\text{need109}$  falls  $179.59 \rightarrow 0.0174$ ; the crossing is *measured*: last zone  $\geq 1$  at  $n = 461$ , first below at  $n = 463$  (0.935) — T110’s  $n \approx 170$  was a fit artefact, but the wall is real, and a depth-preserving resolution study makes  $n^* \approx 462$  an *upper* bound. The failure follows the prime branch (prime powers 512/529/625/729 survive longer).



New and harder: the *ladder wall* — the nested step needs a lag gap  $> 2D$ , and the twin-prime pair  $521 \rightarrow 523$  (gap  $0.003831 < 2D = 0.005617$ ) kills the ladder arithmetically (arithmetic scan to 40 000: 4052/4284 pairs endangered; co-moving  $D \sim 1/n$  forces  $h \sim n \log n/4$ ). The no-reserve finding softens ( $f_{\text{crit}} \sim n^{-0.39}$ , the reserve opens with depth); the circle breaks first at  $k = 109$ ,  $n = 463$  — it tears at the ratio, not at a step: all 117 handovers certify at retention 1.000000 (largest single loss  $5.96 \times 10^{-8}$ ), atom entry free 117/117. Three separate walls: margin  $n \approx 462$  (measured), ladder  $521 \rightarrow 523$  (twin primes), requirement  $n = 727$  ( $\omega_{\text{cert}} \geq 1$ , need109 vacuous on 46 zones). Reinterpretation: the operating variable is the window *depth*, not  $n$  (cheek doubling moves the wall  $463 \rightarrow 47$ ); a proof must beat the exponent gap  $-0.974 = -0.681 - 0.293$  and couple  $\delta_0$  to the local prime gap. Follow-up probe T112 (`adaptive_scaling_probe.py`, contract `ADAPTIVE.SCALING`, the  $n$ -adaptively scaled frame) is running. Series counter: 111 probes, 2781/2781 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.

- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T111 (new subsection “The three walls (T111)”); the T110 three-gaps box and the  $n \approx 170$  mentions corrected to the measured  $n^* \approx 462$  (fit artefact noted); reduction-cascade caption, two-doors figure chain, three-way status split and asymptotic-mountain section updated (the three walls as the measured formulation; depth-not- $n$  reinterpretation; T112 running); abstract and probe-index appendix updated (T111 row; series counter 111/2781; T112 recorded as running and deliberately absent).
- **Website.** `/prime-front`: T111 feed entry (`lib/primeFront.ts`, new verdict enum `CROSSING-CONFIRMED`), status line in the two-doors convergence panel (T111 + T112 running), hero counter Teile 11–111 / 2781 sandbox checks, program-status callout sentence for T111.

### 2026-07-27 (III) (prime front: T110 margin propagation closes the measured-zone circle — MARGIN-PROPAGATES; paper + diary + website sync; NO marker moves)

- **Diary Teil 110 (sandbox).** `experiments/tfpt-discovery/margin_propagation_probe.py` (contract `MARGIN.PROPROPAGATION`, 28/28 checks, 7.6s, verdict `MARGIN-PROPAGATES` with a precision statement): the induction circle closes end to end on the measured zones — the base case  $n = 2$  is certified by an explicit Cholesky at three resolutions ( $\lambda_{\min} \geq 6.93 \times 10^{-2}$ , factor 88–180 over need), the graded Loewner minorant holds all 15 handovers with retention 1.0000 (the strictly scalar variant closes 0/15), and the atom entry costs nothing — it *raises*  $\lambda_{\min}$  on 15/15 (first-order Kato term with helping sign; at  $n = 29$  the atom-free window is not even positive,  $-9.2 \times 10^{-3}$ ). Margin map:  $m_k \geq \text{need109}$  on 16/16 zones (ratio 2.09–179.6); the growth step splits exactly (embedding error  $\leq 2.6 \times 10^{-14}$ ;  $\max |\mu N| = 0.0$  on 15/15); scalar routes killed (bordered Weyl 0/15; Schur/Friedrichs cap vacuous 15/15). Three sharp gaps remain: no reserve ( $f_{\text{crit}} = 1.00$  at the first handover; retention  $1 - 2 \times 10^{-7}$ ), no scalar step law (structurally excluded by the boundary layer), no  $k$ -uniformity (trend  $m_k \sim n^{-1.93}$  against  $\text{need109} \sim n^{-0.98}$ , extrapolated crossing  $n \approx 170$ ). Follow-up probe T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, the deep ladder to  $n \sim 200$ ) is running. Series counter: 110 probes, 2758/2758 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T110 (new subsection “The margin propagates: the circle closes on the measured zones” with the three gaps as a key box); reduction-cascade caption and two-doors figure chain extended to T110; kill list extended by the T110 scalar-step-law kill; three-way status split and asymptotic-mountain section sharpened (the three gaps as the precise formulation; the  $n \approx 170$  warning; T111 running); abstract and probe-index appendix updated (T110 row; T111 recorded as running and deliberately absent); symbol-table row for  $m_k/\text{need109}$ .
- **Website.** `/prime-front`: T110 feed entry (`lib/primeFront.ts`, new verdict enum `MARGIN-PROPAGATES`), status line in the two-doors convergence panel (T110 + T111 running), hero

counter Teile 11–110 / 2758 sandbox checks, program-status callout sentence for T110.

**2026-07-27 (II) (prime front: T109 boundary certificates close (R) conditionally — BOUNDARY-CERTIFIED; paper certification arc T102–T109 + diary + website sync; NO marker moves)**

- **Diary Teil 109 (sandbox).** `experiments/tfpt-discovery/boundary_decay_probe.py` (contract BOUNDARY.DECAY, 29/29 checks, 47.2s, verdict BOUNDARY-CERTIFIED): both remaining scalars of (R) are certified —  $\omega$  *unconditionally* via a graded matrix cap in PSD order (one level per soft direction, Cholesky-verified,  $\omega_{\text{cert}} = 0.2561\text{--}0.7569$  against measured  $0.2366\text{--}0.7531$ , factor  $1.004\text{--}1.124$ , no hypothesis input; the T108 compression Schur distance shrinks  $0.18\text{--}0.52 \rightarrow 0.0024\text{--}0.039$ ), the boundary value via a residual / goal-oriented certificate (Prager–Synge) that *carries* the cancellation (sharpness  $1.0000\text{--}2.4937$ , 16/16 in budget). Mechanism finding: no decay — *cancellation* ( $44\text{--}4.6 \times 10^4$  on 15/16 zones); the Combes–Thomas route is refuted with the exact Green row (1/16). The full chain closes 16/16 (margin  $1.3\text{--}10.4$ ,  $M$ -stable to 3000) conditional on exactly *one* input: the strict positivity margin  $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}\text{--}7.5 \times 10^{-3}$ , i.e.  $10^2\text{--}10^6$  below  $\mu/2$ . Follow-up probe T110 (`margin_propagation_probe.py`, contract MARGIN.PROPGATION) is running. Series counter: 109 probes, 2730/2730 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T109 (new subsection “Boundary certificates: cancellation, not decay”); the reduction cascade figure now ends at the one strict-positivity margin input; two-doors figure chain extended to T109; kill list K8 extended by the T109 Combes–Thomas kill; three-way status split (certified / measured / classical) updated; abstract and probe-index appendix updated (T109 row; T110 recorded as running and deliberately absent); symbol table rows for  $x/h/x_0$  and  $\omega/\omega_{\text{cert}}$ .
- **Website.** /prime-front: T109 feed entry (`lib/primeFront.ts`, new verdict enum BOUNDARY-CERTIFIED), status line in the two-doors convergence panel, hero counter Teile 11–109 / 2730 sandbox checks, program-status callout sentence for T109.

**2026-07-27 (I) (new companion `tfpt_prime_front` — the prime / zeta line documented end to end as RESEARCH DOCUMENTATION; NO new claims, NO marker moves)**

- **New companion paper `tfpt_prime_front` (P).** 22 pages, 9 sections plus two appendices, 8 TikZ figures. It documents the complete prime / zeta line of the project from the first  $\mu_4 \Rightarrow \chi_4$  observation to the current residual — explicitly a *documentation of a research route*, not a results paper. Registered in `tex-artefacts/tfpt_docset.tex` as the fifth companion.
- **Framework interpretation stated once, up front.** Section 3, “The big picture”, collects the reading that motivated the line and labels it as interpretation: the  $E_8$  shell count  $N(2n) = 240\sigma_3(n)$  is the Dirichlet series  $240\zeta(s)\zeta(s-3)$ ; the  $\mu_4$  glue carries  $\chi_{-4}$ , so the  $4k+1 / 4k+3$  split is a fibre property of the compiler and the signed channel is the only pole-free one; Hecke structure is emitted by lattice adjacency (`v535`) rather than put in;  $E_8$  is unimodular, so Poisson summation gives the theta sum an exact functional equation whose fixed axis is the critical line — the same reflection that reappears as  $J_1 \circ J_{1/2} = e^{\pm u}$  (the transport operator, T83) and as the parity split of the Weil pole (T106). Explicitly typed as a picture, not a reformulation, and not used as a premise anywhere.
- **Verified layer restated, not extended.** Sections 4.1–4.7 restate the seven load-bearing modules `v535–v541` with their ledger status markers quoted verbatim (Hecke from geometry, Eichler trace layer, half-integral bridge, relative-trace identity, Weil structure family, amplitude linear carrier, matching-lemma / transport ledger). The `\veri` citations in this paper are the only citations it makes; no ledger row is added, moved or upgraded.

- **Sandbox layer marked as sandbox throughout.** The localization program (I5, the Connes–Consani dictionary, the tightness map, the Sonin crossing, the blind prime demo), the handover induction and the certification arc are carried as clearly tagged *sandbox* content with a per-section provenance marker; every quantitative statement carries its probe label (Tn) and is reproduced verbatim from the research diary. All measured quantities are declared as measured, all window results as window results, all hypothesis inputs as hypothesis inputs.
- **Kill list as a first-class section.** Section 8 records the refuted readings and dead routes of the program (essential-singularity reading, the  $C/g$  proxy law, the naive margin route, the density chain, the Landau–Widom anchor, the accumulated invariant with self-propagation, the symbol route for the odd channel, three certificate candidates for the avoidance norm), each with the probe that killed it.
- **Explicit RH fence.** The paper states in the abstract, in the scope-and-status statement, in the reduction figure and in the status section that *no progress on the Riemann Hypothesis is claimed*; the  $I5 \Leftrightarrow$  Weil positivity  $\Leftrightarrow$  RH relation is carried, as in the ledger, as an equivalence *typing* only. ZETA.HP.CARRIER untouched.
- **Appendices.** Probe index T11–T108 (98 parts, 99 runs, 2509 green checks; the diary’s own series counter at T108 reads 108 probes / 2701 checks including the ten parts that predate the current diary file) with contract, verdict and check count per probe, plus a symbol table for the sandbox objects used in the figures. T109 (`boundary_decay_probe.py`) was running when the document was written and is deliberately absent.

## 2026-07-25 (XIX) (v541 — matching lemma and transport ledger package: two named limits as content; NO marker moves)

- **New module v541 (RTF.GNS.LEDGER.01).** Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 into one load-bearing module (33 checks,  $\sim 10$ s); every core check is *recomputed* from scratch, not cited. Companion to v538/HECKE.GEOM.RTF.01, v539/RTF.GNS.WEIL.01 and v540/RTF.GNS.AMP.01. Classical Gronwall 1913 / Robin 1983 *unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet  $L(1, \chi)$  / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri / Alaoglu–Erdős named classical — NEW is the compiler-native consolidation and the machine-checked window proofs.
- **(A) Window proof [E].** The matching lemma is *proved exact-integer on*  $[4, 10^6]$ :  $7S(j) < 40A(j)$  at every atom (0 violations over 939 870 clash atoms; full enumeration, no-overflow proven, big-integer rechecks); exact margin  $X = 22242575693/270725400000 = 0.082159$  at  $j^* = 554400$ ;  $\rho_{\text{crit}} = 1.1440 > 21/20$ ; maximal-greedy identity  $\Rightarrow$  uniform in the target set and the greedy weights. Structure laws at 0 tolerance: 8-law ( $n \leq 250000$ ), seed-tower +  $\psi$   $w$ -table ( $n \leq 50000$ ), Cohen seeds  $\{-48, -80, -8, -8\}$  ( $s \leq 2000$ ), bracket  $\Theta \leq 2|\psi| \leq 3\Theta$  on  $10^6$ .
- **(B) Transport ledger [E].**  $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$  closes with  $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$  *proven* (sympy kernel collapse; quadrature  $2.8 \times 10^{-16}$ ; exact scale invariance); ledger residual  $4.4 \times 10^{-16}$  on 24 battery rows; the odd-prime side of the Weil functional equals the certified plus combination  $P_\zeta(g_-) + P_\zeta(g_+)$  exactly (rel  $3.9 \times 10^{-16}$ ).
- **(C) Character-exact signed envelope [E].**  $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$  coefficient-exact on all odd  $n \leq 50000$  (both builds); signed certificate 0 violations, margin factor 8.86 $\times$ ; confinement *set equality* on  $10^6$ : zero-credit clash atoms =  $\chi_{-4}$ -coherent odd composites.
- **(D) Archimedean term internal [E].** Legendre duplication bridge  $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$  (sympy +  $1.8 \times 10^{-41}$ ); kernel identity  $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$  pointwise  $7.9 \times 10^{-31}$  on 51 points;  $\Delta_{\text{arch}}(h) = A_{\text{fam}}(h) - A_{\text{shift}}(h)$  on a 10-row battery subset at rel  $9.9 \times 10^{-16}$ .
- **(E)  $\lambda$ -equivariant channel [E].** CM carrier  $g_\lambda = \sum_{\mathfrak{a}} \lambda_1(\mathfrak{a})q^{N_{\mathfrak{a}}}$  exact in two independent routes ( $\mathbb{Z}[i]$  lattice vs Cornacchia reconstruction, 0 mismatches  $n \leq 10^4$ ); CM laws exact;

$\text{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$  as a set equality; canonical phase mean  $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$  exact;  $\lambda$ -window certificate 0.0653 vs 0.2365 unlifted ( $3.6\times$ ), exact Fraction rechecks.

- **(F) Frontier facts [C].** Unlifted coherent chain crossing  $k^* = 14$  (exact Fractions;  $\log_{10} N^* = 23.1$ ); absolute Robin tail factor 6.16 at  $J = 10^6$  and divergent — the constant route stays open (window constants with declared classical all- $n$  extension).
- **Named limits as load-bearing content.** (i) *One* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped  $\neq$  formal proof. (ii) I5 in *one-family* form ( $Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \geq 0$ ) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity  $\iff$  RH (an equivalence *typing*, no progress claim). NOT “almost RH”; ZETA.HP.CARRIER untouched; no marker moves.
- **Ledger/papers/website.** New ledger row RTF.GNS.LEDGER.01; \veri citations in tfpt\_1, tfpt\_3, tfpt\_research\_contracts; Wolfram extension mirrors the exact algebraic identities; website mirrors (/prime-front feed, verification DAG, papers data) updated in the same change.

## 2026-07-25 (XVIII) (v540 — amplitude route and positive linear carrier: open boundary $\lambda^*$ as content; NO marker moves)

- **New module v540 (RTF.GNS.AMP.01).** Promotion of the discovery amplitude chain T67/T68/T69/T70/T71/T72 into one load-bearing module (34 checks,  $\sim 3$  s). Companion to v538/HECKE.GEOM.RTF.01 and v539/RTF.GNS.WEIL.01: the machine-checked route out of the square plane behind the two v539 obstructions. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Möbius square sieve / Shimura / Shintani / Weil 1952 / Farkas named classical — NEW is the compiler-native consolidation.
- **(A) Amplitude Dirac [E].**  $D = \begin{pmatrix} 0 & V \\ V^\top & 0 \end{pmatrix}$  with  $V[d, m] = \sqrt{w_d} b(d) \chi_d(m)$ :  $D = D^\top$ ;  $D^2 = \text{diag}(VV^\top, K)$ ,  $V^\top V = K$  (rel  $2.7 \times 10^{-16}$ ); spectrum =  $\pm$  singular values; Hecke intertwining exact ( $p = 3, 5, 7$ ); Shimura  $b(dp^2) = b(d)(a_p - \chi_d(p)p)$  integer-exact on 644 pairs.
- **(B) Geometric polarisation + Cohen seed [E].**  $b = N_+ - N_-$ ,  $\Theta = N_+ + N_-$  lattice-exact;  $N_\pm \geq 0$  and  $b^2 = \Theta^2 - 4N_+N_-$  for all  $n \leq 50000$ ;  $\Theta$  is a *pure* Siegel–Weil Eisenstein  $\sigma_3$ -eigenform ( $p \leq 13$ , cusp component zero); Cohen seed  $\Theta(d) = -48 L(-1, \chi_d)$  exact-rational on 159 live  $d$ .
- **(C) Deletion universal = square-class double counting [E].** Cauchy–Littlewood window lemma (free Satake symbols,  $k \leq 6$ ): numerator  $1 - stuv X^2$ , hence  $1 - p^6 X^2$  pair-independent at  $st = uv = p^3$  (five channels); and fam  $[1, 0, 2, 0] + 2b[0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$  generating-function exact (Möbius square sieve exact  $n \leq 2000$ ).
- **(D) Positive linear carrier + plus balance [E].**  $\ell^2(d, \mu)$  with  $\mu = 48 |L(-1, \chi_d)| |d|^{-a}$ : full weights  $\lambda_k = 1 + p^{3k} - (\chi p)^k$  (layer  $[1, 1, 1, 1]$ );  $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$  (rel  $2 \times 10^{-15}$ ); the  $b$ /doubling kernel enters *with plus* (sign inverted vs the v539 square-plane minus).
- **(E) Functional equation exact [E].**  $\Theta(i/(8y)) = 8y^{5/2} \Theta^\dagger(iy)$  (rel  $6 \times 10^{-41}$ , Fricke closed);  $\Lambda_\Theta(s) = 8^{1-s} \Lambda_{\Theta^\dagger}(5/2 - s)$  in the strip (rel  $1.4 \times 10^{-40}$ , independent split points); poles  $5/2$  /  $0$  swapped; centre  $5/4$ , plus-locus offset exactly  $1/4$ .
- **(F) Cone facts + named open boundary [C].** Guaranteed cone and library hull FE-self-dual, gap functional  $\lambda^*$  FE-covariant (36/36 sampled); mirror sign law  $\text{sign } \psi(n) = (-1)^{\lfloor n/2 \rfloor + 1}$  exact ( $n \leq 50000$ ); constraint class  $C_{L3} = \{n \equiv 6 \pmod{8}\}$ ;  $\lambda^*$  exists with one exact Farkas witness ( $\lambda_{L3}^* = 0.0511$  at  $n^* = 6$ ); coverage saturates — the residual distance to the Weil cone *is*  $\lambda^*$  on  $n \equiv 6 \pmod{8}$ .
- **Fences.** Euler-region positivity only (edge  $L$ -values, not the central line); not “almost RH”;  $\lambda^*$  is the named open boundary *inside* the claim; ZETA.HP.CARRIER untouched; no marker moves.

- **Surfaces.** Ledger RTF.GNS.AMP.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (613  $\rightarrow$  618); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-25 (XVII) (v539 — Weil structure of the compiler family: two isolated obstructions; NO marker moves)

- **New module v539 (RTF.GNS.WEIL.01).** Promotion of discovery probes T55/T61/T62/T63 (Route-B chain) into one load-bearing module (25 checks,  $\sim$  6s). Companion to v538/HECKE.GEOM.RTF.01: the compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* (obstructions are prominent verified content, not a footnote). Classical Weil 1952 / Gelfand / direct integrals / SU(2)–Chebyshev / Sato–Tate /  $2^\omega$ -series / GNS named classical — NEW is the compiler-native identification and the machine-checked isolation of the two obstructions.
- **(A) GNS fibre structure [E].** Metric  $\ell^2(d, b^2/|d|)$  diagonal; fibres  $\sigma(d) = (\chi_d(3), \dots, \chi_d(13))$  partition the live family (1191 fund.  $d \leq 12000$ );  $\Lambda_{\text{fam}}$  exact constant on each fibre; twist-mix per fibre  $\leq 5.037 \times 10^{-16} < 10^{-12}$ .
- **(B) Isotype / GL(1) core [E].** Character expansions  $\Phi_1 = \chi_0 + \chi_2$ ,  $\Phi_2 = \chi_4 - \chi_2$ ,  $\Phi_3 = 2\chi_0 - \chi_4 + \chi_6$ ,  $\Phi_4 = \chi_8 - \chi_6$ ; core series  $c_{k,0} = [1, 0, 2, 0, 2, 0]$ ;  $G_0 = (1 + Y)/(1 - Y) = \zeta_p(w - 3)^2/\zeta_p(2w - 6)$ ; global  $\zeta(u)^2/\zeta(2u) = \sum 2^{\omega(n)}n^{-u}$  (Dirichlet exact through  $n \leq 2000$ ).
- **(C) Weil linear relation [E].** Prime<sub>F</sub> =  $2P_\zeta(g) - 2P_\zeta(g_b)$  (max rel  $9 \times 10^{-16}$ );  $Q_{\text{fam}} = Q_F + \text{Corr}$  (rel  $2 \times 10^{-16}$ ).
- **Obstructions as verified content [E].** (1) Doubling enters with a *minus* (family positivity does *not* imply  $Q_\zeta \geq 0$ ). (2) Plancherel Corr non-automorphic: polar/arch absorption FAIL; finite Euler factor FAIL;  $e^{-\sum p^{-u}}$  PASS.
- **Finite-class positivity [C].**  $Q_{\text{fam}} \in [4.369, 11.486]$  on the finite test class; dense-class positivity open / RH-adjacent — *not* claimed.
- **Fences.** Not “almost RH” — the claim structurally shows why family positivity does not imply RH; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger RTF.GNS.WEIL.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (608  $\rightarrow$  613); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-25 (XVI) (v538 — compiler relative-trace identity: three projections of one finite RTF; NO marker moves)

- **New module v538 (HECKE.GEOM.RTF.01).** Promotion of discovery probe T53 (relative\_trace\_compiler\_probe.py) into one load-bearing module (18 checks,  $\sim$  14s). Companion synthesis of v535/HECKE.GEOM.01, v536/HECKE.GEOM.EICHLER.01 and v537/HECKE.GEOM.HALFINT.01: these are three *projections* of one finite relative-trace identity  $[E_8/\text{glue orbital counting}] = [\text{Newform/Waldspurger spectral sum}]$ , not three separate results. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical — NEW is the compiler-native realisation and the machine-checked bilateral independence.
- **(R1) Trace identity, both sides independent [E].**  $\text{Tr}_V(\nu_p \circ \pi)$  geometric ( $a_p = -c(p)/8$  from the glue census, no modular input) equals spectral ( $a_p$  from  $\eta(2\tau)^4\eta(4\tau)^4$ , no geometry) for all  $(p, \pi) \in \{3, 5, 7\} \times \{\text{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ . Anchors (Id, Eis, cusp,  $E_4$ ,  $f_8$ ):  $p=3$ : (6304, 5600, 704, 1120, 352);  $p=5$ : (105848, 98280, 7568, 19656, 3784);  $p=7$ : (727680, 688000, 39680, 137600, 19840).

- **(R2) Orbit dictionary [E].** The Eichler split  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is the RTF orbit decomposition (principal orbit  $a \cdot \text{Id} \rightarrow \lambda_{\text{Eis}} = L - \sigma_3^2$ ; elliptic  $b \cdot T_p \rightarrow a_p^2$ ; cross terms cancel). Anchors  $\lambda_{\text{Eis}} = 336/3780/19264$ ,  $a_p^2 = 16/4/576$ .
- **(R3) Period side is lattice counting [C].**  $g$  is the quaternary form  $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$  ( $x, y$  odd; sign  $(-1)^{|u|+|w|}$ );  $b_{\text{rep}} = b$  exactly through  $n \leq 200$ ; then  $[\text{signed lattice count}]^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$  with  $R = 23.1873585645$  (rel  $\sim 10^{-12}$ ). Verdict **ONE-FORMULA**.
- **Fences.** Identity is *finite* ( $p \in \{3, 5, 7\}$ , finite  $d$ -windows); the *infinite* RTF (sum over all  $p$  and  $d$  + archimedean term) is the named open step; no RH; not **ZETA.HP.CARRIER**-relevant; no marker moves.
- **Surfaces.** Ledger **HECKE.GEOM.RTF.01**; `\veri in tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (603  $\rightarrow$  608); website **papers.ts** + **VerificationDag E8** + **primeFront.ts** + **ScriptIndex** via **bash build.sh gen**.

**2026-07-24 (XV) (v537 — half-integral bridge: unique Shimura preimage +  $T(p^2)$  Hecke equivariance + Kohnen scope fence + Waldspurger/Baruch–Mao constancy; NO marker moves)**

- **New module v537 (HECKE.GEOM.HALFINT.01).** Consolidated promotion of discovery probes T38/T41/T44/T45 into one load-bearing module (20 checks,  $\sim 90$  s). Companion to v535/HECKE.GEOM.01 and v536/HECKE.GEOM.EICHLER.01 without duplicating weight-4 Kneser/Eichler checks. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical — the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant  $R$ .
- **(A) Unique Shimura preimage at scale  $-8$  [E].** In the 70-element compiler weight-5/2 theta-monoid exactly one object satisfies  $\text{Sh}_{t=2, \psi=1}(g) = -8 f_8$  (witness  $g = \vartheta_2(q^2)^2 \vartheta_3(q^2) \vartheta_4 \vartheta_4(q^2)$ , verified through  $n \leq 120$ ); three further monoid elements lift to scales  $+8, -16, +16$ .
- **(B) Hecke equivariance [E].** Exact  $T(p^2)$ -eigenform, trivial nebentypus, with  $\lambda = a_p(f_8)$  for  $p = 3, 5, 7, 11, 13$  ( $-4, -2, 24, -44, 22$ ); trial FE  $N = 512$ ,  $\varepsilon = +1$  (centre  $5/4$ ).
- **(C) Kohnen scope fence [E].**  $U_4(g) = 0$  (mass mod 4 on  $\{1, 2\}$ ); level  $32 = 4 \cdot 8$  outside Kohnen 1982 — plus-route structurally closed (negative result part of the claim).
- **(D) Waldspurger/Baruch–Mao constancy [C].** AFE twist  $L$ -values ( $\varepsilon_d = \chi_d(8)$ );  $R(d) = 23.1873585645 \dots$  constant on ten fundamental  $d \equiv 1 \pmod{8}$  (spread  $\leq 10^{-12}$ , including  $d = 89$ );  $d \equiv 5 \pmod{8}$  vanishes.
- **Fences.**  $\text{GL}(2)$  centre  $s = 2$ , not the  $\xi$ -line; no RH; **ZETA.HP.CARRIER** untouched; no marker moves.
- **Surfaces.** Ledger **HECKE.GEOM.HALFINT.01**; `\veri in tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (598  $\rightarrow$  603); website **papers.ts** + **VerificationDag E8** + **ScriptIndex** via **bash build.sh gen**.

**2026-07-24 (XIV) (v536 — Eichler trace layer at the  $E_8$  lattice: Witt  $\lambda_{\text{Eis}}$  + two-sided Eichler identity + Type-A/B local densities + signed  $a_p$ ; NO marker moves)**

- **New module v536 (HECKE.GEOM.EICHLER.01).** Consolidated promotion of discovery probes T33/T36/T37/T42 into one load-bearing module (23 checks,  $\sim 30$  s). Builds on v535/HECKE.GEOM.01 (Kneser counts,  $\nu_p$  law, oldforms) without duplicating those checks. The claim is *not* RH-related: the Eichler/Witt package of the  $\mu_4$ -glue census is lattice-native and two-sided in-suite.

- **(A) Witt layer [E].**  $\lambda_{\text{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$  identically (sympy) and for all 24 odd primes  $p \leq 100$ ; anchors  $\lambda_{\text{Eis}}(3, 5, 7) = (336, 3780, 19264)$ .
- **(B) Eichler identity [E]/[C].**  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ ; type-(i)  $N_{\perp} = (p^6 - 1)/(p - 1)$  live constant on all 240 roots at  $p = 3, 5$ ; type-(ii) live at  $p = 3$  ( $352 = 364 - 12$ ) with Kneser filter = identity;  $p = 5$  T36-validated freeze ( $3784 = 3906 - 122$ ,  $R = 4$ ; Shell(25) re-enumeration budget-gated); assembler  $R(p) = \sigma_3 - 1 - c(p^2)/8 = a_p^2$ ;  $R(7) = 576$  prognosis (Shell(49) out of scope).
- **(C) Local densities [E].** Type-A = isotropic cosets representing the integer  $p$ ;  $N_A = \min(240(1 + p^3), p^7 + p^4 - p^3 - 1)$ ,  $N_B = (\#\text{iso} - 1) - N_A$ ; B empty iff  $p^4 < 241$  (only  $p = 3$ );  $p = 7$  live 82560/743040; injectivity at  $p = 11, 13$ ; fibre refinement scoped to  $\{5, 7, 11\}$ .
- **(D) Signed extraction [E].**  $a_p = -c(p)/8$  from geometric Shell( $p$ ):  $(-4, -2, +24)$  at  $p = 3, 5, 7$ ;  $b = \sigma_3 + a_p \in \{24, 124, 368\}$  (hooks the  $\nu_p$  law of v535).
- **(E)  $O(1)$  assembler [E].**  $\lambda_{\text{geom}}(p)$  and  $R(p) = a_p^2$  for all odd  $p \leq 31$  against the eta product  $f_8$  (Ramanujan bound included).
- **Fences.** Classical Eichler/Siegel/Witt named classical — the claim is in-suite two-sidedness (mod- $p$  geometry left, eta product right) and the closed compiler densities; no RH statement; Shell( $p^2$ ) for  $p \geq 7$  out of scope; weight-4  $\rightarrow \text{GL}(1)$  stays [O]; no marker moves of any pre-existing contract.
- **Surfaces.** Ledger HECKE.GEOM.EICHLER.01; \veri in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension mirrors (exact Witt/density/assembler/ $a_p$  identities); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

**2026-07-24 (XIII) (v535 — Hecke from geometry: Kneser correspondence + affine neighbour-sum + 2-adic oldform recovery on the  $E_8$  census channels; NO marker moves)**

- **New module v535 (HECKE.GEOM.01).** Consolidated promotion of discovery probes T27/T31/T32 into one load-bearing module (25 checks,  $\sim 11$  s). The claim is *not* RH-related: the Hecke structure of the  $\mu_4$ -glue census is lattice-native.
- **(A) Kneser carries Hecke [E].**  $\#\text{iso\_pts} = p^7 + p^4 - p^3$  and  $\#\text{iso\_lines} = \sigma_3(p) \#\mathbb{P}^3(\mathbb{F}_p)$  identically and by full enumeration at  $p = 2, 3, 5, 7$  (135/1120/19656/137600 lines) — the Eisenstein eigenvalue  $\sigma_3(p) = 1 + p^3$  is an exact factor of a lattice-native correspondence count.
- **(B) Neighbour-sum = affine Hecke element [E].**  $\nu_p = a \text{Id} + b T_p$  exactly (overdetermined, residual 0) with  $b = \sigma_3 + a_p$ ,  $a = \#\text{lines} - b\sigma_3$ ; cusp rule  $a_p = b - \sigma_3$  recovers  $a_3 = -4$ ,  $a_5 = -2$  ( $p = 3$  live orbit-reduced;  $p = 5$  T31-validated freeze of  $S[1]$  with  $\lambda_{\text{odd}} = 3784$ );  $p = 7$  consistency  $\lambda_{\text{odd}} = \#\text{lines} - \sigma_3^2 + a_p^2 = 19840$ ; placebo fails.
- **(C) Redundancy = 2-adic oldform structure [E].**  $\dim_{\mathbb{Q}} V = 7 = 5$  ( $E_4$ -copies, levels 1, 2, 4, 8, 16) +2 ( $f_8$ -copies, levels 8, 16); every  $\Theta_j$  an explicit  $\mathbb{Q}$ -linear combination; recovery  $\pi_{\text{cusp}} = (28 - T_3)/32$ ,  $\pi_{\text{Eis}} = (T_3 + 4)/32$ , newform via  $(1 - V_2 U_2)$ ; new quotients each dimension 1; level census purely 2-adic.
- **Fences.** Classical theorems (Kneser, Hecke, Atkin-Lehner, multiplicity one) named classical — the TFPT content is the in-suite mechanics. No RH statement. Weight-4 restriction named:  $\text{GL}(1)/\text{weight-drop transport}$  stays [O]. No marker moves.
- **Surfaces.** Cited in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; ledger row HECKE.GEOM.01; Wolfram extension +5 exact mirrors ( $588 \rightarrow 593$ ); website sync.

**2026-07-24 (I) (v534 — the straddle cone executed: THE FIRST DYNAMICAL SELECTION OF THE ALIGNMENT BIT — reflection positivity keeps exactly**

ONE member of the interacting mark family alive,  $\delta = \pi/2$  with positive coupling; the literal leading-order cone formalization dies; NO marker moves)

- **New module v534 (SEAM.STRADDLE.CONE.01).** The 2026-07-23 review-round contract (recorded “not promoted” in entry (X)) is *executed*, double-verdicted. **V2 = ERFOLG** — **the selection law (the central datum):** with independent couplings  $\lambda_q$  on the four mark quartets of  $M(\delta = m\pi/8)$  and the equivariant (uniform) direction, reflection positivity holds on *every* admissible cut over the whole grid  $g \in \{1/32..8\}$  for *exactly one* member  $\times$  sign:  $\delta = \pi/2$  ( $m = 4$ ) with *positive* coupling — all four Grams PD (37,0,0) *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ( $1.78 \cdot 10^{-6} \rightarrow 5.5 \cdot 10^{-5}$  near  $g = 1/4$ : the  $\pi/2$  interaction pushes *away* from the boundary), Gibbs  $\beta = 2$  identical [E]. The asymmetric members die:  $m = 2/6$  with  $s = +$  already at  $g = 1/32$ , with  $s = -$  from  $g = 1/8$ . The v529 straddle law *refines*: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it.
- **V1 = KILL (the literal cone formalization).** The leading-order soft-block cone  $K_{\text{RP}} = \text{Fix}(G_\delta) \cap \bigcap_c \{\lambda : G_c^{\text{lead}} \succeq 0\}$  is *empty* for every member and both signs (first-order PT via the exact reduced resolvent, Hermitian to  $10^{-15}$ , central-difference regression  $2 \cdot 10^{-8}$ ) — the linear pencil cannot carry the selection; the protection at  $\pi/2$  is *nonperturbative*. Recoverable leading-order content: the straddled-cut drift signs are exact ( $\pi/2$ :  $+1.4 \cdot 10^{-4}$  for  $s = +$ ,  $-3.7 \cdot 10^{-2}$  for  $s = -$ ; asymmetric members reversed) — necessary, not sufficient.
- **The exact geometry behind it [E].**  $\delta = \pi/2$  is the *unique symmetric-straddle member*: stabilizer = the full clock+mirror group (rotations  $\{0, 4, 8, 12\}$ , reflection axes  $\{3, 7, 11, 15\}$  = four admissible bond cuts, straddled  $\{7, 15\}$ ), the four quartets *partition* the 16-circle, and every straddling quartet is reflection-*closed* across its cut with exact sign  $+1$ ;  $m = 2/6$  straddle asymmetrically; odd  $m$  carry no bond axis (the typed v521/v525 placement artifact).  $\text{Fix}(G_\delta)$  is 1-dimensional throughout: uniform on  $m = 2..6$ , NS-wrap-twisted  $(1, -1, 1, 1)/(-1, -1, -1, 1)$  on the tight members  $m = 1/7$ . Typed cross-link: the RP-selected member  $m = 4$  is exactly the carrier of the v528 twist-class order parameter  $O = 1/2$  — the dynamical selector and the kinematic definition point at the *same* member.
- **Honest fences [C].** One toy, one interaction class (FK quartets), flat-band parent,  $\deg \leq 2$  OS bases, finite coupling grid — toy-level evidence *for* the bit’s dynamical origin, *not* a derivation; nothing about the continuum  $\mathcal{A}_{\text{hol}}$  is claimed; WOIT.OS.TWISTOR.01 stays Open, NO marker moves. Transparency: the run-1 preregistered verdict logic conflated formalization and substance (UNENTSCHIEDEN, mismatch typed); run 2 split V1/V2 with no tolerance change and no number changed — note kept verbatim in the probe. Discovery provenance: experiments/tfpt-discovery/seam\_straddle\_cone\_probe.py (15/15, 2026-07-24).
- **Wolfram mirror.** Four exact mirrors (stabilizer census,  $\pi/2$  partition + straddle incidence, reflection-closure signs, equivariant Fix rays incl. the wrap-twisted  $m = 1/7$  rays) in wolfram/tfpt\_readouts\_extension.wl, verified engine run 2026-07-24, 584  $\rightarrow$  588 (the 256-dim Fock spectra, PT corrections and finite- $g$  scans stay Python-only, flagged); suite 527  $\rightarrow$  528 modules.

2026-07-23 (X) (v530–v533 — the external review round promoted: center quotient compiler, Coxeter Stage A tensor clock, dual-degree modular checksum 744, sheet-transfer disc  $-7$  norm line; every identity machine-confirmed exactly before promotion (probe 76/76); NO marker moves)

- **New module v530 (DIAMOND.CENTER.QUOTIENT.01).** The Sheet-Diamond center  $C = M(1, 0)$  carries the fixed primitive line  $Cv = v$  ( $v = (-1, 1, 0)$ ), and the induced operator on  $\mathbb{Z}^3/\mathbb{Z}v$  is *exactly* the atom matrix  $A = \begin{pmatrix} 8 & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} \text{rank } E_8 & |\mathbb{Z}_2| \\ g_{\text{car}} & N_{\text{fam}} \end{pmatrix}$  [E] with  $\text{tr } A = 11$ ,  $\det A = 14$ ,  $\text{disc} = 65 = 5 \cdot 13$ ,  $\text{perm } A = 34 = p_5(a)$ ,  $\prod A_{ij} = 240$ ,  $\det(A - I) = 4$ . The center *self-*



*codes*: row sums  $C\mathbf{1} = (7, 11, 13) = (\det A/|\mathbb{Z}_2|, \text{tr } A, \text{disc}/g_{\text{car}})$ ; with the fixed eigenvalue 1 this is  $(1, 7, 11, 13) =$  the CP-halved  $E_8$  Coxeter exponents. The **v414** resolvent ladder factorises:  $\det(C+kI) = (k+1)\det(A+kI)$  identically, quotient ladder  $(14, 26, 40, 56)$ , differences  $(12, 14, 16)$ , endpoint  $224 = 248 - 24$ . *Honest correction of the review*: eigenvalue 1 is NOT unique to  $C$  ( $K, Q$  share it) — uniqueness holds for the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays **[O]**. New paragraph in **tfpt\_2** (Sheet Diamond section).

- **New module v531 (COX.STAGEA.TENSOR.01)**. The **tfpt\_1** two-stage Coxeter audit (Stage A: a natural order-30 operator with  $\chi = \Phi_{30}$  on the even-flip atlas *or its 8-dim primitive-character space*) is *closed on the character-space alternative*:  $T_{30} = \text{Comp}(\Phi_5) \otimes \text{Comp}(\Phi_6)$  has exact order 30 and  $\chi_{T_{30}} = \Phi_{30}$  **[E]** (CRT,  $\gcd(5, 6) = 1$ : carrier clock  $\times$  family clock = Coxeter clock as *one integer operator*); Ramanujan trace tomography  $\text{tr}(T_{30}^k) = c_{30}(k) = c_5(k)c_6(k)$  for all  $k$ , signed table  $(-1, 1, 2, 4, -2, -4, -8, 8)$ , atom readout  $(2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \text{rank } E_8)$ ; honest kill scope ( $\Phi_{15}$  passes the unsigned triple — the *signed* table discriminates;  $\Phi_{16}/\Phi_{20}/\Phi_{24}$  die on the triple). The flip-atlas intertwiner (COX.FLIP.INTERTWINER) and Stage B stay **[O]**; **tfpt\_1** Stage A prose updated.
- **New module v532 (E8.DEGREE.MODULAR.01, audit-only)**. The dual products of the  $E_8$  invariant degrees,  $(60, 192, 240, 252)$ , satisfy simultaneously  $\sum = 744 = 3 \cdot 248$  (= the  $j$ -function constant, recomputed from the  $q$ -series  $E_4^3/\Delta$ ),  $\prod = |W(E_8)|$ ,  $\gcd = 12$ ,  $\frac{1}{12}(\cdot) = (g_{\text{car}}, \dim S^+, \det L, 7N_{\text{fam}})$  **[E]**. *Mandatory fence*:  $D_{16}$  passes the same checksum ( $1488 = 3 \cdot 496 = 2 \cdot 744$  — heterotic, NOT  $E_8$ -exclusive); six non-heterotic controls fail. *Prime-totative fingerprint (audit)*:  $\{2, 3, 5\} \cup (\text{Exp}(E_8) \setminus \{1\}) =$  all primes below 30, and 30 is the *largest* integer whose totatives are all 1 or prime (scan to  $10^4$ ) —  $h_{E_8}$  as the maximal prime-totative Coxeter number; per the no-free-pattern rule nothing here is load-bearing. New keybox in **tfpt\_3**, cross-note in the **tfpt\_1** 496 box.
- **New module v533 (FTR.DISC7.NORM.01)**. The branch-preserving transfer path  $J, K, C, F = M(1, -2..1)$  is a *norm line*:  $\det M(1, t) = 4t^2 + 14t + 14 = ((4t+7)^2 + 7)/4 = N(\alpha_t)$  with  $\alpha_t = (4t+7+\sqrt{-7})/2$  integral in  $\mathbb{Z}[(1+\sqrt{-7})/2]$ , the path one translation  $\alpha_{t+1} = \alpha_t + 2$ , norms  $(2, 4, 14, 32)$ , constant discriminant  $-7$ , vertex  $7/4 > 0$  **[E]**; 2 splits in  $\mathbb{Q}(\sqrt{-7})$  ( $-7 \equiv 1 \pmod{8}$ , class number 1); controls: winding axis linear, anchor block disc +105. **[C]/[O]** preregistered next test: do the four external transfer solvers share *one* fractional-linear action on  $\alpha_t$  (kill: four incompatible actions)? GATE.FTRANSFER unchanged. New paragraph in **tfpt\_2** + frame note in **tfpt\_research\_contracts**.
- **Wolfram mirror**. All four modules mirrored exactly in **wolfram/tfpt\_readouts\_extension.wl** (20 new checks, verified engine run 2026-07-23,  $564 \rightarrow 584$ ); suite  $523 \rightarrow 527$  modules.
- **Not promoted (honest fences)**. The review's remaining proposals stay research contracts, not verifications: the straddle-cone selector on the interacting FK quartets (the highest-leverage attack on the alignment bit after **v529**) and the metrology-torsor formalisation of  $v_{\text{geo}}$  are recorded as next steps in **experiments/next.txt**; the primes-below-30 identity stays an audit fingerprint exactly as the no-free-pattern rule demands.

**2026-07-23 (IX) (website entry redesign: narrative home, honesty band, proof strip, guided-tour route — editorial only, NO new modules, NO marker moves)**

- **Website home**. Slim one-viewport hero (proof strip: 523 machine-checked modules · Wolfram mirror · Lean carrier · 0 external replications yet; copyable **python run\_all.py** → ALL CHECKS PASSED; CTA trio film / browser check / kill board). Narrative order: Hero → IntroVideo (**#intro-video** kept) → In one breath → WhyThisMatters (plain-English column) → Trust-Contract → ClaimStack → HonestyBand (honest negatives: RXTE/NICER nulls, 2DCS/v529, ten bit tests) → Safeguards (four core stats) → Overview (TheoryUnpacking/ThreeDecoderMap behind progressive disclosure) → PathChooser → Downloads teaser.

- **Archive routes.** Full PapersSection  $\rightarrow$  /papers; PredictionsSection  $\rightarrow$  /predictions; ReconstructionChain + HorizonStory + Gravity  $\rightarrow$  /architecture; RosettaLexicon  $\rightarrow$  /orientation; OpenGates  $\rightarrow$  /verification; EcosystemSection off home (footer adjacent link). Legacy hash redirects on home for old anchors.
- **Nav / design.** Primary nav: Start · Tour · Verification · Kill board · Replication · Papers · FAQ (More dropdown for secondary). SuiteBadge (static per release from `suite.ts/version.ts`). Design pass: drop purple-gradient pills/glow in hero and accent tokens; lab-notebook proof strip as above-the-fold visual.

**2026-07-23 (VIII)** (prominence pass — the v519–v529 breakthroughs lifted into the FRAME texts of the papers and the website data layer; purely editorial, NO new modules, NO marker moves — (1) `tfpt_5_redteam` gains the seam round Target G: the ten-test side-blind scoreboard on the alignment bit (v512/v521/v525/v529) and the first *firing* kill test of the programme — the interacting straddle law (v529), documented as a named threat AND the first hard selection principle for  $\mathcal{A}_{\text{hol}}$ , fenced at toy level; (2) the bit-reduction statements in `tfpt_1`, `tfpt_2` and `origin_theory` extended from the v512 state to the current state (ten side-blind tests, the twist-class definition with gauge-robust order parameter, “in principle interferometrically readable”, v528 — the bit stays formal input); (3) the thermal seam lifted into the `origin_theory` horizon chapter (new keybox:  $T_{\text{seam}} = 4c_3$  measured, temperature the third leg of  $c_3$ , v526) and into the `tfpt_horizon_readouts` head keybox; (4) `tfpt_3` title/keybox now carry the thirty-step story and “the measure chain is derived” (v520/v523) prominently; (5) `tfpt_research_contracts` title/intro keybox now show the executed stages of CELEST.SEAM.01 (M1–M3,  $\delta$ -chains decided,  $w_m$  derived) and WOIT.OS.TWISTOR.01 ( $\alpha/\beta_1/\beta_2$  executed,  $\beta_3$  next under the straddle constraint); both contracts stay Open)

- **Papers.** `tfpt_5_redteam` (Target G + method/outcome/title honesty pass); `tfpt_1`, `tfpt_2` and `origin_theory` (bit-reduction statements  $\rightarrow$  ten tests + twist-class definition); `origin_theory` (thermal-seam keybox, horizon chapter); `tfpt_horizon_readouts` (head keybox third leg); `tfpt_3` (title + keybox: thirty-step story, measure chain derived); `tfpt_research_contracts` (title + intro keybox: executed contract stages).
- **Website.** `papers.ts` (redteam Target G, horizon third leg, audit/contracts abstracts, architecture reduction state); `orientation/StatusMatrix.tsx` (v512  $\rightarrow$  current state); `glossary.ts` (new terms: twist-class choice, thermal seam, straddle law, side-blind scoreboard); new visual components `ThermalSeamLegs` (three legs of  $c_3$ : geometry v58 + anomaly/modular v208 + temperature v526) and `WoitBridgeProgress` ( $\alpha/\beta_1/\beta_2$  done,  $\beta_3$  open, straddle warning) on the home and verification pages.
- **Counters.** Predictions counter made consistent with `predictions.ts` (27 falsifiable test surfaces): `README.md` badge/hero/prose and the intro-film description (23  $\rightarrow$  27; the film narration already says twenty-seven).
- **Honesty.** No marker moves anywhere; SEAM.EQUIV.01 and WOIT.OS.TWISTOR.01 stay Open; the bit stays input; all fences (“derived at probe level”, “at the free level”, “toy model”) preserved verbatim.

**2026-07-23 (VII)** (v529 — the interacting Fidkowski–Kitaev seam toy promoted as the ledger row SEAM.INT.FKTOY.01: the FIRST firing Kill-Test-2 shadow — triple verdict  $F1 = \text{KILL}$  /  $F2 = \text{KILL}$  /  $F3 = \text{ERFOLG}$ ; the celestial/twistor story in `tfpt_3` gains Step 30 and the header goes twenty-six  $\rightarrow$  thirty steps — (1)  $F1 =$

**KILL [E]:**  $\Theta$  exists exactly on the interacting model ( $U_r^2 = 2^8$ ,  $\Theta^2 = +1$ , clock-tower inversion,  $[\Theta, H_g] = 0$ ) so **Kill-Test 1** does *not* fire; but **Kill-Test 2** *fires at toy level*: RP breaks for every  $g > 0$  — inertia ladder  $(37, 0, 0) \rightarrow (33, 4, 0) \rightarrow (31, 6, 0) \rightarrow (29, 8, 0)$ ; Gibbs  $\beta = 2$  same ladder; mechanism = interference; **STRADDLE LAW** (24/24); [C] fence one toy / one interaction class / flat-band parent — narrows the Woit route, does *not* close it; **WOIT.OS.TWISTOR.01** stays Open with the toy shadow documented as a named **Kill-Test-2** threat; (2) **F2 = KILL [E]**: the **TENTH** side-blind test ( $9 \rightarrow 10$ ) —  $\Theta_{15}$  maps  $H(m) \rightarrow H(8-m)$  **EXACTLY** despite  $\delta$ -seeing spectra; (3) **F3 = ERFOLG [E]**: grading +  $\mathbb{Z}_8$  tower + **FK** anchor + constructive OS quotient at  $\pi/2$ ; Wolfram extension  $552 \rightarrow 564$  (verified engine run); suite  $519 \rightarrow 523$ ; **NO marker moves**)

- **Suite.** New module `v529_seam_interacting_toy_fk.py` (24 checks,  $\sim 97$  s); registered in `run_all.py` + `script_registry.csv`; suite 523 modules green.
- **Ledger.** New row **SEAM.INT.FKTOY.01**; **WOIT.OS.TWISTOR.01** text-precised (Kill-Test-2 toy shadow + straddle law); scoreboard rows text-precised ( $9 \rightarrow 10$ ); no status moves.
- **Papers.** `tfpt_3` Step 30; `tfpt_research_contracts` (Kill-Test-2 toy threat + bit scoreboard 10); introduction (tenth side-blind test).
- **Wolfram.** Three exact mirrors ( $\mathbb{Z}_8$  identities; straddle combinatorics; FK binomial ladder).
- **Website.** `papers.ts` (story 30 steps + scoreboard 10), generated indices via `build`; `README.md`, `zenodo_description.html` (523 modules, highest ID v529, Wolfram 564/564), `next.txt` notes.

**2026-07-23 (VI) (v528 — the alignment bit as an exact measurable TWIST-CLASS choice promoted as the ledger row SEAM.BIT.TWISTCLASS.01: the physical definition — verdict SUCCESS; the celestial/twistor story in tfpt\_3 gains Step 29 — (1) Facets #14 ( $\eta$ -flip  $\Leftrightarrow \delta = \pi/2$ ) and #15 (twist-frustration  $\Leftrightarrow \delta = \pi/2$ ) both directions exact [E] (cardinality lemma  $|A_+| = |A_-| \Rightarrow m = 4$ ;  $g$ -class census 224 axes: flip class exactly at  $\{15, 31\}$ ); (2) order parameter  $O = 1/2$  at  $m = 4$  else 0 [E], gauge-robust; (3) unification #14=#15=flag-transitivity = ONE  $\mathbb{Z}_2$  invariant [E] (v512 web 13  $\rightarrow$  15); Kramers = POSITION half (derived), twist class = MODULUS half (the input); (4) [C] measurement sketch (interferometric in principle); the bit remains formal input; **NO marker moves**)**

- **Suite.** New module `v528_seam_bit_twist_class_definition.py` (18 checks,  $\sim 46$  s).
- **Ledger.** New row **SEAM.BIT.TWISTCLASS.01**; **SEAM.TAU.FLAG.01** text-precised (web 13  $\rightarrow$  15); no status moves.
- **Papers.** `tfpt_3` Step 29; contracts + introduction (“symmetry-lift bit” now also “the twist-class choice, in principle interferometrically readable”).
- **Wolfram.** Three exact mirrors (cardinality lemma; 224-axis census;  $O$ -values).

**2026-07-23 (V) (v527 — the silver clock eigenvalue EXPLAINED and DEMYSTIFIED, promoted as the ledger row SEAM.CLOCK.SILVER.01: double verdict STRUCTURE = SUCCESS / CONNECTION = KILL (anti-numerology style); the celestial/twistor story in tfpt\_3 gains Step 28 — (1) STRUCTURE SUCCESS [E]:  $\{1, \sqrt{2} - 1\}$  closed via diagonal Hankel pencil at  $N = 8$ ,  $\tan(\pi/8) = C(3)/C(1)$ , doubling = exact second quantisation, carrier = silver-midpoint axes, gap =  $\text{arcsinh}(1) = \text{gd}^{-1}(\pi/4)$ ;  $N$ -scaling = pure  $N = 8/16$  lattice datum (not continuum); positive clock only for  $N \equiv 0 \pmod{8}$ ); (2) CONNECTION KILL [E]: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null;**

placebos comparable; (3) cross-link v526:  $\sqrt{2} - 1 =$  thermal kernel ratio; *NO marker moves*)

- **Suite.** New module v527\_seam\_clock\_silver\_spectrum.py (22 checks,  $\sim 10$  s).
- **Ledger.** New row SEAM.CLOCK.SILVER.01; no status moves.
- **Papers.** tfpt\_3 Step 28; contracts thermal/silver follow-on.
- **Wolfram.** Three exact mirrors ( $\tan(\pi/8)$  pencil;  $N$ -scaling witness; gap =  $\operatorname{arcsinh}(1)$ ).

**2026-07-23 (IV) (v526 — the thermal seam promoted as the ledger row SEAM.THERMAL.KMS.01: KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain — the THIRD LEG of  $c_3$ ; verdict SUCCESS; the celestial/twistor story in tfpt\_3 gains Step 27 — (1)  $\beta$  MEASURED =  $N$  at both levels [E] ( $N = 8$ :  $\omega = \operatorname{arcsinh}(2^{-3/4})$ ,  $\kappa = q^{-8}$ ;  $N = 16$ : pairing exponent  $(N/2) + 1 = 9$ ; wrong  $\beta$  excluded); (2)  $\beta_{\text{angle}} = 2\pi$  EXACT =  $1/(4c_3)$ ,  $T_{\text{seam}} = 4c_3 =$  Bisognano–Wichmann/Hawking normalisation (v208) [E]; (3) modular Hamiltonian closed [E] ( $\nu_1 = \cos(\pi/24)$ ,  $\nu_2 = \cos(7\pi/24)$ ; Tomita KMS- $\beta = 1$  on all 256 pairs); (4)  $c_3$  third-leg class-1 selection [E]/[C] (Hawking/Nariai chain; v104 cross-link); (5) entropy  $S(8/16/32) = 0.52187/0.63510/0.75006$ ,  $\Delta S/\ln 2 \rightarrow 1/6$ , horizon fractions do NOT transfer [E]; (6) [C]-5 silver =  $C(3)/C(1)$  thermal kernel ratio; non-compact  $\Rightarrow T = 0$ ; *NO marker moves*)**

- **Suite.** New module v526\_seam\_thermal\_kms\_nariai\_bridge.py (22 checks,  $\sim 8.4$  s).
- **Ledger.** New row SEAM.THERMAL.KMS.01; no status moves.
- **Papers.** tfpt\_3 Step 27; tfpt\_horizon\_readouts (third leg of  $c_3$ ); tfpt\_research\_contracts (thermal follow-on); introduction.
- **Wolfram.** Three exact mirrors (Q-polynomial/ $\beta$  measurement;  $2\pi = 1/(4c_3)$  chain;  $\nu$  char-poly).

**2026-07-23 (III) (v525 — the twist-state attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.TWISTBLIND.01: “mark-decorated states do not see  $\delta$ ” — an honest, decided NEGATIVE result (v508 style), the NINTH side-blind test on the v512 scoreboard ( $8 \rightarrow 9$ ); the free-plus-twist class (everything Wick-computable) is EXHAUSTED; the celestial/twistor story in tfpt\_3 gains Step 26 and the header goes twenty-three  $\rightarrow$  twenty-six steps — (1) ALL LIFTINGS WELL-DEFINED AND BLIND [E]: (a) the  $\sigma$ -gauge arc decoration ( $\mathbb{Z}_2$ ) satisfies  $\sigma \circ r = \sigma$  on the swap axes  $\Rightarrow$  the twisted Gram is a diagonal unitary congruence  $DMD$ , spectra IDENTICAL to untwisted (dev 0.0); (b) the Kadanoff–Ceva 4-twist insertion  $\omega(D \cdot)/\omega(D)$  is well-defined on the  $N = 32$  ladder ( $\omega(D)$  real nonzero; the  $N = 16$  odd- $m$  degeneration is the  $C(\text{even}) = 0$  checkerboard artifact) and its RP SPECTRA are the FIRST free-class data of the programme that depend on  $\delta$  (pairwise up to 1.05) — yet the INERTIA is  $(16, 0, 0)$  PD with the same  $\eta = +i$  for EVERY  $m = 1..7$ : the decoration sees  $\delta$ , positivity does not; (c) the  $\mu_4$  twist at  $\beta = \pi/2$  is pure plane gauge; (c') the genuine Bogoliubov defect at  $\beta = \pi/4$ :  $\Theta$ -compatibility selects exactly 2 of 16 sign patterns (NS-wrap-forced), the winner genuinely non-monomial — still PD, spectra identical; STRUCTURE THEOREM (mechanism exact): the defect planes never straddle a cut bond, so every  $\Theta$ -compatible quasi-free mark decoration is RP-spectrally INVISIBLE; (2) SUB-RESULTS [E]: the defect energy  $E_{\text{def}} = 1.2752872$  for EVERY  $m$  (spread  $4 \times 10^{-40}$ ) —  $\delta$ -blind; the 4-twist Casimir  $\ln |\omega(D)|$  measures  $\delta$  ( $-0.890.. - 1.348$ ) but**

exactly mirror-symmetric under  $m \leftrightarrow 8-m$  — a family gauge, never a side selector; Kramers  $\langle \Gamma \rangle_{\text{tw}} = +1$  for every  $m$ ; (3) the HARVEST [E]: two bit-presupposing  $\pi/2$  structures (the  $\eta$  flip  $+i \rightarrow -i$  on the mark-fixing axes via  $\sigma \circ r = -\sigma$ ; the twist frustration of the mark-fixing mirror,  $(4, 4, 0)$  indefinite for both  $\eta$  — the lattice shadow of the Ising twist field’s mirror-oddness) — both formulable only on the fixing axes existing iff  $\delta = \pi/2$  (the v512 facet class); (4) NEGATIVE CONTROLS [E]: the twist-free limit = v519/v521 exactly; the site cut keeps failing; the v512 counterwitness is arc-equivalent at  $N = 16$  to the blind member  $\pi/4$  — no state selector can exclude it; a single twist has  $\omega(D) = 0$  identically; [O]: the door narrows to a genuinely INTERACTING  $\mathcal{A}_{\text{hol}}$  whose OS data are NOT Pfaffian-reducible to the chiral vacuum; Wolfram extension (with v523/v524)  $540 \rightarrow 552$  (verified engine run 2026-07-23, 552/552 + base 116/116); suite 516  $\rightarrow$  519 modules; *NO marker moves — the alignment bit remains genuine discrete input*)

- **Suite.** New module v525\_seam\_bit\_twist\_blind.py (20 checks,  $\sim 80$  s; Pfaffian-exact insertion machinery cross-validated three ways, 40-digit certificates); registered in run\_all.py + script\_registry.csv; suite 519 modules green.
- **Ledger.** New row SEAM.BIT.TWISTBLIND.01 (KILL, honest negative, preregistered); SEAM.BIT.RPBLIND.01 and SEAM.TAU.FLAG.01 text-precised (scoreboard  $8 \rightarrow 9$ , “the free-plus-twist class is exhausted”); no status moves.
- **Papers.** tfpt\_3 Step 26 + the story header (twenty-six steps) + the Step-22 gap sentence; tfpt\_research\_contracts (the QGEO.REALIZE.01 reduction paragraph  $8 \rightarrow 9$  and the WP2 relocation note extended by v525); introduction (the ninth side-blind test added to the reduction-state sentence).
- **Wolfram.** Three exact mirrors (the  $N = 32$  integer frame + the  $\sigma \circ r = \pm\sigma$  congruence mechanism; the  $\pi/2$  existence solveset + the  $(3 + 4i)/5$  root-of-unity census; the  $\delta \leftrightarrow \pi - \delta$  mirror equivariance); the Pfaffian spectra, the 2-of-16 pattern census and  $E_{\text{def}}$  stay Python-only.
- **Website.** papers.ts (tfpt\_3 story mirror 26 steps + scoreboard 9), generated indices (ScriptIndex.tsx, changelog.ts, maps) via build; README.md, zenodo\_description.html (519 modules, highest ID v525, Wolfram 552/552), next.txt notes.

2026-07-23 (II) (v524 — the WOIT- $\beta_2$  milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.BETA2.OS.01: “the OS quotient of the free system made explicit” — verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the celestial/twistor story in tfpt\_3 gains Step 25 — (1)  $\mathcal{H}_{\text{phys}}$  EXPLICIT [E]:  $N = 16$  ( $\deg \leq 2$ ): the  $37 \times 37$  bond-cut OS Gram exactly Hermitian, parity-block-diagonal, inertia  $(37, 0, 0)$  PD (min eigenvalue  $1.7801 \times 10^{-6}$  at 40 digits), null space  $\{0\}$ ,  $\dim 37 = 29 \oplus 8$ ;  $N = 8$  (complete half algebra):  $(16, 0, 0)$  PD (min  $3.3471 \times 10^{-3}$ ),  $\dim 16 = 8 \oplus 8 = 4^2$  — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate  $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$ ; (2) the TRANSFER SEMI-GROUP (Klein–Landau) [E]:  $\tau_k$  exactly Hermitian on all domains ( $k = 1..7$ , dims 29, 22, 16, 11, 7, 4, 2), chain identities exact, vacuum fixed; the positivity pattern = the site/bond dichotomy: even  $k$  PSD via the exact square identity  $T(2j) = A_j^* A_j$  ( $((22, 0, 0)/(11, 0, 0)/(4, 0, 0))$ , odd  $k$  indefinite  $((10, 12, 7)/(5, 6, 5)/(2, 2, 3))$ ) — the one-step transfer is NOT positive (the chirality datum, kill-3 shadow sharpened), its even powers are; (3) THE CLOCK =  $T^{N/4}$  [E]:  $N = 16$ :  $\tau_4$  PD  $(11, 0, 0)$  (min  $6.6821 \times 10^{-5}$ ), compressed spectrum  $[0.0194517, 1.64144]$ , vacuum eigenvalue exactly 1, spectral projections  $\sim 10^{-40}$ ;  $N = 8$ : spectrum EXACTLY

$\{1, \sqrt{2} - 1\} = \{1, 1/\delta_{\text{Silber}}\}$  in both parity sectors — the silver axes of the **v519** torsor return as the clock eigenvalue; the reconstructed rotation group  $U(s) = e^{isH}$  unitary with the group law; (4) the **v522 RESOLUTION [E]**: the census (745, 96, 0) reproduced; **EVERY** anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is **EXACTLY** the domain/wrap artifact; on the quotient the clock has spectral calculus; pre-registered KMS deviation: no contraction on the compact circle ( $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$  exactly,  $\det(G - \tau_4) < 0$ ); (5) **GSO/ $\Theta$  [E]**: the grading survives; the perpendicular torsor mirror descends anti-unitarily ( $\eta_\perp \in \{\pm i\}$ ),  $\Theta_{\text{phys}}^2 = +1$  on every sector (Kramers-free),  $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$  exactly; (6) **CONTROLS [E]**: site cut (3, 3, 1)/(2, 2, 4) — the kill branch fires there; family **A** (4, 4, 0); anti-chiral (8, 8, 0) no quotient; wrong direction non-Hermitian (all mismatches wrap-overlap); **KILL TESTS**: (1)/(2) strengthened, (3) shadow sharpened, none fires, all seven stay live on  $\mathcal{A}_{\text{hol}}$ ; **TRANSPARENCY**: run 1 = 18/20 (a sympy simplify failure on a **TRUE**  $\pi/16$  product formula, fixed by a Laurent-polynomial certificate; no criterion changed); **NO marker moves** — *WOIT.OS.TWISTOR.01 stays Open,  $\beta_3$  is next*)

- **Suite.** New module `v524_woit_beta2_os_quotient.py` (20 checks,  $\sim 5$  s; exact sympy + Pfaffian-Wick + 40-digit inertia certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row `WOIT.BETA2.OS.01` (SUCCESS, [C]-typed per precision (iii)); `WOIT.OS.TWISTOR.01` claim/canonical status extended ( $\beta_2$  executed,  $\beta_3$  next); no status moves.
- **Papers.** `tfpt_3`: Step 25 plus the Woit-bridge paragraph with the  $\beta_2$ -stage status. `tfpt_research_contracts`: a full “ $\beta_2$  stage — executed” block with `\veri`; the  $\beta/\gamma$  roadmap entry marked executed; the scope-fence status sentence extended.
- **Wolfram.** Four exact mirrors (the  $\alpha_1^N = (-1)^F$  wrap arithmetic + the certificate identity; the Klein–Landau one-particle transfer forms with odd- $k$  zero diagonal and the even- $k$  square identity; the exact  $N = 8$  clock spectrum  $\{1, \sqrt{2} - 1\}$  + the transparency  $\pi/16$  product formula certified symbolically; the KMS silver witnesses); the  $\mathcal{H}_{\text{phys}}$  Grams, inertia spectra and the rotation group stay Python-only.
- **Website.** `papers.ts` (`tfpt_3` + contracts mirrors), generated indices via `build`.

**2026-07-23 (I)** (**v523** — the constructive  $w_m$  derivation executed and promoted as the ledger row `CELEST.WP5E.WM.01`: “ $1/\det_j$  is the Atiyah–Bott / zeta-determinant fixed-point factor, **COMPUTED** from three independent sources” — verdict **ERFOLG** per the frozen preregistration; the residual [O] of **v516/v520** is closed at the fibre-zero-mode level and narrows to the **GLOBAL BCOV** integral on  $\mathbb{P}T/\mathbb{Z}_4$  beyond the fibre zero-mode factor; the celestial/twistor story in `tfpt_3` gains Step 24 — (1) **ATIYAH–BOTT (route i) [E]**: the equivariant mode ledger of the fibre  $\mathbb{C}^2$  equals the closed form  $1/((1 - q i^j)(1 - q i^{-j}))$  exactly in  $\mathbb{Q}(i)$  for all  $n \leq 120$ ; **REGULAR** at  $q = 1$  with Abel value  $(1/2, 1/4, 1/2) = 1/\det_j$ ,  $\det_j = \det(1 - g^j) = (2, 4, 2)$ ; ( $C, 2$ ) Cesàro convergence in exact arithmetic; fixed-point splitting at **EVERY** truncation level  $d \in \{7, 12, 25\}$  for the phased (skeleton) **AND** the **v520**-forced unphased (completion) trace — the Abel-limit  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  is the **v516** contact vector **COMPUTED**, not declared; (2) **ZETA/QUILLEN (routes ii/iii) [E]**:  $\det_\zeta$  of the  $g^j$ -twisted circle Laplacian  $= 4 \sin^2(\pi j/4) = \det_j$  **EXACTLY** (Lerch + reflection formula symbolically, Hurwitz certificates  $\sim 10^{-41}$ ); Quillen split  $\det \Delta = \det_j^2$ ,  $|\det \bar{\partial}| = \det_j$ , the real positive holomorphic section **UNIQUE** via the  $SU(2)$  conjugation pairing; the  $\delta_{1f} f_1 f_3$  block at the  $(0, b)$  nodes carries the exact constant term  $1/\det_b$  (deviations  $\leq 5.2 \times 10^{-28}$ ); (3) the **CONSISTENCY**

**CHAIN [E]:**  $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$ ;  $T_5 = 0$  at every scale;  $h_2 : h_1 = 4 : 3$ ; the  $T_3$  budget  $32 - 8c$  forces  $c = 4 = |\mu_4|$  uniquely (leftovers  $24/16/8/0/-8$ ); per-sector Okubo squares  $(18, 9, 18)\langle x, x \rangle^2$ ; total  $45\langle x, x \rangle^2$ ;  $\Phi_{T_3} : 32 \rightarrow 0$ ,  $\Phi_P : 72 \rightarrow 0$ ;  $\psi(A_{\text{fix}}) = +64$ ,  $\psi(\text{contact}) = -64$ ; **(4) NEGATIVE CONTROLS [E]:**  $1/\det^2$  gives  $w' = (0, 5/8, 1, 5/8)$ , leftovers  $(2, 12)$ , Quillen contradiction;  $1/|1-i^j|$  gives  $w''_1 = 1 + \sqrt{2} \notin \mathbb{Q}$ ; the  $\mathbb{Z}_2/\text{Eguchi-Hanson}$  anchor **PASSES** with the same derivation ( $\det = 4$ ,  $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ,  $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $c = 2 = |\mathbb{Z}_2|$ ); the  $SO(16)/D_8$  **KILL** fires;  $\text{diag}(i, i)$  breaks the  $\text{zeta} = \text{AB}$  identity itself;  $k = 3, 5$ :  $w_m = m(k - m)/2 = k h_m$  — the weights wander correctly, only  $k = 4$  carries the  $E_8$  chain; **TYPED PREMISES [C]** (the mandatory fence): **TP-REG** (Abel/Cesàro/zeta regularisation = the definition of the equivariant character), **TP-Q** (Quillen structure of BCOV  $F_1 = \text{v520 TP-4}$ ), **TP-NUM** (unphased numerator = **v520 ERFOLG-A**), **TP-CH** (insertion channel  $(0, j)$ ); **[O]:** the global BCOV integral beyond the fibre zero-mode factor, the resolved-ALE route not built, **SEAM.EQUIV.TWISTOR.01** stays Open; *NO marker moves*)

- **Suite.** New module `v523_celestial_wp5e_wm_derived.py` (26 checks,  $\sim 1$  s; exact  $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{2})/\text{cyclotomic arithmetic} + 40\text{-digit Hurwitz/block certificates}$ ); registered in `run_all.py + script_registry.csv`.
- **Ledger.** New row **CELEST.WP5E.WM.01** (ERFOLG); the **v516** row **CELEST.WP5E.M2.01** and the **v520** row **CELEST.WP5E.MEASURE.01** text-precised (“the  $w_m$  normalisation is now DERIVED at probe level; the residual **[O]** narrows to the global BCOV integral beyond the fibre zero-mode factor”); no status moves.
- **Papers.** `tfpt_3` Step 24 + the “What remains, honestly” narrowing; `tfpt_research_contracts` (the WP5e measure-question block extended by the executed derivation with `\veri`, the M2 fence, the celestial summary blocks and the section status list); **SEAM.EQUIV.TWISTOR.01** stays Open.
- **Wolfram.** Five exact mirrors (the mode ledger = closed form with Abel value; the zeta/Quillen route with the symbolic reflection formula and the  $\text{diag}(i, i)$  control; the derived-weight contact vectors + success table +  $\mathbb{Z}_2/\text{EH}$  anchor on the 240 glue roots; the consistency chain + wrong-weight controls; the  $k = 3, 5$  cyclotomic wandering controls); the Cesàro/Hurwitz/stripped-block numerics stay Python-only.
- **Website.** `papers.ts` (`tfpt_3` + contracts mirrors), generated indices via `build`.

**2026-07-22 (XIII)** (HFQPO scorecard extension — the NICER/MAXI J1820+070 third-tooth null and the AstroSat/LAXPC block integrated as a new evidence-scorecard row (search target, never a claim): the preregistered single-QPO extension of the H3 ladder discriminator (frozen `hypotheses/hfqpo_ext_v1.yaml` BEFORE any photon download) ran on 6 NICER ObsIDs (13.22 GB,  $2013 \times 16$  s = 32.2 ks stack at 19.05 kc/s) — sanity gate PASS (the published 55.12 Hz anchor reproduced at 55.03 Hz, 0.94% rms,  $3.8\sigma$ ), injection calibration PASS ( $\geq 90\%$  recovery at 0.93% rms), the geometric tooth at 82.68 Hz NOT detected ( $0.69\sigma$  single-trial;  $3\sigma$  upper limit 0.75% rms BELOW the detected anchor strength  $\Rightarrow$  `null_with_sensitivity`), the integer line at  $\sim 110.6$  Hz at  $3.82\sigma$  after trials — BELOW the preregistered  $4\sigma$  threshold, documented as a sub-threshold excess consistent with the weak second harmonic of ATel #11951, NOT a hit; AstroSat/LAXPC (GRS 1915+105, scientifically the strongest channel) `infrastructure_blocked` (ISSDC login-gated); the RXTE + NICER nulls now cover five sources on two instruments — the tooth channel is recommended DORMANT; anti-double-counting: the parent RXTE row and the new extension row

share `independence_group hfqpo_ladder_tooth` (one hypothesis, two instruments, never two independent nulls); scorecard regenerated (121 rows), `predictions.txt` HFQPO row extended (“Kanal ruhend”); experiment-only integration — no verification claim, no marker moves)

- **Scorecard.** New row “BH HFQPO third-tooth extension on non-RXTE archives” (NICER/MAXI J1820+070 plus AstroSat/LAXPC; `hfqpo-ladder` extension) in the generator `experiments/build_evidence_scorecard.py` (status null, stage `search_target`, signature `S1e`); `OVERRIDES` give both HFQPO rows the shared `independence_group hfqpo_ladder_tooth`; generator re-run: `evidence_scorecard.json` 121 rows, README stats block refreshed.
- **Experiment surfaces.** `experiments/hfqpo-ladder/extension-nicer-laxpc/` (preregistration, frozen manifest, download log, per-ObsID caches, stack/sanity/injection/scan reports, `results_ext.json`, README) — executed 2026-07-22; `predictions.txt` HFQPO row updated with the NICER finding and the dormant-channel recommendation; `next.txt` round notes.

**2026-07-22 (XII)** (v522 — the  $\text{WOIT-}\beta_1$  milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed and promoted as the ledger row `WOIT.BETA1.GSO.01`: “the clock is time-like, GSO is the gauge datum” — a TYPED result (verdict UNDECIDED per the frozen preregistration, the mechanism identified exactly, gauge-fixed RP established under the corrected typing; the celestial/twistor story in `tfpt_3` gains Step 23) — (1) the HERMITICITY WITNESS [E]: the  $\mu_4$  clock average violates Hermiticity EXACTLY (the one-step insertion  $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$  has entry  $-i/(8 \sin(5\pi/16))$  where the conjugate transpose demands  $+i/(8 \sin(5\pi/16))$ ; entrywise census 745 Hermitian-matching, 96 ANTI-matching, 0 violations — the pairing is complex-SYMMETRIC, strictly weaker than Hermitian: “positivity after gauge fixing” is not well-posed for the clock reading); (2) TIME-LIKE TYPING [E]: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ( $RSR^{-1} = S^{-1}$ ), NONE commutes — the  $\mu_4$  clock IS Woit’s euclidean rotation; the gaugeable ( $\theta$ -compatible) part of the  $\mathbb{Z}_8$  Fock tower is EXACTLY its 2-torsion  $\{1, (-1)^F\}$  = the GSO/fermion-parity  $\mathbb{Z}_2$  (Hermiticity census over all powers;  $\alpha^4$  = the full wrap = the  $2\pi$  rotation of the seam = fermion parity); (3) the POSITIVE PART [E]: GSO-fixed RP holds EXACTLY —  $N = 16$  even  $\deg \leq 2$  parity-projected bond-cut Gram  $(29, 0, 0)$  PD (min eigenvalue  $1.78 \times 10^{-6}$  at 40 digits),  $N = 8$  complete half algebra: all odd rows/columns killed exactly, even block  $(8, 0, 0)$  PD ( $3.35 \times 10^{-3}$ ), no degree truncation; the site-cut defect SURVIVES gauge fixing  $((7, 9, 6)$  — the bond placement is gauge-invariant information); family A keeps failing  $((17, 12, 0)$  indefinite vs family D’s  $(29, 0, 0)$ ); Kramers:  $\Theta_t^2 = \text{id}$  on  $\mathcal{A}^G$ ,  $\Theta_{\text{Fock}}$  restricts with  $\Theta^2 = +1$ ; Ramond control: the wrap  $+1$  “true  $\mathbb{Z}_4$ ” lift fails state invariance — the NS/ $\mathbb{Z}_8$  tower is forced; (4) KILL-TEST BOOKKEEPING: kill test (2)’s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction), kill test (1) stays discharged also gauge-invariantly; both stay formally live on  $\mathcal{A}_{\text{hol}}$ ; (5) TRANSPARENCY: the first frozen run scored 6/13 (the clock-reading checks failed exactly as the mechanism predicts) and the preregistered  $\deg\text{-}2$  invariant dimension 30 was a combinatorial error (correct census  $28 + 4 = 32$ ) — documented in full, the verdict by the frozen [C-3] logic (Hermiticity failure  $\Rightarrow$  UNDECIDED); CONTRACT PRECISION (iii) (prose, no marker): “gauge-invariant subalgebra” at the free level means the GSO/fermion-parity  $\mathbb{Z}_2$  — the only  $\theta$ -compatible part of the  $\mu_4$  tower; the clock is the euclidean rotation itself



(timelike), its equivariant statement moves to  $\beta_2$  (OS quotient first, then the clock as reconstructed transfer operator); Wolfram extension (with v520/v521) 522  $\rightarrow$  540 (verified engine run); *NO marker moves* — *WOIT.OS.TWISTOR.01 stays Open*,  $\beta_1$  *proper re-routed via  $\beta_2$* )

- **v522\_woit\_beta1\_gso\_gauge.py** (WOIT.BETA1.GS0.01, 16 checks,  $\sim 5$  s; exact sympy Pfaffian–Wick + exact Cl(16) Fractions; definiteness via 40-digit spectra, tolerance  $10^{-25}$ ; **deterministic**). G1 the gauge datum (Z8/NS tower, Ramond control, invariant dimensions  $120 > 64 > 32$ ); G2  $\Theta$  normalises the tower and restricts as an involution; G3 the  $(-1)^F$  class does not penetrate  $\mathcal{A}^G$ ; R1 the tower average is OS-symmetric, the odd sector dies; M1 the time-like census; M2 the exact Hermiticity witness; M3 the GSO 2-torsion census; R2 GSO-fixed RP ( $N = 16/N = 8$ ); C1–C3 site-cut/wrong-projection/family-A controls; V1 the frozen verdict. Registered in `run_all.py` (suite 516 modules, green), `script_registry.csv` (cluster *frontier*), ledger row WOIT.BETA1.GS0.01 (new).
- **Contract updated (markers unchanged)**. WOIT.OS.TWISTOR.01: the  $\beta_1$  stage moved to *done* (typed UNDECIDED) with its own block, precision (iii) added, the  $\beta_1$  roadmap bullet annotated *executed*, the closing status line carries the  $\beta_1$  state; ledger WOIT.OS.TWISTOR.01 text extended (beta-1 executed via v522).
- **Paper integration**. `tfpt_3` (new Step 23 “the clock is time-like, GSO is the gauge datum”; the Woit-bridge paragraph extended by the  $\beta_1$  status); veri citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram**. Six exact mirrors (the Z8/NS tower  $S^4 = -1 +$  the Ramond witness  $1/(4 \sin(5\pi/16))$ ); the invariant-dimension chain by exact ranks; the time-like reflection census; the exact Hermiticity witness with the full 745/96/0 census via exact Pfaffian–Wick; the GSO 2-torsion census; the family-A tower-normalisation failure) added to `tfpt_readouts_extension.wl`; the GSO-fixed RP inertia certificates stay Python-only, flagged in the `.wl` comment.
- **Provenance**. Discovery probe `woit_os_beta1_gauge_invariant_theta_probe.py` (16/16,  $\sim 3$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

**2026-07-22 (XI) (v521 — the RP/ $\Theta$  attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.RPBLIND.01: “free OS positivity does NOT see  $\delta$ ” — an honest, decided NEGATIVE result in the v508 style (the celestial/twistor story in `tfpt_3` gains Step 22): does free reflection positivity (v519) or anti-linear  $\Theta$  existence break for  $\delta \neq \pi/2$  on the deformed mark family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$  — which would DERIVE the v512 alignment bit from OS positivity? NO — (1) the CENSUS [E] (exact solvesets, symbolic  $\delta$ ): for EVERY  $\delta$  exactly the two mark-SWAPPING reflections  $\Theta_{\pm e^{i\delta}}$  exist (axes  $\delta/2, \delta/2 + \pi/2$ ; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at  $\delta = \pi/2$  (solveset  $\cos \delta = 0$ , lost immediately at  $\pi/2 - \varepsilon$ ); the v510/v512 counterwitness  $\{\pm 1, \pm(3+4i)/5\}$  carries the swap pair exactly; (2) RP IS  $\delta$ -BLIND [E]: bond-cut Gram inertias  $(8, 0, 0)$  PD for all tested  $\delta$  with the full sorted spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO  $\delta$  information; the  $N = 16$  odd- $m$  failure (site axes,  $(3, 3, 1)$ ,  $\det = 0$ ) is a lattice-PLACEMENT artifact: at  $N = 32$  ALL  $m = 1..7$  have mark-compatible BOND axes with PD inertia  $(16, 0, 0)$ ; continuum: the OS kernel in axis-adapted coordinates is EXACTLY  $1/\sin((s+t)/2)$  — the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram  $(6, 0, 0)$ ); (3)  $\Theta$  EXISTENCE IS  $\delta$ -BLIND [E]:  $\Theta^2 = +1$ , deck normalisation and Fock implementability ( $U^2 = +2^7/2^8 > 0$ , exact Cl(16)) for every**

$\delta$ ; the ONLY  $\delta$ -sensitive v519 clause ( $\Theta\rho\Theta = \rho^{-1}$ ) is for  $\delta \neq \pi/2$  not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff  $\delta = \pi/2$ ) — it PRESUPPOSES the bit; same for the clock-tower clause; (4) the battery HAS TEETH at every  $\delta$  [E]: family A/antipode fails RP structurally (zero diagonal, (4, 4, 0)), the chirality- $\eta$  pinning persists ((0, 8, 0) ND) — RP separates the two real-structure FAMILIES, never the SIDES; (5) precondition controls: hexagonal (marks not concyclic — no seam circle), silver (deck not free), lifting audit (B = index relabelling,  $\delta$ -blind by construction; B' fails the  $\pi/2$  negative control — discarded); [C] FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is  $V_4$  for every  $\delta < \pi/2$  and closes to  $D_4$  exactly at  $\delta = \pi/2$  ( $\Theta_i \circ \Theta_1 =$  the clock) —  $12 \rightarrow 13$  faces, a typed REFORMULATION of the bit, not a derivation; [O] the honest gap: marks decorating the STATE (defect/twist insertions, interacting  $\mathcal{A}_{\text{hol}}$ ) = exactly v519's [O] scope; VERDICT: KILL exactly as preregistered — the free RP/ $\Theta$  battery is the EIGHTH side-blind test on the v512 scoreboard ( $7 \rightarrow 8$ ); the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; *NO marker moves — the bit stays the one [C] carrier input*)

- **v521\_seam\_bit\_rp\_blind.py** (SEAM.BIT.RPBLIND.01, 25 checks, ~25 s; exact sympy solvesets + exact Gram entries + exact Cl(16) Fractions; inertia via 40-digit spectra, tolerance  $10^{-25}$ ; deterministic). S the anti-linear census (family D/A, counterwitness, face #13 closure, cut incidence, side conditions); L the lattice  $\delta$  ladder (frame,  $\pi/2$  reproduction,  $\pi/4$  PD, odd- $m$  artifact,  $N = 32$  refinement, site-axis control, spectra blindness, family-A rejection, chirality pinning); C the continuum axis elimination + counterwitness; LB the lifting audit; T Fock implementability + the clock-tower clause; W scoreboard + hex/silver controls + the preregistered KILL verdict. Registered in run\_all.py, script\_registry.csv (cluster core), ledger row SEAM.BIT.RPBLIND.01 (new).
- **Ledger/paper text precisions (markers unchanged).** SEAM.TAU.FLAG.01: scoreboard text extended ( $7 \rightarrow 8$  side-blind tests, face #13, web  $12 \rightarrow 13$ ); tfpt\_research\_contracts (the v512 reduction paragraph now carries the v521 kill and the honest [O] gap; the WP2 residual sentence extended), introduction and tfpt\_3 Step 2 ( $12\text{-fold} \rightarrow 13\text{-fold}$ , the eighth side-blind test); veri citations in tfpt\_research\_contracts, introduction and tfpt\_3.
- **Wolfram.** Six exact mirrors (the census solvesets incl. the exact-rational counterwitness; the  $V_4 \rightarrow D_4$  closure / face #13 as symbolic composition identities; the 20 incidence solvesets + the eps jump + the integer lattice frame; the continuum axis elimination + Cauchy–Stieltjes minors + the counterwitness cut ordering; the clock-tower clause + hex/silver controls; the  $\delta$ -blind side conditions) added to tfpt\_readouts\_extension.wl; the 40-digit lattice inertia spectra stay Python-only.
- **Provenance.** Discovery probe seam\_alignment\_bit\_rp\_selector\_probe.py (25/25, ~24 s) in experiments/tfpt-discovery/ (2026-07-22); website mirrors regenerated; next.txt round notes.

2026-07-22 (X) (v520 — the WP5e BCOV measure question DECIDED at probe level and promoted as the ledger row CELEST.WP5E.MEASURE.01: “single-valuedness is derived, the completion reading wins” — the  $\delta_{1e} + \delta_{1f}$  consolidation in the v518 style, ERFOLG-A on the preregistered decision layer (the celestial/twistor story in tfpt\_3 gains Step 21): which of the two exact, contradictory measures on  $\mathbb{P}T/\mathbb{Z}_4$  is the true BCOV measure — (A) the DECLARED completion reading (v516) or (B) the DERIVED chiral measure (v518)? — (1) the INVARIANT SUBSPACE [E]:  $\dim_{\mathbb{Q}}\{v : \rho(T)v = v, \rho(S)v = v\} = 8 = 4 \times 2$  in the dressed 16-component Weil system (exact rational nullspace); EVERY kernel vector lies

in the  $\mathbb{Q}(\zeta_8)$ -span of  $\{e_H, e_{H'}\}$  — the two Lagrangian gluings, both  $\theta = E_4$ , level-1 quartic  $= Q^{(0)}$  = the UNPHASED trace; strict trivial-character solutions at physical weight  $(m, k) = (0, 0)$ : ZERO for every dressing (11 combinations); (2) SOURCE 1 — the F-LEMMA [E]: the only strict chiral families are  $\chi_4$  (dims  $(3, 3)$ ) and  $\chi_{10}$  (dims  $(1, 1)$ ), both with nontrivial  $T$ -character EXACTLY; the canonical column-matched member obeys  $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$  pointwise (certificate  $3.9 \times 10^{-16}$ ,  $|G| \geq 59.6$ ); EXACT change of variables  $\Rightarrow$  fundamental-domain independence forces  $f = 0$  for EVERY chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is SHARPENED — its unfolded values are convention bookkeeping); (3) SOURCE 2 — the QUILLEN PAIRING [E]:  $|\bar{t}t - 1| < 8.3 \times 10^{-40}$  on all 15 pairs and both generators;  $|\chi_4|^2 = 1$  exactly vs one-sided  $\chi_4^2 = e(2/3) \neq 1$  ( $T$ -fixed:  $640 \neq 0 \bmod 960$ , 15/15 defective); the doubled 256-dim hol $\times$ antihol system closes STRICTLY at trivial character and physical weight (nullspaces  $(28, 17) \geq 11$ ), contains the unphased diagonal ( $\sim 2 \times 10^{-15}$ ) AND the physical columns  $w_b \otimes \bar{w}_b$  ( $\sigma$   $4.9/8.3 \times 10^{-16}$ ); negative control hol $\otimes$ hol: EMPTY nullspace  $(0, 0)$ ; (4) the CONSISTENCY CLOSURE [E]: the forced unphased numerator in the v518 scaffold reproduces  $\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x \rangle^2$  EXACTLY on every level  $n \leq 8$ ;  $J$ -weighted spreads  $\sim 5 \times 10^{-41}$ ,  $T_5/T_3 \sim 7 \times 10^{-42}$ , zero negative cells;  $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$  — the  $J$ -weighted mirror image of  $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$ ; linearity  $4.6 \times 10^{-41}$ : the circle to v516 is CLOSED; (5) D3 — the  $\mathbb{Z}_2/\text{EH}$  ANCHOR [E]: only the declared reading hits ( $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $\psi = -8$ , scale tooth  $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ); all three derived instantiations fail (spreads 1.12/0.895/0.516,  $\psi$  sign wrong); (6) D1 — COLUMN CANONICALISATION [E]: the physical AB column  $\sum_m i^{bm} e_{(m,m)}$  is contained EXACTLY in the  $\chi_4$  family ( $\sigma$   $1.8/1.7 \times 10^{-16}$ , one scalar per orbit — canonicalising  $(N_1, N_2)$ ); the canonical member STILL fails every tester; NEGATIVE CONTROLS:  $SO(16)/D_8$  ( $T_5 = 20/T_3 = -40$  exact,  $\mu_2$ -only), wrong form ( $|\text{Gauss}|^2 = 64$ ), wrong weight (dev 0.78), shuffle ( $T_5 = -14/T_3 = 36$ ), planted family detected ( $1.5 \times 10^{-16}$ ), hol $\otimes$ hol empty; [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source  $\psi = 64$ ), TP-3 (v518 kernel convention), TP-4 (Quillen structure of BCOV  $F_1$ ); [O] the constructive BCOV derivation of the  $w_m$  normalisation itself; WIRKUNG (text only, no marker moves): the v516 fence and the v518 tension now read “decided at probe level for the declared reading under TP-1..TP-4 (v520)”; SEAM.EQUIV.TWISTOR.01 stays Open; VERDICT ERFOLG-A per the preregistered criteria — both constructive sources fire independently, the consistency closure carries)

- v520\_celestial\_wp5e\_measure\_decision.py (CELEST.WP5E.MEASURE.01, 39 checks,  $\sim 20$  s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/\text{sympy}$  rational nullspaces + 1/960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic). S0 exact foundations (roots, ledgers, A-core +  $\mathbb{Z}_2$ -anchor targets, Weil relations, characters); S1 transports + derived families + the  $k = 0$  slice; S2 D1 column canonicalisation + the sharpened kill; S3 D2 the exact invariant subspace; S4 D3 the  $\mathbb{Z}_2/\text{EH}$  anchor at the integral level; S5 source-1 (the F-lemma); S6 source-2 (the Quillen pairing, doubled 256-dim system); S7 the point-4 consistency closure; S8 negative controls; S9 the ERFOLG-A verdict + typed premises. Registered in run\_all.py, script\_registry.csv (cluster frontier), ledger row CELEST.WP5E.MEASURE.01 (new).
- Ledger/paper text precisions (markers unchanged). CELEST.WP5E.M2.01 (the v516 fence: the completion reading is now DERIVED at probe level from two independent constructive sources under TP-1..TP-4; residual [O] = the constructive  $w_m$  derivation), CELEST.WP5E.DELTA1.01 (the v518 kill sharpened, the tension decided); tfpt\_research\_contracts (the WP5e measure-question prose updated throughout: object

diagram, WP4 arc,  $\varepsilon_1$ /M2 fences, the measure-question target block now carries the executed decision, the closing status lines); `tfpt_3` (Step 17/19 fences updated, new Step 21, the “what remains” block narrowed to the constructive  $w_m$  derivation); `tfpt_4_frontier` (celestial keybox updated). `veri` citations in `tfpt_research_contracts` and `tfpt_3`.

- **Wolfram.** Six exact mirrors (the 16-component Weil data with both Lagrangian gluings; the invariant-subspace forcing  $\dim = 2 = \text{span}\{e_H, e_{H'}\}$  by exact NullSpace + rank; the character arithmetic of the two sources  $|\chi_4|^2 = 1$  vs  $\chi_4^2 = e(2/3)$  with the exact grid defects; the A-core +  $\mathbb{Z}_2$ /EH-anchor targets with the scale teeth; the level-1 Okubo identity of BOTH gluings (240 norm-2 vectors each, summed quartic  $36\langle x, x \rangle^2$ ); the `sol6/shuffle/diag(i, i)` controls) added to `tfpt_readouts_extension.wl`; the Harvey–Moore integrals, SVD memberships and  $J$ -weighted closures stay Python-only, flagged in the `.wl` comment; combined engine run with the v521/v522 mirrors: extension 522  $\rightarrow$  540 (verified, Wolfram Engine 14.3; GATE.WOLFRAM.02, Wolfram README, `tfpt_safeguards`, website counters and `zenodo_description.html` synced 116/116 + 540/540).
- **Provenance.** Discovery probes in `experiments/tfpt-discovery/` (2026-07-22): `celestial_seam_wp5e_delta_e_bcov_measure_decision_probe.py` (28/28,  $\sim 15$  s) and `celestial_seam_wp5e_delta_if_single_valuedness_probe.py` (23/23,  $\sim 17$  s); suite 513  $\rightarrow$  516 modules (green); website mirrors regenerated (ScriptIndex, DAG  $qft/P2$  nodes, `papers.ts` story counters, glossary, changelog mirror); `next.txt` round notes.

**2026-07-22 (IX) (v519 — the WOIT- $\alpha$  milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.THETA.FREE.01: “the real structure exists — and free reflection positivity picks the same family”;** the celestial/twistor story in `tfpt_3` is now TWENTY steps — (1) **THETA CLASSIFICATION COMPLETE [E]:** exactly TWO families of anti-linear structures on  $\mathbb{C}^2$  normalise the clock  $\rho = \text{diag}(i, 1)$  — family D ( $z \mapsto \mu\bar{z}$ ,  $|\mu| = 1$ ):  $\Theta\rho\Theta = \rho^{-1}$  EXACTLY at matrix level,  $\Theta^2 = +1$  (−1 IMPOSSIBLE:  $MM = \text{diag}(|\mu|^2, 1)$  — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ( $z \mapsto \mu/\bar{z}$ ): centralises the clock projectively (NEVER inverts it),  $\Theta^2 = \pm 1$  per  $\mu$ , fixed set  $|z| = 1$  resp. empty; **ROLE SEPARATION:** family A DEFINES the euclidean section (Woit’s  $\rho_{\text{tw}}$  side:  $\rho_{\text{tw}}^2 = -1$ , no real points — replicated exactly on  $\mathbb{C}^4$ ), family D REFLECTS it (the OS conjugation,  $\sigma_{\text{std}} = \text{conjugation with real points } \mathbb{RP}^3$ ); (2) **MARK PINNING [E]:**  $z \mapsto \mu\bar{z}$  preserves the  $\mu_4$  marks iff  $\mu \in \mu_4$  — a 4-element  $\mu_4$ -TORSOR (2 mark axes, 2 silver-midpoint axes), clock conjugation  $\rho\Theta_\mu\rho^{-1} = \Theta_{-\mu} \Rightarrow 2$  clock orbits; (3) **SPIN/FOCK [E]:** the  $\mathbb{Z}_8$  plane has no phase leaks ( $\bar{\zeta}_8^k = \zeta_8^{-k}$ ); exact  $\text{Cl}(16)$ :  $\Theta_{\text{Fock}} = U_r \circ K$  with  $\Theta_{\text{Fock}}^2 = 2^7 > 0$  (normalised +1), inverts the clock tower ( $V \mapsto 4096 V^{-1}$ ) and normalises the deck; the **KRAMERS DICHOTOMY:** the deck-induced candidate  $\Theta_t = \tilde{U} \circ K$  has  $\Theta_t^2 = (-1)^F$  — *the v510 split/nonsplit dichotomy IS the  $\Theta^2 = +1$  vs  $(-1)^F$  dichotomy*; the deck does NOT furnish  $\Theta$ , the seam-circle REFLECTION does; (4) **KILL TEST 1 (free level) [E]:** does NOT fire —  $\Theta$  exists with  $\Theta^2 = +1$  AND  $\Theta\rho\Theta = \rho^{-1}$  and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock); (5) **FREE RP [E]:** the NS vacuum  $C(d) = (2/N)/\sin(\pi d/N)$  has  $C^2 = -1$  exactly; preconditions exact ( $RCR^T = -C$ ,  $SCS^T = C$ ,  $RSR^{-1} = S^{-1}$ ; the deck FAILS anti-invariance); **BOND CUT:** one-particle Gram (8, 0, 0) PD (min eigenvalue  $1.888 \times 10^{-3}$ , 40 digits), even  $\deg \leq 2$  sector (29, 0, 0) PD,  $N = 8$  complete half-sided algebra PD (even + odd each (8, 0, 0)) — RP with NO degree truncation; twist  $\eta = +i$  FORCED ( $\eta = 1$  non-Hermitian); **CONTROLS:** the site cut fails exactly ( $\det = 0$ , inertia (3, 3, 1); continuum control: Cauchy–Stieltjes strictly positive — a lattice-PLACEMENT artifact, RP selects the site placement

relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally  $((4, 4, 0))$  — *RP and  $\Theta\rho\Theta = \rho^{-1}$  select the SAME family*; bonus: the anti-chiral state flips the odd sector to  $(0, 8, 0)$  ND (the free shadow of kill test 3); [C]/[O] NOT shown: the interacting algebra  $\mathcal{A}_{\text{hol}}$  (kill test 1 stays live there), gauge-fixed RP (kill test 2), OS reconstruction, chirality without mirrors (kill tests 3/6),  $\mu_4$  incidence (kill test 7) — the WOIT- $\beta/\gamma$  roadmap named  $(\beta_1 \Theta$  on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant  $\text{SDYM}(E_8)$  sector;  $\beta_2$  the OS quotient of the free system explicit;  $\beta_3$  the  $\text{PT} \leftrightarrow \text{PT}^*$  duality vs  $\sigma_{\text{std}}$  typed;  $\gamma$  the chirality theorem + the mark incidence); TWO CONTRACT PRECISIONS (prose, no marker): (i) “ $\Theta$  induced by the seam reflection” means the seam-circle REFLECTION, not the deck — the deck furnishes the  $(-1)^F$  Kramers class (the v510 bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle; Wolfram extension  $516 \rightarrow 522$  (verified engine run); *NO marker moves* — *WOIT.OS.TWISTOR.01 stays Open: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra*)

- **v519\_woit\_theta\_rp\_free.py** (WOIT.THETA.FREE.01, 34 checks,  $\sim 20$  s; exact sympy + exact  $\text{Cl}(16)$  Fractions; RP definiteness via 40-digit spectra of exact Hermitian matrices, tolerance  $10^{-25}$ ; deterministic). Part W: W-S1 the two-family classification (symbolic solve); W-S2 family D (relations, Kramers-free, mark pinning, seam action, torsor); W-S3 family A (equatorial/ $\rho_{\text{tw}}$ ); W-S4 the  $\mathbb{C}^4$  twistor level ( $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ); W-S5 the  $\mathbb{Z}_8$  spin level; W-S6 the  $\text{Cl}(16)$  Fock level ( $\Theta_{\text{Fock}}^2 = 2^7$ ,  $V \mapsto 4096 V^{-1}$  via the implementer/centre route — a consolidation of the 105-s discovery probe to  $\sim 20$  s with identical claims and tolerances — deck normalisation, Kramers dichotomy,  $\tilde{U}V = V\tilde{U}$  control); W-S7 verdict. Part R: R1 the exact preconditions; R2 the site cut fails; R3 the bond cut passes (incl. the complete  $N = 8$  half-algebra); R4 the continuum control; R5 controls (twist forced, antipode fails, chirality flip); R6 verdict. Registered in `run_all.py` (suite 513 modules, green), `script_registry.csv` (cluster *frontier*), ledger row WOIT.THETA.FREE.01 (new).
- **Contract updated (markers unchanged).** WOIT.OS.TWISTOR.01: the  $\alpha$  stage moved to *done* with the core numbers and `veri{v519}`; the two precisions added as their own sentences (the input definition of  $\Theta$  now names the seam-circle reflection, with the deck carrying the  $(-1)^F$  Kramers class; the marks at the bond midpoints of the 16-Majorana circle); the  $\beta/\gamma$  roadmap bullets added with success and kill criteria; the object-diagram OS row and the closing status line carry the  $\alpha$ -stage state. Ledger: WOIT.THETA.FREE.01 new; WOIT.OS.TWISTOR.01 text extended (“alpha stage executed via v519 ... kill test 1 does NOT fire at the free level and stays live on the interacting algebra; beta/gamma milestones named”). NO marker moves anywhere.
- **Paper integration.** `tfpt_3` (new Step 20 “the real structure exists — and free reflection positivity picks the same family” — the celestial/twistor story is now twenty steps; the Woit-bridge paragraph extended by the  $\alpha$  status: “the missing global object now exists at the free level; the interacting statement stays [O]”). `v519_woit_theta_rp_free.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the two-family classification; mark pinning + the  $\mu_4$  torsor with the clock-orbit map and the cut-point census; the  $\mathbb{C}^4$  level  $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ; the  $\mathbb{Z}_8$  tower conjugation; the  $\text{Cl}(16)$  dichotomy  $2^7$  vs  $(-1)^F$  on explicit Jordan–Wigner gammas incl.  $U_r V U_r V = 2^{19}$  and  $\tilde{U}V = V\tilde{U}$ ; the free-RP preconditions + site-cut degeneration + the Cauchy–Stieltjes continuum control) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $516 \rightarrow 522$ ; the 40-digit PD inertia certificates of the bond-cut Grams stay Python-only, flagged in the `.wl` comment; GATE.WOLFRAM.02, Wolfram README and the website counters synced ( $116/116 + 522/522$ ).

- **Provenance.** Discovery probes `woit_os_theta_realstructure_probe.py` (21/21,  $\sim 105$  s) and `woit_os_theta_rp_freelevel_probe.py` (13/13,  $\sim 5$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v519, `papers.ts` “20 steps” + the WOIT-contract mirror, glossary  $\Theta$ /Woit-bridge entries, changelog mirror); `next.txt` round notes.

**2026-07-22 (VIII) (v518 — the WP5e  $\delta_1$  chain ( $\delta_{1b} + \delta_{1c} + \delta_{1d}$ ) DECIDED and promoted as the ledger row CELEST.WP5E.DELTA1.01:** “the derived chiral measure kills the  $\delta_1$  route” — an honest, decided **NEGATIVE** result in the v508 style, with the v516 tension stated: does the Harvey–Moore/BCOV fundamental-domain integral on  $\mathbb{PT}/\mathbb{Z}_4$ , evaluated under a measure **DERIVED** from blockwise  $SL(2, \mathbb{Z})$  covariance of the dressed 16-component Weil system (solved for — not declared, not scanned), deliver the twisted contact vector  $-A_{\text{fix}} = -(9, -30, -15, 0, 32)$  or the  $\psi = 64$  slice? **NO** — (1) the **16-COMPONENT WEIL COMPLETION** ( $\delta_{1c}$ ): the discriminant module of  $D_5 \oplus A_3$  is  $\mathbb{Z}_4 \times \mathbb{Z}_4$  with  $q = (5x^2 + 3y^2)/8$ , Gauss sums  $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$  (signature  $0 \bmod 8 = \text{rank } E_8$ ), the Weil relations hold **EXACTLY** in  $\mathbb{Q}(\zeta_8)$  ( $S^2 = C$ ,  $(ST)^3 = S^2$ ,  $S^4 = 1$ ,  $T^8 = 1$ , tensor split  $S = S_{D_5} \otimes S_{A_3}$ ), the diagonal  $H$  and anti-diagonal  $H'$  are the two invariant Lagrangians (both counts =  $E_4$ : two  $E_8$  gluings),  $P_H S P_H = \frac{1}{4}J$  with  $3/4$  of every  $S$ -image weight on the 12 shifted classes, and the 16-vector theta  $S$ -covariance **CLOSES**: the  $\delta_{1b}$  **E1.5** residual  $2.91 = O(1)$  (naive 4-character rule) drops to  $\sim 10^{-39}$ ; (2) the **OBSTRUCTION IS A FINITE  $\mu_4$  CHARACTER, IDENTIFIED** ( $\delta_{1d}$ ): the dressed gauge blocks  $M = G[a, b] \eta^{-8}$  carry constant transport  $(t_p, s_p)$  (28+ digits, recognised as exact  $1/960$ -grid phases at  $10^{-25}$ ), the  $T$ -fixed pairs  $(0, b)$  carry  $t = e(-1/2) = -1$  **EXACTLY** (the naked level-matching defect), the transport-groupoid loops land in  $\Gamma_1(4)$  resp.  $\Gamma_0(2)^\pm$  with joint holonomy eigenspaces  $\dim 8/16$  resp.  $6/16$  and every eigen-phase of order  $\leq 4$ ; the **KOBOUNDARY TEST**: the  $SL(2, \mathbb{Z})$  relation defects  $c_{S^4}$ ,  $c_{(ST)^3 S^{-2}}$ ,  $c_{[S^2, T]}$  are  $(1, 1, 1)$  **EXACTLY** on all 15 pairs — the multiplier system is a **CHARACTER** of the orbit stabilisers, **NOT** a genuine 2-cocycle, and on  $\Gamma_1(4)$  it is  $\lambda(\gamma) = i^{2B+C/4}$ ; (3) the **CANCELLATION EXISTS AND IS THE TWISTED FIBRE BLOCK  $f_1 f_3$ , NOT THE THREE SPHERE AXIONS**:  $G[a, b] = f_1 f_3$  is an exact identity (30+ digits; the axion  $T$ -phases are v502 vacuum-energy data), the  $T$ -fix mechanism is one line of exact phase arithmetic —  $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$  at every  $T$ -fixed node — and the exact **CANCELLATION TABLE** reads: none  $\rightarrow$  residual order 4,  $f_1 f_2 f_3$  (the three sphere axions)  $\rightarrow 4$ ,  $f_2 \rightarrow 6$ ,  $f_1 f_3 \rightarrow 1$  (hol **AND** anti), wrong weights  $(1, 1, 2)/(2, 2, 2) \rightarrow 4$ ; the strict solutions are exactly  $\chi_4$  (dims  $(3, 3)$ ) and  $\chi_{10}$  (dims  $(1, 1)$ ) at  $\eta^0$ , certified on the dressed functions at a fresh  $\tau$  ( $10^{-15}$ , **NO** residual multiplier); (4) the **DECISION — ALL THREE PREREGISTERED TESTERS FAIL**: under both derived solutions the integral (30-digit kernels, negative-exponent cell audit, cut-6 vs cut-8 scan) fails  $T_5(V) = 0$  (fractions  $0.5/0.83$ ), fails  $W_2:W_{13} = 4:3$  ( $W_{13} = 0$ ), fails  $V = -A_{\text{fix}}N$  (spreads  $1.05/1.47$ ) and the  $\psi = -64N$  slice, and the honest  $(N_1, N_2)$  orbit rebalance rescues nothing in the positive cone ( $\chi_4$ :  $T_5 = 0$  only at  $\psi > 0$ ;  $\chi_{10}$ : only at  $N_2/N_1 = -4 < 0$ ) — a **GENUINE KILL** on the derived surface; (5) **NEGATIVE CONTROLS**:  $\mathbb{Z}_2/\text{Eguchi–Hanson}$  (gauge-only residual order 2 — the  $\mu_4$  defect is  $\mathbb{Z}_4$ -specific — and the **SAME** fibre dressing cancels it to 1),  $SO(16)/D_8$  (only even sectors, all axion  $b$ -characters in  $\mu_2$ : the order-4 supply structurally absent), wrong form  $(x^2 + y^2)/4$  ( $|G|^2 = 64 \neq 16$ , its  $S$ -rule fails at  $O(1)$ ), wrong signature ( $S^2 = e(-1/2)C$  exactly); [C] the  $f_1 f_3 = \text{KS-weight identification}$  (deck weights  $(1, 3) = \text{the twistor fibre weights forced by the v514 incidence ledger}$  — exact match, **NOT** a complete

BCOV derivation) + the kernel-family convention + the discrete  $\chi_k$  choice (both evaluated, both fail); the MANDATORY TENSION FENCE: the DECLARED completion reading (v516/M2) delivers the  $\psi = 64$  slice and cancels  $32T_3$ ; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure — neither result is hidden behind the other; [O] the BCOV-integral derivation on  $\mathbb{PT}/\mathbb{Z}_4$  that decides between the two measures; Wolfram extension  $510 \rightarrow 516$  (verified engine run); *NO marker moves — the  $\delta_1$  route is decided negative under the derived measure, the cubic-GS question stays dissolved, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched*)

- **v518\_celestial\_wp5e\_delta1\_derived\_kill.py** (CELEST.WP5E.DELTA1.01, 30 checks,  $\sim 20$  s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/1\text{-}960\text{-grid}$  phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic). S0 conventions + 16-class ledgers + targets; S1 the 16-component Weil system (exact) + the E1.5 closure; S2 gauge transport + the  $\mu_4$  multiplier system (koboundary test, character identified); S3 the cancellation ( $G = f_1f_3$ , T-fix mechanism, exact table); S4 the strict solutions (constructive scan,  $m \in \{0, 8, 16\}$ ); S5 the decision (function certificates, integrals, three testers,  $(N_1, N_2)$  scan); S6 negative controls; S7 verdict + the tension fence. Registered in `run_all.py` (suite 512 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.DELTA1.01 (new — “DECIDED: kill under the derived measure, tension with v516 stated”).
- **Contract updated (markers unchanged).** CELEST.SEAM.01: the  $\delta_1$  strand moves from “UNDECIDED at exploration level” to DECIDED by v518; the named remaining target is now the BCOV MEASURE QUESTION (the declared v516 completion reading vs the derived v518 chiral measure — both exact, which is the true BCOV measure is the named [O]); the cubic-GS question stays dissolved (v518 does not revive the  $d$ -channel). SEAM.EQUIV.TWISTOR.01 dependencies extended by v518 (preparation range now v492–v518; the route stays Open). CELEST.WP5E.M2.01 cross-referenced (its [O] pointer “ $\delta_{1b/1c/1d}$  territory” is executed and decided). NO marker moves anywhere.
- **Paper integration.** `tfpt_research_contracts` (the  $\delta_1$  stage moved to *done*, DECIDED with the four exact results and the mandatory tension fence as its own paragraph; “WP5e proper” renarrowed from the  $\delta_{1d}$  derivation to the declared-vs-derived measure question; the object-diagram row, the honest status line and the per-ID status list extended by CELEST.WP5E.DELTA1.01); `tfpt_3` (new Step 19 “the derived measure decides — and disagrees with the declared one” — the celestial story is now nineteen steps; the Step-16/17 fences and “what remains” updated to the measure question); `tfpt_4_frontier` (keybox synced). `v518_celestial_wp5e_delta1_derived_kill.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil system with Gauss sums and tensor split; the two invariant Lagrangians +  $P_HSP_H = \frac{1}{4}J$  + the  $3/4$  shifted-class weight; the koboundary test  $(1, 1, 1)$  on all 15 pairs from the recorded exact transport phases; the  $\Gamma_1(4)$  character formula  $\lambda(\gamma) = i^{2B+C/4}$  on the recorded loops; the T-fix mechanism  $(-1)e(-1/6) = e(1/3) = \chi_4(T)$  uniform for  $f_1f_3$  vs inconsistent  $\{\chi_3, \chi_4\}/\{\chi_5, \chi_6\}$  for  $f_1f_2f_3/f_2$ ; the wrong-form/wrong-signature controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $510 \rightarrow 516$ ; the Harvey–Moore integrals, SVD contraction solves, theta covariance certificates and block transport measurements are numerical and stay Python-only, flagged in the `.wl` comment; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 516/516).
- **Provenance.** Discovery probes `celestial_seam_wp5e_delta1b/1c/1d_*.py` — the Harvey–Moore unfolding (30/30), the Weil-16 completion (31/31) and the order-4 multiplier cancellation (36/36) — in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated

(ScriptIndex, DAG *gft* node + v518, papers.ts “19 steps”, changelog mirror); next.txt round notes.

**2026-07-22 (VII) (v517 — WP5e back-reaction milestone M3 “the  $a_0$  uplift”** executed (SUCCESS on the preregistered v514 S8.3 criterion), promoted as the ledger row CELEST.WP5E.M3.01: the  $(4, \pm 4)$  GH multipole (= the  $a_0$  direction) is uplifted to the twistor-potential kernel  $\chi = \log P_4$  — the LOG of the v515 family polynomial — and the induced Kähler-potential correction is a log-type correction COUPLED TO THE CENTRE COUNT ON FOUR SCALES — (1) the **KERNEL BRIDGE**: the  $O(2)$  section is a NULL coordinate ( $\Delta_{3d}G(\eta_p) = 0$  identically for an ARBITRARY kernel  $G$ : the uplift is well-typed), the **RESIDUE IDENTITY**  $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$  holds EXACTLY (residue  $1/(2r)$ ), so the GH potential  $V = \sum_p 1/r_p$  IS the  $x$ -derivative of the contour transform of  $\chi$ ; flux  $-4\pi$  per centre, source charge  $4 = |\mu_4|$  (the  $V$ -ledger matched); (2) the **ASYMPTOTIC KERNEL LOG**:  $\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$  with LOG COEFFICIENT  $4 = |\mu_4|$  = centre count (the CPS “ $N \log$ ” analogue at kernel level); at the clock-fixed seam fibre  $\zeta = 0$  the  $a_2$  slot closes and the first correction is EXACTLY  $a_0/\eta^4$  = the  $(4, \pm 4)$  multipole in kernel clothes; exact  $m$ -grading ( $m = \pm 4$  pieces PURE  $a_0$ ,  $m = 0$  piece  $-6t_0^4 \varphi_0$ -free: the  $U(1) \rightarrow \mathbb{Z}_4$  breaking sits exclusively in the  $a_0$  pieces); NO  $\log \times$  power terms (the v514 honest zero preserved — the  $a_0$  correction is power-law at infinity); the once-integrated GLT tower  $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + p_8/(56\eta^7) + \dots$  with  $p_n = 0$  unless  $n \equiv 0 \pmod 4$  and  $p_{4k} = 4(-a_0)^k$ : EVERY correction carries the prefactor 4 = centre count; (3) the **EXCEPTIONAL-LOCUS LOG**:  $\chi(\eta = 0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$  ( $t_0$ -coefficient  $4 = |\mu_4|$ , clock phase  $4i$ ); the modulus response  $a_0 \partial_{a_0} \chi = a_0/(\eta^4 + a_0)$  equals 1 AT the exceptional locus and falls off as  $a_0/\eta^4$  at infinity (no constant, no log): the  $a_0$ -log lives at the exceptional locus, exactly as v514 S6.3 demanded; (4) the **PERIOD RESPONSE**:  $\Pi_j(\lambda)$  is LINEAR in  $t_0$  (uniform first-order shift: lockstep + phase ratios  $(1, i, i^2)$  preserved exactly),  $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$  uniformly, and the integrated shift around the  $a_0$  circle is the monodromy  $e^{2\pi i/4} = i$  = ONE Coxeter clock step (the v493 monodromy REPRODUCED from perturbation theory; the finite monodromy is the cyclic centre permutation  $q_p \rightarrow q_{p+1}$ ); the node  $\lambda$ -support is  $t_0$ -FREE (the  $\mathbb{Z}_8$  seam bridge is  $a_0$ -rigid), all 6 pair discriminants  $\sim t_0^2 \neq 0$  ( $K_4$  connectivity: the forced uniform flux vector persists), the BM kernel is modulus-free with  $\int_{S^3} K = (2\pi i)^2$  (the quantised fluxes cannot move); (5) **NEGATIVE CONTROLS**: the  $(4, 0)$  multipole is  $\varphi_0$ -independent in  $V$  AND in the kernel (clock-invariant, breaks nothing); the  $\mathbb{Z}_2$ /Eguchi–Hanson analogue reads  $2 = |\mathbb{Z}_2|$  on EVERY dial ( $(2, \pm 2)$  amplitude  $\frac{3}{2} \sin^2 \theta$ , kernel  $\log 2$ , exceptional  $\log 2 \log t_0$ , response  $1/2$ , monodromy  $-1$ ); wrong centre counts  $k = 3, 5$ : kernel log coefficient  $k$ , modulus slot  $O(6)/O(10) \neq O(8)$  — the observable MOVES with the centre count; forbidden family  $Z^4 - Z$ :  $p_3 = 3 \neq 0$  (selection rule broken),  $e_4 = 0$  (no  $a_0$ -log) — fails BOTH uplift dials; [C] the GLT dictionary ( $\eta_p \log \eta_p$  = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O] the full nonlinear Kähler potential of the resolved  $A_3$  ALE; with v516 the v514 fence M1–M3 is FULLY WORKED OFF; Wolfram extension  $504 \rightarrow 510$  (verified engine run); NO marker moves — the KILL (decoupling from the centre count) does NOT fire, SEAM.EQUIV.TWISTOR.01 stays Open (preparation, not construction), SEAM.EQUIV.01 untouched)

- v517\_celestial\_wp5e\_m3\_a0\_uplift.py (CELEST.WP5E.M3.01, 23 checks,  $\sim 7$  s; exact sympy, no floats; deterministic). S0 replication (GH multipole ledger + the twistor family



at general orbit phase, v515 slice); S1 the kernel bridge (null coordinate, residue  $V$ -ledger identity, source flux); S2 the uplift (asymptotic  $\log 4$ , seam-fibre correction  $a_0/\eta^4$ , exact  $m$ -grading, GLT tower + selection rule); S3 the exceptional-locus log; S4  $\Omega_N$  consistency (period response  $1/4$ , monodromy  $i$ ,  $a_0$ -rigid nodes and fluxes); S5 negative controls; S6 verdict. Registered in `run_all.py` (suite 511 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M3.01 (new, SUCCESS).

- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M2+M3 executed via v516/v517 — the v514 fence M1–M3 is fully worked off: the twisted KS measure (declared completion reading,  $\text{ch}_2$ -weighted) cancels the  $32 T_3$  residual and delivers the  $\psi = 64$  slice without the cubic  $d$ -channel, and the  $a_0$  uplift couples to the centre count on four scales; what stays [O]: the BCOV-integral derivation of the completion reading ( $\delta_{1d}$ ) and SEAM.EQUIV.TWISTOR.01. SEAM.EQUIV.TWISTOR.01 dependencies extended by v516/v517 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M3 bullet moved to *done* with the four coupled scales and the GLT fence; “what remains” narrowed to  $\delta_{1d}$  + the BCOV derivation); `tfpt_3` (new Step 18 “the  $a_0$  uplift: four coupled centre-count scales” — the celestial story is now eighteen steps; “what remains” updated). `v517_celestial_wp5e_m3_a0_uplift.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (multipole ledger + family at general phase; kernel bridge (null coordinate, residue  $1/(2r)$ , flux  $-4\pi$ ); asymptotic  $\log + m$ -grading + GLT tower; exceptional-locus log; period response  $1/4 + \text{monodromy } i + a_0\text{-rigidity}$ ; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $504 \rightarrow 510$ ; GATE.WOLFRAM.02, Wolfram README and the website counters synced ( $116/116 + 510/510$ ).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_a0_uplift_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + v517, `papers.ts` “18 steps”, changelog mirror); `next.txt` round notes.

2026-07-22 (VI) (v516 — WP5e back-reaction milestone M2 “the twisted Kodaira–Spencer measure” executed (SUCCESS on the preregistered v514 S8.2 criterion, ON THE DECLARED COMPLETION MEASURE — verdict B), promoted as the ledger row CELEST.WP5E.M2.01: the one-loop box coefficient per sector on  $\mathbb{P}T/\mathbb{Z}_4$  under the completion contact term  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  — the twisted-sector LOOP contributes the UNPHASED sector trace with the  $g^j$  zero-mode normalisation  $1/\det_j$ , of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part — (1) the COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class  $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$  EXACTLY ( $h_m = m(4-m)/8$  the v502 sector Casimir energies,  $\text{ch}_2(T_m)$  the McKay/Kronheimer charges, v505 S4.3 index bridge): THE THREE SPHERE AXIONS PAIR THROUGH THEIR OWN  $\text{ch}_2$  CHARGES — an exact identity, NO free scale, no fit; (2) PARAMETER-FREE LOCKS:  $T_5 = 0$  for ANY scale  $c$  (the  $\text{ch}_2$  pattern alone kills  $T_5$ ); ratio  $h_2 : h_1 = 4 : 3 =$  the  $\delta_1$ -forced leading ratio (REPRODUCED, not fitted); the  $T_3$  budget fixes  $c = 4 = |\mu_4|$  uniquely (leading system  $\det 256$ , unique weights  $(\frac{3}{2}, 2) = (4h_1, 4h_2)$ ); (3) the SUCCESS TABLE:  $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x \rangle^2/\det_j$  — a PERFECT Okubo square ( $T_5 = T_3 = 0$ ,  $\tilde{\lambda}^2 = 36$  in EVERY twisted channel:  $(18, 9, 18) \times \langle x, x \rangle^2$ ); total  $A_{\text{fix}} + \text{contact} = 45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo} =$  the UNIQUE quartic-free weighting of v505 S3.8 (the rigidity theorem is exactly what the measure lands on); both v508 certificates KILLED ( $\Phi_{T_3}: 32 \rightarrow 0$ ,  $\Phi_P: 72 \rightarrow 0$ ); the v511 slice  $\psi = 64$  SUPPLIED EXACTLY ( $\psi(\text{contact}) = -64$ ) — NO cubic  $d$ -channel needed ( $c_d$  free = 0); remainder inside the exchange span ( $45\langle x, x \rangle^2 = \frac{45}{16}E_{22} = \frac{1}{80}E_{00}$ ,  $\Phi_P = 0$ ): the BULK axion cancels every channel with the SAME  $\lambda^2 = 36$ , the twisted propagator carrying the AB zero-mode factor  $1/\det_j$ ; the v514 measure chain stays  $\mu$ -blind; (4) NEGATIVE CONTROLS: wrong scale  $c \in \{1, 2, 3, 5\}$  leaves  $T_3 = (24, 16, 8, -8)$  — only  $c = 4 = |\mu_4|$  clears the budget; the  $h_1 \leftrightarrow h_2$  shuffle breaks  $T_5 = -14$  AND  $T_3 = 36$ ;  $SO(16)$  glue keeps  $T_5 = 20$  /  $T_3 = -40$  — the KILL branch FIRES there (Okubo failure inherited:  $E_8$  doubly special);  $\text{diag}(i, i)$  degenerates to the  $\mathbb{Z}_2$  target and breaks the conjugation pairing + the index bridge; the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor PASSES with the same mechanism ( $A_2 + \text{contact} = 9\langle x, x \rangle^2 = \frac{1}{4} \times 36\langle x, x \rangle^2$ , coupling scale  $2 = |\mathbb{Z}_2| =$  centre count); (5) the MULTIPLIER NOTE (cross-reference to the parallel  $\delta_{1d}$  strand): the contact weights  $4h$  are REAL RATIONALS with no multiplier system of their own; the  $\delta_{1c}$  residual phases are the SQUARES of the  $A_3$  discriminant  $T$ -phases  $e(3a^2/8)$  — orders  $(1, 8, 2, 8)$  squaring to  $(1, 4, 1, 4)$ , a finite  $\mu_4$  system; Gauss sums  $2e(3/8)$ ,  $2e(5/8)$ , full Weil sum 4; [C] the MANDATORY FENCE: the completion reading (the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation) is DECLARED, supported by the  $\delta_1$  modular-completion finding, NOT derived from the BCOV integral; [O] the BCOV-integral derivation ( $\delta_{1b}/\delta_{1c}/\delta_{1d}$  territory); Wolfram extension  $498 \rightarrow 504$  (verified engine run); NO marker moves — the KILL (leftover independent quartic in ANY sector) does NOT fire, the completion reading stays [C]-declared, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched)

- v516\_celestial\_wp5e\_m2\_twisted\_ks\_measure.py (CELEST.WP5E.M2.01, 23 checks,  $\sim 1$  s; exact sympy/Fraction, no floats; deterministic). S0 replication (roots, class quartics,  $A_{\text{fix}}$  two routes, bulk Okubo); S1 the index bridge + the completion weights (no free scale, parameter-free locks); S2 the contact term (two forms, genuine-contact certificate); S3 the success table (per-sector Okubo squares, Dedekind  $\times$  Okubo, certificate kills, exchange span,  $\mu$ -blind chain); S4 negative controls; S5 the multiplier note (cross-reference to  $\delta_{1d}$ ); S6 verdict. Registered

in `run_all.py` (suite 511 modules together with v517, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M2.01 (new, SUCCESS on the declared measure).

- **Contract updated (markers unchanged).** CELEST.SEAM.01: the M2 bullet moves to *done*; the v511 burden paragraph is closed out — the  $\psi = 64$  slice is DELIVERED by the declared KS measure (v516); the cubic  $d$ -channel is not needed — its physical-justification question dissolves unless  $\delta_{1d}$  revives it. The completion reading itself stays [C]-declared; its BCOV-integral derivation stays [O] ( $\delta_{1d}$ ).
- **Paper integration.** `tfpt_research_contracts` (the M2 bullet moved to *done* with the weight identity and the success table; the v511 fence paragraph extended); `tfpt_3` (new Step 17 “the twisted KS measure: every sector the same Okubo square” — on the way to the eighteen-step story). `v516_celestial_wp5e_m2_twisted_ks_measure.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (completion-weight identity + index bridge; root/quartic replication; parameter-free locks; contact term + success table + certificates +  $\mu$ -blind chain; negative controls; multiplier note) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $498 \rightarrow 504$ ; GATE.WOLFRAM.02 and the Wolfram README synced.
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_twisted_ks_measure_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v516, changelog mirror); `next.txt` round notes.

2026-07-22 (V) (v515 — WP5e back-reaction milestone M1 “the  $A_3 \Omega_N$ ” executed (SUCCESS on the preregistered v514 S8.1 criterion), promoted as the ledger row CELEST.WP5E.M1.01: the back-reacted holomorphic 3-form  $\Omega_N$  is CLOSED FORM on the  $A_3$  twistor family, every  $S^3/\mathbb{Z}_4$  period is  $(2\pi i)^2$ -integral with the lockstep flux vector, and the lens geometry FORCES the source charge  $4 = |\mu_4|$  — (1) the RESIDUE FORM DERIVED: on  $F = XY - P(Z)$  all three ambient representatives satisfy  $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$  exactly and pull back to  $\omega_2 = dX \wedge dZ / X$  (Poincaré residue, derived not assumed); orbifold-cover pullback at  $a_0 = 0$ :  $\omega_2 = 4 dz_1 \wedge dz_2 = |\mu_4| \times (\text{flat form})$  — the residue normalisation CARRIES the source charge 4; clock  $\omega_2 \rightarrow i \omega_2$  (v492 S5 replicated); weight ledger  $4+4+2-8 = 2$ ,  $\Omega \in O(4) = -\deg K_{\mathbb{P}^T}$ ; (2) the TWISTOR FAMILY CLOSED: from the four  $O(2)$  centre sections  $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$  exactly  $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$  ( $e_1 = e_3 = 0$  identically;  $\lambda = 0$  gives exactly the v493 family;  $a_2 \in O(4)$  opens only off-seam), CY-compatible clock lift  $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$ ,  $\gamma^4 = 1$ ,  $\Omega \rightarrow +\Omega$ ; (3)  $\Omega_N$  CLOSED + PERIOD TABLE:  $\Omega_N = \Omega_0 + \sum_p N_p K_p$  with CPS/Bochner–Martinelli kernels on the centre twistor lines,  $dK = 0$  off-source; boundary  $S^3/\mathbb{Z}_4$  at the orbifold point  $(2\pi i)^2 \times 1$  (minimal invariant charge  $4 = |\mu_4|$  — the lens fundamental domain FORCES  $N \in 4\mathbb{Z}$ ), deformed phase  $(2\pi i)^2 \times 4$ , linking  $S^3$  uniformly  $(2\pi i)^2 \times N$ , per-sphere flux vector  $(2\pi i)^2 \times N(1, 1, 1)$  (forced uniform: clock orbit +  $K_4$  connectivity), seam-fibre phase vector  $2\pi i t_0 (i - 1)(1, i, i^2)$ ;  $\Omega_0$  carries ZERO 3-flux (all flux sourced);  $(2\pi i)^2 = \text{fibre residue} \times \text{transversal residue}$  (density  $-\sin 2\theta$ ); (4) CLOCK COVARIANCE:  $\Pi_{j+1}(-i\lambda) = i \Pi_j(\lambda)$  for ALL  $\lambda$ ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ( $8 = 2|\mu_4|$ , the  $\mathbb{Z}_8$  bridge), clock orbits  $4+4+4$ , honest conifold points (Hessian  $\det 512t_0^6$ , quadric rank 4); (5) the HONEST KILL PROBE: forbidden family  $Z^4 - Z$ :  $0/24$  lockstep (vs  $8/24$ ),  $\lambda$ -support 12 twelfth roots  $\neq \mathbb{Z}_8$ , no antipodal pairing — BUT the  $(2\pi i)^2$  quantisation holds there TOO: integrality alone is NOT the discriminator, the discriminator is the lockstep phase structure + the clock forcing; the v509 dial survives; (6) ANCHORS + CONTROLS: EH/ $\mathbb{Z}_2$  (charge  $2 = |\mathbb{Z}_2|$  forced), CPS original (every integer  $N$  allowed),  $\text{diag}(i, i)$  ( $\Omega_0 \rightarrow -\Omega_0$ , collapse at step zero), fractional charges  $N = 1, 2, 3$  excluded; [C] global kernel patching / Hitchin small resolution / CPS brane dictionary; [O] M2 (twisted KS measure), M3 ( $a_0$  uplift), the quantised BCOV coefficient; Wolfram extension  $490 \rightarrow 498$  (verified engine run); *NO marker moves — M2/M3 and BCOV stay [O], SEAM.EQUIV.TWISTOR.01 stays Open (v515 is preparation, not construction), SEAM.EQUIV.01 untouched*

- `v515_celestial_wp5e_m1_omega_n.py` (CELEST.WP5E.M1.01, 30 checks,  $\sim 1$  s; exact sympy, no floats; deterministic). S1 the residue form (three routes, cover factor 4, weight ledger, EH anchor, wrong-weight control); S2 the twistor family (closed form,  $O(2)$  reality, clock lift, section-level dichotomy); S3 periods (generic reduction, lockstep, covariance, collision ledger on  $\mathbb{Z}_8$ ); S4  $\Omega_N$  (BM anchor, lens descent, source curve, conifold nodes, M1 criterion); S5 the KILL probe (honest); S6 negative controls; S7 fences + verdict. Registered in `run_all.py` (suite 509 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M1.01 (new, SUCCESS).
- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M1 executed via v515 — the back-reacted  $\Omega_N$  is closed-form on the  $A_3$  twistor family, all  $S^3/\mathbb{Z}_4$  periods are  $(2\pi i)^2$ -integral with the lockstep flux vector and clock covariance, and the lens geometry FORCES the source charge  $4 = |\mu_4|$ ; honest fence: integrality alone does not discriminate — the discriminator is the lockstep phase structure ( $0/24$  on the forbidden family); M2/M3 stay [O]. SEAM.EQUIV.TWISTOR.01 dependencies extended by v515 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M1 bullet moved to *done* with the core

numbers; the object-diagram row  $\mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{P}\mathbb{T}/\Gamma$  extended by v515 with the M1 kill retired to  $\delta_1/\text{M2-M3}$ ; WP4 handover, WP5 narrative, honest status line and status inventory synced); `tfpt_3` (new Step 16 “the back-reacted  $\Omega_N$ : closed form, integral periods, forced charge 4” — the celestial story is now sixteen steps; “what remains” updated: M2/M3 are the remaining preregistered milestones); `tfpt_4_frontier` stage list synced. `v515_celestial_wp5e_m1_omega_n.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (residue-form identities; twistor family closed; period reduction + lockstep + covariance; collision ledger on  $\mathbb{Z}_8$ ; BM anchor + lens forcing  $N \in 4\mathbb{Z}$ ; source curve  $K_4$ ; conifold nodes; KILL probe +  $\text{EH}/\text{diag}(i, i)$  controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 490  $\rightarrow$  498; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 498/498).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m1_a3_omega_n_backreaction_probe.py` (30/30) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v515, papers.ts “16 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (IV) (external review integrated: route renamed, object diagram, group-extension definition, SEAM.EQUIV route split (MMST/TWISTOR ids, no status change),  $A_3$  role table, WOIT.OS.TWISTOR.01 contract — a pure structure/text integration, NO new physics claims, NO marker moves, suite unchanged at 508)**

An external expert review of the celestial/twistor branch confirmed the substance and asked for *semantic* precision — role separation, status clarity, and the exact naming of what is missing. Six changes plus one new central research contract, all editorial/structural:

- **(Ä1) Route renamed + the Woit bridge.** The `tfpt_3` celestial section and the contract heading are now “**The celestial and twistor continuum route**” (the old “from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow” survives as the narrative-arc subtitle). A new paragraph block in `tfpt_3`, “*The Woit bridge: chiral Wick rotation and the real structure*” [O], introduces Woit’s Euclidean Twistor Unification (arXiv:2104.05099) as a *named external programme* (chiral Wick rotation problematic for purely chiral theories; Euclidean-holomorphic theory on  $\mathbb{P}\mathbb{T}$ , Lorentzian theory via conjugation structure/boundary values), lists TFPT’s existing ingredients (RP v379, OS reconstruction,  $\rho$  = clock v492,  $\sigma$  = deck/seam reflection v510), and names exactly what is missing: a global anti-linear  $\Theta : \mathcal{A}(\mathbb{P}\mathbb{T}/\Gamma) \rightarrow \mathcal{A}(\mathbb{P}\mathbb{T}/\Gamma)$  with  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , plus RP of the interacting BCOV+SDYM functional — “until that exists, the Woit connection is a strong analogy, not a mathematical integration”. No namedropping as confirmation.
- **(Ä2) The object diagram.** `tfpt_research_contracts` (CELEST section, up front) now opens with the chain  $(\Sigma, \mu_4, \rho, \Theta) \rightarrow \mathbb{P}^1 \setminus \mu_4 \rightarrow \mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{P}\mathbb{T}/\Gamma \rightarrow \mathcal{A}_{\text{hol}} \rightarrow_{\text{OS}} \mathcal{A}_{\text{Mink}}$  as a table with one row per arrow (claim id, status, load-bearing modules v492/v493/v497–v514, kill test); the last two arrows are honestly [O] with WOIT.OS.TWISTOR.01 as the target — where mathematics ends and physics begins is visible at a glance.
- **(Ä3) The group extension, exactly.** A definition block:  $\Gamma_{\text{ALE}} = \langle \text{Deck} \rangle \cong \mathbb{Z}_4 \subset SU(2)$  (triholomorphic, preserves  $\Omega$ ; the quotiented  $\mathbb{Z}_4$ );  $\rho$  = Clock =  $\text{diag}(i, 1) \in U(2)$  (Kähler, *not* triholomorphic,  $\det = i$  rotates  $\Omega$  by  $i$  — v492 S5; normalises, is not quotiented); the global object is cyclic  $\langle \rho \rangle$  with  $\rho^2 = \text{Deck}$  (projectively  $\mathbb{Z}_4$ , spin lift  $\mathbb{Z}_8$ ; v492/v506/v507), plus the action table on  $K_{\mathbb{P}\mathbb{T}}$ ,  $\Omega$ , BCOV fields, open-string fields and boundary states, each entry [E] (arithmetic) or [O] (quantisation question); “ $\mathbb{P}\mathbb{T}/\mathbb{Z}_4$ ” is defined once for the whole contract (quotient by  $\Gamma_{\text{ALE}}$ , normalised by  $\langle \rho \rangle$ ) with a cross-reference sentence in `tfpt_3`.
- **(Ä4) SEAM.EQUIV route split (ledger restructuring, no status change in substance).** New ledger rows SEAM.EQUIV.MMST.01 (the conditional scaling-limit route — closed

modulo cited theorems, typing taken verbatim from the pre-split rows/prose; residual  $[O]$  = the cited continuum existence, v336) and SEAM.EQUIV.TWISTOR.01 (the open Costello–Li construction,  $[O]$ ; dependencies on CELEST.SEAM.01/WP modules and WOIT.OS.TWISTOR.01). SEAM.EQUIV.01 *remains* as the parent target with the OR logic as claim text: *closed if MMST.01 OR TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open  $\Rightarrow$  parent stays  $[O]$  as an unconditional claim.* Every surface that said “SEAM.EQUIV.01 closed modulo cited theorem” (status cards in introduction, tfpt\_1/3/4, the central status card, README, origin\_theory, tfpt\_2, tfpt\_5, website StatusPyramid/OpenGates/StatusMatrix/papers.ts/glossary/VerificationDag) now attaches the statement to the MMST route id — the assertions are unchanged, only the ids are clean.

- **(Ä5) The  $A_3$  role table.** In the contract:  $A_3^{\text{family}}$  (lattice family factor,  $\mathfrak{su}(4)$  flavour),  $A_3^{\text{ALE}}$  (ADE type of  $\mathbb{C}^2/\mathbb{Z}_4$ ), the three exceptional spheres (Coxeter/PL structure),  $D_5^{\text{carrier}}$ , and — as external reference rows — Woit’s  $SU(3)_{\text{color}}$  and  $SU(2)_{\text{weak}}$ : per pair either the connecting functor with its vN witness (McKay type-level; Kronheimer–Nakajima quiver v479) or an explicit “*not identical — no identification claimed*”; in particular  $A_3^{\text{family}} \not\cong SU(3)_{\text{color}}$ , and the Woit rows are pure programme references.
- **(Ä6) Anti-number-hunting programme note.** experiments/next.txt (dated 2026-07-22 (IV)): the next real gain lies at positivity, incidence, locality and the  $\Theta$  real structure — *not* at further factorisations; recorded as working discipline per the review.
- **New central contract WOIT.OS.TWISTOR.01 (Open, research contract; new ledger row).** In tfpt\_research\_contracts after CELEST.SEAM.01: input  $X_+ = (\mathbb{PT}/\Gamma)_+$ ,  $\mathcal{A}_{\text{hol}}$  (interacting open+closed twistorial algebra),  $\rho$  (order-4 clock),  $\Theta$  (seam-reflection-induced anti-linear real structure),  $\mu_{\text{BCOV}+\text{SDYM}}$ ; target theorem  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , RP of the gauge-invariant algebra, OS quotient  $\Rightarrow (\mathcal{H}, \Omega, U(\mathcal{P}_+^\uparrow), \mathcal{A}_{\text{Mink}})$  with positive Hilbert metric, positive energy, local causality, *one chiral generation without mirror doubling*, the TFPT charge lattice, internal  $SU(3) \times SU(2) \times U(1)$  action, and the  $(E_8)_1$  seam net as an *actual* boundary net; *seven* preregistered kill tests ( $\Theta$ -clock incompatibility; RP failure after gauge fixing; vector-like mirror generation; Penrose transform only free/self-dual; right character but wrong OPE/net equivalence;  $SU(2)$  stays a spacetime factor; the four  $\mu_4$  marks not incidence-compatibly extendable); typing “this is the actual bridge from compiler to physics — everything in CELEST.SEAM.01 is preparation for it”; honest scope fence (self-dual  $\neq$  full 4D physics: both helicities, generic amplitudes, local matter, full Einstein dynamics, EWSB, confinement are explicit non-claims of the celestial branch). Mirrored on the website (papers.ts section, glossary entries “Woit bridge” and “Real structure  $\Theta$ ”).
- **No marker moves; no new vN modules; suite unchanged (508 modules, green).**

2026-07-22 (III) (v514 — WP5e- $\varepsilon_1$  “the  $O(-2)$  bulk-axion slot” executed (verdict B), promoted as the ledger row CELEST.WP5E.EPS1.01: the  $s = 0$  tower field of the celestial route is now a CONSTRUCTION,  $\tilde{\lambda} = 6$  is pinned three ways, and the first honest back-reaction step is taken in Gibbons–Hawking form — (1) the EQUIVARIANT  $O(-2)$  SLOT: the pushforward ledger  $\pi_* O(-2)$  on  $\mathbb{PT}' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)$  gives per degree  $d$  exactly  $(d+1)^2$  classes = the exact wave-operator nullspaces (1, 4, 9, 16, 25, 36 for  $d = 0..5$ ), the v492 action on the fibre coordinates  $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$  with incidence-forced column weights  $(+1, +1, -1, -1)$  closes the equivariant bookkeeping BLOCK BY BLOCK (all  $d \leq 6$ , all 4 characters); character series  $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots$ ,  $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots$ ,  $P_2 = 6t^2 + 10t^4 + 28t^6 + \dots$ ; the  $d = 0$  slot has multiplicity (1, 0, 0, 0) — THE BULK AXION SURVIVES THE PROJECTION; Molien invariant ring  $= (1 - t^8)/((1 - t^4)^2(1 - t^2))$  —  $Z \in O(2)$ ,  $X, Y \in O(4)$ , relation degree 8 = the  $a_0$  weight (v492/v493 bridge); E3 bijection sharpened (twisted minimal content  $(2t, 6t^2, 2t) =$  the Coxeter eigenvalues,  $\det(A - 1) = -4$ ); graviton control:  $O(+2)$  invariant only from fibre degree 4 with multiplicity  $3 = \{X, Z^2, Y\}$ ; (2)  $\tilde{\lambda} = 6$  TRIPLY PINNED: Okubo  $\text{Tr}_{\text{adj}} X^4 = 36\langle x, x \rangle^2 = (6\langle x, x \rangle)^2$  on the 240 glue roots ( $36 = h^\vee + 6$ ,  $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\text{fam}}$ ); the measure chain cancels  $\mu = 1/4$  EXACTLY (vertex<sup>2</sup>  $\mu^2$ , propagator  $1/\mu$ , anomaly  $\mu$ ;  $\tilde{\lambda}_{\text{eff}}^2 = 36$  for arbitrary  $\mu$ ; wrong bookings EXCLUDED: anomaly-only  $\Rightarrow 3$ , missing propagator renormalisation  $\Rightarrow 12$ ); flux (minimal CPS quantum  $N = 1$ , single channel iff  $k = 1$ ,  $(\kappa/c_3)^2 = 12$  exact); (3) the GH/ $A_3$  BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches, the free orbit re-derives the v493 family ( $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$ ,  $a_0 = -z_0^4$ , monodromy = one clock step), the charge- $k$  GH point is exactly  $\mathbb{R}^4/\mathbb{Z}_k$  ( $k = 4$ : seam boundary  $S^3/\mathbb{Z}_4$ ), periods  $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$  LOCKSTEP (v509 counter-check), Coxeter clock from geometry ( $\Pi \cdot A = i\Pi$ ), source charge  $4 = |\mu_4|$ , EH asymptotic log coefficient EXACTLY 0 (the CPS log is an exceptional-locus statement; Burns contrast  $\det g \neq 1$ ), multipole rule  $m \equiv 0 \pmod{4}$  with  $(4, \pm 4)$  carrying  $-a_0$ ; controls  $\mathfrak{so}_8$  ( $\tilde{\lambda}^2 = 12$  irrational, squares only  $\{9, 36\}$ ),  $\text{diag}(i, i)$  (Veronese),  $k = 2$  (three channels); [C] conditional on Costello’s flat- $\mathbb{PT}$  matching; [O] the quantised BCOV coefficient and the twisted channels ( $32 T_3$ ) — M1–M3 preregistered (the  $A_3 \Omega_N$  / the twisted KS measure / the  $a_0$  uplift, each with success + kill); Wolfram extension  $482 \rightarrow 490$  (verified engine run); *NO marker moves* — WP5e stays [O], SEAM.EQUIV.01 untouched, CELEST.SEAM.01 stays Open)

- v514\_celestial\_wp5e\_eps1\_axion\_slot.py (CELEST.WP5E.EPS1.01, 34 checks,  $\sim 4$  s; exact sympy/Fraction, no floats; deterministic). S1 Penrose accounting; S2 equivariant ledger + character series + Molien + E3 sharpening + graviton control; S3 the  $\tilde{\lambda} = 6$  residue (Okubo / measure / flux) with honest gaps; S4–S5 GH centre dichotomy, periods, Coxeter clock; S6 the CPS “ $N$  log” ledger (kernel charge, multipoles, EH log); S7 negative controls; S8 the M1–M3 roadmap; S9 verdict. Registered in run\_all.py (suite 508 modules, green), script\_registry.csv (cluster frontier), ledger row CELEST.WP5E.EPS1.01 (new, verdict B).
- Contract updated (markers unchanged). CELEST.SEAM.01 now reads:  $\varepsilon_1$  executed via v514 — the slot is a construction,  $\tilde{\lambda} = 6$  pinned by three exact ledgers, the GH/ $A_3$  step re-derives the v493 family and the Coxeter clock from geometry; the quantised BCOV coefficient and the twisted channels stay [O] (M1–M3 preregistered).
- Paper integration. tfpt\_research\_contracts (the  $\varepsilon_1$  stage moved to done with its own stage block; the named remaining targets now carry the M1–M3 milestones verbatim; WP4 handover and honest status line synced); tfpt\_3 (new Step 15 “the bulk axion is a construction;  $\tilde{\lambda} = 6$  is pinned three ways” — the celestial story is now fifteen steps; “what remains” updated).

v514\_celestial\_wp5e\_eps1\_axion\_slot.py is cited via veri in both.

- **Wolfram.** Eight exact mirrors (equivariant Penrose ledger; character series + bulk-axion survival; Molien hypersurface + Veronese control; Okubo perfect square + `so8` control; measure cancellation excluding 3 and 12 + flux dial; GH centre dichotomy; period lockstep + Coxeter clock; multipole rule + EH log ledger) added to `tfpt_readouts_extension.wl` and verified with the active engine, 482  $\rightarrow$  490; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 490/490).
- **Provenance.** Discovery probes `celestial_seam_wp5e_eps1_o2_bulk_axion_probe.py` (20/20) and `celestial_seam_wp5e_eps1_gh_a3_backreaction_probe.py` (14/14) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node, `papers.ts` “15 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (II) (v513 — the  $c_d$  negative certificate, promoted as the ledger row CELEST.DTERM.NONDERIV.01 (anti-numerology certificate in the v508 style, honest rigid NEGATIVE result): can geometry derive  $c_d = 1920$ ? NO — the v511 [C] fence “ $1920 = |W(D_5)|$ ” does NOT survive the honest four-route battery with its PRE-DECLARED look-elsewhere ledger: (1) the CONVENTION TABLE gives  $c_d \in \{1920, 120, 256, 32, 69120\}$  across five normalisations (probe `tr-fund/ $e_8$ -Killing`; Gell-Mann; `su(4)-Killing`; plain trace; `ch3-normalised  $d/6$` ) — only 1 of 5 hits  $|W(D_5)|$ , while the convention-STABLE content is the factorisation  $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$ , both factors [E] (Killing propagator 60 from  $\sum_\alpha \langle \alpha, e \rangle^2 = 120$  over the actual roots;  $T_3$ -slot fixing via  $\det(\mathbf{P}\text{-block}) = -64 \neq 0$ ); (2) K1 DEAD: the charge-legal flux pairings (1, 3)/(2, 2) give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit  $64 \cdot 60/2$  is the FORBIDDEN (1, 2) pairing (charge 3 mod 4, v511 innerness); look-elsewhere: 924 expressions from 21 [E] numbers, 11 hits on 1920, controls 1800  $\rightarrow$  8, 2048  $\rightarrow$  5, 2016  $\rightarrow$  0; (3) K2: quantisation delivers only lattices ( $60\mathbb{Z}$  with 1920 the 32nd multiple, no minimality; strict `ch3` units  $2160\mathbb{Z}$  EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, 3/8 on weights), the integer 32 has 12 factorisations; (4) K3 PROVENANCE CLASH (the death blow): cubic indices  $A_m = (0, 16, 0, -16)$  with  $\sum_m i^m A_m = 32i$  numerically  $= \Phi_{T_3}(A_{\text{fix}})$  — but the quartic class decomposition (10, -4, 30, -4) comes from classes 0/2 vs the cubic 16+16 from classes 1/3: same number, disjoint mechanism = coincidence; the Weyl side is WRONG ( $d$ -channel is  $A_3$ ,  $|W(A_3)| = 24$ ;  $D_5$  has no cubic;  $|W(E_8)|/|W(A_8)| = 1920$  a second accidental 1920); (5) K4 controls ( $SO(16)$ ,  $k = 2$  doubling 7680, false  $g_0$  with six cubics, `so(10)`) separate consistently; VERDICT: NUMEROLOGY as a derivation claim — the v511 [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is  $c_d = 32 \times 60$ ; Wolfram extension 476  $\rightarrow$  482 (verified engine run); *NO marker moves — the fence stays [C], the physical generation of the 32 stays [O], CELEST.SEAM.01 stays Open*)**

- `v513_celestial_dterm_nonderivation.py` (CELEST.DTERM.NONDERIV.01, 24 checks,  $\sim 2$  s; exact `sympy/Fraction`, no floats; deterministic). N0 baseline (240 roots (52, 64, 60, 64),  $A_{\text{fix}} = (9, -30, -15, 0, 32)$ ,  $Q_{dd}$  two routes,  $c_d$  pinned by the  $T_3$  slot alone); N1 convention table + the stable factorisation  $32 \times 60$ ; K1 flux products + look-elsewhere ledger; K2 quantisation lattices; K3 cubic/quartic provenance clash + Weyl data; K4 negative controls. Registered in `run_all.py` (suite 507 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.DTERM.NONDERIV.01 (new).
- **Fence sharpened (text only, marker unchanged).** CELEST.WP5E.DELTA2.01 and the CELEST.SEAM.01 contract prose now carry: “the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded (11/924 vs. 8/924 for the control target 1800) and convention-contingent — v513”; the



honest remaining target of the cubic-GS item is the physical generation of the factor 32, not the Weyl fingerprint.

- **Paper integration.** `tfpt_research_contracts` ( $\delta_2$  fence environment, named remaining targets, status list — new CELEST.DTERM.NONDERIV.01 status line); `tfpt_3` (Step-14 paragraph + the “what remains” sentence); `tfpt_4_frontier` (celestial-route clause). `v513_celestial_dterm_nonderivation.py` is cited via `veri` in the  $\delta_2$ /fence environments.
- **Wolfram.** Six exact mirrors (propagator anchor +  $T_3$ -slot pinning; five-way convention table; flux enumeration + look-elsewhere counts; K2 lattices; K3 provenance clash + Weyl data; K4 controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 476  $\rightarrow$  482; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 482/482).
- **Provenance.** Discovery probe `celestial_seam_wp5e_dterm_coefficient_derivation_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 24/24); website mirrors regenerated (ScriptIndex, DAG *qft* node, `papers.ts`, changelog mirror); `next.txt` round notes.

**2026-07-22 (I) (v512 — the flag-transitivity equivalence web promoted as SEAM.TAU.FLAG.01: the  $\tau = i$  attack executed on the free RP seam circle — BARE MARK-TRANSITIVITY IS FALSIFIED as a selector (on the test family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ ,  $\delta \in (0, \pi/2]$ , the symmetry group is the pair-exchanging  $V_4 = \{z, -z, b/z, -b/z\}$  for EVERY  $\delta$ , simply transitive on the four marks; complete census: deck-commuting Möbius maps are diagonal/antidiagonal; hardening: ALL undirected per-mark data are identical for every  $\delta$  — no local jet sees the bit), and THE SHARP SELECTOR is FLAG transitivity (marks AND their two sides indistinguishable)  $\iff \delta = \pi/2 \iff \tau = i \iff$  the clock — stabiliser jump  $1 \rightarrow 2$ , flag/arc orbits  $2 \rightarrow 1$ , order-4 rotation ( $b^2 = -1$  solveset  $\{\pi/2\}$ ), mark mirror  $z \mapsto 1/z \iff \cos \delta = 0$ ; welded into a 12-FOLD EQUIVALENCE WEB (every link exact): clock rotation,  $D_4$ , mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split  $= (2/\pi) \ln \cot(\delta/2)$ , zero only at  $\pi/2$  — counterwitness exactly  $(2/\pi) \ln 2$ ; arc-Laplacian degeneracy  $4/\pi$ ; the v215-K3 indicator in CLOSED FORM  $(2\sqrt{2}/\pi) |\ln \tan(\delta/2)|$  — the v503 lattice curve analytically),  $\rho$ -twist existence  $((m^2 - n^2)(t - 1/t) = 0 \iff t = 1$ ; Atiyah–Bott rationality  $\delta$ -BLIND —  $\text{Tr}(\sigma e^{t\Delta}) = 1$  for all  $\delta$ ,  $j_{\text{cw}} = 148176/25$  rational non-CM: rationality selects NOTHING), cusp-4-cycle implementability (H4 rigidity:  $V_4$  with orders  $\{1, 2, 2, 2\}$  cannot induce it; centraliser scalar,  $U^2$  nonscalar  $\Rightarrow$  no projective involution — projective order exactly 4,  $\mathbb{Z}_8$ -shadow consistent v510), harmonic,  $j = 1728$ ; NEGATIVE CONTROLS: hexagonal (not concyclic, fails preconditions), silver (harmonic but topologically dead), 3 marks (impossible on the free circle: 0 vs 28 invariant subsets; bare  $n = 3$  transitivity already forces equal spacing — “4 counts”); [C] THE REST: the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ( $V_4 \rightarrow D_4$ : side-indistinguishability of the four equally-decorated marks), v216 local equality does NOT deliver it, its negation is now concretely measurable; scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones; Wolfram extension 470  $\rightarrow$  476 (verified engine run); *NO marker moves — the bit stays [C], QGEO.REALIZE.01 and P2.TYPING.01/AX.P2.01 text-precised only, SEAM.BIT.FREEDOM.01 cross-referenced)***

- `v512_seam_tau_flag.py` (SEAM.TAU.FLAG.01, 29 checks,  $\sim 6$  s; sympy-exact + mpmath 30 digits + one numpy lattice commutator; deterministic). S0 family frame (counterwitness inside the family:  $\lambda = 5/4$ ,  $j = 148176/25$ , cross-ratio  $-1/4$ ); H1 complete symmetry census (census basis, rotation solvesets, stabiliser jump, flag/arc orbits, v216 separation, five-member

table); H2 spectral criteria in closed form; H3 heat-trace rationality ( $\delta$ -blind) + twist existence; H4 monodromy rigidity (Schur); NC negative controls; SYN the 12-fold web + counterwitness scoreboard + the typed sentence. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.TAU.FLAG.01` (new).

- **Reduction state re-stated (text only, markers unchanged).** `AX.P2.01` and `P2.TYPING.01` synthesis notes now read: “four marks + one discrete symmetry-lift bit (flag transitivity  $\iff \tau = i) + \pi$ ”; `QGEO.REALIZE.01` precision: “the remaining input can equivalently be stated as the flag-transitivity postulate — `v512`; the search space for a future `[E]` closure is narrowed to the RP positivity of the odd sector”; `SEAM.BIT.FREEDOM.01` cross-referenced (the modulus half gains the discrete form).
- **Paper integration.** `tfpt_1` (the P2 reduction keybox re-dated 2026-07-22 + Theorem-7 skeleton keybox), `introduction` (narrative chain, consolidated reduction state, status-card addendum), `tfpt_3` (celestial Step 2 + the honest-status reduction bullet), `tfpt_research_contracts` (QGEO kill-test and marks-derived blocks, WP2 note); secondary mirrors `tfpt_2/origin_theory` aligned. `v512_seam_tau_flag.py` is cited via `veri` in all four primary documents.
- **Wolfram.** Six exact mirrors ( $V_4/D_4$  census table incl. flag/arc orbit counts; solveset equivalences + family identities; spectral faces in closed form; twist mode identity + AB  $\delta$ -blindness +  $j$ -rationality control; H4 Schur rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 470  $\rightarrow$  476; the `mpmath` heat-trace numerics and the `numpy` lattice commutator stay Python-only (flagged).
- **Provenance.** Discovery probe `seam_tau_i_established_selector_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 29/29); website mirrors regenerated (`ScriptIndex`, DAG *P2/masses* nodes, `papers.ts` reduction-state mirrors, `StatusMatrix`); `next.txt` round notes.

2026-07-21 (XXIII) (v511 — WP5e- $\delta_2$  of CELEST.SEAM.01 “the full-tensor ledger” promoted as CELEST.WP5E.DELTA2.01 (intermediate verdict, exact): the named escape route of the v508 exchange no-go executed on the FULL adjoint tensor structure of  $\mathfrak{e}_8$  — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  semisimple with NO  $\mathfrak{u}(1)$  (class-0 roots 40+12, root span rank 8),  $h$  in the Cartan OF  $\mathfrak{g}_0 \Rightarrow$  every invariant tensor carries total charge 0 mod 4: the bilinear Hom table is nonzero ONLY at  $j' = -j$  (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have Hom = 0), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ( $\{0, 0, 0\}: 3 = 1$  sym + 2 antisym,  $\{0, 1, 3\}: 2$ ,  $\{0, 2, 2\}: 2$ ,  $\{1, 1, 2\}: 1$ ,  $\{2, 3, 3\}: 1$ ; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir,  $\text{Inv}(S^3 45) = 0$  protects  $T_5$ ), whose exchange quartic  $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$  (TWO independent routes, Killing propagator  $2h^\vee = 60$  from the roots) has  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$  — the v508 master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$  ( $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  rank 3, augmented 4), and becomes EXACTLY solvable relaxed:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$  (residual 0); [C] fences  $64 = \dim \mathfrak{g}_1 = 2^6$ ,  $1920 = |W(D_5)| = 8 \cdot 240$ ; controls:  $SO(16)/D_8$  has NO symmetric cubic and no  $A_3$  block ( $E_8/\mu_4$  doubly special), false  $\mathfrak{g}_0$ /sector assignments separate; the  $\delta_1$  burden SHARPENED to the  $\psi = 64$  slice or the selection-rule relaxation — and the  $\delta_1$  exploration run itself stands UNDECIDED at exploration level (strict holomorphic  $q^0$  reading REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore  $\tau$ -integral with forced leading  $(T_5, T_3)$  ratio 4:3; massive coset levels  $n \geq 2$  mandatory); Wolfram extension  $463 \rightarrow 470$  (verified engine run); *NO marker moves — WP5e proper stays [O], SEAM.EQUIV.01 stays [O]*

- `v511_celestial_wp5e_delta2_tensor_ledger.py` (CELEST.WP5E.DELTA2.01, 41 checks,  $\sim 4$  s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic). D0/D1 structure: 240 roots, class split (52, 64, 60, 64);  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  SEMISIMPLE without  $\mathfrak{u}(1)$  (40+12 class-0 roots, no mixed root, root span rank 8); sectors  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$  as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ( $h = (2, 2, 2, 2, 2; 0^4)$  in the  $\mathfrak{g}_0$  Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at  $j' = -j$  (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims (3, 2, 2, 1, 1) with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\text{Inv}(S^3 45) = 0$ ,  $\text{Inv}(S^3 \mathfrak{a}_3) = 1$ ). D5 the new channel:  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  by Killing-propagator contraction AND the hand route  $|x^2 - (\text{tr } x^2/4)\mathbf{1}|^2/60$ ;  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ ; brackets/tadpoles/ $d \times f$  die exactly; no  $T_5$  analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate  $\psi(A_{\text{fix}}) = 64$ ; relaxed reading EXACTLY solvable with  $c_d = 1920 = 32 \times 60$ . D7 controls: Killing control = v508 replication ( $A_{\text{fix}}$ ,  $E_{13}$ ,  $E_{22}$ ,  $E_{00} = 225E_{22}$ );  $SO(16)/D_8$  bilinears (1, 0, 1), trilinears (1, 0, 1, 0),  $\text{Inv}(S^3 \text{adj}_{D_8}) = 0$ ; false  $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$ :  $(5, 2, 2) \neq (2, 1, 1)$ ,  $\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$ ; false sector content  $(16_s, \bar{4})$  kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py` (41/41), same day.
- The  $\delta_1$  fence (exploration-level, NOT promoted — verdict UNDECIDED). The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence

text: (i) the strict holomorphic  $q^0$  reading is REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading  $(T_5, T_3)$  ratio is 4:3 (sector 2 : sectors 1+3); (iii) the  $SL(2, \mathbb{Z})$  orbit decomposition  $15 = 12 + 3$  identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type  $\tau$ -integral), not a free number; (iv) massive coset levels  $n \geq 2$  are mandatory. The remaining  $\delta_1$  target is the  $\tau$ -integral, evaluated against the  $\delta_2$ -sharpened criterion ( $\psi = 64$  slice or selection-rule relaxation).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.DELTA2.01 ([E] Hom tables/innerness/ $Q_{dd}$ /certificate; [C] the number fences; [O] arity  $\geq 4$ , cubic GS term,  $\delta_1$ ); CELEST.SEAM.01 updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic  $d$ -symbol channel opens  $T_3$  with the weaker certificate  $\psi = 64$ ; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed  $\delta_2$  stage + the  $\delta_1$  exploration fence, WP5e-proper targets renamed ( $\tau$ -integral, cubic GS term,  $\varepsilon_1$ , back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 “the full-tensor ledger: the collapse is real, and one cubic door opens”) with the “What remains” block updated. Wolfram: seven mirrors (innerness count +  $\mathfrak{g}_0$  structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the  $SO(16)$ /false- $\mathfrak{g}_0$ / false-sector controls;  $Q_{dd}$  by two routes; the  $\psi$  certificate + the relaxed solution  $(-1, 8, 2; 1920)$  verified with the active engine the same day ( $463 \rightarrow 470$ , ALL WOLFRAM EXTENSION CHECKS PASSED). Registry + `run_all` (505 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves): the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ( $\tau = i$  — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v491/v499/v506/v507/v510) and  $\pi$ , plus the [C] identifications R1/R2 of v499; AX.P2.01 stays a DECLARED axiom and QGEO.REALIZE.01 stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)**

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (v491:  $4 = 2+1+1$  unique  $\Rightarrow g_{\text{car}} = e_2 = 5$ ), the weight typing (v499: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , residuals R1/R2 [C]), the residual convergence (v503: QGEO-R1  $\equiv$  P2-R1), the clock rigidity and bit origin (v506/v507: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (v510: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler’s dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ( $\tau = i$ , equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and  $\pi$  — plus the [C] identifications R1/R2 of v499.* No new claims; conservative wording; no “axiom abolished” language anywhere.
- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox “The current reduction state of P2” next to the foundational-postulates keybox, with v491/v499/v506/v507/v510 citations. `tfpt_3`: mirror bullet “The current reduction state” in the honest-status section. **introduction**: the anchor paragraph gains the consolidated one-sentence state and the status card row  $a = (1, 1, 2)$  now names the reduction state. Ledger: SYNTHESIS NOTE appended to AX.P2.01 (“reduction state 2026-07-21: four marks + one square-modulus datum ( $\tau = i$ ) +  $\pi$ ”), marker *Axiom unchanged*. Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

**2026-07-21 (XXI) (v510 — the freedom attack on the v506/v507 alignment bit promoted as SEAM.BIT.FREEDOM.01: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the  $\mathbb{Z}_2$  seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in  $D_{16}$ , exactly 1 free;  $\text{Aut}(C_{16}) = D_{16}$  complete census, brute force 8! at  $N=8$ ; quotient topology circle/antipode = CIRCLE  $\chi=0$  vs circle/reflection = INTERVAL  $\chi=1$  — the  $6\pi$  Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT  $\iff$  FREE over the complete 17-involution census in exact  $\text{Cl}(16)$  (free  $\Rightarrow \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F$  nonsplit with 2 roots = the  $\mathbb{Z}_8$  tower; ALL 16 reflections  $\Rightarrow U^2 = \text{scalar } \{2^7, 2^8\}$ , split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two;  $\mu_4 \rightarrow$  equatorial, deck FREE; silver  $\rightarrow$  real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal  $\rightarrow$  NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE  $\Rightarrow \mu_4$  exactly ( $b = \pm i$ ); honest counterwitnesses (“nonsplit  $\Rightarrow$  clock” is FALSE:  $\{0, 1, 8, 9\}$  and  $\{\pm 1, \pm(3+4i)/5\}$  with  $j = 148176/25 \neq 1728$ ); the alignment bit REDUCES from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; Wolfram extension 457  $\rightarrow$  463 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays  $[C]/[O]$ , AX.P2.01 stays an axiom, P2-R1 halved)**

- **v510\_seam\_bit\_freedom.py (SEAM.BIT.FREEDOM.01, 30 checks, < 5 s; sympy + exact  $\text{Cl}(16)$  Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in  $D_N$  on the refined  $2N$ -point seam circle ( $N = 16$ : 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness  $N = 8, 12$ ; odd control  $N = 9$ : ZERO free — an even carrier count is needed, consistent with  $16 = 2^{g_{\text{car}}-1}$ ); the census is COMPLETE because  $\text{Aut}(C_{16}) = D_{16}$  exactly (brute force over all 8! = 40320 permutations at  $N = 8$ , propagation at  $N = 16$ ); quotient topology: circle/antipode = CIRCLE ( $\chi = 0$ , closed), circle/reflection = INTERVAL ( $\chi = 1$ , two boundary corners) — a fixed-point deck would break the  $6\pi = 3 \times 2\pi$  closed-circle count, so “deck free on the seam circle” is exactly the established “the seam is a closed circle”, not a new premise; the complete Fock census: ALL 17 involutions implemented in exact  $\text{Cl}(16)$  — NONSPLIT  $\iff$  FREE with no exception, root dichotomy 2 vs 0; the chain: free  $\Rightarrow$  shift 8  $\Rightarrow$  crossing (all 28 invariant 4-sets)  $\Rightarrow U^2 = (-1)^F \Rightarrow$  central class  $\Rightarrow$  the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch  $\mathbb{Z}_2$  for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the  $\mathbb{Z}_2$  seam fibre; one-sidedness TYPES the  $\mathbb{Z}_2$  as the free antipode (quotient  $\mathbb{RP}^2 = \text{the } |\mathbb{Z}_2| \text{ of } c_3 = 1/(|\mathbb{Z}_2| 2\pi\chi)$ ); the deck-commuting real structures are EXACTLY two families ( $\mu\bar{z}$ : fix line through the poles, NOT free;  $\mu/\bar{z}$ ,  $\mu > 0$ : fix circle  $|z| = \sqrt{\mu}$ , FREE;  $\mu < 0$ : empty); the per-configuration table ( $\mu_4$  equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to  $b = \pm i$  (on free circles the silver family “harmonic AND edge” is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete  $\{0, 1, 8, 9\}$  and the continuous  $\{\pm 1, \pm(3+4i)/5\}$  (free, central,  $j = 148176/25$ , no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck  $\Rightarrow$  free [standard topology; the one-sidedness strand `origin_theory/v456`]; (P-ii) marks on the  $\Theta$ -fix seam circle = existing REALIZE [C] content (QGEO.REALIZE.02 hypotheses, v264) — NO new input. Audit finding documented: “circle-freeness” was nowhere typed before (only mark-freeness, v216); v507 F1.1 explicitly admitted the reflection reading of the edge class — v510 closes that door (a

reflection is not a covering deck). The [C] rest, verbatim: *the alignment bit reduces from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.FREEDOM.01 ([E] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [C] the square-modulus datum); TEXT precisions (markers unchanged) in SEAM.CLOCK.RIGIDITY.01, SEAM.BIT.ORIGIN.01, QGEO.REALIZE.01 (bit  $\rightarrow$  square-modulus datum; edge class topologically excluded) and P2.TYPING.01 (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended (“the position half is topology”), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where v506/v507 are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on  $256 \times 256$  Jordan–Wigner gammas; the real-structure table; harmonic + free  $\Rightarrow \pm i$ ; the counterwitnesses) verified with the active engine the same day (457  $\rightarrow$  463, ALL WOLFRAM EXTENSION CHECKS PASSED; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves.

**2026-07-21 (XX) (v509 — WP5e- $\varepsilon_2$  of CELEST.SEAM.01 “the CPS level-from-flux dial” promoted as CELEST.WP5E.EPS2.01, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism (“level = flux quantum  $\times$  Dynkin index”) executed on the lockstep spheres — the CPS SKELETON exact ( $S^3$  period  $(2\pi i)^2 N$  with pullback density  $-\sin 2\theta$ , exceptional-sphere Kähler flux  $2\pi N$ , boundary level magnitude  $2N$  by the symplectic-boson contraction,  $T(\text{vector}) = 1$ ); the PAIRING MATRIX pinned, not postulated (ch<sub>2</sub> ledger + integrality + unimodularity + effectivity  $\Rightarrow P_{mi} = \delta_{mi}$  by complete enumeration: rows 8/6/8, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  with uniformly ONE quantum per charged current; “level = total flux” KILLED by the lockstep test itself ( $F_i$  unequal = flavour multiplicity), the per-current reading (1, 1, 1) anchored by the lattice current count (240, 0, 0, 0) and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors:  $\mu = 16 = \text{ord}^2$ , Lagrangian  $\mu_4$  glue, KLM 16  $\rightarrow$  1) with the NEW sector-counter dial  $\#_{\text{prim}}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$  — exactly ONE sector iff  $k = 1$ ; the LOCKSTEP THEOREM lifts the v505 “one scale  $\Rightarrow$  one level” to a THEOREM of clock invariance ( $|\Pi_j| = \sqrt{2}t$  equal, monodromy = pure phase  $i$ ,  $\det(A-1) = -4$ : no invariant flux vector; NEW falsifier  $Z^4-Z$ : 0/24 orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls  $k=2/SO(16)/A_2$  separate honestly; Wolfram extension 450  $\rightarrow$  457 (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on  $\mathbb{P}^T/\mathbb{Z}_4$  typed [C], the type-I B-model back-reaction [O]), SEAM.EQUIV.01 stays [O])***

- `v509_celestial_wp5e_eps2_level_from_flux.py` (CELEST.WP5E.EPS2.01, 28 checks,  $< 1$  s; sympy/Fraction exact, no floats; deterministic). Target: derive the boundary level  $k = 1$  from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL 130, 061602; arXiv:2306.00940). *CPS skeleton replicated exactly*: the  $S^3$  period of the brane back-reaction 3-form is  $(2\pi i)^2 N$  (their eq. 3.33; large-gauge invariance quantises  $N$ ), the exceptional-sphere Kähler flux is  $2\pi N$  exactly ( $K = |u|^2 + N \log |u|^2$ , the flat part dies at  $t \rightarrow 0$ ), and the boundary level magnitude is  $2N$  via the symplectic-boson double contraction ( $N = 1, 2, 3$ ; their eq. 9.4), Dynkin normalisation  $T(\text{sog vector}) = 1$ ,  $T(\text{adj}) = 6$ . *Pairing matrix pinned*: the v505 ch<sub>2</sub> ledger ( $f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$ )

+ integrality + unimodularity + effectivity force  $P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of  $PC^{-1}P^T = C^{-1}$ ; with  $P \geq 0$  exactly 2: identity +  $A_3$  diagram flip); fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ , total  $188 = 248 - 60$ , flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: “level = total open-string flux” is KILLED by the lockstep test itself ( $k_i = (64, 60, 64)$  unequal —  $F_i$  is the FLAVOUR multiplicity, the brane matter content); what works:  $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$ , anchored non-circularly by the lattice current count  $(240, 0, 0, 0)$  for  $k = 1.4$  (v505 S4.8) and the embedding index  $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$  ( $60 = 2h^\vee$ ). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ( $\mu = \det(D_5) \det(A_3) = 16 = \text{ord}^2$ ; the  $\mu_4$  glue diagonal isotropic + Lagrangian; KLM  $16 \rightarrow 16/4^2 = 1$ , zero residual fractional flux); NEW DIAL:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1.6$  from the affine marks — exactly ONE sector iff  $k = 1$  (independent of the CPS transplantation). *The lockstep theorem* (the lifting of v505 S4.10): clock invariance  $P(iZ) = P(Z)$  forces  $a_3 = a_2 = a_1 = 0$  sharply, the four branch points are ONE  $\mu_4$  orbit,  $|\Pi_j| = \sqrt{2}t$  EQUAL on all three spheres, the monodromy is the pure phase  $i$  and  $\det(A-1) = -4 \neq 0$  (no invariant flux vector) —  $k_1 = k_2 = k_3$  IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family  $Z^4 - Z$  (smooth, disc  $-27$ , branch points  $\{0, 1, \omega, \omega^2\}$  not a  $\mu_4$  orbit): over ALL 24 orderings ZERO chains with three equal period moduli ( $1$  vs  $\sqrt{3}$ ); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value ( $1$  from the counters and the CPS minimal quantum  $N = 1$ ). *Negative controls*:  $k = 2$  (current count 0, residual 360, prefactor  $-31/48$ ,  $\#\text{prim} 3$  — while the lockstep dial honestly does NOT fire);  $SO(16)$  (fluxes  $(0, 60, 0)$ : spheres  $1/3$  FLUX-DARK, defect  $-30 \neq -78$ , condensation stuck at  $4 = \det \text{Cartan}(D_8)$ );  $A_2$  ( $\det 3 \neq 4$ , glue diagonal  $23/24 \notin \mathbb{Z}$ ,  $\mu = 12$  no perfect square: NO Lagrangian subgroup, Coxeter  $3 \neq 4$ ). Typed honestly: the CPS dictionary itself on  $\mathbb{PT}/\mathbb{Z}_4$  stays [C] (their geometry is the  $\mathbb{C}^2$  blow-up with one  $(-1)$ -sphere and  $\mathfrak{so}(8)$  at negative level — not the  $A_3$  ALE); the type-I B-model back-reaction stays [O] (the same fence as v502/v505).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.EPS2.01 ([E] CPS skeleton/pairing/lockstep/sector arithmetic + [C] CPS dictionary + [O] back-reaction); CELEST.SEAM.01 updated (“ $\varepsilon_2$  executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier  $Z^4 - Z$ ), level = 1 carries the new sector-counter dial  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ; the  $\mathbb{PT}/\mathbb{Z}_4$  back-reaction stays [O]”); the v505 “one scale  $\Rightarrow$  one level” sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the  $\varepsilon_2$  block (done, verdict B; roadmap now  $\delta_1/\delta_2/\varepsilon_1$  + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 “one level is a theorem; the sector counter pins  $k = 1$ ” and the “What remains” paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the  $2N$  contraction + Dynkin normalisation; the pairing enumeration  $48 \rightarrow 2$ ; the fluxes  $(64, 60, 64)$ ; the  $\#\text{prim}$  table  $(1, 3, 5, 10, 15, 27)$ ; the lockstep periods + the  $0/24$  negative probe; the  $SO(16)/A_2$  controls) verified with the active engine the same day ( $450 \rightarrow 457$ , ALL WOLFRAM EXTENSION CHECKS PASSED). **VerificationDag**: v509 on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XIX) (e8-ladder-bed v2 — the 2DCS extension executed with an honest preregistered kill\_2d verdict at the DETECTOR level: the known-answer map recovers 8/8 rungs incl.  $m_7/m_8$  via F3 difference coordinates (MC  $p = 2 \times 10^{-4}$ ), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO 8/8,  $\text{CoNb}_2\text{O}_6$  6/8) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/e8-ladder-bed v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at  $B_y = 0\text{--}3\text{ T}$ , not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of  $\text{CoNb}_2\text{O}_6$  exists anywhere* (web-checked). amendment\_v2 was frozen in hypotheses/e8\_ladder\_bed\_v1.yaml BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); amendment\_v2\_1 discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (results/results2d.json, deterministic): known-answer KA2D 8/8 rungs with  $m_7/m_8$  recovered ONLY via F3 difference coordinates ( $\chi^2/\text{dof} = 1.30$ , MC  $p = 2.0 \times 10^{-4}$ , 0/5000) — the mechanism that pulls  $m_7/m_8$  inside a low-pass window works; the NEG2D localized-limit map (the paper’s own non-E8 closed forms) is correctly rejected ( $p = 0.029$ ); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at  $54/200 = 27\%$  and `jitter2d` shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level)  $\Rightarrow$  preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The  $\text{CoNb}_2\text{O}_6$  leg stays 6/8 ( $p = 0.0038$ ) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row “E8 mass-ladder blind recovery bed”: `data_value` extended (“ $\text{CoNb}_2\text{O}_6$  6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl.  $m_7/m_8$  via F3,  $p=2\text{e-}4$ ; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)”) and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, experiments/next.txt note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at  $B \parallel b$ , 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.



2026-07-21 (XVIII) (v508 — WP5e- $\gamma$  of CELEST.SEAM.01 “the sphere-axion pairing check” promoted as CELEST.WP5E.GAMMA.01, an honest rigid **NEGATIVE** result: the VERTEX-SPACE THEOREM — the  $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are **EXACTLY**  $\text{span}\{s_5, s_3\}$  (dim 2) and the quartics **EXACTLY**  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$  (dim 5), with the **PRODUCT THEOREM** as the master kill (any two invariant quadratics multiply into  $\text{span}\{P_1, P_2, P_3\}$  — zero  $T_5/T_3$  content, so  $\Phi_{T_3}$  annihilates **EVERY** exchange image while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ ); the strict two-index  $\mathbb{Z}_4$  rule **COLLAPSES** (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and the annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ ; side discovery  $K^{(0)} = -15 K^{(2)}$ ; naturalness **DISSOLVES** (ch<sub>2</sub>-natural and AB-weight couplings both certify (0,0) vs required (32,72) — scale-independent); controls:  $SO(16)$  has no sphere partners **AND** uncanceled  $T_5$ ,  $D_8$  no  $T_3$  structure; the O-fence is **SHARPENED**: the exchange sub-branch is dead, the 32  $T_3$  residual **MUST** be carried by twisted BCOV contact terms ( $\delta_1$  binary: coefficient exactly (9, -30, -15, 0, 32), computed not fitted; escape  $\delta_2$ : the full-tensor ledger); Wolfram extension 444  $\rightarrow$  450 (verified engine run); ***NO marker moves — CELEST.SEAM.01 stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, SEAM.EQUIV.01 stays [O]***

- **v508\_celestial\_wp5e\_gamma\_pairing.py** (CELEST.WP5E.GAMMA.01, 27 checks,  $\sim 4$  s; **sympy/Fraction exact, no floats; deterministic**). Question: can the three twisted sphere axions  $\eta_m$  (Coxeter eigencharacters  $\{i, -1, -i\}$  on  $H^2(\text{ALE})$ , v505 S5.8) cancel the rigid 32  $T_3$  residual of the Atiyah–Bott ledger ( $A_{\text{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32T_3$ , v505 S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem:* exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees (1,0,1)) and quartics dim 5 (bidegrees (2,0,1,0,2)) — the vertex basis is **COMPLETE**; *product theorem* (the master kill):  $(as_5 + bs_3)(cs_5 + ds_3) = acP_1 + (ad + bc)P_2 + bdP_3$  identically — every exchange image (any axion count, any rule, any couplings) is annihilated by  $\Phi_{T_3}$ , and  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . *Selection enumeration:* the strict two-index reading collapses ( $C_{j,j'} \neq 0$  only for  $j' = -j$ , forcing  $m = 0$ : 16 raw triples, 4 nonzero, all  $m = 0$ ); twist insertion:  $\eta_1 \leftrightarrow K^{(3)}$ ,  $\eta_2 \leftrightarrow K^{(2)}$ ,  $\eta_3 \leftrightarrow K^{(1)}$ ,  $\eta_0 \leftrightarrow K^{(0)}$ ; channels  $E_{13} = (16, -96, 144, 0, 0)$ ,  $E_{22} = (16, 32, 16, 0, 0)$ ,  $E_{00} = 225 E_{22}$ ; **SIDE DISCOVERY:**  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are **PARALLEL** — this collapse drives the  $\Phi_P$  obstruction). *Unsolvability with certificate (the core):*  $M$  is  $5 \times 2$ , rank 2, kernel 0;  $\text{rank}([M | A_{\text{fix}}]) = 3$ ; annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions (32  $T_3$  structural, rule-independent;  $\Phi_P = 72$  selection-rule-driven);  $\ker M = 0$  forces the unique non-disturbing coupling  $c = 0$  (the sphere axions are **RIGIDLY SILENT**);  $\Phi_P(Q^{(0)}) = 0$  (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the  $P$ -block ( $\det = -64$ ) but  $T_5 = T_3 = 0$  persists. *Naturalness resolution:* ch<sub>2</sub>-natural couplings (9/64, 1/4) give (25/4, -11/2, 97/4, 0, 0) and AB-weight couplings (1/2, 1/4) give (12, -40, 76, 0, 0) — **BOTH** with certificate (0,0) vs required (32,72): the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”). *Negative controls:* invariant-only = v505 S3.9 replicated + sharpened; flipped rule  $m+j-j'$ :  $\Phi_{T_3}$  kill flip-invariant,  $\Phi_P$  rule-dependent;  $SO(16)$ : odd classes **EMPTY** and AB sum (15, -18, -15, -4, 40) —  $T_5$  does not even cancel ( $E_8$  doubly special);  $D_8$ :  $16T_8 + 12s_8^2$ , no  $T_3$  structure — the question cannot even be posed. Number fences **[C]** only:  $32 = (h, h) + (h', h') = 20 + 12$ ,  $72 = 2\tilde{\lambda}_{e_8}^2$ .
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.GAMMA.01 (**[E]** vertex space/selection/certificate + **[C]** fences + **[O]** contact terms); CELEST.SEAM.01 stays Open with the WP5e targets **UPDATED** (“the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms ( $\delta_1$ , binary test) and the full-tensor ledger ( $\delta_2$ )”). Papers: **tfpt\_research\_contracts** WP5e section extended by the  $\gamma$  block ( $\gamma$

done — negative result with certificate; the  $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$  roadmap with success/kill criteria); **tfpt\_3** celestial section becomes a TWELVE-step story with Step 12 “the exchange no-go and the contact-term criterion” and the “What remains” paragraph updated. Wolfram: six mirrors (the vertex-space dims 2/5 by generator-wise nullspace arithmetic; the twisted quadratics with  $K^{(0)} = -15 K^{(2)}$ ; the product theorem, symbolic; the rank-3 certificate  $(0, 32, 72)$  with channels and naturalness images; the  $SO(16)$  AB sum  $(15, -18, -15, -4, 40)$ ; the  $D_8$  quartic  $16 T_8 + 12 s_8^2$ ) verified with the active engine the same day ( $444 \rightarrow 450$ , ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag: v508 on the QFT/celestial node; registry + run\_all (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

**2026-07-21 (XVII) (v507 — the tautology attack on the v506 alignment bit, executed and refuted, promoted as SEAM.BIT.ORIGIN.01: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods  $\Lambda/2\Lambda$  of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  the free involutions are exactly the three  $\sigma_c$ : the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic  $\lambda$ ): a marks-preserving root of  $\sigma_c$  exists  $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$  (one per  $c$ ; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit  $\iff j = 1728 \iff \tau = i$ ; the EQUIVALENCE TETRAD “Uhr existiert  $\iff (\tau=i$  AND deck = CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift NONSPLIT”); the FERMION DICHOTOMY (the new physical face): central  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$  nonsplit vs edge  $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$  SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension  $438 \rightarrow 444$  (verified engine run); *NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01 text precised only)***

- **v507\_seam\_bit\_origin.py** (SEAM.BIT.ORIGIN.01, 26 checks,  $\sim 2.6$  s; **Part A** sympy-exact torus/Möbius arithmetic + **Part B** exact  $Cl(16)$  Fractions; deterministic). Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double  $T_\tau^2 \rightarrow S^2$ , v214/v260), so the common origin would force the central alignment automatically and the v506 bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of  $w \mapsto -w$  on  $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =$  the half-periods  $\Lambda/2\Lambda$  (a  $\frac{1}{3}$ -coefficient never qualifies);  $t_c$  descends iff  $2c \in \Lambda$  — exactly the three half-period translations (control  $c = \frac{1}{3}$  fails); the descended maps are the EXPLICIT rational involutions  $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$  with the exact curve-lift identity  $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$ ,  $\psi_i$  RATIONAL (the  $\wp$ -addition formula as pure algebra), closing to Klein  $V_4$ ; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic  $V_4$   $(4, 3, 3)$ , harmonic  $D_4$   $(8, 5, 3)$ , hexagonal  $A_4$   $(12, 3, 3)$ ) are EXACTLY the three  $\sigma_c \Rightarrow$  every mark-free collar deck (v216) IS a half-period translation. *Core solve (symbolic  $\lambda$ )*:  $\sigma_c$  admits a marks-preserving  $PGL_2$  root  $\iff \lambda = -1$  ( $\sigma_1$ ),  $2$  ( $\sigma_2$ ),  $\frac{1}{2}$  ( $\sigma_3$ ) — per half-period exactly ONE solution, intrinsically “the deck pairs separate harmonically” (pair cross-ratio  $-1$ ); generic modulus:  $V_4$  abelian, ALL  $\sigma_c$  central yet clockless (“central alone is empty” — the bit is the CONJUNCTION  $D_4 \wedge$  central); at  $\tau = i$  the stabiliser jumps to  $D_4$  with  $\sigma_1$  central (2 roots = the clock pair),  $\sigma_2/\sigma_3$  edge-type (0 roots); the v506 seam deck lands on the central  $\sigma_1 = -1/x$  under all 8 mark bijections; CM dictionary: mult-by- $i$  fixes exactly  $c^* = (1+\tau)/2$  (mult-by- $\rho$  3-cycles all three), the clock lift needs the CM unit  $i$  ( $\psi = 2i/(x+1)^2$ ,  $t^2+1$  irreducible over  $\mathbb{Q}$ ) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists  $\iff (\tau=i$  AND deck = the CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift nonsplit; the v493 converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by

8):  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ , nonsplit  $\mathbb{Z}_4$ , 2 roots in the 32-element dihedral lift (the  $\mathbb{Z}_8$  tower); edge arrangement (reflection  $j \mapsto 4-j$ , fixed sites = non-marks = the silver  $\{0, \infty\}$  between neighbour marks): the implementer EXISTS but  $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$ , the extension SPLITS, 0 roots, no  $\mathbb{Z}_8$ ; one-particle  $(\lambda^2+1)^8$  vs  $(\lambda^2-1)^8$ ; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*:  $\{\pm 1, \pm(3+2\sqrt{2})\} = “\tau = i$  with deck = a non-CM-fixed half-period translation” (all 8 bijections  $\rightarrow \sigma_2/\sigma_3$ , never  $\sigma_1$ ) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; “no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION”. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.ORIGIN.01 ([E] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: SEAM.CLOCK.RIGIDITY.01 (“the bit is now fully characterised as a tetrad ... and carries a physical face: the non-split NS Fock extension class — v507”), QGEO.REALIZE.01 (“the class of the deck IS derived (half-period translation, v507); only its position stays the input”, stays [C]/[O]), P2.TYPING.01 (origin cross-reference on the common R1 bit). Papers: `tfpt_3` celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates where v506 is cited. Wolfram: six mirrors (half-period descent  $2c \in \Lambda$  with the rational  $\sigma_i$  lift identity; the core solve  $\lambda \in \{-1, 2, \frac{1}{2}\}$ ; the stabiliser counts  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  via the complete 24-triple enumeration; the CM fixed point  $c^*$  + the lemniscatic  $\psi = 2i/(x+1)^2$ ; the Cl(16) dichotomy  $256 \gamma$ -monomial vs scalar  $2^7$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ( $438 \rightarrow 444$ , ALL WOLFRAM EXTENSION CHECKS PASSED); the v493 audit direction and the 28/21 arrangement census stay Python-side (flagged). `VerificationDag`: v507 on the P2/carrier and QGEO nodes; registry + `run_all` (501 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XVI) (v506 — seam clock rigidity, Route B of the v503 classification, promoted as SEAM.CLOCK.RIGIDITY.01: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck  $z \mapsto -z$  has EXACTLY TWO square roots  $z \mapsto \pm iz$  in  $\text{PGL}_2(\mathbb{C})$ , both *automatically* order 4, and a marks-preserving root exists  $\iff$  the deck is the CENTRAL involution of the mark- $D_4$  (counterexample  $\{\pm 1, \pm(3+2\sqrt{2})\}$ : harmonic with full  $D_4$  but deck non-central  $\Rightarrow$  0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 —  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$ , i.e.  $U^2 = (-1)^F$  EXACTLY, a nonsplit  $\mathbb{Z}_4$ , with the  $\mathbb{Z}_8$  tower  $V^2 \propto U$ ,  $V^4 \propto (-1)^F$  ( $8 = 2|\mu_4|$  fermionically explained) and the R-sector control split ( $U_R^2 = +1$ ); the one [C] rest: “the collar deck is the central involution of the mark- $D_4$ ” — the fermionic  $\mathbb{Z}_4$  forces the ORDER, not the spatial alignment; Wolfram extension  $434 \rightarrow 438$  (verified engine run); *NO marker moves — QGEO.REALIZE.01 stays [C]/[O] (text precised), P2.TYPING.01 R1 mapped onto the same bit)***

- `v506_seam_clock_rigidity.py` (SEAM.CLOCK.RIGIDITY.01, 34 checks,  $\sim 3.5$  s; **Part A sympy-exact Möbius/divisor arithmetic + Part B exact Cl(16) Fractions; deterministic**). v503 localised the one remaining identification input (“why does the raw seam carry the order-4 clock”, QGEO-R1  $\equiv$  P2-R1); v506 reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give  $g^2 = \text{deck} \Rightarrow \text{tr } g \neq 0$  forces  $g$  diagonal with  $(a/d)^2 = -1$ , and  $\text{tr } g = 0$  forces  $\det g = 0$  (impossible) — exactly two roots  $z \mapsto \pm iz$ , both projective order 4, mutually inverse ( $\text{Aut}(\mathbb{Z}_4)$ ), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set  $\{u, -u, w, -w\}$  admits a marks-preserving root  $\iff$

$w = \pm iu \iff \text{marks} = \text{one } \mu_4 \text{ orbit} \Rightarrow \text{cross-ratio class } \{2, \frac{1}{2}, -1\}, j = 1728, \tau = i, \text{ free } 4\text{-cycle}, \text{ deck} = \text{clock}^2, \text{ only the scale free; group form: } |\text{Stab}(\mu_4)| = 8 = 2|\mu_4| \text{ with order profile } \{1:1, 2:5, 4:2\} = D_4, \text{ center} = \{\text{id}, \text{deck}\}, \text{ exactly 2 order-4 elements} = \text{the clocks} \text{ — “deck central in the mark-}D_4\text{”} \iff \text{“clock exists”}; \text{ the counterexample } \{\pm 1, \pm(3+2\sqrt{2})\} \text{ is deck-free AND harmonic } (j = 1728, \text{ cross-ratio } \frac{1}{2}) \text{ with full } D_4, \text{ but the deck sits non-centrally (the central involution crosses the } \pm \text{ pairs)} \Rightarrow 0 \text{ roots; the RP filter passes BOTH roots (the } \text{Aut}(\mathbb{Z}_4) \text{ ambiguity is irreducible, v492 NC3). } B3/B4 \text{ (} n\text{-arithmetic + controls): free orbit } \Rightarrow n \mid 4; \text{ transitivity } \Rightarrow n = 4; \text{ faithfulness kills } n \geq 5 \text{ (kernel argument through } S_4); n = 3 \text{ fixes a mark; } n = 6 \text{ dies independently at the cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); } n = 8 \text{ impossible on 4 marks; hexagonal stabiliser } A_4 \text{ (0 order-4 elements, squares only orders } \{1, 3\}); \text{ generic stabiliser } V_4 \text{ (exponent 2); stabiliser trichotomy } V_4/D_4/A_4 \text{ with order-4 counts } (0, 2, 0). \text{ Theorem B (Fock side, the core result): } S^2 \text{ has } b_1 = 0 \Rightarrow \text{unique spin structure} \Rightarrow \text{NS/bounding on every circle (established P1 datum); one-particle: } D^2 = -1 \text{ (charpoly } (\lambda^2+1)^8), \text{ quarter-shift } C^2 = D, C^8 = 1 \text{ (charpoly } (\lambda^4+1)^4, \zeta_8 \text{ spectrum); Fock (exact Cl(16)): } \tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8}) \text{ implements the deck on all 16 generators and } \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F \text{ EXACTLY; } (cU)^2 = c^2(-1)^F \text{ is never 1 (center of Cl(16) = scalars): the } \mathbb{Z}_4 \text{ lift is FORCED, } 1 \rightarrow \mathbb{Z}_2^{(-1)^F} \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2^{\text{deck}} \rightarrow 1 \text{ nonsplit; the quarter-shift lift } V \text{ (unique even block implementer) has } V^2 \propto \tilde{U}, V^4 \propto (-1)^F, \text{ projective order } 8 = 2|\mu_4| \text{ — the v492 } \mathbb{Z}_8 \text{ spin bridge realised on the 16-Majorana Fock space; R control } U_R^2 = +1 \text{ (split — the forcing rests on NS = established geometry); 16-fold way } \theta_\nu = 1 \text{ at } \nu = 16; \text{ robustness: the forcing already holds at 8 sites (16 = the carrier count } 2^{g_{\text{car}}-1}, \text{ not a tuning). The one [C] rest (verbatim): “the collar deck is the central involution of the mark-}D_4\text{” (} \iff \text{square-root existence on the mark data) — ONE alignment bit; the fermionic } \mathbb{Z}_4 \text{ forces the ORDER, not the spatial alignment. No overclaim.}$

- Ledger + papers + Wolfram + website.** Ledger: new row SEAM.CLOCK.RIGIDITY.01 ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: QGEO.REALIZE.01 (“the rest is now ONE bit: the central position of the deck in the mark- $D_4$ ; order/uniqueness/orbit/ $\tau=i$  follow — v506”, stays [C]/[O]), P2.TYPING.01 (R1 mapped onto the same bit, fermionic halving), QGEO.EMERGE.LIGHT.01 context sharpened. Papers: `tfpt_3` celestial Step 1 (fermionic  $U^2 = (-1)^F$  sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates at the v499/v503 citation sites. Wolfram: four mirrors (the complete square-root solve;  $D_4$  center + order-4 counts + the silver counterexample; the Cl(16) identity  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the split R control; the  $n$ -arithmetic + generic kill) verified with the active engine the same day (434  $\rightarrow$  438, ALL WOLFRAM EXTENSION CHECKS PASSED); the 24-triple stabiliser enumeration and the  $A_4$  profile stay Python-side (flagged). **VerificationDag:** v506 on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (v505 — WP5e- $\beta$  of CELEST.SEAM.01 “the equivariant anomaly ledger on twistor space” promoted as CELEST.WP5E.BETA.01: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators  $(2, 4, 2)$  with Dedekind sum  $\frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ , equivariant characters  $(248, 0, -8, 0)$  by two routes, invariant Okubo  $36\langle x, x \rangle^2$  with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual  $32T_3$ , the index bridge  $f(m) = \text{ch}_2(T_m)$  exact with glue defect  $-78$  both routes, the level dials  $k = 1$  geometric (current count  $240/0/0/0$ ), and the honest E3 REFUTATION:  $a_0$  fills the GRAVITON slot  $O(2)$ , NOT the axion slot  $O(-2)$  — the three sphere classes fill the three twisted axion slots instead; the preregistered kill (“inflow demands a level  $\neq 1$ ”) does NOT fire on the equivariant skeleton; WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed, the global BCOV quantisation stays [O]; Wolfram extension  $429 \rightarrow 434$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- **v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py** (CELEST.WP5E.BETA.01, 47 checks,  $\sim 1.3$  s; sympy/Fraction exact, no floats; deterministic). The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $E_8$   $\mu_4$ -glue monodromy. *AB ledger*:  $\det(1 - g_j^{-1}) = (2, 4, 2)$ ;  $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ ;  $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$  two routes; invariant average  $60 = \text{carrier}$ ; Frobenius  $61568$ ;  $\text{tr}_{g_2} = -8 = 120 - 128$ . *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ( $Q^{(0)} = 36\langle x, x \rangle^2$ ,  $36 = \tilde{\lambda}_{e_8}^2$  — v495 re-derived through the glue grading); the twisted sectors do NOT reduce ( $T_5$ :  $(-8, 32, -8)$ ,  $T_3$ :  $(48, -64, 48)$ ); the AB sum cancels the  $D_5$  quartic exactly and leaves the rigid  $32T_3$  (the only quartic-free weighting is the  $\mathbb{Z}_4$  projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities ( $\text{Tr}_{120}, \text{Tr}_{128}, \text{Tr}_{E_8} = 9(\text{tr}^2)^2$ ) exact. *Index bridge*:  $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$  EXACT (E1 = E2 arithmetic); glue  $\text{ch}_2$  defect  $-78$  both routes (classes  $(-24, -30, -24)$ ); Killing sums  $(320, 320, 240, 320)$ , total  $1200 = 2h^\vee(h, h)$ .  $k = 1$  *geometric*: lattice current count  $240$  at  $k=1$ , exactly  $0$  at  $k = 2, 3, 4$ ; embedding residual  $47k^2 + 219k - 266 = (0, 360, 814, 1362)$ ; integrality alone fixes nothing (honest, as v502); one scale  $\Rightarrow$  one level (v493 periods). *E3 refutation (prominent)*:  $a_0 \in O(8)$  fills the GRAVITON slot  $O(2)$ , NOT the BSS axion slot  $O(-2)$  ( $X = -2 \neq +2 = Y$ ; mismatch  $4 = |\mu_4|$  as a [C] coincidence) — “the theory brings its GS axion as  $a_0$ ” is REFUTED; instead the three  $H^2(\text{ALE})$  classes carry exactly the Coxeter eigencharacters  $\{i, -1, -i\} =$  the three twisted slots (bijection); slot count  $1+3 = 4 = |\mu_4|$  soft. *Negative controls*:  $\text{diag}(i, i)$  breaks the ledger four ways ( $\sum 1/\det = \frac{1}{4}$ ,  $\omega$ -weight  $-1$ , 5 generators);  $SO(16)$  glue: characters  $(120, 0, +120, 0)$ , defect  $-30 \neq -78$ , bulk Okubo fails (independent 4th Casimir);  $k = 2$ : current count  $0$ ,  $c$ -prefactor  $-31/48 \neq -\frac{1}{3}$ . [C] dictionaries ( $\text{CS}(\rho) \bmod 1 = \text{discriminant weight}$ ;  $32 = (h, h) + (h', h')$ ;  $-30 = -h^\vee$ ;  $\Omega$ -contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the  $O(-2)$  bulk axion, CPS level-from-flux, the pairing  $32T_3 \leftrightarrow$  sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.BETA.01; CELEST.SEAM.01 updated to “WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed; the global BCOV quantisation stays [O] — with the rigid  $32T_3$  twisted residual and the sphere-axion pairing as the named remaining targets” at *unchanged* Open status. Papers: contracts WP5e bullet  $\rightarrow$  “ $\alpha + \beta$  done” with the  $a_0$  refutation as an honest finding (kill-test and honest-success paragraphs consistent); **tfpt\_3** celestial section becomes the *eleven*-step story (new Step 11 “the equivariant anomaly ledger on twistor space”, “what remains” = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge +  $-78$  defect; the per-sector Okubo sums over the  $240$  glue roots with the  $32T_3$  residual; the lattice current counts + embedding residuals; the

axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429  $\rightarrow$  434, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag qft node gains v505; registry + run\_all (499 modules at this step; 500 with v506, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIV) (v504 — WP5d- $\beta$  of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ( $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ ), GF2 spans full, rank 32/32; disjoint exactly index 2 = the v501  $\ln 2$  bit), the entropic touching defect BOUNDED  $< \ln 2$  with the Ising  $\frac{1}{4}$ -exponent approach ( $p = 0.2444$ ) while the preregistered  $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate  $\pi K(1-x)/K(x)$  (1.3–2.0%) with EXACT orbifold inheritance ( $C \rightarrow -C$  + the  $\Lambda^2 C$  compound — Longo heredity), and Pimsner–Popa  $E(a) - a/2 = PaP/2$  identically ( $\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$ ,  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ,  $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$  — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension 424  $\rightarrow$  429 (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)**

- **v504\_celestial\_wp5d\_klm\_completeness.py** (CELEST.WP5DB.01, 37 checks,  $\sim 12$  s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval  $\mu$ -index; v501 (WP5d- $\alpha$ ) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the  $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd  $\otimes$  odd is the v501  $\ln 2$  bit, localised at the split point); the neutral  $U(1)$  algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law  $((c/3) \ln 2$  increments, 0.115525, dev  $1.2 \times 10^{-6}$  — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the  $\mathbb{Z}_2$ /orbifold deficit rises 0.4045  $\rightarrow$  0.5899 ( $N = 64 \rightarrow 4096$ ), stays  $< \ln 2$ , and  $(\ln 2 - \Delta_2) \sim N^{-p}$  with  $p = 0.2444$  vs  $2\Delta_\mu = \frac{1}{4}$  (Ising disorder exponent); split-point scan reflection-symmetric to  $3.9 \times 10^{-14}$ ; ED vN same direction; gap continuity to the v501  $x \rightarrow 1$  endpoint; honest note: “defect  $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded  $\Leftrightarrow$  finite index (Longo–Xu) is the correct statement; the  $U(1)$  control bursts  $\ln 2$  from  $L = 128$  and grows as  $\frac{1}{2} \ln \text{Var } Q_A$  (Klich–Levitov slope 0.10134 vs  $1/\pi^2 = 0.10132$ ). *Split/nuclearity*: coupling-block  $\sigma_k$  fall exponentially at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at  $d = 16/64/256/512$ );  $\sigma_1 \sim x^{0.5044}$  (vs  $\frac{1}{2}$ ); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*:  $P_A$  flips  $C \rightarrow -C$  ( $\sigma_k$  identical,  $8.3 \times 10^{-17}$ ) and the even-bilinear coupling Gram is the second compound  $\Lambda^2 C$  (singular values  $\{\sigma_a \sigma_b\}$ ,  $2.8 \times 10^{-14}$ ) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*:  $E(a) - a/2 = PaP/2$  identically; attainment in integers (all 16384 (even, odd) monomial pairs at  $N = 8$ : 8192 extremal with  $M^2 = 4$ ,  $\text{Tr } M = 0$ ,  $PMP = -M$ , 0 violations; 2048/2048 at  $N = 12$ ; pencil min  $\lambda = 0.5000000000$ );  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ; index consistency  $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$  and  $e^{2\Delta_\infty} = 4 = \mu$ ;  $U(1)$ :  $\lambda = 1/(m+1) \rightarrow 0$ . *Controls + honesty*: dimerised phase trivial (drift  $3.6 \times 10^{-15}$ ); wrong prescription  $\{1, P_{AB}\}$  deficit exactly 0;  $\mathbb{Z}_2$  finite-size honesty (14.9% below

$\ln 2$  at  $N = 4096$  — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu's* (cited, not claimed), the concrete seam quotient net and the condensed  $(E_8)_1$  net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d:  $\alpha$  ( $\mu$ -index) +  $\beta$  (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (*Xu's* theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet  $\rightarrow$  “both stages done” with the  $\beta$ -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group  $\lambda_{E_4} = \frac{1}{4}$  + index consistency; the  $\Lambda^2/\text{Minors}$  compound with Cauchy–Binet and  $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$ ; the  $U(1)$  dephasing  $\lambda = 1/(m+1)$  family) verified with the active engine the same day (424  $\rightarrow$  429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field's global  $\det'$  winner is the *hexagonal* torus,  $\tau=i$  is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff*  $\tau=i$ , an isolated zero); and the residual *converges*: QGEO-R1  $\equiv$  P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420  $\rightarrow$  424 (verified engine run); *NO marker moves* — QGEO.REALIZE.01 stays  $[C]/[O]$ , AX.P2.01 stays a declared axiom)**

- `v503_qgeo_emergence_light.py` (QGEO.EMERGE.LIGHT.01, 17 checks; mpmath 40 digits + v280-style lattice DtN + sympy exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*:  $\log F = \log(\tau_2|\eta|^4)$  (Osgood–Phillips–Sarnak) is exactly  $S$ -invariant ( $\text{dev} < 10^{-35}$ ), so  $t=1$  is symmetry-forced critical on the rectangular family  $\tau = it$ ; there  $g''(1) = -1.298 < 0$  (strict maximum) but the  $\tau_1$ -direction has  $h''(0) = +0.298 > 0$ :  $\tau=i$  is a *saddle* of the full moduli space; the only critical points are the CM points ( $E_2^*(i) = E_2^*(\rho) = 0$  at 40 digits, generic  $0.163 \neq 0$ ); the *global*  $\det'$  winner at fixed area is the *hexagonal* torus ( $\log F(\rho) = -1.03352 > \log F(i) = -1.05469$ , diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to  $8 \times 10^{-15}$  exactly at  $\tau=i$  and rises strictly monotonically both ways (min  $8.4 \times 10^{-2}$  away) — an isolated zero;  $R^2 = \mathbb{Z}_2$  exact as permutation matrices (the v492  $\mathbb{Z}_8$  bridge on the lattice);  $j(it) \geq 1728$  with equality only at  $t=1$ . *The  $\mathbb{Z}_2$  chain is modulus-blind*: 4 marks +  $[\mathbb{Z}_2, \Delta_t] = 0$  on *every* torus (v216 confirmed without mark input); pillowcase  $\det' = \frac{1}{2}$  torus  $\det'$  exactly ( $\pm$ -pair structure) — only the order-4 clock distinguishes  $\tau=i$ ;  $\text{Aut}(E_\rho) = \mathbb{Z}_6$  has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ( $j = 1728$  identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on  $H^1$  is a Coxeter element of  $W(A_3)$  (char poly  $\lambda^3 + \lambda^2 + \lambda + 1$ , order 4 =  $|\mu_4| = N_{\text{fam}} + 1$ ) = the P2 cusp-class carrier datum of v499 R1  $\Rightarrow$  QGEO-R1  $\equiv$  P2-R1, one common residual; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input  $[[C]]$ , rest = its dynamical realisation [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data  $\Rightarrow$  order-4 clock”

is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [C]/[O]; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt\_3 Step 2 (light-converse sentence), origin\_theory necessity paragraph, introduction/tfpt\_1/tfpt\_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities  $E_2(i) = 3/\pi$ ,  $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$  exact,  $j/\text{DedekindEta}$  special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock  $\Rightarrow$  square + Coxeter) verified with the active engine the same day (420  $\rightarrow$  424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run\_all (497 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XII) (v502 — WP5e- $\alpha$  of CELEST.SEAM.01 “prefactor and level bookkeeping” promoted as CELEST.WP5E.ALPHA.01: the  $q^{-1/3}$  prefactor of  $E_4/\eta^8$  IS exact  $\mu_4$  vacuum-energy bookkeeping (the clock is inner  $\Rightarrow$  shift orbifold  $\theta = 0^8$ , common  $-c/24 = -\frac{1}{3}$ ; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and  $k = 1$  is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial  $31124 - 27000 = 4124$ ); honest sharpening: glue- $h$  integrality holds for ALL  $k$  and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided ( $\alpha$  done, WP5e proper open); Wolfram extension  $416 \rightarrow 420$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v502\_celestial\_wp5e\_prefactor\_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor:* Hurwitz- $\zeta$  exact ( $\zeta(-1, \theta) = -B_2(\theta)/2$ ;  $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ ; Ising shift  $\frac{1}{16}$  per Majorana); the clock is *inner* ( $h = (2, 2, 2, 2; 0^4)$  and  $h' = (0^5; 1, 1, 1, -3)$  both read the class mod 4 on all 240 roots;  $(h, h) = 20$ ,  $(h', h') = 12$ , sum 32)  $\Rightarrow$  twist  $\theta = 0^8$  in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so  $E_{\text{osc}} = -\frac{1}{3} = -c/24$  is sector-independent ( $c = 8 = \text{rank} = \text{Sugawara } 248/31$ ; one common  $\eta^8$ ); sector ground exponents  $-\frac{1}{3} + (0, 1, 1, 1)$ ; *discriminant form = Casimir energy* two ways (spectral flow  $j^2(h, h)/32 \bmod 1$ ; free fermions exact:  $D_5 = 10$  Majorana  $\rightarrow \frac{5}{8}$ ,  $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$ , glue diagonal  $= 16 = 2^{g_{\text{car}}-1} = \text{the WP5d seam carrier} \rightarrow 1$ ); the rotation reading *fails* (deck  $\frac{3}{16} \neq \frac{3}{8}$  — “clock  $\neq$  deck” at the Casimir level); the four sector characters sum exactly to  $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$ . *Part B — the level:*  $k = 1$  forced three independent ways — (i) the current condition  $h(J) = k = 1$  (adjoint closure  $60+64+60+64 = 248$  at spin 1); (ii) the conformal embedding ( $47k^2 + 219k - 266 = 47(k-1)(k+266/47) = 0$ ;  $D_8: 128k(1-k) = 0$ ); (iii)  $c = 8 \Leftrightarrow 240(k-1) = 0$  (= the prefactor); plus the v498 singular-vector dial ( $\|(E_{-1}^\theta)^m 0\|^2 = m!k(k-1)\cdots(k-m+1)$ , singular vector at grade  $k+1$ ;  $31124 - 27000 = 4124$  at  $k=1$  only,  $\langle s|s \rangle = +4$  at  $k=2$ ). *Honest sharpening (mandatory):* glue- $h$  integrality  $h(J^j; k) = k(0, 1, 1, 1)$  is integer for ALL  $k = 1..8$  — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls:*  $D_8/SO(16)_1$  (same  $c = 8$ , SAME  $q^{-1/3}$  prefactor, but  $h = (0, \frac{1}{2}, 1, 1)$ : the discriminator is glue- $h$  integrality, not the prefactor;  $\Theta_{D_8} + \Theta_s = E_4$ );  $\mathbb{Z}_2$  rotation ( $E = +\frac{1}{6}$ ,  $h = \frac{1}{2}$ );  $\mu_4$  rotation (sector-dependent  $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$  — the common prefactor breaks); wrong  $k \in \{2, 3, 4\}$  fails all five dials coherently (residuals  $360/814/1362$ ;  $\langle s|s \rangle = 4/12/24$ ;  $31124 \neq 4124$ ). [[C]] Kac generation for  $k \geq 3$  ( $k = 1, 2$  machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion  $\lambda_g/(8\pi\sqrt{3})$  on  $\mathbb{P}T$ , the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level  $\neq 1$ ”) is not triggered



on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- $\alpha$  (done) [E]/[C]: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper [O]: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- $\alpha$  landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status  $\rightarrow$  “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]”. Wolfram: four exact mirrors (Hurwitz- $\zeta$  vacuum energies; discriminant = Casimir with the deck failure; the  $k$ -fixing equations incl. the “fixes nothing” integrality table; the character sums +  $D_8$  control) verified with the active engine the same day (416  $\rightarrow$  420, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

**2026-07-21 (XI) (v501 — WP5d- $\alpha$  of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the  $\ln 2$  offset at machine precision,  $\mu_{\text{gauged}} = 4 =$  the v490 census, and the condensation arithmetic  $16 \rightarrow 4 \rightarrow 1$  closes with  $\mu = 1$ ; the preregistered KILL (“ $\mu$ -offset  $\neq 0$  after condensation”) does NOT fire; contract now “WP1–WP5d executed; WP5e the single remaining milestone”; Wolfram extension 412  $\rightarrow$  416 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator —  $(E_8)_1 \Rightarrow \mu=1$  (Haag-dual two-interval net, one sector) vs  $SO(16)_1 \Rightarrow \mu=4$  — executed at the  $\alpha$  (lattice-index) stage on the v367 seam edge (16 critical Majorana layers,  $c_- = 8$ , v450-style ring; Peschel-exact, every Gaussian formula ED-validated to  $\leq 1.5 \times 10^{-15}$ ; the superselection identity  $[\rho, P_{AB}] = 0$  machine-exact). *The  $\mu = 1$  reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* —  $c$  fit 0.5000,  $\max |F(x)| = 5.97 \times 10^{-5}$  at  $N=512$ , falling monotonically to  $1.3 \times 10^{-6}$  at  $N=1024$  (Casini–Huerta extensivity as a lattice witness). *The  $\mu$ -offset*: Rényi-2 exactly  $\Delta_2 = -\ln[(1+R)/2]$ ; the 16-layer seam carrier saturates  $\ln 2$  at machine precision ( $|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$  at  $N=512$ ;  $R \sim N^{-0.500}$ ); index reading  $[F : F_{\text{even}}] = 2.000000$ ,  $\mu_{\text{gauged}} = 4.000000 =$  the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs  $16 \ln 2$  (the offset measures  $\ln[\text{index}]$ , not a blob). *Haag-duality indicator*: fermionic  $S(E) = S(E')$  to  $8.2 \times 10^{-12}$ ; the orbifold *breaks* complementarity ( $D(E) - D(E') = 0.427$  at  $a=64$ ); the complementary-pair sum is exchange-symmetric and  $\leq \ln 4 = \ln \mu(SO(16)_1)$  with max 1.2637 at the self-dual  $x = \frac{1}{2}$  — a *double-limit* statement (the first naive fixed- $x$  check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*:  $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$ ; KLM/Longo–Rehren  $16/4^2 = 1$  and  $16/2^2 = 4 \rightarrow 4/2^2 = 1$  (both routes exact);  $\Sigma d_i^2 = (4, 4, 1)$ ;  $\theta_v = 1$  exactly at  $\nu = 2c_- = 16$ , rivals  $\neq 1$  — so  $\mu = 1$  after condensation ([C] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*:  $\nu=1$  single Majorana (offset non-removable,  $\theta_v = e^{i\pi/8} \neq 1$  — the discriminator has teeth); trivial phase (all  $\leq 10^{-7}$ ); wrong sector sum  $\{1, P_{AB}\}$  (offset exactly 0). [O] WP5d- $\beta$  (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet  $\rightarrow$  “ $\alpha$  stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece”. **tfpt\_3** celestial section: new Step 8 (one paragraph, conservative); **tfpt\_4** keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + [[C]] dictionary); CELEST.SEAM.01 status → “WP1–WP5d executed; WP5e [O] (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients;  $\Sigma d^2$  + index-reading arithmetic; 16-fold-way  $\theta_v$ ) verified with the active engine the same day (412 → 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). wolfram/README.md, GATE.WOLFRAM.02, tfpt\_safeguards Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + run\_all (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family  $\omega_w$  exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 =  $V(2\theta)$  weight by weight;  $|s\rangle$  IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v500\_celestial\_wp5c\_gns\_limit\_state.py** (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*.  $\omega_w$  is quasi-free: loop sector = the affine  $k=1$  vacuum  $n$ -point functions via the *machine-determined* compact anti-involution  $\theta(e_\alpha) = -e_{-\alpha}$  (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings  $x^{wr}$  (the exact Gibbs regulator,  $x$ -adic). *Convergence sharp*:  $\omega_w = \omega_\infty \bmod x^{N+1}$  for all  $w \geq N+1$  ( $N=8$ ), not yet at  $w = N$  — the WP5a threshold  $w = n+1$  holds *identically at the state level*; positivity of every  $\omega_w$  checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the  $E_8$  Cartan matrix,  $\det 1$ ), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel =  $V(2\theta)$  *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at (0,  $\omega$ ,  $\theta$ ,  $2\theta$ ); direct ideal checks  $F_i|s\rangle$ ,  $F_j F_i|s\rangle$  norm 0.  $\mu_4$ -equivariance:  $\omega_w \circ \text{clock} = \omega_w$  for all  $w$ ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) =  $\Theta_{C_j}/\eta^8$  at  $q^2$  — the two-routes identity at the *state level*. *Resolution of WP5b S4.10*:  $|s\rangle$  is the zero vector of  $\text{GNS}(\omega_\infty)$  — the twisted two- $C_1$ -mode floor (6 quarters =  $q^{3/2}$ ) concerns only pre-limit objects (GNS has no state there,  $6 \bmod 4 \neq 0$ ). *Negative controls*:  $k=2$  ( $\langle s|s\rangle = +4$ , dominant blocks full rank — no ideal in the kernel);  $k=0$  (level-1 Gram  $\equiv 0$  — no 248 layer without the central extension);  $D_8$  (rank 2076 = 7380 – 5304, kernel =  $V(2\theta_{D_8})$  PSD, but  $h = (0, \frac{1}{2}, 1, 1)$ : one block of four — one-block completeness stays  $E_8/\mu_4$ -specific); wrong family  $E'_w = mw+r$  (the 248 layer erased,  $710955 \neq 248$ ); no damping ( $897266 \neq 248$ ); *CCR obstruction*: at fixed radial pairing  $c > 0$  positivity forces  $\omega(bb^*) \geq c$  — no  $w$ -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. [C] quasi-free extension to the full  $*$ -algebra; Kac positivity beyond level 2. [O] local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website**. **tfpt\_research\_contracts**: the WP5c bullet → done with the core numbers (state-level threshold  $w = N+1$ , the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing v500\_celestial\_wp5c\_gns\_limit\_state.py; the WP5b handover marked answered. **tfpt\_3** celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity [E] + [C] identifications + [O] net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with  $\dim V(\omega) = 3875$  by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial +  $D_8$  census) verified with the active engine the same day (408  $\rightarrow$  412); the exact Gram/PSD computation stays Python-side (flagged). VerificationDag qft node gains v500; registry + run\_all (494 modules at this step, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor  $a = (1, 1, 2)$  typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the tfpt\_2 “balanced” selection typed as stability; Wolfram extension 403  $\rightarrow$  408 (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])**

- **v499\_p2\_weights\_deligne\_bg.py** (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank  $3 = \# \text{marks} - 1$ , v137/v228; cusp-class local monodromy  $\lambda^3 - 1$  and exact irreducibility  $|\langle U, M_0 \rangle| = 24 = |S_4|$ ,  $\sum_g |\chi|^2 = 24$ , v117; unitarity (U)), the Deligne canonical extension  $E$  of the flavor connection satisfies:  $\deg E = -4$  as a theorem (residue exponents  $\{0, \frac{1}{3}, \frac{2}{3}\}$  the unique lift to  $[0, 1)$ , trace 1 per mark; par-deg 0;  $\infty$  apparent, monodromy product = 1 exactly);  $h^0(E) = 0$  as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary  $h^0=0 \Leftrightarrow a_i \geq 1$ ); integrality = Birkhoff–Grothendieck. Hence  $\{a\} = \{1, 1, 2\}$  unique and  $g_{\text{car}} = e_2 = 5$ . *Selector typed*: the Schur–Horn permutohedron has integer points exactly  $\{(2, 2, 0), (2, 1, 1)\}$  (the two options of the tfpt\_2 “Global computation” theorem) and  $h^0=0$  kills  $\{0, 2, 2\}$  — the “balanced/minimal-variance” heuristic is now stability/positivity. *The  $h^1$  discrepancy, real and resolved*:  $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ , so “generations =  $H^1$  of the rank-3 bundle” is FALSE (negative control: forcing it demands  $\deg -6$  and loses uniqueness,  $e_2 \in \{9, 11, 12\}$  never 5); correct two-object typing:  $L = \mathcal{O}(-D)$  carries the generations ( $h^1(L) = 3 = N_{\text{fam}}$ , the v137 basis =  $H^0(\Omega^1(\log D))$ ),  $E$  carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator  $3 = N_{\text{fam}}$  forces the  $\mathbb{Z}_3$  cusp cover  $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$  (genus 2; NOT the pillowcase double, genus 1,  $j = 1728$  — the deck side);  $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ ; the regular-representation model lands on  $\{0, 2, 2\} =$  the *unstable* Schur–Horn companion ( $h^0 = 1$ ): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity  $\rightarrow$  4/31 solutions;  $\deg -3/ -5$  breaks;  $n$ -mark scan  $e_2 = (n-2)(n+1)/2 = 5$  ONLY at  $n = 4$  (the value, not the uniqueness, is the  $n=4$  content); MS-rational  $e_2 = \frac{16}{3}, \frac{44}{9}$ . *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s diag  $A_0 = a/4$  input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499” (v491 stays active, no supersede). Registry + run\_all (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file tfpt\_readouts\_extension.wl (partition enumeration under the full typing + naive-typing control +  $h^1$  discrepancy; Deligne residue-trace degree with the  $A_0^*$  witness char poly; Schur–Horn lattice points; Biswas  $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double  $j = 1728$ ; the  $n$ -mark formula), verified with the active engine the same day (403  $\rightarrow$  408, ALL WOLFRAM EXTENSION CHECKS PASSED); wolfram/README.md + GATE.WOLFRAM.02 + tfpt\_safeguards Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): introduction anchor paragraph, tfpt\_1 (5,3) keybox, tfpt\_2 H2 winbox and the “Global computation” theorem (the balanced selection typed as stability), origin\_theory (5,3) keybox — all citing v499\_p2\_weights\_deligne\_bg.py; no status card changes. Website: VerificationDag P2

node gains v499; generated indices via build; no marker moved anywhere.

**2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **tfpt\_3\_e8\_audit\_bootstrap: new section “The celestial route: from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow”.** The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible  $\pi$ ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the  $E_8$   $\mu_4$ -glue as a flat  $\mathbb{Z}_4$  monodromy on the  $A_3$  ALE space,  $\text{clock}^2 = \text{deck}$  ( $\text{spin } \mathbb{Z}_8, 8 = 2|\mu_4|$ ) [E] (v492); (2) deformation —  $XY = Z^4 + a_0$  pure scale,  $j = 1728$  frozen,  $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$  [E] (v493); (3) consistency honestly demoted —  $(\kappa/c_3)^2 = 12$  exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the  $(E_8)_1$  character is not a jet-grading conformal block (growth  $n^{2/3}$  vs  $n^{1/2}$ ) but the  $\mu_4$  period sum 248 holds exactly: a boundary-limit shadow of SEAM.EQUIV.01/MMST shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold  $w = n+1$ ,  $27000 = (h^\vee)^3$  derived, quotient = sector sum = 4124,  $SO(16)_1$  contrast [E] (v497); (6) the deleting operator explicit —  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ ,  $J_1^a|s\rangle = 0$  on all 248 (tally 190/57/1), level dial  $2(1-k)$ , one-block closure  $E_8/\mu_4$ -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close SEAM.EQUIV.01; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. **tfpt\_research\_contracts:** a reader’s note at the head of the CELEST.SEAM.01 contract cross-references the new Paper-3 section (the contract stays the full technical reference). **tfpt\_4\_frontier:** one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

**2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v498\_celestial\_wp5b\_singular\_vector.py (CELEST.WP5B.01, 53 checks, exact integer/Fraction, deterministic, no floats).** Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction:*  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$  (weight  $2\theta$ , level  $2 = 8$  quarter units =  $q^2$ , an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the  $[e_\alpha, e_{-\alpha}]$  sign *forced* by the Jacobi identity ( $\text{SGN} = -1$ , not chosen), the invariant form  $\kappa$  derived ( $\kappa(\theta^\vee, \theta^\vee) = 2$ ), Jacobi exact on all 57600  $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496  $\theta$ -adjacent, all  $N_{\alpha\beta} = \pm 1$ . *Singularity (machine):*  $J_1^a|s\rangle = 0$  for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ( $a = F^\theta$  via the central-term cancellation — the *only* case that sees the level  $k$ ); plus  $J_2^a|s\rangle = 0$ ,  $E_0^a|s\rangle = 0$ , exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial:* the  $F_1^\theta$  coefficient

is  $2(1-k) \rightarrow 2/0/-2$  at  $k = 0/1/2$ : without the central extension the deletion operator does not exist.  $\mu_4$  data: glue class  $j(\theta) = 1$  (machine-built  $h$ -adapted chamber,  $\langle \theta, h \rangle = 5$ , height  $29 = h^\vee - 1$ ), clock phase  $i^{2j} = -1$ , class  $(2\theta) = 2$  sheet-even; the deleting  $\theta$ - $\mathfrak{sl}_2$  crosses the sheet-odd classes  $(1, 3)$  while its square is sheet-even; cross-table  $\theta$ -grading  $\times$  glue class consistent (rows  $(1, 56, 126, 56, 1)$ , columns  $(52, 64, 60, 64)$ ); 8 quarters  $= q^2$  via the per-period dictionary. *Level-2 match*: weight  $2\theta$  has multiplicity 1 in the level-2 Fock, so  $U(\mathfrak{g})|s\rangle$  is the  $27000 = (h^\vee)^3$ ; the direct  $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of  $V(2\theta)$  exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient  $31124 - 27000 = 4124$ ; lattice reading  $(2\theta, 2\theta)/2 = 4 > 2$ :  $|s\rangle$  is the kernel generator of  $\text{Fock} \rightarrow E_4/\eta^8$ . *Negative controls (honest separation)*: the level-1 current state  $E_{-1}^\theta|0\rangle$  (one witness,  $\rightarrow -k|0\rangle$  — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at  $k=2$  the *cube* is (generic Kac  $(E^\theta)^{k+1}$  mechanics, honestly typed);  $SO(16)_1$  has the same singular vector but keeps three extra level-1 primaries ( $h = \frac{1}{2}$  breaks one-block fusion;  $5304 \neq 14^3$ ) vs  $E_8$  (comarks all  $\geq 2$ , glue weights  $(0, 1, 1, 1)$  integer) — the *one-block closure* is the  $E_8/\mu_4$ -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- $C_1$  modes never sum to 8 quarters (minimum  $6 = q^{3/2}$ ) — the per-period dictionary (v496), not the per-slot identification, carries  $|s\rangle$  to  $q^2$ : exactly the WP5c GNS/limit-state question; WP5c-e stay open; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally  $190/57/1$ ; level dial  $2(1-k)$ ;  $j(\theta) = 1$ , clock phase  $-1$ ; orbit BFS through depth 4; the  $D_8$  contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O]”). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O] (roadmap)”). Wolfram: 5 exact v498 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $398 \rightarrow 403$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors  $116/116 + 403/403$ . Website: `papers.ts` contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c-e roadmap”), DAG qft node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b-e [O] (roadmap); Wolfram extension  $394 \rightarrow 398$  (verified engine run); no gate closes, no marker moves, SEAM.EQUIV.01 untouched)**

- `v497_celestial_wp5a_null_ideal.py` (CELEST.WP5A.01, 34 checks, exact integer/Fraction, no floats). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family  $\chi_w$  of graded Fock characters on the chiral jet generators (loop energy  $m$ , radial number  $r$ ;  $E_w = m + wr$  in quarter units  $u = q^{1/4}$ ) contains the chiral jet grading as its  $w=2$  member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its  $u^n$  coefficient equals the quarter-moded loop Fock for *all*  $w \geq n+1$ , strictly larger for every  $w \leq n$  (verified  $n \leq 8$ ,  $w \leq 10$ ) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit  $\varepsilon = 1/w \rightarrow 0$  with explicit threshold  $w = n+1$ , not a slice (convention clarified:  $897266 =$  the quarter-moded loop Fock at  $u^4 = \text{integer level } 1$ ). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give  $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$  with residual exactly zero (independent Weyl-orbit cross-check);  $27000 = 30^3 = (h^\vee)^3$  exactly;  $C_2 = (124, 96)$ , Dynkin indices  $6750/750$ ; level-1 integrability ( $\langle 2\theta, \theta^\vee \rangle = 4$ ,  $\langle \omega, \theta^\vee \rangle = 2$ ) leaves

the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient  $31124 - 27000 = 4124 = 1 + 248 + 3875$  exactly. *Two routes, one number*: the  $\mu_4$  theta-split sector sum at  $q^2 = (1036, 1024, 1040, 1024)$  gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space ( $q^3$ :  $34752 = \chi_3$ ). Negative controls:  $SO(16)_1$  through the same pipeline gives *four* components ( $5304 + 1820 + 135 + 1$ ),  $5304 \neq 14^3$  (the  $h^{\vee 3}$  identity is not generic), quotient  $2076 = \Theta_{D_8}/\eta^8$  at  $q^2$  (the recipe validated on a second algebra), block weights  $(0, \frac{1}{2}, 1, 1)$  break one-block fusion (vs  $E_8$ 's integer  $(0, 1, 1, 1)$ ); only  $P = 4 = |\mu_4|$  periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ( $64 \neq 60$ ). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced. `tfpt_research_contracts`:** WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold  $w = n+1$ ;  $27000 = (h^{\vee})^3$  derived; quotient = sector sum = 4124;  $SO(16)_1$  contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel  $\supseteq$  ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function =  $E_4/\eta^8$  incl. the  $q^{-1/3}$  prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $394 \rightarrow 398$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors  $116/116 + 398/398$ . Website: `papers.ts` contract mirror (“WP5a done, WP5b–e roadmap”), DAG `qft` node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak  $F_0 = 11.19275$  Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute  $\varepsilon$ ), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)**

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs  $\rightarrow$  491 usable  $\rightarrow$  119 segment-coherent  $\nu(t)$  points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection”  $F_0 = 11.19275$  Hz is corrected on record (true Vela  $F_0 = 11.1861692$  Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear*  $\delta\nu(\tau)$  residual ( $\mu$ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches  $\varepsilon_{90}$  below  $\varepsilon = 1$  ( $\varepsilon_{50} = 0.55$  in 2019 only)  $\Rightarrow$  no absolute per-channel  $\varepsilon$ -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude  $\sqrt{2g} = 0.40$  (leg gains 0.046/0.114, min  $p = 0.014$  — the 2019 audit-level chance feature: Bonferroni  $\times 10 = 0.14$ , shuffle  $p = 0.40$ ).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute  $\varepsilon$ ); all-channel universal-DSI 95% UL  $\varepsilon < 0.11954$  ( $\approx 12.0\%$ ; HKSJ  $t_2 = 2.92$ ,  $\rho = 0.3$ -inflated 0.137;  $6.91\times$  the predicted  $\varepsilon = 0.0173$ ), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5%  $\rightarrow$  recover 4.8%, UL 5.5%;

coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list, PG.06b FULL bullet), results/results.json, scorecard row data\_value/source via build\_evidence\_scorecard.py (status stays data\_limited; counters unchanged, 120 rows), experiments/README.md catalog row. No claim; firewall intact.

**2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)**

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** wolframscript -file verification/wolfram/tfpt\_readouts.wl → 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; tfpt\_readouts\_extension.wl → 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** verification/wolfram/README.md (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); tfpt\_safeguards Layer-6 count + website papers.ts mirrors (116/116 + 394/394); v479 registry/run\_all deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

**2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v495\_celestial\_wp3\_gs\_alignment.py (CELEST.WP3.01, 25 checks, exact Fraction/sympy).** The pre-registered WP3 “alignment test only”: the Okubo coefficient  $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$  is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ( $\text{sl}_2, \text{sl}_3, \text{g}_2, \text{so}_8, \text{f}_4, \text{e}_6, \text{e}_7, \text{e}_8$ ;  $\text{e}_6/\text{e}_7$  carved out of the v492  $E_8$  root set; negative control:  $\text{sl}_4$  fails,  $5/32$  vs  $3/32$ ); the closed form  $\tilde{\lambda}^2 = 10h^\vee/(\dim + 2) = h^\vee + 6$  holds exactly across the Deligne series;  $E_8$ :  $\tilde{\lambda} = 6$  (unit-trace) resp.  $\lambda_{\text{fund}} = 1/10$  (adjoint-trace) exact, so  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$  resp.  $1/300$  — with  $\kappa/c_3$  itself *irrational* ( $2\sqrt{3}$ ). Byproduct: the printed  $\lambda_{\text{so}_8}^2 = 3/2$  in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity),  $\tilde{\lambda}$ -integrality  $2/8$  (shared with  $\text{sl}_3$ ), only  $g_{\text{car}}=5 \mid h^\vee$  is  $\text{e}_8$ -selective ( $1/8$ ), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not  $\text{e}_8$ -selective evidence. Verdict B.
- **v496\_celestial\_wp4\_character\_shadow.py (CELEST.WP4.01, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the  $(E_8)_1$  character  $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$  is *not* a conformal block of the glue-equivariant  $E_8[\mathbb{C}^2]$  S-algebra in its own jet grading — localised three ways: (a) spin  $1 - d/2$  unbounded below and 248 never a tower dimension ( $d \leq 100$ ); (b) growth  $n^{2/3}$  vs  $n^{1/2}$  (quadratic cumulative count  $31s^2 + 92s + 60$ ,  $31 = k + h^\vee$ ;  $f_n/\chi_n$  strictly increasing  $n = 1..12$ ); (c) the level-2 null ideal deletes exactly  $27000 = 30^3 = (h^\vee)^3$  out of  $\text{Sym}^2(248) = 1 + 3875 + 27000$  — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes =  $60 = C_0$  currents, jet slice cycles (60, 64, 60, 64), one  $\mu_4$  period sums to 248, glue weights (0, 1, 1, 1) integer; free loop Fock  $897266 \gg 248$ , so the truncation must come from a limit) — the same MMST

scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray  $\nu(t)$  leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored)  $\rightarrow$  491 usable  $\rightarrow$  240 per-obs detections  $\rightarrow$  **119 segment-coherent  $\nu(t)$  points** (108 fully coherent), median  $\sigma_\nu \approx 0.6 \mu\text{Hz}$ ; barycentring validated against PINT to  $3.4 \mu\text{s}$  — the *first X-ray  $\nu(t)$  probe* of the search program. Sanity gate PASS:  $\Delta\nu/\nu$  2019 =  $2.39 \times 10^{-6}$  vs  $2.5012 \times 10^{-6}$  published, 2021 =  $1.272 \times 10^{-6}$  vs  $1.26 \times 10^{-6}$ ; the 2024 glitch is not reducible (only  $\sim 150$ -s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ( $\omega = 2.5827$ ,  $\varepsilon_{\text{pred}} = 0.0173$ ): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods ( $<$  the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw  $p = 0.014$  has the kernel as smallest- $p$  battery member BUT Bonferroni  $\times 10 = 0.14$  and shuffle  $p = 0.395 \Rightarrow$  chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021  $p = 0.731$ ;  $\ln \tau$  stack  $p = 0.049$ /shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise:  $\varepsilon_{50} \approx 0.55$  (2019 only),  $\varepsilon_{90}$  never reached below  $\varepsilon = 1$ , 0% power at the predicted  $\varepsilon = 0.0173$  — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag:  $\lambda = e$  in the 2021 leg (raw  $p = 0.002$ , Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider:  $<3$ -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs “pipeline proven” detection at  $F_0 = 11.19275 \text{ Hz}$  ( $H = 18.4$ ) was a noise peak; the true Vela frequency at that epoch is  $F_0 = 11.1861692 \text{ Hz}$  ( $H = 99.7$ , consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray  $\nu(t)$  (PG.06b FULL)” (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.



**2026-07-21** (the two 2026-07-20 prediction-of-record rows promoted to the suite: **v494** —  $\alpha_s$  running +  $\mu$ -distortion branch discriminator machine-checked, ledger rows **COSMO.RUNNING.01/COSMO.MUDIST.01**, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494\_cosmo\_running\_mu\_record.py (23 checks, all green)**. The  $\alpha_s$ -running and  $\mu$ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact):  $\alpha_s = dn_s/d \ln k = -2/N_\star^2 = -r/6$ ,  $\beta_s = -4/N_\star^3$  (NOT  $-8/N_\star^3$ ); LO band  $N_\star \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ . [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check  $< 1\%$ ):  $\alpha_s(51.4) = -7.09 \times 10^{-4}$ ,  $\beta_s(51.4) = -2.70 \times 10^{-5}$ . [C] data: Planck 2018  $(-0.0045 \pm 0.0067) + 0.57\sigma$ ; record leg P-ACT-LB  $(+0.0062 \pm 0.0052) - 1.33\sigma$  (watch channel: positive central value; kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch). [C]  $\mu$ -distortion (Chluba window, +21% calibrated on the  $\Lambda$ CDM anchor): sharp branch  $N_\star=51.4$   $\mu = 1.6 \times 10^{-8}$ , profiled record  $N_\star=56.1$   $\mu = 1.94 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; branch split  $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$   $A_s/N_\star$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; PIXIE baseline  $9 \times 10^{-8}$  does not reach; kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ ). [E] comb blindness (honest S15): the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is structurally invisible in the  $\mu$  window ( $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage-2050). Fences:  $N_\star$  stays an external reheating input (GATE.NSTAR.01); the frozen band  $[50, 60]$  stays the surface of record (v84/v86). Ledger rows **COSMO.RUNNING.01** and **COSMO.MUDIST.01** added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. `predictions.txt` gains the two rows ( $\alpha_s$  [C] with the  $-1.33\sigma$  P-ACT-LB pull and the positive-running kill;  $\mu$ -distortion [C] data-limited with the Voyage-2050 branch split); website `lib/predictions.ts` gains the matching cards `alpha-s-running` and `mu-distortion` (confrontation + experiment blocks pointing at `experiments/tfpt-discovery`); the cosmology DAG node lists **v494** and states the running/ $\mu$  readouts.
- **Paper bodies**. `tfpt_1` (Inflation interface) gains the running consistency-relation paragraph ( $\alpha_s = -2/N_\star^2 = -r/6$ , exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; `tfpt_2` (inflation-pivot keybox) gains the  $\mu$ -distortion branch-decider sentence (sharp  $1.6 \times 10^{-8}$  vs profiled  $2.0 \times 10^{-8}$ ,  $3\sigma$  at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

**2026-07-20 (VIII)** (E8 mass-ladder blind-recovery bed **EXECUTED**: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov–E8 spectra of  $\text{BaCo}_2\text{V}_2\text{O}_8$  (8/8) and  $\text{CoNb}_2\text{O}_6$  (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8\_ladder\_bed\_v1.yaml)**. The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset:  $\text{BaCo}_2\text{V}_2\text{O}_8$  (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**,  $\chi^2/\text{dof} = 0.35$ , MC  $p = 0.0056$  (singles-only run  $p = 0.0004$ ;  $\sigma$ -variant robust);  $\text{CoNb}_2\text{O}_6$  (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**,  $\chi^2/\text{dof} = 0.52$ ,  $p = 0.0038$  ( $m_7/m_8$  lie outside the 0.6-THz low-pass window; the broad 0.43-THz ( $m_1+m_2$ )-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of  $2I - A(E_8)$  equals the Zamolodchikov ladder to  $1.5 \times 10^{-15}$  ( $m_2/m_1 = \varphi$  to  $2.2 \times 10^{-16}$ ; lowest Cartan

eigenvalue  $2 - 2\cos(\pi/30)$  to  $4.5 \times 10^{-14}$ ).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset  $\pm\{0.3, 0.5\}m_1$ : 0/8 detections; jitter dose-response  $\leq 5\%$  at the destructive 30% level; wrong ladders on the real data (harmonic 1..8,  $A_3$ -PF,  $D_8$ -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates,  $\chi^2$ -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked  $\rightarrow$  consistent (stage not\_applicable, tier analog\_positive\_control, leakage detector\_control — the executed-analog basket of qc-recovery-kernel and qgeo-eit-soff); counters 120 rows: 49 consistent / 3 parked; experiments/README.md catalog row + stats (generator); website predictions.ts EXPERIMENTS\_AUDIT counters; experiments/next.txt. This validates the ladder-detector pipeline — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [E], no \veri, no ledger row.

**2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation  $XY=Z^4+a_0$  is a PURE SCALE with the  $\tau=i$  pillowcase frozen, the  $\mu_4$  clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$  with no sector leak; verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O])**

- **v493\_celestial\_wp2\_clock\_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the experiments/tfpt-discovery/probe celestial\_seam\_wp2\_clock\_deformation\_probe.py. *Geometry:*  $P(iZ)=P(Z)$  forces  $a_3=a_2=a_1=0$  sharply and two-sided ( $\prod_k (Z - i^k w) = Z^4 - w^4$ );  $XY=Z^4+a_0$  is smooth iff  $a_0 \neq 0$  (disc =  $256 a_0^3$ ); the branch points  $Z^4 = -a_0$  are one free  $\mu_4$  orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of  $w^2=Z^4+a_0$  are  $I=12a_0$ ,  $J=0$  *identically*, so  $j=1728$  and the  $\tau=i$  pillowcase shape (v168/v214) is *frozen* for all  $a_0$  —  $a_0$  is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:*  $\Pi = t(i-1)(1, i, i^2)$ , all three vanishing spheres carry the same volume  $\sqrt{2}t$  (lockstep); the clock acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3+x^2+x+1$ , eigenvalues  $\{i, -1, -i\}$  = the  $A_3$  exponents,  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ), fixes no cycle ( $\det(A-1)=-4$ ), and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$ , *not* the naive invariant diagonal; the versal weights  $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$  are the regular  $\mu_4$  representation. *Algebra (BHS  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ , arXiv:2305.09451):* mode dictionary  $\{w[m, n] : m \equiv n \pmod{4}\}$ ; the fibre bracket  $-4\text{Nambu}(XY - Z^4 - a_0)$  anchored at the  $a_0=0$  orbifold ( $\{X, Y\}=16Z^3$ , Jacobi + Casimir exact, uniform clock-degree  $-1$ ); S-algebra corrections exactly *linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade — no sector leak, the v492 equivariant sector deforms consistently;  $a_0 \leftrightarrow$  BHS’s weight-0  $c^2$  slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/ $\mathbb{Z}_2$  middle node,  $4=2 \times 2$ ). *Negative controls:*  $a_1 Z \Rightarrow j=0$  (hexagonal); pure  $a_2 Z^2$  stays singular ( $A_1$ );  $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$ ; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications ( $-4\text{Nambu}$  = the  $k=4$  CCA bracket; period = root difference; seam-scale reading via  $\text{clock}^2=\text{deck}$ ); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** tfpt\_research\_contracts CELEST.SEAM.01 section: WP2 bullet “finite, next”  $\rightarrow$  *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction =  $\chi_1$ -diagonal,  $a_0$  pure scale); kill-test paragraph K1

“survived by WP1 and WP2”; status line CELEST.WP2.01 [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02 (four exact v493 mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as v491); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). SEAM.EQUIV.01 stays [O]; no marker moves anywhere. Exact (sympy).

**2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data\_limited→null, no marker moves)**

- **H3 stage 2 executed** (`experiments/hfqpo-ladder/results/archival/`). The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms  $4.4\sigma$ , 280.1 Hz at 3.71%  $9.9\sigma$ ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth  $\nu_3 = 1.5\nu_u$  (661.5/414/252/363 Hz) nor the integer control line  $4\nu_0$  (588/368/224/322.7 Hz) is detected in any source ( $\sim 0\sigma$  single-trial; trials correction  $N=8$  as preregistered).  $3\sigma$  rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) `data_limited→null` (120 rows: 26 null / 33 `data_limited`), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); website `predictions.ts` (hfqpo-tooth entry + EXPERIMENTS\_AUDIT counters); `predictions.txt`; `experiments/README.md`; `experiments/next.txt`. No verification/ module, no ledger row, no marker move (the search target stays [O]).

**2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant celestial chiral algebra on the  $A_3$  ALE space, with the exact correction  $\text{clock}^2=\text{deck}$  (spin bridge  $\mathbb{Z}_8$ ,  $8=2|\mu_4|$ ); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])**

- **v492\_celestial\_z4\_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128  $\mu_4$ -glue grading of  $E_8$  ( $240 = 52+64+60+64$ , additive on all 6720 root pairs) is *inner* —  $h = (2, 2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$ , an order-4 inner torus automorphism = a flat  $\mathbb{Z}_4$  monodromy in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the same class = the diagonal (1, 1) of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  (the v92/v125 Lagrangian glue).  $\mathbb{C}^2/\mathbb{Z}_4$  is verified as the  $A_3$  singularity  $XY=Z^4$  (Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ , Molien exact); the glue-equivariant SDYM( $E_8$ ) sector *closes* with graded dimensions  $60(d+1)/64(d+1)$  — possible only because  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$  — with zero modes = the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$  and density  $1/4 = 1/|\mathbb{Z}_4|$ ; the four glue-sector characters sum exactly to  $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$  (v377), with sector weights = the discriminant form  $(5x^2+3y^2)/8$  and integer glue-diagonal  $h = (0, 1, 1, 1)$  (= locality of the  $(E_8)_1$  extension). *Critical correction*

(*verdict B*): the  $A_3$  deck acts on the celestial sphere as  $z \mapsto -z$  (order 2), *not* as the order-4 clock; the clock is the  $U(2)$  phase  $\text{diag}(i, 1)$  (normalising the deck,  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator), with the exact spin bridge (spin clock)<sup>2</sup> = deck ( $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$ , the  $c_3 = 1/(8\pi)$  winding integer). Clock-invariance selects the 1-parameter  $A_3$  deformation  $XY=Z^4+a_0$  whose four branch points are one  $\mu_4$  orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action  $\text{diag}(i, i)$  killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity =  $\text{Aut}(\mathbb{Z}_4)$ .

- **New research contract CELEST.SEAM.01** in `tfpt_research_contracts` (own section, the fourth contract alongside `Uwall/Gmetric/CONTRACT.F.01`): hypothesis (seam = glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the CCA on the  $A_3$  ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum  $(E_8)_1$  existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (`papers.ts` section, script index). SEAM.EQUIV.01 stays [O]; no marker moves. Exact (sympy).

**2026-07-20 (IV) (evidence scorecard: five new typed rows — the  $\alpha_s$  running prediction of record, the  $\mu$ -distortion  $A_s/N_\star$  branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)**

- **$\alpha_s$  running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index:  $\alpha_s = -2/N_\star^2 = -7.6 \times 10^{-4}$  at  $N_\star=51.4$  (exact slow roll  $-7.09 \times 10^{-4}$ ; band  $N_\star \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ ) and  $\beta_s = -4/N_\star^3$  (leading order; NOT  $-8/N_\star^3$ ) — the identity  $\alpha_s = -r/6$  holds only at leading order ( $\sim 7\%$  off exact). Data: Planck 2018  $-0.0045 \pm 0.0067$  ( $+0.57\sigma$ ), P-ACT  $+0.0060 \pm 0.0055$  ( $-1.22\sigma$ ), record leg P-ACT-LB  $+0.0062 \pm 0.0052$  ( $-1.33\sigma$ , Calabrese et al. 2025 JCAP);  $\beta_s$  (Planck run+runrun)  $0.010 \pm 0.013$  ( $-0.77\sigma$ ). Forecast: LiteBIRD+CMB-S4  $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$  ( $0.14\sigma$ ); CMB-S4+DESI+Euclid statistical  $\sim 3.3 \times 10^{-4}$  ( $2.2\sigma$ , systematics-limited). Same `N_star_reheating` independence group as the  $n_s/r/A_s$  rows (anti double-counting: never an independent hit); kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **$\mu$ -distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch  $N_\star=51.4$  ( $A_s=1.76 \times 10^{-9}$ )  $\mu \approx 1.6 \times 10^{-8}$ , profiled record  $N_\star=56.1$  ( $A_s=2.10 \times 10^{-9}$ )  $\mu \approx 2.0 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; the 23% branch split ( $\Delta\mu = 3.0 \times 10^{-9}$ ) gives a  $3\sigma$   $A_s/N_\star$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; the PIXIE baseline  $|\mu| < 9 \times 10^{-8}$  does NOT reach — factor 4–7 above the band; FIRAS  $|\mu| < 9 \times 10^{-5}$ ). Honest S15 typing: the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is STRUCTURALLY BLIND in the  $\mu$  window — window attenuation  $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two  $A_s$  rows (one branch question, several readings). Kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ . Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (parked\_analog; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall:  $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$  with no charge Hall (seam edge  $c_-=8$ , v450/v451/v489); no engineered material yet ( $\alpha\text{-RuCl}_3$   $c_-=1/2$  contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated  $\kappa_{xy}$ , PRX 12, 021025). (ii) E8 mass-ladder validation bed ( $\text{CoNb}_2\text{O}_6/\text{BaCo}_2\text{V}_2\text{O}_8$ ): all 8 one-particle masses published (Zou et

al. 2021 PRL 127, 077201, NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz  $m_1$ – $m_6$ ; Coldea et al. 2010 Science  $m_2/m_1=\varphi$ ) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning:  $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$  at  $M/m = 8.67$  (2 resonant pairs); best candidate  $^{52}\text{Cr}_2$ – $^6\text{Li}$  ( $M/m=8.635$ ,  $s_0=1.2890$ ,  $\lambda=11.44$ ,  $+0.44\%$  vs  $(3/2)^6$ , gap deviation  $+0.18\%$ ); alternatives  $^{39}\text{K}_2$ / $^{41}\text{K}_2$ – $^6\text{Li}$  ( $\lambda = 11.59/11.20$ , needs double Feshbach control),  $^{176}\text{Yb}_2$ – $^{23}\text{Na}$  ( $+5.9\%$ ); solver validated on the Naidon–Endo anchors (homonuclear 22.69,  $\text{Cs}_2\text{Li}$  4.877); first end-to-end HARDWARE injection of the comb detector at tunable  $\lambda \approx 11.39$ ; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

**2026-07-20 (III) (v491 —  $g_{\text{car}}=5$  as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely  $\{1, 1, 2\}$ , sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)**

- `v491_p2_partition_corollary.py` (P2.PARTITION.01). Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE —  $\{1, 1, 2\}$  — so  $e = (e_1, e_2, e_3) = (4, 5, 2)$  and hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = e_2 - e_3 = 3$  are *corollaries*, not inputs (consistency fixed point  $(2^4 - 1)/5 = 3 = \text{rank } H^1$ ). [E] negative controls, each assumption load-bearing: drop positivity  $\rightarrow$  4 solutions,  $e_2$  ambiguous  $\{0, 3, 4, 5\}$ ; sum 3  $\rightarrow e_2=3$ ; sum 5  $\rightarrow$  not unique; 2/4 parts  $\rightarrow e_2 \in \{3, 4\}/\{6\}$ ; Mehta–Seshadri RATIONAL weights  $\rightarrow e_2 = \frac{16}{3}$  resp.  $\frac{44}{9} \neq 5$  (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins  $(1, 1, 2)$  with ALL THREE targets  $(4, 5, 2)$ ; the partition needs only  $e_1=4$ . [E] sum rule:  $4 \times \frac{N-1}{2} = 4$  iff  $N=3$ ; the v115 route  $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$  agrees. Independence chain: #marks = 4 from v216 (P1-side, no  $g_{\text{car}}$  input); rank  $H^1 = 3$  pure topology (v137/v228);  $|a|_1 = 4$   $g_{\text{car}}$ -free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity  $h^0=0/\text{sum}$  “given (U)”); AX.P2.01 stays a declared axiom, and the  $N_{\text{fam}}$  direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

**2026-07-20 (II) (v490 — the  $T^2$  spin-structure parity census:  $\mu_{\text{pre}}=4$  witnessed on the lattice, the 16-fold-way bridge  $\theta_v=1$  on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)**

- `v490_seam_parity_census.py` (SEAM.MU.01). The  $T^2$  ground-state fermion parity of the v367  $p+ip$  collar, resolved per spin structure ( $AA, AP, PA, PP$ ) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial  $M=3$  phase, with the twisted real-space spectra matching the Bloch bands to  $9 \times 10^{-15}$ . [E] per layer  $(+, +, +, -)$  (the  $\nu=1$  Ising signature, TRIM  $\equiv$  Pfaffian in every sector); [E] the 16-layer carrier has parity<sup>16</sup> = + in ALL FOUR sectors  $\Rightarrow$  gauged torus GSD = 4 =  $\mu(\text{SO}(16)_1) = |\det \text{Cartan}(D_8)|$  — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$  before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers:  $\theta_v = e^{2\pi i \nu/16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489), rivals  $\nu = 1, 2, 8$  not holomorphically condensable; [E]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH  $M=1$  and  $M=3$  ( $\gamma_f=0.000001/0.000000$ ) — it does NOT discriminate  $(E_8)_1$  vs  $SO(16)_1$ . [C] the KLM step  $\mu : 4 \rightarrow 4/2^2 = 1$  stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$ . [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact  $\theta_v$ ).

**2026-07-20 (v489 — the chiral central charge  $c_-=8$  measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); *no gate closes, no marker moves*)**

- **v489\_seam\_modular\_commutator.py (SEAM.CCC.01).** The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367  $p+ip$  seam collar model, entirely from the Peschel Majorana covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ,  $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$ ), on a disk about the torus centre cut into three  $120^\circ$  wedges. [E] ground-state purity  $\text{eig}(i\Gamma) = \pm 1$  to  $6 \times 10^{-15}$ ; [E] per-layer  $c_- = 0.499998$  at  $L=32$  (convergence  $0.499793$  at  $L=16 \rightarrow 0.500000$  at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ); [E] controls: trivial  $M=3 \rightarrow 0.000000$ , orientation flip  $M=-1 \rightarrow -0.499986$ ; [E] layer additivity exact ( $1.4 \times 10^{-14}$ ); [E] the 16-layer carrier lands on  $c_-=8=g_{\text{car}}+N_{\text{fam}}$  (direct  $7.9967$  at  $L=16$ , additivity route  $8.0000$ ); [E] clip invariance  $5 \times 10^{-8}$ . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ( $c_-=8$ ) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

**2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; *no result, no status change*)**

- **introduction.tex — new section “TFPT in the light of algorithmic physics”.** Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.

**2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$  Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)**

- **v488\_majorana\_clebsch\_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung  $M_R = c_3^3 \bar{M}_{PI}$ : a third-generation Majorana factor  $r \approx 1.6696$ , 0.18% from  $g_{car}/N_{fam} = 5/3$  — declined for lack of mechanism. The mechanism search is now EXECUTED: [E] NO RENORMALISABLE CHANNEL —  $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$  and the  $\mathbf{126}$  is NOT in the  $E_8$  hull  $\{1, 10, 16, 45\}$  (v247), so  $M_R$  only via the  $d=5$  operator  $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$ ; channels  $\mathbf{16} \times \mathbf{16} = \mathbf{1} + \mathbf{45} + \mathbf{210}$  with only singlet +  $\mathbf{45}$  admissible. [E] CHARGE TABLE:  $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$ ,  $\text{Tr}_{16} Y^2 = \frac{10}{3}$  (v485). [E] THE  $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero  $\nu^c$  channel weight  $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$  is  $\{2, 3\}$ -smooth, and the multiplicative closure of  $\{2, 3\}$ -smooth rationals never contains  $\frac{5}{3}$  (no factor 5); the UNIQUE natural 5 of the embedding,  $k_Y = \frac{5}{3}$  (Ginsparg, v470), is a full-multiplet trace whose direction has  $Y(\nu^c) = 0$  EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: Clebsches are family-space scalars;  $\text{diag}(1, 1, \frac{3}{5})$  cannot arise from group theory by construction. [O] compiler-flavor inventory typed and declined: exact  $\frac{3}{5}$ 's only in wrong slots ( $K$  lepton gen2/gen1;  $R + L$  lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate ( $M_R$  free inside the PS window, the  $\Sigma m_\nu = 0.059$  eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk's one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)**

- **v487\_transfer\_clock\_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$  vs uniform  $(\frac{1}{3}, \frac{1}{3})$  is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: [E] RUNG READING — the lazy rule's one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124:  $\lambda_n^{1/p_2} = 1 - n/N_{fam}$ , pole at  $n=N_{fam}$ ):  $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$ , while v327's uniform rule puts its second pair mode ON the wall ( $0 = 1 - 3/3$ ) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. [E] DECK PARITY = RUNG INDEX:  $[B, \sigma] = 0$ ; the  $\sigma$ -even pair mode carries rung  $n=1$  (survival  $\frac{2}{3}$ ), the  $\sigma$ -odd mode rung  $n=2$  (survival  $\frac{1}{3}$ ) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment ( $\omega_1$  even-channel,  $\omega_2$  odd-channel) that `experiments/frb-parity-comb` tests. [E] FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) +  $\mathbb{Z}_2$  equivariance + non-negative rates forces UNIQUELY both the split  $(\frac{1}{2}, \frac{1}{6})$  AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs  $h = -\frac{1}{6} < 0$ ); the “sheet-symmetric choice” candidate is superseded by this proof. [E] BEND COROLLARY:  $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$  — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] what remains open is exactly what was open before v486: the arity  $\{2, 3\}$  (anchor input) and the clock's semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (v486 — generator hunt: the FULL transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  from ONE forced local rule — the unique lazy  $\mathbb{Z}_2$ -pair walk (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$ ; the previously generator-less  $(1/3)^6$  mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)**

- **v486\_transfer\_full\_rule.py (HYP.REWRITE.02).** The verified seam transfer spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  (v54/v221), but the only local generator in the suite (v327) produces  $\{1, \frac{2}{3}, 0\}$  — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUE-NESS: for the symmetric 3-channel rule  $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$  (one absorbing family channel +  $\mathbb{Z}_2$  pair) the survival spectrum is  $\{s+h, s-h\}$ ; demanding the PHYSICAL pair  $\{\frac{2}{3}, \frac{1}{3}\}$  forces UNIQUELY (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$  — the lazy  $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum  $\rightarrow$  rates). [E] SIX-HAND GENERATION: over the order-6 =  $2N_{\text{fam}} = |R^+(A_3)|$  hand,  $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$  exactly — both decay gaps now have one recursive generator:  $6 \ln \frac{3}{2}$  (recovery, v76) and  $6 \ln 3$  (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its  $\lambda_2$  arity story ( $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ) is unchanged. [O] the arity  $\{2, 3\}$  stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay =  $1/|\mathbb{Z}_2|$  as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits  $a^T K a = 41$ , Koide 53, and the recorded-only  $1^T Q a = 16$  — and the lepton-exponent observation  $(5, 3, 2) = K\text{-row} = \text{atom triple}$ ; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)**

- **v485\_contact\_diagonal\_closed.py (SEAM.CONTACT.UNIT.02).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of  $|D|$  on circumference  $\ell$  is  $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$  EXACTLY (symbolic limit + 30-digit checks); it vanishes iff  $\ell = 2\pi = 1/(4c_3)$ , the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure:  $F_{U(1)}$  carries exactly ONE log term, the transport  $8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$ , while the  $\varphi_0^{\text{ret}}$  split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly —  $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ ,  $u = \varepsilon c_3 \ln 2$ , all orders, the linear term ABSENT precisely because the diagonal vanishes ( $\text{Tr } C = 0$ ); the series matches v484's  $\text{Tr}(C^k) = 4^k + 2(-2)^k$  term by term. [E] WEIGHT DICHOTOMY: the SAME  $48 = 3 \times 16$  admissible states carry BOTH multiplicities — counted flat they give the  $\varphi_0^{\text{ret}}$  puncture count  $48 = \Omega_{\text{adm}}$ ;  $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget  $41 = 10b_1 = 40 + 1$  ( $\text{Tr}_{16} Y^2 = \frac{10}{3}$  exact,  $\nu^c = 0$ ; Ginsparg  $(3/5)(41/6) = 41/10$ , v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHIO.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam  $\zeta$ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update



notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

**2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ $c_3$  per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHI0.PUNCTURE.01 is the KMS seam unit  $2\pi = 1/(4c_3)$  with  $\frac{1}{4} = 1/|\mu_4|$ , derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)**

- **v484\_seam\_contact\_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight  $c_3$  per boundary contact insertion”: the  $F_{U(1)}$  ladder  $\{0, 3, 6\}$ , the  $\varphi_0^{\text{ret}}$  puncture  $\{1, 4\}$ ). This module shows the shared rule is NOT a new unknown: [E] the unit chain  $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$  — one *bare* boundary insertion divided by the orbit order  $|\mu_4|=4$  is exactly  $c_3$ ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function  $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$  (polylog identity, 40 digits) takes INTEGER multiples of  $c_3 \ln 2$  at the  $\mu_4$  separations ( $G(\pi/2) = -4c_3 \ln 2$ ,  $G(\pi) = -8c_3 \ln 2$ ; mark matrix  $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$ , spectrum  $\{4, -2, 0, -2\}$ ), so the  $\mu_4$ -invariant (orbit-averaged) insertion carries  $-(c_3 \ln 2)C$  exactly and the log-det contact expansion is exactly graded in  $c_3$  per insertion ( $\text{Tr } C^k = 4^k + 2(-2)^k$ ). [E] the grammar is suite-wide: the  $\Lambda$  prefactor  $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$  (v60) carries the SAME multiplicity  $\Omega_{\text{adm}}=48$  as the  $\varphi_0^{\text{ret}}$  puncture at TWO insertions; the  $g-2$  seam loop divides by the same  $2\pi = 1/(4c_3)$ . [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ( $48 = \Omega_{\text{adm}}$ ,  $41 = 10b_1$ ); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

**2026-07-14 (II) (v482–v483 — the deep-dive round: the  $\nu$ -scale decision computation EXECUTED (the integer  $c_3^3$ -rung excluded unstructured, the  $5/3$ -proximate structure escape recorded and declined) and the  $\varphi_0^{\text{ret}}$  puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are  $t$ -independent rationals, so the  $\pi^{-4}$  must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)**

- **v482\_seesaw\_rung\_decision.py (FLAV.NUSCALE.03).** The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung  $M_R = c_3^3 \bar{M}_{P1}$  needs a rescue factor  $\times 1.670$  in  $m_3$  (0.0301 vs 0.0503 eV), while  $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$  ( $> 3\sigma$ ),  $\alpha_3^{-1} \mp 0.10$  ( $> 3\sigma$ ), a  $\pm 10\%$  band on the ADKLR  $\kappa$  run-down and a PS-leg Yukawa-beta envelope  $\times [0.5, 1.5]$  move it by at most  $\times 1.10$  singly and  $\times 1.165$  ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise  $y_\nu = y_t$  (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor  $r \sim 1.67$ ; its post-hoc 0.18% proximity to  $g_{\text{car}}/N_{\text{fam}} = 5/3$  is recorded and DECLINED per v354/v355 (no forcing mechanism; the  $t_0 \sim 30$  discipline of v478). [O] the v481 one-parameter window candidate (NO floor  $\Sigma m_\nu = 0.059 \text{ eV}$ ) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` ( $\nu$ -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483\_pillowcase\_exact\_traces.py (HYP.PHI0.GEOM.01).** The GEOMETRIC side of the  $\varphi_0^{\text{ret}}$  puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat  $\tau=i$  pillowcase is a  $t$ -INDEPENDENT RATIONAL —  $\text{Tr}(\gamma e^{t\Delta}) = 1$  for

$\gamma = \sigma, \rho, \rho^3$  at every tested  $t$  (position-space Gaussian image sums  $< 10^{-9}$ ; mode-space proof: only  $n=0$  is  $\gamma$ -diagonal), matching the Atiyah–Bott fixed-point counts  $4 \times \frac{1}{4}$  ( $\det = 4 = |\mu_4|$ ) and  $2 \times \frac{1}{2}$  ( $\det = 2 = |\mathbb{Z}_2|$ ); the pillowcase contact term is EXACTLY  $\frac{1}{2} = 4 \times \frac{1}{8}$  per mark and the clock-equivariant trace exactly 1, for all  $t$ . THEREFORE the  $\pi^{-4}$  in  $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$  cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ( $4 = |\mu_4|$  marks of weight  $c_3$  give  $c_3^4$ , multiplicity  $\Omega_{\text{adm}} = 48$ ; the bare  $1/(2\pi)$ -propagator 4-cycle differs by the exact rational  $\frac{3}{16}$ ). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight =  $c_3$ ” (the same residual class as the EM-Ward  $c_3$ -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows HYP.PHIO.GEOM.01 + HYP.PHIO.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

**2026-07-14 (website + figure refresh — the /verification journey views become live HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)**

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“~295-script journey”, “260 scripts” → the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ~13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (AuditRound: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (**break-words**) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long ALPHA.QUILLEN residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the SEAM.EQUIV.01 bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\ctthree`, ...) rendered as red KaTeX errors on the site (e.g. the  $\varphi_0^{\text{tree}}$  icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero **katex-error** nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable Expandable component clamps long mirror prose with a fade + “Show more” toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

**2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four  $\mu_4$  marks realised as the exactly solvable *four-interval* multilocal modular geometry with  $\omega \circ \rho = \omega$  manifest; and the absolute  $\nu$ -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation  $y_\nu = y_t$  — *no gate closes, no marker moves*)**

- **v479\_kronheimer\_quiver\_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the  $\det K=1$  keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  are the null vector of the affine- $E_8$  Cartan matrix AND the Perron vector  $A\delta=2\delta$  of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the  $\mathbb{C}^2/\Gamma$  geometry;  $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$  forced by the null identity); the gauge datum is  $|2I|=120=|\mu_4|h(E_8)$ ,  $h(E_8)=30=2 \cdot 3 \cdot 5$ ; the resolution carries the unimodular  $-E_8$  form ( $\det=1$ , SNF = identity — the Poincaré link) with  $3 \cdot 8=24$  deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the  $(2, 3, 5)$  cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over  $\mathbb{Z}$ ,  $\sum \delta_i^2 = 120 = 4 \cdot 30$ ,  $30 = 2 \cdot 3 \cdot 5$ , dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).

- **v480\_multilocal\_four\_interval.py** (QGE0.MULTILOCAL.01). The seam circle minus its four marks IS the  $n=4$  multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic  $\Leftrightarrow$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  EXACTLY (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$ , each sector a single-interval quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1}=C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at  $10^{-2}$  (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in **tfpt\_research\_contracts** (marks keybox + rigidity contract + v478 leg A), **origin\_theory** (BW paragraph), **tfpt\_4\_frontier** (v358 companion note); Python-only by nature.
- **v481\_seesaw\_carrier\_ladder.py** (FLAV.NUSCALE.02, **CANDIDATE** class, **genre v467/v468**). FLAV.NUSCALE.01's  $y_\nu=1$  probe was not the carrier normalisation: one  $SO(10)$  **16** per family + the minimal Yukawa sector ( $\mathbf{10}_H$  / PS  $(1, 2, 2)$ ) forces  $y_\nu=y_t$  at the matching scale (named premise). With explicit 1-loop RG (gauge/ $y_t/\lambda$  up; ADKLR Weinberg-operator running down, hep-ph/0108005) the  $(y_\nu, M_R)$  trade-off collapses to  $M_R$  alone: the observed  $m_3=0.0503$  eV demands  $M_R=9.3 \times 10^{13}$  GeV — INSIDE the PS window  $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$  (v249) at  $\log_{c_3}(\bar{M}_{\text{Pl}}/M_R)=3.15$  (the  $y_\nu=1$  inversion gives the structureless 2.58). [X] honesty gate: the integer rung  $M_R=c_3^3 \bar{M}_{\text{Pl}}$  predicts  $m_3=0.030$  eV — 40% low at 1-loop ( $\kappa$  run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band  $m_3 \in [0.002, 0.115]$  eV and the NO floor  $\Sigma m_\nu=0.059$  eV; frozen record untouched. Integrated in **tfpt\_2\_standard\_model** (new paragraph after v272), **introduction** status card (third named candidate), **tfpt\_5\_redteam**; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the  $\phi_0$  puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights  $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$  verified) but does NOT force the value  $48c_3^4$  (pure rationals vs  $\pi^{-4}$ ) — recorded in **experiments/theory-contracts/phi0\_donnelly\_equivariant\_probe.py**; the target stays [O].

**2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy  $c_{\text{est}} \rightarrow 1$ , CHM parabola, even bands exactly zero) — the bridge’s modular side meets TFPT’s Einstein-derivation input; and the measure condition reduced to one exact KMS time  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$  with the  $h(E_8)=30$  near-miss explicitly declined; both legs stay [O], *no status moves*)**

- **v478\_entropic\_continuum\_legs.py** (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has  $S(L)=(c/3) \ln L + k$  with  $c_{\text{est}}=1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1=2 \times \frac{1}{2}$ , the v450 reading; gapped control  $\sim 6 \times 10^{-8}$ , area law); the Peschel modular Hamiltonian  $K=\ln((1-C_A)/C_A)$  organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr 0.87 $\rightarrow$ 0.99, shrinking plateau drift; constants deliberately not asserted), the even-range bands

vanish identically (particle-hole,  $\sim 10^{-36}$ ) and the odd tail is 1.8% with parabolic envelope — the state-side  $\Delta_\Sigma$  (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT's Einstein derivation consumes (v323/v358): the bridge's modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary —  $d\mu=\delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2=e^{t_0}$  exactly, so the v477 one-moment condition forces  $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$  (one dimensionless KMS time, closed form); the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

**2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving  $R^2$  route typed as *one moment condition*  $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$  with the exact closure  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$  — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)**

- `v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`). [E] The Frullani identity upgrades to a scale-flow representation:  $-\ln a = \int \frac{dt}{t}(e^{-ta}-e^{-t}) = 2 \int \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$  (symbolic +  $10^{-20}$  quadrature) — Bianconi's log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT's  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight  $w(t) dt/t$  gives  $3\beta R \mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2$ ; the flat weight has  $\mu_1=\mu_2=1$  and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding  $m^2=c_3^7 \bar{M}_{\text{P}1}^2$  forces exactly  $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$  — the v475 kill-test number IS a moment ratio — and the closure identity  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$  holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency:  $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\text{P}1}^2=c_3^7$  — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein +  $\Lambda$  level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the  $R^2$  gate and the v359 typing stay [O], no marker moves.

**2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)**

- `v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`). The last open work package of the v473 round, exercised on a gapped staggered-mass chain ( $N=16$ ). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to  $10^{-15}$ ; thermal boundary:  $\text{spec}(PCP) \subset (0,1)$ , Peschel  $K=\ln((1-C_A)/C_A)$  finite,  $S>0$  — v155 in the bridge context). **The reading is decided:** on the pure bulk  $f(x)=x/(1-x)$  is singular on  $\text{spec}(C)=\{0,1\}$ , so the literal operator-side reading  $Pf(C)P$  of the v473 conjecture is *ill-posed* while the state-side  $f(PCP)$  is finite — the compression identity can only mean “build  $\Delta_\Sigma$  from the *compressed* relative metric”, which matches Bianconi's own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic):  $[C^2]_{AA}-(C_{AA})^2=BB^t$ ,  $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$  — first order in the cross block vanishes identically,

so for any analytic  $f$  the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity  $f(C_T)=e^{-h/T}$  holds to  $10^{-11}$  relative; the mismatch obeys the quadratic law  $d \leq x^2$  ( $d/x^2 \in [0.4, 1.0]$ ) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays [O], no marker moves.

**2026-07-13 (v475 — the  $R^2$  kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron  $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$  ( $\approx 51.7\bar{M}_{\text{Pl}}$ , trans-Planckian)  $\Rightarrow$  the light-trace-mode reading is dead, KMS renormalisation the only surviving  $R^2$  route; plus the explicit Lorentzian-positivity witness; *no status moves*)**

- **v475\_entropic\_scalaron\_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. [E] On a maximally symmetric background the relative curvature operator  $\tilde{\mathcal{R}}\tilde{g}^{-1}$  has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (Bianconi's Appendix-B flattening verified symbolically, incl. the  $6 \times 6$  two-form block), giving the exact vacuum action  $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) (= the v473 pin),  $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$  is the exact tensorial  $R^2$  factor. [E] With the pinned  $\beta' = c_3/6$  both mass routes (Starobinsky matching and  $f'/3f''$ ) agree: the raw entropic scalaron is  $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ , i.e.  $m_{\text{raw}} \approx 51.7\bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$  GeV — TRANS-PLANCKIAN, not light; v473's interpretation 2 (the TFPT scalaron as the light  $\mathcal{G}$ -trace mode) is *dead in naive form*; a viable mechanism must supply exactly  $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup> (the kill-test number now WITH tensorial factors). [E] Domain: first log pole at  $R=1/\beta=384\pi^2\bar{M}_{\text{Pl}}^2$  (trans-Planckian) — the obstruction is the mass, not the domain. [E] Lorentzian-positivity witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), spacelike/Euclidean stay positive — the v473 [C] caveat exhibited, the OS route sturdier. [C] Verdict: interpretation 1 survives (Einstein +  $\Lambda$  level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving  $R^2$  route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

**2026-07-13 (v474 — the operator level of the entropic-action bridge: the  $D_5$  Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the  $5 \rightarrow \bar{5}$  conjugation, and the v473  $Q$ -target *decided* — supports exactly  $\{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , the naive pair-block reading killed; *no status moves*)**

- **v474\_entropic\_hodge\_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4 algebra of the v473 bridge, machine-checked on the finite carrier Fock space  $\Lambda^\bullet \mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}} = 32$ ). [E] CAR  $\Rightarrow$  Clifford: the ten  $\Gamma$ 's from  $(a_k^\dagger, a_k)$  satisfy  $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$  exactly (all 100 pairs, integer matrices); the 45 bilinears  $[\Gamma_a, \Gamma_b]/4$  commute with  $\gamma = (-1)^N$ , span exactly  $45 = \dim \mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). [E] the Hodge–Dirac symbol:  $c(\xi) = a^\dagger(\xi) - a(\xi)$  satisfies  $c(\xi)^2 = -|\xi|^2 \text{Id}$  for symbolic  $\xi$  — Bianconi's  $D = d + \delta$  is Clifford multiplication at symbol level. [E] the fold is the  $5 \rightarrow \bar{5}$  conjugation: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16 = 1 + \bar{5} + 10$  (the SM carrier decomposition). [E] honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant ( $\Lambda^2 \rightarrow \Lambda^3$  leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). [E] the  $Q$ -target enumeration (the v473 [C] conjecture decided at the algebraic level):  $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$  with integer  $k$  has exactly the supports

$(d, k) = (2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8, 2^{g_{\text{car}}} = \dim \Lambda^\bullet\}$  — minimal uniform solution  $q = c_3^2 = 1/(64\pi^2)$  per Fock mode, bonus identity  $\text{Tr } Q^2/\delta_{\text{top}} = 32/48 = 2/3 = \dim \Lambda^\bullet/\Omega_{\text{adm}}$ ; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers  $\sqrt{16/5}, \sqrt{2}$  — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the  $R^2$  mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

**2026-07-13 (v473 — the entropic-action bridge: Bianconi’s *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned  $\beta'_B = c_3/6 = 1/(48\pi)$ , the  $R^2$  gap  $3(8\pi)^9$  pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])**

- **v473\_entropic\_action\_bridge.py (GRAV.ENTROPIC.ACTION.01).** TFPT’s classical field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$  (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via  $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$ ) is a local covariant action whose variation gives Einstein + an emergent nonnegative  $\Lambda$  at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her  $\Omega^{0,1,2}$  fiber on the five-slot carrier  $\mathbb{C}^5$  counts  $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$ , while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling  $3\beta R$  to  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,  $\beta'_B = c_3/6 = 1/(48\pi)$ ; [E] her  $\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point, and with  $\varepsilon_\star = e^{-\alpha^{-1}} Q$  it reproduces the closed v60 branch  $(3/4\pi^2)e^{-2\alpha^{-1}}$  with the exact target  $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  ([C] until  $Q$  is constructed); [E] the  $R^2$  kill test — the raw log expansion sits *exactly*  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$  below the TFPT Starobinsky coefficient  $1/(12c_3^7)$ , so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT’s OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

**2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient  $g_{a\gamma\gamma} = -4c_3$ , the free-seam Higgs band  $m_H \approx 129\text{--}134\text{ GeV}$ , and the never-run BH HFQPO  $\times 1.5$  ladder-tooth search; no new vN, no status moves)**

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ,  $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$  (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$  — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction  $\lambda(\bar{M}_{\text{Pl}}) = \beta_\lambda(\bar{M}_{\text{Pl}}) = 0 \Rightarrow m_H = 133.5\text{ GeV}$  (band 129–134,  $m_t/\alpha_s$ -sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO  $\lambda(M_{P1}) = -0.0143 \pm 0.0057, 2.5\sigma$ ) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- $(m_t, \alpha_s)$  RGE pull off the surface at  $> 5\sigma$ ). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at  $\nu_3 = (3/2)\nu_u$  (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (experiments/hfqpo-ladder, preregistered): added as an [O] search-target bullet to the tfpt\_horizon\_readouts “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 +9.2 $\sigma$ , GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (predictions.ts 23  $\rightarrow$  25 cards, papers.ts frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

**2026-07-03 (v472 — the determinant line over the  $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whiteness probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)**

- **v472\_quillen\_detline\_moduli.py (ALPHA.QUILLEN.DETLINE.01).** ALPHA.QUILLEN.EXACT.01 names the  $\alpha^3$  level as the curvature of the *determinant line over the  $U(1)$  seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ( $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$ , v367/v470) in real space on an  $L \times L$  torus with *twisted* boundary conditions — the twist torus  $(\theta_x, \theta_y)$  is the moduli space of flat  $U(1)$  connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus:  $\oint = 2\pi \cdot 1$  exactly (integer to  $10^{-9}$ ) at  $M=1$  for  $L=4$  AND  $L=6$  (size-independent); controls: trivial collar  $M=3 \rightarrow 0$ , orientation flip  $M=-1 \rightarrow -1$ ; the twist-moduli integer equals the Bloch-BZ integer for all three  $M$  (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap ( $= 2.0$ ) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the  $U(1)$  moduli = inflow level  $k_0 = |C| = 1$  — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” holds *verbatim at the finite/model level*, where  $\log \det$  of the occupied frame replaces  $\log \det_\zeta$ . **Not closed [O]:** the *continuum*  $\zeta$ -determinant identification on the abstract seam (= the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: tfpt\_1 (sixth honest step in the EM closure), tfpt\_research\_contracts (residual certification), tfpt\_safeguards ( $\alpha$ -form lever), root README, zenodo\_description.html. Exploration original: experiments/tfpt-discovery/quillen\_detline\_twist\_moduli.py (same day).
- **Bird’s-eye whiteness rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted).** (i) **electron\_sector\_cabibbo\_probe.py**: the  $\alpha$  metrology triangle (TFPT  $\alpha^{-1} = 137.0359992168$  vs Rb-2020  $+0.99\sigma$  / CODATA22  $+1.90\sigma$  /  $a_e$ -route  $+3.39\sigma$  / Cs-2018  $+6.33\sigma$  — TFPT takes the Rb side of the  $5.5\sigma$  metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of  $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$  is excluded by the electron at  $\sim 2 \times 10^4\sigma$ ; linear scaling at  $\sim 10^2\sigma$ ; quadratic survives — the [C] bridge must scale at least quadratically in  $m_\ell$ ); the Cabibbo-angle anomaly as a third dissolution watchdog ( $\lambda_C = 0.2243762$  exact sits  $+0.08\sigma$  on the PDG-2026 rescaled kaon average, between  $K_{\ell 3}$  and  $K_{\mu 2}$ ; exact unitarity puts  $V_{ud} = 0.97450$  — superallowed is the outlier at  $+2.6\sigma$ , all neutron/pion routes agree). (ii) **kernel\_calderon\_uv\_universality.py**: the raw-collar



Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol  $mI'_\nu(mR)/I_\nu(mR)$ , Richardson residual  $3 \times 10^{-4}$ ;  $N_\theta$ -independent;  $|k|$  principal symbol;  $\mu_4$  block structure at machine precision; both spec- $T$  masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays [O]). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure  $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$  [frozen],  $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$  [the 2026-07-02 numerology-flagged Nebenbefund],  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  [frozen],  $\Sigma m_\nu = 0.0588$  eV [v468 route]] + spatial flatness determines the whole flat  $\Lambda$ CDM budget:  $H_0 = 67.15$  km/s/Mpc ( $-0.39\sigma$  Planck18,  $-5.67\sigma$  SH0ES22 — the parameter-free Planck side of the Hubble tension),  $\Omega_m = 0.3133$  ( $-0.27\sigma$ ),  $t_0 = 13.86$  Gyr; honest stresses recorded:  $\omega_b -2.02\sigma$  (BBN-only  $-0.73\sigma$ ),  $\omega_c -1.19\sigma$ , DESI-DR2  $H_0 -2.16\sigma$  (one correlated low- $h$  direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the  $\rho_\Lambda$  leg the  $\alpha$  engine),  $\Omega_c$  leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

**2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)**

- **v471\_seam\_horizon\_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope  $2C(\gamma)$  on deficit cones (0.01–0.05%) and replica sheets  $n=2$  (0.8%),  $n=3$  (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the  $\zeta$ -scale  $\mu_{\text{lat}}$  drifts — the v152 anchor exhibited on data, with *no* canonical  $\ln(m/\mu)=3/4$  emerging (recorded honestly); the BFK/Schur split is exact ( $\sim 10^{-16}$ ) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying  $2C_{\text{cone}}$  via Kac doubling; the v302 transfer masses  $6\ln(3/2)$ ,  $6\ln 3$  sit on the calibrated EH line ( $\leq 0.05\%$ ); and the Perron/attractor mode ( $m=0$ ) is IR-divergent — **the recovery gap  $\Delta=6\ln(3/2)$  is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes  $1/G$  finite). SEAM.THEOREM.01 stays [O]; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at  $f_0/(1+z)$  — all templates moved to detector-frame  $M(1+z)$  (before the fix GW190521,  $z=0.56$ , was filtered  $\sim 1.6\times$  too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked  $p=0.262$ , kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni  $\times 12$ ): ratio semantics (amplitude  $(2/3)^6$  vs *energy reading*  $(2/3)^3$  vs step  $2/3 \times \mu_4$  per-bounce phases  $\{0, \pi/2, \pi, 3\pi/2\}$

(the boundary-birefringence analogue — path birefringence from parity violation  $c_- = 8$  cancels in the inter-echo ratio; only a per-reflection phase does not)  $\times$  lags 0.5–350 ms (ECO through Planckian window), with joint (2, 2, 0)+(2, 2, 1) subtraction and a template-agnosticism control: NO\_VARIANT\_ECHO on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

## 2026-07-02 (v295 robustness patch — external-review follow-up: the numeric $a_2$ -kernel tolerance must dominate its own finite-difference floor)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01). Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference  $a_2$  kernel is compared against an  $\varepsilon = 10^{-3}$  second finite difference whose *own* truncation error is  $O(\varepsilon^2) \sim 10^{-6}$ , yet the tolerance was  $10^{-6}$  — on his machine  $k=3$  landed at  $1.1 \times 10^{-6}$  (ours:  $2.9 \times 10^{-7}$ ). Fix: tolerance widened to  $5 \times 10^{-6}$  so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

## 2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the $\alpha^3$ level = the *computed* bulk Chern invariant + the seam $F$ -normalisation = the embedding index $k_Y = 5/3$ ; both targets stay [O])

- **v469\_seam\_crossedproduct\_route.py** (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the  $SO(16)_1$  spinor has  $h_s = N/16 = 16/16 = 1 \in \mathbb{Z}$ , so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1 \rtimes \{1, s\}$  is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B) = 4/2^2 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4  $\mu_4$  glue runs on the same integer:  $h(J^k) = \{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1)) = k^2$  at net level),  $\mu = 16/4^2 = 1$ , same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_- = 8$  [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern  $|C| = 1$  vs 0 (control),  $\nu = 16 \times |C| = 16$ ,  $c_- = \nu/2 = 8$  — and  $\nu \equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of  $c_-$  is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening**. TfptCarrier/SeamResidualAxiom.lean gains the *parallel* peer-reviewed derivation `seamResidualClosed’`: the invariant-level axiom `collar_invariants` (R1’) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` ( $h_s = 16/16 \in \mathbb{Z}$ ), `glue_locality_integers` ( $\mu_4 h(J^k) \in \mathbb{Z}$ ), `klm_holomorphic` ( $4/2^2 = 1$ ,  $16/4^2 = 1$ ), `sixteenfold_pin` ( $\nu = 16 \equiv 0 \pmod{16}$ ,  $c_- = 8$ ), bundled as `crossedproduct_arithmetic` ([propext] only); `#print axioms seamResidualClosed’` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; lake build OK (3285 jobs).
- **v470\_alpha\_inflow\_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left

on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ( $\alpha^{-1}=137.0359992168$ ; Quillen split  $<10^{-35}$ ). [C] *the level is computed, not minimal*: the  $a^3$  coefficient  $k_0=1$  equals the FHS bulk Chern invariant  $|C|=1$  of the same p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016)  $\eta$ =CS part of  $\delta \log \det$ ; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), replacing v435’s minimality assumption. [E]  $k_Y = \text{tr}(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$  (Ginsparg, Phys. Lett. B197 (1987)):  $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$  — the “GUT 3/5” is the inverse embedding index;  $\frac{41}{6} = \frac{20}{3} + \frac{1}{6}$  (carrier decomposition). [C] the seam  $F$ -normalisation (the one [C] of v434) is *retyped*: once the seam net is  $(E_8)_1$  level 1,  $\langle JJ \rangle = k/(z-w)^2$  is fixed by  $k_Y$  — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses —  $c_- = 8$  (gravitational, feeds SEAM.EQUIV.01/S3) and  $C=1$  ( $U(1)$ , feeds EM1) — so closing R1’ feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E].

- **Integration.** Both registered in `run_all.py` + `script_registry.csv` (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the `seamResidualClosed'` route); the exact joints Wolfram-mirrored (365  $\rightarrow$  368: the LR/glue locality integers, the KLM  $\mu$ -arithmetic, the sixteen-fold pin, the  $k_Y = 5/3/b_1$  identities); cited in `tfpt_1` ( $\alpha$  fixed point + EM-Ward residual prose), `tfpt_research_contracts` (seam residual + Lean section), `introduction` (closure ledger), `tfpt_5_redteam` (Target A residual wording) and the README/website status surfaces; exploration provenance `experiments/tfpt-discovery/closure_route_*.py` (2026-07-02 note in `experiments/next.txt`). Suite ALL CHECKS PASSED through v470. Curated version bumped to **v5.4** (Zenodo deposit updated).

**2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three  $\sim 2\sigma$  mixing tensions as *one* third-generation  $-\varphi_0$  pattern, and the splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$ )**

- **v467\_thirdgen\_mixing\_pattern.py (FLAV.THIRDGEN.PATTERN.01).** The suite’s three  $\sim 2\sigma$  mixing tensions ( $\sin^2 \theta_{13} + 2.0\sigma$ ,  $\sin^2 \theta_{23} + 1.76\sigma$ ,  $|V_{cb}| + 1.79\sigma$ ; v307) are exactly the three *third-generation* channels and share *one* relative deviation  $-5.36\% \pm 1.67\%$  — non-zero at  $3.2\sigma$ ,  $-0.03\sigma$  from  $-\varphi_0$ . [E] vocabulary identities:  $\lambda_C^2 = \varphi_0(1-\varphi_0)$  (the frozen Cabibbo);  $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$ ; the suppression size  $\varphi_0 = 5.3\%$  sits inside the seam-gapped band  $(2/3)^6 = 8.8\%$  (v393). [N] the ladder-base forms  $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2 / (1+\lambda_C) = 0.041119$  drop the joint four-observable  $\chi^2$  from 10.4 to 0.11 with *zero* new parameters;  $|V_{ub}|$  (no bare  $\varphi_0$ ) and the 1–2 channels stay untouched and correct (dressing  $\lambda_C$  would fail at  $-24\sigma$  — scope checked, not chosen); Daya Bay alone already prefers the pattern form ( $\Delta\chi^2 = 4.3$ ). [C] **honest**: post-hoc (5/16 equally-simple dressings fix the two PMNS tensions;  $(1-\varphi_0)/(1-c_3)/\text{resolvent}$  degenerate  $<1.4\%$ ). [O] no seam-transport derivation (the Bernoulli-variance reading  $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$  is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric:  $\theta_{13}$  stable at 0.0231 kills the pattern, at  $\sim 0.0219$  (sub-1%) kills the record’s bare- $\varphi_0$  reading, exclusive  $|V_{cb}| \sim 0.0395$  kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468\_dm2\_ratio\_jarlskog.py (FLAV.DM2RATIO.01).** The splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2$  was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270,  $\delta = 4\pi/3$ ) has one canonical derived CP invariant, and [N]  $|J_{\text{PMNS}}| = 0.029653$  meets the NuFIT 6.0 measured ratio  $0.029805 \pm 0.000792$  at  $-0.19\sigma$  (equivalently  $m_2/m_3 = \sqrt{|J|} = 0.17220$  vs  $0.17264(229)$ ) — zero

dials. [N] the informal `tfpt_2` table heuristic  $m_2/m_3 \sim \pi\varphi_0 = 0.16704$  (a “ $\sim$ ” entry, never frozen) is now  $-2.44\sigma$  and is *superseded as the comparator*. [C] spectrum consistency:  $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$  eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the  $\mu\tau$  texture gives  $\theta_{13}=0$  at leading order). [X] JUNO shrinks the window  $\sim 4\times$ ; adopting the v467 pattern would shift  $|J| \rightarrow 0.028849$  ( $-1.21\sigma$  today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the  $m_2/m_3$  table row retyped to the  $|J|$  comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328’s “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

**2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  back-solves the same axiom seed to  $-0.11\%$ , extending the empirical over-determination to *five* sectors)**

- **v466\_seed\_leptonmass\_channel.py** (`SEED.LEPTONMASS.01`). The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  is a sixth independent readout of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$  to  $10^{-9}$ . [N] **new sector (headline):** back-solving  $\varphi_0$  from the measured *pole* ratio  $m_e/m_\mu = 0.004836$  (PDG 2024) gives  $\varphi_0 = 0.053113$ , only  $-0.11\%$  from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound  $(2/3)^6 = 8.78\%$ , v393) but largely cancels for  $m_e/m_\mu$ , so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306’s five + this) has  $\chi^2/\text{dof} < 1$  and the error-weighted mean stays at the axiom to  $< 0.1\%$ ; with the corroborating heavy-quark  $m_t/m_b$  ( $-0.23\%$ ) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster  $\chi^2$  by  $> 100\times$ , and the scheme-ambiguous  $m_c/m_s = (34/47)/\varphi_0$  (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence;  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

**2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within  $0.01\sigma$ , the  $\theta_{13}$ -independent externally-runnable slice of v306)**

- **v465\_seed\_crosssector\_joint.py** (`SEED.CROSSECTOR.01`). The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation  $\beta$  (a CMB *EB/TB* polarisation observable) and the Cabibbo angle  $\lambda_C = |V_{us}|$  (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $\lambda_C$ ,  $\sin^2 \theta_{12}$ ,  $\beta$  to  $10^{-9}$ . [N] **cross-sector (headline):** back-solving  $\varphi_0$

*independently* from  $\beta$  (CMB; ACT DR6 + Planck PR4 combined) and from  $\lambda_C$  (quark; PDG 2024) gives the same seed within  $0.01\sigma_{\text{comb}}$ . [N] **one-parameter fit:** the error-weighted mean of the three back-solved  $\varphi_0$  reproduces the axiom to 0.06%; the joint  $\chi^2$  at the axiom (zero free parameters, 3 dof) is 0.007,  $\chi^2/\text{dof} < 1$ , combined pull  $0.08\sigma$ . [N]  $\theta_{13}$  **exclusion declared:** adding  $\sin^2 \theta_{13}$  raises  $\chi^2$  by  $\sim 4$  (the documented  $\sim 2\sigma$  pressure point, v328/v338), so this slice is the honest  $\theta_{13}$ -independent statement, not a hidden dilution. [N] **power:** a data $\leftrightarrow$ formula shuffle explodes the cluster  $\chi^2$  by  $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306);  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.CROSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

**2026-06-30 (v463/v464 + Lean + Wolfram — the  $c=8$  holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)**

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a  $c=8$  holomorphic limit must be  $(E_8)_1$ , and (A) the *realisation* input  $R_1$  that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463\_seam\_c8\_holomorphic\_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The  $c=8$  holomorphic-uniqueness pin. [E]  $c=8$  is NOT unique at level 1:  $c = \dim \mathfrak{g}/(1+h^\vee)=8$  has THREE simple solutions  $A_8$  ( $\dim 80$ ),  $D_8=\mathfrak{so}(16)$  ( $\dim 120$ ),  $E_8$  ( $\dim 248$ ), so the  $\det K=4$   $SO(16)$  rival is a genuine same- $c$  competitor. [E] HOLOMORPHY FORCES  $\dim V_1=248$ : the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3}=E_4/\eta^8$ ,  $q^1$  coefficient 248 (the 1-dim candidate space, vacuum  $a_0=1$ ). [E]  $\dim 248+$  level-1  $c=8 \Rightarrow E_8$  uniquely ( $248/(1+30)=8$ ); tower  $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$ . [C] Dong–Mason/Schellekens: the holomorphic  $c=8$  VOA is unique  $= V_{E_8}$  (even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt) — so the identification half is classification-forced.
- **v464\_seam\_oneparticle\_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking  $R_1$ . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol  $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with  $P=$  projection onto  $K<0$  ( $P^2=P$  to  $< 10^{-12}$ ),  $K$  fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ( $\max |P_{2N}-P_N| \sim 1/N \rightarrow 0$ ). [E] the limit is  $c=1$  Dirac: entanglement  $S(L)=(c/3) \ln L$  gives  $c \rightarrow 1$ , 16 copies  $\Rightarrow c_-=8$  (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ( $A_8/D_8/E_8$  each  $\dim=8(1+h^\vee)$ ) and `holomorphy_selects_e8` (forced  $\dim V_1=248=E_8 \neq 80, 120$ ) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the  $E_4/\eta^8$  tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package

(MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

**2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)**

- **v461\_seam\_strict\_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367  $p+ip$  collar. [E] CHERN INTEGER: the FHS Chern is  $C=+1$  (chiral  $M=1$ ), 0 (trivial  $M=3$ ),  $-1$  (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding =  $|C|=1 \neq 0$  and  $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$  (the same 8 as  $c_3$ ’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0,  $C=0$ ; since  $C=1 \neq 0$  NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **v462\_seam\_spinor\_continuum.py (SEAM.EQUIV.SPINOR.01).** The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  exhibited three ways. [E]  $(E_8)_1$  TOWER FROM  $SO(16)_1$  SECTORS: the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  gives  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  — the holomorphic  $(E_8)_1$  character IS the  $SO(16)_1$  vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL  $120 \rightarrow 248$ :  $\chi_o$  gives 120,  $\chi_s$  gives 128, summing to the exact tower  $\{1, 248, 4124, 34752\}$  ( $248=120+128$ ). [E] FINITE 16-MAJORANA FOCK:  $C(16, 2)=120$  NS bilinear currents and  $2^{16/2}=256$  Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$  CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives  $c_- \rightarrow 8$  and the lowest scaled NS gap  $x_1 \rightarrow 1$  with  $|x_1 - 1| \sim 1/L^2$  ( $L=64..1024$ ). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact ( $\theta$  identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite- $L$ , Python-only).
- **Lean (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01).** The two former cited axioms `mmst_existence`  $\circ$  `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding =  $|C|=1 \neq 0$ ,  $c_- \neq 0$ ) and `fock_sectors` (v462:  $C(16, 2)=120$ ,  $2^8=256$ , spinor 128,  $120+128=248$ ); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram. tfpt\_readouts\_extension.wl** adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462  $\theta$ -identity +  $120+128=248$  sector / Fock counts.

**2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the  $\mathbb{Z}_2$  reflection and the  $\mu_4$  clock)**

- **v460\_seam\_s3\_lto.py (SEAM.S3.LT0.01).** NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a  $\mathbb{Z}_2$  reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- $N$  tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367  $p+ip$  collar. [E] FLAT-BAND PARENT: the flattened Bloch  $Q=h/|h|$  has  $Q^2=I$  (flat  $\pm 1$  bands) and  $P=(I-Q)/2$  is an exact projector EQUAL to v367’s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a  $k$ -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel  $|P(n,0)|$  decays rapidly ( $\xi \approx 1.14$  at  $M=1$ ,  $\exp R^2=0.995$ ,  $|P(20,0)| < 10^{-6}$ ) and  $\xi$  grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law  $|R|^{-2.1}$  ( $R^2=0.999$ ) — locality needs the gap. [E]  $\mathbb{Z}_2$  REFLECTION +  $\mu_4$  INHERITED: the flattened modular Hamiltonian  $K_{\text{flat}}=\text{sign}(K)$  inherits  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  ( $u_\Theta=J$ ) and  $[\rho, K_{\text{flat}}]=0$  EXACTLY (sign is odd;  $\rho$  commutes with any  $f(K)$ ), with side-even and off- $\mu_4$  neg controls firing. [C][O] the chiral  $c_-=8 \neq 0$  (FHS Chern  $|C|=1$ ) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756;  $C \neq 0$  forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in run\_all.py, the registry, the ledger and cited in tfpt\_research\_contracts (route (xxvii) of the convergent attacks).

**2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, chirality forced by  $P_1$ , an  $(E_8)_1$  character kill test, the exact MMST citation audit and the complementary lattice-VOA route)**

- **v454\_seam\_edge\_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits  $c_{\text{Dirac}}=1.000$  ( $\frac{1}{2}$  per Majorana,  $L=64..1024$ ); [E] the lattice  $U(1)$  current closes its affine level exactly ( $\langle 0|J_n J_{-n}|0\rangle=n$ ,  $n=1..8$ ; a gapped control vanishes); [E] the level-1 Sugawara  $c=\dim G/(1+h^\vee)=8$  for both  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31). [C]  $16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$  is the level-1 current-algebra  $c_-$ ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455\_seam\_bw\_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- $N$  Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition  $u_\Theta=J$  — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E]  $\|\Theta K \Theta + K\|=0$  exactly at every  $N$  ( $N=8..128$ ); [E] an  $N$ -independent gap  $6 \log \frac{3}{2}$ ; [E]  $[\rho, K]=0$  exactly at every  $N$ ; [E] an  $N$ -independent neg-control margin  $2\|K\|$ . [C][O] the inductive limit inherits  $u_\Theta=J$ ; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456\_seam\_chirality\_from\_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ( $c_- \neq 0$ , the S3 phase) is forced by the one-sidedness that defines  $c_3=1/(8\pi)$  (P1) (G6). [E]  $c_3$ ’s “8” is the one-sided count  $8=|\mathbb{Z}_2| \cdot (f(K/\pi)=2 \cdot 4)$ ; [E] a reflection sends the Chern integer  $C \rightarrow -C$  ( $+1 \rightarrow -1$  on the  $p+ip$  model); [E] a two-sided boundary forces  $C=-C=0$ . [C] the one-sided seam (no reflection) forces  $C \neq 0$ , magnitude  $c_-=8=\text{rank } E_8$  — S3 is a consequence of  $P_1$ ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.
- **v457\_seam\_character\_killtest.py (SEAM.EQUIV.KILLTEST.01).** An explicit charac-

ter/sector kill test of  $(E_8)_1$  against  $SO(16)_1$  (G7). [E] the  $(E_8)_1$  character  $\chi=E_4/\eta^8$  with tower degeneracies  $\{1, 248, 4124, 34752, \dots\}$ ; [E] the weight-1 count  $248=120+128$  (not the free 120); [E] the sector count  $|\det \text{Cartan } E_8|=1$  (not  $|\det \text{Cartan } D_8|=4$ ). [X] a measured 120, four sectors, or a wrong coefficient would *falsify*  $(E_8)_1$ ; the data give  $248/1/\{1, 248, 4124\}$  (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.

- **v458\_seam\_mmst\_citation\_audit.py** (SEAM.EQUIV.MMST.AUDIT.01). The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy  $c=\frac{1}{2}$  (Thm 4.11), [C] H4 decoupled additivity to  $c=8$ , H5 the WZW range  $8 \leq 8 \leq 16$  with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  ( $\det K 4 \rightarrow 1$ ), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459\_seam\_lattice\_voa\_route.py** (SEAM.EQUIV.LATTICEVOA.01). The second, independent route to  $(E_8)_1$  supplying exactly the residual of v458 (G5). [E]  $E_8$  even unimodular ( $\det=1$ ); [E] the weight-1 space  $248=8+240$ ; [E] the 240 roots split 112 integer + 128 half-integer (the spinor,  $248=120+128$ ). [C] the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar =  $A_{E_8}$ , tied to  $P_1$  by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **Lean** (FORM.SEAM.RESIDUAL.01, `TfptCarrier/SeamResidualAxiom.lean`). Kernel-checks the whole G-block arithmetic (the one-sided  $8=|\mathbb{Z}_2| \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the MMST range  $8 \leq 8 \leq 16$ , decoupled additivity,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K 1$  vs 4) by `decide`, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST v458, AGT v459); lake build OK.
- **Wolfram + integration**. Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara  $c=8$  for 120/15 and 248/31; the one-sided  $8=2 \cdot 4$  and reflection  $C=-C \Rightarrow 0$ ; the  $(E_8)_1$  tower  $\{1, 248, 4124, 34752\}$  with  $248=120+128$  and  $|\det \text{Cartan}| 1$  vs 4; the root split  $112+128$  and  $248=8+240$ ) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

**2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- $N$  convergence, three independent reads of  $c_-=8$ , the Ising operator tower, the  $(E_8)_1$  modular data, and  $\mu_4$ -symmetry derived from the four marks)**

- **v449\_seam\_mmst\_uniform.py** (SEAM.EQUIV.MMST.UNIFORM.01). Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A  $Q(1, 2, 4, 7)/12 \rightarrow 2+\sqrt{3}$ , the deviation shrinking monotonically; [E] an *independent* cross-ratio B  $Q(1, 3, 5, 9)/12 \rightarrow$  its own value 2, so convergence is not a tuned point; [E] a uniform  $1/N$  rate  $|Q(N)-Q_\infty| \cdot N \leq 4$  for *both* observables at every  $N_x \in \{36, 48, 60\}$  (one  $N$ -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450\_seam\_edge\_entanglement.py** (SEAM.EQUIV.EDGE.ENTANGLEMENT.01). A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives  $c_{\text{Dirac}}=1.000$  for the critical free-fermion chain; [E] so  $c$  per Majorana =  $1/2$ ; [E] a gapped control saturates ( $c=0$ ). [C] bulk–edge:  $c_-=N_{\text{Maj}} \cdot \frac{1}{2}=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ . SEAM.EQUIV.01 stays [O].
- **v451\_seam\_edge\_cardy\_tower.py** (SEAM.EQUIV.EDGE.CARDY.01). Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension  $h_\sigma=1/16$  (the



periodic–antiperiodic ground-energy split) and [E] the energy dimension  $h_e=1/2$  (the lowest NS pair excitation), both [E]  $L$ -independent (conformal  $1/L$  scaling). [C]  $\{c, h_\sigma, h_e\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the chiral Ising (free-Majorana) minimal model  $M(3, 4)$ ; 16 copies give  $c_-=8$ . SEAM.EQUIV.01 stays [O].

- **v452\_seam\_e8\_modular.py** (SEAM.EQUIV.E8.MODULAR.01). The  $(E_8)_1$  torus modular signature, extending v156's  $q$ -expansion to the transformation data. [E] the character  $\chi=E_4/\eta^8$  is  $S$ -invariant  $\chi(-1/\tau)=\chi(\tau)$  (one primary,  $S=[1]$ ); [E]  $T$ -phase  $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$  ( $c=8$ ); [E] leading power  $\chi q^{1/3} \rightarrow 1$  so  $q^{-c/24}=q^{-1/3} \Rightarrow c=8$ ; [E] holomorphic  $c=8 \equiv 0 \pmod 8$  with  $T$ -phase order  $24/\gcd(8, 24)=3$ . [C] the  $(E_8)_1$  identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453\_seam\_mu4\_from\_marks.py** (SEAM.RIGIDITY.MU4FROMMARKS.01). Folds the last structural premise of the rigidity chain (v446's residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are*  $\mu_4$ = roots of  $z^4-1$ ; [E] the order-4 clock  $z \rightarrow iz$  is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio  $\lambda=2$  (in the  $D_4$  stabiliser, v168); [E] the form basis  $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$  is  $\mu_4$ -graded (weights  $(1, 2, 3)=A_3$  exponents). [C] so QGEO.SYM.01 follows from marks= $\mu_4$  + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].
- **Lean: TftCarrier/SeamEdgeChern.lean** (FORM.SEAM.EDGE.CHERN.01). Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers  $(+1/0/-1)$ , the bulk–edge channel count  $|C|=1$ ,  $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$ , the assembly  $2c_-=N_{\text{Maj}} \cdot |C|$  (so  $c_-=8$ ),  $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$ , holomorphic  $c \equiv 0 \pmod 8$  and the order-3 modular  $T$ -phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror**. `tft_readouts_extension.wl` gains nine exact checks (v452 modular data  $\times 4$ , v450 entanglement assembly, v451 Ising Kac weights, v453 marks= $\mu_4 \times 3$ ): 345  $\rightarrow$  354 checks, all passing (GATE.WOLFRAM.02).  $E_4$  is evaluated via its  $q$ -series since `EisensteinE` stays symbolic for complex  $\tau$ .
- **Papers**. The adversarial audit (`tft_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge  $c_-=8$  from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the  $(E_8)_1$  torus modular data (v452), and the  $\mu_4$  premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

**2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from  $\mu_4$ -symmetry, an independent topological reading of the edge  $c_-=8$ , and pinning the one external MMST fact at four-point order)**

- **v446\_seam\_clock\_invariance.py** (SEAM.RIGIDITY.CLOCKINV.01). Closes v445's single residual one notch further: the operator condition  $[\rho, \Lambda_\Sigma]=0$  (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer  $\Lambda=C(1)C(0)^{-1}=e^{-h}$  (fixed by RP+reflection, v54) of a  $\mu_4$ -invariant RP covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ); [E] a  $\mu_4$ -breaking covariance gives  $[\rho, \Lambda] \neq 0$  (iff-sharp neg control); [E]  $\Lambda$  is a positive contraction (spectrum in  $(0, 1]$ ); [E] the free modular Hamiltonian  $K=\log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki), broken control  $\neq 0$ . [C] so with v445 (commute  $\Leftrightarrow$  block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal  $\Lambda_\Sigma$ ; the residual reduces from an operator commutation to “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].
- **v447\_seam\_edge\_chern.py** (SEAM.EQUIV.EDGE.CHERN.01). An *independent* (topological)

reading of the edge central charge, corroborating **v444**'s correlator route from a different observable. [E] the  $M=1$  collar has integer Chern number  $C=+1$  (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E]  $C=0$  in the trivial phase and  $C=-1$  at  $M=-1$  (opposite chirality); [E] the open strip carries exactly  $|C|=1$  chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at  $1/2$  only in the topological phase. [C] bulk–edge:  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the same  $c_- = 8$  as **v444**. SEAM.EQUIV.01 stays [O].

- **v448\_seam\_edge\_fourpoint.py** (SEAM.EQUIV.EDGE.FOURPOINT.01). Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector  $G^2=G$  ( $\sim 10^{-13}$ ), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio  $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement; [E]  $Q$  converges to the free-fermion conformal cross-ratio  $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$ ; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (**v444**) and four-point order. SEAM.EQUIV.01 stays [O].
- **Lean: TftptCarrier/SeamRigidityForcing.lean** (FORM.SEAM.RIGIDITY.FORCING.01). Kernel-hardens **v445** (and **v442**'s uniformity): the entrywise forcing iff  $\text{clock4 } a = \text{clock4 } b \Leftrightarrow a=b$  (i.e.  $i^a=i^b \Leftrightarrow a \equiv b \pmod{4}$ , residue-only hence uniform in  $N$ ), the four distinct eigenphases, the exact commutant dimension  $\sum n_s^2=64<256$  and the order discriminator ( $128>64$ ) are all proved by kernel **decide** (**forcing\_kernel\_facts**, no domain axioms); only the operator-level lift to the full  $L^2$  is the named premise. lake build OK, #print axioms clean.
- **Wolfram mirror. tftpt\_readouts\_extension.wl** gains three exact **v445** checks (entrywise forcing for all  $a, b \in 0..31$ ; commutant dimension  $64<256$ ; order-2 commutant  $128>64$  swept  $N=4..64$ ):  $342 \rightarrow 345$  checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (**tftpt\_5\_redteam**, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling **v444**, forcing converse **v445** + Lean, derived clock-invariance **v446**, independent topological  $c_- = 8$  **v447**, four-point pinning **v448**) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (**v336**). The research-contracts note lists the same as “eleven convergent attacks”.

**2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, upgrading the rigidity step from “permit” to “force”)**

- **v445\_seam\_rigidity\_forcing.py** (SEAM.RIGIDITY.FORCING.01). The rigidity route (**v398/v442**) showed a  $\mu_4$ -block-diagonal  $K$  *commutes* with the order-4 clock  $\rho = \text{diag}(i^n)$ . This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff):  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \not\equiv j \pmod{4}$ , verified entrywise and swept  $N=4..128$  — commuting *forces*  $\mu_4$ -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$  (the  $\text{ad}_\rho$  nullspace), a proper subspace of the full  $N^2$  algebra ( $N=16$ :  $64<256$ ) — a genuine quantitative restriction. [E] OFF-BLOCK  $\Leftrightarrow$  NON-COMMUTING: every off-block matrix unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2}>0$  (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ( $N=16$ :  $64$  vs  $128$ ) — the four Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ ; **v281/v154/v351**). [E] NONDEGENERACY: exactly 4 distinct eigenvalues  $\{1, i, -1, -i\}$  for all  $N \geq 4$ . [C] SYNTHESIS: commuting with the order-4  $\mu_4$  clock is *equivalent* to  $\mu_4$ -block-diagonality, so RP-definability (**v54/v398**) + clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$  — **v442**'s “permit” upgraded to “force”.

[O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

## 2026-06-30 (Lean kernel-hardening of two of the seven SEAM.EQUIV.01 routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build` OK; `#print axioms` clean. Tracked as two new FORM.\* ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over  $\mathbb{Z}$  are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom):  $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$  (and  $\text{tr } K=0$ ). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ( $8 \leq 8 \leq 16$ ) and `npw_outside_bucket` ( $\det K=1 \neq 4$ ) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

## 2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c=1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368  $p+ip$  collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), `\veri`-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444\_seam\_edge\_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral  $c=1/2$  Majorana CFT: [E] a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2=0.9999$ , with *opposite* per-edge

velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$ , versus the *exponential* bulk row and the exponential trivial-phase ( $M=3$ , no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ( $\eta=1.001 \pm 0.007$ , successive profile spread  $0.028 \rightarrow 0.013$  across  $N_x=36..60$ ) — the numerical hallmark of an existing scaling limit. [C]  $\eta=1 \Leftrightarrow$  one chiral Majorana ( $c=1/2$ ) per edge, so the 16-copy carrier has a chiral  $c_-=8$  ( $E_8$ )<sub>1</sub> edge CFT ( $c_-=g_{\text{car}}+N_{\text{fam}}$ ). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.

- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

**2026-06-30** (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the  $\beta$ -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends ALL CHECKS PASSED), and are \veri-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438\_seam\_hsmi\_borchers.py** (SEAM.EQUIV.BW.HSMI.01) — **intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation  $P \geq 0$ , boost  $K$  with  $[K, P]=P$ , reflection  $J$  with  $JKJ=-K$ ,  $JPJ=P_-$ ), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact,  $sl(2, \mathbb{R})$  positivity (min eig  $> 0$ ); [C] synthesis; [O] the continuum standard pair.
- **v439\_seam\_lieb\_robinson.py** (SEAM.EQUIV.CONTINUUM.LR.01) — **continuum motor.** The explicit gapped  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440\_seam\_lto\_rp\_beta.py** (SEAM.EQUIV.LTORP.BETA.01) —  **$\beta$ -homotopy.** The (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K]=0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta=1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441\_seam\_applicability\_ledger.py** (SEAM.EQUIV.APPLICABILITY.01) — **audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.
- **v442\_seam\_rigidity\_uniform.py** (SEAM.RIGIDITY.UNIFORM.01) — **uniform rigidity.**

The band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$ : exact 4-sector commutation to  $N=256$  (not merely the  $N \leq 64$  sweep), a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]).

- **v443\_seam\_e8\_discriminator.py (SEAM.E8.DISCIMINATOR.01) — kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$ : torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots;  $c=8$  alone cannot decide (v277), only  $\det K=1$  (the genus-1 statement, v90) selects  $E_8$ . A seam with  $GSD>1$  would falsify  $(E_8)_1$ . [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

**2026-06-30 (shadow-export coherence — the tfpt5-shadow mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)**

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5-shadow.zip` the independent surfaces reproduced (`build.sh notes` → 11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **make\_docs\_map.py is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **audit\_sync.py is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean AUDIT OK on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says ALL CHECKS PASSED while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or vN module changed — this is an export/infrastructure fix only.

**2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)**

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`,

`version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.

- **robots.txt** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the `README` gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (E8.DEGREE.JOINT.01) — a consolidation of v6/v66/v355/v431 that derives more from the  $E_8$  Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of  $E_8$  is fixed by its degree multiset.** From  $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$  alone:  $\text{rank} = \# \deg = 8$ ,  $h = \max \deg = 30$ ,  $\# \text{positive roots} = \sum \exp = 120$ ,  $\# \text{roots} = 240$ ,  $\dim E_8 = 248$ ,  $|W(E_8)| = \prod \deg = 696,729,600$ ,  $\sum \deg = 128 = \dim S^+$ .
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers ( $g_{\text{car}}, N_{\text{fam}}$ ) are the *unique* root pair of  $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ : sum of roots =  $\text{rank } E_8 = 8$  (the rank-fill, v6), product =  $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$  ( $2g_{\text{car}}N_{\text{fam}} = h$ , v431);  $\{3, 5\}$  is the unique positive-integer factor pair of 15 summing to 8. So  $g_{\text{car}} = 5$  and  $N_{\text{fam}} = 3$  are forced *together* by two  $E_8$  degree-invariants, not chosen one at a time. The three primary-readout degrees  $\{2, 8, 30\}$  are exactly  $\{\min, \text{rank}, \max\}$ .
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (E8.DEGREE.JOINT.01); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339  $\rightarrow$  342). Exact integer/lattice; no new physical number, no status change.

**2026-06-29 (v436 — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the  $\alpha^3$  Chern level)**

- **New module v436 (OVERDET.FLOOR.02) — it *hardens* OVERDET.FLOOR.01 (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432’s nominal  $\sim 10^{-10}$  multiplies the counting census by hand-chosen plausibilities (input-“8” independence  $0.1^3$ , seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.
  - **[E] assumption-minimal floor.** The  $\alpha$  cubic-class census (v100 layer D) is *pure counting*: of  $N = 94500$  complexity-matched  $F_{U(1)}$  variants only *one* lands in the  $4 \times 10^{-8}$  CODATA window — the TFPT instance, reproducing  $\alpha^{-1} = 137.0359992$  — so  $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$  with *no* subjective probability, prior or multi-observable grammar.

- **[E] monotone concession ladder.**  $-\log_{10} p$  from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$  scorecard-only  $\sim 25.8 \rightarrow \alpha$ -census-only  $\sim 4.98$ ; each rung uses strictly less evidence and the “not chance ( $> 4\sigma$ )” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ( $\sim 1.8 \times 10^{-6}$ ) leaves the floor at  $4.40\sigma$  — they only sharpen it.
- **[C] honest  $5\sigma$  gap.** The conventional  $5\sigma$  line ( $p = 5.7 \times 10^{-7}$ ) is crossed unconditionally by the census times exactly *one* further independent factor  $\leq 0.054$  (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the  $\alpha^3$  Chern-level reading of v435 in named theorems — Dai–Freed ( $\eta$ -invariants and determinant lines, the exponentiated  $\eta$ -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the  $\alpha^2$  Calderón side) and Redlich/Stora–Zumino (the cubic  $\text{Tr } F^3$  anomaly descends to a Chern–Simons level, quantised to  $\mathbb{Z}$ ). It records that the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains **[O]** external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

**2026-06-29 (v435 — a fourth honest step on the  $\alpha^{-1}$  Quillen target: a  $\pi$ -power test isolates the cubic  $\alpha^3$  as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)**

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining **[O]** left after v434 (residual (2), the cubic  $\alpha^3$  Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays **[O]**;  $\alpha^{-1} = 137.0359992$  stays **[E]**.
  - **[E]  $\pi$ -power / metric-independence test.** The three EM-closure coefficients carry  $\pi$ -powers exactly  $\{0, 3, 6\}$ , equal to their  $c_3$ -orders: Maxwell  $k_0 = 1 \rightarrow \pi^0$ , Calderón  $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$ , transport  $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$ . So the Maxwell  $a^3$  is the *unique*  $\pi^0/c_3^0$  (metric-independent) term. A heat-kernel boundary coefficient always carries the  $(4\pi)^{-d/2}$  measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level ( $a^3$ ) + *two* heat-kernel boundary terms, confirming v342/v434’s “ $a^3$  is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
  - **[C] conditional level quantisation.** If the  $a^3$  term is the seam  $U(1)$  Chern–Simons/ $\eta$  boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to  $\mathbb{Z}$ , so  $k_0 = 1$  is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
  - **[O] not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that  $\partial_\tau \log Z_{\text{Maxwell}} = a^3$  — stays external math (type A, v384); the seam  $F$ -normalisation stays the one **[C]** (residuals 1 & 3, v434);  $\alpha^{-1}$  stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside v433/v434; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-28 (v434 — a third honest step on the  $\alpha^{-1}$  Quillen target:  $b_1$  is the  $a_4$  heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])**

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03)** — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems. The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O];  $\alpha^{-1} = 137.0359992$  stays [E].
  - [E]  $b_1$  *is* the  $U(1)$   $a_4$  heat-kernel coefficient (the  $\beta = a_4$  theorem). The one-loop  $\beta$  function of a gauge coupling *is* the  $a_4$  (Seeley–DeWitt  $a_2$  in  $d=4$ ) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ( $\frac{2}{3}$  per Weyl fermion,  $\frac{1}{3}$  per complex scalar),  $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$  (SM) and  $(3/5) \cdot \text{that} = 41/10 = b_1$  (GUT). So the  $b_1$  in the counterterm  $8b_1c_3^6$  is the  $a_4$  coefficient of the carrier  $U(1)_Y$  content (the same number as the  $\beta$ , v159/v246), the *same kind* of heat-kernel object as the  $\alpha^2$  Calderón  $a_4$  (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off:**  $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$ , with  $\text{rank } E_8 = 8$  the  $\chi=2$  seam boundary count (v216) and  $c_3^6$  the six boundary insertions (the  $\pi$ -power test, v342).
  - [C] **residual (1)  $\equiv$  residual (3).** With  $b_1$  and the  $c_3$ -powers heat-kernel-forced, the only freedom left in the exact seam  $a_4$  value is how the gauge curvature couples to the boundary measure — the seam  $F$ -normalisation. So “the actual seam  $a_4 = 8b_1c_3^6$ ” and “the seam  $F$ -normalisation” are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed):** the determinant-line variation factors  $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$ ;  $a^2$  is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading  $a^3 = a \cdot a^2$  is the *first moment* of the Chern density — a topological level at boundary order  $c_3^0$ , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
  - [E] **net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the  $\alpha^3$  Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside v433; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-27 (v433 — a second honest step on the  $\alpha^{-1}$  Quillen target: a solvable 4D model reaches the  $a_4$  order, connecting v391 and v342)**

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02)** — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (v382); it *grounds and connects*, it does *not* close it. The target stays [O] (external math, type A, v384); the value  $\alpha^{-1} = 137.0359992$  stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose  $\zeta$ -determinant variation reached only the low Seeley–DeWitt orders  $a_0, a_1$ ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient  $a_4$  ( $30/360 = \frac{1}{12}, \Omega^2/12$ ). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the  $a_4$  order:** the flat operator  $\Delta = -\partial^2 + m^2$  on a 4-torus of volume  $V$  has heat trace  $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$ , with Gilkey coefficients  $a_0 = V/(4\pi)^2$ ,  $a_2 = -Vm^2/(4\pi)^2$  and  $a_4 = Vm^4/(2(4\pi)^2)$ ; the  $t^0$  (conformal-anomaly) order  $a_4$  is *non-zero* — exactly the order v391’s 1D circle toy could not reach.
  - [E] **it is the order v342 uses:** so v391 (“a  $\zeta$ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey  $a_4$ ”) are unified — the variation is heat-kernel data *at the  $a_4$  order*, not only the low orders v391 reached.



- **[E] the measure is  $c_3$ -built:** the universal 4D heat-kernel measure  $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$  in the seam’s one-sided 2D-boundary normalisation  $c_3 = \frac{1}{2}(4\pi)^{-1}$ ; and the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ .
- **[O] not closed:** computing the *actual* seam  $U(1)$   $\zeta$ -determinant  $a_4$  coefficient ( $= 8b_1c_3^6$ ) on the  $\mu_4$  seam moduli, and the origin of the cubic  $\alpha^3$  Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam  $F$ -normalisation fixing the coefficient *value* stays **[C]** (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.02); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

## 2026-06-27 (v431, v432 — the “unmapped” $E_8$ degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped”  $E_8$  Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430):  $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \det\text{-ladder}\{8, 14, 20\})$ . **[E]** the two-family split is *forced* by  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  — the exponents are the  $\varphi(30) = 8$  totatives of 30, hence coprime to 6, hence  $\equiv \pm 1 \pmod{6}$ , so the degrees occupy only the residue classes  $\{0, 2\} \pmod{6}$ . **[E]** the  $6k$  family  $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$  is the v91 spine; the  $6k+2$  family  $\{8, 14, 20\} = (\det R, \det C, \det L)$  is the winding line  $\det M(s, 0) = 6s + 8$  of the flavor diamond (v135). **[E]** “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to  $E_8$  among the simply-laced exceptionals ( $E_6/E_7/D_5$  fail). **[C] honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$  Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays **[C]** (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the  $L1 \leftrightarrow L2$  bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness  $\alpha^{-1} \sim 137$ . **[C]** multiplying only across disjoint pieces (input over-determination  $\times$  the  $\alpha$  census  $1/94500 \times$  the  $\varphi_0^{\text{ret}}$ -seed cluster counted once  $\times$  one downstream witness) gives a floor  $\leq 10^{-6}$  across a grid of generous assignments, nominal  $\sim 10^{-10}$ . **[E]** that floor sits  $\sim 20$  orders *above* the v100 conditional  $10^{-30.7}$ ; the gap is exactly the evidential value of proving the forms forced, so the  $\alpha$  form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. **[C] honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

## 2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from $E_8$ ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five  $E_8$  Casimir degrees  $\{12, 14, 18, 20, 24\}$  with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided  $\mathbb{Z}_2$  collar / conjugate half-spinor  $S^-$ ), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes  $|\mathbb{Z}_2| = 2$  (the  $\frac{1}{2}$  in  $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$ , v58/v216), and  $2 = \min \deg(E_8)$  is the quadratic/metric — one of the *matched* primary readouts  $\{2, 8, 30\}$ , never one of the unmapped degrees. [E] the two sheets are the 128-spinor:  $\dim S^+ = \dim S^- = 16$ , the spinor block  $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$ , the spinor half of  $248 = 120 + 128$ ;  $S^-$  enters 248 as the conjugate  $(\overline{16}, \overline{4})$  ( $128 = 2 \cdot 64$ ), not a spare singlet. [E] forced-disjoint: the sheet/deck set  $\{2, 16, 32, 128\}$  has empty intersection with  $\{12, 14, 18, 20, 24\}$ , its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits  $\geq 2$  readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the  $S^-$ -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier  $E_8$  scan (tfpt\_4\_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run\_all.py, script\_registry.csv, status\_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt\_5\_redteam, the magnitude/phase keybox of tfpt\_3\_e8\_audit\_bootstrap, the WIMP-no-go passage of tfpt\_4\_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

## 2026-06-26 (v429 — the ‘unmapped’ golden $E_8$ structure is the geometry of the one external input $\theta_i$ )

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle  $E_8$  structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio  $\varphi$  is *unmapped* (it appears only internally — the affine- $E_8$  attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude  $\varphi$  cannot be numerology. [E] this sharpens that statement: because  $N_{\text{fam}} = g_{\text{car}} - 2$ , the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is  $\theta_i = \pi N_{\text{fam}} / g_{\text{car}} = (g_{\text{car}} - 2)\pi / g_{\text{car}}$ , which is *exactly* the interior angle of the regular  $g_{\text{car}}$ -gon — for  $g_{\text{car}} = 5$  the pentagon,  $\theta_i = 3\pi/5 = 108^\circ$ . [E] its cosine is golden,  $\cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$  (partner  $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$ ,  $\varphi = 2\cos(\pi/5)$ ). [E] the golden character is *unique* to  $g_{\text{car}} = 5$ : among the small regular  $n$ -gons only  $n = 5$  has an irrational interior-angle cosine ( $n = 3, 4, 6$  give  $1/2, 0, -1/2$ ), so  $\theta_i$  is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts —  $\varphi$  is the unmapped  $g_{\text{car}} = 5$  signature (v354/v313) and  $\theta_i = 3\pi/5$  is the one external cosmological input (v211) — and *refines* v354 from “ $\varphi$  in no physical readout” (it is the unmapped  $g_{\text{car}} = 5$  signature, v313) to “ $\varphi$  touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop  $\theta_i \approx 170^\circ$  (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite- $T$  solver that decides the branch (v185/v211). It does *not* derive  $\theta_i$ , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

## 2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** (“External works built upon”), `\input` into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

## 2026-06-25 (External AQFT citations + the four-fold $\mu_4$ identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold  $\mu_4$  identity made explicit (v422).** v422 already proved that the seam deck/divisor  $\mu_4$ , the carrier Galois group  $\text{Gal}(\mathbb{Q}(\zeta_5))$ , the discriminant  $\text{disc}(D_5)=\text{disc}(A_3)=\mathbb{Z}/4$  and the  $(E_8)_1$  simple-current glue are ONE cyclic  $\mathbb{Z}/4$  (with the cyclic-vs-Klein negative control). It is now named the *four-fold  $\mu_4$  identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses — in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

## 2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the  $(2, 3, 5)$  singularity is the ONE generator behind the order-30 clock (Milnor  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ ,  $30=2\cdot 3\cdot 5=h(E_8)$ ,  $\varphi(30)=8$ ), so

each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or  $E_8$ ): all seven collapse to one generator  $\Rightarrow$  *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in  $c_3=1/(8\pi)$  from rank  $E_8$ ,  $h(D_5)=2(5-1)$ ,  $\varphi(30)$  and the Milnor number — plus the foreign witness  $\alpha^{-1}\approx 137$ . [C] refines and partly walks back v427, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), tfpt\_safeguards Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (papers.ts, Safeguards.tsx, release.ts, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

## 2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (make\_figures.py, in manifest.sha256): the layered defense, the  $\alpha^{-1}$  uniqueness scan (one  $F_{U(1)}$  variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs  $\leq 5/13$ ).
- **tfpt\_safeguards expanded.** Layer 5 now states the null model explicitly ( $\prod p_i=10^{-25.8}$ ,  $2\times 200k$  pseudo-theories  $\leq 5/13$ ,  $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$ , shuffle 1.2) and the  $\alpha$  uniqueness ( $\leq 10^{-30.7}$ ,  $\sim 102$  bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading  $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$  MeV, a Belle II kill test. No new module.

## 2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of  $F_{\text{transfer}}$ .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap  $6\ln(3/2)$ ) restricted to each sector. [E] the v99 Koide generator linearised at  $q=2$  has rate  $-\Delta$ , time-1 contraction  $(2/3)^6$ ; Boltzmann washout is that contraction (CP phase  $4\pi/3$  native, v320); QCD rate  $b_3=-7$ . [C] only the anchors ( $v_{\text{geo}}$ ,  $C_p$ ,  $M_R$ ) stay external;  $F_{\text{relic}}$ ’s cosmological IC  $\theta_i$  is the one wall. A reduction (extends v383 static $\rightarrow$ dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible  $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds by direct Tomita–Takesaki;  $|\det \text{Cartan}(E_8)|=1$  (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss  $N(3+2i)=13$ , Eisenstein  $N(3+2\omega)=7$ , cyclotomy  $N(3+2\zeta_5)=55$ , Galois  $|(\mathbb{Z}/5)^\times|=4$ ,  $|\det \text{Cartan}(E_8)|=1$ , Pascal

$\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$  ( $2^4 = 16$ ), Coxeter  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  with  $\varphi(30) = 8$  — and the anchor  $a = (1, 1, 2)$  is the compression generator. Extends **v305**.

- **New paper `tfpt_safeguards` (companion **S**).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (**v427**), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the  $F_{\text{transfer}}$  and keystone sections), `tfpt_safeguards`; manifests refreshed.

## 2026-06-25 (The (LTO-RP) reduction of the keystone **SEAM.EQUIV.01: v424**)

- **New module **v424** (**SEAM.EQUIV.LTORP.01**).** An honest MMST-style reduction of the one open keystone — **SEAM.EQUIV.01**’s intrinsic Bisognano–Wichmann lemma (the **v308/v309** residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick ([arXiv:2605.10693](#), [NPW26]), with the central mapping *corrected*. [**E**] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K] = 0$ , **v309/v199**), not the flow —  $\sigma^\psi = \rho$  is type-wrong (a one-parameter group versus a finite order-4 element). [**E**] NPW26’s axiom (LTO-RP) means  $u_\Theta = J$  (reflection = modular conjugation) = the intrinsic BW condition  $\theta K \theta = -K$ , *distinct* from  $\omega \circ \rho = \omega$ : a positive character-block-diagonal  $K$  has  $[\rho, K] = 0$  yet never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [**O**] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code  $|\det K| = 4$  via a trace; Levin–Wen anyonic), while the seam is invertible ( $|\det \text{Cartan}(E_8)| = 1$ , no anyons) and a  $\beta = 1$  KMS state (not a trace) — in neither proved bucket. [**C**] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; **SEAM.EQUIV.01** stays [**O**]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via **v89/v281**).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 ([arXiv:2605.10693](#)), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf ([arXiv:2203.07421](#)).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

## 2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from **QG.AMB.01: v423**)

- **New module **v423** (**QGAMB.NODEP.01**).** Sharpens the **v369/v384** prose redundancy (“**QG.AMB.01** is a [**C**] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [**E**] gap-decoupling margin  $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$  ( $31 = 2^{g_{\text{car}}} - 1$ ), susceptibility  $\chi = 729/665$  finite (negative control gap  $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$ ); [**E**] **QG.AMB.01** is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while  $\alpha^{-1}$ , the mixing angles and cosmology are each *computed* by a compiler script, **v3/v9/v268/v7**); [**E**] the one transitive link is architectural — of the frozen readouts only  $\theta_{13}$  references it, via the 4d-QFT contract spine (**PS.DIRAC**  $\rightarrow$  **CONTRACT.QFT4D**  $\rightarrow$  **QG.G2**  $\rightarrow$  **GATE.METRIC**), a coherence edge, not a numerical input. [**C**] verdict: **QG.AMB.01** is gap-decoupled certification — **v369** as a DAG/script fact;

honest scope — this does *not* discharge SEAM.EQUIV.01, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).

- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open SEAM.EQUIV.01 premise is the *holomorphic*  $|\det K|=1$  point that selects  $(E_8)_1$  — the central charge  $c=8$  alone does *not*, being shared by the non-holomorphic  $SO(16)_1$  counterexample (`v277/v281`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

## 2026-06-25 (The Galois $\leftrightarrow$ Net bridge: `v422`)

- **New module `v422` (SEAM.GALOIS.NET.01).** The seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$  is the SAME cyclic  $\mathbb{Z}/4$  as the  $(E_8)_1$  simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of `v419` to  $G_{\text{net}}/\text{SEAM.EQUIV.01}$ . [E]  $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$  (one Smith invariant factor  $4 = \text{cyclic, det } 4$ );  $D_n$  disc is  $\mathbb{Z}/4$  for  $n$  odd,  $\mathbb{Z}_2 \times \mathbb{Z}_2$  for  $n$  even, so the carrier  $D_5$  (rank  $5 = g_{\text{car}}$ , odd) is cyclic while  $D_8$  (rank  $8$ , even) is Klein; [E]  $(\mathbb{Z}/5)^\times = \langle 2 \rangle$  is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in  $\mathbb{Z}_4 \times \mathbb{Z}_4$  ( $16 = \dim S^+$ ) the glue  $(1, 1)$  is order 4 with Lagrangian quotient  $16/4^2 = 1 = (E_8)_1$  (`v89/v154`); [E] negative control: the Klein / order-2  $\langle (2, 2) \rangle$  gives  $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16)$  (det 4), NOT  $E_8$  (det 1) — cyclic  $\mathbb{Z}/4 \rightarrow E_8$ , Klein  $\rightarrow D_8$ . [C] the bridge: one cyclic  $\mathbb{Z}/4$  threads the seam deck  $\mu_4$  (`v419`),  $\text{Gal}(\mathbb{Q}(\zeta_5))$  and the  $(E_8)_1$  glue — the Galois gearbox  $(\mathbb{Z}/30)^\times$  IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored (`324 \rightarrow 327`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (`327/327`) and the website verification count; manifests refreshed.

## 2026-06-25 (F\_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: `v420`)

- **New module `v420` (FR.POLE.QED.01).** The honest external-physics follow-up to `v93` and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide  $Q$  is rescaling-invariant ( $Q(cm) = Q(m)$ ), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit  $\frac{2}{3} - Q_{\text{src}} = 0.00220$  is one-loop-EM-sized ( $= \varphi_0/24$  to 0.6%,  $= \alpha/\pi$  to  $\sim 5\%$ ), the PDG pole sits at  $2/3$  (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$  per-lepton-log estimate shifts  $Q$  by EM-loop order toward  $2/3$  (the `v82` direction); [O] the exact flow time  $t \approx 2.84$  (`v93`) is scheme/scale-dependent external QED, only sized. [C] verdict:  $F_{\text{pole}}$  is a one-loop EM effect — the lens forces the rate  $(2/3)^6$  (`v327`), external EM supplies the flow; **F\_pole** stays [C], firewalled (`v187`), so  $F_{\text{transfer}}$  is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like `v93/v371`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

## 2026-06-25 (The seam $\mu_4$ is the carrier's Galois group: `v419` + figure)

- **New module `v419` (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (`v409`, RES.COXETER.SYMMETRY.01): why the seam  $\mu_4$  sits on the Galois side, not on the cyclic  $\zeta_{30}$  dial. [E]  $h(E_8) = 30$  is squarefree, so  $\mathbb{Z}/30$  has element orders  $\{1, 2, 3, 5, 6, 10, 15, 30\}$  — *no* order 4, so the seam  $\mu_4$  cannot be a rotation hand and is forced onto the Galois side; [E] by CRT  $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$  (order  $\varphi(30) = 8 = \text{rank } E_8$ ),

and the  $\mu_4$  ( $\mathbb{Z}/4$ ) factor IS ( $\mathbb{Z}/5$ ) $^\times$  (the carrier prime 5), the  $\mathbb{Z}_2$  IS ( $\mathbb{Z}/3$ ) $^\times$  (family 3 = CP conjugation, v316); [E] so the seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ , realised on the carrier clock  $C_5$  (v418) by the Frobenius  $\zeta_5 \mapsto \zeta_5^2$  — the explicit order-4 operator  $G$  with  $GC_5G^{-1} = C_5^2$ ,  $G^4 = I$ ,  $G^2C_5G^{-2} = C_5^{-1}$ . [C] this is the positive content of v409: the  $2^2=4$  is the order of ( $\mathbb{Z}/5$ ) $^\times$ , and “no contraction rate” is because the seam is an *automorphism*, not a hand; so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants (cyclic  $h=30$ , Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321  $\rightarrow$  324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ( $\mu_4 = \text{Aut}(\zeta_5)$  the seam,  $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$  the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

## 2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split (3,2) read in the three atom-rings gives (7,13,55)=(scalaron,  $\Delta_Q$ , quark numerator), and the carrier-5 clock missing from v417 is found as the  $4 \times 4$   $\Phi_5$ -companion  $C_5$ . [E]  $C_5^5=I$ ,  $\text{tr } C_5=-1$ ,  $\det C_5=1$  (the four primitive 5th roots), with the golden ratio as its real-part data ( $2 \cos 72^\circ=1/\varphi$ ,  $2 \cos 144^\circ=-\varphi$ ; output-side, v313/v349; dim 4, not 3 — this refines v417’s gap from “no operator” to “no  $3 \times 3$  operator”); [E]  $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5 \cdot 11=g_{\text{car}} \|\text{Pl}(K)\|_1$ , the  $c_u/c_d$  numerator (v410/v411); [E] the triple  $\det(3I+2\text{Comp}(\Phi_n)) = (7,13,55)$  for  $n=3,4,5$  over  $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$  (atoms 3,2,5, v416); [E] NEG control: the anchor (5,4)  $\rightarrow$  (21,41,461) reaches named values only in the  $\omega$ - and  $i$ -rings ( $21=N_\omega(5,4)$ ,  $41=10b_1$ , v222) but 461 (prime) in the carrier ring, so the carrier ring separates (3,2)  $\rightarrow$  55 from (5,4)  $\rightarrow$  461. [C] honest: (3,2) is the *forced* carrier split (v14), not the unique clean one — a scan finds  $\{(2,1) \rightarrow (3,5,11), (3,2) \rightarrow (7,13,55)\}$ , a small ladder. The sharper discriminator is *cross-sector*: only (3,2) spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while (2,1) is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318  $\rightarrow$  321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

## 2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the  $\tau=\omega$  flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation  $P = (1\ 2\ 3)$  is the order-3 Eisenstein deck ( $P^2+P+I = \text{ONES}$ , so  $\omega^2+\omega+1=0$  on the plane  $\perp (1,1,1)$ ), the order-3 dual of  $J^2=-I$ ; [E] it realises the hex CM norm as  $(3I+2P)(3I+2P^2) = 7I+6\text{ONES}$ ,  $\text{Spec}\{7,7,25\}$  ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega)=\text{scalaron}$ ,  $25=g_{\text{car}}^2$ ) — the dual of v415’s  $\{13,13,9\}$ ; [E] the CP clock  $W=-P^2$  ( $\rho$  of v233) has  $W^2=P$ ,  $W^3=-I$ ,  $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , giving  $\delta_{\text{CKM,lead}}=\arg \zeta_6=\pi/3$  and  $\delta_{\text{PMNS}}=\arg \zeta_6^4=4\pi/3$  (v316/v320); [E] all flavor clocks are powers of  $\zeta_{30}$  ( $\zeta_{30}^{15}=-1$ ,  $\zeta_{30}^{10}=\omega$ ,  $\zeta_{30}^6=\zeta_5$ ,  $\zeta_{30}^5=\zeta_6$ ) while the seam  $\mu_4=i$  is the Galois side ( $\mathbb{Z}/30$ ) $^\times$  of order  $\varphi(30)=8=\text{rank } E_8$  — so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants ( $h=30$  and  $\text{rank}=8$ ). [C] honest gap: the carrier  $\mu_5$  has no  $3 \times 3$  operator (the golden  $\varphi$  is an output-side trace, v313/v349); negatives: no  $U, V$ -diamond operator carries  $\zeta_6$ ,  $[D, P] \neq 0$  (no  $\mu_{12}$ ),  $N(3+2\zeta_6)=19 \neq 7$ . Cited in `tfpt_2`; Wolfram-mirrored (315  $\rightarrow$  318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

## 2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the  $\mu_4$  deck becomes an intrinsic diamond object. [E] the integer commutator  $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$  has  $\text{Spec}(J) = \{i, -i, 0\}$ ,  $J^2 = -I$  on the rank-2 image and  $J^3 + J = 0$  — a  $\mu_4$  quarter-turn born from the binary  $V$  and ternary  $U$  compilers (v410/v95); [E]  $3I + 2J$  and  $5I + 4J$  have spectra  $\{3+2i, 3-2i, 3\}$ ,  $\{5+4i, 5-4i, 5\}$ , so the Gaussian integers  $3+2i$  (norm  $13=\Delta_Q$ ) and  $5+4i$  (norm  $41=10b_1$ ) are *literal eigenvalues*; [E]  $(aI+bJ)(aI-bJ)$  reads  $(a^2+b^2)$  on the  $\mu_4$ -plane and the real part<sup>2</sup> on the kernel ( $\{13, 13, 9\}$ ,  $\{41, 41, 25\}$ ); [E] the intrinsic order-4 deck  $D = -I + J - J^2$  ( $D^4 = I$ ,  $\chi_D = (x+1)(x^2+1)$ ) has  $\chi_2$ -line  $\ker[U, V] = a - 1$ . [C] the geometric-deck identification sharpens (does not close) the v140  $\mathbb{Z}_3$  residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms  $\{2, 3, 5\}$  are ramified/inert/split in the two exceptional CM rings. [E] in  $\mathbb{Z}[i]$  (square/seam)  $2=|\mathbb{Z}_2|$  ramifies,  $3=N_{\text{fam}}$  is inert,  $5=g_{\text{car}}$  splits as  $(2+i)(2-i)$ ; [E] in  $\mathbb{Z}[\omega]$  (hex/flavor)  $3=N_{\text{fam}}$  ramifies,  $2, 5$  are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets  $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$  of  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$  (v403) ramify over exactly one atom each, product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ , two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension 308  $\rightarrow$  315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns  $-\chi$  for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

## 2026-06-24 (The sheet generator $V$ as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis  $V = Q \text{diag}(0, 1, 1)$  of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form  $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$  (proved by the inductive step  $F(n)V=F(n+1)$ ), so  $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; [E] the four bilinear families  $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$ ,  $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$ ,  $a^T V^n a = 11 \cdot 2^{n-1}$ ,  $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$  give  $(6, 7, 9, 11)$  at  $n=1$  and  $\mathbf{1}^T V^2 \mathbf{1} = 13 = \Delta_Q$ ,  $\mathbf{1}^T V^3 \mathbf{1} = 27 = \mathbf{1}^T R a$ ,  $\mathbf{1}^T V^4 \mathbf{1} = 55$ ,  $\mathbf{1}^T V^4 a = 56 = \dim \mathbf{56}_{E_7}$ ; [C] the binary spine is FORCED by  $\text{Spec}(V) = \{0, 1, 2\}$ , so the carrier/Lie matches (rank  $E_8=8$ ,  $\dim S^+=16$ ,  $248/8=31$ ) and the theta cross-link ( $\sigma_3(2)=9=a^T V \mathbf{1}$ ,  $\sigma_3(3)=28=\mathbf{1}^T V^3 a = \det(I+R)$ ,  $\sigma_3(5)=126$ ) are audit-typed (the v95  $G_2$ -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure  $V$ -power readout,  $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1}) / ((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$ , identical to the established  $g_{\text{car}} \|\text{Pl}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$  (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the  $u/d$  identification stay conditional, coupled to the  $(U_{\text{wall}})$  readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the  $\mathbb{Z}_2$  wall ( $t=-2$ ,  $\det=2$ , v135),  $J=M(1, -2)=C-2V$ , completing the sheet axis  $J, K, C, F=M(1, -2..1)$ . [E]  $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)$ ;  $a^T J a = 30 = h(E_8)$ ,  $\det(I+J)=12 = \dim \mathfrak{g}_{\text{SM}}$  and  $\det(2I+J)=40=|R(D_5)|$ ; [C] the Lie readings audit-typed like the  $G_2/F_4$  shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On  $M_t=C+tV$  the elementary-symmetric coefficients encode the atoms as difference orders:  $e_1=3t+12$ ,  $e_2=2t^2+15t+25$ ,



$e_3=4t^2+14t+14$  give  $\Delta e_1=3=N_{\text{fam}}$ ,  $\Delta^2 e_2=4=|\mu_4|$ ,  $\Delta^2 e_3=8=\text{rank } E_8$  (the  $e_3$  curvature is the v218/v224 sheet quadratic; the full triple is new); [E] the anchor energy  $a^T M_t a = 52 + 11t$  ( $30 \rightarrow 41 \rightarrow 52 \rightarrow 63$ , step  $11 = a^T V a$ ); [C] atom/Lie readings audit-typed.

- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center  $C=M(1,0)$  is a resolvent portal: [E]  $\det C=14=\dim G_2$ ,  $\det(I+C)=52=\dim F_4$ ,  $\det(2I+C)=120=|R^+(E_8)|$ , the exceptional shell ladder  $G_2 \rightarrow F_4 \rightarrow E_8^+$  as  $\det(C+kI)$ ,  $k=0,1,2$  (with  $\text{tr } C=12=\chi_{E_8}(i)$ ,  $\sum C=31=2^{g_{\text{car}}}-1$ , v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \veri citations) and the  $E_8$  theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror  $303 \rightarrow 308$ .

## 2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHI0.PUNCTURE.01).** Names the  $\varphi_0$  puncture term  $48c_3^4$  as the order- $|\mu_4|=4$  local boundary heat-kernel (Seeley–DeWitt) contact term of the  $\mu_4$  puncture divisor, summed over  $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$  admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split  $\varphi_0=1/(6\pi)+48c_3^4$ , the puncture count, the closed form  $3/(256\pi^4)$  and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the  $c_3$  Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*:  $\sigma=\text{diag}(1,-1,-1)$  has spectrum  $\{\pm 1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0,1\}$ , none in  $(0,1)$ , so prime-2 carries no contraction rate, whereas the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0,1)$  and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ( $|\mu_4|=2^2$  the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O]  $\rightarrow$  [E]/[C].
- **Axion branch consistency (tfpt\_4\_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- $T$  result already stated later in the same section: the robust dominant-DM branch is the *spine* angle  $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$  (untuned, in band), while the seam hilltop  $\theta_i \approx 170^\circ$  over-produces ( $\sim 5.4\times$ ) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ( $r^2=1$ ) is  $\pm 1$ , a projector eigenvalue ( $r^2=r$ ) is 0 or 1, neither lies in the open interval  $(0,1)$ , while  $2/3$  does — so *no prime-2-only attractor exists*. Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms = propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant  $11=\|Pl(K)\|_1=\det(1|a|\sigma)$  is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

## 2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain

v114–v117 keeps .06–.09 (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal `depends_on` carried along). No claim content changed.

- **Website.** `papers.ts` document count  $8 \rightarrow 9$  (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E*  $\rightarrow$  *A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7*  $\rightarrow$  *Doc 8* for the research contracts and the residual interfaces re-typed  $v_{\text{geo}}$  [O],  $G_{\text{net}}$  [C],  $F_{\text{transfer}}$  [C]; *objections Paper 1*  $\rightarrow$  *Doc 1*.
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [E]  $\rightarrow$  [E]/[C] (given RP) and  $\theta_{12}$  “genuinely open”  $\rightarrow$  “conditionally derived”; `tfpt_2` Starobinsky  $A_s$  [E]  $\rightarrow$  [E]/[C] ( $M$  exact,  $A_s$  a band over  $N_\star \in [50, 60]$ ) and the Part-II abstract “opens the one remaining gap”  $\rightarrow$  “closes the last structural gap”; `tfpt_research_contracts` the QFT4D “nonperturbative measure stays open” line and the “only  $U_{\text{point}}$  stays open” headline reconciled with the [C] redundancy /  $v_{\text{geo}}$  reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [C] redundancy; `introduction` ( $U_{\text{wall}}$ )  $\rightarrow$   $v_{\text{geo}}$ . Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ( $v_{\text{geo}}$  [O];  $G_{\text{net}}$ /SEAM.EQUIV.01,  $F_{\text{transfer}}$ ,  $N_\star$ , QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the `qft_skeleton` “Stelle ghost” open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

## 2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] `v405_seam_equiv_omega.py` (SEAM.EQUIV.02): the *dual keystone* at  $\tau=\omega$  – the family/flavor sector as the order-3 Eisenstein/ $A_2$  face of  $E_8$ , dual to the seam  $= (E_8)_1$  at  $\tau=i$  (v404). A named contract (genre of v286/v398), NOT a closure. [E]  $\chi_{E_8}(\omega) = 0$  (the family CM point degenerates  $E_8$ ) vs  $\chi_{E_8}(i) = 12$ ; [E]  $A_2$  is the  $\tau=\omega$  Eisenstein lattice (Cartan  $\det = 3 = N_{\text{fam}}$ ,  $h = 3 = N_{\text{fam}}$ ); [E]  $R$  lives in  $E_6 \times A_2$  ( $248 = 78 + 8 + 2 \cdot 27 \cdot 3$ ,  $\det R = 8 = \dim A_2 = \text{rank } E_8$ ); [E]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  CM phase; [E] the cusp weights  $\{0, 1/3, 2/3\}$  are the order-3 deck  $= N_{\text{fam}} = \det Q$ . [C] the family sector is the order-3 Eisenstein/ $A_2$  deck; the two keystones sit at the only two elliptic points of  $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical  $\mathbb{Z}_3$  deck map (v140) + the  $v_{\text{geo}}$ -class scale. Cited in `origin_theory`; Python-only.
- [E]/[C]/[O] `v406_pmns_phase_origin.py` (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the  $\tau=\omega$  keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one  $v_{\text{geo}}$ -class unit, so PMNS has *zero free flavor dials beyond*  $v_{\text{geo}}$ . [E] the three angles (v9/v268)  $\rightarrow$  unitary  $U_{\text{PMNS}}$  (v270); [C]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C]  $J_{\text{PMNS}} = -0.0297$  derived; [E] the absolute scale = one seesaw ratio =  $v_{\text{geo}}$ -class (v272/v153). [O] residual = the seesaw operator realisation ( $M_R$ , v184) + the  $\Sigma m_\nu$  floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- [E]/[C]/[O] `v407_dn_pairings_omega.py` (FLAV.SELECTOR.CLOSE.01): the  $R4'$  selector residue folds into the  $\tau=\omega$  keystone – the three  $(d, n)$  pairings ARE the  $\tau=\omega$  family-slice atoms. [E] the frame  $(1, a, \sigma)$  has  $\det = 11 = \|Pl(K)\|_1$  and  $n = (5, -9, 6)$  is the unique covector with  $n \cdot 1 = 2$ ,  $n \cdot a = 8$ ,  $n \cdot \sigma = 121 = 11^2$  ( $\Rightarrow (d, n) \Rightarrow R$ , v136/v139); [E]

$n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$  (the  $A_2$  block reading  $R$  in  $E_6 \times A_2$ , v405); [E] the pairings  $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$ . [C] deriving  $(d, n)$  folds into SEAM.EQUIV.02 ( $d$  anchor-derived, v139); [O] the lone residue “why  $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (The arithmetic hull + the $E_8$ -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ . [E] the squarefree parts  $-1, -3, 5$  are independent in  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so  $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$  with Galois group  $(\mathbb{Z}/2)^3$  (order 8); [E] exactly 7 quadratic subfields  $\{-1, -3, 5, 3, -5, -15, 15\}$ , splitting 4 imaginary  $= |\mu_4|$  (CM/dynamic) + 3 real  $= N_{\text{fam}}$  (RM/static); [E] ramified *only* over  $\{2, 3, 5\}$  = the atoms = the primes of  $h(E_8) = 30$ ; [E]  $2^{\omega(h)} = \text{rank}$  is  $E_8$ -special (the  $E_7$  control  $h = 18$  gives  $2^2 = 4 \neq 7$ ). [C]  $K$  is the abelian  $(\mathbb{Z}/2)^3$  *arithmetic* hull dual to the  $E_8$  *lattice* hull; the 7 quadratic subfields are a candidate for the 7  $E_8$  audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] `v404_e8_char_cm_duality.py` (E8.CMDUAL.01): the  $(E_8)_1$  vacuum character  $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$  read at *both* CM points, extending v282 ( $\chi_{E_8}(i) = 12$ ) to the second elliptic point. [E]  $\chi_{E_8}^3 = j$  (q-series); [E] SEAM  $\tau=i$  ( $\mathbb{Q}(i)$ , order 4): cross-ratio  $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$  ( $E_8$  present, v214/v282); [E] FAMILY  $\tau=\omega$  ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic  $\lambda$  ( $\lambda^2 - \lambda + 1 = 0$ )  $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$  ( $E_4$  vanishes, v220/v267); [E]  $\tau=i, \tau=\omega$  are the *only* elliptic points of  $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where  $E_8$  is present ( $\chi = 12$ , the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where  $E_8$  degenerates ( $\chi = 0$ :  $U_{\text{wall}}, v_{\text{geo}}$ , the CP texture), a candidate dual keystone at  $\tau=\omega$ . Cited in `origin_theory`; core values  $j = 1728/j = 0$  already mirrored (v214/v220/v267/v282), Python-only.

## 2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] `v398_seam_state_rigidity.py` (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer  $T=e^{-H}$  from RP + reflection alone, v54, quasi-free v155, no  $\mu_4$  normal form); [E] band-limited character-block-diagonal core  $[\rho, |k|]=0$  swept  $N=4.64$  (beyond the 3-dim  $H^1$ ); [E] holomorphy  $\Rightarrow (E_8)_1$  ( $c=8$ ,  $|\det \text{Cartan}|=1$  vs 4, index  $4=|\mu_4|$ ). [O] the one residual:  $[\rho, \Lambda_\Sigma]=0$  on the full  $L^2$  (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- [E]/[C]/[O] `v399_gravity_complete.py` (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton  $8\pi$ ,  $\Lambda$ , scalaron  $c_3^7$ , horizon  $1/|\mu_4|$ , recovery  $(2/3)^6$ ) are all boundary/seam/gap, so  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ ; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- [E]/[O] `v400_grav_nonlocal_action.py` (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local  $R+R^2$  as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire  $a=e^u \Rightarrow$  one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces  $R+R^2$  with the atom-fixed scale  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$  (v36/v253). [O] the entire- $a$  assumption + perturbative-only.

- **[E]/[O] v401\_metrology\_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set  $\{a, \pi, v_{\text{geo}}\}$  certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping  $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$ ; **[E]** the  $G$ -vs- $\Lambda$  over-determination 0.11% (v274); **[O]**  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ ).
- **[E]/[C] v402\_ftransfer\_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor  $F_{\text{transfer}}$ , the composition  $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$  (consolidating the **FR.\*.SOLVE** solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

## 2026-06-24 ( $\varphi_0$ leading term from icosahedral combinatorics: v396)

- **[E]/[C] v396\_phi0\_icosahedral.py** (HYP.PHI0.01): the tree-level seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi) = F/(|\text{Aut}| \pi) = (F/(g_{\text{car}} N_{\text{fam}})) c_3$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ );  $F/h = |Z_2|/N_{\text{fam}} = 2/3$  (the same survival ratio as v327). **[E]** Gauss–Bonnet consistent with  $c_3 = 1/(|Z_2|4\pi)$  (v216/v342). **[E]** honest negative: the puncture  $48c_3^4$  is still analytic, not a hypergraph fraction. **[C]**  $\varphi_0^{\text{tree}}$  is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

## 2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- **[E] v395\_hypergraph\_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) unifies v299 (carrier growth), v327 (branching rule  $M$ ) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column +  $M$  on each node’s family vector ( $T_{\text{micro}} = T_{\text{net}} \otimes M$ ); joint attractor marks  $\otimes e_0$ ; one clock hand (6 family steps) gives recovery  $(2/3)^6$ ; v324’s  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis. **[O]** the 3-arm seed ( $P_2$ ) and analytic  $\varphi_0$  remain irreducible (v312).

## 2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — do the “external” frontier scheme-numbers land on the  $\{2, 3, 5\}$  clock? A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). **[E]** the  $\eta_B$  decuple  $A_\Lambda = 10 = |Z_2|g_{\text{car}}$  is ON the clock (v212); **[E]** the Koide rate  $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$  is ON the clock (v303); **[E]** the axion spine angle  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  is ON the clock (v211/v373); **[E]** the proton  $C_p \sim 2.83 \pm 0.15$  is *not* a discriminating clock value — within its  $\pm 5.3\%$  band several simple values fit  $(2^{3/2}, 17/6, 14/5)$ , so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number  $C_p$ . Cited in **tfpt\_4\_frontier**; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

**2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)**

- **v394 (COXETER.ATOMS.01) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird’s-eye [E] re-reading tying v390 (the prime facets) to the anchor  $a = (1, 1, 2)$  and v319 (the  $5 \times 6$  clock). [E]  $a = (1, 1, 2)$  has  $e_3=2=|Z_2|$ ,  $p_0=3=N_{\text{fam}}$ ,  $e_2=5=g_{\text{car}}$ , product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  — the clock’s primes *are* the anchor’s atoms. [E] atoms = facet fields (v390):  $2 \rightarrow \mathbb{Q}(i)$ ,  $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ ,  $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] the QG susceptibility is pure  $\{2, 3\}$ :  $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$ ,  $(2/3)^6 = 2^6/3^6$ , exponent  $6 = 2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2:  $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$  with  $|\mu_4|=4=2^2$ . [E]  $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$  (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01 — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector’s gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383’s “every sector is the same gapped operator” to “the same *seam-driven* operator” (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

**2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)**

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction’s correction, this gives the magnitude, so a parameter-free central value can be quoted as “value  $\pm$  computed first correction”. [E] FIXED-POINT band (the  $\varphi_0$ -derived  $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$ ) = the explicit  $\varphi_0$  puncture term  $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$  (exact;  $= \theta_{12}$ ’s  $\varepsilon$  first-correction,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ); [E] SEAM-GAPPED band (Koide  $F_{\text{pole}}$ , recovery)  $= (2/3)^6 \approx 8.78\%$ , the QG-decoupling bound capped by  $\chi - 1 = 64/665 \approx 9.62\%$  (v337); [E] EXACT-IDENTITY band = 0 structurally ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ); [E] EXTERNAL-RATE band = external physics, quoted ( $m_p/m_e \pm 7.5\%$  v374;  $n_s/r$  the  $N_\star \in [50, 60]$  band;  $\eta_B 1.07 \times \text{v212}$ ), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

**2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)**

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01’s one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope  $c_1(L)$  at  $L = 66, 130, 258, 386, 514$  is a monotone Cauchy-like sequence approaching 1 ( $|c_1(L) - 1|$  strictly decreasing,  $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$ ); [E] a  $1/L$  Richardson fit extrapolates to  $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar  $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$  is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin  $(E_8)_1$  over the same- $c$   $SO(16)_1$  (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01 — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): [E] `component_tendsto` (a non-dominant eigencomponent  $|r| < 1$  decays,  $r^n c \rightarrow 0$ ); [E] `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [E] `deviation_unique` (two starts give deviation vectors both  $\rightarrow 0$ , so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [E] `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

## 2026-06-24 (Website: an animated “Universal Gap Lab” visualising v383–v391)

- **Website — the UniversalGapLab animation on /verification.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction  $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$  racing to its attractor at  $r = (2/3)^6$  (seam-gapped, v303/v387/v388) or the golden  $(\varphi + 2)/4$  (v312), with a gapless  $r = 1$  track that never forgets its start (the Lean theorem FORM.SPECTRALGAP.01) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio  $e^{-u}$  falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

## 2026-06-24 (Honest attempt at an external target: v391 reduces it, not a closure)

- **v391 (ALPHA.QUILLEN.PROGRESS.01) — an honest attempt at the external target ALPHA.QUILLEN.EXACT.01 (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] a solvable model  $\det_\zeta(-\partial_\tau^2 + m^2)$  on a circle of circumference  $\beta$  equals  $(2 \sinh(\beta m/2))^2$ , whose  $m$ -variation  $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$  is a closed heat-kernel/Laurent series (the  $m^3$  coefficient  $-1/360$  verified). [E] so a  $\zeta$ -determinant’s coupling-variation is structured heat-kernel data, and v382’s counterterm  $8b_1 c_3^6$  (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ . [E] the reduction: the open step now reduces to the named sub-target “the seam  $U(1)$  operator’s Seeley–DeWitt coefficient  $= 8b_1 c_3^6$ ”. [O] *not closed*: the actual seam-operator  $\zeta$ -determinant proof stays open, and SEAM.EQUIV.01 continuum existence (MMST, v336) stays the other external [O]; the value  $\alpha^{-1}$  stays [E]. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01 — a Lean 4 proof of the gap  $\Rightarrow$  unique-attractor  $\Rightarrow$  parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction  $x \mapsto x_\star + r(x - x_\star)$  with multiplier  $r = \lambda_2/\lambda_1$ . [E] `iter_sub`: the closed form  $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$  (general  $n$ ); [E] `iter_tendsto`:  $0 \leq r < 1$  (the gap)  $\Rightarrow \text{iter}_n \rightarrow x_\star$  (the unique attractor);

[E] **starts\_agree**: two starts differ by  $r^n(a - b) \rightarrow 0$ , so the attractor is independent of the initial state (parameter-freeness); [E] **no\_gap\_no\_forget**:  $r = 1$  (no gap) leaves  $\text{iter}_n - x_\star = x_0 - x_\star$  constant — the initial condition is then a free parameter, so the gap is exactly what removes it; [E] the seam rate  $(2/3)^6$  and the golden  $(\varphi + 2)/4 = (5 + \sqrt{5})/8$  both lie in  $[0, 1)$ . Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in **origin\_theory**; ledger row **FORM.SPECTRALGAP.01**.

## 2026-06-24 (Completing the Coxeter clock: v390 the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam) facet of $30 = 2 \cdot 3 \cdot 5$ )

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  — the golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (prime-5, carrier) and  $(2/3)^6 \in \mathbb{Q}$  (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field  $\mathbb{Q}(i)$ , i.e. the CM point  $\tau = i$  and the order-4  $\mu_4 = \mathbb{Z}/4$  *square* seam deck (v282/v288/v333). [E]  $h(E_8) = 30$  factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime:  $\text{disc } \mathbb{Q}(i) = -4$  (2 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$  (3 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$  (5 ramifies); the ramified primes are  $\{2, 3, 5\}$  with product  $30 = h(E_8)$ . [E] the prime-2 (Gaussian) facet is the  $\mathbb{Z}/4$  seam deck: min. poly  $x^2 + 1$ ,  $j(i) = 1728 = 12^3$ ,  $\tau = i$  fixed by the modular  $S$  of order  $4 = 2^2$ ,  $|\mu_4| = 4 = 2^2$  — the static, even, seam-orientation facet. [C] so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set  $\{2, 3, 5\}$  is forced by  $h(E_8) = 30$ , not chosen. Cited in **origin\_theory**; a synthesis, no new number; Python-only.

## 2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the loop level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [E] super-polynomial falloff:  $p^k e^{-p^2/M^2} \rightarrow 0$  for every degree  $k$ . [E] one dressed line turns a quadratic divergence into the finite  $(M^2/2)e^{-p_0^2/M^2}$ , while the GR  $\int p^3/p^2 dp$  diverges quadratically. [E] the superficial degree  $\rightarrow -\infty$  at every loop order  $L$  (the form factor multiplies the integrand by  $e^{-(\#\text{internal})p^2/M^2}$ ; dressed cutoff-ratio  $\rightarrow 1$  vs GR  $\rightarrow 4$ ). [E] the regulator is the seam scale  $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  (v253/v259), not a dial. [C][O] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in **tfpt\_4\_frontier**; Python-only.

## 2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (v387) is a typed budget.** A machine-checked answer to “does the  $(\lambda_2/\lambda_1)^n$  correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading  $\lambda_2$  exists, so the prediction surface splits into four classes. [E] *seam-gapped* (Koide  $F_{\text{pole}}$  v303, recovery v221, the QG bound capped by  $\chi = 729/665$  v337): rate  $(2/3)^6$ ; the discrete compiler: golden  $(\varphi + 2)/4$  (v312). [E] *exact identity* ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$ ): no  $\lambda_2$ , so the correction is 0 structurally and a band would contradict the [E] status. [E]

*external-rate* ( $\eta_B$  washout, axion freeze,  $m_p/m_e$  RG): the gapped *shape* with an external rate — only  $F_{\text{pole}}$  has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g.  $\sin^2 \theta_{12}$ ,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ). [E] the Koide first correction (door 2):  $F_{\text{pole}}$  is the one seam-gapped prediction, rate  $(2/3)^6 \approx 0.0878$  (v371), the source quotient 0.33% below  $2/3$  — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

## 2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ( $\sin^2 \theta_W = 3/8$  vs 0.231); [C] the optional carrier-PS branch sits at  $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$  GeV ( $\kappa \in [1.0, 1.2]$ , v253); [E] minimal PS gauge bosons are  $SU(4)$  leptoquarks that mediate rare LFV ( $K_L \rightarrow \mu e$ ), *not*  $p \rightarrow e^+ \pi^0$ , so the branch clears the binding bound by  $M_{PS}/M_{\text{bound}} \sim 10^7$  – proton-safe, with *no* fake  $\tau_p$  window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2$  has its only pole at  $p^2 = 0$  (residue  $1 > 0$ , ghost-free), is exponentially softened in the UV (the ratio  $e^{-u}$  falls faster than any power) and reduces to GR + a calculable  $-p^2/M^2$  correction at low energy ( $M = M_{\text{scal}}$ , v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering  $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$  is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] flavor, horizon recovery and the QG bound all share the  $(2/3)^6$  rate (the family-3 facet; QG susceptibility  $\chi = 729/665$ ), while the discrete compiler transient decays at the golden  $(\varphi + 2)/4$  (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

## 2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for  $F_{\text{transfer}}$ ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer  $\{1, (2/3)^6, (1/3)^6\}$  with gap  $6 \ln(3/2)$  (v56/v82); the affine- $E_8$  adjacency with Perron eigenvalue 2 (Kac marks) and subleading  $2 \cos(\pi/5) = \varphi$ , gap  $2 - \varphi = 1/\varphi^2$  (v312/v313); the decoupling margin  $\Delta_{\text{eff}} \approx 1.648$  (v76/v337). [E] the two number-field facets (v314): golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (carrier-5, static) and  $(2/3)^6 \in \mathbb{Q}$  (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ . [C] the other sectors are the same structure — gravity ( $\delta S=0$  equilibrium, v358/v359), the QG measure (Gaussian



saddle, v365), the boundary QFT (holomorphic  $c=8$  RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221),  $\tau=i$  (v333). [C] the meta-theorem: gapped  $\Rightarrow$  unique attractor  $\Rightarrow$  parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden ( $v_{\text{geo}}$ , the No-Unit theorem v153/v364), [E] (C) external physics ( $F_{\text{transfer}}$ , v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

**2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for  $S_{\text{pert}}$ , and v382 the Quillen determinant-line variation as a tracked target)**

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix  $S_{\text{pert}}$  (the previous documents had the  $S_{\text{pert}}$  skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order  $\omega \leq 4$  and the counterterm space is finite at every order; [E] BRST nilpotency  $s^2=0$  verified exactly via the Jacobi identity for  $su(2)$  and  $su(3)$  (the carrier weak+colour content); [E] the seam gap  $\Delta = 6 \log \frac{3}{2} > 0$  (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order  $T_n$  existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the  $R^2/\text{Weyl}^2$  gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and  $S_{\text{pert}}$  is not the ambient measure QG.AMB.01 (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this*  $F_{U(1)}$  functional” obligation (previously tracked only as EM.WARD.01’s residual, v341/v342) is elevated to its own citable target:  $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$  on the determinant line over the  $\mu_4$  seam moduli. [E] the Quillen split holds at the  $\alpha^{-1}=137.0359992$  root with every coefficient a named atom (index  $8b_1$ , heat-kernel  $c_3^{3,6}$ , discriminant  $q(D_5)+q(A_3)=2$ ); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of SEAM.EQUIV.01 (v336) — the seam  $U(1)$  determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

**2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)**

- **v369 (QGAMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy  $\Rightarrow$  DHR( $(E_8)_1$ ) = Vec: #primaries = |det Cartan| = 1 for  $E_8$  vs 4 for  $SO(16)$   $\Rightarrow$  no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ( $|\det K| = 1$ ); [E] finite Petz recovery rate  $(2/3)^6$  (v221); [E] the gap-decoupling margin  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$  is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)**

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: SEAM.EQUIV.01 is *closed modulo a cited theorem* (lattice v367/v368 + the  $S_3$  stack v376–v379 + Lean FORM.SEAM.MMST.01; residual [O] = the cited MMST/OS existence only); QG.AMB.01 is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; AUDIT OK.

**2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)**

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. [E] the seam KMS cutoff  $f(u)=e^{-u}$  (v259) gives the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$  with  $a(u)=e^u$  entire and nowhere zero (only the  $p^2=0$  pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of  $a$  and carries a Stelle-ghost zero ( $T_1=1+u \rightarrow u=-1$ ;  $T_2 \rightarrow -1 \pm i$ ); the entire  $a$  has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order  $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$  — the ghost decouples, “entire” = “do not truncate”. [C]  $\Rightarrow$  perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

**2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — FORM.SEAM.MMST.01)**

- **FORM.SEAM.MMST.01 — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms):  $D = 16$ , rank =  $c = 8$ ,

the range  $8 \leq c \leq 16$ ,  $c_- = 8 \neq 0$ , and  $\det K = 1$  vs  $SO(16)$ 's 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion  $(E_8)_1$  follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376–v379`; so `SEAM.EQUIV.01` is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

**2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge  $c=8$  from the lattice (`v376`), the  $(E_8)_1$  character (`v377`), the genus-1 torus count = 1 (`v378`), and reflection positivity (`v379`); only the abstract continuum existence (cited MMST) stays external)**

- **`v376 (SEAM.S3.CENTRALCHARGE.01)`**. The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives  $c = 1.0000$  per complex mode (Peschel correlation-matrix EE; the  $L=4m+2$  sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , chiral ( $c_- = 8$ , `v367/v368`). Numerical  $c=8$  from the lattice. Python (numpy).
- **`v377 (SEAM.S3.E8CHARACTER.01)`**. The exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary); the  $\mu_4/\text{GSO}$  promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, 4 primaries). Python (sympy q-series).
- **`v378 (SEAM.S3.MODULAR.01)`**. The genus-1 discriminator: KLM condensation  $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$ ,  $|\det \text{Cartan}(E_8)|=1$  vs  $D_8=4$ , and the single character's modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 (holomorphic =  $(E_8)_1$ ) — the “hard part” `v344` isolated. Python (sympy).
- **`v379 (SEAM.S3.RP.01)`**. Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With  $c=8 + (E_8)_1$  in hand, the OS inputs are complete and `C7/QG.AMB.01` is then discharged by the redundancy theorem `v369`. Python (numpy).
- **Net**: S3 is the master key — on the explicit lattice model the target net  $(E_8)_1$  is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem `v336`), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

**2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (`v375`))**

- **`v375 (OBSERVATORY.REGISTRY.01)`**. A CI over `freeze_file.csv`: [**E**] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [**E**] the headline values re-derive from  $\{\varphi_0^{\text{ret}}\}$  ( $\sin^2 \theta_{12} = 1/3 - \varphi_0^{\text{ret}}/2$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\beta_{\text{rad}} = \varphi_0^{\text{ret}}/(4\pi)$ ); [**C**] a live scorecard records  $\theta_{12}$  vs the JUNO first run ( $0.28\sigma$ , compatible),  $\theta_{13}$  vs NuFIT 6.0 NO ( $2.0\sigma$ , the flagged most-tensioned core prediction),  $\beta_{\text{rad}}$  vs ACT DR6 ( $0.37\sigma$ , a hint),  $r < 0.036$  (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

**2026-06-23 (Track 3 — the  $F_{\text{transfer}}$  functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371),  $\eta_B$  Boltzmann (v372), axion finite- $T$  relic (v373),  $m_p/m_e$  uncertainty budget (v374))**

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes  $Q$  above  $2/3$ ; the required  $\epsilon$  is negative while QED's is positive), so the transfer is the non-QED  $53/54$  operator /  $(2/3)^6$  Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen  $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$  (v212) gives  $\eta_B$  within a factor  $\sim 1.1$  of  $6.1 \times 10^{-10}$  with no  $M_R$  dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- $T$  axion misalignment ODE decides the branch: the frozen spine angle  $3\pi/5$  gives  $\Omega_a h^2 \sim 0.12$  (in  $[0.08, 0.16]$ , lands on  $\Omega_{\text{DM}}$  untuned) while the hilltop  $170^\circ$  over-produces ( $\sim 5.4 \times$ ); replaces the v326 toy. [C]/[X] (lattice  $\chi(T)$  input).
- **v374 (FR.QCD.BUDGET.01).** The  $m_p/m_e$  uncertainty budget: with  $b_3 = -7$  [E] and the two declared externals ( $\alpha_s(M_Z)$ ,  $C_p$ ,  $\sim 5.3\%$  each), the central  $\sim 1824$  (v262) carries a propagated  $\pm 7.5\%$  band  $[\sim 1690, \sim 1960]$  CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

**2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ( $p+ip$ ) lattice model with numerical Chern  $|C|=1$  realises the  $c_-=8$  edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))**

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the  $p+ip$  Bloch model is gapped at  $M=1$  with a numerically computed Fukui–Hatsugai–Suzuki Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ); [E]  $N_{\text{Maj}} = \dim S^+ = 16$  copies give  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). [C] the  $\mu_4$  condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ( $\min_{k_x} |E| \sim 0$ ) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ( $c_- = 8$ ). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

**2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))**

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar  $1/(p^2(p^2+M^2))$  pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ( $\text{tr } P_2 = (d-2)(d+1)/2$ ,  $\text{tr } P_1 = d-1$ ,  $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$  sum to  $d(d+1)/2$ ;  $d=4$  split  $5+3+1+1=10$ ), the EH propagator is  $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$ , and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor  $e^{p^2/M^2}$  (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour

(v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304's entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE:  $\text{tr } P_2=5$  is dimension counting, not  $g_{\text{car}}$  (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

**2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is  $(E_8)_1$  (v366))**

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is  $1/c_3 = 8\pi$  (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped  $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337)  $\Rightarrow$  mode-by-mode convergence, finite model  $\log\text{-det } \text{Tr}' \log M = 6 \log \frac{9}{2}$ ; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ( $\chi = 729/665$ , v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E]  $D = \dim S^+ = 16$  Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8$ ,  $\text{rank } E_8 = 8$  so  $\text{rank} \leq c \leq D$  is  $8 \leq 8 \leq 16$  (in range); [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ( $c_- = D/2 = 8 \neq 0$ ). [C] the MMST scaling-limit theorem then applies, and [E] the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K = 1$  vs  $SO(16) \det K = 4$ , v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)**

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the  $R + R^2$  regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit  $v_{\text{geo}}$ . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume  $\Rightarrow$  Einstein tensor, Lovelock  $\Rightarrow$  conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 +  $v_{\text{geo}}$  residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of  $c_3$  converging to  $1/(8\pi)$ ; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

**2026-06-23 (Direction 6,  $v_{\text{geo}}$  sharpened: after the parameter-free gravity, the gravity sector adds NO new dimensionful input either, so  $v_{\text{geo}}$  is the SINGLE dimensionful input of the complete theory — final tally  $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials: v364)**

- **v364 (VGE0.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made  $v_{\text{geo}}$  a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  reduce to  $v_{\text{geo}}$  times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)  $\times$  (a power of  $v_{\text{geo}}$ ); TFPT's free content is  $\{v_{\text{geo}}, \pi\}$  with ZERO dimensionless dials, a  $\sim 26 \rightarrow 1$  reduction vs the SM. [O]  $v_{\text{geo}}$  stays irreducibly primitive (No-Unit;  $1/G$  UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced  $c_- = 8$ , and the matter vacuum-energy backreaction is finite ( $\rightarrow$  the forced  $\Lambda$  from  $\alpha$ , not  $M_{\text{Pl}}^4$ ) — the loop closes; no new SM-quantum-number read-out (declined): v361)**

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced:  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$  (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE  $\rightarrow \Lambda$  from  $\alpha$ : the matter vacuum energy through  $G_{ab} = c_3^{-1} T_{ab}$  is gap-suppressed (Decoupling Thm v337) to  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$  (v60), the 123 orders =  $119.03 + 3.92$  — *not* the  $M_{\text{Pl}}^4$  catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit:  $\delta\langle K_B \rangle = (8\pi^2 R^4/15)\delta\langle T_{00} \rangle$ , carrier  $T_{ab}$  = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond  $c_- = 8$  and  $\Lambda$  — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

**2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known  $R^2$  scalaron — already falsifiable via  $n_s/r$ ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)**

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap  $\{1, (2/3)^6, (1/3)^6\}$  yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known  $R^2$  scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the  $R^2$  scalaron (spectral-action  $a_4 = R^2/72$ , v36;  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ , exponent  $7 = g_{\text{car}} + N_{\text{fam}} - 1$ ) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law  $\rightarrow f(R) = R + R^2$ , v28/v359). [E] it is ALREADY falsifiable via  $n_s = 1 - 2/N_*$ ,  $r = 12/N_*^2$  (live CMB-S4 kill test). [E] the gap  $\Delta = 6 \ln \frac{3}{2}$  is a dimensionless rate, so a distinct gap-induced curvature<sup>2</sup> term sits at  $\xi v_{\text{geo}} \sim$  the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent  $7 = g + N - 1$  and the gap exponent  $6 = 2N_{\text{fam}}$  are different;  $c_3^7$  and  $(2/3)^6$  unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

**2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with BOTH coefficients TFPT-fixed ( $8\pi = 1/c_3$  from v358,  $\Lambda$  from  $\alpha$  via v60); B6 closed at the local level: v359)**

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR  $G_{ab}$  (not just  $R_{ab}$ ) — trace  $g^{ab}G_{ab} = (1 - d/2)R = -R$  in  $d=4$ , the structure that carries  $\Lambda$ . [E] Lovelock:  $G_{ab}$  is the unique symmetric divergence-free second-order metric tensor, so  $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$  forces  $\nabla^a T_{ab} = 0$  — conservation is an output. [E] coefficient 1:  $8\pi G \rightarrow 8\pi = 1/c_3$  fixed (v358). [E] coefficient 2: Jacobson leaves  $\Lambda$  free, TFPT fixes it —  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions  $\rightarrow$  the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate, gap-decoupled);  $v_{\text{geo}}$  the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the  $\Lambda$  here matches the  $\alpha$ -readout.)

**2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law  $\delta S = \delta\langle K \rangle$  with TFPT’s atoms gives the LINEARISED Einstein equation with  $G$  from  $c_3$ ; the thermodynamic and geometric origins of  $c_3$  coincide via  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ ; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)**

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement  $\Rightarrow$  Einstein equation” with TFPT’s atoms. [E] PARAMETER-FREE COEFFICIENT:  $\delta S = \delta\langle K \rangle$  gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  fixed; all four  $c_3$ -roles consistent (Unruh  $1/(2\pi)=4c_3$ , EH  $1/(16\pi)=c_3/2$ , Einstein  $8\pi=1/c_3$ , Bekenstein  $1/4=1/|\mu_4|$ ) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law’s coefficient  $2\pi/\eta$  equals the geometric  $|\mathbb{Z}_2|2\pi\chi$  iff  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$  (v216) — the thermodynamic and geometric origins of  $c_3$  are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux  $\delta\langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta\langle T_{00} \rangle$ . [E] ENTROPY DENSITY ATOM-FIXED:  $1/4=1/|\mu_4|$  (v57) + central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  (form v59). [O] residual: linear  $\rightarrow$  full non-linear (the original B6) + the  $v_{\text{geo}}$  unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

**2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)**

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ( $\Delta=6\ln(3/2)>0$ , v302) quasi-free 16-Majorana collar (v155/v160)  $\rightarrow$  **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly

reduces the v336/v351 “does the collar have a chiral edge?” residual  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range rank  $\leq c \leq D$ ,  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3=1/(8\pi)$  a one-sided Gauss–Bonnet ( $|\mathbb{Z}_2| \nmid K$ , v58). [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs  $\Phi=0$ ) and  $\det R=8$  is read off — not a PDE. Only the absolute amplitude  $U_{\text{point}}$  (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

**2026-06-22 (bandwidth search of the unmapped  $E_8$  region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ( $\sum \deg = 128 = \dim S^+$ ,  $\prod \deg = |W(E_8)|$ , exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)**

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like  $E_8$  returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about  $E_8$ ), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number):  $\sum \deg = 128 = \dim S^+$ , the spinor half of  $248 = 120 + 128$  (theorem:  $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$ ) – the invariant budget of  $E_8$  equals the fermion content the carrier reads;  $\sum \exp = 120 = \# \text{pos roots}$ ;  $\prod \deg = |W(E_8)| = 696,729,600$ ; exponents = the  $\varphi(30)=8$  totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set;  $\max \deg = 30 = h(E_8)$ . [C] SUGGESTIVE, not a theorem:  $12=h(E_6)$ ,  $18=h(E_7)$ ,  $30=h(E_8)$  are all degrees of  $E_8$  (the  $2T/2O/2I$  McKay ladder, v219), but “ $\deg(E_8)$  contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches  $14=\dim G_2$ ,  $20=\det L$ ,  $24=|W(A_3)|$ ,  $12=|R(A_3)|$ ,  $18=p_4(a)$  (v66 already flagged 12, 18 as fittable) each admit  $\geq 2$  readings, and the lone extra prime in  $|W(E_8)| = 2^{14}3^55^27$  “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

**2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of  $E_8$ ’s 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)**

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an  $E_8$  structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much  $E_8$  structure



maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of  $E_8$ 's eight Casimir invariant degrees  $\{2, 8, 12, 14, 18, 20, 24, 30\}$  (exponents  $+1$ ), exactly THREE feed a readout — 2 (metric), 8 (rank  $\rightarrow c_3=1/(8\pi)$ ), 30 (Coxeter  $\rightarrow g_{\text{car}}=5$  and the order-30 cycle) — and FIVE carry no primary readout  $\{12, 14, 18, 20, 24\}$  (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8,  $h=30$ ,  $\det K=1$ ,  $D_5 \oplus A_3$ , the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of  $E_8$  untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED  $\Rightarrow$  NOT NUMEROLOGY:  $\varphi = 2 \cos(\pi/5)$  appears only in internal structure (the affine- $E_8$  attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation,  $\varphi$  is fitted to nothing — it is the algebraic signature of the 5-fold ( $h=30=2 \cdot 3 \cdot 5$ ), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD ( $E_8$  is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

**2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system;  $\pi$  is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)**

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms  $\rightarrow$  theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop:  $g_{\text{car}}=5 = \max \text{ prime of the output } h(E_8)=30$ , and the 8 in  $c_3 = \text{rank } E_8$  – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E]  $\pi$  is a FIXED geometric constant (the boundary 2-sphere,  $\oint K=4\pi$ ), not a dial. [C] this refines v352's “framework +  $\pi$ ”:  $\pi$  is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

**2026-06-22 (two consequences: the  $c=8$  which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework +  $\pi$ : v351, v352)**

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$  ambiguity” ( $E_8 \det 1$  vs  $SO(16)=D_8 \det 4$ , both  $c=8$ , flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4  $\mu_4$  clock: the index-4 simple-current extension of  $D_5 \oplus A_3$  is  $E_8$  (v154), the index-2 would be  $SO(16)$ , so the order-4 clock (the four Gauss–Bonnet marks, v216) selects  $E_8$  (full  $\mu_4$  condensation  $16 \rightarrow 4 \rightarrow 1$ ) not  $SO(16)$  (partial  $\mathbb{Z}_2$   $16 \rightarrow 4$ ). The bootstrap agrees ( $h(E_8)=30$  max prime  $5=g_{\text{car}}$  vs  $h(D_8)=14$  max prime 7). [E] so  $\det K=1$  is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E]  $g_{\text{car}}=5$  is forced three ways (v6); the 8 in  $c_3$  four ways (rank  $E_8 = h(D_5) = \phi(30) = \det R$ , v54); the  $E_8$  hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles

are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental  $\pi$ . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework +  $\pi$ . Wolfram mirror +2 (297  $\rightarrow$  299).

**2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms –  $g_{\text{car}}=5$  IS the icosahedral 5 of  $h(E_8)=30$ , and the golden ratio is emergent from the  $\mu_4$ -glue, not external: v350)**

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6):  $g_{\text{car}}=5$  is over-determined three ways – rank-fill ( $g_{\text{car}}+N_{\text{fam}}=8$ ), Coxeter-match ( $g_{\text{car}}= \max \text{ prime of } h(E_8)=30$ ), Pascal ( $2^4=C(5,0)+C(5,1)+C(5,2)$ ). [E] so  $g_{\text{car}}=5$  IS the icosahedral 5-fold (the 5 in  $h(E_8)=30=2\cdot3\cdot5$ ), not the  $D_5$  rank; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue:  $D_5$  ( $h=8$ )/ $A_3$  ( $h=4$ ) are non-golden alone, but  $D_5\oplus A_3 \rightarrow E_8$  ( $\mu_4$  index-4, v154) gives  $h(E_8)=30$  (golden) – the seam’s own  $\mu_4$  produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296  $\rightarrow$  297).

**2026-06-22 (the honest test – does the raw seam carry  $\varphi$ ? – answer NO: the bare carrier gives  $\sqrt{2}$ , golden is the icosahedral input; the keystone reduces to “is  $g_{\text{car}}=5$  a pentagon or a count?”: v349)**

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio  $2\cos(\pi/5)$  is the signature of a 5-fold rotation ( $5 \mid h$ ); the carrier  $D_5$  has  $h=8$  (eigenvalue  $\sqrt{2}$ ) and  $A_3$  has  $h=4$  – no 5-fold; the dynamical data (cusp  $\{0, 1/3, 2/3\}$ , recovery  $(2/3)^6$ , seed  $\varphi_0$ ) are rational/transcendental. [E]  $5 \mid h$  holds only for  $A_4=SU(5)$  ( $h=5$ ),  $H_3$  ( $h=10$ ),  $E_8$  ( $h=30$ ) – the output/icosahedral side, so  $\varphi$  is the icosahedral INPUT. [E]  $\varphi$  alone would not suffice ( $2I \neq C_5 \times \mu_4$ ,  $20 \neq 120$ ). [O] the resolution: the keystone reduces to one question – is  $g_{\text{car}}=5$  a PENTAGON (a 5-fold rotation  $\Rightarrow$  golden  $\Rightarrow$  icosahedral  $2I = E_8$ ) or just a COUNT (the rank of  $D_5 \Rightarrow \sqrt{2}$ )? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of  $g_{\text{car}}=5$ , not derivable from  $g_{\text{car}}=5$ -as-a-count); there is no hidden  $\varphi$ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295  $\rightarrow$  296). A NEGATIVE/clarification; prevents a false closure.

**2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)**

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form  $z \mapsto iz$  (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians =  $2I$ , golden-ratio quaternion norm) IS the rank-8  $E_8$  lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders  $\{1, 2, 3, 4, 6\}$  (the 5-fold absent), so  $\varphi = 2\cos(\pi/5)$  is exactly what extends the crystallographic  $\mu_4$  to the non-crystallographic icosahedral  $2I$  (the quasicrystal). [O] so

the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes,  $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$  and rigidity fixes the rest. [O] honest subtlety: TFPT’s  $\varphi$  is currently  $E_8$ -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries  $\varphi$  non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294  $\rightarrow$  295). An analysis/reduction; closes nothing.

**2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom  $P_3$ : v347)**

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ , v216) while the  $2I/E_8$  structure is the Seifert base over the *spherical*  $S^2(2, 3, 5)$  ( $\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$ ), so L2 bridges two geometric types – why it needs the carrier  $(2, 3, 5)$  input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom  $P_3$  (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293  $\rightarrow$  294). An analysis/roadmap; closes nothing.

**2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam  $\Rightarrow \det K=1$  is a six-link chain, five theorems + one open orbifold realisation, the group forced by the  $(2, 3, 5)$  atoms: v346)**

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam  $\Rightarrow \det K=1$  as six links and shows exactly one is open. [E]/[C] L1 raw seam  $\rightarrow \mu_4$  + four marks (Gauss–Bonnet, v216); [O] L2  $\mu_4$  + the  $(2, 3, 5)$  atoms  $\rightarrow$  the  $2I$  orbifold  $\mathbb{C}^2/\Gamma$  (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3  $2I \rightarrow$  affine  $E_8$  McKay (v219); [E] L4  $2I \rightarrow E_8$  du Val (v232); [E] L5  $\mathbb{C}^2/2I \rightarrow$  Poincaré link  $H_1=0$  (v345); [E] L6  $H_1=0 \rightarrow \det K=1$ . [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional  $SU(2)$  axis signatures  $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$ ;  $(2, 3, 5)$  unique to  $2I$ , and  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$  ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point  $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

**2026-06-22 (executing route R3: the  $(2, 3, 5)$ -rewrite attractor link computed to be the Poincaré homology sphere ( $\det K=1$ ), the combinatorial core complete, only the geometric bridge open: v345)**

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has  $H_1 = \text{coker}(\text{Cartan})$ . [E] the Smith normal form of the  $E_8$  Cartan matrix is the identity, so  $\text{coker}(E_8) = 0$  – the  $E_8$ -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of  $\det E_8=1$ ); among ADE only  $E_8$  does this ( $A_n \rightarrow \mathbb{Z}/(n+1)$ ,  $D_n \rightarrow \mathbb{Z}/4$ ,  $E_6 \rightarrow \mathbb{Z}/3$ ,  $E_7 \rightarrow \mathbb{Z}/2$ ). [E]  $\pi_1 = 2I = SL(2, 5)$  is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so  $H_1 = 0$  and the link is THE Poincaré sphere,  $|2I| = 120 = |\mu_4| h(E_8)$ . So R3’s combinatorial half is

[E]-complete: the  $E_8$  attractor link is forced to be the Poincaré sphere ( $\det K=1$ ), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this  $E_8$  du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292  $\rightarrow$  293).

**2026-06-22 (the one keystone bit  $\det K=1$  mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)**

- **v344 (SEAM.DETK.01).** A bird’s-eye, full-spectrum map of SEAM.EQUIV.01’s only open residual, the holomorphy bit  $\det K=1$ . [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link  $H_1=0$  / one 1-dim irrep /  $\det \text{Cartan} = 1$  / full  $\mu_4$  condensation /  $\tau=i$  CM), unified by  $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$  across ADE ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ) – only  $E_8$  gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v37);  $\det K$  is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the  $(2, 3, 5)$  rewrite attractor is the affine- $E_8$  graph (v312) and the  $E_8$  du Val link is the Poincaré sphere (v232), so  $\det K=1$  becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

**2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)**

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter  $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$  finite,  $\chi = 729/665$ ) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ $c$  ( $c=8$  consistent, but the Carlip ambiguity and no holomorphy  $\det K=1$ ) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

**2026-06-22 (the heat-kernel origin of the  $\alpha$  terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)**

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the  $F_{U(1)}$  determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E]  $c_3 = 1/(8\pi)$  is the boundary Gauss–Bonnet coefficient ( $\oint_{S^2} K = 2\pi\chi = 4\pi$ ,  $\chi=2$ , v216). [E] the  $\alpha^2$  (Calderón) term *is* the Gilkey gauge-curvature coefficient ( $30\Omega^2/360 = \frac{1}{12}\Omega^2$ ,  $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$ ), so it is forced, not an ansatz. [E] the  $c_3$ -orders  $\{0, 3, 6\}$  are an arithmetic ladder and  $2c_3^3 = 1/(256\pi^3)$  carries  $\pi^3$  (three boundary insertions). [C] the exact coefficient needs the seam  $F$ -normalisation. [O] the cubic  $\alpha^3$

Maxwell moment (a topological/Chern level) + the full “this is the seam  $U(1)$   $\zeta$ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01.  $\alpha^{-1}$  stays [E] (v3). Wolfram mirror +1 (290  $\rightarrow$  291).

**2026-06-22 ( $\alpha$  as the stationarity of a  $U(1)$  determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)**

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the form of  $\alpha$ , not more decimals”. The EM closure  $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\text{seam}}) = 0$  is rewritten as a *stationarity* of the boundary  $U(1)$  determinant line:  $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$ . [E] every ingredient is a NAMED object, not a knob:  $M=41$  is a  $U(1)$  index ( $\frac{4}{5}M = 8b_1$ ,  $b_1=41/10$  the SM  $U(1)_Y$  beta, EM Ward v48);  $c_3=1/(8\pi)$  the Gauss–Bonnet boundary coefficient ( $8=\text{rank } E_8$ );  $-5/4 = -q(D_5)$  a discriminant form and  $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$  carries the other,  $c_3/\varphi_{\text{base}}=3/4=q(A_3)$ ;  $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$ . The Quillen split is verified at the  $\alpha^{-1}=137.0359992$  root. **No new gate:**  $\alpha^{-1}$  stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that  $F_{U(1)}$  is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289  $\rightarrow$  290).

**2026-06-22 ( $A_s$  reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)**

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have  $A_s$  / the old versions derived it” and “can a ratio define mass itself”. [E]  $A_s$  *is* predicted and *dimensionless*:  $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2c_3^7/(24\pi^2)$  with  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$  a pure ratio, so  $A_s$  pins the e-fold count  $N_\star$ , NOT  $\bar{M}_{\text{Pl}}$  — it cannot define an absolute mass (No-Unit, v153, stands). [C] the  $-11.4\sigma$  at the slow Higgs channel ( $N_\star=51.4$ ) is the slow-vs-fast reheating dichotomy; the  $A_s$ -preferred  $N_\star=56.2 \sim$  the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean  $A_s = 2.124 \times 10^{-9}$  using the algebraic  $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$ ; at that  $N_\star$  the current normalisation gives  $2.047 \times 10^{-9}$  ( $-1.9\sigma$ ) — the old clean  $A_s$  is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic  $N_\star$  with one derived from reheating physics and exposed honestly that  $A_s$  requires fast (pre)heating. Record band  $N_\star \in [50, 60]$  (v84) untouched.

**2026-06-22 (the first-principles boundary, mapped: can  $M_{\text{Planck}}$  be computed? can  $F_{\text{transfer}}$  be first-principles? both NO, with the exact reason: v339)**

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* —  $v_{\text{geo}}/M_{\text{Planck}}$  is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E]  $F_{\text{transfer}}$  *cannot be made first-principles* either: for  $m_p/m_e$  the structure is in place (v262, carrier  $b_3=-7$  + closed electron) but two inputs stay external. [X]  $\alpha_s(M_Z)$  is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). The lattice  $C_p \sim 2.83$  is parameter-free but lattice-only (v35);  $\eta_B$ /axion ride on cosmological initial conditions; Koide  $Q=2/3$  is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

**2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the  $\theta_{13}$  budget: v336, v337, v338, FORM.CARTAN.DET.01)**

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur  $\rightarrow$  Virasoro, correct  $c$ ); WZW range  $\text{rank} \leq c \leq D$ , and  $(E_8)_1$  ( $c=8$ , rank 8,  $D=16$  Majoranas,  $SO(16)_1 \rightarrow (E_8)_1$ ) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by  $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ . [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic  $(E_8)_1$  net” ( $\det K=1$ ), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ; finite susceptibility  $\chi = 1/(1 - (2/3)^6) = 729/665$  gap-suppresses the ambient contribution (margin  $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ , with  $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$  exactly), so no readout depends on the ambient measure  $\Rightarrow$  QG.AMB.01 decoupled; negative control (gap closed  $\Rightarrow \chi$  diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the  $\theta_{13}$  budget, fenced.** The  $+2\sigma$  pressure point (v328) quantified against *both* NuFIT 6.0 NO variants:  $+2.0\sigma$  vs the no-SK central (0.02195) but  $+1.7\sigma$  vs the with-SK best fit ( $0.02215 \pm 0.00057$ ), a  $\sim 0.3\sigma$  systematic spread; [E] 0.023108 lies inside the with-SK  $3\sigma$  band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a  $\sim 6\sigma$  kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit  $D_8 = SO(16)$  Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator:  $D_8$  is even and symmetric just like  $E_8$  (same  $c=8$ ) but  $4 = \det \text{Cartan}(D_8)$  is genuinely not a unit while  $\det \text{Cartan}(E_8)$  is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

**2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)**

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a **COROLLARY** of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations  $U(1)$  are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the  $\mu_4$  clock = rotation by  $\pi/2$  ( $(e^{i\pi/2})^4=1$ ) fixes it  $\Rightarrow \omega \circ \rho = \omega$ . So SEAM.EQUIV.01  $\Rightarrow$  QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ( $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ , gap  $6 \ln \frac{3}{2} > 0$  derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] **QG.AMB.01 decoupled, not a gate:** margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76/v311/v330/v332)  $\Rightarrow$  no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items

$2 \rightarrow 1$ ; live residual =  $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance (=QGEO.SYM.01)` follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, `FAQ`, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

**2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)**

- **v333\_cm\_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual:  $\tau=i$  (the square) is the *unique* order-4 modulus —  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $S^4=I$  and fixes  $i$ , and an order-4 deck selects  $j=1728$  (vs the order-3 hexagon  $j=0$ ) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal  $\pi/2$  spacing). So given the  $\mu_4$  deck,  $\tau=i$  is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGEO.SYM.01). NOT a closure.
- **v334\_conformal\_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation  $\phi \rightarrow i\phi$  flips the wrong-sign kinetic term, making the divergent conformal Gaussian  $\int e^{+|c|\phi^2}$  into the convergent  $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$ ; the entire IDG form factor  $e^{p^2/M^2}$  dresses the propagator pole-free (the polynomial  $R+R^2$  ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended (FORM.CARTAN.DET.01, [E]).** The Conway–Sloane lattice side: the  $E_8$  Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ( $|\det|=1$ ) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness:  $E_8$  is the only even unimodular lattice in  $\dim < 16$ ) is the cited classification residual.

**2026-06-22 (endgame round: the necessity of  $H$  reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem,  $|\det E_8|=1$  and the  $\mu_4$  commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)**

- **v331\_necessity\_of\_H.py (QGEO.NECESS.01, [E]/[O]).** The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam  $\Lambda = |k| + M_f$  commutes with the  $\mu_4$  clock *iff* the four marks sit at the square ( $\tau=i$ ) — moving a mark off the square breaks  $\mathbb{Z}_4$  and  $[\rho, \Lambda] \neq 0$ . The square has cross-ratio 2 ( $j=1728$ ), the unique 4-config with an order-4 automorphism (v214/v267), so  $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$  and “necessity of  $H$ ” sharpens to: the raw

dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **v332\_qgamb\_metric\_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient  $-(D-1)(D-2)/4$  ( $= -\frac{3}{2}$  at  $D=4$ ), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial  $R+R^2$  ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator  $|\det \text{Cartan}(E_8)|=1$  ( $E_8$  has one anyon, vs  $D_8$ ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the  $8 \times 8$  PRODUCT gives `IsUnit (det E8)` via  $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$  — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers  $\mathbb{Z}[i]$ :  $i^4=1$ ,  $i^2=-1$ , and (on concrete  $6 \times 6$  matrices) an offset-4 coupling commutes with the  $\mu_4$  clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

**2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)**

- **v329\_os\_gap\_reduction.py** (QGEO.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization  $dT(-\log T)$ , whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy  $= 6 \ln \frac{3}{2}$ , *independent of system size*. Together with  $H \Rightarrow \omega \circ \rho = \omega$  and  $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ , this exhibits the rotation-invariant flat-pillowcase premise  $H$  as the *single sufficient premise* of both bedrock IDs; residual = the necessity of  $H$  on the raw collar.
- **v330\_qgamb\_admissible\_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ( $C(n)/C(n-1) \rightarrow (2/3)^6$ ), with finite susceptibility  $\chi = 729/665$  so  $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$  converges  $\Rightarrow$  uniform tightness (Prokhorov)  $\Rightarrow$  the projective limit exists; OS reconstruction gives  $H_{\text{OS}} \geq 0$  with gap  $6 \ln \frac{3}{2}$ . The full  $\text{QG}_{\text{amb}}$  (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom):  $|\det \text{Cartan}(A_3)|=4=|\mu_4|$  ( $3 \times 3$ , `det_fin_three`) and  $|\det \text{Cartan}(D_5)|=4$  ( $5 \times 5$ , `decide`), giving the carrier  $|\det \text{Gram}|=4 \cdot 4=16$  (the v281 anyon count). The rank-8  $E_8/D_8$  case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.



**2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the  $F_{\text{transfer}}$  solver suite, the cusp weight derived from a minimal rewrite, and the  $\theta_{13}$  pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)**

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum`  $\rightarrow$  `GeometricModularFlow`  $\rightarrow$  `Mu4ModularSymmetry`  $\rightarrow$  `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a decide-proven discriminator  $(1 \mid 4) \wedge \neg(4 \mid 1)$  (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325\_pillowcase\_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat  $\tau=i$  pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks,  $\mu_4$  deck,  $\omega \circ \rho = \omega$ , holomorphic  $c=8 \Rightarrow (E_8)_1$ , with the recovery gap as input); all pieces machine-checked (incl. the pillowcase  $\chi_{\text{orb}}=0$ ), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326\_ftransfer\_suite.py** (FR.SUITE.01, [E]/[C]). The four  $F_{\text{transfer}}$  readouts productised into one runnable harness, each a gapped relaxation to a unique attractor:  $F_{\text{pole}}$  (Koide, seam rate  $(2/3)^6$ ),  $F_{\text{Boltzmann}}$  ( $\eta_B$ , H-theorem),  $F_{\text{relic}}$  (axion adiabatic invariant),  $F_{\text{QCD}}$  ( $m_p/m_e$ , RG to the UV fixed point), plus the  $F_{\text{RareKaon}}$  bridge. Only  $F_{\text{pole}}$  carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327\_hypergraph\_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight  $2/3$  is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum  $\{1, 2/3, 0\}$ , so  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$  emerges from the rule arity and  $(2/3)^6 = 64/729$  is the recovery gap; with a PROOF that  $2/3$  is not a root of the affine- $E_8$  charpoly ( $p(2/3) = -3520/19683 \neq 0$ , sharpening v312). v324’s injected datum is reduced to the anchor arity  $\{2, 3\}$ .
- **v328\_theta13\_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified:  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$  is  $+2.0\sigma$  above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG < 1%, exponent 5/6 exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon  $F_{\text{transfer}}$  bridge row.
- **Wolfram mirror** 285  $\rightarrow$  288. v325 ( $\chi_{\text{orb}}=0$ ) and v327 (rewrite spectrum  $\{0, 2/3, 1\}$ ,  $(2/3)^6 = 64/729$ , and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

**2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)**

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ” framing was replaced by the current  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  with the two structural

items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/ 212→v324/ 322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

**2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)**

- **Bug fix —  $\delta_{\text{CKM}}$  display value.** The predictions table (`tex-artefacts/predictions.tex`) showed  $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ , but that formula evaluates to  $68.65^\circ$  (the canonical frozen `DELTA_CKM_RAD` = 1.198232 rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to  $68.65^\circ$ . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202\_rare\_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  ( $\sim 1.2\sigma$  above) to the full 2016–2024 combination  $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$  (La Thuile 2026), on which the prediction  $9.45 \times 10^{-11}$  lands at  $+0.08\sigma$ . Framed honestly: a strong *consistency* hit for the closed CKM point, but an  $F_{\text{transfer}}$  bridge — *not* a unique TFPT-vs-SM discriminator (the SM  $(8.6 \pm 0.4) \times 10^{-11}$  is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new  $F_{\text{transfer}}$  bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ( $\pi/3$ ) component of the quark phase”: the lock is to the structural  $\pi/3$ , not the full measured  $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$ , so  $\delta_{\text{PMNS}} = 240^\circ$  (not  $248.7^\circ$ ). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305\_witness\_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule:  $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$  is the *static* carrier ( $\mathbb{Z}/5$ ) field, every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6$  hand), so a dynamic rate explained by  $\phi$  is a cross-field false friend (v314/v315).
- **v203\_eht\_achromatic.py — the  $\mu_4$  Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at  $m \equiv 0 \pmod{4}$  (the  $\mu_4$  fingerprint of the four marks, v201); a  $\mu_4$ -orbit profile has  $\sim 0$  off-mode power while a  $\mathbb{Z}_2$  control leaks at  $m = 2$ .

- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed  $\varphi_0$ , the error-weighted mean matching the axiom to 0.04%, with  $\sin^2 \theta_{13}$  the  $\sim 2\sigma$  pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

**2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)**

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks,  $275 \rightarrow 285$ , *all passing*): v313 (the affine- $E_8$  characteristic polynomial with the golden factor  $x^2-x-1$ ; the  $(2, 3, 5)$  atoms as  $2\cos(\pi/k)$ ; the 31/30 selection), v314 (the rational kernel vs  $\mathbb{Q}(\sqrt{5})$  dynamic-rate split), v315 ( $30=5\cdot 6$ ,  $\varphi(30)=8$ , the Galois group  $\mu_4 \times \mathbb{Z}_2$ , the  $\sqrt{5}$  Gauss sum), v316 ( $\rho=\zeta_6=\zeta_{30}^5$ ,  $\rho^4=-\rho$ ,  $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$ ), v317 ( $N_{\text{fam}}=3$  as the  $\mu_3$  orbit, the 1+2 Galois split), v318 ( $\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$  a pure function of  $\pi$ ) and v320 (the CP lock  $\delta_{\text{PMNS}}=240^\circ$ ). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322\_cp\_lock\_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes  $\arg \rho^k = k \cdot 60^\circ$ , and *which* one is selected is the deck order:  $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$  (node 4); [C] the sub-leading correction is bounded by the quark analogue  $3\lambda^2 \approx 8.7^\circ$ , so the prediction is the band  $240^\circ \pm \sim 9^\circ$ ; [C] against NuFIT 6.0 (NO,  $\delta_{\text{CP}}=212_{-41}^{+26^\circ}$ ) the leading value is  $+1.08\sigma$ , the band edge  $+0.74\sigma$  – compatible today; [X] the nearest wrong node is  $60^\circ$  away, far beyond the budget, so DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323\_bw\_geometric\_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the  $\mu_4$  clock is a geometric rotation  $\rho=\text{diag}(i^n)=\exp(i\frac{\pi}{2}L)$ ,  $\mu_4 \subset U(1)_{\text{rot}}$ ; [E] for a rotation-covariant covariance  $C=f(L)$  the modular Hamiltonian  $K=\log((1-C)/C)=g(L)$ , so  $[K, L]=0$  (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric,  $[K, L] \neq 0$ ); [C] given the seam is the  $(E_8)_1$  chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ( $\omega \circ \rho = \omega$ ) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **v324\_hypergraph\_fiber.py (HYP.FIBER.01, [E]/[C]).** The constructive v312 follow-up: the *minimal* substrate that carries both the  $E_8$  skeleton and the recovery gap is a *product*. [E] the carrier network  $T_{\text{net}}=(A+2I)/4$  (attractor = Kac marks = skeleton) fibred by the 3-node family cusp  $T_{\text{cusp}}=\text{diag}((1-w)^{2N_{\text{fam}}})$ ,  $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$  (spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ); the fibred  $T_{\text{net}} \otimes T_{\text{cusp}}$  has top eigenvalue 1 (eigenvector marks  $\otimes (w=0)$ ) and  $(2/3)^6$  as a genuine eigenvalue (marks  $\otimes (w=\frac{1}{3})$ ) – the  $N_{\text{fam}}=3$  cusp is the minimal fibre and the substrate = carrier  $\times$  family = the  $30=g_{\text{car}}(2N_{\text{fam}})=5 \times 6$  of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of  $2/3$  from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

**2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)**

- **v321\_killtest\_board.py (KILL.BOARD.01, [E]).** The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO  $\sin^2 \theta_{12}=0.306747$  (fastest), #2 CMB-S4/LiteBIRD  $r$

(kill  $r > 0.01$ ), #3 DESI/Euclid  $w$  (kill  $w \neq -1$ ), #4 DUNE/Hyper-K  $\delta_{\text{PMNS}} = 240^\circ$  (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering +  $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  now), #8 LiteBIRD  $\beta$ . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

## 2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes  $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamlsE8}$  assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Muger / Kawahigashi–Longo–Muger holomorphy; Conway–Sloane  $E_8$  uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a `decide`-proven discriminator  $\text{primariesE8} = 1 \neq \text{primariesD8} = 4$  (the  $|\det \text{Cartan}|$  that separates the holomorphic  $E_8$  from the same- $c$  rival  $D_8 = SO(16)$ ). Built end-to-end (lake build, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

## 2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ : v320)

- **v320\_galois\_cp\_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit  $\rho = \zeta_6$  of the family Galois factor:  $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$  ( $60^\circ$ ),  $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$  ( $240^\circ$ ), and  $\rho^4 = -\rho$  locks them as  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  (the  $\mathbb{Z}_2$  sheet flip); [C] this upgrades the previously assigned  $\delta_{\text{PMNS}} = 240^\circ$  to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a  $\delta_{\text{PMNS}}$  robustly incompatible with  $240^\circ$  ( $> 3\sigma$  at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation;  $240^\circ$  is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

## 2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319\_translation\_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314);  $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  and  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (two coprime hands, v315). [E] the *dynamic* hand is  $\mathbb{Z}/6 = 2N_{\text{fam}}$  (ticks  $0, \dots, 5$ ): the recovery exponent is exactly  $6 = 2N_{\text{fam}}$ , rate  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ , order-6 generator  $c^5$  ( $\zeta_6 = \zeta_{30}^5$ , v316). [E] position 0 is the *conserved law* — the resummed clock rate( $n$ ) =  $-p_2 \ln(1 - n/N_{\text{fam}})$  (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from  $n=1$ , and the  $E_8$  live phases (totatives of 30) start at 1 ( $\{1, 7, \dots, 29\}$ ), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is  $\mathbb{Z}/5 = g_{\text{car}}$  (ticks  $1, \dots, 5$ ): the golden field  $\mathbb{Q}(\sqrt{5})$  ( $\varphi = 2 \cos(\pi/5)$ ), field-obstructed (no rational dynamic image, v314) — static-only;  $\zeta_5 = \zeta_{30}^6$  carries no phase. [E] one full turn

$= \text{lcm}(5, 6) = 30 = h(E_8)$ , rank  $E_8 = 8 = \varphi(30)$  live phases pairing into  $|\mu_4| = 4$  invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring)  $\times$  (dynamic 6-ring), the dynamic hand carrying the rate (exponent  $6 = 2N_{\text{fam}}$ ), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation\\_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin\\_theory](#); ledger row TRANSLATE.CLOCK.01.

**2026-06-21 (capstone: the SM structural sector is  $\mathbb{Q}(\zeta_{30})$ + Galois  $\mu_4 \times \mathbb{Z}_2$ , and the complete input is  $\{a, \pi, v_{\text{geo}}\}$  — zero dimensionless free parameters: v318)**

- **v318\_arithmetic\_capstone.py (ARITH.CAPSTONE.01, [E]/[O]).** The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field  $\mathbb{Q}(\zeta_{30})$  with Galois group  $\mu_4 \times \mathbb{Z}_2$  (degree  $\varphi(30) = 8 = \text{rank } E_8$ ,  $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$ ), forced by the anchor atoms  $\{2, 3, 5\} = e(a)$ ; [E] the magnitude seed  $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$  ( $c_3 = 1/(8\pi)$ ) is a pure function of  $\pi$  (no free parameters, anchor-derived coefficients), so the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ ) are  $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters,  $\pi$  the unique transcendental primitive (ARCH.CORE.01),  $v_{\text{geo}}$  the unique dimensionful input (No-Unit Theorem) — complete input  $\{a, \pi, v_{\text{geo}}\}$ . [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin\\_theory](#); ledger row ARITH.CAPSTONE.01.

**2026-06-21 (are the 3 generations a  $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)**

- **v317\_galois\_family.py (GALOIS.FAMILY.01, [E]/[C]).** The v316 follow-up to the generation structure. [E] the three cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$  (with  $\text{Spec}(Q_+) = \{1, 2, 3\}$ , the  $A_3$  exponents) are the cube roots of unity  $\{1, \zeta_3, \zeta_3^2\}$  under  $w \mapsto e^{2\pi i w}$ , and the cyclic family symmetry  $w \mapsto w + \frac{1}{3}$  is a *transitive*  $\mu_3$  orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues  $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$  attach to the weights (attractor = eigenvalue 1 at  $w=0$ ); [E] the family Galois  $\mathbb{Z}_2$  ( $\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$ , = the v316 CP conjugation) refines the orbit to 1+2 — fixing  $w=0$  (rational) and swapping  $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$ ; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a  $\mu_3$  orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in [origin\\_theory](#); ledger row GALOIS.FAMILY.01.

**2026-06-21 (does the Galois  $\mu_4 \times \mathbb{Z}_2$  organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois  $\mathbb{Z}_2 = \text{CP conjugation}$ : v316)**

- **v316\_galois\_readout.py (GALOIS.READOUT.01, [E]/[C]).** The v315 follow-up to phenomenology: does the Galois coupling  $\mu_4 \times \mathbb{Z}_2$  reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit  $\rho = \zeta_6 = \zeta_{30}^5$  is exactly the order-6 =  $2N_{\text{fam}}$  factor  $c^5$  of v315 (the carrier being  $\zeta_5 = \zeta_{30}^6$ ); [C] the leading CP phases are its cyclotomic arguments  $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$  and  $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$  (power 4 =  $|\mu_4|$ ), with  $\zeta_6^4 = -\zeta_6$  the  $\mathbb{Z}_2$  sheet flip and  $\mu_6 = \mu_3 \times \mu_2$  (inherits v231/v233); [E] the family Galois  $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$  ( $\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$ ) *is* CP conjugation  $\delta \mapsto -\delta$ ; [E] the carrier factor  $\zeta_5$  (golden  $\sqrt{5}$ ) carries no phase ( $\mu_4$  static), and the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ , transcendental  $\varphi_0$ ) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in [origin\\_theory](#); ledger row GALOIS.READOUT.01.

**2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois =  $\mu_4 \times \mathbb{Z}_2$ : v315)**

- **v315\_coxeter\_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  couple the carrier (5-fold, golden,  $\mathbb{Q}(\sqrt{5})$ ) and family ( $6 = 2 \cdot 3$ , rational) sectors, or just factorize? [E] the cyclic group *factorizes*:  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (direct product, group-decoupled, in line with the v314 field obstruction); [E] the  $E_8$  exponents = totatives(30) split as totatives(5)  $\times$  totatives(6),  $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$ ; [E] but the *field* couples:  $\mathbb{Q}(\zeta_{30})$  is the compositum of the carrier  $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$  and the family  $\mathbb{Q}(\zeta_3)$ , degree  $8 = \text{rank } E_8$ ; [E] the Galois group is exactly  $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$  (order 8, max element order 4), so the  $\mu_4$  seam clock *is* the Galois group of the carrier's golden field and  $\mathbb{Z}_2$  that of the family field (refining v223); [E]  $\sqrt{5} = \phi + 1/\phi$  is the quadratic Gauss sum in  $\mathbb{Q}(\zeta_5)$ . [C] verdict:  $30 = 5 \cdot 6$  does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in *origin\_theory*; ledger row COX.COUPLE.01.

**2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split  $\mathbb{Q}$  vs  $\mathbb{Q}(\sqrt{5})$ : v314)**

- **v314\_rate\_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- $E_8$  network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate  $(2/3)^6$ ) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part  $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$  plus the golden part  $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$ , while the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles  $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$  and  $2 \cos(\pi/3)=1 = N_{\text{fam}}$  build  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$ , same field, same atoms), but [E] the carrier 5-fold (golden,  $\mathbb{Q}(\sqrt{5})$ ) has NO rational dynamic image – a field obstruction ( $2/3$  is neither an adjacency nor a lazy-update eigenvalue; heat-kernel  $t = \Delta/(2 - \phi) \approx 6.37$  not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , and the  $\phi$ -quasicrystal  $E_8 = H_4 + \phi H_4$  ( $240 = 2 \cdot 120$ ) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in *origin\_theory*; ledger row RATE.TRANSLATE.01.

**2026-06-21 (the golden ratio made precise: the  $(2, 3, 5)$  atoms are the affine- $E_8$  network spectral angles,  $\phi = 2 \cos(\pi/5)$  the  $g_{\text{car}}=5$  signature: v313)**

- **v313\_golden\_atoms\_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- $E_8$  adjacency characteristic polynomial factors as  $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , so the golden minimal polynomial  $x^2-x-1$  divides it and  $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$  is an *exact* eigenvalue; [E] the non-trivial eigenvalues are  $2 \cos(\pi/k)$  for exactly  $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$  ( $2 \cos(\pi/2)=0$ ,  $2 \cos(\pi/3)=1$ ,  $2 \cos(\pi/5)=\phi$ ,  $2 \cos(2\pi/5)=1/\phi$ ), so the golden ratio is *specifically* the  $g_{\text{car}}=5$  (5-fold) signature and  $g_{\text{car}}=5$  gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the  $(2, 3, 5)$  spherical excess  $1/2+1/3+1/5 = 31/30 > 1$  (the condition selecting the finite  $2I \Rightarrow E_8$ ) ties the v63/v76 gap-margin numerator  $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$  to the Coxeter  $30 = h(E_8)$  — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null:  $\phi$  is a structural lattice/network quantity, not a phenomenological readout. [C] direction:  $\phi$  is the  $E_8$ -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in *origin\_theory*; ledger row GOLD.ATOMS.01.

**2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)**

- **v308\_seam\_equiv\_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic  $c=8$   $(E_8)_1$  net” assembled into ONE well-typed logical chain  $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$ , with the exact discriminators  $|\det \text{Cartan}(E_8)|=1$  vs  $|\det \text{Cartan}(D_8=SO(16))|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$  and  $c=g_{\text{car}}+N_{\text{fam}}=8$  all machine-checked; the input gap is the derived recovery gap  $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309\_modular\_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock  $\omega \circ \rho = \omega$  via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so  $\mu_4$  is derived; the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  round-trips to the seam operator and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310\_carrier\_sm\_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  is exactly one anomaly-free Standard-Model generation ( $SU(5) \subset SO(10): 10+\bar{5}+1$ ), with  $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$  (exact) and the one-loop  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts  $b_1$ , so  $N_{\text{fam}}=3$  pins the RG. Closes the spectrum/anomaly/ $\beta$  sub-question; the interacting Epstein–Glaser  $S$ -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311\_gap\_decoupled\_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  gives geometric clustering at rate  $(2/3)^6$  (correlation length  $1/(6 \ln(3/2))$ , finite susceptibility  $729/665$ ), and the v76 margin  $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$  decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312\_hypergraph\_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- $E_8$  Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5) =$  the golden ratio, but the recovery rate  $(2/3)^6$  is *not* any eigenvalue and  $\varphi_0$  is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and the analytic seed  $\varphi_0$ . The open question is precisely delimited, not closed. Cited in `introduction`; ledger row HYP.INJECT.01.

**2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)**

- **v305\_witness\_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h(E_8)=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the single anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive (13 atoms from 2 free primitives,  $6.5 \times$

compression made explicit), so the recurring  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio; the  $E_8$  spine is overdetermined ( $\geq 3$  structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by v100, never multiplied); the negative control shows  $\alpha^{-1}$ 's “137” is foreign to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, v3). Cited in the `tfpt_5_redteam` body; ledger row WITNESS.ECON.01. Python-only.

- **v306\_seed\_crossval.py (SEED.CROSSVAL.01, [E]).** A leave-one-out cross-validation turning the “many observables, one seed  $\varphi_0$ ” worry into a falsifiable out-of-sample test: back-solving  $\varphi_0$  from each of the five seed-grammar observables ( $\sin^2 \theta_{12}$ ,  $\sin^2 \theta_{13}$ ,  $\beta$ ,  $\Omega_b$ ,  $\lambda_C$ ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  out of sample (most-tensioned =  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the  $\chi^2$  by  $> 100\times$ . Cited in `tfpt_5_redteam`; ledger row SEED.CROSSVAL.01. Python-only.
- **v307\_data\_watchdog.py (DATA.WATCHDOG.01, [E]).** The operational decision pipeline over the frozen registry (v84): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live  $n_\sigma$  against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are  $\alpha^{-1}$  ( $1.9\sigma$ ),  $s_{23}$  ( $1.8\sigma$ ) and the most-tensioned core prediction  $\sin^2 \theta_{13}$  ( $2.0\sigma$ , the honest current pressure point);  $\sin^2 \theta_{12} = 0.306747$  is compatible ( $0.02\sigma$ , the JUNO falsifier), the  $r \in [0.0033, 0.0048]$  band is below the BK18 limit and the kill, and  $w=-1$  is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row DATA.WATCHDOG.01. Python-only.
- **Target-A holomorphy kill test (freeze\_file.csv).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ( $|\det K| > 1$ , the same- $c$  rival  $SO(16)_1$ ) would break holomorphy and kill “seam net =  $(E_8)_1$ ” — since holomorphic  $c=8 \Leftrightarrow E_8$  is machine-proven (v83/v143) and  $\det K=1 \Leftrightarrow$  invertible/SRE (v237/v301/v302).

## 2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of $c$ ” framing of SEAM.EQUIV.01: v304)

- **v304\_idg\_nonlocal\_ghost.py (QGAMB.IDG.01, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F/v278` Stelle-ghost attack on the perturbative gravity sector. [E] the local  $R+R^2/\text{Weyl}^2$  four-derivative propagator  $1/(p^2(p^2+M^2))$  carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight  $f(u) = e^{-u}$  (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close QG.AMB.01: the trace regulator is not yet shown to resum to an entire  $a(\square)$ , and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row QGAMB.IDG.01.
- **Double-pushout figure (tfpt\_1\_architecture\_e8).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order- $|\mu_4|=4$  glue is *one* operation in two categories: the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  (v1) and the conformal-net / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$  (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (introduction, tfpt\_research\_contracts).** SEAM.EQUIV.01 is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of  $c$  plays in relativity (cf. v267). Prose sharpening; no status change.



2026-06-19 (the discrete→dynamic lens on **F\_transfer**: one shape, one seam-rate, v303)

- **v303\_fttransfer\_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the  $E_8$  marks, gap  $6\ln(3/2)$ ) — is also the shape of the four **F\_transfer** instances, or an unrelated bolt-on. It is the *same* shape: [E] **F\_pole** (Koide) *is* the seam dynamics (Möbius multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, v82/v54); [C] **F\_Boltzmann** ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0,1)$ , H-theorem); [C] **F\_relic** (axion) freezes at an adiabatic fixed point; [E] **F\_QCD** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3=-7$  (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; **F\_pole** has the seam rate  $(2/3)^6$  *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F\_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory’s simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite v303 5/5, AUDIT OK.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap  $6\ln(3/2)$ , v302)

- **v302\_seam\_gap.py** closes the chain on the single statement v301 had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is  $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\text{fam}}\ln(3/2) \approx 2.4328 > 0$  (multiplicative gap  $1 - (2/3)^6 = 665/729$ ;  $(2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, **SEAM.EQUIV.01**’s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap  $6\ln(3/2)$ ) — *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite v302 5/5, AUDIT OK.

2026-06-19 (Route A’s last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301\_route\_a\_invertible.py** attacks the single remaining *unconditional* gap of **SEAM.EQUIV.01**. Since v300 collapsed Route B into Route A’s rationality, that gap is Route A’s open hypothesis (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev’s periodic table, class D). [E] the carrier is 16 Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); [E]  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) vs 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. [O] LIT-A’s ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of **SEAM.EQUIV.01** reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not

manufactured. [E] suite v301 6/6, AUDIT OK.

## 2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$ , v300)

- **v300\_flataway\_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum (“the spectral mismatch grows with  $\varepsilon$ ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint —  $\text{spec}(|D_\theta|)$  is the integer ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (sympy  $2 \times 2$   $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , + an  $O(g^2)$  common shift), validated against the full Toeplitz operator; [E] any  $g_k \neq 0$  lifts the resonant degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every  $g_k = 0 \Rightarrow f = 0$  — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* [C] the  $(E_8)_1$  character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients (248 at level 1;  $E_8$  even unimodular  $\Rightarrow$  integer weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam, i.e. the external pin *is* the  $(E_8)_1$  rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

## 2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches $E_8$ as the last finite point, v299)

- **v299\_hypergraph\_0d\_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ( $|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$ ; the McKay group is the  $SU(2)$  binary lift  $2I$ , a *different* order-120 group whose McKay graph is affine  $E_8$ ; 30 edges  $= h(E_8)$ ); [E] the  $1 \rightarrow 4$  subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the  $2I$  fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine  $E_8$  McKay graph; [E] Smith ( $\rho \leq 2 \Leftrightarrow$  affine ADE) + Collatz–Wielandt ( $\rho \leq 2 \Leftrightarrow$  a local witness  $Am \leq 2m$ ); [E]\* the autonomous rule (diffusion-generated witness + *local* gate  $Am \leq 2m$ , Perron only as audit) runs  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ ; [E] the negative controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$  (closed form), so neither family ever reaches  $E_8$ . Honest type: [O] the  $(2, 3, \cdot)$  seed is the single irreducible input = P2 — *the rule reaches  $E_8$  as the last finite point on this seed; it is not a derivation of P2.* [E] build green, AUDIT OK.

## 2026-06-18 ( $E_8$ as a network attractor — the Wolfram-program reading, v298)

- **v298\_e8\_network\_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- $E_8$  Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4m$  on the  $(2, 3, 5)$  McKay graph (converges from any positive start;  $Am = 2m$ , top eigenvalue 2); the spherical tower terminates at  $E_8$  and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ $E_8$  is the universal attractor of a minimal  $(2, 3, 5)$  network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives:  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

## 2026-06-18 ( $a_2$ closed form + Route A literature stack, v296–v297)

- **v296\_flataway\_a2\_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact  $a_2$  coefficient in *closed form*: the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ) to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle; e.g.  $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ . Manifestly positive, small- $t$  law  $W_k = t/2 + O(t^3)$  (the  $t^2$  term cancels). Reproduces v295 to machine precision.

- **v297\_route\_a\_literature\_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta \text{Tr} \geq 0$  for all deformations,  $= 0$  iff  $f = 0$ . The exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (Duhamel), validated to rel. err  $< 10^{-6}$ .
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal  $\Delta=0 \Leftrightarrow$  flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes  $a_2$ ) remains. [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292\_flataway\_heat\_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation  $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a *positive-definite* quadratic form in the off-mark curvature ( $c_k(t) > 0$  for every  $\mathbb{Z}_4$  mode, positive on all combinations), leading invariant  $\|f\|_{L^2}$  — so Flat-Away reduces to the single fact “the seam fixes the  $a_2$  heat coefficient”.
- **v293\_flataway\_spectral\_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$  space. Mode  $4k$  splits the degenerate Steklov level  $|2k|$  at first order; the Hessian of the spectral mismatch over modes  $\{4, 8, 12\}$  is positive-definite (eigenvalues  $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294\_flataway\_rp\_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles  $\pi$ , Gauss-Bonnet  $4\pi$ ), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy  $\int f^2$  is strictly convex (Hessian  $2\pi I$ ) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290\_z4\_smooth\_curvature\_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that  $\mathbb{Z}_4$  block-diagonality is *not* mark-locality: a smooth  $\mathbb{Z}_4$  off-mark curvature  $f = \varepsilon \cos(4\theta)$  [E] passes the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , same as mark-local) yet [E] shifts the DtN spectrum ( $\max |\Delta\lambda| = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not  $[\rho, \Lambda]=0$  — which is exactly why Flat-Away cannot be read off the commutator.

- **v291\_flataway\_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family — deviation zero only at  $\varepsilon=0$ ) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289\_marklocal\_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks  $\Rightarrow$  curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on  $4\mathbb{Z}$  (v264/v288), L5 full-operator  $[\rho, \Lambda]=0$  (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- $L^2$  QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk  $\Rightarrow$  single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

## 2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286\_seam\_equivalence\_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ $E_8$  into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible  $\Rightarrow \mu=1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary* (the AQFT form of  $\det K=1$ ). 4/5 discharged.
- **v288\_full\_l2\_subprincipal\_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ , so  $\Lambda = |n| + M_f$  commutes with  $\rho = \text{diag}(i^n)$  on the full operator,  $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$ ; generic curvature  $\rightarrow 4.44$ ), lifting v201/v284 to  $L^2$ . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

## 2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284\_route\_i\_rp\_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP  $\rightarrow$  DtN (v194), L2 marks  $\rightarrow$  pillowcase (v195/v216/v214), L3 flat Steklov  $= \sqrt{-\Delta_{\text{flat}}}$  (verified here), L4 covariance from DtN (v258), L5  $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov ( $\chi_{\text{orb}}=0 \Rightarrow$  unique flat

$\tau=i$ ) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.

- **v285\_route\_ii\_seam\_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E]  $3/4$  discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$ ; the tower  $16 \rightarrow 4 \rightarrow 1$ ), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the  $(E_8)_1$ -at- $\tau=i$  object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$ . Build green, AUDIT OK.

## 2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280\_pillowcase\_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat  $\tau=i$  pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281\_holomorphy\_modular\_data.py** (QGAMB.MODULAR.01): #anyons =  $|\det \text{Gram}|$  ( $E_8 \rightarrow 1$ ,  $SO(16) \rightarrow 4$ ), the anyon-condensation tower  $16 \rightarrow 4 \rightarrow 1$ , Gauss–Milgram  $\rightarrow c=8$ ; holomorphy ( $\det K=1$ ) is the single Tier-B discriminator.
- **v282\_e8\_tau\_i\_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* —  $\chi_{E_8}=j^{1/3}$  gives  $\chi_{E_8}(i)=1728^{1/3}=12$ , so  $\tau=i$  is both the order-4 CM point (QGEO) and the  $(E_8)_1$  modulus (QG.AMB): the two obligations are two faces of one object (the seam is  $(E_8)_1$  at  $\tau=i$ ), **obligation count**  $2 \rightarrow 1$ .
- **v283\_live\_killtest\_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$  floor 0.0586 eV; proton decay  $1.55 \times 10^{35}$  yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

## 2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.04)

- **v279\_qgeo\_obligation\_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat  $\tau=i$  pillowcase metric* (Trojanov-unique,  $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$ ), then  $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow \text{mark-locality} \rightarrow E_8$ . [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal,  $\rho$  commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.04: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

## 2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278\_lszy\_bridge\_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG  $T$ -products of  $S_{\text{pert}}$  (v269) are the OS Wick-rotated correlators

(v240), so amputate+on-shell-residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ; [C] the gap  $\Delta=6\log(3/2)>0$  gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity  $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$  *exactly* = the two-body phase space, so  $2\text{Im } M = \sum \int d\Pi |M|^2$  (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt\_F) is non-unitary  $\rightarrow$  QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

## 2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277\_seam\_calderon\_e8\_match.py** (QGAMB.TIERB.01) reduces Tier-B to ONE bit. [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$ ,  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248 = \dim E_8$  weight-1 currents (240 roots). [E] the discriminator is holomorphy — the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (non-holomorphic), so the one condition is  $|\det K|=1$ . [C] the seam side matches via v276 (flat  $\tau=i$ ) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

## 2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276\_qgeo\_flat\_closes\_commutator.py** (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate. The state-level residual the whole bedrock rests on is the operator equation  $[\rho, H]=0$ . [E] if  $H = f(\Delta_{\text{flat}})$  and the order-4 deck  $\rho (z \mapsto iz)$  is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (verified on the discretised square torus:  $\rho^4=I$ ,  $[\rho, \Delta]=0$ ,  $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full  $H$ . [C]  $\tau=i$  is forced by order-4 ( $j=1728$ ; hexagonal  $j=0$  is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat  $\tau=i$  pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

## 2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt\_F)

- **redteam/rt\_F\_qft4d.py** (REDTEAM.F.01) stress-tests the just-published round (the perturbative  $S_{\text{pert}}$  layer, the PMNS Jarlskog, the  $\nu$ -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity  $R^2/\text{Weyl}^2$  sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator  $1/(p^2(p^2 + M^2))$  has a negative-residue massive spin-2 mode), so  $S_{\text{pert}}$  is unitary for the matter+gauge sector only — which is exactly why the non-perturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the  $\Lambda$ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

## 2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser  $S_{\text{pert}}$  skeleton v269, the

concrete one-loop quartic v271 and gauge  $\beta$ -coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute  $\nu$ -mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, run\_all.py green, AUDIT OK, npm build clean. The Zenodo description (zenodo\_description.html) and the public website were synced in the same release.

## 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272\_nu\_mass\_scale.py (FLAV.NUSCALE.01) types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{EW})^2/M_R$  (one UV input, like  $v_{geo}$ , v153); [C] the observed  $m_3=0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (v249) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; [O]  $M_R$  is not a clean  $\bar{M}_{Pl}$  power, so the absolute value stays open; [X] the NO floor  $\Sigma m_\nu=0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **v273\_eg\_gauge\_running.py (QFT4D.SPERT.03) derives the SM one-loop  $b_i$  via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- **v274\_scale\_overdetermination.py (SCALE.OVERDET.01) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to  $\bar{M}_{Pl}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda/\bar{M}_{Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV) =  $2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- **v275\_qgamb\_roadmap.py (QGAMB.ROADMAP.01) gives the open gate QG.AMB.01 a module.** [E] Tier A gap decoupling ( $\Delta_{eff} = 1.648 > 0$ , v36/v76); [E]/[C] Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, run\_all green, AUDIT OK, build green.

## 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **v271\_eg\_oneloop\_quartic.py (QFT4D.SPERT.02) takes the v269  $S_{pert}$  skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $sd=2$  in  $d=4$ , the one-loop bubble has  $sd=4=d$ , so  $\omega=sd-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(UV) = 1/(16\pi^2)$ . [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270\_pmns\_jarlskog\_assembly.py (FLAV.PMNS.03) assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\delta=4\pi/3$ ) **into one exactly-unitary**

**complex PMNS matrix and derives the leptonic CP strength.** [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a *derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\text{max}}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6$ /triality phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py (QFT4D.SPERT.01) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\dim \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py (FLAV.TH13.01) closes the  $\theta_{13}$  exponent origin.** The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py generates verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv/script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py (QGEO.SYM.02) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”



( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio  $2 \Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

**2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed,  $+(15, 1, 1)$  survives, Hyper-K decisive, v266)**

- **v266\_ps\_threshold\_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the v265 fork. The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed  $(15, 1, 1)$  [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be*  $+(15, 1, 1)$ ), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

**2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)**

- **v265\_qft4d\_fork\_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

**2026-06-18 (Three open-residual advances: the  $F_{QCD}$  solver,  $M_\nu$  from  $D_F$ , and the raw-seam  $\Rightarrow$  mark-locality reduction, v262–v264)**

- **v262\_fqcd\_mp\_me.py** (FR.MPME.02) implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z)=0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py** (FLAV.PMNS.02) makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric;

[C] it reproduces  $\theta_{23}=45^\circ$  and  $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13}=0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264\_raw\_seam\_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho=\omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] AUDIT OK, build green.

**2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)**

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

**2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)**

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

**2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)**

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian

$H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum); [E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [K_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .

- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).
- **v260\_k3\_kummer\_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes =  $\dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated \subsection\* entries, itemize items, and the inline markup \textbf/\emph/\texttt/\vref, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx + website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

**2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)**

- **v257\_quasiorient\_modpd.py (PS.NCG.ORIENT.02) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

**2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)**

- **v254\_inner\_fluctuations.py (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation.** The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)/\mathbf{126}$ . [E] the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E]  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . [C]  $\sigma = \text{scalaron}$ .
- **v255\_spectral\_action\_expansion.py (PS.SPECACT.01) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : [E] structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); [E]  $b/a^2 = 1/3$  (top-dominated); [E]  $\sin^2 \theta_W = 3/8$ ; [E] the  $R^2$  spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; [C]  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; [O] exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py (PS.NCG.ORIENT.01) — honest status, no faked proof.** KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$  to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; [E] the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

**2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and  $\kappa$  pinned to  $O(1)$ , v252–v253)**

- **v252\_full\_finite\_triple.py (PS.DIRAC.02) advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ( $J^2 = +1$ ,  $JD = DJ$ ), grading ( $\gamma^2 = 1$ ,  $[\gamma, a] = 0$ ,  $J\gamma = -\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$ . [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.

- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) **discharges residual 6(b) of v249.**  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the **126**-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

**2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$  — the CCvS Pati–Salam mechanism, v251)**

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) **advances the Dirac contract v250 from a contract to an exhibited mechanism.** On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2=+1$ ,  $\gamma^2=1$ ,  $D\gamma=-\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

**2026-06-17 (Carrier-native Pati–Salam program:  $E_8$  forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)**

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247\_e8\_branching\_no126.py** (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\bar{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one  $45 \Rightarrow$  proton-marginal). **v248\_ps\_algebra.py** (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). **v249\_ps\_unification.py** (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . **v250\_dirac\_triple\_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

**2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)**

- **v246\_unification\_data\_crosscheck.py** (QFT4D.RGTEST.01) — **the first hard data confrontation of the spectral-action prediction.** TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft\_skeleton.pdf**, **figures/qft\_unification.pdf**) added via **make\_figures.py** (PDF + website PNG) and embedded in **tfpt\_research\_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

**2026-06-17 (Spectral-action matter content: the  $SO(10)$  16 = one anomaly-free SM generation, v245)**

- **v245\_spectral\_matter\_content.py** (QFT4D.MATTER.01) **discharges the computable core of the 4d contract v244.** The carrier half-spinor (the  $SO(10)$  16) is verified, in exact rational arithmetic, to be one Standard-Model generation:  $\mathbf{16} = SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of **CONTRACT.QFT4D.01** is now [E]; only the seam→Dirac realisation and the run-down couplings stay [O]. Wired into **run\_all.py**, the ledger, the registry, cited in **tfpt\_research\_contracts.tex**, mirrored to the website.

**2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)**

- **Six emergent-QFT skeleton scripts (v238).** **v239\_kms\_thermal\_time.py** (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. **v240\_gns\_os\_reconstruction.py** (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the v64 mass gap). **v241\_dhr\_sectors.py** (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237). **v242\_spin\_statistics\_modular.py** (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . **v243\_haag\_ruelle\_braiding.py** (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. **v244\_spectral\_action\_contract.py** (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into **run\_all.py**, the ledger and registry, cited in **tfpt\_research\_contracts.tex**, mirrored to the website.

**2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)**

- **New v238\_modular\_lindblad\_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). [E] keystone: the dissipative gap  $= -\log((2/3)^6) = 6 \log(3/2)$  = the OS mass gap  $\Delta$  (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into **run\_all.py**, **status\_ledger.csv**, **script\_registry.csv** and cited in **tfpt\_research\_contracts.tex**; mirrored to **website/public/verification**.

2026-06-17 (Experiments: black-hole-cosmology signatures, the  $\eta_B$  Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the problem\_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ( $w_{\text{in}}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{\text{in}} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (**alternative\_group w\_de\_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai  $Q_{\text{geom}}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a [C] structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES  $\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale (experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); tfpt\_4\_frontier Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (**internal\_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** experiments/evidence\_scorecard.json now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website EXPERIMENTS\_AUDIT mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone experiments/ surfaces (not in the verification suite or ledger); the tfpt\_4\_frontier/tfpt\_horizon\_readouts and papers.ts/predictions.ts edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to tfpt\_5\_redteam (rt\_A\_e8net.py, redteam\_table.txt, redteam/README.md, infrastructure).** The Python red-team surface still emitted the

historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level (v143) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , v148), while the  $q(A_3)$  normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays  $[C]/[O]$  (reduced, not closed).

- **Shadow export now mirrors figures/ but not website/ (scripts/export-shadow.sh, verification/make\_script\_index.py, infrastructure).** export-shadow.sh ships figures/\*.pdf (so make\_manifest.py --check passes literally on the exported tree) while still excluding website/ and the compiled paper PDFs; the SHADOW\_MIRROR.md table is corrected accordingly. make\_script\_index.py now detects a missing website/ and skips its ScriptIndex.tsx mirror, so build.sh notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet (v231\_cp\_mu6\_phases, tfpt\_5\_redteam,  $[E]/[C]$ ).** Sharper than v220/v225 (which only *located* the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}}^{\text{leading}} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the v88 misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays  $[C]$  (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity (v232\_e8\_kleinian\_seam, origin\_theory,  $[E]/[C]$ ).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise QGEO.REALIZE.01 a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves = rank  $E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; TOP0.E8.01, bedrock stays  $[O]$ ).
- **CP is the universal family/triality phase, only sheet-split (v233\_cp\_triality\_phase, tfpt\_5\_redteam,  $[E]/[C]$ ).** One step past v231 in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (v234\_seam\_holomorphy\_selection, tfpt\_research\_contracts,  $[E]/[O]$ ).** The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; v232), (iii) exactly one 1-dim irrep (v219). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ( $[O]$ ):  $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$



(v160)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (v235\_seam\_chern\_simons, tfpt\_research\_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\#\text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic  $= E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).
- **The (2, 3, 5) Brieskorn singularity is the one generator of the skeleton** (v236\_brieskorn\_capstone, origin\_theory, [E]). Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms ( $|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE** (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2, 3, 5\}$**  (v219\_icosahedral\_mckay, origin\_theory, [E]).  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built from the group: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1, 2, 3, 4, 5, 6, 10\}$  — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep is the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222\_cm\_norm\_duality, origin\_theory, [E]). The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index is the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3, 2) split generates (5, 6, 7, 13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle** (v223\_coxeter\_totative\_clock, origin\_theory, [E]).  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h)=\text{rank}$  and  $h=2 \cdot 3 \cdot 5$  (COX.CLOCK.01).

- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120=|R^+(E_8)|$  (magnitude channel), degrees sum  $128=\text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D=4=|\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0=5=g_{\text{car}}$ ,  $\text{rank } H_1=3=N_{\text{fam}}$ ,  $h^0+H_1=8=\text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGEO.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+=16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]).** The seam deck is the square modulus  $j=1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j=0$  ( $\mathbb{Z}/6$ ,  $\arg \rho=\pi/3$ ). The frozen  $\delta=\pi/3+3\lambda_C^2=68.654^\circ$  has leading term  $\pi/3=\arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame  $(d, n)$  with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).**  $d=a^\top R^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}$  Nariai;  $n=(5, -9, 6)$  reads  $\det$  on the first-generation column;  $\det(\mathbf{1}, d, n)=21=3 \cdot 7$ ;  $\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- **$F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).**  $M(s, t)=R+Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source→pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).**  $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau=\frac{8}{3}=|\mathbb{Z}_2|c_\mu$ , product  $\frac{32}{9}$ ;  $A=\mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).**  $C=R+Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$  (hex),  $13=N_{\mathbb{Z}[i]}(3+2i)$  (square),  $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C+xU) = 14+6x$  (slope  $6=|R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C+yV) = 14+14y+4y^2$  (2nd difference  $8=\text{rank } E_8$ ); the anchor block has  $\det B(C+xU) = 2 \det(C+xU)$  and sheet curvature  $6=|R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28=2 \dim G_2$ ,  $52=\dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*,

not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6)$ ,  $(2, 4, 4)$ ,  $(3, 3, 3)$ ,  $(2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling `v217_marks_emergence_scan` ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension  $\rightarrow 248/248$ ); v217 numerical (Python-only). Suite  $\rightarrow 217$  modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ ,  $\text{Aut} = \mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via v214/v54/v56). Suite  $\rightarrow 215$  modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (v168)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ ; all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$ )

on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values mpmath-numerical. Suite  $\rightarrow$  214 modules.

## 2026-06-15 (the $F_{\text{transfer}}$ functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (v213\_ftransfer\_functor, tfpt\_research\_contracts, [C]). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F) = (1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1 = 11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3 = -7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

## 2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, [C]). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]**  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuplet branch** (v212\_leptogenesis\_decouple, tfpt\_4\_frontier, [C]). Both Boltzmann inputs share the decuplet  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184’s  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, [O]). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho : z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents  $= \text{Spec}(Q_+)$ , rank  $3 = N_{\text{fam}}$ . The MODULE *parity* is also proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^* \omega_1 = \omega_3$ ,  $\sigma^* \omega_2 = \omega_2$ ,  $\sigma^* \omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in AuditContract.lean. The geometric COHO.M/MODULE-parity nodes are now **[E]**; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay **[O]**.

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (Möbius Uniformisation.lean, ledger FORM.QGEO.02, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the clock  $\rho : z \mapsto iz$  has  $\rho^4 = \text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma\rho\sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1, -1, swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in AuditContract.lean. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.

## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (SeamDeckClosure.lean, ledger FORM.QGEO.01, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no sorry/admit, no domain axioms — #print axioms = propext, Classical.choice, Quot.sound only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) markLocal\_blockDiagonal: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed structure SeamDeckPremise (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as CalderonProjector encodes the Paper-1 analytic input; (v) SeamDeckPremise.clock\_invariant: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (v210\_mark\_local\_dtn, tfpt\_research\_contracts, [O]). A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (v204\_muon\_seam\_g2, tfpt\_4\_frontier, [C]). The carrier’s second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.
- **An independent gravitational  $3/4$**  (v205\_xi\_threequarter, tfpt\_4\_frontier, [E]/[C]).  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$ ) — the same

$3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.

- **Quantitative  $H_0$  from the  $\Lambda$  branch** (v206\_h0\_lambda\_branch, tfpt\_4\_frontier, [C]).  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1/3}} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route** (v207\_asymptotic\_safety, tfpt\_4\_frontier, [O]). A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam** (v208\_bh\_thermodynamics, tfpt\_horizon\_readouts, [E]/[C]). Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point** (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]). No singularity ( $\varphi \rightarrow \varphi_\star$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

#### 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure** (tfpt\_3, [O]). Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

#### 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to  $c_3$**  (tfpt\_4\_frontier, [C]). Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\hat{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

#### 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01** (v203, [E]/[C]/[X]). The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test** [X]: the residual intercept

must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_{\pm}$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62  $\text{BR}(K^+)$  outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II  $\text{BR}(K_L)$  off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, v198); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \pmod{4}]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.
- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py)**. A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185

docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py** [E]/[O] (claim QGEO.STATE.01, work package A). Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200\_seam\_variational\_scan.py** [C]/[O] (claim QGEO.VARI.02, work package B). The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i$  = the (1, 2, 3) grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198\_modular\_commutator\_reduction.py** [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- **v195\_marks\_lefschetz\_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So



MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.

- **v196\_seam\_energy\_functional.py** [E]/[O] (**claim QGEO.VARI.01**). The failure functional  $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197\_rr\_carrier\_clifford\_d5.py** [E]/[C] (**claim ARCH.RRCAR.02**). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (**claim QGEO.DTN.01**). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^*w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (**claim QGEO.ENERGY.02**). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the (1, 2, 3) grading on  $H^1$  (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7)**. `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

**2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)**

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side  $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by  $\text{rank}(D_5) = 5$ . The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known  $2/3$  ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192\_qgeo\_energy\_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

**2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)**

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F\_transfer-firewall round (v184–v188): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

**2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)**

- **v184\_etaB\_anchored\_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_\Lambda$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_\Lambda = 10$  atom, in the strong-washout window) [C]; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{\text{P1}}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so  $M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **v185\_axion\_relic\_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive.

Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.

- **v187\_ftransfer\_laws.py** [E] (**claim FR.TRANSFER.01**). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be [E], the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188\_frontier\_wording\_guard.py** [E] (**claim AUDIT.WORDING.01**). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  v184 [C]; the old “open relic scale” dark-matter line  $\rightarrow$  v185; the old “near, not exact” Koide line  $\rightarrow$  v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).
- *Review framing points (P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183\_koide\_f\_corner\_transfer.py** [E]/[C] (**claim FR.KOIDE.07**). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T Ra = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (v80). Hard negative control:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182\_reviewer\_residual\_map.py** [E]/[C] (**claim REVIEW.MAP.01**). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count v174 + holographic

reduction), B ( $\equiv$  QGEO.SYM.01), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator  $= F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.

- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , JUNO;  $r \approx 0.004$ , CMB-S4) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

**2026-06-14 (external-review response: CSV fix, G\_net 3-level, QGEO.SYM.01 page, claim detox)**

- **Ledger CSV bug fixed + guarded.** Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

**2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)**

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284.

This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual\_chain, script\_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make\_figures.py; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py [E]/[O] (claim QGEO.SYM.01).** The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→...→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at QGEO.SYM.01.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180\_clock\_is\_mobius.py [E]/[O] (claim QGEO.ISO.01).** Cracks QGEO.CONF.01 one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence QGEO.CONF.01 is *implied* by the milder premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **v179\_conformal\_realisation.py [E]/[O] (claim QGEO.CONF.01).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise QGEO.CONF.01: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **v178\_marks\_kernel\_attempt.py [E]/[O] (claim QGEO.REDUCE.01).** An honest attempt at QGEO.MARKS.01 and QGEO.KERNEL.01. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee–Uhlmann/v156; no drift v158).

Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM [E]` — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168’s cross-ratio to the full uniformisation); `COHOM [E]` —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE [E]` — a *unique*  $\mu_4$ -equivariant iso (multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); `NET [E]` (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar’s boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and

the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175\_net\_existence\_full\_cone.py** [E]/[O] (claim QFT.NETRP.01). The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$* : the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every  $m$*  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2)  $A_2$  *assembled + verified*: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 v156 check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net**: net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in **tfpt\_research\_contracts** (premise (A) closure) and **tfpt\_5\_redteam** (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174\_seam\_fock\_readings.py** [E] (claim QFT.FOCK.01). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in **tfpt\_2** (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170\_diamond\_slice\_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in **tfpt\_3** (slice atlas). Exact; reading is audit (anti-numerology).



- **v171\_os\_moment\_cluster.py** [E]/[C] (**claim QFT.OSMOMENT.01**). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in **tfpt\_research\_contracts** (G5).
- **v172\_trace\_anomaly\_seed.py** [E] (**claim SEED.ANOMALY.01**). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2 / \dim(D_5)$  factorisation. Cited in **tfpt\_2** (seed).
- **v173\_pfaffian\_cp\_car.py** [E] (**claim FLAV.STRONGCP.02**). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in **tfpt\_2** (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Also fixed **build.sh zenodo** on macOS bash 3.2 (empty-array expansion under **set -u**).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (**claims QGEO.RIGID.01, QGEO.REALIZE.01**). Hardens the *finite* half of the one remaining Tier-1 hinge **GATE.QGEO** (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (**QGEO.RIGID.01**, [E]). The seam-collar realisation stays the one premise (**QGEO.REALIZE.01**, [C]). Mirrored in the Wolfram extension (226/226). Cited in **tfpt\_research\_contracts** (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (**claim FR.ETAB.02**). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron 28/79), fed by TFPT’s normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in **tfpt\_4\_frontier** (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166\_higgs\_free\_seam.py** [E]/[C] (**claim HIGGS.FREESEAM.01**). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in **tfpt\_4\_frontier**.
- PyR@TE-sourced: the reproducer’s SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (**verification/pyrate/sm\_higgs\_betas.txt**). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163\_rg\_stability\_flavor.py** [E]/[C] (**claim FLAV.RGSTAB.01**). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in **tfpt\_2** (solar/reactor angle box).
- **v164\_qcd\_scale.py** [E]/[O] (**claim QCD.LAMBDA.01**). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in **tfpt\_4\_frontier**.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ ’s, **verification/pyrate/sm\_yukawa\_betas.txt**). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module v159\_pyrate\_gauge\_crosscheck.py** [E]/[C] (**claim EM.BUDGET.02**). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (**arXiv:2007.12700**, **github.com/LSartore/pyrate**) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,  $\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5} Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of **tfpt\_2**. Cited there with **\veri**; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.
- **PyR@TE reproducer (not vendored)**. **verification/pyrate/reproduce.sh** clones PyR@TE on demand into **experiments/pyrate** (git-ignored), neutralises its interactive

pdf<sub>latex</sub> end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

## 2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F\_transfer + irreducible core, v167)

- **New module v167\_inventory\_post\_closure.py** [E]/[C]. The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (GATE.METRIC.16–19) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + GATE.QGEO (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1=N_{\text{fam}}=3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (v152) and the ambient measure into  $G_{\text{net}}$  (v76). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger SYNTH.INVENTORY.03). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

## 2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module v165\_premise\_a\_closure.py** [C]/[O]. Pins the two residual gates of the v160–v162 (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (v125),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $\text{DtN} = |k|$  (v156/v157).  $R_1$  (**seam clock**) **reduces to GATE.QGEO** [C]: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , v115/v116), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary =  $\mathbb{P} \setminus \mu_4$ ” = GATE.QGEO (cohomology established, v137,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger GATE.METRIC.19). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

## 2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module v162\_seam\_transport\_identification.py** [E]/[O]. Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced:** a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,

$\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (v115) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (v137).  $(\alpha)$  is the rank-8 Calderón polarisation of  $\omega_k=|k|$  (v113/v156). **Closure:** with v160+v161, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, HOR.CLOCK). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger GATE.METRIC.18). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)**

- **New module v161\_one\_particle\_reduction.py [E]/[O].** *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger GATE.METRIC.17). Python-only (numpy + stdlib).
- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)**

- **New module v160\_seam\_gaussianity\_from\_pf.py [E]/[O].** Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment  $\leftrightarrow$  cumulant inversion on a concrete pure Majorana covariance (the v113 “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only  $\text{gap} > 0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual:** v56’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the

readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tftpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

### 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

### 2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral  $c=8$  fixed point is provably stable (v158, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (v157, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee-Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel-Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual:* exact given

(i) the free-bulk premise ( $\text{DtN} = |k|$ , v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the  $\text{DtN} = |k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **$G_{\text{net}}$ 's last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression *PTP* of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **$v_{\text{geo}}$  re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger ANCHOR.VGEO.02).
- **$G_{\text{net}}$  stated as the central closing theorem (v154).** The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger GATE.METRIC.11).
- **Status surfaces re-typed (review):** the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  ( $[\mathbf{E}]/[\mathbf{C}]$ ), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/origin\_theory/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (v68). So  $v_{\text{geo}}$  (v78),  $1/G$  (v68) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (rev `{git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`).  $\sim 270$  files /  $\sim 3$  MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** origin\_theory from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class `[O]`/`[E]`.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. SEAM.THEOREM.01 narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays `[O]` (ledger SEAM.EHMODEL.02).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).



- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2-1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m/12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0$  = the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$  ( $4+5=9$ ,  $\sigma \perp w_0(a)$ ) and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line (=  $T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p_2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue,  $2/3$  motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes  $[E]/[C]/[O]$  instead of the legacy  $[A]/[I]/[L]/[B]/[P]/[R]$  badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to  $[E]/[C]/[O]/[X]$  (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{net}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  error was in fact a correct-but-treacherous notation switch (see above).

## 2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+=1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q_+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading =  $Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion =  $\mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_b r_c = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays  $[C]$  with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its  $[O]$  typing untouched.

- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

## 2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$  rows are made consistent with `v86/COSMO.NSTAR.01`:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_\star \in [50, 60]$  with the scalaron-reheating conditional point  $N_\star=51.4$  ( $n_s=0.9611, r=0.0045$ ), replacing the stale point values  $N_\star=55$  (predictions table) and  $N_\star \simeq 57$  (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger `[P]` typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.2 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents”

wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record  $0.306747$  with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with  $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt}_N$ ). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is

now \input by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue  $Q$ -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\det'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+)$  = the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the R4' residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the `GATE.QGEO` residue collapses to one  $\mathbb{Z}_3$  choice.

- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

## 2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank =  $c$ ) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- $U_{\text{wall}}$  **made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  (diag =  $a/|\mu_4|$ , off-weights  $(8, 0, 5)/144$ ); resonance theorem ( $M_\infty = \mathbf{1} \Leftrightarrow a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad  $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ ; the 121 audit lemma; the lepton words are the compiler atoms  $(8, 5, 3)$  with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and  $\det 8$  — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).



## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.